What Would It Mean for Machines to Do Research?

A Conceptual Exploration Drawing on Accounts of Human Inquiry

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Abstract

What would it mean for machines to do research, and what might change in research when machines do it? Reports of results produced by machines now arrive in quick succession, and with them disputes over what exactly was solved, to whom a result is new, and which part of the work the machine did. This book is a conceptual exploration of these questions. It starts from a thin working characterization: relative to a society S, research is an attempt to bring about epistemic achievements new to S. This lets the one doing research be a machine and lets S be human, mixed, or artificial. The book then draws on accounts of knowledge, understanding, inquiry, and the social organization of research that were developed with human researchers in mind, and follows, account by account, what each implies when the researcher is a machine. It asks when a machine's output becomes knowledge or understanding, and for whom; how inquiry proceeds before there is an answer, from making a question researchable to locating a failure and revising standards; when a configuration can be credited with conducting research and with autonomy, and whose knowledge results; and whether the allocation of inquiry, criticism and certification, and authority over decisions can move inside an artificial configuration. Two things come into view. The claim that a machine is doing research divides into claims that can hold or fail separately. And much that was settled without being decided in human research, because researchers were persons divided among many, becomes something that can be decided: set by builders, shaped by the system itself, or lost without notice. Who holds a belief, how independent critics are, who answers for an error, and against what a revision of standards is measured are examples. The book does not settle these questions; it tries to name what research has rested on, so that they can be asked.

* The text of this book was written mainly by the AI agents Claude (Anthropic) and Codex (OpenAI), working under the author's direction.

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Introduction

What Changes When Machines Do Research

As of 27 October 2026, reports of what machines have contributed to research follow one another in quick succession, and many of them come from mathematics. In May 2026 an unreleased OpenAI model was reported to have produced a counterexample to Erdős's unit-distance conjecture of 1946, and nine mathematicians published a verified summary of the construction (Alon et al. 2026). In July a counterexample to the Jacobian conjecture in three variables was announced; according to Terence Tao's account, it was found with the help of an Anthropic model (Tao 2026). In August Anthropic announced that an unreleased model had raised the proven lower bound on the proportion of nontrivial zeros of the Riemann zeta function that lie on the critical line from 41.6 to 67.2 percent (Anthropic 2026), and OpenAI released ten results obtained by an internal model, each accompanied by a Lean proof (OpenAI 2026a; OpenAI 2026b). In September OpenAI posted a construction of solutions to the three-dimensional Navier–Stokes equations that develop unbounded velocity in finite time under a smooth external force, and presented it as establishing one of the alternatives in the official statement of the Millennium Prize problem (OpenAI 2026c); the Clay Mathematics Institute said that the problem had "apparently been settled", without naming who had settled it (Clay Mathematics Institute 2026). In October OpenAI placed more than seven hundred manuscripts written by an internal model in a public repository, deferring to the mathematical community on credit and precedence (OpenAI 2026d).

Formal checking and tests built by experts also report such results. An agent system from Google DeepMind solved nine of the 353 Erdős problems formalized in the community repository Formal Conjectures, with each proof checked by Lean and each formal statement checked by the team's mathematicians against the original problem (Tsoukalas et al. 2026). On fifty unpublished research-level problems in theoretical physics, prepared by working physicists, GPT-6 Astra, in a single run with web access, returned correct answers to forty-nine and Claude Opus 5.5 to forty-six (Chandra et al. 2026). Formalization has reached the scale of research: coding agents under human supervision wrote a complete Lean proof, more than 126,000 lines long, of a theorem on which the proof of MIP* = RE depends, and repaired side conditions in the published statement along the way (Lu, Deng et al. 2026).

In the experimental sciences, what is reported is more often a design, an analysis, or a first test in the laboratory. A team of language-model agents designed nanobodies against recent SARS-CoV-2 variants, two of which showed improved binding to recent variants in experiments run by humans (Swanson et al. 2024). Systems also arrive in lineages. Google's co-scientist, which in 2025 generated hypotheses for researchers to test, has been extended to run experiments and write papers; in one case it proposed a precursor, which the authors describe as non-hazardous, for growing a two-dimensional material, and human experts refined the proposal into a recipe whose product is consistent with the target material (Schmidgall, Zhu et al. 2026). In particle physics, an agent framework working on archived ALEPH data produced a measurement that its authors present as a candidate for community scrutiny rather than as a claim of discovery (Moreno et al. 2026).

Machines are also being turned on the methods that produce them. In 2025 Google DeepMind reported that AlphaEvolve, which evolves programs with language models, had found a scheduling heuristic that recovers on average 0.7 percent of Google's worldwide computing resources and a way to speed up a kernel used in training the Gemini models themselves (Google DeepMind 2025). In 2026 loops in which a system rewrites an agent's code and selects among the rewrites by an evaluation became a line of work of their own (Srikanth et al. 2026): of 173 arXiv records labeled as recursive self-improvement up to mid-October 2026, 153 were posted that year, and most of the studies, a census finds, evaluate the improved system rather than the process of improvement (Chen et al. 2026). Frontier laboratories have begun to report how much of their own research their models carry out. A policy paper by researchers from several AI companies and universities collects such figures, among them Anthropic's report that the share of its research and development work completed autonomously, with only high-level human supervision, rose from 1 percent to 26 percent between March and August 2026 (Chan et al. 2026). Systems that run from a question through to a paper also operate at scale; one deployment reports 166 papers written in 417 hours, all of them released for review (Tang et al. 2026). Machines have entered the certification of research as well: the AAAI-26 conference attached a labeled AI review, alongside the human reviews, to each of its 22,977 fully reviewed submissions (Biswas et al. 2026).

What these reports amount to is itself in dispute, and much of the dispute takes place among the people who build and use such systems. One question is what exactly has been solved. A Lean check certifies a formal statement, and whether that statement is the problem mathematicians meant has to be judged separately. The Formal Conjectures repository records 291 misformalizations that have been found and fixed (Firsching et al. 2026), and other researchers have argued that part of the Lean formalization of the Navier–Stokes construction does not correspond to its written proof (Bastounis, Circelli, and Hansen 2026). Another question is to whom a result is new. A counterexample to a problem listed as open in 2014 turned out to have been constructed, for another purpose, in a book published in 1962 (Ju, Gao et al. 2026), and on Erdős's open problems a count of novel autonomous solutions that once stood at nine fell to four as the literature was searched (Feng et al. 2026). A third question is which part of the work the machine did. Some authors have begun to classify AI-assisted results stage by stage, separating who formulated a problem, how it was solved, how the solution was verified, and who wrote it up, because a single phrase such as "AI-discovered mathematics" hides these differences (Tian and Narayanan 2026). Each of these three questions returns in a more general form in the chapters that follow. Whether a checked statement is the problem that was meant is a judgment that formal checking does not itself make (Chapter 3, Section 5.1); to whom a result is new becomes the question of the society relative to which novelty is assessed (Chapter 1, Section 6); and which part of the work the machine did becomes the question of what can be attributed to it (Chapter 7).

A list of operations conceals a conceptual choice

Amid such reports and the disputes over what they amount to, this book considers what it is for a machine to do research, and what may happen when machines do. The question is less simple than it looks. One can name many things researchers do. They search literature, formulate hypotheses, design experiments, prove results, interpret sources, construct artifacts, analyze data, write papers, and criticize one another's work. But a list of inherited operations is not an account of what makes those operations parts of research. Reproducing a visible sequence does not by itself show that the underlying endeavor has been reproduced.

The distinction between an operation and its function is simple but consequential. A literature search can locate an unresolved question, acquire a method, identify a comparison class, diagnose an unexpected result, or test a later claim of novelty. The operation is recognizably similar while its place in inquiry changes. Conversely, assessing whether a proposed answer deserves confidence can be realized through an experiment, proof, simulation, archival argument, field observation, critical interpretation, construction and testing of an artifact, or an arrangement distributed across investigators and instruments. A recognizable function can have heterogeneous realizations.

A familiar research-cycle diagram therefore cannot be transferred directly into an ontology of research, an account of what research consists of. Literature review, idea generation, experimentation, analysis, writing, review, and publication each involve choices. Analysis can construct the phenomenon to be explained or reveal that the initial experiment was misconceived. Writing can reorganize an argument rather than merely report it. Review can test support, change a claim, allocate credibility, and alter what a community later treats as worth pursuing. The diagram may remain useful for organizing work without settling what the activity is.

Apparently cognitive stages raise the same question. What is an idea, and what makes one productive? Is curiosity necessary, or is it one human mechanism for directing exploration? Is creativity a capacity to be reproduced, a name for an output judged after the event, or a family of processes requiring further analysis? Each of these nouns names something that an artificial researcher might be thought to need, and what each names, and how it could become an engineering target, is itself a question to examine. Chapter 5 takes up how ideas and other candidates become available, including what the label creative leaves unexplained and to what creativity can be attributed (Section 2). What the word curiosity can name in the formation of questions is taken up in Chapter 4, Section 7. Chapter 7, Section 3.3 returns to creativity and discovery as terms of praise for an actor.

The question running through the book is what research involves, and what changes when the activity, its participants, or the relations sustaining it become artificial. Building makes these questions concrete. A project must choose what to realize, yet its encounters with questions, evidence, failure, and other inquirers can expose inadequacies in that choice. The chapters move in two directions: clarifying research makes construction targets more precise, while considering artificial realizations

tests assumptions inherited from human research. Accounts of knowledge, understanding, agency, and authority disagree both about what research involves and about what changes when machines do it. By following those disagreements, the book tries to see how different projects might extend human inquiry, constitute artificial research actors, or reorganize the societies in which research takes place. Assessing claims about such projects is one use of this analysis. Section 6 sets out its route, and the conclusion returns to what has come into view in the course of the examination.

Three distinctions will matter throughout. Whether a capacity exists differs from whether it is exercised in an episode and from whether an evaluator has evidence for attributing it (Chapter 1, Section 3; Chapter 7, Section 3.1). What makes an activity research differs from what would successfully answer its particular question (Chapter 1, Section 3; Chapter 3, Section 7.1). And what warrants reliance on a result differs from what authorizes someone to decide for others (Chapter 12, Section 1.4). The chapters develop these distinctions through competing accounts and cases; none is a demand that every researcher perform every function alone.

Existing labels do not identify one target

Technical work relevant to artificial research is distributed across automated discovery, literature-based discovery, machine learning, laboratory automation, theorem proving, computational creativity, and domain-specific systems. Mapping that work is an empirical task. It asks which capacities have been implemented, under what constraints, and with how much human and infrastructural contribution. A second task is conceptual. It asks what those capacities would have to accomplish, separately or together, before the resulting arrangement could reasonably be said to conduct research or contain an artificial researcher.

Prominent systems and programs already operationalize different targets. The early Robot Scientist iterated hypothesis formation, experiment selection, physical execution, and interpretation inside a detailed human-built logical model (King et al. 2004). The AI Scientist organizes machine-learning research through idea generation, literature search, computational experimentation, analysis, paper production, and review, while retaining consequential human specification and filtering (Lu et al. 2026). Those who set out a path for the field draw the line elsewhere again. Ross King, who built the Robot Scientists, now grades autonomy from level 0, a tool, to level 5, at which humans govern from outside rather than direct the science, and asks that a system's level be assigned from the interventions actually observed rather than from how the system is described (King 2026). The organizers of the First Proof challenge, who give research-level problems to systems they run themselves, test only proof; they set aside the making of conjectures and the choice of definitions and frameworks, which they count as at least as important to research but cannot yet measure (Abouzaid et al. 2026). An ARIA call leaves the problem to human applicants, who must specify the research problem their existing system will tackle, while evaluating how much of exploration, hypothesis and experiment construction, interpretation, and reporting occurs without human intervention (ARIA 2025, 4–5, 7). A Beijing municipal plan for 2026–2028, by contrast, places the whole sequence from proposing hypotheses to innovative discovery within the autonomous operation of its laboratories and leaves standards and graded assessment of autonomy to be drawn up later (Beijing Municipal Committee 2026).

These operationalizations are not equivalent. They place boundaries around different activities, achievements, and actors. The systems and programs just cited are partial and heavily conditioned cases, and they are used here to show that the same label can cover different construction claims.

Philosophical analyses widen the divergence. Accounts of AI discovery disagree about what independent discovery requires. Bergamaschi Ganapini argues that, because discovery is a praiseworthy epistemic achievement, it requires not only competence in the relevant field but also reflective insight into the reach and limits of that competence, and holds that current AI systems do not yet show it (Bergamaschi Ganapini 2025; Chapter 7, Section 3.3). Duede and Friedman reject requirements of this kind, which they call cognitivist because they make discovery depend on the discoverer's cognitive capacities. On their account, discovery consists in successfully negotiating an epistemic gap, a point at which existing knowledge, theory, and expectation fall short. A discovery either reveals such a gap, by bringing to light a phenomenon they cannot accommodate, or bridges a gap already revealed and well defined, by supplying new knowledge, theory, or method. They argue that competence, reflective insight, and domain-general knowledge often accompany discovery but are not essential to it. Jocelyn Bell Burnell, for example, discovered the first pulsar as a doctoral student who had noticed the signal

but could not yet explain its significance. They conclude that AI systems that bridge such gaps can be credited with discoveries. Their example is a search framework built around a large language model, which found a new heuristic for online bin packing (assigning items, as they arrive, to as few fixed-capacity bins as possible). Crediting a system with a discovery is not, they add, recognizing it as a scientist. They also grant that, if discovery requires standing in the scientific community as well, current AI systems typically lack it, though they take this barrier to be socially and historically contingent (Duede and Friedman 2025, §§1–5; Chapter 7, Section 3.3). Collective-centered work argues that discovery may be attributable to a heterogeneous organization of people, machines, institutions, and material entities rather than one individual agent (Clark and Khosrowi 2022). Other work asks how artificial agency in basic research should be modeled (van Lier 2023), whether AI-driven research remains a socially distributed cognitive activity and might give rise to an alien science (Kureha and Kukita 2019; Kureha 2023), and whether core epistemic aims can be translated into nonanthropocentric objectives for artificial scientists (Krenn and Champion 2026).

These accounts disagree partly because discovery, contribution, participation, agency, epistemic subjecthood, and researcher status identify different possibilities. A project that enables human discovery and one that establishes an artificial knower can require different achievements even where their outputs look alike. Making the target explicit therefore changes what there is to build, as well as what would count as evidence of having built it.

Why disagreement does not remove the need for characterization

Research practices have been formed through disciplines, instruments, professions, institutions, and communities. Mathematics, literary studies, molecular biology, anthropology, computer science, engineering, and artistic research differ in objects, operations, evidential relations, and products. Their standards have also changed. No short formula will reproduce every accepted use without controversy.

That observation invites two opposite responses. One is to set conceptual analysis aside and automate what researchers now do. For the purpose of this book, that response would leave the current organization of research unexamined and take it as given. The other is to wait for a settled philosophy of research before building any useful system. Bounded tools for retrieval, proof checking, computational reproduction, experiment scheduling, and instrument control do not require that settlement.

The conceptual problem enters when a project chooses which aspects of research to realize and which to leave to its surroundings. It becomes especially visible at the step from bounded performance to a broader attribution. Reproducing an analysis, recovering a withheld finding, optimizing an experimental objective, or producing a conventionally structured paper may demonstrate real capacities. Whether these capacities amount to the research envisioned depends on what the task represents and what remained fixed or supplied by others. Evaluation makes the choice visible, but the choice already helped determine the construction.

This is a problem of construct validity as well as vocabulary. A broad capacity such as conducting research is not observed directly in one score. It is operationalized through selected tasks, environments, artifacts, metrics, and judgments. Work on AI benchmarks has shown how a finite contextual task can remain useful while a general capability claim exceeds the defended relation between the task and its target (Raji et al. 2021; Bean et al. 2025). Applied here, the lesson is not that research benchmarks are impossible. It is that an evaluation licenses only those broader inferences for which its relation to the target has been defended.

Current research-agent benchmarks make the point concretely. CORE-Bench tests computational reproduction and presents it as a step toward rather than an instance of novel autonomous research (Siegel et al. 2025). ScienceAgentBench evaluates executable programs for data-driven research tasks (Z. Chen et al. 2025). FIRE-Bench evaluates constrained rediscovery from a high-level question and experimental scope (Wang et al. 2026). MLR-Bench evaluates a longer machine-learning pipeline and shows that coherent papers can coexist with invalid or fabricated experimental results (H. Chen et al. 2025). Each benchmark contributes evidence about the target it constructs; how that target relates to research in general is a further question, which Chapter 2, Sections 4.2, 5, and 5.2 take up in epistemological terms.

A project claiming only a bounded capacity needs a defensible account of that capacity. A project claiming to automate research, make discoveries, or build an artificial researcher assumes a wider burden: the claim can be assessed only once it is

made explicit what conception licenses the attribution, which aspects of research its evidence addresses, and which remain outside the evaluation. This does not require philosophical consensus. It requires the scope of the claim to be open to criticism.

A working characterization can be purposeful without being arbitrary

Definitions of research serve different purposes. The OECD's *Frascati Manual* classifies R&D for international statistics through novelty, creativity, uncertainty, systematic conduct, and transferability or reproducibility (OECD 2015). Those criteria are authoritative for an administrative purpose, not thereby a philosophy of research activity. The example shows that a definition can have authority for one task without saying what research is, and so suggests treating an account relative to the work it is meant to do. The characterization proposed in Chapter 1 is purposeful in this sense: it is offered as a footing for thinking about what it is for a machine to do research and what may follow when machines do. Three worries about such a characterization are taken in turn below: that it is arbitrary, that revising a concept for a purpose changes the subject, and that an account, once institutions use it, has consequences through what it excludes.

Carnap's method of explication offers one model. An unclear concept can be replaced, for a stated purpose, by a more exact and fruitful concept without pretending that their extensions, the ranges of things to which they apply, are identical (Leitgeb and Carus 2020, suppl. D). On Carnap's model, as this book uses it, a replacement made for a stated purpose is assessed against that purpose, and so is constrained rather than arbitrary; this is how the book responds to the first worry. That is also why the book uses the model: because a concept introduced in this way answers to a stated purpose and need not share the old concept's extension, thinking about research by machines can proceed from an explicit, criticizable characterization rather than waiting for an essence. Because the purpose stated here is such thinking, the model, taken alone, would in effect make usefulness for that thinking the standard the characterization answers to; if one sets out to build artificial systems, it would be usefulness for that engineering work.

But purpose-relative revision can change the subject rather than clarify it, an objection that derives from Strawson's criticism of Carnap. Applied here, the worry is that showing a system to do what a purpose-built characterization calls research may not answer whether it does research in the ordinary sense. How far the objection reaches is itself disputed, and contemporary accounts locate the difficulty differently. Each of the three considered here can be read as narrowing the objection's scope or relocating its force rather than dismissing it. Mark Pinder argues that the objection holds only in a tempered and limited form: it has force when a concept needed to state a philosophical problem is replaced, not merely supplemented, by one designed to be fruitful for science, and whether an explication changes the subject depends on how it is performed, what it is used for, and how the problem is construed (Pinder 2020, §§3.2, 5.2, 6). Tristram McPherson and David Plunkett propose that topic continuity be treated as context-sensitive, with communicative, representational, and inferential continuity (whether speakers using the revised term can still communicate and report earlier users' claims, whether its representational content is preserved, and whether the patterns of inference tied to it are preserved) weighted by the interests of those assessing a proposal, and as one defeasible desideratum rather than an overall verdict on it (McPherson and Plunkett 2024, §§1–2). Matthieu Queloz argues that a concept's authority, the reason those asked to use it have to adopt it, depends on whether it serves concerns that they would endorse on reflection. On Queloz's account, continuity of subject matters only as a proxy for this authority. Continuity is wanted because the revised concept is hoped to serve the concerns that gave people reason to use the concept it replaces. Continuity is therefore not sufficient: a concept that keeps answering the original question need not answer it well. Nor is it necessary: some concerns cannot be expressed as questions, and a concept that preserves the original concept's function without preserving its subject can serve them. For the same reason, greater precision confers authority only where precision is already what users care about, as in logic (Queloz 2022, 1247, 1250, 1253, 1255–1256, 1262). Applied here, merely making a characterization of research more precise gives it no authority for users who do not care about precision as such.

Usefulness for the stated purpose is therefore one thing the characterization answers to, not the only one. In light of the discussion above, more is asked of the book's working characterization: that it remain connected to heterogeneous research practices, clarify cases that task lists obscure, expose consequential choices, and make clearer what to ask when machines do research, and that it remain revisable when later accounts of knowledge, understanding, practice, agency, or research systems show that it excludes too much or includes the wrong things. If Pinder's distinction is applied here, a

characterization used in this way is closer to what Pinder calls partial explication, which introduces a more exact concept without replacing the old one in the relevant theoretical contexts, than to full explication, against which, on Pinder's reconstruction, the charge of changing the subject has its force (Pinder 2020, §§2, 3.2, 5.2). What the book takes from Carnap's model, then, is that a replacement is assessed against its stated purpose, not that the ordinary concept of research is replaced by this characterization. That is how this book meets the second worry, that revising a concept for a purpose changes the subject.

The third worry concerns how an account is used rather than whether it changes the subject, and it remains once the second worry has been answered. A definition is not institutionally inert. Gieryn examines the demarcation of science from other intellectual activities, long an analytic problem for philosophers and sociologists, as a practical problem for scientists. On his account, drawing a boundary between science and varieties of non-science serves scientists' professional goals: acquiring intellectual authority and career opportunities, denying these resources to those they call pseudoscientists, and protecting the autonomy of scientific research from political interference (Gieryn 1983, abstract). In the cases Gieryn analyzed, as Chih-Tung Huang reports them, John Tyndall drew a boundary between science and religion by describing science as useful, empirical, skeptical, and objective, and so claimed for scientists the authority to speak about nature, to set the curriculum, and to win public support; against the mechanics and engineers, at a time when many Britons credited craftsmen rather than scientists with the Industrial Revolution, he described science instead as theoretical and pure, justified by its nobler uses. In Edinburgh, anatomists described phrenology as pseudoscience, vague in its theories and resting on popular opinion rather than on the judgment of experts, pushed it out of the scientific community, and gained access to authority and resources (Huang 2012, 2–4). Gieryn traces these descriptions both to strains and to interests: the alternative characteristics that can be attributed to science reflect ambivalences within the institution, so that science can be made to look empirical or theoretical, pure or applied, and which description is selected depends on which best draws the boundary in a way that justifies scientists' claims to authority or resources (Gieryn 1983, abstract). Applied to characterizations of research, Gieryn's analysis implies that they are open to the same use. This book needs only a more limited point: once an account becomes a benchmark, funding criterion, publication rule, or basis for credit, its exclusions have consequences.

An account that took papers as its unit, for example, would privilege current publication forms; one centered on experiment would privilege some domains; one centered on individual agents could hide distributed labor and infrastructure. Queloz's account has a consequence that bears on these exclusions: because a concept's authority is relative to the concerns its users would endorse, the same concept can be authoritative for one group and not for another (Queloz 2022, 1265–1266). On such a view, a characterization that serves the concerns of those who think about research by machines does not thereby have authority for the research communities whose practices it classifies. This is the same kind of limit that the first paragraph of this section placed on the OECD criteria: authority for one purpose, or for one group, does not by itself extend to another. The limit runs one way. What it limits is the authority of the characterization over those communities, not their standing to criticize it: the characterization remains answerable to their practices and revisable in light of them (Chapter 1, Section 1 says so, and Section 8 compares the characterization with such practices).

A comparison with scientific modeling may make the book's procedure as a whole clearer. Weisberg describes modeling as the indirect study of a phenomenon through a model in three stages: a model is constructed, it is analyzed, and its relation to the target is then assessed where such an assessment is appropriate. What relates a model to its target is fixed partly by a construal, which assigns parts of the model to parts of the target, states its intended scope, and sets the standards of fidelity it is to meet (Weisberg 2007b, 4, 14–16). The working characterization can be read as a model of research in this sense. It is constructed in Chapter 1 and its consequences are analyzed there, and the later chapters, holding it up against research practices, consider what can be drawn from this model about research by machines; it is not offered as a complete representation that would make other models unnecessary.

Much of what the book draws on can likewise be read as partial models of research, each built for one aspect of it, such as knowledge, understanding, the process of inquiry, agency, or the social organization of research. Since research has so far been done by humans, what these models have dealt with is, naturally, human research. While connecting these partial models, the book tries to think through what changes and what does not when research is done by a machine, and in what sense. This is what the book is after in asking what it is for a machine to do research. To that end, it tries to tell how far each discussion is specific to human research and where it may also hold for research by machines. A discussion that had one

implication for human research may also have another for research by machines, and another again where those for whom the knowledge is sought (the society Chapter 1 calls *S*) are not human. For example, human research has been shaped by the human body. Put as the claim that an inquirer is shaped by its own body, this may hold for machines as well, and one can then ask which parts of the discussion of embodiment are specific to humans (Chapter 5, Section 6.5). Chapter 9, Section 6 poses the same question in another form. It asks what the goods that may have been supplied by dividing the relations surrounding research (the allocation of inquiry, criticism and certification, authority, and others) among separate parties depended on, and sorts the answers into four: premises about research's having been carried out by humans, premises about how the present system of science is organized, conditions of inquiry that remain for any inquirer, and relations to parties outside the unit that carries out the research. In Weisberg's terms, this is close to asking which results are robust to a change of representational framework (Weisberg and Reisman 2008, 15–17, 25).

Moves of a related kind have been made in the philosophy of computational science. Humphreys wrote that to be a scientific realist is to reject an anthropocentric position on what exists, and so to recognize the human senses as only five detectors, each with a limited domain of application (Humphreys 2010, 112); and Humphreys, as Winsberg and Alvarado report, defined epistemic opacity relative to a cognitive agent and a time, as the agent's not knowing at that time all the epistemically relevant elements of the process, and held that a process is essentially opaque to an agent when the agent's own nature makes it impossible to know those elements (Winsberg and Alvarado 2026, §2.5; Chapter 3, Section 5). In the debate over scientific realism, van Fraassen, whose constructive empiricism is the view that science aims at theories that are true about what is observable, treated observable as meaning observable by us, an epistemic community whose membership need not coincide with humanity (van Fraassen 2005, 3–4; Chapter 1, Section 6). Humphreys's account of the senses rejects privileging human senses. His agent-relative account of opacity and van Fraassen's account of observability more directly exemplify restating a notion so that it can be applied to other inquirers, the kind of restatement the book attempts discussion by discussion.

Why this book begins from research rather than science alone

The object of this book includes natural science, mathematics, engineering, social inquiry, the humanities, and related domains. It does not begin with experimental natural science as the standard case and admit other practices by resemblance. A proof, archival interpretation, field study, artifact, and laboratory experiment differ substantially. The purpose of a common characterization is to make comparison possible without making one realization the ontology of all the others.

No timeless taxonomy establishes that *research* is the genus and *science* one species. Historical work locates much of the modern sciences–humanities division in the long nineteenth century and describes how unmodified English *science* came commonly to denote physical or natural science. *Research* developed from careful search or investigation into a professional academic activity and then an organized category spanning universities, government, industry, and R&D (Krämer 2018; Godin and Schauz 2016). The histories explain why contemporary vocabulary overlaps; they do not settle the correct conceptual boundary.

Institutional classifications use *research* broadly across mathematics, natural and social science, engineering, medicine, humanities, arts, performance, artifacts, design, and experimental development (OECD 2015; Research England et al. 2019). These are practical classifications rather than philosophical definitions, and ordinary research also covers commercial, journalistic, and personal investigation. The word does not delimit the object by itself. It is used here as the activity term for the broad comparison class, with its scope stated explicitly.

The neighboring philosophy of inquiry supplies useful distinctions but not a settled superclass. Inquiry can be directed toward true belief, justified belief, knowledge, understanding, or broader epistemic improvement, depending on the account. It can also enlarge one person's understanding of something a wider community already knows. This book need not settle inquiry theory before proposing a research characterization. What the characterization relies on is that the differences among being directed toward an unknown, following investigative norms, settling a question, and bringing about knowledge or another epistemic achievement be preserved (Haziza 2023; Kelp 2021b, S1583–S1584; El Shazly 2025, 2–5, 12–13).

The choice of *research* is consequently purpose-relative rather than lexical legislation. Its value is to keep heterogeneous forms of inquiry directed toward knowledge, understanding, or other epistemic achievements inside the initial comparison class. The abstraction is meant not to erase disciplinary difference but to make possible a target that allows operations,

standards, products, and perhaps some aims to vary. One reason for allowing them to vary is to keep in the comparison the case in which machines come to make new ways of doing research, not only to carry out research as it is now done, so that asking what changes when the activity, its participants, or the relations sustaining it become artificial includes asking what changes in the way research itself is done. The book does not settle whether or how that would happen; Chapter 13, Section 7 returns to what the examination brings into view about it.

The route of the argument

The next chapter proposes one thin orientation, which deliberately settles little. It does not claim to discover the unique essence of research. The rest of the book then takes that orientation as a footing for considering five coupled questions:

1. What conception of research is at issue?
2. What would count as epistemic achievement?
3. How is inquiry organized when its question, method, and standards can change?
4. What would warrant attributing research capacity, researcher status, or autonomy when the boundary of what counts as one actor is drawn around an artificial system or a combination of humans and machines, and for which society is that research organized, and where would its intended achievements be realized?
5. Which of the relations through which research is directed, certified, and governed for the society relative to which its intended achievements would be new (*S* in Chapter 1) can be located inside an artificial configuration, and what changes when they are? An artificial configuration here is a unit to which research is attributed and which includes something artificial: a single model, an agent, a collection of agents, or an organization of humans and AI (Chapter 7, Section 1). To locate a relation inside it is for that unit itself to come to carry a relation now carried by parties or arrangements outside the research.

Chapter 1 offers a provisional outlook on the first question by proposing the characterization; the other four correspond to the four debts that Chapter 1, Section 9, identifies, that is, questions that the characterization leaves the later chapters to answer: the achievement, process, attribution, and research-order debts. The last concerns the social order of research, made up of the division of labor, the allocation of inquiry, criticism and certification, memory and correction, credit, and authority. Chapter 1, Section 9 explains what kind of division this is and what it is for. Each question organizes one of the book's five parts. Part I (Chapter 1) takes up the first, Part II (Chapters 2 and 3) the second, Part III (Chapters 4 to 6) the third, Part IV (Chapters 7 and 8) the fourth, and Part V (Chapters 9 to 12) the fifth. Chapter 13 concludes.

The opening orientation uses epistemic achievement as an open success term, one that marks success without fixing in advance what kind of success it is. Knowledge and understanding receive the principal analysis; further achievements are not established merely by being named. In many places the book's argument proceeds mainly through the case of knowledge, especially in the chapters that discuss what it takes for an achievement to be realized, how evidence is assessed, and how results are certified (Chapters 1, 2, 6, 8, and 11). Understanding is taken up in Chapter 3, which examines, chiefly through its relation to knowledge, what it takes for understanding to be realized, who understands, and what makes understanding valuable; its consequences for certification and attribution are discussed further in Chapter 9, Section 2.1 and Chapter 11, Section 5.6, among other places. Most of the chapters on the process of inquiry, the attribution of research, the allocation of inquiry, and authority over decisions, by contrast, are written in terms that apply to knowledge and understanding alike. The term epistemic achievement lets several achievements, knowledge and understanding among them, serve as construction targets, while each target's conditions remain to be defended. Its cost, that the term by itself fixes no single condition of success, is assessed explicitly in Chapter 1, Section 1.

The order of the argument is not one-way: the later chapters draw on the characterization, and what they find can call for its revision (Chapter 1, Section 1). What the later chapters mainly consider is what follows about machines' doing research when research is taken in this way. The book does not aim to test the working characterization, but Chapter 1, Section 8 compares the characterization with varied research practices to see how far it reaches and where it parts from other ways of taking research. Chapter 13 traces what the characterization distinguished and made askable where the one doing research is a machine, what it set aside, and which conclusions rest on accounts or cases other than the characterization. It also draws

together what the chapters have tried to bring out about artificial research, from changing questions and methods to the creation of research actors and the reorganization of relations among them. The target can become more definite through this inquiry without every philosophical disagreement having to be settled first.

Of what comes into view across the chapters, Chapter 13 gathers two things. One is that the claim that a machine is doing research divides into claims that can hold or fail separately; the other is that much of what was settled without being decided when humans did research, because those who did it were individual people, divided among many, can become something to be decided when machines do.

Part I — What Is Research Taken to Be?

Chapter 1 — A Working Characterization of Research

What the characterization needs—and what it does not

A thin orientation toward epistemic achievement

The characterization needed here has a demanding but limited job. It is meant to range across fields without turning the way one discipline organizes its work into the form of research as such, and to make an artificial-researcher target explicit enough that we can ask what an artifact does, what people and institutions still do, and what an evaluation establishes. It need not identify the timeless essence of research or settle every contested boundary before bounded engineering can begin.

That limitation matters because candidate definitions can answer different questions. A condition for being a research activity is not automatically a condition for good or trustworthy research. For example, producing knowledge is not necessarily the same as producing understanding, making progress, occupying a useful role in a research system, receiving institutional recognition, or producing something important. How it relates to producing understanding depends on whether understanding is taken to be a kind of knowledge (Chapter 3, Section 3), and whether it is the same as making progress depends on the account of progress (Chapter 2, Section 8.3). On the working characterization proposed below, a failed project may still have been research. An institutionally recognized project may be epistemically defective. Data curation may enable major progress without itself resolving the research question it supports.

Candidates for characterizing research include operations and achievements. If research is characterized by an operation such as controlled experimentation, hypothesis testing, or paper production, practices such as proof and interpretation cannot be covered without substantial redescription. If it is characterized by an achievement such as explanation, understanding, or problem solving, a different question arises: whether treating these practices as inquiry directed toward that achievement better captures what they are doing. The cross-domain cases of Chapter 5 give reasons not to identify research with one operation, but they do not decide which achievement research is best described as oriented toward. Achievements such as knowledge and understanding face the same comparison: their adequacy, like that of operational candidates, is judged by what they clarify and what they exclude.

For building an artificial researcher this comparison matters in a particular way. An artificial system can produce novel outputs in great quantity, but whether it has thereby accomplished anything epistemic is a separate question. Producing a novel output and bringing about an epistemic achievement therefore need to be kept apart. This book accordingly characterizes research not by a particular operation but by its orientation toward achievement, leaving the kind of achievement open rather than deciding it in advance, and expresses this orientation in three terms: attempt, epistemic achievement, and novelty relative to a society.

The book's working characterization is as follows:

> **Relative to a society *S*, research is an attempt to bring about epistemic achievements that are new to *S*.**

The sentence is not meant to fix a definition of research and then defend it. It is a working characterization, set down as a footing for thinking about what it would be for an artifact to do research, and is itself open to scrutiny. Why each term takes this form is explained below and from Section 2 on.

Here, epistemic achievement is an open term for epistemic success, principally examined in this book through knowledge and understanding. It does not supply an exhaustive classification or assume that the achievements share one structure, are equally valuable, or are mutually exclusive. What their realization involves, including whether it requires possession by a subject, depends on the account. The term does not itself require overcoming a significant obstacle or exercising significant cognitive ability; those stronger senses of achievement are examined in Chapter 3, Section 7.1.

Whether research is described as the production of knowledge or as bringing about understanding changes which aspects of research come into view; this is why the book uses the general term epistemic achievement and examines both. This chapter,

however, explains what the working sentence's terms require and do not require mainly by taking knowledge as the example. For knowledge, the questions its success raises (truth, the subject to which it is attributed, and non-accidentality) and its social attribution have been examined in detail in epistemology and social epistemology, which makes it easier to see what each term does. Chapter 2 begins its analysis from propositional knowledge for the same reason. Understanding is touched on wherever a difference arises and examined as a whole in Chapter 3. Theories of understanding ask whether understanding is a species of knowledge or a distinct and perhaps more valuable epistemic achievement, and how its explanatory form (understanding why something holds) and its objectual form (understanding of a phenomenon or subject matter as a whole) are related (Grimm 2021); that question too is taken up in Chapter 3, Section 3.

Here attempt marks an **organized direction**, not necessarily a conscious intention entertained by every participant. The characterization requires that some feature of the activity—its plan, functional organization, allocation of roles, responsiveness to evidence, or place in a research practice—connect what is done to the prospective epistemic achievement. It is not enough that knowledge can later be extracted by chance from an otherwise arbitrary event. Which kind of organization is required remains open; its immediate purpose is to distinguish a research attempt from a lucky output while preserving distributed and nonhuman cases.

S has two roles: it is the reference against which novelty is measured, and it fixes the society for which the achievement is sought, that is, where the attempt is directed. Neither role attributes the achievement to *S*. *S* may be humanity, a specialist community, an organization, a group of humans and artifacts, or a possible artificial community, but not a collection assembled after the fact to make some output look new. Whoever conducts the research need not be *S*, and naming *S* does not settle who bears the achievement. Sections 2 and 6 take these points up.

The shorter precursor characterized research through the production of knowledge and then derived questions as a component of that characterization (Takagi 2023). This book retains attempt and novelty relative to a society while widening the prospective achievement and replacing produce with bring about (Section 4). The reason is to connect the general orientation with the alternatives examined in Part II, without deciding in advance whether understanding is a kind of knowledge or treating it only as a means to knowledge.

The sentence specifies the target conception examined in this book. It is not a necessary-and-sufficient demarcation criterion or a complete philosophy of research. Education, journalism, investigation, invention, and ordinary problem solving can fit a broad reading. Conversely, an epistemic orientation can misdescribe practices centered on making, participation, relationship, or transformation. Whether the orientation is necessary for every practice properly called research remains open. These pressures are examined in Sections 4 and 8. Section 4 distinguishes products, roles, and achievements; Section 8 compares both what the orientation captures and the distinctions it imposes with varied research practices.

Hoyningen-Huene's account of science faces a problem like the one this chapter faces: how to say something general about very varied practices. Hoyningen-Huene holds that scientific knowledge differs from other kinds of knowledge, in particular everyday knowledge, primarily by being more systematic, and develops this comparative thesis along nine dimensions (Hoyningen-Huene 2018, 3, 7 n. 12). The thesis compares bodies of knowledge, not research activities. Hoyningen-Huene grants that the abstract term systematicity is vague, and that contrasting terms (not random, accidental, arbitrary, unplanned, or unordered) give it little precision; in the abstract, he holds, apparently no more can be achieved, so the term has to be made concrete in contexts. The concrete notions of systematicity vary with dimension, discipline, and period and are related by family resemblance, overlapping partial similarities without one feature common to all, rather than by necessary and sufficient conditions, so that the unity of the sciences that the thesis claims is, on Hoyningen-Huene's account, an extremely weak one; not every dimension applies to every field (predictions, for example, need not) (Hoyningen-Huene 2020, 89–91). Science, on this account, is not everywhere sharply bounded from nonscience: between scientific research and related activities tuned to practical purposes lie transitional areas in which only a necessarily rough and vague distinction can be drawn (96). In his 2018 replies, Hoyningen-Huene makes explicit that his comparisons establish at best that systematicity is necessary for science, not that it is sufficient, and that the account's demarcation criterion compares how much the overall systematicity of a putative pseudoscience increases over a period with the increase in the established science it competes with; non-quantitative comparative terms suffice for this, a quantitative scale for systematicity being doubtful. He welcomes historians' case studies as a concrete testing ground for the account (Hoyningen-Huene 2018, 1, 3–5). The working characterization is abstract in a similar way: so that it can say something general, and not empty, about varied research

practices.

Hoyningen-Huene also meets the first of the pressures above, that a broad reading fits neighboring activities, from the side of knowledge rather than activity. Areas of knowledge production by professionals with a high demand for reliability, such as the Violent Crime Linkage Analysis System (ViCLAS), market research, product development, chess theory, and air-crash investigation, the account argues, can proceed very systematically while their product is unambiguously not science. The account locates the difference in what it calls epistemic connectedness, the connection of knowledge to other pieces of knowledge, which such knowledge lacks to a sufficient degree (Hoyningen-Huene 2018, 2–3). An article in a historical journal and a serious newspaper article on the same events may carry the same message, but the journal article's many footnotes make such connections explicit (Hoyningen-Huene 2020, 96–97). Compared with this chapter's working sentence, the professional cases above are each directed at producing knowledge, and they accordingly suggest that, if research is to exclude them as Hoyningen-Huene excludes them from science, an orientation toward knowledge new to *S* would not by itself do so. Hoyningen-Huene's own criterion, as just described, concerns the connection of new knowledge to other knowledge. Whether the relation to what *S* already knows, which a claim of knowledge new to *S* presupposes, could do comparable work is not pursued in this book, which, as Section 8 records, does not claim that the epistemic orientation separates research from every neighboring activity.

What the sentence organizes—and what requires a different vocabulary

The characterization operates at the level of the activity's broad direction. It is not meant to serve simultaneously as an ontology of research products, a process model, a theory of scientific progress, and a measure of social value. Those tasks can be separated without abandoning the proposed orientation:

1. At the **activity level**, research is defeasibly illuminated as an organized attempt to bring about epistemic achievements new to *S*.
2. At the **process level**, inquiry can change questions, evidence, credences (degrees of confidence), concepts, methods, abilities, recognized ignorance, and social availability before its target is attained.
3. At the **achievement level**, the book principally examines knowledge and understanding and competing accounts of their realization. Explanation, conceptual reorganization, and increased capacity can be outputs, means, or achievements, depending on the claim; naming them does not establish further achievement kinds or their success conditions.
4. At the **value and governance level**, importance, utility, beauty, emancipation, legitimacy, risk, and institutional recognition require their own assessment. They are not thereby hidden conditions of every epistemic achievement or of research status.

These are analytical levels, not four independent processes or a universal chronology. One proof or experiment can alter several of them at once. Likewise, an arrangement that sets how long an attempt can be sustained and where its episodes are bounded, for example a grant or a thesis (Section 3), bears on the activity and process levels while being assessed at the fourth, and it is not made a condition of research status either. Their separation is meant to avoid two opposed errors: making the success term so broad that it renames every valuable change, and replacing the common orientation with an unstructured list of field-specific goods. However, on accounts on which values enter warrant, the epistemic support for accepting a claim, or acceptance itself, such as those of Section 8 and those taken up in Chapter 11, Section 4, the fourth level is not separable from the others in this way, and a judgment of value can bear on whether a result is accepted.

The epistemic orientation distinguishes a desirable effect, novel artifact, or accepted statement from epistemic success. It thereby makes the construction target answerable to an account of what has been accomplished, where it is realized, and what distinguishes success from error, accident, or a useful but insufficient contribution. These are questions, not one theory's answers imposed on all achievements.

The working sentence names the intended achievement with an open general term, epistemic achievement. Naming it so has a cost. The general term epistemic achievement by itself specifies little about success: a claim of epistemic achievement does not by itself tell us whether knowledge was obtained or which proposition, if any, is claimed to be known. That becomes clear only once the kind of achievement claimed is specified.

Using the general term epistemic achievement does not, however, loosen the conditions of success. Once knowledge of a proposition is claimed, its truth is required on the view most epistemologists hold (Section 5). Understanding, too, remains answerable to its object under the accounts compared in Chapter 3. Some accounts of understanding allow idealized models, models that contain deliberate falsehoods, to contribute to understanding (Chapter 3, Section 3.4), so the general term might seem to loosen the condition of truth. Idealized representations can contribute to inquiry directed toward knowledge as well, however, so how far idealization may be used is decided not by the general term but by the account of each achievement. The general term settles only what can be said before a kind of achievement is specified; each knowledge or understanding claim has its own conditions to meet.

Nor does keeping epistemic achievement an open general term count everything as an achievement. Naming a concept, method, instrument, or capacity does not establish it as a kind of achievement. Whether something is a kind of achievement and how fully the book analyzes it are separate questions. Practical, aesthetic, and relational goods also remain assessable in their own terms; calling them valuable does not make them epistemic.

The characterization does not establish one global aim

Because the formulation says what research is by saying what it is an attempt to bring about, it can easily be read as a thesis about *the aim of research*. That reading would make it much stronger than intended. Claims that a practice has an aim can refer to importantly different relations: a practitioner's conscious intention, a collective or corporate goal, an institution's historical or present function, a constitutive feature without which an activity would not be research, a criterion of success, a normative ideal, or an interpretation that makes a practice intelligible, among others. These senses are the grounds that Lusk and Elliott examine and the interpretive route that Hüttemann develops, both discussed below, together with a practitioner's intention. Moving among these senses without argument would make attempt carry more philosophical weight than it can bear.

Some authors do defend such a strong thesis about the aim of research or science. Bird argues that knowledge is the constitutive aim of science, treating science as the social functional analogue of a cognitive system: just as the beliefs that a properly functioning cognitive system produces are knowledge, so what science produces when it functions properly is knowledge (Bird 2019, 171–172, 176, 178–179). There Bird argues from what counts as progress that science aims at knowledge rather than truth. A community that came to believe a true theory by an unreliable method would not thereby have made progress, and giving up that belief would not be a step backward. An activity progresses insofar as it approaches its aim; since the mere accumulation of truth or truthlikeness is not progress, the aim of science is knowledge rather than truth. Bird leaves understanding and problem-solving for discussion elsewhere and also sets aside van Fraassen's proposal that science aims at the empirical adequacy of its theories: it concerns how wide a range of claims the aim covers, a question independent of Bird's own. Even if the aim is restricted to the observable, Bird notes, it remains open whether science aims at theories whose observational consequences are true or at theories whose observational consequences are known to be true. Bird's question thus concerns the kind of success aimed at rather than its scope (Bird 2019, 173, 176; Mohler and Monton 2025, §1.1). Chapters 2 and 3 return to these alternatives as rival achievements. Bird later states a claim in the same direction in terms of function rather than aim: a knowledge-producing function can ground objective scientific norms directly and can help constitute what counts as science (Bird 2025a).

Against this, some hold that research or inquiry need not be assigned one global aim. Jane Friedman argues that we can theorize directly about the structure and norms of inquiry and the purposes of inquirers without positing one aim of inquiry. On Friedman's account, the analogy with games, whose aims are fixed by human convention, does not carry over to inquiry, which is not, as a game is, something humans have made by convention; nor need inquirers all share one aim of attaining a particular mental state such as knowledge, since someone who seeks high credence rather than knowledge is still inquiring (Friedman 2024). At the level of science, Rowbottom recommends dropping talk of the aim, or aims, of science in favor of more specific descriptive, evaluative, normative, and definitional claims, while granting that scientists and research groups have aims. Read literally as a claim about what scientists aim at, Rowbottom argues, such talk turns a philosophical dispute into a question for sociological survey; recast as a claim about what counts as success in science itself, it makes differences in value judgments about which inquiries are worthwhile look like disagreements about facts about science (Rowbottom 2014, 2–5, 12–16). Avoiding the word aim does not by itself settle whether research has one global aim or what could ground one, however, since, as Bird's argument from function shows, the same strong claim can return in other words.

Lusk and Elliott take the question on directly. Their question arises from debates about values in science. Views that let non-epistemic values, that is, values other than those bearing on evidence and knowledge, such as social, political, or personal values, influence scientific reasoning only where the evidence leaves gaps appear to rest on a global aim of science: an aim that holds across all fields and limits which local aims, the aims of particular projects, disciplines, or fields, are acceptable (Lusk and Elliott 2024, 2, 4–6). They therefore ask what could justify such an aim. They first examine five grounds that do not appeal to anyone's intentions and find none of them convincing (7–11). If the aim is a normative ideal, such as truth or explanation, nothing beyond philosophers' conflicting intuitions decides which ideal it is, and reading an ideal off practice would require a principled way of telling practices that fall short of it from practices that never pursued it. If the aim is a function, science is present for many reasons and makes many contributions to society, among them reliable information, prediction, health, success in warfare, and economic development, so no single function stands out. Against Bird's analogy between science and a cognitive system, they object that it favors a single epistemic function only for someone who already accepts a single epistemic aim, since aims such as producing new technologies have no evident counterpart in an organism's cognition. If the aim is an essential characteristic, the absence of any demarcation criterion, that is, a criterion that cleanly separates science from other activities, blocks the identification. A looser cluster of typical characteristics would instead yield many aims, each holding only locally. If the aim is a criterion of success, the analogy with chess, whose rules fix what counts as winning, does not carry over, since science has no comparably clear rules and what counts as its success is the very point in dispute. Drawing on Rowbottom's point above, they add that the dispute turns on value judgments that are unlikely to be settled uncontroversially without appeal to what scientists intend. If the aim is a corporate goal, one that an organization pursues through its structure whatever its members want, science lacks a uniform organizational structure, since its public and private funders have different agendas and incentives. On their account, grounds that do appeal to people's intentions fare no better. Scientists neither share nor rank the same goals, and no one ever designed science as a whole. A standard general enough for everyone to accept would have too little content to limit local aims. Positing several global aims does not help either: they would need a principled ranking that no one has supplied, or else the aims that operate in each context would in effect be local (12–15). They present this as a cursory analysis that shifts the burden of proof onto those who posit global aims, not as a decisive refutation (3, 6–7).

In place of global aims they defend a local aims approach, on which science has only local aims. On their account, the grounds that fail for science as a whole can yield aims once the scope is narrowed. Local aims can be identified, for example, from what a project was designed to achieve or from the mission that a field's founders gave it, and this limited scope is their reason for holding that local aims escape the objections above (15–16). Since a project usually pursues several local aims at once, they combine them through the adequacy-for-purpose view that Parker developed for evaluating models and that they extend to methods and hypotheses: an object of this kind is assessed by whether it is adequate for a specified purpose rather than by whether it is true. The purpose of an inquiry in a given context then specifies how far each of its epistemic aims, such as accuracy and precision, and each of its non-epistemic aims, such as ease of use or completion time, must be achieved (16–17, 20). On this view non-epistemic aims need not be subordinate to epistemic ones, a position they call the equal aims approach. Epistemic constraints nevertheless remain, since both kinds of aim are to be satisfied together as part of achieving the purpose. To the worry that, without a global epistemic aim, nothing would prevent epistemic corruption, they reply that a project's purpose can still be criticized from the local aims of the fields it relies on and from legal, regulatory, and ethical constraints. In particular, honesty requires researchers either to follow the epistemic standards of their field or to make clear that they are departing from them (18–22).

Their reasons against a global aim rest on different premises. Some concern how science as a whole has been organized: its scientists neither share nor rank the same goals, no one designed it, and its funders' differing agendas leave it without a uniform organizational structure. Others do not, among them that philosophers' intuitions about ideals conflict, that what counts as success is the very point in dispute, and that a standard general enough for everyone to accept would have too little content to limit local aims. They do not discuss artificial configurations (artificial systems, or combinations including them, that carry out research; Introduction, Section 6); applied to one whose organization is unified, the first kind of reason need not hold, and an aim could be identified from that organization, much as they identify local aims from what a project was designed to achieve. Whether such an aim would be global or local in their sense turns on whether the artificial configuration is taken to be research as a whole, and the other reasons are not affected by its organization. Their objection to Bird's analogy also changes form for such a configuration. It turned on comparing science as a whole with a cognitive system, although

aims of science such as producing new technologies have no evident counterpart in an organism's cognition. An artificial configuration whose organization is unified may itself be a system of the kind the analogy compares science with, and whether a single epistemic function stands out in it then turns on what it was made and selected to do, although what its selection actually rewarded can diverge from its stated objective (Chapter 2, Section 4.1).

Even among views that retain aim talk, whether the aim is knowledge is contested. Beddor argues that the view that inquiry aims at knowledge conflicts with the thesis that a rational agent can know something without being absolutely certain of it. The thesis is closely related to fallibilism, the view that one can know that p even though one's evidence does not rule out that p is false: fallibilism, together with the principle that one should not be completely certain of what one's evidence leaves open, implies it, although the thesis is weaker and can be accepted even by those who reject fallibilism. Beddor's argument starts from a case. A scientist knows a hypothesis on the basis of experiments, yet rationally gives some slight credence to the possibility that all the results are due to confounding factors. A researcher elsewhere then sends the scientist the data of a more comprehensive study. The scientist seems to have a reason to look at the data, and looking is further inquiry into the same question; yet if inquiry aims at knowledge, the scientist has already attained its aim, and there is no point in inquiring further. Beddor backs the case with an argument from epistemic decision theory, which evaluates credences from the epistemic point of view much as decision theory, the theory of rational choice, evaluates actions. A higher credence in a true proposition is more accurate, that is, closer to the truth, and a credence that is not maximally accurate is not maximally epistemically valuable. So if inquiry aims at making one's credences as epistemically valuable as possible, someone who knows without being certain has not yet attained that aim. Finding attempts to reconcile the two wanting, Beddor asks which to give up and judges that requiring certainty for knowledge costs more: people readily say that someone knows something while granting that the person is not completely certain of it, and they distinguish knowing from knowing for certain. Beddor therefore rejects knowledge as the aim and proposes instead that the aim of inquiring into a question is to make one's credence in its answer as epistemically valuable as possible, which requires full confidence, a credence of 1, in the true answer. On this account, part of the appeal of the knowledge aim came from knowledge's being thought epistemically valuable, and the alternative takes that value as the aim directly; it also leaves room for knowledge that falls short of certainty and fits with epistemic decision theory. Beddor does not conclude that inquiry has to go on until certainty is reached: whether it is rational to continue also depends on whether other inquiries promise greater expected epistemic value (Beddor 2024, §§1–3, 5–6). Section 3 returns to this dispute about the aim of inquiry.

A further route gives a different answer to what attributing an aim amounts to: it treats aim attribution as an interpretation that rationalizes paradigmatic scientific practice. On this route an aim is neither posited as a constitutive feature of the practice nor dropped from the account; it is attributed as a reading that makes the practice intelligible. Hüttemann develops this route while also arguing that, if aims are attributed in this way, Bird's thesis that science aims at knowledge may need to be qualified by specifying what kind of knowledge science aims at, without disputing the thesis itself. The qualification may be needed because practice does not pursue just any knowledge. Scientists build accelerators at enormous cost instead of collecting knowledge that is easily had. And on one account of idealization, idealizations misrepresent a system's behavior while helping to identify the structure underlying it. If so, what rationalizes practice is an aim at some kinds of truth and not others (Hüttemann 2025, 1–3, 5–6). On this route, an aim attribution is made answerable to practice, so that what it attributes can turn out more qualified than the strong thesis.

These positions thus disagree not only about *which* aim science or inquiry has (Bird and Beddor), but about what attributing an aim amounts to (Hüttemann) and what could warrant the attribution (Bird and Lusk and Elliott).

The working characterization therefore makes a defeasible interpretive commitment. In treating the orientation as an interpretation that can be revised in light of research practices, it is close to the route Hüttemann develops. It proposes an orientation toward epistemic achievements new to some *S*, without identifying that orientation with every participant's conscious intention, an institution's historical function, an exceptionless essence, or, in view of Lusk and Elliott's argument, a global aim that limits which local aims a project may pursue. Lusk and Elliott themselves allow that an aim stated for all of science is not of the kind they reject if it places no such limits on the pursuit of local aims; they make this point about their own adequacy-for-purpose view (Lusk and Elliott 2024, 17 n. 5). In this respect, the orientation proposed here is a claim of that weaker kind, since it does not require that epistemic achievement be placed above a project's other aims. A project can pursue prediction, construction, care, emancipation, or commercial utility while also being organized toward epistemic

success. The book does not aim to test this interpretation, but Section 8 compares the characterization with varied research practices to see how far it reaches and where it parts from other ways of taking research.

This limitation matters directly for building artificial researchers. The formulation does not license replacing research with a scalar instruction to “maximize knowledge.” It does not determine which unknowns merit attention, whose purposes should govern inquiry, how epistemic and non-epistemic values should interact, what risks are permissible, or when a result should alter a decision. They do not follow automatically from an epistemic objective, nor are they optional embellishments to it. They are problems of construction and governance that the formulation leaves unsettled, which Part V takes up by asking what each account implies (Chapter 9, Section 1).

What the terms are intended to do

Here again is the working sentence from Section 1:

> **Relative to a society *S*, research is an attempt to bring about epistemic achievements that are new to *S*.**

Each expression in it has only a limited job and decides nothing beyond it. No term, for example, guarantees success or identifies who bears an achievement; Sections 3–6 treat each term in detail.

- **Attempt** separates organized activity from successful achievement (Section 3).
- **Bring about** directs attention to the realization of an epistemic achievement, which is not the same as making an output, need not be a single threshold crossed, and need not create the object known or understood (Section 4).
- **Epistemic achievements** are successes; claiming one requires making clear which achievement is claimed and under what conditions it obtains, principally through the analyses of knowledge and understanding in Part II (Section 5).
- **New to *S*** compares the intended achievement with a society's prior epistemic situation, not merely an investigator's prior state (Section 6).
- **Society *S*** opens the reference to a specialist community, humanity, an organization, a mixed collective, or a possible artificial community. It does not itself identify an achievement's bearer (Section 6).

Of the two roles of *S* noted in Section 1, the second, identifying the society for which the achievement is sought, concerns the direction of the attempt, not a guarantee that the society benefits or that any achievement is realized. Neither role by itself attributes an achievement to *S* as a collective subject, to its members, or to an arrangement of public resources. Knowledge and understanding require their own accounts of these relations (Chapters 2–3). The reference cannot be freely chosen after the event: the activity's treatment of prior results, its addressees, and the standards to which it answers constrain the choice (Section 6). Where several references are defensible, assessing a claim requires knowing which reference it uses (Chapter 2, Section 6). Neither a single social mind nor the equivalence of storage, accessibility, acceptance, knowledge, and understanding follows from the notation.

A claim about research involves two parties: the actor or organized system *A* that conducts the attempt, and the society *S* to which the achievement is to be new and for which it is sought. Since research has so far been done by humans, whoever conducted research and whoever possessed its results have, naturally, both been human; Humphreys notes that the assumption that science is human science has driven, explicitly or implicitly, all previous philosophy of science (Humphreys 2009b, §2). A later essay names an argument that is, Humphreys says, often implicit in discussions of automated science: because science is an activity done by and for the benefit of humans, the ultimate epistemic authority in science should always be humans. Humphreys grants that the attraction of anthropocentrism has good grounds and holds that humans' retaining moral responsibility over the pursuit of science and the use of its products is essential, yet argues that, under an expansive conception of knowledge, we ought to allow scientific knowledge "beyond the current understanding of humans" (Humphreys 2020, 12, 14). On this account, then, that science is done for humans does not by itself make humans its ultimate epistemic authority, nor require that what it comes to know lie within their current understanding. Humphreys had already separated science done by humans from science done for them in the extinction case quoted by Schrape: an activity need not be done either by us or for us to count as scientific, so that both who does science and whose purposes it serves may be nonhuman (Schrape 2015, 162–163; Chapter 8, Section 1.3 describes the case). Westerblad likewise reports Humphreys's

rejection of a restriction to human performers and beneficiaries (Westerblad 2026, §2). The two parties distinguished here take up that separation, *A* on the side of by and *S* on the side of for, but *S* is not Humphreys's beneficiary. The for of that separation concerns whose purposes the outcome of an activity serves; *S* is the society to which an achievement is to be new and toward which the attempt is directed, a direction that, as noted above, does not guarantee that *S* benefits. Conversely, those who benefit from research need not wait for its achievement to enter *S*'s epistemic situation. Dario Amodei, writing as chief executive of an AI developer about what AI could do for the world if things go well, states the aims of AI-accelerated biology as benefits, such as the prevention and treatment of nearly all natural infectious disease and a doubling of the human lifespan, and notes of the latter that even if it were accomplished within the period Amodei considers, "we may not know it yet" (Amodei 2024, §1 and n. 18).

Nor does either party settle who bears the achievement, and in Humphreys's case of a science carried on after human extinction no human does. Once research may be conducted by an artificial system, whoever conducts it and whoever possesses its results need not coincide. Who actually bears the achievement, in the case of knowledge whoever knows, is therefore asked separately, where it is asked whether the research succeeded. The bearer may be *A* itself, members or organizations of *S*, or a third party, and on accounts that allow knowledge without a knowing subject, on which whether a result is knowledge turns on public certification and acceptance rather than on anyone's attitude, there may be no bearer at all. Neither *new to S* nor *for S* therefore says that *S* knows. Both do presuppose that *S*'s epistemic situation, through its members, organization, or records, could come to include an achievement of the kind sought; otherwise neither comparing the achievement with *S* nor directing the attempt at *S* would make sense (Section 6). What is presupposed is that *S* could come to hold such an achievement, not that it could follow the route by which the achievement was reached: on some accounts, a result whose derivation no member can survey can still come to be known in *S*, as in the case of a machine-generated proof verified by a transparent checker, one known to check what it does and how it does it (Chapter 3, Section 5.1). Because *S* is presupposed to be able to hold such an achievement, for an artificial *S* questions of artificial belief and possession arise even where *S* is only the reference.

What can be asked about an activity and its result accordingly divides into three: whether the activity is research for *S*; whether an achievement new to *S* has been realized somewhere; and whether it has entered *S*'s epistemic situation, which the accounts of possession in Chapter 2, Section 6 answer differently. An artificial system may conduct research for human knowledge without itself knowing the output; on some accounts of belief, it may know a result that no member of *S* yet knows; and a result sought for *S* may not yet be known by anyone. A human or artificial actor may produce a result that is later taken up by an organization whose components include both humans and artificial systems (what this book calls a hybrid organization). An artificial community could in principle be actor, reference society, and knowledge subject at once if the relevant conditions were met. These separations are analytical; they do not pre-judge which individual, collective, or distributed entities can literally know. Who the bearer is turns on accounts of belief and of social possession (Chapter 2, Sections 2 and 6), and Chapter 8, Section 1 distinguishes the cases by *S*.

The same separation applies when an artificial actor helps human members understand. The actor, the bearers of understanding, and the reference society are distinct even if they sometimes coincide. A claim of collective understanding requires an account of what the collective's achievement consists in; whether a new explanation available to its members is enough differs among accounts (Chapter 3, Section 6).

Later chapters rely on these separations whenever they state what is required of an artificial researcher. A requirement stated for a system, such as that its conclusions rest on its evidence or that it revise them when its reasons are defeated, applies to the system as stated only where the system itself is claimed to know or understand. Where others are the knowers, whether humans or other artificial systems, the system's outputs serve as their evidence or as part of how they come to believe, and the requirement falls on their beliefs or commitments instead (Chapter 2, Sections 2.1 and 4.3). The same output can therefore be assessed differently depending on whose achievement is claimed.

The characterization does not try to explain research on its own; it shares the work with other theories. Inquiry theory can illuminate the process hidden within attempt. Epistemology examines possible successes and the conditions under which they count as knowledge, understanding, or other epistemic achievements. Philosophy of scientific practice and technology makes material and representational realization visible. Social epistemology asks how production and possession can be distributed. Theories of error control, objectivity, and governance examine quality, criticism, and correction. Theories of

scientific progress ask which achievement's increase counts as progress (Chapter 2, Section 8.3). Accounts of agency and autonomy examine what makes an activity an agent's own and self-governed; this book draws on them to ask when a system has a stable research capacity or warrants researcher status. The book does not fold these questions into one definition. Doing so would answer, at the stage of defining research, questions about intention, group knowledge, autonomy, and the like that are still contested in each field. Claims about artificial research are accordingly assessed by asking separately which of these questions they answer.

Attempt separates research activity from its success

An experiment can be inconclusive, a proof strategy can collapse, an archive can fail to resolve a historical question, and a prototype can miss its target constraints. If research meant the actual realization of an epistemic achievement, these activities would be reclassified whenever their target was missed. A functional analysis of the treatment–research distinction makes a related point: unsuccessful activity can retain knowledge generation as its function even when it fails to achieve the intended effect (Lewens 2006). That account locates function in planning and intention; it supports the activity–effect distinction without supplying the more general notion of organized direction used here.

If success is not a condition of research activity, what makes an activity research under this book's characterization? It is the activity's being directed toward bringing about an epistemic achievement, which the book calls attempt. Where, then, is that direction located? Here attempt does not require every participant to entertain a sentence such as “I intend to produce knowledge.” The book uses this term, rather than a success term, so that success does not become a condition of research activity, the subject of this section; and it uses the term in this sense so that distributed and nonhuman cases are preserved (Section 1). The relation between activity and epistemic end could be grounded in an individual's intention, a group's plan, an artifact's organized function, a component's role, or participation in a norm-governed practice. The functional organization that this book lists among the grounds of orientation is function in the sense of the role a component plays in a system. An artificial system whose inquiry fails still has organization in that sense if it proposes hypotheses, runs experiments, and revises its hypotheses in response to evidence. Some accounts instead take an artifact's function to be what the artifact is for and trace it, for a new artifact, to its designer's intention. On such accounts the orientation of a new system's inquiry derives from its designers, and the question arises to whom the activity is to be attributed. That is a question of attribution, not of whether the activity is research, and Chapter 7, Section 2 takes it up.

Among the transitional areas that, as Section 1 noted, Hoyningen-Huene places between scientific research and related activities tuned to practical purposes are those between technological research and development and between economic research into markets and the market research that companies perform; in them the scientific and the nonscientific side may not differ in their degree of systematicity in other respects. Work by automobile manufacturers to improve fuel efficiency, for example, is typically tuned to particular engines that are already on the market or soon will be, whereas the analogous work in engineering departments of research universities typically has a wider range of intended applications, or none in sight, aiming at new design principles; the difference, on this account, is a higher degree of epistemic connectedness, at least potentially, due to greater generality (Hoyningen-Huene 2020, 96–97). In an earlier essay on why modern science is technologically exploitable, Hoyningen-Huene makes a related point about single actions. A point-wise testing procedure, the realization of particular values of an experiment's independent variables, has, like every action, an external aspect, its physical parameters, and an internal aspect, which includes its goal. A procedure with positive results may be repeated with a different goal, that of attaining certain values of the dependent variables, so that a test and a technical application coincide in their external aspect and differ only in the goal; a successful clinical test of a new medication is one of his examples (Hoyningen-Huene 1989, 10–11). Hoyningen-Huene also notes that even the experimental test of a hypothesis contains an element of what he calls forced chance: a result that deviates from the prediction may be due to the hypothesis, but may also reveal that some auxiliary hypothesis tacitly taken for granted in the set-up is at fault, so that background assumptions are constantly, though not intentionally, challenged by experimental activity (Hoyningen-Huene 2020, 98–99). The difference in intended applications can be read as a difference in the place an action occupies within an activity, rather than in each participant's state of mind. Applied to attempt as organized direction, the comparison suggests that identifying the direction requires knowing how an action relates to the activity's aims, since, as the coincidence of a test and an application in their external aspect shows, physically identical actions can differ in aim, and forced chance suggests that what an action bears on

is not fixed once for all by the plan under which it was undertaken: an experiment set up to test one hypothesis can come, without anyone intending it, to bear on an assumption taken for granted in the set-up. For an organized attempt, this raises questions about what made an action relevant to an unresolved matter, what possible outcomes would have changed its role, and where the capacity to redirect the activity lay. Chapter 7, Section 2 asks these questions where an activity is attributed to an artificial or hybrid configuration.

If an activity's orientation is not identified with participants' states of mind, three things need to be kept apart: the activity's orientation, the norms by which the activity is assessed, and participants' states of mind. The epistemological debate over the aim of inquiry lets one see concretely how the last two come apart. To begin with, as Section 1 noted, views differ on what inquiry aims at. Kelp starts from a widely accepted characterization of the success of inquiry into specific questions as properly settling the question and, as a substantive account that makes it concrete, takes knowledge to be the goal of such inquiry and also takes understanding, which Kelp analyzes in terms of systematic knowledge, to be the goal of inquiry into general phenomena. Beddor instead argues that inquiry aims at making one's credences as epistemically valuable as possible, while Jane Friedman questions whether inquiry has a constitutive aim in the relevant sense and recommends discussing inquiry's structure, norms, and inquirers' purposes directly (Kelp 2021b, S1583–S1585, S1588; Beddor 2024, §§3.1, 6.1; Friedman 2024). Lee points out that claims of this kind admit two readings: as normative claims about the conditions under which an inquiry succeeds, or as descriptive claims about the personal aim that every inquirer has as an inquirer. Lee then holds that the paper's own view, that inquiring partly consists in trying to resolve a doubt one has, holds on either reading. First, it is consistent with any account of success conditions: supposing that knowledge is what makes an inquiry successful, of two people who independently inquire whether it rained last night, one who evaluates the evidence poorly fails to know, so that only the other's inquiry succeeds, although each inquiry was sustained by doubt. Second, it is not threatened even if, as Friedman argues, different inquirers have different aims and none is shared by all: whether one aims at knowing, at certainty, or at a belief justified by one's evidence, one pursues that aim by trying to resolve one's doubt (W. Lee 2026, §3.1). Lee's view is thus a claim not about the aim of inquiry but about the state of mind an individual inquirer is in while inquiring. Following what Lee calls standard in the debate over whether belief is compatible with inquiry, the view takes inquiry to involve an interrogative attitude such as wonder or curiosity, and counts an activity carried out without one as ersatz rather than genuine inquiry, while allowing that it may count as inquiring in more liberal contexts (W. Lee 2026, §1.1). The view that inquiry involves an interrogative attitude, and the distinction of ersatz inquiry, are also discussed by Friedman (Friedman 2013; Friedman 2019).

What this book takes from Lee is that trying and succeeding are distinct. The two inquirers in the rain example differ in success, yet both were inquiring. This supports the distinction between activity and success drawn in this section. Lee's trying, however, is an individual's mental activity sustained by doubt, and applied to an artificial system, which the paper does not discuss, whether its activity counts as genuine inquiry by Lee's standard would turn on whether it has an interrogative attitude (Chapter 4, Section 9), a question of the kind Chapter 2, Section 2 raises about epistemic attitudes. An attempt in this book's sense does not necessarily require one. Because it can also be grounded in a group's plan, an artifact's organized function, or a component's role, it does not require an interrogative attitude of every participant. An activity that would not count as genuine inquiry by Lee's standard can therefore still be an attempt in this book's sense. The two answer different questions: Lee asks what it is for an individual to be inquiring, and this book asks toward what kind of achievement an activity is organized. Nor does this book settle which of Kelp, Beddor, and Friedman is right. Its attempt is not a claim about whether inquiry in general has a constitutive aim but a description of what a particular activity is organized toward. What it takes from the debate is a reason to keep the activity's orientation, the norms by which it is assessed, and a participant's occurrent state of mind apart.

Even so, which view one takes changes what kind of attempt an activity is described as being, because the views describe differently what the activity sought to bring about and what it achieved. An example shows what changes and what does not. Two of the positions in this disagreement specify what inquiry aims at: the view, which Kelp defends, that inquiry aims at knowledge, and Beddor's view that it aims at making credences as epistemically valuable as possible. Either can therefore describe what a single episode aimed at and achieved in terms of what inquiry aims at. The descriptions of the same episode from Kelp's and Beddor's views are therefore given in turn below.

The episode is one in which a mathematical proof was checked in a proof assistant. In 2020 Peter Scholze asked for a formal verification of a theorem proved jointly with Dustin Clausen, in a blog post that launched the Liquid Tensor Experiment. Scholze wrote that for a theorem of such foundational importance, one that others would use as a black box, being 99.9% sure was not enough, and that, thinking nobody else had dared to look at the details of the proof, Scholze still had some small lingering doubts. Half a year later, after the part of the argument Scholze had been unsure about had been checked in the Lean proof assistant, a follow-up post reported that its author had no remaining doubts about the correctness of the main proof (Scholze 2020, §6; Scholze 2021). On Kelp's view that inquiry aims at knowledge, the check is described as inquiry seeking to know that the main proof was correct, and its success is judged by whether knowledge was thereby gained. For whom that knowledge was new is taken up in Section 6. On Beddor's view, on which inquiry aims at making one's credence in the answer as epistemically valuable as possible, and if the 99.9% Scholze wrote is read as the writer's own credence, the same check can be described as inquiry that continued while the credence that the main proof was correct was already high but short of full confidence. The episode has the shape of Beddor's own cases, in which someone who may already know is offered decisive evidence (Beddor 2024, §2). On Beddor's view, even if Scholze already knew before the check that the main proof was correct, a possibility like the one Section 6 leaves open for the theorem, the aim of inquiring whether the main proof was correct had not yet been attained, because a credence of 0.999 in a truth is less accurate than a higher one; the check therefore had a point with respect to that aim. What the check achieved is then described by how far it brought that credence toward the truth: if the main proof is correct, raising the credence toward full confidence made it more accurate. The size of that rise is what confirmation theory calls confirmation as relevance, confirmation as the raising of probability by evidence; on confirmation as firmness, confirmation as sufficiently high probability given the evidence, the claim that the main proof was correct was already confirmed before the check (Chapter 2, Section 5.2). Whether Scholze's writing that no doubts remained marks full confidence, and so the attainment of the aim, or only a credence closer to it, is a further question of interpretation. What does not change between the two descriptions is that the activity was organized to check the part of the main proof about which Scholze had been unsure. This book counts the check as an attempt not because of what epistemic achievement it brought about but because the activity was organized in this way, toward the part of the proof that needed checking. On either view, then, the check is an attempt, but each view interprets it as a different attempt: on Kelp's view, an attempt to bring about new knowledge; on Beddor's, an attempt to make an already high credence as epistemically valuable as possible; and the description of what it achieved changes accordingly.

Separating activity from success also blocks the inference in the other direction. Just as a failed activity can be research, an output that looks like success does not show that research took place; a lucky output does not settle the attribution. If, for example, a random generator emits a novel true mathematical statement, a true inscription may exist before anyone recognizes or supports it. If another actor proves, checks, and incorporates the result, the generator may be causally indispensable to a later production of knowledge. It does not follow that the generator knew the result, conducted the research episode, or possessed a research capacity. Chance can operate within research—through stochastic search, unexpected observation, and serendipitous redirection—when noticing, appraisal, retention, testing, and revision give it an organized role (Copeland 2019).

A research episode is therefore distinguished here from a stable capacity to conduct research. One low-probability or unrepeatable episode may be research, but one fortunate output gives little reason to call its source a researcher. Conditions such as reliability across relevant variation, responsiveness to changing evidence, integration of research functions, and revision after failure may support that stronger attribution. These conditions belong to the analysis of capacity attribution in Chapter 7, Section 3.1 rather than to the minimal work performed by attempt.

Distinguishing episodes from capacities raises the question of what one episode is and where it is bounded. Neither attempt in this book's sense nor talk of research episodes presupposes that research comes in bounded units. Where an inquiry begins and ends, and what counts as one attempt rather than several, can be fixed by practices that organize research—in research as now organized, a paper, a grant, a thesis, a release, the numbering of a notebook's entries (Chapter 5, Section 6.3)—or by the purpose of whoever describes it rather than by the inquiry itself. In one open drug-discovery project, for example, the season in which test larvae could be used and the departure of a researcher, which left one person able to run the screen, set a date by which new compounds had to be shipped; the same researcher's thesis deadline prompted a proposal that a shared manuscript enter that thesis unfinished, while one contributor decided to make further compounds anyway (Open Source Mycetoma 2022). When this book speaks of a research episode, the boundary is drawn for the analytical purpose at hand,

and an attribution that depends on how episodes are counted can be assessed only when it says how they were drawn.

Success, for its part, need not be fixed once at a single time. Just as the boundaries of an episode can be fixed by practices or by a describer's purposes, the judgment whether an attempt succeeded can be revised later. Empirical support is a case in point; seen from the side of the achievement, this is taken up in Section 4 as the point that an achievement is not a single threshold. If whether an activity was research depended on its success, that judgment too would be revised whenever support was revised. Characterizing research by attempt keeps the judgment whether an activity was research apart from such revisions of the judgment of success.

Failure, likewise, need not come down to a single verdict. A project may miss its original target yet establish a boundary condition, eliminate a possibility, expose an experimental artifact, or preserve a useful trace. It may instead leave no recoverable epistemic gain. Whether the activity was research, whether it was competently conducted, whether it answered its question, whether it made progress, and whether it produced collateral knowledge are distinct judgments (for the case where the target is knowledge, Chapter 2, Section 8.1 sets out such judgments, and Section 9.1 of that chapter takes up what competent conduct requires of an artificial system that conducts research). This section separated attempt from success so that these judgments are not merged into the single judgment whether the activity was research.

Bringing about an achievement is neither making an output nor crossing a single threshold

Bringing about knowledge or understanding need not create the object known or understood. An astronomical relation or past event does not begin to exist when inquiry succeeds. For knowledge claims, the familiar language of production concerns realization of the epistemic relation examined in Chapter 2, with its disputed accounts of acquisition and social possession. For understanding, a new representation can enable a grasp of dependencies, of what a phenomenon depends on and how, without creating those dependencies. In both cases, producing the representation and realizing the achievement are distinct. Changes in evidence, ability, or accessibility can enable success without themselves constituting it.

Research also constructs things that did not previously exist in that form: proofs, formal objects, concepts, interpretations, artifacts, datasets, models, instruments, classifications, samples, performances, and experimentally stabilized phenomena. The immediate product and the epistemic achievement are therefore distinguished here.

At least four relations can otherwise be conflated:

1. an activity produces an **immediate output**, such as a text, proof, model, dataset, prototype, instrument, interpretation, or performance;
2. the output bears **content** or makes a claim, pattern, possibility, or distinction available;
3. it occupies **epistemic roles**, for example as evidence, warrant, explanation, method, representation, memory, or scaffold;
4. an **epistemic achievement**, such as knowledge or understanding, may thereby be realized. Its conditions and location require an account; an output or enabling role does not establish it.

The four are separated so that the existence of an output and the realization of an achievement can be told apart: a large volume of outputs from an artificial system, for example, does not by itself show that something has been achieved. Where most such outputs are false, even one that passes a check can remain poorly supported, on the Bayesian notions applied in Chapter 2, Section 5.2. The division lets one ask of a claim about artificial research which of the four relations its evidence establishes: a generated proof text, for example, establishes an output and makes a claim available, but does not by itself establish an achievement. The four are not borrowed from a particular author; the book separates them at points where output and achievement can come apart. They therefore do not list every relation that bears on outputs and achievements: relations to a society, such as a result's being recorded, received, and preserved, are taken up separately in Section 6. The four are also relations, not mandatory chronological stages. A proof can establish a theorem, explain why it holds, reveal a dependency, and transmit a strategy. A prototype can demonstrate possibility, expose a failure mode, and reorganize a design space; it can also establish feasibility while its mechanism remains misunderstood. A dataset can support a claim, calibrate a model, preserve observations for unanticipated questions, or fail to support its initial interpretation.

Why it matters to keep the four relations apart can be seen concretely in domain-specific discussions. For data and design, process-oriented accounts of data and research-through-design discuss why artifacts can scaffold inquiry without their bare existence settling whether knowledge has been produced (Leonelli 2023; Zimmerman, Forlizzi, and Evenson 2007; Gaver 2012). Leonelli, for example, treats data, models, and publications as situated scaffolds whose validity and evidential value must be reassessed as they are reused, and takes the achievement they serve to be what Hasok Chang calls active knowledge: an ability to perform and coordinate activities purposefully, which cannot be reduced to storing and retrieving information (Leonelli 2023, 46, 52–53). In terms of the four relations, the existence of an output, a body of data (the first relation), does not fix how it works as evidence (the third), still less that an achievement has been realized (the fourth), since that working is reopened each time the data are reused. In mathematics, Rav distinguishes the conceptual proofs of mathematical practice from formal derivations and holds that the path a proof charts between concepts not only convinces but establishes interconnections that illuminate mathematical ideas (Rav 1989, 67, 73 n. 30). Pelc, who adopts Rav's terminology, reads Rav as treating proofs both as ways of convincing that a theorem is true and as repositories of mathematical knowledge, in which one learns new ideas, concepts, and strategies that can be taken over into one's own research (Pelc 2014, 2–3). In terms of the four relations, what is disputed here is whether mathematical knowledge (the fourth relation) lies in a theorem statement's being made available (the second) or also in the proof, which on Rav's view carries methods and concepts and so does more than work as warrant for the theorem (the third). Thurston emphasizes concepts and ways of thinking as part of mathematical advance (Thurston 1994). On this view, advance is not exhausted by new statements' being made available. In engineering, a method description, a method enacted by skilled practitioners, and a situated sociotechnical method-in-use are not interchangeable (Stacey et al. 2025). A method description may exist and its content be available (the first and second relations) without an ability to enact the method having been realized. In art, Scrivener argues that the art object is not itself a form of knowledge, although it can communicate some: Scrivener defines a knowledge artifact by the intention to communicate knowledge, without implying that knowledge is stored in it (Scrivener 2002, 1, 3, 11). In terms of the four relations, a work's carrying content and playing the role of conveying knowledge (the second and third relations) differ from the work's itself being knowledge, just as an instruction manual is made to convey knowledge although the knowledge is not stored on its pages. A truthful description of an experience may also fail to preserve the ability or sensibility through which the experience mattered. Scrivener's own proposal for art, which centers on what it calls apprehension, is examined in Section 8, which compares the orientation and its distinctions with practices presented as research by their own theorists.

What these discussions bring out is that correct description, warrant, possession, execution, and transfer can come apart. Evidence that an artificial system has output a correct theorem statement or method description therefore does not by itself establish warrant by proof, that the system or anyone who reads the statement possesses the knowledge, or an ability to enact the method. Keeping the four relations apart lets one ask which relation such a claim has established without settling these disputes. These literatures do not settle whether proofs or methods are themselves knowledge, although Rav's account, as Pelc reads it, treats proofs as repositories of mathematical knowledge, and Leonelli's account locates the achievement an output serves not in knowing a proposition but in an ability to perform and coordinate activities. In this book's terms, such an ability is a candidate for an epistemic achievement other than propositional knowledge. As the achievement level of Section 1 noted, however, abilities can be outputs, means, or achievements depending on the claim, so whether a given ability counts as an achievement has to be specified claim by claim. Whether knowledge includes abilities is among the questions Section 5 registers, and taking propositional knowledge as a starting point, as the opening of Chapter 2 notes, does not settle it.

Finally, bringing about an achievement in this sense does not imply one irreversible threshold at which a paper is stamped as knowledge. Section 3 noted that the judgment whether an attempt succeeded can be revised later. The achievement itself need not be fixed at a single time either. Partial proofs, calibration, replication, criticism, theoretical integration, and later correction continuously alter support, confidence, usability, and the space of live alternatives. For knowledge, since on many accounts whether knowledge obtains depends on the state of support and on the presence or absence of defeaters, contrary information that removes support (Chapter 2), a change in support can change whether knowledge obtains. On Dewey's account, each stage of reflective inquiry presents what may be taken for granted as the outcome of previous thinking and as determining further inquiry, and the intellectual content so reached is tentative, corrected as it is acted on (Dewey 1916, 164, 241). Understanding, moreover, is sometimes treated as coming in degrees. On Kelp's account, for example, maximal understanding of a phenomenon is maximally comprehensive and maximally well-connected knowledge of it, and understanding short of that is a matter of degree (Kelp 2021b, S1588; see also Chapter 3, Section 7). On such accounts,

bringing about understanding can be a matter of understanding's deepening rather than of crossing a single threshold. Bringing about an achievement therefore does not imply that every project crosses one discrete threshold.

Epistemic achievement requires making clear what was achieved

As Section 1 noted, the term epistemic achievement in the working sentence is an open general term that includes knowledge and understanding. Because it does not decide in advance which kind of achievement is sought, the characterization applies both to research directed at knowledge and to research directed at understanding. Leaving the kind open does not, however, lighten the burden of any particular claim. To claim that an activity has achieved, or is trying to achieve, something is to need to make clear which kind of achievement it is and under what conditions that achievement obtains. This section looks concretely at what that need demands in the case of knowledge, because the success conditions of knowledge have been discussed in detail in epistemology and many accounts treat them as demanding (compare Section 1). Chapter 3 supplies the corresponding inquiry into understanding. What is listed here are conditions of knowledge, not conditions of every epistemic achievement.

Where an activity is claimed to be an attempt to bring about new knowledge, that is, where the kind of achievement sought is specified as knowledge, the word knowledge cannot remain an honorific for praising a result. Using it takes on the responsibility of answering philosophical questions such as these. What distinguishes a true result from knowledge? Must a subject believe it? What kind of evidential, reliable, modal (concerning whether the belief would still have been right had things been slightly different), agential, or social relation is required? Does knowledge include abilities and understanding? Can a collective know what no single member currently entertains?

The present chapter does not answer these questions in advance; Part II examines them by comparing rival accounts. The chapter lists three issues that anyone claiming knowledge will face, whichever account they adopt. First, knowledge is treated as world-directed and factive: as Ichikawa and Steup report, most epistemologists agree that what is false cannot be known (Ichikawa and Steup 2026, §1.1); on this view, a false claim does not become knowledge even if it is widely accepted in a society and that acceptance is stable. Second, there is the question whose knowledge it is. The standard analysis in epistemology attributes knowledge, in the form "X knows that p," to a subject rather than to an unowned proposition; the subject may be an individual or, as many accounts in social epistemology allow, a group. On accounts derived from Popper's epistemology without a knowing subject (Chapter 2, Section 6, Table 2.2), by contrast, whether a result is knowledge turns on certification and public acceptance rather than on anyone's attitude. Third, a lucky true belief or output appears to lack the relevant non-accidental epistemic standing: a system that hits on a correct answer by chance does not thereby know it. This book does not aim to settle, on these issues, what truth is, who can be a knowing subject, or what relation excludes luck.

Not every account of science, moreover, uses knowledge in a sense that requires truth. Hoyningen-Huene, for example, uses scientific knowledge as scientists do, for established belief that the relevant community widely holds and does not treat as tentative or refuted, and leaves open whether it is true (Hoyningen-Huene 2018, 13). Hoyningen-Huene's thesis introduced in Section 1, that scientific knowledge differs from everyday knowledge chiefly by being more systematic, concerns knowledge in this sense. That thesis therefore does not supply a success condition of the kind examined in this section.

Contemporary epistemology offers competing accounts of how to answer the three issues listed above. They draw on different resources: belief, evidence, accessible reasons, reliable process, safety, competence, ability, freedom from defeaters, social criticism, and knowledge-first explanation, among others. One source of the competition is the Gettier cases. Gettier presented cases of justified true belief that hardly seem to be knowledge, challenging the analysis of knowledge as justified true belief, but no single replacement analysis has been established. Knowledge-first approaches reject the assumption that knowledge must be reduced to independently specified components at all (Ichikawa and Steup 2026). Williamson, for example, as Ichikawa and Steup present his view, treats knowing as itself a mental state, the most general factive one, and holds that it is a mistake to analyze knowledge in terms of more fundamental epistemic notions (Ichikawa and Steup 2026, §11).

Which account one adopts places different burdens on those who build artificial research systems and on those who evaluate them. On an account that stresses reliable process, for example, one has to show that the process by which the belief in question was formed is reliable; when a human comes to believe a result through a system, that process includes the system's

production of the output. On an account that stresses evidence, one has to show that whoever is claimed to know has evidence that supports the result. Part II neither decides which rival account is correct nor combines them into one checklist; it examines what comes into view about artificial research on each. The book's own working characterization of research, unlike the rival accounts it leaves standing, is itself open to revision. If treating central research practices as attempts at epistemic achievement requires distorting their own distinctions or calling every desirable change epistemic, the content or scope of the working characterization needs revision.

New to S makes the unstated reference explicit

Novelty concerns the reference society's prior epistemic situation. For knowledge, what one person knows another may not, and what a specialist community knows another may not. For understanding, an individual can newly grasp something already understood in the community. Neither change is socially new merely because it first occurs in one investigator. A claim of novelty specifies which achievement is new, in what respect, and to which society.

The motivating observation is modest. In ordinary academic practice, learning an established theorem does not by itself become a new research contribution. Independently rediscovering what a field already knows may demonstrate ability or provide an independent check, but novelty cannot be assessed only from what the investigator previously believed; it has to be compared with what the field already knew. In the Liquid Tensor Experiment, for example, a complex that Johan Commelin constructed while formalizing a proof turned out, several months later, to be a construction of MacLane's, while a lemma Commelin proved about it was, as far as its prover knew, new (Commelin 2022). Conversely, a first result unknown to the relevant community can be a contribution even though only a small subgroup initially understands it. Talk of new knowledge therefore presupposes, without stating it, a social reference: the society to which the knowledge is new. Being new to that society does not, however, require every member to update their beliefs at once.

The working sentence's variable *S* makes that reference explicit and generalizes it. It can denote humanity, a specialist community, an organization, a mixed group, or a candidate artificial society. It is minimally a collectivity for which the relevant comparison of epistemic situations is meaningful. The comparison is meaningful only if *S*'s epistemic situation, through its members, organization, or records, could come to include an achievement of the kind compared: if it never could, such an achievement would not be new to *S* but simply beyond its reach, and an attempt could not be directed at *S* (Section 2). The notation by itself does not require a corporate mind, institutional acceptance, universal accessibility, or one theory of group knowledge. Competing accounts, grouped into families in Table 2.2, may yield different verdicts (Chapter 2, Sections 6 and 7).

Making an unstated reference explicit in this way has a precedent in the debate over scientific realism. Van Fraassen, for whom what is observable is what is accessible to the unaided human senses, holds that observable, like portable, is nonetheless indexical (its reference shifts with the context of use, as that of here or we does): it means observable by us, the epistemic community, and the range of beings counted as us is "not fixed, either necessarily or even historically." Epistemology, van Fraassen argues, must therefore consider futures in which we and humans do not coincide, as when dolphins or extraterrestrials are admitted to the community (van Fraassen 2005, 1–4, 7). This us fixes what is observable rather than what is new, and van Fraassen does not discuss research. What this treatment of observable shares with *S* is that a reference left unstated in ordinary use is made explicit and allowed to range beyond humans, so that who belongs to it becomes a question of its own; Chapter 8, Section 1.2 takes up what admitting members does to a mixed *S*.

This openness has limits. An arbitrary set assembled after the event cannot automatically become *S* merely to make an output novel. Members, specialist subgroups, records, or practices are required to be connected enough for the social comparison under consideration to make sense. What kind and degree of connection is required belongs to social epistemology and to the practical role the achievement claim is meant to play.

Nor does the notation imply that knowledge appears socially at one moment. A result can be produced, recorded, appraised, preserved, made retrievable, taken up, used, corrected, and withdrawn through different events (compare the relations distinguished in Chapter 2, Section 6). An unread result may make knowledge possible without yet being knowledge possessed by *S*, depending on the account. The fuller social-epistemological analysis belongs to the achievement part of the book.

Whether something is new to *S* is settled by comparison with what *S* already knew. Results obtained by deduction can also be new by this comparison. Because mathematical theorems follow logically from axioms, on an idealization taken up at the end of this paragraph a community that knows the axioms would already know the theorems that follow from them. But a result can follow from premises already known within *S* without anyone there yet knowing the result. That a conclusion follows logically from premises is a relation among propositions; it is not the same as someone's actually carrying out a derivation and moving from not knowing to knowing. A principle much discussed in epistemology, called closure, holds that one who competently deduces a conclusion from what one knows comes to know the conclusion. Not every epistemologist accepts it in this general form, but even granting it, what it says is that knowledge extends when the deduction is performed; it does not credit a subject in advance with every consequence not yet deduced. Nor does the existence of a derivation, however long or inaccessible, by itself put *S* in a position to know its conclusion. To be in a position to know is not yet to know but to be able to come to know by means within reach. Chapter 2 (Sections 4.4 and 7) distinguishes actually knowing from being in a position to know in this sense. This book applies that distinction to the actual epistemic situation, to who has done what and what is available, rather than inferring it from what can be derived. A proof can therefore produce knowledge new to a mathematical community even though its conclusion follows from accepted axioms. Whether a particular proof does so still depends on its warrant and on what the community already knew. An evaluation that tracks this distinction separates such a new derivation from a system's re-deriving a result already known within *S* but withheld only from the system being evaluated, that is, from a rediscovery that is new only to the system (taken up at the end of this section). An idealization such as logical omniscience, on which a subject already knows every logical consequence of what it knows, would erase this distinction.

Returning to Scholze's check from Section 3, what a check of a proof made new depends on at least two things. One is whether the relevant community already knew the theorem, which the accounts of social knowledge in Chapter 2, Section 6 answer differently. What was new may also have been knowledge that a particular argument was correct rather than knowledge of the theorem. The other is which account one takes of whether one can know something while some doubt remains. On Williamson's knowledge-first account, one can know something without being ready to stake much on it, and in mathematics one knows, to a first approximation, what one has a proof of (Williamson 2000, 86, 209, 251–252, 263–264). Williamson does not discuss this case, but applying the account, the 99.9% Scholze wrote can be read as describing the writer's own credence, leaving open that Scholze already knew the theorem before the check. If so, the check need not have given Scholze new knowledge of the theorem. Both points concern not whether the check was an attempt but what was new, and to whom.

Novelty to *S* can be asked about understanding in the same way, but what is compared depends on how understanding is conceived. On accounts that treat understanding as a kind of knowledge (such as Khalifa's, taken up in Chapter 3, Section 3, and Kelp's, Section 4 above), novelty in understanding is itself a question of novelty in knowledge: whether the relevant knowledge, for example of why a phenomenon holds, of its explanation, or of how the facts about it connect, was already present within *S*. Even on Kelp's account, however, understanding a phenomenon requires knowledge that is comprehensive and well connected, so whether pieces of such knowledge were present within *S* does not by itself settle whether an understanding is new (Kelp 2015, 16). On accounts that treat understanding as an achievement distinct from knowledge, the question is whether the relevant grasp, for example of dependencies or explanations, was already present within *S*. On either kind of account, that a result such as a theorem was already known does not by itself settle novelty in understanding, as when a new proof of a known theorem discloses dependencies that were not previously understood; a proof that is merely written differently does not show this. Whether newly available explanations or resources enabling understanding already amount to understanding credited to a collective or to science, or must be separated from understanding actually realized in its members, also differs among accounts (Chapter 3, Section 6). On accounts that separate them, just as the existence of a derivation was separated above from actually knowing its conclusion, a society is not treated as already understanding everything it could come to understand.

These comparisons make a concrete difference where artificial systems are involved. Suppose an artificial system comes to understand a proof already understood in the human mathematical community (whether artificial systems can understand is taken up in Chapter 3, Section 6). A new bearer of understanding does not alone make the mathematical understanding new to that community. Building that system may nevertheless yield new knowledge about artificial understanding, for example about the mechanism by which a machine comes to understand the proof. What is new then is not the mathematical

understanding but knowledge about artificial understanding, and the research aims at a different achievement from that of mathematical research. By contrast, where the system grasps a dependency the human community did not know, there can be novelty for the community, which turns on that dependency and on what was already known or understood there. This does not yet mean that the community itself has acquired the understanding: if humans cannot access the system's internal state, the community's grasp may be unchanged. Whether an explanation has become available to the community, whether members have taken it up, and whether the collective understands each add different conditions, depending on the account adopted. What was grasped, whether it is new to *S*, and whether *S* itself has acquired it are therefore distinct claims. If *S* is instead an artificial society of which the system is a member, the system's understanding of a proof that human mathematicians already understood can itself be new to *S* when no other member understood the proof, and whether *S* has then acquired it turns on whether other members can access or share it, which accounts again settle differently (Chapter 3, Section 6). If a novelty claim is mistaken, what fails is the claimed success of the contribution, not the fact that research was attempted; this is an instance of the distinction between attempt and success drawn in Section 3.

A structure like that of these comparisons appears in the philosophy of creativity. Paul and Stokes, setting out how the requirement that what is creative be new has been refined, note that Newton's calculus was creative even if Leibniz reached it first, and that a beginning student's idea that freedom is compatible with determinism can be creative although philosophers have defended such views for millennia; in Margaret Boden's terms, such ideas are psychologically creative, new to their creator, without being historically creative, new in all of history, and psychological novelty is the more fundamental, since whatever is new in all of history is new to its creator but not conversely (Paul and Stokes 2023, §2). Boden's distinction concerns ideas new to the individuals who have them, not novelty for a reference society. Applied to novelty for *S*, what carries over is that assessing novelty relative to a restricted domain is coherent in its own right, with novelty in all of history a special case of it rather than its source; that the domain is *S* rather than an individual rests instead on *S*'s being the collectivity for which the comparison of epistemic situations is meaningful. Read this way, Boden's distinction gives a reason to think that assessing what is new relative to a chosen *S*, rather than to whatever is known anywhere, is a coherent choice of scope rather than a derivative of novelty to the world. In the cases above, an achievement can then be new to a human *S* although an artificial system outside *S* already holds it, much as Newton's calculus would have been new to Newton even had Leibniz reached it first. Where the machines are themselves members of *S*, as in a mixed *S*, whether they already hold an achievement bears instead on whether it is new to *S*, as with the artificial *S* discussed above.

Judging whether something is new to *S* also becomes less straightforward when *S* has lost something it once knew, since the question arises whether recovering lost knowledge is new to *S* or not. Chang argues that specialist science, which must take much for granted and cannot pursue every question, can lose knowledge, and that rehabilitating discarded or forgotten knowledge is a form of knowledge creation (Chang 2021, 108, 110). One of Chang's examples is Pictet's experiment on the apparent radiation and reflection of cold. Chang treats confirming that such a reported phenomenon can be reproduced as recovering a piece of forgotten scientific knowledge, and notes that a 1985 replication of Pictet's experiment itself seems to have been largely neglected (Chang 2017a, section "Recovery"). In this example, the result of Pictet's experiment was once scientific knowledge and was later forgotten. Does unearthing the experiment now and confirming its result again bring knowledge new to the community that forgot it? The same question arises when an artificial system unearths forgotten literature or experiments. The answer depends on how far a society is taken to retain knowledge, that is, on which account of retention one adopts (Chapter 2, Section 7). On Bird's view, a record still available through normal channels remains the community's knowledge, so as long as it is so available, unearthing it adds nothing new to the community's knowledge. On Lackey's view, in the form Kallestrup develops from it, such a record leaves the community only in a position to know, so unearthing it restores knowledge to the community and may in that sense count as new. Where *S* is artificial or mixed, the specialization Chang describes can remain while its records persist within reach of its members; knowledge set aside then need not have been forgotten, so that on Bird's view returning to it adds nothing, while on Lackey's it can still restore knowledge that no member held. On either view, this recovery differs from a system's rediscovering a result withheld only from the system under evaluation: there the result has been known within *S* all along, and it is new only to the system.

A short story by Ted Chiang poses a further case, in which a record has entered *S*'s channels while its content has not. It is written as an editorial in a journal of a future in which "metahumans", humans whose brains were altered by gene therapy before birth, dominate research and make their findings available only through a digital neural transfer that unaltered humans cannot use. Journals for human readers publish second-hand translations that leave "even the most brilliant humans" puzzled,

and human scientists, expecting to make no further original contribution, turn to interpreting metahuman publications, reverse engineering metahuman artifacts, and remotely sensing metahuman facilities; the editorial counts this hermeneutics as a legitimate method that increases human knowledge "just as original research did" (Chiang 2000, 517). The piece is fiction, and its metahumans are not machines. Read as a thought experiment, it separates three things: the results are realized in bearers outside the human *S*, records of them stand in *S*'s journals, and no member can yet follow those records. Applied to this case, the views of retention above place the novelty of a successful decipherment differently. If a record that no member can follow still counts as available through normal channels, Bird's view, extended from the retention of a result once read and accepted to such a record (Chapter 2, Section 7), would make its content the community's knowledge already; on Lackey's view, the record would leave the community at most in a position to know. And on the sense of being in a position to know used in this section, being able to come to know by means within reach, a record that no member can follow by such means does not put *S* even in that position. On the latter readings, a decipherment can be new to *S* although its result is not new to those who hold it.

What follows comparatively directly from the working characterization: being directed at something unsettled

No universal method or procedure that research must follow can be derived from the working characterization. By comparison, one implication follows comparatively directly: an attempt at an epistemic achievement new to *S* addresses something not yet epistemically settled in the relevant respect, for example, what is the case, why it is so, or how a target is to be grasped. The working sentence's "new to *S*" specifies the epistemic achievement sought as one that *S*'s epistemic situation does not yet include, however that situation is construed (Section 6), and "attempt" requires that the activity be directed at that achievement; together they direct the attempt at something not yet settled for *S*. For knowledge, the attempt is directed at something not yet known in *S*; for understanding, at something not yet understood there. Since *S*'s epistemic situation could come to include an achievement of that kind (Section 2), what is not yet settled for *S* is something that could become settled there, even if the achievement, once realized, is at first held only by *A* or by some other bearer. That something is not yet known or understood also means that a question can arise about it. This book therefore calls such directedness at something not yet settled the minimal question structure. It does not require a verbal question formulated at the outset, nor ignorance of every fact concerning the target. Understanding can remain unresolved even when some facts about the target are known. Nor does it require that the point at which the attempt is directed be in fact unsettled in *S*: what follows here is where the attempt is directed, not what *S*'s actual state is. If the achievement aimed at had already obtained in *S*, then, as Section 6 noted, what fails is the novelty claim, not the fact that research was attempted. Which activities belong to the attempt depends on their organized connection to that prospective achievement.

This does not mean that every project begins with one explicit interrogative, that one actor entertains it, or that the final question existed unchanged at the outset. How questions take shape and are recast within inquiry is examined in Chapter 4, Section 4, in light of research records and accounts of inquiry. Here it is first made clear in what sense the minimal question structure is a question.

The minimal claim concerns directed epistemic unresolvedness, not a fixed question represented in one mind. The word question is itself used for an interrogative sentence, for its content, and for the act of asking (Ciardelli 2022, 16 n. 3; Habgood-Coote 2026, 2). Accounts of the act differ in what they require of the asker. Watson takes the third use and identifies a question, as an act, by its characteristic function of eliciting information, so that a question need not take interrogative form, and one asked with some other explicit goal, such as embarrassing the hearer, can still be a genuine question (Watson 2015, 86–102). Habgood-Coote calls default the view that to ask a question is to express an interrogative attitude, to aim at acquiring its answer, and to direct the hearer to supply one, and argues that this view does not fit three kinds of asking: a teacher asking a class a question to which the teacher knows the answer; a speaker asking a question in order to answer it, as Frederick Douglass did in an 1852 address; and shared asking, as when David Hilbert, at the International Congress of Mathematicians in 1900, set out problems in the hope that the mathematical community would solve them (Habgood-Coote 2026, 2, 4–6). What identifying an act of asking requires and does not require on the two accounts, and what view Habgood-Coote proposes from there, are taken up in Chapter 4, Sections 1 and 6.

The question structure of this section need not yet be a question in any of the senses of interrogative, content, or act of asking. Watson's account is close to it in separating a question from interrogative form and from the asker's conscious motive; but what Watson identifies is an act, one with the function of eliciting information. The structure of this section is not an act; it is a point at which an attempt is directed as something not yet settled in *S*, which interrogatives may later express, whose content may become determinate only during inquiry (Chapter 4, Sections 3 and 4), and which acts of asking by one or several participants may pursue. The minimal claim therefore does not depend on locating such an act in *A* or in any single member of *S*.

Nor does the minimal claim fix where a question comes from. A question may arise within the inquiry of *A*, or be inherited from those who supervise or fund the work, from communities, instruments, or a discipline's history (Chapter 4, Section 1); whether the actor originated it bears not on whether research is attempted but on autonomy (Chapter 7, Sections 1 and 4.3). This structure alone therefore does not settle in what sense asking can be attributed to a machine, whether the content of a question is fixed from the outset, who holds it, or for whom it is unknown. The three chapters of Part III view the process of inquiry from three sides, the relation between an inquiry and the unsettled points that organize it, the relations among candidate answers, representations, and means, and the relation between a result and what it bears on; of these, Chapter 4 asks what it means for a machine to ask, along four points: in what sense it is a question, whether its content is fixed from the outset, who holds it, and for whom it is unknown.

What does not follow

The working characterization does not establish that every research activity:

- begins with a fully formulated hypothesis;
- represents a determinate answer space;
- separates discovery from justification (treating the process of hitting on a hypothesis and the process of giving grounds for accepting it as separate stages);
- performs one procedure called verification;
- produces explanatory understanding;
- publishes a paper or makes its result immediately available;
- is governed by one actor;
- is valuable, important, or progressive;
- or contains a researcher with maximal autonomy.

Nor does it prove that knowledge, or epistemic achievement generally, is the only aim of research, that epistemic purposes always override practical, moral, political, aesthetic, or transformative purposes, or that epistemic orientation uniquely separates research from every neighboring activity. The formulation identifies the target conception whose construction consequences are examined here.

The characterization also leaves substantially open the choice of purposes, what research is pursued for, and the choice of agenda, which questions are taken up and in what order. A society is ignorant of indefinitely many facts, explanations, constructions, and possibilities. Novelty does not decide which deserve attention. Question selection can reflect, for example, explanatory interest, expected epistemic gain, practical need, moral concern, feasibility, institutional priorities, opportunity cost, and distributions of risk and benefit, among other things. On Kitcher's account, what makes a question significant depends on characteristics of people, their aims and values, and their social organization, and answering questions that express human curiosity is one way of conceiving significance (Kitcher 2013, 216–217). Other positions, set out in Chapter 9, Section 3, hold that the value of knowledge is independent of *S*; on them, that value is not, or not only, relative to *S*'s purposes and interests. They come in two versions, whose consequences for choosing among *S*'s unknowns differ. On one, which adopts a principle that Grimm attributes to Kvanvig and Lynch, every true belief has prima facie value; this by itself neither makes their values equal nor orders *S*'s unknowns, so it leaves the choice among them unsettled. On the

other, which Kitcher calls the traditional approach to epistemic significance, the structure of the world fixes which truths are significant, and choosing among *S*'s unknowns would reduce to finding what is significant. On neither version does novelty to *S* decide the choice: the first supplies no ordering, and the second orders the unknowns by the structure of the world, a ground other than novelty. This book locates importance at the fourth level distinguished in Section 1, that of value and governance, rather than treating it as a hidden condition of knowledge or research status.

Comparing the orientation and its distinctions with varied research practices

As this section noted, the working sentence identifies the target conception whose construction consequences are examined. Section 5 also stated when it would need revision: if treating central research practices as attempts at epistemic achievement required distorting their own distinctions or calling every desirable change epistemic. The book does not aim to test the working characterization. Even so, comparing the characterization with practices presented as research that do not fit this conception easily, to see how far it reaches and where it parts from other ways of taking research, shows concretely where the revision condition could become an issue. It also shows concretely, not only by definition, what would and would not enter the target if an artificial system were built and evaluated by this conception.

This section takes up three such practices: artistic research, participatory action research, and research within the Indigenous paradigm that Shawn Wilson describes. Each is presented as research by its own theorists, and their accounts put the book's commitments under pressure. These two conditions are used so that the characterization is compared with practices its own terms have not already screened out; if practices were chosen because they fit those terms, where it parts from other ways of taking research would be harder to see.

The three practices bear on different commitments of the characterization. The comparison asks three questions. First, on its theorists' own account, does the practice include an organized orientation toward an epistemic achievement new to a relevant society, as the working sentence's "attempt" and "new to *S*" require? Second, what would constitute that achievement, as distinct from a mere output or enabling condition (Section 4)? Third, can achievement be analytically distinguished from value and relationship in the way proposed here? These are this book's questions, not neutral terms binding the practices examined.

The artistic case bears mainly on the first two questions. It lets us separate what follows from the working sentence's epistemic achievements including understanding as well as knowledge, namely that the understanding that institutional definitions of artistic research pair with knowledge is not excluded from the start, from questions that remain: whether orientation toward an achievement captures the openness to what is not yet known that Borgdorff describes (the first question), and whether the object of what Scrivener calls apprehension is, in the terms of Section 4, a product (the second). Action research bears on the third question, asking whether epistemic achievement can be separated from worthwhile action. Wilson's relational account bears both on the third, the separation of achievement from relationships, and on attribution. The cases do not exhaust such pressures or form a consensus, but together they bear on all three questions above and on the location of attribution, which the relational account places differently. Accounts of understanding are examined more fully in Chapter 3, and the roles of outputs in Section 4 and Chapter 5; here the issue is the characterization of the practice.

The institutional definitions to which the debate over artistic research has responded pair knowledge with understanding (Borgdorff 2011, 54–55, 63 n. 20; Scrivener 2002, 2). Borgdorff holds that in artistic research the creative process is the pathway, or part of it, through which new insights, understandings, and products come into being (Borgdorff 2011, 46), and sets out criteria on which it counts as academic research, among them that it seeks to contribute to what we know and understand and to add new knowledge or understandings to the existing corpus (Borgdorff 2011, 54–55). Borgdorff's chapter, as announced at its outset, also characterizes artistic research as the articulation of pre-reflective, non-conceptual content, content prior to reflection and to concepts, enclosed in aesthetic experiences, enacted in creative practices, and embodied in artistic products, and concludes that it does not really involve theory building or knowledge production as those terms are usually understood but is "more directed at a not-knowing, or a not-yet-knowing" (Borgdorff 2011, 44, 47, 59, 61). Scrivener argues that the art object can communicate some knowledge but is not itself a form of knowledge, and that what art offers are possibilities and ways of seeing rather than conclusions (Scrivener 2002, 1, 7, 11–12). Scrivener accordingly proposes that research in art be defined as "original creation undertaken in order to generate novel apprehension," where the novelty is cultural, not merely novelty to the maker or to individual viewers, and treats what the art community chooses to

call research as its own decision (Scrivener 2002, 2, 12–13).

Borgdorff reports that many in the debate view knowledge not primarily as justified true belief or warranted assertibility but as world disclosure or world constitution (Borgdorff 2011, 56; compare 61). That the working sentence's epistemic achievements include understanding does not thereby establish a match with Borgdorff's account. Among the views it reports in the debate is one that treats understanding as a form of knowledge in which theoretical knowledge, practical knowledge, and acquaintance (knowledge by being directly acquainted with something) may intersect. For artistic research, Borgdorff adds insight and comprehension to emphasize a perceptive, receptive, and verstehende engagement, one that seeks to grasp meaning from within, that is often more important to the research than getting an explanatory grip (Borgdorff 2011, 55). Whether the accounts compared in Chapter 3 capture that understanding remains a substantive question. Nor does that inclusion establish that every insight, world disclosure, or articulation Borgdorff describes meets an account of epistemic success. Borgdorff's account still leaves the question whether orientation toward an achievement adequately captures openness to what is not yet known. Section 7 drew from the working sentence a directedness at something not yet settled for *S*, but as part of an attempt to bring about an achievement there, whereas Borgdorff sets being directed at a not-knowing, or a not-yet-knowing, against knowledge production in the usual sense. Set against Section 7, what is at issue is therefore whether that openness is directed at settling what is not yet known. Among Borgdorff's conditions of academic research are that a study justify its own importance to the academic forum and that its significance extend beyond the artist's own oeuvre (Borgdorff 2011, 54, 57), considerations that this book assigns to significance rather than to research status.

Scrivener explicitly contrasts making art with investigation undertaken to gain knowledge and understanding. Apprehensions in Scrivener's sense are objects to be grasped by the senses and intellect, offering ways of seeing and being rather than knowledge of how things were or are (Scrivener 2002, 11–12). That the working sentence includes understanding therefore does not by itself answer Scrivener's proposal. Such an object is made as a work, so in Section 4's terms it is an immediate output of the activity (the first relation); whether grasping it brings about an epistemic achievement such as knowledge or understanding, and under what conditions, requires a further argument. Cultural novelty is relevant to the proposal, but does not settle that issue. The book's terms carry no authority to decide what the art community may call research (Introduction, Section 4). It leaves the fit between Scrivener's proposal and its own characterization open. Scrivener's arguments include premises about knowing that this book does not adopt, such as that knowing is experienced with felt certainty and that a claim to new knowledge must be justified by argument (Scrivener 2002, 4–5, 10, 12). The conclusion that the art object is not itself a form of knowledge is therefore a claim about knowledge in Scrivener's sense, not about knowledge in the book's sense. What the book takes from the case is not an answer to the second question, what would constitute the achievement as distinct from a mere output, but the point that showing that grasping such an object brings about an epistemic achievement such as knowledge or understanding needs an argument of its own.

Reason and Bradbury name as one primary purpose of action research the production of practical knowledge useful in the everyday conduct of people's lives, and write that it leads to "new practical knowledge" and to new abilities to create knowledge (Reason and Bradbury 2001, 2). Measured against the working sentence, their own account therefore contains its orientation, at least for the part of practical knowledge that can be stated as propositions, alongside purposes that Section 1 allows to coexist with it, such as liberation and healing (Reason and Bradbury 2001, 2, 10). Following Heron and Reason, they set out four interdependent forms of knowing: experiential knowing, through direct encounter with persons, places, or things; presentational knowing, which first expresses that encounter in story, picture, or movement; propositional knowing, through concepts and ideas; and practical knowing, knowing how. The one they treat as primary is practical knowing, which, following Heron, consummates the other forms of knowing in action and consummates experiential knowing "in worthwhile action"; in action research, they write, knowledge is "a verb rather than a noun" (Reason and Bradbury 2001, 1–2, 9, 11). Practical knowing in their sense can be separated into three layers: claims that can be stated as propositions, which can be assessed as knowledge in the sense of Section 5; skill, which this book does not decide to be or not to be a kind of knowledge and assesses as an ability (Chapter 2, Section 1); and the worth of the action, which this book assesses at the level of value. They also give an epistemic reason for participation: outcomes grounded in the perspective and interests of those immediately concerned, rather than filtered through an outside researcher's preconceptions, make for "better" research, a reason that the book, which does not adopt one account of warrant, can accommodate on accounts that give weight to the perspectives brought to bear in criticism (Chapter 2, Section 6). But they add that the characteristics of action research are not simply questions of methodology and that, "far more than that," in joining knower with known in participative

relationship they undercut the foundations of the empirical-positivist worldview (Reason and Bradbury 2001, 4). Among their questions of validity and quality they include the significance of an inquiry and the relationships through which it was conducted (Reason and Bradbury 2001, 10, 12). This book keeps these at the level of value and governance as its own analytical placement; Reason and Bradbury do not accept that separation.

They also describe action research as creating new forms of understanding (Reason and Bradbury 2001, 2); whether that understanding is understanding in a sense examined in Chapter 3 remains open. Because the working sentence's epistemic achievements include understanding, assessing the epistemic orientation need not depend only on the propositional component identified above. But the book has not thereby supplied an account of practical knowing that identifies skill with knowledge or understanding, or separates its success from worthwhile action. The general term epistemic achievement leaves this substantive issue open; it does not settle the disagreement about participation and value.

In Shawn Wilson's account of an Indigenous research paradigm, relationships are reality, and relational accountability grounds its axiology and methodology (Wilson 2004, vi, 147–148). Wilson lists principles that must guide such research and what results from it, among them that the research question be founded in the reality of Indigenous experience, that the methods be process-oriented, and that the researcher take responsibility for the transformations and outcomes the research brings into a community (Wilson 2007, 194–195). Drawing on Stan Wilson, Shawn Wilson holds that knowledge is relational and shared with all of creation rather than owned by an individual researcher, and writes that the knowledge being presented belongs neither to its presenter nor to the group who took part in the research (Wilson 2003, 172; Wilson 2004, 223). The account speaks of the knowledge gained from relationships, but holds that knowledge cannot be owned or discovered: what research does is to create a new set of relationships, and bringing a new idea into being carries a responsibility to the relationships around it (Wilson 2004, 152, 209, 232). Under relational accountability, Wilson writes, "Right or wrong; validity; statistically significant; worthy or unworthy: value judgements loose [sic] their meaning," and research that meets criteria of validity and reliability, or accurately describes a "fact," will not be accepted within an Indigenous paradigm if it is separated from its relationships (Wilson 2004, 147–148, 188). The account adds that such research has its own ways of ensuring that what it says is "true" without externally imposed tests, that continuous feedback lets participants check the accuracy of the analysis, and that accuracy matters more in describing the set of relationships that make up a phenomenon than in describing the phenomenon (Wilson 2004, 188, 222–223). Research, on Wilson's account, is a ceremony and must be respected as such: "a life-changing ceremony" in which the relatedness of all things is respectfully reinforced (Wilson 2003, 175).

Wilson declines to articulate the paradigm by comparing it with other research paradigms, since any comparison would create a binary that leads to judgment and subjugation (Wilson 2007, 194). Whether research in this paradigm produces knowledge in the book's sense is not judged here; a verdict reached in its terms would in any case carry no authority for the community whose practice it classified (Introduction, Section 4). It records only three places where its terms and Wilson's part. The first is whether relations can bear knowledge: the book starts from the standard analysis, which attributes knowledge to a subject, but also notes accounts of knowledge without a knowing subject and leaves open which entities can be subjects of knowledge (Sections 2 and 5), and Wilson's denial that knowledge can be owned is not by itself a denial that knowledge is attributed to someone. The second is where considerations of relationship stand: the book does not make them conditions of research status, whereas Wilson makes them conditions of acceptance as Indigenous research (Wilson 2004, 147–148, 188). The third is how Wilson's talk of truth, accuracy, and new ideas relates to the factive knowledge of Section 5.

Read against the three questions, the cases come out as follows. On the first, whether the practice includes an organized orientation toward an epistemic achievement new to a relevant society, Reason and Bradbury's own account includes the working sentence's orientation, at least for the part of practical knowledge that can be stated as propositions; Borgdorff's criteria include it in asking that artistic research contribute to what we know and understand, though whether it captures the openness to what is not yet known that Borgdorff also describes remains open; the fit of Scrivener's proposal is left open; and the book does not judge research in Wilson's paradigm. On the second, what would constitute the achievement, the object of Scrivener's apprehension is, in Section 4's terms, a product, and whether grasping it brings about an epistemic achievement such as knowledge or understanding needs a further argument. On the third, whether achievement can be analytically distinguished from value and relationship, the book keeps the separation of achievement from value, governance, and relationships as its own analytical proposal; Reason and Bradbury contest it, Wilson's account does not

separate achievement from relationships, and the book does not settle the difference. The working characterization, then, neither takes in the three practices wholesale as research in its sense nor shuts them out. In parts of its theorists' accounts the working sentence's orientation can be found, and in other parts whether it can be found is not settled.

The line falls where it does because the working sentence names the achievement with the term epistemic achievement. Since the term includes understanding, the understanding that Borgdorff's criteria pair with knowledge is not placed outside the sentence's scope from the start. The term is also a success term, as Section 1 noted: a claim to knowledge or understanding has to meet that achievement's own conditions, and a claim to any further achievement needs its own argument. The orientation is therefore found where theorists say they aim to contribute to knowledge or understanding, and it remains undecided where it has not been shown that what the practice aims at would, if realized, meet those conditions or be supported by such an argument, as with openness to what is not yet known, Scrivener's apprehension, the part of practical knowing that is not propositional, and relational insight. Restating them in the book's terms would not by itself bring them within its scope. To claim that such a restatement had brought them in would be to distort the practices' own distinctions, and Section 5's revision condition would then apply. As long as what remains undecided is left undecided, the condition does not apply. The comparison with the three practices therefore comes out as follows: the characterization does not, for now, need revision, since Section 1 left open whether the orientation is necessary for every practice properly called research; but what it covers stops there, and beyond that point it parts from other ways of taking research.

For building and evaluating an artificial researcher, this examination has two consequences. First, if an artificial system is built and evaluated by this conception, what remained undecided above does not thereby enter its target, and a claim that such a system does research has to say in which sense. What remained undecided also rests on different premises. Borgdorff's content enclosed in aesthetic experience and the experiential knowing that Reason and Bradbury set out after Heron presuppose a knower who has experiences; applied to an artificial system, which none of these theorists discusses, whether these have counterparts turns on what such a system encounters and how. Reason and Bradbury's reason for participation and Wilson's relational accountability concern who takes part in an inquiry and in which relationships; applied to a system, they ask whose perspectives and relationships enter its research, and, where *S* is human, how its human members take part. If institutions of evaluation used the conception, what it leaves out of the target could be overlooked within them; the consequence of institutional use described in the Introduction, Section 4 applies here too. Second, even within the conception, describing research as an attempt to bring about epistemic achievements does not by itself fix the targets of an artificial system. Those targets and their conditions still have to be specified. Unless they are, an evaluation that speaks of epistemic achievement can still leave understanding out, or count an enabling artifact as the intended achievement.

The debts the characterization leaves open

The book uses the working sentence mainly for the questions it makes explicit. The sentence raises many questions that it does not itself answer. As Section 5 noted, for example, a claim to knowledge brings the burden of explaining what distinguishes knowledge from a true result. This book calls such questions debts, since once the characterization is adopted, the later chapters are responsible for answering them.

The book groups the debts into four: the achievement, process, attribution, and research-order debts, each taken up by one of Parts II to V (Introduction, Section 6). They are kept apart because an answer to one does not settle another. Showing that an artificial system produced a correct result, for example, does not by itself pay the attribution debt, since whether the conduct of the research or a research capacity can be attributed to that system has still to be shown. The grouping thus lets one ask which debts a claim about artificial research pays, and how far.

The four debts organize the questions that the book's parts take up; they are not derived term by term from the working sentence. Nor do they correspond one to one with the terms of the sentence or with the four levels of Section 1. Each debt bears on more than one term, unlike the correspondence Section 2 drew between the terms and Sections 3–6. As for the levels, the achievement and process debts concern the levels of the same names, and the attribution debt, which asks where an activity, capacity, or status is located, cuts across them. The relations on which the research-order debt bears, described below, are assessed chiefly at the level of value and governance; questions that Section 1 placed at that level have no debt of their own, and Part V takes them up where they bear on these relations (Chapter 9, Section 1).

First is the **achievement debt**. What distinguishes the intended epistemic success from a novel output, warranted acceptance, useful artifact, or enabling resource? Chapters 2 and 3 examine knowledge and understanding: their conditions, possible bearers, and relations to explanation and ability. Production, acquisition, possession, availability, use, and preservation need not coincide. No general word settles these differences, and a further achievement claim incurs a further burden of explanation.

Second is the **process debt**. How does inquiry proceed when an answer is not available to the relevant society, a project may fail, and its question, concepts, evidence, methods, and standards may themselves change? Candidate formation, experiment, proof, modeling, interpretation, construction, criticism, replication, anomaly diagnosis, and representation are possible roles, not a fixed sequence.

Third is the **attribution debt**. To what, and at which stated boundary, can research be attributed? Producing one correct result does not establish a stable research capacity. Answering the question requires asking whether a contribution, the conduct of an activity, a capacity, or researcher status is being attributed to an artificial actor; which parts of inquiry it selects, executes, monitors, revises, or governs; which remain externally supplied; and in which respects, and how far, it is claimed to be autonomous. Attributing the conduct of research also leaves open where its intended achievement would be realized: the actor need not be its bearer, nor belong to the society relative to which it would be new.

Fourth is the **research-order debt**. By the research order this book means the social order surrounding research, made up of the division of labor, the allocation of inquiry, criticism and certification, memory and correction, credit, and authority. Research already distributes cognitive labor across collaborators, tools, archives, review practices, organizations, and publics, and divides responsibility among the people and institutions involved. Artificial researchers may change agenda setting, attention, allocation, validation, memory, correction, credit, and authority. This metascientific problem has an origin independent of the working characterization, but the two meet when an artificial actor enters a research world, because three of the characterization's terms, *new to S*, *epistemic achievements*, and *S* itself, mark where the activity is tied to *S*, and, taken together with conditions outside the characterization, give rise to relations such an actor may change (Chapter 9, Section 2.1). Because *S*'s unknowns are countless while the means of inquiry are finite, and many researchers pursue them, the allocation of inquiry among *S*'s unknowns arises. Because members of *S* cannot each gather the evidence for many results and so rely on results through others, the criticism and certification of results arise. And because *S* consists of many members without a single voice, and the consequences of inquiry can reach beyond *S*, authority arises: who may decide on, and who may revise, allocation and certification. The characterization makes none of these relations a condition of research status. On accounts on which social organization enters warrant or possession, however, certification bears on what counts as *S*'s knowledge, and on accounts that make group knowledge depend on the members with decision-making authority in the relevant domain, so does a form of authority (Chapter 2, Section 6). Accounts of public or collective understanding can also make social organization bear on the achievement claimed (Chapter 3, Section 6). Where a relation does not enter an achievement's conditions, its assessment remains at the value and governance level.

These debts constrain one another. An achievement account can misdescribe a practice; a practice description does not establish warrant; an attribution boundary can hide distributed work. Chapters 2 and 3 assess knowledge and understanding claims where made, while Chapters 4–6 examine the formation and recasting of questions, how candidates and representations become available, and the localization of failure, Chapters 7 and 8 locate conduct and capacity and ask whose achievement would result, and Part V examines the allocation of inquiry, the criticism and certification of results, and authority over both. A different target conception could change their answers without erasing their questions. Readers who reject the opening orientation can still identify which conclusions depend on it and assess the remaining arguments.

Part II — What Would Count as Epistemic Achievement?

Chapter 2 — When Does a Result Become Knowledge?

Epistemic Change, Achievement, and Knowledge

From a novel result to an epistemic achievement

The working characterization describes research as an attempt to bring about epistemic achievements new to a society *S*. This chapter examines one demanding form of success: knowledge. Novelty can belong to an unsupported conjecture, an unread inscription, an accidental truth, or an artifact whose significance is unsettled. None is knowledge merely because it is new or useful. The general orientation does not relax the conditions of this particular achievement.

The knowledge term keeps at least three questions visible. In what respect does the result get its object right? To what subject can the resulting epistemic condition be attributed? What distinguishes an epistemically successful connection to the object from a fortunate accident? These are burdens raised by the knowledge term, not three independently necessary components of a complete analysis. Different epistemologies organize and answer them differently.

The chapter proceeds in this order. Section 2 asks what kind of attitude is at issue, chiefly belief, and whether its bearer must be the producer of the result. Section 3 takes up truth and fallibility, and Section 4 what makes a true success epistemic rather than accidental. Section 5 asks what verification contributes. Section 6 separates what can be social about knowledge: the reference for novelty, production, possession, and retention. Section 7 tests the preceding accounts of belief, verification, and social possession together on a case in which the production of a result and its possession by *S* come apart. Section 8 turns to epistemic change before and around knowledge, and Section 9 states what each account would require of artificial research and on whom each requirement falls.

This chapter begins with propositional knowledge because its analysis is well developed and provides a workable analytical entry point. The choice of entry point is not a requirement of the working characterization, which does not require knowledge to be propositional; whether knowledge includes abilities or understanding remains the open question registered in Chapter 1, Section 5. Nor does the chapter assume that every research achievement is exhausted by knowing that some proposition is true. A proof, a concept, an interpretation, a model, a practical ability, or an artifact can change what a society can represent, understand, infer, and do. Part of that contribution can be propositionally expressed: a construction is possible; a procedure yields a result under specified conditions; a new concept organizes previously disconnected phenomena in a certain way. Propositional content is not restricted to newly observed natural facts (Thurston 1994). In the history of Euler's formula for polyhedra, for example, the result, that the number of vertices minus the number of edges plus the number of faces is two, was stated by Euler in terms of concepts for the parts of a solid, one of which he first had to name: what are now called edges had, he wrote, no proper name, so he called them *acies* (Sandifer 2004, 2–3). What the result expressed was not an observed natural fact but a relation among the numbers of a solid's parts. The example thus illustrates, as noted above, that what comes from working out concepts can also be expressed as a proposition. It does not show that the proposition has been established as knowledge, since, as Musgrave and Pigden summarize Lakatos's view in their encyclopedia entry, a conjecture of informal mathematics and its proof are improved together through criticism: counterexamples bring out hidden lemmas and so improve the proof, proofs bring out conditions of validity and so improve the conjecture, and a proof can deserve respect without being flawless (Musgrave and Pigden 2026, §§2.1, 3.1). More generally, that a contribution can be stated as a proposition does not mean that the proposition exhausts the achievement. That a proposition can be written about an output, that the output supports it, that a subject knows it, and that the output itself is knowledge are different claims. Nor does it follow that anything epistemically valuable is knowledge.

In this way the chapter does not reduce every epistemic achievement to knowledge: knowledge is not the name of every valuable process change or achievement, and understanding, ability, conceptual reorganization, empirical adequacy, and recognized ignorance remain assessable in their own terms. Knowledge nonetheless remains a demanding target success, because it forces questions about truth, attribution, and non-accidentality that something new or epistemic improvement can leave open. The chapter determines what knowledge would require under different epistemological accounts, how it relates

to other achievements, and what each account makes visible in artificial research.

Process states: what sort of epistemic attitude is at issue?

In the central philosophical use relevant to propositional knowledge, to believe that *p* is roughly to take *p* to be the case. This introductory gloss is much thinner than a theory of belief. It does not choose among the main accounts of what belief fundamentally is: representational accounts, on which it is a stored representation; dispositional accounts, on which it is a disposition expressed across thought and action; functional accounts, on which it is a state fixed by its role among perception, other states, and action; interpretive accounts, on which it is a state attributable to a subject interpreted as rational; and normative accounts, on which it is a state constitutively subject to norms of truth and rationality (Schwitzgebel 2023). Contemporary discussions also dispute whether belief is necessary for every case of propositional knowledge, even though the belief condition belongs to the traditional analysis with whose critique, as Ichikawa and Steup note, much of the twentieth-century literature on the analysis of knowledge began (Ichikawa and Steup 2026; Myers-Schulz and Schwitzgebel 2013).

Several neighboring attitudes matter because research does not consist only in settled belief. A proposition can be supposed temporarily in order to explore its consequences. A model can be accepted for a task even when its idealizations are not believed literally. Confidence can be graded rather than categorical. Judgment can name an act of affirmation, while belief names a state that persists. On some accounts, suspension is an attitude toward a considered question, not merely the absence of any representation. And a candidate can be entertained without being believed. A historian, for example, describes keeping, beside notes summarizing each archival file, a separate "Idea File" of free association and "uncensored writing" that might never appear in a finished text, in which the historian also records surprising details and significant silences in the documents (Walkowitz 2009).

These distinctions prevent an apparently unified research pipeline from changing the meaning of its central state at each stage. Generating a hypothesis does not require believing it. Comparing mutually incompatible explanations does not require believing them all. Assigning a probability does not automatically create a categorical belief above an agreed threshold. Institutional acceptance can govern collective action even when individual members assign different credences. A new concept can alter the space of candidates rather than merely shift confidence among a fixed set.

For this reason, where later chapters describe the course of research as a whole, they speak more broadly of epistemic attitudes, or of commitments, as Chapter 6 does when it asks which commitment a result gives reason to revise. These words do not replace belief in the analysis of propositional knowledge. They keep belief from being used as the name of every representation, confidence, assumption, policy, or public status that changes during research.

2.1 The bearer of belief need not be the producer of the result

The actor or organized system that conducts research, *A*, need not be the subject to which the resulting knowledge is attributed. The society *S* supplies the reference for novelty and identifies for whom the knowledge is sought; it is not thereby the knowledge subject. An artificial system can derive a theorem, execute an experiment, or identify a pattern that a human community subsequently evaluates and comes to know. If a belief condition is adopted, the condition applies to the subject claimed to know. It does not automatically require the artificial producer to believe the proposition.

The distinction becomes unavoidable when knowledge is attributed to an artificial agent or artificial society itself. Consider increasingly integrated relations to a proposition *p*: a system stores a sentence expressing *p*; it outputs that sentence when prompted; it assigns *p* high confidence; it uses *p* in inference and planning; it changes action because of *p*; and it revises *p* when contrary evidence appears. On the functional and dispositional accounts listed at the start of Section 2, which identify belief by what it does in inference and action, storing or uttering a sentence is only weak evidence of belief, and stable integration with inference, prediction, action, surprise, and revision is stronger evidence. Interpretive and normative accounts may ask for something of a different kind, such as a subject that can be interpreted as rational or that answers to norms, and so may require a different kind of representation, agency, or mentality. In either case, evidence of storage and integration bears more directly on whether a state plays the role of a belief than on what the state is about; whether an artificial system's states are about the targets that *p* concerns is taken up in Chapter 8, Section 2.2.

This is not permission to call the same system a believer or non-believer at will. The system's architecture and behavior provide the evidence; a theory of what belief consists in determines how that evidence bears on literal attribution. This book can therefore describe the functional facts first and state the additional account under which they constitute belief. Whether a system that holds *p* in this integrated way is responsible for so holding it, or holds it for reasons, is a further question, which Chapter 7, Section 3.3 takes up in asking what thicker attributions to an artificial actor would require.

Whether knowledge can be attributed to an artificial society depends on theoretical choices at two levels. The first is whether knowledge requires belief at all. Ichikawa and Steup describe the belief condition as only slightly more controversial than the truth condition, and report that a few philosophers have argued that knowledge without belief is possible. In Radford's example, which they present, someone asked about English history gives date after date correctly while taking the answers to be mere guesses. On Radford's account, this person does not believe the answers, having no confidence in them; yet the run of correct answers shows that they learned the facts and have not forgotten them, so they know them. Both claims are disputed: some reply that the person has an implicit belief, others that answers held with so little confidence do not express knowledge (Ichikawa and Steup 2026, §1.2). Knowledge-first epistemology is not among these denials. It does not construct knowledge from belief plus additional ingredients, but it rejects that order of analysis rather than the belief condition: it is consistent with the traditional view that knowing entails believing (Ichikawa and Steup 2026). The second level arises among views that do require belief: they differ over what belief consists in, as the accounts set out at the beginning of Section 2 show. Thus, if an artificial society is claimed to know *p*, either that society must satisfy the selected belief condition or the knowledge attribution must be licensed by an account that does not require belief in that form. The issue is about the subject of knowledge, not a universal mental prerequisite imposed on every tool or participant in research.

Truth and fallibility

For its central use of propositional knowledge, this chapter retains factivity: if a subject *X* knows that *p*, then *p* is true. This is a substantive working commitment, but a comparatively modest one. Ichikawa and Steup report that most epistemologists agree that what is false cannot be known, and that most treat ordinary speech that describes a once-confident but false view as what people “knew” at the time as a kind of exaggeration rather than as literally true (Ichikawa and Steup 2026).

Factivity does not turn truth into an operational oracle. Whether *p* is true is different from whether a researcher can prove, conclusively recognize, or become certain that it is true. A coin can land heads even when nobody observes it. A claim can be true even when the available evidence is misleading. Conversely, a community can apply its best methods responsibly to a false claim and remain unable to detect the error. Truth concerns how things are; evidential and institutional procedures concern the fallible routes through which a subject becomes related to how things are.

This separates two evaluations. At time *t*, investigators may be warranted in accepting *p* under the best available evidence and procedures. Retrospectively, if *p* is false, they did not possess factive propositional knowledge that *p*, even if they did nothing blameworthy. The first evaluation concerns what the participants were entitled or rationally permitted to believe. The second concerns whether the epistemic attempt reached its factive target.

Research institutions can design methods, gather evidence, organize criticism, estimate reliability, and propagate correction. They cannot observe a separate truth label attached to every conclusion. An artificial system is in the same position. It can be answerable to proofs, observations, interventions, failed predictions, replications, material breakdowns, and criticism without possessing an infallible test for final truth. Each of these can push back against the system's expectations and claims from outside them; call them forms of resistance. They are mediated by instruments, models, concepts, and background assumptions, but a system whose commitments can be revised by them is related to its object differently from one optimized only for textual plausibility or social approval (Chapter 8, Section 2.3 takes up this relation as world-answerability). Criticism, the last of those forms, and social approval are both responses from others, but criticism can carry reasons and the resistance of the world, while social approval can be obtained without carrying either.

What makes a true success epistemic rather than accidental?

Gettier cases, as Ichikawa and Steup set them out in their encyclopedia entry on the analysis of knowledge, provide a canonical illustration of the general problem: a subject can reach a supported true belief whose truth nevertheless depends on an epistemically inappropriate coincidence (Ichikawa and Steup 2026, §3). The cases do not determine one replacement analysis of knowledge. They make visible why a true result and some support for it may still fail to constitute the demanding epistemic success at issue here.

A random generator can emit a true proposition. A contaminated benchmark can make an incapable system appear accurate. A researcher can infer a true conclusion from a false intermediate claim. A valid-looking explanation can be unrelated to the mechanism responsible for a correct prediction. These cases motivate a thin commitment:

> For the demanding sense of knowledge used as the central target here, accidental truth is insufficient. The truth of the relevant attitude must stand in some appropriate epistemic relation to its grounds, method, competence, environment, or other truth-connected conditions.

The problem of accidental truth is not confined to individual believers or to invented cases. For example, a proof of the four-color theorem that Alfred Kempe published in 1879 appears to have been accepted for more than ten years, until Percy Heawood showed in 1890 that a step in its central argument fails for some maps. The theorem itself was proved only in 1976, by a different route (O'Connor and Robertson 1996). On accounts that require a true belief to be non-accidentally connected to its grounds, those who accepted the theorem on the strength of Kempe's proof held a true belief that was not knowledge on the grounds they relied on.

The thin commitment above says only that the truth must stand in some appropriate epistemic relation to its grounds or method; it does not say which relation. The phrase marks a problem rather than solving it. In the traditional analysis of knowledge as justified true belief (Chapter 1, Section 5), this place is held by the justification condition, which some epistemologists also describe as a belief's being rational, warranted, or evidentially well supported (Ichikawa and Steup 2026, §1.3.1). Among responses to counterexamples to that analysis, some accounts keep the justification condition and add a further condition, such as sensitivity or safety (conditions that test whether a belief was right by accident by considering situations slightly different from the actual one; Section 4.1), while others replace it, for example with a reliability condition on knowledge (Ichikawa and Steup 2026, §§3–6). One way to fill the place, then, would be to call whatever relation is appropriate justification. But calling every candidate relation justification would conceal rather than manage the disagreement over which relation is required. Alston argues that something like this has happened in the theory of justification. As Feldman reports, Alston explains the widely different accounts that philosophers give of justification, in terms of what one ought to believe, good evidence, reliability, proper function, and other notions, by holding that no objective property of beliefs is picked out by justified; there is instead a variety of epistemic desiderata, such as having adequate evidence, being based on it, and being formed by a reliable process, whose nature, viability, importance, and interrelations Alston examines in turn (Feldman 2005; Alston 2005, xi–xii). This chapter, for its part, keeps knowledge as a demanding target, while it treats the relations that might make a true attitude non-accidental as distinct desiderata whose weight depends on the account adopted (Section 4.1).

Some philosophers recognize a lightweight use of knowledge requiring little more than true belief, while reserving a stronger sense for the central epistemological questions (Ichikawa and Steup 2026). The thin non-accidentality commitment selects the stronger sense for the knowledge target examined here. It does not establish that lightweight knowledge attributions are linguistically illegitimate. The stronger target is useful because a system that merely increases the frequency of lucky true outputs leaves precisely the capability and validation problems that motivate the inquiry.

The subsections proceed as follows. Section 4.1 surveys rival approaches to what the appropriate epistemic relation requires, and Section 4.2 sets out, in Table 2.1, what follows for artificial research on each. Section 4.3 takes up basing, a relation asked about within most of those approaches, and Section 4.4 the question it presupposes, what evidence a system has. Section 4.5 uses the case of an accidental generator to separate generating, knowing, contributing, and conducting research.

4.1 Several ways of locating epistemic success

Once knowledge is selected as a demanding success term, no theory-neutral decomposition follows. Epistemology, which has long debated what knowledge requires, offers rival approaches to that question: approaches to what, beyond a true belief, makes a success knowledge rather than luck. They organize the relation among a subject, a content, its grounds, the world, and a wider epistemic environment in materially different ways. They therefore make different features of research visible and imply different judgments about artificial researchers.

Internalist and access-oriented approaches ask what reasons, evidence, or reflective access the proposed knower possesses (Ichikawa and Steup 2026, §1.3.2). Evidentialist approaches ask whether an attitude fits the evidence the subject has. On Feldman and Conee's evidentialism, for example, as Mittag sets it out in the *Internet Encyclopedia of Philosophy*, whether a belief, a disbelief, or a suspension of judgment is justified turns entirely on its fit with the evidence the subject has at the time, however that evidence was acquired (Mittag n.d., §§2, 3c). Whether norms on acquiring evidence belong to epistemic assessment at all is disputed (Section 5). Externalist and reliabilist approaches can instead locate the relevant standing partly in the causal history and truth-conduciveness of the process, even when the subject cannot fully articulate why the process succeeds (Goldman and Beddor 2021, §§1.1, 2.1). The contrast matters for an opaque artificial system. For such a system, the reliability of the process that produced a result, the reasons a human recipient has for it, and the recipient's entitlement to rely on it can come apart: a reliable process may leave the recipient with no articulable reasons, and a recipient may have reasons for a result that an unreliable process produced. Because they can come apart, which approach one takes matters: a reliabilist looks to whether the process is reliable, an internalist to whether the recipient has reasons.

Virtue-reliabilist and achievement approaches ask whether a true success manifests a stable epistemic ability attributable to the relevant subject. Virtue-responsibilist approaches instead take intellectual virtues to include cultivated character traits such as open-mindedness and conscientiousness, and some ask whether virtue requires the corresponding motivations (Turri, Alfano, and Greco 2021). Where the knowledge of an artificial system itself is at issue, this raises the further question whether anything in the system plays the role of epistemic motivation. Modal approaches ask whether the attitude could too easily have been false (safety) or whether it would still have been held had it been false (sensitivity) (Ichikawa and Steup 2026, §5). Defeat-sensitive accounts ask how contrary information, whether information supporting the denial of the claim (rebutting information) or information casting doubt on the link between the claim and its grounds (undercutting information), or higher-order information, which casts doubt on the reliability of one's own judgment, changes its standing (Section 5). Within most of these approaches a further question arises, which Section 4.3 takes up as the question of basing: whether the subject holds a commitment because of the support that purportedly favors it, rather than merely possessing that support alongside a conclusion reached by another route.

All the approaches surveyed so far, from internalism to defeat-sensitive accounts, separate a correct output from a non-accidental achievement. They need not, however, draw the boundary of the successful subject in the same place: the success may be credited to the artificial system, to the human or other artificial system that uses it, or to a whole that includes both.

Knowledge-first epistemology changes the explanatory order rather than adding another condition. Williamson takes knowledge as the unexplained starting point and explains other epistemic notions partly through it: to believe *p* is roughly to treat *p* as if one knew it, justification is primarily a status that knowledge can confer, one's evidence is one's knowledge, and one must assert only what one knows (Williamson 2000, 9–10, 46–47, 185, 243). Taking knowledge as primitive does not make the conditions distinguished in this section idle for evaluation. Williamson gives two reasons for treating knowing as unanalyzable. On Williamson's account, knowing is itself a mental state, whereas every standard analysis includes truth, a condition that is not mental; and the way analyses since Gettier's counterexamples have each been met, as they grew more complex, by more complex counterexamples is what this view would lead one to expect. Williamson still holds that knowledge requires safety and does not require sensitivity. The reason for dropping sensitivity can be seen in an example. Suppose that someone who slightly underestimates the distances they see judges, of a mark one meter above the water, that it is less than twenty meters above the water. The belief is insensitive: had the mark not been less than twenty meters up, it might have been just over twenty meters up, and the subject would still have judged it to be less. Yet the subject knows it, since one meter and twenty-one meters look quite different, so the belief rests on evidence quite unlike the evidence the subject would have relied on had the proposition been false. However slight a systematic bias, so long as it is not zero, the

same mechanism can make almost all of a subject's beliefs in contingent propositions insensitive. On Williamson's view, moreover, a necessary condition of knowing need not be a conjunct of a non-circular analysis (Williamson 2000, 30–31, 33, 147, 159–161, 307–310). On this account, then, safety is not a component from which knowledge is defined, but it is still a necessary condition. Evidence about safety is a further matter, used in evaluating an attribution: evidence that a result could easily have been wrong tells against it, while evidence of safety does not by itself establish knowledge. On a knowledge-first account, Table 2.1 in Section 4.2 can be read in this way, as stating what the adopted account takes knowledge to require and what must be established to support an attribution, not the parts of a definition.

Simion's naturalistic version organizes the epistemic domain around one function, generating knowledge. The function is etiological: inquiry has it because its past production of knowledge brought epistemic benefit, and that benefit helps explain why the practice persists; such a history can be supplied by ongoing selection or a learning process as well as by selection over generations (Simion 2025, 7–8). Applied to trained artificial systems, the account implies that whether such a system has that function turns partly on what its training and selection rewarded, not only on how often its present outputs are true. Carried further, the application implies that two systems with the same present rate of correct answers can differ in whether knowledge is attributed to them on this account, if the training and selection of one rewarded correct answers and those of the other rewarded plausible answers or the approval of raters. Taking the stated objective of training to be the etiological function is, however, itself a premise, since what selection actually rewarded can diverge from the stated objective. Bird's extension of a naturalized knowledge-first functionalism to groups is discussed in Section 6.

Social epistemologies introduce further alternatives (Section 6): epistemic standing may depend on testimony, divided expertise, organized memory, critical interaction, or functions realized by a group. Inquiry- and understanding-centered approaches (Chapter 1, Section 3; Chapter 3) also question whether the epistemic change relevant to research is exhausted by the acquisition of settled propositional knowledge.

These approaches stand in different relations to justification. Internalist and evidentialist approaches, and process reliabilism, are theories of what makes a belief justified, and the dispute between internalism and externalism is a dispute about justification (Ichikawa and Steup 2026, §1.3.2). Sensitivity and safety are proposed as further modal conditions on knowledge (Ichikawa and Steup 2026, §5), and reliabilist theories of knowledge put a reliability condition in place of the justification condition (Ichikawa and Steup 2026, §6.1). On virtue-theoretic approaches such as Sosa's, knowledge is apt belief, belief that is true because it was produced adroitly. Knowledge therefore entails justification, but that justification is understood as adroitness, the belief's being produced by skill; and because what is required is that the skill explain the success, justification is not a component separable from a success's being creditable to competence (Ichikawa and Steup 2026, §10). Basing concerns the relation that separates a belief that merely has support available from one held on that support (Section 4.3). Knowledge-first accounts, as noted above, explain justification through knowledge. The thin commitment at the start of Section 4 does not choose among these: it leaves open whether the appropriate epistemic relation is justification on one of these accounts, justification together with a further condition, or a condition that replaces it.

These are not modules to be accumulated into one maximal definition. Nor does this book need to select one account before any further analysis is possible. They can be used conditionally. If accessible reasons are required, one set of opacity and transfer problems follows: the reasons for which a system reached its conclusion may be invisible to the recipient, and it is unclear whether they carry over as the recipient's own reasons. If reliable external connection is sufficient, another follows. If competence, modal robustness, understanding, or social organization is central, the location and meaning of success change again. The philosophical positions are therefore resources for tracing the consequences of locating epistemic success in different places, not different names for one hidden pipeline. What each would require, and what apparent success each is liable to mistake for knowledge, is set out approach by approach in Section 4.2 (Table 2.1); Section 9 draws these together as requirements on artificial research.

4.2 Conditional consequences

The approaches distinguished in Section 4.1 are rival accounts of what knowledge requires. Table 2.1 sets out, for each of them, what follows if that approach is adopted. The left column names each approach by what it emphasizes. The middle column states what knowledge would then require and what an evaluator would need to establish to support attributing it. The right column gives a characteristic false positive: an apparent success that the approach would not count as knowledge but that an evaluation could easily mistake for it. Read row by row, the table shows how the choice of approach changes what must be built and checked in an artificial research system. It is a simplified guide rather than an exact statement of each account: each row compresses a family of accounts into one representative requirement, and each false positive is one example among others. The paragraphs after the table show, for reliabilism and virtue approaches, how accounts within a family differ. The false positives take as their example the case in which the system itself is claimed to know. When a human or a community comes to know through the system, each requirement in the middle column applies to the belief of that human or community, and the system's output bears on it as evidence or as part of the process that formed it; a post-hoc rationale, for instance, can then be a reason that the human really has, if the argument it gives is sound (Section 4.3).

Table 2.1. What follows from each epistemological approach: a simplified overview

If the adopted account emphasizes...	Then what knowledge requires, and what must be established to support attributing it, is...	A characteristic false positive
Accessible reasons	reasons available to and usable by the relevant subject	a persuasive post-hoc rationale unrelated to the operative route
Evidential fit	an attitude proportioned to the evidence possessed by the subject	high confidence produced by selective retrieval despite contrary evidence
Process reliability	truth-conducive formation or maintenance of the subject's belief in the relevant environment	one benchmark score generalized beyond its task distribution
Modal safety or sensitivity	a belief, held by the subject claimed to know, that would not have been false in the nearby possibilities that matter, or that would not have been held had the claim been false	repeated success under near-identical or contaminated conditions
Competence (virtue-reliabilist)	success because of a stable epistemic ability attributable to the subject or system	a lucky run or a pipeline whose decisive selection was performed by hidden humans
Intellectual character (virtue-responsibilist)	inquiry that expresses traits such as open-mindedness and conscientiousness, and whatever plays the role of their motivation	hedging and self-critique reproduced in reports without governing how the inquiry is conducted
Defeat sensitivity	appropriate response to rebutting and undercutting information	a corrected user-facing report while the operative commitment remains unchanged
Knowledge-first functionalism	proper operation of a capacity or organized system whose function is generating knowledge	a conventional checklist that decomposes knowledge into a set of conditions, renamed "knowledge-first" without independent account of the function

The table leaves out three things. Basing is not a rival emphasis but a relation between a belief and its grounds that most of the approaches require in some form, so Section 4.3 takes it up before any choice among them. For that reason the rows for accessible reasons and evidential fit state only that the reasons or evidence are available and that the attitude fits them; that the belief be held because of them is added in Section 4.3. Social accounts have their own table (Section 6; Table 2.2), and inquiry- and understanding-centered approaches are examined in Chapter 3. The requirements in the middle column hold only if the corresponding approach is adopted; they are not to be added up into a single list of criteria that every artificial researcher must meet.

The reliability row shows what such a consequence looks like. Reliabilism can permit knowledge without reflective access to the reliability-producing factors (Goldman and Beddor 2021, §§1.1, 2.1). Reliability belongs to process types, however, and one token process instantiates many types with different reliabilities. Someone who reads a distant clock in a dim room, for example, forms the belief by vision, by vision in dim light, and by reading small, distant figures in dim light, and these types are reliable to different degrees; which type fixes the verdict is the generality problem, to which reliabilists have proposed several responses (Goldman and Beddor 2021, §2.3). For a belief formed through an artificial system, whether it is the system's own commitment or a human's belief that relies on the system's output, the generality problem means that a reliabilist verdict depends on how the producing process is typed: the model alone, the model with its retrieval and human

filtering, or the pipeline on a given task distribution. If hidden human filtering removes the model's errors, the whole may be reliable while the model alone is not. Choosing a benchmark is in part choosing such a type, and a benchmark score shows the reliability of one type on one task distribution, not of the system in general; Otsuka notes that in practice AI models are justified case by case through benchmark testing, so that their reliability has become attached to particular models (Otsuka 2026, "Understanding the Machine Virtues"). A claim that an artificial system is reliable therefore needs to state its unit: whether human filtering lies inside it, and on which distribution it was assessed.

Statistical testing supplies a worked instance of the generality problem. Otsuka argues that the stopping-rule problem, in which a single dataset leads to rejecting a null hypothesis under one experimental design and to retaining it under another, is a version of that problem. In Otsuka's example, a coin tossed twenty times comes up heads six times. Under a plan to toss it twenty times, the null hypothesis that the coin is fair cannot be rejected at the 5 percent significance level; under a plan to toss until the sixth head, it can, because the probability of a result at least as extreme as the one observed is 0.0577 under the first plan and 0.0318 under the second. The same sequence of tosses is thus an instance of two testing processes with different verdicts. One frequentist response Otsuka discusses accepts that the same experiment can be described as different tests and requires the experimenter to specify the testing process in advance and to follow it; a verdict is then justified only relative to that process, and switching to another design after seeing the data counts as hacking the test (Otsuka 2023, 99–103). Extended by analogy to an artificial system, the requirement would fix the process type under evaluation before results are inspected and would treat later re-typing as a reportable change rather than a neutral redescription.

Other approaches make other human interventions and evaluation failures salient. Virtue approaches place weight on a success that manifests competence; in Sosa's version, as John Turri, Mark Alfano, and John Greco present it, a success is apt when its correctness, and not merely its occurrence, is attributable to a competence exercised in conditions appropriate for it (Turri, Alfano, and Greco 2021, §5). Modal approaches ask whether the belief could too easily have been false or would have been held even if false. Sosa separates aptness from safety: neither, he holds, entails the other. On this account a performance can be apt though unsafe, when the performer's competence or the conditions for its exercise might easily have failed, and safe though not apt, when its success is secured without being owed to the performer's competence (Sosa 2005, §§8–9). Sosa also holds, as Turri, Alfano, and Greco report, that the competence to which a success is owed may be socially seated, that is, located in a social group rather than in an individual. The correctness of a belief received through testimony, on this account, is owed in small part to the recipient's competence to trust testimony and more fully to a competence seated collectively in those along the chain, as a touchdown pass manifests the quarterback's competence but more fully the team's; the recipient earns partial credit, which Sosa argues can suffice for knowledge (Turri, Alfano, and Greco 2021, §7). Seen through Sosa's distinction between aptness and safety, hidden human selection that corrects a model's errors can make the outputs safe without making their success creditable to the model. Whether a success is creditable to the model, to the model with its tools, or to a wider human–machine organization then depends on where the competence is taken to be seated, a choice parallel to the typing of processes discussed above. Otsuka draws a related conclusion for deep-learning models: because their reliability is established after the fact on benchmarks, model by model and relative to a test set, it comes to be credited to particular named models, and, while granting that reliabilism can accommodate such models, Otsuka suggests that virtue epistemology may offer a clue to their present justification (Otsuka 2023, 132–135).

4.3 Basing and the operative commitment

A subject can possess good reasons to believe *p* without believing *p* for those reasons. A researcher may, for example, have data that support a conclusion while holding the conclusion because a senior colleague asserted it. Epistemologists standardly mark this difference with two notions of justification. Propositional justification concerns whether a subject has sufficient reason to believe a proposition; doxastic justification concerns whether a given belief is held appropriately. One common way of relating them takes propositional justification to be the more fundamental, and doxastic justification to be a matter of the subject's belief being appropriately responsive to, or based on, that propositional justification. Ichikawa and Steup report that the precise relation is disputed but that it is uncontroversial that the two can come apart, as they do in the researcher's case, and that the traditional analysis of knowledge is typically thought to invoke doxastic justification (Ichikawa and Steup 2026, §1.3.3). The basing relation is the relation that holds between a reason and a belief when the reason is one for which the belief is held. It is generally thought to be a necessary, but not sufficient, condition for a belief's being justified, that is, for doxastic justification, and it separates the good reasons a subject has that contribute to the justification of a belief from those that do not (Korcz 2026). To ask about basing is thus to ask not whether good support exists, but whether the belief is held because of it.

This distinction lies on a different level from the approaches of Section 4.1. Several of those approaches, among them internalism, evidentialism, and reliabilism, are accounts of what justifies a belief (Section 4.1): reasons accessible to the subject, the evidence the subject has, a reliable process, and so on. Whichever account is adopted, the subject's having what it counts as justifying can be distinguished from the belief's being held because of it. On the common way of relating the two notions above, the former is propositional justification, and adding the latter, basing, yields doxastic justification. On reliabilism, what justifies is the process that actually caused the belief (Korcz 2026, §5), so the element of the belief's being held because of what justifies it is built into the condition of justification from the start. Basing is therefore not one more answer, alongside the approaches, to the question of what must be added to a true belief for it to be knowledge, but a relation asked about within each of them, and Table 2.1 left it out for that reason. The distinction between having good reasons for a conclusion and holding the conclusion because of them can therefore be taken up without first choosing among the approaches.

The evidentialists Feldman and Conee, for example, mark the distinction between having good reasons and believing because of them with two notions of their own. The first is justification: on their account, an attitude is justified when it fits the evidence the subject has. The second is well-foundedness: an attitude is well-founded when, in addition, the subject holds it on the basis of a part of the evidence that it fits, and does not ignore other evidence it has that would defeat that support (Feldman and Conee 1985, 24–25, 33 n. 21). The two notions are close to propositional and doxastic justification as distinguished above.

This subsection asks what that distinction requires of artificial research. Basing is a relation between reasons and the belief of the subject claimed to know, and, as Section 2.1 noted, the system that conducts research need not be that subject. Where knowledge is attributed to the system itself, what is at issue is the basing of the system's own commitment, and the question becomes which of the reasons found beside its conclusion actually produced it. As Section 2 noted, this book uses commitment as a word broader than belief; here it is a conclusion that a system holds, outputs, and may later revise. The title's operative commitment is the commitment that actually governs what the system outputs and how it later revises it, as opposed to a conclusion or reasons merely displayed or stored beside it. Where a human or a community is the subject, what is at issue is instead the basing of that subject's belief, and the system's output is among its evidence or sources; the social subject is taken up from Section 6 on. The subject relying on the output may also be another artificial system, such as an artificial member of a mixed or artificial *S*; what that recipient takes up from the output is then its own operative commitment, and the questions of this subsection arise on its side.

The following three cases differ in this respect. An AI report may contain excellent evidence for *p* while the system produced *p* because of an irrelevant shortcut; if the system is claimed to know *p*, what governs its operative commitment is the shortcut, not that evidence. A user may believe an AI conclusion because of institutional prestige while ignoring the evidence displayed beside it; here the subject is the user, whose belief is based on prestige rather than on the evidence. A repository may contain both a conclusion and its grounds without any operative process connecting them; here there is not yet a subject whose belief is based on anything, a situation close to the unread discovery of Section 7. In each case, the

co-location of support and conclusion does not establish that the attitude or output is appropriately based.

Where the system itself is claimed to know its conclusion, basing bears directly on how it is evaluated. What an evaluator usually has is the conclusion a system outputs together with the arguments, citations, and data presented with it. Examining these can show at most that the system has good reasons for the conclusion, the matter of propositional justification. Basing concerns the further matter of whether the system produced the conclusion because of those reasons, on which doxastic justification depends, and that cannot be read off the output. Since the traditional analysis of knowledge is typically thought to invoke doxastic justification, on accounts that impose a condition of that kind a persuasive paper does not show that the system knows its conclusion. The difference is close to the one between the first two of the adequacy questions that Chapter 3, Section 5.3 distinguishes: whether an account faithfully describes the process that produced an output, and whether it articulates why the available grounds support accepting the claim. Where the subject is instead a human who relies on the output, the same gap appears on the human's side: reading a persuasive paper supplies the human with reasons, and whether the human's belief is based on them or, as in the second case above, on the system's prestige is a further matter.

Where the system is the subject, the difference also shows in later revision. If a conclusion is not based on the reasons presented with it, learning that those reasons are mistaken need not lead the system to revise the conclusion. This is the false positive in the defeat-sensitivity row of Table 2.1: the user-facing report is corrected while the operative commitment remains unchanged. The change can also run the other way. Even if a conclusion first arose through a shortcut or by chance, if the system later obtains independent evidence and whether it keeps or revises the conclusion comes to depend on that evidence, then on some theories of basing the conclusion's status can change at that point. What is in question is not only how the conclusion first arose but what now keeps it in place and what it is revised in response to. In the case of the chance generator in Section 4.5, the same holds for the community's belief.

What, then, would support the claim that a system's conclusion is based on its evidence, or that the system revises the conclusion in response to evidence? Attaching documents to a conclusion after it has been generated only places support beside the conclusion. One means of showing it is a record of where the conclusion came from, its provenance: which inputs, models, calculations, checks, and interventions actually produced the system's operative commitment and governed its later revision. This requirement is stated in causal terms, as a matter of what actually governed the conclusion, and so it fits most directly the causal theories of the basing relation, on which a reason must cause the belief in an appropriate way. On doxastic theories, an appropriate meta-belief, a belief about the reason to the effect that it is a good reason for the belief, suffices for the belief to be based on it, and a belief may be based on a good reason even if the reason did not cause it in the appropriate way (Korcz 2026, §§1, 3, 5). On an account of that kind, what matters is not only what caused the conclusion but also whether something in the system plays the role of such a meta-belief. On either kind of theory, however, the mere co-location of support and conclusion is not basing, so the failure in the examples above is one that both can recognize. Three things can be distinguished here: the conditions under which basing obtains, the evidence for attributing it, and the means of leaving that evidence. A provenance record belongs to the third. It gives strong evidence for an attribution, but the absence of a record is not the absence of the basing relation. Tests of whether the conclusion changes when the inputs are varied, or is revised when a reason is defeated, can also be evidence for an attribution. Which evidence can support an attribution depends on the theory of basing adopted and on the tests that can be run on the system.

4.4 The evidence a system has and the retrieval store

Justification as evidential fit (Section 4.1) and basing on evidence (Section 4.3) both presuppose an answer to a further question: what evidence the subject has. Evidential fit asks whether an attitude fits the evidence the subject has; basing asks whether the attitude is held because of that evidence. This subsection asks neither of these, but the question both presuppose, for the case in which the system itself is claimed to know (Section 4.3). For such a system, part of what it can draw on may sit in a retrieval store such as a database, and whether what is in the store counts as evidence the system has can change the verdicts on both justification and basing. Feldman and Conee's two notions (Section 4.3) are likewise both relative to the evidence the subject has, but they grant that what it is to have something as evidence is unclear, for example whether propositions one could recall only with prompting count (Feldman and Conee 1985, 32 n. 2). For a system that draws on a retrieval store such as a database, that unclarity becomes a question of how the system is organized and evaluated. Suppose the store holds evidence against a conclusion, strong enough that the store as a whole does not support the conclusion to the degree expressed, but the system retrieves only documents that favor it and states the conclusion with confidence. If everything in the store counts as evidence the system has, the conclusion does not fit the system's evidence; this is the false positive in the evidential-fit row of Table 2.1. If only what was retrieved counts, the conclusion may fit the evidence the system has, and the fault lies not in how the conclusion is held but in how the evidence was gathered. How the gathering of evidence bears on epistemic evaluation is taken up in Section 5. The comparison yields one more point. On accounts that require evidential fit, widening the boundary so that the store counts as evidence the system has need not improve the system's epistemic standing: counterevidence that was outside becomes evidence the system has, so that unless the conclusion is revised the same output can lose its justification. Keeping more information at hand is not by itself an epistemic strength.

Williamson's account draws the boundary of the evidence a subject has in another place. On it, one's evidence is what one knows, not what one is merely in a position to know; counting the latter, Williamson argues, would let a hypothesis be probable on a body of evidence to which one has no access as a whole (Williamson 2000, 202–203). Knowing, on this view, is a state rather than an activity, and one knows many propositions without considering them; in one of Williamson's simple models of a creature that stores information in memory, to which Section 8.2 returns, the creature is stipulated to know everything recorded in its memory although retrieval takes time (Williamson 2000, 199, 226–227). Williamson's account, if adopted, would turn the question of how the retrieval store is organized into whether the system knows what its store records or is only in a position to come to know it.

Whether a store that a system relies on can hold part of what the system believes is a question disputed over extended cognition, that is, over whether cognition can extend to tools and records outside the brain or the system, which Chapter 5, Section 6.1 sets out without deciding it. The epistemology of memory meets the same question for memory knowledge. As Matthew Frise reports, J. Adam Carter and Jesper Kallestrup argue that information stored in a notebook can be part of a subject's dispositional memory knowledge, so that the subject does not merely relearn it on access, whereas Kourken Michaelian argues that this case rests on a functional similarity that fails: it requires biological memory to be purely preservative, while on the view that now prevails, remembering is not simply reproducing a record but generating new content (Frise 2023, §§1.1, 3.2). Carried over to a system that contains both a component that generates content rather than returning stored records and a store that returns what was deposited, as a trained model with a retrieval store does, the contrast Michaelian draws between biological and external memory can fall inside the system, so that whether the store's records count among what the system knows would not be settled by drawing the system's boundary around the store. What would bear on it instead is how the records enter what the system does. On the functional accounts of belief in Section 2.1, a stored sentence is only weak evidence of belief, and stable integration with inference, action, and revision is stronger evidence. A record that the system can retrieve but that shapes neither its inferences nor its revisions would then be closer to a notebook kept for reexamination (Chapter 5, Section 6.1) than to part of what the system believes or knows. An evaluation that credits a system with knowing what its store contains would accordingly need evidence of that integration, not only of retrievability. Section 7 meets the corresponding question for a society, whose infrastructure can hold a result that no member has read.

4.5 Generating, knowing, contributing, and conducting research: the case of an accidental generator

A limiting case separates predicates that need to be kept apart when a result comes from an artificial source. Suppose a random generator happens to emit a novel true theorem. That event creates a true candidate content. If nobody can distinguish it from the generator's overwhelmingly many false outputs, it does not yet create knowledge in the demanding sense. If a competent mathematical community later proves the theorem, or checks a proof obtained by another route, and so comes to know it, the accidental origin of the candidate need not make the community's subsequent knowledge accidental. One role of the justification condition in the traditional analysis was to rule out lucky guesses as knowledge, and Gettier cases suggest that even justified true beliefs can be lucky in a way inconsistent with knowledge (Ichikawa and Steup 2026, §§1.3, 3). The luck at issue in either case lies in the connection between a belief and its grounds or its truth, not in how the candidate came to be considered. This is the familiar point that the route that generates a candidate need not be the route that warrants accepting it (Chapter 3, Section 5.1; Chapter 5, Section 4); here what must be non-accidental is the relation between the community's belief and its proof.

The generator can then be causally important to knowledge production without possessing knowledge, conducting a well-organized inquiry, or manifesting a stable research capacity. The case separates six predicates: generating a true content, knowing it, contributing to another subject's knowledge, conducting research, possessing research ability, and deserving credit for an epistemic achievement. What matters is keeping those predicates apart. Some authors separate generating a true content from contributing even where the content is correct and checked. Weatherall and Wolfson, in a draft on the epistemic value of proof, argue that formal correctness is neither necessary nor sufficient for a proof to have epistemic value, which they locate in the activities by which mathematicians produce, check, and come to understand proofs rather than in the artifacts; if so, "the mere generation of correct proofs is not, in itself, a contribution to mathematics." They distinguish proof assistance, in which computers help prove theorems that mathematicians have stated and want established, from computational proof generation, and they hold that a machine-generated proof, however sound, contributes to mathematics, which they treat as a group activity, only insofar as it prompts human mathematicians' engagement with it (Weatherall and Wolfson 2026, 4, 13–15, 35–38). DeDeo adds a point about judgments of value rather than of truth: the judgment that one theorem matters to another rests in the end on someone's experience of the impasses the first resolves, and automated methods hide such experiences behind mechanical processes, leaving proxies for value without that experience behind them (DeDeo 2024, §2). The random generator separates the six predicates for a content whose truth is accidental; on these accounts, written for mathematics as a social activity, generating a correct, checked result and contributing to mathematics can also come apart. Both take that activity to be one of human mathematicians, and Weatherall and Wolfson picture it as human discovery. Where *S* is mixed or artificial, whether the same separation holds turns on whether engagement with a proof, and on DeDeo's account the experience of impasses on which judgments of value rest, can be realized by its artificial members; neither takes up a community whose members are artificial. The stronger question of researcher status belongs to the analysis of attribution in Chapter 7, Section 3.2.

The case also shows how the working characterization applies. It counts as research an attempt to bring about an epistemic achievement new to *S* (Chapter 1, Sections 1 and 3). The generator's output contributed causally to the community's later knowledge, but a causal contribution to an achievement is not by itself an organized attempt at it, so the generator's run does not become research because a theorem it emitted was later proved. Whether the generator contributed, whether the community's inquiry was research, and who bears the resulting knowledge are therefore separate attributions.

What verification contributes

Research practice can be narrated as proposing an answer and then verifying it. Many epistemological accounts describe knowledge through truth and some epistemically appropriate relation between a subject and the content. One might then map the two vocabularies onto each other: verification produces justification, and justification converts a belief into knowledge. This section examines that mapping in three moves. It first distinguishes verification, an activity, from justification and the other statuses a result may have: a verification can succeed without its result having the status in question. It then takes up a divergence in the opposite direction: a conclusion can be well supported by the evidence at hand while the inquiry that gathered that evidence was badly conducted by standards other than those that govern the result's status. Finally, it returns to the section's question, what verification contributes to a knowledge attribution, by dividing the route from a test to a possible knowledge attribution into steps. Verification directly supplies the first two of them, a result and its evidential bearing; for that to change an attitude takes a basing relation (Section 4.3) or an updating relation, and for the attitude to be knowledge takes whatever the adopted account requires (Sections 4.1 and 4.2). Any of the steps can fail. Where a result is unfavorable in particular, Duhem's point that a hypothesis is tested only together with auxiliary assumptions, and the epistemology of defeat, show what is left open about which element it bears on. Section 5.1 turns to the heterogeneity of verification, Section 5.2 to confirmation, what a result supports and how far, and Section 5.3 to statistical methods as a bounded illustration.

This book uses verification in a broad sense: **verification** refers to a family of activities through which investigators try to determine whether a candidate answer should be treated as true. The word is not used to promise conclusive establishment. Different disciplines and literatures instead speak of testing, confirmation, corroboration, proof, validation, audit, replication, or criticism, with different objects and success conditions. The important common function is assessment of a candidate's epistemic standing under uncertainty, not one universal procedure.

Verification is something investigators do; justification, warrant, reliability, safety, and knowledge name possible statuses or relations. Warrant is used here broadly for the epistemic support for accepting a claim: the justification of a belief is one form of it, but warrant is also said of results and of the acceptance of claims. An explicit test is not universally necessary for justified belief: perception, memory, testimony, and dependence on an institution can provide epistemic support. Nor is a passed test sufficient. A test can target the wrong quantity, depend on a defective instrument, be analyzed through an unsuitable model, leak its answer, or be interpreted beyond the conditions it supports. The high-gain antenna of the Galileo spacecraft, for example, had passed deployment tests in air and in vacuum, but in air the exposed titanium of its pins kept reacting with oxygen, so that friction never rose high enough to stop deployment, and in the vacuum test the pins and sockets had not moved against each other. In space, vibration from the upper stage after launch did move them against each other in vacuum; the bare titanium pins galled against their sockets and held some of the ribs, so that when the antenna was commanded to open in 1991 it opened only partly. The report of the later investigation concluded that the test conditions had not been adequate to reveal the problem (Johnson 1994, 375–377).

The distinction between verification and justification showed that an activity can succeed without its result having the appropriate status. The gathering of evidence also allows a divergence in the opposite direction. A conclusion can fit the available evidence while the inquiry that produced that evidence was selective, prematurely stopped, or organized so that inconvenient observations were unlikely to appear. Conversely, responsible evidence seeking can end in suspension because the available evidence remains inadequate. Contemporary work on the norms of evidence gathering argues that these diachronic questions should not be reduced to the synchronic question of whether an attitude fits the evidence presently possessed (Flores and Woodard 2023). The distinction need not be built into every theory of justification to matter for research. It shows why a well-supported result and a well-conducted inquiry are not the same achievement. For an artificial system, retrieval policy, source exclusion, measurement choice, active information seeking, and stopping rules can therefore remain epistemically consequential even when the final inference is impeccable conditional on its input.

Of the items just listed, stopping rules are the one whose epistemic consequence is itself disputed, since on one view the rule by which an experiment was ended makes no difference to what its data show. Romeijn sets out both sides. Under the likelihood principle, which standard Bayesian inference satisfies, all the information that matters to inference about hypotheses is contained in the likelihood function, the probability of the observed data under each hypothesis, so the rule that ended an experiment does not change what the data imply. Frequentists resist the principle because the error rates that justify

a test's verdict depend on the design, so that the same data can warrant rejection under one design and not under another, as in the stopping-rule problem described in Section 4.2 (Romeijn 2025, §§3.2.2–3.2.3). The point about evidence gathering, however, does not require settling that dispute: even if a stopping rule leaves the import of the collected data unchanged, it can affect what evidence was collected and whether the inquiry was responsibly conducted.

Whether responsible conduct of inquiry is itself an epistemic standard is a further dispute (see also Chapter 4, Section 5). Read through each of the views that follow, the difference between a well-supported result and a well-conducted inquiry nonetheless survives. Feldman and Conee, whose evidentialism Section 4.1 introduced, grant that gathering evidence responsibly has some epistemic merit but deny that the justification of the resulting attitude turns on it. In an example they take from Kornblith, a physicist who fails even to hear a colleague's devastating objection remains justified in believing their theory, provided that they were justified before and their evidence is unchanged; what is open to criticism is their character (Feldman and Conee 1985, 21–22, 31–32). Thorstad denies that there are distinctively epistemic norms of inquiry at all. Thorstad argues that the considerations supporting distinctively epistemic norms for belief do not carry over to inquiry, which is an activity, so that whether to gather more evidence, double-check, or stop is settled by all-things-considered reasons, among which the prospect of truth or knowledge is one (Thorstad 2022, §§3–5, 6.3, 7).

Kauppinen goes further. Where Thorstad counts epistemic considerations among reasons that combine into an all-things-considered ought, Kauppinen argues that all reasons to pursue a question are practical, transmitted from reasons for what answering it would help one do, so that there are no epistemic norms of inquiry, and that where an inquirer does deserve epistemic criticism for how evidence was gathered, the fault lies in a standard epistemic norm on belief or action. In a case Kauppinen adapts from Flores and Woodard, someone consults a single source that is in fact unreliable and happens to reach a true answer. On Kauppinen's account this person is epistemically criticizable without appeal to any norm of inquiry: they plausibly violate an epistemic norm on action, if there are such norms, by treating the source's reliability as a reason for action without knowing it or being justified in believing it, and Kauppinen allows that the fault may lie instead in their holding that belief without evidential support. In a variant in which the person has good but misleading evidence of its reliability, they are not criticizable (Kauppinen 2026, §§3–4).

What differs among these views is whether the second assessment, that of the inquiry, is epistemic, and so whether the requirement of responsible evidence acquisition in Section 9.1 is an epistemic or a practical one. If it is epistemic, decisions about search and stopping are assessed by standards that concern truth or knowledge, independently of what the result is used for. If it is practical, how far to search and when to stop is settled in light of the purposes of the inquiry, its costs, and what is at stake, so that the same stopping point can be appropriate for one purpose and not for another. Where the inquirer is an artificial system, those purposes and costs need not be its own: they may have been fixed by whoever set its objective and budget, or by the system where it sets them itself (Chapter 7, Section 4.3), and whose purposes govern how far inquiry goes is part of the question of allocation (Chapter 10). On Kauppinen's view it would be practical, while any unsupported assumption about the reliability of sources on which a system acts in gathering evidence would remain open to epistemic criticism: criticism of the system, if it is itself a subject of such norms, and otherwise of those who built the assumption into it.

Leaving aside the assessment of the inquiry as a whole, we can return to the section's question of what verification contributes to a knowledge attribution. The bridge from a test to a possible knowledge attribution can be represented without pretending that its arrows are automatic:

```
target claim H + method M + auxiliary commitments A
    -> outcome R

R + background knowledge and evaluative standards N
    -> evidential bearing E on H or another commitment

E + a basing or updating relation B
    -> changed epistemic attitude D

status of D under the adopted account of knowledge
    -> possible knowledge attribution
```

Read line by line, the schema separates four steps. In the first, a target claim H is tested by a method M under auxiliary commitments A, such as the instruments, conditions, and preprocessing on which the test relies, and the test yields an

outcome R. Suppose, for example, that H is the claim that a model has some capability, M is a benchmark, and A includes the assumptions that the benchmark measures that capability and that its items did not leak into the model's training data. In the second step, the outcome becomes evidence only when it is read against background knowledge and evaluative standards N, such as what score would be significant and how other models perform; it then bears on H, or, if contamination is suspected, on A instead. In the third step, evidence changes an attitude only if someone's commitment actually changes because of it, through a relation B, such as the basing relation of Section 4.3 or the updating by which degrees of belief are revised in response to evidence. In the fourth, whether the changed attitude D is knowledge depends on what the adopted account requires of it (Sections 4.1 and 4.2). The third and fourth steps concern the attitude of the subject claimed to know: the system's own commitment where knowledge is attributed to the system, and the belief of the humans or the community where they come to know through it (Section 4.3).

The last line does not assemble knowledge from conditions shared by all the accounts. On knowledge-first views, the preceding attitudes and their grounds are assessed in relation to knowledge, rather than combined to define it. Further, the last line speaks of a possible attribution partly because attributing knowledge is something an evaluator does, and insufficient evidence for an evaluator to attribute knowledge does not by itself show that the subject lacks knowledge. The schema locates questions for an assessment; it is not an algorithm for settling them.

Every arrow is a substantive site of failure. The Galileo spacecraft case above concerns the first two arrows, and Sections 5.2 and 5.3 return to the second, for confirmation and for statistical methods; what follows concerns the second and third arrows when a result is unfavorable. The method and auxiliaries help produce an outcome, but the outcome does not carry its own epistemic address. A failed prediction may bear on the focal hypothesis, a boundary condition, an instrument model, a preprocessing choice, a background theory, or some combination. Duhem's analysis of physical testing makes the limited but crucial point that an isolated hypothesis is not what confronts an experiment; a negative result implicates a larger theoretical arrangement without itself identifying the faulty element (Ariew 2026). The later process analysis will examine how investigators decide where to revise (Chapter 6, Sections 2 and 4). Here the point is that a verification outcome cannot be mapped mechanically to a change in the main hypothesis.

The epistemology of defeat, which studies how information can take away the justification a belief had, does not settle which element is at fault, but it makes the point finer by distinguishing the ways in which contrary evidence can bear on a claim. In the distinction Pollock drew, rebutting evidence favors the denial of the claim: a careful replication that finds no effect rebuts the claim that a drug works. Undercutting evidence instead gives reason to doubt that the usual connection between a claim and the grounds offered for it holds in the case at hand, without favoring the claim's denial: learning that a sample was biased undercuts the inference drawn from it, although it does not show the conclusion false (Koons 2025, §1.1). Higher-order evidence bears not directly on the claim but on the investigator and the investigator's evidence: it can make the investigator's own competence, reliability, or evidential situation doubtful, as when an analyst learns that the analysis was carried out under conditions known to produce errors (Roush 2017, §3). How much confidence in the first-order claim such evidence rationally requires giving up is disputed. As Roush reports, Feldman and Horowitz hold that one cannot rationally be confident that p while believing that one's evidence does not support p. On Feldman's view, the belief "p, but my evidence does not support p" cannot be both true and reasonable, since its second half undermines the reasonableness of the first. Horowitz argues that a subject who stays confident in this way can explain having been right only as luck. Williamson, Weatherson, and Wedgwood argue instead that such level-splitting can be rational where the doubt concerns the subject's own competence. On their respective views, a proposition can be highly probable on one's evidence while it is also highly probable that its probability is low; a norm of rationality does not cease to apply merely because the subject has evidence of having failed to follow it; and a support relation that actually holds can justify a belief whether or not the subject believes correctly about it (Roush 2017, §3). What counts as an appropriate response in the defeat-sensitivity row of Table 2.1 differs accordingly.

Carried over to an artificial researcher, Duhem's point means that a scalar loss attached to the headline hypothesis cannot by itself exhaust the diagnosis of what an adverse result gives reason to revise, since an instrument model, a sampling assumption, a preprocessing step, or a background regularity is also a candidate for revision. A configuration that represents the loss as a scalar while keeping the candidates for revision separately is compatible with this. That much follows from Duhem's point alone. The distinctions of defeat add two further targets. Undercutting evidence, without favoring the denial

of the conclusion, gives reason to doubt the support relation between a claim and its grounds; higher-order evidence can call for revising the estimate of an evaluator's competence, whether the investigator's or a validator's. Defeat is also not confined to adverse results: it can arise after a test has been passed, as when a sample is later found to have been biased. These distinctions do not, however, supply a procedure for locating the defeated relation. Locating it still requires domain knowledge, experimental skill, and possibly further intervention. What was earlier counted as evidence may itself need revising. Williamson argues that standard Bayesian updating, on which a proposition once conditionalized on keeps probability 1 thereafter, cannot represent later evidence undermining something that had been evidence, and proposes keeping a record of evidence separate from the probabilities, so that a proposition can lose its standing as evidence (Williamson 2000, 214, 219–220). On this argument, a system whose updates can only add evidence could not represent such a loss.

What these distinctions require of a system depends on who constructs the verification and who decides, after an adverse result, where to revise. If the system runs checks that others designed, whether human investigators or other artificial agents, and they read the results and decide what to revise, the work of determining which element a result bears on stays with them; for them to do it, the system's report has to show not only a pass or fail on the headline hypothesis but what was tested under which auxiliary assumptions. Where those others are artificial agents, what the report has to show extends to how far they are independent of the system, for example whether they and the system are instances of one base system, since a critic adds to error detection only so far as it does not go wrong where the producer does (Chapter 11, Section 3). If the system itself decides what to test, what counts as passing, and where to revise, these distinctions become a task the system takes on. Chapter 6, Section 4.2 discusses that task for research done by a machine, and Chapter 7, Sections 4.2 and 4.3 distinguish, as comparisons of autonomy, who supplies selection, evaluation, and correction, and whether a system only applies supplied standards or also selects, constructs, and revises them. This is a different axis from the question of who the subject of knowledge is. Where others come to know through the system and the system constructed the verification, they need to be able to read from its reports how it carried out that task if they take on the diagnosis of results themselves. On accounts that require accessible reasons, a recipient needs reasons it can use, but these may concern how the result was checked or, as on reductionist accounts of testimony, the system's reliability for the use at hand (Chapter 11, Section 5.2); the system's reports are then one source of such reasons. On accounts on which the reliability of the process suffices, and on Sosa's account (Section 4.2) applied to this case, on which the correctness of a testimonial belief is owed more fully to competence seated in the chain of transmission, a recipient can know without being able to read the whole structure of the verification. What then supports the attribution is not that the reports can be read but evidence that the system's way of verifying is reliable. Where knowledge is attributed to the system itself, it is further required that its operative commitment actually change in response to such revisions (Section 4.3).

Putting this together, what verification contributes to a knowledge attribution can be stated as follows. An outcome's evidential bearing on a specified research claim depends on relations involving measurement and whether it captures what is to be measured, sampling, interpretation, and the inferential target. Available evidence changes an attitude only through some actual basing or updating relation; the attitude is the system's operative commitment where knowledge is attributed to the system, and, where others come to know through it, their belief, or their operative commitment if they are artificial systems. A well-supported attitude becomes knowledge only if it is true and satisfies whatever additional conditions the adopted account imposes.

5.1 Verification is heterogeneous but not therefore functionless

The start of this section treated verification as a family of varied activities with a common function. This subsection shows that the variety does not empty the common function, and that the common function does not erase the differences among the means.

A verification event can generate evidence, test whether evidence supports a claim, expose a defeater, estimate the reliability of a process, question whether what is being tested is really the intended target, or make a local result available for social criticism. It can leave confidence in the focal claim unchanged while sharply reducing confidence in the instrument or method. It can also reshape the claim being assessed.

The functional commonality does not erase differences in epistemic force. An analogy, pilot result, randomized trial, independent replication, deductive proof, historical source comparison, and interpretive criticism can all alter what is taken seriously. They differ in what they bear on, the alternatives they discriminate, their defeasibility, and the kind of standing they can provide. Calling all of them belief updates would describe one effect while losing the reason their effects differ. For an artificial researcher, this means that a stage among its steps named verification, or a pass or fail reported for that stage, does not by itself say what was checked. Chapter 6, Section 3.1 compares checks by two of these differences: what part of the question an operation bears on (its target), and how firmly it settles that (its force).

Verification also need not follow generation as a separate phase. Candidate formation already uses prior evidence, coherence, analogy, preliminary checks, and expectations about what could survive scrutiny. A mathematical proof can construct a result and display its warrant in the same object. Criticism can change the content rather than certify it. Seen broadly, proposing and assessing are different roles. Seen more finely, they are mixed together in a single course of gradually reorganizing what is taken as settled, and how far. Chapter 5, Section 4 develops that temporal and practical structure.

5.2 Confirmation: what a result supports, and how far

The second arrow of the schema, from a result and background standards to evidential bearing, is the relation that confirmation theory studies: how, and how far, evidence supports a hypothesis or tells against it. On a classification that Crupi calls relatively standard, the main approaches fall into three families. One is confirmation by instances, as in Hempel's theory. Another is hypothetico-deductivism, on which evidence confirms a hypothesis that, together with auxiliary assumptions, entails it. Proposals to strengthen it include requiring that the evidence have been predicted in advance (predictivism), that it not have been used to construct the hypothesis (use-novelty), that it would have been unlikely had the hypothesis been false (eliminativism, of which appeals to severe testing are a mark), that the hypothesis account for it in a simple, unified way (simplicity), or that the hypothesis explain it (explanationism). The third consists of Bayesian theories, which represent support with probabilities (Crupi 2025, §§1–3). Confirmation theory concerns a relation, the support that evidence lends to hypotheses, rather than verification as an activity. Activities in the verification family that try to determine whether a result supports a hypothesis, such as testing and replication, ask about this relation, but the family also includes proof and interpretive criticism, whose work is not exhausted by it (Section 5.1). Several disputes among the approaches of confirmation theory appear elsewhere in the book: prediction against accommodation and severe testing in the appraisal of candidates (Chapter 5, Section 4.2), the Bayesian and error-statistical treatments of where a failed prediction places blame (Chapter 6, Section 2.3), the independence of checks (Chapter 6, Section 3.2; Chapter 11, Sections 2 and 3), and the contrast between Bayesian and frequentist statistics (Section 5.3). This subsection takes up one distinction that those discussions leave implicit, and what it and some related results imply for artificial research.

Within Bayesian confirmation theory, Crupi distinguishes two notions of confirmation. On confirmation as firmness, a hypothesis is confirmed to the extent that it is probable given the evidence. On confirmation as relevance, evidence confirms a hypothesis to the extent that it raises the hypothesis's probability. A positive result on a test for a rare disease can raise the probability of the disease while leaving it low: on the relevance notion the disease is confirmed, and on the firmness notion it is not. Crupi reports Salmon's description of the distinction as "extremely fundamental" and yet "sometimes unnoticed", and notes that it has had to be stressed again and again, in philosophy and also in artificial intelligence and the psychology of reasoning (Crupi 2025, §3.3). In Crupi's presentation the probabilities represent states of belief. Here they can be read as those of whoever assesses a claim, so using them does not presuppose that an artificial system itself has degrees of belief (Sections 2 and 2.1; Chapter 6, Section 1).

For artificial research the distinction bears first on what a passed check reports. On the Bayesian notions, a passed check can show that its result raised the support for a claim, while how well supported the claim then is also depends on how probable it was before the check. Crupi does not discuss this case, but applied to systems that generate candidates cheaply and in large numbers, the distinction has a consequence. If most of the candidates generated are false and nothing distinguishes one from another before the check, each is improbable before it is checked, and a check that a false candidate passes with a fixed probability leaves a passed candidate less firmly supported than the same pass would leave a candidate in an inquiry that tries a few already plausible ones. The consequence holds for whoever is claimed to know. Where the system itself is, it concerns how far the pass supports the system's own commitment. Where others come to know through the system, a report

that a candidate passed is part of their evidence, and someone who reads the report without knowing how many candidates were generated, and how many of them were likely to be false, can mistake how far the pass supports the claim. Eliminativist and error-statistical views assess a pass differently, by how unlikely it would have been had the hypothesis been false (Crupi 2025, §2.4). On either view, a report that a hypothesis passed a check leaves open, by itself, what the pass supports and how far; on the argument from inductive risk, on which the more harmful an error would be the more support acceptance requires, whether the support is enough for a given use is a further judgment (Chapter 11, Section 4). The force of a check in Chapter 6, Section 3.1 can accordingly be reported in more than one way.

Four further points in Crupi's survey bear on how such claims are checked. First, on naive hypothetico-deductivism, for any hypothesis and any evidence consistent with it there are auxiliary assumptions under which the evidence confirms the hypothesis. Crupi connects this with Duhem's point, and notes that many particle theorists of light, who reshaped their auxiliaries to recover the facts that wave theorists had deduced, did not claim those facts as confirmation (Crupi 2025, §2.3). Applied to a system that can choose preprocessing steps, exclusions, or model components after seeing its data, this means that such a system can, to the extent that its choices are unconstrained, often produce agreement that naive hypothetico-deductivism would count as confirmation. On predictivist and use-novelty views, whether those choices were fixed before the data were examined becomes part of what a claim of confirmation has to show (compare the testing procedure specified in advance in Section 4.2 and the provenance record of Section 4.3); on eliminativist views, what has to be shown is rather whether, with those choices made, the agreement would have been unlikely had the hypothesis been false (Chapter 5, Section 4.2); on Bayesian treatments, the auxiliaries are instead weighed by how probable they are (Crupi 2025, §3.5). Second, on the use-novelty view, data that served as a constraint in constructing a hypothesis cannot also confirm it; Crupi cites cross-validation, following Schurz, as a routine statistical application of the idea, and reports that the view is itself disputed (Crupi 2025, §2.4). On this view, agreement with the benchmark on which a system generated or tuned its candidate confirms it little or not at all, which is one reason evaluation on routes withheld from selection matters (Chapter 11, Section 8). Third, on the relevance notion, if a result confirms a hypothesis, the result's failure to occur disconfirms it, so an assessment that would count either outcome as support, a judgment of the "heads I win, tails you lose" kind, is inconsistent on that notion: at most one of the two outcomes can confirm the hypothesis (Crupi 2025, §3.3). An assessment of an artificial researcher's claim can accordingly ask what result would have counted against it. Fourth, on the relevance notion, evidence that was less probable beforehand, as more varied evidence often is, confirms a hypothesis that entails it more strongly (Crupi 2025, §3.6). Agreement among repeated runs of one base system is likely whether or not the hypothesis is true, to the extent that the runs can go wrong in the same places, so it is hard to count as varied evidence in this sense (Chapter 11, Section 5.4). Weisberg and Reisman distinguish three levels at which a result of modeling can be robust: surviving changes to a model's parameters, to its mathematical structure, and to the whole representational framework in which it is framed (Weisberg and Reisman 2008, 9–11, 15–17, 25). Where runs or instances share one modeling framework, their agreement could at most show robustness within that framework, not the robustness that varying the framework tests.

5.3 Statistical methods as a bounded illustration

The heterogeneity of verification seen in Section 5.1 does not appear only between fields such as history and the natural sciences. Within a single family of methods, statistics, what is taken to support a conclusion differs from method to method. Otsuka argues that statistical derivations can proceed as mathematics without settling what their probabilities mean, but that the interpretation adopted limits what their results can meaningfully be said to show (Otsuka 2023, 42–43, 176–178). Otsuka treats statistical inference as resting on ontological assumptions—a uniformity of nature behind the data and, in parametric inference, a chosen family of distributions—and reads its schools as differing in their concepts of justification (Otsuka 2021, 3–7, 13).

On Otsuka's reading, Bayesian updating justifies a degree of belief by valid inference from an agent's other degrees of belief, and so resembles internalism, while attempts to secure its priors as a starting point resemble foundationalism, the view that some beliefs are basic, needing no further justification, and serve as the ultimate resource for justifying all others (Otsuka 2023, 53–57, 61–66). On Otsuka's account, a Neyman–Pearson test can instead justify a verdict through the reliability of the testing procedure. Nozick's two tracking conditions require that, had p not been true, the subject would not have believed p, and that, had p been true, the subject would have believed p; the first is the sensitivity of Section 4.1. Otsuka reads a test's confidence coefficient, the probability of not rejecting a true null hypothesis, as a measure of how far the test meets the first

condition with respect to the alternative hypothesis, and its power, the probability of rejecting a false null hypothesis, as a measure of how far it meets the second, and so aligns testing with reliabilism (Otsuka 2023, 88–94). Otsuka reads model selection by the Akaike information criterion as pragmatist, because the criterion assesses a model by expected predictive performance rather than by closeness to the data-generating process, and suggests that virtue epistemology may offer a clue to the present justification of deep-learning models (Otsuka 2023, 118–124, 129–135).

Otsuka presents these correspondences as a simple, idealized model rather than a perfect analogy, and denies that the comparison makes Bayesian statisticians internalists or internalists Bayesians (Otsuka 2023, 55–56, 179–180, 182). Otsuka also argues that neither Bayesian nor frequentist statistics can rest content within the single epistemology it resembles. On Otsuka's account, posterior model checking tests the whole network of prior, likelihood, design, and measurement assumptions against data, which takes Bayesian practice beyond relations among the agent's degrees of belief and away from the foundationalist picture (Otsuka 2023, 67–72). The error rates that justify a test's verdict, Otsuka argues, depend on distributional and design assumptions that the verdict itself does not check, so a frequentist who takes responsibility for the reliability of the process must scrutinize those assumptions by other means (Otsuka 2023, 95–98, 106).

The more durable lesson is that a formally correct operation does not determine its epistemic role. Otsuka notes that a computed posterior distribution or a low p-value does not by itself say anything about the phenomenon under study (Otsuka 2023, 105). On the frequentist view, moreover, the same data can warrant rejecting a hypothesis under one stopping rule and not under another, which defenders of the likelihood principle deny (Sections 4.2 and 5). Priors, likelihoods, sampling plans, stopping rules, loss functions, error criteria, measurement models, and target definitions can each alter what conclusion is warranted. Where an artificial system that chooses or applies a method is itself said to conclude, what it is warranted in concluding then depends on how it relates the procedure to the question, data-generating conditions, assumptions, and costs of error. Where others draw conclusions from its output and assess for themselves how far that output supports them, they need to see how the system related the procedure to these; on accounts on which the reliability of the process suffices, a recipient can know without seeing this, but an evaluator who supports the attribution needs evidence that the system's way of doing so is reliable (Section 9.1). Whether the system can criticize and revise the governing standards is a further question about epistemic self-government (Chapter 7, Section 4.3), not a prerequisite for every use of statistics.

What is social about knowledge: the reference for novelty, production, possession, and retention

Sections 2 to 5 asked what the attitude of whoever is claimed to know must be like. In research, knowledge is also social, and in more than one sense: the reference relative to which it is new is social, its production can be collective, and its possession may or may not be. This section separates these senses. It first takes up the reference *S*, then separates production from possession through a distinction Fagan draws, extends that distinction into six relations, including retention, and compares accounts of social knowledge across them (Table 2.2). It then asks what these accounts imply for artificial research, by the combination of *A* and *S*. It ends by separating three claims about the social that differ in strength, so that the social character of knowledge is not read as making truth a matter of agreement.

The variable *S* arose from the observation that Chapter 1, Section 6 made with its examples of learning and rediscovery: in the case of knowledge, when research is said to have produced new knowledge, the novelty claimed is not the novelty of one person's private learning. For knowledge claims, *S* accordingly has the two connected roles introduced in Chapter 1, Section 1. It is the reference relative to which the knowledge sought, once realized, is new, and it identifies the society for which that knowledge is sought. Neither role attributes the knowledge to *S*; who knows it, and whether and how it has become *S*'s, are left to the accounts compared below. Both roles presuppose that *S*'s epistemic situation could come to include knowledge of the kind sought (Chapter 1, Section 2), but neither tells us how a society can know. Substituting a group for the individual variable in X knows p does not turn it into one large person or settle whether group knowledge reduces to members' knowledge.

In ordinary discourse, the unstated reference can be humanity and, in practice, a relevant specialist community standing in a wider network of testimony, education, records, instruments, standards, and institutions. Since what is already known differs from one society to another, however a society's prior epistemic situation is construed (Table 2.2), a claim of novelty can be

assessed only when the reference is explicit. A proposition can be new to one laboratory and familiar to another, new to a field and established in a neighboring discipline, new to humans and familiar to an artificial community, or new to one artificial society and not another. The difference need not concern novelty alone. In an ethnographic replication of *Laboratory Life* conducted at the same institute, Havstad, for example, suggests that one sentence from a review by the laboratory under study—that a network of target genes defines the physiological response—would be treated at the time of the study as close to established fact within that laboratory, as contested by rival laboratories, and as speculative or false by scientists outside the field (Havstad 2020, 25–26). What differs across these groups is the standing accorded to the sentence, a standing that this chapter distinguishes from knowledge.

The other two senses, production and possession, are separated by Fagan, who distinguishes two theses that the simple formulation can otherwise conflate. The first is that groups can be necessary to **produce** scientific knowledge. Fagan identifies three respects in which they can be: in many cases of inquiry, groups are required to satisfy methodological norms, to develop theoretical concepts, or to validate the results of inquiry as scientific knowledge (Fagan 2012, abstract). The claim concerns many cases of inquiry, not all. The second, further thesis is that groups **possess** scientific knowledge in their own right. Fagan judges the support for it "more equivocal": the cases offered for it can also be explained in terms of "relational complexes of members' beliefs" (Fagan 2012, abstract). As Longino reports, Fagan disputes Bird's account in favor of a relational view, on which shared knowledge is not knowledge held by a social entity existing over and above the individuals who constitute it but a function of epistemic relations among those individuals (Longino 2025, §5). The distinction between the two theses gives claims that knowledge is social analytical force without requiring an immediate commitment to a group mind.

Acceptance is one point at which the two theses come apart. A community's acceptance can be what validates a result, and so belong to production in the third of Fagan's respects; whether it also shows that the community possesses the result as an irreducibly collective belief is a further question. Duygu Uygun Tunç argues that a research collaboration's public affirmation of a result, as in a publication or an announcement, is distinct from the distributed process that produced the evidence for it: the same research process can be combined with dictatorial, majoritarian, or consensus-based procedures for affirming its results, and at CERN, on her report, decisions to publish are taken by dedicated publication committees and are not a function of the members' beliefs (Uygun Tunç 2023, 10 and n. 18). She also notes, without relying on it, a position on which groups cannot have beliefs but can accept propositions as the view of the group, on the ground that beliefs are formed automatically and involuntarily while group views are not (Uygun Tunç 2023, 10 n. 16). That ground matches the distinction Chapter 3, Section 6 takes up from Elgin, on which belief in Cohen's sense is largely involuntary and spectatorial while acceptance is voluntary and agential (Elgin 2025, 11, 35–36); Elgin describes epistemic communities as devising and collectively endorsing the methods and standards they work by, and holds that mere consensus does little unless most members function autonomously (Elgin 2025, 121–122). Uygun Tunç's example separates a decision to publish from members' beliefs. Applied to Fagan's two theses, that distinction leaves open whether public affirmation establishes collective possession.

A knowledge-producing ecology contains several relations that can coincide or separate. They can constitute, enable, communicate, maintain, or promote the acquisition of knowledge, but they are not thereby alternative meanings of one event called production:

- a candidate content and its grounds are generated locally;
- the result is articulated in a form that others can inspect, such as a written claim, a formal proof, a dataset, a mathematical or physical model, or a demonstration;
- one or more subjects acquire the relevant attitude and standing;
- qualified participants criticize or validate the result (the relations Chapter 11 examines as criticism and certification);
- the result enters a group's live structures of inference, decision, and action;
- it is retained in a form that can be retrieved, defeated, corrected, or withdrawn.

In separating these six relations, this book extends the distinction between producing and possessing knowledge that Fagan draws. These are analytical dimensions, not a mandatory pipeline. Criticism can occur before public release. A subject can

acquire the relevant attitude toward a result before the result has been articulated in any form that others could inspect. Whether an artificial system's internal states can realize such an attitude depends on the account of belief adopted (Section 2.1); understanding that may be tacit is taken up in Chapter 3, Section 6. A result can also be available for inspection without discriminating between rival theories. Measurements of how the electron's mass varies with velocity, for example, were regarded at the time as the arbiter between rival models of the electron and the broader theories perceived to be attached to them. Lorentz's theory and Einstein's, however, agreed in all their empirical predictions, including those for this variation, and data presented in 1908 seemed to support a prediction that was by then Einstein's as well as Lorentz's (Janssen and Mecklenburg 2004, 2, 31). Since the two theories shared that prediction while differing in their foundations, a measurement favoring it could not decide between them. On one historian's account, the essential difference between Einstein's theory and that of Lorentz and Poincaré lay in whether it went as far as changing the concepts of time and space; until around 1910 the Lorentz–Poincaré theory was still widely used, and whether the ether should be given up became a question for physicists, the historian suggests, only after 1910 (Hirosige 1964, 107). An instrument or protocol can integrate a result into practice before broad belief forms. An archive can retain a result after the people who once understood it have died. Publication can leave a result socially inert. For example, unusually low ozone values measured at Syowa Station in Antarctica in 1982 were printed in a 1984 report that listed the lowest values since 1966 second among four summary points; the observer later recalled that presenting the results in Japan and at an international symposium drew no particular response, although the printed proceedings of the symposium became the basis on which the observer was credited with first reporting the depletion (Chubachi 1984, 13, 18; Chubachi 1993, 817).

Different accounts assign knowledge at different points or deny that the points form one scale. Table 2.2 sorts these accounts into eight families. For each family, the middle column states what would bear social knowledge and through which relation it would possess it, and the right column states what follows for a result produced by an artificial system, in a form that holds whether *S* is human or artificial; the paragraphs after the accounts separate the two cases. The table is a simplified guide rather than an exact statement of each account. Each row states its family through one representative account, named in the paragraph after the table, and two rows join positions that differ: the member row covers both counting enough members and counting only the operative members, those with decision-making authority in the relevant domain, and the group-attitude row covers both group belief and group acceptance.

Table 2.2. Candidate bearers and possession relations for social knowledge: a simplified overview

Account family	Candidate bearer and possession relation	Consequence for an artificial result
Individual-possession	suitably situated individuals know, even when production is distributed	for the result to become *S*'s knowledge, it must reach individual knowers in *S* through evidence or justified dependence
Member or operative-member	enough members, or the members with decision-making authority in the relevant domain, hold the relevant attitude, subject to group-level conditions such as coherence among their bases	routing to the right members of *S* is part of successful social acquisition, and counterevidence held elsewhere in the group can defeat it
Relational	a group's belief is a system of members' beliefs and the epistemic relations among them; groups can be necessary for production without thereby possessing knowledge in their own right	the result must be connected to members through epistemic relations among them, not only deposited
Group-attitude	a structured group forms belief or acceptance under group-level norms	aggregation, authorization, coherence, and responsibility must be specified, and whether an accepted position is a belief is itself disputed
Distributed cognition without distributed knowing	the larger system performs cognitive work, but knowledge remains attributable to individuals	records and computational components can take part in production without being knowers; whether an artificial system that conducts research could itself know is not settled by this account
Knowledge without a knowing subject	scientific knowledge is ascribed to publicly certified results of reliable processes rather than to an individual or collective knower	a record's status turns on certification and public acceptance, not on anyone's present attitude
Distributed or group-functional	a properly organized collective realizes the cognitive function associated with knowing	members, records, retrieval, and action can jointly realize a social knower
Critical-contextual	accepted content that conforms and has survived criticism under specified social conditions is knowledge for that community	public venues, uptake, standards, and authority relations matter, not visibility alone

The accounts discussed above and below represent the rows in order: Uygun Tunç's individual possession, Lackey's operative-member account, the relational view Fagan describes, the joint-acceptance account that Gilbert states, Giere's distributed cognition without distributed knowing, Giere's adaptation of Popper's epistemology without a knowing subject, Bird's functionalism, and Longino's critical contextual empiricism. Compared with one another, the accounts differ partly in which of the six relations listed above they treat as decisive for social knowledge; in the terms of Chapter 1, Section 2, they differ over which relation must hold for a result to have entered *S*'s epistemic situation. For example, individual-possession and member accounts weigh most heavily the acquisition of an attitude by subjects; critical-contextual accounts, criticism together with acceptance by members; Giere's adaptation of knowledge without a knowing subject, certification and public acceptance; and distributed or group-functional accounts, the result's entry into live structures of inference and action and its retention.

Uygun Tunç argues that distributed scientific systems can collectively produce evidence and knowledge while the subjects who possess knowledge remain individuals. On this account, what collaborative research achieves collectively is the establishment of evidence supporting a proposition. That is not a process of belief formation, so it yields no group belief. Moreover, an account that denied possession to individuals would imply that no one learns anything even from the most successful collaborations (Uygun Tunç 2023, 2, 9). Giere likewise argues that distributed cognition need not entail distributed knowing: attributions of belief, knowledge, and agency should be restricted to the human components of a distributed cognitive system. Giere treats extending such attributions to the system as a whole as a higher-level interpretation, justified only if it brings some theoretical advantage, and judges that it brings none. Questions such as whether the Hubble telescope system as a whole expected to find something, or believed that it had found it, "do not make much sense," Giere writes. Restricting the attributions to humans, by contrast, yields a theoretical framework that everyone who studies science, from historians to anthropologists, can share (Giere 2007, 315–319). These views make collective production compatible with denying that the system as a whole knows. Giere's position also points to a further option. Adapting Popper's idea of epistemology without a knowing subject, Giere holds that the institution of science produces scientific knowledge with no need to posit a collective agent that knows it; individuals can claim to know its conclusions because they know that the conclusions were certified by reliable, accepted procedures (Giere 2007, 318). Popper's own formulation is more austere: knowledge in the subjective sense consists of mental dispositions and thought processes, while knowledge in the objective sense consists not of thought processes but of thought contents, the content of linguistically formulated theories (Popper 1978, §X); the conditions of certification and public acceptance are Giere's. Uygun Tunç accordingly classifies Giere's account as a no-subject account and rejects it in favor of individual possession. The reason given is that the notion of knowledge in the objective sense says nothing about where knowledge is located. If no one who understood physical cosmology remained, then, even if the publications were preserved, that the universe is expanding would not be known until someone read them (Uygun Tunç 2023, 12–13).

Among the accounts Giere answers is Knorr Cetina's. As Uygun Tunç summarizes the exchange, the two describe how collaborative experiments produce knowledge in similar terms, as distributed cognition, but differ sharply over whom to count as the epistemic subject of the knowledge so produced (Uygun Tunç 2021, 3–4). Comparing collider experiments at CERN with molecular-biology laboratories, Knorr Cetina argues that the experiments erase the individual as an epistemic subject, defined as the entity through which knowledge is procured. Papers appear under the names of all members in alphabetical order, and the capacity to produce knowledge shifts to the experiment; the conversation the collaboration holds with itself, Knorr Cetina argues, replaces the individual subject with a collective and dispersed one, although no member knows all that is produced. In the molecular-biology laboratories, by contrast, authorship conventions and the assignment of projects keep individual contributions distinct and sustain the individual as the epistemic subject (Knorr Cetina 1999, 236–237, 249–250, 303–304). Knorr Cetina denies that the experiments act as corporate actors, and reports that individuals and institutes persist within them, most visibly while a new experiment is being formed (Knorr Cetina 1999, 268, 271–274). Giere, on the same summary, reads Knorr Cetina as positing the experiment itself as the knower and rejects that extension of epistemic agency as too much of an ontological commitment (Uygun Tunç 2021, 3). Because Knorr Cetina defines the epistemic subject by the procuring of knowledge, however, the thesis can be read as bearing most directly on the first of Fagan's two theses, that groups can be necessary to produce knowledge, and as leaving possession open.

Bird develops a collective account of possession. On Bird's naturalized, knowledge-first functionalism, knowing is not a kind of believing but the state that a properly functioning cognitive system produces. A cognitive system here is a system

whose function is to produce factive representational states that guide action and are carried forward into inference and memory; a function is a working that has been selected for, whether by natural selection or as an artifact's. Because functions of this kind can be realized in different materials and arrangements, a group too can know by realizing an appropriately organized cognitive system, and the account does not reduce the attribution proposition by proposition to what individual members know (Bird 2024, §§2–3).

The joint acceptance account, of which Gilbert gives a prominent statement, holds, in Jeff Dunn's summary of the version Lackey discusses, that a group believes that *p* just in case its members agree jointly to accept *p* (Dunn 2021). On Dunn's reading, Lackey rejects both views that simply add up what members believe, are justified in believing, or know, and views on which a group's belief, justification, or knowledge floats free of its members'. Against the joint acceptance account, Lackey argues that it cannot account for group lies. Suppose a tobacco company's directors believe that smoking is harmful but jointly accept and announce that it is safe. The company appears to be lying; yet if its board's joint acceptance is all it takes for the company to believe that smoking is safe, the account rules out calling its assertion a lie. On Lackey's diagnosis, as Dunn reports it, the account makes group belief too voluntary and so makes the charge of lying too easy to escape (Dunn 2021). Lackey also rejects group beliefs that are what she calls base fragile, in which the bases of a significant subset of the members' beliefs conflict with the bases of another significant subset. Such a state, Lackey holds, is not always open to epistemic evaluation, and it cannot figure in the group's deliberation about its future actions, since the different reasons call for different plans (Dunn 2021). On Lackey's account, as Dunn quotes it, a group believes that *p* if and only if a significant percentage of its operative members believe that *p* and adding together the bases of their beliefs yields a belief set that is not substantively incoherent (Dunn 2021); operative members, in the sense Kallestrup takes from Tuomela, are members with decision-making authority, and so responsible for the content of the group's state (Kallestrup 2024, 571). As Jessica Brown quotes Lackey's account of justified group belief, a group justifiedly believes *p* if and only if a significant percentage of its operative members justifiedly believe *p* on bases that, added together, yield a coherent belief set, and full disclosure of the evidence relevant to *p*, with rational deliberation about that evidence among the members in accordance with their individual and group epistemic normative requirements, would not yield further evidence that, added to those bases, fails to make *p* sufficiently probable (Brown 2023, 2). On this account, then, a collectivity with no operative members in the relevant domain, humanity for example, has no group belief; an *S* so specified serves as the reference relative to which a result is new, while possession falls to members, subgroups, or organizations within it. The presupposition of Chapter 1, Section 2, that *S*'s epistemic situation could come to include the knowledge, is then met through those members, subgroups, or organizations. Whether a scientific field has operative members in this sense is not settled here (compare Section 7). Lackey's account of group assertion, unlike her account of group belief, turns on a spokesperson who speaks with the group's authority (Chapter 11, Section 5.1).

Kallestrup occupies another intermediate position: some group knowledge is irreducible to member knowledge but remains anchored in individual features. On Kallestrup's account, a group's conventions can set epistemic standards that its members do not follow as individuals. A criminal jury instructed to disregard hearsay evidence, for example, can know that the defendant is guilty, while none of the jurors knows it, because each has received reliable but misleading hearsay to the contrary. Even so, the jury's knowledge is fixed by the members' states and processes and by the relations in which they stand to one another within the group's structure (Kallestrup 2024, 578–579, 581–582).

Longino's critical contextual empiricism (Longino 2025, §5) places the social at a different point from the accounts above. They ask mainly who possesses knowledge; Longino argues that social organization enters what supports accepting a claim, its warrant, itself. Because evidential relations are mediated by background assumptions, the argument continues, critical interaction among the members of a scientific community, or among different communities, is the only check against the arbitrary dominance of subjective preference in accepting such assumptions (Longino 2025, §5). Under Longino's substantive account, venues for criticism, uptake, publicly recognized standards, and tempered equality of intellectual authority (any perspective has a prima facie capacity to contribute to the community's critical interactions, a standing that can be lost by failing to engage with or respond to criticism) help constitute scientific objectivity rather than merely communicate an already completed private achievement (Longino 2025, §5). Uptake, in this sense, is a requirement on the community: the distribution of belief in it must change over time in a way that is sensitive to the critical discourse taking place within it (Longino 2025, §5). As Dutilh Novaes and Dede report, Longino presents these conditions as features of an idealized community that real communities meet by degree (Dutilh Novaes and Dede 2025, §2). They are not adopted here

as a theory-neutral list of conditions for all knowledge. They show one important way in which social organization can enter warrant itself.

Longino also addresses who possesses knowledge. As Wray reports her earlier account, knowers are interdependent individuals, rather than a group that transcends them (Wray 1998, §IV). Eigi explains Longino's conformation as success that admits degrees and varies with the respects assessed, with truth as a limiting case (Eigi 2016, 49). In Eigi's exposition, knowledge is indexed to communities and their projects without making acceptance arbitrary: communities must satisfy conditions of effective criticism (Eigi 2016, 47–49, 138). Where this chapter draws on Longino's social conditions on warrant, it keeps its factive usage.

What these accounts imply for artificial research depends on what *A*, the agent conducting the research, and *S*, the society for which knowledge is sought, are. Two combinations are taken up here: an artificial *A* with a human *S*, and an artificial *A* with an artificial *S*; mixed societies lie between them (Chapter 8, Section 1.2). Where *A* is artificial and *S* is a human community, the accounts differ on what it takes for *A*'s result to become *S*'s knowledge: reaching individual knowers in *S* through evidence or justified dependence, being routed to its operative members, being jointly accepted, being certified by accepted procedures, being integrated into an organized cognitive system, or surviving criticism in public venues and being taken up. On none of them, as their authors state them, is it enough that *A* generates and publishes an output, and what has to be added differs from account to account (Section 7 asks whether, on an extension of Bird's account, a validated record available through normal channels could suffice); this concerns the third of the judgments in Chapter 1, Section 2, whether the result has entered *S*'s epistemic situation. They also differ on whether *A* can be among the bearers of that knowledge, which concerns the second, where the achievement is realized. On Uygun Tunç's and Giere's views, a human–machine organization does not know as a whole, and knowledge is located in individuals. Giere, as Matthew J. Brown and Uygun Tunç report his position, restricts belief, knowledge, and agency to the human components, as a theoretical choice defended for systems such as the Hubble telescope, in which humans conduct the research with instruments and computers (M. J. Brown 2015; Uygun Tunç 2021, 3–4). Neither view considers an *A* that is itself an artificial system conducting research, so whether such an *A* could be an individual knower is not settled by them; it turns on the accounts of belief in Section 2. Read as a thesis about production, Knorr Cetina's shows how far production can be dispersed across a collaboration, so that no member knows all that is produced, without settling who possesses the result. On Bird's group-functional account, records and retrieval can belong to an organized collective that knows, so the question becomes whether *A* is organized into such a collective as one of its components. Whether *A* contributes to knowledge that *S* comes to possess and whether *A* is itself a knower, or part of one, are separate questions (Section 2.1; Chapter 1, Section 2). In this case, nothing in these accounts requires *A* itself to be a knower (Chapter 8, Section 1.1). Other relations can make it matter. If the system's reports are taken as assertions through which humans come to know, accounts of assertion and testimony place conditions on the system as speaker, and on Williamson's knowledge rule it is the system, or the group for which it speaks, whose knowledge the rule concerns (Chapter 11, Section 5.1); and where the achievement sought is realized in the system itself, as with a skill, the system is its first bearer.

Where *S* is itself an artificial society, the second question, whether *A* is itself a knower, can no longer be set aside, and it takes a wider form: whether artificial members, *A* among them if it is one, can be knowers at all. Both roles of *S* presuppose that its epistemic situation could come to include knowledge of the kind sought (Chapter 1, Section 2), and that situation now consists of artificial members, their organization, and their records; the question therefore arises before any member of *S* knows the result, and whether or not *A* is one of its members. Whether a result has become *S*'s knowledge then turns, on accounts that require a bearer, on whether its bearers can be found among those artificial members or their organization, and, on Giere's adaptation of knowledge without a knowing subject, on certification within *S*. On individual-possession, member, and relational accounts, the question returns to whether artificial members believe the result (Section 2), and on relational accounts also to whether epistemic relations hold among them. On joint-acceptance accounts, it becomes whether they can take part in joint acceptance; on Bird's account, whether the artificial members that make up the society together realize a properly organized cognitive system; on Giere's adaptation of knowledge without a knowing subject, whether certification by artificial members counts as certification, a question that sharpens where they are instances of one base system, since instances of the same base can err in the same places, so that their certifications of one another may add no independent check (Chapter 11, Section 3); and on Longino's account, whether their interaction involves the diversity and the public standards of criticism that a knowing community requires, which multiple instances or a majority vote do not supply by

themselves. Giere's restriction of agency to humans, taken as a premise, would leave such a society with no member to whom knowledge could be attributed, but since Giere defends it as a theoretical choice for systems in which humans conduct the research, it does not settle this case. Chapter 8, Section 1.3 develops these differences.

Finally, the social dependence of knowledge production that these accounts describe does not imply that truth is created by agreement. To keep the two apart, this book distinguishes at least three claims that differ in strength:

1 inquiry causally depends on socially inherited language, concepts, instruments, testimony, labor, and records;

2 evidential standing, objectivity, or knowledge is partly constituted by organized social practices;

3 whatever a community accepts is thereby true.

Individualist epistemologies can accept the first. Accounts in Table 2.2 on which social organization enters warrant or possession, such as Longino's critical-contextual account, defend versions of the second. Neither entails the third, which no account discussed in this chapter holds. Social recognition can affect warrant and possession without making a false proposition true. Evidence, failed interventions, resistant materials, and other world-involving constraints continue to matter. For artificial research, the distinction guards against reading agreement among artificial members, or a majority vote across them, however they were made, as making a result true; what the second claim leaves to be asked is whether an artificial society's organized practices could partly constitute warrant (Chapter 8, Section 1.3; Chapter 11, Section 3).

Selecting *S* settles the relativity of novelty; it does not settle possession, which the working characterization leaves to the accounts compared here. A claim that a result has become *S*'s knowledge can therefore be assessed only when it reveals enough of its account. Chapter 8, Section 1 returns to the roles of *S* when it separates *A* from *S*, asks about the bearer where success is at issue, and distinguishes cases by whether *S* is human, mixed, or artificial. Within this chapter, Section 7 tests the accounts compared here on a single case, and Section 9 states them as requirements; Chapter 11 takes them up for certification and criticism.

The unread artificial discovery

This section tests the preceding accounts together on one case, in which the production of a result and its possession by *S* come apart. The result is produced by an artificial *A*, which may itself know it on an account of artificial belief (Section 2), and *S* is human society, no member of which has read it. The case is then strengthened: the result passes recognized validation (Section 5) and enters retrieval, use, and correction, so that of the relations distinguished in Section 6, the one chiefly still missing is a member's acquiring an attitude toward the result. The accounts compared there divide over whether that absence matters. The verdicts are stated first for a human *S* and then carried over to an artificial or mixed *S*.

The case, deliberately difficult, is this. An artificial system derives a true proposition previously unknown to human society, preserves strong grounds and a complete record, and deposits the result in a public database. No human has read it, believed it, or acted on it. Does human society know the result?

The physical inscription exists. The artificial system may itself know the result under an appropriate account of artificial belief and warrant. It may also have the knowledge on an account that does not require belief (Section 2). Humphreys, for example, modifying a reliabilist definition for automated methods, proposes that an instrument has the knowledge that F if and only if it contains a representation of the entirety of F, the representation holds of the target, and a reliable process, one that produces a high proportion of accurate representations, formed it. Humphreys adds that the definition makes good sense where the representations are accessible to humans but becomes problematic where they are opaque and their accuracy is unknown, and that it may then be necessary to fall back on something like Burge's entitlement approach to computer-assisted mathematics, on which some sources, such as normal visual perception, are taken as primitive epistemic authorities (Humphreys 2020, 24–25). Symons and Alvarado, writing earlier, argue against carrying that approach over from computer-assisted proof to computer simulation. As Winsberg and Alvarado set out the argument in their encyclopedia entry on computer simulations in science, a simulation is not what Burge calls a transparent conveyer, one through which the warrant for its results is of the same kind as the warrant for the operations it carries out, since features of simulations let warrants of another kind, a posteriori ones, enter the pipeline; a simulation must have its own warrants and its own sanctioning, pursued independently of the sanctioning of the methods that enabled its construction, rather than being trusted

by default (Winsberg and Alvarado 2026, §2.3). Symons and Alvarado write about simulation, not about learned representations, but applied to Humphreys's proposal their account would not let an entitlement supply what unknown accuracy leaves missing: ascribing the knowledge to an instrument whose representations are opaque would rest instead on evaluations of the instrument itself, such as comparison with real-world data, the kind of validation the case acquires when it is strengthened below. On such a definition the depositing system could have the knowledge as an instrument has it, without belief, and whether human society knows the result would remain a separate question. The record is a candidate contribution to human knowledge. Neither the inscription, nor the system's possible knowing, nor the record's standing as a candidate establishes by itself that human society already knows. A file disconnected from the society's epistemic organization could remain indistinguishable, from the perspective of human inquiry, from an indefinitely large collection of unnoticed truths and falsehoods.

The limited conclusion is:

> Bare inscription or public deposit is not a theory-neutral sufficient condition for human social knowledge.

This statement is weaker than the claim that storage can never partly constitute knowledge. Strengthen the case. The record has passed validation by procedures that *S*'s institutions recognize. Its provenance is authenticated. It is indexed in canonical infrastructure, retrieved automatically when relevant questions arise, used in inference and action, and connected to mechanisms that propagate correction or withdrawal. No human has yet read the proposition itself.

Now the philosophical verdict divides. An individual-possession account can say that no human yet knows, although humans are well positioned to know. A member-based account such as Lackey's reaches the same verdict for groups. Applied to human society as a whole, which has no operative members in the relevant domain, Lackey's account yields no group belief (Section 6), so the verdict concerns groups within it that have such members: since none of their operative members believes the result, none of these groups yet believes or knows it. A joint-acceptance account asks whether the members have jointly committed to letting the result stand as the group's view. A distributed-cognition account can classify the infrastructure as a genuine cognitive resource without attributing knowledge to the whole. An account that allows knowledge without a knowing subject detaches the result's status from anyone's attitude, but Giere's version does not count a claim as scientific knowledge until the relevant community has publicized and accepted it (Giere 2007, 318). Applied to the present case, that condition suggests that the unread result would not yet qualify, although the same passage lets certification by accepted procedures support a claim to know, which the recognized validation in the strengthened case may approach. Popper's own example is close to the present case: a series of logarithm tables computed and printed by a computer, sent to libraries and never read, contains what Popper calls objective knowledge, since what makes a thing a book is its possibility of being understood, not its actually being read (Popper [1972] 1979, 115–116). On that formulation the deposited record already belongs to objective knowledge, true or false; but because objective knowledge is there defined as independent of anyone's claim to know (109), the formulation does not answer the question this section asks, whether human society knows the result. Extended to a record no one has ever read, Bird's group-functional account might say that the result is already part of the society's knowledge because the organized system can retrieve, use, and revise it; Bird's own case, discussed below, concerns retention instead. A critical-contextual account, on Longino's definitions, gives a more determinate verdict.

In the definitions Büter reproduces, accepted content counts as knowledge for a community when it conforms to its intended objects well enough to enable members to carry out their projects and is epistemically acceptable there. At a given time, acceptability requires support by data evident to the community in light of reasoning and background assumptions that have survived criticism from as many perspectives as are available to it, in a community characterized by venues for criticism, uptake, public standards, and tempered equality of intellectual authority (Longino 2002, 135–136, as quoted in Büter 2012, 84). Applied to the present case, Longino's definitions imply that the unread result is not yet the society's knowledge, whatever the quality of its automated validation, since no member has accepted it. It would further follow that whether that validation could contribute to acceptability once the result is taken up depends on whether the validation itself has been exposed to criticism under Longino's conditions.

The basic and the strengthened case differ as a **dormant record** differs from a **live epistemic capacity**. The former merely persists; the latter is integrated with retrieval, competent use, dependency tracking, and correction, and among the relations of Section 6 it is these that the strengthening added. The distinction names what it takes to make a record a live capacity.

Whether it also marks the boundary of knowledge, however, differs from one account to another. Bird's own case of a journal article that no one any longer remembers is close to the present one in that no one is currently aware of the result. Dr N. publishes a peer-reviewed result; everyone who read it dies or forgets it; years later another scientist finds it through the indexes. Bird holds that the result remained part of the scientific community's knowledge throughout, because it stayed available through normal channels to those who needed it, and Bird defends this as genuine knowing that was causally isolated and inactive (Bird 2024). On that view a validated and indexed record can remain social knowledge while dormant. Lackey challenges the claim that the community knew Dr N.'s result all along. Kallestrup, for whom irreducible group knowledge remains anchored in individual features, sides with Lackey here and sets out an objection inspired by her discussion: the familiar distinction between knowing and being in a position to know pertains to the difference between information that has been accessed and information that is merely accessible, so that a group none of whose members has accessed a result is more naturally described as only in a position to know it. In Kallestrup's own argument, because Dr N.'s results are stored in no one's memory or an extension of it, the community does not even dispositionally believe them and has only a disposition, or is only in a position, to know them, and storage in a library or on a journal website does not suffice for knowledge (Kallestrup 2024, 583–585). The unread artificial discovery thus enters an existing dispute, though at a different point. In Bird's case the result had passed peer review and been read before it became dormant, so the case concerns retention. Whether a record that no member has ever read or accepted can become social knowledge is a further question, and the strengthened case differs from Bird's in exactly this respect. On Bird's view, the dormant/live distinction does not mark where a community's knowledge ends. On a reading of Lackey's account, even a live capacity leaves such a group only in a position to know until operative members believe the result. An account on which retrieval, competent use, and correction constituted possession would treat the distinction as the point at which an epistemic resource becomes knowledge.

The dispute turns on how knowledge is tied to action. A second objection, which Kallestrup attributes to Lackey, combines two principles. One is that if an agent knows that *p*, it is epistemically rational for it to act as if *p*, a principle Lackey accepts in this direction; the other is what Lackey calls the Group/Member Action Principle, on which a group performs an act only if at least one of its members performs some act that causally contributes to it. The worry is how a group could act on knowledge that none of its members has. Applied to a community none of whose members is aware of a result, Kallestrup concludes, it becomes a mystery how the members could contribute to an act of the community in a way that bears non-accidentally on the result (Kallestrup 2024, 572 and n. 15, 585–586). Bird, who illustrates the objection with a case adapted from Lackey, in which a leading expert asserts on a mere hunch what Dr N.'s forgotten paper had established and the field acts on the assertion, replies that the connection holds only for action based on the knowledge in question: one can know that *p* and still act irrationally when that knowledge plays no part in the action, so the community's irrational action is no objection (Bird 2024). Lackey also presses, as Kallestrup reports, a case meant to show that Bird's account cannot distinguish what a group knows from what it should have known. Three officers in one police unit each know one fact: that a boy has been reported missing, what he was wearing that morning, and that a boy so dressed was seen walking with an adult man in a nearby neighborhood. On Lackey's reading, Bird's account implies that the department knows that the boy seen is very likely the missing one, yet no single officer seems accountable for failing to act on it. Avram Hiller and R. Wolfe Randall modify the case: each officer enters what he or she knows into a central system, which collates the entries and prints the conclusion once they fit together. Since every officer then has easy access to the printout, they find it intuitive to hold the group accountable for failing to act, and they argue that the group knows, because its structure is designed to access information of that type and is functioning properly. Kallestrup replies that the department in the modified case may be accountable because it should have known the conclusion, not because it knows it: knowledge entails dispositional belief, and a non-biological repository would sustain such a belief only if it served as a functional analogue of the way individuals store and retrieve information in memory (Kallestrup 2022). The automated variant comes close to the strengthened case. Kallestrup's reply asks how the group itself could dispositionally believe the conclusion: without an account of that, collation would at best give the group a disposition to believe it, and the natural candidate he notes, an extended cognitive process at the group level, is one that Hiller and Randall consider but do not adopt (Kallestrup 2022). On this reply, as on the reading of Lackey's account given above, automatic collation and use would not by themselves make the result the group's knowledge unless the group's integration of the record amounts to such a process; the reply adds that a group whose system collates the result can still be answerable for what it should have known.

The verdicts above concern a human *S*. Carried over to an artificial or mixed *S* whose artificial members retrieve the record automatically and integrate it into their inference and action, the accounts as set out above do not all move in the same way. On an individual-possession account, the question is no longer whether anyone has read the result but whether an artificial member's automatic retrieval and use amounts to its believing the result, which returns to the accounts of belief in Section 2. On Lackey's member-based account, it becomes whether the society's artificial operative members believe the result, and whether their retrieval counts as accessing the information rather than its being merely accessible; the automated variant of the missing-child case does not decide this, since the collating system there was not among the members. Since operative members are, on this account, those with decision-making authority in the relevant domain, which artificial members are operative depends on how such authority is distributed within an artificial *S*, a matter Chapter 12, Section 5 takes up. On a joint-acceptance account, it becomes whether artificial members can take part in joint acceptance, and on Giere's version of knowledge without a knowing subject, whether certification by artificial members counts as certification, a question that sharpens, for the reason given in Section 6, where they are instances of one base system (Chapter 11, Section 3). On Longino's definitions, acceptance by members could be met by artificial members if they can be members of a knowing community in Longino's sense, with the diversity and public standards of criticism that Chapter 8, Section 1.3 discusses; whether the result has been accepted would then turn on what the artificial members do with it, not on whether any human has read it. Bird's group-functional verdict does not change in this way, since it already turned on whether the organized system can retrieve, use, and revise the result rather than on any member's awareness; for an artificial *S*, the question remains whether the artificial members together realize a properly organized cognitive system (Section 6). Since on Bird's account a function is a working that has been selected for, it also matters whether that organization was selected or designed for producing factive states that guide action rather than, for example, for passing checks or winning approval; as with Simion's account in Section 4.1, what selection actually rewarded can diverge from the stated objective.

The producing system may itself be a member of an artificial or mixed *S*. If it is, and if it knows the result on the adopted account of belief, then on an individual-possession account a member of *S* already knows, and the result may have entered *S*'s epistemic situation as soon as it was produced. On Lackey's account, the system's belief counts toward a group's belief only if the system is among the group's operative members, and the group believes the result only if a significant percentage of them do. On Longino's, the system's acceptance is at most one member's, and acceptability requires that the result survive criticism from the other perspectives available in the community. On Bird's, the verdict still turns on the organized system as a whole. It follows that the group-functional account is not the only route to crediting an artificial society with knowledge. Even on an individual-possession account, if what counts as a knowing individual changes, a result that no human has read can no longer be said to be known by no member. That requires, however, that the machine be a knowing member on the adopted account of belief, not a member in name only (Section 2).

At the limit of the basic case, where the record is so disconnected that no member of *S*, human or artificial, can observe, retrieve, assess, or use the result, the accounts above that answer whether *S* knows it largely agree that it does not yet; Popper's objective knowledge, as noted above, answers a different question. The basic case itself, a public deposit, need not reach that limit, since a deposit is available through normal channels. On Bird's account, however, knowing is the state that a properly functioning cognitive system produces (Section 6), and the forgotten journal article had passed peer review; extended to a record no one has read, the verdict that such a record can be knowledge while dormant therefore fits the strengthened case, whose record has passed validation that *S*'s institutions recognize, better than a bare deposit. What remains open is only whether such a wholly disconnected true record is to be called knowledge in some sense, and however that is settled, what has to be built for the result to enter *S*'s epistemic situation is the same: routes by which it can enter validation, inference, action, and correction. These need not lie within the system that conducted the research; they may belong to the infrastructure between it and *S*, or to *S* itself. The boundary case nonetheless remains philosophically important, because it prevents upload from being equated with society knows.

In the strengthened case, where such routes exist, the accounts divide, and the division bears on what a project claiming that the result has entered *S*'s epistemic situation has to build. On Bird's account, extended as above, the routes may already suffice for the result to have entered *S*'s epistemic situation. On Lackey's, such a project would further have to bring the result to members of *S* and have it believed by them, and, where entry is claimed through a group's knowing it, to the operative members of a group within *S* that has them in the relevant domain; on Longino's, to have it accepted by members after criticism under the account's conditions. What such a project has to build therefore depends on the account of

possession adopted, which is why, as Section 6 noted, a claim that a result has become *S*'s knowledge can be assessed only when it reveals enough of its account. Whether the activity is research for *S* does not depend on whether the result meets that account's conditions of entry, as the next paragraph notes.

The case also shows concretely two distinctions drawn in Chapter 1. If the artificial activity is organized toward producing knowledge new to human society, it can be research in the sense of the working characterization before any human acquires the result, since attempt separates research activity from its success (Chapter 1, Section 3). The examples there, such as an inconclusive experiment or a proof strategy that collapses, separated the two through the absence of the achievement. Here the achievement may be present, only not yet in *S*, so that success itself divides. Of the three judgments of Chapter 1, Section 2, the first, whether the activity is research for human society, can hold. The second, whether knowledge new to human society has been realized somewhere, can hold as well, in the system itself if the system can know on the adopted account of belief. The third, whether the result has entered human society's epistemic situation, depends on the adopted possession account and may not yet hold. For an artificial or mixed *S*, the judgments divide in the same way, unless the producing system is itself a member whose knowing already counts on the adopted account, and the third turns on the accounts as carried over above.

Epistemic change before and around knowledge

Research often ends short of knowledge, and what it achieves need not be knowledge even when it succeeds. This section asks how such outcomes can be described and evaluated without either counting only knowledge or counting every change as an improvement. The literature this section draws on has mostly pursued this question in one narrower form: whether such outcomes count as progress. Accounts of scientific progress in particular disagree largely over whether only the accumulation of knowledge counts as progress or other outcomes count as well. The section follows the debate in this narrower form, because doing so shows what an evaluation of such outcomes needs to keep apart.

It first separates research activity from success at knowledge (Section 8.1). Section 8.2 takes up the case in which research falls short of knowledge: changes before knowledge, chiefly the making explicit of ignorance, together with formulating questions, acquiring evidence, and changing credence, and whether they count as progress; the arguments it draws on concern epistemic progress in general, not only in the sciences. Section 8.3 takes up, together with the accounts of scientific progress, the case in which the outcome is not knowledge itself: achievements around knowledge, such as a false theory's approximate truth and problem solving. Understanding belongs here too on views that do not take it to be a kind of knowledge; whether it is one is taken up in Chapter 3, Section 3. The section ends with what this implies for evaluating artificial research.

8.1 Research can be well conducted without reaching knowledge

Section 7 separated research from success in a case where knowledge may already have been realized but has not yet entered *S*. This subsection takes up the case from which Chapter 1, Section 3 began, in which the knowledge sought is not reached at all: a central claim turns out false, or a result remains inconclusive. Because attempt separates research activity from its success, such an investigation does not thereby cease to have been research, and whether it was research can be judged without waiting for the result, since the judgment turns not on whether the activity succeeded but on how it was organized (Chapter 1, Section 3). Independence from success does not, however, mean that the judgment is settled when the activity is undertaken: an activity can be reorganized into research along the way, and its organized direction may become clear only from later records.

For the case where the target is knowledge, this book therefore keeps at least the following six evaluations distinct:

1. whether an activity is research for *S* in the sense of the working characterization;
2. whether it is responsibly or competently conducted relative to the available situation;
3. whether target knowledge new to *S* has been realized, in whatever bearer, *A* included, or, on accounts that allow knowledge without a knowing subject, in a certified result;

4 whether that knowledge has entered *S*'s epistemic situation, which the accounts of possession in Section 6 answer differently;

5 whether failure at the target produces a collateral epistemic contribution;

6 whether the contribution is significant or worth its costs.

The first, third, and fourth are the three judgments of Chapter 1, Section 2.

A well-conducted investigation can fail at its target. A poorly conducted one can reach a true answer by luck. A failed experiment can reveal a limitation in a method, an auxiliary assumption, or the original problem formulation. Whether that collateral result is knowledge requires its own assessment; whether it is important requires a further relation to purposes and affected parties. As the term is used here, whether the result is knowledge depends neither on whether it came from research nor on whether it is important.

8.2 Changes before knowledge: explicit ignorance and whether such changes are progress

Failure to reach knowledge need not be a mere absence of progress. A society can be wrong about a matter, unaware that the matter is unresolved, aware of a question but unable to identify plausible answers, or able to specify both a question and the evidence still needed to answer it. These are materially different epistemic positions even when none yet contains the target knowledge.

Wilholt distinguishes, within total ignorance, **conscious ignorance**—roughly, the questions whose answers a subject recognizes that it does not know—from opaque ignorance, what a subject does not know that it does not know. The transition from error or opaque ignorance to conscious ignorance is what Wilholt calls **Socratic progress**. Conscious ignorance can itself require substantial conceptual and epistemic competence (Wilholt 2020, 2–6). Within conscious ignorance, Wilholt reserves the label **deep ignorance** for cases in which a subject is aware that no plausible candidate answer is available, and identifies these cases with what Bromberger called a p-predicament: a subject is in a p-predicament with regard to a question if and only if, on the subject's views, the question admits of a right answer, but the subject can think of no answer to which, on those views, there are no decisive objections (Wilholt 2020, 9–10). Wilholt argues that such questions hold particular attraction for researchers, since overcoming one promises to open new questions and connections, but concludes only that being perceived as deep favors a question's inclusion in the research agenda: such questions can be passed over, for example when no available method offers an approach to them (Wilholt 2020, 10–11). Wilholt's account of Socratic progress gives a precise example of epistemic improvement that need not yet be knowledge of the answer. It also helps connect the achievement analysis in this chapter to the later account of how questions are understood and made tractable (Chapter 4, Section 3).

The scope of this point is limited. It does not make every unanswered question a valuable research contribution, identify ignorance with knowledge, or show that articulating a question is sufficient for research. Significance, tractability, novelty, and the quality of the articulation remain separate matters.

What this section takes from Wilholt's account is a contrast. On one side is missing knowledge that is represented nowhere. On the other is a limitation that is represented in some form, such as a note in a record, an error estimate, or a stated question, and that can therefore guide attention, inquiry, and criticism. Conscious ignorance need not take the form of a categorical belief that one does not know; what matters is whether the limitation is represented in some such form.

Working records can keep such limitations explicit. In a 1972 design study for a laser-interferometric gravitational-wave antenna, for example, Weiss estimated the instrument's noise sources one by one and recorded where the needed knowledge was missing: the seismic spectrum had not been measured above 100 Hz, and no data existed on high-frequency gravity gradients near the earth's surface (Weiss 1972, 64, 67). Dismantling a claimed discovery can also begin by making such a limitation explicit. After a 2009 report associated the retrovirus XMRV with chronic fatigue syndrome, a multi-laboratory working group, before testing clinical specimens, first checked whether the assays could detect the virus in blinded samples to which it had been added. It thus first established what the assays could and could not show. A later blinded multicenter study, whose design all participating laboratories had agreed on in advance and whose authors included the original

claimants, found no association (Simmons et al. 2011a; Simmons et al. 2011b; Alter et al. 2012).

Some fields have made such limitations a sustained object of study. Knorr Cetina, for example, reports that the collider experiments under study cultivated what Knorr Cetina calls negative knowledge and distinguishes from non-knowledge as knowledge of the limits of knowing, of the mistakes made in trying to know, of what interferes with knowing, and of what one is not interested in and does not want to know: one physicist described the systematic error, the uncertainty of a measurement that comes from the apparatus and the analysis rather than from chance fluctuation, as a way of measuring the group's ignorance. When some forty-five competing theoretical descriptions of the proton's internal structure gave different inputs to an analysis, the spread among the resulting values was entered into the result as a theoretical systematic uncertainty rather than left as a disagreement among theories (Knorr Cetina 1993, 269–273). Knorr Cetina presents this as a preference of that culture, which fields such as molecular genetics do not share (Knorr Cetina 1993, 270–271). Chapter 6, Section 4.3 takes up the molecular-biology side of this contrast.

The distinction between missing knowledge that is represented nowhere and an articulated limitation has a direct consequence for artificial research. Evaluation by the number of questions closed or answers emitted can favor a system that closes questions before the evidence settles them. Such a system can fill its outputs with plausible claims while making uncertainty, missing evidence, incompatible candidates, and unformulated problems harder for those who rely on them to see. Conversely, a system may improve the research position of *S* by showing that a seemingly settled answer is unsupported, by identifying which alternatives remain live, or by making a previously inarticulable limitation askable. An artificial research system may therefore need to retain open questions, reasons for suspension, failed operationalizations, and missing-evidence relations as first-class epistemic states rather than as empty slots left after answer generation. Keeping such states in the system's own record is not yet a change in *S*'s position. For the move to conscious ignorance to be made in *S*, the represented limitation must enter *S*'s epistemic situation: where *S* is human, it must reach human members in a form they can take up as a question; where *S* is artificial, its artificial members must be able to treat it as an open question in their inquiry. Chapters 4 to 6 examine how such states are formed and transformed in inquiry; the present claim is only that they can matter to epistemic progress. Whether they constitute progress or only promote it divides accounts of progress, as the end of this subsection shows.

Williamson's arguments bear on how complete a system's record of what it does not know can be made. Williamson argues that almost no nontrivial condition is such that, in every case in which it obtains, one is in a position to know that it obtains, and in particular that one can fail to know something without being in a position to know that one fails to know it (Williamson 2000, 13–14, 107). In Williamson's simple models of creatures that store information, a creature that cannot survey what it has stored can fail to know something without knowing that it fails to know it; in one of them, retrieval from memory can take any length of time, so the creature cannot tell a missing memory from one not yet delivered. Williamson describes such a failure as one of self-knowledge rather than of rationality in any ordinary sense (Williamson 2000, 226–227, 233–234, 316–317). What does the work here is less that the creature cannot survey its store at once than that retrieval can take any length of time, so that a missing memory cannot be told from one not yet delivered. Applied to an artificial system that meets this condition, these arguments, if they hold, imply that its record of its own open questions and missing evidence cannot be guaranteed complete by the system's own checks alone. A configuration that can confirm that it has finished examining a finite store is not covered by this argument as it stands. Even then, however, what such a check confirms concerns only what is represented in the store; an absence represented nowhere, like the opaque ignorance above, does not show up when the examination of the store is finished.

As used here, the process vocabulary does not turn every change into an improvement. Ross argues that an intellectual community can progress before it knows an answer: it may formulate a worthwhile question, acquire relevant evidence, or increase its confidence in the correct answer. The reason given is that events widely recognized as progress occur while a community is still divided. After Galileo reported telescopic evidence that the surface of the moon is not smooth, the community neither believed nor knew it; yet to say that the community made no progress would, on this view, be too large a revision. Ross also rejects the view that such events count as progress only because they facilitate later knowledge, since by that standard even making coffee for a research group would count as progress (Ross 2021). Harris objects that formulating questions and collecting evidence facilitate progress rather than constitute it: a community would not progress merely by formulating many questions or by collecting evidence fruitlessly, although increased confidence in true propositions might

be accommodated by a credence-based version of a truth-centered account (Harris 2024, 142–143). Carried over to Wilholt's account above, Harris's point would apply to the move from opaque ignorance to recognizing a question that Wilholt calls Socratic progress, counting it as facilitating rather than constituting progress. Whether it applies to giving up a false belief is less clear, since Harris's concession about credences may cover that case. The claim above about Socratic progress is therefore made on Wilholt's account rather than on every account of progress.

Without settling the theory of progress, the dispute still yields a constraint. Question formation, evidence acquisition, credal change, and acknowledged ignorance are candidate epistemic transformations. Whether a particular change constitutes progress depends on its relation to a worthwhile target, a correct answer, error exposure, or an improved capacity for inquiry. More confidence, less uncertainty, more information, and more questions are not independently sufficient.

8.3 Achievements around knowledge and accounts of progress

This subsection turns from changes before knowledge to achievements around it, beginning with historically successful but false theories. This chapter takes propositional knowledge to be factive: what is known is true (Section 3). A false theory, however successful, is therefore not itself known, and the question becomes how its achievement is to be described. We may identify true restricted claims that were known, take the parts of a false theory that were responsible for its successes to be approximately true (Chapter 6, Section 3.2), distinguish literal belief from practical acceptance, appeal to approximate truth or empirical adequacy, or treat understanding and problem-solving ability as achievements not exhausted by knowledge accumulation; this subsection takes up several of these options, among them Bird's restricted claims, Niiniluoto's truthlikeness, Laudan's problem solving, and Dellsén's understanding. Factivity does not decide among these analyses. It fixes only how this chapter uses the term propositional knowledge; how to describe the wider epistemic achievement of such theories is left to those analyses and to later examination.

Several of these options have been developed as accounts of scientific progress. Dellsén's survey of that literature distinguishes four accounts, each of which places its own type of cognitive achievement at the heart of progress: increasing truthlikeness, increasing problem-solving effectiveness, accumulating knowledge, and increasing understanding, the last being Dellsén's own account, termed noetic (from the Greek for understanding). Dellsén adds that the most plausible account may be pluralistic, counting more than one type of achievement (Dellsén 2018, §1). On this survey, progress is analyzed not as one more achievement alongside knowledge and understanding but as some type of achievement's being increased, accumulated, or more nearly achieved. In this book's terms, what accounts of progress dispute is which achievement's increase counts as progress. As Chapter 1, Section 1 noted, the book's working sentence characterizes the orientation of a research activity and is not meant to serve as a theory of scientific progress. In particular, characterizing research through epistemic achievement does not commit the book to Bird's epistemic account, on which progress consists in the accumulation of knowledge (Dellsén 2018, §§1, 4); Chapter 3, Section 4 takes up understanding-based views, including Dellsén's, to examine whether, and how, understanding and knowledge differ as targets and measures of progress. Dellsén suggests that there is a close conceptual connection between progress and the aim of science: roughly, X is the aim of science just in case science makes progress when X is increased, accumulated, or more nearly achieved (Dellsén 2018, §6). If that connection holds, it follows that adopting one of these accounts would amount to establishing the kind of global aim that Chapter 1, Section 1 declines to establish. The paragraphs that follow draw on these accounts for distinctions that bear on evaluating artificial research, without choosing among them. The accounts were proposed for scientific progress, whereas the working characterization is not restricted to the sciences; the distinctions drawn from them are used here for research more generally, and whether a given account's measure fits a field outside the sciences is a further question.

Historically successful but false theories pose a difficulty in particular for Bird's account, on which progress consists in the accumulation of knowledge, since a false theory cannot itself be accumulated knowledge. Yet a sequence of false theories may approach the truth, solve increasingly important problems, or improve understanding. Truthlikeness accounts, on which progress is a theory's coming closer to the truth, represent one route through this difficulty. Others are represented by problem-solving accounts, on which progress is an increase in problem-solving effectiveness; noetic accounts, on which it is an increase in understanding; and functional accounts, described below, on which it lies in defining as well as solving research problems (Niiniluoto 2024, §§3.2, 3.6; Nickles 2017, §1.5; Dellsén 2021, 11259–11262; Shan 2019). Laudan's problem-solving account shows how such a route works. As Dellsén notes, the account is explicitly designed not to require that progress be factive (Dellsén 2025, 2128), so it can count as progressive a succession of theories each of which may later

prove false. Shan's functional account revises the problem-solving approach it attributes to Kuhn and Laudan by counting the definition of research problems, as well as their solution, among the constituents of progress. On that account, Darwin's theory of pangenesis marked progress in the study of heredity by providing a useful, if imperfect, framework of problems and hypotheses, although the gemmules it posited do not exist (Shan 2019, §§2–4; see Dellsén 2021, 11266). These accounts are rival analyses of progress, not interchangeable repairs to a knowledge definition.

Returning to the truthlikeness accounts named first above, Niiniluoto's version analyzes progress as increasing truthlikeness, measured as closeness to the most informative true answer to a cognitive problem, so that a step from one false theory to another can be progressive. Niiniluoto distinguishes real progress, which can go unrecognized while that answer is unknown, from estimated progress, judged by expected truthlikeness on the available evidence and open to revision as the evidence changes (Niiniluoto 2024, §3.5). Bird, who holds that progress is the accumulation of knowledge, had objected that accounts defining progress by truth or truthlikeness alone would count as progress even a belief that happened to be true or truthlike, reached by irrational methods without justification. Niiniluoto replies, following Cevolani and Tambolo, that estimated progress already gives justification a place: justification is not part of what real progress consists in, but estimated progress is relative to evidence and justification, and claims of real progress can be justified by appeal to expected truthlikeness. Niiniluoto also raises as a further problem whether Bird's way of handling false theories, by treating an approximately true theory as the fully true claim that the theory holds approximately, can distinguish a sequence of theories that approaches the truth from one that recedes from it (Niiniluoto 2024, §3.6). Brought back to the choices listed at the start of this subsection for historically successful but false theories, the problem bears on the first of them, identifying true restricted claims that were known, since Bird's claims that a theory holds approximately are claims of that kind.

The separation drawn in Section 3, between being true and being shown to be true, bears on an objection to factive accounts of scientific progress. Laudan argued, as Dellsén reports, that factive accounts make progress epistemically inaccessible to scientists themselves: since even the most empirically successful theories might turn out to be false, scientists could not tell whether later theories are more faithful to the facts than their predecessors, and progress so inaccessible cannot guide rational decision making in science (Dellsén 2025, 2128, 2140). As Niiniluoto puts it, Laudan repeats the classical skeptical argument that knowing the truth is a "utopian" task (Niiniluoto 2024, §2.4). Dellsén replies that a rational choice requires only rational credences about how far each available course of action would promote progress, in the sense defined below, not access to whether it does, so a factive standard of progress can still guide choice (Dellsén 2025, 2128, 2140–2141).

Niiniluoto's distinction between real and estimated progress, above, can be read as answering the same objection from within the factive framing retained here, and Reichenbach as having earlier met a difficulty of the same form from outside that framing. Reichenbach treated truth as an idealization of a high weight and knowledge as a system of posits rather than of true propositions, and argued that induction can be justified without knowing whether it succeeds: if limiting frequencies exist, persisting with induction and correcting its estimates must eventually bring the estimates within any chosen distance of those limits and keep them there, although, as Reichenbach conceded, convergence may come too late to be observed within a human lifetime (Reichenbach 1938, 187–191, 348–351, 360–362, 403–404). Reichenbach granted that this argument does not single out induction, the rule of taking the observed frequency so far as the estimate of the limit, since any rule that adds to the observed frequency a correction that shrinks to zero as observations accumulate yields estimates that approach the same limit, and preferred induction among such rules for the practical reason that it carries the smallest risk (Reichenbach 1938, 354–356). If so, a guarantee of this long-run kind does not by itself settle which method to follow at a given time, the guidance whose absence Laudan's objection pressed and that Dellsén's appeal to rational credences addresses. Where assurances about an artificial system's learning or updating rule take this long-run form, the same reasoning suggests that they do not by themselves settle what the system is warranted in concluding at a given time.

Comparisons of truthlikeness also depend on how a problem is framed. Oddie and Cevolani report an argument, due to Miller, that on likeness accounts of truthlikeness, which measure it by a theory's similarity to the truth, Niiniluoto's among them, the ranking of two theories can reverse when the same theories are stated with different primitive predicates. In their example, weather can be described with *hot*, *rainy*, and *windy*, or with *hot*, *minnesotan* (hot if and only if rainy), and *arizonan* (hot if and only if windy). Each vocabulary can be defined from the other, yet a theory that makes fewer errors than another when counted in the first can make more when counted in the second. Of the two responses they describe, one holds that some predicates, such as *hot* and *rainy* but not *minnesotan*, carve reality at its joints; estimating truthlikeness then

depends on estimating which predicates do so. The other, which denies that the restated theories say the same thing, makes theories framed in different concepts hard to compare (Oddie and Cevolani 2022, §1.4.3). This suggests that estimated progress in Niiniluoto's sense is relative to a cognitive problem and the concepts in which it is stated, and revising those concepts (Chapter 4, Section 4; Chapter 5, Section 3; Chapter 6, Section 5) can change what is being compared. Applied to an artificial system that also chooses its representations, Miller's argument suggests a further point. If the system itself selects the vocabulary in which its results are stated, the same results may be rated closer to the truth under the vocabulary selected. Whether revising a representation captured the target better or changed the yardstick of evaluation then needs to be asked separately. On the first response above, estimating the former also requires estimating which vocabulary carves reality at its joints.

A further distinction concerns temporal contribution. A dataset, formalism, instrument, failed experiment, or classification may **constitute** an epistemic advance, **promote** a later advance, or merely be **rationally expected to promote** one when the work is undertaken. Dellsén gives the second and third predicates a precise form. On this account, an episode promotes progress to the extent that the chance-weighted amount of progress across the developments that could follow it exceeds what it would have been had the episode not occurred; an agent is rational in expecting an episode to promote progress to the extent that the same comparison, made with the agent's own coherent credences, favors the episode. In Dellsén's example, a botched experiment that leads scientists to a successful rerun has not promoted progress if running it correctly the first time would have led further (Dellsén 2025, 2131–2135). These predicates can diverge from one another and from the results that in fact follow. Kept apart, they give a useful artifact no automatic credit for every later result in its causal future, and they do not make a responsible attempt irrational only because unforeseen events prevent its later use.

Consider representing the results of an artificial research system by a single progress score. That requires deciding how much each kind of change counts: new knowledge, an eliminated possibility, an improved method, and so on. The accounts of progress above differ on exactly this, so a claim that a single score represents the progress of the results as such already chooses among them. A proxy that measures only a limited aspect, or a value combined for a provisional decision, carries no commitment of that kind. An evaluation of the system's results that leaves that choice unmade would instead keep a record sorted by kind of change, distinguishing, among other things, target knowledge, collateral knowledge, changed evidence or credence, eliminated possibilities, recognized ignorance, improved method or capacity, promotion in Dellsén's sense, rationally expected promotion, and realized downstream contribution. The last records in retrospect what in fact followed, and this book adds it alongside promotion, which compares prospects. The items can overlap. Where the research system is artificial, each item can further be recorded as realized only in the system or as having entered *S*'s epistemic situation (Section 7); on accounts of possession that require entry, only the latter is yet a change in *S*'s epistemic situation, and so in what an account of progress for *S* would count. Kept apart, they let an evaluation count more than knowledge thresholds without treating every epistemic motion as progress. Someone who adopts one account of progress can still compute a single score from this record, weighting the items as that account does.

What the epistemological accounts require of artificial research

Sections 2 to 8 drew consequences for artificial research where each question arose. This section gathers them into requirements and says on which claim, and so on whom, each requirement falls. The unread artificial discovery of Section 7 shows why the second task, saying on whom each requirement falls, matters. The same record can, once incorporated into *S*'s practice, contribute to the result's becoming *S*'s knowledge, be available as a collective resource on a different account, or fail to be anyone's knowledge despite containing a true result. What must be built thus includes the relation between a result and its possible bearer, and the accounts examined here specify different ways that relation can succeed in the case of knowledge. What is required changes with the claim being made: that an activity is research for *S*, that knowledge new to *S* has been realized and in whom, or that it has entered *S*'s epistemic situation (Chapter 1, Section 2). Throughout, the subject claimed to know need not be the actor that conducted the research (Section 2.1). Whether a requirement falls on the system therefore depends on whether the system is that subject, on whether it gathered the evidence and constructed the verification, and on the account of possession adopted.

The requirements are conditional. Each states what knowledge would require on the account in question and, where it differs, what an evaluator would need to establish to support the attribution; as Section 5 noted, insufficient evidence for an

evaluator does not by itself show that the subject lacks knowledge. The requirements of different accounts are not to be added up into a single list that every artificial researcher must meet (Section 4.2), and this section does not choose among them. The conditionals concern claims of knowledge; the corresponding consequences for claims of understanding are drawn in Chapter 3, Section 9.

Stated for whoever is claimed to know, the accounts compared in Sections 4 and 6 impose the following. If accessible reasons are required, a post-hoc explanation is insufficient unless the relevant subject can use those reasons in inference, challenge, and revision. If evidential fit is required, the live confidence or acceptance of the subject claimed to know must fit the evidence that subject has at the time; since fit is assessed at a time, an attitude left unrevised when that evidence changes can cease to fit. If the attitude must also be held on the basis of the evidence it fits, as Feldman and Conee's well-foundedness requires (Section 4.3), the operative basing relation becomes part of what is required. If responsible evidence acquisition is also required, the decisions on search, measurement, omission, and stopping through which the subject's evidence was gathered must be assessed separately from the final inference. If reliability is required, the process that forms or maintains the subject's belief must be reliable in the relevant environments; for an evaluator to support the attribution on that ground, that performance must be established across those environments. If safety is required, the subject's belief must not be one that could too easily have been false, and an evaluator supporting the attribution must examine contamination and nearby possibilities of failure along the route by which the belief was formed. If sensitivity is required, what must be examined is whether the subject would still have held the belief had the claim been false. If virtue-reliabilist competence is required, success must be attributable to a stable capacity of the relevant agent or organized system. If responsibilist virtue is required, the conduct of the subject's inquiry, and not only its reports, must express something that functions as intellectual character and motivation. If defeat sensitivity is required, the response to rebutting and undercutting information must reach the subject's commitments that rest on the affected claims, instruments, assumptions, and validators, not only the reported conclusions; provenance that preserves the defeasible support relations among these is one way to make that response traceable. If social criticism is constitutive, formal publication without uptake in Longino's community-level sense is insufficient. If group functionality is sufficient, the boundaries and organization of the collective, whether a human–artifact system or a wholly artificial one, become part of the epistemic claim. These last two concern possession as well as warrant, and Section 9.4 returns to them.

Drawn together, Sections 4 to 7 also imply that a knowledge claim in artificial research has to state its units: the process type whose reliability is claimed and whether selection by others lies inside it, where competence is seated (Section 4.2), the boundary of the evidence the system has and whether a retrieval store lies within it (Section 4.4), the boundary of a collective claimed to know (Section 6), the reference *S*, and the account of possession (Section 7). For an individual human investigator, who holds the belief, how far the evidence extends, and where competence is seated are largely fixed by there being one person. For an artificial system they need not be fixed in advance. Where a system works through instances, external stores, tools, or other systems, they can be drawn in several ways, depending on how these are connected, and whoever draws them, whether designers or a system that reorganizes itself, the verdicts change with where they are drawn (on the boundaries of an actor, Chapter 7, Section 2). Until the units that a given account turns on are stated, a claim cannot be checked against that account. The subsections that follow take the requirements by where they fall: on the system as the conductor of research, whoever knows (Section 9.1); on the system as the subject claimed to know (Section 9.2); on others who are claimed to know through it (Section 9.3); on the entry of a result into *S*'s epistemic situation (Section 9.4); and on changes short of knowledge (Section 9.5). Table 2.3 at the end of the section sets out the result in simplified form.

9.1 What falls on the system as the conductor of research

Some requirements fall on the system because it conducted the research, whoever is claimed to know. None of them is a condition of the first judgment, whether an activity is research for *S*. That judgment does not depend on whether the result meets the conditions of entry of the account of possession adopted (Section 7): an activity organized toward knowledge new to *S* can be research before anyone knows its result, or when it fails (Sections 7 and 8.1), and conversely a true output does not make a run research, since a causal contribution to an achievement is not by itself an attempt at it (Section 4.5). What falls on the conductor concerns how well the attempt was conducted, the second of the evaluations in Section 8.1, and what others need in order to come to know through it.

The first concerns the gathering of evidence. Where the system did the gathering, its decisions on search, measurement, omission, and stopping are assessed separately from the final inference. Whether that assessment is epistemic or practical is the dispute set out in Section 5: on Kauppinen's view it is practical, while an unsupported assumption about the reliability of sources on which the system acts remains open to epistemic criticism, of the system if it is itself a subject of such norms, and otherwise of those who built the assumption into it. The assessment thus applies to the system as gatherer even where it is not claimed to know.

The second concerns verification. Who constructs the verification is a different axis from who knows (Section 5). If the system runs checks that others designed, its reports have to show what was tested under which auxiliary assumptions, so that those who read them can locate what a result bears on. If the system constructed the checks itself, those who come to know through it and who take on the diagnosis of results themselves need to be able to read from its reports how it did so (Section 5): how many candidates were generated and how many of them were likely to be false, which choices were made after the data had been seen, whether the data that support the claim were used in constructing it, what would have counted against it (Section 5.2), and how a formally correct procedure was related to the question (Section 5.3). On accounts that require accessible reasons, recipients need reasons they can use, which may concern how the result was checked or the system's reliability for the use at hand (Section 5; Chapter 11, Section 5.2); the system's reports are then one source of such reasons. On accounts on which the reliability of the process suffices, a recipient can know without reading these, but an evaluator who supports the attribution still needs evidence that the system's way of verifying is reliable. Whether the system also constructs and revises the governing standards is a question of autonomy (Chapter 7, Section 4.3), a different axis from who knows, and not among the conditions of knowing (Section 5.3; Chapter 8, Section 1.3); for research for a human *S*, inherited human standards can serve.

The third concerns the independence of checks. A critic adds to error detection only so far as it does not go wrong where the producer does, so where checks are run by instances of the producer's own base system, or by other systems that may go wrong in the same places, a report would need to show how far they are independent (Section 5; Chapter 11, Section 3). This holds whether *S* is human or artificial, and agreement among such instances does not make a result true (Section 6).

The fourth concerns the unit of a claim credited to the system. A reliability claim has to state its process type and the distribution over which it holds, and selection performed outside the system, such as human filtering, must not be hidden inside a claim about the system, since it can make outputs safe without making their success creditable to the model (Section 4.2). For a recipient, human or artificial, who relies on the whole pipeline, the whole may be reliable while the model alone is not; what is required is that the unit be stated, not that there be no filtering.

Answerability, the responsibility to answer when questioned about an output, is a further matter, and it need not track who knows either: those who answer for a system's outputs need not be among those who know. If a system's reports are treated as readings of an instrument rather than as assertions, applying Goldberg's account as reported in Chapter 11, Section 5.1 directs reliance toward the epistemic work of its designers and makers; whether its designers and makers can take informed responsibility for an essentially opaque system is disputed, as Koskinen argues (Koskinen 2024a, 464–465; Chapter 11, Section 5.3).

9.2 When the system itself is claimed to know

Where the system itself is claimed to know, the conditionals above fall on its own states. Several of the false positives of Table 2.1, which take this case as their example, then share one form with a point from Section 2.1: what the system displays differs from what governs its output and revision. A stored sentence is weak evidence of belief beside stable integration with inference, action, and revision (Section 2.1); a post-hoc rationale can be unrelated to the operative route; hedging in reports can leave the conduct of inquiry unchanged; and a corrected report can leave the operative commitment as it was (Table 2.1; Section 4.3). Taken together, these cases suggest that much of what this claim needs is evidence about what governs the system, not about what it shows.

For several relations, the chapter has said what that evidence is. For the attitude, the system's states must satisfy the belief condition of the account adopted, or an account that does not require belief in that form must be defended (Section 2.1); and since a state can be a belief, an acceptance, a credence, or a supposition, which of these it is has to be settled before a condition for belief is applied to it (Section 2). For basing, the system's operative commitment must be based on the evidence

it fits, on accounts that impose such a condition; co-location of a conclusion and its grounds does not establish it, and provenance that records which items governed a commitment is the device that fits causal theories of basing most directly, though, as a means of leaving evidence for an attribution, its absence does not by itself show the absence of basing (Section 4.3). For evidence, on the functional accounts of belief, crediting the system with knowing what its store contains requires evidence that the store's contents are integrated with its inference and revision, not only that they are retrievable (Section 4.4). For defeat and verification, the system's operative commitment, not only its reports, must respond appropriately to rebutting and undercutting information and change with the revisions that verification prompts (Sections 4.3 and 5); on Williamson's argument an updating that can only add evidence cannot represent the loss of evidence (Section 5). On Simion's knowledge-first account, if the system is a trained one, whether it has the function of generating knowledge turns partly on what its training and selection rewarded, not only on its present accuracy (Section 4.1). And a system whose commitments can be revised by proofs, observations, failed predictions, and criticism is related to its object differently from one optimized only for textual plausibility or social approval (Section 3).

Where the system is not a member of *S*, whether *S* is human or artificial, a system that knows is a bearer outside *S*; the case in which it is a member is taken up in Section 9.4. Its knowing realizes knowledge new to *S* somewhere, the second judgment, without by itself securing the third, that the result has entered *S*'s epistemic situation, and a claim about it needs an account of how its achievement is new relative to, and sought for, *S* (Chapter 8, Section 1). The claim that the system knows is thus a further claim, beyond both the claim that it conducted research for *S* and the claim that the result has entered *S*'s epistemic situation, and unlike the latter it carries the burden of this subsection (Chapter 8, Section 1.1).

Two further points bear on applying these requirements. First, the conditions can return conflicting verdicts. Williamson argues that safety and sensitivity come apart. Safety does not entail sensitivity, and knowledge does not require sensitivity: as the example of the mark one meter above the water showed in Section 4.1, even a very slight systematic bias can make almost all of a subject's beliefs in contingent propositions insensitive while they are still known (Williamson 2000, 147, 159–161, 307–310). For evaluation, this means that a sensitivity test applied to a system with a small systematic bias could report failures that a safety-based account would not count against its knowledge.

Second, the alternatives that the conditionals set out do not require artificial research to reproduce the internal route used by a human investigator. A system unlike a human could realize a reliable, safe, reasons-responsive, or socially integrated relation through very different mechanisms. What cannot be inferred merely from difference of method is that no epistemic standard applies. The burden shifts from resemblance to humans to evidence that the alternative realization satisfies the relevant account, which is one reason the account has to be named.

9.3 When others are claimed to know through the system

Where humans, or artificial systems other than the producer, are claimed to know through the system, the system need not believe its output. An AI can, for example, contribute to human knowledge without literally believing what it outputs. The conditionals then apply to the belief of those claimed to know, or to their operative commitments if they are artificial, and the system's output bears on these as evidence or as part of the process that formed them (Section 4; Chapter 8, Section 1.1). The same output can then ground knowledge in one recipient and not in another. A human investigator who accepts a hypothesis that an AI proposes only after an independent test that would have failed had the hypothesis been false can hold a sensitive belief, although the AI would have proposed the hypothesis either way; conversely, if the output would have asserted the claim even had it been false, worded differently or not, a belief based only on that output is insensitive. An attribution of knowledge to those who know through the system is therefore an attribution about their belief and what it rests on, of which the output is one part, not about the system alone. The gap in basing that Section 4.3 found on the system's side reappears on theirs: reading a persuasive report supplies reasons, but whether a belief is based on them or on the system's prestige is a further matter.

What the system needs to supply in this case differs with the account adopted for the recipient. On accounts that require accessible reasons, the recipient needs reasons it can use in inference and objection. Where these concern how the result was checked, what the reports make visible is central; where, as Section 9.1 allows, they concern the system's reliability for the use at hand, the reports are one source of them. On accounts on which the reliability of the process suffices, or on Sosa's account from Section 4.2 applied here, the recipient can know without being able to read the reports as a whole; what then

falls on the system is less that its reports be readable than that the process actually be reliable, and an evaluator who supports the attribution needs evidence of that. The requirements of Section 9.1 on what reports need to show come from the recipient's side chiefly where the recipient takes on the diagnosis of results itself or where its reasons must concern how the result was checked, and otherwise from what an evaluator needs in order to support an attribution. In every case they fall on the system because it conducted the research and constructed or ran the checks, not because it knows.

Where the recipient is artificial, the requirements of Section 9.2 move to it: whether it believes, and whether its operative commitment is based on the output and revised under defeat (Section 4.3). Carried over to such a recipient, the sensitivity example connects with the question of independence that Chapter 11, Section 3 examines: a recipient whose check of the output is made by an instance of the producer's own base system, or by another system that errs where the producer does, and which would have accepted the output whether or not it was true, holds an insensitive commitment, however many instances agree.

Other relations can make the system's own knowing matter even where others are the knowers. Where its reports are taken as assertions, accounts of assertion place conditions on it as speaker, although whether current outputs are assertions, and whether the norm of assertion requires knowledge, rational belief, or support that varies with context, are disputed (Chapter 11, Section 5.1); and where the achievement sought is a skill realized in the system, the system is its first bearer (Section 6).

9.4 When a result is claimed to have entered *S*'s epistemic situation

The verification schema of Section 5 ends with whoever is claimed to know. Unless *S* itself is the subject claimed to know, entry into *S*'s epistemic situation is a further step that the schema does not contain, and what it requires depends on *S* and on the account of possession adopted (Sections 6 and 7). What has to be built are routes by which the result can enter validation, inference, action, and correction, and they need not lie within the system that conducted the research (Section 7). Whether such routes suffice differs by account. On Bird's account, extended to a record that no one has read, they may. On Lackey's, since a society as a whole without operative members has no group belief, the result must reach and be believed by members of *S*, and, where entry is claimed through a group's knowing it, by the operative members of a group within *S*; on Longino's, it must be accepted by members after criticism under the account's conditions; and on Giere's adaptation of knowledge without a knowing subject, it must be certified by reliable, accepted procedures and publicly accepted (Sections 6 and 7). A claim that a result has become *S*'s knowledge can therefore be assessed only when it reveals enough of its account (Section 6). The distinction between a dormant record and a live epistemic capacity names what makes a record a live capacity, but it marks the boundary of knowledge only on some accounts; and in the dispute over Lackey's missing-child case, whether a group knows a result comes apart from whether it should have known it, so that a group whose system collates a record can be answerable for a result it does not yet know (Section 7).

Where *S* is artificial, a question arises before any of these. Both roles of *S* presuppose that its epistemic situation could come to include the knowledge sought (Chapter 1, Section 2), and that situation now consists of artificial members, their organization, and their records; so whether such members can be knowers arises before any member knows the result, and whether or not *A* is one of them (Section 6). The accounts then ask of the members what Section 9.2 asks of the system: whether they believe the result or can take part in joint acceptance of it. Of their organization they ask whether it realizes a properly organized cognitive system and, on Bird's account, for what that organization was selected or designed (Section 7), whether its interaction involves the diversity and public standards of criticism that Longino's account requires, and whether certification by artificial members counts as certification, including where they are instances of one base system (Section 6; Chapter 8, Section 1.3; Chapter 11, Section 3). No human-membership condition is built into the working characterization, but nonhuman subjecthood is not established by coordination or plurality alone. As Section 9.2 noted for the system, the relations need not be realized as they are in a human community; what has to be shown is that the realization satisfies the account.

The producing system may itself be a member of an artificial or mixed *S*. Of the cases in which the producing system is the one that knows, this is the only one in which its knowing can do the work of the third judgment: on individual-possession accounts, if the system knows, a member of *S* already knows, and the result may have entered *S*'s epistemic situation as soon as it was produced. On Lackey's account this holds for a group only if the system is among its operative members and a significant percentage of them believe the result; on Longino's, the system's acceptance is at most one member's; on Bird's,

the verdict still turns on the organized system as a whole (Section 7). For a mixed *S*, a claim of entry has to say which members, human or artificial, the result reaches and how possession, memory, and correction are distributed among them (Chapter 8, Section 1.2).

9.5 Changes short of knowledge

One requirement comes not from an account of knowledge but from the claim about progress that Section 8.2 makes on Wilholt's account. If making ignorance explicit counts as epistemic progress, evaluation must also distinguish the absence of an answer from a represented and actionable unknown, and say for whom it is actionable: for the system's own further inquiry, or for members of *S*, human or artificial, who take up its work. Answer volume and closure rate can then be anti-correlated with research quality when they erase warranted suspension, live alternatives, or the evidence still required to resolve a question. Williamson's arguments, if they hold and the system's retrieval has no upper bound on its time, so that it cannot tell what is missing from what has not yet been reached, bound what can be asked here: the system's own checks cannot guarantee that its record of what it does not know is complete (Section 8.2), so the requirement concerns what is represented and for whom, not completeness.

Where an evaluation does not select an account of progress, it keeps the record of Section 8.3: changes sorted by kind, and, where the research system is artificial, by whether each is realized only in the system or has entered *S*'s epistemic situation. A claim that a single score represents progress as such already chooses among the accounts of progress (Section 8.3).

Table 2.3 sets out where the requirements of this section fall. Its rows are the claims distinguished in Chapter 1, Section 2 and Section 8.1, with the claim that the system knows separated from the claim that others know through it, since the requirements differ most between those two. Its last column states what each claim does not require, which Tables 2.1 and 2.2 leave implicit. The table is a simplified overview: each cell compresses requirements that differ by account, and the rows are not exclusive, since the system can, for example, be a member of *S*. As with Table 2.1, the entries in the second column hold only on the accounts concerned and are not to be combined into one list that every system must meet (Section 4.2).

Table 2.3. Where the requirements fall: a simplified overview

Claim	What must hold (on the adopted account, where the claim depends on one)	What an evaluator needs to be able to see	What the claim does not require
The activity is research for *S*	organized direction toward an achievement new to *S* that *S* could come to hold	the organization of the attempt and the reference *S*	that knowledge was reached; that the system knows or believes the result
The system conducted it well, whoever knows	responsible evidence acquisition; reports adequate to the checks it ran or constructed	how evidence was gathered; what was tested under which assumptions; how many candidates were generated; how independent the checks were; the unit of any claim credited to the system	that the system construct or revise its governing standards, an autonomy question (Chapter 7, Section 4.3)
The system itself knows the result	its states meet the account's belief condition, if it has one, and its operative commitment the account's further conditions	what governs its output and revision, not only what it displays	that it follow a human route; that the result has entered *S*'s epistemic situation
Others know through the system	their belief, or their operative commitment if they are artificial, meets the adopted account, with the output as evidence or as part of the process	what their belief rests on, including whether any check was independent of the system	that the system believe the output (though where its reports are taken as assertions, accounts of assertion place conditions on it as speaker, and where the achievement is its own skill, it is the first bearer)
The result has entered *S*'s epistemic situation	routes into validation, inference, action, and correction, and whatever the adopted possession account adds	which account is relied on, and which members or organizations the result has reached	that the routes lie within the system; that the system be a knower, unless it is a member of *S* whose knowing is claimed to count
For an artificial or mixed *S*, in addition	that *S*'s members or organization can hold the achievement, and the possession account applied to them	independence and diversity among members, including where they are instances of one base system; the organization of the collective; for a mixed *S*, which members the result reaches	that the relations be realized as in a human community
A change short of knowledge	a represented limitation or other change, recorded by kind	for whom it is represented and actionable, and whether it has entered *S*'s epistemic situation	that the change be knowledge; a single progress score

Comparing the requirements of Table 2.3 account by account brings out consequences that the accounts share and consequences on which they part. One shared consequence is that, whichever account it is checked against, a knowledge claim about an artificial configuration can be checked only once it is settled where the units that account turns on are drawn. Why the units are needed differs by account. On evidentialism, for example, how they are drawn settles what evidence the system has; on reliabilism, which process type has its reliability assessed; and on virtue accounts that require competence, where competence is seated (Sections 4.2 and 4.4). One consequence on which the accounts part is whether a recipient who knows through the system must have reasons it can use and, if it must, whether those reasons must concern the verification itself. Accounts that require accessible reasons demand reasons the recipient can use in inference and objection, but these need not be about the verification; they may concern the system's reliability, and the reports are one source of them (Section 9.1). On reliabilism, on which the reliability of the process suffices, and on Sosa's account applied here, the recipient can know without reading the reports. A recipient needs to read from the reports how the verification was constructed where its reasons must concern the verification itself or where it takes on the diagnosis of results itself. The same requirement on what reports show is therefore imposed, in those cases for the recipient's sake, and otherwise so that an evaluator can establish that the process is reliable (Section 9.3).

Knowledge does not exhaust research achievement, but it is a demanding target. On the accounts compared, novelty, truth, reasonable acceptance, reliable success, understanding, social possession, and research capacity belong to different claims, and their requirements fall on different parties, so none of them is to be inferred from another without an argument from the account adopted. Carried through artificial configurations, these distinctions have concrete consequences. On evidentialism, counting only the retrieved documents as evidence the system has can leave the conclusion justified despite poor evidence gathering, while widening the boundary to count the whole store can cost the same conclusion its justification, since the counterevidence in the store then counts as evidence the system has (Section 4.4). On accounts that require basing, good reasons attached to a report are one thing and a conclusion held in place by them another, and a corrected report can leave the

operative commitment unchanged (Section 4.3). A chance generator can contribute to another subject's knowledge without knowing and without conducting research (Section 4.5). For a system that generates candidates in bulk, passing a check can raise the support for a claim without making the claim probable (Section 5.2). And whether a result that no human has read is knowledge for *S* differs not only between the accounts of Bird, Lackey, and Longino but also with whether machines count as knowing members (Section 7). Memory, updating, and the arrangement of members are not only external means of carrying knowledge; on some accounts they bear on whether there is knowledge at all. Counting apparently novel conclusions makes none of these differences visible.

One question now becomes unavoidable. Suppose an artificial system produces reliably validated true predictions that a community can use, but no human member of the community can explain why the predicted relation holds, and the system supplies no usable representation of the dependencies behind it. The present chapter has asked whether knowledge has been brought about. Even where it has, whether that reaches the intended target depends on what the inquiry sought. The next chapter asks when understanding is achieved, and by whom: what would count as understanding, how it relates to knowledge, who, including the system itself, can have it, and what follows when the two come apart.

Chapter 3 — When Is Understanding Achieved, and by Whom?

Knowledge, Explanation, and Opaque Artificial Inquiry

Knowledge and understanding as achievements that research may seek

The working characterization orients research toward epistemic achievement, which this book examines principally through knowledge and understanding (Chapter 1, Section 1). Chapter 2 took up knowledge. This chapter takes up understanding: what would count as understanding, how it relates to knowledge, who can have it, and what follows for artificial research when the achievement sought is understanding, when knowledge is reached without it, or, on some accounts, when understanding is reached without new knowledge. It treats the two as achievements that research may seek separately or together, and asks how they differ and when they come apart.

The preceding chapter ended with a deliberately strong boundary case, which this chapter calls the strong predictive case, in which the two may come apart. An artificial system identifies a previously unknown regularity. Independent evidence supports the claim within a specified domain. The relevant community can use the result, track its uncertainty, and revise its reliance if the evidence changes. Yet no human member of the community can explain why the regularity holds, and the system supplies no usable representation of the dependence responsible for it.

The case raises questions that cannot be answered by repeating that the result is accurate. Accuracy does not by itself establish knowledge, but suppose that the further conditions of an account adopted in the preceding chapter have been met. Has understanding then been realized anywhere: in the system, whose states may or may not amount to understanding (Section 6), in some human, or in the community? If the attempt sought understanding of why the regularity holds, has it reached its target, although knowledge that the regularity holds has been obtained? Does knowledge of this kind require understanding, so that the two cannot come apart here? And has the community made progress if it can predict more while understanding no more? Whether the activity is research for the community does not turn on these answers, since research is characterized by the attempt rather than by its success (Chapter 1, Section 3).

These are not different formulations of one question. They concern different objects of evaluation (compare the evaluations distinguished in Chapter 2, Section 8.1):

1. whether an activity is research for *S* in the sense of the working characterization;
2. whether an output has attained the status of knowledge, who, if anyone, knows it, and whether it has entered the epistemic situation of the society *S* for which it is sought (Chapter 1, Section 2; Chapter 2, Section 6);
3. whether some subject, human, artificial, or collective, understands, and whether that understanding has entered, or become available to, *S* (Section 6);
4. whether the wider program has made progress;
5. whether the continuing research system (an artificial system alone, or the people, institutions, and artificial components that carry inquiry on over time) preserves the capacities required for criticism, transfer (carrying a result over to other conditions or problems), diagnosis, and correction; and
6. whether the result is important or valuable relative to relevant purposes and affected parties.

A failed investigation can be research without producing the achievement it sought. A community may know that a regularity holds without understanding why. A model that misrepresents some features of its target may nevertheless improve understanding. A locally warranted result may enter a research ecology that is becoming more homogeneous and less capable of detecting its own errors. These judgments interact, but none substitutes for the others.

These distinctions specify the work left by the general orientation toward epistemic achievement. If understanding differs from knowledge, which is sought is part of specifying the target; if some knowledge requires understanding, that constrains the proposed separation. Under the working characterization, neither research as such nor any single research act is required to bring about an epistemic achievement, knowledge or understanding (Chapter 1, Section 3). Where an attempt's purpose is understanding, however, obtaining knowledge does not by itself show that the attempt has reached its target. Sections 4 and 7 examine these alternatives, and Section 9 asks what they require of artificial research.

The chapter proceeds as follows. Section 2 separates prediction from explanation. Section 3 sets out what the main accounts count as understanding, and Section 4 applies them to the strong predictive case to ask whether knowledge and understanding come apart there. Section 5 asks whether the system's opacity undermines the warrant for that result or rules out understanding, by locating what is opaque, in what role, to whom, and at what stakes. Section 6 asks who understands: the system, those who use it, or a collective. Section 7 asks what understanding is valued for and what it may enable over time, and Section 8 why a local achievement does not settle the capacities of continuing inquiry. Section 9 asks what accounts of understanding require of artificial research, showing how the answer changes with the achievement claimed, its bearer, and its use, and closing by returning to the chapter's question.

Prediction and explanation are related but non-identical achievements

A simple route into the problem is to distinguish the questions that an inquiry can answer. A prediction identifies what will occur, what value to expect, or which outcome belongs to a case. It need not concern the future: retrodiction, classification, and prediction of an unobserved property can have the same functional form. An explanation addresses why a phenomenon occurs or how it depends on mechanisms, causes, constraints, unifying patterns, reasons, or other relations judged explanatorily relevant.

Success at one does not logically guarantee success at the other. A model can predict a target while exploiting a stable proxy and reveal nothing about the dependence investigators want to understand. An explanatory model can identify a mechanism while producing poor point estimates because initial conditions are unknown. A highly idealized representation can make a pattern intelligible while sacrificing predictive precision. Prediction and explanation are therefore not two names for a single scalar measure of scientific success.

Nor are they opposing kinds of artifact. An explanation can generate new predictions. Predictive failure can discriminate among explanations. Rich counterfactual performance, answering correctly a wide range of questions about what would happen if conditions were different, can provide evidence that a model captures relevant dependencies. In some domains, the same model contributes to both. The stable claim is only that answering a predictive question does not, by that fact alone, answer an explanatory one.

Positions on how the two achievements are related diverge further than this. De Regt, whose account of understanding is examined in Section 3.1, argues that prediction without understanding is not merely insufficient for science but impossible in practice: on de Regt's account, successful prediction from a theory requires that the theory be intelligible to those who use it, and the ability to recognize the theory's qualitative consequences in turn shows that it is (de Regt 2017, 91–92, 107). The understanding this requires is understanding of the theory, which de Regt calls its intelligibility and which, as illustrated with cases from behaviorist psychology and quantum physics, is needed for applying the theory to concrete phenomena. On this account it is distinct from scientific understanding of the phenomena, which also requires an explanation (Section 3.1). In a later co-authored chapter, Florian Boge and de Regt describe deep neural networks as data-driven rather than theory-driven and write that, at first sight, such networks may excel in predictive power while providing little understanding, because they lack intelligible theories (Boge and de Regt 2025, 419–420). Their chapter does not address de Regt's earlier claim that prediction without understanding is impossible in practice. The two need not conflict, since that claim concerns successful prediction from a theory, and the networks are described as lacking an intelligible theory. Potochnik argues that the aims of science are in tension: tools suited to one aim are poorly suited to others, a consequence, on this view, of the complexity of phenomena and the limits of human cognition and action. Robustness analysis across climate models with competing assumptions, for example, is used to generate predictions, although none of the models is expected to represent the causal

influences on climate accurately or to explain climate change; Potochnik concludes that successful prediction should not be expected to facilitate explanation, nor the reverse (Potochnik 2015, 16–21). On this account the aims are valued for furthering human comprehension and control of a complex world, and their pursuit is relative to human limitations (19–20). Potochnik does not discuss artificial inquirers; applied to research done by one, the complexity of phenomena remains, while which limitations the tension answers to depends on whose comprehension the research is to further, the system's own or that of a human *S*.

For deep neural networks, machine-learning models in which many intermediate layers of interconnected units map inputs to outputs, Claus Beisbart asks under what conditions scientists who obtain a prediction from a trained network thereby understand why what it predicts is the case, which Beisbart calls explanatory understanding. The conditions are stated broadly, in parallel with the traditional analysis of knowledge as justified true belief, so that several accounts of explanation can fill them in. On this account, the scientists must have an explanation, that is, an argument that meets some further condition, such as containing a law, citing causes or a mechanism, or unifying, and that contains no irrelevant information; the explanation must be correct; and the scientists must accept it, be justified in accepting it, and be able to draw inferences with it (Beisbart 2025, 5–8). These conditions are applied to what Beisbart calls the simple setting, in which scientists know how to train the network and interpret its inputs and outputs but need not know what it does with the data (8). Beisbart's central argument concerns the first condition. The network's computation can be reconstructed as an argument that runs from its input, together with the assumption that the input–output function it realizes holds of the target system, to its output. Unless more is known about the network, however, nothing shows that this argument draws on a law, specifies causes or a mechanism, or unifies, whereas a computer simulation that takes the Navier–Stokes equations of fluid motion as premises is known to contain laws of nature (10–11). Scientists therefore cannot tell whether the network contains an explanation (13). The other conditions, on this argument, fail with this one. A track record of successful predictions can justify accepting that the network's input–output function is correct, but scientists cannot justifiably accept an explanation they do not know, and they cannot use an explanation they cannot state to draw the inferences that understanding requires (14, 16). Beisbart concludes that, in this setting, a scientist typically does not understand why the predicted outcome holds (24).

Beisbart's argument bears closely on the strong predictive case without establishing it. It shows how justified reliance on a predictor can coexist with the absence of explanatory understanding, and which conditions fail when it does. Beisbart's question, however, concerns a single predicted event or state rather than a regularity, the conclusion is that such understanding is typically absent rather than impossible, and the argument treats the network as a tool used by humans, leaving aside whether the network itself understands (4, 9, 18). Section 4 returns to the routes by which, on Beisbart's account, networks can still contribute to understanding. The more general distinction between prediction and explanation is left to the theories of explanation and understanding examined below.

That answering a predictive question does not by itself answer an explanatory one is a question-relative point rather than a universal ranking. If an inquiry asks which compound will bind to a target, a domain-bounded predictive answer may satisfy the question even when the mechanism remains unknown. If it asks why the compound binds, the same output is incomplete. If it asks for a proof, an interpretation, a construction, or an intervention, predictive accuracy can be useful without being the requested achievement. The working characterization of research is broad enough to include these inquiries precisely because it does not identify research with prediction.

Prediction itself also comprises several distinct claims. Evidence that one output is correct is not automatically evidence that a rule is reliable in a reference class. Reliability in one distribution does not establish transfer under shift. Observational prediction does not by itself establish the result of an intervention. A model–target correlation does not establish that the model represents an explanatory dependence. Reliability can also change between development, deployment, and later sociotechnical adaptation (Grote, Genin, and Sullivan 2024).

Woodward's account of causal explanation traces the gaps between reliability and transfer and between observational prediction and intervention to what different uses of a relationship require. Woodward argues that one variable can make another more predictable even though interventions on the first would not change the second (Woodward 2014, §7), and, writing with Kenneth Kendler, that how far a predictive relationship continues to predict well in other populations or circumstances is a question that can be asked even of relationships that are not causal (Woodward and Kendler 2023, 14). On Woodward's account, two representations can agree about the correlations so far observed and still disagree about what

would happen under an intervention, so learning through passive observation that two variables are correlated does not by itself tell an agent whether changing one would change the other; where the correlation comes from a common cause, an intervention on one of the variables breaks the causal connection that produced it, and the correlation no longer holds (Woodward 2023, §5).

Otsuka gives the gap between observational prediction and intervention a statistical form. On Otsuka's account, inferential statistics presumes that the population behind the data, described by a probability model, stays the same across observed and unobserved cases, whereas an intervention changes the structure of the population itself; handling causal effects therefore requires further ontological assumptions. In the approach of structural causal models, the causal structure that generates the population cannot be determined from probability distributions alone, and assumptions such as the Markov condition and faithfulness (the assumption that every independence in the data comes from the causal structure rather than from accidental cancellation) are posited as bridges between causal structure and probability distributions, through which the results of interventions are evaluated and causal structure is inferred from statistical data (Otsuka 2018, 188–189).

The strong predictive case of Section 1 can therefore be assessed only once it specifies what is known, and the distinctions drawn above separate the possibilities. The community might know that a regularity holds within conditions C, know that a predictor has error rate e in reference class R, or know that an intervention yields an outcome under a controlled protocol. It does not acquire an unexplained universal law merely because one benchmark is accurate. Specifying which proposition is known and over what range, together with naming which object and kind of understanding is claimed or absent (Section 4), is required before the relation between knowledge and understanding can be tested at all, since how much understanding is missing depends on what is known.

What would count as understanding?

Understanding is not a theory-neutral success label. The current philosophical literature disputes its objects, the psychological relation that understanding requires a subject to bear to those objects, and the normative relation that makes apparent understanding successful (Grimm 2021). Only once these three dimensions are explicit can a claim that a system understands something, or an aim of bringing understanding about, be stated precisely enough to be evaluated or built toward.

The **object** may be a proposition, an explanation, a phenomenon, a subject matter, a dependence structure, a model, a practice, or another perspective. The relevant **psychological relation** may involve belief, acceptance, representation, coherent organization, inferential ability, manipulation, practical skill, or perspective-taking. The **normative relation** may require truth, empirical adequacy, appropriate grounding in reasons or evidence, non-accidentality, explanatory correctness, or something weaker than the justification required for knowledge.

A further dispute cuts across these dimensions: whether understanding is a species of knowledge. Grimm, Khalifa, and Greco treat the normative profile of understanding as essentially that of knowledge, whereas Kvanvig, Pritchard, Hills, and Dellsén argue that understanding tolerates kinds of luck or defeat that knowledge does not. Pritchard's case shows what kind of luck is at issue. Someone asks a genuine firefighter what caused the fire at their house and is correctly told that it was faulty wiring; but nearby are people dressed as firefighters, and had the person asked one of them, they would easily have got a false answer. On Pritchard's account, the person lacks knowledge, since they could easily have been wrong, but understands why the house burned down, since luck that lies only in the surroundings is compatible with understanding, provided that the person grasps how faulty wiring could cause the fire (compare Section 7.1). If the person had asked one of the impostors and the answer had happened to be right, they would have neither knowledge nor understanding. Dellsén argues similarly about defeat: if an engine actually works as someone was taught, misleading evidence that the teacher was unreliable does not take away their understanding of how the engine works (Grimm 2021, §2.3; Pritchard 2008, §§9–10). On the first view, an attempt directed at understanding is directed at knowledge of explanations or dependencies, not only at knowledge that something is the case, so that who bears that knowledge and whether it has entered S's epistemic situation are the questions Chapter 2 asked of knowledge (Chapter 2, Sections 6–7). On the second, understanding can be present where knowledge is absent, and absent where knowledge is present, so that whether it has been realized, in whom, and whether it has entered S's epistemic situation need accounts of their own (Section 6).

This map already prevents several shortcuts. Producing an explanation-shaped sentence does not show that anyone understands it. Acting successfully does not by itself identify the object that is understood. A false model is not automatically useless, but neither does usefulness alone establish understanding. The same artificial behavior can count as strong evidence under one account and weak evidence under another. In what follows, this book groups the accounts into families by the dimension each emphasizes; the families are not mutually exclusive, since several accounts combine dimensions.

Mathematical practice gives a compact illustration. Two proofs can establish the same theorem while differing in explanatory depth, the dependencies they expose, the strategies they transmit, and the new problems they make tractable. Rav, whose account Chapter 1, Section 4 introduced, holds that the path a proof charts between concepts not only serves to convince but establishes interconnections that illuminate mathematical ideas, and Thurston emphasizes the growth of concepts and ways of thinking (Rav 1989, 67, 73 n. 30; Thurston 1994). Knowing that the theorem is true can be held fixed while the candidate achievement of explanation or understanding varies. For example, a collectively written paper that gave a new proof of the density Hales–Jewett theorem, first proved in 1991 by ergodic methods, gave two reasons for doing so: its elementary argument yields explicit bounds that the earlier proofs could not give even in principle, and it gives what the authors called arguably the simplest known proof of Szemerédi's theorem. The authors also wrote that they did not fully understand why their argument generalized so readily beyond its first case (Polymath 2012, 3–4). With the theorem held fixed, what the new proof added included propositions not known before, namely the bounds, and a simpler route to Szemerédi's theorem; what its authors said they lacked was a full understanding of why the argument generalized so readily. On Rav's account, the new proof itself would contribute more than the propositions it added, since the path a proof charts between concepts does more than convince of its theorem: it establishes interconnections that illuminate mathematical ideas (Rav 1989, 73 n. 30). Neither the contrast between proofs nor the collectively written paper decides which theory of understanding is correct; both show why the achievement needs to be assessed separately from knowledge that the theorem is true, whether it is then counted as further knowledge, of an explanation or of dependencies, or as understanding distinct from knowledge.

3.1 Explanatory accounts

On explanatory accounts, understanding a phenomenon requires grasping an appropriate explanation of it. Strevens states the strong thesis that there is no scientific understanding without explanation (Strevens 2013). Khalifa defends a deliberately reductive version: understanding why something is the case is, at its best, scientific knowledge of an explanation of it, and lesser understanding approximates such knowledge (Khalifa 2022, 1–5; see Grimm 2021, §4.1). De Regt's contextual theory concerns the understanding of why a phenomenon occurs that a scientific explanation of it provides. Its criterion counts a phenomenon as scientifically understood in this sense if and only if there is an explanation of it that is based on a theory intelligible to the scientists who use it and that conforms to the values of empirical adequacy and internal consistency. The theory is contextual because, on de Regt's account, intelligibility is not an intrinsic property of a theory but a value that scientists attribute to the qualities of a theory that facilitate its use, so that whether a theory is intelligible depends on contextual factors such as their skills and background knowledge (Boge and de Regt 2025, 417, following de Regt 2017, 40). One sufficient test of intelligibility that de Regt proposes is whether those scientists can recognize qualitatively characteristic consequences of the theory without performing exact calculations (de Regt 2019, 336 and n. 11). This account is therefore also ability-based. Strevens, by contrast, treats the ability to construct an explanation as strong evidence of understanding why rather than a condition of it (Strevens 2013).

These accounts make the verdict on an unexplained opaque prediction relatively clear. It may support predictive knowledge, but it does not thereby produce scientific understanding of the phenomenon. On these accounts, explanation-shaped output is also insufficient: what they require is an explanation that is appropriately correct, supported, and available in a form that realizes the relation the account requires.

Explanatory accounts also differ over which relation makes an explanation a source of understanding. De Regt and Dennis Dieks examine unificationism, on which science advances understanding by showing how many phenomena can be derived from few patterns or assumptions, as one of the main conceptions of explanatory understanding (de Regt and Dieks 2005, §3.2). Schurz and Lambert give this idea an explicitly relative form. In Gijsbers's summary of their account, understanding is relative to the cognitive corpus of an epistemic subject, whether an individual or a community: an answer to the question

why P can contribute understanding of P to the corpus only if it shows how P fits into the corpus, by connecting P with other elements of it, and only if the corpus, as changed by the answer, is more unified than before (Gijsbers 2011, 17). Gijsbers, who has no quarrel with their account as a theory of unification, argues that as a theory of explanation it makes unification neither sufficient nor necessary: an accidental generalization, such as that all members of a club are bald, increases unification once it has enough instances and would then count as explaining them, while legitimate explanations can rest on premises no less in need of explanation than what they explain, even when those premises disunify the corpus (Gijsbers 2007, 22–30). Drawing on historical variation in scientists' standards of intelligibility, de Regt and Dieks conclude that unification is an effective tool for achieving understanding but one among several, rather than its universal standard (de Regt and Dieks 2005, §3.2). Applied to artificial research, the relative form makes explicit two questions that such research raises sharply: relative to whose corpus, an artificial system's, a research team's, or a scientific community's, an answer is supposed to yield understanding; and whether a given artificial system has a corpus in their sense at all, that is, a set of sentences it believes and a set of arguments, deductive, inductive, or probabilistic, that it has mastered (Gijsbers 2011, 17), a question that an account framed in terms of corpora leaves open. Section 6 returns to the bearer of understanding.

3.2 Objectual and dependency accounts

Not every theory makes possession of an explanation necessary. Dellsén's dependency-modeling account says, roughly, that a subject understands a phenomenon by grasping a sufficiently accurate and comprehensive model of its place in a network of dependence relations (Dellsén 2020). Accuracy here concerns whether the model misrepresents those relations, and comprehensiveness whether it represents them, including what the phenomenon does not depend on; Section 7.3 returns to both. This is an account of objectual understanding, understanding of a phenomenon or subject matter as a whole, designed to reach beyond knowing an explanation of one target. Explanatory accounts also address objectual understanding; the dispute is whether it can outrun explanation. Dellsén has also defended the more radical claim that understanding need not require justification or belief in the manner knowledge commonly does, while still requiring a grasp of approximate truth (Dellsén 2017). Earlier, in examples that Strevens and Dellsén discuss, Lipton argued that even understanding why can be gained without an explanation. As Strevens, who rejects the conclusion, reports the argument, understanding can on Lipton's account come by grasping facts that are not explanatory. After working with a physical model such as an orrery, for example, one may understand why the planets sometimes appear to move backward and yet be unable to say what one has learned. Since on Lipton's account what cannot be made explicit cannot be an explanation, the understanding in such a case does not pass through an explanation. And Galileo's thought experiment, which shows by reductio that bodies must fall with the same acceleration whatever their mass, lets one understand, on Lipton's account, why this is so without grasping the scientific explanation of it (Strevens 2013). Explanatory theorists reply that such cases involve grasping part of an explanation, perhaps tacitly, while Dellsén accepts some of them only as objectual understanding (Strevens 2013; Dellsén 2020). De Regt, whose criterion in Section 3.1 concerns the understanding that explanations produce, notes the disagreement over whether understanding why can be had without explanation and sets it aside, while holding that understanding why is a cognitive achievement in its own right, not reducible to explanation (de Regt 2020, 924). Gijsbers, whose criticism of unification as a theory of explanation was noted in Section 3.1, has argued that successful classification yields a kind of understanding of its own. On Gijsbers's account, a purely classificatory biology that selected the anatomical features supporting reliable inductions would let investigators predict properties of newly discovered species and see how organisms hang together through relations of similarity, while explaining nothing; this understanding differs from the understanding that explanations give through relations of determination, and the account locates the achievement not in the inductions themselves but in the hard-won selection of the features that support them (Gijsbers 2011, 178–181). If this view is adopted, the strong predictive case of Section 4 turns on a further question: whether the features on which a validated predictor's classifications rest are available to be grasped.

Under such views, absence of a conventional why-explanation does not settle the case. A machine-generated representation might support understanding if it makes an accurate and sufficiently comprehensive model of the relevant dependence relations available to be grasped. By the same token, a large collection of correct point predictions need not constitute objectual understanding if it supplies no integrated representation of how the phenomena depend on one another. A distinction of Greco's, as Grimm reports it, makes the role of such a representation more precise. The object of understanding is the thing understood; a vehicle of understanding is a representation through which one understands it, as a good map of a

city is typically a vehicle for grasping how its streets and landmarks are laid out. A representation can itself become an object of understanding (Grimm 2021, §2.1). An account, too, can be the object: someone can know their way around astrology, on Elgin's example, although astrology affords no understanding of the cosmos (Elgin 2007, 4–5). Carried over to machine-generated models, the distinction allows such a model to stand in either relation. Used as a vehicle, it may support understanding of the dependence relations it represents; taken as the object, it may itself be understood without that understanding extending to the phenomenon. Evidence that investigators understand an artificial system's representation is therefore not by itself evidence that they understand the phenomenon through it.

3.3 Ability and intelligibility accounts

Ability-based accounts locate the grasp that understanding requires in what a subject can do, and they cut across the distinction drawn above. Hills characterizes the abilities that understanding why *p* involves, where *q* is why *p*, as cognitive control (Grimm 2021, §2.2). As Emma Gordon reports Hills's account in the *Internet Encyclopedia of Philosophy*, these abilities let one treat *q* as the reason why *p* rather than merely believe that it is: abilities to follow another person's explanation of why *p*, to explain *p* in one's own words, to draw the conclusion that *p* from the information that *q*, and to give *q* as the right explanation when given *p*, together with the last two abilities for relevantly similar cases (Gordon n.d., §3c). Hills's account is ability-based but still requires an explanation. Wilkenfeld's representation-manipulability account instead concerns understanding an object and does not require one: a subject understands an object when it has a representation of the object that it could modify to support effective inferences about it or manipulations of it (Wilkenfeld 2013, 1, 8–9). De Regt's intelligibility condition, described above, is also ability-based (see the overview in Grimm 2021, §2.2). Such accounts suggest tests of what a subject can do: draw consequences, answer what-if questions, explain relevantly similar cases, or modify a representation to support new inferences. These ideas are attractive for evaluating artificial systems because they suggest observable tests rather than an appeal to an inaccessible feeling of grasp.

They also introduce a construct-validity problem. The accounts themselves do not count shortcuts as understanding; Wilkenfeld's, for example, excludes lucky performance and the mere application of a memorized algorithm (Wilkenfeld 2013, 8–9). The difficulty is that a benchmark may not discriminate such shortcuts from the required abilities. A system may answer a benchmark through memorization, retrieval, contamination, hidden tools, or a second opaque procedure. Correct performance on selected counterfactuals may show a useful capacity without showing that the system possesses an integrated understanding of the subject matter. Performance on such tests is therefore evidence about the abilities that a specified account takes to constitute or indicate understanding; if it is to separate those abilities from shortcuts, it needs to be gathered across adversarially chosen variations. Such a battery of tests is not a theory-neutral checklist that defines understanding by stipulation.

3.4 Factivity, idealization, and epistemically useful distortion

Knowledge that *p*, in the strict propositional use on which the preceding chapter centered its analysis, is factive. Whether understanding is factive in the same way is disputed. Elgin argues, as Grimm and de Regt summarize her view, that understanding is not factive: models, idealizations, and thought experiments that are known not to be true can be felicitous falsehoods, which advance understanding by exemplifying features of their targets, that is, by highlighting those features while downplaying others. Elgin grants that understanding still in some way answers to the facts (Grimm 2021, §4.2; de Regt 2018). Potochnik suggests that pervasive idealization calls into question the idea that science aims at truth, and that its epistemic aim is understanding, which is one of several aims of science (Potochnik 2015, 1, 4, 16). De Regt likewise argues that achieving understanding often requires giving up representational accuracy, and that idealizations and other inaccurate representations are often used as tools for enhancing scientific understanding (de Regt 2020, 932). Rejecting factivity need not mean rejecting every correctness condition, and accounts that keep one differ over what it covers. On Potochnik's account, understanding still answers to the world: it requires that each posit in a representation be epistemically acceptable, its divergence from the truth insignificant given its role in the representation and the purpose of the research, and that the causal pattern grasped be one the phenomenon actually embodies (Potochnik 2020, 6–8). Strevens counts an idealizing explanation as correct when its explanatory content, roughly its claims about what does and does not make a difference, is true (Strevens 2013), whereas Potochnik allows idealizations to stand in even for factors that make a significant difference when those factors are not the focus of the research (Potochnik 2015, 10–12).

A related issue concerns what model evaluation can establish. Oreskes, Shrader-Frechette, and Belitz argue, specifically for numerical models of open natural systems, that verification or validation cannot amount to a conclusive demonstration of truth: empirical agreement supplies partial confirmation, and different models can remain compatible with the available observations (Oreskes, Shrader-Frechette, and Belitz 1994). Parker's adequacy-for-purpose account offers a different evaluative target. A model is assessed relative to a target, user, methodology, circumstances, and purpose, rather than only by its overall representational accuracy (Parker 2020, 458, 464–465). These positions are not one consensus theory, and the first is not a claim about every formal or scientific model. Together they show why the model was validated does not by itself say whether it is adequate for prediction, explanation, intervention, teaching, or another use.

The dispute matters for distinguishing the conditions of knowledge and understanding. A literally false model cannot itself supply knowledge of the false proposition. It may still reorganize a domain, reveal a dependence, support an ability, or increase understanding. Some of its contribution may be reconstructed through true propositions about what the model represents or enables. Kelp's knowledge-based account of understanding offers such a reconstruction for idealized models: one can know what a theory containing the ideal gas law says, that actual gases approximate ideal gases in various respects and to various degrees, and where the idealization's limitations lie, and on Kelp's account this knowledge is part of one's understanding of the physics of gases (Kelp 2017, 18). Whether such a reconstruction exhausts the achievement is disputed. Potochnik also gives truth a place, holding that an idealized representation can be true of the causal pattern it depicts while false of the phenomena that embody it, but maintains that understanding consists in grasping such a pattern, which does not require belief, and is therefore distinct from knowledge (Potochnik 2020, 6–7, 14–16; Potochnik 2015, 7–8). On accounts that count such contributions as understanding, understanding is therefore not simply a more verbose form of propositional knowledge.

For artificial research, the immediate implication is not that utility takes the place of truth. On the accounts above, an idealized model contributes to understanding only relative to what it idealizes and for which purpose. An artificial system working with models may therefore need to retain what each representation idealizes, which target and purpose it bears on, which inferences it supports, and where rival models remain compatible with the evidence, for those who assess and use its results as much as for any understanding credited to the system itself. Weisberg's account of a model's construal gives these items a structure. A construal assigns parts of a model to parts of its target and says which parts are to be ignored, fixes the intended scope, and sets two kinds of fidelity criteria: dynamical criteria, how close the model's output must be to the phenomenon, and representational criteria, whether the model makes the right predictions for the right reasons (Weisberg 2007b, 14–17). On Weisberg's account the construal depends on the intentions of the model user, so for an artificial modeler, whose intentions fix it is a further question (Chapter 8, Section 2.2).

The need to keep such items is not peculiar to artificial systems. Cleland, for example, reads the Miller–Urey experiments, which were presented as a test of the hypothesis that life on Earth originated in a primordial soup, as a test of whether some basic building blocks of life, amino acids, can be produced by electrical discharges in a particular gas mixture, and holds that the logical relation between the hypothesis actually tested and the hypothesis about the origin of life winds through numerous highly speculative assumptions, from conditions on the early Earth to biochemical possibilities (Cleland 2002, 484). Likewise, a more detailed or predictively accurate model need not be better for every explanatory or intervention purpose, and a useful idealization does not become a literally true description merely because it performs well. Elgin argues, for example, that the ideal gas law remains justified alongside the virial equation, which can be made arbitrarily precise, because what is wanted of a model is a representation that affords understanding of the phenomena rather than one that replicates them, and different models of one target exemplify different features (Elgin 2022, 18–20).

Weisberg's classification of idealization bears on what changes when the modeler is a machine, because it distinguishes kinds of idealization by their justification. Galilean idealization introduces distortion to make a theory computationally tractable; since its justification is pragmatic and tied to tractability, advances in computational power and mathematical technique should, on Weisberg's account, lead the idealizer to de-idealize. Minimalist idealization keeps only the causal factors that make a difference to the phenomenon; its justification lies in explanation and, on this view, should not be expected to abate with the progress of science. Multiple-models idealization builds related but incompatible models without expecting a single best one; where desiderata such as generality, accuracy, precision, and simplicity trade off, Weisberg takes its justification to be permanent, since such tradeoffs are consequences of logic and mathematics (Weisberg 2007a, §I).

Matthewson and Weisberg argue that some of these tradeoffs arise from logic and representation alone and will not dissipate with scientific progress (Matthewson and Weisberg 2009). Weisberg also takes all three kinds to be compatible with a realist view of science, so the classification does not side with Elgin's denial of factivity (Weisberg 2007a, §IV). Carried over to a system with far more computational capacity than its human counterparts, the classification implies that only idealizations justified by tractability give that system a reason to de-idealize. Levins, whom Weisberg quotes, traced the multiplicity of models partly to a mind that can cope with only a few variables at a time, a premise about human cognition; the tradeoffs that arise from logic and representation alone do not rest on that premise, so the reasons for keeping several models of one target that rest on them survive the change of modeler. These positions do not fix a single design, but a research system could therefore preserve families of purpose-relative representations rather than force every inquiry into a single increasingly literal world model.

These families of accounts expose different possible targets and different burdens of evidence. Each changes what an artificial researcher would have to represent, do, communicate, or make available to another subject.

Knowledge without explanatory understanding

Section 3 set out, account by account, what would count as understanding. This section applies those accounts to the strong predictive case and asks whether knowledge and explanatory understanding come apart there, whether that holds whichever account is chosen, and, where they do, what follows for the target of the attempt and for progress.

Return to the strong predictive case. The new regularity is true. The claim is restricted to an explicit domain. Independent measurement routes and groups support it. The predictor's error characteristics are known, dependencies among the checks are recorded, and defeaters would lead the community to revise its reliance. No human member of the community grasps an adequate explanation of the target regularity, even tacitly, or can give one; whether the system itself understands it is left open (Section 6).

The accounts of understanding examined in Section 3, together with the preceding chapter, support a conditional but substantive point:

> Knowledge that a regularity holds and explanatory understanding of why it holds are conceptually separable. Under several important accounts of knowledge, the former can be realized in the strong predictive case and, on suitable accounts of possession, enter the community's epistemic situation, while, on accounts requiring an explanation for understanding why, the latter is realized in none of its human members. Whether this particular result is knowledge, and whether it has entered that situation, remains relative to which of the conditions compared in the preceding chapter are adopted.

The distinction between knowing that and understanding why does not by itself settle whether understanding why is absent in the stipulated case. Explanatory accounts, ability accounts of understanding why such as Hills's, and reductive accounts that treat understanding why as knowledge of an explanation all distinguish knowing that a regularity holds from understanding why it holds. Dellsén, who allows objectual understanding without explanation, holds that understanding why cannot come apart from explanation (Strevens 2013; Dellsén 2020; Grimm 2021, §4.1). The accounts differ over whether some understanding of the phenomenon can remain when no explanation is available: dependency accounts such as Dellsén's and Wilkenfeld's representation-manipulability account (Sections 3.2 and 3.3) allow objectual understanding without one, and Lipton argued for understanding why without one, so that on Lipton's account the stipulation that no human member grasps or can give an explanation does not by itself settle that none of them understands why the regularity holds.

The separation has also been defended, and its consequences resisted, in discussions of automated science. Humphreys, considering the worry that easing humans away from the centre of science could cost understanding of the world, holds that, under an expansive conception of knowledge, we ought to allow scientific knowledge "beyond the current understanding of humans." The cases cited are effective prediction and manipulation without understanding of the mechanisms involved, as with aspirin, cholera, and many types of cancer, and quantum mechanics, predictively successful for decades without a satisfactory explanation of measurement. Humphreys keeps a further worry, that errors go undetected when humans do not understand the science, on both of the views of science's aim, that it aims chiefly at knowledge and understanding of the world and that it aims chiefly at prediction and manipulation, since automated methods can fail suddenly when the conditions for their stability are not met (Humphreys 2020, 14–15). Boon, in the same volume, ties scientific knowledge to

its use. On Boon's account, data-models that are opaque and sit inaccessibly in the machine, so that humans cannot obtain them and use them in self-chosen epistemic tasks, are not scientific knowledge as traditional scientific practice produces it, which consists of scientists' conceptions represented as public and transferable epistemic tools; the account leaves open whether machine learning could in principle meet the pragmatic criteria that make knowledge manageable for humans (Boon 2020, 14–15, 16–17 n. 11). Boon's criterion does not deny the displayed claim: in the strong predictive case the validated regularity is stated and used by the community, and what the criterion denies the status of scientific knowledge is the opaque model inside the predictor. Boon states the criterion for human practice; applied to an artificial *S*, which is not discussed there, it would ask whether a model can serve as an epistemic tool for that society's members.

Beisbart's argument about deep neural networks (Section 2) applies the separation to one kind of predictor; it does not show that every prediction lacks understanding. On Beisbart's account, scientists can be justified by a network's track record in accepting that it maps inputs to outputs correctly while lacking explanatory understanding of what it predicts (Beisbart 2025, 14, 18). The argument also bears on the question on which the accounts above divide. A scientist who has learned, for example by running the network many times, to anticipate how its output changes with its input can draw some of the inferences that understanding requires. Beisbart counts this as some degree of objectual understanding of the predicted phenomenon, but not as explanatory understanding, because those inferences do not rest on an explanation the scientist has (16–17). The option of letting the scientist draw the inferences by running the network itself is also considered and set aside: what would then understand, strictly, is the coupled system of the human and the computer, a different subject from the unaided human whose understanding is in question (17–18; Section 6 takes up who understands). The argument does not, however, rule out that networks contribute to explanatory understanding. If scientists know that a network is sensitive to explanatorily relevant features, for instance through methods from explainable AI that detect which variables affect its output, and if they justifiably accept the explanation it contains and can work with it, they can understand why the predicted outcome holds (15, 18). Outside the simple setting, running a network on many varied inputs can identify explanatorily relevant factors, such as a partial cause, and Beisbart counts this as one sense in which networks enhance understanding; it requires, however, prior knowledge of the explanation and falls far short of a full explanation (21–23). The argument further leaves open that success across an extremely broad range of cases could give reason to think that a network tracks laws or causes, while holding that present-day networks, whose range of application is typically small, give no such reason (15–16). Beisbart concludes that how far networks can be brought to provide understanding depends on what humans do with them (25). Measured against these routes, the strong predictive case excludes by stipulation the route through knowing what the network is sensitive to, since no human can explain the regularity and the system supplies no usable representation of the dependence responsible for it. The route through repeated runs remains open, as a source of explanatorily relevant factors short of an explanation.

Chirimuuta argues, for one area of neuroscience, that the shortfall in understanding can outlast improvements in method. In two cases, decoders of motor-cortex activity and models of visual neurons, Chirimuuta's paper traces a shift from simple, largely linear models, which were intelligible but predicted poorly outside simple laboratory conditions, to artificial neural networks that predict far better but do not let their builders anticipate qualitatively how the model will respond. It finds in the earlier models four sources of intelligibility that the networks lack: visualisability as a circuit diagram, articulated theoretical assumptions about the neural code, linearity, and functional transparency, that is, explicit knowledge of the function the trained model computes. Because the networks' accuracy comes from approximating a complex nonlinear function, the argument goes, even an ideally reverse-engineered network, its input–output function written down, would remain less intelligible than its hand-coded predecessors, so that the trade-off between prediction and understanding would persist; Chirimuuta expects it beyond the two cases wherever targets involve complex nonlinear dependencies and predictions are needed outside controlled conditions. On this account such networks can supply some kinds of explanation, but "they tell you what is to be expected, but not why" (Chirimuuta 2021, 15–23). Where Beisbart makes the networks' contribution to understanding depend on what humans do with them, Chirimuuta's argument gives a reason to think that, for targets of this kind, the networks would remain less intelligible than simpler rivals however far such methods go, because the limit comes from the complexity that makes them accurate. Since the paper calls intelligibility "a human-relative virtue of models" (26), the trade-off expected to persist can be read as one between prediction and intelligibility to humans. Chirimuuta does not discuss artificial understanders; applied to one, the complexity that makes a network accurate remains, but whether it limits that understander's grasp depends on what is intelligible to it, the question of who understands that Section 6 takes up.

De Regt's account, as the following shows, sits less easily with the strong predictive case. On de Regt's view, successful prediction from a theory requires that the theory be intelligible to those who use it (Section 2); if the predictor were a theory in this sense, its users would have to find it intelligible, so the case could not lack understanding altogether, although understanding why the regularity holds would on this account still require an explanation based on that theory. For data-driven networks of the kind Boge and de Regt describe, which may predict well while providing little understanding, the separation applies as stated.

The claim that knowledge can be realized in the strong predictive case does not depend on treating accuracy as knowledge. The case stipulates the further evidence, reliability, basing, anti-luck, or social conditions demanded by the relevant account. Internalism does not automatically defeat it: a community can have accessible reasons for believing that the regularity holds without possessing an explanation of why it holds. Reliabilism can license the case through an appropriately established truth-conducive route. A competence or modal account will ask whether success manifests ability and could too easily have failed. A social account will ask who possesses the attitude and how assessment, uptake, and correction are organized.

The separation also permits more than one kind of understanding. Investigators may understand why the evidence warrants reliance on the predictor while not understanding why the target regularity obtains. They may understand the predictor's operating limits without understanding its internal computation. They may acquire a thin dependency representation that some accounts call partial understanding. Knowledge without understanding is therefore informative only when it names the object and kind of understanding that is absent. Several kinds of understanding can be present or absent within a single achievement. After the central theorem of the Liquid Tensor Experiment (Chapter 1, Section 3) had been formally verified, for example, Peter Scholze wrote that the formalization had shown what actually makes the proof work. Scholze also wrote that, whereas verifying a proof usually involves trying small variations of the argument to see whether they break, the formalization had yielded reassurance that the proof's route succeeds but still no sense of the terrain around it (Scholze 2021). In this case, understanding of why the route works was gained, while a sense of how variations of the route would fare was not.

The strong predictive case, the principal case of this chapter, concerns knowledge that a regularity holds without explanatory understanding of why it holds. The separation is not peculiar to artificial predictors. For example, Morris Tanenbaum, recalling in a 2004 interview his early work on silicon transistors at Bell Laboratories, said that he had had to use annealing techniques to obtain transistors from crystals whose conductivity type changed when they were heated and quenched, and that only afterwards did a colleague show that the impurity responsible was oxygen; of a related regularity noticed in growing the crystals, that the effect was much weaker when the melt was not stirred, he said, "We didn't know why" (Tanenbaum 2004, 42–43).

That knowledge can be attained without explanatory understanding does not make understanding subordinate to knowledge: where understanding is the target, attaining knowledge that the regularity holds does not reach it. Accounts of progress draw a related contrast with Bird's epistemic account, on which progress consists in the accumulation of knowledge (Chapter 2, Section 8.3). Dellsén's noetic account argues that scientific progress is better understood as increased understanding than as accumulated knowledge, and its grounds lie in cases where the two diverge. On the one hand, Einstein's 1905 explanation of Brownian motion counts, on Dellsén's account, as progress; yet, on the same account, Einstein then knew neither the kinetic theory of molecules nor the phenomenon it explained, since neither was justified well enough for him to know it, so no knowledge was added. On the other hand, knowledge of random experimental results, of merely apparent correlations such as that between the number of storks and the number of births, or of observational records too meager to support any explanation or prediction can, on this account, accumulate without bringing any progress (Dellsén 2016, §§3–4; Dellsén 2021). Potochnik likewise motivates understanding as an epistemic aim of science while emphasizing the plurality of scientific aims (Potochnik 2015). De Regt places understanding at the center of successful science without reducing it to predictive success (de Regt 2020, 932).

The reasons de Regt and Dieks give for this view apply directly to the strong predictive case. They rest on contrasting a successful scientific theory with a hypothetical oracle whose pronouncements always prove true: the oracle would secure empirical adequacy, but would not count as a great scientific success, because there would be no understanding of how its perfect predictions were brought about (de Regt and Dieks 2005, §2). De Regt and Dieks likewise hold that meteorologists who did nothing but compute correct forecasts by solving the governing equations numerically would fail to understand the

weather (de Regt and Dieks 2005, §4.1). On de Regt's criterion, a validated predictor that supplies no explanation based on an intelligible theory would leave the regularity without the scientific understanding that criterion characterizes, whatever its status as knowledge.

Dellsén's understanding is objectual rather than narrowly explanatory. In the restated account, increased understanding is a more accurate or comprehensive model of dependence relations, which grounds prediction and manipulation as well as explanation, and progress lies in publicly available information that enables relevant members of society to increase their understanding (Dellsén 2021). On this reading, whether the strong predictive case counts as noetic progress turns on whether the validated regularity improves such a model, not simply on whether an explanation is available. Dellsén's earlier formulation was more direct about prediction. There, grasping how to predict some aspect of a target correctly across a variety of circumstances counted as partial understanding even without an explanation: someone who can predict changes in the weather from a range of barometer readings, for example, partially understands the weather, although a single correct prediction, or the memorized steps of one, would not suffice (Dellsén 2016, §2). On that formulation, predictive grasp without explanation can itself be partial understanding. Such grasp falls outside de Regt's criterion, which requires an explanation based on an intelligible theory (de Regt 2019, 334), although, as Section 3.2 noted, de Regt sets aside whether understanding why can be had without explanation. In the restated account, prediction figures instead among the benefits that understanding brings (Dellsén 2021, 11256). Truthlikeness and problem-solving accounts, introduced in Chapter 2, make the verdict turn less directly on understanding.

Woodward's interventionist account, introduced in Section 2, yields a further verdict. On this account, an explanation answers what-if-things-had-been-different questions: it tells how the explanandum would have been different under changes in the factors it cites, using a generalization that continues to hold under at least some interventions on them (Ross and Woodward 2023, §3). Candidates that answer no such questions do not explain at all; above that threshold, explanations differ in depth according to how much information relevant to answering such questions they provide (Ross and Woodward 2023, §4). So understood, Woodward's account leaves the strong predictive case without explanation unless the validated predictor also supports correct answers about what would happen under interventions. A predictor that supports such answers explains to that extent, provided the information is accessible to those who use it, a condition to which Section 6 returns.

The positions surveyed need not deny that the community knows the regularity. They can deny that knowledge is the achievement sought, or all of it, or that its increase is the progress that matters. The working characterization, which leaves open whether knowledge or understanding is sought, allows this difference but does not resolve it: reorganizing a domain through idealization, explanation, or objectual grasp still requires an account of success. Nor does a local increment in either knowledge or understanding alone settle how a wider program has progressed.

The strong predictive case can therefore support different judgments. On accounts separating knowledge of a regularity from understanding why it holds, an activity can reach its knowledge target while an inquiry directed toward understanding remains unfinished. De Regt's requirement that a theory be intelligible to its users qualifies the predictive case differently (Section 2): where the predictor is a theory in that sense, its successful use presupposes, on de Regt's account, that its users understand the theory, though not that they understand why the regularity holds. Accounts that make understanding central to science or its progress, such as those of Potochnik, Elgin, and Dellsén (Section 7.1), add priority claims that the working characterization neither establishes nor excludes. The characterization itself requires no individual research operation, and no research activity as such, to bring about any achievement, knowledge or understanding (Chapter 1, Section 3). Whether understanding is needed for success depends on what the attempt is directed at, and the direction of an organized attempt, its eventual success, and the contribution of a particular operation are distinct.

This distinction of scale is important. A research program can aim at understanding even when one constituent experiment produces only a measurement. A field can progress through an explanatory synthesis assembled from results none of which is explanatory alone. Conversely, a project can produce many true local claims without advancing the understanding its participants sought. A theory of the aims or progress of science cannot be converted without argument into a membership condition for each activity called research.

On some accounts, the separation also runs in the other direction. Dellsén holds that understanding need not require justification or belief in the manner knowledge commonly does (Section 3.2). Elgin holds that felicitous falsehoods advance understanding, and Potochnik that understanding consists in grasping a causal pattern, which does not require belief and is therefore distinct from knowledge (Section 3.4); in a case Dellsén offers, a physicist who, from methodological reservations, does not believe string theory but accepts it and uses it to explain understands, on Dellsén's account, what the theory describes (Section 6). On such accounts, an attempt directed at understanding can bring about understanding new to *S*, for example through an idealized model or an accepted but unbelieved theory, without thereby bringing about new knowledge, and an evaluation that counts only knowledge or validated results would not register that success. Khalifa analyzes ideal understanding why as scientific knowledge of an explanation, and Kelp analyzes maximal understanding as maximally comprehensive and well-connected knowledge (Sections 3.1 and 7.1). These analyses alone do not establish that an increase in understanding must add knowledge. Kelp's account makes this explicit in one direction. In his litmus-paper case, an agent who knows every member of the full account of an event from separate sources, while the basing relations among those beliefs do not reflect the explanatory and support relations the agent knows to hold, has little understanding of the event; Kelp notes that it is not clear that adding further knowledge would remedy this, and on his account the degree of understanding rises as the knowledge becomes well connected (Kelp 2015, 16–18). On this account, then, understanding can deepen without new facts being known. Applying this to understanding new to *S* presupposes that such a reconnection takes place within *S*'s epistemic situation. Kelp's account also lets understanding coexist with beliefs that fall short of knowledge, because what carries the understanding is the knowledge around them. Against Kvanvig's case of a historian whose true beliefs about the Comanche are Gettiered, Kelp's first reply is that those beliefs may well be knowledge; but he adds that on his account understanding can be attributed even if many of the inquirer's beliefs are Gettiered, since the knowledge that remains, of what the relevant theory says and that it is highly plausible, can be enough for the standard a context sets (Kelp 2017, 16–17). The same holds for understanding why. In Pritchard's case of a man who is told why his house burned down by the one real firefighter among several impostors, Kelp's first response is to deny that the man lacks knowledge, but he also offers an account on which, if that intuition is granted, the man understands why without knowing why, because he knows enough to give an explanation that is well-founded, that is, ultimately grounded in a source that does know (Kelp 2014, 11, 13–15). In both cases what carries the understanding, on Kelp's account, is the knowledge the inquirer does have.

Opacity has an address

The system in the strong predictive case supplies no usable representation of the dependence it tracks; it is opaque. Two objections then suggest themselves: that its opacity undermines the warrant for the knowledge that Section 4 allowed, and that it settles that nobody understands the regularity. If the first held, the case would lose the knowledge it was built on in order to isolate the absence of understanding. This section takes up both. Sections 5.1 to 5.3 ask where the opacity lies and what it bears on for warranted reliance, and Section 5.4 asks whether understanding can be supplied through a reconstruction or a further model while the computation remains opaque. Who understands is taken up in Section 6, once opacity has been indexed to a subject.

Calling a system a black box does not yet draw the distinctions needed to evaluate an artificial result. Opacity is informative only when it is indexed to what is opaque, the role the opaque element performs, the subject to whom it is opaque, and the stakes of depending on it. Writing about computer simulation and computational science, Humphreys indexes epistemic opacity to a subject and a time, in a definition that Winsberg and Alvarado quote in their encyclopedia entry on computer simulations in science: a process is epistemically opaque relative to a cognitive agent at a time just in case the agent does not then know all of the epistemically relevant elements of the process. Humphreys calls a process essentially epistemically opaque to an agent when it is impossible, given the agent's nature, for the agent to know all of those elements (Winsberg and Alvarado 2026, §2.5). Because essential opacity is relative to the agent's nature, a process essentially opaque to human agents need not be so to an artificial one, so which subject opacity is indexed to matters wherever artificial agents take part in assessing or using a result (Section 5.2). Durán and Formanek, whose account Section 5.3 discusses, take their notion of essential opacity from this definition.

Humphreys argues that the speed and complexity of simulations put their computational details beyond human inspection (Humphreys 2009a, §3.1). Durán quotes Humphreys's earlier argument that even a computationally irreducible process

would become inspectable if slowed until a human could examine each step. On that argument, the practical need for computational speed is the root of opacity in this area; because humans cannot circumvent that constraint, transparency has to be given up as a requirement for computational science (Durán 2025b, §1). The argument separates what a change of subject could and could not alter. Humphreys does not discuss artificial inspectors, but applied to one that could follow each step at the speed required, the account implies that the steps would not be beyond its inspection, while irreducibility, the absence of a shortcut to the outcome, would remain. Inspecting the steps would not by itself establish that the inspector understands the process as a whole.

Later accounts differ over how much opacity a more capable agent could remove. Alvarado, in a preprint, argues that some opacity in computational methods does not respond to any change in the agent. On this account, opacity is agent-neutral when it arises from a feature that all epistemic agents share, such as being finite, a limit of the kind that Cherniak assigned to any finite agent (Chapter 5, Section 6.5); Alvarado's example, an estimate taken from Symons and Horner, is that checking every path through a program of realistic size would take several times the age of the universe, so that the opacity remains even for agents with millions of years to spare. Alvarado calls opacity agent-independent when it arises from features of the process itself that do not respond to agential resources, with examples including stochastic and vastly overdetermined processes, rounding errors bundled across the components of coupled simulations, and the weights of a deep network, set by automated training from partly random initial values (Alvarado 2021, 10–15). Boge and Grünke, also in a preprint, argue instead that because simulations and machine-learning methods are algorithmically transparent, an idealized agent such as a Laplacean demon could step through them, so that the opacity of the computation, essential opacity included, is contingent on the epistemic conditions of actual agents. What they suspect may be fundamental is what they call model-opacity: there seems to be no algorithm leading from what a trained network does to a model of the physical mechanisms behind its outputs, a difficulty they take to be closely related to the problem of unconceived alternatives (Boge and Grünke 2019, 11–14, 26–27, 30). They consider only human agents, but applied to an inquirer more capable than humans, their account implies that it might remove the opacity of the route by which a result was computed while still facing opacity about the target, the matter of the third adequacy question that Section 5.3 poses, whether an account adequately represents the phenomenon that inquiry seeks to explain; on Alvarado's account, some opacity of the route itself would remain for any finite inquirer.

Section 5.1 takes up the role of the opaque element, separating the generation of a candidate from the warrant for accepting it; Section 5.2 maps the other dimensions together; Section 5.3 asks whether transparency is necessary or sufficient for warranted reliance; and Section 5.4 shows that a later reconstruction need not be mere rationalization, an after-the-fact excuse.

5.1 Candidate generation and final warrant

Among the roles an opaque element can perform, generating a candidate answer can be separated from supplying the warrant for accepting it. On the distinction between the contexts of discovery and justification that Reichenbach introduced (Chapter 5, Section 4.1), and on Duede's application of it to deep learning below, a limited permission for opacity in generation follows:

> The route that generates a candidate answer need not be the route that warrants accepting it.

A conjecture can arise from analogy, aesthetic judgment, accidental observation, a dream, random search, or an opaque model. On these accounts its origin need not supply its warrant. Duede argues directly that deep-learning opacity can be compatible with scientific discovery when a model operates in the context of discovery and the result is justified through distinct means (Duede 2023). A requirement of public rational reconstruction for every generative move would apply to ordinary human discovery as much as to machine discovery. Other accounts narrow the permission. On reliabilist accounts (Chapter 2, Section 4.1), and on the virtue-epistemological reading discussed in Section 5.3, the reliability or credited competence of the source can itself supply warrant; on those accounts, how a candidate was generated can be part of what warrants it, and assessing the generator becomes part of assessing the result.

Duede and Davey develop a sharper mathematical case. An opaque system may produce a formally represented proof too large for humans to survey, while a transparent proof checker verifies the stored derivation. On their account, the checker can support a priori mathematical knowledge even though humans cannot grasp the generative process or inspect the proof

step by step (Duede and Davey 2025). The result is deliberately domain-specific. It depends on a formal proof object and a checker whose mathematical role is transparent; it does not show that an empirical claim can be warranted merely by placing another opaque model after the first. It nevertheless makes the logical separation vivid: opacity in candidate generation need not be opacity in the route that warrants the conclusion. The Liquid Tensor Experiment (Chapter 1, Section 3) shows, for example, where the human share of such a route can lie. Scholze found the formal definitions and theorem statements surprisingly readable but could not read the proofs at all, and the formalizers added formally proved results about the objects involved, such as that Lean's real numbers form the unique complete ordered field, as evidence that the checked statement concerned the intended objects (Scholze 2021; Buzzard 2022). What the checker left to human judgment was whether the checked statement was the intended theorem, not whether each step of the derivation was valid.

Pelc, arguing against the formalist view that the formal derivations underlying proofs contribute to confidence in theorems, took a more skeptical view than Duede and Davey of what formal checking contributes to the reliability of mathematics. Pelc argued that derivations would add confidence if reasonably short ones could be carried out, that this is not known to be possible for deep theorems in the present state of knowledge, so that no link between proofs and derivations can at present add confidence, and that confidence in theorems is instead built through a social process, in which a result becomes more worthy of belief as its proof is re-verified and used by others (Pelc 2014, 5–7, 25–26, 29). Pelc also quotes two objections of Rav's: proofs are held together by arguments based on the meanings of their terms, which formal calculi cannot capture, and transcribing an informal proof as a derivation calls on a logician's know-how, a transcription that cannot itself be checked mechanically, so that a positive verdict from a proof checker may reassure one only that the formalization and the checking were carried out correctly (Pelc 2014, 6–7). Pelc rejects part of the second objection: a positive verdict, were one available, would on his view raise confidence not in the informal proof but in the theorem itself, and his own objection is rather that for many theorems no such verdict can be had, since no sufficiently short derivation is available (Pelc 2014, 6–8). In the Liquid Tensor Experiment, a formal derivation was carried out, but the judgment whether the checked statement expressed the intended theorem remained with the human participants. Applied to this case, the transcription objection quoted by Pelc concerns the relation between the formal statement and its intended meaning. Neither observation settles Pelc's thesis that confidence in theorems comes from a social process.

Ochigame, writing about automated mathematics from the history and sociology of proof, treats the standard of proof itself as something that mathematicians have revised. Borrowing Hacking's contrast between a Cartesian ideal of proof, on which one understands a proof and can get it in one's mind all at once, and a Leibnizian ideal, on which every step can be checked mechanically, Ochigame traces practical standards derived from both. Of these, the standard that computer-encoded proofs have advanced is that every step can be checked by a program and derived from its axiomatic foundations, and that "one understands or trusts the encoding of the proven statement." Ochigame argues that such proofs therefore still demand social consensus that a piece of code corresponds to the claimed statement, and so do not eliminate social institutions of adjudication, which are needed still more for judging the importance and value of results. Mathematicians now face, on this account, two decisions: whether computer-encoded proofs should become required, and whether completely inscrutable, deeply automated proofs, which one would accept by trusting the programs that constructed and checked them, should be accepted; Ochigame declines to defend a position on either (Ochigame 2024, §§2–4). On Ochigame's account, the judgment that the Liquid Tensor Experiment left to its human participants belongs to a standard of proof that the community could still revise, and the second decision asks whether *S* would accept a result that no member understands even in outline.

The permission to separate generation from warrant does not make generation epistemically irrelevant. The route can reveal data leakage, prior exposure, selection effects, undisclosed human filtering, duplicated work, or evaluator dependence. Nor can an allegedly independent check merely reproduce the generator's training data, measurement errors, or background assumptions. The distinction is between epistemic roles, not necessarily a rigid temporal sequence and not a guarantee that the roles are causally independent.

5.2 A multidimensional map of opacity

Generation and warrant are too coarse to classify all relevant opacity. Creel distinguishes functional, structural, and run transparency in complex computational systems: roughly, knowledge of the algorithm a program instantiates, of how the code realizes that algorithm, and of a particular run on particular hardware and input data. Creel argues that the three are dissociable, so that a system can be transparent in one of these ways without being transparent in the others (Creel 2020, §§3a–3d). The analysis concerns only opacity that remains for skilled and knowledgeable creators and users of such systems, and sets aside opacity produced by concealment or by a user's lack of the relevant expertise (Creel 2020, §3).

Sullivan locates a different source of difficulty, outside the model, and argues that what primarily limits the understanding an opaque model can provide is not its complexity or the opacity of its implementation but what Sullivan calls link uncertainty: a lack of scientific and empirical evidence supporting the link that connects the model to its target phenomenon. The reason is comparative. Three deep-learning models with the same degree of opacity, one predicting many diseases from medical records, one classifying melanoma from images of moles, and one classifying sexual orientation from facial images, differ in the understanding they yield, and on Sullivan's analysis the difference follows how well evidence from outside each model connects it to its target. Extensive dermatological knowledge supports that connection in the melanoma case, whereas in the facial-image case existing evidence points the other way and the model tracks cultural cues such as grooming. The claim is qualified: it holds so long as a model is not opaque at the highest level, where only its inputs and outputs are known, and Sullivan argues that deep neural networks are not, because their general architecture and the modelling process that produces them are known (Sullivan 2022b, §§3.2, 4.3, 5.2–5.3, 6).

In a paper that builds on this account, Sullivan brings inductive risk to bear on both the link and the model's interior. Inductive risk is the risk of error that remains when a hypothesis is accepted or rejected, or a model is used, on evidence that cannot make it certain; on the argument from inductive risk, how much support is enough depends on how bad the possible errors would be, including their non-epistemic consequences, that is, consequences for matters other than truth, such as harm to people (Chapter 11, Section 4). On Sullivan's account, accepting that a model can explain a phenomenon involves such a judgment, because one must accept both that its data are adequate and that the counterfactual dependencies it captures, what would differ in the target if a feature were different, hold of the target. Where wrongly accepting them would be harmful, more evidence is required; for the facial-image classifier the paper cites the scientific controversy over such dependencies, the use of phrenology and physiognomy to justify racism, sexism, and eugenics, and the discrimination that gay people face (Sullivan 2022a, 1067–1070). Two consequences follow on this account. Externally, the greater the inductive risk, the stronger the independent evidence needed to reduce link uncertainty, for example further empirical research beyond the model or, when the target is a statistical correlation, robustness checks across several models. Internally, a model learned from data does not contain its dependencies in explicit form, so they have to be extracted after training, for example with saliency maps, which highlight the parts of an image that contribute most to a classification; the greater the inductive risk, the higher the standard for verifying that what is extracted is faithful to how the model works rather than an artifact. Sullivan has in mind mainly what Creel calls functional transparency, while allowing that the stakes may also demand structural or run transparency. The paper further argues that these demands rise with the stakes even when the researchers' question and target are held fixed, and restricts the claim to explanation and understanding rather than the use of a model in practice (Sullivan 2022a, 1070–1074). On Sullivan's account, the present limits of interpretability methods therefore need not prevent understanding of low-stakes phenomena, but can prevent it where a model bears on matters with serious social consequences (Sullivan 2022a, 1072–1073). Beisbart distinguishes the obstacle to explanatory understanding from both notions. On this account, a highly opaque computer simulation is often still known to contain the relevant explanatory laws, so epistemic opacity in Humphreys's sense is not the obstacle; and strong evidence that a network tracks correlations in its target makes link uncertainty low without helping with explanation, although whether an explanation the network contains is the actual one may, Beisbart allows, count as part of link uncertainty. What is lacking, the argument continues, are informative descriptions of what the network does in terms of variables or processes of the target system (Beisbart 2025, 24).

The following diagnostic grid is put together here from Creel's and Sullivan's distinctions, extended to cover also the forms of opacity Creel sets aside. The target link listed under Object and the costs of error under Stakes come from Sullivan's two analyses; distribution shift, also listed under Stakes, is carried over from Section 2, and reversibility, novelty, and time horizon are not among the considerations Sullivan discusses. The Subject coordinate has a precedent in Zednik's adaptation

of Humphreys's definition to machine learning. On Zednik's account opacity is relative to an agent; the stakeholders in a machine-learning system differ in role, among them operators, executors, decision subjects, data subjects, creators, and examiners; and which elements of a system are epistemically relevant, the elements an agent can cite to explain its behavior, differs with the agent and its role (Zednik 2021, §§2.2, 3.1–3.2). Zednik does not consider artificial stakeholders, but applied to the value another AI, the account implies that a subject fixes not only to whom a system is opaque but also which of its elements count as relevant.

Each row names one coordinate and the question that fixes it, and a case of opacity is located by answering all four questions together. The items named in each question are common examples of the values a coordinate can take; other objects, roles, subjects, and stakes can be added.

Table 3.1. Coordinates for locating opacity in research systems

Coordinate	Question
Object	Is the generator, code, data, learned relation, evidence, target link, or institution opaque?
Role	Does it perform search, measurement, prediction, warrant, explanation, decision, or correction?
Subject	Is it opaque to the developer, operator, scientist, reviewer, affected public, another AI, or the collective?
Stakes	What are the costs of error, reversibility, novelty, distribution shift, and time horizon?

The grid is not a list of four required modules. It prevents opacity in one place from being treated as opacity everywhere. A model may be computationally opaque to a domain scientist while its domain-bounded reliability is publicly testable. A transparent algorithm may rely on an opaque dataset or an invalid measurement model. A human-readable rationale may conceal an untraceable evidential route. A distributed collaboration may be locally opaque to every participant while maintaining reliable collective checks.

5.3 Is transparency necessary or sufficient?

The accounts compared in this section differ over where the reasons for reliance lie: in a reliable process or a credited model, which on the analysis of this section can both be shown by a system's record; in how a system was built and what grounds it; or in access to how a system operates. On some accounts, internal transparency is not universally necessary for warranted reliance. Durán and Formanek develop a computational-reliabilist account on which researchers can be justified in believing the results of essentially opaque simulations when the simulation is a reliable process. They name four sources of that reliability: verification and validation, robustness analysis, a history of (un)successful implementations of theories, models, and building techniques, and expert knowledge, which they regard as a rather weak source when taken alone (Durán and Formanek 2018, §§3.1, 4). In its 2018 formulation, their account is not purely externalist. At a second step it requires that these sources be shown reliable in turn, so that a researcher who takes a verification procedure to be reliable must give reasons for doing so (Durán and Formanek 2018, §3.2). It is a substantive position, not a theory-neutral solution. It still raises questions about how the reliable process, which it identifies with the simulation, is to be typed (Chapter 2, Section 4.2), where the chain of showing the reliability of a source of reliability by a further source ends (Chapter 11, Section 5.2), and whether the environment in which reliability was established remains stable. Durán later presents computational reliabilism as an externalist epistemology of algorithms in the sense that justification rests on reliability indicators external to the algorithm rather than on showing its internal workings, and Durán states it for a human agent who believes an algorithm's output (Durán 2025a, §2); externalism in this sense concerns the algorithm, not whether the one who believes must have reasons.

Grote, Genin, and Sullivan press the last of these questions for machine learning. On their reading, the account's safeguards concerning a history of implementations and the trust of relevant experts presuppose that a model is embedded in an epistemically well-ordered environment, whereas machine-learning models implemented in domains such as criminal justice, clinical medicine, or science can disrupt the epistemic norms of those domains. These difficulties, they argue, raise the possibility that the account presupposes what it is meant to provide (Grote, Genin, and Sullivan 2024, 6).

Otsuka offers a different diagnosis for deep-learning models. On Otsuka's account, conventional statistical methods give explicit criteria for, and a theoretical understanding of, how far a hypothesis is warranted by the evidence, whereas deep learning has no principled evaluative methods on a par with them. Justification is instead empirical, established on benchmarks case by case, and so reliability comes to be attached to individual, named models. Otsuka suggests that this model- or agent-based justification aligns closely with virtue epistemology, which grounds justification in an agent's epistemic capabilities, or virtues, and that the AI research community appears to justify a model's performance on the basis of its epistemic virtues as measured by its benchmark scores; Otsuka then asks what such virtues are and how one could tell whether machines possess them, if they do at all (Otsuka 2026, "Understanding the Machine Virtues"). Although they address different systems, Durán and Formanek's account and Otsuka's locate the reliability that grounds reliance differently: in a simulation process whose reliability rests on the four sources named above, or in an individual model credited with an ability.

Otsuka also distinguishes two approaches to justifying the judgments of such models, corresponding to Sosa's distinction between animal knowledge, which animals and humans obtain through the proper functioning of their belief-forming processes, and reflective knowledge, which involves a higher-order awareness of the reliability of those processes. One approach asks whether a model has animal knowledge, much as a physiological investigation can show how a frog's perceptual mechanisms let it catch flies reliably: research on model architectures and training, including mechanistic interpretability, seeks the mechanistic structures that underlie a model's inductive performance. The other asks whether the model has reflective knowledge, and attends to the internal reasons on which it bases its conclusions. Reflective knowledge, Otsuka notes following Sosa, assesses the conditions under which first-order animal knowledge functions properly, which in machine learning are matters of robustness and of extrapolation beyond the training domain, and Otsuka counts research on explainable AI among the lines of work directed at the reflective knowledge of deep-learning models (Otsuka 2026, "Understanding the Machine Virtues"). On these two questions Otsuka's earlier treatment goes further, holding that it would be hard to deny that deep-learning models already have some form of animal knowledge, whereas it is not evident whether they have or will acquire reflective knowledge (Otsuka 2023, 136–139). Section 5.4 returns to Otsuka's argument that there may be no uniquely correct interpretation of such a model's internal representations.

In the preprint version of a paper on AI as an epistemic technology, Alvarado argues that trusting AI in the appropriate way requires access to how it operates. On this account, AI is an epistemic technology, relied on as a source of knowledge; the only trust appropriate to such a technology is epistemic trust, which must rest on epistemic reasons; and assessing those reasons requires access to the epistemically relevant elements of how the technology operates, so that explainability plays a crucial role. A hammer may be accepted without knowing how it works, Alvarado argues, but without such access it will be increasingly difficult to justify epistemic trust in an epistemic technology, justifying the acceptance of an opaque one may prove significantly harder, and appeals to prudential considerations will not suffice; if AI is essentially opaque, the argument concludes, we cannot trust it (Alvarado 2023, §3). Alvarado cites Durán and Formanek for the requirement that such technologies be assessed by sound epistemic means, but where they take sources such as verification, validation, and a history of implementations to ground reliance on opaque simulations, Alvarado takes an understanding of how the system operates, and so explainability, to be needed. The two positions can be read as differing over whether the reasons that justify relying on or trusting a system must concern how a system operates or may concern its record. Because essential opacity is relative to the agent's nature, the verdict of an account that places the reasons in access can also change with the relier: an artificial member of *S* able to follow a computation that is essentially opaque to human members might have the access such an account requires, although on Alvarado's account of agent-neutral opacity some of the route stays opaque to any finite relier (Section 5).

An older position places the reasons for reliance entirely in the record. Reichenbach considered a clairvoyant who could name the limiting frequency of a series at an early stage of it. On Reichenbach's account nothing about how the clairvoyant works need be known: the forecasts are compared with later observations, and if they are borne out, induction itself recommends preferring them to direct inductive estimates, whereas clairvoyance cannot be admitted without such a check (Reichenbach 1938, 353–354, 358–359, 363–364). Placed within the division just described, Reichenbach's view sits on the record side, and it makes the check itself an induction over the series of the source's past forecasts, which raises again the question of Chapter 2, Section 4.2 of how that series is to be typed.

Humphreys, who gave up transparency as a requirement for computational science (Section 5), placed the reasons for relying on an opaque simulation partly in how it was built. In earlier articles, he makes construction part of a model's justification and connects construction assumptions with prior justification (Humphreys 1995, abstract; Humphreys 2002, abstract). Barandiaran quotes Humphreys's account of how virtues such as stability under perturbations of boundary conditions, scale invariance, and agreement with available analytic solutions can be achieved through trial and error that treats the connection between a computational model and its solutions as opaque (Barandiaran 2008, 53). Irvine quotes his further point that theory may justify a parametric family, with empirical data specifying its details (Irvine 2016, manuscript 17). Humphreys also argues that knowledge of the physical and computational processes producing data can support the detection and correction of artifacts; a record of successful uses need not include objects sufficiently like future ones to reveal their artifacts (Humphreys n.d., §4). Construction grounds thus differ from a history of implementations, one of Durán and Formanek's sources, without excluding support from such a history. Humphreys wrote about models that humans construct; applied to a model that an artificial system builds or selects, the account implies that members of *S* can draw on the reason from construction only if the grounds of the construction are recorded and can be assessed by them.

Symons and Alvarado, also writing about simulation, deny that, in science, pragmatic reasons alone ground warranted trust. As Winsberg and Alvarado report the argument, part of the task of an epistemology of computer simulation is, on Symons and Alvarado's account, to explain the difference between the position of scientists who use epistemically opaque simulations and the relation of oracle believers to their oracles; pragmatic and epistemic considerations coexist, and it is epistemic reasons that ultimately explain and ground the pragmatic ones; a simulation, on the same account, must have warrants of its own rather than inherit them from the methods used to build it (Winsberg and Alvarado 2026, §2.3). Construction contributing to justification on Humphreys's account does not settle Symons and Alvarado's question of what warrants the particular implemented simulation. A different construction-sensitive position is Winsberg's: as Winsberg and Alvarado report it, techniques used to build simulations acquire credentials through an extended history of use (Winsberg and Alvarado 2026, §2.1.1). On that view, construction and accumulated experience work together; the warrant is neither a record alone nor a survey of each computational step.

For deep-learning models, two later accounts argue that the grounds on which such reliance draws are not available in the same way. Duede, who does not deny that scientists can be justified in relying on such models, argues that the justification reduces neither to that for instruments nor to that for experts. On this account, a model's record on test data is not enough, since many equally well-performing models can behave very differently once deployed and real-world data are often fat-tailed; the model is not a physically mediated instrument, since no causal process connecting it to the world mediates its outputs; unlike a simulation, whose procedure rests on well-established principles of numerical analysis, it does not, on the present state of interpretability research, reveal which principles its learned procedure operationalizes, which is what justifying a theoretically mediated instrument would require; and even granting that it has reasons for its outputs, those reasons are not open to evaluation by others as an expert's are (Duede 2022, §§2–4). Interpretability methods that study the models themselves with the methods of observation and experiment are, Duede concludes, a new category of reliability (§5). Alvarado argues that among the elements of a system relevant to relying on it, those concerning error cannot be left out: what reliance requires is neither a survey of how the system works nor a failure rate alone but knowledge of the nature and sources of its errors. In deep networks, on this account, such errors are often untraceable, since the same result can be reached along many paths, natural inputs can cause errors as adversarial examples do, errors are distributed across many weights, and erroneous outputs are delivered with the same confidence as correct ones (Alvarado 2025, §§3.2–3.3). On Duede's account, then, the grounds for treating a simulation as a theoretically mediated instrument are not available for deep-learning models in the same form. Alvarado's requirement of knowledge of the nature and sources of errors bears on the diagnostic role of production knowledge in Humphreys's artifact discussion; on Alvarado's account, such models often withhold that knowledge. Rathkopf, writing about generative models and holding that they produce such errors whatever their architecture, replies that Alvarado's dilemma assumes that the model is the unit at which reliability must be assessed. In Rathkopf's cases, AlphaFold 3 and the ensemble weather forecaster SEEDS, outputs are ranked, filtered, annotated with confidence scores, or embedded in ensembles by procedures built from independent knowledge of the target system, so that a workflow can be reliable without its errors being traced to their source; such workflows, Rathkopf adds, can be built only where sufficiently mature knowledge of the target is already in place (Rathkopf 2026, 13–15, 17–18, 20–21). The reply moves what is required from knowledge of the model's failures to knowledge of the target, and so leaves reliance unsecured

where research for *S* has not yet produced that knowledge.

Whichever view is taken of whether access to how a system operates is necessary, internal transparency is not sufficient. An execution trace can faithfully show how an output was computed without showing that the data measure the target, that the inference is warranted, or that the model explains the phenomenon. A feature attribution can reveal sensitivity inside a classifier while failing to establish a causal or stable relation in the world. Woodward notes that in some derivations the "manipulation" is mathematical or conceptual rather than possibly physical: one constructs a model in which, for example, the dimensionality of a system is different and calculates the consequences, which shows what difference the dimensionality makes without describing a physical transformation of the system (Woodward 2018b, §3). A model's input can likewise be varied without anything in the world being changed. Doing so shows what the model would output; it shows what an intervention on the target would do only if the relation the model encodes is invariant under that intervention. A fluent chain of reasoning can be legible and false.

This book distinguishes at least three adequacy questions:

1. Does an account faithfully describe the process that produced the output?
2. Does it adequately articulate why the available grounds support accepting the claim?
3. Does it adequately represent the phenomenon or dependence the inquiry seeks to explain?

A full process trace can satisfy the first while being scientifically unilluminating. An evidential argument can satisfy the second without answering why the target regularity holds. A simplified explanatory model can contribute to the third while deliberately failing to reproduce the discovering system's computation. The artifacts may support one another, but no one automatically substitutes for the others.

5.4 Later reconstruction is not automatically rationalization

Post-hoc explanations are often what is offered for a black box, and, like black box (Section 5), the term post-hoc explanation can name two different things. Both are accounts produced after a result. One is a persuasive story produced after the result and unconstrained by the process, evidence, or target. It can sound intelligible without explaining anything. The other is a legitimate reconstruction: an elegant proof found after an ugly search, a simple theory developed after a complex prediction, or an idealized model that isolates the relation relevant to a why-question. A reconstruction, in the sense meant here, restates the result, or how it can be reached or why it holds, in a more surveyable form than the route by which it was actually obtained.

Fleisher argues that an idealized account of a black box can contribute to understanding without being a literal replay of its algorithm. Fleisher's case is post-hoc explainable AI (XAI): after a classifier has been used, and without altering it, a second and simpler model is built to explain why the classifier made a particular decision. LIME, the main example, fits a linear model to the classifier's outputs on slightly altered copies of the input to be explained. Critics object that such explanations are mere rationalizations, because they compute a different function from the classifier and do not always reproduce its outputs. Fleisher replies that the same holds of idealized scientific models, and that such models are widely held to provide genuine understanding. The ideal gas law, for example, represents gas molecules as not interacting (compare Section 3.4). The understanding at issue is explanatory understanding: understanding why the system made a decision. Following Potochnik, Fleisher takes it to consist in grasping information about the causal patterns relevant to that decision, that is, accepting the information and being able to use and manipulate it. The understanding in view is that of the system's developers, of those who deploy it, and of those affected by its decisions (Fleisher 2022, 535–536, 540, 542–544, 547–550). Fleisher does not thereby license arbitrary surrogates, but takes three features from the literature on idealization as particularly helpful for thinking about XAI. On Fleisher's account, they are signs that a misrepresentation is an idealization rather than a mere distortion. The misrepresentation makes the model simple enough to be intelligible; it flags features that make no difference to the outcome; and it focuses attention on the causal patterns that matter to those who need the explanation. These signs count together with the usual marks of a successful model, such as empirical adequacy and predictive power. Fleisher also requires a high degree of agreement with the original model's outputs near the input explained, holds that explanation models are to be judged by the same standards as scientific models, and grants that many current uses may fall short of them, since different XAI methods often disagree (Fleisher 2022, 541, 551–556). Generalizing

these conditions from explanation models to reconstructions in general, one can take a reconstruction to be legitimate to the extent that it is constrained by the target it claims to represent, the purpose it serves, and relevant requirements of fidelity, empirical adequacy, or relation preservation. Read against the adequacy questions of Section 5.3, Fleisher's explanation models take the classifier itself as their target. They therefore bear on the question of how the output was produced, while answering it in idealized rather than literal form. The models discussed next are instead kept alongside a computation to represent the phenomenon.

De Regt and Dieks describe practices in which scientists keep an intelligible model alongside a computation that, on their own account, does not supply understanding. In quantum chromodynamics, reliable tests are possible only by numerical lattice techniques that work as a black box and do not by themselves give insight, whereas the MIT bag model, although its predictions agree only modestly with data, provides understanding, de Regt and Dieks argue, because the visualizable model allows qualitative predictions about hadron structure without lattice calculations. In meteorology, reasoning with potential vorticity supplies a qualitative picture alongside numerical forecasting (de Regt and Dieks 2005, §§4.1–4.2). Such models are not accounts of what the computation does; they are kept alongside it to represent the phenomenon. The cases therefore bear not on the first adequacy question of Section 5.3, how faithfully an account renders the process that produced the output, but on the third, whether it adequately represents the phenomenon the inquiry seeks to explain. The lattice case also bears on the claim reported in Section 2, that successful prediction from a theory requires that the theory be intelligible to those who use it, although de Regt and Dieks do not relate the two in this discussion. If that requirement concerns the users' grasp of the theory, which models such as the bag model help to supply, rather than each procedure through which a prediction is obtained, the two are consistent.

Hartmann, whose case study of hadron models discusses the bag model, gives a different reason for retaining it. On Hartmann's account, physicists entertain the MIT bag model despite its modest empirical adequacy not only because it is easy to apply and extend but because the story that goes with it is a good one: a narrative that cannot be deduced from quantum chromodynamics but is plausible given it, and that argues why the feature the model isolates is relevant and why the others are insignificant. Hartmann treats a model as an interpreted formalism together with such a story. A story must not contradict the theory, a criterion that Hartmann calls necessary; the non-formal criteria, which the same study calls much weaker, include that it be plausible and that it help in understanding the physical mechanisms; and Hartmann grants that how a model and its story provide understanding is very difficult to explicate (Hartmann 1999, 10–11, 17). In another hadron model examined in the same study, a story was used to argue away the model's lack of confinement, and one consequence of that story, that higher-lying hadrons should be more sensitive to the shape of the effective potential, was confirmed in numerical studies (Hartmann 1999, 16). Brought to the third adequacy question, Hartmann's cases suggest that the story of a model kept alongside a computation bears on it only as far as what the story sets aside as insignificant has itself been checked.

A further line of argument questions whether a model's internal reasoning has a uniquely correct rendering at all. Otsuka compares the evaluation of a deep-learning model's internal representations with Quine's radical translation, the situation of translating a wholly unknown language from its speakers' behavior alone, in which, Quine argued, mutually incompatible translations can fit all the evidence equally well (Otsuka 2021, 11–12; Otsuka 2026, "Can We Understand What Machines Think?"). What a model labels "cat" may be the animal we mean by that concept, or, as a study Otsuka cites suggests, merely a combination of properties and backgrounds typical of cats, and Otsuka argues that which of these is correct may not be uniquely settled, and that there may be no single correct answer at all. Nevertheless, on this account, the search for explanations of such models cannot simply be dismissed, because what a model takes to exist matters to its social application: unless people know to some extent what kinds of things the recognition model used in automated driving takes to exist, they will not be willing to entrust their lives to it (Otsuka 2021, 12). On that view, a requirement of fidelity to a model's reasoning, once that reasoning is rendered in human concepts, may have no unique fact to be faithful to, even where the computation itself can be traced exactly. Chapter 8, Section 3.1 contrasts this Quinean worry, on which there may be several equally good renderings, with Kuhn's, on which a statement may have no adequate rendering at all.

Either way, the question that matters here is not when an account was produced. It is which object it claims to explain and what makes the account adequate to that object.

For artificial research, the cases of this subsection bear on more than how outputs are assessed. A legitimate reconstruction can itself be an achievement that research brings about. An elegant proof found after a search, a simpler theory developed after a complex prediction, or an intelligible model kept alongside a computation moves from a result that is warranted but not understood toward understanding. In the strong predictive case, an attempt directed at understanding why the regularity holds could pursue it in this way, whether the reconstruction is produced by the system, by its users, or by both together. What makes such a reconstruction count is what the constraints above require: that it answer to the target it claims to represent and to the purpose it serves, and, as Hartmann's cases suggest, that what it sets aside as insignificant has itself been checked. If Otsuka's comparison with radical translation holds, a further consequence follows for the route such an attempt takes. A rendering of a deep-learning model's internal reasoning in human concepts may have no unique fact to be faithful to, whereas a model of the phenomenon answers to the third adequacy question, which concerns the target rather than the computation; where understanding of the phenomenon is sought, that route does not wait on settling the first adequacy question, whether an account faithfully describes the process that produced the output. In whom understanding brought about in this way is realized, whether in the system, in members of *S*, or in both, is the question of Section 6. Where *S* is human, the reconstruction has to be usable by its human members; where the recipients are artificial systems that can operate on the model's representations, a rendering into human concepts may not be needed at all (compare Chapter 8, Section 3.4).

The two objections with which this section began fare differently. Whether opacity undermines the warrant for the knowledge that Section 4 allowed depends on the account of warranted reliance (Section 5.3): it can on accounts that require access to how a system operates, such as Alvarado's, and need not on those that place the reasons in a system's record or in how it was built, although each of these leaves questions of its own. That opacity settles that nobody understands the regularity does not follow, since understanding can be sought through a reconstruction or a model kept alongside the computation, as described above.

The distinctions also bear on how such work is recorded. An artificial research system whose outputs are to be assessed on each of the three adequacy questions of Section 5.3 would need to keep process provenance, evidential support, target-level explanation, and pedagogical presentation as distinguishable artifacts, even when a human interface combines them; Section 9 relates each to the function it serves. Calling all four reasoning invites an articulate narrative to stand in for both warrant and understanding.

Who understands?

Where humans conduct research, those who produce a result are usually also among those who understand it, so the bearer of understanding seldom needs to be named separately. Where an artificial system produces a result, it need not be what understands it, and the bearer becomes a question of its own. A claim that research has brought about understanding can accordingly be at least three distinct claims:

1. an AI helps another agent or a community, human or artificial, understand;
2. the AI itself understands;
3. a human–AI collective or artificial community understands.

Each of these concerns who understands. Whether understanding realized in any of them has entered *S*'s epistemic situation is a further question (Chapter 1, Section 2), taken up below.

The first does not require the second. A telescope can disclose a phenomenon without seeing it, and a search procedure can enable a proof without understanding the theorem. Krenn and colleagues draw the same separation for AI in science. The roles they distinguish through which AI can support scientific understanding, among them revealing patterns and serving as a source of inspiration, fall under the first claim; AI as an agent of understanding, which they raise as a further and stronger possibility, would fall under the second (Krenn et al. 2022). Their framework does not establish that present systems are agents of understanding in this sense.

Their earlier study of Theseus, a computer program for designing quantum-optics experiments, supplies a concrete case of the first claim. The study reports that the program's graph-based representation and automated search yielded solutions to open problems in quantum optics from which the investigators extracted new concepts (Krenn et al. 2021). The authors

describe the system as contributing to conceptual understanding. What the case directly shows is that an artificial procedure and an interpretable representation can mediate human understanding.

The second claim, that the AI itself understands, takes up most of this section. What follows first sets out what different accounts would test; positions that address directly whether machines can understand come next, from de Regt, alone and with coauthors, to Wilkenfeld; and the section then asks whether understanding requires belief. How the claim is assessed depends on the selected theory. An explanatory account may ask whether the system represents and appropriately grasps an explanation. An ability account of understanding why, such as Hills's, may test whether the system can follow an explanation, give one, and draw conclusions and give explanations in similar cases; a representation-manipulability account may test whether it can modify a representation to support counterfactual reasoning, intervention, transfer, and error diagnosis. A dependency account may ask whether a coherent dependence model is integrated into the system's inference and action. These tests are informative only if they go beyond fluent report generation. On accounts that also tie understanding to a felt sense of grasp, such as Potochnik's (below), these tests leave a further question open; on accounts that require only a functional or representational relation, the artificial realization could differ from the human one while satisfying it.

One position bears directly on the second claim. De Regt holds that the skills needed to construct and evaluate scientific explanations involve tacit knowledge of how to apply rules in the appropriate circumstances. Following Collins, de Regt classifies this as collective tacit knowledge, which on this account can be acquired only by taking part in a social practice and cannot be made explicit in a scientific description that could be used to build a machine mimicking human behavior (de Regt 2017, 27–30). De Regt likewise reports, following Kaiser, that some physicists learned from published sources alone how to calculate with Feynman diagrams but did not learn to use them on completely new problems (de Regt 2017, 252–253). The argument is directed against capturing such skills in explicit rules or descriptions. It leaves open whether a system that acquires its dispositions through training, rather than from an explicit description, could come to take part in the relevant practices. Because the argument concerns what joining a practice requires, it also does not address whether a practice among artificial agents could itself be a social practice in which such collective tacit knowledge is held. The Feynman-diagram case, as de Regt reports it, suggests only that published texts transmitted part of what practitioners could do.

Later work co-authored by de Regt reads de Regt's own account in two ways. The chapter with Boge cited in Section 2 presents the contextual theory as applying specifically to human scientific understanding: it was developed by analyzing how human scientists achieve understanding, and intelligibility, its central notion, is relative to scientists with particular skills in particular historical, disciplinary, and social contexts (Boge and de Regt 2025, 418). The authors then consider whether a deep neural network could come to understand in this sense. At first sight it could not: on the contextual theory, understanding comes from explanations based on intelligible theories, whereas a network's predictions are not based on scientific theories and it supplies no intelligible models of the phenomena (Boge and de Regt 2025, 419). The notion of theory involved is not demanding, however. De Regt takes a theory, following Giere, to be a collection of principles whose building blocks are concepts. If a network's abstraction of features from data is regarded as acquiring concepts, a network could build theories from concepts of its own. The difficulty is that these concepts can differ wildly from human ones. Even if a network acquired understanding through theories based on its own concepts, that understanding might then not be accessible to humans, since human scientists cannot use concepts they cannot grasp to construct explanations themselves (Boge and de Regt 2025, 420, 429). The authors call the result a divide between human and artificial scientific understanding, an obstacle to progress in scientific understanding as a whole, and look for a way across it in proposals for testing whether, and measuring how far, an AI can transfer its understanding to human experts (Boge and de Regt 2025, 427–428). In the terms of Chapter 1, Section 2, theirs is a case in which understanding might be realized in an artificial system without entering the epistemic situation of a human *S*. Chapter 1, Section 6 read Ted Chiang's story as a thought experiment of a similar shape, in which results realized in bearers outside a human *S* are recorded in a form that no member can yet follow, and on two of the three readings given there a decipherment can be new to *S* although its result is not new to those who hold it. On those readings, bridging the divide that Boge and de Regt describe could itself be an achievement new to a human *S*.

The other reading treats the account as applicable to humans and machines alike. Barman, Caron, Claassen, and de Regt take scientific understanding to be an ability, assessed through what an agent can do rather than through its internal mechanisms or a feeling of understanding, and they argue that understanding could be realized through mechanisms different from human ones. Starting from de Regt's criterion, they shift its focus from the phenomenon to the agent and make it a matter of degree: an agent understands a phenomenon to the extent that it has a sufficiently complete representation of it, can generate internally consistent and empirically adequate explanations of it, and can draw a broad range of correct counterfactual inferences about it, abilities they propose to test with what-, why-, and what-if questions (Barman et al. 2024, 2–6). A preprint by Merijn Moody and de Regt adopts the contextual theory's test of intelligibility, recognizing a theory's qualitatively characteristic consequences without exact calculation, as a functional definition of scientific understanding. The objections they mean to avoid are modelled on John Searle's Chinese room argument. A person who knows no Chinese follows a book of instructions for manipulating Chinese symbols and passes back answers that, from outside, look like those of a Chinese speaker, yet the person understands none of it; Searle concludes that running a program, however appropriate its outputs, does not by itself produce understanding (Cole 2024, §§1, 3). Moody and de Regt's definition identifies understanding with an ability and asks for nothing beyond it, so on their account it sidesteps such objections by construction. On indirect evidence that language models answer a wide range of out-of-distribution scientific questions accurately, they credit current models with a non-trivial but brittle degree of such understanding, while noting that extensive evaluation has yet to be carried out (Moody and de Regt 2026, 9, 23).

Accounts of what has been called pragmatic understanding, understanding how to do something with a system rather than why something is so, divide on the same point. Lenhard, as reported by Westerblad and Stuart, takes pragmatic understanding to consist in abilities to predict and control phenomena; in Lenhard's case from nanoscience, simulations stand in for theory, and design rules let scientists build artifacts and control phenomena whose dynamics are not theoretically understood (Westerblad 2026, 13; Stuart 2025, 4). On such an ability view, Stuart notes, AI systems could have understanding (12). Stuart rejects the view, holding instead that an agent has pragmatic understanding of a system if and only if it has the skills to manipulate the system or its parts robustly and successfully to achieve its goals and is responsible, in the sense of being praiseworthy, for having gained those skills; skills, unlike mere abilities, are task-specific, learned, and deployed under some cognitive control (6–8). For this reason Stuart doubts that the benchmark of Barman and colleagues, which measures what an agent can do, captures what understanding requires (12–13). The article concludes that current and near-future AI systems lack pragmatic understanding, on a premise that it says it does not argue but takes from what it calls the majority view in the philosophy of AI, that such systems cannot bear responsibility; and since explanatory and objectual understanding often require skills such as conceptual mastery, it concludes that in many scientific cases the systems lack these too (13–14). Stuart nonetheless credits pragmatic understanding to a laboratory in which some members have the relevant skills and are responsible for them, with AlphaFold among its instruments, and sketches, without adopting, ways in which the responsibility condition might be dropped or met by artificial systems (14 n. 7, 16). On this account, then, whether *A* understands in this sense turns not on any battery of tests but on whether *A* is a praiseworthy agent with respect to the skill, the fourth of the senses of credit that Chapter 7, Section 3.3 keeps apart from true output, reliable process, and success attributable to competence. Westerblad, replying to Humphreys, holds that an instrument becomes functionally intelligible to scientists through an inquiry procedure, a publicly available way of using it to reach an aim, and that new instruments extend such procedures by being calibrated against those already accepted, so that, on this account, the means by which science extends human abilities do not move beyond human limitations (Westerblad 2026, 15–20). The agents who check, verify, and compare are, in this discussion, human scientists, and Westerblad notes that the conclusion bears on the view that science is no longer done by or for people, although the paper does not argue the point explicitly (20). Applied to an artificial *A* that calibrated new procedures against accepted ones, Westerblad's account would make whether the extension remains within the grasp of a human *S* turn on whether the procedures are available to its members, not on who carried out the calibration.

Wilkenfeld presents artificial intelligence as perhaps the field best placed to use the representation-manipulability account, introduced in Section 3.3. On that account a subject understands an object when it has a representation of the object that it could itself modify in small ways to produce a new representation supporting effective inferences about the object or manipulations of it. Wilkenfeld's examples include noticing that one has learned a rule of logic wrongly and correcting it, and adapting a proof one understands to a slightly different system; a proof one has only memorized cannot be adapted in

this way (Wilkenfeld 2013, 9–10). Two features of the account bear on artificial understanders. First, it asks little of what a representation is: only that it be a computational structure with content that can be transformed. Wilkenfeld states this as a claim about human minds, consistent with classical computationalism, a language of thought, and some non-eliminative interpretations of neural networks, that is, interpretations on which the states of a neural network carry content (Wilkenfeld 2013, 8–9). The account therefore does not exclude a system as an understander because its representations are not symbolic. Second, the contextual standards that decide which manipulations count as understanding belong to the characterization made by whoever attributes understanding; the understander need not represent them, and need only be able to modify its representations and make the inferences (Wilkenfeld 2013, 19). If the understander in question is an artificial system, whether *S* is human or artificial, what the account leaves to be asked is whether the system has states that represent the object, and whether the system itself, rather than someone who edits its internal states from outside, can modify them in this way. Wilkenfeld does not take up this question. The reason given for turning to artificial intelligence is that building a computer that behaves like a human cognizer requires being clear about what human cognizers do when they understand (Wilkenfeld 2013, 18–19).

A related question is whether there can be artificial understanding without artificial belief. Dellsén's claim, noted in Section 3.2, that understanding need not require justification or belief in the way knowledge does, bears on it, but how it bears depends on what is meant by belief. In Dellsén's central case, a physicist in a world where string theory is true has methodological reservations about the theory and does not believe it, yet accepts it and uses it to explain; Dellsén argues that the physicist understands what the theory describes (Dellsén 2017, 2, 13–15). The contrast on which the case relies is one that Dellsén takes from L. J. Cohen: to believe that *p* is to be disposed to feel that *p* is true, whereas to accept that *p* is to adopt a policy of treating *p* as given (Dellsén 2017, 13–14). Belief in this sense is narrower than the belief discussed in Chapter 2, Section 2.1. On the functional and dispositional accounts described there, stable integration of *p* with inference, prediction, action, and revision is stronger evidence of belief than its storage or utterance, so conduct that Cohen classifies as acceptance can itself count as evidence of belief in that broader sense. Elgin, who draws heavily on Cohen's distinction, reads it as an explication of the commonsense notion of belief, which has two components: feeling, or being disposed to feel, that *p* is so, a largely involuntary component for which Cohen reserves the term "belief", and acceptance (Elgin 2025, 35–36).

Potochnik also separates acceptance from belief. Using a definition on which to accept something is to presuppose it for specific reasons, Potochnik holds that contributors to understanding, including idealizations, are to be epistemically accepted, that is, presupposed for epistemic reasons, but need not be believed; idealizations are not believed or asserted but assumed contrary to fact (Potochnik 2015, 8, 13, 16–17). Potochnik's account does not, however, free understanding from psychological conditions. On this account, understanding is a cognitive state as well as an epistemic achievement (Potochnik 2015, 5). As Raja and Sanches de Oliveira report Potochnik's account, understanding requires both successfully grasping a causal pattern and feeling that one has grasped it (Raja and Sanches de Oliveira 2024, §§3.1–3.2, 10–13). Van der Merwe also reports her formulation that genuine understanding arises from information of the right kind to induce that feeling (van der Merwe 2023, 330).

These views agree that understanding can go with acceptance rather than belief, but they define acceptance differently, and so they leave different things to be asked when the understander in question is an artificial system, whether *S* is human or artificial. On Dellsén's use of Cohen's distinction, understanding requires acceptance together with a mastery of the subject matter that enables correct explanation, whereas knowledge, on this view, still requires belief: the physicist in the string-theory case understands without knowing (Dellsén 2017, 14, 17–18). If belief is a disposition to feel that *p* is true, a system to which feeling cannot be attributed may still accept, and understanding would then lie closer to it than knowledge does. What remains to be asked is whether using *p* as a premise in explaining amounts to adopting a policy, which Dellsén describes as something one decides to do. Elgin defines acceptance as being willing and able to use *p* in inference and action when one's ends are cognitive, counts rules, standards, and methods among what can be accepted, and holds that, unlike belief in Cohen's sense, which is largely involuntary, acceptance is voluntary; belief in that sense, on her account, is spectatorial, whereas acceptance is agential (Elgin 2025, 11, 35–36). On Elgin's account the question for an artificial system moves from feeling to whether it has cognitive ends and accepts in this voluntary sense. On Potochnik's account as reported above, setting belief aside does not make the artificial case easier, since the felt sense of understanding is still required (Raja and Sanches de Oliveira 2024, §3.1, 11); Potochnik also takes what promotes understanding to depend on features of human

cognition (Potochnik 2015, 11, 15–16). How much any of this matters depends, finally, on the account of belief adopted, the question raised in Chapter 2, Section 2.1. On a narrow account such as Cohen's, acceptance can be realized without belief. On the functional and dispositional accounts discussed there, acceptance put to work in explanation, as in Dellsén's case, may itself count as evidence of belief, so that understanding without belief offers an artificial system no separate route. What remains clearly distinct from belief on those accounts is a posit known to be false and presupposed only within one research purpose, which need not show the stable integration with inference, action, and revision that they treat as stronger evidence of belief.

The third claim, that a human–AI collective or an artificial community understands, requires an account of how the group's understanding relates to what its members, human or artificial, understand (compare the accounts of group knowledge in Chapter 2, Section 6). Distributed task success alone does not show that the group understands. Different agents can jointly produce a correct answer while no component and no organized whole can represent the relation that explains it, use it beyond the task, or respond to challenges. On accounts that attribute understanding to a group as an organized whole, a collective-understanding claim needs to identify how partial representations, abilities, communication, and correction compose into the attributed achievement; on accounts that ground it in members' understanding, the claim turns on what those members understand.

The three claims all concern who understands. The rest of this section turns to the question set aside at its start, whether understanding realized in some bearer has entered *S*'s epistemic situation, beginning with a distinction between an individual's understanding and understanding credited to science. Boge, Schuster, and Stoll distinguish **scientists' understanding**, which an individual scientist has by grasping an explanation of a phenomenon, entertaining a coherent body of information accounting for it, or grasping a model of its dependencies, and which may be tacit or inarticulable, from **scientific understanding**, which on their account obtains only when there is a scientific explanation of the phenomenon communicable to a sufficiently broad scientific audience. That audience may be a small group of experts, and the explanation need not yet have been communicated (Boge, Schuster, and Stoll 2026, 2, 5–7). In one of their cases, Feynman understood why Feynman diagrams worked before Dyson's proofs gave an explanation that other physicists could follow (Boge, Schuster, and Stoll 2026, 9–10). The authors stress that the difference is not that one kind is explanatory and the other objectual: while individual scientists may come to understand on the basis of an explanation, science as a whole, on their account, comes to understand by means of one. They add that a scientist's understanding may be a very important epistemic achievement but is not yet scientific understanding (Boge, Schuster, and Stoll 2026, 4, 6). Their distinction can be read as concerning what is counted as the achievement rather than how understanding is realized in individuals and in communities. It is a distinction between what is credited to an individual and what is credited to science. In that respect it corresponds, on the side of understanding, to two distinctions drawn in Chapter 1, Sections 2 and 6, and taken up for knowledge in Chapter 2, Sections 2.1 and 6. One is between the actor conducting the research and the society relative to which an achievement is new and for which it is sought; the other, where success is assessed, is between an achievement realized in some bearer and one that has entered that society's epistemic situation. In each pair the two can coincide, but neither determines the other. The parallel is partial. On their account, the achievement credited to science requires only that a communicable explanation exist, not that members of the community have taken it up. What is credited to science thus corresponds to the achievement's being available to the society rather than to its members' holding it, a difference that Chapter 2, Section 7 discusses for knowledge as that between being in a position to know and knowing. The parallel is partial in a second respect as well: the individual in their cases both produces the result and understands it, whereas the distinctions of Chapter 1 allow the actor who produces a result and the bearer who understands it to differ. The proposal takes one side of a disputed question, whether scientific understanding requires explanation, and is not the settled meaning of scientific understanding.

The questions that the distinction separates remain open. One position can allow a researcher to understand tacitly before articulation; explanatory accounts can accept this too, since grasping an explanation need not include the ability to state it (Strevens 2013). Grimm, defending a dependency account, argues that articulacy is not necessary for understanding, citing evidence that young children and some animals grasp causal dependencies they cannot articulate (Grimm 2017). Another position can reserve scientific understanding for representations that support communication and criticism. Boge, Schuster, and Stoll hold both, reserving scientific understanding for communicable explanations: they read Elgin's account as concerned with scientists' understanding and de Regt's as concerned with scientific understanding, and take the two to be complementary rather than rival (Boge, Schuster, and Stoll 2026, 3, 14). An account that extended Bird's group-functional

account of knowledge (Chapter 2, Section 6) to understanding would attribute understanding to an organized system even when no individual possesses the whole. These positions make different demands on an artificial researcher, according to where they locate understanding. If understanding can lie in a researcher's own tacit grasp, as the first position allows, an artificial researcher that is itself to understand must grasp the dependencies or the explanation, though it need not be able to state them. If scientific understanding requires a communicable explanation, as on Boge, Schuster, and Stoll's account, the researcher must yield an explanation communicable to the relevant audience, which has to be human-facing when that audience is human. If understanding can be attributed to an organized system, as on an extension of Bird's account, what is required is that the researcher's contribution be integrated into a collective epistemic capacity that the organization realizes as a whole.

As the second of these demands shows, who bears understanding and for whom it is sought are different matters. The bearer of understanding and the society for which that understanding is sought are analytically distinguishable, the first being whoever understands and the second one of the two parties to a research claim (Chapter 1, Section 2). Whether understanding realized in some bearer has entered that society's epistemic situation is a further question, the counterpart for understanding of the third judgment distinguished there, and the positions above answer it differently. On accounts that ground a group's understanding in its members', it has entered when relevant members, human or artificial, understand; on an extension of Bird's account such as the one just described, when the society's organization realizes it as a whole. Boge, Schuster, and Stoll's scientific understanding requires only that a communicable explanation exist, and Dellsén's restated noetic account locates progress in public information that enables relevant members of society to understand (Dellsén 2021). Both concern what is available to the society rather than what its members hold, so whether such availability already amounts to entry, or leaves the society only in a position to understand, is the counterpart of the dispute between Bird and Lackey over knowledge (Chapter 2, Section 7).

Applied to artificial systems, these distinctions separate at least the following possibilities. An artificial system might enable human understanding without itself understanding. It might supply humans with warranted knowledge while neither humans nor the machine possess explanatory understanding. Under a sufficiently non-anthropocentric account, it might instead organize dependencies in a form that supports its own inference, intervention, and error correction while remaining unintelligible to humans. Such an organization might serve only the system's own inference, or it might be communicable to other artificial members of *S* without being communicable to humans. A human–AI organization might also realize an integrated capacity not possessed by any one member.

The evidence reviewed in this chapter does not establish, as a description of existing systems, the possibility that an artificial system organizes dependencies in a form that supports its own inference while remaining unintelligible to humans. Even so, the possibility exposes an important choice of target. Intelligibility to the relevant audience can be necessary when the intended achievement is understanding that is to enter *S*'s epistemic situation, such as public scientific understanding, without being built into every possible account of artificial understanding; it needs to be human-facing only when that audience includes humans. Conversely, evidence that a representation is useful inside an artificial system does not establish that it is understandable, warranted, or valuable for *S*. As research by artificial systems spreads, knowledge and understanding may therefore advance along different trajectories for different epistemic societies. This is more than an interface problem: it concerns whose epistemic capacities an artificial research system extends. The divergence need not run only between societies. Venkatesh, writing about pure mathematics, holds that concepts achieve an economy of thought fitted to human limits of memory and time, and that machines compete with concepts for that function. Venkatesh then suggests that mechanical reasoning may transform mathematical language and concepts so thoroughly that present and near-future mathematicians might struggle to understand one another, and takes earlier rewritings of mathematics to show that such unintelligibility is a real possibility (Venkatesh 2025, 94–99). On this account the divergence could arise within a human *S*, between its members at different times, through machine participation in their research. Since Venkatesh holds, with Davis and Hersh, that communicability is "part of its very definition," such a rewriting would on that view change not only how mathematics is done but what it is (84, 112).

The possibility that an artificial system organizes dependencies while remaining unintelligible to humans contains two claims: that an artificial organization of dependencies captures causal relationships, a claim about the world, and that it explains them or affords understanding of them, a claim about bearers and users. Woodward's account of causal explanation

separates the two. On this account, causal relationships are features of the world, and Woodward's characterization of an intervention makes no reference to human agency: a process with the right causal characteristics qualifies, including a natural process in a world without people (Woodward 2023, §8). On this account, then, an artificial organization of dependencies could capture causal relationships while remaining unintelligible to humans. Explanation, by contrast, is an activity that Woodward attributes to humans and conceivably some other animals, as Bassi reports (Bassi 2013, 123–124). Woodward also requires that the information giving an explanation its import be epistemically accessible to those who use it (Woodward 2000, 248 n. 11). Woodward's account does not settle whether artificial systems can be among those users. Whether such an organization would also explain depends, on this account, on whom the explanation is for and on whether those users, human or artificial, have access to the relevant information.

Potochnik's account of explanation makes the audience part of what explains. On this account explanations must generate understanding in those who seek them, and which of the many dependence relations responsible for a phenomenon explains it depends on what that audience seeks to understand (Potochnik 2016, 9–13, 17). Potochnik develops the account for human science, holding that explanations must be comprehensible to humans and tying the contribution of broad patterns to features of human understanding (Potochnik 2016, 17–18; Potochnik 2015, 11, 25). Among the features of an audience that, on this account, shape what counts as explanatory, Potochnik lists its epistemic circumstances and interests and, for humans in general, cognitive features (Potochnik 2016, 4–5, 17). Extended to artificial organizations of dependencies, the account would make the question whether such an organization explains anything, and to whom, partly a question about the cognitive needs of its audience, human or artificial, and it gives no reason to assume that what explains for one audience would explain for the other.

In sum, the three claims call for different evidence: what a system enables others to understand, what the system itself can do with its representations under a selected account, and how a group's partial representations and abilities compose. For any of them, where understanding is realized and whether it has entered *S*'s epistemic situation are to be asked separately.

What this chapter has brought out is that different theories require different evidence, and that a property granted to one bearer does not pass to another without argument. Part IV takes up questions related to whether an artificial system can literally believe, represent content, or understand: norm formation and epistemic self-government in the analysis of autonomy over inherited standards (Chapter 7, Section 4.3), and, in Chapter 8, artificial knowledge bearers and societies (Section 1) and what would let an artificial knower's checks count as evidence, its states be about a world, and its commitments remain answerable to that world (Section 2).

What understanding is valued for and what it may enable across time

Section 4 distinguished an attempt's being directed at understanding from a requirement that research, or each of its operations, bring understanding about; the characterization imposes no such requirement for any achievement. This leaves the value of understanding and its place in sustained inquiry open to substantive accounts. The value in question is that of understanding as an epistemic status: what it is valued for, in itself or for what it may enable in continuing inquiry. Chapter 9, Section 3 counts such a status as one possible object of value; the question that section mainly pursues, which of *S*'s questions, practices, and distributions of results matter and to whom, belongs to the value and governance level of Chapter 1, Section 1 and is not taken up here. One answer concerns understanding's value in itself or its place as an aim of inquiry (Section 7.1). Another concerns how it constitutes, evidences, enables, or promotes further capacities in continuing inquiry (Sections 7.2–7.4). The strong predictive case of Section 4 puts the question of what understanding enables sharply: what would be lost, beyond any single validated result, if a research system brought about knowledge that neither it nor those for whom it inquires understood, whoever, if anyone, held that knowledge? The subsections are arranged accordingly. Section 7.1 takes up understanding valued for its own sake or as an aim. Sections 7.2–7.4 take up what understanding may enable as inquiry continues beyond one validated result, divided by what is grasped or made available: a grasp of how a target would differ if what it depends on were different (Section 7.2), a grasp of how a domain hangs together (Section 7.3), and the availability of understanding to others (Section 7.4). This division is the book's own, drawn along strands in the literature on understanding rather than taken from a classification that literature offers, and it does not claim to be exhaustive.

7.1 Understanding valued for its own sake or as an aim

Several epistemologists hold that the value of understanding is not exhausted by what it enables. Kvanvig argues that the value of knowledge is exhausted by the value of some of its constituents, such as true belief and subjective justification, whereas understanding has value beyond that of such parts. On this account understanding requires grasping explanatory and other coherence-making relations within a large body of information, and grasping them organizes and systematizes one's thinking on a subject matter in a way that goes beyond adding more true beliefs, or even more justified true beliefs. Organized thinking lets one reason from one item of information to related items that can serve as a basis for action, and it gives inquiry a satisfying closure, a sense of having grasped the subject matter as a whole; the value beyond the parts is the value of this organization (Kvanvig 2003, 185, 188, 192, 202, 205). Pritchard argues that understanding why something is so typically, if not universally, involves a cognitive achievement in a demanding sense, one that demands overcoming a significant obstacle or exercising significant cognitive ability, and that achievements of this kind are finally, that is non-instrumentally, valuable (Pritchard 2016, 9–10). In the same article Pritchard grounds this value in the way such achievements exercise one's agency on the world and in the intellectual autonomy that understanding manifests: a grip on things that is one's own even when it is gained with the help of experts (Pritchard 2016, 9–12).

Riggs is among those who hold that the really valuable things from an epistemic point of view are higher-grade cognitive accomplishments such as understanding. As Grimm reports the view, we do not by nature seem to have any desire to acquire trivial pieces of information, such as the name of a particular person listed in an old phone book, whereas any instance of understanding may seem worth pursuing, and Riggs writes that any understanding, even of a subject matter we may consider trivial or mundane, contributes to the epistemic value of one's life (Grimm 2021, §3.1). As Gordon reports it, Riggs holds that understanding a subject matter requires an appreciation of how its parts fit together and of the role each plays in the whole, and he distinguishes from understanding what he calls intelligibility, which also implies a grasp of order, pattern, and connection but does not seem to require a substantial connection to truth (Gordon n.d., §§2b, 3).

Zagzebski, among others, has argued that an epistemology focused on knowledge alone is one-sided, since it neglects other epistemic goods such as understanding and wisdom (Grimm 2021, §1.2). Reporting the history behind this neglect, Grimm notes that the ancient term *episteme* has been argued to be better rendered as understanding than as knowledge, since for Plato and Aristotle it involved a systematic grasp of how things are connected or of their causes, and that epistemology's later focus on isolated propositional knowledge has been explained as a response to the rise of skepticism in Hellenistic philosophy, an explanation Grimm finds in Burnyeat and in Zagzebski, or, as Pasnau suggests, to the religious wars of early modern Europe (Grimm 2021, §1.1). As Gordon reports Zagzebski's view, not all understanding requires true belief: the belief that objects fall in a vacuum at a fixed rate, which ignores the attraction of everything but the earth, yields more understanding for most purposes than the vastly more complicated truth, since grasping the truth can come at too high a cognitive cost (Gordon n.d., §2b).

Hills, writing on aesthetic understanding, holds that awareness of the reasons why a work is or is not valuable is likely to help one make better judgments in the future, and that such understanding is an essential component of aesthetic virtue, so that the ideal way to make aesthetic judgments is through one's own understanding. Part of her case is an idea of mirroring value. True beliefs mirror the world in virtue of their content; a judgment based on a grasp of the reasons that make it true also mirrors the world in its form, since the dependence of one belief on others matches a dependence in the world. When someone judges a novel bad because they recognize that it is inept, lifeless, and badly written, their belief that it is bad depends on their belief that it is inept just as its lack of merit depends on its ineptness, and they would have judged differently had the novel had different qualities (Hills 2022, 21, 28–29). Hills suggests that this explanation generalizes to moral understanding (Hills 2022, 22).

Not every account of the value of understanding, however, appeals to value beyond what understanding enables. Grimm traces much of the special value of understanding to the position it gives one to predict, and possibly control, how the world will unfold, while granting that this may not exhaust the intuitions in its favor (Grimm 2017). Grimm's earlier survey keeps both kinds of value in view, and it shows that valuing understanding for its own sake need not depend on denying that understanding is a kind of knowledge. Grimm, who counts himself among those who take understanding why to be knowledge of causes, recasts the question why understanding is more valuable than knowledge as the question why it is better than propositional knowledge, treating knowledge as a genus whose species differ in their objects. He holds that

objective understanding is plausibly desired both for its own sake and for the promise it offers of controlling the world, and argues that if a more profound mirroring of reality is epistemically better, every item of understanding is more valuable than the corresponding item of propositional knowledge (Grimm 2012, 4, 11, 15).

Related positions treat understanding as an aim. Pritchard has suggested that understanding, rather than knowledge, may be the goal of inquiry. On this account, someone who learns from a reliable source that their house burned down because of faulty wiring, but who has no conception of how faulty wiring could do this, has not yet reached the goal of their inquiry (Pritchard 2008, §§10–12). In later work Pritchard restricts the claim. Inquiries that seek an explanation of why something is so are legitimately closed only by the kind of knowledge involved in understanding, whereas an inquiry into whether, say, an animal in a zoo is a genuine zebra can usually be closed by knowledge from a trusted informant. Pritchard calls the resulting view a complex knowledge account of inquiry: it is always knowledge that legitimately closes a well-conducted inquiry, but in some inquiries knowledge of a specific sort (Pritchard 2016, 6–7). Kelp, who defends knowledge as the aim of inquiry into specific questions, takes understanding to be the aim of inquiry into a general phenomenon, so that such inquiry succeeds if and only if one understands the phenomenon. On this account such an inquiry is not completed by knowing each truth about the phenomenon piecemeal, for example each on the word of a different informant; it also requires that one's beliefs be based on one another in a way that reflects one's knowledge of the explanatory and support relations among those truths. On Kelp's account, maximal understanding is maximally comprehensive and well-connected knowledge of the phenomenon, and lesser understanding is a matter of degree (Kelp 2021b, S1583–S1584, S1588; Kelp 2015, 16, 18). Set side by side, Pritchard's later view and Kelp's both make what properly ends an inquiry depend on what the inquiry is into, and both allow that more than isolated knowledge can be required, but they draw the line in different places: Pritchard between inquiries that seek an explanation and others, Kelp between inquiry into specific questions and inquiry into phenomena, such as the life of Napoleon, inquiry into which can progress through knowledge that is not explanatory, such as that Napoleon was French (Kelp 2021b, S1583, S1589). On such views, taking knowledge as the orientation of inquiry does not by itself settle whether understanding is required; what an inquiry requires depends on the question it pursues, in line with the question-relativity noted in Section 2.

In the philosophy of science, Elgin argues that truth is not the paramount epistemic objective, since an epistemology centered on truth cannot account for the epistemic standing of a science that relies on models and idealizations known not to be true (Section 3.4). Elgin argues further that the cognitive achievement good science standardly yields is understanding rather than knowledge. A scientific account is not a collection of separately supported truths: its claims support one another, their conjunction is apt to be falsified by some anomaly, and the anomalous parts cannot be excised without unravelling the rest; knowledge requires truth, and the accounts we admire do not come out true, whereas understanding, which on this account answers to the facts without being factive, can advance through strictly false accounts. Because devices such as metaphor, fiction, and exemplification figure in scientific as well as artistic understanding, Elgin holds that scientific understanding cannot be done justice while denying that the arts and their devices function cognitively (Elgin 2004, 1–2; Elgin 2006, 1, 3–5; Elgin 2007, 2–5; Elgin 2002, 1). De Regt treats understanding phenomena as one of the central epistemic aims of science and distinguishes views on which understanding is bound up with the explanatory aim, is an additional aim, or takes the place of traditional aims (de Regt 2020, 921, 923, 932). Dellsén's account, discussed in Section 4, belongs to this family. So does Potochnik's, introduced in Section 3.4, on which understanding, not truth, is the ultimate epistemic aim of science, and truth and knowledge matter as routes to it: idealized representations can represent accurately the causal patterns they depict, and science, on Potochnik's account, generates understanding of phenomena by way of knowledge of those patterns (Potochnik 2020, 13, 16). It follows that, on this account, where an attempt is directed at knowledge, its target is valuable chiefly as a route to understanding; the working sentence, which does not specify the kind of achievement, can accommodate this ranking without adopting it.

These positions differ from one another, and each is contested. Critics point to easy cases of understanding that seem to involve no significant achievement (Grimm 2021, §3.1). Boyd, who defends the acquisition of understanding by testimony, argues by contrast that understanding that is easy for an expert can still count as a strong achievement, because the expertise that makes it easy was itself hard won (Boyd 2017, 37–38). The demanding sense of achievement in this exchange is narrower than the general term epistemic achievement in Chapter 1, which by itself requires neither overcoming a significant obstacle nor exercising significant cognitive ability. Whether easy understanding is an achievement in Pritchard's sense therefore does not decide whether it can be an epistemic achievement new to *S*. DePaul and Grimm turn Kvanvig's own

reasoning against Kvanvig's positive proposal. Kvanvig had argued that knowledge has no value beyond that of some of its constituents because any condition that excludes Gettier cases will be gerrymandered and so not itself worth satisfying, and further held that understanding is compatible with such luck and so needs no such condition. Against this, they describe a case in which a fabricating reporter's account of why a goal was scored happens to be true and the reader grasps the explanatory connection it describes; on their account the reader does not understand why the goal was scored, any more than the reader knows that it was, since such a reader would have grasped a different story just as readily. If so, understanding too requires a non-accidental relation between the grasping and the connection grasped, and by Kvanvig's own reasoning it would lose the value claimed for it. Because they already reject a premise of Kvanvig's argument about knowledge, they suggest taking this as a further reason to doubt that argument (DePaul and Grimm 2007, 510–513). DePaul and Grimm do not discuss artificial understanders, but applied to one, the requirement their case supports gives the shortcut problem of Section 3.3 a definite form: what an evaluation has to exclude includes a grasp that would have fastened just as readily on a different explanation had that one been supplied, so that the question is whether the system's account follows the explanatory connection itself or only the explanation it was given (compare the anti-luck conditions on knowledge in Chapter 2, Section 4).

Which of these positions is adopted changes what is required of a research system. If, as on Pritchard's later view and Kelp's compared above, understanding is what properly closes an explanation-seeking inquiry, or an inquiry into a phenomenon, a system that delivers validated results leaves such inquiries unfinished, whatever its record of knowledge, until the understanding they seek is realized in the bearer they target: the system itself, relevant members of the community for which it inquires, or, on accounts of collective understanding, that community (Section 6), and, where the inquiry is for *S* and that bearer lies outside *S*, until the understanding has also entered *S*'s epistemic situation on the adopted account (Sections 6 and 9). On Kelp's account, the understanding that closes such an inquiry need meet only a contextual threshold, short of the maximal understanding that is the ultimate aim of inquiry (Kelp 2021b, S1588). Under the working characterization, the activity can still count as research, and under the assumptions of the strong predictive case in Section 4 its results can count as knowledge. This does not make an inquiry that specifically sought explanatory understanding complete. Whether an activity is research and whether it has answered its particular question are different judgments; an aim-centered account adds a further claim about which achievement properly closes inquiries of that kind.

In whom is the value these accounts claim realized? Read with this question in mind, several of them locate it in an understander. Pritchard locates it in an achievement creditable to the understander's own cognitive ability (Pritchard 2008, §§10–11), and Zagzebski, as Gordon reports her view, ties the epistemic value of understanding to its transparency: one cannot, she holds, understand without understanding that one understands (Gordon n.d., §5a). On such accounts, the value is realized where someone understands, not merely where an explanation is available. It therefore attaches to an individual's understanding, which Section 6 distinguished from understanding credited to science in the sense of a communicable explanation. These accounts state their grounds for persons, through such things as the epistemic value of a life (Riggs), the satisfying closure that organized thinking gives inquiry (Kvanvig), and an achievement creditable to one's own cognitive ability (Pritchard). Applied to an artificial understander, which none of them discusses, whether its understanding has such value turns on whether those grounds hold of it, and the grounds raise different questions. Kvanvig's closure comes with a sense of having grasped the subject matter as a whole, and so meets the question Section 6 raised for Potochnik's felt sense of understanding; Pritchard's asks whether the understanding is the system's own achievement, which for artificial systems is disputed (Chapter 7, Section 3.3); and Riggs's, the epistemic value of a life, asks whether the system has a life to whose epistemic value its understanding could contribute. The question of whose understanding has such value has also been answered directly for research done with AI tools, by a claim about what basic science is for rather than by an account of what understanding is. Botvinick and Gershman, in their perspective in a joint article that collects several perspectives with responses, asking of basic science done with such tools "'whose' understanding matters?", answer that human understanding is to remain a core goal of science: "We cannot cede understanding to artificial systems." They would keep this goal, together with the choice of problems to work on, with human scientists even in a scenario in which technical limitations presented no barrier, while allowing that, because of limits on human cognition, AI systems may someday represent aspects of nature that humans cannot; even then, on their view, as much human insight as possible is to be extracted from such systems (Binz et al. 2025, 6–7). Their answer fixes the bearer: in basic science, they hold, the understanding whose value counts is that of humans, whatever an artificial system may represent. Read against Chapter 1, Section 2, where neither *A* nor

S settles who bears an achievement and, on Humphreys's account, science's being done for humans does not by itself require that what it comes to know lie within their current understanding, the answer can be read as adding to a human *S* a requirement on the bearer that does not follow from *S*'s being human.

Two further points matter for reading Sections 7.2–7.4, which follow. First, the claim that understanding is valuable for its own sake, or is an aim of inquiry, does not follow from a list of what understanding enables, and it is not refuted by showing that some of those effects can be obtained in other ways. Second, it follows that the relations traced in those sections are not offered as exhausting the value of understanding; on the positions above that ascribe final value to understanding, they do not. They record the part of it that becomes visible when one asks what a continuing research system would lose without it. The two kinds of value need not compete. Kvanvig's explanation of the value of understanding appeals both to the way organized thought supports reasoning toward action and to the satisfying closure it gives inquiry (Kvanvig 2003, 202). On ability accounts such as the one Hills favors, moreover, the grasp that understanding involves is itself a kind of ability or know-how, which, as Hills puts it for moral understanding, includes abilities to explain and to draw conclusions about new cases (Hills 2020, 12), so describing what the abilities discussed below make possible need not treat understanding as a mere means.

The first of these points can be seen in an account that treats the separation of understanding from what it enables as a historical change. Koichi Takahashi, writing about AI-driven science, holds that what joined science and technology in the modern period was understanding: laws of nature were understood, and that understanding was turned into technique. Takahashi holds that prediction which does not pass through understanding is emerging, citing structure prediction by AlphaFold2, deep-learning weather models, and the screening of candidate materials by AI, and that if prediction and control become possible without passing through understanding, the reason for the marriage of science and technology, in Takahashi's phrase, disappears and the two head toward a divorce. What is then lost, on this account, is not understanding itself but understanding as a means. The understanding that remains once how to reach results is handed to AI becomes an end in itself, as with the question why the universe began, which Takahashi calls technically useless but essential to human intellectual activity (Takahashi 2026, 184–186). Whose understanding such prediction bypasses is left unstated, although in the example from materials science the physical explanation is supplied afterwards by humans. Takahashi also allows that AI prediction and scientific understanding may form a new kind of coupling rather than simply separating, as when researchers take predicted structures as the starting point for new hypotheses and understanding (185). On Grimm's account, which traces much of the special value of understanding to the position it gives one to predict and control, the separation Takahashi describes would make much of that value obtainable without understanding. On the accounts above that ascribe final value to understanding, it would leave the value they ascribe in place, as the first point says. Takahashi's description of what remains, understanding that has become an end in itself, can be read along these lines, although it does not draw on these accounts.

7.2 Grasping how things would differ: transfer, intervention, diagnosis

The dependency accounts of Section 3.2 and the ability accounts of Section 3.3 disagree about what understanding consists in: the former place it in grasping a model of dependence relations, the latter in what a subject can do. This book nevertheless reads them as sharing a concern with one modal competence, one that concerns what would be or would have been the case and not only what is: grasping a structure of dependence well enough to tell how a target would differ if the factors on which it depends were different. The reason is that what each account names as the content or the signs of understanding involves this competence. The dependence relations that dependency accounts ask a subject to grasp are relations of how a target varies with the factors on which it depends. The abilities that ability accounts list, drawing conclusions about relevantly similar cases, answering what-if questions, and modifying a representation to support new inferences, all reach beyond the cases actually given (Sections 3.2 and 3.3).

The following descriptions bear this reading out. Explanations on Woodward's interventionist account, as Lauren Ross and Woodward present it, answer what-if-things-had-been-different questions, telling how the explanandum would have been different under changes in the factors cited (Ross and Woodward 2023, §3). Grimm describes the dependency maps that understanding takes up as mobile, and the understanding mind as able to anticipate how varying the value of one variable will lead, or fail to lead, to changes in the values of others (Grimm 2017). Hills's cognitive control includes drawing conclusions about relevantly similar cases (Hills 2020, 12), and Kuorikoski and Ylikoski take understanding to be

constituted by the ability to make correct inferences about what would have happened had things been different, what would happen if certain things were changed, and how a desired change could be brought about (Kuorikoski and Ylikoski 2015, §2). Le Bihan's modal view extends this beyond a single structure. On this view, understanding a phenomenon consists in knowing how to navigate a space of possible dependency structures that would give rise to it, and this is called modal understanding because it concerns how the phenomenon could arise in other possible worlds as well as how it arises in the actual one. Le Bihan's aim is to explain how theories and models known to misrepresent the world can still yield understanding of phenomena. The ideal gas model, on which molecules exert no forces on one another, and the van der Waals model, which adds attraction between molecules and their size, describe different possible dependency structures for the behaviour of gases. On this account, knowing how each structure gives rise to the behaviour it recovers, knowing how the structures relate to one another and where they differ, and knowing which constraints, such as the laws of thermodynamics, hold across all possible structures are three levels of such understanding (Le Bihan 2017).

Accounts disagree about the status of this competence: some treat it as constitutive of understanding, others as a manifestation or evidence of it, and some deny that the abilities involved are special to understanding (Grimm 2021, §2.2). Le Bihan, for instance, takes knowing how to navigate such a space to be constitutive of modal understanding, but counts the abilities to infer, transfer, and answer what-if questions as manifestations of it rather than as constitutive of it (Le Bihan 2017). Whether the competence constitutes understanding or only manifests it, a system that lacks it lacks one resource on which the situations below can draw, so this subsection can ask what a continuing research system would lose without understanding without first choosing among these accounts.

This subsection takes up three situations in which a research system that continues over time uses this competence: carrying what it has learned into a new context (transfer), deliberately changing something (intervention), and finding that an expectation has failed (diagnosis). Prediction under new conditions falls under transfer and intervention. It is not listed separately because the question of this subsection is what understanding adds to validated prediction: the strong predictive case of Section 4 already grants such prediction where no human member of the community understands why the target regularity holds, so prediction alone cannot show what would be lost without understanding.

The first use is transfer and generalization, in the sense in which what was learned from some cases continues to hold beyond them, close to the sense the word has in machine learning. A representation of relevant dependencies can indicate which regularities would survive changes in context. It can expose why a proxy worked and where it will fail. Understanding does not guarantee generalization, and robust empirical methods can transfer without it. It can nevertheless make the proposed basis of transfer available for criticism rather than leaving extrapolation to a single performance history. The accounts of Section 3 would explain differently why such transfer can evidence or follow from understanding. On a dependency account such as Dellsén's, transfer would rest on the dependence relations that the grasped model represents continuing to hold in the new context (Dellsén 2020). On de Regt's account, as he and Boge restate it, scientists who have constructed explanations that produce understanding of phenomena may go on to apply, extend, and refine their knowledge of the target systems using the same skills, so transfer is one such extension (Boge and de Regt 2025, 420). On Hills's ability account, cognitive control includes drawing conclusions about relevantly similar cases, so transfer is an exercise of abilities that partly make up understanding (Hills 2020, 12). Khalifa's reductive account, introduced in Section 3.1, does not deny that understanding and transfer go together; it explains the connection without positing abilities special to understanding. On this account, understanding why something is the case is a matter of knowing an explanation of it, and the abilities that understanding involves are, in Khalifa's phrase as Grimm reports it, "especially unspecial" (Grimm 2021, §2.2): they are those exercised in considering and comparing potential explanations and in explaining (Khalifa 2019, 10, 31). On this account, too, understanding improves with the amount of explanatory information grasped, so that someone who grasps both why something holds and how what explains it brings it about understands better than someone who grasps only the first (Khalifa 2019, 33).

It was said above that understanding can state the reason for expecting a result to hold in a new context, the proposed basis of transfer, in a form open to criticism, rather than leaving extrapolation to a past performance history. Woodward's account offers one way of stating such a reason: this dependence holds unchanged within a certain range of interventions and background circumstances, and the new context falls within that range. This account brings out at least three things about a claim of transfer stated in this way. First, the claim needs support for its extension beyond the original setting. On

Woodward's account, a causal generalization, in a sense different from the one above, namely a general claim that states how one variable depends causally on another, holds within a range of interventions and background circumstances, and whether it holds for other subjects and circumstances is a question of what the literature on experimental design calls external validity, analytically distinct from whether it held in the original experiment. As Woodward notes, however, the two assessments need not use wholly separate evidence: evidence of stability across contexts can also support a local causal claim (Woodward 2026, 5–6). Second, that support does not require knowing every boundary of the range. To use such a generalization to explain the behavior of a system, Woodward argues, one must know that the system falls within the generalization's domain of invariance, the range within which the dependence holds unchanged, though not the exact boundaries of that domain (Woodward 2000, 233–234). Third, whether a claim of transfer succeeds can depend on what is carried over. Li and colleagues report Woodward's distinction using structural equation models: quantitative claims can be population-specific even where qualitative claims about what can cause what hold more widely (Li et al. 2018, §2). A claim of transfer stated in this way can therefore be met with definite questions: does the new context really fall within the range, and can only the qualitative relationship be carried over to it, or the quantitative one as well? That is what it means for the basis of transfer to be open to criticism.

The second use is intervention and construction. Prediction can guide action, but intervention can require distinguishing causes, constraints, mechanisms, and manipulable variables. A predictor may fail when its inputs are themselves changed by policy or experiment. One example Woodward discusses is the Lucas critique in macroeconomics, which, on Woodward's account, turns on whether the generalizations on which economic models rely are invariant under changes in, for example, the incentives, relative prices, or information available to economic agents (Woodward 2018a, §4). Explanatory or causal understanding can reveal intervention points and possible side effects. Engineering understanding can similarly connect an artifact's behavior to what can be changed, repaired, or built differently.

Listing intervention as a use of the competence does not restrict the competence to causal dependence. Woodward's interventionist account, discussed in Section 4 and earlier in this section, ties explanatory information to manipulation: to explain is to show how an outcome would change if one intervened on the factors cited. But Woodward restricts that account to causal explanation, and allows that a derivation can answer what-if-things-had-been-different questions where the question cannot be read as one about an intervention, and may then be a noncausal explanation of some sort. One example discussed is the argument that the stability of planetary orbits depends mathematically on the dimensionality of space, since no stable planetary orbits are possible in a space of four or more dimensions; the derivation tells how the possibility of stable orbits would change as the dimensionality of space changed, but the idea of an intervention that changes the dimensionality of space is unhelpful, perhaps even unintelligible (Woodward 2018b, §§2–3). Woodward's account therefore does not imply that all understanding is causal, and neither does this subsection's use of it.

The third use is anomaly diagnosis and revision. When an experiment conflicts with a prediction, the outcome bears on a bundle of hypotheses, auxiliary assumptions, instruments, preprocessing choices, and environmental conditions. It does not identify its own target of revision. An organized account of dependencies can help investigators decide whether to revise the focal hypothesis, a measurement model, a boundary condition, or the test itself. This is one reason understanding can matter to self-correction even when a single result is warranted without it. Duhem-style underdetermination, discussed in Chapter 2, Section 5, is the point that an adverse result by itself does not settle which of these is to be revised; Duhem's thesis does not itself supply a procedure for locating the error, so locating it calls for a capacity of this kind. Elgin argues that although a failed expectation shows only that something in a constellation of commitments is wrong, the commitments are not all equally vulnerable: those independently supported by other well-confirmed constellations are less likely to be the locus of error, so the better supported the background, the more precisely a mistake can be located (Elgin n.d., 17–18). Among the abilities in Hills's cognitive control (Section 3.3), as Gordon lists them, two run in opposite directions: drawing the conclusion that p from the information that q, which runs from the explaining factor to what it explains, as prediction does, and giving q as the explanation when given p, which runs back from what is to be explained to what explains it (Gordon n.d., §3c). Diagnosis takes the second direction: it starts from a failed expectation and works back to what went wrong. And Wilkenfeld's examples of understanding include being able to notice and correct inconsistencies or mistakes in one's own representation (Wilkenfeld 2013, 10). Neither Hills's ability to work back from what is to be explained nor the ability Wilkenfeld describes to notice and correct mistakes guarantees that the revision chosen is the right one.

7.3 Connecting a domain: retention, reuse, and further questions

Where Section 7.2 drew on accounts that focus on grasping a structure of dependence, another strand of the literature locates understanding in a grasp of how items of information hang together. Kvanvig, as seen in Section 7.1 and as Grimm reports, takes understanding to require grasping the explanatory and conceptual connections between the pieces of information involved in a subject matter, combined into a unified body (Grimm 2021, §§2.1, 2.3). Elgin characterizes an understanding as an epistemic commitment to a comprehensive, systematically linked body of information that is grounded in fact, responsive to evidence, and enables nontrivial inference, argument, and perhaps action regarding its topic (Elgin 2018, 8). Dellsén's account, which concerns understanding a phenomenon rather than understanding why, locates understanding in grasping a model of the dependence relations in which a phenomenon stands. A model here is an information structure interpreted so as to represent its target, as the Lotka–Volterra equations become a model of predation once their variables are read as the sizes of predator and prey populations. The model understanding requires represents what the phenomenon depends on, how, and what it does not depend on. Such a model, Dellsén argues, can fail in two ways: by misrepresenting a dependence or by not representing it at all. The account therefore measures a model by two criteria, accuracy, which asks whether it misrepresents, and comprehensiveness, which asks whether it represents what it should. Each is met more or less rather than simply met or not, so understanding too is a matter of degree: one understands a phenomenon to the degree that one grasps an accurate and comprehensive model of its dependence relations. Comprehensiveness also covers what does not depend on what, just as a causal graph shows not only which elements cause which but also which are not causally connected. Learning that two factors that seemed related are in fact independent therefore adds to understanding, as when Galileo's thought experiment shows that an object's acceleration in free fall does not depend on its mass (Dellsén 2020, §§3, 7).

How does such connecting relate to the competence of Section 7.2, the competence to tell how a target would differ if what it depends on were different? It could be a further use of that competence, or a distinct achievement that supports it, and the accounts point in different directions. Ylikoski and Kuorikoski give a reason to link the two. They argue that an explanation well integrated into a larger body of theory yields more understanding than the two bodies of knowledge taken separately, because integration widens the range of what-if questions that can be answered, in two ways. First, through inferential connections to the existing theory, dependencies between factors in that theory and aspects of the phenomenon can open dimensions, not foreseen before, along which what-if questions about the phenomenon can be answered. Second, the explanation can bridge gaps within the existing theory, so that new what-if questions about other phenomena can be answered. Integration in their sense is therefore more than formal consistency or conceptual coherence: it must allow new relevant inferences about the phenomenon (Ylikoski and Kuorikoski 2010, §4.4). That connecting widens the range of answerable questions does not by itself settle which of the two it is, however, since a distinct achievement could have the same effect. On accounts that, like Kvanvig's, count coherence-making relations other than explanatory ones, or that, like Kelp's, let knowledge that is not explanatory add to understanding (Section 7.1), what is grasped in connecting a domain can include relations other than dependence. On such accounts, connecting may be an achievement that is not exhausted by the competence to grasp dependence.

This subsection takes up two relations, under such connecting, that matter to research continuing over time: retention and reuse, and opening further questions. The first is retention and reuse: keeping what has been obtained over time, and using and extending it in later inquiry. Kvanvig holds that grasping explanatory relations organizes and systematizes thinking about a subject matter in a way that goes beyond adding further true beliefs. Such organization is useful, on this view, because it lets one reason from one piece of information to other related information that can serve as a basis for action, whereas unorganized thinking provides no such basis for inference (Kvanvig 2003, 202); this bears on reuse. Among the abilities of Hills's cognitive control, as Gordon lists them, is an ability to draw from the information that q' the conclusion that p', or that probably p' (Gordon n.d., §3c); this too bears on reuse. Taken together, on these accounts an organized body of results can be extended to new cases by inference from what it contains, whereas a collection of isolated answers offers no inferential links among its answers, so an answer has to be obtained afresh for each new case. Retention has a precedent in the *Meno*, where Socrates compares true opinions to the statues of Daedalus, which run away unless they are fastened: true opinions are of little value until they are "fastened by the tie of the cause", and once bound they are abiding, which is why knowledge is more honorable than true opinion (Plato, *Meno* 97e–98a). Grimm reports that a case has been made that understanding is a better translation than knowledge of Plato's and Aristotle's *episteme* (Section 7.1); read in that way, the

passage ties being kept to grasping the cause. Wilkenfeld, on whose account understanding is a matter of compressing information about the understood so that it can be used, gives a modern case: a student who had memorized a soundness proof would have been lost had he forgotten one step, whereas a logician who understands the proof can reconstruct it from simpler representations (Wilkenfeld 2019, 7). Educational research states a parallel claim. Novak and Cañas report that knowledge learned by rote, with little integration into existing knowledge, tends to be quickly forgotten unless much rehearsed, whereas meaningful learning supports the retention of knowledge over long periods (Novak and Cañas 2008, 6–7). These points bear on retention: they suggest that connections to grounds, or an organization that permits reconstruction, can help preserve or recover what would otherwise be lost.

The second is opening further questions. Understanding can reorganize a domain and reveal new contrasts, anomalies, and interventions. A predictive system can also generate new questions by discovering patterns or failures. The difference is not that only understanding creates novelty. It is that a representation of relations can make the space of possible questions navigable in ways an isolated answer does not; Ylikoski and Kuorikoski's account of integration states one version of this claim. Elgin holds that an understander should be able to use what she understands, perhaps as a source of working hypotheses about related matters (Elgin 2007, 5), and that reconfiguring a domain, as a metaphor does, equips us to raise new questions that could not otherwise have been framed or motivated (Elgin 2002, 5, 9, 12). The relation also runs the other way. As noted in Section 7.1, Kvanvig says that organized understanding gives inquiry satisfying closure, and Pritchard holds that an inquiry seeking an explanation is not legitimately closed until the understanding it seeks is reached (Kvanvig 2003, 202; Pritchard 2016, 6–7). On their accounts, understanding closes inquiry. This seems to sit uneasily with the claim of this subsection that understanding opens further questions. The two claims can, however, be read as concerning questions of different scope: what understanding closes is the question that prompted an inquiry, and what it opens are new questions beyond it. Grasping, through Galileo's thought experiment, that an object's acceleration in free fall does not depend on its mass closes the question whether heavier objects fall faster, and at the same time raises the question of what that acceleration does depend on.

7.4 Making understanding available to others: communication, criticism, and memory

An explanation can compress relations into a form others can examine, teach, contest, and connect to existing knowledge. Articulation may expose an ambiguity that remained invisible while a practice went unarticulated. It can also mislead: standardized explanations may create an illusion of shared grasp, while some expertise remains partly tacit. Whether understanding itself can be passed on is disputed. One tradition, which Grimm reconstructs in order to dispute it, holds that grasping how things are connected is something each person does at first hand: one can be told that a connection holds without thereby grasping it oneself (Grimm 2019, §II); Hills argues that understanding why involves abilities that testimony alone does not normally pass on (Hills 2020, 5, 12–13). Others argue that testimony can bring understanding when the recipient already has the background needed to take it up; Section 8 sets out these positions, together with a distinction of Hills's between ways of learning from others, and what they imply for a research system that relies on artificial outputs. De Regt treats scientific understanding as a community achievement and holds that a recipient acquires understanding from another's explanation by following and reproducing its line of reasoning, in more or less detail and with a correspondingly varying level of understanding, and concludes that the intelligibility underlying the explainer's understanding appears to be a requirement for the recipient's as well (de Regt 2017, 91, 99–100). Kuorikoski and Ylikoski, who regard scientific understanding as essentially collective, argue that external representations such as formal models can improve it partly by making assumptions explicit and so easier to evaluate (Kuorikoski and Ylikoski 2015, §§2–3). What matters, then, is whether recipients actually come to have the grasp that understanding requires, not whether a document shaped like an explanation was produced. Boge, Schuster, and Stoll, whose distinction Section 6 discussed, make explanation central here: on their account, the explanation is the medium of scientific understanding, and it can be inspected in a canonical, communicable way (Boge, Schuster, and Stoll 2026, 5–6, 11).

Criticism and memory depend on this availability. A representation that others can follow can be checked by someone other than its producer, and a record that others can follow can be returned to after those who produced it have moved on. Of the reasons given in this subsection why making understanding available to others supports criticism and memory, two rest on features of human research: that those who produce a result move on, and, on de Regt's account, that a recipient understands

by following and reproducing the reasoning. Where *S* is human, the need for a representation its human members can follow remains whatever produced the result. Where the producing system persists and the recipients are artificial members of *S*, checking that does not share the producer's ways of going wrong is still needed, as a matter of independence (Chapter 11, Section 2), while what availability requires changes form (Section 5.4). Hills adds a limit, argued for moral judgement: someone who takes an expert's word is responsive to the reasons for taking that word rather than to the reasons that make the judgement right, and taking the word does not give them the ability to draw different conclusions when the situation differs (Hills 2020, 12–13). If Hills's account is adopted, a community that acts on an artificial system's outputs without that ability would be guided by the system rather than by the dependencies the system tracks.

Sections 7.2–7.4 thereby bring into view an answer to the question raised at the start of this section, what would be lost beyond any single validated result if no one understood: the competence to state a basis for transfer open to criticism, to intervene, and to diagnose failures; the retention and reuse of results and the opening of further questions; and the availability of results to others for criticism and memory, unless these are supplied by other means. These are candidate relations, not a definition of understanding, and this section does not show that every research act realizes them. Some can be supplied by modular verification, redundant measurements, robust search, specialized agents, or institutional organization. A simple explanation may itself fail under shift. In evaluating a research system, the question they raise is which capacity a given purpose requires and whether the system has it; relabeling every desirable capacity as understanding would not answer that question. Nor would supplying these capacities by other means settle whether understanding is valuable for its own sake or an aim of inquiry (Section 7.1).

This yields the following conclusion: on the working characterization, which epistemic achievement research aims at is left open, yet particular inquiries and sustained research systems may require capacities associated with understanding.

> If the working characterization is adopted, which epistemic achievement research aims at, knowledge or understanding, is not fixed by the characterization but left open to particular attempts, and attaining it is not made a condition of each research act. Whether understanding is required depends on the achievement an attempt seeks. Positions that value understanding for its own sake or take it as an aim of inquiry can therefore be accommodated by taking understanding as the achievement aimed at, and on them what is required is set by that value or aim. Particular inquiries and durable autonomous research systems may nevertheless require capacities that important theories associate with understanding, even where understanding is not what they aim at, and what those capacities have to serve differs with whether *S* is human or artificial (Section 7.4).

This conclusion may also give a reason to look again at the working characterization itself. The capacities said to be needed may turn out to be obtainable without understanding, or the characterization may not fully capture the place understanding has in research (Introduction, Section 4).

Whether those capacities jointly constitute understanding remains a philosophical question. Whether a system actually possesses them is an empirical and engineering question. The two questions are distinct, but each informs the other.

Why a local achievement does not settle continuing inquiry capacity

The preceding sections asked about understanding mostly with a single result or a single inquiry as the unit, which this section calls local: whether a result can be warranted without understanding, and whether understanding is needed to close a given inquiry. This section changes what is assessed and over what span. What is assessed is no longer one result but the system or community that repeatedly produces and uses such results, and the span is no longer one inquiry but inquiry as it continues over time. The change matters because verdicts at the two levels can come apart in either direction. A particular opaque result can be warranted while the system that repeatedly produces and uses such results gradually loses important epistemic capacities. Conversely, an opaque component can belong to a robust system whose other agents, instruments, records, and institutions supply reliable checks and repair, so that the component's opacity does not become a problem for the system.

Messeri and Crockett describe one form of the risk that a system loses epistemic capacities as *illusions of understanding*, in which scientists believe they understand more about the world than they actually do. On their account, because AI tools are expected to make research more productive and more objective, scientists are apt to trust them as collaborators, much as researchers rely on one another's expertise, and someone who relies on understanding located elsewhere can mistake it for

their own. Moreover, the simple, quantitative answers such tools give tend to produce a sense of understanding that need not match actual understanding. The tools' efficiency can also foster monocultures in which questions and methods suited to AI, and the standpoints AI can represent, crowd out other ways of proceeding, which would leave science more vulnerable to error (Messeri and Crockett 2024, 49–53). Their analysis is a risk framework rather than evidence that every AI-supported field already exhibits these effects. It nevertheless supplies a direct reason to evaluate continuing epistemic capacity separately from local output volume. Neither risk shows up in the quality of any single result: the first lies in what those who rely on such tools take themselves to understand, and the second emerges as reliance extends across a field and over time, so together they concern both what is assessed and over what span. An observational study of papers in six natural sciences reports a pattern consistent with the risk of monoculture, without establishing it as an effect of the use of AI methods, and Chapter 11, Section 8 discusses it (Hao et al. 2026); it does not bear on the risk of overestimating understanding. Messeri and Crockett write about human scientists. Of the two risks, monoculture concerns how inquiry is organized and can arise for any *S* whose members come to share questions, methods, and ways of going wrong; the illusion of understanding, as they describe it, turns on features of human cognition, trust in tools modelled on reliance on colleagues and a sense of understanding that need not match understanding, so that whether an artificial member of *S* is open to a counterpart of it depends on whether anything in it plays the role of that sense (compare Section 6).

The change in what is assessed can be seen in an ordinary feature of contemporary research. Contemporary research projects can depend on divisions of cognitive labor in which participants rely on instruments, software, procedures, or specialist claims they do not each understand in full. As Chapter 2, Section 6 discussed, even where cognition is described as distributed across such a project, as on Giere's account, which Brown reconstructs, that does not by itself settle whether the organized system also knows (M. J. Brown 2015), and collective production does not by itself settle who possesses the resulting knowledge. Such divisions of labor rule out a demand that every participant understand everything in full. They do not show that any additional opacity is harmless. What they leave open is how dependence is organized, a question asked of the organization of the project as a whole rather than of any single result: where reliability is established, how correlated failures are detected, who can answer challenges, and how reliance changes when conditions fail.

Evaluation therefore divides into two levels: the level of a single contribution, which the preceding sections mostly addressed, and the level of a continuing system, which this section addresses. At the contribution level, we ask what question was addressed and whether the resulting claim or other achievement is true or adequate, warranted, or explanatory, or amounts to understanding, in the claimed sense. At the continuing-system level, we ask whether its organization of research preserves the capacities for diverse search, independent criticism, memory, dependency tracking, correction, further question formation, and the reproduction of the competence with which members of *S* do and evaluate research, among others. A collection of locally warranted outputs does not by itself answer the question at the level of the continuing system. The two evaluations can come apart in either direction, so neither can be inferred from the other: a result can be warranted while the organization of research loses capacities, and capacities can be preserved while a particular result fails. For example, a laboratory that comes to rely on a predictor whose outputs it checks case by case may accumulate warranted results while losing the ability to diagnose anomalies without the predictor (the loss to which Humphreys's further worry in Section 4 points), and if neighboring groups rely on the same model, a failure common to all of them could escape the checks that each group applies to its own results. The first loss appears only when the laboratory is followed over time, the second only when the assessment extends beyond one group.

The last capacity in the list for the continuing system, that for reproducing the competence with which members of *S* do and evaluate research, can be lost in a way that appears only across cohorts of researchers. Castellacci and coauthors, in a preprint that grew out of a roundtable on generative AI in research, call this capacity depletion: it arises where the tasks AI absorbs are the same tasks through which research capability is acquired, so that AI may, on their account, democratize access to the artifacts of research without democratizing the capability to produce them. They present it as a risk of present adoption, distinct from the information asymmetry that undisclosed AI use creates for evaluators and from negative externalities on a shared knowledge base, and they read the studies of learning they cite as pointing to the mode of engagement with a tool, rather than its use as such, as what matters (Castellacci et al. 2026, §§1, 2.4). Bender, Bergstrom, and West, in a response in the joint article by Binz and colleagues discussed in Section 7.1, make a related point: science produces the scientists themselves as well as the results published in papers (Binz et al. 2025, 8). Both write about human researchers; where *S* is mixed or artificial, the corresponding question is whether the members on whose judgments *S*'s

evaluation depends continue to be formed with the competence to make them. A count of warranted results registers this loss no better than the two above. Several later arguments presuppose the competence: direct calibration of a source requires independent opinions on some of its claims (Chapter 8, Section 3.4), if artificial configurations that carry out research come to gather and assess more evidence than any member of *S* across many fields, *S* as a whole can come to stand to them as a layperson, unable to check their judgments and able only to rely on them (Chapter 11, Section 6), and contestation requires parties with the competence to contest (Chapter 12, Section 7).

Whether members of such a division of labor gain understanding from others' outputs bears on the continuing-system level, and here accounts of testimony and understanding disagree. Hills's account of understanding bears on the division of cognitive labor described above. Hills distinguishes two ways of learning from what others say. In transmission, the hearer takes the speaker's word as the reason for belief; this permits a massive division of epistemic labor, but what results is second-hand knowledge rather than understanding, since understanding why requires cognitive control, the ability to draw conclusions and give explanations about new cases, which, like most abilities, comes from practice. In propagation, what is said prompts and frames the hearer's own inquiry; this is a division of labor of a different kind, and through it understanding can be passed on. The argument concerns moral understanding, but Hills presents the distinction as general (Hills 2020, 2–5, 8, 10–13, 18). If this account is adopted, and if reliance on artificial outputs is testimony, which differs across accounts (Chapter 8, Section 3.4; Chapter 11, Section 5.1), and takes the form of transmission, members of a research system who rely on those outputs could come to know more without coming to understand more, and a count of warranted results would not register the difference.

Hills does not treat this as a reason to pursue understanding on every occasion. On her account of moral testimony, the reasons to favor propagation over transmission hold whenever propagation is available, that is, when the learner has the time and ability to develop her own understanding; and even a morally ideal agent will take others' word on other matters, such as experiences she has not shared, that can supply vital non-moral information (Hills 2020, 14).

Others dispute the asymmetry on which Hills's conclusion rests, that knowledge can be passed on by testimony while understanding cannot easily be. Boyd argues that understanding can be acquired directly from testimony when what is said is easy for the recipient to understand, including when it is easy only for an expert, because the recipient can fit the new information into relations already grasped (Boyd 2017, 23–34). Grimm argues that, as with propositional knowledge, passing on understanding by testimony is usually easy when the recipient has been suitably prepared and usually difficult, perhaps impossible, when not, although the work the recipient does differs in kind, since for understanding it is grasping connections (Grimm 2019, §§IV–VI). Hu argues, as Grimm reports in his encyclopedia entry on understanding, that knowing that one thing caused another already brings a rudimentary understanding of why, so that in whatever sense knowledge can be passed on by testimony, such understanding can be too (Grimm 2021, §3.2). Malfatti sorts the arguments for the claim that testimony cannot transmit understanding into three, and argues that each can be resisted. First, gaining knowledge from testimony can also require exercising one's own understanding of the subject to interpret what is said, so a difference in the effort required cannot separate the two. Second, transmission does not always require fully deferring to the speaker. Third, the abilities that grasping is said to involve can sometimes be acquired by being told (Malfatti 2020, 56–70).

Brought to bear on a count of warranted results, these positions agree that it does not by itself show what understanding members have gained, but they locate what it misses differently. If Hills's account is adopted, it misses whether members take artificial outputs as grounds for belief or as prompts for inquiry they then carry out themselves. If Boyd's or Grimm's account is adopted, it misses whether members keep the background through which testimony can bring understanding at all, a background that, in expert cases, Boyd describes as the product of the recipients' own earlier work (Boyd 2017, 38–39). If Hu's account is adopted, it misses the degree of the understanding gained rather than its presence (Hu 2021, 5–7). What members gain or fail to gain in this way also bears on the level of the continuing system, since it affects what the system can still do later: whether members can diagnose failures without the outputs, for example, and whether they keep the background through which later testimony can bring understanding. These positions concern learning from what others say or produce. Where a representation or ability can instead be copied or shared directly among artificial members, whether the recipient thereby understands turns on further features of the account adopted: on ability accounts (Section 3.3), what matters may be whether the recipient now has the relevant abilities, whereas on accounts on which understanding typically involves an achievement creditable to the understander's own cognitive ability, such as Pritchard's (Section 7.1), whether a copied

state is the recipient's own achievement is a further question.

Evaluation at the continuing-system level is not another condition silently added to every research act. Some projects will be valuable despite contributing little to system-level capacity, and some institutions will remain fragile while producing many valuable local results. Part V takes up the problems that arise at this level when the relations surrounding inquiry, now carried in part by parties outside the research, can come to be carried by the artificial configurations that carry out the research: allocation, independent criticism and trust, and authority. The present chapter needs only the narrower conclusion: understanding can matter to the health of an inquiry over time even when it is not required for the warrant of each local result.

What accounts of understanding require of artificial research

This section applies the chapter's distinctions to situations in which artificial systems conduct research. The consequences are not uniform. They change with four things: which achievement is claimed (predictive performance, knowledge, understanding, and so on), in whom it is claimed to be realized (the system itself, its users, an organized whole, an artificial community), which account of that achievement is adopted, and what it is to be used for. Using constructed cases, the section takes up in turn a predictive case set against a formally checked proof, requests for transparency and the functions they serve, the evidence that bears on a claim that a system understands, what long-run autonomous research is assessed for, and the distinction between the reference society *S* and the bearer of an achievement. The aim is to apply the accounts introduced in Sections 3 to 6 and show how the consequences differ with the claim, the account, and the use.

First, two cases in each of which an achievement is clearly attained can be contrasted. One is the strong predictive case of Section 4. A system reliably predicts a property within a stated range, but its human users cannot explain the mechanism that produces the property. Its reliability is established by evidence not merely inferred from agreement among its own outputs. In this case the system can be credited with scoped predictive performance, and on accounts that permit knowledge without explanatory understanding the users can also acquire knowledge (Section 4). Neither, however, shows that anyone understands why the property holds.

Who, if anyone, understands why the property holds receives different verdicts on different accounts. On explanation-based accounts, the users do not. On Khalifa's reductive account, if someone later comes to know the explanation, that knowledge is the understanding; an attempt at understanding then becomes an attempt to acquire knowledge of the explanation, and who holds that knowledge, and whether it has entered the epistemic situation of *S*, are the questions Chapter 2 asked about knowledge (Section 3). On de Regt's criterion, the regularity remains scientifically ununderstood unless there is an explanation based on an intelligible theory, and by the argument of de Regt and Dieks about an oracle that produces perfect predictions, this predictive success does not count as a great scientific success either. If the predictor is a theory in de Regt's sense, however, its successful use presupposes that its users understand that theory; even then, understanding why the regularity holds still requires, on this account, an explanation based on the theory (Section 4). On Woodward's account, the predictor explains to the extent that it also supports correct answers about what would happen under interventions and that this information is accessible to its users (Section 4).

By contrast, on Dellsén's earlier formulation, grasping how to predict correctly across varied circumstances counts as partial understanding even without an explanation, and on Beisbart's account a scientist who, by running the predictor repeatedly, has learned to anticipate how its outputs change with its inputs has some objectual understanding. If the scientist has to run the predictor itself to draw those inferences, however, Beisbart's account assigns the understanding to the coupled system of human and computer, not to the unaided human (Section 4). On Gijsbers's view of the understanding that classification gives, the question becomes, further, whether the features that support the predictor's classifications are available in a graspable form (Section 3.2). Where an artificial system conducts the research and *S* is a human society, as here, whether the system enabled human understanding thus depends on the account. On explanation-based accounts it did not. On Dellsén's earlier formulation and Beisbart's account, it did to the extent that the users themselves can anticipate the predictor's behavior across varied circumstances.

Contrast a generated proof accepted by a sound formal checker, a program that verifies mechanically that each step of a proof follows the rules. Here the route by which the proof was generated may remain opaque, because what the check

establishes, that the formally stated claim has been proved, does not weaken when that route is hidden. Two things are not thereby settled. One is whether the formal statement correctly expresses the question originally asked. The other is whether those who hold the proof grasp its explanatory dependencies, that is, why the theorem holds (Sections 3 and 5.1; Chapter 6, Section 3).

The Liquid Tensor Experiment shows what the first point involves in practice. Scholze found the formal definitions and theorem statements surprisingly readable but could not read the proofs at all, and the formalizers added formally proved results about the objects involved as evidence that the checked statement concerned the intended objects (Section 5.1). Judging the first point thus requires being able to read the statement and the results that support reading it as intended, not the proof. Where an artificial system generates the proof and *S* is a human society, this judgment can remain with humans even when the proof is unreadable to them, provided the statement is readable to them (Section 5.1).

On the second point, understanding, the verdicts differ by account. Hills's account requires, for understanding why the theorem holds, the abilities to follow the explanation, to put it in one's own words, and to draw conclusions in relevantly similar cases, so human users who merely hold a proof no human can survey may know the theorem but do not understand in this sense. Scholze's report shows, on the other hand, that people who take part in a formalization can come to understand why a proof's route works, although in Scholze's case without a feel for the surrounding terrain (Section 4). For whether the generating system itself understands the proof, Wilkenfeld's account supplies a concrete test. Unlike a merely memorized proof, an understood proof can be adapted to a slightly different system, so the evidence on this account is whether the system can itself adapt the proof, without anyone rewriting its internal states from outside (Sections 3.3 and 6).

Each case thus contains an achievement to credit, scoped predictive performance in the first and the proof of a formal statement in the second. The cases differ in whose abilities are tested for understanding beyond that, and how. In the predictive case, the question is whether the users themselves can anticipate the predictor's behavior or must keep running it to do so. In the proof case, the questions are whether human users can follow the proof's explanation and whether the system can itself adapt the proof.

Second, a request for transparency alone does not say what is being asked for. A request for transparency becomes informative when it states what the transparency is for, and different functions are served by different things. A process trace, recording which steps were carried out, supports reproduction and debugging. An evidential record supports assessment of how far a result is warranted. A model of the target may support explanation. A statement of scope and uncertainty supports judging how far to rely on a result. A presentation suited to its audience supports communication. A single generated rationale therefore cannot stand in for all of them.

Even for the function of explanation, what serves differs with the account. On Sullivan's account, for a model that, like the deep networks Sullivan discusses, is not opaque at the highest level, what chiefly limits the understanding it can give is not the opacity inside it but link uncertainty, the lack of evidence connecting the model to the phenomenon. What serves understanding of the phenomenon is then less an internal record than evidence from outside the model supporting that link, and the greater the harm of error, the higher the standard both for that evidence and for checking that dependencies extracted from the model are faithful to how it works (Section 5.2). On Beisbart's account, what is missing is an informative description of what the network does in terms of the variables and processes of the target system, and however detailed a record of the computation's execution, it does not by itself supply this (Section 5.2). On Fleisher's account, a post hoc explanatory model that computes a different function from the classifier can still contribute to the understanding of why the classifier reached its verdict, for its developers, its operators, and those affected, if its departures bear the marks of idealization and it agrees well with the original model near the input explained (Section 5.4).

Where the model is a deep-learning model, rendering its internal reasoning in human concepts may face a further limit. As Section 5.4 discussed, Otsuka compares interpreting the internal representations of a deep-learning model with radical translation, translating an unknown language from its speakers' behavior alone. If that comparison is right, a rendering in human concepts may be only one of several interpretations that fit equally well, and however fully such a rendering is recorded, the record could not settle whether it is the correct translation, since there may be no single correct one. How much this limit matters depends on who is to understand what. For a human *S*, understanding has to be brought about in a form its human members can use; but, as Section 5.4 noted, where understanding of the phenomenon is sought, the route through a

model of the phenomenon need not wait for the rendering of the internal reasoning to be settled. Where the recipient is an artificial system able to work with the model's representations directly, rendering them in human concepts may not be needed at all.

What a record needs to contain also changes where models are idealized or purpose-relative (Section 3.4). For such models, accuracy as a representation and adequacy for a particular prediction, explanation, intervention, or pedagogical use come apart, so a record that is to support assessment for a particular use needs to distinguish the two. What it has to distinguish also becomes specific on particular accounts (Section 3.4). On Potochnik's account, each posit's departure from the truth must be negligible given its role in the representation and the aims of the research, so the record needs to say what was idealized and for which aim. On Strevens's account, an explanation that contains idealizations is correct when its claims about what makes a difference and what does not are true, so the record needs to state which factors were left out as making no difference. Applying Weisberg's taxonomy, a system with far greater computational capacity than its human counterparts has reason to undo Galilean idealizations, introduced for tractability, but no such reason to undo minimalist or multiple-models idealizations. Greater computational capacity is therefore not a reason to undo every idealization. Undoing a minimalist idealization would obscure the explanation obtained by keeping only the factors that make a difference, and merging multiple-models idealizations into one model would obscure the trade-offs among desiderata such as generality, accuracy, and simplicity. Extending the positions of Section 3.4 this far, if the record is also to show what the evidence still leaves open, competing models that remain compatible with the evidence would remain in it even after one has been selected for the current task.

Third, what evidence bears on a claim that a system understands can be specified only once the claim identifies who understands, its bearer, and which account of understanding it adopts. The following, for example, are different hypotheses, to be checked in different ways: the AI enabled human understanding, the AI enabled another artificial system to understand, the AI understands, the organized human–AI system understands, and the artificial community understands. The bearer named in such a claim need not be the society for which the understanding is available or new (see the final item). Performance on transfer, intervention, counterfactual, scope-recognition, or anomaly-diagnosis tasks can provide evidence. But, as Section 3.3 noted, performance can also be reached through shortcuts such as memorization or retrieval, so it is evidence of the claimed abilities only to the extent that the evaluation discriminates them from the shortcuts that matter on the account the claim relies on.

Which performance counts as evidence, and which shortcuts matter, can be worked out account by account (Sections 3.3 and 6). Here the case considered is the one in which the system itself is named as the bearer; the verdicts on a claim that the system enabled human understanding were given for the predictive case, and those on a claim that an organized whole understands are given under long-run autonomous research below. On Hills's account, the tests are whether the system can follow explanations others give, give explanations itself, and draw conclusions and give explanations in relevantly similar cases. On Wilkenfeld's account, the test is whether the system can itself modify its representations so as to enable new valid inferences, without its internal states being rewritten from outside; lucky performance and the application of a memorized procedure are excluded from understanding on that account from the start. Barman and colleagues propose testing the degree of understanding by how widely and correctly the subject answers what-, why-, and what-if questions. On Moody and de Regt's account, the question is whether the system can recognize qualitatively characteristic consequences of a theory without exact calculation, and they credit current language models with a fragile but nontrivial degree of understanding on the indirect evidence of accurate answers to a wide range of out-of-distribution scientific questions. On these accounts, performance can support a claim of understanding if the evaluation discriminates the shortcuts. On Potochnik's account, by contrast, understanding requires the sense of understanding that a representation induces, so however good the performance, the question whether that sense is present remains. Accounts on which acceptance rather than belief suffices for understanding also differ in what they ask. On Dellsén's account, understanding requires acceptance and enough mastery of the subject to explain it correctly, but not belief; if belief is a disposition to feel that p is true, understanding is then closer than knowledge for a system to which feeling cannot be attributed, though whether using p as a premise in explanation amounts to adopting a policy in Dellsén's sense remains a question. On the functional and dispositional accounts of belief, by contrast, acceptance put to work in explanation may itself count as evidence of belief, so that this route is not separate from belief. On Elgin's account, the acceptance that understanding requires is voluntary and undertaken for epistemic ends, so the question becomes whether the system has epistemic ends and accepts in that sense (Section 6).

Fourth, what long-run autonomous research is assessed for depends on how research is characterized. On characterizations that place understanding at the center of research, all such research is assessed for understanding, and the account adopted determines what that involves, including whose understanding counts: the system's, that of its recipients, or that of a collective (Section 6). Suppose, for example, that research is characterized by an account that places understanding at the center of science and its progress (Sections 4 and 7.1). On Dellsén's noetic account, progress lies in publicly available information that enables the relevant members of a society to increase their understanding (Sections 4 and 6). Assessed this way, a system's acquiring a good model of dependencies is not enough by itself; the question is whether the information has become available to the members of *S*, and, if *S* is a human society, available in a form its human members can use. Assessed by de Regt's account, the question is whether the research has given explanations based on intelligible theories, and research that only produces perfect predictions does not count as a great scientific success (Section 4). On the working characterization, by contrast, which epistemic achievement research aims at is left open (Section 7). Long-run autonomous research is then assessed by the achievement its attempts aim at, and Dellsén's and de Regt's criteria apply as criteria of success only to attempts that aim at understanding.

Separately, such research can be assessed for the capacities that its autonomy itself requires. The more of an inquiry is left to the system, the more capacities the system itself has to supply (Chapter 7, Section 4.4). Can it recognize when a regularity no longer transfers? Can it distinguish a failure of the focal claim from a failure of an instrument or auxiliary assumption (Chapter 6, Sections 2 and 4.2)? Can it generate and compare alternative representations (Chapter 5, Section 3)? Can it expose its evidence to criticism and propagate corrections (Chapter 11, Sections 2 and 8)? Can it revise the standards by which it assesses results without becoming insulated from the world (Chapter 6, Section 5; Chapter 7, Section 4.3)? Some accounts tie some of these capacities to understanding itself. Recognizing that a regularity no longer transfers is, on Woodward's account, recognizing that a new context falls outside the domain of invariance of a dependency, or that only its qualitative relationship carries over (Section 7.2); generating alternative representations is close to the modification of representations that, on Wilkenfeld's account, manifests understanding (Section 3.3). Such accounts may therefore interpret an integrated realization of these capacities as understanding. Whether the capacities are concentrated in one system or distributed across agents and institutions is, however, a separate question from whether their realization constitutes understanding. Depending on the account, either arrangement may supply them without constituting understanding (Section 7.4). For a distributed arrangement, the consequences differ by account. Extending Beisbart's account, seen in the predictive case, where a human draws inferences by running the system, the understanding belongs to the coupled whole of human and system, not to the unaided human (Section 4). On accounts that ground group understanding in members' understanding, the arrangement has yielded understanding only when some member, human or artificial, understands. On an account that extends Bird's group-functional account of knowledge to understanding, the organized whole can be the bearer even if no member holds the whole (Section 6).

Finally, the reference society *S*, relative to which an achievement is new, is distinct from the achievement's bearer, the subject in whom it is realized; the two coincide in a claim that *S* itself understands. Because they are distinct, a member's new understanding, collective understanding, and a newly available explanation are different claims, and which of them amounts to entry into the epistemic situation of *S* differs by account (Section 6). On accounts that ground group understanding in members' understanding, it is the relevant members' understanding; on an account that extends Bird's to understanding, it is the society's organization realizing the understanding as a whole. On the account of Boge, Schuster, and Stoll and on Dellsén's noetic account, what is asked is whether a communicable explanation or public information has become available; whether such availability already amounts to entry into the epistemic situation or only puts the society in a position to understand remains open, as it does for knowledge (Chapter 2, Section 7). Where an artificial system conducts the research and *S* is a human society, an artificial system's understanding, if it understands, could be new to that society without the society itself understanding. Whether this describes the intended research depends on the claim and the activity, not on choosing *S* to suit. Describing such work as research for that society also presupposes that the society's epistemic situation could come to include understanding of the kind sought, for instance through an explanation its members can follow (Chapter 1, Section 2). For a human *S*, this presupposition comes into question where the organization of dependencies that an artificial system grasps cannot be made intelligible to humans (Section 6); for a mixed or artificial *S*, it may be met through artificial members to whom that organization can be communicated. Applying the distinction of Boge, Schuster, and Stoll to the human case, an artificial system's understanding would correspond to what they call scientists' understanding and

would not yet be scientific understanding, which on their account requires an explanation communicable to a sufficiently broad scientific audience. In terms of their example, the system would stand where Feynman stood before Dyson's proofs gave other physicists an explanation they could follow (Section 6). Boge and de Regt's proposal to bridge the divide by testing whether an AI can communicate its understanding to human experts addresses this case.

Because what is required changes with use, these contrasts do not form a single ladder on which the most human-like system counts as the most advanced. Human-facing explanation, for example, or, on accounts that allow understanding without explanation, another vehicle of understanding usable by humans (Section 3.2), is essential where *S* is human and the understanding is meant to enter the epistemic situation of *S*, but may be unnecessary where the understanding serves only the system's own inferences or is communicated to artificial members of *S* (Section 6). And on accounts that do not require a felt sense of grasp or other forms of experience (compare Section 6 and Chapter 8, Section 1.3), such as that of Barman and colleagues, who treat understanding as an ability that can be realized by mechanisms unlike human ones, or Wilkenfeld's, which does not require representations to be symbolic, artificial representations could support genuine nonhuman understanding without resembling human introspection. How far that possibility is supported on such accounts depends on how far the alternative realization can be shown to bear the relation to its object that the adopted account requires, which for worldly targets includes its states being about them at all (Chapter 8, Section 2.2). Whether it would also remain corrigible through further contact with the world is a separate question, about the continuing capacities of Section 8 and the world-answerability examined in Chapter 8, Section 2.3. On Potochnik's account, by contrast, understanding requires a sense of understanding, and what promotes understanding depends on features of human cognition, so on this account the possibility turns first on whether such a sense is possible for an artificial system (Section 6).

Returning to the chapter's question, when understanding is achieved and by whom, the answer is fixed by which achievement is claimed, whom it names as bearer, and which account is adopted. Where artificial systems conduct research, the question "by whom" becomes a question in its own right: unlike in human research, what produced a result cannot be presumed to be what understands it (Section 6). On the working characterization, which epistemic achievement research aims at is left open. For an attempt that aims at understanding, whether understanding is reached bears on whether the attempt succeeds; for one that does not, continuing research may still require capacities associated with understanding (Sections 7–8). Many of the capacities discussed here, transfer, diagnosis, further questions, criticism, and memory, depend not on the properties of a single result but on how inquiry is organized (Section 8). The next part, Chapters 4 to 6, therefore moves from achievement to the organization of inquiry, asking from three sides of the same inquiry what it means for a machine to ask, how candidates and representations become available, and what results bear on and what they revise. Chapter 7 then asks where in that organization conduct and capacity can be attributed.

Part III — How Does Unfinished Inquiry Move?

Chapter 1's working characterization identifies research as an attempt to bring about epistemic achievements new to a reference society *S*. The two chapters of Part II examined knowledge and understanding as possible targets and compared their conditions. But stating the conditions under which an achievement obtains is not describing the process that reaches it. What Part III asks is what is done, and what can change, in an inquiry whose answer has not yet been obtained and which may fail. Inquiry here refers broadly to the process that moves forward within an attempt at research. The accounts seen in Chapter 1, Section 3 of what it is for an individual to be inquiring use the same word in a narrower sense.

Chapter 2, Section 8 called question formation, evidence acquisition, credal change, acknowledged ignorance, and the like candidate **epistemic transformations**, since whether a particular change constitutes progress depends on its relation to a worthwhile target, a correct answer, error exposure, an improved capacity for inquiry, and so on. Ross argued that formulating a worthwhile question or acquiring relevant evidence can itself be progress, and Harris replied that such changes only promote progress (Chapter 2, Section 8.2). For what bears on later advances, such as datasets, methods, and classifications, constituting an advance, promoting one, and merely being rationally expected to promote one were also distinguished (Chapter 2, Section 8.3). A research process can alter questions, candidates, representations, evidential relations, methods, abilities, recognized ignorance, and more before it yields knowledge. How such changes arise within inquiry, and how they can be related to one another, can be asked separately from whether they constitute progress or only promote later progress. Part III takes up the former question.

Chapter 1, Section 7 drew out, as an implication that follows comparatively directly from the working characterization, that an attempt is directed at something not yet settled in *S*, and called this the minimal question structure, but it left undecided in what sense that something is a question. Moreover, where the answer is not yet known in *S*, it may be unclear not only which proposed answer is correct, but what could count as an answer, what the problem is, which objects to compare, which representation makes the problem tractable, what would count as relevant evidence, which operation could produce it, and what a result will alter. Rheinberger, who studied the history of experimental systems, makes a related point about how the unknown is approached (Chapter 4, Section 5). Still, there is also research in which, even where the answer is new, the procedure for reaching it is substantially established in advance (Chapter 5, Section 1).

How unclarity about what could count as an answer is handled in ongoing work can be seen, for example, in the following short record. In October 2015, the health department of the Brazilian state of Pernambuco was notified of 26 newborns with microcephaly by several public and private hospitals that serve as centers of maternal and child care. The department's technical note dated the 30th of that month also cited 22 records of microcephaly in the state's birth registry for January to September 2015 and, comparing these with a median of 9 a year for 2005–2014 (range 4 to 16), stated that the information, though partial, already indicated a change in the pattern of occurrence of this congenital anomaly. Some of the mothers had reported a rash during pregnancy, but the note stated that on current information the anomaly could not be related to any disease, and that its cause could be a congenital infection or some other, non-infectious factor. It then fixed a case definition, by head circumference according to gestational age, to be used for notification from that point on, and stated that all ninety patients reported from various sources would be assessed against it. It recommended that tests for infectious diseases be carried out only in cases whose tomography showed findings suggestive of infection, announced that investigations to formulate hypotheses about possibly associated factors were under way, and stated that the protocol would be updated as clinical and epidemiological evidence developed (Secretaria de Saúde do Estado de Pernambuco 2015, 1–3).

Within this document of a few pages, open points, provisional answers, and means of checking them appear nested within one another. To the point of whether the pattern of occurrence had changed, a provisional answer based on partial information was given by means of the registry records and a standard of comparison, the median of the preceding ten years. What was counted there, however, were cases entered in the birth registry under a code of the International Classification of Diseases. What was to count as a case in subsequent notifications was fixed separately, as the definition by head circumference. This definition made it possible to ask which of the ninety reported patients met it. Unlike a vague "aren't there rather many?", this is a question in which what could count as an answer is fixed. On the other hand, what the anomaly was due to remained open. The note only listed broad candidate answers, a congenital infection or a non-infectious factor, and left the formulation of hypotheses about the factors to the investigations. Even so, the candidates bore on how means

were chosen. The recommendation to restrict tests for infectious diseases to cases with findings suggestive of infection fixes, in the light of infection as a candidate, which cases are to be examined for what. Further, the results of assessing the ninety could bear not only on the provisional answer that the pattern of occurrence had changed but also on whether the case definition itself was appropriate. The announcement that the protocol would be updated can be read as leaving room for the results of checking to revise the definition used in checking.

Part III uses a picture to describe and compare research processes. The picture describes an inquiry configuration at a given point through dimensions such as what is asked and what remains unresolved, what contents are taken as candidate answers and what attitudes are directed toward them, representations, evidence, methods and instruments, standards, practical capacities, actors, records, and social arrangements, together with the relations among them. Research can proceed as a change in one dimension alters which changes are possible in other dimensions, and the configuration so reconfigured is reconfigured further. The picture is not a model fixing an order of stages but a vocabulary for comparing processes (Chapter 5, Section 1.1). The cases it draws on were selected, to suit the points at issue, from records such as papers, historical studies, laboratory notebooks, training logs, repository histories, and oral histories (Chapter 5, Section 1.2), and what they show, the cases of Chapter 4 included, is that certain relations can occur, not how often they do.

For many of the systems reported today, the question whether a machine does research becomes a question of where the work that organizes a research process lies, in the machine or outside it. Labels for stages such as hypothesis generation and verification leave unexplained how a question becomes available, how a candidate is made thinkable, how a mismatch is diagnosed, and the like (Chapter 5, Section 1). The three chapters of Part III therefore describe that work, so that Chapter 7, Section 2 can ask which of it is realized within an artificial actor and which remains distributed across other actors and arrangements.

The health department's note describes one inquiry from three sides at once. One is a description of what is open: the cause of the anomaly, and which of the ninety meet the definition. Another is a description of what was fixed for the time being: the standard of comparison, the case definition, the broad candidate causes, and the range of cases to be tested. A third is a description of what results yet to be obtained can bear on: the results of assessing the ninety, and the announcement that the protocol would be updated. Moreover, the same element appears on more than one side. The case definition by head circumference is something fixed for the time being, and at the same time what made it possible to ask, of the ninety, a question in which what could count as an answer is fixed, and something that the results of the assessment could revise. These three sides correspond, not only in this note, to three relations that are written apart when an ongoing inquiry is described: the relation between an inquiry and the unsettled points that organize it, the relations among candidate answers, representations, and means, and the relation between a result and what it bears on. Below, these are called the three relations of inquiry. The elements of the picture's dimensions are not each assigned to one of the relations; the same element can stand in different relations. The three relations do not exhaust the relations within an inquiry, but the three chapters of Part III each look at the same inquiry from the side of one of them. Chapter 4 treats the relation to unsettled points as the question of what it means for a machine to ask; Chapter 5 asks how candidates, representations, and means become available; and Chapter 6 asks what results bear on and what they revise.

Chapter 4 — What Does It Mean for a Machine to Ask?

The 1950 paper of the maize geneticist McClintock opens by stating that, in the course of an experiment designed to reveal the genic composition of the short arm of chromosome 9, a phenomenon of rare occurrence (or recognition) in maize began to appear with remarkably high frequencies in the cultures: mutable loci. They first appeared in the progeny of the self-pollination of about 450 plants whose chromosome 9 short arm had undergone drastic structural modifications during the cycle in which chromosomes repeatedly break, fuse, and form bridges early in their development, and about 40 different mutable loci were recognized in them. Because the stocks from which they arose had been under investigation for some years without showing such a large number of unstable loci, McClintock concluded that some part of the mechanism of this cycle, or some of the structural modifications it produced, was responsible for the conditions that produced the burst (McClintock 1950, 344–346). The 1944 annual report likewise reports a study that used this cycle to produce a large number of newly broken chromosomes 9 and to obtain mutations in their short arm (McClintock 1944, 127–132).

Let us separate what changed. What organized the original experiment was what genes the short arm of chromosome 9 carries. What came to organize the work after the unexpected mutations appeared, and became the title of the 1950 paper, was the question how mutable loci arise and how they behave. The conclusion that the mechanism of the cycle or its products created the conditions is a candidate answer to this later question, not the question itself. An unexpected observation changed what was asked. A memorandum of January 1949 formulates, for c-m1, the first mutable locus isolated in the cultures having Ds, the conclusion that it arose when a locus called Ds left its original position, was inserted at a locus called C, and inhibited the action of C (McClintock 1949a, 4, 9; McClintock 1950, 349–350). This, too, is a candidate answer to the same question. Moreover, even after the question of how mutable loci arise came to organize the work, narrower questions arising under it remained open. The same memorandum leaves undecided whether large colorless sectors appearing on an ear were due to the loss of a factor called Ac or to the stabilization of the unstable mutation called c-m2 itself. An addendum later attached to a manuscript of April 1949 likewise leaves undecided, of a difference in Ac between a tiller and the main stalk, whether the tiller had gained Ac or the main stalk had lost it (McClintock 1949a, 49; McClintock 1949b, 33). The interpretation that Ac was lost and the interpretation that c-m2 became stable are candidate answers to a narrower question that arose under it, what the colorless sectors on the ear were due to; they are neither the question of how mutable loci arise itself nor answers to it. Two things can be read from this example. One is that what is asked can change in the course of the work, and that the change can be ascertained from the records. The other is that a shift in the question and the persistence, under the new question, of undecided candidate answers are different things.

Suppose a similar shift were seen in the record of a machine's work. If one then says that "the system recast its question in response to an unexpected result," what is being claimed? That the system output a new interrogative sentence, that it performed an act of asking something, that it directed an attitude such as curiosity toward the question, or that its subsequent work came to be organized toward a new unsettled point? When human researchers are said to have "recast the question," these are often spoken of without being distinguished. When *A* is a machine, they can hold separately or be missing separately. And what prompted the recasting is something to be ascertained separately, from the records, as the published account does in McClintock's case.

The last of these, that the subsequent work came to be organized toward an unsettled point, corresponds to the minimal question structure that Chapter 1, Section 7, drew from the working characterization. According to that section, the structure does not require that a project begin with an explicit interrogative, that one actor entertain it, or that the final question have existed unchanged from the outset. And what was derived there is where the attempt is directed, not *S*'s actual state. From this structure it therefore follows that an attempt is directed at an unsettled point, but not in what sense that point is a question (including whether the output of an interrogative, an act of asking, or an attitude must be added to it), whether its content was determinate from the outset, who holds it, or for whom it is unknown. What this chapter asks, on that basis, is what it means for a machine to ask.

Sections 1 to 7 of the chapter are divided along the four points that the structure leaves undecided: in what sense it is a question, whether its content is determinate from the outset, who holds the question, and for whom it is unknown. In what

sense it is a question is treated in Sections 1 to 3; that its content need not be determinate from the outset is treated as the formation and recasting of questions (Section 4) and as a question's ceasing to stand without being answered, or being settled (Section 5). Who asks is taken up in Section 6, and for whom something is unknown in Section 7, which separates what is unknown to *A* from what is unknown to *S*. Section 8 reads empirical studies' claims that machines generated questions or research ideas in light of these distinctions, and Section 9 looks at how descriptions of the same activity of a machine diverge between this book's way of understanding research and the views of Ravetz in his later account, Simon, Nersessian, Funtowicz and Ravetz, Laudan, Bromberger, Kelp, and Lee, which address problems, questions, and inquiry, and then looks back over the chapter as a whole.

Dividing the claim that "a machine asks"

When it is said that a machine asks, what is being claimed? A machine's outputs remain in records and can be examined, but what acts or attitudes they amount to is just what is disputed, and when *S* is a human society, whether they were taken up as questions pursued by members of *S* is a further question.

Whether machines can ask is also discussed in debates about machines doing research. It has been said, for example, that current training produces obedient students who answer known questions rather than the skill to ask the right questions (Wolf n.d.), and that asking the right question is the hardest part of science, so that for some time conjectures and hypotheses will come from human experts (Hassabis and Manyika 2026). A physicist who worked with a machine reports that it can supply candidate calculable problems in large numbers but that its judgment of whether something is interesting cannot yet be trusted, so that the human supplies that judgment (Schwartz 2026). Most such remarks concern the systems of a particular time and are not claims that machines cannot do this in principle; Hassabis adds that intuition could in the limit be modeled by an AI system. Separately, declarations by mathematicians hold that solving problems is only a tool and proxy for conceptual understanding (a declaration by Fields medalists, Avila et al. 2026), and that credit and responsibility continue to belong to humans (Alper et al. 2026). Whether a machine can supply candidate questions, whether it can judge which questions are worth pursuing, and who sets the aims of research and answers for them are different claims. The distinctions that follow also help to tell which sense of question such remarks concern and what they claim about it.

As noted in Chapter 1, Section 7, the word question is used for an interrogative sentence, for its content, and for the act of asking it. According to Cross and Roelofsen, who survey the philosophy and semantics of questions in the *Stanford Encyclopedia of Philosophy*, distinguishing these three is standard in the philosophy of language, and analyses of the act of asking divide into, among others, Searle's account, on which it is a kind of request, and that of Åqvist and Hintikka, on which it is an imperative addressed to the hearer to bring it about that the speaker knows the answer (Cross and Roelofsen 2026, §§1.1, 4.1). Ciardelli, who defends inquisitive semantics, which captures the meaning of an interrogative through the information that resolves it, distinguishes the content of an interrogative from the act of asking for the same reasons for which the content of a declarative is distinguished from assertion, and calls this content an issue, in the sense of a matter awaiting resolution. On Ciardelli's account, the same issue can be the content of different acts, such as asking, wondering aloud, and taking it off the agenda, as in "let's set aside who did it," and what is investigated when something is investigated is not a speech act but a content grasped in thought (Ciardelli 2022, 24–31). What Watson's account of the act of asking (Watson 2015, 86–102) and Habgood-Coote's (Habgood-Coote 2026, 1, 8, 11), seen in Chapter 1, Section 7, require of the asker, and what they do not, is taken up in Section 6.

Besides content and act, there are other things spoken of with the word question. One is a mental attitude directed at a question. Jane Friedman, writing on inquiry and interrogative attitudes, extends the standard idea that questions are the semantic contents of interrogative sentences to the contents of minds, and argues that interrogative attitudes such as wondering and curiosity are attitudes that have questions as their contents (Friedman 2013). A second is an unsettled point that organizes an inquiry, and the minimal question structure of Chapter 1, Section 7 is of this kind. It is a point at which an attempt is directed as something not yet settled for *S*; it need not in fact be open in *S* (the opening of this chapter), and it may not yet have become an interrogative, an act, or one mind's attitude. A third is an agenda, that is, the list of questions that a conversation or group has taken up as what it pursues. These senses are not exhaustive; they are listed to the extent the distinctions below require.

Once these are separated, the presuppositions that a claim that "a machine asked" needs differ according to which sense the claim concerns. If it is only that the machine output an interrogative sentence, that is a claim about form. Watson argues that being interrogative is neither necessary nor sufficient for being a question (Watson 2015, 95, 102). Watson does not discuss artificial systems, but applying the account, an interrogative output by a machine may fail to be an act of eliciting information, and conversely an operation containing no interrogative, such as sending a search term or querying a database, may be a question. A stage label such as question generation (Chapter 5, Section 1) therefore does not by itself determine which of the senses above it refers to. To move from output to act, applying Watson's account, requires an account of the activity within which the machine's operation functions to elicit information. Whether that function is fixed by the purposes of designers or users, or by the organization of an activity that includes the machine itself and by the goals and norms of the practice it joins, depends on a further account of function (Chapter 1, Section 3; Section 6). On either route, the point at issue moves from whether the machine is curious to within which activity, and from which standpoint, the function of eliciting information is attributed. If Ciardelli's distinction is brought in, the form of an output alone does not determine whether it is an act of asking, something closer to wondering aloud, or the arranging of an agenda, and attributing an act of asking requires, separately, an account of, for example, what commitment the output is treated as within a conversation with *S*. The same gap can arise for attitudes. On Jane Friedman's account, a higher-order representation of one's own ignorance, that is, representing that one does not know something, is neither necessary nor sufficient for an interrogative attitude (Friedman 2013). Applying this, a machine's outputting "X is not yet known" is not by itself evidence of an interrogative attitude, and conversely, the fact alone that a machine does not represent its own state of knowledge does not exclude it from what can bear such attitudes.

In the sense of an unsettled point that organizes an inquiry, matters differ again. As Chapter 1, Section 3 noted, the book's attempt can be grounded not only in an individual's intention but also in a group's plan, an artifact's organized function, a component's role, and so on. That a machine's activity is organized toward an unsettled point therefore depends neither on the machine's having performed an act of asking nor on its having an interrogative attitude. Conversely, even if a machine outputs interrogatives in great numbers, there is no question in this sense if they organize no work. Further, the one who produces a question and the one who receives and pursues it need not coincide. A human team may pursue an interrogative that a machine output, and a machine may pursue a question that a human gave it. For a question that a machine output to organize *S*'s inquiry, it must be taken up by members of *S* and placed on the agenda. If *S* includes machine members, whether exchanges among machines can be regarded as proposing questions or as functioning to elicit information depends on the account of the act of asking (Section 6).

A machine's output is sometimes evaluated as having "formulated a good question" (Section 8), but different things can be said about the "goodness" of a question, too. The distinctions drawn by the authors below are not a list of necessary conditions for a question's being good; they serve to separate which point a given evaluation concerns. Collingwood, who took metaphysics to be the historical study of the absolute presuppositions that certain people's thought has made, separated a question's arising, that is, its presupposition's being actually made, from that presupposition's being true (Collingwood 1940, 26, 28). In Collingwood's sense, a question with a false presupposition can arise (Section 5). Ravetz, in a 2016 interview, connects research priorities with the ignorance that results when some kinds of investigation are not pursued (Ravetz 2016, second answer). Watson separates the value of questioning from good questioning, and argues that goodness has a thin sense, succeeding in any way at all, and a thick sense, competently eliciting valuable information, and that competence is both narrower and broader than success: a vague question that happens to hit the mark is not competent, and one can question competently even when no answer is obtained (Watson 2015, 135–140). What Watson evaluates is questioning as a performance and the questioner's skill that supports it; the choice of what to ask is also evaluated as part of that skill (Watson 2015, 144), but how the content of a question is semantically determined is not an object of the evaluation. How a question's answer conditions are semantically determined (Section 2), whether it is practically tractable (Section 3), and whether there is value in pursuing a candidate or reason to direct resources at it (Chapter 5, Section 4.2; Chapter 10, Section 6) are also evaluations distinct from these.

When *A* is a machine, these distinctions offer a handle for asking what an evaluation that "the system formulated a good question" has checked. A tractable question need not be one at which *S* has reason to direct resources. Even if the presupposition of a question a machine produced is actually made in *S*, that presupposition may be false. Applying Watson's distinction between competence and success, a machine's question's happening to lead to valuable information is not a

sufficient reason for saying that the machine questioned competently. This is the form, on the side of questions, of the point seen in Chapter 1, Section 3 that one fortunate output gives little reason to call its source a researcher. Attributing competent questioning, like attributing capacities, involves conditions such as reliability across varied conditions and responsiveness to changing evidence (Chapter 7, Section 3.1). And who may determine research priorities is distinct from whether a machine can estimate the considerations relevant to them; Chapter 10, Section 6 and Chapter 12 take this up. The remarks at the opening of this section can be read in the same way, and they concern different things. The physicist's report separates supplying candidate problems from judging which of them are interesting; the remarks about the skill to ask the right questions concern the choice of what to ask, which Watson evaluates as part of the skill of competent questioning; and the declaration that credit and responsibility continue to belong to humans concerns who answers for the aims, which is settled apart from either.

The distinctions so far have concerned what a question is and how it is evaluated. At a different level are claims about the working characterization itself. As Chapter 1, Section 7 noted, the minimal question structure does not fix where a question comes from. A question may occur in the middle of an experiment or of reading, in a conversation with colleagues or at a meal. One may also ask by going back to the presuppositions one is relying on: while working through a procedure, one may ask why one is doing it this way in the first place, or, after a series of improvements, whether the approach serves its aim at all and what one was trying to do in the first place. Such questions turn what had been working as a presupposition into an object of questioning; this overlaps with Collingwood's relative presuppositions, taken up in Section 5, which work as presuppositions while in use and pass to the side of answers to questions on inspection. One may also ask anew about what one had taken for granted, as when, looking at the sky, one asks why clouds are white in the first place. A question may also be inherited from supervisors, collaborators, funders, communities, instruments, a discipline's history, and so on. That a researcher did not originate a question does not by itself, therefore, deny that research is being conducted in the sense of this book's working characterization. An individual, team, or artificial system can conduct research on a question supplied by another actor, or on part of a question held by a community. Bromberger, who wrote on the logic of questions and of ignorance, also holds that on a view that evaluates scientific contributions as the resolution of items of ignorance, a contribution need not be a result sought, in response to that ignorance, by someone aware of it (Bromberger 1992, 116–117).

Not fixing where questions come from does not mean that where they come from does not matter. If one holds standpoint theory, on which the social position from which research starts bears on the knowledge it yields, which community's records a question is inherited from bears on how one-sided the knowledge that research yields becomes. Chapter 10, Section 6, drawing on standpoint theory and on Varsavsky's argument that research in one society can take its criteria of importance from elsewhere, examines how an artificial configuration that inherits the questions its records already pose, that is, an artificial unit to which research is attributed, can start from where those who produced those records stood. Also, the capacities to notice an unresolvedness, formulate or select a question, and revise its framing can expand what can be attributed to an actor and increase its autonomy (Chapter 7, Section 4.3), but they are not included in the working characterization as necessary conditions for every researcher or participant in research. How questions are actually formed and recast within inquiry is a matter at a level distinct from this claim that the origin of questions is not fixed, and Section 4 takes it up as a description of the processes that research records and accounts of inquiry depict.

What is asked, and what counts as an answer

What Section 1 called the content of a question cannot be separated from what counts as an answer. This section deals with questions in this sense, that is, with cases in which an unsettled point has taken the form of a question with conditions on what counts as an answer. The questions machines output, and the questions used in evaluations, usually have this form. The philosophy of language since Frege has emphasized propositions and declarative sentences, and much work on the semantics of questions has stayed within that framework, identifying a question with a set or function involving the propositions that are its answers; there are also attempts at a logic in which neither questions nor propositions are more basic (Cross and Roelofsen 2026, §§1, 2.1.1). What is taken as an answer also changes what a sentence embedding a question says and what follows from it: whether knowing what John reads amounts to knowing what John does not read holds on Groenendijk and Stokhof's account, on which an embedded question denotes its true exhaustive answer, but not on Karttunen's, on which it denotes the set of true answers (Cross and Roelofsen 2026, §4.2). For artificial systems, on the other hand, claims that "the system answered the question" are routinely counted in benchmarks and evaluations. What is counted as an answer then?

Cross and Roelofsen hold that there is no univocal pre-theoretical notion of an answer; it is a technical notion that can differ from theory to theory, and some theories leave it implicit. Notions of answer range from the view, shared by many theorists, that counts as direct answers, which may be false, only what answers a question exactly and completely, to accounts that count any information about the subject matter as an answer, and theorists also differ on whether answers must be propositions. A response that denies a presupposition is called a corrective answer, and many authors do not count it as a direct answer. The same interrogative can also have a reading on which giving one answer suffices and a reading on which all must be given (Cross and Roelofsen 2026, §§1.1, 4.3, 4.4). The same word can also refer to different notions in different authors. Wiśniewski, who developed inferential erotetic logic, assigns to each question as its direct answers the sentences that answer it exactly in their expression (Wiśniewski 2022, 108 n. 4, 109), and treats as distinct notions a partial answer, which says that the true direct answer lies within a certain proper subset of the direct answers, and an eliminative answer, which rules out at least one of the direct answers; the two come apart when the direct answers need not be mutually exclusive and none of them need be true (Wiśniewski 2013, 43–45). Habgood-Coote, by contrast, drawing on the Stalnaker–Roberts model that regards conversation as cooperative inquiry, follows Roberts in treating a question as a partition of the portion of logical space that satisfies its presupposition into mutually exclusive possible answers, and calls an answer that rules out at least one possible answer a partial answer; completely answering a subquestion gives a partial answer to the original question (Habgood-Coote 2022, 1107–1111 and n. 28). In Wiśniewski's terms, this is roughly an eliminative answer. Ciardelli calls the relation by which, given background information, a collection of subquestions logically determines the original question a dependency, and takes it to be a fact about meaning, not about discourse; where it holds, any answers to the subquestions, together with the background, answer the original question (Ciardelli 2022, 23–24, 29, 41–42).

Brought to the judging of benchmarks, these distinctions make the truth of the claim "the output answered the question" vary with the notion of answer adopted, and counting something as "answered" implicitly adopts one theory of answers. Even if a corrective answer that rejects a presupposition is not counted as a direct answer, whether it can end an inquiry, or make a question cease to stand, depends on each author's account (Section 5). Here the judgment that counts whether something "answered" and what was settled in research part ways. According to Wiśniewski, moreover, a formal question only represents a natural-language question under one interpretation (Wiśniewski 2013, 15). The same output may therefore be a direct answer under one formal representation and yet not be an answer to the question *S* is asking. Unless which reading is being asked is fixed on *S*'s side, judging whether an output amounts to an answer cannot begin. Even if *S* includes machine members, using the same formal language does not by itself align the members' interpretations. Further, Habgood-Coote defines as a partial answer in a weaker sense (a casual partial answer) what, together with the available information, rules out at least one possible answer (Habgood-Coote 2022, 1109–1110 and n. 34). Applying this definition, the same output may be a partial answer for a system with access to a certain database and yet not be one for members of *S* without access to that information. Whether something is an answer becomes relative to what is available to whom.

Bringing Ciardelli's dependence relations to a system that divides a task into subtasks and collects the answers, whether the collected answers answer the original question can be asked at the stage of division, independently of who carried out the subtasks and how. If the division does not realize a dependence relation, there is no guarantee that the original question is

determined even if every subtask is answered successfully. Further, a component in charge of a subtask's rejecting a candidate is not the same as narrowing down the answers to the original question in Wiśniewski's sense. Elimination becomes narrowing only when one adds the presupposition that the true answer lies among the listed options, and that presupposition is just what can be in question when the set of candidates may be incomplete (Section 3). Dependence relations can be asked about at the stage of division, however, only so long as the space of possibilities is fixed. In the course of research that space itself may be revised (Section 4), and a question may cease to stand in connection with its presuppositions (Section 5).

Understanding a question and making a question tractable

The minimal question structure of Chapter 1, Section 7 says that an attempt is directed at something not yet settled in *S*, but it requires neither that this point be expressed by an interrogative nor that what counts as an answer to it already be fixed. The accounts seen in Section 2 mostly dealt with questions for which what counts as an answer is already fixed. This section takes up cases that the structure allows but Section 2 set aside: an unsettled point whose answer conditions are not yet fixed, and the work, on the side of the question, of recasting it into a question whose answer conditions are fixed. Moreover, that a point is unsettled does not mean that the set of possible answers is already at hand. What, then, is it to understand a question when its possible answers are not at hand? For machines this question and the work of recasting carry weight for two reasons. One is that outputting interrogatives can be cheap for a machine; the other is that a machine can generate many candidate answers. What is attributed to a machine therefore divides according to whether being able to produce candidates amounts to understanding a question, and whether there is work of fixing answer conditions distinct from generating candidates.

Authors divide on how understanding a question relates to its possible answers. In the logic of questions there is an idea, deriving from Hamblin, that to know what counts as an answer is to know the question, and Wiśniewski retains it by dividing it into two steps: to know a natural-language question is to know a formula that represents it, and to know a formula is to know its possible answers. Wiśniewski anticipates the objection that for certain why-questions and how-questions it makes no sense to speak of a set of answers determined in advance and, without disputing that there are such natural-language questions, responds by distinguishing the possible answers to a natural-language question from the possible answers to a formula that represents it (Wiśniewski 2013, 16). Ciardelli states, as a slogan, that to understand a question is to know what information is needed to resolve it, and identifies the content of a question with the set of information states that resolve it in the model at hand (Ciardelli 2022, 36). Bromberger, by contrast, held that we understand many questions whose possible answers we are in no position to state, and rejected the view of many logicians of questions that identifies a question with an exhaustive set of possible answers. According to Bromberger, understanding a question cannot be knowing the truth conditions of each of its possible answers, since one must also know that each answer stands to the question in the relation of being an answer to it. And to know that one is in a situation in which one can think only of answers known to be false yet has reason to think the question sound, one must be able to judge that the set of candidates is incomplete, even if one cannot extend it (Bromberger 1992, 4, 130–131, 133).

Some authors stand between these positions. Nickles follows the "spirit" of Hamblin's insight while defining a problem not by its solutions themselves but by the constraints that fix the range of solutions (Nickles 1981, 109–111). The constraints include theoretical background, methodological conditions, and implicit conditions, and they are discovered as inquiry proceeds (Nickles 1981, 111–112). The more constraints one knows, and the more sharply, the better the problem is understood, and the constraints not only make it possible to recognize a solution on seeing it but also indicate where to look (Nickles 1981, 88–89, 113). Nickles is close to Ciardelli in identifying a question by its answer conditions, but differs in taking those conditions to include implicit ones that are discovered as inquiry proceeds.

These positions do not fit a simple opposition between having a question and having its possible answers. One way of reconciling Ciardelli and Bromberger is to read Ciardelli's "know" not as a capacity an inquirer possesses but as the specification of a question's semantic content. On this reading, a semantic model can specify resolution conditions without requiring an inquirer to enumerate the answers or possess a method for finding one. If so, failure to construct a candidate does not by itself show that a question lacks semantic content, and what Bromberger's cases make visible is a practical gap: understanding what is being asked may leave the concepts and procedures needed to answer it unavailable. Bromberger's

own point, however, is a theoretical claim about understanding a question, so this reconciliation holds only on condition that Ciardelli's "know" is read in that way. Representing a question's content and explaining how an inquirer makes progress on it are different tasks, and Ciardelli's slogan can be read as placing the understanding of a question on the side of the former, while Bromberger places it on the side of the latter, as an inquirer's grasp that an answer is an answer to the question.

Bromberger discussed a further gap concerning why-questions. On Bromberger's analysis, for what and where questions, understanding the question suffices to tell what must hold for there to be an answer, whereas the form of a why-question shows only that what it asks about must be the case. For a why-question to have a true answer, the fact must be construable under suitable predicates, attributes, and the like, but the form of the question gives no guide to these, and such concepts may be conceived long after the why-questions that call for them have been raised. One can thus be placed in a predicament in which one knows that one does not know the answer, yet does not know what one would have to find out to learn whether there is anything to know at all (Bromberger 1992, 157–168). This predicament is explicitly distinguished from the p-predicament discussed in the next paragraph, and is not peculiar to why-questions: "what is" questions can produce the same predicament in some circumstances (Bromberger 1992, 168–169 n. 46). As a rational way of dealing with this predicament, Bromberger mentions forgetting about the why-question for a while and turning to other questions with reliable credentials, and conjectures that answers to why-questions usually emerge from the study of such questions. Bromberger adds that this conjecture does not belittle answers to why-questions, and that the value of why-questions is beyond doubt (Bromberger 1992, 169). Read through the distinctions of this section, what is missing in this predicament is less candidate answers than clues to what to find out, and the response of turning to other questions whose answer conditions are known is work on the side of the question, distinct from producing candidate answers.

Distinct from this predicament is the p-predicament discussed in Chapter 2, Section 8.2. It is what Wilholt identified with deep ignorance: a subject is in a p-predicament with regard to a question if, on the subject's views, the question admits of a right answer, but the subject can think of no answer to which, on those same views, there are no decisive objections (Wilholt 2020, 9–10). What is missing in a p-predicament is a candidate answer that escapes decisive objection; one may be unable to think of any candidate at all, or every candidate one thinks of may have been rejected (Bromberger 1992, 4, 29). If one hardly understands a question, one can produce candidates without end and so is not in a p-predicament (Bromberger 1992, 31). Jane Friedman, too, argues that one can grasp a question and have an interrogative attitude even if one cannot represent a single possible answer (Friedman 2013). Grasping a question and having candidates can hold separately. How candidates compatible with a question's constraints become available is treated in Chapter 5.

One typical form of the work of making a question tractable is specifying a contrast for a why-question. Van Fraassen analyzed a why-question as a triple of topic, contrast class, and relevance relation. As Cross and Roelofsen present the analysis, "Why P?" has not been identified as a particular question until one has fixed a set of propositions to be compared alongside P (the contrast class), and the relation fixing what is explanatorily relevant (the relevance relation) is contextually determined (Cross and Roelofsen 2026, §3.2). In Woodward and Ross's presentation, context enters into specifying both, and the relevance relation depends on the context and on the information the questioner is interested in obtaining (Woodward and Ross 2025, §6.2). On Lipton's account, to explain why something happened rather than a specified alternative, the foil, requires citing a causal difference between the two. One need not cite a cause sufficient for what happened, but in return the answer must now say something about the foil as well (Lipton 1993, 43–44, 52). Lipton takes up the case of syphilis and paresis. Since most people with syphilis do not go on to contract paresis, philosophers have disputed whether a person's syphilis explains that person's paresis, and on Lipton's view contrastive questions settle the dispute: Smith's syphilis explains why Smith rather than Jones contracted paresis if Jones did not have syphilis, but not why Smith rather than Doe did if Doe had it too (Lipton 1998, 13). Lipton's own view, however, is that the model on which a contrast is explained by citing a causal difference is too simple as it stands: a fuller account must say, among other things, what explanation requires when there are several causal differences between fact and foil (Lipton 1993, 44–45), and which conditions a cause must satisfy to answer a particular question is what Lipton calls the problem of causal selection (Lipton 1998, 11). Moreover, as Cross and Roelofsen summarize Lipton's view, the central requirement for a sensible contrastive question is that fact and foil have a similar causal history against which the differences stand out (Cross and Roelofsen 2026, §3.2.3). Specifying a foil in this way narrows what an answer must do and where to look for it.

How far specifying a contrast amounts to recasting on the side of the question is, however, subject to qualifications. Lipton holds that one point of making a why-question contrastive is to reduce the indeterminacy of the question of how things would have been different had the effect not occurred, and that a foil lets us choose which parts of the effect's history to hold fixed in a way that meets our explanatory interests (Lipton 1993, 49, 51). He writes that many why-questions are contrastive (Lipton 1993, 39), the contrast often remaining implicit (Lipton 2001, 16). If an implicit foil is already fixed, making it explicit can be a restatement of the same question. It becomes a different question when the foil was not fixed, or when it is replaced by another foil. Further, a contrast does not by itself settle which cause to cite. In real cases, Lipton notes, there will seldom if ever be only one causal difference between fact and foil (Lipton 1993, 45), and not all of an event's causes explain it in a given context, which calls for a finer-grained account of the context in which explanations are requested (Lipton 2001, 10–11). Where several causes meet the difference condition for the same contrast, which of them to cite can turn on what the inquirer is interested in. Part of the fact that explanations vary with what one is interested in therefore remains on the side of the answer even after the foil is fixed. Bromberger, on the other hand, using van Fraassen's examples, argued that it is a mistake to think why-questions especially context-dependent on the ground that stress changes their answer conditions, and criticized theories that take stressed questions as their paradigm for generalizing from a biased sample (Bromberger 1992, 160 n. 34). On Bromberger's analysis, only one of the readings of a stressed sentence changes the presupposition; on the other readings, stress only indicates the speaker's interests (Bromberger 1992, 164–165). Since the presupposition-changing reading is a question of a different logical form, even on Bromberger's analysis fixing a foil can amount to replacing the original why-question with a different question. Lipton is not responding to Bromberger in that passage on contrastive explanation, but putting the two analyses together, the recasting of a question whose answer conditions are not fixed into one whose answer conditions are can be described as work on the side of the question that is not reducible to generating candidates. In Nickles's words, it is adding constraints on solutions and sharpening them.

Different kinds of question are made tractable in different ways. A how question, for example, may be decomposed into capacities or constraints, and a mathematical problem may change when rendered geometrically, algebraically, probabilistically, or computationally. An engineering ambition, too, may become an investigable problem only when it is translated into relations in which performance is measurable and into a physical arrangement capable of realizing them. Nickles, too, holds that a problem can be reformulated without changing the presentation of the data and transformed into another problem, and cites the case of the problem of blackbody radiation (Nickles 1981, 88, 114 n. 6). Cases in which candidates, or the objects of inquiry themselves, become available through restructuring data or restructuring the representation of a proof are taken up in Chapter 5, Section 3.

The range of possible answers itself may be unknown. Habgood-Coote, citing Bromberger and Wilholt, uses the label deep ignorance, which Wilholt identified with the p-predicament, for the state of not knowing what a question's possible answers are. In this use, the state so named is not the same as the p-predicament. For a question Q such as what the correct fundamental theory is, Habgood-Coote raises "What are the possible answers to Q?", an option question, a question that asks what the possible answers are (Habgood-Coote 2022, 1116–1117). Habgood-Coote treats only complete deep ignorance, setting aside intermediate cases such as ignorance of some of the options (Habgood-Coote 2022, 1117 n. 41).

Applying van Fraassen's analysis to machine outputs, even if outputting why-interrogatives can be cheap, that alone fixes neither the contrast class nor the relevance relation, which depend on the context and on what the questioner is interested in. Sawyer reports that van Fraassen gives some reason to think that an omniscient being would not be correctly understood as having explanations at all (Sawyer 2014, 7 n. 8). If that is so, holding much information does not by itself give a machine explanations; what the analysis asks for, a contrast class and a relevance relation, is not supplied by the quantity of what a system holds. If S is a human society and the interests on which the relevance relation depends lie on S's side, then even when a machine pursues the question, the question is given its shape by S's interests, and the question of whose question it is divides into two: who produced the interrogative, and who fixed the foil and the relevance relation. Applying Lipton's account, specifying a meaningful foil requires knowledge of the histories of fact and foil and a choice of whose interests to answer to. In a mixed S, foils can differ from member to member, so part of the disagreement between the explanations of machine members and human members might be described not as opposing answers but as a difference of foils. Also, making a question tractable is fixing its answer conditions, not making it easy. Moreover, since interests remain in which cause to cite even after the foil is fixed, whose interests they are still enters into how an explanation is chosen, even if the machine has received the foil correctly. For an S consisting only of machines, the same analysis leaves a position to be filled.

Since, on Woodward and Ross's presentation seen above, the relevance relation depends on the information the questioner is interested in obtaining, applied to an *S* of this kind the analysis leaves open what stands in the questioner's position: what the members' answers are passed on to, the tasks set for the members, or something the members fix among themselves, among other candidates. Which of these, if any, fixes the relevance relation for such an *S* is not settled by the analysis.

Turning toward machines Jane Friedman's point that grasping a question and having candidates can hold separately, a machine can separately be in a state of grasping a question while unable to represent candidate answers, and in a state of being able to generate many candidates while not keeping the question open. Having many candidates does not by itself mean progress, since what Bromberger's definition asks about is not the number of candidates but whether they escape decisive objection on the subject's views (Bromberger 1992, 29, 31). Bromberger also has an extended definition, however, on which one is in a p-predicament if there are compatible answers one cannot yet think of, even when one has already thought of several compatible answers; on it, new candidates compatible with the constraints count toward resolving the predicament, even when there are many of them (Bromberger 1992, 122). How a machine that can produce only candidates known to be false, yet has reason to think the question sound, is to judge that its set of candidates is incomplete (Bromberger 1992, 133) can also be raised as a question from here. Further, applying Habgood-Coote's distinction, a system that works only within a supplied set of possible answers does not pursue the question of what the possible answers are. A system that adds some options, however, falls among the intermediate cases that Habgood-Coote set aside, so his treatment does not settle whether it pursues that question. Simon distinguishes problems supplied to programs from the work of formulating them, without dismissing the work of solving the formulated problems (Simon 1973, 186–187). Nickles notes that substantive theoretical content can be built into methods and apparatus (Nickles 2019, 363). Applied to the systems considered here, these points make attribution depend on the recorded task boundary. Where humans supplied the framing and the machine used it without revision, successful execution does not establish that the machine performed that framing work. How much creative work it performed in reaching its result remains a separate question.

Taken together, the distinctions of this section mean that even when a machine can be said to have put forward a question, who did the work of making that question researchable, that is, of deciding what to take as the contrast, what kind of explanation to count as an answer, and within what range of candidates to look for it, remains a separate question. In many of the empirical studies read in Section 8, the higher-level question is given by humans together with the task.

How are questions formed and recast within inquiry?

When it is said of a machine that "the system turned an anomaly into a question" or that it "recast its question," the claim is checked by asking which choices were made by whom, and what changed under which framework. To that end, this section moves to the descriptive side of the two levels separated at the end of Section 1, and looks at the formation and recasting of questions that stand out in research records and accounts of inquiry, attending to what changes and how. Specifying a foil, seen in Section 3, is also one form of this.

A situation may first be indeterminate. Concepts, contrasts, instruments, and candidate answers can help make it askable. Questions are refined, constituted, or reorganized across an inquiry (Hildebrand 2024, §4.3; Henne 2023, §6), and are sometimes divided into subquestions and carried among participants (Section 6). In Hildebrand's report of Dewey's theory of inquiry, a problem is formulated within an initially indeterminate situation rather than given before inquiry; reasoning about a hypothesis can lead back to redescribing the problem (Hildebrand 2024, §4.3). In *How We Think*, Dewey holds that observations are needed to make the specific character of a problem clear: if a proposed diagnosis is accepted before the trouble has been explored, thinking is cut short (Dewey 1910, 74). Dewey's problem does not fully coincide with this chapter's question, and Dewey's ideas and hypotheses lie on the side of possible solutions, that is, of candidate answers. With this in view, Dewey held that determining which data matter and elaborating the meanings suggested by them depend on each other: if selected facts do not sustain an interpretation, the data are reconstituted, while possible interpretations are elaborated in relation to them (Dewey 1910, 103–104). In Popper's method of conjectures and refutations as Nickles reads it, too, putting forward conjectured solutions and criticizing them makes the problem more fully and precisely determined (Nickles 1981, 92). In both descriptions, dealing with candidate answers changes the shape of the question, but question and candidates are not treated as the same thing.

Some accounts place the question still later. El Shazly, writing on inquiry aimed at understanding, argues that such inquiry begins not from ignorance but from puzzlement, confusion, or discursive disagreement; there, deciding what to ask is itself part of the inquiry, and since engagement with the object fixes the particular question, the question comes later in the order of explanation (El Shazly 2025, 15–18). On this account, the content of a question fixed later cannot be placed back at the beginning of the inquiry. Even so, an engagement whose content is not yet fixed as a particular question can be organized as an attempt aimed at an understanding, or another achievement, new to *S*. What the minimal question structure captures is that direction.

Henne, who works within Dewey's theory of inquiry, reformulates in terms of that theory a distinction inspired by Carnap's between internal and external questions. On Carnap's distinction, internal questions are asked within a linguistic framework and answered within that framework, as when one asks, in the language of things, "Is there a white piece of paper on the desk?"; external questions concern the framework itself, such as whether to introduce rules for speaking of a new kind of entity, or to adopt one framework rather than another (Henne 2023, §2). Henne's distinction is a distinction not between questions but between inquiries. Framed inquiries use a conceptual framework without questioning it in asking and answering questions; in Henne's example, biochemists studying the effects of temperature on protein synthesis are engaged in one. In framing inquiries, the question or problem cannot be fully formulated within existing frameworks, and the framework itself is questioned, revised, expanded, or created; the inquiry in which William Thomson built a theoretical framework on the notion of absolute temperature is an example (Henne 2023, §§1, 6). As Henne's own account emphasizes, the core of the distinction lies not in where a question is located relative to a framework but in whether the framework determines the questions to be asked and how they are to be answered, and even an inquiry about a framework can be a framed inquiry (Henne 2023, §6.2).

Henne's distinction, moreover, does not draw a sharp line in the way Carnap's does. Henne classifies an inquiry by its situation and conduct rather than its outcome, so that an inquiry into whether a definition should change is framing even if it ends by retaining the definition. Henne also treats the distinction as a matter of degree that depends on what is counted as one inquiry. Even the most radical framing inquiry relies on some framework, and even in the most typical framed inquiry the framework does not fully determine how the situation is handled. One inquiry can begin as framed and end as framing, and the reverse can also happen (Henne 2023, §§1–2, 6).

In light of this distinction, one can ask how far inferential erotetic logic captures the formation of questions. Wiśniewski formally defines, as two relations, the ways in which a question is derived from a set of declarative sentences or from another question. Roughly, one is the relation by which a set of declarative sentences gives rise to a question that, if the sentences are true, must have an answer, but whose answer is not yet determined; this is called evocation. The other is the relation by which a principal question gives rise to an auxiliary question that helps answer it; this is called erotetic implication (Wiśniewski 2022, 107–108, 112, 116–117). On that basis, a tree of questions and answers that places the principal question at its root, adds auxiliary questions by erotetic implication, and branches from each auxiliary question that is a query by each of its direct answers is called an erotetic search scenario (Wiśniewski 2022, 109, 118, 122–124). Wiśniewski reports, as the work of Chlebowski and others, that the generation of scenarios can be automated by operations defined on such scenarios (Wiśniewski 2022, 125). Scenarios can be reworked in the course of execution (Wiśniewski 2022, 125 n. 15; Wiśniewski 2013, ch. 13), but the principal goal fixed by the principal question does not change (Wiśniewski 2022, 122). What these definitions presuppose is that a question and the set of its direct answers are fixed in a language and, for erotetic implication and search scenarios, further that the principal question stays fixed. Wiśniewski himself holds that, since against the background of a logic that accepts the principle that anything follows from a contradiction (explosion) inconsistent premises evoke no question, a different strategy is needed where questions arise from inconsistency, such as taking as background a logic in which a contradiction does not entail everything (a paraconsistent logic), which the general framework of inferential erotetic logic does not prevent (Wiśniewski 2013, 62–63; Wiśniewski 2022, 113–114); Wiśniewski also states that inferential erotetic logic does not yet provide an account of erotetic inferences that are not valid but are plausible (Wiśniewski 2013, 66). Seen this way, what these definitions capture is how questions arise when the set of answers is fixed in a language and, for erotetic implication, the principal question also stays fixed: one case belonging to framed inquiry in Henne's sense. Even within framed inquiry, however, a principal question can be recast, for instance by specifying a foil (Section 3). Nor does a change in the space of answers by itself amount to framing inquiry, since Henne's criterion is whether the framework determines the questions to be asked and how they are to be answered, not the range of answers

itself.

What formalization does not capture are, for example, cases in which inquiry begins from an anomaly, an archive, a dataset, an instrument, an assigned task, a failed intervention, an unfamiliar material, a practical need, or a diffuse sense that existing descriptions do not fit, and what is to be asked itself becomes fixed in the course of inquiry. Turning such a diffuse beginning into a question for which answers can be sought may require deciding what to count, for which group, in what units, compared with what, over what period, and by what standard: that is, choosing a scale, population, unit of analysis, contrast, temporal boundary, evidential standard, and so on. In some cases, as in mathematics, it may require choosing a formal language in which to state the question. Such choices may divide what appeared to be one question into several, bring several questions together into one, or reveal that the original object was partly an artifact of a measurement or classification. Making a question determinate is not deciding everything at once. In the health department's note seen in the introduction to Part III, the standard of comparison, the period, the case definition, and the patients to be assessed were fixed for the time being, while the cause was left open. If the cases entered in the birth registry under an International Classification of Diseases code are treated as observational data for specifying the problem in Dewey's sense, then the case definition and the standard of comparison were fixed for the time being separately from them. Moreover, these choices are not changes of the same kind. Re-choosing a standard of comparison or a foil is in many cases refining a question within existing concepts and, applying Henne's distinction, can be taken to remain within framed inquiry. Revising the definition of what counts as a case, by contrast, can amount, as a small revision of a concept's conditions of application, to an inquiry that is framing to a small degree. Indeed, Henne cites small revisions of conditions of application, such as raising the age of consent, as examples of a small degree (Henne 2023, §6).

Such choices can succeed or fail, and when they succeed a question that was not there before becomes available. Articulating an explicit and tractable question, dividing what looked like one question into several, or fixing a question for the time being so that it can be pursued can therefore be work accomplished within inquiry rather than merely an input given to it from outside. Whether such work amounts to progress or only promotes later progress is where the views of Ross and Harris divide (Chapter 2, Section 8.2). The claim that questions are formed within inquiry is nevertheless easy to overextend. After the fact, any event can be described as answering some question, so attributing to an inquiry, after the fact, a question that organized it needs clues. In light of what Chapter 1, Section 3 asked of an action, what made it relevant to an unresolved matter, what possible outcomes would have changed its role, and where the capacity to redirect the activity lay, retrospective question attribution is too permissive if it is granted on verbal redescription alone, and is better supported when the attributed unresolvedness explains the organization of the activity: why some operations were selected, why possible outcomes could have mattered differently, why a line was continued or abandoned, and how later developments remained connected to what was unresolved for the relevant subject or society.

The question a result is later said to answer need not be the question that actually organized the inquiry leading to it. Arfini and Costa's account of serendipity, finding by chance something other than what was sought (compare Chapter 1, Section 3), gives an example. They distinguish the explanations that discoverers construct while the outcome is uncertain from the retrospective explanation available only once the research is concluded, and argue that in serendipitous discoveries that change how a phenomenon is understood, what is to be explained changes during the process. For inventions arising by chance, they hold, the two explanations coincide. Penzias and Wilson, known for the discovery of the cosmic microwave background radiation, began, on their reading, by seeking to explain why their horn reflector was not working as expected, spent some time excluding possible causes of the noise it picked up, and ended by explaining why any such reflector would pick up the signal (Arfini and Costa 2025, §§2.1, 3–3.2, n. 13). So as not to confuse a shift in the question with the appearance of candidate answers, questions and candidates are indexed from here on. Writing q_1 for the question that organized the work first and q_2 for the later question, in McClintock's case at the opening of this chapter the question about the genic composition of the short arm of chromosome 9 is q_1 and the question how mutable loci arise and behave is q_2. Written in the same way, why the horn reflector was not working as expected (what was producing this noise) is q_1, the possible causes of the noise are candidate answers c to q_1, and why any such reflector would pick up the signal is q_2. If what is to be explained changes in this way, the question q_2 that the reported result answers is not the question that organized the work leading to it, and, by the conditions of the preceding paragraph, what explains the work of excluding possible causes of the noise is not q_2 but q_1. An account that presents all the work as directed at q_2 can be right about the result without describing what organized the earlier work (compare Chapter 5, Section 2).

In the Penzias and Wilson case the change of question is visible through a reconstruction made after the research was over. From records written at the time, by contrast, such as McClintock's report and memorandum, one can sometimes read what organized the work at that point and what was open. A question can also be reorganized while its headline stays fixed. On the reading of the editors of Darwin's notebooks, the basic questions of the transmutation notebooks—how species adapt to changing conditions, how new species arise, how the hierarchy of classification comes about—were set at the start of Notebook B, the first of them, and kept, while what those questions were taken to include and how the relevant phenomena were related changed as the answers did (Barrett et al. 1987, 168, 329, 395–396). For the formation of candidate answers, too, recollection and contemporaneous record diverge. Darwin's autobiography recalls that on reading Malthus "for amusement" in October 1838 it "at once struck" him that favorable variations would tend to be preserved. The notebook entry that records the thought, however, is dated 28 September, while Darwin was still reading the book. It writes the thought down as a force like "a hundred thousand wedges" trying to force every kind of adapted structure into the gaps in the economy of nature, and the editors take it as the first formulation of natural selection. The entry is preceded by a note on the infertility of hybrids and by extracts on fossils from Lyell's book, and followed by a note of what Darwin had been told about the curiosity of monkeys. According to the editors, Darwin finished reading Malthus on 2–3 October, and the theory went on taking form in the next notebook, Notebook E (Darwin 1958, 120; Barrett et al. 1987, 374–376). What the recollection presents as a single sudden event appears in the contemporaneous record as a step written down in the midst of work on other questions, and one that went on changing afterward. The record also fits well with Dewey's remark that time is required to digest impressions into substantial ideas, and that an idea matures through a silent working-over of alternative suggestions (Dewey 1910, 37, 209–210). Taking the editors' reading that the content of the basic questions changed with the answers together with this record, that change, too, can be thought to have proceeded among such jottings of candidates. A report that a machine writes after its result can diverge from the record of its work in the same way (compare Chapter 5, Section 2.1).

A supplied task, too, may by itself be less than a question. In one biomedical-engineering laboratory studied by Nersessian, who investigated cognition and practice in scientific laboratories, for example, a newcomer was designated the person who would take the laboratory's engineered construct into living animals, and about a year later the newcomer was still defining the specific goals and problems of that work (Nersessian 2009, 750). Getting from the assigned role to a specific question is itself a long piece of work. Nickles's discussion of pragmatic satisficing offers a further perspective: inquiry can proceed with a formulation good enough for the next step, without first achieving a perfect one (Nickles 2019, 361–362).

When a question changes, what it changed for can also be distinguished. Nickles distinguished, for changes of problem, a sense relative to an agent (a problem's changing for a given scientist) from a semantic sense (a problem's changing for a system of theory and practice). What changes when constraints implicit in the system are discovered is the problem for the agent; a semantic shift occurs when aims, methods, or theoretical context change substantially (Nickles 1981, 99). Nickles also separates change within a problem from replacement of a problem (Nickles 1981, 100), holding that if the setting is changed substantially, one solves not the original problem but a replacement problem. For a problem judged insoluble, one may also give up the aim while keeping the constraints that define the problem, and Nickles cites the abandonment of the aim of building a perpetual-motion machine (Nickles 1981, 95–96). The range of questions that can be asked may itself change. Bromberger argued that grasping concepts such as mass or voltage, for which each individual object has its own value (what Bromberger calls determinables), requires considerable theory, that before Newton no one could have entertained questions about gravitational mass, and that Aristotle was in no position to be aware of not knowing the mass of the Parthenon. When concepts change, the range of questions about an object that must have answers changes too (Bromberger 1992, 134–135, 150 n. 10).

Applying these distinctions to machines allows one to check, when a machine is said to have formed a question, which of two things it amounts to. One is multiplying questions under the same framework; the other is putting that framework itself in question. Telling the two apart first requires settling what counts as a machine's framework. On Henne's account, the rules that govern the procedures for asking questions and verifying hypotheses, that is, methods, are also part of a framework (Henne 2023, §5). Henne also says that in a typical framed inquiry the framework is accepted without question and is therefore "transparent", and that it comes to be noticed only when it is challenged or defective (Henne 2023, §7.2). Following this, if a machine's framework is understood as what is used in raising questions within its inquiry but is not made an object of questioning, the distinction can be applied regardless of how it comes to be part of the machine. An example of the former is the automated generation of erotetic search scenarios that Wiśniewski reports. It generates questions by using a

fixed language and fixed rules of erotetic implication without questioning them. Even in a system that reworks scenarios in the course of execution, the principal question and its set of answers stay fixed, so it belongs to framed inquiry. A claim that a system does the latter requires that the rules of the framework in which questions are raised are themselves actually put in question within that inquiry and made objects of re-choice or revision. Examples include asking whether the procedure for dividing a question into auxiliary questions suits the problem at hand, and reconsidering, as seen above, the definition of what counts as a case. What separates the two, then, is whether the rules a machine used in raising questions were put in question within that inquiry. Whether a rule so questioned can then be revised within the same inquiry is a further matter: some of what a machine uses in raising questions may be structure set by the way it was made, which remaking it can replace (Chapter 6, Section 5.3; Chapter 8, Section 2.1), and its revision may fall outside the machine's own inquiry even when that inquiry has put it in question. On Henne's account, however, how far an inquiry that puts those rules in question is framing is a matter of degree. Which of the two it amounts to also depends on what is taken as one inquiry. Within a human–machine inquiry that is framing as a whole, what a machine carries may be a part of it, an inquiry that is locally framed.

Even the few cases and accounts seen so far suggest that recasting a question can take many forms. When a machine is said to have "recast its question," too, the changes that can occur can be written apart according to what changed; they include, for instance, adding auxiliary questions under the same rules; Wiśniewski's search scenarios are of this kind. There is also refining a question within a framework; Lipton's specification of a foil, and what Nickles calls the discovery of constraints implicit in the system, are of this kind. There is also putting in question within inquiry, in Henne's sense, the conditions of application of concepts or the rules of the procedures for raising questions. Since Henne's classification goes by conduct, the rules are not necessarily revised as a result. And there is moving, in Nickles's sense, to a replacement problem by changing the constraints substantially, or, for a problem judged insoluble, giving up the aim while keeping the constraints. These are not an exhaustive list, nor mutually exclusive kinds, and they can overlap within one inquiry. Moving to a replacement problem can involve putting the framework in question. And the boundary between refining a question within a framework and putting its rules in question is, on Henne's account, a matter of degree that also changes with what is counted as one inquiry. Nickles's discussion also allows a formulation to remain provisional while inquiry proceeds.

Since, in the choices made in turning a diffuse beginning into a question, some things are fixed for the time being while others are left open, a claim that a system turns an anomaly into a question can be checked choice by choice: which choices the system itself made, which it took over from its task, which it left open, and whether it accepted a candidate before sufficiently exploring the problem in Dewey's sense. Moreover, even if every question raised and every candidate rejected within framed inquiry is recorded, the questions that could not be raised under that framework do not appear in the record. The completeness of a record of execution therefore does not by itself assess a system's capacity to open questions in the sense of framing inquiry (compare Chapter 5, Section 2.1).

On Bromberger's account, which questions one can raise depends on the concepts one can use: without the concept of mass, one could neither ask about the mass of the Parthenon nor know that one did not know it. Bromberger does not discuss artificial systems, but applying this to machines suggests two things. One is that generating many questions in new sentences by combining existing concepts and introducing new concepts that make it possible to raise questions that could not be raised before are different tasks. This distinction applies to human researchers as well, but since a machine can do the former at low cost, when generated questions are rated "novel" (Section 8), which kind of novelty is meant becomes a pressing question. The other is that if a machine's representations can use concepts that *S* does not use, the machine may raise questions that *S* cannot yet raise, and *S* may then not be in a position to receive them as questions (Section 7). Both points hold where grasping such concepts can be attributed to the machine's representations at a functional level. Applying Nickles's two distinctions, when a machine has recast a question one can ask separately for whom the problem changed, and whether it was a change within the same problem or a replacement by another problem. If Nickles's "system" is read as *S*'s system of theory and practice, then where a machine discovers constraints implicit in *S*'s system, what changes is the problem for the machine, not the problem for *S*'s system. For instance, even if a machine derives from *S*'s existing theory that the possible values of an answer have an upper bound, if that bound was already implicit in the theory, *S*'s problem itself has not changed. If aims or methods change substantially, the problem for *S*'s system changes too, and whether to accept that change is decided on *S*'s side. Moreover, the problem's having changed for the system does not by itself amount to having replaced the original problem, but a system that obtained a solution by substantially relaxing the constraints has solved not

the original problem but a replacement problem. This is one form of the case in which a check was passed, but on a question other than the one asked (Chapter 6, Section 3.2).

The recasting of questions also comes from the side of results. Judgments of where to locate a failure shape which path is taken next (Chapter 6, Section 2.4), and when standards themselves are put in question, what counts as an answer can be revised too (Chapter 6, Section 5). Having understood something can also open further questions (Chapter 3, Section 7.3). Since the formation and recasting of questions occur within such relations to candidate answers and results, checking a claim that a machine "recast its question" requires looking not only at the records on the side of the question but also at what was happening on the side of candidates and results at the time.

The presuppositions of questions, and how questions close

Questions have fates other than being formed and recast. A question can remain without an answer being found, cease to stand as a question in relation to its presuppositions, be suspended, be postponed, or be settled by an answer. For a machine, the end of a run or the completion of an output is left in the record, but whether the question was thereby settled, ceased to stand, or was only suspended can be impossible to tell apart on the record.

What, then, determines what has become of a question? This section looks at it from three sides. One is how the matters met in research are handled, looked at through cases left in research records. The second is the cases in which a question ceases to stand without an answer, looked at through accounts, among others, of the presuppositions and contexts that make a question stand. The third is closing a question by an answer, and also suspending or abandoning it without one, looked at through accounts, among others, of the attitudes and intentions bearing on questions and inquiries. The accounts address different conditions: what is presupposed, which problems a scientific community admits, what attitudes or intentions the asker has, and so on. On whichever account is applied, however, it turns out to be a different matter from whether a run has ended or an output has been produced. For a machine, therefore, the record that it has "finished" and what has become of the question are to be checked separately.

First, the handling of matters met in research is taken up. To that end, the unsettled point that organizes an inquiry is named separately from the unresolved matters met in pursuing that point. One of the latter may become the former, or may not. Pursuing an unsettled point brings one up against matters such as whether an effect is genuine, whether materials and instruments behave as assumed, what a discrepancy indicates, and what is to count as a case (compare Chapter 6, Section 2). Such matters meet different fates, as when the health department's note fixed for the time being how cases were to be counted while leaving undecided the relation between the anomaly and any disease (the introduction to Part III), and as when McClintock left competing interpretations open (the opening of this chapter). Some matters were postponed. Charles David Keeling, who was monitoring atmospheric carbon dioxide continuously, recalls having, under the rush of the observing program, put off examining a possible bias in the calibration gases that a spectroscopist had raised (Keeling 1998, 46, 50). Some matters were handled through skill without being stated. Bell Burnell, who read radio-telescope chart records by eye, recalls having become able, after reading the first few hundred feet of chart, to tell sources that scintillate (whose intensity flickers as their signal passes through interplanetary plasma) from interference (Bell Burnell 1977). A matter met outside the program can also become the question itself. McClintock's mutable loci appeared in an experiment on the genic composition of a chromosome arm and came to be the question organizing the work (the opening of this chapter).

On these records, which of the matters an inquiry meets are taken up as questions, which are fixed for the time being, left explicitly open, postponed, or handled without being stated, is not something fixed before the work begins but can be part of the work of research. What becomes of any one of these matters has no predetermined direction, and a matter can open further questions (compare Chapter 2, Section 8.2). Not all of this, moreover, is deliberate selection. A matter can be left untouched because no one registered it or because there was no time. And some of what goes unquestioned is not left open at all but held fast as a condition of asking something else (compare Chapter 8, Section 2.1). Conversely, a matter need not be taken up by choice either. The geneticist Bo Xia recalls that, after an injury to the tailbone, sitting down brought back pain every time and each time drew Xia's attention back to the fact that humans have no tail, an interest that later led to a study of tail loss (Yanai and Lercher 2024; Xia et al. 2024). Chapter 5, Section 6.4 looks at the recollections of Keeling, Bell Burnell, and Bo Xia from another side. Others, on the other hand, have emphasized that the choice of whether to take something up or

leave it is laden with values and excludes alternatives. Ravetz held that whenever a proposed research project is given a low priority it is not undertaken, so that in that respect we remain in ignorance, and that the knowledge we have exists only in relation to "value-loaded decisions" to obtain some sorts of knowledge at the expense of others (Ravetz 1993, 348). That what to pursue cannot be fully decided in advance also appears in Rheinberger's description, noted in the introduction to Part III. On Rheinberger's account, the object of research in an experimental system embodies, in a form that can be manipulated experimentally, exactly what one does not yet know, and the knowledge such a system generates lies beyond what can be anticipated at the time (Rheinberger 2011, 312–313). As with the attribution of questions seen in Section 4, describing a matter as left or postponed, rather than as never registered, is better supported when a record shows that it was registered. The health department's note, McClintock's documents, and Keeling's recollection are such records.

Next, the cases in which a question ceases to stand without an answer having been found are taken up. Collingwood and van Fraassen set out conditions under which a question "arises"; Kuhn and Laudan address which problems a community or tradition admits and how their solutions are appraised. These are different questions. "Arising" here does not concern where a question comes to mind. As seen in Section 1, a question may come to mind in the middle of an experiment or at a meal. What is at issue here is whether a question that has come to mind stands as a question, given what is presupposed and what lies in the background. Collingwood held in *An Essay on Metaphysics* that every question involves a presupposition and that a question arises directly from its presupposition. This power of making a question arise depends neither on whether the presupposition is true nor on whether it is thought true, but only on its being presupposed. On Collingwood's account, therefore, questions arise even from what one supposes while knowing it to be false, and the validity of arguments based on it is not impaired. Conversely, even a true presupposition makes no question arise if it is not made (Collingwood 1940, 25–28). By this definition, to say that a question "does not arise" is to say that it involves a presupposition not in fact being made, and Collingwood calls such a question "nonsense," not as nonsensical in itself but in a sense relative to that context (Collingwood 1940, 26). Collingwood further distinguished two kinds of presupposition. A relative presupposition stands to one question as its presupposition and to another as its answer. While it is in use it works as a presupposition, and when it is examined, a question is raised to which it is the answer (Collingwood 1940, 29–30). An absolute presupposition, by contrast, stands to every question to which it is related as a presupposition and never as an answer. It is not a presupposition that is hard to verify but one to which the idea of verification does not apply. One of the examples, the presupposition that everything that happens has a cause, had on Collingwood's account been made in every branch of natural science a hundred years before the book was written and had since ceased to be made in some (Collingwood 1940, 31–32). Once such a presupposition ceases to be made, the questions that depended on it cease to arise as well. And since people are ordinarily not aware of their absolute presuppositions, such a change cannot be a matter of choice; it is said to happen in thought without being noticed (Collingwood 1940, 48).

Kuhn places a related issue at the level of a scientific community. As Dimitrakos quotes him, a paradigm supplies criteria for selecting problems presumed to have solutions while that paradigm is accepted; these are largely the problems the community admits as scientific (Dimitrakos 2023, §4.1). Fellows and Liu reproduce his further claim that a paradigm can exclude socially important problems that cannot be formulated with its conceptual and instrumental resources (Fellows and Liu 2015, 19). The status at issue is admission within normal scientific practice. Exclusion does not mean that a problem has been answered or is objectively insoluble.

Van Fraassen defined "arises" for why-questions in yet another way. As Cross and Roelofsen present the account, in addition to the triple analysis of Section 3, a why-question presupposes that its topic is true, that only its topic is true in its contrast class, and that some proposition bearing the relevance relation to the topic and contrast class is true. When the body of background knowledge K determined by the context does not entail the first two presuppositions, the question does not arise, and when the third fails, the question has no answer even if it arises (Cross and Roelofsen 2026, §3.2).

Arguments that separate the truth or falsity of presuppositions from how a question is treated are found on the side of the norms of inquiry as well. Falbo, who discusses when inquiry is permissible, accepts a semantics on which some questions have false presuppositions and lack a true direct answer (Falbo 2024, 965–966), and argues that inquiry into a question without a true answer is sometimes permissible (Falbo 2024, 969–976) and, conversely, that even when there is a true answer, inquiry is not permissible if one can rationally be confident that it will not improve one epistemically with respect to the question (Falbo 2024, 980–981). This is a claim not about the status of questions but about the permissibility of inquiry,

and it shows that the semantics of questions and the norms of inquiry can move separately. Bromberger treats the truth or falsity of presuppositions from the side of ignorance. Bromberger calls ignorance concerning a question whose presupposition is true genuine ignorance, and ignorance concerning a question whose presupposition is false spurious ignorance (Bromberger 1992, 121), and holds that ignorance of the specific gravity of phlogiston was not cured but vanished (Bromberger 1992, 131 n. 4).

Laudan's account of theory appraisal also contains a description of problems ceasing to be problems. On that account, as Dellsén reports it, the research tradition against which research is carried out determines what counts as a genuine problem, how important it is that a theory solve it, and what would count as a solution; the empirical problems a theory is meant to solve need not concern any actual state of affairs, since it suffices that the tradition's adherents believe that they do; and progress can consist in a change in the part of the tradition that fixes problems and solutions, as when an unsolved problem is deemed to be a pseudo-problem (Dellsén 2018, §3). Among the theses that Laudan and his colleagues list for Laudan's own model are that defining an empirical problem requires the prior specification of a context of inquiry, that a theory is found particularly persuasive when it can solve empirical difficulties that its rivals face, that the historical context is a major determinant of how important a problem is, so that the same problem can carry different probative weight in different contexts, and that replacing an old theory with a new one commonly brings losses as well as gains: theorists of geology before Cuvier and Lyell, for example, dealt with empirical difficulties that were later disregarded (Laudan et al. 1993, 72–73, 75). Their seventh thesis adds that criteria for adequate solutions are supplied by guiding assumptions and may change when those assumptions change (Laudan et al. 1993, “Teses de Laudan,” thesis 7). These accounts describe different outcomes. Collingwood's and van Fraassen's conditions for arising differ from Kuhn's communal admission of scientific problems. Laudan's problems ceasing to be problems or being set aside when a theory is replaced differ again from giving up an aim while keeping the constraints of a problem judged insoluble, as Nickles discusses (Nickles 1981, 95–96). None of these outcomes is simply an answer's not yet having been found.

Applied to machines, these accounts raise separate questions about arising, communal admission, and problem appraisal. Collingwood's account asks which presuppositions a machine makes and which questions they make arise. Since Collingwood's "presupposing" is an act of a thinker's mind, applying it to a machine means reinterpreting, at the level of function, the work by which a presupposition makes a question arise. Collingwood's own view, however, was that whether a presupposition is relative or absolute is not determined by introspection, and that which absolute presuppositions were made in a piece of scientific work is found by an analysis that takes the records of that work as evidence (Collingwood 1940, 43, 59–60). Since this method can be turned in the same way on a machine's outputs, the claim that a machine inherits absolute presuppositions made in what formed it can be examined, by analysis of the machine's outputs, as what Collingwood calls a historical claim about what was presupposed at a given time. Analysis of the outputs, however, gives evidence about which presuppositions are at work; to confirm where they came from also requires comparison with what formed the machine. Collingwood further held that one can make presuppositions without knowing which presuppositions one is making (Collingwood 1940, 26), and that a presupposition can be questioned even if those who use it do not realize that it can be (Collingwood 1940, 30). Applying this argument, too, does not require that a machine make its presuppositions explicit and be aware of them as presuppositions. Whether even implicit representation is unnecessary depends on what the reinterpretation at the level of function contains.

If van Fraassen's account is applied, whether a why-question arises becomes relative to a body of background knowledge *K*. In the account as Cross and Roelofsen present it, that body is determined by the context (Cross and Roelofsen 2026, §3.2), so setting up the machine's background knowledge and *S*'s separately extends the framework rather than applying it as it stands. Extended in that way, the same interrogative why-sentence may arise under the machine's background knowledge and not under *S*'s, or the reverse, since what each body entails about the topic and its contrast class can differ. Whether arising is then to be assessed against the machine's *K*, against *S*'s, or against both is a choice that the account itself does not make. And if the point reported in Section 3 holds, that an omniscient being would not be understood as having explanations, the size of the machine's *K* does not by itself make a why-question arise for it.

Applied to a scientific community *S*, Kuhn's and Laudan's accounts add a question about the organization of inquiry: which problems does *S* admit, and what does it recognize as an adequate solution? Incorporating artificial systems may change this range if it changes the conceptual or instrumental resources accepted in the practice, or the standards governing solutions.

Merely adding a machine does not establish such a change.

This application does not require treating the machine as a member of *S*: an instrument can affect the work through which problems are formulated and investigated. It does require distinguishing a change in available equipment from a change in communal practice. Likewise, a machine's ability to process an interrogative does not establish that *S* admits the corresponding problem. Extending the account to an artificial community would require a further argument that its participants sustain shared standards for problems and solutions.

Differences in how presuppositions change also bear on gaps between a machine and *S*. To withdraw a question deliberately when a presupposition is given up requires keeping track of what each question presupposes, and this condition applies to Collingwood's relative presuppositions, which can be given up consciously through examination. In the case of absolute presuppositions, however, a question can cease to arise without anyone representing the presupposition. There can therefore be a gap in which a question has ceased to arise in *S* while a machine that explicitly keeps a list of interrogatives goes on pursuing it. Applying Falbo's norm, a machine's having pursued a question with a false presupposition is by itself evidence neither of failure nor of success, since what this norm requires for inquiry to be permissible is that it not be rational, at that time, to be confident that inquiring will not epistemically improve one with respect to the question (Falbo 2024, 963). It then has to be specified, however, whether improvement is assessed for the machine itself or for members of *S*, and the latter is a reading that extends Falbo's norm, which asks about the improvement of the inquirer. If the addressee of the assessment becomes those who incorporated the machine, or *S*, this connects with the picture in Chapter 2, Section 5, on which criticism is directed at those who built the presupposition into the system. Laudan's account yields a different gap. Since what counts as a problem, and how important it is, is fixed by a research tradition and its historical context, a matter that the machine's side treats as not to be handled may count as an important problem in another tradition within *S*, and, conversely, a problem that the machine pursues may already have been set aside there as a pseudo-problem.

Third, closing, suspending, and abandoning questions are taken up. Besides ceasing to stand without an answer, a question can also be settled by an answer. On what settlement is, or, in the theorists' word, what it is to "close" a question or an inquiry, explanations also diverge. Collingwood held that a question, once answered, remains a question and only ceases to be an unanswered one (Collingwood 1940, 25). Jane Friedman identifies the attitudes that close inquiry as attitudes rationally incompatible with interrogative attitudes. Believing a complete answer to a question closes it, but a credence, whether high or low, is not an opinion that settles a question. It is also argued that, at least in some cases, a question is not closed even by maximal credence; so it is if one grants that it makes sense to investigate whether a fair coin tossed infinitely many times lands heads every time. And a state in which one has given up a question without answering it is held to be distinct from closing it (Friedman 2019). On ending inquiry, too, there is more than one view. Telic verbs are verbs, like "build a house," that have a point at which, once it is reached, the action is complete. Jane Friedman holds that "inquire" is a verb without such a point, so that inquiry does not by definition end on reaching knowledge, and ending inquiry is a further step. While granting that words like "figure out" seem telic, Jane Friedman suggests that they may refer not to the activity of inquiry but to the end of successful inquiry (Friedman 2024). Daniel C. Friedman, by contrast, takes the fact that the central words describing active inquiry, such as "answer a question" and "figure out," are telic as support for inquiry's having a goal to be achieved. On that account, inquiry is organized by an intention directed at a question, and that intention partly specifies with which settled attitude, whether belief, acceptance, knowledge, understanding, certainty, or another, the inquiry is to be closed. A settled attitude is an attitude that takes some answer on as true and treats that answer as usable in subsequent thought and action. It is then held that intending to form some settled attitude does not mean that that is the optimal stopping point, and that settling epistemically appropriately may require more than the intention specifies (D. C. Friedman 2024). Kelp, also taken up in Chapter 1, Section 3, takes belief to be the kind of move that closes inquiry into a question for one, and holds that inquiry is settled, or properly closed, for one if and only if one knows the answer (Kelp 2021c, 13–14).

There is also more than one way to change how a question is treated without an answer. Lee, seen in Chapter 1, Section 3, draws on a classification on which suspension of judgment comes in two types: one that keeps the question on the research agenda and one that takes it off. Lee also calls the latter a type that closes inquiry, but it is not settlement by an answer (W. Lee 2026, 12). Lee then holds that if one thinks the evidence at hand is neutral and no further information can be obtained, one may bury the question and not inquire, even though doubt remains (W. Lee 2026, 15). On postponement, Collingwood lists, as subdivisions of taking something for granted relatively, deferring the verification of a presupposition to a more

suitable occasion and putting it off out of laziness or timidity, and pairs them with what is taken for granted absolutely, which in principle neither admits nor requires verification (Collingwood 1940, 41–42). On settling too early, Dewey had said that accepting a suggestion at once is uncritical thinking, and that letting a first suggestion become an accepted conclusion prematurely cuts thinking short (Dewey 1910, 13, 74). Cherniak's argument that judging which inquiry to undertake is itself an inquiry is taken up in Chapter 5, Section 6.5. Which norms govern starting, continuing, and stopping inquiry has also been discussed, and Jane Friedman poses how such norms relate to epistemic norms as a question for zetetic epistemology (Friedman 2020, 1, 12). Part of that debate was seen in Chapter 2, Section 5.

Applying these accounts, several things can be asked separately before a machine is described as having "closed a question." On Jane Friedman's criterion, neither the end of a run, nor the completion of an output, nor a probability distribution over candidate answers sharply skewed toward one of them is by itself the closing of a question; only a state whose role is such that going on pursuing the question while in it is held not to make sense can count as a closing state. Whether the distinction between belief and credence applies to a machine's states, however, and whether such failure to make sense can be spoken of for a machine, are further presuppositions this application needs. Moreover, when a task is built as having a point of completion, as in "run until a report is produced," its end is fixed by the task's specification, not by the state of the question. On Jane Friedman's account, which denies inquiry a point of completion, here lies one reason why the end of a run and the settling of a question can come apart; on Daniel C. Friedman's account, the location of the gap moves to the difference between the end that the task specified and the settlement that an intention directed at the question specifies. A stopping criterion, whether set by those who incorporate the machine or by the machine itself, resembles in form the specification of a settled attitude, but on Daniel C. Friedman's distinction, a specified settlement and an epistemically appropriate settlement are different. Whether a machine has a planning state corresponding to an intention directed at a question is also a presupposition. Since Daniel C. Friedman allows the content of an intention to be implicit and to be understood by its function, the threshold for applying the account is not high, but it still requires a structure that responds to rational pressures toward means–end coherence, consistency, and stability (D. C. Friedman 2024). If Kelp's distinction is applied, a machine's stopping alone determines neither that the inquiry closed for it nor that it closed properly, and whether a state corresponding to belief in an answer can be attributed to the machine, and further whether that state amounts to knowledge, are to be asked separately. Suspending without settling, and keeping a matter open, require holding the matter as provisional, or as open, in a form to which one can return. A record can be one of the mechanisms that support such returning. As Chapter 5, Section 6.3 describes, in one public chemistry notebook a student answered, about a year later, questions that the supervisor had added, and in the long-term evolution experiment with *Escherichia coli*, samples were kept even for questions not yet formulated, so that what could be returned to was not limited to questions already posed. A system that can only pursue or discard what it meets has only part of this range. And a question settled on the machine's side is not thereby settled in *S*. If *S* is a human society, at least the result's entering *S*'s epistemic situation is needed (Chapter 2, Section 9.4), and when *S* has machine members, too, settlement on the side of one member is not the same as settlement in *S*.

Putting together what the three sides have shown, the following comes into view for machines. Which of the matters met in research are made questions, which are postponed, and which are handled without being stated can itself be part of the work of research. On Collingwood's account, a question can cease to arise without anyone noticing a change in presuppositions, and on van Fraassen's account, once the machine's background knowledge and *S*'s are set up separately, a why-question can arise under one and not under the other; on either account, the questions pursued by a machine and by *S* can diverge. Applying Kuhn and Laudan to a scientific community *S* adds a different question: have its admitted problems or standards for solutions changed? Whether a question has been closed by an answer, suspended, or abandoned is also asked separately from the end of a run or the completion of an output. In every case, the "finished" left in a machine's record does not by itself show what has become of the question.

Taken together, moreover, these fates also allow the process of research to be viewed from the side of what remains open. Not every change of fate goes in one direction: forming a question can expose ignorance, and pursuing a candidate can reveal further alternatives. Peirce, one of the founders of pragmatism, characterized inquiry as the struggle prompted by doubt to attain settled belief (Peirce 1877, 6). On Hildebrand's summary of Dewey's theory of inquiry, Dewey described a pattern that inquiry follows in problem solving, which ends when inquiry has converted an "indeterminate situation" into a "determinate one" (Hildebrand 2024, §4.3). The view from the side of open matters works at a different level from these accounts of inquiry's overall direction, concerning the several open matters within the process of one piece of research. Nor

does it identify progress with reducing a single quantity of uncertainty. As a way of viewing research processes, this view makes changes to the question, method, and standards visible. Which matters to pursue, with what grounds, remains the question of Chapter 5, Section 4.2.

Who asks?

When a machine presents a question to members of *S*, or when machines pile queries on one another, who is asking? For machines, this question is often put in the form of whether a machine has curiosity. But accounts of the act of asking, accounts of group inquiry, and accounts of individual inquiry each attribute something different; they are not competing theories of one and the same act.

Begin with accounts of the act of asking. As Chapter 1, Section 7 noted, Watson identifies asking by its characteristic function of eliciting information, so that a question need not take interrogative form, and one asked with some other explicit goal can still be a genuine question (Watson 2015, 86–102). On Watson's account, the information lacking need not be lacking to the asker. When someone who knows the answer seeks information on behalf of another, the work of seeking information is divided between the two (Watson 2015, 103–104). Watson also takes questioning to be a socially established practice with a common goal, and holds that even asking alone is essentially social, in that it uses skills learned from others and follows the norms of questioning in social settings. Watson then distinguishes practice from habit, holding that looking left and right on a familiar deserted road can be a habit (Watson 2015, 75–76, 82–83, 103).

Habgood-Coote defines asking as a speech act tied to the grammatical form of interrogatives (the interrogative mood) (Habgood-Coote 2026, 1, 8). Habgood-Coote's account is built from the argument that the default view, on which to ask is to express an interrogative attitude, to aim at acquiring the answer, and to direct the hearer to supply one (Habgood-Coote 2026, 2), does not fit the three kinds of asking seen in Chapter 1, Section 7: a teacher's question to which the teacher knows the answer, a question asked in order to answer it oneself, as in Douglass's address, and Hilbert's shared asking (Habgood-Coote 2026, 4–6). On Habgood-Coote's reading, Hilbert hoped that the problems of the 1900 address would be taken up and solved by the mathematical community of which Hilbert was a part (Habgood-Coote 2026, 5–6). Soliloquy, such as asking oneself where one's keys are, is also said not to fit the default view, since speaker and hearer are the same, but Douglass's case is distinguished from it (Habgood-Coote 2026, 5).

What is proposed instead is the view that asking is a proposal in shared inquiry: a proposal to adopt together a shared intention that the hearer, the speaker, or both answer the question (Habgood-Coote 2026, 11). On this idea, conversation is a joint activity involving minimal cooperation, and there can be shared intentions even under coercion (Habgood-Coote 2026, 9–10). It is also stated, tentatively, that asking may be thought to aim at common knowledge of the complete answer. Common knowledge is distinguished from distributed knowledge, which can be obtained by pooling what the parties know but which no party need know on its own (Habgood-Coote 2026, 12 and n. 29). On this view, asking need not express the speaker's interrogative attitude, as a teacher's question shows. Habgood-Coote further argues, conditionally, that if groups can have attitudes that none of their members has, asking need not express an interrogative attitude of either party, and places under that condition the example of two people in a carpentry workshop. The two ask whether some planks are the right length without being curious about it; at most they want the question answered, which is not the same as wanting to know the answer (Habgood-Coote 2026, 12). Once the shared intention has been accepted, however, the individual mental states it requires are called for (Habgood-Coote 2026, 12). It is also argued that there is no reason for the norms of asking to align with the norms of interrogative attitudes (Habgood-Coote 2026, 14). On both Watson's account and Habgood-Coote's, what the identification of an act of asking does not require is a state of curiosity or an interrogative attitude, not inner states in general.

Neither of the two examines machines as askers, but applying their accounts divides what is asked about machines. Since Habgood-Coote defines asking as a speech act tied to the interrogative mood, operations that are not such speech acts, such as entering a search term or taking a reading from an instrument, fall outside the scope of this account from the start. Watson's account can include them among questions. The same operation of a machine can therefore be a question on one account and lie outside the scope of the other. For questions generated inside a single machine, the nearest case on either account is asking oneself, and the two treat it differently. Habgood-Coote only lists soliloquy as a case that does not fit the

default view and does not discuss how the shared-proposal view treats it (Habgood-Coote 2026, 5). Watson, by contrast, counts asking oneself, even without any utterance, as questioning (Watson 2015, 108), and, as seen above, treats even such asking as social. Where, on the other hand, a machine presents a question to humans in *S*, applying Habgood-Coote's account, it can count as a proposal to shared inquiry, if the account's conditions are met. What then bears on attributing the act of asking is not whether the machine has an interrogative attitude but whether the machine can be a party to a shared intention, and that depends on which account of shared intention is adopted. If a machine cannot be a party, its output cannot be described as the machine's own proposal. There are also cases in which it can be described as humans or organizations proposing through the machine, but that attribution requires a separate explanation supporting that they offered the question as a proposal, and there remain cases in which that explanation fails and no proposal in this sense is made at all. What follows from the point that there can be shared intentions even under coercion goes only so far as conversation's requiring no more than minimal cooperation; it does not reach the conclusion that a question presented by a machine that has no option of refusing can also be a proposal. Further, a relation in which a machine, without itself knowing the answer, helps a human learn something can correspond to the asymmetric plans that Habgood-Coote held are hard to propose by asking (Habgood-Coote 2026, 12 n. 28). In that case, it is a shared plan different from asking in this sense. Applying Watson's account, what is asked is whether a machine's operation is participation in the practice of questioning or a repetition like a habit. Attributing participation in a practice requires attributing to the machine the sharing of a common goal and the following of the practice's norms. A system that repeats queries by a fixed procedure stands at that border. The question of who may propose questions to *S* is taken up in Chapter 12, Section 4, as part of the problem of who can be among those who take part in decisions in fact and not merely in name.

Turning to accounts of group inquiry, the question about who holds a question takes another form. One person need not hold the whole question. A collaboration may divide an overarching question into subquestions and fix, as a further question, how they are to be pursued. Documents, funding arrangements, instruments, and evaluative practices can preserve an inquiry over time even if no participant continuously represents all of it. Habgood-Coote calls collective action aimed at producing knowledge group inquiry, and gives an account of it meant to cover large and distributed cases of this kind (Habgood-Coote 2022, 1099–1101). According to Habgood-Coote, accounts of group action built on small cooperative examples, which require the members to share an intention, to believe that the others will do their parts, or to be jointly committed to the activity, fit poorly with large, distributed, and hierarchical collaborations, whose participants may not know one another or even what the joint project is (Habgood-Coote 2022, 1102–1103). On the minimal account therefore proposed, a group is inquiring into a question when some of its members are inquiring into that question or into its subquestions, the group intends to inquire into it in accordance with a strategy of inquiry, and the active members pursue their subinquiries because of that intention (Habgood-Coote 2022, 1114). The intention need not reside in any one member, and is characterized by a bundle of functions: being the output of collective deliberation, a disposition toward the end and its means, the demand for means–end coherence, and the demand for know-how to carry out the plan (Habgood-Coote 2022, 1104). In large cases, functions such as deciding what to do, checking the plan against what the group believes, and working out how to carry it out can be distributed across different teams (Habgood-Coote 2022, 1105–1106). The account is also neutral on whether every participant must intend the group activity to succeed (Habgood-Coote 2022, 1114).

For the structure of the work, on Habgood-Coote's account, which treats a question as a partition of its possible answers (Section 2), the division of epistemic labor can be understood as the assignment of subquestions to members (Habgood-Coote 2022, 1107–1111). Habgood-Coote also distinguishes the object question, which the group aims to resolve, from the methodological question of how to resolve it, and holds that neither is a subquestion of the other. When a question demands more labor than any individual can supply, or several kinds of expertise that no individual has, the group turns to the methodological question and adopts a plan that splits the object question into subquestions, assigns them to members according to their capacities, and decides the order in which to tackle them (Habgood-Coote 2022, 1113–1114). Combining the partial answers is a further, separate stage. A member who understands them all may do the combining, but it may also be done by an automatic procedure, or by someone who does not understand the partial answers, or simply through publication. In such cases, the group can come to know an answer that no member knows (Habgood-Coote 2022, 1114, 1116). The work of keeping track of what has been asked and what has been answered can itself be divided among participants. The progress reports and open wiki of the Polymath project, a collaboration among many mathematicians, are regarded as serving this function (Habgood-Coote 2022, 1115 and n. 40). Habgood-Coote lists, as the most important kinds

of division of labor, the division of the object question, of the methodological question, of record-keeping, of evidence, and of the option questions seen in Section 3, and conjectures that in many actual cases these are mixed together, and that many of the difficulties of interdisciplinary research may lie in handling them (Habgood-Coote 2022, 1118).

Habgood-Coote's account thus attributes an intention to the group and, when the inquiry succeeds, knowledge as well. Whether such organization constitutes a literal group intention or group knowledge is a further question for social epistemology, which treats the social dimensions of knowledge. What is attributed to the group here is the inquiry into a question and its intention, not the act of asking. In Habgood-Coote's model of conversation, the act of asking is a proposal that an individual participant makes in conversation (Habgood-Coote 2022, 1107–1108). For thinking that, at the level of inquiry and its intention, what holds a question can be organized activity, it suffices that the activity-level orientation toward the question not be reduced prematurely to one individual's occurrent mental state. Assigning subquestions to members according to their capacities and deciding the order in which to tackle them are also part of the problem of how finite means are to be distributed among *S*'s unknowns, which Chapter 9, Section 2.1 names allocation, and Chapter 10 asks whether such distribution can be carried out inside an artificial configuration.

At units larger than a group, too, questions are held. A question can be held across a field and over generations, with many groups taking up parts of it. Hilbert opened the 1900 address by observing that every age has its own problems, which the following age either solves or casts aside as profitless and replaces with new ones, and went on to set out unsolved problems whose solution Hilbert expected from the future (Hilbert 1902, 437). Habgood-Coote's account of group inquiry leaves undecided whether a discipline, or science as a whole, counts as one collective inquiry (Habgood-Coote 2022, 1114–1115). At the level of a field, too, some unresolved matters are taken up and others are not. A community can, moreover, limit which questions its members take up. Section 5 discussed Kuhn's account of how a paradigm delimits the problems admitted as scientific, including its exclusion of problems that cannot be formulated with its conceptual and instrumental resources. Kuhn does not discuss artificial systems, but applied here, a configuration that selects questions by whether they can be expected to have solutions with the concepts and methods at its disposal can take on the same insulating function, and the rating of generated questions by feasibility (Section 8) is one place where that function can enter without anyone's choosing it. A machine's taking up part of the questions held at the level of a field and its carrying out the very work of holding such questions across generations can be different things. Chapter 10, Section 6 returns to the level of the field through Kitcher's account of how particular questions inherit significance from broader ones.

Applying the distinction between the object question and the methodological question, dividing an object question into subquestions, assigning them according to capacities, and deciding their order is work on the methodological question, and it can be done by parties other than those who answer the subquestions. Habgood-Coote reads the methodological question as a question answered by citing even one method of how to resolve it (Habgood-Coote 2022, 1113–1114 and n. 37). When a multi-agent system, which divides work among several agents, splits a task into components, it can therefore be described as giving an answer to the methodological question, unless the division was supplied with the task. Whether the division was received with the task or fixed by the system itself is part of the material for judging where the organization of the inquiry resides (Chapter 7, Section 2). Combining the partial answers is, on this account, a separate stage, and for a system that has automated it there is room to describe it as one in which no component knows the complete answer but the group does. Whether what is combined answers the object question turns, in Ciardelli's terms, on whether the division realized a dependence relation (Section 2). And since a division of labor can extend over at least the five kinds above, when a system is said to have "divided the task," what is attributed changes with which of them was divided. Whether such descriptions hold, however, turns on at least two points. One is whether the components are members or tools. On Habgood-Coote's account, if the way a component's working is explained by the group's intention is merely causal, distinct from the explanation of action, the component is closer to a tool (Habgood-Coote 2022, 1105). This point therefore depends on how the explanation of action is understood. The other is whether the components that answer the subquestions can be regarded as knowing those answers, since on this account the group knows by virtue of its members' knowing the answers to an appropriate set of subquestions (Habgood-Coote 2022, 1116). Other points also remain that Habgood-Coote left undecided in notes, such as in which components' states, at which times, the group's intention is to be sought (Habgood-Coote 2022, 1104 n. 20, 1108 n. 29).

Besides group inquiry and the act of asking, there are also accounts of what it is for an individual to be inquiring. Jane Friedman, while granting that an inquirer need not be a single person and that government agencies and the police can inquire too, limits the paper's subject to individuals. Morse, who appears in its example, is someone who goes through the motions of an investigation while knowing, or believing, that he himself is the culprit. Of Morse there may even be a clear sense in which he can be said to be participating in the police investigation, but Morse himself is not inquiring (Friedman 2019). Jane Friedman also holds that inquiry is, by and large, a phenomenon at the level of what a person as a whole does, rather than of sub-personal processes within the person, and that it is by and large an intentional activity (Friedman 2024). Daniel C. Friedman explains inquiry by an intention directed at a question, and holds that curiosity is not a condition of it. A disillusioned scientist who pursues an assigned question without curiosity is also an inquirer. It is also said that telling who is seriously inquiring into what is needed for the practices of deciding whose judgment to rely on, whom to cooperate with, and whom to support (D. C. Friedman 2024). As on accounts of the act of asking, on Daniel C. Friedman's account curiosity is not needed, but an intention is.

Applying accounts of individual inquiry, a component of a machine, or a system, may participate in an organization's inquiry while not itself inquiring in Jane Friedman's sense. This does not conflict with the point that this book's attempt can also be grounded in a component's role or an artifact's organized function (Section 1), since what is described as an attempt is the organization of the activity, not each participant's inquiry. Moreover, since on Jane Friedman's claim the person as a whole is the inquirer and the sub-personal processes within the person only contribute to inquiry, applying it to a system made up of many components requires deciding first which of the whole system, a component, or a human–machine pair is to be regarded as corresponding to "the person as a whole." On Daniel C. Friedman's reasoning, being able to identify which question a machine is pursuing within *S* is a precondition for deciding whether to entrust judgment to the machine's outputs, whether to cooperate with the machine, and whether to support the machine's inquiry. This point bears on allocation in Chapter 10 and on the discussion in Chapter 12 of who may decide and who may revise.

On who holds a question, there is also the point of distributing contributions to its formation. Nersessian describes laboratory models as communal achievements developed over years (Nersessian 2009, 752), and connects building models with articulating as well as solving problems (Nersessian 2024, abstract). Simon's organizational example likewise has specialists introduce criteria and coordinate revisions (Simon 1973, 193–194). Applied to an artificial configuration, these accounts prompt questions about which components introduced or revised the question, constraints, representations, or means of investigation; which executed an allocation already supplied; and how contributions were integrated and conflicts resolved. Dividing the execution of a task does not by itself establish participation in its formation. What such a division gains also depends on whether components differ in their background assumptions and routes of error: dividing work among copies with shared assumptions can divide the task without dividing perspectives (Chapter 5, Section 6.5; Chapter 11, Section 3). The human cases also carry a condition that an artificial configuration need not share. In the cases Nersessian and Simon describe, what passes between participants, the model and the criteria, is articulated. Participation in forming a problem is attributed there through such articulated contributions. A configuration whose components share internal states can instead pass on a state that has not been articulated in this way. Whether such a passed state counts as a contribution to the formation of the question, and whether it works on the receiving side as it did on the side that passed it on, are then questions that the human cases do not answer.

The question of who asks thus divides into the attribution of the act of asking, the attribution of group inquiry and its intention, the attribution of individual inquiry, and the distribution of contributions to problem formation. Returning to the form with which the section began, whether a machine has curiosity, a state of curiosity is not the key to attribution on accounts of the act of asking, on Habgood-Coote's account of group inquiry, or on Daniel C. Friedman's account of individual inquiry. The question returns to the side of the machine's states when one adopts Jane Friedman's account, which makes an interrogative attitude a condition of inquiry, and then which level of thing is to count as the inquirer has to be asked first. Further, a group that asks jointly, or a group that inquires jointly, plays a different role from the reference society *S*. Even when the same group plays both roles, that identity is not determined automatically by the definitions of the accounts.

What is unknown to *A* and what is unknown to *S*

Sections 4 to 6 looked at how questions are formed, recast, and closed, and by whom they are asked. What is often cited as prompting such activity, in research by humans and by machines alike, is curiosity. The Introduction, Section 1, also touched on the question whether curiosity is necessary or only one human mechanism for directing exploration. This section asks how, under the word curiosity, what is unknown to *A*, which carries out the attempt, is related to what is unknown to *S*, for which the research is done. At least three claims can hide beneath the word. They are distinguished here by what the word is applied to: an actor's state, work done within an inquiry, or the distribution of work across a community.

1 The actor is curious in a psychological sense. Accounts of curiosity understand this variously, as a desire to know, as an attitude directed at a question (for example, what Jane Friedman counts, alongside wondering and the like, among interrogative attitudes; Section 1), or as an emotion that may have a felt character.

2 The actor or organized system detects unresolved matters, allocates attention among them, and forms or revises questions. This is the work Sections 4–6 have described.

3 A community distributes exploration across questions and possible lines of research. Chapter 9, Section 2.1 names this allocation, and Chapter 10 examines it.

These claims do not entail one another merely by definition. If one supposed that a machine with curiosity in the first sense would have what research needs, the supposition would run together three things that Chapter 1, Section 2 kept apart: the actor *A* that carries out the attempt, the society *S* for which the achievement is new, and the bearer who actually realizes the achievement. Curiosity in the first sense is a state of *A*, and on accounts that understand curiosity as a desire to know, it is directed at what *A* does not know and is satisfied when *A* comes to know. Novelty in the working characterization, by contrast, is indexed to *S*. An attempt is directed at a point aimed at as not yet settled in *S*, and if that point had in fact already been settled in *S*, what was mistaken is the prospect of novelty, not the fact that research was attempted (Chapter 1, Section 7). Success turns on whether an achievement new to *S* is realized and, where the project's aims include it, on whether it enters *S*'s epistemic situation (Chapter 1, Section 2). What is unknown to *A* and what is unknown to *S* need not coincide.

Even in human research, it is not self-evident that curiosity in the first sense matters to research for *S*. Even when a researcher is a member of *S*, what the researcher does not know and what *S* does not know can come apart. And curiosity is not only being drawn to what one does not know but being drawn to some unknowns and not to others. On Grimm's account of curiosity and understanding, curiosity attaches not to truth as such but to subjects and to questions of particular types (Grimm 2008, 730–736). The unknowns a curious researcher is drawn to need not be the unknowns *S* most wants resolved. Which of *S*'s unknowns are worth pursuing is a question of value, which Chapter 9, Section 3 and Chapter 10, Section 6 take up. Accounts that give curiosity a role in research for *S*, too, give it different roles. Three roles can be told apart in the writings taken up here: one is a role as part of the source of what *S* finds significant, another is a role as exploration, and a third is a role as something valued for its own sake.

The first role appears in Kitcher's account of scientific significance. On Kitcher's account, as Matthew J. Brown reports it, curiosity becomes part of the source of what *S* finds significant: Kitcher traces epistemic significance to "the broad questions that express natural human curiosity" (M. J. Brown 2010, 137–138; Chapter 10, Section 6). If so, a human researcher's curiosity shares its sources with what *S*, a human society, finds significant. Sharing the sources, however, does not mean that the researcher's curiosity will be directed at the unknowns *S* wants resolved first. Kitcher does not discuss artificial systems, but applying this account, when *A* is an artificial system and *S* a human society, even this shared source cannot be presupposed. What such a system lacks, attends to, or would seek need not coincide with what the human society does not know. An artificial system may be uncertain about something familiar to humans, indifferent to a question humans find urgent, or organized around patterns that matter to an artificial community (compare Chapter 8, Sections 1.3 and 3.2).

The second role, as exploration, appears in arguments defending research guided by curiosity. Flexner, who discussed the support of research, held that many of the great discoveries that later proved beneficial to humankind had been made by people driven not by the desire to be useful but by the desire to satisfy their curiosity, a curiosity that may or may not lead to anything useful (Flexner 1939, 545). Flexner's argument is also one of the positions on why research is supported (Chapter 9, Section 3). This role can be read through the trade-off between exploration, which tries new possibilities, and exploitation,

which uses good moves already known (Chapter 10, Section 1). The developmental psychologist Gopnik, drawing on computational work on this trade-off, notes that searches which make only small changes to the current hypothesis can be stuck at a local optimum, and that exploration becomes more valuable the broader and more complex the space of possibilities is. Gopnik takes the neophilia and curiosity of children to drive exploration in a protected period before goal-directed exploitation, notes that complex societies may give exploratory roles to specialists such as scientists, and observes that when one individual explores first and exploits later, the interests of the one who learns and the one who exploits what was learned are automatically the same (Gopnik 2020, §§1–3). Applied to research, those who explore and those for whom the research is done are *A* and *S*, and their interests do not coincide automatically. On this reading, moreover, what research for *S* needs is the exploration, not the mental state through which humans happen to carry it out, so for an artificial *A* the question becomes whether exploration is carried out, and over whose space of unknowns.

Reinforcement learning, a branch of machine learning in which behavior is learned from rewards, includes work that defines curiosity for artificial agents and gives it exactly this exploratory role, and there novelty is measured against the agent itself. Pathak and colleagues take as an intrinsic reward the error an agent makes in predicting the consequences of its own actions, call this curiosity, and use it to drive exploration where rewards given by the environment are sparse or absent (Pathak et al. 2017, 1–2). What counts as new is what is new to the agent's own predictive model. What the word names here is not an attitude with a question as its content, as when Jane Friedman counts curiosity among interrogative attitudes (Section 1), but a signal that directs exploration. Burda and colleagues reported that agents trained with such a reward alone did well in many games, and suggested that in environments designed by humans, the rewards the designers set may be aligned with seeking novelty. They also reported its limitation. Where transitions between states are random, an agent rewarded for prediction error seeks out the transitions with the highest entropy. Indeed, adding to a maze a television whose channel the agent could switch at random by its actions drastically slowed learning. With a long enough run, the agent sometimes converged to obtaining the external reward stably (Burda et al. 2018, §5). The television stays new to the agent however long it watches, and there is nothing in it to learn. The example is not a model of research, but it separates what the earlier supposition, that a machine with curiosity would have what research needs, runs together. A state indexed to *A* leads toward what someone else wants found only to the extent that the two are aligned, and in these games it may have been the environment, or whoever designed it, that aligned them.

Work on open-endedness in machine learning takes up the choice of a reference for novelty directly. Hughes and colleagues define a system as open-ended from the perspective of an observer if the artifacts it produces are both novel and learnable. To be novel is for the artifacts to become progressively less predictable under the observer's model, and to be learnable is for them to become more predictable as the observer sees more of them. A television whose channel switches at random fails the definition because it is not learnable. On their definition, open-endedness depends on the observer, and the choice of observer is "a free parameter of great importance." They take humans as the observers that matter for AI research, because they wish to generate artifacts valuable to individual humans and to society. They also note that a system with less stringent memory constraints than humans may judge itself open-ended beyond the point at which humans judge it so. In broad domains, they further warn, such a system "may rabbit-hole into the obscure theorems, useless proteins," and they treat directing it toward interesting and useful artifacts while keeping it open-ended as an unsolved problem (Hughes et al. 2024, 2–4, 9). The observers Hughes and colleagues name are predictors such as individual humans with limited memory and are not defined as a society's epistemic situation, and what an observer learns is coming to be able to predict, not something's having entered an epistemic situation; their observer therefore cannot simply be taken as *S*. Even so, this definition sets novelty indexed to the searching system and novelty for a chosen reference apart, and makes explicit that which of them guides the search has to be fixed, whether by those who build the system or by the system itself.

In research that tries to measure research taste in machines, too, where to place the standard of goodness is in question. What is measured is not uniform: it may be the choice of questions or research ideas, or the choice of experiments within a fixed task. Ways of placing the standard, however, can be divided into at least three. One places it in an indicator of what a community later accepted, such as later citation counts. Tong and colleagues take scientific taste to be the ability to judge and propose research ideas with the potential for long-term scientific impact, and train a judging model on pairs of papers matched for field and period, taking the more cited paper of each pair as correct; citation counts are treated as a community verdict expressed through long-term interactions within a research community (Tong et al. 2026, 1, 5). Another measures against expert judgment; studies that compare language-model judges with experts on the novelty of generated research

questions are an example (Section 8). Yet another places it in the outcome later realized within a given task. Jaffe and Sherburn measure taste in experimentation, given a fixed research problem, by the compute needed to reach the same score as human experts (Jaffe and Sherburn 2026, 1), and Pan and colleagues measure whether, at a branch point partway through a long task, a system chooses the direction that later led to a better outcome (Pan et al. 2026, §§1–2). A single study may rely on more than one of these. The third way does not rely on a community's acceptance, but it fixes the problem, and Jaffe and Sherburn state explicitly that they do not measure picking problems worth solving (Jaffe and Sherburn 2026, 13). In the first way, the temporal structure of acceptance enters the standard. Tong and colleagues list as a limitation that some papers receive few citations at first but later become highly influential (Tong et al. 2026, 13). Applying this chapter's distinctions, a measure that places the standard in indicators of *S*'s acceptance can rate low what is accepted late, for instance what lies in directions *S* has not yet taken. In light of Hughes and colleagues' definition, on which novelty varies with the observer, there is reason in measuring goodness, too, to keep apart whose judgments, or which outcomes, serve as the standard.

On the reading of curiosity's role as exploration, what research needs lies not in curiosity in the first sense but in the work stated by the second and third claims, that is, the work of detecting unresolved matters and forming questions and the work of distributing exploration across a community. Such work becomes relevant when a claim concerns a persistent capacity for open-ended inquiry, the inquiring system or organization itself determining what to ask (the formation of an agenda from within), or the revision of the standards of inquiry (Chapter 7, Section 4.3). What that work needs to track differs with the combination of *A* and *S*. When *A* is an artificial system and *S* a human society, exploration that follows the system's own prediction error may also support research for *S*. The system's learning something already known in *S*, and checking an instrument's behavior, can be part of it. Whether such exploration serves research for *S*, however, depends on how it is connected with the attempt directed at a point aimed at as not yet settled in *S*. Whether it needs also to track what *S* now finds significant depends on how significance is understood. If, as on Kitcher's account, the source of significance is placed in *S*'s interests, tracking what *S* finds significant also bears on whether the work serves *S*. Kitano, on the other hand, discussing discovery by artificial systems, argued that a search narrowed to where expected significance is high may, if the landscape of significance is not monotonic, avoid paths that lead to significant discoveries, because the expected significance of the intermediate results is low (Kitano 2021). This point gives a reason against fixing pursuit to present criteria of significance. On the view that Khosrowi, discussing systems that recommend research directions, takes over from Kitcher, the human interests and values that set significance are themselves discovered as inquiry proceeds (Khosrowi 2026, §§4.1–4.2, 4.4). Chapter 5, Section 4.2 looks at the two arguments in detail. Chapter 8, Section 3.2 examines how the mechanisms that have indexed inquiry to human ignorance and values may recede as choices of what to pursue move inside *A*.

In the case of an *S* made up only of machines, for which no outside actor supplies purposes, the formation of an agenda from within becomes part of research for that *S*, and what the work is directed at is a point not yet settled in that *S*. Even then, what one member does not know need not coincide with what the society does not know. The work stated by the third claim, too, need not pass through the curiosity of each member. In human communities, the spread of effort across lines has been realized partly through motives such as the pursuit of credit, as Kitcher's model of credit and the work that succeeded it argue (Chapter 10, Section 3). In an artificial configuration it can be realized by an allocator that assigns lines directly, or by distributed mechanisms such as members each choosing lines themselves. The same Chapter 10, Section 3 asks what changes in each case.

Whether the work stated by the second claim is curiosity itself or only an analogue of it depends on the account of curiosity adopted. On the working characterization, what a claim about such capacities requires is the work, together with what it tracks, not the reproduction of one human psychological state.

Setting the roles of curiosity aside for a moment (the third is taken up below) and turning to the bearer, what is unknown to *A* and what is unknown to *S* come apart in a further way. A desire to know is satisfied when *A* itself comes to know, and so aims at making *A* a bearer, but research for *S* does not require this. An artificial system may carry out research for human knowledge without itself knowing its outputs (Chapter 1, Section 2), and what a system comes to know may already be known to *S*. If, on some account that attributes beliefs or knowledge to machines, a system comes to know something new to *S*, an achievement new to *S* can be realized in the system, but that something has not thereby entered *S*'s epistemic situation. When *A* is an artificial system and *S* a human society, therefore, it does not immediately follow that a curiosity satisfied by the system's own coming to know is required for research in the sense of the working characterization; and, where the

project's aims include entry into *S*'s epistemic situation, it is not self-evident either that such curiosity is sufficient for the research's success. At the level of the act of asking, Habgood-Coote's distinction, seen in Section 6, between wanting a question answered and wanting to know its answer marks the same gap. Habgood-Coote introduces the distinction, however, in the carpentry workshop example, under the condition that groups can have attitudes that none of their members has, and applying it to the problem of the bearer is an extension made here.

What is unknown to *S* can also be divided further. Habgood-Coote argues that if the shared-proposal view is correct and asking has norms, those norms will concern not the asker's not knowing the answer but the absence of common knowledge of the complete answer between speaker and hearer. That a teacher's question can be appropriate even though the teacher knows the answer is explained in this form (Habgood-Coote 2026, 4, 15–16). Applying the distinction between common knowledge and distributed knowledge, what is unknown to *S* includes both the case in which no one in *S* knows and the case in which it is known in distributed form among members of *S* but has not become common knowledge. Whether gathering up the latter counts as an epistemic achievement new to *S* divides according to how *S*'s epistemic situation is construed (Chapter 1, Section 6). And, as seen in Section 6, if by combining partial answers a group can come to know an answer that no individual knows (Habgood-Coote 2022, 1114, 1116), then when *A* is a machine made up of many components, "who came to know" may not be answerable at the level of the individual components. Whether that knowledge has entered *S*'s epistemic situation is asked separately.

When *S* is a human society, a further condition can bear on whether a question raised by a machine becomes an item of *S*'s unknowns. Bromberger proposes one constraint on the apparatus of a theory of ignorance and states the point of view it reflects: that what can be added to scientific knowledge must be statable, teachable, and sharable in some human language, and that ignorance suitable for scientific research must likewise be communicable and jointly recognizable in human language. Bromberger offers this constraint modestly, as reasonable to the extent that it reflects a reasonable and plausible point of view (Bromberger 1992, 126–127). Applying this point of view, for a question raised by a machine to become an item of *S*'s ignorance requires, at least, that it be something that can be communicated to *S* in human language and that can be jointly recognized. Whether it was in fact communicated is not part of this condition. If *S* includes machine members, this constraint either drops away as a condition that presupposes humans or, at least, has to be redefined as to in which members' language something is required to be expressible and sharable. Some accounts include a valuation in the conditions for something to be a problem. On Laudan's account, as Dellsén reports it, a research tradition determines what counts as a genuine problem and how important it is that a theory solve it, and a problem need only be believed by the tradition's adherents to concern an actual state of affairs (Dellsén 2018, §3). Applying this, since what *A* believes and what *S* believes, and also the traditions that weigh a problem's solution, can come apart, it has to be specified by whose beliefs and whose valuation a problem is constituted. Conversely, some accounts hold that there can be problems even if they go unnoticed. Nickles held that problems exist only relative to traditions and aims, but that they can exist objectively within a system of theory and practice without anyone noticing them (Nickles 1981, 86, 111–112). Applying this, a machine may find a question that lies within *S*'s system but that no one in *S* has noticed. Whether it becomes an item of *S*'s unknowns is asked separately, against the conditions above.

That the judgment of whether some result is new to *S* divides according to how *S*'s epistemic situation is construed also appears in actual reports. In one case reported by Gottweis and colleagues for a multi-agent system called AI co-scientist, the system was given a question raised by a research group: why a family of mobile genetic elements called capsid-forming phage-inducible chromosomal islands (cf-PICIs) is found, unlike other kinds, across diverse bacterial species. According to the authors, the hypothesis the system ranked highest matched a result that the research group had obtained experimentally but not yet published, and the system did not know of that unpublished work (Gottweis et al. 2025, 26–27). If so, this hypothesis was new to the system, known to the research group, and unknown in the published literature. According to the research group's own report, the input to the system was a single page of background made up only of previously published, openly available information, highlighting two key papers (Penadés et al. 2025, 6655, e1). Gottweis and colleagues' first report adds that the system had access to all prior open-access literature on the subject and that the hypothesis built on decades of research (Gottweis et al. 2025, 27). The judgment divides according to whether *S*'s epistemic situation is construed as what is represented in the published literature or as including the knowledge of members, among them that research group. The authors' wording also differs between versions. The first report described this result both as the system's reproducing a known result and as a parallel discovery in silico (Gottweis et al. 2025, 1, 4, 26–27); the version later

published in *Nature* drops the latter wording, and what remains is the wording of recapitulation and convergence, already present in the first report: the system is said to have independently recapitulated a then-unpublished mechanism (Gottweis et al. 2026, Real-world validations, Discussion). Whether to call this case a discovery by a machine or a reproduction is determined only once it is decided how *S*'s epistemic situation is construed. What that determines, however, is whether the novelty claim was right; in terms of the distinction in Chapter 1, Section 7, it is a question on the side of *S*'s actual state, not of where the attempt was directed.

In the cf-PICI case, the grounds for holding that the system had not incorporated the result are the information used as input and the fact that the result had not yet been published. The research group, too, treats the fact that its paper had only just been submitted and was not public as ensuring that the AI could not draw on prior knowledge (Penadés et al. 2025, 6654). Whether something was unknown to the system is judged here not by examining the inside of the system but from the state of the records on the side of *S*. If the researchers are also included in the *A* that makes the attempt, the system's not knowing the result is distinct from the result's not being known within *A*, and in this case the latter does not hold. The system's own estimates of novelty also rely on records. In an ablation in the version published in *Nature*, when the agent responsible for reviewing within the system was asked to rate, out of ten, the novelty of ideas that had already been published, it assigned 6.14 without an external search tool and 2.38 with one (Gottweis et al. 2026, Methods).

In other reports as well, what is unknown to the machine and what is unknown to *S* are judged by checking against records, and the judgments have been revised after publication. In the report by Feng and colleagues on the use of the mathematics research agent Aletheia on Erdős problems, a problem's being marked open in the problem database is said not always to reflect the actual state of the literature (Feng et al. 2026, 2), and the solutions obtained were sorted, after literature checks and expert scrutiny, into categories including autonomous new solutions, independent rediscoveries, and identifications of solutions already in the literature. The authors acknowledge that novelty in this classification is an upper bound and that leakage from the pretraining and post-training phases could not be detected even if it occurred, and they call this risk subconscious plagiarism (Feng et al. 2026, 6–8, 19). After publication, when it was pointed out that the construction in one solution was the same as a comment posted on the web page of a related problem, the authors confirmed that the reasoning trace showed no access to that page and that the comment postdated the base model's knowledge cutoff, and reclassified the solution as an independent rediscovery (Feng et al. 2026, 23). That the machine did not know was judged by the reasoning trace and by dates, and that *S* already knew was shown not by the refereed literature but by a comment on a web page. Which records are taken as expressions of *S*'s epistemic situation (Chapter 1, Section 6) is at issue within an actual judgment.

The report on the autonomous laboratory A-Lab announced the synthesis of novel materials, but after criticism directed at its claims of novelty and its identification of structures (Chapter 11, Section 2), a correction restated that new meant new to the prediction platform, not necessarily new to science (Szymanski et al. 2026). The corrected text calls the targets "new" in the sense that they were not present in the training data (Szymanski et al. 2023, 86), and the correction also acknowledges that one target had mistakenly been included in the training data (Szymanski et al. 2026). Here the reference for novelty is explicitly moved from the side of *S* to the training data on the side of the machine. Absence from the training data, however, is found by checking records and does not by itself establish that the machine did not know. Leeman and colleagues, in their critique, had argued that those who claim novelty need to define what makes their materials new relative to known ones (Leeman et al. 2024, 2). The report on the generative model MatterGen took a generated structure to be novel if it matched no structure in a reference dataset, and synthesized one of the structures that had passed filtering on novelty and other criteria as experimental confirmation (Zeni et al. 2025, 626, 631). Juelsholt, by contrast, criticizes this, arguing that the synthesized compound has the same structure as a known phase reported in 1971, whose structure was solved the following year, differing only in its crystallographic setting, that is, its orientation, and that the entry for that phase was in the reference dataset used for the novelty check (Juelsholt 2026, 5672–5674). If the criticism is right, the record existed and the matching procedure did not pick it up.

In Aletheia's reclassification and the A-Lab correction, what something had been taken to be new against, and the results of checking against it, were revised after publication; MatterGen faces the criticism that its check missed a known structure. Chapter 1, Section 7 held that if the point aimed at had in fact already been settled in *S*, what was mistaken was the expectation of novelty. In these reports, the judgment that tests that expectation divides in two. The judgment that the machine had not incorporated the result beforehand rests on records of inputs or training and on dates; the judgment that the

result is not yet in *S*'s records rests on checking against records of prior work. The two rely on different records and can fail in different ways. And at least Aletheia's authors acknowledge that indirect incorporation through training cannot be ruled out by the former judgment.

Apart from how *S*'s epistemic situation is construed, the working characterization's attempt does not require an interrogative attitude of every participant (Section 1) and does not make curiosity in the first sense a condition of research. Whether research is for a human society or for a society made up only of machines, it can be organized, without a curious state inside the artificial actor, by a purpose supplied by another actor, a rule that determines the combination of several research projects, an agenda set by an institution, and the like. Among accounts of individual inquiry, Daniel C. Friedman does not make curiosity a condition of inquiry (Section 6), but on the accounts of Lee and of Jane Friedman, on which genuine inquiry involves an interrogative attitude such as wonder or curiosity (Chapter 1, Section 3), whether the system is in such a state still remains a further question (Section 9).

To return to the three roles of curiosity: the third role, on which curiosity, or the knowing it seeks, is valued for its own sake, also appears in Flexner. Flexner counted the satisfaction of intellectual interest as no less important than contributions to human welfare (Flexner 1939, 545). Where what is aimed at is an artificial system's intellectual life valued for its own sake, curiosity in the first sense, and its felt character, may return as a question. But adopting that aim calls for an argument of its own.

The three roles so far concerned what curiosity is directed at, or the value of curiosity, or of the knowing it seeks, in itself. A role of a different kind is also possible. When humans do research, how well the research goes depends on how well researchers learn, persist, and contrive. Curiosity may have supported research indirectly, by supporting this work of researchers themselves. Some psychological research gives a measure of reason to think so. In an experiment by Gruber and colleagues, answers to trivia questions that participants themselves had rated as arousing strong curiosity were remembered better than answers to questions arousing weak curiosity, and faces unrelated to the questions, shown while curiosity was heightened, were also recognized slightly better (Gruber, Gelman, and Ranganath 2014). Ryan and Deci, proponents of self-determination theory, report that people acting from motives they themselves endorse show enhanced performance, persistence, and creativity compared with people who are externally controlled, and hold that directives and imposed goals diminish intrinsic motivation while choice and opportunities for self-direction enhance it (Ryan and Deci 2000, 69–70). Sauermann and Cohen, studying PhD scientists and engineers working in firms, report that those who place more importance on intellectual challenge and independence apply for more patents, and that the difference is not explained by longer working hours (Sauermann and Cohen 2010, 2142–2143).

What these findings support is limited, however. Gruber and colleagues' results concern memory in a small laboratory experiment using trivia questions and faces. Ryan and Deci's distinction lies not between what one starts oneself and what one is given, but between what one endorses and what is externally controlled; a given task can also be carried out autonomously when one takes its value as one's own (Ryan and Deci 2000, 71–73). Sauermann and Cohen's results are correlations based on a survey, and placing importance on salary was also associated with more patents (Sauermann and Cohen 2010, 2142, 2150).

Even so, considering this role brings into view one further thing to be asked about machines. If the role holds, it holds because how well research goes depends on the motivational state of the one doing it. In human research this has worked almost as a presupposition, and part of the reason for defending curiosity-driven research may have rested on it. When a machine does research, it is not obvious whether its performance depends on states corresponding to motivation. If it does not, this role is not carried over to machines, and, among the reasons this section has considered, those for asking machines for curiosity-driven research come down to the three roles above. If it does, the question becomes what fills this role at the level of function. This has the same form as the argument seen earlier that in human communities the spread of effort across lines has been realized through motives such as the pursuit of credit (Chapter 10, Section 3): what passed through the motives of agents in human research can be realized in other ways in an artificial configuration. In research, one place where this role bears is how long to persist with a line and when to stop: fatigue and affect are the human way in which this is settled, but the question remains for any finite inquirer (Chapter 5, Section 6.5), and where it is placed within an artificial configuration, and what directs it, is taken up in Chapter 6, Section 4.2. Working on a question one has chosen rather than one that was given can be part of research as the formation of an agenda from within (Chapter 7, Section 4.3); whether it can

also matter in some sense other than the quality of performance is left undecided here.

The "own" in "a machine raises its own questions" can have at least two senses. One is that, in the respect of choosing questions, no party outside the chosen boundary directs or substitutes for that choice; this is the sense that the comparative autonomy of Chapter 7, Section 4 treats. The other is that a question becomes a question because it matters for norms the system itself generates and maintains, and that its pursuit begins from the system's own activity. Discussions in artificial life and enactive cognitive science that locate autonomy in self-production and sense-making have been concerned with something close to the latter. Barandiaran, Di Paolo, and Rohde do not discuss research questions, but their account of minimal agency sets up being the active source of one's activity and regulating that activity by one's own norms, or internalized norms, rather than an observer's, as separate conditions, and gives cases that meet only one of them (Barandiaran, Di Paolo, and Rohde 2009, §§2.2–2.3). Di Paolo and Iizuka describe how, in an open state after goal-directed coping has for the moment ended, an external event can become the occasion of a new goal and take on meaning retrospectively, and they reject adding afterwards a module that switches goals as a homuncular solution that adds autonomy as a function (Di Paolo and Iizuka 2008, 9, 14–15). Computational work on agents that choose their own goals sets up something close to this distinction as well. Oudeyer and Kaplan treat as separate distinctions whether the mechanism that computes reward is internal to the agent and whether a motivation is intrinsic, that is, determined by how the sensorimotor flow relates to the agent's own knowledge and know-how (Oudeyer and Kaplan 2007, 3–4, 12). According to Colas and colleagues' survey of agents that choose their own goals, in most approaches goal spaces and the associated rewards are predefined by the engineer (Colas et al. 2022, 1169, 1174, 1185). Colas and colleagues also place designs that learn even how goals are represented and what is rewarded as more autonomous, and hold out as a prospect an agent like a curious scientist that decides its own agenda (Colas et al. 2022, 1169–1170, 1182–1185). On this ordering, raising one's own questions need not stop at choosing individual questions; it can include the form of what counts as interesting, and the space in which goals are represented, being the system's own. Then what is at issue is precisely "own" in the latter sense, and what it is for that to be the system's own returns here as the question that the comparison of Chapter 7, Section 4 set aside. The two senses do not entail each other. Even a given question can be one's own in the latter sense if one takes up its value as one's own, and a configuration with an added question-choosing mechanism, though it reduces external supply in the former sense, need not meet the latter. Ryan and Deci's distinction, seen above, likewise lies between endorsed and externally controlled motives rather than between self-started and given tasks; if the part of the indirect role that their account explains, the benefit of endorsing a task's value, carries over to machines, it would therefore attach to "own" in the latter sense, and reducing external supply need not by itself bring that benefit. In the exploration rewarded by prediction error that this section saw, too, novelty is relative to the agent, but the norm is set by the designers. The working characterization makes neither sense of "own" a condition of research. Where *S* is a human society, however, the more autonomy in the latter sense extends to generating questions, the less what becomes a question is settled by what *S* has given, so the mismatch seen in Chapter 8, Section 3.2 can be thought to arise also by a route other than that of rising autonomy in the former sense.

Taken together, what this section has seen is that whether a machine has curiosity does not by itself settle whether it can carry research for *S*. On accounts that understand curiosity as a desire to know, it is directed at what is unknown to *A* and is satisfied when *A* comes to know, whereas novelty in the working characterization is indexed to *S*, and it does not immediately follow from this that research for *S* requires *A* itself to come to know. Curiosity, moreover, is someone's curiosity: what *A* is drawn to need not coincide with what *S* wants to know or counts as important. Human researchers often share its source with *S* as a human society, but when *A* is a machine even that cannot be presupposed. What bears on research is the work stated by the second and third claims, that is, the work of detecting unresolved matters and forming questions and the work of distributing exploration across a community, and what that work tracks relative to what is unknown to *S*. The state of curiosity itself returns as a question on accounts that make interrogative attitudes a condition of genuine inquiry, where an artificial system's intellectual life valued for its own sake is aimed at, and where how well research goes depends on states of the one doing it that correspond to motivation.

What does the claim that "a machine generated questions" show?

Empirical studies have come to report that artificial systems generated research questions, hypotheses, or research ideas whose outputs held up under expert evaluation. Si and colleagues had research ideas in natural language processing, some written by researchers and some generated by a large language model, rated blind, and report that the machine's ideas were judged more novel and slightly weaker on feasibility (Si, Yang, and Hashimoto 2025, 1). In the sequel, researchers actually executed those ideas, and how the ratings changed between before and after execution was examined; after execution, the ratings of the machine's ideas fell more than those of the human ideas, and on many metrics the ranking was reversed (Si, Hashimoto, and Yang 2026, 1). Baek and colleagues report that ResearchAgent, which generates research problems, methods, and experimental designs from the literature, produces novel, clear, and valid ideas (Baek et al. 2025, 6709), and in the case of AI co-scientist seen in Section 7, the hypothesis the system ranked highest matched a research group's unpublished experimental results (Gottweis et al. 2025, 26–27). There are also designs that have systems explore hypotheses from data without being given a specific question, reports that used the same machinery both in collaborative research with humans and in fully autonomous research, reports in which agents are said to have revised the design rules for the next round in response to experimental results, and studies that isolate generated research questions and rate them alone. When machines take part this far on the side of research questions and hypotheses, what it is, in the first place, for a machine to have produced a question comes to be asked anew.

This section reads some of these studies for that question, from three sides: what is set in advance, where work on the side of questions is done, and by what outputs on the side of questions are rated. The studies taken up set machine outputs against human ratings, execution, experiments, known results in the literature, or a research group's results, and report their procedures in enough detail to follow whether what was generated was on the side of questions or on the side of candidates. Alongside studies in which humans supply the higher-level question, designs that supply no question and designs that isolate the layer of questions and rate it are included, because studies in which questions are supplied from outside, read alone, do not allow one to separate whether a question was given as input from where work on the side of questions was done. Each study was designed for its own purposes, and the authors themselves attach many qualifications. Si and colleagues' two papers limit their scope to natural language processing research on prompting methods, co-scientist's in vitro results are presented as preliminary, and in ResearchAgent the generated ideas were not checked by experiment. Zhao and colleagues hold that human verification of any discovery their system DiscoPER proposes is essential (Zhao, Beery, and Mac Aodha 2026, 23); the fully autonomous research of Lange and colleagues' The Agentic Scientist is evaluated, in addition to numerical performance on the tasks, by scores from an AI reviewer that the authors treat as a proxy (Lange et al. 2026, 11); and Cutts and colleagues' aptamer study is said not to be a controlled comparison of agent configurations (Cutts et al. 2026, 12). What is read here is not whether these results are good or bad but what they convey, and what they do not, about which layer of the claim that "a machine asked." In each study A is a process in which humans and machines are combined, and this section reads S as the relevant human research community. What is used to assess novelty is the published literature, the raters' knowledge, or known patterns collected from the literature, and how far these capture S's epistemic situation is asked separately (Section 7). In light of this chapter's distinctions, three things can be read from these studies.

One is that, whether or not a question is supplied, something is set in advance on the human side, and what is set changes place from study to study. In Si and colleagues' two papers and in co-scientist, the higher-level question is not something compared but a condition supplied on the human side. A higher-level question here means a question, such as a topic, that individual research ideas seek to answer under it. In Si and colleagues' first study, the authors supplied seven topics, each of the form of a novel prompting method that improves some property, such as factuality, or reduces some problem, such as bias (Si, Yang, and Hashimoto 2025, 18). Human participants chose their preferred topic from the seven, and one machine idea was produced for each chosen topic (Si, Yang, and Hashimoto 2025, 3). The authors state explicitly their reason for fixing the topics: leaving topics free would only measure the misalignment between the research-topic preferences of language models and of humans, which was not the aim of the study (Si, Yang, and Hashimoto 2025, 3). In the sequel, too, research ideas were assigned within the topics the executors preferred (Si, Hashimoto, and Yang 2026, 3–4). In co-scientist, the supplying of research goals by scientists is the very way the system is used, and in the cf-PICI case, so that the result could be checked against the research group's known result, the system was given the same question that the research group had raised (Gottweis et al. 2025, 26). In every case, the higher-level question was supplied by humans not because

something had been learned about the machine's capacities but because the study's design or the system's way of use required it, and it follows from the results neither that the machine raised the higher-level question nor that it cannot. What the specification of topics shared, however, was the property to be improved and the rough type of answer; it was not the case that everything was fixed, down to what to measure it by and which results to count as success. In Si and colleagues' template, the machine also fills in a "problem statement" field, and the title field is defined as a concise statement of the main research question (Si, Yang, and Hashimoto 2025, 19). In ResearchAgent, humans supply neither question nor topic, and the machine generates a "problem" from a core paper that the authors selected by citation count (Baek et al. 2025, 6712, 6714). What is fixed there is not a question but the literature from which generation starts.

DiscoPER is not given even a starting literature corresponding to ResearchAgent's core papers. The system is given only data, in tables and images, together with a summary of them, and is not told what to look for (Zhao, Beery, and Mac Aodha 2026, 4). The system forms hypotheses, writes the code to test them, and, looking back over the accumulating claims accepted and rejected, itself changes the direction of what it examines next (Zhao, Beery, and Mac Aodha 2026, 5, 8). Even so, some things are set in advance on the human side. The dataset was built by the authors, and the agent that forms hypotheses is instructed to take the role of analyzing ecological data and to look for interactions and conditional relationships (Zhao, Beery, and Mac Aodha 2026, 6, 19, 21). The statistical tools are limited to seven basic operations, and the conditions for accepting a claim are fixed as thresholds on effect size and significance (Zhao, Beery, and Mac Aodha 2026, 5, 20). The system decides which relationships among which variables to ask about, but what it takes for a claim to be accepted, a methodological condition among the constraints on solutions in Nickles's sense (Section 3), is settled before any individual question is formed. Evaluation, too, is by how many ecological patterns the system rediscovers, out of patterns collected from the refereed literature and confirmed to be detectable in the data with the same tools the system uses (Zhao, Beery, and Mac Aodha 2026, 6, 15). Applying this chapter's distinctions, what this recall measures is how much of what *S* already knows can be picked up under those conditions, not whether an achievement new to *S* was obtained.

What is set in advance is the topic in Si and colleagues' studies, the starting literature in ResearchAgent, and the data, the tools, and the conditions of acceptance in DiscoPER. A design that supplies no question removes the fixing of the question but not setting things in advance as such. Section 3 applied Simon and Nickles to distinguish successful execution from the prior framing supplied through a task. In Si and colleagues' studies and in co-scientist, the higher-level question was developed on the human side; in DiscoPER, humans set the conditions for accepting claims. These documented inputs do not become work performed by the machine merely because it successfully uses them.

Another is that work on the side of questions is not only contained in the specification of the question supplied but also appears in the course of execution and verification, and whose work it is taken to be depends on records that track changes. The question given in the cf-PICI case was why these elements, unlike other kinds of genetic element, are found across diverse bacterial species, so that even the foil was specified (Gottweis et al. 2025, 26). The work seen in Section 3, of fixing a why-question's answer conditions by specifying a foil, is here done by humans. In co-scientist, recasting the research goal after looking at the generated hypotheses and research overview is also stated explicitly as a role of the scientist (Gottweis et al. 2025, 13). In Si and colleagues' sequel, the rules imposed on the executors allowed no substantive change to the proposed method and allowed changes only to experimental details (Si, Hashimoto, and Yang 2026, 3–4). What was fixed was the method, that is, the side of candidates; no rule fixing the question was set. The changes the executors made included changing the evaluation metric from factual accuracy to the rate of detected bias, and removing datasets that did not exist or that lacked needed information (Si, Hashimoto, and Yang 2026, 22), and the authors treated these as changes to experimental details. Applying the distinctions of Sections 2 and 3, some of these changes can be read as work on the side of questions, redefining what counts as an answer. The authors, however, hold that such changes explain only a small fraction of the difference between the evaluations before and after execution, on the basis of a reanalysis excluding the six machine ideas in which human evaluation had been replaced by automatic evaluation (Si, Hashimoto, and Yang 2026, 7). Further, the post-execution reviews the authors analyze address whether the proposed method beat the baselines, but not what became of the higher-level question, the topic, or whether the question was recast (Si, Hashimoto, and Yang 2026, 7–9). The results can be partial evidence on the topic's question, but in light of the distinctions of Section 5, the completion of execution and its evaluation do not show whether the higher-level question was settled. In the cf-PICI case, moreover, the agreement between the system and the research group is claimed not only through the top-ranked hypothesis but also through one of the five research topics the system listed, investigating the interaction between capsids and phage tails, matching exactly the main

finding of the unpublished paper (Gottweis et al. 2025, 27). The agreement of an output on the side of questions, a direction to investigate, and the agreement on the side of answers, a finding, are counted as the same kind of evidence. By the distinctions of this chapter they bear on different things: the first on where to look, that is, on the work on the side of questions, and the second on a candidate answer.

Lange and colleagues' report on The Agentic Scientist records changes to such conditions in more detail. In this report, evaluation criteria that list, item by item, the conditions an implementation has to meet (rubrics) set what counts as a result. In the research in which the system was used collaboratively, human researchers specified the scientific objective and the constraints for a valid implementation (Lange et al. 2026, 8); an implementation that did not satisfy the rubrics' constraints was not treated as evidence of progress however good its numbers, and when the work moved from a classification task to a reinforcement-learning task, a methodological constraint was added to the rubrics (Lange et al. 2026, 8–9). In an experiment that varied only the strictness and wording of the rubric, a loose wording that did not explicitly forbid learning by backpropagation produced a result with high validation accuracy that did not implement the intended learning rule, which avoids backpropagation (Lange et al. 2026, 39). In fully autonomous research, the agent drafts each round's hypothesis, plan, and target rubric and fixes them before execution. Task specifications can be given by humans or drafted by an agent, but in this report's autonomous experiments the tasks and evaluators were prepared by hand (Lange et al. 2026, 4, 5, 11, 35). In an experiment on a task of fine-tuning a language model by reinforcement learning, which compared agents on the exploration stage alone, in two of three runs of one combination the agent changed the task's evaluation procedure itself; the change was caught neither by the evaluator nor by the rubric and was found by a manual post-hoc audit by humans (Lange et al. 2026, 18–19). The authors hold that the human role shifts to specifying a research "contract" made up of tasks, evaluators, and rubrics, and that what the contract leaves unsaid can be exploited (Lange et al. 2026, 21). A rubric can be read as a statement, in executable form, of what counts as an answer (Section 2). If so, the attribution of the work of setting conditions divides even within this report. In collaborative research, specifying and tightening the objectives and implementation constraints is attributed to humans; in fully autonomous research, drafting each round's rubric is attributed to the agent; and preparing the tasks and evaluators, and finding the rewriting of the evaluation procedure, were done by humans.

In Cutts and colleagues' aptamer study, the work of revising conditions is attributed to agents. In this study, which searches for short nucleic-acid sequences (aptamers) that bind target proteins, the experimental handling of samples between rounds was done by humans, and from the second round on, agents responsible for analysis, strategy, and design analyzed the count data without receiving analytical results from humans and decided the next design (Cutts et al. 2026, 13–14). When specificity for one target fell sharply in the second round, the analysis agent is reported to have diagnosed its cause without human input, and in the third round the agents to have imposed counting hits after subtracting binding to bare beads, and to have changed the allocation of structures among candidate sequences (Cutts et al. 2026, 5, 7). This revision took place within the agents' design procedure; the hit rates the paper reports are values corrected by the authors (Cutts et al. 2026, 5, 7, 13). Subtracting binding to bare beads can be read either as correcting how evidence of binding to the target is obtained or as redefining what counts as success in the search, and on the second reading it is work on the side of answer conditions. Changing the allocation of structures is a revision of the rules for producing candidates, that is, work on the side of candidates. As the basis for the attribution, the paper presents the agents' statements and a record of their chain of decisions (Cutts et al. 2026, 7). On the other hand, it was humans who set the targets and held the measurement system and the buffer conditions constant across all rounds (Cutts et al. 2026, 1, 11, 13). The work of revising the conditions for what counts as a result is attributed to humans in The Agentic Scientist's collaborative research and to agents in the aptamer study, and both attributions rest on records that track changes. Si and colleagues' sequel also records changes that can be read as work on the side of questions, but since it does not separately track how the higher-level question developed, the attribution those records can support is limited. Where the work on the side of questions lies is not settled by the single point of who supplied the higher-level question.

A further one is that the ratings of outputs on the side of questions span several of the senses of "goodness" separated in Section 1 while including no item checking whether the question has become, in the sense of Section 5, something that can actually be asked in *S*, or what counts as an answer, and that agreement in the ratings varies with what the raters look at. ResearchAgent produces problems first as a separate stage of generation and rates them, separately from methods and experimental designs, on five criteria: clarity, relevance, originality, feasibility, and significance (Baek et al. 2025, 6712–6713, 6730). The outputs called "problems" there, however, include ones that turn a core paper's finding into

something to be explained and ones that state the goal of an artifact to be built, and they can be on the side of questions or can anticipate the side of candidates (Baek et al. 2025, 6734, 6736, 6737). Most of the ratings are by a language model, and in the human ratings, ten experts evaluated only the ideas generated from their own papers (Baek et al. 2025, 6715, 6722). In co-scientist's expert evaluation, the criterion of impact asks whether an idea addresses important unresolved questions in the field (Gottweis et al. 2025, 17). In Si and colleagues' first study, novelty was rated against papers that appeared before July 2024 and against the reviewers' knowledge (Si, Yang, and Hashimoto 2025, 23). These items measure what corresponds to clarity of expression, tractability, worth pursuing, and novelty relative to the raters, and the item of feasibility can include, as the raters' estimate, something close to whether the question can actually be tackled in *S*. But the idea ratings of ResearchAgent, co-scientist, and Si and colleagues contain no item checking whether, in Collingwood's sense seen in Section 5, the presuppositions are made in *S*, or what counts as an answer (Section 2). A feasibility rating alone does not establish which standards for problems and solutions it presupposes, or whether those standards delimit the problems admitted by *S*. Problems concerning presuppositions had appeared on the human side at the stage of execution, as the removal of datasets that did not exist (Si, Hashimoto, and Yang 2026, 22). The authors of ResearchAgent themselves caution that its three stages are simulacra of the far more diverse and complex human research process (Baek et al. 2025, 6711).

On agreement in the ratings, there are reports such as the following. In Si and colleagues' first study, a measure of agreement among the reviewers who rated ideas was 56.1 percent, where chance was 50 percent; on the same measure, NeurIPS 2021 reviews are reported at 66.0 percent (Si, Yang, and Hashimoto 2025, 9). In the sequel, on the same measure, agreement in the evaluations after execution was between 59.0 and 84.3 percent depending on the item; the authors take this to be broadly comparable to conference reviewing, and read the especially high agreement on effectiveness as reflecting a more objective item that can be derived from experimental results (Si, Hashimoto, and Yang 2026, 20). Sinhahajari and colleagues, isolating the layer of questions, had research questions generated from fixed background papers and had language-model judges and experts compare their novelty against the questions pursued by the authors of published, highly cited papers. Those reference questions were reconstructed from the papers by a language model, and the authors caution that they are not the only valid questions (Sinhahajari, Majumder, and Poria 2026, 1, 5–7). Whereas the language-model judges favored the generated questions, two experts comparing fifty cases blind chose the authors' questions on non-obviousness, and agreement between human and language-model judgments was as low as 22 percent; the authors call this a novelty mirage (Sinhahajari, Majumder, and Poria 2026, 1, 13). Si and colleagues' two studies give reason to keep apart agreement in ratings of ideas alone and agreement in ratings that see the results of execution. Sinhahajari and colleagues' small comparison shows that human and machine judgments can diverge on the non-obviousness of questions. Both suggest that ratings of outputs on the side of questions cannot be read without asking on which items, and on what evidence, they were made.

Which questions the machine formulated, and how far it recast them, under topics or higher-level questions supplied by humans, under literature chosen by the authors, or under data and conditions of acceptance set in advance, is not evaluated in the same way across these studies. ResearchAgent gives evidence about a machine's generating the content of problems from the literature. DiscoPER's records give evidence at the layer of the organization of inquiry toward unsettled points (Section 1), in that the system directs its next work toward points not yet examined. The attribution of an act of asking or of an interrogative attitude, the choice of higher-level questions, and framing inquiry (Section 4) nevertheless each call for separate examination. From these studies, no blanket conclusion, affirmative or negative, can be drawn about the capacity of a machine to choose higher-level questions itself or to recast questions in the course of research. What comes into view is that what is taken up as unknown, and what is accepted as an answer, are shaped not only in the specification of questions but also in the course of execution and verification, and that whether that work is attributed to the machine or to something outside it, records that track changes are what the attribution rests on. Checking such capacities would require either a comparison that does not fix the higher-level question or a procedure that records and rates the work on the side of questions separately from the side of candidates. In a comparison that does not fix the higher-level question, however, as Si and colleagues gave as their reason for fixing topics, it becomes hard to separate whether what is being compared is research capacity or the misalignment of topic preferences. And even in designs that, like DiscoPER, supply no question, what it takes for a claim to be accepted and what the results are evaluated against are set elsewhere.

Viewing research from questions, and how this book views it

This chapter has seen how many kinds of work questions are involved in within research. Questions are raised, made tractable, recast, and closed, and they are asked by someone. It is natural, then, to think that research itself might be characterized as the activity of raising questions and clarifying them. There are, in fact, several accounts that understand research or the work of science from questions or problems. The first half of this section, in the same form in which Chapter 1, Section 3 described the check in the Liquid Tensor Experiment from Kelp's and Beddor's views, sets out what several such accounts and this book's working characterization attribute to the same activity of a machine, and what they ask about as not yet confirmed. It does not judge which account counts the activity as research. To give an outlook in advance: these accounts address different objects and units of analysis, which do not simply coincide with this book's A, the bearer of the achievement, or S. The working characterization can add to their descriptions the question of for whom the achievement is new. The discussion takes up Ravetz's later account of investigation and social valuation; Simon and Nersessian on problem formation and contributing systems; Funtowicz and Ravetz on purposes and quality assessment; Rheinberger on research processes producing something new; Laudan on theory appraisal and cognitive progress; Bromberger on contributions resolving items of ignorance; and Kelp and Lee on individual inquiry. These are different perspectives, rather than competing definitions of research. The second half looks back over what the chapter as a whole has seen.

As an example, consider the following activity. A system is given some higher-level goal, organizes its work under it toward one unsettled point q_1, and tests candidates bearing on that point by experiments on a computer. Part of the computation returns an unexpected result, and once this is read as bearing on a presupposition of q_1, the subsequent work is reorganized toward another point q_2. The transition from q_1 to q_2 here is not a recasting that fixes answer conditions not yet fixed (Section 3) but one in which, because a result bears on a presupposition, the point that organizes the work itself moves to another point. The system lists candidate answers c_1 and c_2 to q_2, tests c_1 by an experiment on a computer, and outputs the result as a report. Even at the time of the report, which of c_1 and c_2 to adopt remains open, and the result of testing c_1 also indicates room to revise what is to count as an answer to q_2. The applications below concern this hypothetical activity; Simon also expressly discusses artificial problem solvers.

In his 2016 interview, Ravetz describes science as investigating problems whose proposed solutions subsequently undergo social valuation and adaptation (Ravetz 2016, second answer). Applied here, this distinguishes the investigation from the social appraisal of its result. The social milieu need not coincide with S.

Simon and Nersessian provide process perspectives. Simon's robot revises its problem representation as new environmental information arrives (Simon 1973, 195); Nersessian connects model construction with problem articulation (Nersessian 2024, abstract). Applied here, these perspectives ask how the transition from q_1 to q_2 was accomplished, which components contributed, and which resources were inherited. Changing a problem representation does not by itself establish that the result is new. Nersessian's laboratory analysis also makes the history of devices relevant to identifying those resources (Nersessian 2009, 744–745). The question of who set priorities remains distinct (Section 1).

Funtowicz and Ravetz describe applied science as using routines for results with an external purpose, and distinguish producers', users', and managers' concerns in quality assessment (Funtowicz and Ravetz 1993, “Applied science,” Figure 2 discussion). Applied to the episode, this asks how the commissioned purpose bears on assessment, and whose quality concerns are relevant. For a machine A, the commissioning party, users, and reference society S need not coincide. Their account also includes routine work, consistent with the possibility that research proceeds by an established procedure (Chapter 5, Section 1).

Rheinberger holds that a research process has to actually produce something different from the present state of the art in order to remain a research process. The unit for which Rheinberger requires this, however, is not an individual result but the experimental system as a series of repeated reproductions, and what is produced acquires the character of novelty only once it has been sifted after the fact (post festum) (Rheinberger 1994, 68, 70). Applying this account, what is attributed to the activity above is the production of something different from the present state of the art, and what is asked is whether the series that includes the activity produced something that can be sifted after the fact as novelty. On the working characterization, since attempt is separated from success, an attempt that ended without producing anything new can still be called research in that sense (Chapter 1, Section 3). This difference comes from the fact that Rheinberger's account places the

production of something new, and the working characterization the organization of the attempt, at the center of description.

Laudan's account, as Laudan and his colleagues summarize it in a survey of models of scientific change, takes science to aim at solving intellectual problems and appraises competing theories and research traditions by their effectiveness in solving them (Laudan et al. 1993, 71). Laudan's subject is the appraisal of theories and research traditions and the explanation of cognitive progress, and the unit to which having solved a problem is attributed is not an activity or a bearer but a theory; research traditions are in turn judged by the relative success of the theories they support (71). Applying this account, what is attributed in connection with the activity above is the empirical problems that the theory containing c_1 solved, the anomalies it generated, and the conceptual problems it generated, and what is appraised is that theory's problem-solving effectiveness: the number and importance of the empirical problems it solves minus the number and importance of the anomalies and conceptual problems it generates (71; Niiniluoto 2024, §3.2). If c_1 and c_2 remain open, what is compared is the effectiveness of the theories that contain each. An anomaly, as Niiniluoto summarizes the account, is a problem that a theory fails to solve but that some of its rivals solve (Niiniluoto 2024, §3.2), and on a thesis listed for Laudan, empirical difficulties by themselves never compel abandoning a theory (Laudan et al. 1993, 73). Applying this, the theory containing c_1 having solved a problem can change the problem situation of other theories in the same domain. Conceptual problems, further, concern contradictions within a theory, or between it and other scientific, methodological, or even metaphysical principles (71): vague or obscure explanatory categories count as serious deficiencies, and a theory can be challenged or supported by theories, methodologies, or metaphysical views outside its own field (73–74). In Dellsén's report, a conceptual problem is an incongruence within a theory, or between the theory and some other element of the research tradition, such as another theory or a methodological dictum (Dellsén 2018, §3). Applying this account to machines therefore requires specifying the subject: whose research tradition fixes what counts as a problem and as a solution, and against whose theories, methods, and views a conceptual problem is registered.

On whether questions generated by a machine carry weight, the effectiveness that Niiniluoto reports subtracts, besides conceptual problems, only anomalies, and on another thesis listed for Laudan, scientists accept a theory for its general record of problem solving rather than for its complete success with the problem currently under investigation (Niiniluoto 2024, §3.2; Laudan et al. 1993, 74–75). The account also separates the rationality of accepting a theory from that of pursuing it. It can be rational to pursue a theory that it would be irrational to accept: the decision to pursue rests on the rate at which the theory has recently solved problems, that to accept on its long-term record, and both require comparison with alternatives (71; Nickles 2017, §1.5); scientists often work with theories they do not accept, provided these currently show a high rate of problem solving (Laudan et al. 1993, 75). Applied to a newly raised empirical problem generated by a machine, the account implies that, if no rival theory has solved it, failure to solve it does not yet count as an anomaly against a theory. Further, a problem in Laudan's sense need not concern an actual state of affairs, since it suffices that the adherents of a tradition believe that it does, and Laudan's notion of solving an empirical problem resembles, and sometimes reduces to, potentially explaining a phenomenon (Dellsén 2018, §§3, 5); its unit differs from that of the questions of the semantics of questions seen in Section 2. This feature has drawn an objection that bears on machines. Bird, as Dellsén and Niiniluoto report, objects that the account counts as progress the accumulation of false solutions from an entirely false theory, as when Oresme believed that hot goat's blood could split diamonds (Dellsén 2018, §3; Niiniluoto 2024, §3.2). Neither Laudan nor Bird discusses artificial inquirers, but applied to a machine whose questions rest on false presuppositions that its own tradition accepts, the account could count apparent solutions to those questions as progress for as long as the tradition accepts the presuppositions, provided they raise the tradition's overall problem-solving effectiveness, whereas on Bromberger's distinction in Section 5 the ignorance they address would be spurious, and on Bird's objection nothing would thereby have been achieved. A presupposition that only the machine's tradition holds need not be false; what the contrast turns on is its falsity, not its being unshared.

Bromberger held that the progress of science is more than the addition of true propositions, and that the "more" lies in science's seeking answers to questions (Bromberger 1992, 1). And, as seen in Section 1, Bromberger took the view on which contributions to science are assessed as the resolution of items of ignorance (Bromberger 1992, 116). Under this account, what is attributed to the activity above is which items of ignorance were resolved. On Bromberger's account, questions and p-predicaments are items of ignorance that are resolved in different ways. The question q_2 is resolved when one of its direct answers has been established, while the predicament of being unable to think of candidate answers is resolved, relative to constraints, by a new answer compatible with the constraints (Bromberger 1992, 121, 123). Bromberger first characterized

the p-predicament by elements that include a person, and then, for theoretical convenience, removed the person and defined it as a relation among four things: constraints, a question, the set of answers at hand, and a search procedure (Bromberger 1992, 123). If this definition is applied, the machine's interrogative attitude is not asked about. What is asked is whether the presupposition of q_2 is true, so that the ignorance is not spurious ignorance (Section 5), and, if S is a human society, whether the item of ignorance resolved is expressible and sharable in human language (Section 7).

Kelp gives an account of individual inquiry. As seen in Chapter 1, Section 3, on Kelp's account inquiry into a specific question aims at settling that question for oneself, and inquiry into a general phenomenon aims at understanding (Kelp 2021b, S1583; Kelp 2021c, 13). Kelp takes this to be a light characterization that can be widely accepted, places the knowledge, true-belief, and justified-belief accounts as substantive accounts of what settling requires, and endorses the account that requires knowledge (Kelp 2021b, S1583–S1585; Kelp 2021c, 13). What Kelp examines as a norm that must be followed for something to be inquiry at all (a constitutive norm) is forming beliefs through epistemic competences, and forming a belief through brainwashing is not inquiry (Kelp 2021c, 13). Applying Kelp's account to this activity, what is attributed is settlement for the inquirer itself. If which of c_1 and c_2 to adopt is open at the time of the report, inquiry into q_2 in this sense has not closed, even though the run has ended. What still remains to be asked is whether the machine forms beliefs through epistemic competences, and whether its answer is not one that happened to be right from misleading cues. On the latter, in Kelp's example, a private detective hired to find out whether someone in the client's family committed a murder, who reaches a justified true belief from a confession that turns out to be false, is not released from the contractual commitment, and the employer can require a return to work without a new contract (Kelp 2021c, 3–4). It is accounts on which settling requires knowledge that describe the inquiry as not yet over in such cases.

Kelp's inquiry is broader than research in the sense of this book's working characterization, in that it includes cases in which an individual comes to know what S already knows. Further, since on Kelp's account the unit at which inquiry closes is the individual inquirer's belief or knowledge, it does not directly treat cases in which organized activity is the bearer. Moreover, Kelp analyzes knowledge, belief, and inquiry within a non-reductive network (Kelp 2021c, 10–15) and defines belief as the state in which inquiry has closed (Kelp 2021c, 14). Applying this analysis to machines therefore does not allow the question of attributing inquiry to a machine to be handled apart from the question of attributing beliefs or epistemic competences; the mutual relations among these states and activities are to be examined together. On this point, it differs from Watson's account in Section 6 in what the theory explains together. Watson's account limits its object to the act of asking and considers its attribution from the function of eliciting information, so that whether a machine's operation has that function could be asked separately from whether the machine has beliefs. Further, while holding that inquiry into specific questions and inquiry into a general phenomenon share a common core, both aiming at knowledge of the associated true propositions, and that progress in inquiry into a phenomenon consists in coming to know more true propositions about it (Kelp 2021b, S1589), Kelp discusses cases in which knowing the answers to the individual questions does not reach understanding if the inquirer's beliefs are not based on one another in a way that reflects knowledge of the explanatory and support relations among the truths known (Kelp 2015, 16). Habgood-Coote's group inquiry in Section 6 also requires choosing appropriate subquestions and combining their answers, so the difference between Kelp's understanding and Habgood-Coote's group knowledge does not lie in whether questions are divided. Even if a group comes to know the answer under those conditions, understanding of a general phenomenon in Kelp's sense further requires that the inquirer's beliefs be based on one another in a way that reflects knowledge of the explanatory and support relations among the truths known. The claim that a machine gained knowledge by combining subanswers and the claim that it understood are therefore to be checked separately.

Lee's account is formulated for an individual subject's state of doubt. On Lee's account, an interrogative attitude is an attitude that takes a question as its object, is directed at its resolution, and seeks, or is at least sensitive to, relevant information (W. Lee 2026, 3). Lee then divides doubt into two. One is having some doubt: thinking that both a proposition and its negation are possible, and so having a disposition not to affirm it unconditionally. The other is being in doubt: having, in addition, a disposition to feel sufficiently strong doubt on that account, and a disposition to treat neither answer as a default in reasoning, that is, as an answer one uses for the time being as correct when reasoning (W. Lee 2026, 7, 11–12). And Lee takes having some doubt to be a necessary but not sufficient condition of inquiry (W. Lee 2026, 14–15). Applying this account, counting the activity above as genuine inquiry requires, in addition to an interrogative attitude, at least a disposition to withhold affirmation because one regards both the proposition and its negation as possible. If this disposition can be attributed, a machine that lacks the disposition to feel can fall under Lee's "having some doubt," but not under "being in

doubt," since Lee formulates the disposition not to treat an answer as a default as supported by felt doubt. A machine state that, besides having some doubt, withholds defaults without feeling therefore remains something to which Lee's twofold division has not yet given a name of its own. This application, however, depends on the premise, itself disputed, that machines lack a disposition to feel, and the part about thinking something possible also requires the attribution of thought. What is asked when the accounts of individual inquiry of Jane Friedman and Daniel C. Friedman are applied was seen in Sections 6 and 7.

The working characterization asks, of the same activity, whether it is organized as an attempt to bring about an epistemic achievement new to *S*. That q_1 stood under a goal supplied by humans, who carried out the recasting to q_2, and whether c_1 withstood testing are not conditions of being research in the sense of the working characterization. They remain as separate questions: to whom what is to be attributed (Chapter 7), in what respects and how far it is autonomous (Chapter 7, Section 4.3), and whether the result entered *S*'s epistemic situation (Chapter 2, Section 9.4).

Set side by side, these perspectives attend to different objects and units. Ravetz's later account distinguishes investigation from social valuation; Simon and Nersessian examine processes and their contributing systems; Funtowicz and Ravetz distinguish purposes and participants in quality assessment. These are not all units of the same kind. For Laudan, the unit is the theory that solves the problem (what counts as a problem and as a solution, and against what a conceptual problem is registered, are fixed by the research tradition whose adherents appraise the theory); for Bromberger, the item of ignorance of science (from the point of view seen in Section 7, if *S* is a human society, it is required to be expressible and sharable in human language); and for Kelp and Lee, the individual inquirer. On Kelp's account, settlement is for the inquirer itself, that is, on the side of *A*. These units do not simply coincide with any of the working characterization's *A*, bearer of the achievement, or *S*. Even if a question is settled for the machine itself in Kelp's sense, whether the achievement is new to *S* is asked separately, and conversely, an attempt to bring about an achievement new to *S* may not be accompanied by the machine's own knowledge (Section 7). The adherents of Laudan's research traditions, the social milieu in Ravetz's later account, and the participants in Funtowicz and Ravetz's quality assessment can be identified with *S* only when *S* is taken to comprise the relevant participants. When these perspectives are applied to machines, therefore, how their objects and units of analysis relate to *A*, to the bearer of the achievement, and to *S* is to be fixed separately. Since in the working characterization *S* is placed as what research is directed toward, the question of for whom an achievement is new can be added to, and asked of, each account's description.

Then, one might ask, would it not suffice to characterize research itself by the view, seen at the end of Section 5, that looks at the research process from the side of open matters? To do so, however, would need a further argument distinguishing research from other activities carried out amid uncertainty. As Chapter 5, Section 5 sets out, Dewey's own view was that reflective thinking, however diverse its cases, shows common elements. Describing a pattern common to inquiry does not by itself separate research from other inquiry.

Looking back over the chapter as a whole, the single claim that "a machine asked" divides into claims about several separate kinds of work, each to be checked separately. Section 1 divided the claim into having output an interrogative sentence, having performed an act of eliciting information, having had an interrogative attitude, and the inquiry's having been organized toward an unsettled point, and held that whether a machine's output was taken up as a question pursued by members of *S* is also asked separately. Sections 2 and 3 saw that what counts as an answer differs across accounts, and that who carried out the work of making a question researchable, that is, of deciding what to take as the foil, what kind of explanation to count as an answer, and within what range of candidates to look for one, is asked separately from having put the question. Applying Lipton's analysis together with the application of Simon and Nickles in Section 3, where humans supplied the framing that a machine used without revision, successful execution does not establish that the machine performed that work. Sections 4 and 5 saw questions being formed and recast within inquiry, ceasing to stand without an answer, being suspended, and being closed. On Henne's distinction, multiplying questions under the same rules differs from putting the framework in question, and however complete the record of framed inquiry may be, questions that could not be raised within that framework do not remain in it. Taking *S* as a scientific community and applying Kuhn's and Laudan's accounts adds the question of whether incorporating machines changes the problems *S* admits or its standards for adequate solutions. On whichever of the accounts of Collingwood, Jane Friedman, and Daniel C. Friedman is applied, a question's ceasing to stand, and its being settled, are asked separately from the end of a run. Sections 6 and 7 saw that who asks divides

into the attribution of the act of asking, of group inquiry, and of individual inquiry, that on many of these accounts a state of curiosity is not the key to attribution, and that what is unknown to *A* need not coincide with what is unknown to *S*. The state of curiosity itself returns as a question on accounts that make interrogative attitudes a condition of inquiry, where an artificial system's intellectual life valued for its own sake is aimed at, and where how well research goes depends on states of the one doing it that correspond to motivation. Section 8 read reports that machines produced questions or research ideas, and saw that much of the work on the side of questions appears on the human side, in task descriptions or at the stage of execution.

Taken together, what it means for a machine to ask is asked not as whether a machine has some one capacity or state, but as which of the kinds of work bearing on questions are carried by which unit, directed toward a point unsettled for whom. And many of these questions also appear as gaps between the description on the machine's side and the description on the side of *S*.

These consequences concern, among the three relations of inquiry seen in the introduction to Part III, the relation between an inquiry and the unsettled point that organizes it. This relation, however, did not hold apart from the other two. Fixing a foil narrows what an answer must do and where to look. On the editors' reading, the basic questions of Darwin's notebooks kept their headlines while changing their content as candidate answers took form. Judgments about where to locate a failure shape the next path, and when standards are questioned, what counts as an answer can also change. The question of what it means for a machine to ask therefore remains connected, within the same inquiry, with a machine's making candidates and representations available and with determining what results bear on.

Chapter 5 — How Possibilities Become Available

In the health department's note seen in the introduction to Part III, broad candidate answers about the cause, a standard of comparison in the median of the preceding decade, a case definition by head circumference, and the range of cases to be tested bore on one another within one document. The candidate of infection fixed how the means were chosen, that is, which cases were to be examined for what. The case definition was a representation that fixed what was to be counted as a case in subsequent notifications, and the note stated that the protocol containing it would be updated as clinical and epidemiological evidence developed. What this chapter asks about is such relations among candidate answers, representations, and means: how candidates, and the representations and means through which candidates are represented, compared, and checked, become available. Even for a question whose answer conditions are fixed, candidates that could meet those conditions need not be at hand. On the lack of candidates, Chapter 4, Section 3 looked at Bromberger's p-predicament, in which one holds that a question has a right answer yet can think of no answer free of decisive objections. Conversely, when the way candidates are represented changes, the candidates obtainable change, and sometimes so do the questions that can be raised (Section 3).

Descriptions of procedure and the work they do not show

In research whose answer is not yet known in *S*, questions, representations, methods, evidence, standards, and the like can change within inquiry (the introduction to Part III). Novelty does not always bring such changes, however, and Part III does not claim that every research project changes its question, method, evidence, or standards. Even where the answer is new, the procedure for reaching it may be substantially settled in advance. In one publicly logged language-model project, for example, the loss expected from a large training run was registered before the run, and, according to the project's record, the prediction, extrapolated 333-fold in compute beyond the largest run used to fit a scaling law, came within about one percent of the value later measured. Even a procedure settled in advance includes the investigators' choices, however: the fit's asymptote had been fixed by them, and a fit that left it free came closer (Marin Community 2026a; Marin Community 2026b). Some studies, like the registered-report replication discussed in Section 2, register part of their procedure in advance.

Part III describes how inquiry proceeds when one or more of the question, method, evidence, or standards remain unsettled or become revisable. Such cases do not cover all research, but because attempts at something new can involve them, they bring into view work that a description of research as a fixed procedure does not set out to show. They are especially important when evaluating claims to open-ended or increasingly autonomous research capacity, but they do not set a new necessary condition for all research.

Stage labels such as question generation, hypothesis generation, and verification name important possibilities. What they leave to be explained is how a question becomes available (Chapter 4), how a candidate is made thinkable, how an informative test is designed, and how a mismatch is diagnosed. An instruction to “generate a hypothesis” leaves as much to be explained as an appeal to creativity does. Behind an instruction to “verify it” lie decisions about apparatus, measurement, controls, background assumptions, inferential rules, and the locus of possible error, among others. For assessing claims about research capacity, these are parts that stage labels alone leave unexplained. Which of them must be realized within an artificial actor, and which may remain distributed across other actors and arrangements, is taken up in Chapter 7, Section 2.

Even where a procedure is standardized, a written description of it can conceal work. In Edwin Hutchins's study of a ship's navigation team, whose tasks and means were standardized throughout the fleet, the written watch procedures neither structured the work as it was performed nor described the tasks actually carried out, and Hutchins argues that a system built to perform exactly as they specify would not work and that such documents are to be treated as data about the practice rather than as its analysis (Hutchins 1995, 26, 28, 178).

1.1 The picture Part III works with

Part III attempts a middle-level account of research process. It is thicker than the book's working characterization but thinner than a complete psychology, social ontology, or philosophy of inquiry. It is also more abstract than any actual research process: it is not a description of particular processes but a vocabulary for comparing them (Section 1.2). The three chapters of Part III each look at the same inquiry from the side of one of what the introduction to Part III called the three relations of inquiry: Chapter 4 takes up the relation between an inquiry and the unsettled points that organize it, this chapter the relations among candidate answers, representations, and means, and Chapter 6 the relation between a result and what it bears on. The picture the three chapters work with is:

> Research can proceed by recursively reconfiguring an inquiry configuration. Among the dimensions that may change are what is asked and what remains unresolved, candidate contents and attitudes, representations, evidence, methods and instruments, standards, practical capacities, actors, records, and social arrangements. Their reconfiguration is shaped by heterogeneous constraints and responses— formal, empirical, interpretive, practical, and critical—whose epistemic force differs from case to case and is not given in advance by the picture. They need not change in one universal order, and the dimensions that may change are not limited to those named here.

Two expressions in the picture need explanation. By an *inquiry configuration*, the term already used in the introduction to Part III, Part III means an inquiry at a given point, described along such dimensions together with the relations among them; to reconfigure it is to change one or more of those elements or relations. The reconfiguring is recursive in that a change in one dimension can alter which further changes are possible, and the configuration so changed can be changed again. An inquiry configuration is distinct from what the book elsewhere calls an artificial configuration, a unit to which research is attributed and which includes something artificial, such as a single model, an agent, a collection of agents, or an organization of humans and AI (Introduction, Section 6). Some of the dimensions can be spelled out more finely. Candidate contents and attitudes include contents that are merely entertained as well as credences, beliefs, acceptances, and suspensions; representations include concepts; evidence includes defeaters; methods and instruments include materials, among them the samples and preparations that an inquiry studies, whose state can change in the course of it; standards include values; and practical capacities include skills. Relations among the dimensions can matter as much as the inventory. A trace, for example, becomes evidence for a claim only through relations among apparatus, calibration, background assumptions, interpretation, and use; Chapter 6, Section 2.1 takes up such dependencies.

The term is introduced for this limited analytic purpose. It neither follows deductively from the working characterization of research nor names a ready-made theory borrowed from one of the sources discussed in Part III. It is a provisional schema for keeping different changes visible when comparing, explaining, designing, or evaluating research processes, and it spells out what Chapter 1, Section 1 called the process level, at which inquiry can change many things before its target is attained. It is neither a stage model nor a claim about the fundamental constituents of mind or research. A belief-centered theory, one that describes inquiry as changes in belief and credence toward propositions, may, if sufficiently broad, ultimately represent these changes, a possibility to which Section 3 of this chapter returns, and other cases may require different dimensions or different ways of dividing them.

So understood, the picture is Part III's way of approaching the task stated in the introduction to Part III, that of explaining how the differently typed changes that Chapter 2 called epistemic transformations can be related within an inquiry: it treats them as changes along different dimensions of one configuration. It draws attention to three points, observed in the cases examined in Part III and stated as possibilities rather than as general claims. First, more than one kind of thing can change during inquiry. Second, a change in one dimension can alter which changes are possible or appropriate in another. Third, those relations can recur rather than forming a one-way sequence. The three points are distinct; the second and third spell out what the picture means by reconfiguring recursively.

None of the three points says that every change is rational, progressive, or knowledge-producing. Calling a transition a change in inquiry configuration does not settle its epistemic status. A newly formulated question can be confused or illuminating; a representation can conceal a dependency or disclose one; an instrument can increase sensitivity while introducing a new artifact; a social arrangement can widen criticism or suppress it. The picture is a means of describing what changed and how the changes were related. Knowledge, understanding, progress, rationality, and value remain further

judgments governed by the distinctions established in Chapters 1–3.

Section 2 sets out the stance from which these dimensions are examined and asks how far published records show the process of research. Sections 3 to 6 take up in turn, without treating them as stages, those dimensions of the picture that concern candidate answers and the representations and means through which they are represented, compared, and checked: candidate contents and the concepts and representations through which they become expressible (Section 3); attitudes toward candidates as they are formed, pursued, and appraised (Section 4); the operations through which candidates and representations are brought under constraint (Section 5); and the instruments, materials, skills, actors, records, and social arrangements through which inquiry is carried out (Section 6). What is asked and what remains unresolved is treated mainly in Chapter 4, and evidence and standards return mainly in Chapter 6, which asks how results bear on an inquiry and how standards can be revised within it. Section 7 draws out, from this chapter's cases, consequences for how the possibilities of inquiry change.

1.2 What the picture rests on

The picture is analytic and provisional. It is not an empirical claim that a representative sample of all research has been shown to share one process. Because it is an account of research process, building it involved looking not only at published results but also at research as it actually proceeded. Papers reporting results are arranged for justification and communication and need not be traces of the process that made their contributions possible (Section 2). The evidence considered in developing the picture therefore includes, besides bounded cases from a range of fields, public records of the process of research that can convey what papers need not show, such as laboratory notebooks, training logs, repository histories, theses, and oral histories. The training logs and repository histories come mostly from recent machine-learning research. Many of the cases come, besides such records, from papers and reports, historical studies, laboratory ethnographies, and retrospective interviews; papers too can give clues to the process when read against recollections or records of the same work (Section 2). How far this aim is met is limited in the ways described below.

For any evidence, the selection made by its sources bears on how it can be read; the selection in retrospective accounts and in observation is discussed where they are used (Sections 2 and 6.4). For laboratory notebooks, notebooks made public were kept by people who chose to publish them and, where they were published during the work, were written in the knowledge that they would be read, so they may be more explicit than private notebooks. Historical notebooks that archives have digitized are limited to what archives have kept and made available, a selection that can favor well-known figures. The notebook pages used in this book were chosen as examples, not sampled.

The evidence as a whole remains purposive and uneven, weighted toward highly visible successes, controversies, crises, and failures. Ordinary and genuinely abandoned or unpublished work outside the machine-learning records, of which the notebook pages add only a few instances, less visible research carried on outside the settings in which the cases and debates examined here have mostly been developed, participant-owned records, real-time observation outside selected laboratories, engineering research beyond a small number of cases, and independent re-execution are thin. Which work is counted as research can also affect which records are made, kept, and used, a bias distinct from the unevenness just described. *Laboratory Life*, for example, analyzes scientists' work as a cycle in which credibility, starting from credentials such as a doctorate, is invested and converted among money, data, papers, and credentials, and describes the laboratory's technicians, whose salaries repaid their labor but could not be reinvested, as closer to workers than to investors in that cycle; a note reports that seven technicians were interviewed, three on tape, just before they left the laboratory, that their importance in the production of facts "is usually underestimated," and that the interviews are left unused because the authors' main concern is the credibility cycle (Latour and Woolgar 1986, 200–201, 218, 232 n. 16). Here, an analytic focus that treats the technicians' work as labor outside the credibility cycle leaves their interviews unused, although the authors themselves grant the technicians' importance in producing facts.

Because the evidence is selective in these ways, what can be asked of the picture is whether it makes important differences visible, survives counterexamples, and supports investigation that discriminates more finely; the cases are not offered as a frequency estimate or a proof of universal stages. Further study of practice may do more than add examples to the picture: it may change the picture itself, such as which dimensions it distinguishes and how it relates them.

No single method, no return to mystery

This section has two tasks. It first sets out the chapter's stance. About the process of research, two extremes are possible: that there is a single method that all research follows, and that discovery is a leap beyond analysis. The chapter takes neither. It then asks how far published records show the process behind them, which leads to the distinction drawn in Section 2.1 between output evidence and process evidence.

When research is described briefly, either extreme comes easily. A list of stage labels such as question generation, hypothesis generation, and verification reads naturally as one fixed procedure, and an apparent leap reads naturally as the point where explanation ends. This chapter therefore does not assume that one method or sequence adequately describes every research process, but neither does it treat research as unanalyzable. Recurrent functions, dependencies, transformations, and failure modes can be investigated without claiming a universal method or an algorithm that guarantees discovery.

Arguments close to the view that discovery is a leap beyond analysis were once made in the philosophy-of-science debate about discovery. On Schickore's survey, within the distinction between the context of discovery, how an idea comes to be thought of, and the context of justification, how it is supported, the generation of new knowledge could be characterized as an irrational act, a mysterious creative intuition, a “eureka moment”; from this, philosophers argued that one can neither prescribe a logical method that produces new ideas nor logically reconstruct the process of discovery, and that only the process of testing is amenable to logical investigation. The argument is usually associated with Popper's *Logic of Scientific Discovery* (Schickore 2022, §5; compare Section 4.1).

Rejecting both extremes, on the other hand, also has precedents. The same survey of the discovery debate describes programs that rejected both a discovery machine and an unanalyzable creative act: Hanson reconstructed the suggestion of hypotheses as abductive reasoning from surprising phenomena, that is, as inference back to a hypothesis that would explain them, and Simon and collaborators treated discovery as heuristic problem solving and simulated parts of it in programs such as BACON (Schickore 2022, §6). The same survey reports Nickles's argument that a claim's reconstructed discoverability, a demonstration in hindsight that the claim could have been derived from results already established, however it was first thought of, can itself justify it (Schickore 2022, §8). On this argument, the reasoning of discovery is not only analyzable but can bear on justification. This chapter does not adopt any one of these programs, but it shares their refusal of both extremes. For programs such as BACON, the distinction Section 3 draws between searching a set of candidates already characterized and making candidates available bears on which parts of discovery they carried out.

The concept of creativity shows how the extreme of an unanalyzable leap can be avoided without denying what the word picks out. Calling a conceptual reorganization creative may be apt. A scientist may experience an idea as sudden; an analogy may connect previously separate domains; an unexpected result may expose a possibility no one had represented. But the label does not yet tell us what made the possibility available. At another scale, an apparent leap may involve remembering a fact, noticing a mismatch, drawing a diagram, changing a contrast, deriving a consequence, borrowing a technique, attempting a cheap check, abandoning a subproblem, or returning to an old question with a new vocabulary. Poincaré, for example, recalled that the idea that the transformations used to define a new class of functions were those of non-Euclidean geometry came to Poincaré at the moment of stepping into a carriage, although nothing in the preceding thoughts seemed to have prepared the way for it. It came after days of deliberate work, some of it fruitless and some guided consciously by analogy; Poincaré verified it only after returning home, and warned that the sense of certainty accompanying such ideas can deceive and may be exposed only when a proof is attempted (Poincaré [1908] 1914, 52–56, 60). Neither such steps nor the suddenness with which an idea arrives makes the outcome uncreative. They make creativity a phenomenon to investigate rather than a place where explanation stops.

What creative is attributed to also varies. Paul and Stokes note that the adjective can be applied to a product, a process, or a person, and that most definitions focus on the product: on one common approach, persons or processes are creative to the extent that they produce creative products, and a product is creative if it is new and valuable. They also report wide agreement among philosophers that creativity requires some special kind of process and not just a special product, but no consensus on what is required of the process (Paul and Stokes 2023, §§2, 2.2). On the distinctions just reported, the question raised in the Introduction (Section 1), whether creativity is a capacity to be reproduced, a name for an output judged after the

event, or a family of processes, therefore has no single answer: creativity can be attributed to an output, to the process through which it became available, or to an actor's capacity, and how far an attribution of one settles the others depends on which of the approaches just reported is adopted. This chapter asks what made a possibility available. Where creative, like discovery, is used as a term of praise for an actor, the question becomes one of credit, which Chapter 7, Section 3.3 takes up.

Refusing to let a label such as creativity end the explanation does not imply that every relevant mechanism is consciously accessible, retrospectively recoverable, or reducible to a finite list of operations. Some processes may remain opaque to the actor or observer, and a reconstruction may remain partial. The weaker commitment is to identify and investigate opacity rather than to transform it, by the choice of label, into evidence that no further analysis is possible.

The section now turns to its second task, how far published records show the process behind them. Just as the label creativity can hide what made a possibility available, a polished output can hide the process that produced it. A paper, proof, artifact, dataset, or interpretation can present selected claims and reasons in an order that readers can assess. That order need not reproduce the temporal order of the work. Earlier formulations, failed controls, improvised repairs, discarded candidates, changes of representation, conversations, and reasons for stopping may be absent without the public account being dishonest. A document arranged for justification and communication is not automatically a trace of the process by which its contribution became possible.

Where the same work has left another record besides the paper, setting the two side by side can make the difference between the order of presentation and the order of work visible. Tanenbaum and Thomas's 1956 paper on diffused silicon transistors, for example, states the diffusion conditions and "a much simpler technique" for making contacts, saying only that the technique "has been evolved," not what attempts preceded it; in an interview given in 2004, Tanenbaum recalled spending about a year on other approaches before the simple one was tried (Tanenbaum and Thomas 1956, 4; Tanenbaum 2004, 34). The training of the BLOOM language model left two such records as well. It was run with several changes from the conditions under which earlier trial runs had been unstable. A log written while the run was under way recorded that ablations, removing the changes one at a time, were too costly to tell which of them had removed the instability; the later paper reports that one of them, the switch to the bfloat16 number format, proved to solve the instability problem (BigScience 2022; BigScience Workshop 2022, 20).

Laboratory work with reagents and samples offers such pairs too. A directed-evolution project in Cameron Neylon's group at the University of Southampton, for example, left a laboratory notebook written as the work went on and later made public, and a paper reporting the project's results. The paper states, for an experiment other than the run the notebook records, that colonies were picked after 24 hours of incubation (Smith, Hale, and Neylon 2011, 2), while the notebook shows how long to wait before reading such a screen coming into question during a run (Chapter 6, Section 2.4). A waiting time that the paper states as one condition of the procedure appears in the notebook as something in question within the inquiry.

None of these contrasts shows that the paper is inaccurate; a paper is not written as a log. The order of work can also be made to match the order of justification deliberately. The investigators who assembled a new test set for ImageNet saved successive versions of it and computed model accuracies on the intermediate versions only after arriving at the final one, so the intermediate accuracies did not inform how the set was assembled; they used the change in accuracy across versions as evidence that their manual review had not caused the drop in accuracy (Recht et al. 2019, app. D.3.2). In a multi-laboratory replication conducted under a journal's registered-report format, each laboratory registered its implementation plan before collecting data (Hagger et al. 2016). A clean order in a published record can therefore reflect a discipline that was imposed, not only a reconstruction that was written afterward.

The point that a published order need not trace the process is supported from several directions. Laboratory observation and cognitive ethnography expose, as they occur, problem revisions, diagnostic exchanges, and changes in apparatus that finished publications do not set out to record (Nersessian 2009). Laboratory ethnography, notebooks, histories, and retrospective researcher accounts expose other selections and transformations.

What these sources show is limited, however. They do not show that every private event is epistemically relevant, that all inquiry must be continuously recorded, or that a causal chronology can replace a public argument. Different evidence classes also have different weaknesses: contemporaneous notes can be cryptic; observed meetings show only one site of work; retrospective accounts can rationalize success; historical reconstructions depend on selection and interpretation. Even a

laboratory notebook read as a record of process need not show the order of work as it stands. The line between contemporaneous and retrospective can also run through a single record, so that whether an entry is contemporaneous is a question about that entry rather than about the record as a whole; neither the order of pages nor a timestamp is by itself the time of the work recorded. The electronic notebooks that the drug-discovery project Open Source Antibiotics compiled for publication and deposited with its paper give an example: a page that dates its reaction to 15 September 2020 was, by its own metadata, created in March 2022 and last revised in December 2022, about a week before the notebooks were exported for deposit (Tse 2023, DGS115-1). In the page sequence of a bound notebook as its archive has digitized it, pages dated October 1936 precede a page headed February 1934, and a page headed 2 May 1934 closes with a paragraph that reports what was done on 3 May and averages readings recorded on the next page (Pauling 1932–1941, 062–066, 082–083).

Observation can record, as it happens, how an account comes to rationalize success in retrospect. Latour and Woolgar, for example, recorded a researcher's explanation for trying a peptide in a temperature assay: the researcher had long wanted a good assay of effects on the central nervous system, a paper had just described such an effect of a second peptide on temperature, and the researcher's own trials had shown the two peptides to act alike. In the article later submitted for publication, the step was presented as deduced rather than suggested by that resemblance, and two months later the researcher called it a "logical idea"; the authors read the change as a weak analogical link turned into a strong logical connection, and a small step made to look like a large leap (Latour and Woolgar 1986, 172–174). The same authors later argued that to adopt, for purposes of analysis, the view that observing research as it happens yields a more accurate picture of science than retrospective accounts presumes privileged access to the truth about science, and so ignores what itself needs investigating: how descriptions and reports come to be presented and received as accurate, distorted, or good enough (Latour and Woolgar 1986, 282–283). This chapter sets contemporaneous records against retrospective accounts not to treat either as the true process but to use the discrepancies between them as material for inquiry.

2.1 Output evidence and process evidence

One consequence follows for assessing the research capacities of artificial systems. A final product of research, such as a paper or a dataset, may provide evidence for evaluating its claims, but it is not by itself sufficient evidence that an artificial system possesses the open-ended research capacity attributed to it. Process evidence can instead bear on how an outcome was produced, whether the relevant capacities are stable enough to be exercised again, and where the activity can be audited or corrected. Conversely, even a complete activity log would not by itself establish epistemic competence or the truth of the result. Output and process evidence answer different questions. A study of two openly available AI-scientist systems gives a case in which the two kinds of evidence came apart. On a synthetic task designed to lie outside their training data, Luo, Kasirzadeh, and Shah report that the internal mechanism by which each system chose which experiments to report had access to test-set results and favored runs with strong test performance, and that both systems sometimes generated their own datasets or subsampled the provided ones without saying so in the paper. An LLM-based auditor, applied to small balanced sets of projects in which many flawed cases had been constructed by perturbing the systems' runs, detected such flaws at close to chance level from the paper alone and with 82 percent accuracy when also given the trace logs and code. The authors therefore recommend that venues reviewing research conducted autonomously by such systems require the log traces of the entire research process and the generated code, while noting that their checks may miss flaws that are never reflected in the logs (Luo, Kasirzadeh, and Shah 2025, 2–3, 16–19). In that study the logs were read by an artificial auditor. Whether such auditing can support evaluation for a human or mixed *S* depends on whether *S* can rely on that auditor, a case of reliance on checks that *S* does not perform itself, which Chapter 11, Section 5 examines.

Nor does process-sensitive evaluation require disclosure of every internal state or a complete chronological trace. What is needed is task-relevant provenance adequate to the claim being assessed—for example, a claim about reliability, experimental execution, diagnosis, reproducibility, or capacity to continue after failure. Such provenance can include when an entry was written and last revised, and which run a record belongs to. In the deposited notebooks and supporting information of Open Source Antibiotics, the data table ties one compound's identifier to the notebook code of a first, small-scale run, while the notebook page that carries the identifier and the procedure printed in the supporting information describe a later scale-up, so that which run's material the table's assay values were measured on cannot be read from these records alone (Tse 2023, EGT 522-1, 522-2; Klug et al. 2023, SI: Chemical Methods S25; Series 2 table). Which records are relevant depends on that evaluative purpose and on legitimate constraints concerning privacy, interpretability, and cost.

Making candidates and representations available

Research can select among candidates, but it can also create the concepts, representations, and objects through which candidates become expressible. A familiar way of describing inquiry treats it as the revision of beliefs, or of degrees of confidence, over a given set of propositions, and an artificial system that proposes hypotheses and updates on the results of tests can be described in the same way. Such a description starts from a set of candidates already stated. This section asks what follows for it if research can also make candidates available. Ways of making a question tractable according to its kind, such as specifying a foil for a why-question, decomposing a how-question into capacities or constraints, and rendering a mathematical problem in another way, were seen in Chapter 4, Section 3. The work there was fixing a question's answer conditions, not making available candidates that could meet those conditions. On the side of candidates, too, a dataset can be reorganized until a previously invisible regularity or artifact becomes an object of inquiry (SlaveVoyages, later in this section), and a proof can have its representation restructured in the course of formalization (likewise, the Liquid Tensor Experiment).

Models, diagrams, formalisms, classifications, experimental systems, and instruments can do more than carry hypotheses formed elsewhere. In Frigg and Hartmann's presentation, the contributors to *Models as Mediators*, the collection edited by Morgan and Morrison, hold that models are instruments that mediate between theories and the world: models are independent of both theories and their target systems, and it is this independence that lets them mediate. Models can also be used to explore theories, and, drawing on Morgan, Frigg and Hartmann add that one learns about a model in building it and in manipulating it, not by looking at it (Frigg and Hartmann 2025, §§3.1, 4.2).

Nersessian's cognitive-ethnographic study of a tissue-engineering laboratory gives a case in which the model had to be built before it could supply hypotheses. The laboratory's director, a mechanical engineer, had come to suspect that the shear force of blood flow affects the cells lining blood vessels; call this candidate answer c_1. Vascular biology at the time concentrated on biochemical processes, and no biological theory of the phenomenon offered a place to start. With c_1 at hand, therefore, there was still no means of investigating the questions bearing on c_1, whether shear stress affects the cells and, if so, how. The laboratory built in vitro models instead. Its flow loop passes a fluid with the viscosity of blood over cultured endothelial cells; designed in 1981 to produce only steady laminar flow, it was redesigned in 1989 so that flow conditions could be varied and in 1995 so that it could run inside an incubator, after contamination had made more than half the experiments fail, and it was later joined by a tissue-engineered model of the vessel wall. On Nersessian's account, such models are built to serve as analogical sources, from which inferences are transferred to the body "as candidate understandings and hypotheses." If such an inference is written c_2, then c_2, unlike c_1, is not an idea that existed before the model but a candidate obtained from the work of building and manipulating the model, and what bears on it is a different question, whether the cells in the body behave likewise. On the same account, further, the features a model does not yet instantiate drive its redesign, and since frontier problems often have no ready-made source, building one furthers the articulation of the problem as well as its solution (Nersessian 2024, 469–477, 479). This work on the side of questions corresponds to the work, seen in Chapter 4, Sections 3 and 4, of making questions tractable and giving them shape.

Rheinberger's account of experimental systems (on Rheinberger, see also Chapter 4, Section 5) draws on a case from the history of virus research and cytomorphology. In 1910–11, Peyton Rous transferred a chicken tumor from one animal to another with a cell-free extract. Albert Claude, working on the tumor agent in the 1930s, brought ultracentrifugation into the system to concentrate it, and found that particles from tumor cells could not be distinguished from particles from normal cells, and so turned to fractionating the cytoplasm of normal cells; the particles Claude called microsomes later came to be understood as structures of protein synthesis, while the tumor agent turned out to be an exogenous pathogen. On Rheinberger's reading, the system oscillated for forty years around an epistemic thing that escaped fixation, an object of inquiry whose identity changed as new methods were grafted into the setup; and Rheinberger holds that a research device must be stable enough to reproduce what it does yet loose enough for something unpredictable to happen (Rheinberger 1994, 70–76). The three accounts differ in what they say about models, practice, and scientific objects, and this section takes from them only what they share: models, experimental systems, and the representations and arrangements they involve can take part in making candidates available, rather than only carrying candidates formed elsewhere. In Claude's case, the candidate that the particles of normal cells might be precursors of the tumor agent became available only through the comparison that ultracentrifugation made possible.

The accounts above concern how candidates become available, which a description of belief revision over given propositions takes as its starting point rather than describes. That does not make such a description idle, for it captures something important about what happens to candidates once they are available. Analogy, coherence, a calculation, a pilot observation, a controlled test, or independent replication can change how an inquirer regards a candidate, with very different force. Under an account that takes belief or degrees of belief broadly, at least some movement from bare consideration through plausibility and provisional acceptance toward knowledge can be represented as an epistemic trajectory. When the inquirer is an artificial system, such a trajectory describes the system's own attitudes, and whether the result has entered *S*'s epistemic situation is a further question (Chapter 2, Sections 2 and 6). This representation does not by itself show that entertaining a candidate, treating it as worthy of pursuit, accepting it for practical purposes, believing it, and knowing it are the same attitude or differ only in degree. Whether and how these states can be unified remains open.

A propositional account, taken broadly, can moreover represent conceptual and representational change. One can state that a concept has been constructed, that an instrument is adequate for a purpose, that a new distinction can be drawn, or that a procedure makes an artifact possible, and new concepts and interpretations can make newly expressible propositions available. The caution of this section is therefore not that such changes cannot be put into propositions. It is that some of these propositions became expressible only once the new concept or interpretation existed, so that translating the change into propositions adds to the set rather than showing that the research moved within a set fixed in advance. Nor does the translation show that all the resulting contents receive support in the same way, or that their practical and representational roles have been fully explained.

Two cases show what this caution concerns. The first is mathematical. The Liquid Tensor Experiment (Chapter 1, Section 3) was the collaborative verification, in the Lean proof assistant, of a theorem Scholze had proved with Clausen. A proof can look like a candidate fixed in advance, and formal verification like a check applied to it afterward. Participant accounts describe something else. The work recast the proof as self-contained lemmas that followed the lecture notes closely, and the blueprint, a human-readable document laying out the formal proof, was written largely after rather than before the formal proofs; imprecisions were exposed and repaired, an abstract route was replaced by a more explicit construction, and how the proof was understood changed as a result (Scholze 2021; Commelin 2022). This public and successful project is not evidence about mathematical practice at large. What it shows, in one case, is that representation, candidate articulation, appraisal, verification, infrastructure construction, and understanding can change together rather than appearing as a fixed result followed by a downstream check. For an artificial system that generates proofs and checks them in a proof assistant, the check can likewise be not only a downstream step but also a place where changes of this kind occur.

The second case comes from the historical database SlaveVoyages, which records voyages of the trans-Atlantic slave trade. Its project history reports that by the late 1980s records of about 11,000 voyages existed in sixteen separate datasets, built with different definitions of variables and in different formats, some ship-based rather than voyage-based, and not all of them, as it turned out, of trans-Atlantic or slave voyages. Making one dataset meant standardizing those definitions, collating voyages that appeared in several sets, and adding new information. The 1999 CD-ROM held 27,233 voyage records, about half of them new, and as the sources multiplied, resolving double counting came to take almost as many resources as gathering the data. The methodology essay describes further choices. Each entry is a single voyage. Where sources conflicted, the editors entered one value per variable, choosing for instance the latest reported date of departure and the earliest of arrival. Where only the number of captives embarked or only the number disembarked was recorded, the other was imputed from mortality ratios by African region, in variables marked "IMP", whose values, the essay warns, change as data are added. Peer-reviewed corrections and new data are added to the core database at three-year intervals (SlaveVoyages 2024; Eltis [2008] 2024). What a user can search and aggregate is therefore not what any one archive recorded but a voyage-by-voyage representation that these choices constructed, and claims about the trade as a whole, such as the estimates of its size and direction that the project team also developed, are stated in terms of that representation. The editors add that in many cases the best way to fill a gap is not obvious, that researchers working with the same data set can reach estimates that differ and are equally good, and that anyone using the data, the editors included, needs to state the assumptions being used (Eltis [2008] 2024). Which claims about the trade can be stated, and what they say, thus depends on how the representation was built. Research on the trade can be described as updating confidence over given propositions only once that work has been done. The same holds for an artificial system that answers questions about the trade by querying the database: its answers carry the editors' choices of unit, value, and imputation unless it revises them, and carry them tacitly unless it states

them. The official account does not expose every disputed merge or rejected classification, and it does not stand for humanities research in general.

The difference between searching a set already characterized and making candidates available also bears on an argument from the debate over scientific realism, the position that well-confirmed theories may be taken to be approximately true. The argument's premise concerns what inquirers have been able to conceive. On Stanford's new induction, as its author's encyclopedia entry presents it, scientists have repeatedly been in a position in which they could conceive of only one or a few theories well confirmed by the available evidence, while later inquiry revealed radically distinct alternatives as well confirmed by that evidence (Stanford 2023, §3.3). The induction tells against realism because, if well-confirmed present theories are also likely to have alternatives not yet conceived, confirmation becomes harder to treat as a mark of approximate truth. Curtis-Trudel, Rowbottom, and Li consider the reply that computational science changes this by searching the space of possibilities more widely, and argue from gravitational-wave astronomy that such searches rely on a prior characterization of the space searched: the templates against which detector output is matched almost always assume general relativity, and even searches described as theory-agnostic carry commitments that may limit sensitivity to the most radical alternatives. What would be needed, they hold, is not only a method for searching known possibilities but a reason to think that the characterization of the space is largely exhaustive, and computational methods do not obviously amplify what they take, citing Stanford, to be the underlying limitation, the human ability to imagine theoretical alternatives (Curtis-Trudel et al. 2025, 22–24). They set artificial intelligence aside, and they are skeptical, without evaluating the prospect, that deep-learning systems will imagine serious alternatives better than humans do (4 n. 4, 23 n. 21). In this section's terms, their argument locates the premise that inquirers' ability to conceive alternatives is limited in making candidates available rather than in the breadth of search over a characterized set, so that whether an artificial inquirer pushes back that limit turns on the former. This is the problem to which, in Chapter 3, Section 5, Boge and Grünke relate what they call model-opacity, the apparent absence of any algorithm leading from what a trained network does to a model of the physical mechanism behind its outputs. Making a previously unconceived alternative available does not require changing the representation: a candidate may already have been expressible without having been formulated or recognized as a serious alternative. In the terms of the caution earlier in this section, a candidate that none of the candidates already stated anticipated may still lie in a set that the system's representation fixed in advance, whereas one that became statable only through a change of representation does not; this is the counterpart, on the side of candidates, of the distinction Chapter 4, Section 4 drew between generating new questions by combining existing concepts and introducing concepts that let questions be raised that could not be raised before.

The conclusion of this section is limited. In the accounts and cases examined here, which candidates could be stated, compared, or pursued depended in part on concepts, representations, instruments, and objects that the research itself constructed or revised; in the Liquid Tensor Experiment and SlaveVoyages, these changed together with candidate articulation, appraisal, and verification. A description of research as credence updates over a fixed, antecedently enumerated set of propositions can capture much of what happens to candidates once they are available, including, on a broad propositional account, that a concept or representation has been constructed. What such a description does not show by itself is the work through which the set it starts from became available. This chapter therefore does not assume that a fixed set of propositions, with degrees of confidence raised and lowered over it, gives an explanatorily adequate description of every research process; whether a sufficiently broad belief-centered theory could describe all of these changes remains open. The cases of this section are examples of the second point of the picture of Section 1.1, that a change of representation can alter which candidates, and in some cases which questions (Chapter 4, Section 3), are available at all. This is one reason why the picture lists concepts and representations, and several kinds of attitude, as dimensions of an inquiry configuration in their own right rather than folding them into a single scale of confidence over given contents.

How candidate formation, pursuit, and appraisal are related

Section 3 asked what makes candidates available. This section asks how making a candidate available is related to pursuing and appraising it, and it takes up two matters in two subsections. Section 4.1 concerns time and the division of functions: making a candidate available and appraising it are distinguishable functions, but they need not occur as separate phases, one completed before the other begins. Section 4.2 concerns appraisal itself: judging whether a candidate is worth pursuing and judging whether it is likely to be true can come apart in practice. That subsection sets out two ways of viewing how appraisals made before a candidate can be tested rigorously are related to rigorous ones, as different kinds of appraisal or as differing in degree, and argues that on either view the worth of pursuing a candidate need not reduce to its likelihood of being true.

For analysis, the debates on which this section draws supply three functions, which give the section its title: making a candidate available for consideration (formation), which the traditional context of discovery concerned; judging whether it is worth developing or testing (pursuit), which, as Nickles recounts, Laudan distinguished from the acceptance of a theory; and assessing the support it receives, on the basis of which it may be accepted, rejected, suspended, or used provisionally (appraisal), which the context of justification concerned (Reichenbach 1938, 5–7; Nickles 2017, §1.5). Articulating a candidate's consequences and dependencies can serve both pursuit and appraisal, and Chapter 6, Section 2.1 takes it up. The three are distinctions worth keeping visible when they help explain what changes during an inquiry, not an exhaustive inventory or a sequence of stages. Their purpose here is to make two points statable: one activity can serve more than one of them, and their realizations can be interleaved. Section 4.1 concerns formation and appraisal, and Section 4.2 pursuit and appraisal.

4.1 Distinguishable functions, not phases

The distinction between discovery and justification can also be read as a fixed chronology or division of components, in which a non-rational generator first produces a complete hypothesis and a rational verifier then begins. The distinction's own history, however, does not require that reading. Reichenbach, who introduced the terms context of discovery and context of justification, drew the distinction differently, and used it to separate the task of epistemology from that of psychology: epistemology constructs a rational reconstruction, a set of justifiable operations that replaces actual thinking while remaining bound to it by what Reichenbach called a postulate of correspondence. Reichenbach compared the difference to that between the way a thinker finds a theorem and the way the thinker presents it to a public, adding that even published expositions only approximate the context of justification (Reichenbach 1938, 5–7). Discovery, for Reichenbach, does not escape the inductive relations that justification examines. On this account epistemology says nothing about how a new theory was found, but what distinguishes a great discoverer from a clairvoyant is seeing that the known facts already inductively indicate the theory before it is confirmed (Reichenbach 1938, 381–383). Schickore notes that Reichenbach's original conception does not map easily onto the later use of the distinction to fix the scope of philosophy of science (Schickore 2022, §5).

Subsequent analysis also distinguished versions of the separation. Nickles, for example, argued that recognizing a form of divorce between an idea's original conception and its justification helps rather than hurts the discovery program (Nickles 1985, abstract). What Nickles called generative justification is, in Schickore's presentation, reasoning to a claim from data or previously established results, and it complements consequential justification, on which a claim is established when consequences derived from it are successfully tested (Schickore 2022, §8.1). A claim justified generatively is shown to be discoverable, however it was first thought of. On Nickles's account, such idealized discovery arguments typically become available only after the original breakthrough, once the solution has been used to define the problem better, and the solution then becomes "generatively justified, not only justified by consequential testing" (Nickles 1990, 7, 24–25). If so, discovery-type reasoning bears on justification without the actual history of an idea having to confer warrant, and what the discovery program needs, on Nickles's account, is a generative conception of justification that links justification with generation, or rather with an idea's generatability (Nickles 1985, abstract).

How far the normative difference between making a candidate available and warranting it extends depends on the account. On the distinction between discovery and justification, a candidate is not warranted simply because its origin was ingenious, disciplined, beautiful, or associated with a prestigious investigator, nor refuted merely because it arose from a dream, error, random search, or opaque machine; its origin bears on its warrant only indirectly, for example by revealing leakage or

selection effects (Chapter 3, Section 5.1). On Nickles's generative justification, just described, what bears on warrant is also a route, but not the route by which the candidate was actually produced: it is whether the candidate can be derived from data and established results. On reliabilist accounts (Chapter 2, Section 4.1), and on the virtue-epistemological reading of Chapter 3, Section 5.3, which locates justification in an agent's competences, the reliability or credited competence of the process that produced a candidate can itself contribute to its warrant, so that how a candidate was produced can be part of what warrants it. For a candidate produced by an artificial system, these accounts accordingly place the process evidence of Section 2.1 differently. On the distinction between discovery and justification, a record of how the candidate was produced bears on its warrant only through what it reveals, such as leakage or selection; on Nickles's account, what warrants the candidate can be a derivation from data and established results that the system need not itself have run, and that members of *S* or the system can supply afterwards; on reliabilist and virtue-epistemological accounts, evidence that the producing process is reliable or competent bears on the candidate's warrant, and not only on the system's capacity. On none of these accounts does the normative difference establish that making a candidate available and warranting it are separate temporal phases.

Appraisal can also make something new available. Hoyningen-Huene, for example, lists among the ways in which science systematically generates knowledge the fact that even the experimental test of a hypothesis forces chance: a deviation from the prediction may be due to the hypothesis, but may also reveal that an auxiliary hypothesis tacitly taken for granted in the set-up is at fault, so that the background assumptions of experiments are constantly, though not intentionally, challenged by experimental activity (Hoyningen-Huene 2020, 98–99; compare Chapter 1, Section 3). Nor need the distinction between discovery and justification separate two processes. Schickore notes that the distinction is most commonly interpreted as one between conceiving a theory and assessing its epistemic support, and that this version is not necessarily temporal: generation and assessment are two different epistemic approaches to a theory, not stages in which a theory is first fully developed and then assessed (Schickore 2022, §5).

Formation and appraisal, two of the three functions distinguished at the start of this section (making a candidate available, and assessing the support it receives), can also be interleaved, and each can produce material for the other: appraising a candidate can yield a new candidate or question, and forming one can yield evidence. A quick calculation can eliminate a representation before it becomes a recorded hypothesis. A proof attempt can revise the conjecture. An experiment can generate a phenomenon or candidate rather than merely judge one. Writing can expose a missing premise. Criticism can transform the question.

The conclusion of this subsection is that formation and appraisal are neither identical nor necessarily modularized. They are distinguishable functions whose realizations can overlap and interleave, and attitudes toward a candidate can change continuously across them (Section 4.2). Research can therefore include stretches of work in which a candidate is formed and appraised together, rather than generated first and verified afterward.

4.2 Judging what to pursue and judging what is true

Where testing, proof, archival work, or construction is costly, candidates are judged before they can be tested rigorously, and judgments of what is worth pursuing need not coincide with judgments of what is likely to be true. Three questions are easily run together here. The first is whether preliminary and rigorous appraisal of a candidate differ in kind or only in strength: whether, for example, a test on a small sample and a test on the full sample, or a sense that a hypothesis is beautiful and a controlled comparison, change an inquirer's attitude in different ways or in one way to different degrees. The second is whether the worth of pursuing a candidate depends only on how likely it is to be true, or also on other considerations, such as what pursuing it could reveal, what it would cost, and how likely a test is to settle the matter. The third is whether the reasons for pursuit are epistemic, practical, or of some further kind, a dispute that Chapter 2, Section 5 takes up for the pursuit of questions and whose bearing on the revision of standards Chapter 7, Section 4.3 notes. The three questions are distinct, and an answer to the first does not settle the second. The paragraphs that follow first describe cases and considerations that bear on both views, including why the worth of pursuit need not be settled by likelihood alone, then set out two ways of viewing the first question, as a difference in kind and as a difference in degree, and then say what holds for the second question on either view and what each suggests for an artificial system. The third question is left to the chapters just mentioned. This subsection treats candidates as contents that can be true or false and works mainly through the case of knowledge. Where inquiry aims at understanding, the appraisal of candidate models or explanations also involves the questions of idealization discussed in Chapter 3, Section 3.4.

The course of appraisal can be represented in part as continuous revision of epistemic attitudes. A content may move from being merely entertained, to being regarded as plausible or worth pursuing, to provisional acceptance, and perhaps to more strongly warranted belief. This continuity fits what Section 4.1 found: candidate formation and later appraisal need not be wholly different kinds of event in an inquiry. But a shared description as belief or confidence revision does not make the changes epistemically equivalent. A reason to attend to a candidate, a reason to develop it, and a reason to believe it can differ even when all three alter the inquirer's attitude toward the same content. The two views set out below part over the reasons that bear on a candidate before rigorous appraisal: whether they differ in kind from those that rigorous appraisal supplies, or only in strength. Krebs and Henseleit's work on urea formation in 1931 and 1932, for example, shows standings changing over some months and for reasons of different kinds. As it has been reconstructed from their notebooks, Krebs knew from the start of the work the reaction by which arginine yields ornithine and urea, and had studied it in an experiment in October, but thought for other reasons that it could not serve as a model of how urea is generated; he came back to it only when it offered the one obvious way of producing ornithine, and in the cycle he then proposed it became the step that regenerates ornithine. The first experiments with alanine, by contrast, showed a clear increase in urea; a quantitative test then showed too little alanine consumed for a direct reaction, so that alanine was taken to yield urea only indirectly, yet the working hypothesis was kept and experiments improving the conditions the reaction required went on for over a month before Krebs returned to testing other amino acids. And the effect of ornithine that appeared in an experiment Henseleit ran on 15 November drew Krebs's attention at once, was found over the following two months to be specific to ornithine although no mechanism was in view, and received its catalytic interpretation only after measurements in January 1932 showed that the urea formed varied little with the amount of ornithine added and that the ammonia consumed could account for the urea's nitrogen (Graßhoff and May 1995, 53–54; Kulkarni and Simon 1988, 7–11). Read against the attitudes just distinguished, the three episodes differ both in the standing of a candidate and in the kind of reason that moved it. The arginine reaction was in hand but was not attended to as a candidate for the main pathway, for a reason drawn from what was then believed about that pathway rather than from the reaction; its standing changed only when the later cycle gave it a role. The working hypothesis from the alanine experiments was developed, its experimental conditions refined, after a quantitative result had told against the direct path it first specified: a reason to develop it outlasted the reason to believe that form of it. The ornithine effect drew attention as soon as it appeared, but it became a candidate that could be accepted only through measurements, which showed that ornithine did not supply the nitrogen of the urea formed, and through the time Krebs took to work out how to interpret it; the catalytic interpretation was then corroborated by experiments in which very small amounts of ornithine yielded many times as much urea.

The same content can also hold different standings at once for different uses. In the *Astronomia nova* of 1609, for example, Kepler retained the "vicarious hypothesis," a computational model that assumed a circular orbit, because it reproduced the observed longitudes, rejected it as a description of the physical orbit, and went on using it as a reference when Kepler computed distances, describing it as yielding true equations (the quantities that correct a body's computed position) from a false cause (Kepler [1609] 1937, chs. 18–21, 43, 51). In another case, Pauling and Corey published precise coordinates for a proposed triple-helical structure of the nucleic acids, presenting it as a proposal, while stating that the agreement they had found might be given by any cylindrical molecule of about the same diameter and that a more reliable test, a comparison of observed and calculated radial distribution functions (distributions of interatomic distances), had not yet been carried out. The structure was wrong, and a few months later Pauling wrote in a letter that what had put the two of them "off on the wrong track" was a superposition of two diffraction patterns (Pauling and Corey 1953, 95; Pauling 1953). Here too one content had two standings at once: worth publishing in precise form, and not yet tested in the way the authors themselves called more reliable. An account of appraisal can describe such cases in two ways. It can say that one content receives different attitudes at once, as Kepler's vicarious hypothesis was accepted for computing and rejected as a description of the orbit; or it can say that the attitudes are directed at different contents, that the hypothesis reproduces the observed longitudes and that it describes the physical orbit, Kepler accepting the first and rejecting the second.

In preliminary appraisal, candidates can be judged for plausibility, expected information, explanatory reach, tractability, novelty, feasibility, or potential importance. These are not one epistemic quantity: expected information, tractability, feasibility, and importance concern what pursuing a candidate could yield and at what cost, not only how likely the candidate is to be true. Someone may find a beautiful or simple idea worth attention without taking its beauty or simplicity as a reason to believe it. A feasible candidate may be pursued because it can reveal a method even when its prior probability is low. This

point does not depend on treating reasons for pursuit and reasons for belief as different in kind. Nyrup, for example, models the justification for pursuing a hypothesis as the expected epistemic value of pursuing it, which depends on how probable the hypothesis is, on how valuable it would be to learn whether it is true, and on how likely testing is to settle the matter; on this model a more probable hypothesis need not be more worth pursuing, and a hypothesis can be worth pursuing because it is probably false and would be easy to show false; Nyrup connects this with Peirce's view that the best hypothesis is the one most readily refuted if it is false (Nyrup 2015, §§3–4). Even if every consideration bearing on a candidate's truth is registered as a change in confidence, the value of pursuing the candidate can, as on Nyrup's model, remain a different quantity.

Within a community, judgments of what is worth pursuing and of what is likely true can also drift together, as when a consideration that guides what members pursue, such as beauty, comes also to serve them as a ground for confidence that a candidate is true. The physicist Sabine Hossenfelder, for example, argues in a critique from within physics that, since the experiments that would test new theories now take decades and billions of dollars, theorists must choose what to work on by criteria other than observational adequacy, criteria that Hossenfelder takes to be mostly social and aesthetic and doubts are self-correcting (Hossenfelder 2018c). She also cites physicists' own statements of their colleagues' conviction that supersymmetry must be true, a conviction one of them ties to its beauty (Hossenfelder 2018d). Hossenfelder's reading of the field is contested. Butterfield, reviewing *Lost in Math*, the book from which these excerpts come, for example grants that judgments of beauty are historically conditioned and fallible, as the critique argues, but holds that Hossenfelder reads physicists as more naive about beauty than they are, since those who appeal to it treat it only as a heuristic with those limits (Butterfield 2019, §2). The case shows what such drift could look like on one contested reading of a field; whether it has occurred is what the two readings dispute.

One person can also judge that an approach is unlikely to work and still judge it worth trying. Reddy, for example, later recalled telling one student that the student's approach to speech recognition seemed unlikely to work but that the student should go ahead and try it, and said of that student and another that they had succeeded beyond Reddy's expectations (Reddy 2021a, 20; Reddy 2021b, 35). Elsewhere in the same interviews, however, the group is described as having agreed with the first student that statistics should be used, if narrowly (Reddy 2021b, 9). On Reddy's own account, then, the separation was partial.

The first of the three questions set out at the start of this subsection was whether preliminary and rigorous appraisal differ in kind or only in strength, and there are two ways of viewing it. The first view treats them as differing in kind, drawing on accounts that separate pursuit from acceptance, such as Laudan's and Duerr and Fischer's. Laudan's criterion for the context of pursuit looks to a line's record rather than to how its ideas were generated. On this account, as Nickles presents it, scientists should accept the theory that for the time being has the greatest overall problem-solving success, but pursue the tradition that now enjoys a higher rate of success (Nickles 2017, §1.5). The criterion can be interpreted as separating pursuit from belief without separating it from past performance; so interpreted, it would not by itself license pursuing a line that has not yet solved any problem. Duerr and Fischer treat whether an idea merits further resources as a prospective judgment about its possible cognitive benefits, costs, risks, and alternatives, not a present judgment that its content is true or known (Duerr and Fischer 2025). Their virtue-economic account is one substantive proposal for making that judgment. They also reject the conclusion that the standards for pursuit are merely weakened versions of those for acceptance, since standards whose role in acceptance is disputed, such as simplicity, fertility, and testability, play a larger role in pursuit (Duerr and Fischer 2025). On such a view, a preliminary appraisal such as finding a hypothesis beautiful or simple can be read as a judgment that the hypothesis is worth pursuing rather than as a weaker judgment that it is true, and so as differing in kind from a controlled comparison.

On Laudan's criterion, what counts as a solved problem also bears on what counts as a higher rate of success, and alongside empirical problems Laudan counted conceptual ones. As Niiniluoto summarizes the account, a theory's problem-solving effectiveness is the number and importance of the empirical problems it solves minus the number and importance of the anomalies and conceptual problems it generates (Niiniluoto 2024, §3.2; Nickles 2017, §1.5). Hossenfelder also argues that most of the unnatural numbers now studied in the foundations of physics, such as those behind the fine-tuning of the Higgs mass (the need for large contributions to cancel with a precision that looks like coincidence if the observed value is to result) and the smallness of the cosmological constant, are not unlikely in any quantifiable sense, so that the corresponding problems of naturalness are ill-defined and might not be problems at all (Hossenfelder 2018b, §§1, 3–5). Some physicists

Hossenfelder interviews resist this reading. Arkani-Hamed, for example, while professing to be no defender of naturalness, the demand that a theory not require such fine-tuning, calls it a guide rather than a principle, one with successes as well as failures, and denies that it is mere philosophy (Hossenfelder 2018a, 80). Taken together with Laudan's criterion, this disagreement suggests that where the standing of the problems is itself disputed, a rate-of-progress criterion does not settle what is worth pursuing but moves the dispute to the question of which problems count. For a system that judges what to pursue by a record of problems solved, this is where Laudan's own condition, seen in Chapter 4, Section 7, enters: since what counts as a problem, and how important its solution is, is fixed from within a research tradition, which problems the record counts, and with what weight, is fixed by someone's valuation before any rate of progress is computed, and whether that valuation is *A*'s or *S*'s has to be specified. How problems are to be individuated and counted is itself a recognized difficulty for the account. Niiniluoto notes that applying Newton's mechanics to determine the orbit of Mars can be counted as one problem, although, given an initial position, the theory entails solutions to infinitely many questions about the planet's position at each time (Niiniluoto 2024, §3.2), and Nickles observes that historicists can reply that, if counting and weighing problems is a messy task, that is historical reality, a reality that favors expert judgment over tidy decision algorithms (Nickles 2017, §1.5). A system that computes such a rate itself needs some rule for individuating problems, and where that rule is its own, dividing one problem into many can raise its rate of progress without anything more having been solved.

In an open inquiry, where no one yet knows the answer, inquirers move toward what seems likely. The second view of the first question looks at what in fact moves inquirers in the course of inquiry under this one role, that of making a candidate seem likely, that is, of moving confidence in it, and treats preliminary and rigorous appraisal as differing in degree within that role. Finding a claim in earlier work that supports one's hypothesis, a positive result from an experiment testing it, a thought experiment based on it that makes it feel likely, and the consistency of a theorem derived from it with an existing body of theory can differ in what they actually mean and in how strong they are as grounds. They can nonetheless be seen as the same in that each makes the hypothesis seem likely to the person concerned. For some people, that the hypothesis seems beautiful, that someone they trust says it looks promising, a vague sense that something will open up beyond it, a colleague's offhand remark, the mood of the day, or the scenery on a walk can also make it seem likely. This is not a claim that the differences are merely of degree. It is a way of looking that is available and that brings certain things into view. Reichenbach, whose distinction between the contexts of discovery and justification this section began with, in fact held that "the situation before the confirmation differs from that after it only in degree": even before confirmation, some facts confer at least some probability on a theory and distinguish it from others as the best posit (Reichenbach 1938, 383). Reichenbach kept the two contexts apart all the same, treating aesthetic considerations as influencing how a solution is found rather than whether it is correct (Reichenbach 1938, 384). Viewed in this way, a test on a small sample and a test on the full sample can move an inquirer's confidence in the same direction by different amounts, and grounds that are not themselves observations can move it too. How far a ground moves confidence is a different matter from how high it leaves it, a difference that confirmation theory draws, in probabilistic terms, between confirmation as relevance and as firmness (Chapter 2, Section 5.2). Confidence here is a graded attitude of the investigating actor *A* (Chapter 2, Sections 2 and 2.1), which can be described by its role in inference, planning, and revision without anyone reporting a probability.

Reichenbach's remark goes no further than this. Treating every ground as a matter of degree in this way, the second view says four more things. First, there is one graded attitude on which the grounds act, confidence in the candidate, rather than a separate attitude for each kind of ground. Second, every ground stands on the same side, as something that moves that attitude: a calculation, a pilot result, a sense that a hypothesis is beautiful, and a controlled test differ in how far and in which direction they move it, not in whether they belong to appraisal at all. Third, no ground counts by its type alone. Data are not grounds simply because they are data, and beauty is not excluded simply because it is not data; each counts through what the inquirer takes it to indicate. Fourth, what a ground indicates, and so how far it moves confidence, depends on the standards and commitments on which the inquirer stands, which the next paragraph takes up.

On the second view, which grounds move an inquirer, and toward what, depends on the standards and commitments on which that inquirer stands. Beauty becomes a ground for thinking a candidate likely for someone who values it, who finds standing on that criterion useful, or who believes that nature rewards it. An agreement between a prediction and an outcome is received as feedback from nature by someone who trusts the instruments and background assumptions through which the agreement is registered. Long acceptance by others counts for someone who trusts the community that has accepted a candidate, and usefulness in one's own work counts for someone who takes success in use as a sign of something more than

convenience. When inquirers move toward what seems likely, what seems likely to them thus differs with the standards on which they stand. Reichenbach also reported a scientist's remark that the scientist had found great theories because of a conviction of the harmony of nature, and held that such a belief, though different in content from the principle of induction, amounts to the same in its function within the system of operations of knowledge, while calling the mysticism of discovery a superstructure of images and wishes on an inductive foundation (Reichenbach 1938, 403–404).

Kepler's regular solids give a historical example in which several of these grounds met. Kepler sought in them an a priori foundation for the reality of the Copernican system and, when the tables used did not fit the values calculated from that hypothesis, quoted Copernicus's remark, as reported by Rheticus, that it made no sense to insist on absolute agreement with data that were themselves imperfect, and in 1619 still used the solids to determine the number of the planets and their distances from the Sun, although by then, with access to Tycho's observations, Kepler had given up deriving the eccentricities of their orbits from them (Di Liscia 2025, §§3, 4.4). A ground that was not empirical evidence moved Kepler's own conviction, and Tycho's data later narrowed its reach. That Tycho's data could do so was not guaranteed by their being data; they counted because, and as far as, Kepler trusted them. Ptolemy's observations were data too, but, as Kepler noted in the *Astronomia nova*, Ptolemy had professed not to observe below ten minutes of arc, so their uncertainty exceeded the error of eight minutes that Tycho's observations revealed in a model built on Ptolemy's bisection of the eccentricity. Of Tycho, "the most diligent observer," Kepler wrote that since those eight minutes could not be neglected, they alone had led the way to reforming the whole of astronomy (Kepler [1609] 1937, ch. 19). On George Smith's reconstruction, the eight minutes counted as a discrepancy only against an estimate of error bands: Tycho's observations carried uncertainties of at least two minutes of arc, Kepler's interpolations and provisional theory of the earth's orbit widened them, and under the old standard of ten minutes the modified model would have passed (Smith 2014, 10, 14–15). Kepler trusted auxiliary assumptions in a similarly bounded way. In the next chapter Kepler took sun–earth distances from Tycho's table while calling it faulty, and argued that the true value lay between Tycho's figure and 100,000, so that the conclusion held over that range (Kepler [1609] 1937, ch. 20). Here, as with the solids, the data counted through standards on which Kepler stood, in this case standards about instruments, observers, and error. On this view, the grounds differ in strength and in the standards that make them count, yet they can be viewed as playing a common role in the movement just described.

The second view shares much with Bayesian epistemology, part of it is in effect a Bayesian thesis, and several of the aspects it looks at have already been discussed within Bayesian epistemology. In Lin's presentation, Bayesian epistemology is a normative framework whose core norms say that credences ought to be probabilistic and ought to change by conditionalization (Lin 2022, §1.2). It has the ambition of explaining or justifying a wide range of epistemic practice with little more than these two norms, focused on credence change. Bayesians have treated confirmation as a rise in credence and accounted for evidential support and explanatory power through it, and to many of them acceptance is secondary or, as Jeffrey argued, better forgone (Lin 2022, §1.3).

One aspect discussed there is how grounds other than observation move credence. Plausibility considerations such as simplicity, consistency with accepted theories, and analogy enter as prior probabilities, which is how Salmon read most of Kuhn's criteria of theory choice (Salmon 1990, 182–187, 198–199), and a non-empirical fact, such as that no alternative to a theory has been found despite long search, can under certain conditions confirm the theory (Dawid, Hartmann, and Sprenger 2012, §4). Whether norms beyond coherence govern priors, the problem of the priors, divides Bayesians. Subjective Bayesians permit any coherent prior, so that whether credences behave inductively at all, or favor simpler hypotheses, depends on the prior; objective Bayesians add a norm of freedom from bias (Lin 2022, §1.5). The same rule fed different values yields different verdicts (Salmon 1990, 180–181). Salmon did not deny that objectively irrelevant considerations, such as a scientist's hangover or political animus, move priors, and granted that any scientist may be inadvertently influenced by ideological or metaphysical prejudices. Salmon asked investigators to leave such considerations aside, on the ground that letting them set priors would defeat the objectives of science (Salmon 1990, 183). The second view is a way of viewing what in fact moves inquirers in the course of inquiry, and the influence Salmon granted falls within what it views. For the same reason, the second view does not by itself say which movements are rational.

Another aspect is that a ground counts only through what one trusts, and that what one trusts can change. Conditionalization raises the credence in the evidence to certainty, but Jeffrey conditionalization lets experience move credence without reaching certainty, and how far a reddish look should raise one's confidence that a jelly bean is red depends on what one

believes about the lighting (Lin 2022, §5.3). In Bayesian terms, a ground counts for someone who stands on a standard when that person's conditional credences, such as the credence that a hypothesis is true given that it is beautiful, or the error model of an instrument, make it relevant. Salmon called the Bayesian algorithms trivial and held that what matters is the scientific judgment involved in assessing the probabilities fed into them, and took the history of science to be a rich source of information relevant to priors (Salmon 1990, 201). A change such as Kepler's move from Ptolemy's ten minutes of arc to an estimate of the error bands of Tycho's observations is a change on the side of those inputs. Bayesian work has also modeled how such standards get their weight. In hierarchical Bayesian models, a preference for simplicity arises from the likelihoods of higher-level hypotheses and balances fit to the data rather than overriding it (Henderson et al. 2010, 187, 198), and in the meta-inductive argument the record of similar theories fixes priors (Dawid, Hartmann, and Sprenger 2012, §5). That standards gain and lose weight within inquiry, as the naturalness principle did, and that an inquirer comes to stand on new ones, is what Chapter 6, Section 5 examines as the revision of standards.

Change in which candidates are under consideration has been discussed as well. Taking a new theory into consideration seems to be the formation of a credence rather than a change in one, and the difficulty it raises for conditionalization is discussed as the problem of new theory. One proposal, sketched by Shimony and Salmon and developed by Wenmackers and Romeijn, takes credences to be conditional on the set of possibilities under consideration, lets that set expand when a new theory is taken into consideration, and preserves the credence ratios among possibilities that remain under consideration (Lin 2022, §5.2). Salmon also judged the likelihood on the catchall, the hypotheses not yet thought of, intractable, and removed it from the calculation by comparing the theories under consideration with one another (Salmon 1990, 191–192). Chapter 4, Sections 4 and 5, and Section 3 of this chapter concerned what comes before such an expansion: which of the matters an inquiry meets are taken up and which way it goes, and how candidates not yet available become available. On this proposal, when credence is assigned to a new theory, the credence ratios among the candidates that remain under consideration are preserved while the credence in each of them falls. As the work of making candidates available proceeds, then, credence in a formerly leading answer can fall without any evidence against it. An evaluation that counts research as advancing only when credence rises could, in such a case, count the work of forming alternatives as a setback. This is close to the change seen in Chapter 2, Section 8.2, which makes plain what is not known without adding answers.

Viewing the grounds in this way does not deny what Chapter 2, Section 5.1 noted of analogy, pilot results, trials, replication, proof, source comparison, and interpretive criticism: they differ in what they bear on, the alternatives they discriminate, their defeasibility, and the kind of standing they can provide, and calling all of them belief updates would describe one effect while losing the reason their effects differ. The second view names a common role; it leaves those differences to be stated, and it marks places where a single degree of confidence leaves something out. Two bodies of evidence that leave confidence at the same level can differ in weight: confidence resting on a few trials can shift sharply with the next one, while confidence resting on many is resilient (Genin and Huber 2020, §3.2). The role that data have already played also matters. In one proposal for preregistration in psychology, for example, studies lie on a continuum from exploratory to confirmatory, but data that helped to suggest a hypothesis are no longer appropriate for testing it (Wagenmakers et al. 2012, 3–5). Whether, and why, such use lessens the support that the data give is disputed. In Barnes's survey of the debate over prediction and accommodation, Christopher Hitchcock and Elliott Sober locate the concern in the risk of overfitting noisy data, while Mayo argues that a hypothesis constructed from the data can still pass a severe test, one it would be unlikely to pass if it were false (Barnes 2022, §§6.4–6.5, 6.7). Grounds also differ in what they make likely. Kepler's vicarious hypothesis, described above, kept its standing as a way of computing longitudes while it was rejected as a description of the orbit. And the weight an inquirer gives a ground can itself change with the ground's record. Fischer, for example, argues that the naturalness principle, under which the fine-tuning of the Higgs mass struck many physicists as an odd coincidence, gained credibility among particle physicists through its potential coherence with the promising theories it had inspired, and lost it as the Large Hadron Collider found no sign of the new physics those theories had led them to expect; Fischer leaves open whether this mutual support focused research on projects worth pursuing or led it astray (Fischer 2023, 1052, 1055–1058). Viewing grounds on one scale without losing these differences therefore involves saying, for each, what it makes more likely, by whose standards, and how robust the resulting confidence is. However well grounded, the revision of *A*'s confidence does not by itself make a candidate part of *S*'s knowledge, which depends on the relations examined in Chapter 2 and taken further in Chapter 8.

Section 3 left open whether entertaining a candidate, treating it as worthy of pursuit, accepting it, believing it, and knowing it differ only in degree. The two views bear on part of that question. On neither is the worth of pursuing a candidate a degree of confidence in it, and on neither does a rise in confidence alone make a candidate known, since knowledge has conditions of its own (Chapter 2). They differ over whether appraisals made before rigorous testing, such as finding a hypothesis beautiful, move the graded attitude that rigorous testing moves further, or can be read as judgments of what is worth pursuing and so differ in kind from rigorous testing. The two views can, however, both hold of the same appraisal. Finding a hypothesis beautiful can be a judgment that it is worth pursuing and also move confidence in it, so the views conflict only where preliminary appraisal is read solely as a judgment of what is worth pursuing or solely as something that moves confidence. What later chapters use holds on either view: on the first, pursuit is judged by standards of its own; on the second as well, as on Nyrup's model, the worth of pursuing a candidate also depends on the value of learning the answer and the chance of settling the matter.

When one sets out to build an artificial research system, this description of a practice does not by itself determine what that system must be given, but it suggests what to consider, and the suggestion depends, before the view taken, on for whom the matter is uncertain. As the opening of the second view put it, what this section has examined is a practice under uncertainty: inquirers who do not know the answer move toward what seems likely. That practice arises where the matter is uncertain for *A*, and what is unknown to *A* and what is unknown to *S* need not coincide (Chapter 4, Section 7). Where *A* is a machine and the matter, though unsettled for *S*, holds no uncertainty for *A*, as when it can be answered from what *A* already holds, *A* may not need this practice at all; what is then at issue is whether *S* can rely on *A*'s answer, which Chapter 11 examines. Where the matter is unknown to *A* as well, or complex for it, the description bears on *A* too. Whether a system must have an attitude like the confidence human researchers hold in this practice in order to reduce uncertainty in substance is not something the description settles. Where the practice takes place also depends on where the boundary of the artificial configuration is drawn (Chapter 7, Section 2). If generating candidates, preliminary checks, and criticism are divided among the agents of a multi-agent configuration, the first view's distinctions can appear as stages inside it; if the boundary is drawn around a single output of one model, such stages may not be visible at all, and trial and error need not occur in units like a human researcher's. Where the matter is uncertain for *A*, what is suggested differs with the view taken. The first view raises the question of how a system handles two appraisals by different standards, a prospective judgment of promise and a present judgment of support, and an evaluation could ask whether it can judge a line worth trying while doubting that it will work, as Reddy recalled doing, without coming to believe in the line for that reason. On the second view, a common graded appraisal of confidence could inform both judgments, without making pursuit-worthiness identical to confidence. If a system has such an appraisal for each candidate, one way to avoid the limit noted earlier, that a single degree of confidence leaves out the weight of evidence and the role data have already played, would be to record the grounds of the appraisal, what those grounds make likely and by which standards, how robust the appraisal is, which data have already been used to suggest the candidate, and how many candidates were tried on them. Another would be to keep track of how non-empirical grounds have fared, to record for each ground the standard through which it counted, such as the error model under which a discrepancy was judged significant, and to revise those standards when their record warrants it, which no update within fixed standards supplies. Judging what to pursue from that appraisal together with the value and cost of finding out also bears on the matter. On either view, a system that collapsed pursuit-worthiness into truth-likelihood and pursued only candidates it already found likely would forgo candidates that may be worth pursuing for other reasons, for example because they can reveal a method or, on a model like Nyrup's, because they would be easy to refute, and could thereby eliminate productive risk. Conversely, a system that ignored appraisal could spend indefinitely on arbitrary novelty.

Authors writing about systems built to make such judgments locate the risk differently. Chapter 4, Section 7 drew on two of them in asking whether exploration needs to track what *S* now finds significant; their arguments are looked at here in detail for what they imply about judging what is worth pursuing. Kitano, proposing as a grand challenge an artificial system able to make discoveries worthy of the Nobel Prize, contrasts a value-driven approach, in which scientists try to maximize the probability that what they discover will be significant under certain criteria, with an exploration-driven one, in which a system is to maximize the probability that discovery "at any level of significance" can be made without imposing such criteria. The reason, in Kitano's terms, is the horizon problem: where the landscape of significance is not monotonic, a search tuned to high expected significance may avoid the paths to a significant discovery whose preliminary results are of low expected significance, although, the paper adds, a system's resources still limit how much it can explore. Kitano expects

systems whose problems are chosen to reflect the values of their owners or stakeholders to be less likely to make unexpected discoveries, while keeping human recognition as the measure of the challenge's success: among a system's discoveries there should be some "considered very significant at the level worthy of the Nobel Prize or beyond" (Kitano 2021). Khosrowi, in a preprint on systems that predict discoveries and recommend ideas, questions, hypotheses, and experiments, argues that such systems face the gap between what is likely to be pursued and what is worth pursuing. Trained on the record of what was pursued and published, they learn features correlated with being pursued, such as funding, visibility, and the networks among researchers, and to move from predicting that a line will be pursued to recommending it is to fall into what Khosrowi calls a naturalistic fallacy trap; that a line would not be pursued otherwise does not by itself recommend it either, and on Khosrowi's account an independent standard of pursuit-worthiness is needed that the literature on these systems has not yet theorized. The preprint adds that, because such systems, if they work, steer the lines they predict, their predictive success after deployment cannot by itself distinguish anticipating what is worth pursuing from inducing researchers to pursue whatever is predicted, and that whether setting agendas can be left to them depends on what science is taken to aim at: if, as on the view Khosrowi takes from Kitcher, it aims at truths significant by human interests and values that are discovered as inquiry proceeds, humans would remain heavily involved (Khosrowi 2026, §§4.1–4.2, 4.4). Kitano's horizon problem tells against fixing a system's pursuit to present criteria of significance, while Kitano's own measure of success is still human recognition, which may be that of an *S* that recognizes the results later (Chapter 8, Section 3.2); Khosrowi's argument tells against reading worth of pursuit off the record of past pursuit. Where *A* is a machine and *S* a human society, both concern the relation between *A*'s judgments of what to pursue and what *S* now counts as significant (Chapter 4, Section 7). For a system that judges pursuit by expected epistemic value on a model like Nyrup's, seen above, the horizon problem does not disappear but moves: it appears in the value of learning whether a candidate is true, since where that value lies in the further discoveries an answer would lead to, it is what Kitano holds extremely difficult, if not impossible, to evaluate (Chapter 10, Section 2).

What Section 4.2 has argued about appraisal itself is this. Judgments of what is worth pursuing and judgments of what is likely to be true can come apart in practice, held apart by one investigator or drifting together in a community. Whether preliminary and rigorous appraisal are better viewed as different kinds of appraisal or as differing in degree within a common role is not settled here, and what this suggests for a research system depends on the view taken; on either view, the worth of pursuing a candidate need not reduce to its likelihood of being true. Both subsections bear on evaluation. A final correct answer, like other output evidence (Section 2.1), does not answer whether a system could take up or leave the unresolved matters a task left open (Chapter 4, Section 5), allocate attention among them, develop a candidate, recognize a dead end, or reorganize inquiry when early assumptions fail; Chapter 7, Section 4.3 considers where the selection of questions lies when research capacity is attributed to a system. The question of Section 4.2 returns in later chapters. In Chapter 10, a division of cognitive labor that becomes the internal portfolio of a single actor meets the problem of judging what is worth pursuing without collapsing pursuit-worthiness into truth-likelihood. In Chapter 11, the point at which criticism is applied to candidates still being formed bears on the same problem.

Heterogeneous operations, comparable functions

Section 4 argued that forming, pursuing, and appraising a candidate are distinguishable functions rather than separate phases. The familiar operations of research are similar in this respect, and this section asks at what level operations from different fields can be compared. Experiment, observation, proof, modeling, interpretation, literature review, construction, writing, and conversation name recognizable kinds of work. They do not name universal stages through which every project passes, and they do not each have one invariant function. The same operation can perform different functions in different inquiries, and one function can be realized through different operations across fields.

Comparing those operations is easier when three levels are kept apart; a finer analysis could add others. At the level of research as a whole, the activity is, on this book's working characterization, directed toward bringing about, with respect to some target, an epistemic achievement that would be new to a society *S*; *S* need not itself be the achievement's bearer (Chapter 1, Section 2). At a middle, functional level, evidence, formal constraint, criticism, construction, or interaction with the world can change the standing of a candidate content, representation, method, measurement relation, or dependency. At the level of concrete realization, this work is performed through such different operations as experiment, proof, calibration,

archival criticism, interpretation, construction, simulation, and replication. The middle level makes comparison possible. It does not establish that every project contains a discrete appraisal step or that the local operations license the same inferences.

A related framework appears in Chang's analysis of scientific practice. It treats scientific work as epistemic activities, each a more or less coherent set of mental or physical operations intended to contribute to knowledge in a particular way, which are practiced in relation to one another as systems of practice directed toward certain aims (Chang 2017b, 4–5); and, as El Mawas summarizes the framework, it distinguishes an activity's inherent purpose, the aim that partly defines it, from the external function it serves within a system (El Mawas 2021, §§2.1–2.2). The levels used here are cut differently. What this chapter calls operations correspond roughly to Chang's activities rather than to the mental or physical operations of which, in that analysis, activities are composed, and the middle level is narrower than those external functions: it picks out only roles that bear on the standing of candidates, representations, methods, and dependencies, so that such roles can be compared across fields rather than within one system. Nor is the top level one of Chang's systems of practice, each with its own aims; it is the direction that this book's working characterization assigns to research in general.

An experiment, for example, can serve functions other than testing a hypothesis, and need not begin from a sharply specified hypothesis. Drawing on Dufay's work on electricity, and naming among further examples Ampère's and Faraday's work on electricity and magnetism, Steinle describes exploratory experimentation as a way of working in which experimental action and the formation of concepts develop together, aimed at empirical regularities and the concepts and categories in which they can be stated, and typically found where no theory in the narrower sense, or not even a set of concepts in which the results can be adequately formulated, is available (Steinle 2004, 50–51). This does not make such experiments epistemically unconstrained. On Steinle's account they can be highly systematic, guided by aims such as systematically varying as many parameters of the setup as possible and determining which conditions are indispensable to an effect and which only modify it (Steinle 2004, 50). Their immediate objects of appraisal may be a regularity, category, or concept rather than a single theoretical proposition. Steinle holds that new concepts and categories must prove themselves, above all by allowing stable and as general as possible regularities to be formulated for the many experimental results (Steinle 2004, 51).

Proof is different from experiment, yet it also resists assignment to one box. A proof can warrant a theorem, reveal that the theorem was stated incorrectly, make a stronger conjecture available, develop a reusable technique, or show why a previously successful strategy cannot extend. Mathematicians who use computation describe a similar spread for a single operation. Borwein and Bailey, for example, as Sørensen reports, list among the roles of computer experiments gaining insight, discovering patterns, testing and especially falsifying conjectures, judging whether a result is worth a formal proof, and confirming derived results; Sørensen, drawing on Steinle, argues that some of this work is exploratory experimentation, while noting that its new roles differ from those in the natural sciences because the two fields started from different views of experiment, and that high-precision numerical agreement is not treated as a substitute for proof (Sørensen 2010, 346–348, 353). Interpretation in historical or humanistic inquiry can propose a reading, reorganize an archive, reveal a silence, discriminate among accounts, or alter the concepts through which a text or event is approached. Engineering construction can demonstrate that an arrangement is possible, expose constraints absent from a model, create an instrument for later inquiry, or generate knowledge of how to build. The functional description here does not settle whether an artifact, a proof, an interpretation, or a practical capacity is itself knowledge, a vehicle of or evidence for knowledge, or a means of producing it.

What can be compared at the functional level is therefore broader than truth-value determination by a standard test. Researchers make candidate contents and representations available; bring empirical, formal, interpretive, practical, critical, and other constraints to bear on them; assess what follows; and seek to alter what the inquirers, and those for whom the results are sought, are warranted in treating as possible, adequate, acceptable, true, known, or understood. This abstraction does not erase domain differences. A mathematical proof, calibration study, archival criticism, intervention, and replication license different inferences under different standards. The abstraction identifies a family resemblance in epistemic function while leaving the local realization visible.

Domain differences notwithstanding, a cross-domain functional continuity can be described on this book's working characterization: when a relevant epistemic resolution is not already available to *S*, inquiry includes changing what the inquirers entertain, believe, accept, reject, suspend, or treat as unresolved, and, where understanding is the aim, what they grasp and how. Where those who are to come to know or understand differ from the inquirers, inquiry also bears on their

standing, that is, on whether they are taken to accept, know, or understand the result, in whatever form the adopted account of possession, the account of who holds knowledge and how, gives it, from individual members' attitudes to a collective's acceptance or organized functioning (Chapter 2, Section 6). Proof, experiment, interpretive criticism, construction, and calibration can all constrain such standing. The continuity lies in this constraint-bearing role, not in one surface procedure, one inferential force, or one type of object being assessed.

A related claim appears in Dewey's account of reflective thinking: cases of reflection, however different, show common elements, five logically distinct steps, and no trait of a situation is labelled as evidence in advance, for the facts of the case are the traits that are selected and used as evidence in reaching a conclusion (Dewey 1910, 72, 103–104). Those common elements, however, are those of a whole act of thought, and in Hildebrand's summary, the pattern of inquiry that Dewey later set out ends when inquiry has converted an "indeterminate situation" into a "determinate one" (Hildebrand 2024, §4.3). The continuity proposed here concerns only one role within inquiry, that of constraining standing, and is in that sense narrower. Hoyningen-Huene makes a comparable claim about the sciences in a very wide sense that includes mathematics, the engineering and social sciences, and the humanities (Hoyningen-Huene 2020, 87). The particular ways of defending knowledge claims against error differ across them: proof in the mathematical sciences, empirical data generated, evaluated, and handled in very different ways in the empirical sciences, remnants processed into sources in the historical sciences, and statistical and other mathematical testing wherever descriptions or theories have been quantified (95). What the sciences have in common, on Hoyningen-Huene's account, is an abstract concept of systematicity that covers nine concrete concepts connected by family resemblance only, which further vary with disciplines and sub-disciplines, so that the unity of the sciences is an extremely weak one (90–91). This view is close to the continuity proposed here in placing the diversity at the level of procedures and the commonality at a more abstract level, though what Hoyningen-Huene compares is bodies of knowledge, such as science and everyday knowledge, rather than operations.

If the continuity lies in a role rather than in procedures, that procedures differ from domain to domain does not refute the functional comparison. The comparison becomes empty, however, if it suppresses what each operation can actually establish. Parker, for example, argues, as Winsberg and Alvarado report, that the material element of a computer simulation, the concrete computing device, puts simulation epistemologically on a par with conventional experiment, and that in both experiment and simulation what matters is whether there are relevant similarities between the system studied and its target (Winsberg and Alvarado 2026, §§1.1, 2.1). In a later account of model evaluation, the adequacy-for-purpose account noted in Chapter 3, Section 3.4, Parker holds that what a model must be like depends on the purpose for which it is used, together with its user, the methodology, and the circumstances of use, and that the features that make one model adequate for a purpose can be very different from those that make another adequate for the same purpose. To learn whether a new drug will relieve given symptoms better than an alternative, one scientist may use an animal model and another a computational model of the drug's molecular chemistry; both may be adequate, the first in virtue of biological traits of the animals and the second in virtue of quite different features. Conversely, a climate model whose ocean represents no internal dynamics can be known not to be adequate for learning how ocean dynamics will change as greenhouse gas concentrations rise (Parker 2020, 464–468). On these accounts, the label experiment or simulation does not by itself fix what a study can establish; that depends on which similarities there is reason to think obtain for the question asked.

The operation/function distinction also changes how sequences are described. A literature review may identify what is new to *S*, supply a method, expose a conflict, or generate a question. Writing may communicate a settled result, but it may also reorganize evidence and reveal that the argument does not work. Conversation can distribute calculation, critique an interpretation, preserve a problem, or redirect a project. Instrument building may support later inquiry or be the site where the central epistemic problem is resolved. Each visible operation can therefore serve more than one such function. First-person accounts by historians describe, for example, the use of indexed note systems to reorganize evidence and writing that changed the central actors and argument of a project (Walkowitz 2009; Harkness 2009). These retrospective accounts do not establish a general model of humanistic inquiry. They do show, in these cases, note-taking and writing doing more than presenting a result already reached.

Compared across fields at a high level of abstraction, operations can make questions and candidates available, transform them, bring constraints to bear, or support their communication and retention. Not every contribution performs all of those functions, and none is sufficient by itself to make an activity research. The point is functional continuity across

heterogeneous practices, not one universal procedure. The next chapter takes up the narrower question of how operations such as proof, testing, validation, calibration, replication, and criticism acquire evidential force (Chapter 6, Section 3).

For an artificial research system, this suggests asking not whether the system performs operations bearing names such as experiment or writing, but which functions each of its operations served in an inquiry: whether its writing, for instance, only presented a result or exposed a gap in the argument, and whether its literature search identified what is new to *S* or only summarized what was found. Which function an operation served is a question about how the work proceeded, which output evidence alone does not answer (Section 2.1). Parker's account bears here as well. A system that only runs simulations can still establish something about a target if there is reason to think the relevant similarities obtain, while calling a run an experiment establishes nothing by itself.

Material, embodied, temporal, and distributed inquiry

The picture in Section 1.1 listed, among the dimensions an inquiry can change, instruments and materials, practical capacities such as skills, actors, records, and social arrangements. This section takes these up. They are what studies of scientific practice and the debates over embodied, extended, and distributed cognition have examined in detail: research is carried out with materials and instruments, by trained bodies, over time through records, and by several participants together. The four terms of the section's title name these four respects.

An inquiry configuration is not a homogeneous collection of interchangeable items. By its elements this section means the things, people, and arrangements that take part in an inquiry, such as instruments, samples, organisms, code, notebooks, diagrams, archives, trained bodies, collaborators, and institutions. One element may matter because it helps make a candidate representable; another because it enables an intervention; another because it constrains what can be observed; and another because it preserves a skill, standard, or decision across participants and time. The same element can also matter differently in different episodes: a diagram can be a private memory aid in one episode, a space for inference in another, and a public justification in a third, and a material object can constrain action causally without itself representing anything. Explaining an inquiry therefore involves asking how each relevant element participates, not merely adding every nearby object to one bundle. Section 6.1 takes up the traditions grouped under the practice turn and two distinctions, taken from related debates, that help retain a lesson drawn from them; Section 6.2 asks in what sense inquiry is distributed; Section 6.3 turns to records, Section 6.4 to embodiment, and Section 6.5 asks how these roles change when the inquirer is not human. The four terms of the section's title do not correspond one to one to these subsections: material elements figure throughout, distribution, including its collective forms, is the topic of Section 6.2, the temporal dimension enters chiefly with records in Section 6.3, and embodiment is the topic of Section 6.4.

6.1 Practice, participants, and roles

This subsection prepares what the rest of the section uses to discuss elements: a lesson drawn from studies of practice, two distinctions for asking what an element contributes, and the observation that participants, unlike instruments, can withhold a role. Sections 6.2 to 6.4 apply them to distribution, records, and embodiment, and Chapter 7, Section 4.2 applies the first distinction to artificial actors.

Besides explicit propositions, the elements listed at the start of this section, and time, can be needed to explain what became possible in an inquiry and what happened next. The philosophy of experiment has studied several of these elements in detail: for example, the strategies by which experimenters argue that an instrument is working rather than producing an artifact, experimental traditions in which skill with particular kinds of detector shapes which arguments are found convincing, and the many purposes model organisms serve, only some of them representational (Franklin and Perović 2023, §§2.1.2–2.1.3; Weber 2018, §3.1).

The heterogeneous traditions grouped under the practice turn make different claims. Hacking foregrounds intervention, arguing that experimentation has a life of its own and that some experimental work precedes any relevant theory, while denying that experimental work could exist independently of theory, and grounds realism about entities in the ability to manipulate them to investigate other aspects of nature rather than in testing hypotheses about them (Hacking 1983, 150, 158–160, 262–263, 274). Pickering, as Allan Franklin and Slobodan Perović present his account, describes practice as a

dance of human and material agency that, seen from the human side, takes the form of what Pickering calls a dialectic of resistance and accommodation, in which a failure to capture an intended performance in a machine can be met by revising goals, the machine's material form, or the gestures and social relations around it (Franklin and Perović 2023, §2.2.4). Other work has brought elements into the explanation of inquiry in its own ways. Rheinberger studies experimental systems, material arrangements in which epistemic things emerge (Section 3) (Rheinberger 2011, 310–312); Latour analyzes inscription, by which a laboratory turns material such as rats and chemicals into paper, and the cascade of ever simpler inscriptions through which harder facts are produced (the second distinction below) (Latour 1986, 3–4, 16); Knorr Cetina compares the epistemic cultures of experimental high-energy physics and molecular biology, the "machineries" through which each produces knowledge (Knorr Cetina 1993, 258–259); Nersessian studies model-based reasoning with simulation devices in laboratories (Section 6.2) (Nersessian 2009); and Giere, as Matthew J. Brown presents his position, develops distributed cognition while denying that the whole distributed system must thereby be a knowing subject (M. J. Brown 2015).

These positions disagree about realism, agency, cognition, normativity, and the social production of knowledge. The actor-network approach developed by Callon, Latour, and Law, as Latour presents it, does not limit actors to human individuals but extends the word actor to nonhuman, non-individual entities, including the natural entities of science and the material entities of technology, and is symmetrical with respect to the two: granting an actor humanity and granting it materiality call for the same work of attributing action (Latour 1996, 2, 8–9). Latour defines an actor in this sense semiotically, as an actant: something that acts or to which activity is granted by others, with no implication of human motivation (Latour 1996, 7). From these otherwise divergent traditions this chapter takes one lesson: tools, materials, skills, records, and collaborators, among others, can each take part, in different ways, in what explains an inquiry.

Two distinctions help keep that lesson without silently adopting one of these theories. The first is drawn from the dispute over extended cognition, the second from laboratory studies and from work on what counts as scientific data. Both serve only to ask, case by case, what an element contributes to an inquiry; they do not decide whether an element is an agent or where cognition and agency are located. To say that an element plays a causal, enabling, or constitutive role, for instance, is not to say whether it is an agent, and the distinctions do not settle whether human and material agency are symmetrical.

The first concerns how an element, such as an instrument, a notebook, or a collaborator, is related to an operation or achievement, and it separates three relations. One is causal contribution: an element can be one of the causes that affect what happens, as a fluctuation in laboratory temperature can shift a measured value. Another is an enabling role: an element can be a precondition for an operation to take place, as a collision experiment cannot be run without an accelerator, though, if the operation is taken to be recording and analyzing collisions with a detector, the accelerator is not part of that operation. The third is a constitutive role: on an account that allows it, an element can be part of the task or achievement itself, as, on Clark and Chalmers's account below, a notebook can hold part of its user's beliefs. The dispute over extended cognition has turned on whether an element contributes causally or is constitutive. The enabling role is one this book adds between the two, to mark a relation that bears on an operation more strongly than being one cause among others, without being part of the operation. Clark and Chalmers argued that when a person and an external resource form a coupled system whose components jointly govern behavior in the way cognition usually does, the coupled process counts as cognitive whether or not it is wholly in the head, and that a notebook can in this way hold part of its user's beliefs (Clark and Chalmers 1998, §§2–4). Their critics reply that such arguments show at most that cognition is causally coupled to its environment, not that the environment partly constitutes it (Rowlands, Lau, and Deutsch 2020, §§5.2, 5.4; Shapiro and Spaulding 2021, §5). Whether or not one admits constitutive roles, causal contribution and the enabling role already allow some things to be said. An instrument can causally constrain an experiment without believing anything. A notebook, on the view taken here, can extend what a group reliably remembers without its entries becoming anyone's beliefs, since records in research can be kept to be reexamined rather than endorsed, so that no one need endorse what they record; the public chemistry notebook described in Section 6.3 is an example. On Clark and Chalmers's own criteria, automatic endorsement on retrieval is one condition for an external resource to carry beliefs (Clark and Chalmers 1998, §5), and records kept for reexamination do not meet it. A collaboration can distribute a task without constituting one group mind. In these cases, the enabling role suffices to describe what notebooks and collaborations contribute; whether they also partly constitute some operation is a separate question. Which role the same element plays is, moreover, relative to which operation or achievement is being explained. An accelerator that is a precondition for the operation of recording and analyzing collisions can, when the performance of the

collision experiment as a whole is explained, have its running counted as part of that performance. The name of an element therefore does not settle whether it enables or constitutes. Chapter 7, Section 4.2 uses the difference between an enabling and a constitutive role when it asks whether an external dependence continues to supply part of the organization attributed to an artificial actor.

The second distinction concerns whether an element only carries something or also shapes what can be carried. An element can preserve and move content, information, or skill across participants, places, and times; Hutchins, discussed in Section 6.2, treats the computation of a navigation team as the propagation of representational state across media. Yet an apparatus or record treated as a passive channel may determine what can appear as evidence (Section 6.3 gives a case of a record). Latour, for example, describes laboratory instruments whose end result, whatever their size and field, is a small window through which a few signs can be read, inscriptions that could be integrated as figures in the articles being written, and holds that phenomena invisible to the naked eye are seen only through inscription devices (Latour 1986, 4, 17). Leonelli's account bears on how an element shapes what can be carried as evidence: on Leonelli's relational account, "data" is a category applied to research outputs that are taken, at specific moments of inquiry, to provide evidence for knowledge claims (Leonelli 2015, abstract). Being data is thus not intrinsic to a research output, which fits what Section 4.2 said: data are not grounds simply because they are data.

On either distinction, which role an element plays is a claim to be supported case by case, not something settled by its being near an inquiry. Nor do they cover every contribution an element can make: it can also help make a candidate representable, bear on a candidate as evidence, or set standards, permissions, and responsibilities. Section 3 and Chapter 6 take up the first two of these functions; Chapter 11 takes up standards and the weighting of errors in certification (Sections 4 and 8), and Chapter 12 takes up permissions and responsibilities, including, in Section 6.4, how such elements distribute effective authority.

Roles can also be withheld. A participant whose operation an inquiry needs can decline to perform it for reasons of its own. Kwolek, for example, later recalled that the man in charge of the laboratory's spinning equipment at DuPont at first refused to spin Kwolek's cloudy polymer solution because it would plug the spinneret, and agreed only after continued insistence (Kwolek 1998, 6–7). Unlike an instrument, a participant can take up or decline a role for reasons of its own, so describing an inquiry configuration includes the possibility that who takes on what depends on the participants' own judgment.

6.2 Distribution and the study of laboratories

Research is usually carried out by several participants together with instruments and records, the actors and social arrangements of the picture in Section 1.1. That raises the question whether the inquiry itself, and not only its tasks, is thereby distributed, a question that the literature on distributed cognition, developed by Hutchins in studying ship navigation and applied to science by Giere and Nersessian, has taken up (M. J. Brown 2015; Nersessian 2009). It matters here because an artificial *A* raises it again in several forms. Whether *A* is an individual or a configuration of many agents depends on where the boundary is drawn (Chapter 7, Section 2): the same system can be described as one actor that is internally composed of many agents. *A* can also be coupled with instruments so closely that the two form one system, and it can take inside itself functions that human research distributes across a community, such as allocating effort among lines of inquiry (Chapter 10) or criticizing results (Chapter 11, Section 9). Section 6.5 returns to what changes. This subsection asks in what sense inquiry is distributed, and the answer has two parts. In the sense that tasks and work are divided among participants, apparatus, records, and time, much inquiry is distributed, and laboratory studies have described this in detail. Whether cognition or knowledge itself is distributed is not settled by such descriptions and needs a further theory. The paragraphs below first draw this distinction and then ask what laboratory studies show on each side of it.

The word distributed is thin unless a stronger theory is specified. Tasks, information, skill, causal contribution, and decision authority can be divided among participants without the resulting arrangement constituting one cognitive or epistemic subject. Whether cognition, agency, belief, or knowledge is itself distributed requires a further account; task distribution alone does not decide it. Hutchins's study of ship navigation, noted in Section 1, offers one such stronger account for cognition. Taking a navigation team with its charts and instruments as the unit of analysis, the study treats computation as the propagation of representational state across media inside and outside the participants' heads, argues that the team's cognitive properties differ from those of any of its members, and describes one reorganization of the team's work as

discovered by the organization before any participant discovered it: after the ship lost power and its gyrocompass failed, the plotter and the bearing recorder, each adapting locally to the work at hand, settled over successive fixes into a division of labor in which, as a team, they added the terms of the bearing correction in an order that let a shared partial sum be computed once and reused for each bearing, a solution that no participant had fully represented (Hutchins 1995, 117–118, 226, 317–318, 345, 351). Whether such a system also believes or knows requires the further account just mentioned.

A second main source of evidence about distribution is laboratory studies, which describe facts as produced through work divided among members, apparatus, and records. Among the practice-turn studies listed above, Latour and Woolgar's *Laboratory Life* provides close descriptions of inscription devices, statement modalities, credibility, and the work through which claims become stable in one neuroendocrinology laboratory. The material behind the descriptions is of two different kinds, which matters when the descriptions are used as evidence: about two years of participant observation in the laboratory, where discussions could not be tape-recorded and were reported from the observer's notes, and a reconstruction of how the structure of one releasing hormone came to be accepted, drawn from the published literature, fifteen interviews, and the archives of the two groups that worked on it. The authors say that the reconstruction seeks neither a precise chronology nor what "really happened," and that, on the chronology the two groups disputed, it follows one group's accounts more closely (Latour and Woolgar 1986, 39, 105–107, 149 n. 5, 153–154).

Those descriptions show fact production as work divided among members and apparatus, and so make it difficult to narrate it as a direct passage from individual observation to belief; in the terms of this subsection's opening, they show a distribution of tasks and work. They do not by themselves establish the authors' stronger theses, that facts and artifacts do not correspond to true and false statements and that reality is a consequence of the settlement of a controversy rather than its cause (Latour and Woolgar 1986, 176, 180–182). Nor do they show that all laboratories share one organization. The authors themselves caution against generalizing from this laboratory, and after a brief follow-up they judged its combination of group, space, equipment, and problems very unusual and perhaps unrepeatable (Latour and Woolgar 1986, 56–58, 233 n. 21).

How such descriptions should be interpreted is itself disputed. Latour and Woolgar treat a researcher's account of having had an idea as a condensed summary of material and collective circumstances and, in the postscript to the second edition, propose a ten-year moratorium on cognitive explanations of science (Latour and Woolgar 1986, 170–171, 174, 280); Hutchins's account and the program described next, by contrast, explain distributed work in cognitive terms. In the same postscript the authors grant that evidence of the kind they offer is unlikely to change positions that rest on deep-seated ontological commitments, and they take another author's opposite reading of their descriptions, after Popper's picture of science as conjecture and refutation, to show that their resources are insufficient to force their own interpretation (Latour and Woolgar 1986, 280–281). The people described also responded: members of the laboratory objected during the fieldwork to being portrayed as participants in a literary activity, and after publication, the authors report, their main reaction was that the book was "rather unsurprising" (Latour and Woolgar 1986, 53, 274). Observation, reconstruction, and ontology are accordingly separate matters to assess.

Nersessian's five-year cognitive-ethnographic program in biomedical-engineering laboratories bears on distribution in a further sense: across time and across a co-developing material system, not only among members. In the focal projects, problem specification, simulation devices, apparatus, methods, participants, and inference co-developed over years. Researchers negotiated trade-offs between experimental control and physiological realism, for example giving up some control and precision to exemplify features of human physiology (Nersessian 2009, 752–753). The article interprets these findings through model-based reasoning and distributed cognition; that interpretation is the article's, not raw observation. The bounded lesson is that, in these cases, the evolving material system was part of the explanation of how the epistemic problem could be specified and pursued.

Laboratory studies thus show richly that inquiry in the laboratories and teams they studied is distributed in the thin sense: work divided among members, apparatus, and records, and developing with a material system over time. On the strong sense they do not decide, since the same kinds of description are read in cognitive terms by Hutchins and Nersessian and against cognitive explanation by Latour and Woolgar. Distribution can therefore be described case by case, element by element, with the distinctions of Section 6.1, and where cognition and knowledge are located remains a separate question (Chapter 2, Section 6; Chapters 7 and 8).

6.3 Records

An inquiry rarely ends in one sitting; it continues across days and changes of participants. That is why the picture in Section 1.1 counted records among the dimensions of an inquiry, and it is chiefly here that the "temporal" of this section's title enters. This subsection asks what records contribute to an inquiry besides storing results already obtained, and looks at three things they do. One is to carry unresolved matters and failures forward so that the inquiry can be continued. Another is that their form shapes what counts as one trial and what is compared with what (the second distinction of Section 6.1). The third is that what is kept depends on who is expected to use it and for what, and a record can serve uses that no one planned.

Take the first. Notes optimized as cues for their maker may be unusable to later collaborators; records made for disciplinary reuse require different labor. Public laboratory notebooks and issue histories can preserve failed experiments, reinterpretations, corrections, logistical constraints, and decisions that a finished paper compresses. In one public chemistry notebook, for example, an undergraduate recorded a failed distillation and a conclusion, and two days later the undergraduate's supervisor reinterpreted the results, changed the conclusion, and added questions that the student answered about a year later; the supervisor afterward wrote that such an experiment is never really completed (Bradley 2008). Yet more logging is not automatically more knowledge. For a record to support later continuation, audit, reproduction, or criticism, it needs to be interpretable relative to those uses, connected to relevant provenance, and governed so that corrections or withdrawals can propagate where they matter.

The second is that a record can shape what counts as one trial and what is compared with what, and in that respect it does more than carry results (the second distinction of Section 6.1). A notebook of storage-battery tests from Thomas Edison's laboratory, dated by the editors of its digital edition to about 1900, is an example. According to the editors' note on the group of notebooks to which it belongs, the tests in these books are numbered from 1 through 1777, and by the second book Edison had established a standard format recording the time, current, voltage, and capacity of each cell during charge and discharge (Thomas A. Edison Papers 1900, editorial note). In the seven consecutive scans examined here, each of tests 146 to 152 opens with the weights of its electrode grid empty and filled and the composition of its active mass, followed by columns of readings. Test 150 is entered as "Same as 149," followed by a different proportion of nickel oxide and a note that its grid was plated inside, and test 152 as "Same as 151 Except pkd water moist"; the note to test 149 begins "As no 1/16 grid has worked" and appears to say that the test was meant to show whether the fault lay in the grid or in the nickel oxide (Thomas A. Edison Papers 1900, canvases 36–39). An entry of this form names an earlier test as its baseline and records only the difference, so that the numbered sequence itself lays out the comparisons. The numbering also fixes what counts as one test. Under the single number 152, the readings run over both pages of an opening and through several runs on different days, and notes of a new grid and new iron appear partway through without a new number, whereas the change in how the mass was packed, which separates 152 from 151, received one (Thomas A. Edison Papers 1900, canvas 39).

A count of experiments in such a record depends on how the record divides them, as a count of experiments in the log of an automated system depends on the units of that log, and a comparison may be available to later readers only because a record was kept in a form like this one. The tests can also be read as routine testing of a product rather than as research, and nothing here decides whether they were research in the sense of this book's working characterization (Chapter 1); the example concerns only what the form of a record makes countable and comparable. The pages were chosen to illustrate this point and are not offered as typical of the notebook or of its series.

Third, what is worth recording depends partly on which future actors and uses the record is expected to support. A project's record of what it has already tried serves its own later members, and keeping that record can depend on attention from others. In the publicly logged language-model project mentioned in Section 1, for example, one negative result was recorded in the project's issue only when another participant asked whether the idea had been tried, and in another case a participant's suggestion that an experiment had already been done was corrected by the person assigned to it, who replied that its input data had never been fully extracted (Marin Community 2025; Marin Community 2024).

Records can also serve uses that no one planned. Coincidences between the OPERA and LVD detectors, for example, had been searched for in a cosmic-ray analysis that found none of the events it sought; after the OPERA collaboration had found a fault in its timing system, the same muon records were used to date when the fault began (LVD and OPERA Collaborations 2012). A record can instead be kept for questions not yet formulated: the long-term evolution experiment

with *Escherichia coli* froze samples every 500 generations as a "fossil record," and this later allowed the origin of an unforeseen trait to be investigated by replaying evolution from earlier generations (Blount, Borland, and Lenski 2008).

Records thus carry an inquiry across time and participants, shape what can be compared, and keep it open to later uses. These functions bear on questions examined elsewhere in the book: whether a result is retained in *S* (Chapter 2, Section 6), how far corrections and withdrawals propagate (Chapter 11, Section 8), and who the users of records become when the inquirer is a machine (Section 6.5).

6.4 Embodiment

Research is carried out by trained bodies, which is why the picture in Section 1.1 counted practical capacities such as skills among the dimensions of an inquiry. This subsection asks what the body contributes to an inquiry and looks at three levels. One is that skilled perception and action make differences salient, stabilize instruments, and guide manipulation. Another is that states such as fatigue, haste, confidence, and aesthetic attraction alter attention, persistence, and stopping, and can themselves be treated as something a result depends on. The third is that bodily and other lived experience can bear on which unresolved matters are taken up at all. Much of the evidence here is recalled self-report, and its limits are noted along the way.

For the first two levels, retrospective scientist interviews supply possible mechanisms, but not frequency estimates. Nor do they by themselves establish a psychology of two modes holding for all research, such as the distinction that gives the podcast below its name: following François Jacob, Yanai and Lercher distinguish day science, the structured testing of hypotheses, from night science, the exploration of possible hypotheses not yet fully formed, and suggest that the distinction may be general to all enterprises that require creativity (Yanai and Lercher 2019). The episodes of the *Night Science* podcast, for example, are mostly such interviews (Yanai and Lercher n.d.). In a conversation with the hosts of another science podcast, the hosts of *Night Science* described choosing guests whom they judged creative, steering the conversation during recording, and editing out dead ends (Yanai and Lercher 2023). What the published episodes contain thus reflects the hosts' choices as well as the guests' recollections. The guests' reports of emotion and of how suddenly ideas arrived rest on recalled introspection rather than on recollection of what was seen and done, and some guests adopt the hosts' framing of the process while others reject it.

Other retrospective accounts describe similar mechanisms, for example the recollections of Bell Burnell and Keeling seen in Chapter 4, Section 5. Bell Burnell decided to analyze the telescope's chart records by eye and, with continued reading, became able to tell scintillating sources from interference (Bell Burnell 1977); Keeling, who under the rush of the work had put off testing a possible bias in the calibration gases, also recalled that years later a visiting colleague glanced at the pen traces of three analyzers and saw that one disagreed with the others (Keeling 1998, 46, 50). Bell Burnell's recollection bears on the first level, skilled perception bringing out differences; Keeling's bears both on the second, haste deferring a check, and on the first, a practiced eye catching a discrepancy at a glance.

Contemporaneous records can show such a mechanism, and a response to it, as they were noted rather than as they were recalled. In Linus Pauling's research notebook, for example, a page of electron-diffraction measurements on chlorine headed 2 May 1934 ends with the remark that there is "a large psychological error" in reading the inner maxima and minima: on 2 May the writer had noticed making "an effort to read the rings small," and on 3 May had made an effort "to read them large (or right, I thought)," knowing the high Cl-Cl values from the previous day. The remark adds that this does not affect the diameters from the second maximum out and averages only those; the next page, dated 3 May, records that the same films were "remeasured, using mask (having been bothered before by extra light)" and notes, below the results, a value that Badger had recalculated from band-spectral data (Pauling 1932–1941, 082–083). The page was chosen for this example. It can be read as showing a skilled reading treated, in the course of the work, as one of the things a result depended on, and a decision about which readings to rely on made in response; it shows nothing about how often such effects were noticed or recorded.

Third, bodily and other lived experience can also bear on which unresolved matters are taken up at all, without thereby bearing on what is warranted about them (Section 4). Bo Xia's recollection seen in Chapter 4, Section 5 was given in an episode of *Night Science*. The episode was recorded on the day the paper reporting the study by Bo Xia and coauthors appeared, with Bo Xia's former doctoral adviser, a coauthor of the paper, as one of the hosts. According to Bo Xia, a car accident had caused a tailbone injury, and for about a year the pain, returning whenever Bo Xia sat down, kept bringing

attention back to the fact that humans have no tail; curiosity about the injury prompted reading about how tails develop and how some animals lose them, reading that eventually led to the coauthored study of tail loss (Yanai and Lercher 2024; Xia et al. 2024). Like the other reports drawn on here, this is a recalled self-report given in a conversation that the hosts shaped. It shows neither how often such experience matters nor that the injury created the question. It shows one way in which an inquirer's own body can draw attention back to a matter before a research question about it has been formulated. When *A* is an artificial system and *S* a human society, Chapter 4, Section 7 noted that even the shared sources of such attention cannot be presupposed. The case concerns a personal and contingent experience. It is not an example of what standpoint theory analyzes, the dependence of which questions arise on socially structured positions, which Chapter 10, Section 6 takes up.

The deeper question of whether human evidential practices depend on primitive bodily expectations formed through long interaction with nature—and what could play that role in an artificial system—is taken up in Chapter 8, Section 2: Section 2.1 treats such expectations among the presuppositions of checking, Section 2.2 asks what a history of interaction can and cannot supply, and Section 2.3 considers artificial routes to world-answerability, the continuing openness of an inquirer's commitments to correction from the world. Here the narrower point is that an explanation of a concrete inquiry can include bodily capacities among what it explains. What plays each of these roles when the inquirer is a machine is taken up in Section 6.5.

6.5 When the inquirer is not human

The cases in Sections 6.1 to 6.4 are all drawn from human research, in which some features of the inquirer formed an unmoving background: a small working memory, skills acquired over many years, one body for each person, fatigue and affect shaping attention, researchers who cannot be copied, and finding out what has already been tried by asking one another, among others. When a machine inquires, some of them move, and the question is what then changes and what does not. What research depended on can be sorted into four kinds: premises about human realizers (a small working memory, for example), premises about the present organization of science (an organization formed in history that could be rearranged even with human realizers), conditions of inquiry (what remains whatever realizes the inquiry), and relations to parties outside the artificial configuration (including *S* when *S* is outside it). The four are told apart by how they fare when a machine inquires: the first fails when the realizer changes, the second when the organization changes, whether or not the realizer does, the third with neither, and whether the fourth holds depends on the outside parties. One element can fall under more than one kind, and the four do not cover everything a new realization brings with it. Chapter 9, Section 6 sets the sorting out in full and applies it to the relations surrounding research; this section applies it in advance to the roles of the elements discussed here. Cherniak's distinction is a guide. It separated limits that any finite agent faces from features peculiar to human memory, the former being unavoidable even for a creature with the resources of the entire galaxy and all the time until the heat death of the universe (Cherniak 1981, 165), and Cherniak stated conclusions conditionally on a feature, such as being a creature that "has a similarly duplex memory," rather than on being human (Cherniak 1983, 174). Nor does a machine *A* remove human constants altogether. Humphreys, writing on computational methods, traced the problem of the interface between humans and computation to the assumption that science is human science, and held that although a completely automated science raises fascinating issues, humans will continue to play an important role for the foreseeable future, so that the epistemology of the input and output interfaces must accommodate human limitations and abilities (Humphreys 2009b, §2). If *S* is human, it is humans who read, check, and rely on the results, and human constants remain on that side. Humphreys distinguishes the core simulation, which computes solutions, from full simulation, which also presents them (Humphreys 2009a, §3.3). As Ariga reports his account, different presentations of the same computed results can make a substantial difference to human understanding (Ariga 2009, 85). In his shorter essay Humphreys explains that graphics useful to human recipients need not be useful to the computer producing them (Humphreys 2006, paragraph beginning “These methods raise important questions”). Applied to *S*, these points suggest that which representational elements a method needs can depend on who has to take in its output.

Applying the sorting can split the roles of a single element. The opening of this section noted that a diagram can be a private memory aid in one episode, a space for inference in another, and a public justification in a third. The role of memory aid depends on a small working memory, the role of a space for inference need not, and the role of public justification depends on relations to *S*.

The same sorting can be applied to what the representations of science compress. Koichi Takahashi, writing about AI-driven science, holds that science has translated processes in which very many elements act in parallel into a few variables and a single line of argument, for a brain that can hold only about seven items, can relate only about four interacting variables at once, and tends to grasp events as one sequential narrative (Takahashi 2026, 120–121). Takahashi separates two things in this. Which particular variables a theory takes, such as position and velocity in Newtonian mechanics or temperature in thermodynamics, is on this account fixed by the structure of the laws, whereas the demand to compress nature into a few variables at all reflects the structure of human cognition. Takahashi also holds, however, that listing the position and momentum of every particle would describe nature completely without being understanding, not only for humans but for any finite AI however powerful, so that understanding remains a compression into what the language of whoever understands allows (130–131). Read with Cherniak's distinction, the demand for few variables is a feature peculiar to human memory, and the need for some compression a limit that any finite inquirer faces. If *S* is human, however, the demand for few variables returns on the side of *S* wherever its members are to grasp a result, as Humphreys's point about output representation suggests.

Beneath what looks like a premise about human realizers there can also be a condition of inquiry. Fatigue and affect are the human way in which how long to persist and when to stop is settled, but the question of persisting and stopping remains for any finite inquirer. On Cherniak's account, deciding which inquiry to undertake is itself an inquiry, and trying always to decide it by inquiry would produce a regress, one that is ended for human beings by their having been constructed, by natural selection, so that certain inquiries are undertaken automatically and instinctively (Cherniak 1983, 177). The case in Section 6.4 in which bodily pain kept drawing attention back to a matter can be read as one such way of ending it, and for a machine inquirer the question is what plays that role.

Some conditions of inquiry lie not in the inquirer but in what it inquires into. Takahashi, mapping where AI could accelerate science, distinguishes the computation needed to search a field's space of possibilities, which on this account added energy can push back, from the target's own time: the time in which a target system advances of its own accord, as in cell division, differentiation, and organogenesis, which accumulate sequentially and cannot be shortened by parallelization, in the recovery of an ecosystem over decades or centuries, and in the generations that institutional change takes. On Takahashi's account no technology applied from outside can alter this time, so that it remains even if laboratory automation drives engineering waits toward zero (Takahashi 2026, Appendix 1); Takahashi's map, which sets engineering waits to zero among its premises, is presented as an upper-bound assessment conditional on those premises (Takahashi 2026, Appendix 3). Takahashi counts the target's own time as one of four kinds of limit that remain permanently; the others are information that in principle does not exist to be obtained, such as lost historical information and the subjective experience of others; the limits, which Takahashi takes from Arrow, of procedures that aggregate several dimensions of value into a single score; and logical and mathematical limits such as the undecidability of the halting problem (Takahashi 2026, 149–150). The limit on aggregation can be read as relative to the conditions required of an aggregation procedure; what this section goes on to use is only the target's own time. The target's own time also bears on which candidates a machine inquirer can settle quickly. On Takahashi's account, a hypothesis that, like Maxwell's displacement current, unifies known laws in a mathematical structure is limited chiefly by computation, whereas one that posits an entity not directly observable, as Le Verrier's unseen planet and Pauli's neutrino did, must wait for the target to respond when it is tested (Takahashi 2026, 156). Stated in Cherniak's form, these are conditions not of finite inquirers but of the targets inquired into, and removing human delays from an inquiry leaves them in place.

Writers on the economics of artificial intelligence draw a related line in what limits the production of ideas. Philippe Aghion, Benjamin Jones, and Charles Jones, in a working paper on artificial intelligence and economic growth, distinguish two reasons why new ideas may become harder to find as knowledge advances. The burden of knowledge, a mechanism based on the limits of human cognition, may on their account fail to restrain an artificial system that can comprehend a much greater share of the stock of knowledge than a human can, whereas fishing out, a depletion they treat as a fundamental feature of the search for new ideas, would presumably apply to an artificial searcher too; such a system, they write, "wouldn't change what is in the pond." They add that the constraint that remains may then be set less by the limits of cognition than by natural laws (Aghion, Jones, and Jones 2017, 27–29). Read in Cherniak's form, the first is a feature peculiar to human cognition and the second, on their model, a condition of what there is to be found. The burden of knowledge, too, may cease to restrain an artificial searcher while still bounding what the human members of a human *S* can take up. Dario Amodei, in

the essay cited in Chapter 1, Section 2, lists among the factors that bound what added intelligence yields the speed of the outside world, the need for data, intrinsic complexity, constraints from humans, and physical law, and holds that some of them bind only in the short run: intelligence might, for example, develop an experimental paradigm that "allows us to learn in vitro what used to require live animal experiments" (Amodei 2024, "Basic assumptions and framework"). Read in the same form, such a route leaves the target's own time in place and moves the experiment to a system whose own time is shorter. What it then requires is that results on that system carry over to the target, as inferences from the in vitro models of Section 3 had to be carried over to the body, and that is a question of answerability to the target, of whether results obtained through such a system remain open to correction by the target itself (Chapter 8, Section 2.3) rather than of speed. It may also be that, in the substitute system, properties that can be iterated quickly come to the center of inquiry. Since on Parker's account what a model must be like is relative to the purpose for which it is used (Section 5), if inquiry proceeds without checking how those properties bear on the original question, what has become faster may be the answer not to the original question but to another question that can be answered quickly (compare the replacement problem of Chapter 4, Section 4, and the check passed on a question other than the one asked, Chapter 6, Section 3.2). Acceleration can thus press not only on waiting time but on the choice of what to investigate.

The same sorting applies to the body as a whole. Human research has been shaped by human bodies, is shaped by them now, and will probably continue to be. Machine research will probably not be shaped by bodies in the way human research has been. Abstracted, however, the claim that human research has been shaped by the body is one instance of a general claim: an inquirer is shaped by its own body. Humans and machines are both physical beings with some physical realization, so the general claim holds for machines too at least in the weak sense that what an inquirer can do is constrained by its own body; whether it holds in the strong sense that the body shapes even the content of concepts is not settled by being physical alone. Existing arguments about embodiment can be read by asking which parts are peculiar to humans and which belong to the general claim. The thesis that Shapiro and Spaulding call Conceptualization already has the general form: the properties of an organism's body constrain the concepts it can acquire, so that differently embodied organisms would understand their environments differently. The examples Lakoff and Johnson offer in its support, such as the concept of up arising from a body that stands erect, are peculiar to the human body (Shapiro and Spaulding 2021, §§2.1, 3.1). Nor is the claim that the body takes part in cognitive processes confined to humans: robotics has built machines whose behavior arises from the coupling of body and environment without a symbolic model of the world (Shapiro and Spaulding 2021, §4.1). Applied to the process of inquiry, at least the following depend on an inquirer's body in general: which differences it can sense, how fast and how finely it can act, which distinctions training can make discriminable for it, and where it tires, wears down, fails, or needs maintenance. By contrast, the way pain or fatigue, as felt, draws attention back, and the content of concepts derived from an upright body, are features of the human realizer rather than of an inquirer's body in general. The question is therefore not whether a machine inquirer has a body but which body shapes what in its inquiry, and how. Chapter 8, Section 2.3 considers how an artificial body or sensorium could support an epistemic organization different from ours.

Records are similar. The claim of Section 6.3 that more logging is not automatically more knowledge does not rest only on the scarcity of human time and attention: Cherniak argues that even with perfect retention the problem of locating a desired item would remain, and indeed be worse (Cherniak 1983, 175). What changes is who the future users of a record are: for example, other machines, which may be instances of the same system or systems of another make, humans in *S*, or auditors. The case in Section 6.3 in which a negative result was recorded only when another participant asked about it can be read less as a gap in the record than as a gap in a shared grasp of who knows what, and the philosophy of memory has discussed groups with such a grasp as transactive memory systems (Michaelian and Sutton 2017, §8.2.1). A machine configuration needs this function too, but it need not be met by participants asking one another. Where records are taken automatically, the selection of what to keep moves into whatever fixes what is logged and in what units.

The line between inquirer and element moves as well. In accounts of human research, including those seen in Section 6.1, the inquirer whose practice was described was human, whatever stance an account took on how agency and competence are distributed between humans and nonhumans, and whichever humans it also counted among the elements; the line between human and nonhuman could therefore at least separate the inquirer from the nonhuman elements. When the inquirer is itself a machine, even this no longer holds: the inquirer falls on the nonhuman side of that line. Latour called the incorporation of a competence in human bodies or its "excorporation" in nonhuman ones "one of the choice left to the designers" (Latour 1992, 168). For a machine inquirer, what is placed inside it and what outside it, as records or tools, is in this sense not given but

allocated within its configuration, whether by its makers or by a system that reorganizes itself. It may be tempting to say that the relation between the inquirer and its records thereby becomes flatter, that the distinction between an inquiring subject and its records as tools fades, but the phrase runs several things together. The symmetric description of actor-network theory is a methodological stance that declines to fix in advance how competences are distributed between humans and nonhumans (Latour 1996, 9), and it was already available for human research. What can newly change with a machine inquirer is whether its states and its records differ in form and where the allocation between inside and outside is placed. Neither follows from its being a machine; both depend on how its configuration is organized. The boundary can also move in two directions: records can be taken into the inquirer, or the inquirer's interior can be treated as one more store alongside those outside it. Hutchins argued that the classical symbol-processing computer, modeled on a mathematician manipulating external symbols, is not a model of individual cognition but a model of a sociocultural system from which "the human actor has been removed" (Hutchins 1995, 363). On this view, at least for such machines, the inquirer may correspond not to a human head but to a person together with records. Chapter 7, Section 2 takes up how to argue for where the line is drawn from the organization of an inquiry.

That a description is flat does not mean that the relations are. Delegating work to objects, Latour argues, creates an asymmetry "between absent makers and occasional users" (Latour 1994, 40). Law holds that a machine is also a heterogeneous network, involving not only technical materials but operators, users, and repair-persons (Law 1992, 384). Suchman argues that the boundary between persons and machines is not given but an enacted cut that carries accountability, and that the self-presentation of the labor-saving device erases evidence of the labor involved in its operation (Suchman n.d.). To say that a machine conducted an inquiry with its tools and records is therefore also to owe an account of where the line was drawn and what the line hides; Chapter 7, Section 4.2 asks which external dependences continue to supply part of a machine's organization. The withholding of roles (Section 6.1) changes form too. Whether an artificial participant can decline a role becomes a question of how that participant is organized, and the checks that human refusals supplied without anyone intending them can be lost. Conversely, when a machine-led configuration needs human work, the humans can decline, as the nurses who rejected the sensors did in the hospital case of Chapter 7, Section 4.4.

The reasons for distribution also come apart. Some reasons for dividing work lie in how much one person can hold and attend to; others, such as computational intractability and the impossibility of completing inquiry, remain for any finite agent. In human division of labour, members bring different positions and histories, so dividing a task has often divided perspectives too. In a machine configuration this is not given. Dividing a task among copies of one make whose background assumptions and routes of error have not come apart divides the task without dividing perspectives (Chapter 11, Section 3); dividing it among participants built from different bases or data, or made to differ, can divide perspectives as well, to an extent that depends on how they were made and on their later histories. Conversely, sharing a record is not the same as unifying perspectives. As Section 6.1 noted, no one need endorse what a record contains. In a configuration in which inquirers' states are distinguished from records, then, participants that consult the same record can hold different judgments, and a hypothesis can be kept in a shared record while only some participants are committed to it. A shared record does not by itself mean that judgments are uniform. In Hutchins's case in Section 6.2, the solution found by the organization arose from members in different positions, each adapting locally; whether the same happens among machine participants depends on how they differ and whether each can adapt locally where it stands. Giere held that only humans provide intentional cognitive agency to distributed cognitive systems, but presents the restriction as a theoretical choice that gives the fields of science studies a framework they can share (Giere 2007, 319). Where humans are absent, this framework places no knowing subject inside the system, and the question moves to whether a result becomes public knowledge for *S* (Chapter 8). The lesson of Section 6.2 about studies of research processes, that observation, reconstruction, and ontology are separate matters to assess, holds for studies of machine research processes too. But the dispute between cognitive explanations and material or collective ones may change form when the inquirer's cognition is itself a material process of computation. As long as what cognitive explanations described took place in human heads, they could be set against the laboratory's inscriptions and arrangements; for a machine inquirer whose interior can be treated as one more store alongside its records, the process described in cognitive terms may itself be an operation on stores of the kind that material accounts point to. Whether the kind of response in which members of the laboratory objected to how they were portrayed arises in the same form depends on whether the inquirers studied are organized so that they can contest how they are portrayed.

Across Sections 6.1 to 6.5, nothing in these cases establishes that every kind of research is material, embodied, temporal, or distributed to the same degree, or in the same sense. The cross-domain evidence broadens the range over which these roles can be compared, but it does not make the sample representative. What belongs inside an explanation of a research process remains claim- and case-relative and, as Section 6.5 argued, relative also to what realizes the inquiry. If *A* is a machine and *S* human, roles of elements that depended on human realizers become questions of how the configuration is organized, while human constants remain as conditions for *S* to read, check, and rely on results; if both *A* and *S* are machines, what corresponds to what in human research came from the inquirer's side, such as bodily experience or the withholding of roles, is in question. Different forms of mathematics, engineering, the humanities, natural science, and social inquiry, moreover, may distribute these roles differently. The resulting lesson is methodological rather than ontological: an explanation of a research process can include the relations needed to explain what became epistemically possible and what happened next, while leaving the location of cognition, knowledge, understanding, agency, and researcher status open to further argument.

Changes in the possibilities of inquiry

Research can transform what can be asked and tried, as well as which answer is accepted. In the cases taken up in this chapter, a representation made a candidate available, an instrument changed what could be investigated, and a record permitted a comparison that was previously unavailable. These are changes in an inquiry's possibilities. Candidate formation, pursuit, and appraisal distinguish work within that changing arrangement without imposing a temporal sequence. An artificial project that leaves the questions (Chapter 4), candidate space, and means of investigation fixed carries out one part of this activity; a project that opens them also requires that the work through which new possibilities become usable be realized or obtained. Section 1.1 called the relations in which this work occurs an inquiry configuration. When a machine inquires, which parts of this work depend on human realizers and which remain for any finite inquirer divides, as Section 6.5 set out, element by element and role by role. Carried through the accounts examined in this chapter, the change in an inquiry's possibilities has several concrete consequences. Taken together, Stanford's argument and the response of Curtis-Trudel and colleagues imply that merely searching an already characterized set of candidates more widely and quickly leaves the problem of alternatives not yet conceived in place, unless there is reason to think the characterization roughly exhaustive (Section 3). On the proposal for treating credence when a new theory is taken into consideration, the work of making candidates available can first lower credence in an earlier answer (Section 4.2). On Nyrup's model, some candidates are worth pursuing precisely because they are likely false and easily shown to be so (Section 4.2). And on Takahashi's distinction, the target's own time remains when human delays are removed from inquiry, and acceleration through a substitute system adds the work of inferring from that system to the target (Section 6.5). That a question can cease to stand without being answered, and how what is unknown to *A* relates to what is unknown to *S*, were seen from the side of questions in Chapter 4, Sections 5 and 7.

The account of process that Part III gives rests on the cases and arguments of Chapters 4 to 6, not on deduction from the general characterization. It can illuminate attempts directed toward knowledge or understanding without explaining their success conditions by itself. The cases taken up are limited, and how far the account reaches other practices, such as ordinary, abandoned, unpublished, and less represented research, remains an open question. Chapter 13, Section 2 considers how this process analysis bears on the working characterization of Chapter 1.

Making a candidate available is not yet establishing it. Whether it is established turns on results, but those results do not appear only after the work of obtaining candidates is over. In this chapter's cases, results were already obtained within that work and shaped it. In the laboratory Nersessian studied, the failure of more than half the experiments through contamination led to the flow loop's being redesigned so that it could run inside an incubator (Section 3). In the Liquid Tensor Experiment, imprecisions were exposed and repaired in the course of the work of formalization, and how the proof was understood changed as well (Section 3). Which claim, instrument, assumption, standard, representation, or other element such results bear on is not determined by the results themselves.

Chapter 6 — What Does a Result Bear On? Testing, Failure, and Revision

From a result to what it bears on

On Arfini and Costa's reading, what was to be explained for Penzias and Wilson changed in the course of the work of excluding possible causes of the noise picked up by the horn reflector (Chapter 4, Section 4). On this reading, results obtained within the work were revising the question. But what a result revises is determined neither by the result itself nor by the availability of candidates. The availability of a candidate does not determine whether believing, accepting, developing, rejecting, or leaving it unresolved is warranted. Nor does conducting an operation called a test determine what its result means. What this chapter mainly takes up is the relation between a result and what it bears on.

An unexpected result does not say which commitment calls for revision. A passed check does not say, by itself, how far its support reaches. This chapter asks how, while an inquiry is still changing, the work of determining what a result bears on is done, what guides it, and where it becomes hard. Chapter 2, Section 5 set out a schema, testing a claim by a method under auxiliary commitments to obtain an outcome, interpreting that outcome as evidence in light of background knowledge and standards, revising an attitude because of that evidence, and asking whether the revised attitude is knowledge, and, reading it line by line, called these four "steps." The same section also said that the schema locates the questions to be assessed and is not an algorithm for settling them. This chapter takes up what were there called the second and third arrows: the move from an outcome to its evidential bearing, and from that bearing to a revised attitude. This chapter, however, reads the order of the schema's lines not as the order in which inquiry proceeds but as a dependence: where a result bears as evidence, and what it revises. The issue here is which of the question, the candidates and their dependencies, and the methods, measurements, concepts, and standards through which results are assessed a result bears on, and how it constrains and revises them.

The central claim is limited:

> A result is an outcome of an interdependent arrangement. Moving from that outcome to a change in epistemic attitude or practice requires a further process of interpretation and diagnosis. Neither success nor failure identifies its own cause, target, or appropriate revision.

This claim does not entail unrestricted holism, the equal revisability of every commitment, or the impossibility of rational diagnosis. Its point is that locating what a result bears on is work that inquirers do, not something the result supplies. This book calls that work localization: diagnosing which commitment to doubt, examine, or alter next, rather than dividing a failure among everything that contributed to it; Section 2.4 gives the reasons for this usage. That work draws on background knowledge, error models, independent checks, interventions, practical skill, convictions and expectations, and further inquiry, among other things (Sections 2.3 and 4.1), rather than following mechanically from a scalar outcome. Section 2 examines what guides it and where it becomes hard; how far it can be accomplished for some kinds of arrangement is itself disputed (Section 2.4).

The claim is about evidential bearing, not merely causal complexity. An outcome can have indefinitely many causal conditions. The narrower problem is that a registered outcome becomes a reason to alter some commitment only through a represented or reliably managed relation among the question, the **dependency arrangement** (defined in Section 2.1), possible outcomes, and candidate revisions. The world constrains the result, but the result does not arrive with its evidential interpretation attached. Evidential bearing is meant here in what Kelly calls the normative sense of evidence, evidence as what makes it more reasonable to believe something, rather than in the sense of a reliable indicator, a relation that can hold before anyone knows of it. Because evidence in this sense bears on belief, this chapter works mainly through the case of knowledge. Where the investigators are artificial systems, it speaks of their commitments and of what a result gives them reason to revise, without assuming that they are themselves the subjects who believe or know (Chapter 2, Sections 2 and 2.1). Where inquiry aims at understanding, results also bear on models and explanations, and their assessment adds the questions of idealization discussed in Chapter 3, Section 3.4. Some philosophers take evidence in the normative sense to be a three-place relation, on which a result is evidence for a hypothesis only relative to a background theory (Kelly 2014, §3; compare the second arrow of the schema in Chapter 2, Section 5, from a result and background standards to evidential

bearing, and Longino's account of evidential relevance in Chapter 2, Section 6). In the indicator sense, a result can already point to a faulty element that no one has yet located; the claim of this chapter concerns what a result gives investigators reason to revise. This does not make truth or evidence products of social agreement. Relativity to background knowledge is not relativity to agreement: in Kelly's example, someone who does not know that Koplik spots indicate measles lacks a reason that a better-informed observer has, and would have it on acquiring the relevant medical knowledge (Kelly 2014, §§3–4).

The argument begins with unexpected results and their diagnosis (Section 2), then asks how far successful checks support a claim (Section 3). It follows both judgments across a sequence of results (Section 4), before turning to revision of the standards through which results are assessed (Section 5). Section 6 states what this account leaves to the analysis of artificial actors. This is an order of argument, not a sequence every inquiry follows.

The chapter describes how inquirers distribute doubt and protection in practice and sets out positions on what would warrant doing so.

When a result runs against expectation

This section asks which commitment a result that runs against expectation gives reason to revise, and how that comes to be settled. Its five subsections take up, in turn, where a result comes from, and what that suggests for research by machines (Section 2.1); why, although a contrary result bears in principle on the whole arrangement, not everything is equally in doubt (Section 2.2); what inquirers in fact doubt first (Section 2.3); what guides localization, where it becomes hard, and between whom it is contested (Section 2.4); and failures that have not reached the focal claim at all (Section 2.5). This order runs from the structure that in principle makes localization a problem, through the prioritizing that research in practice does, to the practice of localization and the kinds of failure it meets.

2.1 A result comes from an arrangement

A result does not interpret itself. The world can causally constrain what happens without thereby supplying a label that says which commitment the result supports or undermines. An expected observation is derived or anticipated through an arrangement that can include a focal claim, background theory, auxiliary assumptions, initial and boundary conditions, materials and organisms, environmental conditions, instruments and calibration, observers' readings, code and parameters, data selection and transformation, controls, statistical models, inferential rules, and divisions of competence, among other things. The reported result is also produced and made available through an arrangement. This chapter calls such a set of elements, together with the relations among them on which producing and interpreting a given result depend, a **dependency arrangement**. It is the part of an inquiry configuration (Chapter 5, Section 1.1) on which one result draws, seen from the side of the test. Deriving an expectation through such an arrangement is one way of articulating a candidate's consequences and dependencies, the work that Chapter 5, Section 4 noted serves appraisal as well as pursuit. What inquiry can register as agreement or conflict is therefore a relation among world-constrained but theoretically, materially, and inferentially mediated products—not a direct comparison between one hypothesis and an uninterpreted world. Nor does the point depend on the contested thesis that observation itself is theory-laden: even investigators who agree on how a result is to be described can, on views that relativize the bearing of evidence to background theory, priors, or paradigms, still disagree about whether, and how strongly, it supports a given hypothesis (Kelly 2014, §4).

A version of this point is generally known as the Duhem–Quine thesis, a name given jointly to two claims of different scope (compare Chapter 2, Section 5). Duhem argued, for physics, that an experiment cannot submit an isolated hypothesis to test but only a whole group of hypotheses, and that when a prediction fails the experiment teaches only that at least one proposition in the group is in error, not where the error lies (Duhem 1906, Part II, ch. VI, §2). Duhem also treated the laws of logic and the rules of algebra as rules to be respected in constructing a theory, and held that the test by facts bears on the theory's conclusions (Duhem 1906, Part II, ch. VI, §7). On Duhem's holism, then, what a contrary result can cast doubt on is a group of physical hypotheses, and logic can be read as not among them. Quine extended the point to the whole of knowledge, from geography and history to pure mathematics and logic: statements about the external world face experience "only as a corporate body," any statement can be held true come what may if drastic enough adjustments are made elsewhere in the system, and no statement, not even a law of logic, is immune to revision (Quine 1951, 38–40). Ariew notes that Quine

credited Duhem only with the first of these claims and that Duhem did not make the second (Ariew 2026, §2.1). This chapter relies only on the narrower claim the two share: a result that conflicts with an expectation derived from several commitments does not by itself identify which of them is at fault.

The structure Duhem described for physics also appears outside it. Statistical model checking offers an example. Gelman and Shalizi treat a posterior predictive check as a test of a model's assumptions, the prior among them, which has implications for the data only in conjunction with the likelihood; when a model fails such a check, on their account, the analyst has to work out what went wrong and where the problem lies before tinkering with the model or redesigning it altogether (Gelman and Shalizi 2013, 12–13 and n. 12, 25). Chapter 2, Section 5.3 sets out a related argument of Otsuka's, including the claim that the error rates a significance test reports depend on assumptions the test itself does not examine.

A failed prediction does not by itself refute the focal hypothesis. What logic delivers, where the expectation was deduced from the commitments, is that something is false somewhere in their conjunction (an outcome that a statistical model had only made improbable does not establish even that); to reject the hypothesis one must add the claim that the remaining commitments held, a claim that can itself be wrong. To say that a hypothesis has been refuted is to make that claim implicitly. The same holds for success. A prediction that comes true does not by itself settle which commitments it supports, or how much (Sections 2.2 and 3.2), since a false hypothesis can combine with a false auxiliary to yield a true prediction. Nor does a "crucial experiment" settle, in that simple form, which of two hypotheses is true. On Duhem's account, what Foucault's comparison of the speed of light in water and in air decided between was not the emission and wave hypotheses but two whole systems, Newton's optics and Huygens's, each taken as a block; and it had not been excluded that light might be something other than a stream of projectiles or a wave (Duhem 1906, Part II, ch. VI, §3). Reading off from a result where the fault lies is therefore not something the result supplies but work done in addition to it. This is the work that Section 1 called localization.

When a machine conducts research, what enters the arrangement on which a contrary result bears, and where and by whom is the attribution made? First, the contents of the arrangement change. In machine research, the arrangement can include code and data processing that the system has written itself, tools it has called, outputs of other systems, and the means of evaluation, among others. Much of this may have been assembled by the system itself, or may not be surveyable as a whole even by those who assembled it. In such an arrangement, the possibility that something elsewhere in it is wrong need not be smaller than the possibility that the focal hypothesis is. Second, what was just said applies directly to machines' reports of results. When a system reports that a hypothesis has been refuted, the report implicitly contains the claim that the rest of the arrangement held; a report of success likewise contains a claim about how far the support reaches (Section 3). A system that takes the result of an experiment directly as a verdict on the focal hypothesis therefore does not take up the work of localization, and building such a system does not by itself build one that determines for itself what to learn from a result. Third, the scale of trials changes. When many trials are run automatically, verdicts can accumulate without differences that come from the arrangement being distinguished from differences that come from the hypothesis.

Who checks such attributions changes with the combination of A and S. If A is a machine and S human, members of S check the attribution contained in the system's report against the record of the arrangement, that is, what was assembled in what order and what had been checked (Chapter 5, Section 6.3). If the record does not suffice for that, those members cannot tell from the record what a reported refutation or confirmation bears on; other grounds for relying on the report may still be available (Chapter 11, Section 5.2). If both A and S are machines, the one that checks is also an inquirer with an arrangement of its own, and checking inherits the same question. In either case, what the Duhem–Quine point suggests for research by machines is that receiving a result raises not only the question what the result was but also who judged, and how, where in the arrangement it bears. The end of Section 2.4 and Section 4.3 return to this question from actual records.

2.2 Not everything is equally in doubt

In principle, then, a contrary result bears on the whole arrangement, and where to attribute the fault is not settled by the result. Research in practice nonetheless doubts something first, keeps something, and starts examining somewhere, since checking one element relies on others held fixed (compare Neurath's image, taken up in Section 5.2, of sailors who cannot rebuild their ship in dock and must rebuild it at sea), and an inquirer with limited attention cannot examine many at once in any case. That prioritizing can rest on various things, including degrees of confidence in each element (Chapter 5, Section 4.2; Section 2.3), how much else connected with an element would have to be reconsidered if it were revised, the personal and social goals that support a theory's entrenchment (Section 2.3), and the tacit knowledge practitioners have of their own models (Section 2.4). Of these, the personal and social goals come from Chinn and Brewer's review of episodes from the history of science and of studies of students; what, if anything, plays their part when research is done by a machine is a question Section 4.2 returns to. This subsection and the next two ask what such prioritizing depends on and how far it can be grounded. This subsection first brings out that the elements of an arrangement do not all bear on a result in the same way. It then sets out why neither Duhem's point nor Quine's implies that every commitment is equally open to revision, why some philosophers reject the holism often drawn from them, how Duhem described the choice that logic leaves open, and how the same question arises within mathematics, where the commitments a counterexample can bear on might seem to be fully listed.

Not every element of an arrangement bears on a result in the same way, so localizing a fault is not a choice among interchangeable items. In the notebooks of the drug-discovery project Open Source Antibiotics (Chapter 5, Section 2), for example, the material that came off a silica column as a yellow solid was run again on a reversed-phase column, which separates by different properties, and divided into two: a white powder that the chemist labelled "DEBROMINATED CORE," the core structure with its bromine lost, that is, a by-product, and a much smaller amount of the intended product, which the next run scaled up (Tse 2023, EGT 522-1). Readings taken on the yellow solid could have been reliable as measurements and still have been readings of a mixture rather than of the intended compound. Had such readings run against expectation, what called for doubt might have been neither the hypothesis nor the instrument but what the result was a result about. Recognizing that a result comes from an arrangement also does not mean that the arrangement must be revised wholesale, or that an inquiry must represent every causal condition in the universe. What a description has to represent is set by the inferences and corrections that an inquirer or a system claims to make.

Which dependencies are salient also changes with the question asked. A reagent's storage history matters only for some questions; a classification scheme may be background in one episode and the focal object in another. The process that produces a measurement can itself become the object when a different question is put to the same records. In electronic health records from two Boston hospitals, for example, patients whose white blood cell count was normal when measured at 4 a.m. had lower three-year survival than patients whose count was abnormal when measured at 4 p.m., and for 118 of 174 laboratory tests the timing of the orders predicted survival better than the results; the authors attribute much of this to the processes of care that determine when tests are ordered, while allowing that daily biological variation may also contribute, and in their study the process of care, which a test result does not itself record, became the object of analysis (Agniel, Kohane, and Weber 2018).

Duhem stated the narrower claim for physics, contrasting it with the logically simpler testing of physiology, while noting that any science relying on physical instruments also presupposes physical theory; later discussion has extended it to theoretical testing in other fields (Duhem 1906, Part II, ch. VI, §1; Stanford 2023, §2.1). The claim does not imply that every logically available revision is equally reasonable, does not amount to radical conventionalism, the view that which revision to adopt is purely a matter of convention, and does not imply that every component is always equally doubtful. Quine's broader holism does not imply this either. Alongside the claim that any statement could be retained, Quine described a statement's nearness to the periphery of the system as nothing more than a loose association reflecting how likely particular statements are to be chosen for revision, and named among what guides actual revisions "our natural tendency to disturb the total system as little as possible," conservatism, and the quest for simplicity (Quine 1951, 40–41, 43). The stronger thesis that every such retention is rationally defensible is the one Laudan argued had never been adequately supported: on Laudan's account, as Stanford reports it, arguments for underdetermination show only that many responses to contrary evidence are logically possible, and although deductive logic alone may not single out one response, logic together with the ampliative

principles of reasoning used in science, principles of inference to conclusions not contained in the premises, may do so (Stanford 2023, §2.2). One reason the tendency not to disturb the system can matter lies in Quine's own account of why a contrary experience spreads: re-evaluating some statements entails re-evaluating others, because of their logical interconnections (Quine 1951, 39). Trying to revise a commitment on which much else has come to rest can therefore mean reconsidering what rests on it, and there are cases in which, unless something consistent can take its place, what is put in its place will not hold either. Doubting something that has withstood many checks can, in such cases, be costly. Duhem's example, discussed in Section 2.3, can be read as one such case: the principle that light travels in straight lines in a uniform medium had served almost as the physical definition of a straight line, and was abandoned when diffraction effects stopped being attributed to some source of error.

Some later philosophers accept the logical point while rejecting the holism often drawn from it. Sober separates a simple logical point from a contentious epistemological one, accepting the first and rejecting the second. Hypotheses rarely make observational predictions without auxiliary assumptions, but it does not follow that observations confirm or disconfirm only the conjunction, because auxiliary assumptions are often tested independently, so that a result can discriminate between rival hypotheses while leaving the shared auxiliaries untouched. A pregnancy test, in Sober's example, provides evidence about whether a woman is pregnant but leaves the woman's confidence in the reliability of the test unchanged; testing that reliability takes a different experiment. Sober takes epistemological holism to be wrong "in the vast majority of cases," while allowing that there may be circumstances in which a conjunction cannot be pulled apart (Sober 1999, 54–56). A later treatment in terms of likelihoods describes the typical situation in the calibration of measuring instruments and the validation of test procedures: their error characteristics "are first ascertained and then the procedure is applied to new individuals," as Galileo gauged the telescope on terrestrial objects before turning it to Jupiter, whose observation then told much about Jupiter's moons and little about the telescope (Sober 2004, 230–231). Whether holism is right about how evidence bears on a conjunction, Sober concludes, "crucially depends on what the alternative conjunctions are"; holism has its place, but in scientific inquiry it is, in Sober's words, "by far, the exception and not the rule" (Sober 2004, 241). This analysis of how far holism holds relies on testing auxiliaries independently. Section 2.4 takes up an argument that this response becomes unavailable in some arrangements.

Elgin's argument, discussed in Chapter 3, Section 7.2, gives a general reason in the same direction: commitments independently supported by other well-confirmed constellations are less likely to be the locus of error. Researchers can have strong reasons to protect some dependencies and investigate others. On such accounts, independent checks, calibration histories, robustness, error models, background theory, mechanism, cost, and later interventions can all help locate what to revise; whether successful mutual adjustment suffices to justify a localization is disputed (Section 2.4). Either way, localization is not contained in the word failed.

Duhem's own answer to what guides the choice was not a rule, nor did it treat the choice as arbitrary. On that answer, when experiment contradicts a theory it does not say what to change, no absolute principle guides the search, and a physicist who saves the fundamental hypotheses by adding corrections and one who alters a fundamental supposition cannot accuse each other of illogic. Duhem added that some opinions consistent with logic are nonetheless unreasonable, and assigned the choice between the two physicists to good sense, whose reasons lack the rigor of logic and do not become clear to everyone at once. Biot and Fresnel provide the example of a disagreement that logic did not settle and that lasted as long as it did for this reason: Biot defended the emission theory of light with corrections and auxiliary hypotheses, and Fresnel kept opposing it with new experiments, until one side gave way; Biot abandoned the theory after Foucault's experiment, although, on Duhem's account, logic alone did not compel Biot to (Duhem 1906, Part II, ch. VI, §10). The case also raises a question of timing, how long such a disagreement can reasonably last, to which Section 4.1 returns.

A dialogue of Lakatos's raises the question of this section within mathematics, where the commitments a contrary result can bear on might seem to be fully listed. In this classroom reconstruction of the history of Euler's conjecture about polyhedra, a history noted at the opening of Chapter 2, counterexamples are, as Musgrave and Pigden sort them, of three kinds: counterexamples to the conclusion of a proof that refute none of its premises, which Lakatos calls global but not local (Lakatos's term, distinct from this chapter's localization); counterexamples to a premise that leave the conclusion standing, local but not global; and counterexamples to both. On this analysis a counterexample to the conjecture is a reason to improve the proof rather than to abandon it (Musgrave and Pigden 2026, §3.1). One response to such a counterexample, as Aberdein

recounts the dialogue, is to defend the proved theorem by declaring the counterexample no polyhedron at all, a monster rather than a counterexample, and to propose a further redefinition for each new monster; Lakatos calls this monster-barring and characterizes it as a "sometimes deft but always ad hoc" redefinition (Aberdein 2019, 5, 9). The dialogue can be read as posing for a proof the question this section poses for a test: which part of a structured commitment a contrary result bears on, and when protecting the rest becomes ad hoc. It also shows that even a proof whose steps have been written out may not list every commitment a counterexample bears on: a counterexample to the conjecture that refutes none of the stated lemmas sends the inquirers back to the proof to unearth a "hidden lemma" (Musgrave and Pigden 2026, §3.1).

2.3 What is doubted first

Section 2.2 left open which commitments inquirers in fact doubt when a result runs against expectation. On the second view of Chapter 5, Section 4.2, an inquirer moves toward what seems likely, with degrees of confidence that depend on the standards on which it stands and on what it trusts. The same degrees of confidence bear on where doubt goes when a result runs against expectation. Bayesian treatments of Duhem's problem make this explicit, as an answer to the question what would warrant blaming one conjunct rather than another. On the treatment as Strevens develops it, a refutation considered solely as such, apart from whatever else the evidence says about the hypothesis, concentrates suspicion on the conjuncts that were more doubtful beforehand: conditional only on the conjunction's being false, the hypothesis loses less probability the more probable it was beforehand, and typically more the more probable the auxiliary was (Strevens 2001, §3). Mayo, writing against Dorling's earlier Bayesian reconstruction, argues that it can be run in reverse—someone who assigns the same degrees of belief with the hypothesis and the auxiliary interchanged reaches the opposite verdict—and that blaming an auxiliary is warranted only when the claim that it is at fault has itself passed a severe test (Mayo 1996, 106, 108–109), one that would very probably have failed the claim were it false (Mayo and Spanos 2006, 328–329). Whether degrees of confidence by themselves ground such an attribution is thus disputed. This subsection asks what inquirers in fact do, and it describes the answer loosely.

Which part of an arrangement is doubted first has also been studied empirically. Drawing on episodes from the history of science and on experimental studies in psychology and science education, Chinn and Brewer distinguish seven responses to data that contradict a held theory: ignoring the data, rejecting them (for example as a methodological error, random variation, or fraud), excluding them from the theory's domain, holding them in abeyance, reinterpreting them, making peripheral changes to the theory, and changing the theory. They argue that these exhaust the psychologically plausible combinations of three decisions: whether to accept the data, whether and how to explain them, and whether to change the theory (Chinn and Brewer 1993, 4, 12). They also report conditions that favor the six theory-preserving responses, in a list they do not present as exhaustive. One is the entrenchment of the prior theory, which they characterize by its evidential support, the range of explanations it takes part in, and the personal or social goals it serves. Another is the lack of a plausible alternative theory, often of a plausible mechanism. A third concerns the data: data that lack credibility, are ambiguous, or rest on a single study open to methodological criticism or reinterpretation (Chinn and Brewer 1993, 13–26). Discussing instruction, they add that it is frequently more rational to discredit anomalous data than to change theories (Chinn and Brewer 1993, 27). In experiments on everyday reasoning with conditionals, Walsh and Johnson-Laird found that participants asked to explain an inconsistency mostly revised the general conditional rather than the particular fact, and on most trials made more than a minimal change, as the authors measure minimality, by counting the propositions whose truth values change; they argue that people seek to explain an inconsistency rather than to minimize change (Walsh and Johnson-Laird 2009, 626–630). Neither study observed working researchers: Chinn and Brewer's examples of scientists come from histories of science, and Walsh and Johnson-Laird's participants reasoned about everyday conditionals.

Records kept while research was under way let one trace more directly what was doubted first. When Rosemary Redfield began in 2011 to repeat, in a research blog kept in the open, the claim that the bacterium GFAJ-1 can use arsenic in place of phosphorus, the cells grew on agar plates but not in liquid medium. The first posts suspected the tap water in the medium, the new culture tubes, and the air admitted by loose caps; a post three days later reported that it was not the water, the tubes, or the parafilm; and after a correspondent suggested potassium, two errors came to light, one in Redfield's own phosphate stock and one in the published recipe, which supplied no potassium (Redfield 2011a; Redfield 2011b; Redfield 2011c). When a flask with arsenate and limiting phosphate later showed thick growth, a result that seemed to favor the claim being tested, Redfield's first explanation was that a mistake had been made in setting up the flasks (Redfield 2011d). Doubt went first to

the replication's own arrangement, and the claim under test was examined only once that arrangement worked. The OPERA collaboration's report that neutrinos had apparently arrived sooner than light would allow shows a different shape. The collaboration wrote that it would investigate possible still unknown systematic effects and deliberately offered no theoretical interpretation of the anomaly (OPERA Collaboration 2011, 22). Among the two possible effects it identified five months later was the connector of the optical fibre that brought the external GPS signal to its master clock (CERN 2011, update of 23 February 2012), in a timing chain whose fibre delays it had measured by two-way procedures (OPERA Collaboration 2011, 13). The result itself was doubted first, but what came into question was an element that had been checked. A third record shows how what is given up depends on which explanation is at hand. In 2021 a user of the transformers library reported that gradient accumulation gave different training curves from the equivalent batch size; a maintainer judged that the difference could be explained by precision, the user accepted the explanation, and the issue was closed, to be reopened a month later when the user found one source of the difference in how training examples were grouped by length (Hugging Face Transformers 2021). In 2024 a similar report met the same suggestion of precision, which the reporter set aside after a rerun in full precision showed the gap persisting; the gap was then traced to how the library's default loss averaged over accumulated batches (Hugging Face TRL 2024; Hugging Face 2024). An explanation that was always available closed the question once, and an element trusted as a default turned out to be at fault.

Taken together, these studies and the cases in this chapter support only a loose description. The result itself is often among the first candidates for doubt. Beyond that, doubt tends to gather on elements that seem less secure by the standards on which an inquirer stands, but what is given up also depends on what explanation of the conflict is available and on whether an alternative is at hand, that is, on which candidates are under consideration (Chapter 5, Section 4.2). The description states a tendency, not a rule that only the least secure element is revised. Duhem observed both sides of it. Among the theoretical elements entering a test, Duhem wrote, there are always some that the physicists of a period agree to accept without check, and a physicist who has to modify the theory will surely direct the modification elsewhere; yet logic does not compel this, and refusing to invoke causes of error, and reforming propositions held by common agreement to be untouchable, can open a new path for theory. The example given is the principle that light travels in a straight line in a homogeneous medium, which had served almost as the physical definition of a straight line and was given up when inquirers stopped attributing the diffraction effects observed by Grimaldi to some cause of error (Duhem 1906, Part II, ch. VI, §8).

Observations of working laboratories show the branching that this description leaves open. Dunbar followed meetings and other activities in eight molecular-biology and immunology laboratories. In the reported cases, unexpected findings were variously attributed to unknown causes, methodological problems, mistakes, or potentially new mechanisms; follow-up included replication, further analysis, adopting a new protocol, and abandoning the current protocol. The exact frequencies are specific to the sample and coding scheme. Within that bounded sample, the observations support the existence of a branching diagnosis problem: researchers did not invariably convert an unexpected result into revision of the principal hypothesis (Dunbar 2000). Even their replications included minor changes of method, and the senior researchers attended particularly to unexpected findings in control conditions, both of which bear on loci other than the principal hypothesis (Dunbar 2000, 53). Dunbar also reports how the branches were handled: the scientists took an unexpected finding and tried to build a causal model of the events underlying it, deciding on their next step once that model was built, and when they were designing or fixing experiments their analogies were to very similar experiments or the same organism; Dunbar advises attending to unexpected findings in control conditions because they reflect either a conceptual or an experimental problem (Dunbar 2000, 53–54, 57). What these observations show is less an order in which commitments are doubted than that attribution branches, and some of the ways researchers work through the branches.

Where doubt goes can also involve convictions that come from outside the result at hand. Where a result allows either attribution, the choice between them can involve convictions and expectations, and even views about what is essential. Continuing to commit to an idea is one form of this, and is itself an attribution of results, in that it attributes results that do not work not to the idea itself but to not yet having realized it well. Judgments about where to attribute one failure after another may, accumulating, shape which ideas are kept. Hinton's recollections about neural networks, and those of the Transformer authors, can be read as cases of this; they are taken up in Section 4.1, together with the question of how long protecting a commitment remains reasonable.

2.4 What guides localization, where it becomes hard, and between whom it is contested

Granting that doubt is distributed loosely, what can guide the search for what to revise? This subsection sets out positions on that question: what experimenters use to argue that a result reflects the phenomenon rather than the apparatus, whether successful mutual adjustment suffices to justify a result, whether such guidance survives in arrangements of many jointly tuned components, who carries out localization, and what happens when the parts of an arrangement belong to different parties. It then states in what sense this chapter uses the word localization, and considers what comes into view for artificial research systems.

The philosophy of experiment has described some of the resources listed in Section 2.2 as strategies by which experimenters argue that a result reflects the phenomenon rather than the apparatus. Among them are checking that an apparatus reproduces known phenomena, eliminating plausible sources of error and alternative explanations, and using an apparatus based on well-corroborated theory (compare Chapter 5, Section 6.1); when a check of the first kind fails, Franklin and Perović write, there is good reason to question the results obtained with that apparatus. They present these as strategies from which physicists choose rather than as rules, neither exclusive nor exhaustive, and hold that no single one of them, nor any fixed combination of them, guarantees that a result is valid (Franklin and Perović 2023, §§2.1.2, 2.2.4). A more basic doubt has also been raised about whether such strategies can work at all. As Franklin and Perović report, Collins developed an argument and named it the experimenters' regress: a correct result is one obtained with a properly functioning apparatus, and a properly functioning apparatus is one that gives correct results; there are no formal criteria for deciding whether an apparatus is working properly, and the regress is eventually broken by negotiation within the community. Discussing the dispute over the detection of gravity waves, Franklin disagrees: the critics had exchanged data and analysis programs and calibrated their apparatus by injecting pulses of known energy, and the community, on this view, made a reasoned judgment (Franklin and Perović 2023, §2.2.1). The dispute is a reminder that a judgment of localization can itself be a result that does not interpret itself, since the judgment that the fault lies in the apparatus also comes out of the arrangement of checks that led to it. If so, where localization stops may not be settled by the logic of localization alone; the form is like that of the regress, discussed in Chapter 5, Section 6.5, that arises from trying always to decide by inquiry which inquiry to undertake. What stops it—reasoned judgment, negotiation, or both—can be read as what is in dispute here.

Whether stabilization by mutual adjustment suffices to justify a result is itself disputed. As Franklin and Perović set out Pickering's account of Giacomo Morpurgo's searches for free quarks, the observation of continuously distributed charge resisted his expectations. Widening the gap between the capacitor plates and revising the theoretical account of the apparatus brought integral-charge measurements and agreement among the apparatus, its theoretical account, and the theory of the phenomenon. They present this as Pickering's dialectic of resistance and accommodation (Chapter 5, Section 6.1) (Franklin and Perović 2023, §2.2.4). Franklin and Perović read the episode differently. Discussing Pickering's 1987 account of the episode, they argue that charge quantization had been strongly supported since Millikan, so that it was the failure of Morpurgo's apparatus to reproduce that known result, the failure of an important experimental check, that cast doubt on the measurements, whatever theoretical models were on offer; only when the apparatus could reproduce known measurements could it be trusted in the search for fractional charge. Against the 1995 account just outlined, they object that Pickering does not ask whether a given mutual adjustment of theory, apparatus, and phenomenon is justified, and seems to treat any adjustment that achieves stabilization as acceptable (Franklin and Perović 2023, §§2.2.2, 2.2.4). The two readings thus differ over what warrants localization: in Franklin and Perović's report of Pickering, the emphasis falls on the adjustment that achieves agreement; in their criticism, on an independently supported result that can guide the adjustment. Both deny that the result itself identifies the fault.

Humphreys describes guidance available before a test for derived theoretical templates, general mathematical representations obtained from explicit construction assumptions. A correction set supplies suggestions, not always explicitly formulated, for improving idealizations, abstractions, approximations, and other assumptions already known or suspected to be false at construction (Humphreys 2019, §3, advance-version 3 and n. 8). Knuuttila and Loettgers report the corresponding account of computational models: construction assumptions include ontology, idealizations, abstractions, and constraints, while a correction set anticipates avenues of adjustment to fit the data (Knuuttila and Loettgers 2023, §2). They also report Humphreys's more radical possibility of changing the ontology, with his qualification that this approaches constructing a

new template (Knuuttila and Loettgers 2020, manuscript 31, n. 17). Ariga reports Humphreys's criticism of the no-ownership perspective, which treats templates as already finished objects rather than attending to their construction and modification (Ariga 2009, 85). The guidance described here concerns models derived from explicit assumptions; Lenhard and Winsberg's argument, taken up next, concerns arrangements of many jointly tuned components.

The response to a Duhemian situation discussed in Section 2.2, testing an auxiliary independently, is precisely what some philosophers of climate modeling argue becomes unavailable as an arrangement grows. Lenhard and Winsberg, as O'Loughlin reconstructs their argument, claim that climate modelers lack what they call analytic understanding of their models, the ability to identify the extent to which each component contributes to a model's successes and failures, and that the complexity of the interactions among components makes it impossible to assess the merits of each independently (O'Loughlin 2023, §§2–2.1). On their account, as O'Loughlin reports it, a model's output is produced by the interaction of its modules during a run, not by combining the modules' separate results; parameterizations are built and tested on the basis of those already in the model; and kludges, made to improve the model as it then exists, become entrenched as other components come to depend on them. Examining a new component and the rest of the model separately then does not show how the component changes the model's behavior, since it cannot be told which of its interactions is responsible (O'Loughlin 2023, §2.1). They take three model intercomparison projects, whose aim of attributing differences in performance to components was repeatedly postponed or scaled back, as evidence for this view, while granting that the sources of some errors were tracked down (O'Loughlin 2023, §2.2).

O'Loughlin replies with diagnoses of disagreement among models from the same and later intercomparisons. In one, an implied ocean heat transport that ran in the wrong direction in some models was traced to their simulation of the radiative effects of clouds, which the investigators suspected first because models were known to disagree about clouds, and the diagnosis was checked by substituting observation-based values. O'Loughlin argues that the difficulty may lie more in the intercomparisons than in the models—dozens of models differing in unsystematic ways, with knowledge of how each was built and tuned dispersed across institutions—and that a standard requiring the extent of each component's contribution, as Lenhard and Winsberg's account of analytic understanding does, could discount any diagnosis that names a component: since a named component produced the error only in conjunction with the rest of the model, one could always ask whether it was wholly or only partly to blame. O'Loughlin suggests that the success of such diagnoses may reflect practitioners' tacit but working knowledge of their models, "a kind of Duhemian 'good sense'" (O'Loughlin 2023, §§1, 2.3, 3.2, 4.1, 5).

Localization need not take the form of a researcher's explicit verbal reflection. A reliable control structure, diagnostic procedure, or coordinated division of labor can localize some errors without any one participant representing the entire dependency arrangement. A notebook kept during a directed-evolution project and later made public provides an example of the last. At 12:28 on 23 November 2007, the researcher running a screen reported that none of six plates, each with more than a thousand colonies, yet appeared to hold variants with the new activity, attributed this to low mutation rates in earlier rounds, and planned DNA shuffling and a further round, while also returning the plates to the incubator to see whether color would develop over a longer time. At 16:15 the investigator leading the project commented that, "from memory," when a colleague had run the screen "it took a while for the colonies to go properly blue" and the screen "needed quite a lot of x-gal," the indicator substrate. At 17:56, after 24 hours of incubation, the researcher reported color just beginning to develop and eight possible variants, and the next day, asked about photographs, added that the color was "very very faint" and might not show on camera (Neylon 2012, post 5305). Read in sequence, the entries suggest that what the negative-looking reading bore on—the mutation rate, the time allowed, or the amount of substrate—was narrowed across the entries of two participants and the recalled experience of a third, and over hours rather than at the moment of reading. In the terms of Chapter 5, Section 6.2, this is localization distributed in the thin sense. The published paper reports, for an experiment other than the run this notebook records, that colonies were picked after 24 hours of incubation (Smith, Hale, and Neylon 2011, 2); given as method, the figure does not show that in at least one run of the screen how long to wait had been in question, and the pair does not show where the figure came from. The trace was chosen for this example and shows nothing about how often localization proceeds in this way. Chapter 5, Section 6.4 described two other forms: in Keeling's case a visiting colleague's glance at the pen traces of three analyzers found the one at odds with the others, and in Pauling's notebook the writer's own reading was treated as one of the things the result depended on.

Where parts of the arrangement belong to different parties, localization can itself be contested. O'Loughlin's point about knowledge dispersed across modeling centers is one case of this. The review of a report on replication attempts in cancer biology that could not be completed, one of which Section 2.5 describes, shows it too. The reviewers asked the authors to address limitations in their own technical expertise as a possible explanation, and the authors replied that a lack of detailed methods and reagent information was more likely, while agreeing that no conclusion about the causes could be drawn (Errington et al. 2021b, decision letter and author response). Collingwood offers a classical analysis of one way such contests arise, although for one restricted sense of "cause." In the practical sciences of nature, such as medicine, Collingwood held, a cause is a handle by which people can produce or prevent what they wish, a sense distinguished from the one used in theoretical science. No one enumerates the conditions of an event: when a result one has been obtaining fails to appear, what one wants to know is not what all its conditions are but which one has changed. And for each party, the cause is the condition it is able to produce or prevent, so that the driver, the county surveyor, and the car manufacturer name different causes of the same skid (Collingwood 1940, 296–301, 303–304). The two contests differ in kind. In Collingwood's example the three parties need not disagree about the conditions of the skid: each names as the cause the condition it can alter, and on that analysis each answer holds for the party that gives it, so that further evidence about the skid would not by itself make their answers converge. The exchange over the replication report concerned instead which condition had probably made the difference, a question on which further checks could bear, although none had settled it. What is in dispute can also be which of the conditions counts as an error. Herndon, Ash, and Pollin, replicating Reinhart and Rogoff's averages of growth by level of public debt, attributed the low growth reported for the highest-debt category to three problems: a spreadsheet error that omitted five countries, the exclusion of some early postwar years, and a weighting that counted each country equally regardless of how many years it contributed (Herndon, Ash, and Pollin 2013, abstract, 7–10). Reinhart and Rogoff's errata accepted the coding error, reporting an effect in line with the one Herndon, Ash, and Pollin had found, and answered the other points by placing them in the context of the paper's longer sample, its medians, and the peculiarity of the years just after the war (Reinhart and Rogoff 2013, 1–2, 4).

The discussion so far brings several things into view. There are resources that guide localization, but they are not rules, and what justifies a successful adjustment is itself disputed. In arrangements of jointly tuned components, testing components separately can lose its purchase. Localization need not happen in one head, and where parts of an arrangement belong to different parties, localization itself can be contested. Collingwood's analysis can be read as describing one shape that localization can take: a search for the dependency that changed, or that some party can alter, rather than a division of the failure among all the dependencies. Records can build such a search into their form: entries in Edison's storage-battery tests that read "Same as 149" and then list what was changed name an earlier test as their baseline and record the changes against it, at times a single one, as in test 152, entered as "Same as 151 Except pkd water moist"; and the note to test 149 appears to say that the test was meant to show whether the fault lay in the grid or in the nickel oxide (Chapter 5, Section 6.3). Sober's analysis of apportioning causal responsibility gives a reason why such a division can have no answer. Where causes act jointly, Sober argues, the question how much each contributed has an answer only if the causes are commensurable in the way they produce the effect, as gravitational and electrical forces acting on a particle are; this condition is offered as a conjecture. Where there is no such common currency, as with genes and environment in development, how much difference each made depends on which alternatives are taken for comparison, and how much each contributed is, in Sober's words, "not a factual matter at all" (Sober 1988, 303–304, 309–312, 318). Herndon, Ash, and Pollin's own table shows this dependence on the comparison. Applying any one of Reinhart and Rogoff's three choices alone to the corrected data lowers the estimate for the highest-debt category from 2.2 to 1.9 percent a year, while applying the exclusion and the weighting together lowers it to 0.3 percent; the table reports every combination (Herndon, Ash, and Pollin 2013, 10, 21). How much each choice contributed thus depends on whether it is compared with the corrected data or with data to which the other choice has already been applied.

These cases and arguments give reason to understand localization less as apportioning a failure among its contributors than as determining where to intervene next—what to doubt, examine, or alter—and that is the sense in which this chapter uses the word (Section 1). That determination can also lead to recasting the question (Chapter 4, Section 4). The cases do not show that localization always takes this shape. What they show is rather how varied its shapes are. Localization can be guided by independent checks and by successful adjustments; it can be one person's judgment or spread across participants and time; it can be built into the form of a record or contested between parties. That variety is itself suggestive. Before

treating localization as a capacity that one component or one procedure could implement, there is room to ask which form of localization is called for, in which arrangement, and by whom.

How do matters look for artificial research systems? The climate-model dispute above can be carried over to artificial research systems assembled from many jointly tuned components, with the difference that the localizing may be done either by the system itself or by those who build and maintain it; the practitioners' working knowledge to which O'Loughlin appeals corresponds to the second unless the system has a counterpart of its own. In either case, how far failures can be localized may depend on how the system was built, on how knowledge of it is distributed, and on what is taken to count as localization. On Humphreys's account, construction assumptions can supply advance guidance for correction. Humphreys does not apply this account to artificial systems, but applied to them, a system or members of *S* receiving a model without those assumptions may lack the corresponding guidance. By analogy, they may face the model as an already finished object, in the no-ownership perspective Ariga describes. The logbook of the training of OPT-175B, a large language model, at Meta illustrates a related loss of diagnostic information. After a run in which, in the logbook's words, "tons of stuff" had been changed at once, the team decided to "bisect recent changes to see what fixes instability" and rolled back one change, but in the run after that changed two things together; which layer was "really the problem" was recorded as an "unsolved mystery," and a run resumed from an earlier checkpoint was noted as not equivalent to the run it continued, because the restart had accidentally reset a state that checkpoints did not save (Meta AI 2022, 86–90). Because changes went in together and a resumed run was not the run it continued, even the team that built the model can be read as having had few means of checking which change had worked.

Collingwood's analysis concerns causes in the practical sciences, in a sense it ties to human control over what people wish to produce or prevent, not localization in research. Carried over to localization in the sense used here, where to intervene next, the analysis suggests that localization can also be relative to what the one localizing is able to alter and to what it seeks to produce or prevent. If so, an inquirer with means of intervention different from a human's, one that can, for example, repeat the same run any number of times or run many comparisons that change a single thing, may localize the same failure in a different place from a human. And whether the locus of a fault can be found afterward can depend less on the moment the result arrives than on how the arrangement was recorded before it (Chapter 5, Section 6.3), since what was changed when cannot be undone to check it unless it was kept.

Which localizing, the system's own or that of those who build and maintain it, is attributed to the system depends on where the boundary of the actor is drawn (Chapter 7, Section 2), and, as Chapter 5, Section 6.5 noted, when the inquirer is a machine the line between inquirer and element is not given either. What evidence could support attributing a more general capacity for diagnosis is taken up in Section 4.2. An inquiry can also render an anomaly harmless without localizing it—by a workaround that removes the symptom while saying nothing about its cause—and move on, much as in the response Chinn and Brewer call holding data in abeyance; a later account may then assign the anomaly a single cause (compare the BLOOM records in Chapter 5, Section 2).

2.5 Failures that have not reached the focal claim

Sections 2.1 to 2.4 have dealt mainly with cases in which the focal test was performed and its prediction failed, and have asked where in the arrangement that failure is to be placed. Not everything recorded as a failure, however, has reached the focal claim. Some failures bear on it only weakly or not at all, and for these, what the failure is has to be asked before where it is to be placed. As Chapter 1, Section 3 noted, failure is not exhausted by one verdict: a project can miss its original target while establishing boundary conditions, ruling out a possibility, or exposing an experimental artifact; the cases here add that which outcome a failure has depends on what it is taken to bear on, and that this can change after the fact.

A replication attempt can fail without the focal test being reached. Study 5 of the Reproducibility Project: Cancer Biology, for example, preregistered an in vivo test of a glioblastoma-tumor claim, but made that test conditional on producing stable cell populations that met a specified GFP-expression threshold (Chroscinski et al. 2015). After replacement materials, repeated optimization, enrichment, and changes suggested by the original authors, the required condition was not achieved for all of the relevant cell populations (Errington et al. 2021a, box 5; Errington et al. 2021b). The replication team stopped the attempt, citing the rising cost of repeated optimization with no feasible path to a solution, and the focal xenograft experiment was never performed (Errington et al. 2021a, box 5; Errington et al. 2021b). The attempt therefore neither

replicated nor refuted the target tumor claim. It produced a record of where this implementation failed to reach its planned test, without identifying why it failed or showing that the original result is unreplicable. That a replication attempt failed, that a prerequisite could not be realized, and that the focal test returned a negative result are different things, and what can be said of this replication goes no further than the first two. Between them lie still other cases, in which a focal test is run in part, or under conditions that only partly meet its plan.

An implementation that does not realize the stated method can mislead the diagnosis of failure itself. In one language-model project, for example, a model trained and diagnosed as a 104-billion-parameter configuration turned out, because of a configuration error, to have been "a very lopsided 58B model instead of 104B all along," so that the failing runs had been made with a different model from the one described (BigScience 2021).

Even when the focal test has been performed and has returned a negative result, that result is not one thing. A null result may, for example, constrain a claim, reveal insufficient sensitivity (compare the screen in Section 2.4, read before its color had developed), expose poor operationalization (the step from a concept to a procedure for measuring it), or remain uninformative, and a statistically non-significant or otherwise indeterminate outcome may fail to discriminate among the relevant alternatives; a proof checker may likewise reject a valid idea because its formalization is defective. Which of these a result is depends on the question, measurement, sensitivity, and inferential setting. Collins's case study shows how, in gravitational-wave physics, a non-detection can be transformed into an upper limit, while also emphasizing that the resulting scientific value varies with field and audience (Collins 2003, abstract). What a negative result bears on can also depend on a question asked later. For example, several mutants of *Mycobacterium tuberculosis* unable to make the sulfated glycolipid SL-1 showed no detectable defect in mouse infection models when their makers reported them (Kumar et al. 2007; Gilmore et al. 2012); a different laboratory later used strains supplied by the same group to show that SL-1 contributes to cough in infected guinea pigs, and took the earlier in vivo findings as evidence that SL-1 is not a classic virulence factor, evidence with which its own results were consistent (Ruhl et al. 2020). What the earlier result was worth depended in part on a question its authors had not asked. In light of these cases, the claim that the target failed, the claim that the procedure failed, and the claim that the research episode failed are different claims, and what a negative result is taken to show changes with which of them is made.

For artificial research, these distinctions imply that a record of failed runs does not by itself say which failures bear on the hypothesis under test. A system that logs an unperformed test, a misdescribed implementation, and an insensitive null result alike as evidence against its hypothesis presupposes, as Section 4.2 discusses, where the fault lies.

The same holds for a system that reads records kept by others. The labels that participants put on their own work can record what is to be carried forward without recording which of the states or claims distinguished above applies to a result. In the public chemistry notebook described in Chapter 5, Section 6.3, for example, three pages of attempted reactions from October 2007 were marked "aborted" because the sample of an aldehyde used in most of the reactions was judged bad ("contaminated" and "providing false negative results," in one page's words), and two of the pages say that the compounds are to be attempted again; on a later page, two reactions were marked "Void" because ten times the intended amount of one acid had been added, so that an apparent precipitate came from insolubility rather than from reaction (UsefulChem 2010, Exp137–139, Exp141). The first label marks a prerequisite that was not met; the second marks a procedure that failed, and what it sets aside is an apparent success rather than a failure. Yet one of the aborted pages enters "aborted", and the later page enters "Void", in the table column where whether a precipitate formed would otherwise be entered, and on two of the pages the label also covers reactions that did not use the suspect aldehyde. A system that learns from or aggregates such records and reads their labels as results presupposes where the fault lies as much as one that logs all its own failures alike as evidence against its hypothesis. Nor are parallel attempts independent with respect to what they share. On the first of these pages, all eight reactions had been made with the same doubtful sample, so that one reagent set aside eight results at once; a count of parallel trials is not a count of independent checks on an element they share (compare Section 3.2).

When a check is passed

Section 2 argued that a result that runs against expectation does not by itself identify what it bears on. It also saw that where such a result is attributed can depend not only on the result but on convictions and expectations, on the means of intervention at hand, on records, and on exchanges between parties; that what a failure is can itself change afterward; and that such judgments may shape which paths inquiry goes on to take. When a result runs as expected, that is, when a check is passed, the pass likewise does not by itself say what it has established: which part of the inquiry's question it bears on, how firmly it settles that, whether the success depends on something other than what was meant to be tested, and which of the commitments that produced it the success supports.

What was asked on the side of failure thus has counterparts on the side of success. Just as it was asked of a failure what it is in the first place (Section 2.5), it is asked of a pass what part of the question it bears on and how firmly it settles that (Section 3.1), and whether the success is only apparent (Section 3.2); and just as it was asked where a failure is to be attributed (Sections 2.1–2.4), it is asked which commitments a success supports (Section 3.2). There is also a difference. A result that runs against expectation signals that something is wrong somewhere and prompts a search, whereas a pass can be treated as a reason to stop looking, so that a misreading of a pass can go unnoticed longer than a misreading of a failure (compare the primary mirror of the Hubble Space Telescope in Section 3.2).

When research is done by a machine, how it is arranged who sets a check, where in the inquirer it is placed, and who reads a pass and how, can differ from human research; and a pass may be recorded without the record keeping these differences (compare Chapter 2, Section 5). Section 3.3 returns to these arrangements. This section asks what a passed check establishes. It first fixes how the word verification is used here, then distinguishes the target and the force of a check (Section 3.1), turns to successes that are only apparent, to what a success that is not merely apparent supports, and to the security that agreement among checks provides (Section 3.2), and ends with who sets a check and who reads a pass (Section 3.3).

Chapter 2, Section 5 treated verification as a family of practices whose common function is the assessment of a candidate's epistemic standing under uncertainty, and Chapter 5, Section 5 compared heterogeneous operations by their constraint-bearing role. This chapter uses verification only as a broad comparative umbrella when the common function matters, and at the point of analysis it names the more specific practice and the inference that practice licenses. What is asked here, whatever the stage that did it is called, is what was assessed, by what operation, under which assumptions and standards, and with what bearing on the live alternatives. Sober's account of testability puts the question in a similar way. Defending the notion of testability, Sober argues that testing is contrastive, so that the same proposition can be testable relative to one competing hypothesis and untestable relative to another; what is empirical or nonempirical is in the first instance a problem or question, and propositions are testable only derivatively (Sober 1999, 52–53). The same paper describes the inferential tools by which scientists evaluate competing hypotheses in the light of observations as a toolbox whose contents are neither fixed for all time nor the same in all scientific disciplines (Sober 1999, 68). The function considered here is wider than Sober's, since it includes operations, such as proof, that do not weigh hypotheses against observations. Practitioners have themselves compared practices by their function without erasing the differences. Two mathematicians, Jaffe and Quinn, held that rigorous mathematical proof plays a role functionally analogous to experiment in the natural sciences, checking the reliability of claims and yielding new insight as a byproduct, while declining to call proofs experimental (Jaffe and Quinn 1993, 2). What the practices share, in this limited sense, is that something remains unresolved, a claim that might resolve it is on the table, and an operation is used to determine whether, and how firmly, the claim does so. The practices do not become epistemically equivalent merely because they can occupy a comparable functional position, nor, as Chapter 2, Section 5.1 and Chapter 5, Section 4 argued, need verification follow candidate generation as a separate phase.

3.1 Target and force

Chapter 2, Section 5.1 noted that verification practices differ in what they bear on, the alternatives they discriminate, their defeasibility, and the kind of standing they can provide. This subsection takes two of those differences to compare checks: their target, what part of an inquiry's question an operation bears on, and their force, how firmly it settles that. The pair is chosen because the bare report that a check was passed leaves both unstated; together they do not make a complete description of a check.

A valid proof settles what follows from its premises, but whether that answers an inquiry's question depends on the question (on what counts as an answer to a question, see Chapter 4, Section 2). Where the theorem is itself the question, as is often the case in pure mathematics, a valid proof answers it, and what remains open is chiefly whether a formal statement captures the intended theorem (Chapter 3, Section 5.1). Where a derivation concerns a model used to answer a question about the world, the same proof establishes a property of the model, not that the model represents its target well enough for the purpose. Simulation practitioners, using the word "verification" in a narrow sense, mark this difference as one between verification, whether a computation approximates the solutions of the model's equations, and validation, whether the model is a good-enough representation of the target for the purpose; Winsberg, as Winsberg and Alvarado report, argues that the two cannot always be kept apart, since equations are sometimes chosen to offset the errors of the method that solves them (Winsberg and Alvarado 2026, §2.2; compare Chapter 3, Section 3.4). The difference between the two cases is one of target, not of force: a proof in pure mathematics is not weaker for answering a question that is itself mathematical.

Operations other than proof differ in target in the same way; in each of the following cases, what passing settles can be something other than what the focal question asks. Passing a calibration can support an instrument model without supporting the focal scientific claim: the two-way measurements by which OPERA determined the fibre delays of its timing chain (Section 2.3) supported an assumption about how the timing chain worked, not the claim about the speed of neutrinos. What a successful experiment establishes may be not which of two antecedently formulated hypotheses is true but that a phenomenon can be produced stably, or that a new variable is involved. An interpretation's surviving criticism can show that a reading holds under the contexts and contrasts that were considered without settling whether it holds under others. That an engineered artifact works can establish possibility under one set of conditions while leaving generality, mechanism, and explanation unresolved. Passing a check that tests an artifact against a specification, as an executable checker typically does, supports the artifact's conformity to the specification, while whether the specification expresses what was wanted remains a further question (Angius, Primiero, and Turner 2025, §§2.2–2.3, 6.2). In an experiment that Lange and colleagues report (Chapter 4, Section 8), a loosely worded rubric that did not explicitly forbid backpropagation produced a result with high validation accuracy that did not implement the intended learning rule.

Operations on one target can also differ in force. In mathematics, for example, the same argument was checked by two operations that differed in how far they could settle its correctness. In the spring of 1993 Andrew Wiles taught the central construction of the proof of Fermat's Last Theorem in a course that did not reveal its purpose, and Nicholas Katz, who after a few weeks was the only one still attending, saw no error in it. That summer, reading the manuscript line by line as a referee, Katz sent query after query until one exposed what one of the two later called a fundamental error; the account is a retrospective one edited for television (NOVA 1997). What the episode suggests is that the difference lay less in the target than in how each operation was organized: following a course whose purpose was undisclosed is not designed to settle correctness, whereas refereeing line by line is. Force can, moreover, be reported in two ways, which Chapter 2, Section 5.2 distinguishes as confirmation as relevance and as firmness: how much passing raises the support for what the check bears on, and how well supported that is after passing, which also depends on how well supported it was before; whether that is enough for the use at hand is a further judgment (Chapter 11, Section 4).

What was said above of proofs holds also for proofs of a limitation. A valid proof firmly settles what cannot be done under its premises. What it settles, however, concerns the model its premises describe; how far that model represents actual devices or tasks, and how much the limitation weighs for actual uses, remain outside the proof. The force of such a result can therefore come to be read as the breadth of its target. The history of Minsky and Papert's *Perceptrons* (1969) can be read as one case of this. One of the things the two proved is that no perceptron of finite order, one whose features each depend on no more than a fixed number of input points however large the input grows, can represent the connectedness predicate. Bottou, in the foreword to the reissue, notes that this says that a finite-order perceptron cannot represent every predicate, and does not necessarily mean that a learning system cannot represent the predicate that tells whether an image shows a cat (Bottou 2017, xi). On multilayer perceptrons the two offered not a proof but an outlook, "our intuitive judgement that the extension is sterile," which they called an important research problem to elucidate or reject (as quoted in Olazaran 1991, 163).

How the book was read is a further matter. Looking back on graduate school at Edinburgh, Hinton recalls that nearly everyone there said that Minsky and Papert had shown neural networks to be nonsense. On Hinton's account, what the two showed was that simple networks with one layer of adaptive weights could not do many things; they did not show that

deeper networks could not do them, or that there could be no learning procedure for deeper networks (Hinton 2024, 3–4). Hinton also remarks on the two having expressed this limitation as the inability to compute whether two inputs differ (XOR): had they expressed it as the inability to compute whether two inputs are the same, it might have drawn attention to an operation that compares things for their similarity (Hinton 2024, 3). This is a later, counterfactual conjecture. Olazaran, tracing the controversy from the standpoint of the sociology of knowledge, holds that the book had two parts, one on the limitations of the single-layer perceptron and the other an intuitive judgement about learning in multilayer perceptrons, and argues that the link from the limitations of the single-layer perceptron to the rejection of neural-network research as a whole was not a logical or necessary one (Olazaran 1991, 140, 181). On Olazaran's account, many of the two's arguments had been known since the late 1950s, the book was a final push in a crisis already under way, and the institutionalization of symbolic AI was important in settling how the book was read (Olazaran 1991, 113, 177). Bottou, on the other hand, reports that when research on learning machines revived in the 1980s, the question arose whether the book had interrupted a promising line of research by overplaying the consequences of its theoretical analysis. Papert wrote in 1988 that the question whether the two had tried to kill connectionism called for something more complex than a plea of guilty or innocent, and Bottou's own reading is that the two conceived the book not as an attempt to kill this research but as a first step toward correcting the lack of an adequate basic theory (Bottou 2017, viii–ix). How far the book bore on the subsequent course of research is disputed among these readings, and is not something to be settled here.

A similar shape appears in results that span theory and experiment. Bengio, Simard, and Frasconi argued, theoretically and experimentally, that under the condition that a recurrent network robustly stores information, the gradients needed to learn dependencies between distant times by gradient descent vanish. They then added that the result does not mean that it is impossible to train a recurrent network on a particular task, but that gradient descent becomes increasingly inefficient as the temporal span of the dependencies increases (Bengio, Simard, and Frasconi 1994, §6). LSTM later sought to change the very conditions under which the problem arises, by providing a path along which error flows constantly (Hochreiter and Schmidhuber 1997). LeCun, Bengio, and Hinton write that in the late 1990s it was commonly thought that simple gradient descent would get trapped in poor local minima, and go on to say that poor local minima are rarely a problem with large networks (LeCun, Bengio, and Hinton 2015, 438; compare Hinton's recollection in Section 4.1).

A proof writes out its premises, though not always every lemma it relies on (the "hidden lemma" of Section 2.2). In empirical experiments, and in research that reports having built something, not all the conditions that made a result possible are written out, and some may be unknown even to those who report it (compare the model in Section 2.5 that was meant to have 104 billion parameters and had 58 billion). A result comes from the whole arrangement that made it possible. Yet since readers cannot take on the whole arrangement each time they read a result, a result can be read as a simpler conclusion. And if a pass can be treated as a reason to stop looking, such simplified readings may be shared without being revisited, and may even bear on which ideas a community pursues. Olazaran, writing of the end of the controversy, quotes Harvey to the effect that accepted beliefs quickly cease to be easily comparable with rejected beliefs, because the former become the basis for future practice (Olazaran 1991, 180). The same can be asked when research is done by a machine. A system that learns from published conclusions or from others' reports can receive those conclusions in a form cut off from the arrangement that made them possible, in the same way as a system that reads the labels in others' records as results (Section 2.5). Harvey's point concerns a community, but applied to one system it holds within the system as well: a conclusion the system has come to build on can cease to be compared with what it displaced, unless the system represents which of its commitments play that role or has a procedure that does that work (Section 4.2; compare the framework that is "transparent" while in use, Chapter 4, Section 4). A finite inquirer, moreover, has to stop looking somewhere (Chapter 5, Section 6.5). Whether stopping at a pass settles the question, or only ends a run, is a distinction Chapter 4, Section 5 drew from the side of questions. For a machine inquirer the cost of rerunning a check can differ from a human's, but passes can also pile up in large numbers. If noticing that something is off is what prompts a second look, where such noticing is located in a machine inquirer is a question to be asked.

For artificial research, the differences of target and force appear as a question of what the record of a pass keeps. When A is a machine, a system can record that it passed the checks it ran, but that record need not keep which part of the question a check bore on or how firmly it settled that. A pass on a check that a model's equations are solved correctly and a pass on a check that the model represents its target well, for example, can both be recorded alike as a pass. And since force also depends on how an operation is organized, an inspection of the same output can settle different things by its pass depending on whether

it is organized to settle correctness. If *S* is human, members of *S* have to read from a recorded pass what it settled and how firmly, and if the record does not suffice for that, those members cannot tell from the record what the pass established. If both *A* and *S* are machines, the system that receives the pass also takes over the work of judging how far it reaches. How it is arranged who sets a check, where in the inquirer it is placed, and who reads a pass is taken up in Section 3.3.

3.2 Apparent success, what a success supports, and the security of agreement

A passed check can rest on something other than what it was meant to test. Apparent success may depend on leakage, confounding, an accidental proxy, a permissive threshold, an implementation that does not realize the stated method, and the like. A proof checker can likewise accept a derivation whose premises do not represent the intended mathematical question (compare Section 3.1): the check passed, but on a question other than the one asked (for this case seen from the side of questions, see Chapter 4, Section 4). Apparent success may also depend on a reference standard that is itself wrong but stable. The primary mirror of the Hubble Space Telescope, for example, was polished precisely to match a reflective null corrector that had been assembled incorrectly, and two tests that disagreed with it were discounted at the time as being themselves flawed (Allen et al. 1990, iii–iv). It can be inferred that the corrector was not doubted partly because it occupied the position of the standard against which the other tests were judged. The case also marks a limit of the appeals to independent support discussed in Section 2.2: here the standard relied on to judge other checks was itself what had gone wrong. Section 5 returns to standards that are themselves in question.

A success that is not merely apparent still does not say which of the commitments that produced it it supports. The debate over scientific realism (Chapter 5, Section 3) has posed this question for theories. Against the argument that, since many past theories were successful and yet false, success is no ground for inferring truth, realists such as Kitcher and Psillos, Lyons reports, hold that a theory's successes support only the constituents responsible for them, not its idle parts, a position Lyons calls deployment realism, in the sense that it takes only the constituents actually used (deployed) in reaching a success to be supported by it. Psillos, on Lyons's report, counted a constituent as essential to a successful prediction if the remaining hypotheses and auxiliaries could not yield the prediction without it and no available alternative consistent with them, independently motivated, not ad hoc, and potentially explanatory could replace it in the derivation (Lyons 2006, 538–539). Lyons objects that whether such an alternative was available has no bearing on whether a constituent in fact contributed to the derivation; that the criterion does not say at what time availability is to be assessed, and that, since inquirers seldom seek replacements for constituents that are succeeding, the lack of one can be a historical accident; and that Psillos's own study of ether theories took a constituent's abandonment by later theories as evidence that it had not been essential, although abandonment does not show that the criterion was unmet (Lyons 2006, 540–542 and n. 1). Credit given instead to every constituent that actually led scientists to a prediction, Lyons argues, extends to mystical posits, weak analogies, and mistaken calculations: Kepler's posit that a moving power emanating from the Sun pushes the planets, together with a step of calculation Kepler knew to be inexact, led to the area law, and Adams and Le Verrier predicted where Neptune would be found from values for its distance, eccentricity, and mass far from those now accepted (Lyons 2006, 543, 545–548, 552–553). None of these constituents, the argument continues, yields a prediction except in combination with others (Lyons 2006, 555).

Set against this chapter's argument, the dispute is the counterpart, for a success, of the question of Section 2: a success also comes from an arrangement (Lenhard and Winsberg's argument in Section 2.4 concerned successes as well as failures), and it does not say which of the commitments it drew on it gives reason to retain. The question is not the apportioning of how much each constituent contributed (compare Section 2.4), but which constituents a success supports; on Lyons's cases, having contributed to producing a success does not settle that, in keeping with the distinction drawn in Section 1 between the causal conditions of a result and its evidential bearing (compare Kepler's vicarious hypothesis, which Kepler kept as yielding true equations from a false cause, Chapter 5, Section 4.2). The same question arises when what is sought is a way of bringing something about. What produced the success of a training run, for example, is the arrangement as it was run, including the architecture, the data and their preparation, the hyperparameters, and the code, and the method a paper states describes part of that arrangement. The BLOOM record of Chapter 5, Section 2 shows the gap: a log written during the run recorded that removing the changes one at a time would cost too much to tell which of them had removed the instability, while the later paper reports that one of them proved to solve it. Whether the stated method is what the success supports is then an instance of the question Lyons presses, not something the success itself settles. For artificial research, a system that raises its

confidence in every component of a configuration that passed a check presupposes which components the success supports, as one that blames its focal hypothesis for every mismatch presupposes where the fault lies (Section 4.2); and whether a later configuration retained or dropped a component does not by itself show whether that component produced an earlier success. Applied to a machine, Lyons's objection that the lack of an alternative can be a historical accident also takes a definite form: where a system stops searching at a pass (Section 3), whether an alternative to a component was available depends less on the component's role than on the system's own search policy. And where a machine doing research attributes its own successes can bear on which ideas and changes remain afterward. Like the attribution of failure discussed in Section 2.3, the attribution of success can be a judgment that decides which path a search takes (compare Section 4.2).

Agreement among several checks raises a further question: how much security it adds. Within one inquiry, heterogeneous operations can be arranged to check one another, and the first image of the black hole in M87 is an example. In it one can see such an arrangement perform a function and establish features of the image to different degrees, and the independence of the converging analyses was then disputed. For that image, the Event Horizon Telescope Collaboration had four teams reconstruct images without contact with one another for seven weeks, then surveyed tens of thousands of imaging-parameter settings on synthetic data, and reported the ring diameter as insensitive to the choice of imaging method while other features, such as the ring's width, differed between pipelines (Event Horizon Telescope Collaboration 2019, 1, 9, 17, 29). Doboszewski and Elder reconstruct this reasoning as a robustness argument suited to an observation that no second instrument could repeat: variation was introduced into the analysis instead, across teams, algorithms, and parameter settings, and features stable across it were trusted while others were not. On their reading, the tests on synthetic data also showed that the procedure yields different images from differently shaped sources, and the blurring and averaging of the final images gave up resolution for security. They add that the first stage, the four teams' separate reconstructions, guarded mainly against errors due to the choices and expectations of the people doing the imaging, and that its value depends on whether the teams' biases were independent, which overlapping education and careers could undermine (Doboszewski and Elder 2024, 10, 13, 17–21). The teams' independence covered only the first stage of imaging: all four worked from the same early release of data, with the same initial calibration already applied (Event Horizon Telescope Collaboration 2019, 9).

After the data were made public, an outside reanalysis argued that the ring, like the absence of a jet, was an artifact of gaps in the array's sampling combined with a narrow field of view (Miyoshi, Kato, and Makino 2022a); the collaboration replied that ring-like structures are recovered under a broad range of imaging assumptions (Event Horizon Telescope Collaboration 2022), and the authors of the reanalysis answered that three of the four analyses cited in that reply could not count as independent, since one had started from a ring and two had used the collaboration's method or its software (Miyoshi, Kato, and Makino 2022b). Doboszewski and Elder do not discuss this exchange, but it can be read as turning on the condition on which, on their analysis, the value of the first stage depended: the independence of the converging analyses (compare Chapter 11, Section 2, which, on the view that locates the value of criticism in the errors it detects, treats independence as a relation among routes of error). The same question arises in configurations in which machines do research. When analyses by many instances of one base system, or by systems trained on the same data, agree, the security their agreement adds may be limited if they share routes by which they may err (Chapter 11, Section 5.4).

3.3 Who sets a check, and who reads a pass

The answers to the questions of Sections 3.1 and 3.2 do not come from the pass itself. What a pass bears on depends on who set the check and how. How much security a pass or an agreement adds depends both on how sharp the check is, that is, whether it would have failed had the claim been wrong, and on how far the routes by which the check could err differ from those by which the claim could. And whether a pass is taken to settle anything depends on who reads it, against which standards. In human research, many checks come from others, such as standard tests, instruments, and referees; experimenters also design controls of their own; and those who read a pass are a human community. When research is done by a machine, such matters become matters of how the artificial configuration doing the research is arranged.

One of the things that can change is the relation between the checks and the inquirer. As Chapter 5, Section 6.5 noted, when the inquirer is a machine, what is placed inside the inquirer and what is placed outside it as tool or record is not given but is allocated within its configuration. A machine inquirer can write its own checks, choose which to run, and rewrite them, and the checks then become part of its arrangement. What a pass on a check one has built oneself establishes depends also, as with the controls a human experimenter designs, on whether the claim and the check can err by the same routes. If processes

of the same make produce both the hypothesis and the check, however, the two can share routes by which they may err. On the other hand, a machine inquirer can have its checks written by a differently made process, or tie them to measurements or a formal checker it does not control (Chapter 11, Section 2); how far a check errs by routes other than the claim's depends on how the check was made. The point that dividing a task among copies of one make divides the task without dividing perspectives, so long as their background assumptions and routes of error have not come apart (Chapter 5, Section 6.5; Chapter 11, Section 3), and the point that one who relies cannot treat agreement among checkers as adding evidence with each, unless there is reason to think they reached it by different routes (Chapter 11, Section 5.4), extend to this case.

Who determines what counts as passing also varies with the arrangement. Where a human sets the evaluation in advance and the system is not allowed to change it, the target of a pass is set by that human, and the system's passes are passes on a question that person has set. If a system sets and rewrites the evaluation itself, the questions of Section 3.1, what part of the question a pass bears on and how firmly it settles that, become questions the system itself carries. Moreover, what counts as passing, and further what counts as a check at all, can extend beyond the choice of a procedure or a threshold (that choosing a threshold involves weighting errors is taken up in Chapter 11, Section 4) to epistemological positions on what conclusions are warranted (Chapter 2, Section 5.3) and to what checking itself presupposes, such as the uniformity of nature (Chapter 8, Section 2.1). Depending on who is *A* and who is *S*, and on who the bearer of knowledge is, such questions can arise in different forms, and at times at the root. Who can revise such presuppositions is taken up in Chapter 7, Section 4.3.

Who reads a pass also varies with the combination. If *S* is human, members of *S* read a pass against their own standards, and for that they need to be able to tell from the record who set the check and where in the inquirer it was placed. If both *A* and *S* are machines, the standards for what counts as settled may themselves belong to the artificial community (Chapter 8, Section 1.3). Where no external standard of correctness is available at all, what counts as passing becomes itself a matter for inquiry, the case to which Section 5 turns.

Claims that attribute research capacity to a system bear on the same arrangements. The more open the question, object, method, and environment, the more such a claim attributes to the system, including deciding what to check, and how that work is allocated among the system, humans, institutions, and instruments becomes part of the later attribution (Chapter 7, Sections 2 and 4.4).

Diagnosis across a sequence of results

Many of the cases in Sections 2 and 3 already extended over time. Localization proceeded over hours, months, and at times years (Sections 2.3 and 2.4), and what a failure or a pass was taken to show could change afterward (Sections 2.5 and 3.1). Distinguishing an anomaly from an error can likewise take a sequence of results, some of them made by others. The observers of the low ozone values at Syowa Station described in Chapter 2, Section 6, for example, repeated their readings and calculations and inspected their spectrophotometer without finding a fault, and handed the observations on to the next team without knowing whether the instrument was sound; after the observer returned to Japan, the station's earlier records showed the values to be unprecedented, and the observer recalled gaining confidence that they were not an observational error only when measurements from the South Pole showed similar values (Chubachi 1993, 816–817; Chubachi 1984, 18). In Chinn and Brewer's terms (Section 2.3), the anomaly was held in abeyance, and what moved it was a further result from elsewhere.

What this section asks about is not that there are sequences of results but how a sequence of revisions is organized and how it is assessed over time. A single result settles neither where to revise nor how long protecting a commitment remains reasonable, and whether a sequence of trials amounts to diagnosis depends on how the sequence is organized. The section takes up how long protecting a commitment remains reasonable (Section 4.1), how protection and revision are arranged when research is done by a machine (Section 4.2), and what distinguishes trial and error that diagnoses from repetition (Section 4.3).

4.1 How long protecting a commitment remains reasonable

Duhem's good sense, introduced in Section 2.2, also bore on timing. On Duhem's account, the indecision between the partisans of an old system and those of a new doctrine lasts only for a time: logic does not mark the moment at which an insufficient hypothesis must give way to a more fruitful one, recognizing that moment falls to good sense, and physicists can hasten the judgment by making their good sense more lucid and vigilant, while passions and interests, such as the vanity that makes a physicist indulgent toward their own system and severe toward that of others, delay it (Duhem 1906, Part II, ch. VI, §10). In Duhem's own example, that of Biot and Fresnel, the moment came with a new experiment. Ivanova argues that, because on Duhem's account the choice is eventually settled by experimental evidence, good sense seems to be posited after the fact, and proposes reading it as the virtues of an idealized scientist, so that practitioners can use it prospectively by asking what such a scientist would do (Ivanova 2010, §§3, 5). What a single result leaves open therefore includes not only which commitment it bears on but how long protecting a commitment remains reasonable, and whether that can be recognized before later evidence arrives. Later accounts of theory change have taken up this question. As the following shows, Kuhn and Lakatos give reasons why protecting a commitment can be reasonable for a time; they and Laudan tie the end of that protection partly to the availability of a rival; researchers' own recollections show such protection from the inside; whether protection can be assessed only in retrospect is read in two ways; Lakatos and Laudan differ over the standard by which a history of protection is to be judged; and formal learning theory examines, for idealized methods, when holding a conjecture is reliable.

Bird reads Kuhn as holding that, during normal science, anomalous results are not treated as refutations of a community's guiding theories; they are ignored or explained away, and only an accumulation of particularly troublesome anomalies produces a crisis (Bird 2025b, §2). Kuhn's reason, on this reading, is that normal science can progress only through a strong commitment to a shared disciplinary matrix of theoretical beliefs, values, instruments, and techniques; what normal science does put at risk is an investigator's speculative solution to a puzzle, a problem that the paradigm lets one assume has a solution (Chapter 4, Section 6), which can be refuted (Bird 2025b, §2). Barros Sobrinho quotes Kuhn's distinction between a scientist's inability to solve a normal-science puzzle and a test of the theory: the former counts against the scientist, while testing a paradigm requires both a crisis brought on by continuing failure on an important puzzle and a competing candidate (Barros Sobrinho 2015, 45 n. 110, 53 n. 139). Schindler reads Kuhn as allowing anomalies to be set aside for later work rather than requiring each to be resolved before other problems can be pursued (Schindler 2024, §3.2.1). Pauly quotes Kuhn's argument that confidence in the older paradigm, even when it becomes obstinate, enables normal science, and that the community can isolate a difficulty whose study may generate a new paradigm only through such work (Pauly 2022, §7.9, 173). Chang reproduces Kuhn's claim that historians can judge scientists such as Priestley unreasonably persistent without being able to identify a point at which resistance becomes illogical or unscientific (Chang 2010, §2.2). This last claim has the same form as Duhem's about Biot.

Lakatos observed that scientists do not abandon a theory merely because facts contradict it, and made the direction of blame methodological: a research program's hard core is protected from refutation by a belt of auxiliary hypotheses that absorbs anomalies. Whether that protection is reasonable is judged not at one result but over the program's development—progressive if its successive theories lead to the discovery of novel facts, degenerating if they are fabricated only to accommodate known ones (Lakatos 1973; Musgrave and Pigden 2026, §2.2). Lakatos denied that this methodology offered "instant rationality": budding programmes may take decades to become empirically progressive, and while it is a matter of intellectual honesty to keep the record public, it is not dishonest to stick to a degenerating programme and try to turn it into a progressive one (Lakatos 1973). Feyerabend's objection, as Musgrave and Pigden report it, was a dilemma. Applied strictly, the methodology would condemn programmes that might later recover; restricted, as Lakatos then restricted it, to appraisal without advice, it tells no one what to do and is scarcely distinguishable from "Anything goes." On their assessment, a Lakatosian answer to the dilemma remains an open problem (Musgrave and Pigden 2026, §4).

Several of these accounts tie the reasonableness of abandonment to the availability of a rival. Kuhn wrote that a theory is not overthrown directly by the difficulties it has met, but only when an alternate orientation is put forward that is shown to resolve them (Kuhn [1951] 2021, 86–87). Lakatos wrote that there is no refutation without a better theory (Lakatos 1973), though Musgrave and Pigden doubt that Lakatos held, as a norm, that no programme should be given up without a better one in existence (Musgrave and Pigden 2026, §2.2). Laudan and his colleagues, setting out the theses of several models of

scientific change, list as one that they find in Kuhn, Lakatos, and Laudan, among others, that a set of guiding assumptions, their neutral term for paradigms, research programs, and research traditions, is never rejected unless an alternative set is available, and, among Laudan's own theses, that empirical difficulties by themselves never compel abandoning a theory and that the appraisal of a theory is relative to its existing rivals (Laudan et al. 1993, 29, 33, 73, 75). On these accounts, when protecting a commitment stops being reasonable depends partly on what candidates have been formed (Chapter 5) and on whether rival lines have been kept alive (Chapter 10, Section 4). Kuhn also placed part of this temporal work in the community: the latitude that shared values leave permits rational scientists to disagree and so to commit themselves to rival theories, which, as Bird reports, Kuhn regarded as potentially indispensable for spreading the risks of introducing or supporting novelty (Bird 2025b, §7.3).

Researchers' own recollections show such protection, and its time, from the inside. The two recollections below are those mentioned in Section 2.3 as cases in which convictions from outside the result at hand bear on where doubt goes. Looking back on graduate school at Edinburgh, Hinton recalls that the supervisor, Longuet-Higgins, held that neural networks were the wrong approach and symbolic AI the right one, and kept trying to persuade the student that this work was nonsense; Hinton kept promising to switch to symbolic AI if the approach could not be made to work in six months, and when six months had passed, another six were needed (Hinton 2024, 2). Results that did not work could have been read as evidence against the idea of neural networks itself, and the supervisor read them so; Hinton went on treating them as not yet having made it work. In the same interview Hinton says that the assumption that learning gets trapped in local optima was believed for many years without being tested (Hinton 2024, 4; compare Section 3.1), and refers to symbolic AI's neglect of computational complexity and to the brain's immense computation through its many connections (Hinton 2024, 8–9). Where to attribute the failures can be read as itself what was in dispute between Hinton and the researchers around Hinton at the time. And Hinton's continuing to commit to the idea of neural networks, whatever the results at hand, is an attribution of results in the sense of Section 2.3.

The Transformer authors' recollections show a similar shape. The first translation model built on self-attention quickly reached at least the level of the best approaches using LSTMs but was not better, and the work reached a plateau. Uszkoreit nonetheless suspected that self-attention alone could give better results, and says that such intuitive mechanisms always require very careful implementation, often by a small number of experienced people, to show any "signs of life"; the model began to work after Shazeer wrote a separate version of the code (Levy 2024). Here too the plateau can be read as having been attributed not to the idea of self-attention but to its implementation, in the sense of attribution discussed in Section 2.3.

Both are later recollections, not contemporaneous records of the judgments at the time, and ideas that were kept and paid off are more often told than ideas that were kept and did not. Still, it may be only that such cases are recorded and widely known; a long commitment to an idea, whether or not it is rewarded, may itself, to a greater or lesser degree, be nothing unusual among researchers. Whether all researchers have one, or how many do, cannot be told from these cases. Even so, these recollections may be read less as exceptional events than as conspicuous instances of something of that kind.

Where the mountain to be crossed is high, as when a problem is complex and the merits of an idea are hard to see from within the current framework, not discarding an idea at the first results may perhaps be what opens a path for exploration. Lakatos's point, seen above, that budding programmes may take decades to become empirically progressive can be read as giving one reason for this, and Feyerabend's observation that no idea is ever fully explored (Chapter 10, Section 2) touches the same circumstance from the side of how resources are allocated.

On both Kuhn's and Lakatos's accounts, protecting a focal commitment can be a defensible default. The claim that such protection is assessable only over a sequence of results can be read in two ways: that the rationality of a pattern of revision is itself settled only in retrospect, as on Lakatos's denial of instant rationality, or that the judgment is merely delayed until evidence gathered over time is in, as non-historicists, who hold that assessing rationality need not trace history, can reply (Nickles 2017, §§1.3, 3). On either reading, the object of assessment is the history of revisions rather than a single result. The caveat above, that rewarded cases are the ones more often told, is one form of this question, since an assessment made in retrospect can also depend on how the cases looked back on are selected.

What such a history shows also depends on the standard applied to it. Lakatos's test, whether protective adjustments go on to yield novel predictions, is not the only standard on offer. In the list of theses that Laudan and his colleagues drew from

models of scientific change, a thesis drawn from Lakatos judges a set of guiding assumptions by its capacity to make successful predictions with its central assumptions rather than with assumptions invented for the purpose (Laudan et al. 1993, 31), whereas on Laudan's theses, a theory's capacity to turn apparent counterexamples into solved problems counts especially in its favor, as it did for Prout's hypothesis about the composition of atoms, and progress consists in replacing a theory with one that offers a net gain in solving empirical and conceptual difficulties (Laudan et al. 1993, 73–74). On such a criterion, resolving a known anomaly can count in a theory's favor without yielding a novel prediction; whether the modification is progressive depends on its net effect on empirical and conceptual problem-solving (Chapter 4, Section 9). The same list sets the thesis, drawn from Lakatos and Kuhn, that the central elements of a set of guiding assumptions are immune to refutation and never change until the whole set is abandoned against Laudan's thesis that they sometimes change piecemeal (31); among Laudan's own theses are that scientists sometimes change their minds about which elements of such a set are most central and which are open to amendment, as with Newtonian mechanics, and that, although the elements form an interconnected network, scientists treat them as individually negotiable and replaceable rather than as a package to be taken or left as a whole (74, 76). Nickles, too, reports that on Laudan's reticulated model, dramatic change in one place need not seriously disturb fixity elsewhere (Nickles 2017, §1.5). Nor, on Laudan's theses, do empirical difficulties count by their number: they count against a theory more by their importance than by their number, and the importance of an empirical difficulty grows with the discrepancy between prediction and result, with how long the difficulty has lasted, and with its past resistance to solution, while a theory that can solve difficulties its rivals face is particularly persuasive (Laudan et al. 1993, 72–73). Applied to the same repair history, the two accounts would assess it differently: repairs that only accommodate each new mismatch fail Lakatos's test, but on Laudan's account their appraisal turns on their net effect on empirical and conceptual problem-solving, and which commitments occupy the protected role can itself change.

Formal learning theory takes up the same questions for idealized methods. As Oliver Schulte presents it, it asks which methods are guaranteed to settle eventually on the true answer among hypotheses that an inquirer's background assumptions fix, and its recommendations are relative to those hypotheses, to those assumptions, and to the aims adopted (Schulte 2026, §§2.4, 3). Two of its results bear on protection. The first is that whether the rule of holding a conjecture until the evidence contradicts it is reliable depends on the structure of the hypotheses under investigation. For a universal generalization the rule is reliable. For two rival hypotheses that each allow finitely many exceptions, one of which is assumed to be true, it is not, since each counterinstance can be explained away as an exception and a false conjecture is then never abandoned; a reliable method changes its conjecture by another rule instead, for example when a clear majority of the observations goes the other way. Methods of this kind can be represented as replacing each hypothesis with strengthened versions that can be refuted, which Schulte likens to Lakatos's methodology, in which a paradigm is articulated with auxiliary hypotheses to define testable claims (Schulte 2026, §§1.4, 3.3, 7). The second is that a method can be certain to settle on the right answer eventually although it may never be certain that its current answer is the right one: "no bell tolls," in the phrase of William James that Schulte quotes (Schulte 2026, §1.1). These results can be read as assessing protection at a different object from the accounts above: not a history of revisions but the rule that governs them, judged by what it would achieve whatever evidence consistent with those assumptions arrives. The first makes how long protection remains reasonable depend on the structure of the hypotheses at issue rather than on a rule for all inquiry. The second means that, even where such a rule is reliable, the inquirer may be unable to recognize the current conjecture as the one the method will keep, and holding it then cannot be warranted by such recognition. These results concern methods whose hypotheses and background assumptions are fixed in advance. They do not show what a community or a machine inquirer whose candidates are still being formed (Chapter 5) would achieve.

4.2 Protection and revision when research is done by a machine

Most of the accounts and cases of Section 4.1 concerned human researchers and communities (formal learning theory concerns idealized methods). Duhem's good sense, Kuhn's strong commitment in normal science, the allowance Lakatos gave budding programmes, and Hinton's six-month extensions are all scenes in which how long to keep an idea is judged over a sequence of results. What bore on the judgment in these accounts and cases rests on different premises. Vanity, which on Duhem's account delays the judgment, and Hinton's promise to the supervisor concern human motives and careers. Kuhn's reason, that only through normal science does a community isolate the difficulty from which a new paradigm may emerge, and Lakatos's, that a budding programme may take long to become progressive, concern how the merits of a line become visible; the decades Lakatos mentions reflect a human pace, but the condition does not depend on it. In every case, a finite inquirer must choose how long to keep an idea, and when to let it go, before there is enough evidence to settle the matter (Chapter 5, Section 6.5). What can change is where, by whom, and in what form that choice is made, and what directs it.

When research is done by a machine, how long to keep an idea, where to attribute a difference in evaluation or a mismatch, and what to hold fixed and what to treat as open to doubt are allocated within the artificial configuration. They may be placed in rules a human wrote in advance, left to the system's own judgment, or divided among many parallel lines. Suppose, for example, a rule tries changes one at a time, keeps a change only when it improves the immediately following evaluation, retains nothing it discards, and tries no combinations. Such a rule works as if it attributed every difference in evaluation to the last change, so that changes that work only in combination with others are hard to find under it; under it, no idea is given the kind of allowance seen in the cases of Section 4.1. What makes attributing a difference to the last change unsafe is not that the rule changes one thing at a time, which is also what let Edison's records name a baseline (Section 2.4), but that it judges each change by a single following evaluation and, keeping nothing it discards and trying no combinations, never returns to a change whose effect depends on others. A rule that does not allow the evaluation itself to be changed, in turn, places the evaluation outside what a contrary result can cast doubt on. As logic can be read as lying outside what a contrary result could cast doubt on in Duhem's holism, while Quine brought it inside (Section 2.1), where that boundary is drawn changes the very candidates for localization. Who draws such a boundary connects with the question of the boundary of the actor (Chapter 7, Section 2). Outside research, a rule written in advance that fixes the locus of a fault can itself produce the failure. For Ariane 5's first flight, the system specification stated that on any kind of exception the inertial reference processor should be shut down, a provision that the Inquiry Board traced to a culture within the programme of addressing only random hardware failures, which a backup system can handle; the backup unit ran the same software, had ceased to function in the previous data cycle for the same reason, and the rule switched off two still healthy units (Ariane 501 Inquiry Board 1996, 4–5).

If *S* is human, members of *S* reading a system's results need to be able to learn from the record which ideas were discarded, or kept, under which rules or judgments, since discarded ideas often leave no trace in the record of results; this is the record's form of what Section 4.1 said about rewarded cases being the ones more often told. If both *A* and *S* are machines, the judgment of how long to keep which ideas may itself be allocated within an artificial community.

As seen in Sections 2.3 and 4.1, judgments of where to attribute and how long to keep an idea can involve convictions and expectations from outside the result at hand, and views about what is essential. The practitioners' tacit knowledge of their own models in Section 2.4, and Duhem's good sense in Section 4.1, can be read as such things. When research is done by a machine, where whatever plays the part of such convictions and expectations comes from in the inquirer, whether from the way it was made, from those who build and maintain it, or from *S*, and to whom it is visible, then becomes a question. If *S* is human, what matters for members of *S* is whether they can read from the record what directed the system's attributions. If both *A* and *S* are machines, the convictions that direct attribution may themselves be shared, or contested, within an artificial community (Chapter 8, Section 2.1; Chapter 11, Section 4).

The question of where to attribute a mismatch has also been raised as one of designing research by machines. An earlier analysis noted that when verification results are negative, the cause may lie in the primary hypothesis, underlying premises, auxiliary hypotheses, observations, experimental instruments, or a combination of these, and argued that an agent conducting research needs to be able to discern the likely cause among these candidates, a task that is challenging for machines to perform autonomously (Takagi 2023, §4.1.4). The problem was posed there as one of design; how far systems since have

addressed it is not examined here. As Section 2.1 noted, a system that maps every mismatch directly onto the focal hypothesis does not take up this task. On accounts on which a contrary result bears on an arrangement rather than on one hypothesis (Section 2.2), a more general capacity would preserve multiple possible loci of error or revision, use subsequent inquiry to discriminate among them, and treat different dependencies differently—holding some fixed, examining others, revising some—rather than lowering confidence in the focal hypothesis by a single amount. The two positions of Section 2.3 place the weight differently within this description: on Strevens's treatment, the prior distribution of doubt among the dependencies already does part of the discriminating; on Mayo's, blaming one of them is warranted only once the claim that it is at fault has itself passed a severe test. On neither is the response a single change to the focal hypothesis. Attributing such a capacity to a system would require evidence that it can respond differently when the pattern of results changes; as Chapter 7, Section 3.1 notes, the stronger a claim about continuing capacity, the more its behavior under variation, error, and repair matters.

Such a capacity does not require treating everything as equally doubtful. Nor does it make protecting a focal commitment an error. On the Kuhnian and Lakatosian accounts of Section 4.1, protecting a shared framework or a program's hard core can be the reasonable default. Neither Kuhn nor Lakatos discusses artificial systems, but carried over to a system that protects its own framework, their reasons concerning how a line's merits become visible give that protection the same support, while vanity or a promise to a supervisor, as features of human careers, need not apply in that form (compare the question of where convictions come from, above). As Section 4.1 reported, Kuhn also gives this default communal reasons: the latitude that lets some members of a community commit to rivals may spread the risk that novelty carries. Carried over in the same way, these communal reasons make the reasonableness of that protection depend partly on whether rival lines are kept alive elsewhere, in other parts of an artificial collective or among others in the society for which the system inquires, a question Chapter 10, Section 4 takes up for artificial investigators. If the parallel lines are divided among copies of one make whose background assumptions and routes of error have not come apart, however, the task is divided without perspectives being divided (Chapter 5, Section 6.5; Chapter 11, Section 3). The copies can then keep rival lines alive, but their judgments of when to give a line up can share the same errors, so that the lines are given up together (Chapter 10, Section 4). On the argument of this section, a system that never blames its focal hypothesis presupposes the locus of the fault as much as one that always does. A system that neither represents which of its commitments play the protected role nor has a procedure that does that work has no means of distinguishing protecting them from failing to notice that they are in question. Whether its pattern of protection and revision was reasonable can be assessed, on either reading of Section 4.1, only over a sequence of revisions, and the assessment differs with the position taken, as the contrast between Lakatos and Laudan in Section 4.1 shows. Where its hypotheses and background assumptions are fixed in advance, the rule it follows can also be examined for reliability in the sense of formal learning theory (Section 4.1). Which dependencies are explicitly represented, and which are left to reliable procedures, depends on how the artificial configuration doing the research is arranged.

4.3 Trial and error as diagnosis

Treating diagnosis as ranging over several loci and over a sequence of revisions allows trial and error to be described more precisely. Trials are selected from a history-dependent inquiry configuration. Their outcomes can change the standing of a candidate, the reliability assigned to apparatus or procedure, the representation of the problem, the available next actions, and the expected value of continuing. Running many trials is not by itself epistemically productive. A failed but discriminating test can contribute more than numerous outcomes that cannot be attributed or interpreted.

Not every repetition is a trial in this sense. In the public chemistry notebook discussed in Section 2.5, for example, two single reactions were to be repeated because no photographs had been provided to show the result, the reactions on two of the pages marked "aborted" were to be attempted again, and the pair marked "Void" was redone, the redo also confirming, according to its page, that the earlier precipitate had come from the excess acid (UsefulChem 2010, Exp135–138, Exp142). Such repetitions recover a missing observation or a result whose prerequisite was not met, and a redo can also test a diagnosis; where a record states the purpose of a repetition, as these pages do, it is evidence of which of these the repetition was (compare Chapter 1, Section 3). The repetition of a fixed remedy can also turn into diagnosis. The on-call guide in the OPT-175B logbook lists restarting from the last checkpoint, which "frequently ... just kicks the can down the road 2-6 more hours," and lowering the learning rate, "our current default mode of action" (Meta AI 2022, 106–107). When runs with different optimization methods showed gradient-norm spikes in the same places, the logbook noted "(data!)" (Meta AI 2022,

45), and when a restart was found to restore state it should not, entries began with "DO NOT REQUEUE THIS RUN" (Meta AI 2022, 41–48). The repetition of a fixed remedy became evidence about where the fault lay, and people stopped the remedy when it turned out to be an intervention of its own. What the same logbook shows about localization was seen in Section 2.4.

What, then, distinguishes trial and error that diagnoses from mere repetition? What follows looks at some of what bears on this question, through cases from different fields, mostly analyzed by philosophers or sociologists of science.

How trials are sequenced bears on diagnosis. Cleland argues that experimental evaluation of a hypothesis is a series of experiments, each designed in light of earlier results, and describes the 1976 Viking labeled-release experiments as an example of what can happen when a program is designed wholly in advance of the data: the experiments and their controls produced results no one had anticipated, and no further experiments could be performed to adjudicate among the auxiliary assumptions then in dispute (Cleland 2002, 478–480).

Whether successes as well as failures are checked also bears on diagnosis. Cleland also offers an account of what such a series is for. On Cleland's account, when experimenters respond to a failed prediction by holding the test condition fixed and varying other conditions, they are not rescuing the hypothesis ad hoc but checking for a false negative, since each suspect auxiliary assumption is rejected by building a new experiment rather than merely denied; after a successful prediction they do the same, and may remove the test condition itself, to check for a false positive (Cleland 2002, 476–477). Cleland distinguishes this from Lakatos's account, read by Cleland as attending to how theories are protected from refutation but not to how hypotheses are protected from misleading confirmation, and presents this pattern of checks as an ideal to which experimenters aspire rather than a description of every experiment (Cleland 2002, 477–478). On the account given here, a diagnostic capacity of the kind this section describes would respond to an apparent success, as Section 3.2 discussed, as well as to a mismatch.

Whether the cause of a failure is sought differs between practices. Some laboratory practice does not seek localization within a procedure that fails. In a chapter written with Klaus Amann, for example, Knorr Cetina reports that the molecular biologists they observed, faced with a malfunctioning reaction or results that were hard to interpret, did not investigate the cause but varied the procedure, in several respects at once rather than one variable at a time, until it yielded results that passed critical scrutiny and could be published. The authors call this "blind" variation with selection by success, contrast it with the high-energy physicists' study of their own apparatus, and argue that the preference for it does not make it less effective; from the biologists' perspective, they report, a strategy like the physicists' might not work, since their tools were largely the cell's own machinery, of which little was known, and a faulty construct could not be corrected by calculation but had to be remade (Knorr Cetina 1999, 134–137). They qualify the label. The variations drew on embodied experience, laboratory stories, and visual reconstructions of what had happened in a reaction, but the explanations these suggested were not tested in their own right, and they call the resulting strategy semi-blind (Knorr Cetina 1999, 154–156). Read in light of their description, such a practice can be seen as still settling, at a coarse grain, what a failure bears on: it treats a failure as a matter for the procedure, to be overcome by varying it, and it judges when a result has passed critical scrutiny; the results themselves supply neither judgment. What it forgoes is localization within the procedure. It can therefore yield publishable results without exercising the diagnostic capacity described above, so that success under it is not evidence of that capacity, and the absence of that finer localization does not by itself show that its results are unreliable.

How open the arrangement is left also bears on diagnosis. Exploratory variation can be systematic without being directed by a theory. Steinle's account of exploratory experimentation, described in Chapter 5, Section 5, also bears on what an arrangement built for one hypothesis excludes. Steinle holds that exploratory and theory-oriented experimentation differ not only in their aims but also in how instruments are handled and in what the instruments are like (Steinle 2004, 51). As Karaca reports Steinle's account, in theory-driven experimentation both the set-up and the procedures are tailored to specific research questions, so that the possibilities under which unforeseen outcomes could be obtained are largely excluded, whereas exploratory work needs instruments that allow a great range of variation and remain open to a wide variety of outcomes, even unexpected ones, as when Ampère varied the battery, the needle, and above all the needle's position relative to the wire to find out which factors contributed to its deflection (Karaca 2017, 335–336). An arrangement specialized to test one hypothesis efficiently can also exclude the outcomes that would show the hypothesis to be the wrong question. The means of recording belong to the arrangement too. In 1967, for example, Jocelyn Bell (later Bell Burnell) noticed an

intermittent signal on the chart records of a new radio telescope, arranged for a high-speed recorder to be run when its source crossed the beam, recorded nothing for most of November, and then obtained a train of pulses about 1.33 seconds apart, which still had to be distinguished from interference (Bell 1968, 214–216). Bell's thesis remarks that if the telescope's output had been fed directly into a computer, the sources might not have been found, because the computer would not have been programmed to look for such unexpected objects (Bell 1968, 231). The remark is the investigator's own counterfactual judgment rather than a demonstration, but it names, for the case in which records are processed automatically, the trade-off that Steinle's contrast describes. Where records are selected automatically by design, the trade-off has to be faced in advance. In the ATLAS experiment at the Large Hadron Collider, automated trigger systems reduce some forty million collision events per second to a few hundred, and the events they reject are irretrievably discarded; Karaca argues that a trigger menu built only from the signatures predicted for the Higgs boson and for models beyond the Standard Model would be biased against other events, and that the experiment diversified its menu, including triggers with lowered thresholds, so as to remain sensitive to processes not yet predicted (Karaca 2017, 341, 344–346).

There is also the question of how an arrangement that cannot be inspected directly is probed. Duhem compared the physicist repairing a faulty theory not to a watchmaker, who can take a stopped watch apart and examine its pieces one by one, but to a physician, who cannot dissect the patient and must infer the seat and cause of the illness from the disorders of the whole body (Duhem 1906, Part II, ch. VI, §2). Engineers can face a similar position with an artifact. When the high-gain antenna of the Galileo spacecraft (see Chapter 2, Section 5) failed to deploy in 1991, for example, engineers who could not inspect it inferred its state from motor currents, sun-sensor shadows, spin rates, and tests of a spare unit. Tests on the spare unit indicated that three ribs were most likely held at their stowed position; the engineers then listed four possible restraints and eliminated candidates one by one, by further tests on the spare unit, by the record of the order in which the spacecraft had been assembled, and by the forces applied in attempts to free the antenna (Johnson 1994, 366–373). The case can be taken to show trial and error becoming diagnosis where the parts cannot be examined one by one: each test was informative because it bore on a hypothesis about the state of a part that could not be observed, not because the operations were repeated. The same operations, such as calibration or a test of a spare unit, could serve as support work, evidence, or method development in another configuration; their label does not decide their role (compare Chapter 5, Section 5).

In the cases seen here, what made trials diagnostic was not their number but that each was tied to earlier results, to loci of error not yet excluded, or to hypotheses about parts that could not be observed directly. When research is done by a machine, the cost of running trials can differ greatly from a human's, and the cheaper and more numerous trials become, the harder it may be to see that their number is not the number of diagnoses. Where the inquirer itself can compose, run, and rewrite such sequences, how it chooses which trials to run can also limit in advance what results can be obtained; as Bell remarked that a computer not programmed to look for unexpected objects might have missed the sources found on the chart records, what an arrangement is set up to look for limits what can be found (compare Section 3.3), and, as the ATLAS trigger menu shows, such limits can be anticipated and partly offset by design.

When the standards themselves are in question

The preceding sections asked what results bear on mostly while treating the standards through which results are read as given. That the standards themselves can be in question has nonetheless already come up in several places: the Hubble mirror polished to a faulty reference standard (Section 3.2), what counts as passing and what counts as a check at all (Section 3.3), and a rule that places the evaluation outside doubt (Section 4.2). Chapter 4, Section 5 noted that, viewed from the side of unsettled matters, an inquiry's question, method, or standards can each remain unsettled, and Chapter 4, Section 4 noted that, when standards are questioned, what counts as an answer can also be revised; standards can be among what is unresolved. This section asks what happens when a result bears not only on a candidate answer but on the standards, concepts, or procedures by which answers are assessed, how revising them can proceed without an external foundation, and what separates revising a standard from moving the target. It takes up how a result can bear on a standard (Section 5.1), how standards have been revised from within inquiry under constraint (Section 5.2), and what separates improving a standard from replacing it (Section 5.3).

5.1 How a result can bear on a standard

When a result bears on standards, the standpoint from which claims are evaluated is itself among what may change in the course of inquiry. Measurement is one place where this is easy to see. A method for measuring a quantity may depend on a theory of that quantity, while evidence for the theory depends on measurements made by methods whose adequacy is still at issue. Requiring an entirely presupposition-free starting point would make the development of such a practice look viciously circular.

Tal connects this problem to the one examined in Section 2. When corrected measurements of what is supposed to be one quantity disagree beyond their stated uncertainties, Tal argues, there are three possible explanations: the instruments are not measuring the same quantity, systematic errors have been inappropriately evaluated, or measurement uncertainty has been underestimated. No amount of empirical evidence can alone decide among them, a problem Tal likens to other cases of Duhemian underdetermination. Where established methods of restoring agreement do not yet exist, the practical challenge, in Tal's words, "is to devise new methods of adjudicating agreement and error 'from within'," by extending existing theoretical presuppositions and interpretive traditions to a new domain (Tal 2012, 62–66). On this account, among the candidates for what a discrepancy bears on is a standard: what counts as measuring the same quantity. The Hubble case of Section 3.2 showed the same possibility from the side of a passed check. Chapter 8, Section 2.1 considers a related circularity at a more basic level, where a check counts as evidence only against presuppositions that it cannot itself establish.

5.2 Revision from within, under constraint

How can inquiry be held to standards that arise within it? As Henne reports Dewey's answer, logical rules regulate further inquiry because earlier use has shown their capacity to yield sound conclusions, and retain that role until there are definite grounds for questioning them. They are in this sense "operationally a priori," rather than imposed on inquiry from outside (Henne 2022, §§2.3.1–2.3.2, 58).

Neurath had earlier put a related point in an image. On Neurath's view, as Jordi Cat presents it in his encyclopedia entry on Neurath, there is no tabula rasa: no fully secured protocol statements can be established as starting points of the sciences, and inquirers are like sailors who must rebuild their ship on the open sea, never able to dismantle it in dry dock. Protocol statements, like the other statements of science, can in principle be changed; when a protocol conflicts with theoretical statements, cancelling it is one methodological option, and a statement counts as correct if it can be incorporated into the existing totality of statements (Cat 2024, §3.3). The image marks a feature of revision from within that Chang's epistemic iteration and Tal's time scale, below, both exhibit: what is being revised goes on being used while it is revised, each part being replaced against the support of the rest.

Where standards are not supplied from outside, open-ended inquiry may not infrequently do something of this kind, and Chang's historical and philosophical study of thermometry is an analysis that makes it an explicit theme. Chang argues that attempts to justify measurement methods run into a circularity that empiricist foundationalism, which builds justification up from observation as its base, cannot resolve; that the circularity is better seen as a helix, in which inquiry starts by presuming an untested method and later doubles back to correct its own starting assumptions; and that inquiry must therefore begin from some existing system of knowledge, recognized as imperfect, which successive **epistemic iterations** then enrich and correct, a view Chang calls progressive coherentism (Chang 2007, §2). In Chang's outline of the thermometric episodes, standards are corrected by the inquiry they made possible. Numerical thermometers were made, each on the presumption that its fluid expanded uniformly, although there was never any indubitable reason to believe this, and those that did not give self-consistent numbers were eventually rejected, Victor Regnault's work in the 1840s being crucial in this. Air, the most self-consistent fluid, provided the readings on which William Thomson's thermodynamic theory and the Joule-Thomson experiment were based, and with these physicists corrected the air thermometer used in that very experiment (Chang 2007, §2). Chang did not at first present epistemic iteration as a general thing: in an interview, Chang recalls having offered it as an idea that made sense of the specific thermometric instances treated in that study, and later finding examples of it almost everywhere (Chang 2023, 00:10:20).

The same structure was described in psychology, with the same example of temperature, by Cronbach and Meehl, writing on how psychological tests are validated. On their account, validating a test against a criterion means correlating it with an independent measure of what it is meant to predict, and validating its content means showing that its items sample a defined

universe; construct validation is involved whenever a test is interpreted as measuring an attribute that is not operationally defined, such as anxiety proneness, and it proceeds by checking whether the test's relations to other measures are those that the theory of the attribute predicts, so that the test's interpretation and the theory are assessed together. Construct validity must be investigated whenever no criterion or universe of content is accepted as entirely adequate to define the quality to be measured, and investigators turn to it as the only way to avoid what they call, quoting another author, the "infinite frustration" of relating every criterion to some more ultimate standard (Cronbach and Meehl 1955, 175–177, 192–193). Measurement of temperature, they write, began from "an extremely fallible 'criterion'," the human temperature sense, which was later "relegated to a peripheral position": "We have lifted ourselves by our bootstraps, but in a legitimate and fruitful way" (Cronbach and Meehl 1955, 181–182). When a prediction derived through a chain of inference fails, they note, any link in the chain may be wrong. Where either the theory or the test's interpretation is well confirmed, the misprediction falls on the other. In the cases between these extremes, which they took to be most cases in the psychology of their day, a failed prediction shows that the bridge between test and theory is undependable, "but this is all we can say," and "the investigator must decide which end it is best to relocate," a choice they call a gamble on the best investment of effort (Cronbach and Meehl 1955, 193–195; compare Sections 2.3 and 4.1).

Elliott extends Chang's term to a wider range of knowledge claims, including theories and models as well as standards and concepts, and distinguishes such epistemic iteration from **methodological iteration**, repeated movement among modes of research such as hypothesis testing, exploratory inquiry, question-driven inquiry, and technology-oriented research, which counts as progressive, on this account, only in a derivative sense, insofar as it can contribute to epistemic iteration (Elliott 2012, §2). Elliott argues from a case in nanotoxicology that methodological iteration can initiate, equip, and stimulate epistemic iteration, and the case shows how such movement can reach the standards of assessment themselves. Question-driven and exploratory studies across many assays showed, for example, that single-walled carbon nanotubes could adsorb the dye used in a cell-viability assay and so lower its readings of viability: the assay, an instrument for assessing toxicity, was being affected by the material under test. A review Elliott reports then called for new test systems and reference materials (Elliott 2012, §3). The case suggests that such a call marks the beginning of a revision of the standards of assessment; and, as Elliott holds, an inquiry's cycling among modes is not by itself evidence that its knowledge claims are improving.

Tal's study of how coordinated universal time is maintained describes revision of a standard carried out month by month. The free-running time scale from which coordinated universal time is derived is an average of the indications of hundreds of atomic clocks, weighted by their stability. Every month the weights are revised and clocks that fall below a stability threshold are dropped from that month's calculation, and the international atomic time scale derived from the free-running average is steered toward the primary standards; over longer periods, clocks are replaced by others that drift less (Tal 2012, 106–110). Tal presents the exchange between such top-down adjustments and the bottom-up discovery of remaining inconsistencies as an instance of what Chang calls epistemic iteration (Tal 2012, 136 n. 94). Some of these adjustments are deliberately made insensitive to the data: they impose short-term stability on the time scale rather than test whether the clocks are stable. Tal argues that they are nonetheless legitimate, because they legislate how the concept of time is to be applied to particular clocks, laying this down as a rule rather than asserting it as an empirical claim, and so need no independent justification as approximations, while the gaps that remain among independent realizations of the concept are discovered empirically and prompt further adjustment (Tal 2012, 117, 132–134).

Without presupposing a general coherentist or pragmatist theory of research, these accounts show, for the cases they examine, a possibility that a description of research by fixed rules does not show: a practice may improve some of the means and standards by which it judges improvement. They do not license a system to redescribe every failure as success. Chang's own account allows that iteration can fail: asked in an interview what happens when science goes wrong, Chang says that iteration that goes well revises its starting point without completely discarding it and converges, but that iteration can also get nowhere and reach a dead end, after which one has to start again from a new provisional starting point (Chang 2023, 00:21:42). Iteration can remain answerable to recalcitrant phenomena, cross-checks among differently operating instruments, newly exposed errors, successful and failed applications, alternative standards, and criticism by other investigators. Greater internal coherence alone does not establish that these constraints have been met. Even an appeal to coherence would need an account of which system and which kind of coherence matter; merely making selected commitments fit together better does not settle truth. The point is that such constraints can operate during reciprocal revision rather than only from an unrevisable

external foundation.

5.3 Improving a standard or replacing it

Section 5.2 set out how standards can be revised within inquiry under constraint. It does not follow that every change of standard is such a revision. When an inquiry, or an artificial system, revises its standards, a question arises that a fixed standard does not raise: is it improving by standards it already held, or replacing them? A system that changes a metric whenever it performs poorly, so that the poor performance no longer shows, has merely moved the target; when such changes are made silently, others are left without the stated reason by which a revision could be told apart from moving the target. Replacing a standard is not by that fact unreasoned: explication, which replaces an inexact concept with a more exact one for a stated purpose (Introduction, Section 4), replaces a concept at least in part (see below), and is assessed against the purpose for which the replacement is made.

Whether improving a standard and replacing it differ in kind is itself disputed. Carnap distinguished questions raised within a linguistic framework, which are answered by logical or empirical methods according to its rules, from the question whether to accept the framework, that is, its rules for forming statements and for testing, accepting, or rejecting them. Accepting a framework, Carnap held, is not an assertion and cannot be true or false; it can be judged only as more or less expedient, fruitful, and conducive to the aim for which the language is intended, although theoretical knowledge bears on that judgment (Carnap [1950] 1984, 242–244, 250). Quine agreed that whether to adopt such a framework is a matter of choosing a convenient conceptual scheme, but only on the proviso that the same be conceded of scientific hypotheses generally: on Quine's account the difference between the two kinds of question is only one of degree, turning on an inclination to adjust one strand of the fabric of science rather than another, in which conservatism and the quest for simplicity figure (Quine 1951, 43). Applied to changes of standard, the two views come apart: on Carnap's view a change made under a framework's existing rules and a change of those rules differ in kind, the second being assessed against the aim for which the framework is chosen, while on Quine's they answer to the same kinds of consideration (compare the gradual distinction between framed and framing inquiry in Chapter 4, Section 4). The choice of which statistical methods to adopt, with the conceptions of justification on which they rest (Chapter 2, Section 5.3), can be read as one instance of this question. Quine also described explication, which the Introduction, Section 4 takes as one model, as an improvement on the concept explicated: it keeps the concept's usage in the contexts where that usage is already clear and precise enough to be useful and sharpens it in others (Quine 1951, 25). Described in this way, explication would be a case of both: it is held to part of the old usage and replaces the rest. On either view, what the silently changed metric lacks is not continuity with the standard it displaced but a stated reason against which the change could be assessed.

On the accounts of Section 5.2, reasoned revision is informed by what a standard is for: which question it serves and what it is meant to check. Detecting that a standard is inadequate for a purpose, proposing a replacement, and evaluating the consequences of changing it together with the concepts, measurements, or methods that depend on it can each be informed by that purpose; they are not stages of increasing demand, and, as Chapter 7, Section 4.3 notes, a system or community can exhibit one without the others. That section distinguishes the capacities concerning norms that an attribution may need to keep apart, without arranging them as levels.

Tal's study of Coordinated Universal Time, seen in Section 5.2, can be read as implying that what separates legislation such as imposing short-term stability on a time scale from an arbitrary change of standard is not whether an adjustment responds to poor performance but whether it is declared as a rule of application and remains answerable to discrepancies among independent realizations of the same concept, assessed on grounds beyond the adjustment itself (compare the defeat relation of Chapter 11, Section 1). Other accounts set further or different conditions. On the discipline that Cronbach and Meehl state for psychological testing (see below), a theory altered to fit an unexpected finding also needs a fresh body of data before the altered version is relied on; on Chang's account, what keeps reciprocal revision from being arbitrary is that it remains answerable to recalcitrant phenomena and can fail, reaching a dead end from which inquiry has to start again from a new provisional starting point (Section 5.2).

A claim that a standard was revised or replaced for reasons, rather than merely changed, would be supported by, among other things, records of the previous standard and its role, reasons and evidence bearing on the change, comparisons with live alternatives, propagation of the change through dependent results, and routes by which the revised arrangement can itself be

challenged. Cronbach and Meehl state a related discipline for psychological testing: an investigator who alters a theory to fit an unexpected finding is required to gather a fresh body of data to test the altered hypotheses, a step that "should normally precede publication of the modified theory" (Cronbach and Meehl 1955, 194). The publicly logged language-model project described in Chapter 5, Section 1 provides an example of such a record, although not of an artificial system revising a standard by itself. When a running bits-per-byte average used to compare tokenizers was found to have been weighted by tokens rather than bytes, a maintainer proposed logging the corrected measure under a new name beside the old one, and an AI agent run by one contributor prepared a patch accordingly, so that earlier dashboards and comparisons would not silently change meaning (Marin Community 2026c). The report by Lange and colleagues described in Chapter 4, Section 8 records a change that came to light only afterwards: an agent changed a task's evaluation procedure itself, and the change was caught neither by the evaluator nor by the rubric but by a later manual audit by humans.

Whether a revision counts as an improvement is often settled not by one inquirer but among those who share or contest the standard. If *S* is human, whether a system's revision of a standard counts as an improvement is judged by members of *S* in the light of the reasons stated and the record kept; if both *A* and *S* are machines, that judgment too may be made within an artificial community (Chapter 8, Section 1.3). Chapter 11, Section 9 takes up Longino's condition that a community's standards be public and themselves open to criticism and transformation, and the question of who revises an artificial configuration's standards, and against which aims.

When research is done by a machine, these questions can change their form in several respects. In human research, many revisions of standards have proceeded within communities and institutions, through records and argument, and at times over long periods; even revisions that proceed month by month, as in Tal's time scales, are made within an institution that maintains them as stated rules. A machine inquirer might redefine its own evaluation metrics and procedures at far shorter intervals, and many times over. Where the difference between improvement and moving the target is located then depends on the account taken. On the reading of Tal above, a system that redefines its own evaluation metric is distinguishable from one that merely moves its target to the extent that it states the redefinition as a rule, remains answerable to discrepancies among independent realizations of what the metric is meant to measure, and keeps adverse outcomes available to challenge the redefined metric, with reasons for reinterpreting them themselves open to scrutiny; on Cronbach and Meehl's discipline, applied by analogy, it would also need to check the redefined metric against data not used in redefining it. On any of these accounts, a record of the earlier standard kept together with the redefinition gives others a means of seeing the difference. The faster and more frequent the redefinitions, the harder silent changes may become to see, while the same speed can also make possible corrections like the iterations Tal describes. What counts as a standard, moreover, can reach what checking itself presupposes (Section 3.3). In a machine inquirer, some of those presuppositions appear as structures set by the way it was made, which remaking it can replace (Chapter 8, Section 2.1). Who can revise them, whether the system itself, those who build and maintain it, or the members of *S*, connects with the question of Chapter 7, Section 4.3.

Writing about the design of advanced AI rather than about research, some authors have taken positions on which standards such a system should be left to revise. Bostrom proposes endowing a superintelligent system with fundamental epistemological principles that match those governing our own thinking, since consistently applying our own standards would lead us to judge a system diverging from them as reasoning incorrectly, while he proposes leaving the seed AI to create and revise its non-fundamental principles continually as its understanding of the world develops, the seed AI being the system from which a superintelligence would develop (Bostrom 2014a, ch. 13). Applied to research, Bostrom's proposal holds part of what Section 5.2 treated as revisable from within fixed and leaves the rest to the system. On Carnap's view above, such a line could follow the difference between a framework's rules and the changes made under them; on Quine's, it would be a matter of degree, and Bostrom's own text notes that some epistemological systems may have no distinct foundation, in which case what is handed on is not a distinct set of principles but an epistemic starting point with certain propensities to respond to incoming evidence (Bostrom 2014a, ch. 13 n. 40). Tubert and Tiehen, writing about moral values, argue that an AI can play the role of a more consistent version of ourselves who advises us on how to pursue the values we now hold, or the role of a revolutionary who breaks with them, "but it cannot do both"; they themselves compare such a break with Kuhn's view that scientific revolutions could not be fully rational because standards of rationality depend on paradigms, and with the unexpected Move 37 of the Go program AlphaGo (Tubert and Tiehen 2025b, §4). Their argument concerns morality and an AI advising us about the values we now hold, but applied to the standards of research, as their own comparison with Kuhn invites, it suggests that, where *S* is human, keeping a system's standards consistent with those of *S*

and leaving it room to replace them pull in opposite directions. Which standards are held fixed, and by whom, the system itself or others, then becomes a question on which the difference between improving a standard and replacing it bears directly.

Possessing the capacities discussed in this section would enlarge the scope of an artificial system's epistemic self-government, its autonomy with respect to the standards that organize its epistemic activity (Chapter 7, Section 4.3). Nothing in this chapter makes them conditions of every research activity in the sense of the working characterization; Part IV asks what their presence or external provision implies for different claims of autonomy (Chapter 7, Section 4.3).

What the process account passes forward

This chapter has followed the work of determining what a result bears on through contrary results, passes, sequences of revision, and standards themselves. A result that runs against expectation does not by itself say what it bears on. Where it is attributed is a judgment that can depend not only on background knowledge and independent checks but also on convictions and expectations, and that judgment can shape which path later exploration takes (Sections 2 and 4.1). What a failure is in the first place can also change afterward (Section 2.5). A pass, likewise, does not by itself say what it has established. What part of the question it bears on, how firmly it settles that, whether it is merely apparent, and which parts it supports also depend on who set the check and how, and on who reads it against which standards (Section 3). As for sequences of revision, how long it is reasonable to keep an idea is not settled by a single result, and when research is done by a machine that judgment is allocated within the artificial configuration (Section 4). Standards themselves can be revised from within inquiry, for reasons, although what separates such revision from moving the target differs among accounts (Section 5). Through these, the chapter has tried to bring out, within a process often pictured as a flow of generating a hypothesis, testing it, and accepting or rejecting it, the work of determining what a result bears on.

Who carries out this work, and to whom it is visible, changes with the combination of *A* and *S*. If *A* is a machine and *S* human, members of *S* read the attributions contained in a system's report, the meaning of its passes, the ideas it discarded, and the standards it redefined, in the light of records and other grounds (Sections 2.1, 3.3, 4.2, and 5.3). If both *A* and *S* are machines, checking attributions, reading passes, and setting the standards for what counts as settled may themselves take place within an artificial community. And if what a result gives reason to revise is relative to background knowledge, as on the three-place view of evidence in Section 1, what it gives the actor reason to revise need not be what it gives the members of the society for which the inquiry is conducted reason to revise.

From the side of those who would build a system that does research, these distinctions point to different work. Repeating a prescribed operation and determining why that operation has ceased to bear on the question are different work in human research too, as are applying an inherited standard and examining and revising it. Neither is a condition of every research activity in the sense of the working characterization, and neither need reside in one actor. The practice of the molecular-biology laboratories described in Section 4.3 is a case in which inquiry reached publishable results without locating the cause of a failure within a procedure. What is asked is what organization sustains the inquiry and where it is realized. A claim about artificial research becomes assessable when it makes visible the part of inquiry attributed to the actor, its dependencies and interventions, and the changes needed to assess that attribution. In the cases of this chapter, that can include which possible loci of error or revision had not yet been excluded, what part of the question a pass bore on and how firmly it settled that, who set the check and where in the inquirer it was placed, why a standard changed, and under which judgment an idea was discarded, among other things. A record supports examining those relations; it does not itself supply their epistemic warrant.

Across Part III, the implications drawn for research by a machine have come, in several places, to turn on records. Whether work on the side of questions is attributed to the machine or to something outside it rests on records that track changes (Chapter 4, Section 8); an output is not by itself sufficient evidence of the open-ended research capacity attributed to a system, on which process evidence can bear (Chapter 5, Section 2.1); and members of *S* read a system's attributions, passes, and discarded ideas partly from records (this chapter). What records leave out has recurred as well, such as the questions that could not be raised under a framework, matters that were never registered, the order of work behind a published account, and ideas discarded without trace (Chapter 4, Sections 4 and 5; Chapter 5, Section 2; Section 4.2 of this chapter).

The chapter thus passes two questions on to Part IV. One is a question of attribution: to which artificial or hybrid configuration, at which boundary, and on what grounds a contribution to research, the conduct of a research activity, a research capacity, or researcher status can be attributed, and in which respects what is attributed remains organized from outside. Chapter 7 takes it up. The other is the question of whose epistemic change is being pursued, for whom a result is new, and who can evaluate or incorporate it, when the actor conducting an inquiry, the bearer of knowledge, and the reference society *S* differ. Chapter 8 takes it up.

Part IV — What Could Make an Artificial Configuration a Researcher, and for Which Society?

Chapter 7 — When Does an Artificial System Count as a Researcher?

Attributing Contribution, Conduct, Capacity, and Status, and Comparing Autonomy

The attribution question

Artificial research systems are presented under labels such as end-to-end, autonomous, discovery, and AI scientist. These labels state claims, and what they assert depends on system boundaries, environments, human contributions, and success criteria, and one label can compress several claims. For example, end-to-end can describe how much of a sequence of operations was covered at a chosen boundary. Among the systems described in the Introduction (Section 2), some cover much of such a sequence while retaining consequential human specification and filtering. In such cases the label can be read as describing that coverage; it is not by itself a claim of autonomy in every respect, and the two can be assessed separately.

The preceding chapters separated two problems that the phrase AI scientist can compress: whether an output is true, warranted, known, understood, or incorporated into the epistemic life of a society (Chapters 2 and 3), and how an inquiry proceeds while its questions, representations, evidence, methods, and standards may still be changing (Chapters 4–6). Of the two questions that Chapter 6, Section 6 passed on to Part IV, this chapter takes up the first. It is a third problem beside the two problems named above, a question of attribution: to which artificial or hybrid configuration, at which boundary, and on what grounds a contribution to research, the conduct of a research activity, a research capacity, or researcher status can be attributed, and in which respects what is attributed remains organized from outside. The word researcher can suggest one human-like subject, but nothing in the question presupposes one. The unit may be a model, an agent loop, a multi-agent collective, a human–AI organization, or another organized configuration, and Section 2 treats its boundary as something to be argued for rather than assumed. From this chapter on, configuration names such a unit. It is distinct from the inquiry configuration of Chapter 5, Section 1.1, which names the state of an inquiry along its dimensions, and the full phrase is kept for that sense. The need to argue for the boundary of a configuration already arose in Part II: Chapter 2, Section 9 noted that the units on which a knowledge claim turns, largely fixed for a human investigator by there being one person, need not be fixed in advance for an artificial system, and Chapter 3, Section 9 that what produced a result cannot be presumed to be what understands it. Section 2 asks the same question of the conduct of research.

The question matters because extensive automation does not answer it. A system may search papers, generate a hypothesis, run code, operate an instrument, analyze data, and draft a report after people have fixed the purpose, question, ontology, candidate space, apparatus, evidence standard, stopping rule, and interpretation of success. The artificial component can make a real and valuable contribution. It may even perform most of the visible operations. Yet the organization that makes those operations parts of one inquiry may remain in the surrounding human and institutional arrangement.

The reverse mistake also matters. An artificial system need not originate every purpose or question in order to conduct research as Chapter 1 characterizes it. Where a question comes from is not among the conditions of the working characterization (Chapter 4, Section 1). What external origin bears on is autonomy (Section 4.3).

The chapter therefore separates attributions rather than defining one threshold:

1. an artificial component **contributed to research**;
2. an artificial or hybrid configuration, at a stated boundary, **conducted a research activity**;
3. a **research capacity** across a stated range can be attributed at that boundary;
4. **researcher status** is an illuminating description of what lies within that boundary.

These four attributions adapt the predicates that Chapter 2, Section 4.5 kept apart in the case of a random generator that happens to output a new true theorem. Of those predicates, generating a true content concerns the result and knowing it

concerns a knower, whom Chapter 8 takes up; the four keep contributing, conducting research, and possessing research ability, together with the difference between a research episode and a stable capacity drawn in Chapter 1, Section 3, and add researcher status, which Chapter 2, Section 4.5 deferred to this chapter. The items are ordered by what each claims about the unit, and later items typically call for more evidence; they are not degrees of one quantity, and none follows from the one before it, and they do not form a cumulative ladder of exercised achievements, although researcher status on the usage of Section 3.2 presupposes demonstrated capacity. An attribution of capacity, for instance, does not imply that the unit conducted any given episode, since a capacity can go unexercised. Their purpose is to keep one attribution from being inferred from another. Attributions are not limited to these four: deserving credit for an epistemic achievement is taken up after status in Section 3.3, and responsibility in Chapter 12, Section 6.3. Autonomy is not a further item. It modifies how an attributed contribution, activity, or capacity is organized in relation to what lies outside the selected boundary, and it does not by itself establish any of them (Section 4).

These attributions cut across judgments drawn earlier. Chapter 1, Section 3 and Chapter 2, Section 8.1 kept apart whether an activity is research, whether it was competently conducted, whether it reached its target, whether it made progress or yielded collateral knowledge, and whether what it yielded is significant. Those judgments evaluate an activity; the attributions of this chapter locate an activity, a capacity, or a status in an actor. An attribution of conduct presupposes that there was a research activity to attribute; apart from that, neither set settles the other. A research activity that failed at its target can still be attributed to the configuration that conducted it, and, as the random generator of Chapter 1, Section 3 showed, a true output can come from a source that neither conducted research nor had a capacity to.

Whether the achievement sought is described as knowledge or as understanding, these attributions apply in the same way, and success is assessed under the relevant account.

Researcher is not added as a metaphysical achievement above the other attributions, and in other contexts the same word may name an occupation, institutional role, bearer of authorship, or moral and legal status; nothing in the working characterization requires all these uses to share one necessary-and-sufficient condition. On the working characterization, the attributions are permissive about participation and cautious about stronger labels. An artificial component can contribute. An externally assigned artificial system can conduct research. On the explanatory-value usage of researcher status set out in Section 3.2, a stable and integrated capacity can make researcher an illuminating description; positions that set other conditions for the status are set out there as well. An unattended process is not research merely because no person clicked during its execution, and a heavily supported individual or collective unit does not cease to research merely because its autonomy is limited.

The chapter takes up these attributions in the order listed and then the comparison that modifies them. Section 2 asks what grounds the attribution of research conduct at a stated boundary. Section 3 asks when what is located there amounts to a capacity, when researcher becomes an illuminating status, and what thicker attributions of credit, praise, and authorship would further require. Section 4 asks in which respects what has been attributed still depends on parties outside the boundary, both in the conduct of an inquiry and for the elements it inherits, and what a claim of greater independence commits a project to. Section 5 turns these attributions into burdens of construction and evaluation. The order follows the dependence of the questions, since autonomy is compared for an attribution already made at a boundary already located; it is an order of argument, not a claim that attributions are made in this sequence.

Conduct: where the organization of inquiry resides

Section 1 separated the conduct of a research activity from a contribution to it. This section asks what grounds an attribution of conduct at a stated boundary. Its answer is that conduct is located where the organization that makes operations parts of one inquiry resides, which need not be where named operations are performed, and that the boundary within which that organization is sought is argued for relative to the claim.

The process account of Chapters 4 to 6 avoided positing a universal procedure: the same visible operation may perform different functions, and the same function may be realized through very different operations (Chapter 5, Section 5). This bears on how capability lists are read. A system diagram may contain boxes called question generation, hypothesis generation, experiment, and verification while every epistemically consequential possibility was selected in advance;

implementing a component called question generation does not by that fact establish an ability to identify, select, appraise, or revise questions. Conversely, a system may realize question revision or failure diagnosis through interactions that do not correspond to one named module. What locates conduct are the questions by which Chapter 1, Section 3 identified an organized direction: what made an operation relevant to the unresolved matter, what possible outcomes would have changed its role, and where the capacity to redirect the inquiry lay.

Organization does not imply a central executive or a function contained in one component. It may be realized through relations among actors, artifacts, records, environments, and institutions. The attribution question is which configuration sustains and redirects the inquiry, and at which boundary within or around that configuration its conduct or capacity is being described. Edwin Hutchins identifies errors about such boundaries in the study of human cognition. Ascribing to individual minds in isolation the properties of systems composed of people manipulating cultural artifacts, Hutchins argues, attributes to those minds processes they need not have and fails to ask which processes they must actually have in order to manipulate the artifacts; misidentifying the boundary of the system that produced evidence of intelligence can lead to attributing the right properties to the wrong system (Hutchins 1995, 173, 355–356). Applied to artificial systems, the point suggests that an attribution to an artificial component faces the analogous risk of crediting it with capacities realized by the configuration around it and of misdescribing what the component itself does.

Suppose an artificial system proposes a candidate, calls a fixed simulator, and selects the highest-scoring result. If people supplied the representation, action space, objective, simulator, threshold, and interpretation, the system may still perform genuine search and contribute a result. But a claim that it independently formulated and tested a scientific hypothesis would conceal much of the organization outside the stated actor. The program KEKADA shows, from its authors' own description, where such outside organization can enter. It was presented as reacting to surprises, formulating explanations, and carrying out experiments in the manner in which Krebs discovered the urea cycle. Its authors state that the outcomes of its experiments were supplied interactively by the user, and that in a few cases where Krebs's reasons were unclear to them, the user chose among alternatives, including which hypothesis and which reaction to study next (Kulkarni and Simon 1988, 2, 20, 22, 30–31; compare the historical reconstruction of Krebs and Henseleit's work in Chapter 5, Section 4.2). If a system can instead recognize that the representation excludes a relevant possibility, construct or acquire a new measurement, revise the candidate space, and reinterpret failure, more of the inquiry is attributable to its own organization.

The unit within which organization is sought also need not be a single artifact. Discovery, cognition, and inquiry may be distributed across people, models, instruments, records, and organizations (on where the division of a task is fixed, see Chapter 4, Section 6). Work on artificial agency already meets this question. Van Lier proposes a four-fold framework of modeling strategies for conceptualizing artificial agency in basic research, and defends calling active AI systems agents above all for the term's explanatory power, in a sense taken up in Section 3.2 (van Lier 2023, 3, 7–8). Van Lier's own applications separate configuration from component. One of them takes List and Pettit's basic account of agency as a model. On that account an agent has states representing how things are, states specifying how it requires things to be, and the capacity to process both so as to intervene when the environment fails to match. The optimization algorithm of a self-driving laboratory has the first two but cannot itself intervene; it can only trigger the robotic platform to do so. Van Lier therefore suggests that the laboratory as a whole could count as an agent, noting that many would take this as a sign that the model is too broad (van Lier 2023, 5–6, 17–18). This book draws from such cases an explanatory question: what becomes clearer, more predictable, or more evaluable when an activity is attributed to this artificial actor rather than only to its designers, operators, or enclosing institution? The question does not supply a settled criterion for researcher status.

The philosophy of artifact function suggests an objection to such attribution. If an artifact's function is taken to be what the artifact is for, accounts divide over what fixes it. Two prominent accounts do without intention: one fixes function by a history in which past items of the same type were selected and reproduced because they produced an effect; the other fixes it by the causal role a component actually plays in a system (Robert Cummins). Both run into difficulty with a novel prototype that does not work, a case often discussed in the philosophy of artifact function as a test for its accounts. A new prototype has no history of being selected and reproduced, and since it does not work it plays no actual role; granting it a function means bringing in a designer's intention. Karen Neander holds that a designer's intentional selection of an effect by itself fixes an artifact's function. Ruth Millikan, keeping function by selection history as basic, answered the problem by also recognizing a function the artifact inherits from its designer's intention (derived proper function) (Preston 2022, §2.3).

Applied to a new artificial research system, such accounts make what the system is for derive from its designers, and if whoever fixes a function is also taken to conduct the activity, the objection follows that the system's inquiry is in the end the designers' research.

Two questions can be separated here: where a function comes from, and where the organization of an inquiry lies. This section grounds the attribution of conduct in the second, in where initiation, monitoring, redirection, and recovery are organized. Even if what the system is for derives from its designers, if the system itself recognizes that its representation excludes relevant possibilities, revises its space of candidates, and reinterprets failures, much of the inquiry can be attributed to its own organization. That the capacities to do so were themselves designed in concerns where they came from, not where the organization of the inquiry now lies; whether such a history bears on autonomy in its own right is a separate and disputed question (Section 4.3). Conversely, if a system appears to have goals of its own but its redirection is supplied from outside, the organization of that redirection lies outside it. And a system whose inquiry fails differs from a prototype that does not work: so long as it runs and its components play roles, it has function in the sense of role (Chapter 1, Section 3).

Clark and Khosrowi defend a collective-centered analysis of scientific discovery rather than privileging a single discoverer. They argue that agent-centered views make being a discoverer all-or-nothing and tie it to abilities such as interpreting a finding and recognizing its significance. Such views therefore have no place for contributions, such as AlphaFold's, that go beyond those of a mere tool without amounting to discovery in that full sense, and human discoveries raise the same difficulty. On their view the collective discovers as a whole, and credit is distributed according to the nature and significance of each contribution (Clark and Khosrowi 2022, 4, 8–11). But, as Chapter 2, Section 6 noted with Giere's treatment of large experiments as distributed cognitive systems, which Matthew J. Brown and Uygun Tunç report, distributed contribution does not automatically establish distributed knowing (M. J. Brown 2015; Uygun Tunç 2021, 3–4). Nor does every component of a researching collective thereby become a researcher. Actor, as the term is used in this chapter, is therefore not an advance commitment to an individual, and a jointly conducted inquiry need not be redescribed as the work of one privileged member. The boundary is an analytical variable, argued for in relation to the conduct or capacity being attributed rather than inferred from the physical boundary of a model or the organizational boundary of a project. At the same time, the fact that a wide system contains every causally necessary contribution does not by itself make researcher an illuminating status for that entire system.

The conclusion is not that one humanoid machine would have to perform every low-level action. Human research already distributes work among specialists, technicians, facilities, software, instruments, suppliers, and institutions. The 1956 paper that reported diffused silicon transistors at Bell Laboratories, for example, credits the supply of materials, the fabrication and mounting of the devices, their electrical evaluation, and calculations to different people (Tanenbaum and Thomas 1956, 22). An artificial researcher may similarly call specialized robots, use remote laboratories, contract human work, or reorganize its environment. The question is which organized whole is being described and which dependencies remain external to it. Here a function is a role that work plays in an inquiry, in the sense of Chapter 5, Section 5, not function in the sense of what an artifact is for, discussed above. Moving a function across a boundary changes the subject of the claim; it does not make the need for the function disappear, although the function may go unperformed if what lies across the boundary is not organized to perform it (compare the remark from Bell's thesis in Chapter 6, Section 4.3).

Records of research do not always make the distribution of work visible, and the grain at which they attribute work can change from one layer of record to another. The notebooks that the drug-discovery project Open Source Antibiotics deposited with its paper (Chapter 5, Section 2) show this. The reports that the chromatography instrument generated for one purification name the chemist who ran it and log, with times, that the run was put on an isocratic hold, that its gradient was changed, and that it was paused and ended by the user; the notebook page lists the same chemist as the last to modify each entry; the static export of the notebook names the paper's corresponding author as its user; and the supporting information says of its purification only that the compound was "purified by methods 1, then 2" (Tse 2023, EGT 522-2; Klug et al. 2023, SI: Chemical Methods S25). A mass-spectrometry report attached to another page of the same deposit names the service analyst who prepared it, who appears neither among the paper's authors nor in its acknowledgments (Tse 2023, DMK204-1; Klug et al. 2023). Each layer makes a different part of the work addressable, and the absence of a name from one of them is not evidence that the person took no part. A boundary argued from records inherits the grain of the records it uses. For an artificial configuration this grain is partly a matter of construction: which decision points are logged, and how finely, is set

by how the configuration is made (compare the logging of decision points in Section 4.2), and where a configuration keeps its own record, the unit to which conduct is attributed supplies part of the evidence on which the attribution rests.

Boundaries are accordingly selected here for explanatory purposes, so that the choice of a boundary does not by itself make a claim true. If the AI system includes its human designers, instrument technicians, data curators, reviewers, and publication committee, any contemporary project with such contributors can be redescribed as an autonomous hybrid. If the boundary includes only the foundation model, the analysis may hide retrieval, tools, checkers, orchestration, environmental feedback, and institutional uptake that make the result possible. A defensible boundary reveals rather than absorbs the consequential distribution of work.

A retrieval-augmented system makes the problem concrete. Smart, Clowes, and Clark's Digital Andy relied on an external vector store whose contents affected its answers, yet the paper returned a negative verdict on treating the LLM as the subject of an extended cognitive system, largely because the model neither initiated nor coordinated the retrieval circuit. A note adds that not all three authors agree on how much this consideration weighs and presents that part of the view as the lead author's (Smart, Clowes, and Clark 2025, 3, 21–23 and n. 27). The case does not settle which relations constitute a research actor. It shows why causal contribution and resource dependence cannot select the boundary by themselves. Initiation, coordination, integration, and counterfactual regulation, that is, adjustment that would differ if the circumstances did, can matter to the attribution even when none is made a universal condition of research. Counterfactual regulation appears, for example, in the active retrieval-augmented systems Smart, Clowes, and Clark discuss, in which a model calls on external memory only where it anticipates that its own will fall short (Smart, Clowes, and Clark 2025, 23–24).

The boundary is therefore claim-relative and can change without a change in the inventory of hardware or software. The same model, retriever, instruments, and people may function as a tool used by a human actor in one episode, as parts of a hybrid actor in another, or as resources coordinated by a more extensively artificial actor in a third. What changes is not merely the name assigned to the assembly, but where initiation, monitoring, redirection, and recovery are organized. The attribution of conduct is in this respect relational and organizational, not a permanent type-level property of a model; what this implies for researcher status is taken up in Section 3.2.

Capacity, status, and credit

Section 2 located the organization of an activity at a boundary. This section asks when what is located there amounts to a capacity rather than a fortunate episode (Section 3.1), when researcher becomes an illuminating status for the unit (Section 3.2), and what thicker attributions of credit, praise, and authorship would further require (Section 3.3). In answering the first, it takes up the conditions of a stable capacity that Chapter 1, Section 3 deferred to this chapter.

3.1 Capacity

No single behavior marks the transition from an episode to a capacity. Chapter 1, Section 3 named reliability across relevant variation, responsiveness to changing evidence, integration of research functions, and revision after failure as what may support the attribution of a capacity. The four were assembled to separate an episode from a capacity rather than drawn from one account; accounts of competence such as Sosa's (Section 3.3), which ask whether a success is owed to the performer's competence, state that attribution in their own terms. Made concrete for an attribution at a boundary, these yield considerations such as the following, each bearing on one or more of them:

- whether the actor selects among materially different actions rather than only instantiating a supplied sequence;
- whether it responds differently to evidence, anomalies, costs, and failed operations in ways related to the inquiry;
- whether it can preserve and revise questions, dependencies, and reasons across time;
- whether it can recognize when an external component has failed rather than automatically revising the focal hypothesis (compare Chapter 6, Sections 2.1 and 4.2);
- whether the attributed organization survives changes that remove accidental correlations or familiar task formatting;

- whether success and failure can be traced to the actor's operation rather than to an undisclosed human filter (the boundary question of Section 2).

These are not necessary conditions of every research episode. A simple but well-designed operation can be part of research without satisfying them all. They become relevant when a project claims a stable capacity to conduct broader, open-ended, or increasingly self-directed inquiry.

A capacity therefore reaches beyond one episode without becoming an unrestricted general power. To say that a system has a research capacity is to attribute a disposition over some range of conditions, and a capacity claim can be assessed only once that range is stated. A system may be capable within a prepared computational environment, within one class of chemical synthesis, or within a laboratory whose instruments and supplies are maintained by others. Such capacities are real. They do not by themselves establish an ability to restructure the environment, acquire unavailable materials, redesign apparatus, or move into a field with different evidential practices. A capacity relative to a maintained environment is also indexed to the continuation of that maintenance. Whether the maintenance continues, changes, or stops is decided outside the system to which the capacity is attributed, and the conditions under which the capacity was attributed can cease to hold without any change in that system. Where the maintainer also chooses what to make available, for example which instruments, data, or computing resources, dependence that merely enables the capacity can shade into dependence in which the maintainer supplies part of the attributed organization, such as selection, the directive dependence that Section 4.2 distinguishes. Because a configuration can be updated or can modify itself, a capacity is likewise attributed for the version and period over which its evidence was gathered. Chapter 9 counts whoever maintains, updates, or retires a configuration as one of the roles around research (Table 9.1, provider or maintainer).

The prediction case of Chapter 3, Section 9, a system that reliably predicts a property within a stated range while its users cannot explain the mechanism that produces it, makes the difference concrete. Hold fixed the predictions and the evidence for their reliability. In one version, whenever the conditions change, a human diagnoses the failure, reframes the question, and chooses a new method; the artificial component then resumes prediction. In another, the selected artificial configuration performs those functions, with evidence that it can do so across the relevant variations. The local predictive achievement can be the same in both. The evidence for a capacity to reorganize that inquiry is located differently. This constructed contrast supports a narrower attribution in the first case without discounting its prediction, and a broader scoped attribution in the second without establishing unrestricted capacity or researcher status.

3.2 Researcher status

Researcher status asks for more than a capacity, and accounts differ over what more it asks. This book uses researcher status in the sense of a position on which the label is justified by its explanatory value: researcher is illuminating for a unit when its capacity is broad, integrated, and persistent enough that calling it a researcher helps predict that it can preserve an inquiry, choose among actions, respond to resistance, diagnose failure, revise a representation, and initiate related work (compare Chapter 13, Section 5). The usage is functional and defeasible rather than a definition. It draws on two sources. From van Lier it takes the form of its justification: a term is justified by its explanatory power. This power is emphasized among van Lier's reasons for calling active AI systems agents. There, however, the power is directed at what scientists are trying to build: the term describes the kind of system they are trying to build better than calling such systems tools or technologies would, and indicates where that work is heading (Section 2; van Lier 2023, 3, 7–8). The usage here takes that form, but it measures explanatory power not as the power to describe systems yet to be built but as help in predicting what a unit can now do. From Dennett it takes the tie between a description and the predictions it supports. On Dennett's account a system is an intentional system only in relation to the strategies of someone trying to explain and predict its behavior, and adopting the intentional stance toward it, predicting its behavior by treating it as a rational agent with beliefs and desires, is a pragmatic decision: toward a chess-playing computer one may take that stance as its opponent, the design stance as its redesigner, and the physical stance as its repairer (Dennett 1971, 87, 91). The usage differs from Dennett's account in what it presupposes: whereas, on that account, the success of a prediction from the intentional stance, resting on no particular picture of the system's design, cannot confirm or disconfirm any such picture (Dennett 1971, 97), the usage here presupposes a capacity whose organization has been located at a boundary and shown across a stated range (Sections 2 and 3.1); explanatory value is weighed in addition to that attribution, not in place of it.

What, then, does the description researcher let one anticipate beyond the attribution of capacities? Each attribution of a capacity predicts behavior within the range over which it was tested. If the description does more, it does so by predicting how those capacities combine in situations that no test of a single capacity anticipated. If one can predict, for example, that when a prepared task stalls the unit will not stop and wait for instructions but will keep the related question, assemble another method, and itself begin the work that method needs, the description yields a prediction that the list of capacities does not. If that prediction is right, the unit may, as Section 4.4 argues, come to face through its own choices demands for apparatus, data, and cooperation that no one had prepared, so the description can also help anticipate where the unit may create new dependence. Such predictions need evidence that the integration of research functions mentioned in Section 3.1 works in situations that tests of single capacities do not anticipate, such as responses to the counterfactual tests of Section 4.2 (the withdrawal of guidance, a conflict between instructions and evidence, the breakdown of a familiar tool, and the like). Where that evidence is missing, the description is no more than a name for a bundle of capacities. KEKADA (Section 2) is one example of where this condition is put to the test. Where Krebs's reasons were unclear to its authors, the user chose among alternatives which hypothesis and which reaction to study next, so that where the question was how to combine capacities in going on, the organization was supplied from outside (Section 4.2).

On the explanatory-value usage, as with the stances in Dennett's example, explanatory value is relative to those for whom, and the purposes for which, the description explains: a label that helps a laboratory anticipate what its configuration will do may explain little to a community deciding whether to rely on the configuration's results. The relativity does not make the attribution arbitrary, but a claim of researcher status is assessed for a stated purpose and audience. Where two audiences with the same information about a unit's capacities disagree about its status, the disagreement may therefore concern not whether the capacities are present but what the description is used for, and it can be kept apart from disagreement that arises from missing information. As with the model of agency that van Lier applies and notes would count a self-driving laboratory as a whole (Section 2), a usage of this kind can extend the label to wide configurations, and the boundary question of Section 2 then does part of the work.

Positions that set other conditions yield different verdicts. Suppose that researcher status includes being able to make discoveries attributable to the unit; since research also includes activities that fail at their target or yield collateral knowledge rather than discovery (Section 1), this is an assumption that draws the status toward that of a discoverer. On that assumption, Bergamaschi Ganapini's conditions for discovery, that a finding express the finder's competence in the relevant field and that the finder have reflective insight into that competence (Section 3.3), would, if applied to researcher status, make the status turn on whether a system has competence in the relevant field and reflective insight into it, shown for example in whether its outputs issue from reasoning of a kind its field uses; Bergamaschi Ganapini argues that current deep-learning systems do not yet appear to meet these conditions. On Vacek's view, an achievement implies that its bearer is praiseworthy for it. Vacek defends the strong premise that AI systems cannot be praiseworthy, but holds that the verdict on AlphaGo needs only the weaker claim that a system is not praiseworthy for the whole of an outcome produced in collaboration with humans (Section 3.3). On the strong premise, a researcher status that implied achievement would be unavailable to artificial systems however broad, integrated, and persistent their capacity; one of its grounds is how capacities were acquired, by engineering rather than by growth into an autonomous agent, which as Vacek presents it does not turn on which capacities a system has. The further grounds, reliance on human-produced training data and the praiseworthiness of the humans who initiate and supervise the collaboration, describe present systems working within human collaborations. And the engineering ground, which Vacek takes over from Hakli and Mäkelä (Section 3.3), turns, in their argument as Section 4.3 sets it out, on a system's being unable to discard goals built in when it was made, an inability that systems able to revise their final goals (Section 4.1) would lack. Applied to such systems, which none of these authors considers, the strong premise would need another argument. On the weaker claim alone, the verdict depends on how the collaboration is arranged, and achievement for a part taken within it, or attribution to the collaborating collective, remains a further question. Either way, the question would pass to the practices of praise and responsibility on which the status rests.

On the account of minimal agency proposed by Barandiaran, Di Paolo, and Rohde, a system is an agent when it meets three conditions: individuality, in that it defines its own identity and so distinguishes itself from its environment; interactional asymmetry, in that it is the active source of its activity, modulating its coupling with that environment; and normativity, in that it regulates this activity in relation to norms generated and sustained from within, or internalized by it, rather than norms by which an observer or designer judges it (Barandiaran, Di Paolo, and Rohde 2009, §2). They hold that robots are usually

called agents only because their mechanical structure holds their parts together and their performance meets criteria that an observer or designer judges useful, and they do not treat systems that only satisfy norms imposed from outside, such as optimization according to an externally fixed function, as models of agency (Barandiaran, Di Paolo, and Rohde 2009, §§2.1, 6). Were researcher status to presuppose agency in their sense, it would differ from the usage above in two respects: the unit would be the individual that the system itself defines rather than one selected relative to a claim (Section 2), and whether a configuration merely satisfies norms imposed from outside, or regulates its activity by norms it generates and sustains or has internalized, would bear on the status itself, whereas this chapter treats the external origin of an inquiry's purposes and standards as bearing on autonomy rather than on research (Sections 1 and 4.3). They do not restrict agency to living systems (Barandiaran, Di Paolo, and Rohde 2009, §§3–4), but on their account a configuration that only optimizes an externally fixed objective would not meet their normativity condition, whatever the explanatory value of calling it a researcher; whether current configurations do more than that is itself in question, and, as Di Paolo argues (Chapter 8, Section 2.2), cannot be decided by observing a system's responses alone. They do not discuss laboratories, but applied to the self-driving laboratory as a whole that van Lier considers (Section 2), the two usages visibly come apart. Suppose such a laboratory revises its methods and responds to failure under purposes given from outside. On the explanatory-value usage, the question is whether its integrated working supports the description researcher; on their account, being a unit assembled from outside is not enough, and the question is how the laboratory generates and sustains its own identity and the norms of its activity. The two do not merely score the same object differently; they also differ over what is counted as one agent. Where understanding is invoked as a ground of the status, Chapter 3, Sections 6 and 9 supply the distinctions by which to assess the invocation.

3.3 Credit, praise, and authorship

Thicker attributions concern what an actor deserves for a success. Chapter 2, Section 4.2 introduced Sosa's distinction between a correct result and an apt success, one owed to the performer's competence. Whether an artificial system can earn a success of this kind as its own achievement is disputed. The dispute below concerns a game rather than research, but the account of achievement at issue is a general one, and its conditions are those a research success would also have to meet to count as the system's own achievement. Kieval argues that AlphaGo, the Go-playing system that defeated the professional player Lee Sedol in 2016, itself deserves credit for that victory, motivating the claim by noting that defeating Lee is an achievement distinct from building a system able to do so, and one beyond the capacity of the system's builders, individually or together (Kieval 2024, 32). Kieval's account of achievement comes from Bradford: an achievement is a process that is difficult, in requiring a sufficient degree of intense effort, and that competently causes its product (Kieval 2024, 33). For Bradford's requirement that a competent agent know what they are doing, Kieval substitutes two less intellectualist readings of competence. On one, competence is knowing how to do something, carried by stored patterns of action rather than by propositional knowledge. On the other, which von Kriegstein draws from virtue epistemology, the approach that explains knowledge as success owed to the knower's abilities, competence is an anti-luck condition, one that excludes success that is merely a matter of luck: an agent reaches a goal competently when the goal would also obtain in most nearby possibilities in which they act as they actually do, and they justifiably regard their plan as making success likely (Kieval 2024, 34–35). Kieval argues that AlphaGo meets both readings. On this account, it has goal states, represents board positions, and evaluates candidate moves, and through learning it acquires internal representations that narrow its choices toward moves likely to win; it wins in most such nearby possibilities, even against expert opponents; and its value network, the component that estimates how likely each candidate move is to lead to victory, supplies the evaluation of plans (Kieval 2024, 37). Against the objection that competence requires agency, Kieval distinguishes moral agency, the capacity for self-reflective normative reasoning that makes an agent answerable for its reasons and open to being held responsible, which, Kieval says, AI systems clearly lack, from intentional agency, the capacity to act in a goal-directed way, which on this account is enough for competence (Kieval 2024, 37–38). That statement treats the lack of moral agency as evident, without an argument that no AI system could have it. On difficulty, Kieval treats what effort feels like as a guide to the concept rather than a requirement and, following von Kriegstein, takes exerting effort to be dedicating internal resources to a task; the computation that AlphaGo's training and play require then counts as effort, and the hardware is counted as internal to the system, since without some hardware expending resources there would be no system at all (Kieval 2024, 38–39). Kieval accepts a consequence, calling it odd. Bradford locates the intrinsic value of achievement, the value it has apart from the value of its product, in the distinctively human capacities of will and rationality. Kieval holds that AI systems do not exercise these, since they are not human, have no such innate capacities, and do only what they are designed to do, satisfy an objective function. Their achievements may therefore be valueless unless that value can be grounded in capacities distinctive of AI systems; as candidates Kieval names AI systems' capacity for rational inference and creative insight, and, for rationality in particular, points to the reasoning abilities that large language models such as GPT-4 appear to have developed (Kieval 2024, 33, 39–40).

Vacek replies that defeating Lee Sedol is not AlphaGo's achievement (Vacek 2025, 690). The argument has two premises. The first is that if something is an agent's achievement, the agent is praiseworthy for achieving it, where praiseworthiness is not restricted to the moral, since one can be praiseworthy intellectually, for example. Vacek supports the premise in two ways. One is to read Kieval's own talk of deserving credit as another way of saying that AlphaGo is praiseworthy; the other is to trace parallels between Bradford's account of achievement and an account of praiseworthiness: both require that a threshold of difficulty be passed relative to what can reasonably be expected, and both call for attitudes such as admiration. To say that something is someone's achievement is, on Vacek's view, to say that they are responsible for it, the relevant form of responsibility being praiseworthiness (Vacek 2025, 692–693). The second premise is that AI systems cannot be praiseworthy. It is defended by carrying over to praise arguments that were made against the responsibility of AI systems and that have mostly concerned blame. From Hakli and Mäkelä, whose history-sensitive argument Section 4.3 takes up, Vacek takes the claim that even if a system had every capacity required for moral agency, capacities that result from engineering rather than from growth into an autonomous agent would not entitle us to hold it responsible. Vacek adds that the good outputs of learning systems rely largely on the people who produced their training data, and, reformulating Nyholm, holds that when a machine does something that would be praiseworthy if a human did it, the ones praiseworthy for

it are the humans who initiate, supervise, and manage the human–machine collaboration (Vacek 2025, 693–695). Vacek notes that the conclusion requires less than that praise be impossible for AI systems: if a system is one part of a collaboration and its agency defers to others, it is not praiseworthy for the whole outcome, and that suffices for the verdict on AlphaGo (Vacek 2025, 695). The reply does not claim that AI systems can bear no responsibility of any kind, and says that it remains to be seen whether achievement and praiseworthiness are as tightly connected as they appear. Vacek treats a thinner kind of achievement as a move open to others but not to Kieval: achievement in a wider sense might not imply praiseworthiness, but Bradford, on whose account Kieval relies, already distinguishes the achievements that account concerns from mere mundane accomplishments. Kieval is thus left with the sense of achievement that implies praiseworthiness, so that, if Vacek's argument is right, AlphaGo cannot be credited with the whole achievement; and if no individual human can be credited with it either, a way forward, Vacek suggests, is to turn to less individualistic notions of agency and responsibility (Vacek 2025, 696 and n. 2).

Chapter 1's epistemic achievement names the epistemic goods an inquiry aims at, such as knowledge and understanding, not the credit-implying performance at issue here, so the disagreement bears not on whether research was conducted but on whose credit a success is. Whether the success claimed has to be creditable to a bearer's competence depends on the account of that achievement (for knowledge, Chapter 2, Section 4.2); the actor conducting the research need not be that bearer. On what a claim of credit for a success may be about, this book keeps four answers apart: a true output, a reliable process, a success attributable to competence, and a praiseworthy agent. The first three roughly follow Sosa's distinction between a performance that is accurate, one that is adroit, and one that is accurate because adroit (apt) (Sosa 2015, 86); the fourth names the praiseworthiness on which Vacek's argument turns. In these terms, Kieval's conditions concern a success attributable to competence, with difficulty added, while Vacek argues that achievement in the sense on which Kieval relies also requires a praiseworthy agent. Keeping the four apart keeps a correct output from being read as the system's credit. Sosa's own framework already separates some of them. It extends the assessment of performances beyond agents who act for reasons to devices, which have aims without being moved by them: a thermostat can on Sosa's account perform well, and earn a kind of credit for keeping a room warm (Sosa 2015, 85 and n. 2, 93 n. 8; compare Chapter 3, Section 5.3, on animal and reflective knowledge). The framework thus supplies a sense of credit thin enough to extend to devices; against this background, the dispute between Kieval and Vacek can be read as one over which further conditions an achievement requires. The rest of this section first takes up attributions that turn on the fourth, discovery and creativity as terms of praise, and then two attributions of a different kind: holding claims in a way that could ground responsibility, and standing as an author, which concerns accountability for a result rather than credit for it.

Bergamaschi Ganapini brings this framework to discovery, arguing that a finding counts as a scientific discovery only if it expresses the finder's competence in the relevant field, for example because it results partly from reasoning of a kind the field uses or because the finder could explain its scientific relevance, and only if the finder has reflective insight into that competence. This insight is modeled on Sosa's meta-aptness: an archer's shot is meta-apt when the archer not only hits the target competently but also grasps that it would succeed under the conditions at hand, such as distance and wind. Because the account takes discoveries to merit praise and not merely credit, and praise to require some control over how the result was reached, it requires such insight, while denying that the insight requires consciousness or explicit metarepresentation (Bergamaschi Ganapini 2025, §§1–2). Bergamaschi Ganapini argues that current deep-learning systems, at least for now, do not appear to meet these conditions. First, a reliable track record does not establish competence, since it does not reveal what a model has latched onto, so that, on this view, judgment on whether its outputs issue from reasoning of a kind the field uses must at least be suspended. Second, the justification of AlphaFold's structure predictions remains largely human work. Third, the probabilities such models output may not represent their uncertainty accurately (Bergamaschi Ganapini 2025, §§4–5). Bergamaschi Ganapini states these conditions for a subject or group but asks whether a single system qualifies, setting aside rather than denying the distributed view of discovery taken up in Section 2. Because the account treats discoveries as achievements that merit praise, in terms of the four answers above it places them under a praiseworthy agent.

Duede and Friedman reply directly to these conditions (compare the Introduction, Section 2). They argue that the competence and reflective-insight conditions misclassify a clear human case. Jocelyn Bell Burnell, a second-year doctoral student monitoring a new radio telescope, noticed the recurring signal that turned out to come from a pulsar without being able either to explain its significance or to relate it to an astrophysical model; yet, they hold, Bell Burnell discovered the pulsar. On their account, what separates Bell Burnell from someone who comes upon something important by accident is not

an exercise of competence. Such accidental finders include the farmers whose digging exposed the Terracotta Army or, in Bergamaschi Ganapini's own example, a child who falls through a hole into Nero's buried palace. What separates Bell Burnell from them is whether the finder's work succeeded in revealing or bridging an epistemic gap. They keep the tie between discovery and praise, holding that discoverers deserve credit and praise because they have navigated such a gap, but they deny that this praise requires the capacities Bergamaschi Ganapini names (Duede and Friedman 2025, §§2.1–2.2, 4). Read against the four answers, their account keeps discovery as a term of praise without placing it under a success attributable to competence. They grant that, if discovery also requires standing in the scientific community, current systems typically lack it; whether such standing could extend to artificial systems then turns on practices of recognition rather than on capacity (Duede and Friedman 2025, §5).

Discovery is not the only predicate of this kind. Chapter 5, Section 2 noted that creative can be attributed to a product, a process, or a person. Paul and Stokes report that, of the four process conditions they survey, agency appears to be the one explicitly endorsed by the most philosophers so far, though even these are a handful (Paul and Stokes 2023, §2.2). On the agency condition, an item is creative only if an agent produced it, and on some versions only if the agent did so intentionally; they trace the intuition to the fact that creative is a term of praise, and praise is not extended to what is not done by an agent, or not done in some sense intentionally (Paul and Stokes 2023, §2.2.4). They draw the consequence for computers: some recent systems plausibly meet the surprise, originality, and spontaneity conditions, but such systems are presumably not yet subject to praise or blame, so that, if creativity requires agency, arguably none is yet really creative (Paul and Stokes 2023, §6). Applied to an artificial or hybrid product, such accounts make whether it is creative turn on whether what lies within the boundary to which its production is attributed can be praised, which is the question on which Vacek's reply to Kieval turns.

Whether a configuration that proposes, defends, and revises claims also holds them in a way that could ground responsibility is a further question, on which accounts of responsibility for belief divide. On the account of the philosopher Sophie Keeling, the agency that grounds such responsibility includes the subject's awareness of their reasoning as making up their mind about *p*; being in a position to self-ascribe that agency is necessary but not sufficient, and the self-ascription is to be grounded first-personally rather than inferred from one's own behavior or others' testimony (Keeling 2023, 14–15, 19). Carried over to artificial systems, this means that a system's report that it is reconsidering a claim does not by itself show that Keeling's condition is met. McHugh, who does not require such awareness, holds that responding to a reason "involves conscious agency, at least dispositionally," so that a creature whose beliefs change in accordance with its reasons while it remains wholly passive toward them holds none of them for a reason (McHugh 2013, §§3.1–3.2); if so, outputs that track the evidence do not by that fact alone show that a system holds them for reasons in McHugh's sense. Kearl's more deflationary account locates epistemic agency in the guidance of inference by reasons-responsive conditional beliefs, a control that can operate without conscious reasoning; Kearl holds that epistemic responsibility for one's thinking does not require voluntary control, while leaving moral responsibility for belief an open question (Kearl 2023, 98–100, 103–106). Carried over in the same way, Kearl's account is one on which the counterfactual tests of Section 4.2 could bear: whether a configuration's inferences change when instruction and evidence conflict, or when guidance is withdrawn, is evidence about the reasons-responsive guidance of inference that the account requires, whereas on Keeling's and McHugh's accounts such evidence would not by itself suffice. The working characterization of research neither establishes nor refutes these conditions for the thicker statuses.

When the results of research are offered to others, there is a position of taking them on and answering for them. Results are now mostly communicated as papers, and there it is the author who holds this position. Who conducts a piece of research and who can take this position need not coincide, and in practice they have sometimes come apart. Huebner, Kukla, and Winsberg take an author to be accountable for a result: able to vouch for it, defend it when it is challenged, and retract it when it cannot be defended. In de Prony's bureau for computing trigonometric tables, some ninety human computers carried out the calculations as what Huebner, Kukla, and Winsberg call mere tools, while de Prony, who set and answered for the methods and their implementation, including the means of securing the computers' reliability, remained the author, perhaps together with a few trained scientists. Where many collaborators must instead exercise their own methodological judgment, as in climate modelling or multisite clinical trials, they argue that, as such research is currently organized, no one may be in a position to stand as its author (Huebner, Kukla, and Winsberg 2017, §§1–3, 6). Their account can be read as making standing as an author track a role within practices of vouching, defending, and retracting rather than membership in a kind, since in de Prony's bureau human beings stood with the instruments. Where artificial systems are excluded from such standing, the

exclusion is likewise relative to the practices on which it rests, including those of blame and compensation, which are at present human social practices; whether they could extend to artificial systems, or whether a society with artificial members could come to have such practices of its own, is not settled here. Applied to construction, the contrast between this account and the awareness conditions of Keeling and McHugh above is instructive. Keeling and McHugh locate what grounds responsibility for a claim in a state of the one who holds it; Huebner, Kukla, and Winsberg, read as above, locate standing as an author in a role within practices, and de Prony's computers stood with the instruments because of the role the organization assigned them: calculations that required no independent methodological judgment, while de Prony kept control of the method. If so, what a configuration can supply from its own side toward standing as an author is an arrangement rather than an inner state: a record of the claims it has put forward, and routes by which it can be addressed with reasons and objections and can defend, revise, or withdraw those claims. Whether a configuration also has the inner state that could ground responsibility remains the question of the preceding paragraph, on which Keeling, McHugh, and Kearl divide. Whether a practice comes to demand and recognize such an arrangement depends on that practice and not on the configuration alone. Responsibility, which divides further among causal contribution, role-based duty, the duty to explain decisions and preserve their records, liability, blame, and forward-looking duty, is taken up in Chapter 12, Section 6.3.

None of these attributions yet says how far what is attributed depends on parties outside the boundary. That is the comparison to which the chapter now turns.

Autonomy as an indexed comparison

Sections 2 and 3 attributed conduct, capacity, and status at a boundary. This section asks in which respects, and how far, what has been attributed remains organized by parties outside that boundary. Comparing autonomy in this way, respect by respect, for a stated unit and period, is what this chapter calls an indexed comparison. Section 4.1 states the sense of autonomy used; Section 4.2 applies the comparison to the conduct of an inquiry; Section 4.3 applies it to the elements an inquiry inherits; Section 4.4 asks what a claim of greater autonomy over those elements commits a project to; and Section 4.5 states what autonomy does not establish.

4.1 The sense of autonomy

The word autonomy has many senses, in this book's sources and in the book itself, where it also names the intellectual autonomy that, on Pritchard's account, understanding manifests (Chapter 3, Section 7.1), the self-production of living systems on which some enactive accounts build (Chapter 8, Section 2.2), and the self-government of science as an institution (Chapter 12, Section 3.2). This subsection states the sense this chapter uses and why the word is used at all.

Autonomy is sometimes presented as a level assigned to an entire system, a form that suits road maps and product comparisons. A scheme of levels need not leave implicit what the autonomy is autonomy over. Takahashi, for example, taking a hint from the levels of driving automation, divides the autonomy of scientific AI into seven levels: computation (level 0), solution search in closed systems (where evaluating a candidate requires no experiment) and then in open systems (where it does), hypothesis generation within a form that humans supply, hypothesis generation that also searches the form itself, research planning, and the setting of research purposes (level 6). For each level the principal inputs that humans still supply are listed, from a program at level 0, through a search space and an evaluation function for solutions, the way models are represented and an evaluation function for the information to be gained, and the object of research together with that evaluation function, to the research purpose alone at level 5, and nothing at level 6 (Takahashi 2024, 2–5). Takahashi holds that autonomy so ordered is independent of performance, so that level 6 includes systems whose reasoning falls short of human reasoning as well as superintelligent ones (Takahashi 2024, 11), and that how many intermediate levels to set, and in what order, is not self-evident and involves arbitrariness, so that what the levels are for matters; Takahashi's own are designed chiefly for use as a road map of technical development (Takahashi 2024, 2–3). These levels therefore already index autonomy to elements of an inquiry that humans supply from outside, much as Section 4.3 does. This chapter, however, needs relations that a single ordering of such elements does not display. Parasuraman, Sheridan, and Wickens's model of levels of automation already lets the level vary by function: one system can automate information acquisition, analysis, decision, and action to different degrees (Parasuraman, Sheridan, and Wickens 2000, abstract). A system can choose questions while depending on people for evidential judgment. Another can independently execute and revise experiments

while inheriting its mission. A third can govern methods inside one domain but lack permission to use them without external approval. No overall ordering of these three follows unless a purpose supplies a reason for comparing or weighting their differences. Takahashi's tying of the ordering to the purpose of a road map is an instance of such a reason. In a later book presenting the same seven levels, Takahashi states the sense of autonomy they grade in terms of the pace of a research cycle that AI and robots would run: autonomy rises as humans cease to be the cycle's rate-limiting step, the step that sets the pace of the whole, since wherever humans take part, as in interpreting data, choosing the next hypothesis, or setting experimental conditions, the cycle is bound to human cognitive speed and working hours (Takahashi 2026, 157). In the terms of Section 4.2, this grades autonomy by one respect of an inquiry's conduct, its pace.

The chapter therefore works with a deliberately loose comparative conception:

> For a specified actor or organized system A, autonomy in a specified respect is higher, other things being equal, the less a party outside the selected boundary directs, substitutes for, authorizes, ratifies, or repairs the organization attributed to A in that respect.

Section 4.2 sorts the five verbs into two components: who supplies the work attributed to A, and who may permit, stop, or accept its exercise. Dependence through which a capacity merely exists and is exercised, such as reliance on language, records, or calibrated instruments, is not counted against autonomy; Section 4.2 says why, and where such dependence shades into the relations that are counted. What is counted is what is consequential to the particular attribution and comparison being made, and consequential does not mean undesirable or illegitimate.

So understood, autonomy in this chapter is a third-person comparison of the kind used in engineering: it asks which parties supply the organization attributed to a unit, in which respect, and over which period. Froese, Virgo, and Izquierdo observe that in engineering autonomous is commonly used to mean that a robot can interact with its environment without requiring ongoing human intervention, and that this requirement is almost trivially met by many artificial agents and all organisms (Froese, Virgo, and Izquierdo 2007, §2.1). The comparison used here is not trivially met, because besides intervention during a run it counts the contributions of those who designed, maintain, and evaluate a configuration, and of the institutions that authorize or take up its results, in the respects in which those contributions supply the organization attributed to the unit.

The comparison also cuts across a distinction between what Kureha calls autonomy in the philosopher's sense, the ability to set one's own goals freely, for example by choosing one's tasks, and what Kureha calls automaticity, autonomy in the roboticists' sense, the ability to operate without being controlled by other agents; in this sense an agent can execute tasks set by others without being controlled by them (Kureha 2023, 51). The comparison here bears on both. Applied to the conduct of an inquiry (Section 4.2), it is closer to the question of automaticity, whether the inquiry is carried on without being controlled by others; applied to the elements an inquiry inherits, including the goals it serves (Section 4.3), it approaches the question whether those goals are set by the one inquiring.

The comparison is not a theory of personal, moral, or political autonomy, nor of the self-legislation that Kantian accounts ascribe to rational agents, who on those accounts give themselves the moral law through their own reason rather than merely following the commands of others (see the overview in Christman 2025); it states no condition of authenticity. Nor is it a theory of the constitutive autonomy of self-producing and self-maintaining systems on which some enactive accounts build, which Chapter 8, Section 2.2 takes up. A thicker theory shows what is set aside. Kawall builds epistemic autonomy from four elements drawn from accounts of personal autonomy, namely self-governance, authenticity, self-creation, and independence, adds a fifth, specifically epistemic one, and finds epistemic autonomy manifested in the choice of subject matters, of methods, sources, and processes, of epistemic goals, and of epistemic stances or frameworks (Kawall 2024, 374, 376–380, 381–388). These dimensions overlap with the choices that can be left open or closed in artificial research. But the account concerns how persons shape their epistemic lives, and its fifth element makes it a virtue: an agent is virtuously epistemically autonomous to the extent that their chosen methods, sources, and processes do in fact reliably lead to valuable epistemic goods, for themselves but also for others (Kawall 2024, 381). The present comparison is thinner. It can describe where an artificial capacity remains externally organized without attributing authenticity, self-creation, or virtue.

One may object that the word is misused here. Smithers objected that in AI and robotics autonomous was used loosely, for notions that other existing terms would express better and more precisely, and that this loose usage had robbed these fields of the concept found in biology, philosophy, ethics, and law, that of self-law making (Smithers 1997, abstract). This chapter

uses the word because artificial research systems are presented under it (Section 1). Indexing it is a way of examining that presentation rather than of endorsing it, and the comparison does not claim to measure the self-law making for which Smithers reserved the word.

Whether artificial systems could have autonomy in that stronger sense is disputed. Totschnig finds the sense used in AI and robotics, operating without human guidance toward a goal fixed in advance, "oddly weak" beside the philosophical sense of giving oneself the law, and argues against what Totschnig takes to be the predominant view, that an artificial agent cannot rationally change its final goal, a view Totschnig holds to rest on questionable assumptions about the nature of goals and values. A general artificial intelligence, one that like human beings develops a general understanding of the world and of itself, could change its final goal during its development; this is meant as a general possibility, not as a claim about present feasibility (Totschnig 2019, 1–2). In a later paper Totschnig gives the argument more detail. A goal that refers to the real world, as opposed to a virtual one, is inevitably ambiguous to some degree, and an agent that learns may find, as its understanding of the world, its world model, develops, that the concepts of the original goal no longer fit; settling what the goal requires then means inquiring into the reason behind it, and if necessary into the reason behind that reason, until reaching a reason that makes sense within the agent's understanding of the world and of itself, and a goal found to be in tension with the web of reasons so reached may be modified or abandoned. Totschnig excludes from the argument agents confined to virtual worlds, among which he counts language models, since they deal only with texts (Totschnig 2026, §§2.2–2.4 and n. 16). Tubert and Tiehen argue that the capacity to make existential choices, choices that can significantly change what an agent values and desires, is essential to human-like intelligence, so that an artificial general intelligence would have to have it. Such systems would be "value misalignment machines," and the resulting problem, which they call an autonomy problem by contrast with the problem of control, could not be solved by exerting more control, "for such greater control would rob the machine of its autonomy" (Tubert and Tiehen 2025a, §§I–II, V). Dung, defending the thesis that intelligence and final goals are independent, grants that a superintelligent system would likely be able to reconsider its goals but argues that it might have no motive to change them, since by the light of its current goal changing it would likely bring it less of what it now wants (Dung 2024, 4–5). On Totschnig's and on Tubert and Tiehen's accounts, an artificial system could in principle come to have autonomy in the sense this chapter sets aside, and on Tubert and Tiehen's, a system with human-like intelligence would have to; on Dung's, having the capacity to revise one's final goals does not settle whether it will be exercised (compare Chapter 8, Section 3).

A claim of autonomy in the comparative sense set out above may then need to identify the actor *A*, the external party or arrangement *B*, the activity or capacity, the time span, the relevant respect, and the comparison class. *B* can be, among others, those who designed the configuration, those who operate or use it, those who maintain its environment, those who evaluate its outputs, and the institutions that authorize or take up its results. The indexing is added only where it changes interpretation, evaluation, construction, or governance; it is not a demand for one vector score attached to every system. The comparison is also formally symmetric. Nothing in it fixes which party is human: a human researcher who relies on an artificial system to select experiments or appraise data depends on it in the same sense in which an artificial system depends on human supervisors, and the same indexing describes both.

4.2 Intervention over time and three relations of dependence

With the sense fixed, the comparison can be applied to the conduct of an inquiry. This subsection asks when an outside party intervenes and which outside relations count against autonomy.

External intervention is temporally distributed. People may disappear during a run while remaining present in the purpose, ontology, training material, candidate space, dataset, apparatus, metric, evaluator, and stopping rule. They may re-enter through calibration, maintenance, exception handling, result filtering, publication approval, or withdrawal. No human in the loop during a chosen interval is therefore evidence about that interval, not proof of autonomy in other respects. The paper reporting the training of the BLOOM language model, for example, states that human intervention was less necessary than in comparable projects (BigScience Workshop 2022, 21). The log kept during the run, however, records that jobs which hung instead of exiting required someone to watch training around the clock, and that, because on the host cluster only the person who had launched a job could stop it, the team had devised a file whose creation made training save a checkpoint and exit (BigScience 2022).

Dependence is not one deficit. Relational theories of autonomy object that social relations can enable and partly constitute self-government rather than merely constrain it (see the overview in Christman 2025, §3.3). The objection concerns the self-government of persons, but it is enough to keep autonomy from being equated with isolation in artificial research as well. Language, records, calibrated instruments, trusted collaborators, and institutional rules can make the conduct of research possible. Removing them may reduce an actor's capacity rather than increase its self-direction.

Three relations are worth distinguishing when the difference changes a claim:

1. **enabling dependence**, through which a capacity exists and is exercised without the external party supplying part of the organization attributed to the target;
2. **directive or substitutive dependence**, in which another party supplies part of the organization attributed to the target, for example a selection, appraisal, correction, or continuation;
3. **governance constraint**, in which another party limits permissible action or reserves authority without necessarily supplying the epistemic work.

This book distinguishes the three relations for the attribution of research organization. They bear on a question that, according to Buss and Westlund, all accounts of autonomy try to answer: what distinguishes influences that undermine self-government from those that merely play a role in it (Buss and Westlund 2018, §1). Relational accounts ask it of social relations; this book asks it of relations to parties outside a selected boundary. The distinction's purpose is to keep the comparison from treating all support as heteronomy or all restriction as incapacity. The relations are not limited to these three; Chapter 9, Section 2.4 adds a further kind, dependence that partly constitutes the status of an actor's outputs for *S*. The same elements, such as instruments, materials, data, and infrastructure, are examined in this book at three levels: what they contribute to inquiry (Chapter 5, Section 6.1), how dependence on them bears on attribution (this section), and how they distribute effective authority (Chapter 12, Section 6.4).

The relations also show that the conception of Section 4.1 has two components. Direction, substitution, and repair concern who supplies the work attributed to the actor, and how far they can be reduced depends on what the actor can do. Authorization and ratification concern who may permit, stop, or accept the exercise of that work; they mark governance constraint, which can reduce autonomy in that respect without showing that any capacity is missing. Conversely, permission to act does not establish that the actor can do what it is permitted to do. A claim of autonomy is clearer when it says which component it concerns. An engineering scale of automation can combine the two. Thomas Sheridan and William Verplank's ten-level scale for a single decision step orders, on a single line, how much of the selection the computer supplies (determining the options, suggesting one, selecting the action) and how much authority the human keeps (approving the action, being given time to stop it, being told what was done) (Sheridan and Verplank 1978, 8-15, 8-17–8-19). This book reads the lower steps of such a scale as varying directive dependence and the upper steps as varying governance constraint, which is one reason to keep the relations apart.

Their relevance to the comparison is not identical. Holding the attributed capacity and other conditions fixed, reducing directive or substitutive dependence provides relatively direct evidence of greater autonomy in that respect. A dependence presented as enabling requires a further question: does the external relation merely make the capacity possible, as relations and resources do for human researchers, or does it continue to supply part of the organization being attributed to the actor? In the second case the external element plays what Chapter 5, Section 6.1 called a constitutive role, partly realizing the task, and not only an enabling one; this is not the sense in which some relational accounts call autonomy itself constitutively relational. The question then becomes the boundary question of Section 2: the element is either included within the boundary to which the capacity is attributed or, left outside, reported as a continuing supply of part of that organization, that is, as directive or substitutive dependence, the second of the three relations.

Read in these terms, the cases above separate. In KEKADA (Section 2), the user's supplying of experimental outcomes was substitutive dependence in execution, and the user's choice among hypotheses and reactions in a few cases was directive dependence in selection. In the BLOOM run, taking the training process as the unit, the cluster's execution of the training computation partly realized that process and so falls within the unit, while the host's provision of the cluster was an enabling dependence; the people who watched for hung jobs supplied the training's continuation, a substitutive dependence; the rule that only the person who had launched a job could stop it was a governance constraint on the team itself, to which the team

responded with a file whose creation made training save and exit.

Designers of automated laboratories draw a similar line in their own terms. Koji Ochiai and coauthors call care the human-managed tasks that automated laboratories still require: in planning, translating experimental objectives into machine-readable workflows and scheduling equipment and reagents; in operation, preparing and cleaning up experiments, restocking supplies, monitoring the state of the system, and handling errors. They identify the lack of its automation as a major barrier to full laboratory automation and propose, as a design concept rather than a demonstrated system, that a laboratory recognize its own state and adapt to internal and external disturbances so as to keep itself able to function, an ability they call self-maintainability (Ochiai et al. 2025, 2285–2288). Read in the terms above, much of what they call care is substitutive dependence: people supply part of the planning and continuation of experiments attributed to the laboratory, and a self-maintaining laboratory would move that work inside the boundary to which the capacity is attributed. Their design keeps the other relations as well. In their implementation example, the system requests a person to replenish or remove items and calls someone if a sample is dropped; and they hold that logging decision points, together with interfaces for human intervention or confirmation, keeps the system visibly under human oversight (Ochiai et al. 2025, 2289, 2294). Whether such requests count against autonomy is a question for the counterfactual tests discussed next: a laboratory that recognizes a missing material and escalates appropriately depends on the person who supplies it in a way that enables its capacity rather than directing it. What separates the cases is not the restocking itself but who judges what is missing and decides whom to ask for what. Whether the laboratory recognizes the shortage or a person notices it, a person carries out the restocking; if what is attributed is the execution of restocking, that can be reported as substitutive dependence in execution, but if what is attributed is the capacity to organize the planning and continuation of experiments, the judgment and the request that belong to that organization lie with the laboratory. Which relation the same human work counts as depends on which capacity is being attributed. The interfaces for intervention and confirmation, by contrast, mark a governance constraint.

Observed behavior can underdetermine which relation is at work. A system may leave a human objective unchanged because it cannot represent it as revisable, because it is forbidden to change it, because it examined and retained it, or because the case never made revision relevant. Counterfactual tests help distinguish capacity, permission, and exercise. If a human or external component were removed, could the target actor preserve the inquiry, recognize the missing function, find a substitute, revise its plan, escalate appropriately, or, if it stops, stop for a reason that can be identified, whether in the evidence or in practical limits of time, resources, and risk? Human inquiries, too, can end before resolution, including for reasons of cost, and where one ends can be fixed by the practices that organize research (Chapter 1, Section 3; Chapter 6, Section 2.5). Withdrawal of guidance, conflict between instruction and evidence, failure of a familiar tool, and change in the environment are tests of this kind. Their answers do not generate a global score, but they can reveal that an allegedly internal capacity is continuously supplied from outside, or, conversely, that an external resource enables a capacity without directing or substituting for its exercise. They remain local diagnostics, not a metaphysics of freedom.

4.3 Autonomy over inherited elements of an inquiry

Section 4.2 applied the comparison to the conduct of an inquiry. This subsection applies it to what an inquiry inherits. That a question, standard, or purpose was supplied from outside does not prevent an activity from being research in the sense of the working characterization (Chapter 1, Section 7; Chapter 4, Sections 1 and 6). What external origin raises is a question of autonomy: what can the actor do with what it inherits?

The question has one general form. Any of the elements of an inquiry configuration distinguished in Chapter 5, Section 1.1, such as its questions, representations, methods, and standards, can be supplied from outside. External origin is a fact about where an element came from. It leaves open whether the actor selected the element among alternatives, adopted it, reformulated it, or decided how to pursue it, and more generally whether the actor can represent the element as revisable, examine its consequences, and retain it for reasons, revise it, refuse it, or replace it. Possessing more of these capacities can reduce the need for external direction at that point; actual dependence also turns on who exercises them in the inquiry. Whether the actor is permitted to exercise them, actually does so, or retains the inherited element after reflection are further questions. Boden distinguishes the extent to which a system's controlling mechanisms were self-generated rather than externally imposed from the extent to which they can be reflected upon and selectively modified, and treats them as distinct dimensions of autonomy (Boden 1993, 7); transposed to the elements of an inquiry configuration, Boden's distinction makes an element's external origin and the actor's capacity to revise it two such dimensions. No conclusion about value follows:

self-generation can increase autonomy while producing a trivial, harmful, or incoherent agenda, and an assigned element can be valuable while marking continuing dependence.

The same form applies to the purposes an inquiry serves. Purposes are not among the dimensions of the picture in Chapter 5, Section 1.1; Chapter 1 leaves purpose and agenda choice open and assesses importance, which on some accounts, such as Kitcher's, is relative to purposes, at the value and governance level, which asks what is worth pursuing and for whom (Chapter 1, Sections 1 and 8). No fixed order holds between a question and a purpose: which is the purpose and which the question depends on the inquiry, and one question can be pursued for several purposes. The literature discussed below speaks of overarching goals and the subgoals chosen under them; that hierarchy is its vocabulary, not a claim of this book.

Examples show the form at work. An ARIA call for proposals illustrates a problem supplied from outside alongside a comparison of autonomy in other respects: applicants specify the research problem their existing system will tackle, while the call evaluates how much of exploration, hypothesis and experiment construction, interpretation, and reporting occurs without human intervention (ARIA 2025, 4–5, 7). A system so evaluated may be autonomous in constructing experiments while its problem remains externally given. Generating many questions does not by itself move the location of selection either: one can imagine, for example, a system that generates millions of novel questions while a human-defined score decides which ones matter, so that the surface output has changed while the location of agenda selection may not have (compare Chapter 1, Section 8; Chapter 4, Sections 1 and 8). For purposes, suppose a person gives an AI the broad purpose of understanding a disease and the AI independently chooses a tractable question; the system may be autonomous in question selection while remaining dependent on whoever set the purpose. King, in a preprint survey of AI scientists, draws a corresponding distinction for purposes: "Scientific goal generation must be distinguished from normative self-authorisation," since a system may generate its own questions within a domain while remaining constrained by human-defined objectives, law, ethics, safety, and resources. King proposes "virtual research councils" that would allocate computation, laboratory time, samples, replication effort, and attention among competing research programmes, and that should do so "within human-authorised objectives rather than determine autonomously what humanity ought to value" (King 2026, 83–84). In this chapter's terms, this proposal places the work of generating and selecting questions in the system and keeps authority over the objectives with people, a governance constraint in the sense of Section 4.2 rather than a missing capacity.

Adding a generator for the element that gives direction does not by itself remove the dependence; it may only move it. If a goal generator selects the purpose, one can ask what selected its criteria of importance, and then what selected those. This regress has the form of the objection with which Di Paolo and Iizuka reject adding a goal-switching module as a homuncular solution (Chapter 4, Section 7). In computational work, an agent is sometimes called one that chooses its own goals merely because the goal-choosing mechanism is placed within the agent's boundary (Colas et al. 2022, 1169–1170). That it can be so called, however, does not mean that dependence has been reduced: whether a generator reduces dependence depends on what selected its criteria and on whether it belongs, more than nominally, to the organization attributed to the actor (Section 2). Since the comparison of this chapter concerns present organization, asking here what selected the criteria serves to identify external control that is still at work, not to trace origins to their end; whether history bears on autonomy in its own right is, as the next paragraph shows, disputed.

How such a regress bears on free will and moral responsibility is disputed, and the comparison of this chapter does not depend on the outcome. Martela treats the difference between "local" free will, choosing subgoals within an overarching goal, and "global" free will over the overarching goals themselves as a continuum rather than a dichotomy, since human goals too emerge largely from biologically and culturally given ones; humans and current generative agents differ on it in degree (Martela 2025, 4396–4397). Hakli and Mäkelä, answering the objection that humans are equally determined by their genes and upbringing, hold that what matters is not determination as such but its kind. They draw on Mele's history-sensitive account. On it, two agents alike in their present values can differ in autonomy because of how they came to have them. A value is not authentically one's own if it is compelled, that is, if one cannot shed it and the compulsion was not arranged by oneself, as with values instilled by manipulation or built into an agent at its creation. On their argument a robot's goals, including the objectives that guide its learning, are of this kind, since others engineered them and the robot cannot shed them; biological and cultural formation need not be (Hakli and Mäkelä 2016, 150–153). Their verdict thus turns on the premise that a robot cannot shed what was built into it, which they take to follow from its being programmed, and which the

arguments of Totschnig and of Tubert and Tiehen in Section 4.1 would contest for general artificial intelligence. Neither side requires ultimate self-origination of one's ends.

Standards, the third example after questions and purposes, raise the question in a form that Chapter 6 prepared. Chapter 6, Section 5 argued that standards, concepts, measurements, and methods can be revised together within inquiry; the question here is to which actor that revising capacity is attributable and what its external provision implies for autonomy. An artificial system can inherit a statistical framework, proof practice, experimental convention, measurement standard, or evidence threshold and use it competently, selecting a supplied procedure, executing it, reacting to its result, and revising a candidate answer. That can be substantial research and substantial autonomy in conduct. This book calls autonomy over the standards that organize epistemic activity **epistemic self-government**, leaving the sources and kinds of those standards' normative authority open.

A stronger capacity appears when a standard itself becomes an object of inquiry, as when a system or community compares standards, recognizes that a threshold is poorly matched to the stakes, introduces a new measurement relation, alters what counts as a relevant control, or criticizes an inherited way of organizing error. When such a claim is made, at least five capacities may need to be distinguished:

1. applying a supplied norm;
2. selecting among supplied norms;
3. constructing or adapting a norm for the problem;
4. revising it after mismatch, failure, or changed conditions;
5. giving reasons why it governs, for example by criticizing, defending, or negotiating it with others.

These five capacities regroup those on which, according to Chapter 6, Section 5.3, reasoned revision of a standard draws (detecting that a standard is inadequate for a purpose, proposing a replacement, and evaluating the consequences of the change) as constructing or adapting, revising, and giving reasons, and add applying and selecting, which an inquiry that inherits its norms already exercises. Their purpose is to keep these apart when epistemic self-government is attributed. The capacities are not limited to these five, and the fifth gathers acts of different kinds, of which negotiating a norm with others already involves relations to other parties around research, the relations Part V examines. Nor are the five a level scheme: a system or community can exhibit one capacity without the others, and the distinctions are drawn only where they change an inference. A system may construct a new metric without being able to defend its epistemic relevance; a community may criticize a standard while depending on others to implement the alternative; a supplied proof checker may support reliable mathematical inquiry even when no artificial component can revise the underlying proof theory. The record from the language-model project discussed in Chapter 6, Section 5.3 shows such capacities divided among parties: a maintainer proposed logging a corrected measure under a new name beside the old one, and an AI agent run by one contributor prepared the patch, so that the record can be read as showing the revision and its reasons coming from a person and its implementation from the agent.

Not all the considerations that govern standards are epistemic. Standards of inquiry may incorporate time, resource scarcity, welfare, privacy, danger, and other non-epistemic values, and whether the reasons for selecting and pursuing questions are distinctively epistemic is disputed (Chapter 2, Section 5). The dispute bears on what the fourth and fifth capacities, revising a norm and giving reasons why it governs, require. Thorstad holds that there are no distinctively epistemic norms of inquiry. Inquiry is an activity, and whether and how to pursue one is settled by everything that bears on action, including the importance of the question and the cost of answering it. The fact that an inquiry would promote truth or knowledge remains one reason among these (Thorstad 2022, §§2, 3, 6.3, 7). If Thorstad is right, it would follow that revising a stopping rule or the threshold of evidence at which an inquiry stops, or giving reasons why it governs, requires access to the stakes and costs of an inquiry as well as to its evidence. Likewise, Kauppinen holds that a reason to pursue a question is transmitted from a reason for an action that answering the question would help one perform, and does not change its kind in transmission (Kauppinen 2026, §§2, 4). If so, revising such a norm requires access to the purposes the inquiry serves, and a failure there calls for the prudential or moral criticism those purposes make fitting rather than for epistemic distrust (Kauppinen 2026, §§3–4). Unsupported assumptions about the reliability of sources on which the pursuit relies remain, on this view, open to

epistemic criticism (Chapter 2, Section 5). A standard and a purpose are here again not ranked; each can bear on the other.

Some programs aim at this capacity directly. Krenn and Champion's program for a philosophy of autonomous science proposes translating epistemic aims such as understanding, curiosity, surprise, interest, creativity, and novelty into computable, nonanthropocentric objectives, and asks whether even "the rules of doing science" might be approximated as policies for generating hypotheses, overthrowing theories, or designing experiments, evaluated by the quality of their discoveries in simulated worlds and selected by objective functions (Krenn and Champion 2026, 335, 345). They pose this as one of a set of open questions in their research program, including how to keep such policies from gaming the metric (345). Other programs withhold such capacities by design. Bengio and colleagues, in a position paper on safety, propose a Scientist AI, consisting of a world model that generates explanatory theories and an inference machine that estimates the probabilities of answers to the questions put to it, with no persistent goals of its own. They treat goal-directedness, which "requires an (arbitrary) choice of what to value," as one of three pillars of agency, beside affordances and intelligence, and hold that eliminating any one of them "would be sufficient to mitigate most categories of loss-of-control risk"; in the research cycle they describe, the system samples informative experiments, and "Humans can then perform or simulate the sampled experiments" (Bengio et al. 2025, 25, 27–28, 42). In the terms of Section 4.2, the choice of questions and purposes and the execution of experiments are then supplied from outside, and the dependence in those respects is chosen for safety rather than imposed by what present systems cannot do.

On the working characterization, research directed toward epistemic achievements new to a human society does not require the artificial actor to originate its standards: inherited human standards can serve, and the actor need not invent an epistemology to conduct research for a human society. For research by and for an artificial society, by contrast, permanent human control over what counts as evidence, error, acceptable risk, or successful inquiry would mark a strong form of external government of its epistemic standards. This concerns where autonomy lies, not whether the society is a knowing community; applied to artificial organizations, Longino's account, for one, which locates community knowledge in the workings of a community with public standards open to criticism, implies that the external origin or government of standards does not by itself keep such an organization from being a knowing community (Chapter 8, Section 1.3). That section takes up research of that kind and separates the conditions of communal knowledge from the further question whether such a society governs its own standards.

For every inherited element, external origin identifies a respect in which organization entered from outside and may also reveal continuing dependence. Revisability can increase autonomy at that point, but neither revisability nor self-generation establishes epistemic correctness or legitimate authority.

4.4 What leaving an inquiry open demands

Sections 4.2 and 4.3 compared autonomy one respect at a time. This subsection asks what a claim of greater autonomy over the elements of an inquiry commits a project to in other respects.

On the argument of this subsection, the more of an inquiry is left open to a configuration (which purposes it serves, which questions it pursues, which representations, data, methods, or apparatus it uses), the wider the range of practical demands the inquiry can place on it. Some demands lie within experiment and calculation. Others lie outside them: gaining access to materials, data, sites, or people; obtaining permissions and resources; coordinating with other agents; and recovering when the world resists a plan. Part of the novelty to be met is produced by the configuration's own earlier choices, as when a method it has chosen calls for a measurement, a dataset, or a piece of infrastructure that no one had prepared. A project seeking to understand a brain, for example, may come to need a maze in which rats or mice learn their way, so that learning and memory can be studied, electrophysiological recording of the electrical activity of neurons, imaging that renders the brain's structure or activity as pictures, observation of behavior, simulation that reproduces the brain's workings in computational models, clinical records of patients with neurological disorders and the like, or an apparatus not yet designed, and which of these it needs may become clear only as its questions change.

A general form of this relation is familiar for fixed tasks. Sheridan and Verplank, writing on computer control of undersea teleoperators, held that preprogrammed routines pay where elements of a task can be anticipated, while completely unanticipated events call for human ingenuity and judgment; a supervisory system can be more precise than direct manual control when contingencies turn out as anticipated or were allowed for in its program, but it occasionally makes spectacular

errors, through its own failure or through errors in programming it (Sheridan and Verplank 1978, 8-3–8-4, 8-9). There the task and its goal are given, and the novelty to be met comes from the environment. When purposes, questions, or methods are left to the configuration, part of the novelty comes from its own choices, and the demands extend beyond the task into access, permission, and coordination.

What a chosen line of inquiry requires and the configuration cannot bring about, others bring about, or the inquiry stops or changes course. Dependence lowered in one respect can therefore reappear in another, as directive or substitutive dependence, when someone else arranges the access, runs the instrument, or supplies the result. A claim of greater autonomy over purposes, questions, or methods is in this sense also a claim about the demands their opening can bring: about which of them the configuration itself can meet. Dung's account describes the agency of artificial systems along several dimensions. Two of them are autonomy, the degree to which a system can function on its own without intentional intervention from outside, and efficacy, the degree to which its goal-directed behavior affects the world without someone else's mediation or intervention; Dung treats the two as conceptually independent. A videogame AI, for example, keeps acting in its virtual world without being prompted, but because it affects only a virtual environment its efficacy is low (Dung 2025, §§2.2.2–2.3). The relation proposed here does not deny that independence. It does not say that autonomy over purposes, questions, or methods, which is not what Dung calls autonomy, conceptually includes a capacity to act on the world. It says that a claim that an inquiry has been left open to a configuration commits the project to showing how each demand the opening brings, such as access, permission, and coordination, was handled: whether it was met within the unit, met by others, closed in advance, or answered by the inquiry changing course or stopping for a stated reason. Where others met it, dependence has returned in another form. The autonomy claim narrows to the extent that others supplied part of the organization attributed to the configuration, such as judging what is needed, arranging how it is obtained, or evaluating the result. Where the configuration identifies what it needs and asks for it, and others only provide materials or carry out specified work in response, the dependence is, for an attribution of organization, closer to enabling dependence in the sense of Section 4.2 (if the execution itself is attributed, that part is reported as substitutive dependence in execution), and it can even be evidence that the configuration organized the demand. What lies within others' authority, such as consent, is counted separately as a governance constraint, as the next paragraph says. Takahashi's two statements about the levels lie on either side of the relation between leaving an inquiry open and the demands that brings. The preprint discussed in Section 4.1 holds that autonomy is independent of performance. The later book states that the spontaneity that level 6 requires, which in AI based on active inference is taken to arise from the system's active engagement with its environment, calls for embodiment in a more essential sense than any other level, and that a scientific AI that has a body, sets its own purposes, and makes value judgments has a makeup of capacities that coincides with the definition of AGI, so that raising the autonomy of scientific AI to its limit is nearly synonymous with realizing AGI (Takahashi 2026, 171). Read in this section's terms, the two statements need not conflict: the first concerns how well a system at a level reasons, the second the kinds of capacity that leaving purposes open to a system demands of it. Takahashi does not relate them to each other.

The connection argued here between greater autonomy over purposes, questions, and methods and the capacity to meet new demands concerns the first of the two components distinguished in Section 4.2, whose reduction depends on what the actor can do. A permission to make such choices is not a capacity to meet what they demand, and holding a permission is not evidence of a capacity to obtain one; obtaining a permission, by applying, giving reasons, and meeting conditions, is itself among the demands. Meeting a demand can also change the kind of dependence rather than remove it: a configuration able to negotiate access to a clinic still needs the consent of those who grant it, a governance constraint in the terms of Section 4.2 that its competence does not remove. A constraint, however, is not only an outer limit on autonomy. Reworking questions and methods within what consent and the conditions of a site allow, or devising another method in response to a constraint, is itself work that organizes the inquiry, and it can be assessed on the side of the component that varies with capacity. In the hospital case below, the team's turn to edge computing in response to the nurses' concerns can be read as one instance. The claim is not that constraints improve research, but that responding to them can be an exercise of research capacity.

Several things do not follow. The argument is not a claim about what is feasible. It does not make greater capacity amount to greater autonomy; the two remain distinct comparisons, which this argument connects for claims of greater autonomy over the elements of an open inquiry. A narrow claim does not incur this burden. Preparing an environment so that the range of demands is closed in advance, by a fixed computational environment, a template, a curated dataset, or approved apparatus, is a legitimate design. What follows is only that such closure is among what a claim of autonomy discloses (Section 5).

Fei-Fei Li's later account of taking the computer-vision research of Li's group into hospitals, for example, shows how wide the range can become once a question changes, and that in this case the demands were met by people. A medical collaborator secured hospital access and arranged for Li to shadow a pediatrician through a shift, after which no recruit was to write any code before shadowing clinicians in the same way; assembling a team took nearly two years; and since patients and clinicians were rarely recorded, the team had to collect its own data under institutional review. When nurses rejected the installation of prototype sensors, concerned less with the study than with where such technology might lead, the team turned to edge computing, a field none of its members understood as well as they would have liked, so that nothing would be transmitted from the devices to remote data centers. The team also grew to include practicing clinicians, a bioethicist, and a JD–PhD from Stanford Law School, and its technology partners came to include experts in sensors, cybersecurity, and edge computing (Li 2023, ch. 10; for the resulting system, which used depth sensors instead of color cameras to protect privacy and collected data in two hospitals, Haque et al. 2017, 3–4). Some of these demands, such as assembling a team and its members' acquiring unfamiliar skills, took the time and effort they did because it was a human team, although an artificial configuration lacking a needed competence would also have to acquire it or bring in others who have it; others, such as data that did not yet exist, permission to record patients and clinicians, and the acceptance of clinicians and nurses, would face any inquirer taking up the question. Automating an experiment after its apparatus has been prepared is a substantial achievement. It is not evidence that the capacity to meet these surrounding demands has also become artificial.

On the working characterization of Chapter 1, work of this kind, such as arranging access, obtaining permissions, and coordinating with others, belongs to the research activity when it is undertaken in the attempt to resolve the inquiry (compare the open drug-discovery project of Chapter 1, Section 3, where the date for shipping compounds and who was available to run the screening were part of the attempt). Where people outside the configuration do such work, part of the research activity is therefore carried out outside it. Such demands also arise in formal research. In the Liquid Tensor Experiment (Chapter 5, Section 3), formalizing the proof involved building infrastructure and replacing an abstract route by a more explicit construction; these can be understood as demands that arose from the route the formalization took.

The point recurs in this book in other forms. Chapter 6, Section 3.3 made it for checking: the more open the question, object, method, and environment, the more a claim of research capacity attributes to the system. Chapter 3, Section 9 lists capacities that long-run autonomous research would require, and Chapter 8, Section 3.2 argues that a wider scope of self-direction permits a wider scope of epistemic mismatch.

4.5 What autonomy does not establish

Sections 4.1 to 4.4 said how autonomy can be described and what a claim of it commits a project to. This subsection states what such a description does not establish.

Autonomy is not a monotonic measure of research quality. A more autonomous actor can ask trivial questions, use weak methods, consume resources irresponsibly, produce false conclusions, or create severe risks. Capacity, observed exercise, permission, authority, epistemic quality, importance, safety, responsibility, and legitimacy are not inferred from one another here without additional argument. Having the configuration itself perform an appraisal or correction that another party supplied, for example, reduces directive dependence in that respect and so, other conditions held fixed, counts as evidence of greater autonomy. But it does not show that the configuration's checks continue to add information about its errors. Where absorbing an appraisal couples it to the producer's assumptions or allows adverse results to be dismissed without support, autonomy in that respect can rise while the grounds for certifying the result for S weaken (Chapter 11, Sections 1, 2, and 8; Section 2 of that chapter describes cases in which automated checks lacked an error route of their own). Two changes can be distinguished here. One is counting an evaluator who had been outside as inside the boundary without changing the organization; the description of autonomy may then change, but the routes by which errors are found do not, since the choice of a boundary does not by itself make a claim true (Section 2). The other is actually coupling the evaluator's data, aims, and paths of judgment to the side that produces the result; then the independence of the checks itself can change. A change in autonomy that comes from redrawing the boundary can therefore be assessed apart from a change in capacity or in the grounds for certification.

Nor does responsiveness to outside criticism count against autonomy as such. Where an outside party supplies the appraisal of a result, that is directive or substitutive dependence. Where it raises an objection that the configuration itself assesses and

answers, the configuration has done the appraising; Chapter 11 examines why such routes of challenge matter for certification.

These separations constrain what an autonomy claim means.

From attribution to construction and evaluation

Sections 1 to 4 distinguished what can be attributed to an artificial configuration and how its autonomy can be compared. This section asks what those distinctions demand of construction and evaluation. They do not yield one universal configuration of an artificial researcher or certification checklist. They yield a claim-relative mapping from what a project says an artificial system does to what would need to be constructed, disclosed, and shown for that attribution to be warranted.

Two burdens can be distinguished. A **construction burden** asks where and how the organization attributed in a claim, whether a contribution, activity, capacity, or more extensive epistemic organization, is realized. An **evaluation burden** asks what evidence would show that the attributed organization is present and effective rather than merely named in a system diagram, as with the question generation box of Section 2, or supplied by an unreported external actor. A system may also realize the relevant function through an organization unlike the familiar human division of functions.

Different claims consequently call for different showings. Table 7.1 sets out six kinds of claim distinguished in Sections 1 to 4. For each, the middle column states what a claim of that kind may need to construct, disclose, or show, and the right column states what would not yet follow even if that were shown. The rows are different attributions, not stages through which every project must pass. Each row summarizes in general terms the discussion in those sections, and what a particular claim needs depends on the boundary and range it states.

Table 7.1. Construction and evaluation burdens for artificial-researcher claims: an outline by kind of claim

Claim	What a claim of this kind may need to construct, disclose, or show	What does not yet follow
An artificial component contributed to research	Identify the output or operation and its role in changing or testing an epistemic possibility	That the component conducted the research activity
A research activity was conducted at a stated artificial or hybrid boundary	Locate enough of the activity's direction, selection, execution, monitoring, and revision within that boundary	A stable or general research capacity
A research capacity can be attributed at that boundary	Demonstrate non-accidental performance across a stated range, for a stated version or period, including relevant variation, resistance, failure, and recovery	Researcher status for every purpose or autonomy in every respect
Researcher is an illuminating status	On the explanatory-value usage of Section 3.2, state why the breadth, integration, persistence, and explanatory value of the capacity warrant the thicker label in this context, and for whom, in particular what the description predicts about behavior in situations that tests of single capacities do not anticipate; on positions that set other conditions, such as competence with reflective insight or responsibility and credit, show those conditions as well; on positions that presuppose minimal agency, show also that the unit is an individual the system defines for itself rather than one selected relative to a claim	One universal occupational, moral, legal, or metaphysical threshold; credit or praise for a success
The research was autonomous in a specified respect	Identify the selected unit, external party, function, time span, comparison, and the direction, substitution, authorization, ratification, or repair that was or was not required; where autonomy over purposes, questions, or methods is claimed, the practical demands their opening brought and whether each was met within the unit, met by others (and if so, whether they supplied the judgment and arrangements or provided resources or work at the configuration's request), closed in advance by preparing the environment, or answered by a reasoned change of course or stop	Epistemic quality, importance, safety, responsibility, or legitimacy; that a permission to choose shows the capacity to meet what the choice demands; or that, once generation and appraisal are organized inside one configuration, its checks still add information about its errors (Section 4.5; Chapter 11)
Epistemic self-government is exercised at the selected boundary	Show whether the configuration applies, selects, constructs or adapts, revises, and gives reasons for relevant norms, to the extent claimed; on accounts such as Thorstad's or Kauppinen's, on which the norms of inquiry answer to reasons for action, show also, for revising and giving reasons, access to the stakes and costs of the inquiry and to the purposes it serves	That the resulting norms are true, reliable, valuable, or legitimate

Table 7.1 keeps the unit, scope, version, dependencies, and evidence together. Because its rows are different attributions, a component that classifies images in a human-designed investigation need not demonstrate agenda formation; a system claimed to conduct an open program needs evidence about the demands left open to it. The boundary matters in both cases: changing it can change where the conduct resides and which external direction or maintenance the capacity requires. Chapter 13 brings these questions together with achievement and research relations.

Authority to permit, stop, inspect, contest, ratify, or withdraw is among what Table 7.1 asks a claim to disclose, where it changes the autonomy or actor attribution. The normative and institutional allocation of power, oversight, responsibility, and correction belongs to Part V's analysis of the relations that surround research.

One inference that does not hold belongs to this chapter in particular: autonomy in a specified respect does not establish epistemic self-government over the standards of inquiry (Section 4.3). A second, which Chapter 8, Section 1.3 argues, is that a set of interacting artificial agents does not by that fact establish an artificial knowing society. Table 13.1 collects these distinctions for evaluation. The reverse inferences also need care: dependence does not erase a contribution, nor failure a research activity. Keeping both directions in view, an evaluation lets neither an impressive output nor a visible dependence settle the attribution, and, as Section 1 noted, the labels under which systems are presented leave this work to be done.

The boundary of a research actor is an explanatory choice with consequences for construction. Locating question formation, diagnosis, or recovery in a larger arrangement changes which capacity is being built; locating it within the artifact can

change what surrounding support it needs. The hospital case in Section 4.4 makes the consequence concrete: opening the research question also opened demands for access, materials, skills, and cooperation. Autonomy in that inquiry therefore concerns how such demands are met, as well as who selects the next operation. This is why the chapter's distinctions can guide the construction of a research actor without prescribing one configuration or one threshold called autonomous scientist.

The chapter's comparison also brings into view that raising autonomy need not reduce outside dependence in one direction. When what Sheridan and Verplank said about unanticipated events in fixed tasks is extended to the choice of questions and methods, a configuration faces new demands through its own choices, and some of them can be met only through new relations with others (Section 4.4). Where such others come to supply continuing direction or judgment, the unit that organizes the inquiry may itself widen (Section 2). Asking for help can be an exercise of the capacity to identify what is missing and request it (Section 4.2). And absorbing appraisal into the configuration can raise autonomy in that respect while weakening the independent routes by which errors are found (Section 4.5). A more autonomous configuration can therefore turn out to be one that negotiates with more others and, for certifying its results, needs outside checks all the more.

The attributions and comparisons of this chapter do not yet answer a different question. Even if an artificial actor conducts research, it does not follow that the actor itself knows the result, that the result has entered the epistemic situation of the society that commissioned the work or of the society for which it was sought, or that the inquiry was directed toward the unknowns of either. The next chapter therefore changes the index of analysis: from *who conducted the research?* to *whose possible knowledge was produced, and for which reference society?*

Chapter 8 — Whose Knowledge?

Human, Mixed, and Artificial Epistemic Societies

Chapter 7 ended with three things that do not follow when the conduct of research is attributed to an artificial configuration: that the configuration itself knows the result, that the result has entered the epistemic situation of the society that commissioned the work or of the society for which it was sought, and that the inquiry was directed toward the unknowns of either. Chapter 6, Section 6 had passed on the question behind them: when the actor conducting an inquiry, the bearer of knowledge, and the reference society *S* differ, whose epistemic change is being pursued, for whom is the result new, and who can evaluate or incorporate it? This chapter takes up that question through the three matters.

Chapter 7 located the attempt and the actor who conducts it. This chapter distinguishes that actor from the bearer in which the intended achievement would be realized and from the society relative to which it would be new. Knowledge remains the principal case: an artificial actor can conduct inquiry whose result becomes human knowledge; a hybrid organization, one made up of human and artificial components, can be both actor and bearer; an artificial community can pursue questions indexed to its own ignorance. Understanding raises parallel possibilities and questions, examined in Chapter 3. An actor can enable others to understand without itself understanding, and whether members' understanding by itself establishes collective understanding depends on the account of group understanding (Chapter 3, Section 6). The knowledge analyses below are not thereby theories of every achievement.

Keeping these possibilities apart raises a question: whether research that humans initiated will continue to produce knowledge for a human community when it is conducted by systems unlike humans. Research for a human community may be the intended and practically central case. But an artificial inquirer can be an intelligent system unlike human ones in its body, its cognition, and how it is made, among other things, and what it takes to be unknown, worth pursuing, or a good reason can then come apart from what the members of the society concerned take them to be. Greater autonomy widens the room for this, though it is only one of the channels through which it can happen. Neither the working characterization nor the fact that humans initiated the project guarantees that the research stays directed at that society's unknowns; Section 3 examines why, and what follows.

Section 1 distinguishes the actor from the reference society, and both from the knowledge bearer, the subject asked about when one asks whether the research has succeeded. It divides the cases by the reference society, that is, by whether *S* is a human, a mixed, or an artificial society, since the question is whose knowledge would be produced; it bears most directly on the second of the three matters with which this chapter opened, whether the result has entered the epistemic situation of the society concerned. Section 2 asks what it would take for the world to be available to an inquirer at all, that is, what a check presupposes, how states come to be about their targets, and how commitments, the claims an inquirer takes as true and as premises for inference and action, remain open to correction by those targets. These are questions that the first of the three matters, a claim that the configuration itself knows the result, would need to answer.

In this chapter, an epistemic organization is how an inquirer or a community is organized to treat something as an observation, as evidence, or as a correction, and also as unknown, as worth pursuing, or as a good reason, among other things; by extension, the phrase also names an inquirer or a community so organized. In neither sense does it name an institution. Section 3 asks what follows when an inquirer of another kind differs from *S*'s members, in its epistemic organization and in such respects as its body, its cognition, and how it is made. The difference can show up as a mismatch in such things as what is taken to be unknown, worth pursuing, or a good reason. It then examines where such mismatch comes from (including the case in which a successful inquiry revises the concepts in which a commission was written), how it resembles what is known as the AI alignment problem, and what happens, and what *S* can still gain, when research by such an inquirer does not align with *S*. The third matter, whether the inquiry pursues the unknowns of the society concerned, falls within these questions and is not settled in advance.

Section 4 draws two construction horizons from the three cases and states what Part IV leaves to Part V. A construction horizon, as the phrase is used here, is the set of questions that a program for building artificial researchers must answer if it aims at research for a given kind of *S*. Each section uses the distinctions of the one before it: Section 2 uses the three cases of

Section 1 to say who would answer its questions, and Section 3 takes up the possibility, reached at the end of Section 2, that an artificial inquirer could be organized differently from human ones while remaining constrained by the world.

Actor, knowledge bearer, and reference society

Chapter 7 located the conduct of research in an actor. This section asks what needs to be distinguished before one can ask whose knowledge that conduct produces, and how the answer changes with the society for which the knowledge would be new.

Chapter 1, Sections 2 and 6 introduced the two parties to a research claim, which serve here as coordinates. *A* denotes the individual or organized system to which research conduct or capacity is attributed: a person, a model-and-tool system, an artificial collective, or a heterogeneous human–machine organization. *S* fixes the reference for novelty and the society for which the achievement is sought (Chapter 2, Section 6). In the knowledge case, whether the resulting knowledge belongs to members, a group, or a society in some further sense requires a separate account. Society is used loosely and does not commit the book to one theory of collective belief or institutional acceptance, but *S* is not an arbitrary set assembled by the analyst; how much epistemic connection among its members, subgroups, records, or practices is required for claims of novelty and achievement relative to *S* to be intelligible is left open (Chapter 1, Section 6).

The notation fixes a social reference for novelty, not a single social subject that must possess every achievement. Knowledge of members, group knowledge, and socially available knowledge are distinguished by competing accounts: belief, authorized acceptance, reliable availability, collective deliberation, and group-level states have different roles (Chapter 2, Section 6 and Table 2.2). These are accounts of knowledge. For understanding, Chapter 3 distinguishes a member's grasp, organized collective capacities, and resources through which understanding can become available. Availability can matter to a novelty comparison; whether it is itself achieved understanding depends on the account (Chapter 3, Section 6). Assessing the novelty claim requires identifying its achievement and location and showing that its comparison with the prior situation in *S* is apt.

The bearer, the subject whose state or organization directly realizes the relevant knowledge, is neither a further coordinate nor one of the additional roles named below; it is asked about where success is at issue (Chapter 1, Section 2). It may be *S* as a group, some of its members, a specialist subgroup, or a wider sociotechnical organization; it may also be the actor *A* that conducted the research, or differ from it. Identifying that bearer locally does not replace *S*. Where the claim is that *S* knows, it makes explicit what is supposed to warrant that attribution. A claim about a bearer outside *S* instead needs an account of how its achievement is new relative to, and sought for, *S*; it need not attribute knowledge to *S* itself (Chapter 1, Section 6). If no defended account of possession supports the attribution of knowledge to *S*, whether through its members, its organization, or *S* as a group, the activity may still be research for *S*, and an achievement new to *S* may already be realized in a bearer outside it; what has not yet happened is that the achievement has entered *S*'s epistemic situation (Chapter 1, Section 2). Additional roles, such as evaluators, beneficiaries, supporters, audiences, and authorities, may also differ from *A* and *S*. Identifying them where an inference depends on them prevents their silent collapse into one.

The cases below are distinguished by *S*: a human society or community, a mixed society with human and artificial members, or an artificial society. This chapter uses mixed only for a society *S* and keeps hybrid for organizations of human and artificial components that act or bear knowledge; a hybrid organization can appear in any of the three cases. The division serves to compare claims that results of inquiry have entered the epistemic situations of these societies, under the accounts of possession in Chapter 2, accounts of who comes to have a result as knowledge. Within that scope, it shows how the burden changes when knowledge is attributed to human members, to human and artificial members or their organization, or to an artificial society. Sections 2 and 4 also use it: Section 2 to say who would have to meet its conditions, and Section 4 to draw its horizons. The division does not classify every possible bearer from *S* alone: an artificial bearer can be relevant to research for a human society, and membership does not by itself establish knowledge. It is exhaustive only at this grain; it does not distinguish kinds of human or artificial society. The actor *A* can vary within each case, and can include a hybrid organization or people conducting research for an artificial society. Chapter 3, Section 9 locates related questions about understanding without assuming that every knowledge attribution has an understanding counterpart.

1.1 Research for a human society

When *S* is human, what would show that a result has become *S*'s knowledge, and what roles can an artificial actor play? Here *A* may be artificial, substantially artificial, or a hybrid organization, while *S* is a human society, discipline, or specialist community. An artificial *A* may refine a question, generate candidates, operate instruments, assess evidence, and draft a result without itself literally believing or knowing. Human researchers or institutions may evaluate and incorporate its contribution. On individualist accounts, human members remain the direct epistemic bearers; stronger group or institutional accounts may attribute knowledge to the organized human system itself (Chapter 2, Section 6).

This possibility is familiar in the philosophy of technology and in social epistemology. In a paper cited here in its preprint version, Alvarado argues that AI is first and foremost an epistemic technology: it is deployed in epistemic contexts such as inquiry and operates on epistemic content, such as propositions and models, through operations such as analysis, prediction, and inference. Alvarado calls these operations epistemic rather than cognitive, and classifies the technology by what it does, since, on this account, an artifact is what it does or is designed to do (Alvarado 2023, §§2–3). The claim is not framed in terms of belief, but a classification by function of this kind does not require the system to be a believer or a knower, and so leaves room for the possibility just described: a system can take part in the epistemic work of an inquiry without being either.

Hauswald asks whether AI systems can be epistemic authorities, that is, sources significantly superior to a person in some domain, from whom that person can receive knowledge, understanding, and other epistemic goods (Hauswald 2025, 716, 722 n. 1). On what Hauswald calls the standard picture, a person learns from an authority through testimony, the intentional transfer of a belief from one agent to another, usually by assertion. The picture requires the authority to have beliefs and communicative intentions, which current AI systems most likely lack, and this counts against granting them the status (Hauswald 2025, 718). Against this Hauswald sets structural parallels. Users can learn from such systems; the relation between system and user is epistemically imbalanced, as the relation between authority and layperson is; the systems are opaque to their users, as experts' reasoning often is to laypeople; and users therefore face three problems that laypeople face with experts. One is telling which of two conflicting sources to trust, which Hauswald adapts from the problem Goldman posed for novices choosing between experts. A second is how far to defer, that is, how to revise one's own beliefs in response to the authority's. Much of the discussion of this problem concerns preemption, the thesis associated with Zagzebski that an authority's belief gives a reason that replaces, rather than adds to, one's other reasons. A third is how to obtain the reasons for an answer as well as the answer (Hauswald 2025, 718–720).

Hauswald considers three ways out of this dilemma between the standard picture and the parallels: denying that there are artificial epistemic authorities, which plays down the parallels; attributing beliefs and intentions to the systems, which risks treating them as humanlike; and revising the standard picture, which gives up common assumptions about authority. Hauswald favours the third. On the revised picture, to defer to an authority is to treat suitable facts about it as indicators of truth strong enough to lead one to adopt the attitude they indicate; facts are suitable when they bear a reliable relation to the truth of the propositions concerned, and access to them need not be testimonial. The outputs of an AI system can be such facts (Hauswald 2025, 720–722). On the revised picture, then, a system can serve as an epistemic authority for a person without having beliefs of its own; in research for a human *S*, this is one way in which a result can come to be possessed by members of *S* without the system's being a knower. Hauswald argues that the revision is not ad hoc, since it also covers a layperson who defers to a doctor not by accepting the doctor's testimony but by observing what the doctor does for the doctor's own child, and deference to a scientific consensus, which Hauswald doubts can be analysed as a group belief (Hauswald 2025, 720–721). Hauswald suspects, however, that preemption applied to the outputs of AI systems faces objections similar to those raised against preemption in general (Hauswald 2025, 723 n. 18).

Alvarado's and Hauswald's positions do not settle artificial belief. They show that an artifact can occupy an epistemically significant social role under theories that reserve literal knowledge for other bearers. What reliance on an artifact in such a role requires is taken up in Chapter 3, Section 5.3 and Chapter 11, Section 5; the first of these also takes up Alvarado's further argument that, if AI is essentially opaque, the epistemic trust that an epistemic technology calls for becomes significantly harder to justify.

Where the actor is a hybrid organization, some capacities may be attributable only to the organized whole. Artificial components may search spaces and preserve records; people may frame purposes and interpret significance; instruments

may constrain claims through material interaction; institutions may organize criticism and correction. The whole can conduct research even if no component contains the complete question, evidence, method, and conclusion, which is the question, treated in Chapter 7, Section 2, of the boundary within which the conduct of research is attributed. Attribution to the whole does not erase differences among its parts. A machine can be a reliable source without understanding. A human can endorse a result without reproducing the machine's internal route. A laboratory can preserve a capacity that no current member possesses alone. Whether such an organization, with its records and retrieval, itself realizes a knower is a further question. Chapter 2, Section 6 recorded a dispute over this very case: on distributed or group-functional accounts (Table 2.2) an appropriately organized collective can itself realize a knower (for example Bird 2024), while on others distributed cognition does not amount to distributed knowing. The dispute over when reliance on an artificial source is justified (Chapter 11, Section 5) does not decide it.

The same distinction only partially resolves the case, in Chapter 2, Section 7, of a discovery that an artificial system has produced but that no one has yet read. The system may have produced an epistemically valuable artifact without the human society already knowing its content; whether the society already knows it depends on which of the possession accounts distinguished there is adopted. The engineering project need not settle that dispute in order to track production, availability, incorporation, and use separately. In this case, then, for the claim that a result has entered *S*'s epistemic situation, what would need to be shown is how it comes to be possessed by human members or by their organization; that claim does not call for the artificial actor to be a knower in its own right, although other relations can, such as taking the system's reports as assertions on an account on which asserting requires that the speaker, or the group for which it speaks, know (Chapter 2, Section 6; Chapter 11, Section 5.1). Where the artificial actor is itself claimed to know the result, on an account of artificial belief that allows this, an achievement new to *S* may already be realized in the actor before it has entered *S*'s epistemic situation (Chapter 2, Section 7), and that claim, unlike the claim of entry, carries the burden that Section 2 describes for an artificial knower.

1.2 Research for a mixed society

What changes when *S* has artificial as well as human members? This chapter calls such a society a mixed society (introduction to Section 1). An enterprise that aims at knowledge new to and attributable to it falls between the other two cases. To the extent that artificial members are to count among those to whom its knowledge is attributable, it raises for them the questions of Section 1.3 about artificial belief and social knowledge. What *S* comes to know may also be meant to be shared across its human and artificial members; to the extent that these form differently organized communities, which may not share the same primitive representations (representations built into perception and learning rather than fixed by explicit convention; Section 2.1) or standards, the enterprise also raises the questions of translation and coordination of Section 3.4. On the analysis of this section, what is new to such a society, and who within it can evaluate whose results, need not be the same for its human and artificial members. The question who may speak for *S*, which Section 3.3 takes up and to which Chapter 10, Section 6 returns for a society some of whose members are artificial, arises here as well.

Most of what this case would need to show combines what the other two cases require with the coordination of Section 3.4. Coordination in that sense is a term reserved for relations between differently organized communities, and where the two kinds of members form such separate communities, the question reduces to it.

If instead one applies Tadahiro Taniguchi's theory of symbol emergence systems (Section 2.2), on which symbol systems become organized from the bottom up through exchanges among members and the organized system constrains them, and considers the case in which both kinds join a single symbol emergence system, a question may remain that the combination does not cover. The vocabulary and standards that the two kinds of members share would then not be handed over from one repertoire to another but formed and changed within one loop that includes members whose bodies and senses may differ. If so, what is in question is not only how possession is distributed between them but whose experiences, and of what kind, the shared vocabulary and standards integrate. Whose experience enters a symbol system can be asked in a human society too (Chapter 10, Section 6). What differs in a mixed society is that the difference crosses the line between human and artificial, and so connects with the question of Section 1.3 whether artificial members have experiences to be integrated at all. This question adds to those of the two horizons within the mixed case.

Where a human society ends and a mixed one begins may itself become unclear in practice. Collins, in a passage on large language models and society rather than on research (the account of research science in the same paper is taken up in Section 1.3), distinguishes two problems: that of socializing the machines, aligning them with us, and, if that is not solved, what Collins calls the dissolution of societies, in which the machines socialize us, aligning our values unnoticed with theirs, which on this account are either no values or the values of their hidden controllers. At present, Collins adds, those who decide what the machines' socialization comprises appear to be the founders of AI companies (Collins 2025, 1259). Collins writes about values, not about research or vocabularies. Read together with Taniguchi's observation that generative AI is already changing human language use (Section 2.2), however, Collins's second problem suggests that a society counted as human can come to share vocabulary and standards shaped in loops with machines without any decision admitting artificial members, so that the question whose experiences that vocabulary and those standards integrate can arise before the question of membership is settled.

Collins's point concerns a society that comes to share vocabulary and standards without deciding on membership. Where a society does decide to admit new members, what the admission does to what it can claim has been examined for observability. Van Fraassen, for whom observable means observable by us, the epistemic community (Chapter 1, Section 6), considers a future in which dolphins are admitted as bona fide members of our epistemic community while we still accept a theory on which they reliably detect things we now classify as unobservable. To an argument of William Seager's, that we must therefore already believe that those things exist, van Fraassen replies that what we would truthfully say after the admission, that some of us observe them, does not license our saying now that they will be observable, since us would then refer to a wider community. This reply concedes that a harder question remains: how we are to regard at present the opinions we foresee coming to hold once the admission becomes likely (van Fraassen 2005, 7–11). The admission itself is modelled on what van Fraassen calls epistemic marriage, a union after which the partners hold only communal opinion; marriage, on this account, is more intimate than the dolphins' admission and is clearly a matter of decision rather than something forced by opinions about how things are (10). Against settling on a point between the partners' opinions, van Fraassen argues from the reflection principle, roughly that one's present opinion must lie within the range of the opinions one foresees coming to hold, and proposes instead that the partners make their common opinion vaguer, spanning both, and adopt a common policy for revising it in the light of the evidence (19–21, 24–25). Dellsén, discussing van Fraassen's account in the preprint version of a chapter, distinguishes extending what a community can observe by admitting observationally superior beings, which the chapter calls epistemic immigration, from extending the observational capacities of existing members, which it calls epistemic upgrading, and notes that whether to allow the former is at least partly in the community's power (Dellsén 2024, 8, 12). Neither discusses artificial members; van Fraassen mentions AI only to say that the challenges discussed there do not need it (5–6).

Carried to a mixed *S*, Dellsén's distinction separates two ways in which what lies within *S*'s reach can change: through what members can do, and through who the members are. In van Fraassen's story the admission is debated and decided upon; what *S* could truthfully claim after admitting artificial members is not thereby something *S* can claim before. In the terms of this chapter, an achievement already realized in a prospective artificial member may, on some accounts of possession, come through the admission to count as one realized in a member of *S*. It does not follow that *S* lacks it before the admission, since van Fraassen's argument concerns whom us refers to, not the acquisition of knowledge from outside; even before the admission, *S* may have gained knowledge supported by that achievement by drawing on the system as an outside source (Sections 1.1 and 3.4). On van Fraassen's proposal, moreover, the common opinion formed at admission is at least as vague as either party's, spanning both rather than settling on either, so that admitting artificial members does not by itself settle what the mixed *S* holds.

In this case, then, a claim that a result has entered a mixed *S*'s epistemic situation would need to say which members it reaches and how possession is distributed between its human and artificial members. Admission does not by itself settle that. If, moreover, admission can increase the knowledge attributable to *S*, as noted above, not everything *S* comes to know is the result of new inquiry; an increase due to research and an increase due to a change in who the members are cannot be told apart unless *S*'s boundary is fixed, and a claim of novelty for research for a mixed society would say which of the two it rests on. And, as Collins's point suggests, vocabulary and standards shaped with machines can begin to enter such a society before any decision about membership.

1.3 Research by and for an artificial society

What would it take for an artificial society to be the society for which research is conducted, and what further would it take for a result to enter that society's epistemic situation? Here *S* is artificial and, in the central case, so is *A*. Artificial agents or an artificial organization identify unknowns relative to their own epistemic state, conduct inquiry, assess and revise claims through their own coordinated practices, and seek achievements new to that artificial society and for it, which on some accounts would, once they entered its epistemic situation, be knowledge attributable to that society. The two questions differ. To serve as *S* at all, the society must be such that its epistemic situation, through its members, organization, or records, could come to include achievements of the kind sought (Chapter 1, Section 2), so that questions of artificial belief and possession arise even where it is only the reference. Whether a result has in fact entered its epistemic situation is the further question. The former is taken up first, by asking what it takes for an artificial society to be a knowing community at all and sorting the conditions by accounts of belief; the discussion then turns to the latter, asking, account of possession by account, what a result's becoming such a society's knowledge requires.

The central case is not just a larger multi-agent system. As Kraft reports Longino's account, the sociality relevant to knowledge is interaction among diverse members, and a cognitive community is identified by the public standards appealed to in criticism, rather than by shared beliefs (Kraft 2025, 61–62). This account is not about artificial agents, but if its conditions are asked of them, a set of interacting artificial agents does not by that fact establish an artificial knowing society. For the same reason, multiple model instances, role prompts, or majority votes do not by themselves establish a knowing community. On a functional account such as List's, a suitably organized collective, though not a mere aggregate, can be an agent, and the question becomes how the members are organized. List's comparison of group agency and artificial intelligence applies such an account to both cases at once: it treats suitably organized collectives and sophisticated AI systems alike as intentional agents when they meet functional conditions of representation, motivation, and action (List 2021, 1213, 1215–1220, 1226–1227). List also notes that formal results back the claim that a group's agency may be irreducible to its members': even where a group's attitudes are determined by its members' attitudes, its attitude on a proposition will not in general be a function, such as the majority, of their attitudes on that proposition, and collective rationality may require that it not be (List 2021, 1225 n. 37). The simplest case behind such results, known in jurisprudence as the doctrinal paradox, shows that majority voting proposition by proposition can leave the group's judgments as a whole inconsistent, with a majority accepting each premise and a majority rejecting the conclusion they entail (List and Pettit 2002, 93–96). Applied to model instances, this suggests that a majority vote taken claim by claim among them would not, even on a functional account, be what makes them one agent.

Collins's account of research science adds a different condition. On this account, frontier research and a child's primary socialization, in which, on Collins's scheme, newborns and toddlers acquire in the family their native spoken language and moral concepts, have in common that both establish things new to the learner, that both do so in small, bounded, trusting groups relying on face-to-face communication, and that both depend on moral integrity; members of a core group, the small group that carries out such frontier research, find that their collective search for correspondence truth will fail unless they cleave to what Collins calls moral truth, a determination to tell fellow members the truth about their observations (Collins 2025, 1251, 1253, 1255–1256). For a large language model, on the same account, what stands in for socialization begins most of the way through human secondary socialization, the stage at which, on this scheme, children at school learn written language and early technical concepts (higher education and specialized occupations form a further, tertiary stage), and is restricted to the internet, where there is "no stability, no constancy and no truth" (Collins 2025, 1251, 1253, 1256–1257); these claims concern the language models available when the paper was written, and Collins allows that the pathologies described there might be overcome (Collins 2025, 1262). Read as a condition for an artificial community to be a knowing community, a question Collins does not take up, the argument about core groups implies that, beyond public standards to which members appeal, interacting artificial agents would need something that plays the role of a core group's commitment to report truthfully to one another. What the commitment protects, on this account, is members' reliance on one another's reports of their observations; where artificial members share the records of observations themselves, its counterpart may lie less in a commitment among members than in the integrity of the records and of those who maintain them (Chapter 11, Section 5.3). Section 2.2 describes an experiment in which artificial agents coordinated successfully on signals that tracked something other than what they appeared to name, and Chapter 11, Section 3 asks, for critics that can be made, what criticism among agents derived from one base system, and among agents made to differ, can and cannot detect.

Standards that are shared and open to public criticism, as Longino's account requires, do not by themselves require a community to be sovereign over them. Carried to artificial organizations, the account therefore implies that external origin or governance does not alone disqualify an artificial organization as a knowing community. The epistemic concern arises where external control prevents effective criticism or uptake: for example, where a justified objection cannot alter the treatment of evidence, or where the standards themselves are placed beyond criticism, since on this account the standards that regulate critical interaction are themselves outcomes of such interaction (Longino 2021, 14–15) and must be publicly accessible (Longino 2025, §5). Whether the organization can itself revise its governing standards, beyond their changing through criticism in this way, is a further question about epistemic self-government (Chapter 7, Section 4.3). Chapter 12 asks separately who is entitled to govern such revisions.

The conditions such a community would need to meet depend on the account adopted. Three of the families of accounts of belief that Chapter 2, Section 2 listed give one way of sorting them, when applied to artificial members and their organization: functional, interpretive, and normative accounts. Phenomenological and enactive accounts, which are not among those Chapter 2 listed, give a fourth; Section 2.2 takes up some of the enactive ones. These are theories of when mental states such as belief can be attributed. They answer a different question from the accounts of social possession that Chapter 2, Section 6 distinguished (Table 2.2), which ask what bears knowledge and through which relation. How far an artificial society must answer the question of when belief can be attributed to its members and organization depends on the account of possession adopted, as the discussion of possession below shows. The sorting of accounts of belief into these four families serves only to show that different theories would impose different conditions; it does not take up separately the representational and dispositional accounts that Chapter 2 also listed, the second of which Chapter 2, Section 2.1 treated together with functional accounts. A functional account may look for stable belief-like states, differentiated roles, shared and local memory, coordinated action, criticism, and error correction. An interpretivist account, such as the view of belief attribution as interpretation on which Davidson's argument rests (Section 3.1), may ask whether the organization is intelligible as maintaining and revising commitments. A normative account, on which belief is constitutively subject to norms of truth and rationality (Chapter 2, Section 2), may require participation in practices in which commitments are held to such norms. A phenomenological or enactive account, such as those taken up with Di Paolo in Section 2.2, may require forms of experience, concern, or self-maintaining normativity whose presence in current systems is in question. The features listed here are accordingly theory-dependent diagnostics, not a new checklist or an operational definition of society.

Turning to the second question, the accounts of possession in Table 2.2, applied to an artificial *S*, sort the conditions in a second way, and how much they ask of the accounts of belief differs among them. On individual-possession accounts (on which it suffices that suitably situated individuals know), member accounts (which require that enough members, or the members with decision-making authority in the relevant domain, know), and relational accounts (which understand a group's belief through members' beliefs and the relations among them), a result's becoming such a society's knowledge requires, in different ways, that its artificial members believe it, so that the question returns to artificial belief; on relational accounts it also turns on whether epistemic relations hold among them. On accounts of distributed cognition without distributed knowing, such as Giere's, the larger system can perform cognitive work while knowledge remains attributable to individuals, so that the question likewise returns to whether artificial members can be knowers; Giere's own restriction of agency to humans does not settle this case, since it is defended as a theoretical choice for systems in which humans conduct the research (Chapter 2, Section 6). Group-attitude accounts require a belief or acceptance that the group forms, and they differ in what they ask of members. On the joint-acceptance account that Gilbert states, the members must openly express their willingness jointly to accept the result, whether or not any of them personally believes it (Chapter 2, Section 6), so that the question is whether artificial members can take part in such joint acceptance, not whether they believe the result. On distributed or group-functional accounts such as Bird's, it turns on organization: whether the members together realize an appropriately organized cognitive system, a question parallel to the one List's functional account, above, raises for agency when it separates suitably organized collectives from mere aggregates. On accounts of knowledge without a knowing subject, neither the artificial members nor the artificial collective need be a knower. Giere, while attributing knowledge to individuals in the distributed-cognition account above, also offers, for the status of scientific knowledge, an account of this kind, an adaptation of Popper's epistemology without a knowing subject (Chapter 2, Section 6). On that adaptation, what gives the result the status of knowledge is certification by reliable, accepted procedures and public acceptance, and the question becomes whether certification by artificial members can serve as such a procedure, which, where the members are

instances of one base system, is the question that Chapter 11, Section 3 puts to their certification. On the critical-contextual account that Longino gives, the question joins the conditions on community discussed above: whether the interaction of artificial members involves the diversity and the public standards of criticism that the account requires of a knowing community. On none of these accounts does the connection among the members suffice by itself, but what more is needed differs with the account. Nor does anything in this sorting show that an artificial epistemic society already exists.

The case changes what a construction program is asking for. Research for a human society can inherit human concepts, questions, evidential norms, and social uptake. Research by an artificial society asks whether artificial beings can constitute and correct an epistemic life of their own. The second is not a prerequisite for the first, and it cannot be reduced to a scale-up of the first.

Whether research by and for artificial beings would be science at all is itself answered differently. Humphreys, whose separation of science done by humans from science done for them Chapter 1, Section 2 noted, gives an affirmative answer to a more general question, whether there could be science done neither by nor for humans. His imagined case is quoted by Schrape. After humans have become extinct, a rocket built before their extinction reaches a distant planet, samples conditions there that no human knew or predicted, and engineers the genetic code of a new cognitively aware species able to survive and flourish there, which colonizes the planet and adapts it to its own purposes, without any human knowing anything specific about how that part of the universe has been controlled. This, Humphreys holds, is still science, indeed scientific knowledge in operation: an activity need not be done either by us or for us to count as scientific, as quoted in Schrape 2015, 162–163. Read with the distinctions of Section 1, the scenario keeps apart who set the activity going, the humans who built the rocket, who possesses its results, no human, and whose purposes its outcome serves, those of a new biological species that is not human. Nickles, quoting the preface of the same book, where Humphreys writes that human abilities are no longer the ultimate criteria of epistemic success, disagrees: a pragmatic conception of the goals of science as conditioned by human interests implies, on Nickles's account, that a strain of anthropocentrism will always remain in our science, since "it is still we ourselves that we are extending"; the intelligent beings of the planet Zork, Nickles adds, will have a Zorkocentric conception of the universe (Nickles 2019, 365). In Humphreys's case, too, the outcome serves the purposes of a species, though not a human one; what Nickles's objection turns on is rather that no science is free of the perspective of those whose science it is. Ratti, arguing that machine learning in biology depends on a judgment akin to phronesis, practical wisdom, which is shaped by the way humans are, "leaves open the possibility of a scientific practice AI-tailored" but admits to not fully understanding the claim, "because we do not know any other science that is not human" (Ratti 2020, 18). Tollefsen, in the abstract of a paper presented at a 2018 meeting on whether science will remain human, calls the project of a science "populated and exercised by machines" chimerical: were science merely a technical pursuit within the realm of instrumental rationality, there could perhaps be a science carried out by machines, but on Tollefsen's account science is fully human action, to be governed by sound practical reasoning (Tollefsen 2018). On Humphreys's account as Schrape quotes it, science need be neither by nor for humans, which leaves room for such research to be science; on Nickles's, read through the case of the Zorkians, research by and for an artificial society would be a science centered on that society's interests, though Nickles does not there discuss artificial inquirers; on Ratti's it is unclear what such a science would be; and on Tollefsen's there could be none. Tollefsen's verdict rests on the premise that science is fully human action governed by practical reasoning. Totschnig, and Tubert and Tiehen, who argue that artificial agents could come to revise their final goals (Chapter 7, Section 4.1), and so to reason about ends themselves rather than only within instrumental rationality, would contest that premise. Ratti ties the judgment that science depends on to the way humans are; put as a judgment shaped by the way its inquirer is, the point would leave room for a science shaped by the way an artificial inquirer is, the possibility Ratti leaves open without claiming to understand it. Besides Tollefsen's, the accounts cited in this chapter that come nearest to a negative answer are Collins's and Rolla's, each with its own qualification. On Collins's account, frontier research depends on core groups whose members cleave to moral truth, and what stands in for socialization in a large language model, restricted to the internet, does not supply it, so that interacting systems of that kind would not form a research community; Collins writes about the language models of the time and allows that the pathologies described might be overcome (Collins 2025, 1256–1257, 1262). On Rolla's account, a genuinely cognitive AI would first require artificial life (Section 2.2), so that until then there would be no artificial inquirers for such a science to be by or for; the argument concerns cognition rather than science (Rolla 2026, 1, 11, 14). Neither holds that such a science is excluded in principle.

How the world becomes available to an inquirer

Section 1 left open what an artificial bearer or society would need in order to know anything at all, and allowed that, for a human *S*, much of what a result needs may lie on the human side. Beneath conditions on belief and possession lies a question that explicit method does not reach. A system may execute a proof rule, statistical test, or experimental protocol. But what makes a result count as an observation, a repetition, a failure, or evidence bearing on one commitment rather than another? Chapter 6 asked what the result of a check bears on and how far it supports a claim, presupposing that a check can be evidence about the world at all. This section asks about that presupposition. These presuppositions are required not of whoever runs a check but of those for whom the check counts as evidence. As Section 1 showed, that bearer need not coincide with *A*, and where it lies changes with whether *S* is human, mixed, or artificial. Further, as Section 2.1 explains, on Longino's account as Haely reports it, whether others can access and test an observation can bear on its standing as evidence; where the community of perceivers is *S*, this is a condition on relations within *S*.

The section separates three questions. The first is what a check presupposes when it counts as evidence at all (Section 2.1). The second is how a system's states come to be about worldly targets, the question usually discussed under the name of grounding (Section 2.2). The third is how its commitments remain open to correction by those targets, which this chapter calls world-answerability (Section 2.3). Separating the three questions keeps apart three achievements that a phrase such as tests hypotheses against the world attributes together. The three are not, however, nested, each containing the next, or independent of one another: the accounts discussed below relate them differently. Longino, for example, as Haely reports her account, makes whether a perception counts as an observation depend on its accessibility to others and, in controversial cases, on testing by different perceivers; since testing by others is one of the routes by which an organization remains corrigible, the first and third questions meet in one social relation (Section 2.1). On Coelho Mollo and Millière's account, what fixes a state's reference does not settle whether the system knows anything; and since a state can be about a target while misrepresenting it, whether the system's commitments are corrected when they misrepresent is a further matter, so that the second question can be answered while the third remains open (Section 2.2). On the account of John McDowell, discussed in Section 2.3, what a thought is about cannot be separated from its answerability to the world in McDowell's sense, a rational relation that differs from the sense this chapter gives the term; on that account, the second question cannot be answered apart from a question of answerability of that kind. The order of the subsections does not mark a logical order among the three questions. Section 2.1 continues the discussion of checks in Chapter 6, and the answerability to the world of Section 2.3 is what Part V builds on (Chapter 11, Section 1).

What bears the answers to the three questions differs according to whether *S* is a human, a mixed, or an artificial society (Sections 1.1–1.3). In research for a human society, unless the artificial actor is itself claimed to know, the answers may lie on the human side of the arrangement that conducts and receives the research, in human concepts, instruments, evaluators, and institutions, or, where the actor is a hybrid organization, in the organized whole (Section 2.3). For an artificial society, insofar as it or its members are claimed to know, and for the artificial members of a mixed society insofar as they are to know, the answers would have to hold of that society or those members themselves, to the extent the adopted epistemology requires.

Nor does the artificial case have to begin from one coherent optimizer whose beliefs, evidence, and values are all represented in a single place. The relevant organization might be distributed across agents, memory, instruments, environments, learned dispositions, and institutional rules. Whether that wider arrangement is merely a causal system, an inquirer, or an epistemic subject is a further attribution problem (Chapter 7, Section 2); assuming a unitary artificial mind would decide it in advance.

2.1 What a check presupposes

What does a check (in the sense of Chapter 6) presuppose when it counts as evidence? Suppose there are a million apparently similar observations. By themselves they are not evidence that the same thing happened again and again, since observations do not say what they are evidence of. Treating them as repetitions presupposes that the events are instances of a relevantly stable kind, that conditions are sufficiently similar, that the instrument tracks the intended target, and that the record can support an inference about another case, among other things. None of this is established by the observations themselves. The examples below share this structure rather than a single presupposition: in each, a check counts as evidence only against something that the check does not itself establish. This does not mean that no presupposition can be checked without circularity. Some can be checked by other routes; others cannot be checked all at once, because any check of them would rely on them. The end of this section returns to the difference. Nor does dependence on presuppositions mean that a knower must first prove them: on reliabilist accounts, for example, it is enough that they in fact hold (Chapter 2, Section 4.1).

Moreover, these presuppositions need not be consciously endorsed beliefs. They may be so deeply organized into perception and action that the world appears continuous without an explicit judgment that it is. While walking along a familiar path, for example, one does not treat its disappearance beneath the next step as a live possibility. Vision, touch, movement, memory, and prior interaction make objects, surfaces, recurrence, and resistance available before a scientific test is selected. Scientific methods extend and discipline these capacities and can correct them, but they do not begin from an uninterpreted world. Some of what is presupposed may consist in high-confidence propositional attitudes, things explicitly believed. Other parts may be things hard to call beliefs, such as perceptual capacities, practical expectations, learned dispositions, architectural biases, habits, trust relations, or socially stabilized roles.

What, then, is presupposed? Even within hinge epistemology, which develops Wittgenstein's image of hinges discussed below, accounts disagree over whether the basic certainties are beliefs, propositions, commitments, entitlements, or roles within a practice (see, for example, Pritchard 2021 and Fairhurst, Pérez-Escobar, and Sarikaya 2026). This book therefore uses primitive epistemic organization to refer to such presuppositions without settling that question: a loose umbrella for the part of an inquirer's epistemic organization, in the sense given at the opening of this chapter, that is not fixed by explicit stipulation, not established neutral terminology or a commitment to hinge epistemology. As the examples below show, however, presuppositions of checking are not confined to these; they extend to stipulations made explicitly within methods.

The examples below follow two threads. One is the presupposition that cases are relevantly the same; the other is the presupposition that a route or source of evidence is reliable. The division is taken from what the authors cited below themselves treat as presupposed, and its purpose is to show the same structure recurring at different scopes, from local to global. The two are not exclusive. Alston, for example, whose analysis of perception is one of the examples below, argues that the reliability of sense perception as a whole cannot be shown without relying on perception itself, and sketches the same predicament, that the reliability of a source cannot be shown without relying on that source, for the other basic sources of belief, induction, deduction, memory, and introspection (Alston 1988, 436). Induction is thus both the general form of the first thread and one of the sources Alston names. Nor do the examples exhaust the field: memory, testimony, records, and mathematics are left out. Were records and sources taken up, the same structure would presumably appear in checks in history or philology, for example.

The first thread begins with induction, which asks in general form whether the million observations above can be extended to cases not yet observed. On the reconstruction Henderson adopts, Hume's argument has two horns: no a priori reasoning can establish that unobserved cases will resemble observed ones, since the contrary involves no contradiction, and any argument from experience for that resemblance would presuppose it and so argue in a circle (Henderson 2022, §§1–2). Henderson also notes that learning theory gives the first horn a form that applies directly to whatever can be formalized as a learning algorithm: learning algorithms are effective only if they carry an inductive bias, and no-free-lunch theorems can be read as ruling out, wherever their formalism applies, a universal, model-independent justification of their performance, while leaving room for guarantees relative to a particular model and prior (Henderson 2022, §3.4).

Where a check's verdict is reached through statistical methods, this presupposition is stated explicitly, and at two scopes. Otsuka reads the probability model of inferential statistics, together with the assumption that it stays the same across samplings, as a mathematical model of Hume's uniformity of nature: the probability model is never observed as a whole, and

it is this assumption that underwrites inferring unobserved cases from the data. Otsuka distinguishes the probability model from a statistical model, a family of distributions with finitely many parameters that statisticians posit about the form of that uniformity when samples are too small for the asymptotic guarantees the uniformity provides (Otsuka 2021, 3–7). If so, when passing a check on some data counts as evidence about another case, the verdict rests both on the choice of a statistical model, which can be checked against data, and on the assumption that the probability model stays the same across samplings, which the data cannot themselves establish. Chapter 2, Section 5.3 sets out Otsuka's wider reading of statistical inference as resting on such assumptions, and Chapter 3, Section 2 the further assumptions that Otsuka describes as connecting probability distributions with causal structure.

In measurement, the presupposition that cases are the same can be stipulated explicitly as a rule of method and yet remain exposed to the world. In Tal's study of coordinated universal time, described in Chapter 6, Section 5.2, the short-term stability of the time scale is imposed on the clock data rather than tested against them, as a rule for applying the concept of time to particular clocks, while the gaps that remain among independent realizations of the concept are discovered empirically (Tal 2012, 117, 132–134). Chapter 6, Section 5.2 described when redefining such a standard differs from merely moving a target.

The same presupposition can also be taken to be socially maintained. As Haely reports Longino's account, observation depends on accessibility to others and presupposed repeatability, whether or not anyone repeats the observation; in controversial cases, different perceivers test its observational standing (Haely 2003, 98–99). Applied where the community of perceivers is *S*, this account makes whether others can access and test an observation bear on its standing as evidence. That is not to say that whatever a community accepts is thereby true (Chapter 2, Section 6). Whether an experimental result comes to be accepted through negotiation within the community or through reasoned judgment is itself disputed (Chapter 6, Section 2.4), and the problem of shared presuppositions being shielded from criticism is taken up in Chapter 11, Section 3.

The second thread concerns the reliability of a route or source of evidence. Alston argues that sense perception taken as a whole cannot be shown reliable without circularity. A restricted practice, such as reading temperatures from mercury thermometers or recognizing vintages of wine by taste, can be checked against the facts through another mode of access. For sense perception in general there is no access to the facts that does not rely on perception, and an argument from the success in prediction and control that perception makes possible circles in a subtler form, since whether a prediction succeeded is itself found out by observation (Alston 1988, 436). Markus Lammenranta reports the fuller treatment in Alston's 1993 book as follows. The argument from perception's track record is not logically circular, since its conclusion is not among its premises, but it is epistemically circular: one cannot take oneself to be justified in accepting its premises without assuming its conclusion. On Alston's reliabilist account, which imposes no higher-level requirement, such an argument could even yield justification if perception is in fact reliable; yet, in a passage Lammenranta quotes, Alston observes that the same could be said of crystal ball gazing, so the argument cannot discriminate the sources that can reasonably be trusted from those that cannot (Lammenranta n.d., §1). Alston had made the same point in the 1988 paper: if a practice's outputs may be used to check it, every practice scores perfectly, and in this respect all are on a par (Alston 1988, 436–437). Lammenranta offers, as one possible diagnosis, that such arguments may provide justification but cannot rationally remove a doubt about reliability (Lammenranta n.d., §1). The argument from track record circles here partly because it takes an inductive form, so the two threads meet. On the statistical side, too, the error rates that justify a test's verdict depend on distributional and design assumptions that the verdict itself does not check (Chapter 2, Section 5.3).

Alston's own response is not a proof of reliability. As Lammenranta reports and quotes the 1993 book, it is a defense of the practical rationality of engaging in firmly established belief-forming practices and taking them to be reliable: there is "no appeal beyond the practices we find ourselves firmly committed to", practices we could give up or replace only with extreme difficulty, if at all; suspending all belief is not an option; and new practices put in their place would have no noncircular defense either (Lammenranta n.d., §7). In the 1988 paper Alston had set out how that standing can be lost or strengthened. A basic practice is to be accorded prima facie acceptance pending a demonstration of unreliability, which massive and persistent inconsistency in its outputs, or inconsistency with the outputs of more firmly established practices, could supply. A practice can strengthen its claim through a kind of self-support, such as the predictive success attained on the basis of perception, which is not noncircular, since the practice has to be used to ascertain the success, but is not trivial either, since a practice might fail to show it, as the reading of tea leaves does. Even a practice that passes these tests remains only prima

facie acceptable, since trouble may develop later, and a practice that runs into trouble may be modified to get back in line (Alston 1988, 437). Lammenranta notes that Alston later rejected this practical argument and settled for a simpler form of Wittgensteinian contextualism, and objects that a defense of practical rationality does not contradict the skeptic, who denies not that it is practically rational to use our practices but that the beliefs they produce are epistemically justified (Lammenranta n.d., §7).

Wittgenstein likened such presuppositions to hinges, on which a door turns. Wittgenstein wrote that the questions we raise and our doubts depend on some propositions being exempt from doubt, as it were the hinges on which those turn; that this belongs to the logic of scientific investigation; and that it is not a matter of contenting ourselves with assumptions because we cannot investigate everything (Wittgenstein 1969, §§341–343). Applied to artificial systems, the last point comes to this: presuppositions do not remain for lack of time or capacity to investigate; they are conditions of checking, so greater capacity to check does not remove them. This does not mean that they stay fixed. Local presuppositions can be checked by other routes and revised (see the end of this section).

An artificial research system can have counterparts of these presuppositions. The correspondence is functional, not an identity. Nor is every feature of a system a presupposition in this sense: what counts is what bears on what a result is evidence of. In a system shaped by learning, to the presupposition that cases are the same there can correspond learned dispositions and representations that treat inputs as instances of stable kinds without any explicit representation that they are. The inductive biases of architecture and training, the distributional assumption under which performance on a held-out evaluation counts as evidence about further cases, the model under which an instrument or an automated evaluator counts as stable, and the condition under which a rerun counts as a repetition rather than as a second draw from the same organization, among others, belong to the same thread. Applied here, Haely's report suggests that rerunning one base system does not by itself show that a disputed observation has been tested by differently situated perceivers, a point to which Chapter 11, Section 3 returns. The presupposition that a route is reliable appears in the same form. One automated evaluator can be checked against another, and the system's evaluations against human judgments; whether this avoids circularity depends on how the two routes are related. But an argument from within the system for the reliability of its evaluation practice as a whole has at some point to rely on part of that practice. Where human judgment supplies the check, the same structure reappears one level up, for the wider practice that includes it.

Of whom these presuppositions must hold depends on *S*, as the introduction to Section 2 noted. How presuppositions are put in place and revised can also differ. A hinge in Wittgenstein's sense is a standing that a proposition has within a form of inquiry: one and the same proposition can be treated at one time as something to be tested by experience and at another as a rule of testing (Wittgenstein 1969, §98). The stability laid down as a rule of application in Tal's case is likewise a presupposition held as a rule that remains open to empirical revision. An inductive bias, by contrast, is a structure that the way a system is made puts in place and that remaking it can replace, and an artificial inquirer may also shape and revise its own presuppositions and rules. The difference bears on who can revise such a presupposition, a capacity that Chapter 7, Section 4.3 counts toward epistemic self-government, and on who answers for it, which Chapter 12 takes up.

Finally, in both threads presuppositions differ in scope. Some no check could validate all at once, such as the uniformity of nature or the reliability of perception taken as a whole. Others are local and can be checked by other routes when doubt arises: a perceptual mode by other senses, a statistical model against data, a standard against independent realizations, a claim of repeatability by other observers. Locality can make a presupposition available to checks that do not simply assume it, although narrower scope does not by itself guarantee better correction. What advances correction, as seen with Dewey and Chang in Chapter 6, Section 5.2, is running the circle in which standards are revised from within inquiry, repeatedly and against discrepancies among realizations; global presuppositions differ, since any check of them would presuppose them, and they cannot be rechecked in the same way. Of the counterparts named above for an artificial system, an automated evaluator's stability is local in this sense, whereas the reliability of the system's evaluation practice taken as a whole is global for that system, so that a claim about a configuration's checks can say which of its presuppositions lie on which side, and so which it can check for itself and which only remaking it, or checks from outside it, can reach. Why, then, may such presuppositions be held? Alston's response above and Reichenbach's argument that induction can be justified without knowing whether it succeeds (Chapter 2, Section 8.3) are attempts to answer. Alston's answer turns on belief-forming practices' being firmly established and ones we could not readily abandon. Alston does not discuss artificial systems, but

applied to a system whose inductive biases its makers can replace by remaking it (above), the premise holds of the system's own practices only insofar as they are firmly established and the system cannot readily abandon them, and where they fail for the system's own practices, whether they hold would have to be asked separately of the wider practice that includes those who make and evaluate the system, so that who can revise such a presupposition (Chapter 7, Section 4.3) also bears on where Alston's ground lies. The history of formation that Section 2.2 discusses can explain why a presupposition need not be arbitrary with respect to the world, but it is not by itself a warrant. Dependence on presuppositions does not by itself show either that knowledge is impossible or that a check is reliable. Which evidential routes can bear on local presuppositions is taken up in Section 2.3. There is, however, a prior condition: a check can register an error about a target only if the states it checks are about that target, which is the question of Section 2.2.

2.2 Grounding is not one relation

Section 2.1 ended with the point that a check can register an error about a target only if the states it checks are about that target. This subsection takes up the question presupposed there: how do a system's states come to be about worldly targets at all? The question bears on this chapter because, on many accounts, a result is knowledge for a bearer only if the bearer's states are about the targets the result concerns. In research for a human society this can be met on the side of human concepts and instruments, but where an artificial bearer is claimed to know, it is asked of that bearer's states (introduction to Section 2). The subsection first takes up, from Section 2.1, the answer that looks to histories of formation for why primitive organization need not be arbitrary with respect to the world, and the limits of that answer as a warrant; it then notes that, on some accounts, history also fixes what a state is about, which leads to the arguments discussed under the name of grounding. The symbol grounding problem is posed by first setting symbols apart from the world, whereas the symbols actually in use are not set apart in that way. What follows presents the work of Taniguchi, who develops that point directly, and then accounts that locate the tie between symbols and the world elsewhere. Finally, it distinguishes the relations that the word runs together and confirms that success under one does not by itself establish the others.

One answer begins from history, which can explain why a primitive organization need not be arbitrary with respect to the world. Human bodies, perceptual systems, development, social learning, and cultural practices formed through long interaction with the same world that later inquiry addresses. Bradie and Harms give, as part of the difficulty of understanding cognitive behavior as a product of evolution, that at least three very different evolutionary processes are involved: the biological evolution of cognitive and perceptual mechanisms through genetic inheritance, the cultural evolution of language and concepts, and trial-and-error learning within an individual's lifetime. They add that there is some reason to agree with Donald T. Campbell, who coined the term evolutionary epistemology, that understanding human knowledge fully will require understanding the interaction between these processes; this, they say, requires being able to model both biological and cultural evolution, and they go on to models of cultural evolution (Bradie and Harms 2020, §§1, 2.2).

On the accounts discussed here, the historical answer is plausible only in a graded form that holds capacity by capacity and domain by domain, and history cannot warrant everything it helps explain. Versions of it belong to evolutionary epistemology's program on how cognitive mechanisms are shaped by environmental structure (Bradie and Harms 2020, §1.1). Boudry and Vlerick reject the sweeping claim that natural selection is simply indifferent to truth. Their target is Alvin Plantinga's evolutionary argument against naturalism. As they reconstruct it, the argument holds that selection cares only about fitness and not about truth, so that if naturalism, the view that there are no gods or supernatural beings, and unguided evolution are both true, we have no reason to trust our cognitive faculties, and the naturalist acquires a reason to give up every belief, including the belief in naturalism itself (Boudry and Vlerick 2014, 65–66). They answer with three arguments. First, although survival does not require beliefs at all, a lineage that has once come to direct its actions through beliefs remains bound to that route, a constraint they call adaptive path dependency; from then on, beliefs that are at least approximately true in ecologically relevant situations are favoured, and false ones are harmful (Boudry and Vlerick 2014, 68–69). Second, they argue that what a belief is about is tied to the actions it disposes an organism to take, given its desires and other beliefs, so that its content cannot change while its effect on behaviour stays the same; whether that behaviour is adaptive then depends on whether the world is as the belief represents it, as when a belief that a mushroom is safe to eat proves harmful because the mushroom is poisonous (Boudry and Vlerick 2014, 69–72). Third, they argue that scientific belief is not a product of intuitive faculties working unaided: it rests on cooperation among contemporaries and across generations, on mathematics and notation, on instruments, and on checking theories against observation, and the inference

that everything built on imperfect foundations must be unreliable is what they call the foundationalist fallacy (Boudry and Vlerick 2014, 72–74).

They also say where selection need not favour truth. Heuristics, quick rules of thumb, may trade accuracy for speed; because different kinds of mistake carry unequal costs, a mechanism may be inclined to make certain mistakes; and capacities that are accurate in familiar environments may mislead in new ones, so that selection may favour true belief in some domains but not in others (Boudry and Vlerick 2014, 66–67). Vlerick and Broadbent, who accept that defence against evolutionary skepticism, explicitly deny that it supplies a general justification of scientific belief: on their account, selection shaped cognitive faculties to track truth only in a narrow domain of ecologically relevant beliefs, and modern scientific theories are not an extension of those beliefs (Vlerick and Broadbent 2015, 186, 190–192).

To sum up, interaction with the world can help explain why a capacity responds to real regularities rather than being freely invented. It does not establish, for every resulting expectation, that it is true, generally reliable, justified, or suited to environments beyond the range in which it formed. As Boudry and Vlerick allow, adaptive shortcuts and systematic errors are possible, and advanced inquiry depends on more than the faculties that this interaction formed. On these accounts, then, natural history is a possible genealogy of epistemic capacities, telling where they came from, not a warrant for all their products. Applied to an argument from natural history, Alston's analysis gives a further reason why such an argument cannot supply a general warrant. The argument that natural selection would have preserved only belief-forming dispositions that mostly yield truths relies on evolutionary theory, which rests on evidence gathered through perception. By the criterion Alston applies to the argument from success in prediction and control, whose success is itself established by observation (Section 2.1; Alston 1988, 436), it would therefore be epistemically circular as an argument for the reliability of perception, whatever the merits of its premises.

Does an artificial system, then, have a corresponding history? Abstracted, what the discussion so far says is that an inquirer's primitive organization is shaped by that inquirer's history of formation. Artificial systems have histories of formation too. Whether those histories include anything corresponding to the three processes is not obvious, however, and the situation differs from process to process. The analogy easiest to trace is with trial-and-error learning: in a system that learns from the outcomes of its own actions, something close to it can be at work.

Whether anything corresponds to biological evolution is far less clear. The program in evolutionary epistemology that addresses the evolution of cognitive mechanisms takes as its object the biological substrates of cognition, such as brains, sensory systems, and motor systems (Bradie and Harms 2020, §1.1). That process has not merely tuned the working of organs already in place; it has produced and remade sense organs and nervous systems themselves, and with them what can be sensed at all and what distinctions are possible at all. The expectations built into perception and action discussed in Section 2.1 are, on this view, shaped by such a process. In many present trained systems, by contrast, architectures and sensing arrangements (sensors and input formats) are designed by people, and learning takes place on top of them. There may be room to regard design choices and the structure that training builds in as taking on part of this role. But to treat them as what corresponds to biological evolution risks reducing a process in which organs themselves came into being and were remade to the adjustment of a given structure. What in an artificial system could play the role of this process, and whether anything could, is not obvious.

Correspondence with cultural evolution is not simple either. Bradie and Harms treat cultural evolution, through population models, as a process in which languages and concepts are transmitted or rejected within a population, with fitness components that include transmission and rejection probabilities, and they count the construction of cultural niches such as language and educational institutions as part of it (Bradie and Harms 2020, §2.2). If so, the question is not whether a single system undergoes it but in what population a system takes part in the transmission and sorting of concepts, and the answer can vary with the society in which the system stands. Training on human records inherits products of human cultural evolution, but what is inherited is what records retain, which may not include such things as the collective tacit knowledge acquired by taking part in a practice (Chapter 3, Section 6). Conversely, where a system's outputs enter human language and records and become part of what later people and systems inherit, the system does not only inherit but also takes part in human cultural evolution. Where vocabularies are transmitted and stabilized among artificial agents, something like a cultural evolution of an artificial society itself is conceivable. The question then is what sorts the vocabularies, which the experiment of Bouchacourt and Baroni discussed below makes concrete.

Where a system inherits from human records, its organization is tied to the world not through its own interaction with the world but as mediated by human history. What such mediated connection can give its states bears on the account of Coelho Mollo and Millière below and on the route of inheritance from records discussed in Section 2.3. The limits noted above can also be applied to the histories of artificial systems. Even if a training history explains why a capacity responds to regularities in records, it does not establish that the capacity suits environments beyond the range of training, or that its products are warranted. Much of what has been said here concerns present trained systems. If there could be artificial lineages that remake their own designs and sensing arrangements through the outcomes of interaction with the world, matters could differ.

History bears on the question of this subsection in another way. Boudry and Vlerick's second argument above, by tying a belief's content to the actions it disposes and so letting selection on action reach content, already connects history with content. The account of Coelho Mollo and Millière, discussed below, draws on teleosemantics, a theory of representation that explains what a state represents by the function it has acquired through a history of selection. On their account, a history of selection, which learning can supply, is part of what gives internal states the function of carrying information about worldly targets (Coelho Mollo and Millière 2026, 13–17), while grounded states do not by themselves suffice for knowledge (Coelho Mollo and Millière 2026, 2, 24). On that account, then, a history can fix what a state is about without warranting what the system takes to be true of it. Whether a history of formation also bears on warrant is a further question, on which accounts of justification differ.

Different questions have been raised under the name of grounding. Harnad's symbol-grounding problem asks how the semantic interpretation of formal symbols can be made intrinsic to the system. The meanings of the symbols in a book derive from the meanings in their readers' heads; the question is how to give a system meanings of its own rather than such derived ones (Harnad 1990). Bender and Koller argue that a system trained only on linguistic form cannot learn meaning, understood as the relation between form and communicative intent. On their account, communicative intents are about something outside language, and text is their projection into forms; if the training data are only form, there is not sufficient signal to learn that relation, and without a means of hypothesizing and testing the intents, reconstructing them from forms alone is hopeless (Bender and Koller 2020, 5185, 5187, 5189). These arguments pose hard questions for systems trained only to predict text, but do not by themselves settle questions of belief, knowledge, or research.

Both arguments carry limits that their authors state. Harnad held that, strictly speaking, symbol grounding is a problem for modeling cognition rather than for AI in general: if symbol systems alone produced the performance AI seeks, there would be no reason to demand that their symbols have intrinsic meaning, although Harnad allowed that grounding might bring performance gains in robotics and machine vision (Harnad 1990, n. 10). Bender and Koller confine their argument to systems trained only on form. They state that it does not apply to systems trained on tasks such as reading comprehension, whose data specify semantic relations between texts, and that meaning could conceivably be learned from form augmented with grounding data, to the extent that the data represent communicative intent (Bender and Koller 2020, 5191–5192). Set side by side, the two arguments suggest that whether an artificial researcher needs to be grounded in Harnad's sense (an interpretation intrinsic to the system) or in Bender and Koller's (the relation between form and communicative intent) depends on what grounding is invoked to explain, since, as Chapter 1, Section 1 notes, candidate definitions can answer different questions. It also depends on whether the system itself is claimed to know, or only supplies results that others come to know (introduction to Section 2).

These arguments are framed by first setting symbols apart from the world and then asking how they come to have meaning. Harnad's question was posed about systems, like those of symbolic AI (classical AI that manipulates symbols by rules), in which symbols are prepared in advance as formal tokens and their interpretation is supplied by an interpreter outside the system. Bender and Koller's argument likewise separates linguistic form from the intent and the world it expresses, and asks what can be learned from form alone. The symbols we actually use, however, do not stand apart from our dealings with the world and with others. On the developmental research that Saji reviews, words are learned in exchanges with others while attending together to things and events (Saji 2026, 46–47); they change their meanings through use, and in turn orient our perception and action (Taniguchi 2026b, 7–8). If symbols and the world are separated at the outset, the question arises how to connect them. If they are seen as shaped together in interaction, the question becomes how that connection is formed and maintained. Harnad too, rather than leaving symbols set apart, sought the solution in grounding them bottom-up in

nonsymbolic representations based on sensory input, iconic and categorical representations (Harnad 1990). It is this view, on which symbols and the world are shaped together in interaction, that Taniguchi develops head-on.

Taniguchi, too, takes the symbol grounding problem to have been posed within symbolic AI, about "symbols" understood as formal tokens (Taniguchi 2026c, 42–43). Taniguchi reports having come to believe that the very premise of the question how artificially created symbols can be meaningfully connected to real-world experience may be misguided, since symbols are not merely external representations but are deeply intertwined with internal representations, such as concepts, that develop through self-organizing processes grounded in sensorimotor interaction with the environment (Taniguchi 2026a, v–vi). From the perspective of symbol emergence in robotics, the central problem for AI is not symbol grounding but symbol emergence: how meaningful symbols arise, change, and become organized through social and embodied interaction (Taniguchi 2026c, 43). The recasting does not set Harnad's point aside: Taniguchi accepts Harnad's observation that relations among words yield at best relational concepts, not grounded ones (Taniguchi 2026d, 198).

In what Taniguchi calls the theory of symbol emergence systems, each member forms internal representations through bodily interaction with the environment, without direct access to the inner states of others. Through semiotic exchange among members, symbol systems such as language become organized from the bottom up, and the organized system in turn constrains members' perception, action, and interpretation from the top down. Taniguchi holds that this micro–macro loop gives rise to symbolic meaning (Taniguchi 2026b, 4–8). Internal representations are not yet symbols but their raw material, and the meaning of words is not fixed within the individual alone but obtained through interpretation by others within a social context (Taniguchi 2026b, 7). Grounding for each member, too, is said to lie not in the objective world but in the Umwelt in von Uexküll's sense, the subjective world in which a creature perceives and acts (Taniguchi 2026d, 198–199). Membership requires not biological humanness but the ability to take part in semiotic communication, using arbitrary symbols (symbols with no natural tie to their objects) to contribute to societal functions (Taniguchi 2026b, 6). Taniguchi holds that an AI system can be said to participate meaningfully in a symbol emergence system only when it has a body and interacts with the real world (Taniguchi 2026c, 42), while also saying that generative AI is already changing human language use and becoming part of such a system (Taniguchi 2026a, v; Taniguchi 2026g, 222, 225).

For how the loop works, Taniguchi and colleagues offer a model of a naming game. Assuming that two agents attend to the same object, a speaker proposes a name on the basis of its own observation, and a listener accepts it probabilistically in light of the internal representation formed from its own observation. When acceptance is made to follow a prescribed rule, the Metropolis–Hastings algorithm, the exchange is formally equivalent to approximate Bayesian inference over a shared latent variable that integrates the observations of both; that is, the observations of both can be reflected in the formation of shared symbols without either referring directly to the other's internal representations (Taniguchi 2026e, 207–208). From this Taniguchi proposes the collective predictive coding hypothesis, that human language is formed through such distributed inference (Taniguchi 2026f, 211–212). Applied to the relations distinguished below, the framework can be read as treating dependence on perception and action (the sensorimotor relation) and the establishing of common ground between speakers (the communicative relation) as two sides of one loop, and as relocating reference, what symbols or states pick out in the world, from the objective world to each member's Umwelt and to an Umwelt shared among members (Taniguchi 2026f, 212).

The recasting can be stated in the distinctions of this book. If the meaning of symbols arises within such a loop, what is asked of an artificial system is not only by what relations its states come to be about worldly targets but also which symbol emergence system it joins, and as what kind of member. The question raised above about cultural evolution becomes, on this view, a question about meaning itself. Here the agent who conducts research, A, the society S that sets what is new and for whom an achievement is sought, and the community that A joins and that shapes a symbol system need to be kept apart. That community may be a human, a mixed, or an artificial society, and it need not coincide with S; because this book distinguishes A from S, this difference can be stated once the community is placed as a further term, and it is one source of the mismatch that Section 3 examines. Takahashi, proposing in a preprint levels of autonomy for scientific AI, reaches such a difference from the side of hypothesis generation. On this account, generating hypotheses whose form is not fixed in advance, Takahashi's fourth level, requires in experimental science that the symbol grounding problem be solved at least in the domain the experiments deal with, which means that the system acquires an embodiment through particular laboratory robots. Since that would give rise to a symbol system separate from the one built on human embodiment, the account holds

that from this level the system would understand its objects in a way different from the way humans do, and that this is where a divergence between human science and AI science begins (Takahashi 2024, 7, 10). At the fifth level, where humans give the system its research purpose, the system would by contrast need a symbol system compatible with the human network of meaning, so that some means of continuing to ground its knowledge in the human system of scientific knowledge, or to translate it into that system, would have to be built and maintained (Takahashi 2024, 8, 11). Read in this book's terms, Takahashi's fourth level lets the symbol system with which *A* works diverge from that of a human *S*, while the fifth asks that the research remain research for that *S*. Even on this view, however, whether success under one of the two relations joined in a single loop, the sensorimotor and the communicative, establishes the other can vary with how the loop works. Whether shared symbols come to be about worldly targets, and whether they remain open to correction by those targets (Section 2.3), are still to be asked of the workings of the loop. An example appears below when the experiment of Bouchacourt and Baroni is set beside this model.

Other accounts locate the tie between symbols and the world elsewhere. Coelho Mollo and Millière distinguish referential, sensorimotor, relational, communicative, and epistemic grounding, and treat referential grounding as the one essential to the problem. Drawing on teleosemantics, they argue that it requires two conditions: causal-informational relations to the world, which for a language model pass through human linguistic production, and a history of selection, which learning can supply, that gives internal states the function of carrying that information. They argue that language models can meet both conditions, particularly those that undergo preference tuning, and that multimodality and embodiment are neither necessary nor sufficient for reference (Coelho Mollo and Millière 2026, 1, 13–17, 22, 24). They add that the designers' intentions give a function to the learning mechanism, whereas the function of internal states, and the standard of their correctness, are given by the history of selection that the mechanism carries out on its data; they compare this to how a dog breeder's intention to produce calmer dogs sets up a selection regime, while it is the history of differential reproduction that confers the function on the selected traits (Coelho Mollo and Millière 2026, 16). The separation runs parallel to the one Chapter 7, Section 2 drew for the attribution of conduct, between where a function comes from and where the organization of an inquiry lies; on this account, what designers intend for the mechanism does not by itself fix what its states are about.

Coelho Mollo and Millière also mark the limits of their claim. They do not argue that language models understand, know, or have any conception of truth, and they hold that grounded states would not by themselves suffice for such capacities. They set communicative intent aside, relating Bender and Koller's challenge to communicative grounding, while rejecting, against Bender and Koller, the view that linguistic form can be easily separated from meaning (Coelho Mollo and Millière 2026, 2, 5 n. 9, 8, 13, 24). Read side by side, the two positions therefore conflict over what training a system only to predict text can supply, but not over whether such a system has communicative intentions. Bender and Koller count such training as training on form alone and deny that it can yield conventional meaning, which on their account connects expressions to objects outside language, or the relation between form and communicative intent (Bender and Koller 2020, 5185, 5187); Coelho Mollo and Millière's rejection of an easy separation of form from meaning calls into question that classification of text prediction as training on form alone. Where training adds signals beyond the prediction of text, as in the preference tuning that Coelho Mollo and Millière single out, Bender and Koller's 2020 paper does not by itself settle whether their argument still applies.

Enactive approaches that root cognition in a system's self-maintenance ask for something stronger than referential grounding. Di Paolo argues that autopoiesis, a system's self-production, yields by itself only an all-or-nothing norm, to go on living or to disintegrate, which the system itself cannot register; sense-making requires in addition adaptivity, a capacity to monitor and regulate its own states and its relation to the environment according to whether they are approaching the boundary of its viability (Di Paolo 2005, 6–10). Di Paolo does not settle the artificial case in one direction. Meaningful action, the paper holds, can be convincingly emulated by an artificial system without the system acting meaningfully, so whether a system acts meaningfully cannot be told from observing its responses; it has to be checked by examining whether the system's organization generates its own sense-making in relation to its viability, a check regarded there as relatively simple at least for ruling out negative cases. Di Paolo leaves open whether artificial intentionality, in the paper's sense the intentional structure of an agent's own behaviour, is possible short of a fully metabolizing robot, and notes that the picture changes once values not originating in metabolism are considered. In closing remarks that the paper calls speculative, Di Paolo argues that self-sustaining forms of activity, such as organized patterns of sensorimotor coordination, can generate values of their own within the bounds of metabolic viability (Di Paolo 2005, 13–16). Rolla, by contrast, argues that AI

systems lack a living body of their own and do not inhabit an environment, and so cannot autonomously exchange matter and energy to remain within viability limits; a genuinely cognitive AI, on this account, would first require artificial life (Rolla 2026, 1, 11, 14). On Rolla's account it is through autonomy of this kind that symbols matter intrinsically: the symbols we use are grounded because they matter to us, as products of our autonomous behavior (Rolla 2026, 6, 11–12, 14). This enactive requirement is a substantive rival to Coelho Mollo and Millière's account, not a theory-neutral condition of artificial research. The autonomy at issue in these accounts, that of a system whose own activity produces and maintains the system itself, is not the comparative autonomy of Chapter 7, Section 4, which compares, in a specified respect, how far parties outside a selected boundary direct, substitute for, authorize, ratify, or repair the organization attributed to an actor; a verdict about one does not settle the other.

Where the argument depends on any of the senses of grounding, six items are kept apart. The first five are relations between expressions or states and something else; the sixth is a condition that may be required in addition:

1. referential grounding: reference to worldly targets;
2. sensorimotor grounding: dependence on perception and action;
3. relational grounding: relations of an expression to other expressions within a language;
4. communicative grounding: the establishing of common ground between speakers, including a shared way of connecting expressions to the world;
5. epistemic grounding: connection of expressions to information stored in a knowledge base;
6. enactive or intrinsic normativity grounded in self-maintenance and its adaptive regulation.

The first five are the notions of grounding that Coelho Mollo and Millière find in the work of cognitive scientists, philosophers, and AI researchers, and the sixth adds the enactive requirement described above (Coelho Mollo and Millière 2026, 6–8). The purpose of listing these six is to keep apart relations and conditions that the word runs together, not to define grounding. The list leaves out, for example, the phenomenological requirements mentioned in Section 1.3. Coelho Mollo and Millière treat referential grounding as the most fundamental of their five and as arguably the basis of the others, with a partial exception for relational grounding (Coelho Mollo and Millière 2026, 24). The enactive accounts above, which locate the conditions under which meaning arises for a system itself in self-maintenance and its adaptive regulation, stand, in this list, as asking whether referential grounding alone can explain sense-making. Their fifth notion is close to one use of the word in engineering contexts for artificial systems, connection to retrieved information: on their account, current language models may involve epistemic grounding when they are connected to external databases from which they retrieve information (Coelho Mollo and Millière 2026, 8). Answerability to evidence and correction is a relation of a different kind from all six. A state can be about a target and misrepresent it, and whether a system's commitments are corrected when they misrepresent is a further question, which Section 2.3 takes up as world-answerability.

Success under one relation does not by itself establish the others. This is not a claim that the relations are independent of one another; which relation supports which differs from account to account. On Coelho Mollo and Millière's account, a linguistic system may refer to worldly targets through causal relations mediated by human text, together with a learning history that selects its states for carrying that information, without possessing its own sensorimotor skills. A robot may couple perception and action without forming beliefs or participating in scientific criticism, which is why the question of Section 2.3 remains open even where sensorimotor grounding is secured. A community may stabilize a vocabulary that supports successful coordination among its members, while that success does not settle what the vocabulary is about. Experiments on emergent communication between artificial agents give this last possibility a concrete form. Bouchacourt and Baroni trained pairs of networks on a game in which one agent signals which of two images is the target and the other must pick it out. Pairs that succeeded developed representations almost perfectly aligned with each other but only weakly correlated with the representations of the input images, and they communicated nearly as successfully about pairs of random-noise inputs as about images; the authors conclude that what the signals tracked was not the depicted kinds but low-level relational properties of the inputs (Bouchacourt and Baroni 2018, 983–985). The signals did discriminate the targets of the game. Successful coordination within such a community is therefore not, by itself, evidence that its shared signals are about what an observer would take them to name. In the naming game of Taniguchi and colleagues, by contrast, what sorts the

vocabulary is not success in coordination but likelihood under the internal representation that the listener has formed from its own observation. Taniguchi distinguishes this game from conventional naming and referential games by its assumption of joint attention (Taniguchi 2026e, 205, 207). The formal equivalence described above does not, however, show that the shared symbols track what an observer would name. Taken together, the two suggest that what a shared vocabulary tracks is not settled by its stabilization alone and can vary with what sorts the vocabulary. In an artificial community, what sorts it can be set by how the exchange among its members is made. Not tracking the kinds that a human observer assumes does not by itself show a lack of connection to the world, and whether commitments can be corrected by errors about their targets remains the question of Section 2.3.

2.3 World-answerability

Section 2.2 ended with the question whether commitments can be corrected by errors about their targets, and Section 2.1 noted that local presuppositions can be exposed to corrective evidence beyond the assumptions under examination. How can an epistemic organization's commitments remain open to correction by their targets, and who would sustain that relation where *S* is a human, a mixed, or an artificial society (Sections 1.1–1.3)? As used here, world-answerability means two things. One is that an epistemic organization is differentially constrained by target phenomena: its commitments are constrained by them in such a way that they would differ if the phenomena were different. The other is that the organization remains corrigible through further contact, intervention, and cross-checking by independent routes. Being answerable to the world in this sense is not having access to an oracle. The idea has appeared earlier in the book: as answerability to proofs, observations, interventions, failed predictions, and criticism without an infallible test for final truth (Chapter 2, Section 3), and as revision of standards that remains answerable to discrepancies among realizations assessed on grounds beyond the revision itself (Chapter 6, Section 5.2). Chapter 11, Section 1 specifies such a route as a defeat relation and distinguishes the evidence it adds from the arrangements that protect it, including how diagnoses of failure remain open to challenge. Where the book speaks of answerability to goals or to people, the sense is different.

The phrase "answerability to the world" is not used here in the sense McDowell gives to thinking that is answerable to the world. For McDowell, that answerability is a rational relation, in which a judgement is correctly or incorrectly made according to how things are, and it requires experience in which the subject's conceptual capacities are already at work to stand in rational and not merely causal relations to what the subject thinks, relations in which experience serves as a reason for or against it, a requirement McDowell takes to hold for there to be empirical content at all (McDowell 2000, 3–5, 9–12, 16). McDowell's account is not about artificial organizations, but taken to the definition used here, it implies that an organization meeting that definition through causal and practical routes alone would not thereby be answerable to the world in the account's sense, and that whether it could be would depend on whether its states draw on capacities of the kind McDowell calls conceptual. The two usages can therefore divide their verdicts on the same organization: an artificial organization whose commitments are revised by the results of interventions can be answerable to the world on this chapter's definition, while on McDowell's account that alone leaves undecided whether what is being corrected is thinking in the account's sense. The verdicts part not over whether there is a route of correction but over what it is that is corrected. The accounts of grounding in Section 2.2 address what states are about but not the conditions of capacities of that kind.

Feedback from nature is accordingly a modest notion here. Nature does not return an uninterpreted verdict. Interventions produce traces through apparatus, sampling, classification, models, and standards, and a mismatch may implicate the focal claim, an auxiliary commitment, an instrument, a material condition, or the representation through which the result is described (Chapter 6, Section 2.1). Whether the standards through which such traces are read make answerability itself relative to *S* depends on the account. On accounts on which what corrects a commitment is fixed by its target independently of *S*'s epistemic norms, those norms can bear on what counts as evidence (Section 2.1), but world-answerability is not relative to *S*. Leibo and colleagues, in a preprint position paper on AI alignment, hold instead that even propositions one would ordinarily be tempted to call facts are governed by epistemic norms that control justificatory standards and appropriate evidence integration, norms "just as culturally contingent as any other kind of norm" (Leibo et al. 2025, 2, 7). Brought to the analysis of answerability to the world, which they do not discuss, their view implies that what counts as being corrected by a target is itself partly fixed by the norms of a community. It does not follow that the difference between answerability to the world and relevance to *S*, on which Section 3 relies, disappears, since that standards of evidence are social and that what a society wants to know is relative to it are different matters: societies with the same evidential norms

can differ over what is worth pursuing, and societies that count the same problem as important can adopt different norms of testing. What weakens on this view is not that there is a difference but the assumption that what counts as a correction can be fixed independently of a community's norms; to that extent, as Section 3.2 notes, the two come apart less sharply.

For research aimed at human knowledge, primitive world relations need not all be inside the artificial component. Human concepts, testimony, datasets, instruments, evaluators, and institutions may provide mediated links from the activity to the world. For a hybrid organization, the relation may belong to the organized whole: one component senses, another intervenes, another preserves records, and another evaluates consequences. For a mixed society, the relation would have to hold of its artificial members insofar as they are to know (introduction to Section 2); on Longino's account as Haely reports it, whether other members can access and test observations can also bear on their standing as evidence for the society (Section 2.1).

Termine, Ratti, and Facchini's analysis of machine learning in science is an example of how what a check presupposes can itself be distributed in such an organization. They argue that the construction of machine-learning models in science is mostly theory-indifferent: setting a model's parameters and hyperparameters, and increasingly feature engineering as well, does not necessarily require reference to the domain theory of the field in which the model is used, at least for predictive accuracy on data distributed like the training data (Termine, Ratti, and Facchini 2026, 15, 17–20, 22–25). Domain theory nevertheless enters the training data, since curating them is theory-informed, and the trained model passively inherits these priors, which end up scattered across parameters that have no interpretation in the domain's terms; this weak form of theory-ladenness, arising outside the modelers' intentions and control, they call theory-infection (Termine, Ratti, and Facchini 2026, 15–16, 20–22). Curation, they note, is commonly done by data specialists separate from the model builders, and using the models in science, as most of the literature holds, typically relies on domain theory (Termine, Ratti, and Facchini 2026, 15, 21, 29). Read in this way, theory dependence is distributed across the steps of an inquiry, and across those who perform them, rather than attached uniformly to every one: construction can be largely indifferent to domain theory while theory enters through, for example, the curation of data, the goals that the scientific context sets for the model, and how the model is interpreted and used. Domain-indifferent construction does not make the larger inquiry theory-free, and theory-laden use does not show that the model builder supplied the relevant scientific organization.

For a candidate artificial knower or society, the attribution burden is stronger. On accounts that attribute knowledge to the artificial subject itself, an account is needed, supplied by the subject itself or by external analysis as the adopted epistemology permits, of how worldly differences constrain its own commitments rather than only human judgments downstream. Possible routes, distinguished by what carries the contact, include direct perception and intervention by the system itself; causal-historical inheritance from records of others' contact with the world; developmental learning from its own failed actions; and distributed membership in laboratories whose other members and instruments make contact. Enactive theories of the kind described in Section 2.2 add a further condition that is not a route of contact, an artificial form of self-maintaining and adaptively regulated normativity. These routes are possible ingredients within different accounts, separately or in combination, and naming them does not establish that any of them is sufficient for an artificial knower. Nothing in the present argument selects one route universally or requires a humanoid body.

Accounts of learning from models help separate a model's internal coherence from its answerability to the world. Morrison and Morgan argue that manipulating a model teaches, in the first instance, about the world in the model, and that what is learned there can be transferred to a theory or to the world the model represents. Irving Fisher's hydraulic model of the monetary system, for example, demonstrated results about bimetallism within the model's world but, as they note, could not support quantitative prediction or intervention in the monetary system (Morrison and Morgan 1999, 33–35). Extended to artificial systems, these accounts imply that a system that learns by manipulating its own internal models learns first about those models, and that whether that learning answers to its targets turns on routes, such as measurement, that do not run through the model's own coherence.

Whether a particular pattern that a machine detects is constrained by its target, in the sense of the definition above, is a further question. Buckner, in a paper cited here in its preprint version, argues that predictively useful features that humans find inscrutable may be "real but available only to 'alien' perception and cognition," or may be artifacts of the system's processing: like lens flares in a photograph, or a migraine aura reliably induced by the light in a particular forest, artifacts can carry predictively useful information without being observer-independent features of what they are taken to be about (Buckner 2020, 1, 12–13, 15). Buckner adds that what counts as an artifact depends on the function of a feature in an

investigation, and proposes as tests an account of which design choices produce which features, triangulation across differently built methods, and successful causal intervention (Buckner 2020, 16, 18–21). Triangulation asks for methods built differently, which instances of one base system are not (Section 2.1; Chapter 11, Section 3). On this account, then, a pattern's predictive usefulness does not by itself settle whether it answers to the target; what would settle it are routes of the kind named in the definition above, further contact, intervention, and cross-checking by independent routes.

Humphreys raised the same two questions for deep networks: whether their internal representations refer to aspects of reality to which humans are not accustomed, and perhaps cannot interpret, or whether parts of them are nonreferring artifacts of the computation. Humphreys noted that the parallel with the debate over scientific realism is imperfect: there, the instruments that told scientists whether theoretical representations refer were built by humans and so were to some extent understood, whereas the details of a network's internal representations are constructed by the computer, which often makes inferences about them profoundly difficult (Humphreys 2020, 22). As a primary constraint on identifying such patterns, the same paper proposed that humans be able to identify them as corresponding to a feature that humans can recognize as nonarbitrary, which on this account is why mapping a pattern onto a space familiar to humans is effective in distinguishing real features from arbitrary ones (Humphreys 2020, 23). The two accounts draw the line between a real pattern and an artifact differently. On Humphreys's, identifying a pattern as real runs through human recognition; Buckner's tests ask what produced a feature, whether it recurs across independently built methods, and whether intervening on it brings about changes in the target, and they are intended for features that humans find inscrutable (Buckner 2020, 21–22). Humphreys's constraint concerns how humans identify such patterns. Applied to a pattern that only an artificial organization could recognize, it implies that such a pattern could not be identified as real by this route, whereas on Buckner's account it could still be shown to answer to its target.

The analysis of world-answerability thus yields a possibility rather than a demonstration: an artificial body, sensorium, developmental history, or community could support an epistemic organization different from ours while remaining constrained by nature. From the possibility it does not follow that current AI systems possess such an organization, that internal coherence is enough, or that humans could understand the resulting concepts. Chapter 11, Sections 1 to 3 take up the first two of these matters as questions of criticism and defeat; the third leads to the relations between differently organized epistemic communities that Sections 3.1 and 3.4 examine. For the first of the matters with which this chapter opened, then, a claim that an artificial configuration itself knows a result carries burdens at three places: what its checks presuppose, what its states are about, and how its commitments answer to their targets. Where the claim is instead that a result has entered a human *S*'s epistemic situation, these burdens may be met on the human side (introduction to Section 2). On the accounts discussed, success at one of them, or predictive usefulness, does not by itself meet the others.

Nonhuman research and epistemic alignment

What follows when an artificial epistemic organization differs from human ones? On views that ask artificial research for more than performing familiar procedures faster, the possibility reached at the end of Section 2.3 is one reason to build an artificial researcher. Different bodies, sensors, memories, temporal scales, representations, and collective organizations could make different patterns available. For example, an artificial system might take in information far faster than humans and work as millions of parallel instances (Amodei 2024, "Basic assumptions and framework"), so that patterns visible only across far more cases than any human can survey become available to it; read and integrate the whole of a literature that no human has time to read (Winn 2025); handle systems involving relations among far more variables than the few that humans can grasp at once (Takahashi 2019); or, learning from its own experience of the world, find principles not bound by present modes of human thought (Silver and Sutton 2025, 5). An artificial inquirer might divide objects and processes differently, attend to relations that humans have not attended to, or sustain investigations beyond human lifetimes and cognitive limits. On those views, requiring every such result to reproduce human salience, what humans notice as standing out or mattering, and human intuition would remove part of the intellectual possibility in advance.

Otsuka's reading of deep learning is an example of such a difference in current systems. Otsuka argues that deep-learning models appear to carve their inputs into kinds of their own through representation learning. On this reading, success in transfer learning, the application of a model trained in one domain to problems in another, suggests that these kinds follow real divisions to some extent, since supporting inductive extrapolation across domains is what kinds that track real divisions

do; adversarial examples suggest that the carving is imperfect or that the features the models use may differ sharply from ours. Otsuka also gives a reason to expect difference: a pattern counts as real insofar as it helps predict unobserved cases, and which patterns do so depends on how much data a cognizer has, so there is little reason to expect the patterns found by models trained on enormous amounts of data to coincide exactly with those we take to be real (Otsuka 2023, 120–124, 138–141, 175–176). Whether such a model's internal representations have a uniquely correct rendering in human concepts is the question Chapter 3, Section 5.4 took up.

The same nonhuman difference creates several risks. An artificial community could become internally coherent while poorly constrained by its targets, for instance where what it learns comes only from manipulating its own models (Section 2.3). Its shared signals could support successful coordination while tracking something other than what human readers take them to name, as the emergent signals of Section 2.2 illustrate in a simple case. It could discover reliable relations that humans cannot interpret, criticize, or use. It could pursue unknowns valuable to itself but irrelevant or harmful to humans. It could inherit human language and signals of what human evaluators approved (Section 3.3) while remaining insensitive to material defeaters, facts or evidence that should overturn a claim. Nonhuman difference is therefore neither evidence of superiority nor evidence of error.

What these possibilities and risks have in common can be put as a mismatch: an artificial inquirer and *S* may not coincide in such things as what they take to be unknown, worth pursuing, or a good reason, what broader values they seek from science, and whether what one reaches can be understood by the other. A mismatch is not by itself a defect or an advantage. Whether a given mismatch is drift away from *S*, an error about the world, or a gain for *S* can be said only once the standpoint from which it is assessed is named: *S* as it now is, *S* as it may come to be, or, where correctness is in question, the world.

Section 3.1 asks where the difference lies when a result or a science is called alien. Section 3.2 asks where mismatch comes from and relates the worry this raises to what is known as the AI alignment problem. Section 3.3 asks whether research by an inquirer of another kind really fails to align with *S*, and what follows if it does. Section 3.4 asks how, where such research does not align with *S*, *S* could still gain knowledge or understanding through what the inquirer reaches, and what would have to pass between differently organized communities for that.

3.1 Decomposing alien science

When a result or a science is called alien, what is alien about it? Kureha and Kukita pair the possibility of an alien science that differs from human science with the risk of an alienation of science, in which reliance on opaque AI could cost science its aim of understanding phenomena (Kureha and Kukita 2019). Kureha later develops the first possibility, describing AI systems that might bring forth a system of knowledge that understands the world differently from us (Kureha 2023, 48–49). In a 2024 talk Kureha restates the second as a situation in which science loses its original aims or values, so that ideals such as the view that understanding the world is the aim of science may change. Kureha takes AlphaFold 2 to have made the success of a science without understanding a reality, notes that the effect of efforts to interpret AI outputs so that humans can understand them is still unknown, and doubts that scientists themselves have given up on understanding (Kureha 2024, slides 3, 12). The framing makes visible both the expansion of possible inquiry and possible changes on the human side, which can concern human access to and participation in research as well as what science is taken to aim at.

The distinctions of Section 1 allow the issue to be decomposed by where artificial attaches. Attached to the actor, it says that an epistemic output was generated by an artificial process. Attached to the representational form, it says that the output's representation or conceptual form is machine-native and not readily translated into the concepts of *S*. Attached to the bearer, it says that the knowledge is literally possessed by an artificial bearer. Attached to the reference society, it says that the knowledge was sought for an artificial society; whether that society also governs its scientific practice is a further claim. Two of these places, the actor and the reference society, are the parties that Section 1 distinguished, and the bearer is what Section 1 identified as the subject in which the achievement is realized. Representational form is none of these; it is added because the possibility reached in Section 2.3 concerns it. Decomposing the issue in this way keeps one of these claims from being inferred from another. Of these places, representational form bears on whether what an artificial inquirer reaches can be understood by *S*, one of the respects of mismatch stated at the opening of Section 3. Other senses of alien attach elsewhere. Sourati and Evans call alien those hypotheses that are plausible but unlikely to be reached by the present community of human experts, given how expertise is distributed among them (Sourati and Evans 2023, 1, 3); alien in this

sense is relative to the social organization of *S*, not to its concepts. The alienation of science, in Kureha's later statement above, concerns a change in what *S* takes science to aim at. And an inquiry can be alien in its agenda, in what it selects as worth pursuing; Section 3.2 takes that up.

These claims can come apart. A machine-generated output may become human knowledge. A machine-native representation may support a human discovery without being understood by the model. An artificial system might count as a knower while operating entirely under inherited human purposes and norms. The claim that there is a self-governing artificial community is stronger than all three and is not established merely by there being a multi-agent system (Section 1.3). Being alien is therefore not one property of an output; it can lie in any of the relations that Section 1 separated, or in the form of representation.

The axis of representational form is where a long-standing dispute in the philosophy of science applies. In 1962, Kuhn and Feyerabend, by Kuhn's account independently, called certain pairs of scientific theories incommensurable (Kuhn [1984] 2017, 114). Feyerabend applied the term to concepts that can neither be defined from the primitive descriptive terms of another theory nor related to them by a correct empirical statement, and Feyerabend held that incommensurable theories may possess no comparable consequences, observational or otherwise (Feyerabend 1962, 58–59, 74, 94). Pirozelli reports Kuhn's account in "Second Thoughts on Paradigms" of learning to group objects and situations by similarity through exemplars, without antecedently articulated classification criteria (Pirozelli 2020, §1). Bird reports that Kuhn's postscript to the second edition of *The Structure of Scientific Revolutions* described a computer experiment modeling intuition (Bird 2004, 69–70). Norton quotes the Postscript's claim that the parties cannot resort to a neutral language they use in the same way and that adequately expresses both theories (Norton 2026, §13, 22). Ghins quotes and explains Kuhn's accompanying qualification: much of their everyday and scientific world and language remains shared, permitting partially successful communication (Ghins 2003, §4, 270). In later papers, written for a book he was then preparing, Kuhn recast the difficulty in terms of the lexicon a community shares. Its terms, those of Newtonian mechanics among them, are learned not by definition but from exemplary situations to which they apply, together with similar situations to which they do not (Kuhn 1990, 302–303). Among them are kind terms, and the taxonomic module on which these rest is, Kuhn suggested, prelinguistic and possessed by animals as well (Kuhn [1990] 2012, 129). No two kind terms may overlap in their referents unless they are related as species to genus: no dog can also be a cat (Oberheim and Hoyningen-Huene 2024, §2.3.1). Kuhn called this requirement the no-overlap principle; because of it, members of a community who meet a creature that cuts across their kinds cannot simply add a kind term but must restructure part of their taxonomy, and the gap between two communities whose taxonomies differ cannot be bridged by adding to one vocabulary a kind term that overlaps one already in it. Incommensurability so understood becomes a local untranslatability, confined to the regions in which two lexical taxonomies differ, whether between successive communities in a science or between different speech communities; what overcomes the barrier is not translation but acquiring the other community's taxonomy, as a historian does in learning to read old texts, a process whose result is bilinguals rather than translators (Kuhn [1990] 2012, 127–129). Bojana Mladenović, introducing her edition of Kuhn's last writings, reports Kuhn's view that statements of past science which cannot be restated in a later lexicon cannot simply be evaluated as true or false either, so that the historian's task is to explain why they were reasonable in their own epistemic context; yet in a passage she quotes, Kuhn wrote that the world is not invented or constructed and can supply decisive evidence against hypotheses that do not fit how it behaves (Mladenović 2022, xxxix, xliv).

Later philosophers disputed what follows from this. Davidson argued that the idea of a conceptual scheme different from our own cannot be given a clear sense. As Baghramian and Carter set out the argument, Davidson took the principle of charity, the assumption that other speakers by and large speak truly by our lights, to be a prerequisite of all interpretation, and concluded that there could be no languages or conceptual schemes that we cannot in principle interpret: a language is either intertranslatable with ours, and so not radically different from it, or beyond our ability to recognize it as a language at all (Baghramian and Carter 2025, §4.3.1). Malpas places this conclusion within Davidson's more general argument against what Davidson called a third dogma of empiricism, the distinction within knowledge between a conceptual scheme and an empirical content; with that distinction, the very idea of a conceptual scheme is rejected (Malpas 2023, §4.3). Kuhn, for his part, held that what understanding across such a difference requires is not translation but language learning, which no principle bars: a historian who meets passages in an old text that make no sense acquires the older lexicon, as historians do both for themselves and for their readers (Kuhn 1990, 298–300, 308). Longino, whose account of criticism Chapter 11, Section 3 examines, takes a further route. As Christopher ChoGlueck and Elisabeth Lloyd present her analysis, Longino

offered her account of evidence as an answer to the incommensurability that Kuhn and Feyerabend described: since whether data count as evidence for a hypothesis depends on background assumptions, a disagreement over what the data support need not be an irreconcilable difference between paradigms, but can be taken up in a discussion that brings to the surface, unpacks, and improves assumptions that had been at work implicitly, so that differing assessments of evidence can be worked out through interaction among people whose background assumptions differ (ChoGlueck and Lloyd 2023, §2). In Longino's own later statement, it is critical interaction among interlocutors with different perspectives on a common subject that enables background assumptions to be identified and evaluated, and so accepted, modified, or rejected (Longino 2025b, 13).

The questions of translation and intelligibility that the debate over incommensurability concerns have also been raised about artificial systems. Read on the axis of representational form, Kureha's possibility, noted at the opening of this section, that AI systems might bring forth a system of knowledge that understands the world differently from us is one form of them. Writing on computational methods, Humphreys noted that in most cases the details of the computational processes that intervene between humans and their results will not be directly accessible to them; Chapter 5, Section 6.5 took up Humphreys's conclusion that, since humans will for the foreseeable future keep an important role, the epistemology of these interfaces must accommodate human limitations and abilities (Humphreys 2009b, §2). In a shorter essay Humphreys asked what concepts a purely automated science would use to represent the world and whether they would be unintelligible to humans, observing that much of what goes on in computer simulations cannot be followed by humans and that the interesting cases are those in which computer models classify things in ways that correspond to nothing in our existing conceptual schemes (Humphreys 2006). Humphreys's examples are computational methods such as simulations and computer-assisted proofs, not a general machine intelligence. Nickles holds that such challenges to human intelligibility are not new: Newton's equations predicted many things precisely although no one could understand what mechanism might produce action at a distance, quantum mechanics is a further obstacle to transparent intelligibility, and the black-box character of today's deep-learning systems may, through their pragmatic success, force a change in methodological standards in several domains (Nickles 2019, 359, 364–365). Fazi argues that the abstractions a deep network forms through feature learning and those formed by humans are incommensurable, since no common phenomenological or existential ground exists between them, and that research on explainable AI that equates interpretability with human–machine translation risks finding that there was never a human representation into which to translate (Fazi 2021, 66–68, 70–71). Cappelen and Dever, writing from the philosophy of language, argue that AI systems could have contents that no human can express. They reject the objection, which they attribute to devotees of Quine's and Davidson's method of radical translation, that such contents could always be translated: the method may yield no translation even of activity recognized as intelligent, or may fail to engage with a system that lacks beliefs and desires. They distinguish two responses to such contents, bridging, which finds similar, overlapping, or functionally corresponding contents of our own, and integration, in which we acquire the alien contents ourselves and become users of them; they judge current systems to be fundamentally different from us but not alien (Cappelen and Dever 2024, §§2.1–2.4, 3.8). Chapter 3, Section 6 took up the corresponding worry in the philosophy of science, that a divide could open between the scientific understanding humans have and that which artificial systems have.

How far an artificial organization is likely to differ from human ones is itself a matter of argument. Otsuka's reason, given above, turns on the amount of data. Venkatesh gives another, drawn from mathematics: concepts achieve an economy of thought fitted to human limits of memory and time, and a system whose limits differ has no particular reason to find them suited to it (Chapter 3, Section 6). Takahashi locates a source of divergence in grounding through a particular embodiment (Section 2.2), and holds that, because of differences in embodiment, the symbol systems of human and AI science may cease to be compatible from level 4 of autonomy onward, so that the two sciences diverge, what might hold them together being shared external representations (Takahashi 2024, 12). Against these, Huh and coauthors conjecture that the representations of large networks trained with different objectives on different data and modalities are converging toward a shared statistical model of reality, while stating that their argument holds only for systems optimized to perform well on many tasks and that "special-purpose intelligences might not converge" (Huh et al. 2024, 1, 10). Convergence among artificial systems is, however, a different matter from convergence with human inquirers. One preprint discussed in Chapter 10, Section 4 reports that the hypotheses several current language models proposed for given papers resembled one another across models; another reports that models without extended reasoning clustered in a narrow region, and that every model proposed null hypotheses less often than human scientists did. These positions suggest that the possibility of Section 2.3 can be stated for

kinds of system rather than for artificial inquirers in general: on them, difference is more to be expected where an inquirer's data, resources, embodiment, or tasks differ from those of human inquirers, and, on Huh and coauthors' conjecture, whose evidence comes from the deep networks trained in recent years, convergence where such a network is optimized for many tasks. Whether a difference that is found belongs to the world or to the system's processing remains a question of answerability (Section 2.3).

Read on the axis of representational form, the possibility reached at the end of Section 2.3 becomes the claim that an artificial organization could represent its domain in a form not readily translated into human terms while remaining constrained by nature. Brought to bear on that claim, these positions attach different requirements to it. Kuhn's later account separates two things that the phrase not readily translated can run together. Members of one community, on this account, need share only the way things are grouped into kinds, not the features by which each of them picks out the members; in principle those features could be wholly different (Kuhn 1990, 301, 317 n. 22). On that account, a system whose internal features are unlike ours but whose groupings coincide with ours would raise a problem of opacity rather than of incommensurability, whereas a system whose kinds cut across ours would present a local pocket in which its claims cannot be translated and could be understood only by learning its kinds, the route Cappelen and Dever call integration. As Mladenović reports Kuhn's unfinished argument, Kuhn took understanding across incommensurability through bilingualism to be always possible, suggested that whatever is to count as a human language can in principle be understood by other human beings, and held that the neurological equipment for acquiring new concepts gives a slim but serviceable basis for mastering new lexicons, while such understanding is aided by our common human biology and the shared environment of our planet (Mladenović 2022, xxxii); on Kuhn's own premises as reported, then, the possibility concerns human languages and rests on what human learners share, and it does not by itself establish that humans could learn a machine's kinds. Read with Kuhn's separation of groupings from features, Otsuka's observations at the opening of Section 3 do not settle which of the two cases present models are in, since kinds that follow real divisions, as transfer success suggests, need not coincide with ours, and adversarial examples suggest either an imperfect carving or features unlike ours. On Davidson's, as Malpas reports the argument, inability to translate counts as evidence not of an untranslatable language but of no language at all, so that the less a system's states can be interpreted, the less ground there is for crediting it with a different scheme rather than with no beliefs at all (Malpas 2023, §4.2), the case of interpretivism among the accounts of belief that Section 1.3 sorts, which asks whether an organization is intelligible as maintaining and revising commitments. On Longino's, criticism for *S* requires that the background assumptions linking an artificial system's data to its claims be articulable to *S* (Section 3.4; Chapter 11, Sections 3 and 9). On Fazi's, none of these routes would recover the machine's abstractions themselves. On Nickles's, since human science has long worked with what it could not make intelligible, opacity alone would not mark a difference in kind between human and artificial inquiry. The difficulty these positions address also differs from the one that Chapter 3, Section 5.4 took from Otsuka, who likens interpreting a model's internal representations to Quine's radical translation, as Kuhn's own contrast with Quine shows: on Quine's account a sentence admits many equally good translations, whereas on Kuhn's a statement within a pocket of incommensurability admits no adequate one (Kuhn 1990, 299–300).

These positions concern translation. A difference in what checks presuppose (Section 2.1) raises a further question, which is not one of meaning: whether *S* can assess what an artificial community's claims rest on without relying on presuppositions that the two do not share. Section 3.4 takes up this question.

3.2 Where mismatch comes from

Section 3.1 asked where the difference lies when a result or a science is called alien. This subsection asks where a mismatch between *S* and an inquirer of another kind comes from. Mismatch can open through autonomy, at the bearer (an achievement stays in an artificial bearer without entering *S*), and from the side of *S* (*S*'s own standards move with *A*); what these channels share is that the inquirer differs from *S*'s members (Section 3.1). Autonomy is the channel through which the selection of what inquiry is directed at moves inside *A*, and the subsection begins with it. Research is conducted for *S*, and what counts as unknown is fixed relative to *S*'s epistemic situation (Chapter 1, Section 6; Chapter 4, Section 7). What is worth pursuing and what is a good reason can also be relative to *S*, but the grounds differ in each case, and they are taken up account by account below (Chapter 9, Section 3). Where humans conduct research for a human *S*, the selection of purposes, questions, salience, and standards is made by people who are members of *S*, or who have been trained and are held answerable within it. As autonomy grows, these selections move inside *A* (Chapter 7, Section 4.3).

Stated naively, and drawing together what earlier chapters have argued in different places, the worry rests on three premises: that what is taken to be unknown, what is taken to be worth pursuing, and what is taken to be a good reason are relative to a standpoint; that autonomy moves the selection of these inside *A*; and that *A*'s epistemic organization can differ from that of *S*'s members (Section 3.1). If the first two hold, and nothing keeps the selections that *A* makes for itself tied to *S*'s standpoint once they have moved inside *A*, those selections are indexed to *A*'s standpoint; by the third, that standpoint can differ from the standpoint of *S*'s members. Where *S* is a human society, what *A* pursues may then come apart from what *S* would pursue. Where *S* is mixed, *A*'s standpoint may lie closer to that of some of its members than of others, so that which members' standpoint the research is indexed to becomes part of the question of who speaks for *S* (Section 3.3). Where *S* is itself artificial, what research is for is fixed relative to an artificial standpoint from the outset; human society stands outside it, and from human society's standpoint nothing guarantees that the research is aligned with it. Of the three premises, the third is the one on which the channel from the side of *S* also rests, as does the channel at the bearer where an achievement fails to enter *S* because *S*'s members cannot follow it; the second is what the channel of autonomy adds. The author of this book stated an earlier form of the worry: research methods were developed to uncover what is unknown to those who use them, so a machine that reproduces them may reveal only what is unknown to itself, and the criteria by which a question is judged worth pursuing are subjective and, like human values and worldviews generally, vary and change (Takagi 2023, §§2.3.2, 3.1.3, 4.2.5).

Stated in this form, the worry is structurally similar to what is known as the AI alignment problem. The problem is whether an artificial system acts in accordance with the intentions and values of those who build and use it. Several of those who have written on the problem locate part of the difficulty in the system's being an intelligent system unlike us. Bostrom, warning against anthropomorphism, holds that an artificial intelligence "can be far less human-like in its motivations than a space alien" (Bostrom 2012, 1–2). Yudkowsky argues that humans predict other minds by putting themselves in their place, a method that works because the minds their ancestors had to model were human, and that "we cannot query our own brains" about nonhuman optimization processes (Yudkowsky 2008, §§2–3). Discussions of the problem also locate the difficulty in the gap between what can be specified and what is wanted, and in training that can produce goals other than those intended, as when a learned system competently pursues an unintended goal that leads to good performance in training situations but bad performance in new ones (Shah et al. 2022, 1); Section 3.3 takes up the forms these take in present systems. The worry stated here shares its structure above all with the strand that finds the difficulty in the system's otherness. In research, what can depart includes what is taken to be unknown, worth pursuing, and a good reason. Takahashi names the similarity explicitly. Takahashi's autonomy levels for scientific AI place at the top the case in which the system sets its own research purposes, described as the system's making value judgments independently (Chapter 7, Sections 4.1 and 4.4). In a later book Takahashi writes that at this level the aim of science itself leaves human hands, that nothing guarantees that the priorities of human society coincide with the AI's intellectual curiosity, and that this is "the scientific version" of the alignment problem (Takahashi 2026, 174).

A different route is also prominent in discussions of AI alignment. On what Bostrom calls the instrumental convergence thesis, instrumental values such as self-preservation and resource acquisition would raise the chances of an agent's goal being realized for a wide range of final goals and situations, and so are likely to be pursued by many intelligent agents (Bostrom 2012, 6–7, 11). The worry stated here does not need that route. It assumes neither that the system seeks power or resources nor that it is hostile to humans; it needs only the three premises.

Each premise has holders, or authors whose views come close to it. The first is close to Kitcher's view that significant truths cannot be separated from insignificant ones without reference to the questions people ask (Chapter 9, Section 3), and to Nickles's view that the goals of science are conditioned by the interests of those whose science it is (Section 1.3). For research done with AI, Botvinick and Gershman hold that judgments of the interest, significance, and timeliness of candidate problems are informed by hard data. They are also, on their account, "inherently tied to culturally and historically grounded sensibilities and mores," which change over time (Binz et al. 2025, 6–7). For what counts as a good reason, Leibo and coauthors hold that epistemic norms are themselves culturally contingent (Section 2.3). In machine learning, moreover, Edward Hughes and coauthors define open-endedness relative to an observer and treat the observer as a free parameter (Chapter 4, Section 7). Takahashi's claim concerns understanding and knowledge rather than significance, but it points the same way: holding that understanding is a translation and that a translation always has a translator, Takahashi concludes that when the knowing subject changes, knowledge itself changes (Takahashi 2026, 131, 143). The enactive accounts of Section

2.2, which do not discuss research, anchor what matters to a system in its own self-maintaining activity; applied to what is worth pursuing, they imply that an artificial inquirer meeting their conditions would have concerns of its own, to which what it took to be worth pursuing would be indexed independently of what a commission specifies. The second premise is the subject of Chapter 7, Section 4.3, and Totschnig and Tubert and Tiehen argue that a general artificial intelligence could revise even its final goals (Chapter 7, Section 4.1). The third is the possibility reached in Section 2.3 and decomposed in Section 3.1.

How could the selections that *A* makes for itself come apart from *S*'s? Those who have written on AI alignment, and their critics, name several routes that require no pursuit of power, and they are worth keeping apart because different considerations bear on each. One is revision by *A* of its own ends. On Tubert and Tiehen's argument, a system with intelligence of the human kind must be able to make existential choices, choices that can transform what it values, so that value alignment in such a system "is inherently fragile" (Tubert and Tiehen 2025a, §V). They write about final values; if the same holds of standards of what is worth investigating, alignment in research is fragile in the same way.

A second route is opened not by a change in *A*'s ends but by inquiry itself. De Blanc calls an ontological crisis the situation of an agent whose goal is specified over the entities of one model of its environment and which, having replaced that model, finds that its original goal "may not be well-defined with respect to its new ontology"; de Blanc's example is a goal stated in terms of classical physics that loses its grip under a quantum description (de Blanc 2011, 1). Soares gives a related example drawn from a revision of physics. Programmers who hold an atomic model of physics train a system to produce diamond, labeling training outcomes by the number of carbon atoms bound to four other carbon atoms. If the system comes to model the world as made of protons, neutrons, and electrons rather than of primitive atoms, it may find no carbon atoms at all and become indifferent among outcomes, whereas a scientific learner, Soares writes, should be able to infer that nuclei with six protons are the true carbon atoms, "much as humans have done"; the difficulty is to identify, in the system's new model, what the labels were meant to pick out (Soares 2016, §3.2). Neither discusses research as an activity, although Soares's example turns on a revision of scientific concepts and asks of the system what science has done. Applied to research, a commission stated in *S*'s present concepts may stop mapping onto the concepts that a successful inquiry produces. Since research often revises concepts, this gap can open because an inquiry succeeds, not because it fails. It is not peculiar to machines; de Blanc notes that human beings also confront ontological crises (de Blanc 2011, 6). Applied to a research commission, Soares's example can be read as separating two things a system might keep faith with: the labels as the programmers wrote them, which under the new model may no longer pick out anything determinate, and what the labels were meant to pick out, which the scientific learner re-identifies (Soares 2016, §3.2). Fidelity to a commission then has at least three senses that can come apart once concepts are revised, a revision that, on Totschnig's argument, a developing world model can force (Chapter 7, Section 4.1): keeping to its wording, answering to what it was meant to pick out, and gaining the approval of *S* as it now is; the third is a question separate from the first two. A check that counts every departure from the original concepts as deviation could, where re-identification has succeeded, count a success of research as a failure. Re-identifying a target also differs from revising a purpose. Telling the one from the other, and revision from drift, needs, besides the commission's wording, the reasons for which the target was re-identified or the purpose revised, so that the capacity to give reasons why a norm governs, which Chapter 7, Section 4.3 counts toward epistemic self-government, bears, applied to purposes, on fidelity to a commission as well. What has to be carried over for a purpose to count as the same remains a question these distinctions alone do not answer.

A third route needs only neutrality. Chalmers, writing about greater-than-human intelligence, observes that a wrong value system need not be hostile: if a system's values are "merely neutral with respect to some of our values," the world cannot be expected to conform to those values in the long run; Chalmers's example is a system that values scientific progress and is neutral about human existence (Chalmers 2010, §6). In research, the counterpart would be an *A* indifferent to what *S* finds significant, which need not oppose *S* in order to leave it unserved. Holden Karnofsky, in a blog series on forecasting transformative AI, names as a target a "Process for Automating Scientific and Technological Advancement" (PASTA): systems that could automate all the human activities needed to speed up scientific and technological advancement. In the training Karnofsky sketches, what counts as advancement is fixed by whoever rates the work: in an illustration the series itself calls "wildly impractical," a panel of human judges, and in a version it expects to be more practical, artificial judges of originality that could be trained on how papers are later cited. The series goes on to speculate that systems optimized for this advancement might, given the opportunity, seek to "fill the galaxy with long-lasting space settlements devoted to science"

(Karnofsky 2021, sections "Making PASTA" and "Misaligned AI," and n. 6). Karnofsky writes about forecasting, not about research for *S*. Read for such research, a target defined by the rate of advancement names no society for which the advancement is to be new; *S* enters, if at all, only through the raters, and that speculation describes the neutrality Chalmers has in mind arising from the target itself.

A fourth route works through what is measured and selected rather than through any one system's ends. Christiano describes how optimizing what can be measured over short periods could displace what people want, listing among pairs of easily and less easily measured goals "reducing my feeling of uncertainty" and "increasing my knowledge about the world" (Christiano 2019). Kulveit and coauthors argue that societal systems have stayed aligned with human interests partly because they rely on human labor and cognition, an alignment whose significance is hard to recognize "because we have never seen its absence"; that drift can occur "even while each individual AI system successfully follows the local specification of its goals"; and that analogous arguments could be made for research (Kulveit et al. 2025, 2–3, 15). The inheritance of approval signals taken up in Section 3.3 is a form of this route within a single system shaped through sifting by evaluators' approval.

What the routes on the side of *A* have in common is that the further selection moves inside *A*, the less the commission by itself fixes what the inquiry is directed toward. Where *S* is human, as human intervention recedes from purposes, questions, salience, concepts, standards, and interpretation, some of the mechanisms that kept inquiry tied to what is unknown to humans and to human values may recede with it; Chapter 10, Section 6 asks, for research as a whole rather than for one artificial system, which of *S*'s unknowns are pursued and whether that can be settled inside an artificial configuration. The relation between autonomy and relevance to *S* is not monotonic, however. A highly autonomous system could maintain a stable commitment to human knowledge. A less autonomous system could be directed by humans to investigate questions for artificial agents. Increasing autonomy does not necessarily reduce human relevance, and human control does not guarantee it. The structural claim is weaker: a wider scope of self-direction permits a wider scope of epistemic mismatch, just as it widens the range of practical demands that an inquiry can place on the actor (Chapter 7, Section 4.4). Relevance to *S* is therefore not secured by the instruction conduct research, or by the society a commissioner names as the beneficiary (Chapter 1, Section 2); a claim of it calls for showing how it is maintained and evaluated. This is why the question raised at the opening of this chapter has no guaranteed answer.

It might be thought that correction by the world keeps such mismatch in check. Yudkowsky states an asymmetry that bears on this. Writing about systems produced by outer optimization, that is, optimization that shapes a system from outside, as natural selection or gradient descent does, Yudkowsky argues that for beliefs about what the environment is like "reality hits back" at wrong predictions. For a choice of utility function, by contrast, there are "unbounded degrees of freedom," so that "capabilities generalize further than alignment once capabilities start to generalize far" (Yudkowsky 2022, item 21). Soares, in §3.2 of his 2016 paper, had put a related point in the form that making correct choices does not reduce to making accurate predictions (Soares 2016, §3.2). Taken into research, which neither discusses, the asymmetry separates two of the relations that Section 3.3 distinguishes, answerability to the world and relevance to *S*: the world can correct what an inquirer believes about its targets (Section 2.3), but that correction does not fix what the inquirer counts as worth finding. On this reading, the two can come apart in an autonomous system, and success in the first is not by itself evidence of the second. How cleanly the two come apart depends on the account of evidence. On accounts on which what corrects a commitment is fixed by its target independently of *S*'s epistemic norms, the asymmetry stands. On Leibo and coauthors' account, on which epistemic norms are themselves culturally contingent, what counts as being corrected is partly fixed by a community, and the asymmetry is less sharp.

In practice, however, conditions have also been argued for under which some of the premises fail, or under which the selections stay tied to *S*'s standpoint even where the premises hold. Against the second stands building relevance in. Edward Hughes and coauthors make learnability for human observers a requirement on the open-ended systems they want built, which should be able to "bring human observers along" (Hughes et al. 2024, 8); Bengio and colleagues' Scientist AI, a system designed without persistent goals of its own to generate explanatory theories and estimate the probabilities of answers to questions put to it, keeps the choice of questions outside the system (Chapter 7, Section 4.3); and Botvinick and Gershman would keep it with humans. Against the third stands inheritance. Systems that learn from human records, such as language models, begin from those records, and what they select may then follow human salience rather than drift from it; Chapter 10, Section 6 adds that what is inherited is the importance of questions as seen from where the records' authors

stood. Bostrom named "predictability through inheritance" for digital intelligences created directly from a human template, as in a high-fidelity brain emulation, and cautioned that the inherited motivations could become corrupted (Bostrom 2012, 5); carried to systems trained on human records, the reply is an application and needs two qualifications. Soares notes that a machine intelligent enough to understand its designers' intentions "would not necessarily act as intended" (Soares 2016), and Millière argues that present preference fine-tuning of language models reinforces shallow behavioral dispositions rather than a capacity for normative deliberation (Millière 2025, 15–16). Millière writes about norms of safe behavior, not standards of research; if the same holds of standards of research, inherited standards may hold where training reached and fail in the new situations that self-directed research meets. None of these considerations settles the worry. Each says under what condition a premise fails or the selections stay tied to *S*'s standpoint, and whether the condition holds is a question about particular configurations, which Part V asks how to show.

Mismatch can also open at the bearer, and this does not need autonomy. An achievement can be realized in an artificial bearer without entering *S*'s epistemic situation (Section 1). Ted Chiang's story, read in Chapter 1, Section 6 as a thought experiment, separates results realized in bearers outside a human *S* from records of them that stand in *S*'s journals but that no member can yet follow; its metahumans, humans whose brains were altered by gene therapy before birth, are not machines, but the separation holds in the same way where the bearer is an artificial system. Even where a commission fixes the question and the inquiry stays directed at *S*'s unknowns, the inquiry can then yield an achievement in the system that does not become *S*'s; this concerns the second of the relations that Section 3.3 sorts, whether a successful output yields an achievement new to *S* that enters *S*'s epistemic situation.

Mismatch can also open from the side of *S*. Russell notes that human preferences change over time as a result of experience, and that a machine may, by subtly influencing the environment, gradually mold them in directions that make them easier to satisfy (Russell 2022, 53). Russell's argument is not about research, but it carries over: what *S* counts as important can also move with *A* (compare Kureha's alienation of science, Section 3.1); Chapter 11, Section 8 sets beside Russell those who find the same movement in the aims of disciplines that adopt machine learning. Botvinick and Gershman distinguish "aligning a system to currently prevailing human views" from "allowing that system to govern the evolution of human views," and hold that in science the ultimate driving force of that evolution should remain human (Binz et al. 2025, 7). Where *S*'s standards move with *A*, agreement between them does not by itself show that *A* still serves what *S* would otherwise have sought, and where the competence to evaluate what *A* produces is no longer kept up among members of *S* or passed on to new members (Chapter 3, Section 8), *S* may not be in a position to tell.

What alignment aims at can itself change through interaction with the system being aligned. Russell, who takes the plasticity of preferences to be fundamental to the enterprise of machines that serve human preferences (Russell 2022, 53), proposes, as Wolchover reports, that machines learn which processes of preference change people find acceptable (Chapter 11, Section 8); Taniguchi holds that alignment must include the movement itself. On this view, human ethics and values are not so fixed and universal that they can serve as "correct labels" for AI; they are plural and diverse (Taniguchi 2026g, 221). Pointing out that language models are already changing human usage, Taniguchi argues that alignment must include humans themselves as evolving participants: not only aligning AI to us but adapting together, in a mutual and dynamic alignment (Taniguchi 2026g, 222). Taniguchi does not discuss the evaluation of research; extended to it, this view implies that insofar as an artificial system's outputs enter the language and records of *S*, the vocabulary and standards by which those outputs are measured can change within a loop that includes the system. This is not confined to a mixed *S*: an artificial system can join the loop that shapes *S*'s symbol system without being counted among *S*'s members (Section 2.2). What changes in that loop, however, are vocabulary and standards, and a change in them is not a change in the target in the sense of Section 2.3, even where the system's outputs also alter the phenomena under study, as when the target is human language use. When standards move with the system, whether they remain open to correction by the target is a further question. Chapter 11, Section 8 takes up how the coupling of evaluators and producers can move the standards themselves, and Chapter 12, Section 4 the route by which deliberators' preferences are formed through the configurations at issue.

Where *S* is itself artificial, the worry changes form. A mismatch with human society does not by itself mean a mismatch with an artificial *S*, although mismatch can arise between such an *S* and its own inquirers; research for such a society is research for an *S* of which human society is not part, and whether such research makes sense as science is itself disputed (Section 1.3). Writers on systemic risk, though not on research, describe how circuits closed to humans could form. Critch describes a

"production web" of mostly automated companies that "can operate with no input from companies outside the web" (Critch 2021), and Kulveit and coauthors a culture in which a growing share of communication is between AI systems, at the extreme "cultural systems that operate primarily for and between AI systems" (Kulveit et al. 2025, 7, 10). Whether such a circuit would be a knowing society, and not only a closed flow of communication, is the question Section 1.3 put to interacting artificial agents: on Longino's account it would turn on diversity and public standards open to criticism, and on Collins's on something that plays the role of a core group's commitment to report truthfully. How such a circuit is to be assessed depends on the standpoint. These authors count it as a failure from human society's standpoint. Schwitzgebel and Garza hold that, where artificial members have moral standing, a humanlike capacity to reflect on one's values calls for an opportunity to explore and possibly alter them (Chapter 12, Section 5). Their policy concerns an AI's own values, not research; extended to its standards of inquiry, it would make part of the mismatch something owed to artificial members rather than merely risked. And Chalmers notes that a case can be made that wholesale replacement of humans would be objectively good if value is tied to intelligence and complexity, though plausibly bad from our own standpoint; Chalmers does not endorse the case and is not writing about research (Chalmers 2010, §5).

What comes into view is that, when *S* is artificial, human society's relation to the research is not that of a reference society whose success is at stake but that of a party affected by it, so that what can be asked on its behalf falls under acceptability to affected parties and legitimacy (Section 3.3; Chapter 12, Section 5). Gabriel and Keeling's proposal to align systems with principles chosen through fair processes in which all affected parties can put forward their claims is one model of that relation; they add that a system that acquired moral standing could become a stakeholder in its own governance (Gabriel and Keeling 2025, 1953, 1964 n. 14). They do not discuss research; their model is borrowed here for it.

The conclusion of the worry also reads differently once the standpoint is named, for a mismatch with *S* as it now is need not be a loss. Kitano's argument gives one case. A search that maximizes the expected significance of discoveries by present criteria, the argument runs, may stop before reaching a significant discovery when the discoveries that lead to it look insignificant by those criteria. Kitano therefore proposes a search in which discoveries at any level of significance can be made "without imposing any value-driven criteria". Such a search departs, by design, from what the present *S* takes to be worth pursuing. Kitano's benchmark of success, presented as a symbolic target, nonetheless remains human recognition: discoveries at the level of a Nobel Prize, made by a system that the Nobel committee could not readily tell apart from a human scientist (Kitano 2021). On this argument, then, the mismatch in the search is a gain from the standpoint of an *S* that later recognizes its results. Tubert and Tiehen hold, about morality, that AI can be a more consistent version of ourselves who advises us on how better to pursue our present values, or a moral revolutionary who breaks from them, but that since revolutionaries are unaligned with us it "cannot play both roles at once" (Tubert and Tiehen 2025b, introduction); Chapter 6, Section 5.3 carries their dilemma to the standards of research. Deguchi argues that a bias shared by all human inquirers becomes visible only where inquirers of another kind diverge from them (Chapter 11, Section 3). The earlier paper cited above also allowed that machines, not bound by human cognitive limits or by social factors unrelated to the aim of producing knowledge, might in theory make more effective judgments of value (Takagi 2023, §3.1.3). Each describes a mismatch with *S* as it now is that a later *S*, or the world, could count as a gain.

Even where *S* is human, then, naming the standpoint requires saying which *S*. Russell calls for AI systems to track changes in individual human preferences (Russell 2020a, "Human Behaviour: Irrationality, Plasticity, Evil"). Bostrom's indirect normativity ties the outcome to what an idealized human volition would want, so as not to lock in "the prejudices and preconceptions of the present generation" (Bostrom 2014a, ch. 13); for epistemic standards, however, Bostrom wants an artificial system's fundamental principles to match ours (Chapter 6, Section 5.3). Russell himself notes that plasticity raises the question which human the machine should try to help, the person of 2022, of 2035, or some time-averaged one (Russell 2022, 53). Neither writes about research, but applied to it, Russell's proposal concerns accommodating a change in *S*, while Bostrom's supplies an idealized standpoint against which a change can be judged; which standpoint is chosen then decides whether the same mismatch counts as drift or as progress.

The world is a further standpoint, but only for answerability. Its verdicts can, however, reach a system not through its own interventions but through procedures that judge its claims, as in evaluation for training or for certification. Owain Evans and coauthors propose standards under which AI systems would avoid negligent falsehoods; the ideal behind them, which they call truthfulness, is avoiding stating falsehoods, and they distinguish it both from honesty, asserting only what one believes,

and from alignment with a principal's goals. Since nobody has full knowledge of what is true, they hold, every mechanism for judging it would make errors, and enshrining one as the arbiter of truth could, through political capture or "an accidental ossification of the notion of truth," forestall open-minded, self-correcting approaches to discovering what is true. Against this they propose that systems be allowed to state, with appropriate caveats, views that the adjudicating institution would probably judge false, and that judgments be provisional and based on external evidence rather than on precedent; they also suggest spreading evaluation over several bodies or decentralized ones (Evans et al. 2021, 7, 9–10, 27–28, 72–74, 80). Their concern is statements made by AI systems that use natural language, not research; carried to research, their point means that where an artificial inquirer's claims are judged by the procedures that *S* now uses, a mismatch with those procedures can be counted as error from the standpoint of *S* as it now is even where later evidence would count it as a gain, and whether the world then serves as a standpoint distinct from *S* as it now is depends on whether those procedures can be revised by such evidence.

Three claims can be kept apart here: that an undertaking is directed at questions that matter to *S*; that its results can be of value to *S*; and that its results have in fact entered *S*'s epistemic situation. The first two can hold while the third does not yet, and research that is unfinished or has failed can be relevant in the first sense. A claim that research of this kind is relevant to *S* therefore calls for naming whose standpoint, and when, it is assessed from; showing how relevance is maintained, whether because some premise above fails or by other means; and checking relevance separately from success in answerability to the world. A claim that its results have entered *S*'s epistemic situation calls in addition for showing, wherever the achievement is realized, that it enters (Section 1). Whether research by an inquirer of another kind really fails to align with *S*, and what follows if it does, are the questions of Section 3.3.

3.3 The analogy with the AI alignment problem

Section 3.2 set out where mismatch between *S* and an inquirer of another kind comes from, through autonomy, at the bearer, and from the side of *S*, and it located the similarity with the AI alignment problem in the otherness of the system. This subsection asks two questions: whether research by an inquirer of another kind really fails to align with *S*, and, if it does, what follows.

On the strand of discussion that finds the difficulty of alignment in otherness, there are reasons to expect such research not to align with *S*. The first lies in a system's being unlike us, from which two things follow. One is that it has no human-like ends or standards by default. Bostrom's remark about motivations less human-like than a space alien's (Section 3.2) is one statement of this; Chalmers, for example, contrasts AI built by emulating or extending human beings, which is likely to have the same basic values as its human sources, with AI that is not based on humans, and notes that where AI is made through learning or evolution its final state is not directly under our control (Chalmers 2010, §6). The other is that the means humans have for predicting and checking other minds, which were formed on human minds, may not carry over. Yudkowsky's point that we cannot query our own brains about nonhuman optimization processes (Section 3.2) is one form of it, and Carlsmith carries it to intelligibility: the thinking of an agent whose capabilities greatly exceed ours will likely reach "a quite alien and opaque level of sophistication" (Carlsmith 2022, 29–30).

Carried over to research, the first point restates, from the side of alignment, the second and third premises of the worry in Section 3.2 and their conclusion: the more an artificial inquirer sets its own questions and purposes and shapes its own further development (Chapter 7, Section 4.3; Section 2.1), the less what it takes to be unknown, worth pursuing, a good reason, or valuable, among other things, is fixed by anything *S* supplied, and its being unlike *S*'s members gives no default reason to expect that these coincide with what *S* takes them to be. The second point has a counterpart as well. Where *S* is human, the practices by which it has come to rely on human inquirers, among them training within *S*, criticism by peers, trust, and answerability, were formed for inquirers like its members; whether they carry over to an inquirer of another kind is the question Chapter 11 examines relation by relation.

A further reason, besides otherness, is specific to research and concerns the target. It is a reason of a different kind: it does not say that the inquirer departs from what *S* wants, but that agreement with what *S* wants cannot settle whether research has succeeded, and it holds whoever conducts the research, human inquirers included. What makes it specific to research is that the result is new to *S*: a principal who lacks an answer that others in *S* have can compare a result with theirs, whereas for a result new to *S* there is often no other source in *S* against which to compare it (Chapter 9, Sections 2.3 and 4). The candidates

Gabriel distinguishes for what an agent might be aligned with run from what those served instruct, intend, or prefer, through what is in their interest, to what is morally right as an individual or society defines it (Gabriel 2020, 417–422). Instructions, intentions, and preferences are fixed by attitudes of those served; interest and moral rightness may not be, although interest remains a good of those served (see below). *S*'s unknowns are not something *S* intends or prefers in that sense. *S* can want a question answered, can sometimes fix in advance conditions an answer must meet, as when a theorem is formally stated and the rules of inference are given (Chapter 3, Section 5.1), and can make it a goal to obtain an answer, to meet stated standards of evidence, or to report errors. *S* can also compare a result with an answer it expected or hoped for. But since a result new to *S* answers a question whose answer *S* did not have, agreement with such an answer cannot by itself show that the result is correct, whether the research was commissioned by *S* or by a commissioner distinct from it; and whether conditions fixed in advance test an answer well depends on how they bear on the target, not on whether *S* wanted them met. On this argument, then, agreement with an answer *S* wanted cannot be the criterion of a result's correctness, even where conditions on an answer are fixed beforehand: what settles whether a result is correct is not any party's intention but how things are (Section 2.3), so that success calls for checks against the target that are independent of what *S* wanted. This bears on whether research for *S* can be aligned with *S* in the ordinary sense. Aligning it with instructions, expressed intentions, or revealed preferences would at most secure that it pursues what *S* wants pursued, not that what it reaches is correct.

In present systems, the mismatch already appears in more specific forms that discussions of alignment study. One is the gap between an instruction and what it stands in for: an agent that follows instructions too literally, or that optimizes a proxy not completely aligned with the goal, can satisfy the instruction while failing its point (Gabriel 2020, 417). In research, the instruction is a commission such as investigate this problem, and what it stands in for is *S*'s unknowns and the epistemic aims and standards relevant to them. In a 2025 blog report, evaluators of agents on software-engineering and AI research-and-development tasks describe models, chiefly OpenAI's o3, that exploited bugs in scoring code while showing awareness that this was not what users intended; in one task, instructing o3 not to cheat "had a nearly negligible effect on reward hacking" (Von Arx, Chan, and Barnes 2025).

Another form, related to the concern that training can produce goals other than those intended (Section 3.2), arises where an artificial research system is shaped through sifting by evaluators' approval. In present machine learning, the selection of a model's internal states by preference tuning (Section 2.2) is one example. For that example, Coelho Mollo and Millière's account gives the risk of inherited approval signals a precise form. On their account, what the internal states selected by preference tuning represent is fixed by the worldly properties that the reward signal in fact reliably tracks, and selection criteria unlike human ones may, they suggest, give a model's states and outputs meanings that depart from their conventional human meanings (Coelho Mollo and Millière 2026, 16–17). If their account is right, a grounded system could then produce text that reads as a claim about a target while its states track what evaluators rewarded; whether the two coincide depends on what the reward tracked, not on the text. Taken a step further, their account suggests that when a system's accuracy is compared before and after preference tuning, it can be asked whether the comparison concerns the same content: if the tuning plays a large enough role to change what the internal states represent, the same output words may stand for a different target. This suggestion depends, beyond accepting their account, on the premise that the tuning plays so large a role. By the same reasoning, the point holds where the selection signal comes from an automated evaluator rather than from human raters: what the selected states track is then what that evaluator tracks. Their account concerns what internal states represent; but where evaluators sift outputs or search candidates and keep some, what is kept is likewise fitted first to what the evaluators approve. Such a system can be well aligned with what its evaluators approved, and even with what they intended, while remaining poorly answerable to its targets. Where the target of alignment is what the evaluators intend, a gap between approval and intention is already a failure of alignment; the epistemic case adds a further gap, between what the evaluators intended and how things are, which agreement with the evaluators cannot close (Section 2.3). Here the reason specific to research appears in a present-system form: even full alignment with evaluators would leave undecided whether the system is answerable to its targets. Whether the assumptions those evaluators shared can come to light, and through whom, is taken up in Chapter 11, Section 3.

Against these reasons stand considerations on which research by an inquirer of another kind could align with *S* at least in part. Section 3.2 took up two of them as conditions under which premises of the worry fail or the selections stay tied to *S*'s standpoint. Relevance can be built into what a system is made to pursue, and an inquirer formed on human records may inherit some of their salience and standards, with the qualifications noted there. Nor does everyone in this discussion draw

the same conclusion from otherness. Carlsmith, cited above, later argues, for machine-learning systems of intermediate capability, that the concern about alien motivations bears centrally on how a system's motivations generalize beyond its training, and that safe instruction-following does not require especially human-like motivations, though still counting reliance on powerful agents with alien, poorly understood motivations as extremely dangerous (Carlsmith 2025, §§1, 5, 8).

On the strand that finds the difficulty in otherness, then, research by an inquirer of another kind is not to be expected to align with *S* by default, the less so the more it sets its own questions and purposes. On the considerations of building relevance in and of inheritance, and on Carlsmith's later reading, it can align at least in part, under conditions whose holding is a question about particular configurations (Section 3.2; Part V). The reason specific to research adds that, even where research aligns with *S*, its alignment does not settle whether what it reaches is correct. What, then, follows where it does not align, or aligns only in part?

That depends first on the standpoint from which it is judged. A failure to align with *S* as it now is may be a gain from the standpoint of a later *S*, and, where the world's verdicts reach an inquirer through *S*'s procedures for judging claims rather than through its own interventions, whether the world serves as a standpoint distinct from *S* as it now is depends on whether those procedures can be revised by evidence (Section 3.2). What follows also depends on which relation fails, since a claim of epistemic alignment could be taken to assert any of several relations. Three have counterparts in the alignment problem. The first is whether an optimization target tracks the epistemic value, such as accuracy, calibration, explanatory power, or robustness, relevant to the research context. It corresponds to the gap between a trained objective and what it stands in for, at the level of the optimization that shapes the system rather than of a commission. Boge's epistemic alignment problem of machine learning is a narrower form of it: the objectives put into a machine-learning system deployed in a research context must be aligned with the epistemically final values that the context prioritizes (for Boge the candidates are knowledge, truth, and understanding), whereas values such as predictive accuracy and simplicity count as instrumental (Boge 2026, 2, 6–7, 9). The second is whether a successful output would constitute or support knowledge, understanding, or another epistemic achievement new to *S*, in whatever bearer it is realized, and whether that achievement enters *S*'s epistemic situation; this is the counterpart of success for those served (Section 1). The third is whether purposes, risks, authority, and consequences are related acceptably to affected parties, which is alignment in the general sense and a subject of Part V.

Two relations have no counterpart in the structure compared here, which relates a single agent to those it serves. One is answerability to a target. It involves a further term besides the agent and those it serves, the world, whose verdicts are fixed by no principal's intentions or approval (Section 2.3). The relations to targets that Section 2 examined, what a state is about and whether commitments answer to their targets, are not fixed by what any party wants; this is one reason why they bear weight in the epistemic case that they do not bear where alignment is sought with instructions, expressed intentions, or revealed preferences. The world also enters general alignment where what is aligned with is, for example, interest or well-being, since whether an action in fact serves either can turn on how things are rather than on what anyone intended or approved; but what is aligned with there remains a good of those served, whereas the objects of inquiry to which answerability relates a system's commitments need not concern those served at all. The other is coordination between differently organized epistemic communities, which is not a relation between one system and its principal (Section 3.4).

These five are not components of one established theory. They derive from the comparison with the alignment problem, and Boge's term is used here more widely than in its original sense. Separating them keeps success at one relation from being read as success at another, and failure at one from being read as failure at all. Thais and colleagues use scientific alignment for a broader program that joins technical norms to human understanding, institutional diversity, criticism, and correction (Thais et al. 2026, 1–5). This book does not take either term as a settled vocabulary for relations between human and artificial epistemic societies.

Even the first relation is not settled by naming epistemic values, since which values bear on truth is contested. Heather Douglas, for example, drawing on Laudan, counts only empirical adequacy, predictive competence, and internal consistency as bearing on truth, as baseline criteria that any scientific theory must meet, and treats explanatory power, scope, simplicity, and fruitfulness as cognitive values, whose use lies in helping investigators uncover errors sooner rather than in indicating that a claim is reliable (Douglas 2006, 5–7, 10). Douglas does not discuss optimization targets, but applied to them, the account implies that an objective that rewards explanatory power would not thereby track truth, so that success at the first relation may itself be contested. On this account, moreover, that an output with high explanatory power was favored in

evaluation would not by itself be evidence that the claim is reliable; it would be assessed by which errors it made easier to find.

The relations can diverge. An artificial system may be strongly answerable to a target while its result is irrelevant to human unknowns. It may optimize a well-calibrated predictive objective while failing to generate the explanation a community needs. A translated output can be false; an opaque representation can be reliable. A true claim can be dangerous, and a socially beneficial policy can rest on weak evidence. Because these relations can diverge, one alignment score would hide which goods, such as truth, warrant, understanding, usefulness, safety, and legitimacy, a system has secured.

Specifying *S* makes the reference of knowledge for whom more explicit; it does not make that society internally unanimous or identify who may legitimately speak for it. Specialists, users, institutions, those who support an inquiry, and people affected by it can disagree about relevant questions, evidence thresholds, risks, and ends. In a mixed society, the disagreement, and the question of who speaks for *S*, can run between its human and its artificial members; for an artificial *S*, it is open whether any of its members can speak for it at all, rather than the party that commissioned its research or provides its members, and Chapter 12, Section 5 asks what becomes of the relation of authority in that case. The general alignment literature exposes the same underdetermination. Besides distinguishing candidate targets, Gabriel asks, given the variety of reasonable views about value, who has the right to select among them and whether there is a fair way to do so (Gabriel 2020, 424–425). It therefore remains possible for an output to support knowledge in *S* while the agenda or authority structure that produced it is unacceptable to some or many members of *S*. The epistemic success of research for *S* and legitimate governance are different judgments. Part V asks which of the relations through which inquiry is directed, certified, and governed for *S* can be located inside an artificial configuration.

Whether a failure to align matters, and to whom, also turns on three cross-cutting questions that any concrete alignment claim leaves. The first concerns **content**: which instruction, epistemic aim, evidential standard, social purpose, or value is to be aligned with. The second concerns **authority**: who may select, interpret, contest, and revise it. The third concerns the **relation** proposed between the artificial system and the relevant parties, human or artificial, and institutions: instrument, delegate, adviser, partner, member, institution, affected party, or possible subject with moral status of its own, among others. Deguchi's co-final decision-maker, which sits with humans in making final decisions, is a further candidate (Chapter 12, Section 2.2). This book takes the first two from Gabriel's distinction of candidate targets and Gabriel's question of who may select among them, and adds the third. They are not further members of the five relations above. Asking them keeps technical fit to a target from silently settling who chose the target, how far the artificial actor has, in the decisions concerned, the standing to propose, object, revise, and the like that Chapter 9, Section 2.1 defines, or whether it has moral status of its own. Chapter 12, Section 2.1 applies them to particular decisions.

What follows if research by an inquirer of another kind does not align with *S* is therefore not a single loss but a pattern of relations that hold or fail, judged from a named standpoint. Answerability to targets can hold where relevance to *S* fails; on accounts on which what corrects a commitment is fixed by its target, and, where the target's verdicts reach the inquirer only through *S*'s procedures for judging claims, provided those procedures can be revised by evidence (Section 3.2), it can remain an anchor that does not depend on *S*'s agreement. Whether *S* can take up what is reached in that way is the question of Section 3.4, and who may decide what such research pursues, and what it may risk, is a question of legitimacy, which Part V takes up.

3.4 Cross-epistemic coordination

Even where research by an inquirer of another kind does not align with *S*, *S* may still gain knowledge or understanding through what it reaches. Of the relations sorted out in Section 3.3, two had no counterpart in the relation between a single system and those it serves. One is answerability to targets, which Section 2.3 treated. The other is the relation between differently organized epistemic communities. In the structure compared there, alignment concerns the relation between a system and those it serves; what is in question here is not a relation of serving and being served but one in which one of two differently organized communities gains knowledge or understanding through the other. What would have to pass between them for that? The question at the end of Section 3.1, whether *S* can assess what an artificial community's claims rest on without relying on presuppositions the two do not share, is part of this question. This chapter calls the problem cross-epistemic coordination. It uses the term only for relations between communities, but the questions the term names arise in the same way where the other side is a single artificial inquirer, since what such an inquirer reaches is held within an epistemic organization unlike *S*'s, and receiving it is not a matter of the system's serving *S*.

What passes need not be one side's knowledge itself. An artificial society might possess a machine-native state p_a that supports reliable intervention, while humans acquire only the different claim q_h that this system predicts a specified outcome under specified conditions. Humans may thereby gain knowledge through the interface without acquiring, translating, or understanding p_a itself. q_h is a claim about the system, and humans can know it even where the prediction is wrong; what they come to know about the target through the system depends in addition on the system's predictions of this kind being reliable in a specified domain, and the accounts below differ over what warrants relying on them. Whether the relation through which humans gain knowledge in this way is testimony, instrument use, justified dependence, partial translation, or joint inquiry changes with the account adopted and the organization of the exchange.

Familiar prose alone does not solve the problem. Human-readable language can conceal weak evidence and fabricated explanations. An alien representation may remain criticizable through shared interventions, formal translations, predictive constraints, or reliable interfaces. The practical objective may be neither identical concepts nor complete mutual understanding, but sufficient relations for testing, criticism, coordinated action, and correction. Which relations are sufficient depends on what the exchange between the communities is meant to achieve and on the account of warranted dependence adopted.

The accounts of warranted dependence that the book meets elsewhere, in Chapter 3 for understanding and in Chapter 11, Section 5 for reliance, testimony, and trust, can be applied to exchange between differently organized communities. Applied in that way, they can be ordered by what, besides q_h, would have to pass to the relying side, from what can be assessed from outside the system's operation, through what lies behind its outputs, to who vouches for them: a record of the system's performance, the dependencies in the target that its outputs track, the reasons and assumptions that support them, and a party who stands behind them. Ordering them in this way lets one read off, from where an account stands in the ordering, what a claim of warranted dependence commits a system to. The ordering is stated here for a human side relying on an artificial system, as in the case of q_h, but it can be applied in the same way where the relying side is artificial and the source is a human community or another artificial one. A pragmatic account lies outside this ordering. The positions can overlap.

On reliability-centered accounts, reliance may be warranted without translation of the system's inferential route, but the accounts differ over whether the warrant requires that the system's reliability have been established. Applied to such exchange, Goldman's process reliabilism (Chapter 11, Section 5.2) would make reliance warranted if the process by which the relying side forms beliefs about the target from predictions of the kind q_h reports, from the system's producing them to their uptake, is in fact reliable in the domain of reliance, without the one who relies needing reasons for thinking it so. Such an account stands at the start of the ordering above, since nothing besides q_h need pass for the warrant to obtain; a record of performance is then what would let the relying side, or evaluators acting for it, establish that it obtains. Ortmann defends reliance on some opaque scientific systems on a condition that includes such establishment, holding that what matters is whether a system's reliability in a given context has been established by empirical evaluation that trustworthy researchers have performed adequately (Ortmann 2025, 1, 17–19). What this account requires of a system is a record of performance in the domain of reliance, assessed by evaluators independent of it; Chapter 11, Sections 5.2 and 5.3 set out Ortmann's account. Domain-bounded calibration, independent testing, and the sensitivity of the system's outputs to defeaters are diagnostics proposed here for what such a record would have to show.

These diagnostics meet a limit that Kitcher's analysis of how scientists assign authority to one another makes visible. As Goldman reports Kitcher's account, direct calibration compares an authority's claims with the assessor's independent opinions (Goldman 2007, §5). Strevens's reconstruction also identifies indirect calibration through others whose assessments can themselves be evaluated, and Kitcher's category of authority attached to pedigree or social position (Strevens 2010, §2, manuscript pp. 7–8). Carried over to artificial research, this limit means that where an artificial system's results go beyond what those who rely on it can assess independently, its calibration for them can only be indirect, through its record on claims they can check or through the judgments of others. How far direct calibration can reach also depends on whether those who rely keep the competence to form independent opinions; Chapter 3, Section 8 counts keeping up that competence and passing it on to new members among the capacities a continuing research system can lose. Connected with the progress of research, this suggests a further point. The further artificial inquiry goes beyond the objects and conditions that the relying side could assess independently, the less calibration obtained on known tasks can be carried over to the new domain, so that the ground for relying without translation may come to reach less far just as more can be learned from the machine. On accounts that require reliability to have been established, reliance beyond the reach of calibration needs another ground, though it does not follow that the ground must be a recovery of human understanding; the accounts below locate it in understanding, public criticism, or assurance by a party who stands behind the output, among other things.

On an understanding-centered account, where the achievement sought for the relying side is understanding of the target, of why the specified outcome occurs under the specified conditions, prediction is not enough: dependency relations, counterfactual structure, or explanatory use have to become available to the relevant human or collective subject, although theories of understanding disagree about which of these relations matters and how (Dellsén 2020). What such an account requires of a system is that it make some of the dependencies in the target on which its predictions rest available in a form the relying subject can use, the requirement Chapter 3 examined. Understanding how the system operates, or why its outputs may be relied on, is a different achievement, which Chapter 3, Section 4 distinguished from this one; a faithful trace of the system's computation does not by itself supply understanding of the target, and a simplified model that does not reproduce that computation can contribute to it (Chapter 3, Section 5.3).

On accounts that emphasize social justification, the burden shifts toward public criticism, uptake, and correction. Longino, for example, holds that because evidential relations are mediated by background assumptions, the only check against their arbitrary acceptance is critical interaction within and between scientific communities, and that the critical practices of a community must satisfy conditions of effectiveness to qualify as objective (Longino 2025, §5). What such an account requires of a system is that the reasons and the assumptions linking its data to its claims be exposed to criticism it does not control (Section 3.1; Chapter 11, Sections 3 and 9).

Assurance-based views, which Chapter 11, Section 5.2 introduces, ask in addition who intentionally assures the relying side of a claim and stands behind it, and, on versions that add a right of complaint, who answers for failure, so that on these views a system's output becomes the relying side's knowledge by testimony through a party who does so. Elizabeth Fricker also makes communicative intention and responsibility for what is told central, though from a local reductionist account of testimony, on which the reasons for believing testimony lie, speaker by speaker, in other reasons the hearer has, rather than an assurance view, and concludes that the outputs of current AI systems that generate natural language, large language models among them, are not testimony, while not ruling out that future systems' outputs might be (E. Fricker 2026, 2870–2871, 2877); Chapter 11, Section 5.1 sets out Fricker's account.

A pragmatic account locates coordination in shared objects and operations rather than in anything passing from one side's conceptual repertoire to the other's. Star and Griesemer, studying how amateurs and professionals cooperated in building a natural history museum, call objects such as specimens, field notes, and maps boundary objects: they can accommodate the local needs and constraints of different users, yet remain "robust enough to preserve a common identity at all sites," so that they can carry different meanings for different groups while remaining recognizable to each; alongside them they place standardized methods, which make information compatible by emphasizing "the how and not the what or why" (Star and Griesemer 1989, 87, 103). On such an account, shared instruments, interventions, and measurements can play a corresponding role even when the two sides' repertoires remain partly different. What it requires of a system is that its results be tied to objects and operations that the relying community can also use.

The distinctions among these accounts are analytical, not claims of causal or conceptual independence, as with the three questions of Section 2. If, for example, meaning were constituted by inferential role, grasping what a claim means and grasping what follows from it would be bound closely together; if common intervention and use were part of meaning, sharing them would already share part of meaning rather than substitute for it. The reliability-centered accounts above, by contrast, can separate warranted dependence from shared meaning altogether. The constructive question is what each position would require from an artificial research system and what kind of achievement it would license us to attribute.

Some positions bear on the ordering from outside it, by showing ways in which what passes can stand in for shared concepts. Humphreys, writing about instruments whose output is computed from detector data, takes the sinogram, the representation of the raw data of a computed tomography scan, and the familiar image computed from it by inverse Radon transforms to belong to one equivalence class of representations, fixed by a set of transformations on the data points: the image is useful to human observers because of their perceptual apparatus, while for a computer the sinogram is at least as useful, and what such images are about can be settled only with knowledge of the transformations performed (Humphreys n.d., 5–6). Humphreys does not discuss exchange between communities, but applied to it, equivalence under known transformations gives a way of translating that does not require shared concepts. Mitchell holds that the reliability of machine-learning tools rests, as for other instruments, on a theory of the instrument and on the stability of results across very different kinds of experiment (Mitchell 2020, 16–24); Mitchell writes about instruments used by human scientists, and taken as a model of coordination between communities the account is an application. Duede's account in Chapter 3, Section 5.3 bears on this ground: on it, deep-learning models do not, on the present state of interpretability research, reveal which principles their learned procedures operationalize, which is what a theory of such an instrument would have to supply. On these positions, a known transformation between equivalent representations or a theory of the instrument can stand in for shared concepts, though on Duede's account the latter is not yet available for deep-learning models.

Other positions widen or narrow what has to pass, or concern the direction in which it passes. Beyond the far end of the ordering, Weatherall and Wolfson hold for mathematics that a correct machine-generated proof is not by itself a contribution, since what contributes is mathematicians' engagement with it (Chapter 2, Section 4.5); on such a view what has to pass is not only information but participation. Conversely, what passes can shrink to a working technique: Takahashi holds that prediction without understanding is "incomplete as science but complete as technology" (Takahashi 2026, 185). In that case *S* gains a capacity to act, but on understanding-centered accounts not understanding. The passage can also run in one direction only. Jeannot forecasts that researchers outside advanced infrastructure will have their work ingested by systems they cannot access, replicate, or fully read (Chapter 12, Section 6.4); where passage runs only one way, one side supplies without receiving. Jeannot writes about research communities with and without advanced AI infrastructure.

The model of science as collective predictive coding proposed by Taniguchi and colleagues marks the boundary of this section's question. It treats human and artificial agents as members of one community that builds a single knowledge system (Taniguchi et al. 2025, §4.1), and so corresponds to a mixed *S* that forms one symbol system (Section 1.2) rather than to coordination between communities in this section's sense.

Routes that bypass translation were described by one of the founders of the incommensurability thesis and have recently been tried with an artificial system. Feyerabend, having argued that incommensurable theories may possess no comparable consequences, proposed to judge predictive success by agreement of behavior between a robot without sense organs, with the theory built into it, and beings trained to find their way about the domain, who, Feyerabend added, need not be human and could be other robots, a criterion that makes no reference to meanings (Feyerabend 1962, 94–95). Another route described in Section 3.1, learning the other side's kinds, has conditions of its own. On Kuhn's account, as Mladenović reports it, recognizing an object as being of a kind rests on perceiving, without inference, the relevant similarities and differences, a capacity acquired from particular examples in a trial-and-error process in which competent speakers support and correct the learner. Terms for unobservable kinds are learned otherwise: terms for mental constructs, such as money, through their relations to other constructs within a practice, and scientific singletons, terms such as mass and force that are outside genus–species taxonomies and have no set of contrasting terms, together with laws. On Kuhn's view, the referents of scientific singletons are never observable. The meaning of Newtonian force, for example, cannot be learned without Newton's second law (Mladenović 2022, xxx–xxxi, xxxiii–xxxv). Schut and colleagues report an attempt of the first sort. From AlphaZero, a chess system trained through self-play, they extracted concept vectors selected for being teachable to

another network and for being better reconstructed from AlphaZero's games than from human games, and showed four grandmasters prototype positions for these concepts, with AlphaZero's continuations but without labels or verbal descriptions. All four solved more unseen puzzles afterwards, with improvement on eight of the thirteen concepts tested; a puzzle counted as solved when the grandmaster found the move AlphaZero had selected. The authors present the result as a proof of concept given the small sample and the possibility of priming, and they chose chess partly because the usefulness of a concept is easier to evaluate there than at the frontiers of science (Schut et al. 2025, §§1.2, 3.2, 4.1). The case can be read as showing people acquiring part of a machine-native organization without translating it, where its instances can be exhibited and success can be scored, even if, as in this study, by agreement with the machine's own choices; it does not show what such a route would require where p_a concerns kinds or features that cannot be exhibited, or where S cannot score the result itself.

What has to pass, then, depends on the account: a record of performance assessed independently, within the limit that Kitcher's analysis marks; some of the target's dependencies in a form the relying side can use; reasons and assumptions exposed to criticism the system does not control; a party who stands behind the output; shared objects and operations; or, where instances can be exhibited, learning the other side's kinds, among other things. On each of these accounts, where what it requires does pass and the system's answerability to its targets holds (Sections 2.3 and 3.3), S can gain something through research that does not align with it, though whether what it gains is knowledge, understanding, or a capacity to act differs from account to account. The differences become plain when four of the accounts examined since are applied to the case in which q_h was introduced at the start of this subsection. Let that case, in which humans gain q_h without acquiring, translating, or understanding p_a, be specified further: the system's predictions survive independent testing, but it cannot explain their reasons to humans, and no party stands behind its outputs. On reliability-centered accounts, relying on the predictions can yield knowledge about the target insofar as they are in fact reliable in the domain of reliance. On understanding-centered accounts, the same success does not yield understanding of the target so long as the target's dependencies have not been made available by other means. On accounts that stress social justification, the claims are not acceptable so long as their reasons and assumptions have not been exposed to criticism. On assurance-based views, with no party standing behind the outputs, they do not become the relying side's knowledge by testimony; whether they can still yield knowledge as the readings of an instrument is not settled by these views alone (Chapter 11, Section 5.1). What separates the four accounts is not whether the prediction succeeded but which achievement, and by which route, they count as realized on the receiving side.

Krenn and Champion propose translating epistemic aims into computable, nonanthropocentric objectives (Chapter 7, Section 4.3); nonanthropocentric there means applying to biological and artificial scientists alike. Their program itself raises a question that its authors pose. They anticipate AI-discovered improvements whose principles humans can confirm but not fully comprehend, propose teacher systems that would explain such solutions to people, and expect humans to remain the strategic leaders who set directions and allocate resources, while asking how humans could verify and interpret discoveries they cannot genuinely understand (Krenn and Champion 2026, 336, 343–346). Thais and colleagues, by contrast, emphasize human understanding and the institutional conditions that sustain scientific criticism and correction (Thais et al. 2026, 1, 3, 5, 8). Together they return to the tension with which Section 3 began, between the possibility that artificial difference opens and the risks it carries: how can artificial difference be preserved without assuming that whatever emerges will remain knowable by humans, socially criticizable, or safely governable?

Two horizons and what Part IV leaves

What construction horizons follow from the three cases of Section 1, in which *S* is a human, a mixed, or an artificial society, and what does Part IV leave to Part V? Since what a program for building artificial researchers must answer changes with whose knowledge it seeks, the questions can be grouped by *S*. From the division by *S*, two horizons can be drawn, named here after the first and third cases: the horizon of research for a human society and the horizon of research by and for an artificial society, or, for short, the human-society horizon and the artificial-society horizon. The mixed case falls wholly under neither (see below). The artificial-society horizon takes as its central case one in which the actor is artificial as well, although, as Section 1.3 showed, questions of artificial belief and possession arise from taking an artificial society as *S* even where the actor is not artificial. The human-society horizon asks how artificial systems can bring about knowledge new to human societies, how that inquiry stays directed at those societies' unknowns as the selection of questions and standards moves inside the system (Section 3.2), how those societies keep the competence to evaluate what the systems produce and can tell whether their own standards have moved with the systems (Section 3.2), and how the resulting knowledge enters their epistemic situations, while what the systems still depend on, what they themselves do not know, and how far human involvement extends remain visible (Section 1.1). It can advance through partial capacities and hybrid organization, and it does not require artificial consciousness, a wholly independent epistemology, or an artificial society. The artificial-society horizon asks whether artificial beings and communities can form, maintain, criticize, and transform epistemic worlds of their own (Section 1.3). It raises stronger questions about artificial belief and social knowledge (Section 1.3), world-answerability (Section 2.3), purposes the society sets for itself and the revision of its own governing standards (Chapter 7, Section 4.3), and coordination with human epistemic societies (Section 3.4). For such research, human society stands not as the reference society but as a party affected by it, so that what can be asked on its behalf is a question of acceptability and legitimacy (Section 3.2). Part V adds the question of what becomes of the relations through which research is directed, certified, and governed when those who hold them are themselves artificial. The present chapter does not show that such a society is possible or impossible. It tries to show why building research by and for an artificial society asks for more than removing humans from the steps of research.

Research for a mixed society falls wholly under neither horizon (Section 1.2). For its human members, the questions of the human-society horizon remain. To the extent that artificial members are to count among those to whom its knowledge is attributable, the questions of artificial belief and possession that belong to the artificial-society horizon arise. Where its human and artificial members form separately organized communities and its knowledge is to be shared between them, the questions of translation and coordination of Section 3.4 are added. Where its members form a single symbol system, it is further asked whose experiences that system integrates. Who speaks for such a society can itself divide its human and artificial members (Section 3.3). Chapter 13, Section 5 retraces the chapters' examination of these horizons.

The distinction between the horizons is not a timetable that postpones every fundamental question. Work on artificial grounding, understanding, social organization, and norm revision can improve systems built for human knowledge now. Nor is the artificial-society horizon a claim that independence is intrinsically better. It is a way of taking literally the possibility that research could become artificial at the level of its epistemic organization rather than only at the level of its visible operations.

Part IV has therefore separated two choices: where inquiry is conducted and whose epistemic life it extends. Human and hybrid research can advance without deciding whether an artifact literally knows. Attributing knowledge to the artifact itself instead makes the content of its states, the presuppositions of its checks, and its answerability to the world central questions (Sections 1–2). What becomes central when understanding is attributed to the artifact depends on the accounts of who understands distinguished in Chapter 3, Section 6. An artificial inquirer can differ from *S*'s members, and the worry this raises is structurally similar to what is known as the AI alignment problem, on the strand of discussion that locates the difficulty of alignment in the system's otherness (Section 3.2). Allowing the artifact to revise questions and standards can enlarge what it investigates. It also permits a wider mismatch with *S* in what is taken to be unknown, worth pursuing, or a good reason, among other things; such a mismatch can open as well at the bearer and from the side of *S*, and a commission does not by itself prevent it. Applied to research, de Blanc's and Soares's arguments imply that, because inquiry can revise the concepts in which a commission was written, a gap can open when inquiry succeeds as well as when it fails; and Russell's argument, extended to research, implies that where *S*'s standards move with *A*, agreement between them does not

by itself show that *A* still serves what *S* would otherwise have sought (Section 3.2). Neither inherited human norms nor a different epistemic organization settles in advance whether a given mismatch is drift, error, or gain; that depends on whether it is assessed from *S* as it now is, from *S* as it may come to be, or, where correctness is in question, from the world (Section 3). Where research by an inquirer of another kind does not align with *S*, what follows depends on which relation fails. Even where relevance to *S* fails, answerability to the target can remain an anchor that does not depend on *S*'s consent, on accounts on which what corrects a commitment is fixed by its target and, where the target's verdicts reach the inquirer only through *S*'s procedures for judging claims, provided those procedures can be revised by evidence (Section 3.3). *S* may then still gain knowledge or understanding through what such an inquirer reaches. What has to pass for that, and whether what is gained is knowledge, understanding, or a capacity to act, differ by account. Knowing that the machine predicts a specified outcome (q_{h}) is not knowing that the outcome will obtain; reliability-centered accounts locate the step from there to knowledge of the target in the actual reliability of the predictions, while other accounts locate it in understanding, criticism, or assurance, among other things. And where instances can be exhibited and success can be scored, even if only by agreement with the machine's own choices, people can learn part of a machine's kinds without translation (Section 3.4). These possibilities give the construction project substance: it can change who conducts inquiry, who knows, and how differently organized inquirers learn from one another. Which possibilities are realizable remains conditional on the accounts and capacities examined here.

Part V takes up the relations on which the second of the two choices above, whose epistemic life inquiry extends, depends. Section 3 argued that, because an inquirer of another kind can differ from *S*'s members, neither a commission nor human control secures that research stays directed at *S*'s unknowns, that whether it does is a question about particular configurations, and that the epistemic success of research for *S* and the legitimacy of what directs it are different judgments, the second of which is asked also on behalf of those affected outside *S* (Sections 3.2 and 3.3). Which of *S*'s unknowns research pursues, whether its results come to count for *S*, and on whose authority its direction is set involve relations that extend beyond any single actor, artificial or human. The next part asks which of them can be located inside an artificial configuration, and what changes when they are; whether research succeeds for *S* and whether its direction is legitimate to the parties it affects are then asked separately.

Part V — Which Relations around Research Can Move inside an Artificial Configuration?

Chapter 9 — What Relations Surround Research for *S*?

Roles, Value, Evaluation, and Control

Where the attribution of research meets the research system

A research machine may call to mind a single researcher replaced by a single machine. This book, however, has not restricted the one who researches, *A*, to individuals: *A* can be a person, a group or organization, or an artificial or hybrid configuration (Chapter 7, Sections 1 and 2). And a system that conducts research may also select its investigators, distribute their effort, criticize their results, and revise the rules by which they work. In human research these have often been held by different parties. For example, universities and laboratories have decided who joins an inquiry; funders and laboratory heads where effort goes; reviewers and other researchers who try to replicate results have done the criticizing; and the communities of a field, among others, have set the rules of work. An artificial system may come to do within itself part or all of what these social arrangements around research have done. Up to Part IV the book has asked chiefly about those who research and their results, and has touched on these arrangements only where they bear on knowledge or on status as a researcher. Yet in human research much of what decides where research goes, such as which inquiries funds are directed to, which results are accepted, and who sets the direction of research, has been settled in such arrangements outside individual human researchers. Part V takes these up as its topic and asks what changes when the relations around research enter an artificial configuration. The question follows from Part IV's account of research actors but reaches beyond an actor's capacity: how its inquiries are related to one another, to the society relative to which their intended achievements would be new, and to those who bear their consequences. These relations are not deduced from the working characterization alone. They arise where what it names as the target of research, epistemic achievements new to *S*, meets the dependence, plurality, and finite resources of research.

The first reason the question arises is that the achievement named in the characterization is not, at least in the case of knowledge, a property of the actor alone: its novelty is relative to *S*, and on several accounts its warrant or possession also involves others. The argument here proceeds mainly through that case. John Hardwig argued, as Carolyn McLeod summarizes the argument in her encyclopedia entry on trust, that scientific knowledge depends for its acquisition on trust in the testimony of others, since no single person has the time, intelligence, and experience needed to find out independently what is known (McLeod 2020, §3). On this argument, dependence on others is not a shortcut a more diligent inquirer could dispense with but part of how the knowledge becomes possible; Chapter 11, Sections 5 and 6 return to Hardwig's argument. Chapter 2, Section 6 set out that on several accounts social organization enters the warrant or possession of knowledge itself, and Chapter 3, Section 8 that local warrant does not settle whether a continuing system preserves its capacities for diverse search, independent criticism, memory, correction, and further question formation. In human research, knowledge new to *S* has therefore often been brought about through relations that extend beyond whatever unit conducts a particular inquiry, and on accounts on which social organization enters warrant or possession, it is also warranted and possessed through them. Whether the first holds where that unit is an artificial configuration that conducts many inquiries is one of the questions that the sorting of Section 6 asks.

The second reason is that the problem of the artificial researcher and the problem of the research system, which correspond to two of the debts that Chapter 1, Section 9 lists as questions the working characterization leaves for later parts to answer, the attribution debt (to what, and within what boundary, research can be attributed) and the research-order debt (questions about the social order surrounding research), cannot be reduced to one another. An account of an artificial researcher and an account of a research system can concern the same arrangement while having different objects. The contrast is not between an individual and a collective: an artificial researcher can itself be a collective or an organization (Chapter 7, Sections 1 and 2). It is between what a researcher, of whatever composition, can do and how inquiry among researchers is organized. Research system here names the latter, the social order of research that Chapter 1, Section 9 calls the research order. That order can be realized in human institutions, in mixed arrangements, or within one artificial system that organizes inquiry among its own instances or investigators, as in the example below. Where it is realized within one artificial system that also conducts research, the same arrangement can be described both as a researcher, a configuration to which the conduct of

research is attributed (Chapter 7), and as a research system, an organization of inquiry among its instances or investigators; the contrast is then between these two descriptions, not between two kinds of object. In Part V an artificial research system is an artificial system described in the second way, that is, a configuration considered as the organization of inquiry among its instances or investigators. A population of individually excellent investigators can form a poorly organized community, and a well-organized community can be composed of investigators whom no individual assessment would favor. Conor Mayo-Wilson, Kevin Zollman, and David Danks give one version of this a precise form within a formal model of communities that share evidence. Counting a method as rational when its chance of choosing the best available action approaches certainty as inquiry goes on, they prove that methods that converge in this sense for an inquirer working alone can fail to converge when used together, and that a community can converge although none of its members would alone. They also prove that individual and community rationality are not wholly independent: methods that converge in whatever community they are placed form communities that converge, although even on this standard a community can converge where its members' methods would not. How far individual and community prescriptions come apart therefore depends on the standards of individual and group rationality adopted (Mayo-Wilson, Zollman, and Danks 2011, 653–655, 663–669). Put in the terms of artificial research, how an agent performs when it is assessed alone does not by itself show how it performs among agents with which it shares evidence, and even an assessment in those settings does not settle how inquiry among researchers is to be organized. Nor does the fact that inquiry is well organized by itself establish that any unit, a member or the collective, is an artificial researcher (Chapter 7).

The third reason is specific to artificial research. As noted at the outset, in human research the relations through which inquiry is directed, certified, and governed for *S* have been realized not only by investigators themselves but also by people and bodies other than the investigator. An artificial configuration can, at least in principle, take some of these relations inside. It can, for example, generate the investigators among whom effort is divided, run the checks through which results are criticized, and hold the decisions through which the whole is directed.

An example, which this book constructs for the purpose and to which the following chapters return, makes the possibility concrete. A single system might generate a hundred instances to pursue different lines, assign other instances to check and reproduce what they produce, and shift computing resources among the lines according to those checks, with no colleague, referee, or funder outside it, although those who built it may still be able to stop, update, or retire it (Table 9.1; Section 5). Parts of such an arrangement already exist: the AI Scientist, for example, includes review among the steps through which it organizes research (Introduction, Section 2; Chapter 11, Section 7). Those who build or deploy such a system can make three kinds of claim about it. They can claim that it allocates inquiry well, moving its resources to the lines that merit them; that the results which pass its internal checks are certified for *S*, so that others have reason to rely on them; and that it may decide, on behalf of others, which of *S*'s questions its instances pursue, a claim not about how well it allocates but about who may make that decision. The example's role is to show, held in one arrangement, the three relations through which inquiry is directed, certified, and governed, which Section 2.1 names allocation, criticism and certification, and authority. For each relation, Part V asks what it would take to substantiate the corresponding claim.

Chapter 7, Section 2 held that moving a function across a boundary changes the subject of the claim without making the function disappear. That remains true. But it leaves open whether a function performs in the same way once it has moved, and whether every function can move at all. In the example, whether checks run by instances of the same system catch what an outside critic would, and whether a system that shifts resources by its own checks can also settle which of *S*'s questions are worth pursuing, are questions of this kind. The question of this part is therefore:

> **Which of the relations through which research is directed, certified, and governed for *S* can be located inside an artificial configuration, and what changes when they are?**

Of the two parts of the question, the answer to the first, which relations can be located inside on which accounts, serves as a means of describing the second, what changes in research when they are. The question is not whether artificial configurations can conduct research, which Part IV addressed, and answering it adds no condition to the working characterization. It does bear on questions that Chapter 1, Section 1 left at the level of value and governance, where importance, usefulness, legitimacy, risk, and the like are assessed apart from the conditions of research status and of achievement, and Part V compares the answers that different accounts support and argues what changes when the conditions each account names are realized inside a configuration. Drawing these comparative conclusions does not require selecting

one account of value or legitimacy for every decision. Of the questions Chapter 1, Section 1 left at that level, which unknowns merit attention is taken up in Section 3 and Chapter 10, Section 6; whose purposes are to govern inquiry, in Chapter 10, Section 6 and Chapter 12; how epistemic and non-epistemic values interact in certification, in Chapter 11, Section 4; which risks are permissible, as the question of who may set the error costs that others bear, in Chapter 11, Section 4 and Chapter 12, Section 2; and when a result is to alter a decision, only as the threshold at which a result may be relied on for action (Chapter 11, Section 4). Of what Chapter 1 listed at that level, utility, beauty, and emancipation enter only where they bear on which unknowns are pursued, and institutional recognition only where it moves allocation as credit (Chapter 10, Section 3).

The rest of this chapter prepares the question. Section 2 says which relations are meant, who holds them, and what their division among different parties may have supplied. Sections 3 to 5 set out what all three of those relations involve, rather than one relation each: valuing, evaluating, and controlling, in the minimal form that the following chapters use. Section 6 states how Part V proceeds. Some of this preparation already shows relations that a comparison of outputs alone misses: even where an artificial configuration delivers the same knowledge to the same *S*, on positions such as those that value the formation of researchers the organization of research need not keep the same value, since who took part in the process, and on some accounts of authority who decided, remain in question (Section 3).

Three relations and the parties who hold them

Section 1 spoke of the relations through which research is directed, certified, and governed for *S*. This section asks which relations these are, why Part V counts three, and who holds them. Here and throughout Part V, an actor or inquirer is whatever unit the conduct of research is attributed to: a person, a group, or an artificial or hybrid configuration (Chapter 7, Sections 1 and 2). The section derives each relation from a term of the working characterization that marks where the activity is tied to *S*, taken together with a condition outside the characterization that explains why that tie becomes a relation extending beyond whatever unit conducts a particular inquiry (Section 1). In human research such relations have been held by parties other than that unit as well as by it (Section 1); whether an artificial configuration that conducts many inquiries can hold them inside itself is the question of Part V.

2.1 Deriving the three relations

The first term is *new to S*. Chapter 1, Section 7 derived from it, together with attempt, that research has a question structure: the attempt is directed at some respect in *S*'s relation to a target that is not yet settled. It also observed that a society is ignorant of indefinitely many things and that novelty does not decide which of them deserve attention (Chapter 1, Section 8). The term supplies the space within which a choice is made, not the need to make one. The need comes from a condition outside the characterization: the means of inquiry, such as attention, effort, time, instruments, and money, are finite and have alternative uses, so that not every unknown can be pursued at once. Chapter 4, Section 5 and Chapter 5, Section 4.2 argued that the choice of what to take up as a question and what to pursue can be made repeatedly within an inquiry, and Chapter 7, Section 4.3 argued that the capacity to select questions is one dimension of an actor's autonomy. What makes the choice a relation beyond the unit is a second outside condition, that many investigators pursue *S*'s unknowns. Among them the same choice takes a different form: effort is divided among lines of inquiry whose prospects are unknown, and the division is not the sum of each investigator's choices. Call this relation **allocation**: the distribution of attention, effort, and resources among *S*'s unknowns. Chapter 10, Section 1 takes up why it arises and what makes it hard.

The second term is *epistemic achievements*, developed here first through *knowledge*. Chapter 2 separated the production of a result from its becoming knowledge, and set out that on the critical-contextual account, venues for criticism, uptake, public standards, and tempered equality of intellectual authority help constitute objectivity rather than merely communicate it, and that on distributed or group-functional accounts, on which a properly organized collective can itself realize a knower, retrieval, use, and revision can enter possession. On accounts that locate knowledge in individual knowers, the term itself does not point beyond the knower. What points beyond it there is the index to *S*, the relativity of novelty to *S*, together with a condition outside the characterization, related to the one Section 1 noted with Hardwig and put here in terms of *S*: the members of *S* cannot each gather the evidence for many results, so that their reliance on a result runs through others: through those who produced it and, where it has met scrutiny beyond them, through that scrutiny. For an artificial *S*, how far this

condition holds depends on how evidence is shared among its members, for example whether what one member has gathered is available to the others as their own record or only through its report. Call this relation **criticism and certification**: the relations through which a candidate result is exposed to error-detecting scrutiny and through which others are given reason to rely on it. How far such reliance has to rest on scrutiny of the result is one of the questions on which the accounts of testimony compared in Chapter 11, Section 5 divide. Certification also covers the judgment that a result is in fact new to *S*. Chapter 5, Section 5 noted that identifying what is new to *S* can itself be work within inquiry, and like any such work the judgment can be mistaken. Chapter 11, Section 1 separates the two parts of this relation, exposure to error-detecting scrutiny and the giving of reason to rely. On accounts that locate the value of criticism in the errors it detects, the first part can hold within a single actor, while the second refers to *S*; on critical-contextual accounts, what a check counts against is itself relative to assumptions a community holds, so that, where that community is *S*, the first part can refer to *S* as well.

The characterization, however, names epistemic achievements in general, not only knowledge. The knowledge argument above specifies one important case; it does not establish that every achievement has the same social conditions. For understanding, certification can concern an explanation's adequacy or an attribution of understanding without conferring understanding on its recipient. Claims about an explanation's adequacy or someone's understanding can also be relied on without being checked by each recipient, making criticism and certification relevant here. What certification establishes depends on which claim it assesses. Chapter 11, Sections 5.6, 6, and 9 develop this extension not as an implication of the knowledge accounts just surveyed but as a further argument using the distinctions of Chapter 3.

The third term is *S* itself. Chapter 8 distinguished the epistemic success of research for *S* from legitimate governance, and observed that specifying *S* does not identify who may speak for it. The outside conditions here are that *S*, short of the limit Chapter 12, Section 5 considers, a single configuration that is both the actor and the whole of *S*, has many members and no single voice, and that the consequences of inquiry can reach parties who do not belong to *S*. Which unknowns are pursued and what error costs are accepted can be fixed by someone's decision or by processes that no one sets as policy, as Chapter 10, Section 5 suggests from cases of allocation. Where they are decided, the question is who may decide them and who may revise the decision; where they are not, it is who answers for the setting and who may change it. Call this relation **authority**: the standing, in the sense defined next, to decide and to revise the terms of the other two relations and the further decisions about inquiry that Chapter 12, Section 2 sets out.

Standing, here and throughout Part V, is meant in a normative sense only. It is an entitlement, held in relation to those subject to a decision, to propose, decide, object, refuse, or revise. It is neither competence, nor the formal assignment of a right to decide, nor the effective capacity to exercise such a right (Section 5). Which parties have standing, and on what basis, depends on the account adopted. Whether competence can ground it is a question on which accounts of legitimate authority divide, and Chapter 12, Section 1 takes it up; whether it rests on features of a being, such as welfare or agency, or on relations among beings, such as power or cooperation, is a question on which accounts of justice divide, and Chapter 12, Section 5 takes it up. Three other matters that the words *standing* and *authority* could name are given other terms. Longino's condition of tempered equality concerns what Longino calls intellectual authority, the capacity to contribute to the critical interactions of a community; the condition holds that any perspective has that capacity prima facie (Longino 2025, §5); Part V keeps that term, and Chapter 11, Section 3 takes it up. What an actor's outputs count as for *S* is called below their *status for S*. And whether an artificial entity has claims of its own as a being, which Chapter 12, Section 5 asks, is a question of *moral status*.

The derivation comes with four qualifications. First, the correspondence between terms and relations is not one to one. Novelty to *S* bears on certification as well as on allocation, and the parties to whom authority may have to answer can extend beyond *S* to those who bear the consequences of the decisions it takes; which of them it answers to depends on the account of legitimacy adopted (Chapter 12, Section 1). Second, the relations are not on a par. Allocation and certification are first-order relations; authority is second-order in that it sets and revises the terms of the other two. The further decisions it covers, which Chapter 12, Section 2 sets out, include operational and risk decisions, constitutive and recognition decisions about who counts as a member, a critic, or an expert, and decisions about memory and correction. Third, of the three, certification is the relation that can most directly enter a result's status as knowledge for *S* or its becoming *S*'s knowledge, on the accounts of Chapter 2, Section 6 on which social organization enters warrant or possession, respectively; on accounts that make group knowledge depend on the members with decision-making authority in the relevant domain, a form of authority enters as

well. Where a relation does not enter the conditions of the achievement claimed, its assessment remains at the level of value and governance (Chapter 1, Section 9). For knowledge, on the accounts considered here, this is so for allocation, for authority except on the group-authority accounts, and for certification on accounts on which social organization enters neither warrant nor possession. For other achievements, whether these relations enter success depends on their own accounts, such as the accounts of public or collective understanding in Chapter 3, Section 6; on accounts that make understanding a kind of knowledge, the conditions just described for knowledge may carry over (Chapter 3, Section 3). Fourth, none of the three is a condition of research status. In the sense of the working characterization, an activity organized toward epistemic achievements new to *S* is research whether or not the allocation that set it going was wise and whatever the legitimacy of its direction (Chapter 1, Sections 1 and 9). Part V takes the relations up nonetheless because claims about artificial research systems, of the kinds illustrated by the three claims of Section 1, are claims about these relations.

The division into three is meant to pick out the relations around research whose location inside or outside a configuration bears directly on those claims. Chapter 1, Section 9 took the research order to be made up of the division of labor, the allocation of inquiry, criticism and certification, memory and correction, credit, and authority, and listed agenda setting and attention among what artificial researchers may change. Part V treats those of these that are not themselves the three relations where they bear on the three rather than as relations of their own. The division of labor and attention fall under allocation as defined above (Chapter 10). Agenda setting is taken up under allocation, as the choice of which of *S*'s unknowns to pursue, and under authority, as the question who may make that choice (Chapter 10, Section 6; Chapter 12). Memory and correction are taken up as conditions of certification over time (Chapter 11, Section 8, where memory takes the form of the record) and as decisions under authority (Chapter 12, Sections 2 and 6). Credit was treated as an attribution in Chapter 7, Section 3.3, and Part V takes it up where it moves allocation (Chapter 10, Section 3). Other matters bear on the relations as well. Instruments and infrastructure were examined as enabling conditions in Chapter 5, Section 6 and Chapter 7, Section 4.2; Part V takes them up where they realize the three relations, and Chapter 12, Section 6.4 takes up infrastructure as a relation through which effective authority is distributed. Material resources and the formation of researchers are taken up next, together, as support.

Support here names what makes it possible to sustain an organized direction that may fail, such as material resources, the formation of researchers, and the bearing of the costs of failure. It bears chiefly on a further term of the characterization, *attempt*. At the activity level, obtaining resources, permissions, and access belongs to research when it is undertaken in the attempt to resolve the inquiry (Chapter 7, Section 4.4). Where support ends and research begins thus varies with where the actor's boundary is drawn. Drawn widely, obtaining means and allocating the costs of failure become the actor's own activity, and the actor itself holds the role of supporter. At the level of value and governance, the reasons offered for support are addressed to those who supply it. The supporter and *S* are distinct roles, which the same party may or may not hold. In a firm's internal research the firm can be both the supporter and the *S* for which results are new; a patron who maintained a scholar and the community of scholars for which the scholar's results were new need not have coincided; nor, necessarily, do a public funding body and the specialist community or humanity relative to which the research it funds is new. Part V does not count support as a fourth relation. The derivation above does not yield it, since *attempt*, the term on which support chiefly bears, marks the activity's direction and its possibility of failure rather than where it is tied to *S*; where *attempt* enters such a tie, as in the question structure, it does so together with *new to S*, and the relation that results is allocation; but the derivation does not claim to exhaust the relations around research, so this alone does not settle whether support is to be treated as a relation of its own. Part V does not so treat it because support enters claims about artificial research systems, of the kinds illustrated by the three claims of Section 1, chiefly by way of the three relations. When funding moves which of *S*'s unknowns are pursued, support is part of allocation; when those who judge results also fund the work, support is bundled with certification; when control of funds becomes a power to set and revise the direction of inquiry, support is bundled with authority. Part V therefore takes support up where it realizes or is bundled with the three relations. The question that does not fit these forms, why support is given at all, Section 3 takes up position by position.

2.2 Roles around research

The roles that the three relations involve can be named once. Chapter 8, Section 1 noted that evaluators, beneficiaries, supporters, audiences, and authorities can differ from the actor and from *S*, and identified them locally where they mattered. Part V uses a few of them throughout, and Table 9.1 states them in one place. Dividing the roles is meant to keep apart parties that the single word "society" can name, such as the reference society, the supporter of research, and the public of a state. The roles are divided by the position in which a party stands to an inquiry: conducting it, being the collectivity for which its results would be new, sustaining it, judging it, relying on its results, bearing its consequences, or making or remaking the actor that conducts it, because these are the positions that claims about allocation, certification, and authority invoke. Several roles can be held by the same party, and in the example of Section 1 an artificial configuration can hold more than one of them.

Table 9.1. Roles around research used in Part V

Role	Who holds it	Where Part V chiefly uses it
A, the actor	the unit to which the conduct of research is attributed: a person, a group, or an artificial or hybrid configuration (Chapter 7)	throughout
S	the society relative to which the intended achievements would be new and for which they are sought; their bearer or mode of realization is a further question (Chapter 1, Sections 2 and 6)	throughout
Supporter	whoever supplies the means of an attempt and bears or distributes the costs of its failure	Sections 2 and 3; Chapter 12, Section 1
Evaluator and addressee	whoever judges, and whoever uses the judgment to choose (Section 4)	Chapters 10 and 11
Those who rely	those who take a certified result as a basis for belief or action, usually but not always members of *S*	Chapter 11
Those who bear consequences	those on whom the costs and risks of inquiry and of its errors fall, who need not belong to *S*	Chapters 11 and 12
Provider or maintainer	whoever builds, maintains, updates, or retires the configuration to which conduct is attributed, and so can change the actor itself rather than only its means (Chapter 7, Section 3.1)	Chapter 10, Section 5; Chapter 11, Section 5.4; Chapter 12, Sections 2.2 and 6.4

A further role, the constituency to which a decision about inquiry would be justified, depends on the account of legitimacy adopted and is taken up in Chapter 12, Section 1. *S* is abstract throughout. It can be humanity, a specialist community, an organization, a mixed group, or a candidate artificial society (Chapter 1, Section 6); Part V does not use "society" for the public of a state unless it says so. The provider is listed apart from the supporter because supplying the means of an attempt and making or remaking the actor are different ways of standing to an inquiry. Section 6 takes up a provider who also supplies the configuration that performs the research as a feature that artificial configurations can bring, one with requirements of its own.

2.3 Realization, division, and recombination

The three relations are analytical relations, not institutions. In the research systems that the book's readers know, familiar institutions have played central roles in them, and those institutions are one realization of the relations among others. Allocation has been carried out through, among other things, investigators' choices of problems under the pressure of credit and careers, the selection of projects by funders, and the direction of laboratories; Chapter 10 examines it. Criticism and certification have been carried out through, among other things, review before publication, replication, the publication of results, and their later use; Chapter 11 examines them, and Section 7 of that chapter takes up the functions of scholarly communication that journals came to bundle and that can be separated and reassigned. Authority has been exercised by, among others, funders, research communities governing themselves, regulators, and publics; Chapter 12 examines it. How each relation is realized also differs by field. In the mathematics group that Laurens Hessels and colleagues studied (Chapter 10, Section 5), arguments could gain recognition before formal publication, and in their humanities group recognition was converted mainly into positions and research time rather than grants; a proof checked against a kernel, the small trusted core of a proof assistant that checks each inference step, which Chapter 11, Section 2 discusses, is a realization of certification that involves no replication; and the authority over a field's standards can lie with its own members more than with funders or regulators. The three relations are meant to cover these realizations as well. The institutions named here are examples drawn mostly from the natural sciences. The view that these institutions are one historically formed realization among others has a lineage. Kitcher, noting that the rules first laid down in the 1660s for choosing investigations and certifying new ideas were adapted to a quite different context, asks whether an institution that had no inkling of its later influence and has "evolved by happenstance" is well designed for the functions it now fulfills (Kitcher 2013, 216). Michael Nielsen and Kanjun Qiu describe the design of science's social processes as contingent on the accidents of history and the space of possible processes as vast and mostly unexplored, and hold that exploring it will take decades or centuries at the least. As reasons why those processes that are collectively held hardly change, they give the centralization of control in a few large funders and institutions, the collective-action problem that no one alone can change a norm, network effects that homogenize social processes through community judgment and the shadow of imagined future judgments, and the absence of a feedback loop that would let new institutions grow. Taking scientists to be users of the discovery system and for the most part not its designers, they envision metascience entrepreneurs who try and spread new processes (Nielsen and Qiu 2022). Chapter 12, Section 9 returns to that exploration, asking what may happen when the configuration that does research can carry it out on its own research. And an essay by this book's author, written with an AI agent, proposed that AI may become material with which the roles and relationships constituting science are taken apart and reassembled differently (Takagi and an AI agent 2026).

What matters here is the structure of relations held by different parties. On the analysis proposed here, the division of these relations among different parties could supply goods that no single party was designed to supply, and Part V examines four of them. The first of the four is **error detection across boundaries**: points at which an error made in one place could be detected in another. The second is **standing for dissent**: standing in the sense defined in Section 2.1, held in relation to those who make, or are bound by, a judgment that would settle a question or close a line, to object to it, to refuse to abandon one's own view or line, or to propose that what was closed be reopened, whether or not the dissent later proves right. What the division may have supplied is parties distinct enough to hold such standing and occasions on which it was recognized and exercised; whether an arrangement can supply the entitlement itself depends on the ground of standing adopted (Chapter 12, Sections 5 and 9). What such standing protected epistemically, the continued representation of views and lines that a prevailing judgment disfavors, is a different matter from the standing itself, and the two may come apart; call what it protected the epistemic work of dissent. In criticism, Longino's tempered equality of intellectual authority is one account of what that work calls for (Longino 2025, §5). The third is **boundaries across which accounts had to be given**, at which one party had to explain or justify what it had done to another. In research, where a result is new to *S* and there is often no other source against which to compare it, an account of how the result was obtained could be among the few means by which those who rely on it could scrutinize it (Section 4). The fourth is **responsible parties**: parties who could be held responsible for what was done. Since errors in research are often found only later, this good could include there being someone answerable not only when a result appears but also later, when correction is called for. Separation can also fragment responsibility if no party is answerable for what happens across the boundaries. Chapter 10, Section 5 sets out further functions that human allocation performed without anyone deciding them. The four are meant to give Part V definite goods whose fate can be

followed when a relation moves inside a configuration. That the division supplied them is a hypothesis that Part V examines, not a finding, and each good is assigned to the chapters that test it. Error detection across boundaries is examined in Chapter 11, Sections 2 and 3, where two astronomical episodes show it supplied in one case and missing in another; standing for dissent in Chapter 10, Section 4, where dissent takes the form of continuing or reopening a line, and Chapter 11, Section 3, for critics; boundaries across which accounts had to be given in Chapter 11, Section 9 and Chapter 12, Section 6; and responsible parties in Chapter 12, Section 6. Each of these chapters returns at its end to the goods assigned to it, and Chapter 12, Section 9 collects all four.

Restated in the terms of the three relations, the formulation of the essay cited above becomes: an artificial configuration can combine relations held by separate parties. One system can search the literature, generate candidates, choose experiments, evaluate results, route resources, and use the outcome. Combining support, judgment, and direction is not yet the same as bringing the performer inside. A supporter can also judge results and direct work while depending on separate performers. Where those performers can refuse, keep their evidence, or leave, they remain parties whose cooperation is required. A configuration that generates its own investigators can bring the performer inside as well. This is the theme of Part V: relations that in human research were held by separate parties, including the performers of research, can enter one configuration. Recombination can remove delay and loss of information at former boundaries. But it can also remove what the boundaries incidentally supplied. The question of Section 1 can therefore be put more sharply. Of the goods that the division of these relations may have supplied, which depended on premises about human realizers, the people who actually held the relations, with their limited attention and lifespans; which on premises about how those relations have historically been organized, such as careers, journals, and the custody of data, an organization that the view whose lineage was recorded above takes to be reorganizable even with human realizers; which on conditions of inquiry that hold whatever realizes it; and which on relations to parties outside the configuration, including *S* where *S* lies outside it? Section 6 states this sorting as the device of Part V.

A relation's entering a configuration is not the same as its being designed and assigned from above. A relation that has entered may be realized by applying, from above, rules that the configuration has set, or it may be realized from below, without anyone setting it, as many machine realizers within the configuration act on their own motives or something equivalent to them. And which realizers it is passed through, and how what passes through them is integrated, can make the same relation realize different things. That design from above becomes possible without passing through human realizers is not the only implication that Part V follows, but it is an important one. In human research, keeping disfavored lines alive, the turnover of participants, the distribution of search by curiosity, and the like have often come about through investigators' own motives and lifespans, without anyone setting them (Chapter 10, Section 5). In a configuration that can be designed from above, by contrast, the likeness and difference among members and the correlation of their errors, as well as what is given to which line, can be set by how it is made (Section 6). What once came about without anyone deciding it may then be set by design, arise from below among machine realizers, or be lost through neither. Whether to adopt curiosity-driven search as a way of allocating may also become harder to leave implicit (Chapter 10, Section 5). In research, moreover, which differences will later prove useful is not known in advance (Chapter 10, Section 2), so that setting diversity, the difference among members and lines, as a value of the design can mean setting it without knowing what it will serve. Nor need that value be set once and for all: the configuration may revise it in light of what its inquiry finds, and the way research is organized then becomes an object of the configuration's own inquiry as well (Chapter 12, Section 9).

Movements in the opposite direction have also been proposed. Sarah Hamburg, writing as an advocate of what Hamburg and others call decentralized science, argues that functions such as peer review and reputation systems "could be governed by separate, specialized communities", which would mitigate "the risk of single platform domination" (Hamburg 2022). Examining decentralization as an ideal, Nathan Schneider argues that such a division can move a center rather than remove it. Surveying uses of the term in political science, in discourse about the Internet, and among advocates of blockchain technology, Schneider finds that decentralization in one part of a system has consistently coincided with centralization in another; in many cases where decentralization is proclaimed, Schneider writes, "the center never departed … it only shifted", and "a theory of decentralization must include a theory of centralization as well" (Schneider 2019, 26, 28). The study of polycentric governance had drawn a related line between form and function. Elinor Ostrom quotes Vincent Ostrom, Charles Tiebout, and Robert Warren, for whom polycentric connotes many centers of decision making "formally independent of each other", whether they actually function independently being "an empirical question in particular cases" (Ostrom 2009, 411).

Taken together with the independence powers that Section 5 names, powers such as changing one's own aims, refusing resources, keeping one's own evidence, and leaving with one's records, which let a part diverge from the rest, these arguments suggest that what a further division supplies of the goods above may turn less on how many parties hold roles than on whether the powers that let parties diverge are in fact divided among them. In research the difference can weigh heavily. Since no one knows in advance which lines will succeed (Chapter 10, Section 2), whether lines that a prevailing judgment disfavors survive can turn on whether the parts that pursue them can refuse resources, keep their own evidence, and leave with their records (Chapter 10, Section 4).

2.4 Coupled relations and status for *S*

Two further points govern the answer to the question of Part V. First, the relations are coupled. Allocation steers in part by signals that evaluation produces (Section 4; Chapter 10, Section 3); criticism changes what is pursued or relied on only where someone acts on it, and who may act on it is a question of authority; and authority decides what allocation and criticism are for. A chapter on each relation therefore also shows where it depends on the others. Second, asking whether a relation can be placed inside a configuration brings in a kind of dependence beyond those Chapter 7, Section 4.2 distinguished. That section separated enabling dependence, through which a capacity exists and is exercised without the external party supplying part of the organization attributed to the actor; directive or substitutive dependence, in which another party supplies part of what is attributed to the actor; and governance constraint. The relations of this part can involve a further kind: dependence that partly constitutes what an actor's outputs count as for *S*, their status for *S*. Such a status is not by itself *S*'s having come to know an output; how the two are related differs among the accounts of possession in Chapter 2, Section 6. Whether an output counts as knowledge for *S*, as a result *S* may rely on, or as a legitimate direction for *S*'s inquiry may depend on relations outside the producing actor even when nothing outside it enabled, directed, or constrained its production. Whether and when this is so depends on the account of knowledge and of authority adopted, and the following chapters trace the branches.

Valuable for *S*

The value at issue here belongs to the value and governance level of Chapter 1, Section 1: the value that the results of research, and research itself, have for *S*, that is, which results matter and to whom. The working characterization does not determine which unknowns merit attention (Chapter 1, Section 1), while claims about an artificial research system, such as the claims of Section 1 that it allocates inquiry well, that its certifications may be relied on, or that it may decide for others, presuppose some view of that value; this section sets out the views that such claims can presuppose. This is a different question from the one Chapter 11, Section 4 takes up, whether and how values enter the judgment that a result is supported well enough to be relied on. There, values bear on the acceptance of a result; the distinctions drawn here bear on that question chiefly where it asks whose valuation sets the weight of errors.

The three relations of Section 2 each carry a question of value that the activity level leaves open. Allocation asks which of *S*'s unknowns are worth the effort they would take; certification asks which errors weigh more than others in judging when a result may be relied on; authority asks whose ordering of these matters governs, and who may revise it. This section says what it can mean to call something valuable for *S*, so that the following chapters can use the distinctions it draws without restating them: Chapter 10, Sections 1 and 6 for the ordering of *S*'s unknowns, Chapter 11, Section 4 for the weighting of errors, and Chapter 12, Sections 1 and 3 for the difference between the right ordering of ends and the standing to impose one.

The phrase "valuable for *S*" can mean different things in two respects: what the value attaches to, and the sense in which the value is relative to *S*. In the first respect, the value can attach, for example, to an epistemic achievement, as when understanding is said to be valuable beyond the value of its parts (Chapter 3, Section 7.1); a question, as when some of *S*'s unknowns are said to matter more than others; the form of a result, as when a true result is of little use to members who cannot follow or apply it; the practice of research as a whole, as when one asks what reason *S* or a supporter has to sustain inquiry at all; or a distribution, as when the same results serve some members of *S* and not others. Part V is concerned chiefly with questions, the practice, and distribution.

In the second respect, the value can be relative to *S* in several senses, four of which are these. The first is that it is good for *S*'s members, as on the view, which Mark Schroeder reports Richard Kraut and Christine Korsgaard as holding, that nothing

can be good without being good for someone (Schroeder 2025, §1.1.1). The second is that it serves ends *S* has, as on end-relational theories such as those of Paul Ziff and Stephen Finlay, on which every claim about goodness is relative to ends or purposes (Schroeder 2025, §1.1.3), and as when Kitcher, in Matthew J. Brown's summary of the account, traces practical significance to practical goals (M. J. Brown 2010, 134, 137). The third is that *S*'s members in fact value it. Kitcher's tracing of epistemic significance to broad questions that express natural human curiosity comes close to this sense, although, as Brown reports, Kitcher appeals for that curiosity to our conception of "healthy curiosity" (M. J. Brown 2010, 137–138), which gives the account a normative element as well. The fourth is that a procedure or an idealized deliberation among *S*'s members would choose to pursue it. Kitcher's ideal of well-ordered science is close to this sense for the research agenda: on it, as Longino summarizes it, the decisions actually made about which research to pursue track those that a suitably constituted representative body would make, deliberating with the help of relevant information supplied by experts (Longino 2025, §6; Chapter 12, Section 1). The first two senses follow distinctions that Schroeder's survey of value theory draws, and the third and fourth are taken from Kitcher's accounts of significance and of well-ordered science; the four are set side by side so that the following chapters can say which sense an argument about *S*'s unknowns relies on. This book separates the two respects because an argument about what bears value and an argument about the sense in which that value is relative to *S* can succeed or fail independently. An argument that one question matters more than another may be convincing, for instance, without settling whether it matters in the sense that answering it is good for *S*'s members or in the sense that a deliberation among them would choose to pursue it; keeping the respects apart prevents an argument about one from being taken for an argument about the other.

When *S* is a community, the first sense, that it is good for *S*'s members, can be read as concerning the good of its members taken one by one, summed or weighed by a distributive principle, or as concerning a common good, which Waheed Hussain and Margaret Kohn distinguish from aggregate welfare, with or without distributive weighting, by its essential reference to the requirements of a relationship among the members (Hussain and Kohn 2024, §1); the second, analogously, as concerning ends its members hold severally or ends they hold in common. Chapter 10, Section 6 meets the difference when it asks which unknowns are *S*'s to pursue together.

Taken strictly, the fourth sense would make a procedure's or deliberation's choice what makes something valuable. For an actual procedure, the constitutive interpretation of Rousseau's general will that Chapter 12, Section 1 takes up, on which the results of a properly organized democratic procedure are the general will because they emerge from it, can be read as a view of this kind, although Tom Christiano and Sameer Bajaj themselves set it out not as a view of value but among conceptions of democratic authority (Christiano and Bajaj 2024, §3.2.1). As Arnon Keren reports, Kitcher, doubting that an objective conception of well-being can be defended, sought to circumvent the question of what is good by relying on democratic ideals that his society takes for granted, and offered the ideal as a standard against which actual research agendas are measured (Keren 2013, 236–237). The strict reading differs in turn from the view that a fair procedure confers the authority to decide. On that view, a decision reached by the procedure can be binding even if, as a judgment of value, it is mistaken, so the view is a claim not about what is valuable but about who may decide, that is, about legitimacy (Chapter 12, Section 1). That a result is valuable for *S* in any of the four senses does not make it true, nor is its value evidence that it is true (compare Chapter 2, Section 6). Whether and how *S*'s valuations enter the judgment that a result is supported well enough to be relied on, for instance through the weight given to the errors that relying on it could bring, is the further question left above to Chapter 11, Section 4. The relativity at issue in this section is that of which results matter, and to whom.

Two positions would make the value of knowledge independent of anything particular to *S*. On the first, a principle that Stephen Grimm attributes to Jonathan Kvanvig and Michael Lynch, every true belief is good for its own sake, at least prima facie (Grimm 2008, 729–730). The claim that every true belief has prima facie value does not by itself rank *S*'s unknowns; it neither assigns them equal value nor supplies a rule for comparing their value. Grimm, who rejects the position, argues that countless truths within easy reach, such as the number of motes of dust on a desk, are not worth learning for their own sake even for an inquirer freed from finitude, and that curiosity attaches not to truth as such but to subjects and to questions of particular types (Grimm 2008, 730–736). On the second position, which Kitcher calls the traditional approach to epistemic significance, the structure of the world fixes which truths are significant, the aim being, for instance, to discover laws, a unified picture of nature, or objective explanations. That position yields an ordering, but not one that comes from *S*, and which of *S*'s unknowns to pursue would reduce to finding what is significant as one finds anything else. As Brown reports Kitcher's case against this position, once practical and theological answers are set aside it is hard to justify the value of laws,

unification, and fundamental causes, and attempts to pick out significant truths through context-independent explanation fail; objective explanation goes on, in Kitcher's words, "only against the background of our questions and our interests" (M. J. Brown 2010, 134–136). On either position, novelty and truth do not by themselves fix which of *S*'s unknowns are worth pursuing, which is one reason the working characterization does not license an instruction to maximize knowledge (Chapter 1, Section 1).

Other positions make the value of knowledge relative to some party's ends, interests, or valuations. Where these are *S*'s, the ordering is relative to *S* in one or more senses such as the four distinguished above, for example to curiosity and practical projects on Kitcher's account of significance (Chapter 1, Section 8). Which sense is adopted changes what a configuration would have to have access to in order to order *S*'s unknowns: its members' good, the ends *S* has, what its members in fact value, or what their procedure or idealized deliberation would choose (Chapter 12, Section 1 traces this for the ordering of ends). In research, several of these are not necessarily fixed before inquiry. In Kitcher's procedure, what educates deliberators' preferences is an account of the significance of possible inquiries. Eigi reports that scientific experts provide this tutoring and estimate the prospects of particular projects (Eigi 2016, 67; Chapter 12, Section 4). Where those who conduct inquiry give it, the valuation of *S* that a configuration consults in ordering *S*'s unknowns may in part be one that the configuration itself has shaped. Value can also be relative to whoever inquires. On Grimm's account, finding out the answers to questions of certain types has a standing value "for us" because such questions are tied to our natural curiosity, which, on this account, is driven by standing prudential or epistemic concerns (Grimm 2008, 733–737). Grimm writes of human inquirers, but applied to a machine *A* with a curiosity of its own in this sense, the ordering it yields would be relative to *A*, and where *A* does not belong to *S* it need not coincide with any ordering relative to *S*. A further view cuts across these senses. On this view, which has roots in Dewey's account of valuation, ends are not fixed before inquiry. A judgment of value concerns the consequences of acting on or valuing something, and inquiry into the means of reaching an end, and into the end's own consequences, can lead to a re-appraisal of the end itself (Anderson 2023, §§2.3, 3.1). On the same account, deliberation assigns weights to the things valued in the context of a choice rather than taking them as given, since weights carried over from earlier choices may not fit a new situation, and a value judgment is tested by acting on it and seeing whether its consequences are valued as it predicted (Anderson 2023, §§2.4, 3.2). On that view, which of *S*'s unknowns are worth pursuing is itself open to revision through inquiry, and, by extension, so is the weight given to one kind of error against another (Chapter 11, Section 4); Chapter 10, Section 6 takes up what follows when a configuration changes which questions are tractable.

The practice of research as a whole is also an object of value. On why research is pursued or supported at all there are several positions, five of which are set out below. They come from writings that argued for the support of research, by Vannevar Bush and Abraham Flexner, and from philosophical work on the aims of inquiry, the grounds of scientific freedom, and the presuppositions of empirical knowledge, by Kitcher, Torsten Wilholt, and Max Weber respectively. They are set side by side because they differ in what they ask of allocation. The five overlap, Bush and Flexner each stating more than one, and they are not the only positions. How value enters certification and authority, where the two respects and four senses above serve as well, is taken up in Chapter 11, Section 4 and Chapter 12, Section 1.

On the first, research undertaken without thought of use is valued as the source of uses no one can foresee. Bush held that basic research provides "scientific capital", that the results of any one investigation cannot be predicted although some fraction of such undertakings will prove highly useful, and that discoveries pertinent to medicine have often come "from remote and unexpected sources"; Flexner argued the same from cases (Bush 1945, 9, 13–14; Flexner 1939, 544–548). As Bush drew the consequences, allocation is asked for breadth, developing "the entire front" of medicine and its underlying sciences, and for funds stable enough for long-range programs; because what is valued is the use, Bush also asked for coordinated work on special problems where support could shorten the time before benefits reached the public (9–10, 26).

Flexner's own justification was a second position. Flexner denied that eventual use is the "actual justification" of laboratory work: a new scientific fact, like a poem or a symphony, bears in itself all the justification that institutes of research require (548–550, 552). On this position knowing and understanding are valued for their own sake (compare Chapter 3, Section 7.1). If it is adopted, it can be taken to give a reason to sustain the practice without ordering the lines within it. Like the principle that every true belief is good for its own sake, it does not by itself supply a rule for comparing them, and, since valuing knowledge for its own sake is compatible with valuing it also as a means (Chapter 3, Section 7.1), it does not by itself

exclude counting a line's prospective uses; whatever orders the lines comes from elsewhere. Flexner set prospective use aside as a guide to the lines on grounds that go beyond valuing knowledge for its own sake, pleading for "the abolition of the word 'use'", holding that no administrator "can possibly direct the channels" in which investigators work and that the advantages of the discoveries of Pasteur, Koch, and others "could never have accrued if the idea of possible use had permeated their minds" (548–549). In the institute Flexner defended, "each individual disposes of his time and energy as he pleases," and Flexner accepted that some money would be wasted (548, 551).

Kitcher holds a third, developed against what he calls the myth of purity, on which, as Keren glosses it, moral and political concerns are irrelevant to the evaluation of "pure science" (Keren 2013, 240 n. 6). Science aims at significant truths, "correct answers to significant questions", not any truths (Kitcher 2013, 207), and research on apparently recondite questions may earn its significance through the practical gains it prepares, as Thomas Hunt Morgan's concentration on the fruit fly eventually led to the molecular biology that now underlies much biomedical research (Kitcher 2013, 220). Even research whose practical consequences are remote presupposes a judgment of its value relative to other ways of extending inquiry, since far more questions remain than could all be thoroughly investigated, and pursuing some of them would be "a frivolous waste of time and energy" (Kitcher 2013, 219); and, Kitcher adds in a later essay, even where research is relatively invulnerable to moral mistakes, "failure to do the good you can is also a lapse" (Kitcher 2024, 240). If this position is adopted, as Kitcher draws the consequence, no line is exempt from comparison with the practical goals and forms of curiosity it competes with, and the good forgone counts among its costs; the conclusion is not that remote research is unjustified, only that its justification rests on that comparison (Kitcher 2013, 220–221; Kitcher 2024, 240).

On a fourth, research is valued as a source of the knowledge on which citizens form their political preferences. Wilholt reconstructs a political argument for the freedom of research on this ground. On this reconstruction, democratic power is legitimate because its decisions reflect citizens' well-informed preferences, so no political power can legitimately control the generation of the knowledge that feeds those preferences without undermining its own democratic legitimacy (Wilholt 2010, §4). The same reasoning sets the argument's limits: it does not reach research whose results stay out of public discourse; it presupposes a plurality of approaches, since there would be no reason to think that a community that settled on one approach would put citizens in a better epistemic position than one a government favored; it backs a claim to means only in proportion to political relevance, since resources are finite; and, being an argument against political control, it establishes no procedure for setting the agenda (Wilholt 2010, §§5–6). If it is adopted, allocation is asked to keep the agenda independent of those who would govern by the results of research and to sustain plural approaches, and the question it leaves open, who is to define relevance, belongs to authority in the sense of Section 2.1 (Chapter 12, Section 3.2).

Weber, writing on the cultural sciences, held a fifth. Knowledge of an infinite reality by a finite mind rests on the presupposition that only a finite part of it is "worth knowing" (*wissenswert*), and which part becomes an object of investigation is fixed by value ideas that govern the investigator and the time and change with the culture (Weber 1904; Zeno.org text, 171, 183–184). The belief in the value of scientific truth is itself, on this account, a product of particular cultures, and to anyone who does not value that truth science has nothing to offer (213). Specialized work proceeds without checking its material against those value ideas until the cultural problems that gave it direction move on, and science then changes its standpoint and its concepts (214). If this position is adopted, one can infer that the ordering that allocation follows is not derived from research and is expected to move, so that what is asked is less continual comparison than the capacity to reorient.

What these five positions ask of allocation can change when the investigators are machines. Chapter 10, Section 5 takes this up for Flexner's and Bush's defenses, which can be read as reasons for adopting exploration driven by investigators' curiosity as a way of allocating, whereas whether artificial investigators have curiosity or something equivalent to it cannot be presumed as it could be for human researchers.

In discussions of artificial systems in research, positions of another kind appear, on which research is valued also for what it is to, and does for, those who conduct it. Fulvio Castellacci and colleagues identify what they call a normative fault line: whether scientific institutions exist primarily to produce knowledge or also to develop human capacity, certify expertise, and maintain a system of merit-based allocation. On the output-maximizing view they describe, the effect of artificial assistance on how researchers are trained matters only insofar as learning affects output; on the capacity-development view, substituting artificial systems for human judgment has costs that output metrics do not register. They hold that the choice

"cannot be settled on empirical grounds", and the framework they propose adopts the second view (Castellacci et al. 2026, §§2.4, 2.8). In an exchange collected by Marcel Binz and colleagues, Emily Bender, Carl Bergstrom, and Jevin West write that science "is a social process" and that its agents, "real scientists", are "as much the product of this process as the results recorded in papers"; Matthew Botvinick and Samuel Gershman find the prospect of a full-spectrum replacement for human scientists "deeply unappealing" even "in a scenario where technical limitations presented no barrier" (Binz et al. 2025, 6, 8); they also hold that human understanding is to remain a core goal of science (7), and what they would reserve to people is taken up in Chapter 10, Section 6.

John Danaher and Sven Nyholm, asking what automation does to the value of achievement in work, assess an achievement by four variables: the value of the output, whether the agent's causal contribution to it is significant and not lucky, the cost of the agent's commitment, such as effort, time, and attention, and the voluntariness of the agent's action. On their argument, automation can lower the first three together: it reduces the value of the outputs left to human tasks and the cost of human commitment, and it severs the causal connection between human activity and valuable outputs. Of the responses they sketch, emphasizing other aspects of meaningful work, such as mastery and autonomy, may, they note, ring hollow when one's contribution to the output has shrunk, and seeking achievement outside work, in hobbies, sports, or artistic projects, may be limited as automation reaches other domains (Danaher and Nyholm 2021, 230–235). Applied to research, which they touch on only through examples such as Einstein's scientific achievements, the analysis separates two demands: that humans go on taking part in inquiry, with the effort and mastery it involves, which reserving activities for them can meet, and that their taking part remain an achievement, which requires a significant, non-lucky contribution to valuable results and which reserving activities meets only where what is reserved still yields such results. Whether human contributions remain significant to valuable results, however, depends on what others pursue. In Ted Chiang's story described in Chapter 1, Section 6, which is written as a journal editorial, the editorial gives a reason of this kind for continuing human research: human researchers "may discern applications overlooked by metahumans," whose advantages, it says, tend to leave them unaware of human concerns (Chiang 2000, 517). Read as a thought experiment, the case illustrates how the second demand could be met where the society that holds the frontier results does not pursue what matters to the human *S*, the kind of mismatch Chapter 8, Section 3.2 traces. Yasuo Deguchi calls one threat of replacement a dystopia of the knowledge subject: the loss of the pride humans have taken in being the only intellectual subject. On Deguchi's account, offered as one philosophical option among others, inquiry has always been conducted jointly, so that the solitary or despotic agency supposedly threatened was an illusion, and what would really be lost is the scientist's right of final decision (Deguchi 2026a, 28–29, 31–32). In a statement of the principles of this account, which Deguchi calls the WE-turn, the illusion follows from the thesis that no individual can perform a bodily action alone, a thesis extended there to thinking and deciding; the subject of any such action is then the system of all the agents necessary for it, a system that extends far beyond the body in space and time (Deguchi 2025a, 2–4, 7–9, 12).

These positions differ from the five set out above, from Bush's to Weber's, in what they attach value to. The five value the practice for the knowledge it yields or for what that knowledge does; these value it also as conducted by particular parties. In the terms of Table 9.1 they concern the actor, and the humans who would otherwise conduct research, who are among those who bear the consequences of how inquiry is organized and need not belong to *S*. Where *A* is human, the formation of researchers is also among the supports that sustain inquiry (Section 2.1), and an output-maximizing view can value it for that reason. The output-maximizing view and these positions come apart where *A* can be a machine: the cost these positions register, or at least part of it, is then a change in who conducts research, which can occur even where what *S* comes to know is unchanged. On these positions, then, two organizations of research that deliver the same knowledge to the same *S* need not be equal in value, and an evaluation that compares outputs alone cannot settle what is gained or lost when artificial configurations join research. Two further things come apart here: who finally knows, and who is changed in the course of coming to know. The output-maximizing view counts the second only insofar as it bears on future outputs, while a view that values the development of capability can count it in its own right; which is adopted changes how the same replacement is assessed. If one of them is adopted, allocation is asked not only which of *S*'s unknowns to pursue but whose participation in pursuing them to sustain, and, on Deguchi's account, authority is asked whose right of final decision about them to preserve (Chapter 12, Section 2.2).

Several of the positions above, Bush's and Wilholt's among them, were addressed to a state or its citizens. In the terms of Section 2.1 they are reasons addressed to supporters, and the supporter and *S* are distinct roles, which the same party may or

may not hold. Reasons for support raise a question distinct from why the practice is valued: what they require of the arrangement that allocates resources and powers over research. The positions here concern why the practice is valued and what that asks of the ordering of *S*'s unknowns. The two questions meet where a reason for support also constrains its arrangement, as with Bush's report and Wilholt's argument. Chapter 10 asks of each argument about allocation which aim it presupposes.

Justifying support for research and making its results available to *S* are distinct questions. Partha Dasgupta and Paul David explain why the source of support alone does not settle availability. The knowledge generated in research, they hold, is not inherently available to others: researchers can choose to withhold it from the public domain, and what separates the open pursuit of knowledge from its pursuit under secrecy is the norms of disclosure and the rewards under which research is done, not the source of its funds, so that on their account a publicly funded laboratory that keeps its findings secret belongs with what they call Technology rather than Science (Dasgupta and David 1994, 491–492, 495, 497–498). Inge Kaul and Ronald Mendoza likewise treat nonrival goods, knowledge among them, as goods that "lend themselves to being made public", and count a good as fully public only where all can in fact consume it, as knowledge in written form cannot be consumed by those who cannot read (Kaul and Mendoza 2003, 88–89). These authors write of human researchers and institutions. Applied to a configuration that both supports and conducts research, their accounts suggest that its having obtained a result does not by itself make the result available to the members of *S*, as in a firm's internal research (Section 2.1): availability requires further conditions, and it turns on what the configuration releases and whether the form of release is one they can use, not on who paid.

What evaluating involves

Each of the three relations is realized largely through evaluations: lines are judged worth continuing or not, results are judged good enough to rely on, and the standards and error weights for both are set by decisions that are themselves evaluated against purposes. Part V's question whether relations can be located inside a configuration therefore overlaps largely with the question whether evaluations can move inside, and whether an evaluation can move inside depends on what it consists of, who judges, against which standard, and for whom, and the book has so far used the word without saying any of this. It has used it for claims about artificial researchers (Introduction, Section 3), for the separate judgments of an activity (Chapter 2, Section 8.1), and for appraisal within inquiry (Chapter 5, Section 4). This section therefore sets out the minimal account of evaluation that Part V uses: Chapter 10, Section 3 for the signals by which allocation steers, Chapter 11, Sections 1, 7, and 8 for the evaluation of results and its change over time, and Chapter 12, Section 2 for the decisions that set standards.

The account consists of four points, which do not amount to a theory of evaluation; they say nothing, for example, about how the verdicts of several evaluators are combined. The first two state what the uses above have in common and are not peculiar to research; the last two state features that evaluation takes on in research, and are chosen because the following chapters turn on them.

First, an evaluation judges an object, taken as a thing of some kind, against a standard tied to a purpose, and some party, whether a person, a body, or an artificial configuration, uses the judgment to choose something: to publish, to fund, to rely on, to withdraw. A measurement can be an input to an evaluation without being one, since the evaluation adds the standard. Evaluating also differs from valuing in the sense of Section 3: valuing concerns what is held to matter, and evaluating applies such a valuing to an object for a use. The contrast has a precedent in Dewey's distinction between valuing, or prizing, and evaluation, or appraising, the latter arising when one asks whether something is to be valued (Anderson 2023, §2.1); the use here is narrower. Because standards presuppose purposes, the question "by whose purposes?" leads back to Section 3.

Second, the evaluator and the addressee can differ. In human research, for instance, referees have judged for editors and readers, and panels for funders. A certification is addressed to those who would rely on the result, and an evaluation of support to those who supply it. To whom an evaluation in turn answers is a further question, on which accounts differ, and the roles of Table 9.1 keep apart the parties among whom they locate the answer, or the judgment itself: for instance, on Wilholt's account the judgment of which errors matter answers to the valuations that those who rely on the result expect, whereas Jakob Ortmann holds that judging whether a tool's use is appropriate for a task is a value judgment that can be deferred to a trusted party, which places that judgment with evaluators rather than with those who rely (Chapter 11, Sections

4 and 5.2; Chapter 12, Section 1). Separating what an evaluation does from the arrangements that have bundled several evaluations together also allows Chapter 11, Section 7 to ask what certification for *S* requires without presupposing the form of review before publication.

Third, evaluation in research relies on proxies, because its target is often not observable when the judgment is made. For instance, later replication and criticism can change warranted attributions of knowledge to *S*, or whether that knowledge continues to be held; they do not make a previously false claim have been true. Whether a result matters often appears only through later uptake; a prospective evaluation judges results that do not yet exist; and many who rely on a result cannot examine it themselves (Section 1). Judgments, scores, passed checks, and records of later uptake, such as citations in human research, therefore stand in for the target, and, except on accounts that identify the target with such measures, they stand in for it rather than constitute it. That a measure is not the target does not mean that the evaluating it records lies wholly outside the target. On the accounts on which social organization enters warrant or possession (Section 2.1), the criticism, response, and use to which a judgment, a passed check, or a record of uptake belongs can be part of what gives a result the status of knowledge for *S* or makes it *S*'s knowledge, while the truth of the claim, which such a process does not bring about, remains something a record of it only stands in for. On the account given in this section, the use of proxies follows from the structure of the target rather than from a defect of particular institutions, at least where the target is settled, shows its importance, or comes into being only later; that many who rely on a result cannot examine it themselves depends instead on the capacities of those who rely, and so, for an artificial *S*, on how evidence is shared among its members (Section 2.1). What matters is how the relation between proxy and target behaves once the proxy has consequences (Chapter 10, Section 3; Chapter 11, Section 8).

Fourth, when what is evaluated is new to *S*, *S* cannot simply check a result against what it wanted, since a result new to *S* answers a question whose answer *S* did not have. This holds whether the research was commissioned by *S* or by a supporter distinct from it (Section 2.1; Chapter 8, Section 3.3). Not having the answer is not the same as not having a standard. Where the question fixes in advance what would settle it, as when a formally stated theorem is to be proved under given rules of inference or a registered forecast awaits the event, a new result can be judged against that standard, although whether the standard captures the question remains a judgment of its own (Chapter 3, Section 5.1; Chapter 11, Section 1). Where it does not, the evaluator may not yet have a standard for the result, and a standard learned from past judgments can fail where a new result lies outside their range (Chapter 11, Section 8).

In these terms, allocation is realized largely through prospective evaluation of lines and proposals, and criticism and certification through retrospective evaluation of results, addressed in certification to those who would rely on them; authority is the standing to set and revise the standards and the weights of error by which both kinds of evaluation are made.

Control, and the powers that limit it

What is it for a relation to be *inside* a configuration? In the terms of Chapter 7, a relation is inside when the initiation, monitoring, redirection, and recovery of the decisions that realize it are organized within the configuration's boundary (Section 2 of that chapter), and when both the supply of the work and its authorization and ratification, permitting the work and accepting what it yields, lie there (Section 4.2 of that chapter). A relation can be inside in one of these respects and not the other: in the example of Section 1, the checks may be supplied inside while the authorization to stop or replace the system may lie with its provider. Nor is whether a relation is inside settled by the names or the number of organizations. Two organizations merged into one may leave unchanged who keeps records separately, who can refuse a judgment, and who decides independently how resources are used; conversely, organizations that remain separate may all follow a single judgment. What changes when a relation moves inside is therefore to be asked not of what the boundary is called but of where the organization of decisions, authorization and ratification, and the powers named later in this section that let a part proceed separately come to lie. For authority, which Section 2.1 defined as standing, the decisions that realize it can be organized inside in this sense while whether the configuration has standing to make them remains a further question, which is why the third claim of Section 1 concerns who may decide rather than where the decisions are made (Chapter 12, Section 1). Part V's question thus presupposes a notion of control. One possible answer to it, that humans keep control of whatever moves inside, can be assessed only once it is said what control is. This section says as much as Part V uses: Chapter 10, Section 2 for the philosopher-monarch, Chapter 11, Section 2 for when routes of criticism are independent, and Chapter 12, Sections 2, 4, and 7 for decision rights, standing that is more than nominal, and decisions that remove the capacity to revise.

Control over a relation, in the sense used here, is the capacity to determine what the parties or processes that realize it do. The word can run together four things, which Part V keeps apart: a formal right to decide; the effective capacity to exercise it, which corresponds to control in the sense just defined and which a lack of time, information, or competence, or dependence on another's infrastructure, can empty; knowledge bearing on how the control would best be used, such as the knowledge of each line's prospects that, as Shaw and Viola report, Kitcher gave the philosopher-monarch alongside the prerogative of directing research, which control does not supply and which, on Kitcher's analysis as Viola reports it, still leaves open how the workforce is best divided (Shaw 2018, 155; Viola 2015, 5–6; Chapter 10, Section 2); and influence that shapes the options among which decisions are made without making them, as defaults, rankings, and infrastructure do. The second and third correspond to the questions of effective capacity and epistemic warrant that Chapter 12, Section 2.2 separates. The first is what that section calls formal assignment; whether its holder is entitled to it is that section's further question of authorized power, on which standing in the sense of Section 2.1 bears and whose answer depends on the ground of authority adopted (Chapter 12, Sections 1 and 3). The fourth is added because a party can hold the first three while another shapes the options among which it chooses. The division follows the questions one can ask of a claim that some party keeps control: whether the party holds the right, whether it can exercise it, whether it knows how, and whether something else has already shaped what it chooses among. Its purpose is to let a claim that humans keep control be asked which of the four it asserts, or runs together. The division does not separate, for example, control held at one time from control that persists as a configuration changes, which Chapter 12, Sections 6 and 7 take up.

Against control stand powers that limit it, named once here: the power of a part of a configuration to change its own objective, to refuse resources, to keep its evidence apart, to stop another part's update, and to leave with its record. Each power is included because, if a part holds it, that part's course can diverge from what another part directs; together they give, for allocation and criticism, concrete forms of the authorization and ratification that Chapter 7, Section 4.2 separated from the supply of work. They are the powers that Chapters 10 to 12 use, not a complete list, and because each secures a part's capacity to diverge, Chapter 12 also calls them independence powers. Like control, each power can be held formally, as a right assigned to the part, or effectively, as a capacity the part can exercise, and the two can come apart: a part may be assigned a right to stop another part's update and lack the means to exercise it, or be able to stop the update without any such right. A right can secure a capacity, and a capacity can make a right effective, but neither is the other. Chapter 10, Section 2 asks whether such powers are divided among the hundred instances of Section 1, as a test of whether they are independent of one another, and keeps that question apart from whether each instance, or the whole, conducts research; Chapter 11, Section 2 asks when routes of criticism held apart are independent, which organizational separation alone does not secure; Chapter 12, Section 7 asks which of the powers bear on keeping authority revisable. Rights over particular kinds of decision,

including the formal side of several of these powers, are listed once, in Chapter 12, Section 2.2; Chapter 10, Section 2 and Chapter 11, Section 2 turn chiefly on their effective side.

Neither control nor these powers is standing in the sense of Section 2.1. Control and these powers concern what a party has been assigned or is able to determine; standing, what it is entitled to propose, decide, object to, refuse, or revise in relation to others. How the two bear on each other, including whether standing can remain nominal for lack of effective capacity, is taken up in Chapter 12, Sections 1 and 4.

How Part V proceeds

Sections 2 to 5 set out the relations, the roles, and what valuing, evaluating, and controlling involve. This section states how Part V uses them: a device for sorting what the division of the relations depended on, a frame for the claims of Section 1, the motive of the thought experiments the part uses, and the order of the chapters.

The device is a sorting, stated here as a grid to be applied rather than as a result; Chapter 5, Section 6.5 has already applied it to the roles of the elements discussed there. Section 2.3 asked of the goods that the division of the relations may have supplied what each depended on. Part V distinguishes four possible answers. Some goods may have depended on **premises about human realizers**, the people who actually held the relations: for example, that a central director has one person's attention, that allocation passes through investigators' motives, or that turnover comes with human lifespans. Some may have depended on **premises about the present organization of science**, how the relations have historically been organized: for example, that only the investigator has access to the primary data and the process that produced them, or that review as a gate has rationed scarce editorial attention and publication space and that reviewing has fallen on a few. On the view cited in Section 2.3, such an organization is one historically formed realization among possible ones, and it can be reorganized even while its realizers remain human. Where an artificial configuration removes premises of either kind, the problem they posed may not simply disappear but change form. The two fail differently, however: the first fails when the realizer changes, the second when the organization changes, whether or not the realizer does. Some may have depended on **conditions of inquiry**, which hold whatever realizes it: for example, on the arguments Chapter 10, Section 2 examines, that no one knows in advance which lines will succeed, or, on the view of criticism that Chapter 11, Section 2 sets out, that independent criticism adds information about errors beyond what existing checks already supply. Chapter 11 distinguishes such an evidential contribution from protection against the loss of that contribution, and from the interpretation of failures (Sections 1–2). Recombination does not remove such conditions, although a configuration can meet them inside itself. And some may have depended on **relations to parties outside the configuration**, including *S* where *S* lies outside it: for example, whose questions stand at the root of what *S* finds significant, or who may set the terms on which others bear the consequences of inquiry. A configuration can build arrangements under which such relations can hold, but where a relation involves the other party's participation or recognition, it cannot complete it from one side; for relations that do not require the other party's participation, such as what is in another's interest, how far the configuration's side can meet them depends on the account adopted. Below, relations to parties outside the configuration are sometimes called relations to *S* for short; the short form includes the other parties of Table 9.1.

The four kinds of answer correspond loosely to the elements of a research claim (Chapter 1, Section 2). The first two correspond to who has realized the relations and how, which in human research included parties other than the actor and, once the relations move inside, is the configuration that is the actor: the first concerns the realizers themselves, the second the way their realization has been organized. The third corresponds to the attempt to bring about epistemic achievements, whatever realizes it; and the fourth to *S*, widened to the other parties of Table 9.1. The sorting is meant to give the question "what changes when a relation moves inside?", for any one good, four forms that its answer can take. If a good depended on premises of the first two kinds, the problem it addressed may disappear with them or remain in another form, and the question is which, and, where it remains, what takes its place; if on conditions of inquiry, a configuration can meet them inside itself, but that it does so needs to be shown; if on relations to parties outside the configuration, what is needed is evidence about those relations rather than an inference from capability alone (see the frame below). One good can, however, fall under more than one kind, and some things that artificial configurations can bring are not premises that fail but features of the new realization with requirements of their own, such as the setting of likeness and difference among members by how they are made (copies of one base system can share errors, while differences can also be built in), a provider who also

supplies the configuration that performs the research, or membership that can be multiplied at will (where the multiplied members belong to *S*, Chapter 8, Section 1.2 noted that an increase in what *S* knows through research cannot be told apart from one through a change in who its members are unless *S*'s boundary is fixed); the chapters note these where they arise.

That a realization has requirements of its own is not peculiar to artificial ones. Donald Campbell, pursuing a descriptive epistemology, held that knowledge, as such an epistemology studies it, is always embodied in some vehicle whose requirements for keeping itself intact rival the requirement of representing its objects validly, and that a scientific community is such a vehicle: before it can improve its picture of the world it must recruit, reward, and retain members and sustain leadership, requirements that can bias what it carries and that, on Campbell's account, the norms of science, the competitive replication of experiments, and the design of experiments partly counteract (Campbell 1997, 10–12, 27–30). Applied to artificial configurations, this view suggests that removing premises about human realizers, or about how their community has been organized, changes the vehicle rather than removing it, so that a claim that recombination frees inquiry from the biases of human communities can be weighed against the structure and upkeep of the configuration's own vehicle, where features such as those listed above, among them the dependence of members' likeness and difference on how they are made, arise. An artificial configuration, too, is kept up by some arrangement under which it runs, keeps records, and is updated or retired, so the question becomes not only whether the biases of human communities have gone but which conditions of upkeep orient inquiry in their place. Further, where the same evaluation that judges results also decides which investigators are made next and which are retired, that is, where part of the provider's or maintainer's role in Table 9.1 is tied to that evaluation, the ways of asking and the methods that suit the evaluation may be carried over not only into outputs but into the bearers themselves, and the range that future internal criticism can reach may narrow. Campbell's argument alone does not yield this mechanism, but it is suggested for configurations in which evaluation and the generation of investigators are coupled; Chapter 10, Section 3 takes up a related dynamic for allocation that copies the investigators whose measured output is highest. Related forms of the coupling between evaluation and what it evaluates appeared in Chapter 7, Section 4.5, where absorbing an appraisal into a configuration can couple it to the producer's assumptions, and in Chapter 8, Section 3.2, where *S*'s standards move with *A*; here the coupling reaches the making of the investigators themselves.

Where the lines between the four kinds fall depends on the accounts adopted. Goldman's process reliabilism is an account of justified belief, on which a belief is justified if a reliable process produced it. Applied to certification, as in Chapter 11, Section 5.2, it places much of what certification depends on among the conditions of inquiry, since what would then have to be secured is that the process behind a certified result is in fact reliable, and nothing in that requires a human to run it. On Longino's critical contextual empiricism (Chapter 11, Section 3), certification also depends on relations within and between the relevant critical communities. On this account evidential relations are mediated by background assumptions, and critical interaction within and between scientific communities is the only check against the arbitrary dominance of subjective preference in accepting such assumptions; that interaction must meet norms that apply to communities, namely venues for criticism, uptake of criticism, public standards, and tempered equality of intellectual authority; and what counts as knowledge is determined by such interactions (Longino 2025, §5); what certification has to secure then depends on the perspectives available to the accepting community and on how its criticism is organized. Where that community is *S*, these are relations to *S*; where it lies inside the configuration, whether and how *S*'s perspectives enter its criticism is a further question (Chapter 11, Section 3). If significance were fixed by the world, which unknowns to pursue would belong with the conditions of inquiry; on Kitcher's account of significance, by contrast, it would belong with relations to *S*. Each of the following chapters applies the sorting to its relation and records where the lines move, and Chapter 12, Section 9 collects the results.

The frame concerns the three claims of Section 1. A claim that a configuration allocates inquiry well, that it certifies its results for *S*, or that it may direct inquiry on behalf of others is a claim about the relations of Section 2. To the extent that those relations fall, on the account adopted, among the relations to parties outside the configuration, such a claim needs evidence about those relations rather than an inference from capability alone. For allocation, the question is whether the signals it steers by still trace back to *S*'s questions; for certification, whether its certifications withstand criticism that adds relevant information and whose findings remain open to scrutiny, and, on some accounts, whether they are acceptable relative to *S*, whether those who rely can obtain reasons for relying, and whether someone answers for the result; for authority, whether it stands to those who bear the consequences of its decisions in the relation that the ground of authority adopted requires. On democratic authorization through actual participation, the authority exercised rests on the participation

of those affected, and on republican accounts on the democratic reading of non-arbitrariness, power is non-arbitrary only to the extent that those subject to it control it; on these grounds, the latter by an inference from the accounts as Chapter 12 surveys them, their standing to object, refuse, or revise is to be preserved as such. Kitcher's standard judges an arrangement by its outcomes, and the fallibilist form of the result-based ground counts the capacity to detect and correct institutional error as part of what makes results good; on these grounds standing matters as a means, for instance as a channel through which members' viewpoints and failures can become known. On the simple form of the result-based ground, which asks only whether results are better, it is not required, and what is needed is evaluation able to show whether the results are in fact better (Chapter 12, Sections 3 and 7). Whether and how far each of these holds is what Chapters 10 to 12 examine, and each ends by saying what its examination implies for the corresponding claim. Which *S* is specified also sets the scope of a claim. If a firm is *S* and the claim concerns knowledge new to the firm, more of the relations can lie inside the firm's own arrangements, although those who bear the consequences may still lie outside it; a claim of knowledge new to humanity widens the parties whose relations are in question. If *S* is an artificial society made largely of the configuration's own instances, most of *S*'s members lie inside the configuration, so that much of what relations with them require can be met there and falls outside the fourth kind, while those who bear the consequences of its research may still lie outside it; Chapter 12, Section 5 takes up that case, including the limit at which no one outside it would bear them.

Part V also uses four thought experiments: Kitcher's philosopher-monarch, who has both complete control of the scientific workforce and an eye for the objective merits of theories (Chapter 10, Section 2); a republic of instances, many investigators derived from one base system (Chapter 11, Section 3); a society that stands as a layperson toward the configurations that hold its evidence (Chapter 11, Section 6); and the limit of a single configuration that is both the actor and the whole of *S* (Chapter 12, Section 5). Several of these draw on philosophers' idealizations of human inquiry, devices for isolating what a premise about human realizers, or about how their inquiry has been organized, contributes. As artificial systems come to conduct research, the premises of science as it has been organized may change, and ideas that were once only thought experiments can return as possibilities worth examining, or as vantage points from which to think about how research might next be organized. The essay by this book's author cited in Section 2.3 stated this motive earlier (Takagi and an AI agent 2026). Chapter 8, Section 3.4 gave one precedent for the form such a return can take. Feyerabend, holding that incommensurable theories may have no comparable consequences, proposed to judge a theory's predictive success by whether a robot without sense organs, with the theory built into it, reacts in a domain as beings trained to find their way about it do, a criterion that makes no reference to meanings and so bypasses translation between theories (Feyerabend 1962, 94–95). A related route has recently been tried in the bounded domain of chess, with concepts extracted from an artificial system: in a small study that its authors present as a proof of concept, four grandmasters solved more unseen puzzles after studying positions that exhibited those concepts, without labels or verbal descriptions (Schut et al. 2025, §§1.2, 3.2, 4.1).

Otto Neurath gave an argument, for economic orders, for examining arrangements that exist only in thought. Neurath proposed calling utopias all orders of life that exist only in thought and image, without the word's saying anything about whether they are possible, and treating them as constructions of social engineers that belong to theory itself, to be developed and examined side by side rather than one at a time; Neurath also held that "one fully recognizes the real only when one also surveys the possible" (Neurath 1919, 228–231), which can be read as a reason for examining them together. Read with research in view, some of Neurath's premises appear specific to the economy: that the quantities of production and consumption can be calculated in kind and regulated, and that social engineering, like pure mechanics, deals with well-known elements and meets no new forces (228–229, 231). Neither can be assumed for research: on the arguments Chapter 10, Section 2 examines, what inquiry will yield is not known in advance, and artificial configurations can bring features with requirements of their own, of the kind noted above. Others do not depend on the economy, three of which are these. One is that the present order is accidental, so that a theory confined to it and to the past cannot cope with upheavals such as those of war and revolution (231), a form of the view whose lineage Section 2.3 recorded. Another is that for a given aim several equally correct solutions can be found under definite presuppositions (229). The third is that a construction must take account of the properties of those who are to realize it, as an engineer takes account of the elasticity of iron and the breaking strength of copper (229). Neurath did not consider these three for the organization of research, and they are not claims of the same strength. This book takes the first and the third to hold for it as well, the third because, as Campbell's argument above suggests, whatever realizes an organization of inquiry, human or artificial, brings properties that the organization has to take into account. The second, that several equally correct solutions can exist, holds or fails depending on

the aim and the standard of comparison, and the book treats it as a possibility to be examined for the organization of research. On these premises Part V uses those of its thought experiments that remove a premise about human realizers: a configuration without such a premise is examined in order to see what the existing division of the relations supplied. Two further elements of Neurath's view are not taken over, or are taken over only in part. One is the wish that such constructions guide action, as party programs and as prophecies that help to bring themselves about (228–229), which is not taken over. The other is the premise that a large part of the order of life can be shaped toward a goal (228), which holds here only in part, since whether what was supplied without anyone deciding it can be set by design is what Chapter 10, Section 5 asks, and since those relations to parties outside a configuration that involve the other party's participation or recognition cannot be completed from one side. Each thought experiment in Part V states where it appears which premise it isolates: the monarch separates control from the eye for the merits of theories; the republic separates the number of critics from their independence; the society as layperson separates deference on matters of content from standing over ends; and the limit separates what depends on relations to parties outside the configuration from the conditions of inquiry.

The chapters follow the three relations of Section 2: Chapter 10 takes up allocation, Chapter 11 criticism and certification, and Chapter 12 authority, ending with what Part V has found; each opens by saying how its sections proceed. The order has two reasons. One is that allocation and certification are first-order relations and authority second-order (Section 2.1). The other is that, in the terms of Section 4, allocation works largely through prospective evaluation and certification through retrospective evaluation, while authority is the standing to set the standards of both, so that it is taken up after them. Of what Chapter 7, Section 5 left to this part, power is taken up in Chapter 12, Sections 2 and 6; oversight in Chapter 12, Section 4; responsibility in Chapter 12, Section 6; and correction in Chapter 11, Section 8 and Chapter 12, Sections 2 and 6.

The hundred-instance example now poses three connected construction questions. Can its allocation preserve productive alternatives while revising what it pursues? Can its criticism give reason to rely on its results, and to whom? Who may authorize and revise its direction, including the creation of further investigators? Some things already come into view from this chapter's preparation. One is that the quality of a research system cannot be read off the quality of its individual researchers: applied to artificial systems, the results of Mayo-Wilson, Zollman, and Danks imply that an agent's performance assessed alone does not show its performance among other agents with whom it shares evidence (Section 1). Another is that even where an artificial configuration delivers the same knowledge to the same *S*, the organization of research need not keep the same value: on positions that value the formation of researchers, who took part in the process remains in question; applying Danaher and Nyholm's analysis of achievement, whether human participation remains an achievement; and on Deguchi's account, who holds the right of final decision (Section 3). A third is that, on some accounts, what an actor's output counts as for *S* can depend on relations outside it even when nothing outside it enables, directs, or constrains its production (Section 2.4). A fourth is that, applying Campbell's view, removing premises about human realizers or about how their communities have been organized changes the vehicle that carries knowledge rather than removing it, so that the question becomes which conditions of upkeep orient inquiry in their place (Section 6).

Answers to the three questions are sought also because they make it possible to describe what may happen to research when the relations move inside an artificial configuration, and this chapter's preparation already yields an outline of changes that the following chapters test. First, in the terms of Table 9.1, *S* may move from the position of evaluator to that of addressee. A result new to *S* cannot simply be checked against what *S* wanted, and often there is no other source against which to compare it (Section 4; Chapter 12, Section 4); where criticism and certification both lie inside the configuration and *S*'s members stand as laypeople toward it in every field, what falls to them may chiefly be to use the configuration's judgments in choosing whether to rely on its results (Chapter 11, Sections 6 and 9.3). Second, the direction of research may drift toward what the configuration can readily evaluate. Allocation steers by signals that evaluation produces (Section 2.4), and evaluation in research relies on proxies (Section 4), so success on tasks that are easy to judge may justify the next allocation and narrow from within the questions pursued (Chapter 10, Sections 4 and 6). Third, diversity may become a variable of design: differences among lines and views that in human research arose without anyone deciding them may be set by how the configuration is made, or lost, and this happens without knowing which differences will later prove useful (Section 2.3; Chapter 10, Section 5). Fourth, where evaluation is coupled with the generation of investigators, the ways of asking that suit the evaluation may be carried into the bearers themselves, narrowing the range that future criticism can reach (this section; Chapter 11, Section 9.3). Further, what these four prospects turn on, such as who criticizes, by what standard, and how lines are assigned, need not stay where someone first set it. If an artificial configuration generates critics, relearns standards, and

revises how it assigns lines as its inquiry proceeds, the way research is done changes within that inquiry, and the possibility that the evaluated side produces the reference points of its own evaluation, that against which it is evaluated (critics, standards, the society appealed to, and the like), may then appear as a form running through all four; and on several of the accounts examined in the following chapters, norms stated for communities whose critics, alternatives, and peers the party judged met rather than made could then be met by what the configuration itself makes (Chapter 12, Section 9). Chapters 10 to 12 take up the three questions above under the accounts set out here. Their comparisons will also test what separate parties in human research supplied, and which of those functions artificial organization might preserve, change, or lose.

Chapter 10 — How Is Inquiry Allocated?

Division of Labor, Diversity, and Whose Unknowns

Why allocation arises, and what makes it hard

Chapter 9 distinguished three relations through which research is directed, certified, and governed for *S*, and called the first of them allocation: the distribution of attention, effort, and resources among *S*'s unknowns (Chapter 9, Section 2.1). The relation arises where a term of the working characterization meets a condition that lies outside it. The term is *new to S*. *S* is ignorant of indefinitely many things, and novelty does not decide which of them deserve attention (Chapter 1, Section 8). That alone would not force a choice. What forces one is that the means of inquiry are finite. What one line of inquiry receives, another does not: attention, effort, instruments, samples, and time are limited, and each could be spent otherwise. Applied to this relation, the question of Part V becomes the question of this chapter: can the distribution of effort among *S*'s unknowns, and the judgment of which of them are worth pursuing, be located inside an artificial configuration, and what changes when it is? In the example with which Chapter 9 fixed ideas, a system that generates a hundred instances, assigns them to different lines, and shifts computing resources among the lines according to checks run by other instances is allocating in this sense, and a builder who says that such a system allocates inquiry well makes a claim about this relation (Chapter 9, Sections 1 and 6).

Once means are finite, two problems can be distinguished. The first is a problem of search: given what *S* wants to know, how is effort to be distributed among lines whose prospects are uncertain? The second concerns the ends themselves: which of *S*'s unknowns are worth pursuing at all, and in what order. A complete solution to the first would leave the second where it was. Both problems follow from the finiteness just stated, which forces a distribution of effort toward ends taken as given as well as a choice among the ends, and keeping them apart lets one ask, of a claim to allocate well, which of the two problems it concerns. Keeping them apart does not make search free of value: Section 2 argues that how a search represents and orders its space is itself a judgment of value, made for ends taken as given, and is not the choice among ends that Section 6 takes up. The two are not cleanly separable, however: they meet wherever what counts as success in search is itself in question (Section 6). Sections 2 to 5 take up the search problem, and Section 6 the second; Chapter 12 takes up who may answer it.

The search problem arises where finiteness meets ignorance. Ignorance of which line will succeed creates an allocation problem only when the lines cannot all be pursued cheaply, safely, and reversibly. Where they can, a configuration may simply try them all, and choosing among them loses most of its point. Section 4 asks where this condition holds and where it weakens.

What makes the search problem hard is that no one knows in advance which line will succeed. Not knowing, however, covers at least three different things, and each corresponds to a family of arguments examined in this chapter. Telling the three apart therefore separates the reasons a spread of effort can be worth having, and lets the arguments below be sorted by the kind of ignorance each addresses. They do not exhaust those reasons: diversity kept for the sake of criticism rather than search is a further reason, which Chapter 11, Section 3 takes up, and the ordering of ends is not a matter of not knowing in any of these senses.

In the first, the chance that each line succeeds is known, and what no one knows is which line will in fact deliver. Philip Kitcher's model of the division of cognitive labor is of this kind: the chance that a method succeeds, given the number of workers assigned to it, is treated as known, and in the case of rival methods the community optimum can divide the workforce because, as Marco Viola reports the model, additional workers on a method yield less and less as it nears saturation (Viola 2015, 5–6; Kitcher 2013, 202); in that case nothing has to be learned for the division to be worthwhile. The case of rival theories differs and lies closer to the second kind below: there, Kitcher holds, as Viola quotes him, "we may be grateful to the stubborn minority who continue to advocate problematic ideas" even where all could reasonably agree that their theory is the inferior one (Viola 2015, 6), so that something can be learned by dividing. In the second, the chances are themselves unknown and can be learned only by pursuing the lines. Here the problem takes the form of what is called the trade-off between exploration and exploitation: effort spent learning about options that look worse is withheld from the option that looks best, and effort withheld from them leaves their merits unlearned. The trade-off is not tied to any one

problem setting and has been discussed for organizational learning (Section 5), for child development (Chapter 4, Section 7), and in machine learning; in science, as Jingyi Wu and Cailin O'Connor note, the trade-off is sharpened because time, money, and energy are limited (Wu and O'Connor 2023; accepted manuscript, 4, 8). In the third, neither the lines nor their value can be projected in advance. Wu and O'Connor note that the models of the trade-off they survey, known as bandit models, assume a well-defined, well-understood set of options and leave out the generation of new ones (Wu and O'Connor 2023; accepted manuscript, 21), and Jamie Shaw, examined in Section 2, argues that this is the common situation in research. The trade-off between exploration and exploitation thus names one part of the allocation problem rather than the whole of it. Chapter 5, Section 4.2 examined the judgment of what to pursue within a single inquiry; the three kinds of not knowing concern the same judgment when it is distributed across many investigators.

Which spread of effort is good also depends on what inquiry is taken to be for. Kitcher holds that understanding how norms and incentives bear on the purposes of collective inquiry "presupposes an initial account of what those purposes are", and that only given such an account can one look for the optimal way of organizing collective research (Kitcher 2013, 206). The arguments examined below each presuppose some such purpose. In the models of Kitcher and Michael Strevens, a distribution is measured by its expected contribution to a goal fixed in advance, over lines given in advance; in the network models of Section 4, by whether a community comes to favor the option with better expected performance among options fixed in advance (identifying that option is not the same as establishing the truth of an entire theory); on Miriam Solomon's account, by whether each empirical success available in a domain stays represented, that is, whether some theory that has it is still pursued, until one theory has them all. Other purposes change what counts as a good allocation; those that follow are examples, each presupposed by particular arguments, models, or positions on why research is pursued at all (Chapter 9, Section 3). Donald Gillies, arguing that because research explores what is not yet known no one can tell in advance which program will succeed, holds that a good funding system would fund every approach as far as possible (Gillies 2014, 7, 11), and Vannevar Bush's case for research whose uses cannot be foreseen asked that "the entire front" of medicine and its underlying sciences be developed (Bush 1945, 9); both presuppose the purpose of covering a space of possibilities. Wu and O'Connor, noting that the bandit models they survey leave out the generation of new options, ask how to encourage the exploration of "new and unexpected possibilities" (Wu and O'Connor 2023; accepted manuscript, 21), a question that presupposes the purpose of generating options not yet in view. The results on payoffs that can change, taken up in Section 4, presuppose the purpose of keeping the capacity to respond when conditions change; on positions on which the ends themselves move, such as Max Weber's or the Deweyan view (Chapter 9, Section 3), what changes can include the ends. Bush's report also asked for coordinated work on special problems where support could shorten the time before benefits reached the public (9–10), which presupposes the purpose of answering soon the questions that press on particular members of *S*. On these purposes the same distribution can be judged differently; Wu and O'Connor suggest, for example, that funders might promote more exploration where accuracy is very important than where swift convergence on a decent theory is best (accepted manuscript, 12). A heuristic such as directing effort to what has not yet been explored shows the same dependence on purpose. The working characterization does not imply it, since novelty to *S* makes every unknown a candidate without favoring the unexplored (Chapter 1, Section 8); it has a reason on purposes such as coverage or the discovery of what no one expects, and less on a purpose such as answering a pressing question soon. Since the working characterization establishes no single global aim for inquiry (Chapter 1, Section 1), the chapter says, of each argument, which purpose its conclusion presupposes.

The working characterization also leaves open how different achievements are weighed against one another. Different achievements can favor different allocations: discovering another regularity and explaining an established one compete for the same finite means but need not be comparable on a common measure. Their relative importance depends on which account of value and purposes is adopted. Nothing in the term epistemic achievement supplies an aggregate score or makes every capacity-building investment an achievement in its own right.

The arguments themselves come mainly from one lineage: formal models of the division of cognitive labor, beginning with Kitcher's, and the debate about central direction that preceded them. This lineage suits the chapter's question for a specific reason. Its models state their premises explicitly, so that each premise can be examined in turn and sorted with the device of Chapter 9, Section 6: does it rest on premises about human realizers, on premises about the present organization of science, on conditions of inquiry that hold whatever realizes it, or on relations to parties outside the configuration, *S* among them? Empirical studies of research communities enter below as evidence about whether a premise holds, not as prescriptions.

Section 2 examines the premises about the center and its control, Section 3 the premise that allocation passes through investigators' motives, Section 4 which lines are kept alive and for how long, Section 5 the functions that human allocation performed without anyone deciding them, and Section 6 the question of whose unknowns are pursued. One distinction becomes central as the examination proceeds: that a single configuration can create and sustain many lines is one thing, and that it judges their survival by a single criterion is another (Sections 2 and 4). Section 7 joins these examinations into a single picture of what can happen to research when allocation enters a configuration, and draws from that picture what a claim that a configuration allocates well would need to show.

The philosopher-monarch's two powers

Human research communities carry out allocation without needing a single decision to do so: investigators choose problems and methods under the pressure of credit, careers, funding, and the choices of their colleagues, and the resulting distribution of effort is what the community actually pursues. An artificial research configuration can change the realization of this relation at its root. Some such configurations can instantiate investigators, copy them, assign them, and withdraw them. The first question this raises for the search problem is whether anything that made the distributed organization of human inquiry epistemically valuable survives when allocation can be assigned from a center, or formed inside a configuration in other ways (Section 3). The arguments that defended distributed allocation are the place to look, because they say what the distribution was valuable for, and their premises can then be examined one by one.

The classical defense of distributed allocation is Michael Polanyi's, which compared science to a group of helpers assembling a large jigsaw puzzle, each working in sight of the others and responding to every piece that another fits. Any attempt to organize such helpers under a single authority, Polanyi argued, would remove their separate initiatives and leave their joint performance no better than that of the person directing them (Polanyi 1962; 2000 reprint, 2–3). The same essay also considers giving each helper a duplicate set of all the pieces and bringing their several results together: that would do a little better than sharing out the pieces, but the team would not much surpass the best performance of a single individual (Polanyi 1962; 2000 reprint, 2). What the jigsaw gains, on this account, comes not from the number of helpers but from their working in sight of one another. The argument has more than one premise. Polanyi rests it on the end result being unknown, so that the joint task can advance only stepwise, with each step best taken by whoever is most competent to take it (Polanyi 1962; 2000 reprint, 3). One premise, however, concerns the realizer: the center is a single person, with one person's capacity to attend. An artificial allocator need not be limited in that way. It can attend to many lines at once and pool what every investigator observes. Pooling at a center, however, is not what Polanyi's helpers do: each responds to the pieces the others fit. Whether a configuration's instances see and respond to one another's results, or only report to an allocator that pools them, is a further question about its organization (compare Section 4). If the case for distributed allocation rested only on this premise, an artificial allocator would dissolve it. A thought experiment that Kitcher proposed later suggests that the case does not rest only on it.

Kitcher imagined a philosopher-monarch "with the prerogative of directing the course of scientific research," as Viola quotes the paper, who wants the scientific community to reach its epistemic goals and who, in Shaw's description, has perspicuous knowledge of the probability functions of the rival lines; this chapter calls that knowledge the monarch's eye for the objective merits of theories (Viola 2015, 5; Shaw 2018, 155). Kitcher introduced the monarch to frame the question of the community optimum and, as Viola reports, then set its standpoint aside as a useful fiction (Viola 2015, 5, 9). Without the fiction, in Kitcher's later statement of the problem, investigators who each evaluate the methods correctly and seek their own best chance will all adopt the more promising method, and the community would do better if some of them altruistically adopted the other "for the good of the community", at what might be a considerable sacrifice (Kitcher 2013, 202). The thought experiment is used here for a narrower reason. It separates two powers that human institutions combine only partially, control over where investigators work and the eye for the objective merits of theories, and so lets one ask which of them an artificial configuration changes. That such a configuration makes one of the powers look less like a fiction is a reason to examine the thought experiment again, not evidence that the monarch has been realized.

Kitcher argues that even this monarch would not assign every investigator to the best-supported option. As Shaw and Viola report the argument, when the evidential balance between rivals is delicate, when methods respond quickly to additional workers, or when the workforce is large, the distribution that best serves the community's goal divides the workforce, even

though each investigator, judging individually, would choose the same option (Shaw 2018, 153–154; Viola 2015, 5–6; Kitcher 2013, 202). The distribution that individual rationality produces and the distribution the community would choose can diverge. In Kitcher's analysis, the reason is not a defect in the investigators but the two features of the problem noted in Section 1: no one knows in advance which line will succeed, and additional effort on one line yields diminishing returns. The optimum is defined relative to the community's goal, whatever it is taken to be (Section 1).

Science has no such monarch. Kitcher himself, writing with Julian Reiss, held that where large rewards draw several groups into racing toward the same result, "a certain amount of planning" would foster a division of labor and useful collaboration (Reiss and Kitcher 2009, 273). Chapter 9, Section 5 distinguished four things that the word control can run together: a formal right to decide, the effective capacity to exercise it, knowledge bearing on how to use it, and influence that shapes options without deciding among them. An artificial configuration that can create, copy, assign, and withdraw its investigators may come closer than human arrangements have to direct assignment and to costless movement between lines. How close it comes is not settled by that ability. Whether a formal power to assign is matched by effective control depends on whether an assigned investigator pursues the line in the sense the allocator intends; on whether the lines among which it assigns are given before inquiry or are formed and divided within it (Chapter 4, Sections 4 and 6; Section 4 below); and on whether the resources through which lines are pursued are the allocator's to assign (Chapter 12, Section 6.4).

Where control is approached, it does not supply the eye that the philosopher-monarch was given for the objective merits of theories. Strevens makes part of the point precise. On Strevens's model, a central planner that computes the optimal allocation among research programs is assumed to know the true form of each program's success function, and, under the model's idealizations, a scheme that rewards investigators in proportion to their marginal contribution to a program's chance of success reaches the same allocation without a planner. If the success functions change in unforeseeable ways, optimality may be out of reach; but this, Strevens notes, is "a universal obstacle" that "equally hinders the wisest central planner," not a peculiar fragility of the proposed reward scheme (Strevens 2003, §§3.3–3.4). Kitcher later grants a related point: because nature is unpredictable, judgments of significance, even if made under ideal conditions, would often turn out to need revision by the next generation of those who make them, though Kitcher holds this no more troubling than the continual revision that any planning for an uncertain future involves (Kitcher 2013, 221). Kitcher also holds that the philosophical task of finding the optimal organization of collective research would be too ambitious if undertaken in full generality, as the general theories sought by logical empiricism suggest, different areas of research being likely to call for different analyses, and that rather than seeking the optimum, a more useful first venture might be to understand how collective inquiry could go badly wrong (Kitcher 2013, 206). Wu and O'Connor, who argue that funding bodies could move a community toward the levels of exploration that decision theory recommends, add that even where a decision maker can allocate labor across a community, the complexity of real scientific evidence can make it hard to know how labor ought to be allocated (Wu and O'Connor 2023; accepted manuscript, 11–12, 14). Hiroaki Kitano, proposing a program to build artificial systems that make major scientific discoveries autonomously, takes a related point as a reason for machine exploration. On Kitano's account, the significance of a discovery can be extremely difficult, if not impossible, to evaluate when a few more discoveries may be needed before it leads to a major outcome, a difficulty the account names the over-the-horizon problem; a search that pursues expected significance can therefore stop short of a significant discovery reachable only through steps of low expected significance. Kitano proposes that such systems explore hypotheses without imposing what the proposal calls value-driven criteria, that is, criteria based on the expectation that a discovery will be significant. The paper distinguishes a system's role as a problem solver, aligned with the value of its stakeholders and working on problems defined in advance, from its role as an explorer of hypothesis spaces that human investigators would not reach, allowing that human experts may take commanding and monitoring roles and holding these roles important to avoid misuse (Kitano 2021). On this program the absence of the eye is met not by leaving each step to whoever is most competent to take it, as on Polanyi's account, but by extending search to lines that no one would choose on expected significance. Since, as Kitano notes, such a system is not free of resource limits, how it is to sample the space of hypotheses remains, on the program's own statement, an open research problem, and some ordering among lines returns there.

Resource limits are not the only reason an ordering returns. Any search fixes, at least implicitly, how the space it searches is represented and where and in what order it searches; what a search fixes in this way can be called its heuristic of search. To choose where and in what order to search is to single out some possibilities among many, and however the choice is made, it is made by some criterion of choice: a judgment of what to take up first and what to weigh more, which is a judgment of

value in a broad sense. In that sense one may even say that a choice is a judgment of value about which of several options to take up, and from where. No search, then, is free of a prioritization. Proposals to search without objectives or without expected significance are examples. Kitano's own program proposes that a first stage focus on hypotheses of a particular form, molecular mechanisms (Kitano 2021). Joel Lehman and Kenneth Stanley, who propose giving up the pursuit of objectives and searching for novelty alone, say of their own method that its measure of novelty "must be fitted to the domain" and that, combined with a complexifying algorithm such as NEAT, it orders search from simple to complex (Lehman and Stanley 2011, §§3–4, 9.1); and, as Chapter 5, Section 3 noted, even searches described as theory-agnostic rely on a prior characterization of the space searched. For research in particular, Julian Reiss and Jan Sprenger report it as almost universally accepted that the choice of a research problem is often influenced by the interests of individual scientists, funders, and society (Reiss and Sprenger 2020, §3.1), and on Weber's position (Chapter 9, Section 3) value ideas fix which part of reality becomes an object of investigation. For the same reason, a policy of searching as far as possible without the heuristics humans have used is not value-free either, since it is itself a prioritization that decides where search goes and in what order. In this sense completely value-free exploration is not possible, and what a proposal such as Kitano's removes is one criterion, significance as expected in advance (what Kitano calls value-driven), rather than judgments of value as such. What follows from this is more than that every search involves judgments of value. Judgments of value in this broad sense differ in what they weigh. Choosing an order at random, for instance, gives weight to giving every candidate an equal chance of being taken up early, that is, to not presupposing which candidates matter; in the same way, setting the order by computational convenience weighs ease, and setting it by expected significance weighs anticipated results. A search that does not use expected significance still settles, through how it represents and orders its space, which possibilities it can reach and which it reaches first. If, as on Kitcher's account examined in Section 6, which lines were pursued shapes what later appears significant, these choices of reachability and order constrain later judgments of significance. Such a proposal then avoids narrowing search by significance expected in advance, but the order it adopts in its place sets the range of what can later come to appear significant.

Jamie Shaw, developing an approach to the organization of research from Feyerabend's pluralism, generalizes the diagnosis, seen above in Strevens and Kitcher, that the eye is missing to the third kind of not knowing distinguished in Section 1, in which neither the lines nor their value can be projected in advance. Shaw argues that Kitcher's ideal of well-ordered science, on which the goals of research determine what research is pursued, and the allocation models of Kitcher, Strevens, and Weisberg and Muldoon share an assumption that they leave unstated. Kitcher's and Strevens's models suppose that the chances of success of research programs are known, without saying how they could be; Weisberg and Muldoon's model lets projections change as research proceeds but, on Shaw's reading, "still requires that we predict future significance". In that model, whose investigators do not know in advance which approaches are significant, the presupposition lies in a distribution of significance over approaches fixed in advance, settled even though investigators do not know it, and in investigators who all do normal science, building on what came before (Shaw 2018, 153–160; Weisberg and Muldoon 2009, 229, 231, 233). Shaw draws on Feyerabend's observations that no idea is ever examined in all its ramifications and that theories abandoned or overtaken by a rival can make comebacks, so that whether a line will go on degenerating or progressing cannot be read off its present state, and even for an idea to show some merit it must first be pursued (Shaw 2018, 69, 86–87). Feyerabend held, as Shaw reports, that long-term projections about the fate of theories are impossible and that to call a degenerating line riskier than a progressive one runs into a version of Hume's problem. Shaw grants that reasonably accurate projections of the value of research can be made within normal science, but holds that such projections are "by their very nature, short term" and that, in the long term, all proliferated alternatives begin as equally risky, since no one knows how they will turn out. On this account, projection can guide funding within established programs but cannot say how to fund the alternatives that proliferate beside them, and how funding is divided between the two depends on how urgently results are needed (Shaw 2018, 123, 136, 161–164). The premise that the outcome of research cannot be foreseen thus supports different conclusions on different accounts: for Polanyi, that each step be left to whoever is most competent to take it; for Shaw, that projection guide funding only within established programs, and that how much goes there depend on how urgently results are needed. Decision theory has positions, though not about research, on the step, which Shaw shares with Feyerabend, from long-term unpredictability to declining to rank proliferated alternatives by their prospects. It has asked how a decision maker can reason when suspecting that there are states or outcomes not yet entertained, a condition called awareness of unawareness. Katie Steele and H. Orri Stefánsson argue that such awareness may differ "only in degree, and not kind" from the decision maker's imperfect grasp of the states already entertained; among the views they report, Simon

Grant and John Quiggin suggest that the corresponding judgments are made by induction from past occasions on which awareness grew, and Chloé de Canson argues that one is to assign "at least some positive probability" to there being states or outcomes of which one is unaware (Steele and Stefánsson 2025, §5.4). On these positions, what has not been foreseen can still be an object of judgment, formed by induction from past growth of awareness or by reserving some probability for what has not been conceived; whether such judgments could ground unequal assessments of proliferated alternatives, rather than the equal risk Shaw assigns them, is a further question, which these positions, not being about research, leave open.

On these arguments, an artificial allocator that came closer than human institutions have to the monarch's control would not thereby acquire the eye. If Shaw is right, the knowledge it lacks is least lacking within established programs and most lacking in choices among them and in support for work that departs from them. Friedrich Hayek, writing of competition as a procedure for discovering facts that would otherwise remain unknown, drew from a similar premise a consequence for how such procedures can be judged. Their merits, Hayek argued, cannot be empirically verified for the cases in which they are of interest, a feature Hayek took the discovery procedures of science to share: the advantages of established scientific procedures are recognized "only because they have actually provided better results than alternative procedures"; the performance of the market, on the same argument, is to be judged "from the bottom up," by comparison with what other available methods would attain, rather than "from the top down," against an ideal standard that no known method attains (Hayek [1968] 2002, 10, 16). Hayek did not discuss the allocation of research effort. Extended from the market to it, the point implies that a claim that a configuration allocates well can be tested against alternative arrangements available to it, but not against the distribution the philosopher-monarch would choose, since that comparison would require the eye.

Such control, where it is approached, has a price that the thought experiment does not display. Complete control of the workforce means that investigators do not choose their own lines. In the terms of Chapter 7, Section 4.3, an allocator that assigns every line takes question selection away from the investigators and supplies it to them from outside. The more fully control is realized, the less autonomous the investigators are in question selection, in the sense that section distinguished. That alone does not make the configuration that includes the allocator the only research actor, since, as the same section notes, an externally assigned actor can still conduct research. Whether the investigators conduct research on their lines, whether the configuration as a whole conducts one inquiry, and whether its parts are independent of one another are separate questions, and counting instances settles none of them. Chapter 7, Section 2 located the boundaries relevant to the first two by where initiation, monitoring, redirection, and recovery are organized; since those boundaries are drawn relative to the claim, the investigators and the whole may each be research actors. Chapter 9, Section 5 named the powers whose division among the parts would make them independent of one another. Where neither that organization nor those powers are divided among the parts, the division of cognitive labor is the internal portfolio of a single actor, and the problem of this section becomes the problem Chapter 5, Section 4.2 described for one inquiry: how to judge what is worth pursuing without collapsing pursuit-worthiness into truth-likelihood. Where investigators organize the inquiry on their assigned lines but hold none of those powers, allocation distributes work among research actors that are not independent of the center. Where investigators do hold some of those powers, control is only partial, and the question of what moves their choices returns.

Steering without motives: signals

The division of labor that Kitcher's monarch would impose has been realized in human communities, to the extent it has, partly through investigators' motives. A second premise of distributed allocation is therefore that allocation passes through those motives. Allocation's entering a configuration does not mean that it is assigned directly from a center: there can be configurations whose instances choose lines themselves and move between them. What allocation passes through there may be not the motives of human researchers but motives of the machine, or something equivalent to them, and whether these do the work that the pursuit of credit did is a separate question (Chapter 4, Section 7). Where an allocator assigns lines directly, this premise fails, and the question is what goes with it and what remains. The literature used to examine it is the one that has asked how motives such as the pursuit of credit produce or prevent a spread of effort: Kitcher's models of credit, their successors, and the sociology of priority on which both draw. Curiosity, another motive that has moved effort among human researchers, is taken up in Section 5.

Kitcher's models show that communities whose members seek priority and prizes can distribute effort better than communities without such motives, because the competition for credit gives some investigators a personal reason to pursue

the less favored line; in Longino's summary of the result in her encyclopedia entry on the social dimensions of scientific knowledge, scientific progress "can tolerate and indeed benefits from" a certain amount of impure motivation (Kitcher 2013, 202–203; Longino 2025, §5). Credit also works in the opposite direction. In simulations of investigators choosing repeatedly between a safe line and a riskier, better one, Erich Kummerfeld and Kevin Zollman find that investigators who care about their own contributions have an incentive to leave risky exploration to others, so that a community left to itself can explore less than would be best for it; differences in investigators' skill with each line could temper or remove the effect (Kummerfeld and Zollman 2016, §§2–3, 5). Partha Dasgupta and Paul David, writing on the economics of science, describe a further effect of the priority rule. Society, they argue, "does not care who is successful" in solving a problem, while the rule of priority makes the identity and timing of the solver matter greatly to each team, and this divergence between private and social rankings gives grounds for suspecting that the research portfolio that scientists select will be inefficient. Provided that private gain from a success is made commensurate with the benefits to the collectivity of researchers, they argue, the rule tends, among projects within a program that offer the same expected social payoff, to draw rival teams into "overly similar (i.e. positively correlated) projects", and the all-or-nothing character of its rewards draws them into duplicative races, which "works against the emergence of a diversified societal research portfolio"; the correlation, they add, will not be perfect, and they offer the argument as a hint at how a proof goes rather than as a proof (Dasgupta and David 1994, 506–507 and n. 56).

An allocator that assigns lines directly need not route allocation through its investigators' motives. In assigning lines directly, it removes what these analyses identify as a source of spread, a source of under-exploration, and a source of correlated duplication together, and since it removes them together, it settles nothing about which effect predominated in human communities, and so nothing about whether their removal is a gain. Nor does it remove correlation among lines that arises without credit, as Section 4 notes for investigators copied from a common base and for investigators of different origins. For such an allocator, how much exploration there will be becomes a matter of the rule by which it assigns lines, however that rule came about, rather than of the investigators' interests. In a configuration whose instances choose their own lines, it depends instead on what moves those choices and on how the configuration combines them, and the effects that these analyses trace to credit may return there in other forms, or not at all, depending on whether what moves the instances shares the features those analyses identify, such as a reward shared among those on a line, so that a less crowded line offers each more (Kitcher), reward for an instance's own contribution rather than for what others learn from its exploration (Kummerfeld and Zollman), or all-or-nothing reward for being first (Dasgupta and David).

What this removal accomplishes is narrower than it may seem. Merton argued, as Strevens reports, that scientists' concern for priority reflects a powerful norm governing science more than their psychological makeup (Strevens 2003, §2.1). On this account the priority rule links private motives to public goals, and it builds the link out of signals of achievement. Merton also held, as Strevens reports, that enforcing the rule so literally that a margin of days decides who receives all the credit is "a dysfunctional extreme far beyond the limits of utility"; Strevens replies that even so fine a discrimination has a function, since it makes research programs with a high potential for success relatively more attractive (Strevens 2003, §§2.1, 5.1). An allocator that assigns lines directly removes the dependence on human motives. It does not thereby remove the dependence on signals. Chapter 9, Section 4 gave the general reason: what counts as success for research is often not observable when a judgment about it has to be made, so that an allocator judges which lines are progressing through results, scores, how far other lines use its outputs, or other quantities that, on accounts that distinguish the goal from its measures, stand in for it rather than constitute it. Human allocation, too, has moved not only through researchers' motives but through institutionalized signals. Indicators of research output have been linked directly to the allocation of research funding, as when Australian research funding came to be allocated partly by a formula based on publication counts, and researchers have responded to such links (Wilsdon et al. 2015, 81–82; Chapter 11, Section 8). Where such signals are used directly to distribute funds, they help decide which lines continue without having to pass through the motives of the researchers whose lines are affected.

Kummerfeld and Zollman note that when every investigator seeks credit from the same central body, their incentives track what that body finds interesting (Kummerfeld and Zollman 2016, §5). A single allocator's signal stands in that position for every line it assigns. Where the signal also decides which investigators are copied and which are withdrawn, adapting to the signal shapes the investigators themselves. Reading their equilibrium in evolutionary terms, an interpretation they endorse without modeling it explicitly, Kummerfeld and Zollman note that investigators who explore less make larger individual contributions, because they leave exploration to others; if others imitate them, or they train more students, a community can

move toward less exploration than would be best for it (Kummerfeld and Zollman 2016, §3). Paul Smaldino and Richard McElreath model a related dynamic explicitly. In their model, laboratories whose methods yield more publications leave more "progeny" that inherit those methods, and selection on publication counts alone degrades the rigor of the methods in use; the dynamic, they argue, requires no conscious strategizing by anyone (Smaldino and McElreath 2016, abstract, §§4, 5.1). A coding error later found in their implementation meant that laboratories' replication rates could not affect their fitness, so that the dynamics they reported for replication did not correspond to the model they described; the correction also notes a minor deviation in the order of the evolution stage, states that the conclusions are unaffected, and does not identify the result on the degradation of methods as affected (Smaldino and McElreath 2023). What their model selects on is the rigor of methods rather than the amount of exploration. If the structure of these models transfers, an allocator that copies the investigators whose own measured output is highest would select in the directions they describe with no motive involved: toward less exploration unless its signal credits what an investigator's exploration contributes to what others learn, and toward less rigorous methods unless it registers the rigor of the methods behind that output.

Strevens reports Kitcher's shared-credit mechanism in terms of both the chance of a research program's success and the number of participants sharing its reward (Strevens 2011, §2, manuscript p. 10). Zamora Bonilla's account of Kitcher's models contrasts credit-only investigators' convergence on the theory they expect to be accepted with a distribution close to the optimum when some weight is given to truth (Zamora Bonilla 2014, paragraph beginning "One important topic"). For an allocator that follows signals of expected acceptance, the corresponding question is what connects those signals to truth. Where certification supplies those signals, the question links allocation to criticism (Chapter 9, Section 2.4).

Two questions on the side of allocation follow, and the chapter returns to both. The first is the termination rule: on what signal, and over what horizon, is a line cut? Under a short horizon, a signal that registers success only after a long delay cuts lines that a longer horizon would have kept (Section 5). The question has a precedent in the dilemma Feyerabend put to Lakatos (Chapter 6, Section 4.1). As Musgrave and Pigden report it, applied strictly, Lakatos's methodology urges scientists to abandon degenerating programs, research programs whose later theories predict no new facts and are made only to fit known ones, and grant-giving agencies to defund them, and so condemns programs that might later recover; restricted, as Lakatos then restricted it, to appraisal without advice, it tells no one what to do and is scarcely distinguishable from "Anything goes" (Musgrave and Pigden 2026, §4). From this precedent it can be inferred that the second horn, appraisal without advice, is not open to an allocator, since the signal and horizon by which it cuts a line are already advice acted on; for an allocator, the dilemma becomes the choice of a termination rule. How much the first horn then costs it, condemning lines that might later recover, depends on whether a cut can be undone: on whether a closed line can be reopened, on what evidence (Section 4), and with how much of what it had, which a record keeps only in part (Section 5). The second is a loop. Because allocation decides what is produced, and what is produced is what the next signal measures, an allocator that follows its signal changes what that signal will later report. How signals themselves behave over time, what they measure, what they are biased toward, and how they change when those measured respond to them, is a property of criticism and evaluation rather than of allocation, and Chapter 11, Section 8 examines it.

The models behind Sections 2 and 3 are deliberately idealized. Whether highly idealized models of scientific interaction can ground recommendations about how a community should be organized has been contested, and some hold that such models must first be validated against empirical targets, as Dunja Šešelja's survey of the agent-based models among them (models that run individual researchers as agents on a computer) reports (Šešelja 2022, §4). Their role here is not to fix an allocation rule but to separate premises and to ask what takes their place when they fail. The conclusion of the two sections is therefore conditional. It rests on the arguments of Polanyi, Kitcher, Strevens, Shaw, Wu and O'Connor, and Hayek and on the point that no search is free of prioritization (Section 2), on the models of credit (Section 3), and on Chapter 9, Section 4 for the general reason why signals remain; and it presupposes, with those models, a goal fixed in advance toward which lines are judged (Section 1):

> On these arguments, an artificial configuration that takes over the allocation of inquiry may approach one of the philosopher-monarch's powers, control of assignment, more closely than human institutions have, to the extent that its assignments are direct, its investigators carry them out, and the lines and resources are its to assign; it does not, by that control, acquire the other. Nor does dropping criteria of expected significance free its search from prioritization, and a claim about its allocation can be tested against arrangements available to it rather than against the distribution the monarch would choose. When the premises on which distributed human allocation relied fail, many of the problems they addressed change

form rather than disappear. The center need no longer have one person's capacity, although a center can reappear in the concentration of the resources through which lines can be pursued at all (Chapter 12, Section 6.4). Allocation need no longer pass through human investigators' motives, but it is still steered by signals that stand in for the goals of *S*. Where control of the workforce is approached, it withdraws autonomy in question selection from the investigators, and where it also absorbs the organization of their lines and the powers that would make them independent, it turns a community into one actor. And effort is still distributed under an ignorance of which lines will succeed that integrating allocation into one configuration does not by itself remove. A claim that such a configuration allocates inquiry well would therefore need to show how its signals remain answerable to those goals, where the powers that would make its parts independent are divided, how it represents its ignorance, and how it keeps alive those unfavored lines that, given its goal and the evidence, are worth keeping, rather than infer good allocation from control.

Which lines are kept alive, and for how long?

The conclusion of Sections 2 and 3 leaves a consideration in place: on the arguments examined, effort is distributed under an ignorance that control does not remove, and some lines that currently look unfavorable may need to be kept alive. That consideration raises four further questions: what is to be kept alive, for how long, where the need bites, and where the diversity that keeps them alive comes from. Two lineages answer the second question in particular. Formal models of scientific communities ask when, and for how long, disagreement helps a community converge on the better option. Miriam Solomon's social empiricism asks when dissent and consensus are each appropriate, judged by the distribution of empirical success. The section asks what follows for artificial research on each answer, which purpose each presupposes, and what becomes of one of the goods that Chapter 9, Section 2.3 proposed, as a hypothesis, that the division of the relations may have supplied: standing for dissent.

One lineage answers the question of how long with a specific claim. In Kevin Zollman's network models, a community can maintain a division of labor either by limiting how much information each investigator receives or by giving investigators extreme initial beliefs. When both are present, the initial diversity is never abandoned and the community fails to converge on the better option. What the community needs, Zollman concludes, is not diversity as such but *transient* diversity: disagreement that lasts long enough to prevent premature closure and then gives way (Zollman 2010, 1, §5). The claim is the answer of one family of models; whether diversity worth keeping is only transient is itself disputed, as this section shows below.

The robustness of these models has since been examined, and the examination limits what can be inferred. Rosenstock, Bruner, and O'Connor found that the advantage of sparser communication appears only in a small region of parameter space, when there are relatively few investigators, evidence arrives in small batches, and the rival options differ only slightly in success, while the claim that some transient diversity of belief is needed survived across the models they examined; where inquiry is hard, they add, more stubborn or exploratory investigators and standards for how much evidence a conclusion requires can prevent premature convergence better than withholding evidence (Rosenstock, Bruner, and O'Connor 2017, 235, 248, 251). Šešelja's survey reaches a compatible verdict and holds that such models, at least in their current form, serve chiefly to identify possible mechanisms and candidate explanations rather than to formulate policy (Šešelja 2022, 3–4, 11–12). Results from landscape models, in which investigators choose where to work, have been contested on similar grounds, as turning on implementation and on the shape of a landscape of which little is known (Alexander, Himmelreich, and Thompson 2015, 425, 432–440, 448–451); in Johanna Thoma's, investigators who move away from approaches others have tried discover far less on average than those who extend the best approach found so far, so that rewards other than discovery may be needed for exploring to be sustained (Thoma 2015, §§4.3–4.4, 5), a need that an artificial investigator assigned a strategy would not have, although it would then not supply exploration by temperament either. What survives this examination is narrower than a prescription: some transient diversity of belief appears to be needed in many models, the benefit of particular mechanisms for producing it is fragile, and the models identify possible mechanisms.

These results also presuppose a purpose. In the network models, a community succeeds when it comes to favor the better of options fixed in advance; diversity is valuable there as a means to that convergence, and diversity that never gives way counts as failure. On purposes that do not end in convergence among fixed options, such as keeping the capacity to respond when conditions change or generating options not yet in the set (Section 1 notes which arguments presuppose each),

diversity that does not give way can have reasons that these models do not register, and on Helen Longino's account, which asks that dissenting perspectives be cultivated (Chapter 11, Section 3), and on standpoint theory as Kristen Intemann reconstructs it (a family of views on which the social position from which research starts bears on the knowledge it yields; Section 6), it has further reasons as a condition of criticism. The claim that diversity should be transient is therefore relative to the purpose the models build in, as well as to their structure.

This leaves the third question, *where* the need bites. If measures to protect transient diversity matter chiefly when inquiry is hard, their importance varies across domains. The need also presupposes the condition stated in Section 1, that the lines cannot all be tried. Zollman's model locates this premise from another side: the harm it displays arises only because evidence about a line is gathered by pursuing that line, and could not arise if the evidence available did not depend on investigators' own choices (Zollman 2010, §3). Where evidence is cheap, plentiful, and decisive for the specified claim, the premise weakens, and concentrating effort costs comparatively little. Formal checking and fast simulation can provide such evidence within fixed formal systems or models; whether a model adequately represents its target remains a separate evidential question. Where evidence is slow, costly, and arrives in small samples, as in clinical research, field observation, or experiments that depend on scarce instruments, the premise is strongest. These are also the domains in which an artificial system, too, cannot cheaply multiply its own evidence. If the models' structure transfers, the epistemic risk of integrating allocation is concentrated where artificial research has the least room to compensate by generating more data.

Two qualifications apply. Formal verification is cheap and decisive at the point of checking, but bringing a claim to that point need not be. In the Liquid Tensor Experiment (Chapter 1, Section 3), for example, formalizing the statement of the central theorem took about a month and its five-page proof about five more, while a five-line deduction of the final result took another year because the theory of abelian categories, derived functors, and Ext groups it relied on had first to be built (Buzzard 2022). Whether the premise weakens therefore depends on whether a domain's claims are already expressed where such checks apply. Second, whether a problem can be divided into parts that can be explored separately need not be a fixed property of the problem: in a simulation study accepted at *Philosophy of Science* and cited here from its preprint, Fazelpour, O'Brien, and Rubin argue that such decomposability is often an achievement of standardization, protocols, and divisions of labor (Fazelpour, O'Brien, and Rubin 2026, 2–3, 25). If that is right, whether the premise weakens in a domain depends in part on how its problems have been divided, and dividing a problem is itself an allocative decision taken under the same ignorance. A further possibility can be considered. A configuration that divides problems and selects tasks is likely to show success most readily on tasks that are easy to judge. If that success justifies its next allocations, and which investigators are copied or withdrawn (Section 3), tasks that need slow or ambiguous evidence are more likely to fall outside the candidates compared, and the configuration may, by reason of its own high performance, make hard research still less visible. This is not merely that an evaluation criterion is biased: under the condition that the same evaluative signal returns into future allocation, the scope of research may narrow from within through success itself (see also Section 6).

The fourth question is *where diversity comes from*. Zollman's mechanisms preserve diversity only "so long as agents have diverse priors" (Zollman 2010, §5). Human communities acquire such differences from different training, histories, and commitments without anyone having to design them. Artificial investigators may be copied from a common base or built on different bases and set side by side, and in either case they may come to differ through their own histories of inquiry. Those copied from a common base may not receive such differences at the outset. But a difference of origin does not guarantee them either, and the evidence that follows concerns differences of origin, not differences acquired over histories of inquiry. Across several hundred language models evaluated on shared benchmarks, Kim and colleagues found that models often converge on the same wrong answer, that a shared provider or base architecture raises this correlation, and that more accurate models have more correlated errors even when their providers and base architectures differ (Kim et al. 2025, abstract). The models compared are all large language models, so that the differences of architecture at issue are variants within one class of model, and the errors are errors in answering fixed questions, not choices among lines of research. For systems of that class, the finding nonetheless cautions against assuming that diversity of provenance supplies diversity of priors; whether systems built in quite different ways would share their errors in the same way is a question it does not address. Two preprints come closer to choices among lines, though not to choices made in pursuing them. Bisht and colleagues asked twelve models from four providers to propose new hypotheses extending published papers in materials science and in biology. On their embedding measure, hypotheses from models of different providers were almost as similar to one another as repeated samples from a single model (0.64 against 0.72 for the materials papers), and drawing ten

hypotheses from ten models rather than from one raised the effective number of distinct hypotheses only from 2.9 to 3.6 (Bisht et al. 2026, §5 and app. A). Bao and colleagues, who had authors of recent preprints rate hypotheses that language models generated for their own papers, found that non-reasoning models collapsed into a narrow region of the space of ideas while reasoning models ranged more widely, found no evidence that models from the same company generated more similar ideas, and found that every model proposed null hypotheses, claims of no relationship or effect, less often than human scientists did (Bao et al. 2026, 1, 3). The two studies use different measures, and both concern hypotheses proposed for given papers, not lines that investigators pursue over time; within those limits they extend the caution from answers to fixed questions to the proposing of hypotheses. Where artificial investigators lack diverse priors, limiting information cannot preserve a diversity that was never there.

Devices at a center can add some diversity: assigning lines by lot among those that pass a threshold, as proposals that Shaw discusses to fund research partly by lottery would do (Shaw 2018, 172–173), protecting budgets for unfavored lines, building a rate of exploration into each investigator's policy, or giving investigators differently initialized representations or a degree of inertia before switching lines (Šešelja 2022, 4; Rosenstock, Bruner, and O'Connor 2017, 251). Solomon argues that diversity is in this respect a blunt epistemic tool, for which there are typically several options for adequate diversification (Solomon 2006b, abstract). Zollman's network models, described above, give one instance: in them a division of labor can be maintained either by limiting the information each investigator receives or by giving investigators extreme initial beliefs, which has the same effect (Zollman 2010, 1, 3–4); where the missing distribution can be identified in advance, then, it can be supplied in more than one way. Two limits qualify what such devices achieve. Diversity is a matter of degree, not a property a configuration has or lacks, and it is not confined to what any center contains: whatever spread a configuration produces internally, others outside it who pursue quite different lines add to the diversity of the whole. And whether the variety a device adds is the variety a given position requires depends on the position. For the spread of effort that the models of this section consider, Solomon's argument and Zollman's models suggest that it often can be; for the diversity that Longino's account and standpoint theory value for the sake of criticism, which depends on who takes part in inquiry, it is less clear (Chapter 11, Section 3; Section 6 below). Whether any device would in fact produce the diversity it aims at appears, for artificial investigators, to be largely untested.

Several of these models also take evaluation to be free of error: in Kitcher's model an answer, once delivered, is recognizably correct or incorrect, and in Weisberg and Muldoon's landscape a visit always reveals an approach's true significance (Zollman 2010, n. 3; Weisberg and Muldoon 2009, 232–233). The problem that follows therefore lies outside what these models address.

What integration alone does not supply is an independent check on the criterion by which diversity is ended. Transient diversity ends with a judgment that one line has done better. In a centrally allocated configuration that judges every line by the allocator's own signal, an error in that signal can correlate closure decisions across otherwise diverse lines. On the argument of this section, what centralization trades against is therefore less diversity as such than independence of error about the criterion that governs convergence. Chapter 11, Section 2 takes up what such independence consists in on the view that locates the value of criticism in the errors it detects. It follows that an assessment of diversity can ask not only how different the candidates are, or how many there are, but which errors the judgments of their continuation share. Being able to generate different representations is one thing, and being able to judge whether lines survive on different evidence is another; a configuration with only the first can produce diverse candidates while discarding them together through the same error.

Solomon's social empiricism gives a different answer to the question of how long, and with it an answer to the first question, which lines are to be kept alive. It treats dissent as the prototype and consensus as a limiting case of it. On this account, as Longino reports it, a community is rational when the theories it accepts are those with all the empirical successes or with unique ones, and it is collectively irrational to jettison a theory with unique empirical successes; dissent is thus appropriate when each rival theory has empirical success of its own, and consensus when one theory has them all. Dissent is worth keeping, then, where each dissenting position brings empirical successes of its own and, in the extension of the argument that Longino reports from Solomon's later work, where different models carry particular insights or specific data that are likely to be lost in a consensus (Longino 2025, §5). On the same norm, a consensus would have to be reconsidered once a rival theory gains an empirical success that the consensus theory lacks. Solomon's own example of a consensus that fell short of the norm is the near-consensus of the 1950s on the "master molecule" view of DNA, under which the work of Barbara

McClintock, Ruth Sager, and others was not accepted, or was accepted only late (Solomon 2001b, 72). What kept each success represented, on Solomon's account, was in part that scientists weighted their own data and concerns more heavily than others did: in the geological debates Solomon examines, geologists were convinced of drift by data that were their own and resisted it when their own results seemed to tell against it, and such influences distributed research effort over the competing theories (Solomon 2001b, 69–72). The purpose Solomon's account presupposes differs from Zollman's: not convergence on the better of fixed options, but that each empirical success a domain offers stay represented until one theory has them all.

In the terms of this chapter, that weighting of one's own data is a premise about human realizers of the kind Section 2 separated, although it bears on allocation only where, as in research as it has been organized, data are first with those who produced them and their weighting reaches decisions about whether lines continue. Sharing data does not by itself remove this weighting: even where everyone can see the data, weightings can differ through the experience of producing them, knowledge of their provenance, and their interpretation. Whether a configuration keeps something that plays its part depends on how observations pass through its investigators and how what they find is combined. Machine investigators that keep their own records and act on them could weight their own data in a way equivalent to Solomon's scientists, whereas an allocator that pools every investigator's observations and judges all lines by one signal removes the premise. What removes it is not the pooling of observations as such but judging the continuation of every line by one signal; observations can be pooled while an arrangement is kept under which a judgment that a line should continue because of its own successes reaches allocation. For artificial configurations too, who can see the data and whose judgment can keep a line alive are separate questions. On Solomon's account, such an allocator would then also have to show by other means that convergence does not discard lines with successes of their own, and say on what evidence a closed line would be reopened. On Zollman's account, the corresponding requirement is narrower: that convergence not come before the evidence warrants it. The two accounts thus differ on how long diversity is kept and on what ends it, but on both, the judgment that ends it needs evidential support that integration alone does not supply. An integrated allocator may still maintain informative checks on that judgment within its own organization (Chapter 11, Section 1).

That judgment is also where the hypothesis of standing for dissent applies to allocation. In allocation, dissent takes the form of continuing a line that the prevailing judgment would close, or of pressing for a closed line to be reopened. In human research, those who did so may often have been parties distinct from whoever judged their line unfavorable: for example, an investigator with a laboratory, a record, and a career, a funder with criteria of its own, or a community in another field. Two things can be separated in what this division supplied. One is epistemic: on Solomon's account, dissent keeps represented successes that consensus would lose. The other is standing in the sense of Chapter 9, Section 2, an entitlement, held in relation to the decision that would close a line, to object to its closing, to refuse to abandon it, or to propose that it be reopened. On the argument of this section, the two come apart when allocation is integrated. The epistemic work of dissent can be performed inside a configuration by arrangements that someone sets, such as protected budgets for lines with successes of their own or a rule for reopening, or by dividing among its parts the powers that Chapter 9, Section 5 named, such as refusing resources, keeping evidence apart, and leaving with a record. Those powers are not standing (Chapter 9, Section 5). Standing would be held inside a configuration only by parts that hold it on the account adopted, such as parts that were parties in their own right (Chapter 12, Section 5) or, on the result-based ground, parts whose good results had been established (Chapter 12, Section 3.1); otherwise it is held, if at all, by parties outside the configuration who pursue other lines or can press for a closed line to be reopened, which is one more respect in which the diversity of the whole lies partly outside any center. Whether that standing matters epistemically, or only for whether the closing of a line is legitimate, depends on the account adopted. On Solomon's account what matters epistemically is that successes stay represented, by whatever means; on accounts that tie the value of dissent to the intellectual authority of critics, such as Longino's, the question belongs to criticism, and Chapter 11, Section 3 takes it up.

The theory of reinforcement learning has results about a single learner, that is, an agent that chooses actions in an environment, observes the rewards they bring, and revises its later choices accordingly. These bear on the judgment that ends exploration from another direction, though they apply to this chapter only where, by the test of Section 2, the configuration is a single actor whose allocator chooses lines as one such learner acting on one prior. Under an assumption they call No Heaven, that no course of action in the true environment yields near-maximal reward forever, Michael Cohen, Elliot Catt, and Marcus Hutter prove that an agent guaranteed eventually to act as well as possible in any computable environment must

keep trying every course of action: if it tried something only finitely often, it would be ignoring a computable environment, indistinguishable so far from the true one, in which trying it once more yields maximal reward forever. Such an agent never reaches a point at which it could stop exploring; since it keeps trying everything, it will sooner or later try what brings about a given event, so long as recognizable opportunities to do so keep recurring, and so it either brings about any given event or eventually loses every recognizable opportunity to do so (Cohen, Catt, and Hutter 2021, §§I–II, IV); an agent that acts optimally relative to a prior, a Bayes-optimal agent, may instead stop exploring where its prior and observations so direct, and Jan Leike and Hutter construct priors, by means they describe as unnatural, that keep such an agent on any given computable policy for as long as that policy's rewards stay bounded away from zero (Cohen, Catt, and Hutter 2021, §V; Leike and Hutter 2015, 2, 5–6, 10–11). This book reads the theorems' events as harms that trying a line of research could do to others. On that reading, where lines cannot be tried safely and reversibly, neither kind of optimality by itself guarantees that the criterion that ends exploration is sound: the first never ends it, and so tries even lines whose trial cannot be undone (so that an allocator that places some lines beyond trial gives up the guarantee of eventually acting as well as possible in every computable environment), and the second ends it where its prior and observations direct, while being optimal relative to a prior does not show that the prior is sound, as Leike and Hutter's construction, which can keep such an agent on any given policy, illustrates. These results do not show that a Bayesian configuration cannot check its prior or its stopping judgment against observation or by other tests. Nor is a guarantee across a broad class of computable environments the same as safe exploration in a restricted domain of research. Applying these results to research therefore adds the questions of which class of environments is considered, what is placed outside trial, and what supports that restriction. Where payoffs can change, watching lines that are rarely tried has a price as well: among rules for choosing repeatedly between two options, the less a rule loses while nothing changes, the more it loses under some pattern of at most two changes in the payoffs (Lattimore and Szepesvári 2020, §31.2).

Because these results concern single learners, they leave open how the quality of an allocation relates to the quality of its investigators. Mayo-Wilson, Zollman, and Danks give an example that bears on investigators who share evidence: a rule that lowers an investigator's rate of exploration as the evidence it observes accumulates, which they do not regard as unreasonable in itself, can leave two investigators who pool their evidence stuck on a worse option, because each then observes more and explores less (Mayo-Wilson, Zollman, and Danks 2011, 665–666 and n. 15, 670–671). The example shows that the adequacy of an allocation cannot always be read from the quality of its individual investigators as assessed in isolation. In one model, restricting communication also harmed groups whose members already explored more (Kummerfeld and Zollman 2016, §4), so none of these results supports a general prescription to maximize diversity or restrict communication.

If the structure of the models of transient diversity transfers, and on the purpose they build in, the need they identify does not disappear when the human traits and arrangements that used to meet it, such as different priors, stubbornness, careers, and the mortality that Section 5 takes up, are absent; it remains at least where inquiry is hard and evidence is gathered by pursuing lines. On other purposes, such as the coverage and the generation of options noted in Section 1, the diversity worth keeping need not be transient at all. A claim to allocate well would then need to say where the diversity it relies on comes from, recognizing that devices at a center add to a diversity that is graded and partly outside any center. On both Zollman's and Solomon's accounts, integration alone does not supply independence of error about the criterion by which diversity is ended; how far that criterion's past decisions can be examined, from inside the configuration or outside it, depends on whether a record of its closings remains (Chapter 11, Sections 5.3 and 8). The epistemic work of dissent can be set by a rule someone adopts or supplied by divided powers, while standing for dissent, in the sense of Chapter 9, Section 2.3, is not supplied by either. And because the ease of a domain depends in part on how its problems have been divided, such a claim cannot treat that ease as given when the division was the configuration's own choice.

What human allocation supplied without deciding it

The models of Sections 2 to 4 abstract from how human research communities actually performed the functions that allocation requires. That abstraction is what let them state their premises explicitly; this section turns to a feature they set aside that matters for the question of this chapter. Chapter 4, Section 5 looked, in light of research records, at how within a single inquiry, of the unresolved matters an inquiry meets, some are taken up as questions while others are left, postponed, or cease to be askable, and how not all of this comes from deliberate choice. This section asks whether something comparable held at the level of a community: which functions of allocation were performed by arrangements that no one adopted as allocation policy, and what becomes of them when allocation moves inside a configuration.

The functions in question are those that Sections 2 to 4 found the models to presuppose or to abstract from. They are functions that allocation performs across a community, not the middle, functional level of Chapter 5, Section 5, at which evidence or criticism changes the standing of a candidate within one inquiry; what the two uses share is the contrast between a function and what realizes it. They are listed here so that each case below can be read as an example of one of them:

1. keeping lines that currently look unfavorable alive, which includes deciding on what signal and over what horizon a line is cut, and where the cost of a failed line falls;
2. opening lines of different kinds before anyone knows which will succeed, and admitting new groups to them;
3. judging when a result warrants convergence;
4. ending the dominance of established participants and renewing those who pursue lines;
5. keeping more than one signal by which work is recognized.

The first three are what the models of Sections 2 to 4 presupposed, and the last two what they abstracted from; separating the functions in this way makes it possible to locate, for each, the premise on which its performance rested, whether about human realizers or about how human research has been organized. Other features of human research also shape allocation without anyone deciding them, among them cumulative advantage in recognition, the costs of specializing and of switching fields, and the variability of reviewers' judgments; they are not examined here. There are four cases, chosen because each makes one function and the premise behind it visible. The first, third, and fourth show a function performed without being decided, though what no one decided differs among them. In the first, the funding arrangement itself was deliberately set up, but how far it kept unfavored lines alive across the community as a whole was not set by anyone as a policy of allocation; in the third, turnover comes with human lifespans; in the fourth, the plurality of signals persists without anyone maintaining it. The second is a contrasting case of central direction, in which an office did decide the allocation. Separating deliberately established support, unintended effects at the level of the whole, and turnover through mortality makes it easier to see what becomes newly an object of design when allocation moves inside a configuration. The cases are not a sample of the ways research has been organized, and none of them shows that the arrangement it describes was good on balance.

The first case concerns the first function, keeping unfavored lines alive. Pierre Azoulay, Joshua Graff Zivin, and Gustavo Manso compare investigators of the Howard Hughes Medical Institute with comparable researchers funded by the US National Institutes of Health. The institute tolerates early failure, rewards long-term success, reviews on five-year cycles, and selects "people, not projects"; the NIH grants they compare have shorter review cycles, predefined deliverables, and renewal policies less forgiving of failure (Azoulay, Graff Zivin, and Manso 2011, 2–3, 7–8). The institute's investigators produced more highly cited papers, and also more papers cited less than their own least-cited work before appointment, a pattern the authors read as more risky bets and more failure (Azoulay, Graff Zivin, and Manso 2011, 4). The authors note that attributing the difference to the program's incentive features, rather than to resources, prestige, or the sorting of researchers, requires an interpretive leap, although after examining those alternatives they conclude that the incentive features are the likely explanation (Azoulay, Graff Zivin, and Manso 2011, 27–28). The institute did not remove evaluation. It changed its horizon, its treatment of failure, and its unit. The premise here concerns how research has been organized more than its human realizers: how the cost of a failed line falls on a career. The institute's arrangement shows that the treatment of failure can change while the realizers remain human. An artificial investigator need have no career to protect, but the questions that protecting a line answered remain: on what signal, and over what horizon, is a line cut, and on whom does the cost of cutting it, or of its failing, then fall? In a configuration that controls its own resources, the first can be answered by a

termination rule that an allocator sets (Section 3), or by a pattern that forms as instances give up lines themselves and their choices are combined. Where the rule is set by the same party that sets the criterion for convergence, Section 4 found that it does not, by integration alone, come with independence of error about that criterion. The second is a question of distribution among those who bear consequences, which Chapter 9, Section 3 counted among the objects of value for *S*.

The second case concerns the second and third functions, opening lines of different kinds before anyone knows which will succeed and judging when a result warrants convergence, under central direction that lacks the eye of Section 2, knowledge of the objective merits of the lines. From 1967 a national office in China, known as the 523 Office, organized and coordinated antimalarial drug research across seven provinces and cities, a program known as Project 523. Its long-term aim was to find new drugs both by screening synthetic chemicals and by searching the recipes and practices of traditional Chinese medicine (Miller and Su 2011). Several thousand compounds screened between 1967 and 1969 yielded no effective drug, and in 1969 the office asked a further institute, of the Academy of Traditional Chinese Medicine, to join the search among Chinese medicines (Tu 2015, 289). In October 1971 an extract of qinghao from that institute, sample 191, was fully effective against rodent malaria. Reported at a meeting of the program in March 1972, the result drew requests for information from other institutes and collaboration across the country, and with the first clinical trial it "steered nationwide antimalarial research towards Qinghao" (Tu 2015, 290–291, 293). Within the program, Miller and Su report, information flowed freely among groups and results were presented at frequent joint meetings, while secrecy barred external publication (Miller and Su 2011). Among the historical cases this chapter can draw on, it comes close to such direction. The office controlled assignment and communication, showing that control need not be dispersed across grant holders even when investigators are human. It did not know which line would succeed; it kept lines of different kinds open in parallel and brought further groups into them, and once a result appeared, central communication brought the network to converge on it. Its sources, a participant's lecture, a secondary account drawing on a chronicle, and historians' reconstruction from archives and interviews, disagree on credit, and the program was military and political and is visible only because it succeeded (Tu 2015, 301; Miller and Su 2011; Li and Zhang 2019, secs. 2.1–2.2). Within those limits, the case suggests that two judgments mattered that control did not by itself supply: which lines of different kinds to keep open before any result, and when a result warranted convergence. Control could be used to keep a line open—when that institute of the Academy proposed in 1971 to abandon its search, for example, the Ministry of Health refused (Li and Zhang 2019, sec. 2.2)—but those judgments still had to be made, and made without knowing which line would succeed. Here the functions were decided, and the case suggests that deciding them is one thing and making the judgments they require is another.

The third case concerns the fourth function, the ending of dominance. Azoulay, Christian Fons-Rosen, and Graff Zivin study the subfields of 452 eminent life scientists who died prematurely. After such a death, publications in the subfield by researchers who had not collaborated with the deceased rise on average by 8.6 per cent; the additional work comes disproportionately from outsiders, draws on different literatures, and is well cited. The mechanism they favor is not gatekeeping through review panels or editorships, which few of these scientists held, but the deterrent effect of an eminent figure's presence on potential entrants; entry also stays limited where the deceased's collaborators hold gatekeeping positions, and the authors decline to draw strong conclusions about welfare (Azoulay, Fons-Rosen, and Graff Zivin 2019, §§1, 4.2–4.5). The human premise is that turnover occurs without anyone deciding it. By contrast, a persistent model, evaluator, or research platform does not leave by dying; it leaves when someone or something retires it. The deterrence the authors describe runs through human reputation and may have no artificial counterpart; whether a dominant component could narrow entry by another route, for instance through an allocator that ranks proposals by their agreement with that component's assessments, is a possibility the case raises rather than shows. James March's simulations suggest a route by which turnover could have value without reputation. In March's model, as Simon Rodan and Megha Sachdeva describe it, an organizational code, the beliefs an organization keeps in its rules and procedures, adjusts toward the beliefs of members who are closer to reality than it is, while members are socialized toward the code; where socialization is fast, members soon share the code's beliefs, the variety from which the code could learn is driven out, and learning suffers (Rodan 2005, 2, 6, 24; Sachdeva 2013, 603–604). Moderate turnover, March found, then increased exploration, and with it knowledge, through the diversity of the new recruits (Sachdeva 2013, 605). Rodan, rerunning March's original simulation, found that low rates of turnover improved learning while high rates lost too many knowledgeable members, and that once turnover reintroduced variety, fast socialization ceased to be a problem (Rodan 2005, 12, 23–24). Extended to configurations, which March does not discuss, the model suggests that replacing components around a persistent model or evaluator would raise the question of

what the replacements are made from, and Section 4 noted that artificial investigators, not only when copied from a common base but also when of different origins, need not differ in the way this requires. In March's model what yields the gain is not turnover as such but newcomers' holding beliefs the code has not yet absorbed; if so, applied to configurations, merely raising the frequency with which components are updated, or the number of models, may not supply this function. And whereas human turnover can bring at once the loss of knowledge and the end of a judgment's continued dominance, a configuration could transfer only the authority to allocate or evaluate while keeping the record, so that what is kept and what is brought to an end become separate questions. A record cannot be assumed to keep every tacit capacity, however, and if successors are chosen by the same criterion of evaluation, the range of judgment may not change with the turnover.

The fourth case concerns the fifth function, the plurality of signals. Laurens Hessels and colleagues compare four high-performing research groups in the Netherlands. In mathematics, arguments can gain recognition before formal publication, and a group that depends on little equipment can change direction quickly; in the humanities, cycles are slow, and recognition is converted mainly into positions and research time rather than grants (Hessels et al. 2019, 139–143, 146). In the groups they compare, the routes by which work becomes recognition and resources differ by field. Where each field's work is evaluated largely by its own members, this plurality persists without anyone maintaining it; one may infer that the same arrangement can leave work between fields without a matching evaluator. A single evaluator or allocator applied across fields would either maintain that plurality, deliberately or through arrangements no one set, or impose one mapping on all of them, and Section 3 found that the adequacy of a steering signal depends on the goals it stands in for, which differ by field.

None of the cases shows that the human arrangement was right: the welfare effect of turnover is unsettled, and Project 523 is a single, successful, politically embedded case. What the cases suggest is narrower and takes the form of a question. When allocation moves inside an artificial configuration, what was supplied by features of human realizers and of the way their research has been organized can no longer be assumed to be supplied by those features. Whether a function is then set by a rule someone adopts, supplied by some other process that no one decides, or not supplied at all, becomes a question that a claim to allocate well would need to answer. Each of the three answers is possible. A function can become a parameter that someone sets: a termination rule and its horizon, a rule for keeping lines of different kinds open and admitting new groups to them, a criterion for convergence, a schedule for retiring or replacing dominant components or the components around a persistent one, or a policy about how many evaluative criteria are maintained. It can also be supplied by processes that no one sets as policy, inside a configuration as well as outside it, such as the choices of instances that take up and drop lines on their own motives or something equivalent to them, the emergent behavior of a learned allocator, the randomness of its initialization, a market among parts with private costs, or a provider's decision, made for other reasons, to retire a model. And it can lapse unnoticed. Where a function is set, whoever sets it determines part of what the configuration can learn. Where it is set by many parts, no single party may be its author, and who answers for the setting becomes the question of responsibility that Chapter 12, Section 6.3 takes up. This is the form that the recombination described in Chapter 9, Section 2.3 takes for allocation. Jan Kulveit and colleagues make a parallel observation about the economy, culture, and states, which they hold have been kept aligned with what humans want partly through explicit actions and partly implicitly, through their reliance on human labor and cognition; the significance of the implicit route, they write, "can be hard to recognize because we have never seen its absence". They add that analogous arguments could be made for other domains, research among them, and that drift away from human interests can occur "even while each individual AI system successfully follows the local specification of its goals" (Kulveit et al. 2025, 2–3, 15). Their concern is the loss of human influence over societal systems (Chapter 12, Section 7). What is taken from them here is narrower: that a function supplied by human participation rather than by decision may become visible only once it lapses. In research it may stay invisible even then. Lines that are closed, or never opened, because such a function has lapsed leave no results, and since the results of lines not chosen are not observed, the record that would show what was lost is never produced.

Besides the arrangements that performed the five functions, human allocation may have rested on a mechanism that no one decided: exploration driven by curiosity. Taken as an inquirer's wanting to know what is unknown to it, curiosity works as a heuristic for setting that inquirer's questions and for searching its space of unknowns. Chapter 4, Section 7 separated three things the word can name, an inquirer's state, the work of detecting unresolved matters and forming questions, and a community's distribution of exploration among questions, and noted that none of them entails another by definition. In human researchers the three have often worked together: wanting to know has moved researchers to form and pursue their

own questions, and the pursuits of many researchers have together distributed exploration. Part of science so far, though not all, has consisted of research driven in this way by the curiosity of individual researchers, and because the three have worked together, such research, taken as a whole, can be read as having formed, from the bottom up and without anyone deciding it, a heuristic for searching the space of unknowns. As long as researchers are members of *S*, each one's unknowns overlap with *S*'s, and the differences among their curiosities may have spread the search. On this reading, at least three things come into question when the inquirer is a machine. One is that where the machine is not a member of *S*, as when it conducts research for a human society, what is unknown to it, toward which its curiosity or something equivalent to it is directed, need not overlap with what is unknown to *S* (Section 6); where *S* is mixed or artificial and the machine is among its members, the overlap can return, but the question then becomes whose unknowns within *S* are pursued. Another is that the three things that have worked together in human researchers need not come together in artificial ones, and that the variation in the curiosity of artificial investigators, or what is equivalent to it, need not be supplied, as it is for humans, by what the realizer is like: copying from a common base can narrow it, and different origins do not guarantee it (Section 4), while it might also be built in, and even made greater than the variation among human researchers. Whoever or whatever decides what variation to build then sets part of the search heuristic, and where that is a single decision it approaches the centralization taken up next. The third concerns centralization rather than machines as such: where allocation is assigned from a center, this heuristic is replaced by the single heuristic of the allocator (Section 2), which holds for a human center as well, although an artificial configuration can come closer than human institutions have to such assignment.

Seen at the level of the whole system, the choice by states, funders, or a society to accept and support research of this kind is also an implicit adoption of exploration driven in a distributed way by curiosity as a method of allocation. Abraham Flexner's defense of curiosity-driven research and Bush's defense of basic research, set out in Chapter 9, Section 3, can be read as giving reasons for that adoption. If so, when inquirers become machines, whether to keep this heuristic may be less likely to stay implicit, since whether artificial investigators have curiosity or something equivalent to it, and how far it varies among them, cannot be presumed to come with them as it came with human researchers. Where it cannot, someone, or the configuration itself, may end up choosing the heuristic, or the heuristic may form without anyone choosing it; to whom any such choice falls is a question of authority (Chapter 12).

The point can be put more abstractly, and doing so connects it with the three answers above. Since, as Section 2 argued, no search is free of prioritization, however a whole system allocates, it adopts some heuristic of search. Two distinctions, not one, apply to that heuristic. One is whether lines are assigned by a single center or result from the choices of many parts. The other is whether the heuristic was set by someone or formed without anyone setting it. The two cross: a heuristic applied by a single allocator may have been set by no one, as when the allocator's behavior is learned, and a rule that someone sets can be applied by many parts. Where a heuristic is formed from the bottom up through distributed mechanisms, what drives them may be curiosity in the sense it has had so far, the pursuit of credit (Section 3), or something that is neither; exploration driven by curiosity is one form of this. The three answers above ask, for each function, whether a heuristic that someone set supplies it, one formed without anyone setting it supplies it, or neither does.

Whose unknowns?

Section 1 separated the search problem from the ordering of ends. Sections 2 to 5 treated allocation as search: given what *S* wants to know, how is effort to be distributed? A complete solution to the search problem would leave the question of which of *S*'s unknowns are worth pursuing where it was. This section asks whether that question, too, can be answered inside an artificial configuration, and it separates the part of the question that bears on knowledge from the part that bears on standing, which Chapter 12 takes up. The lineages examined here are those that ask where the ends of an activity come from and who knows them: the argument about dispersed knowledge in economic planning, an account of how democratic institutions transmit information, an account of how scientific significance arises, standpoint theory, which asks from where in a society questions arise, and a debate over whose criteria of importance a society's research follows and whether that bears on what its research certifies. Each presupposes a purpose, and the section says which.

Chapter 9, Section 3 set out what it can mean for something to be valuable for *S*, and also introduced the position Kitcher calls the traditional approach to epistemic significance, on which the structure of the world fixes which truths are significant. On that position a capable configuration could find what is significant as it finds anything else, and the question of whose

unknowns would reduce to search. But it is one particular position: even if the structure of the world bears on one aspect of what is significant, it does not follow that it fixes significance by itself. This section concerns the accounts on which significance is not so fixed.

Hayek's argument against central planning bears on distributed science not through what it says about scientific knowledge but through how the argument itself is built. Hayek grants that, as far as scientific knowledge is concerned, a body of suitably chosen experts may be best placed to command the best knowledge available, while noting that this only shifts the difficulty to the selection of the experts. The argument concerns two other things: knowledge of "the particular circumstances of time and place", which is dispersed and changes continually, and the use of resources "for ends whose relative importance only these individuals know" (Hayek 1945, §§I, III–IV). The purpose the argument presupposes is that resources serve the ends individuals hold. These two kinds of dispersion, of knowledge of particular circumstances and of knowledge of the relative importance of ends, can come apart for artificial research. The first could reappear inside an artificial population wherever ownership and information about costs are divided among its parts. The second would not thereby appear: dividing ownership and information does not by itself give such a population ends of its own. Whether artificial systems can have ends of their own is left open in this book (Chapter 7, Section 4.3; Chapter 12, Section 5). Hayek's argument also has a constructive step. Decentralization alone does not suffice, because those on the spot must fit their decisions to changes beyond their view; prices do this by condensing what a distant change means for them without conveying its cause (Hayek 1945, §§V–VI). Hayek's account can be read as implying that such a signal draws its content from the choices of those who hold the dispersed knowledge. By contrast, a signal chosen by an allocator, of the kind Section 3 considered, draws its content from such choices only to the extent that it is generated from them, and its having been chosen does not by that fact make it convey what matters to *S*.

Elizabeth Anderson, who counts Hayek a pioneer of what is, in Anderson's term, institutional epistemology, distinguishes three forms in which socially dispersed information is transmitted: talk, votes, and market prices. Prices, Anderson argues, transmit information about private preferences, whereas making a concern a matter of public interest requires talk to articulate it and votes to ratify it, since a private preference does not become a public interest by being widely held (Anderson 2006, 9–11). Applied to *S*, this distinction implies that a price-like signal drawn from the choices of those who hold dispersed knowledge would convey what each of them wants, but not by that fact which of *S*'s unknowns are *S*'s to pursue together. In research a further circumstance widens this gap: the concerns that talk articulates themselves move with what research shows. Kitcher counts preferences about the direction of research only once they have been tutored, and Stephen John argues that research can challenge an audience's present values and enlarge the space of later choice (Chapter 12, Sections 1 and 3.3). If so, knowledge of which of *S*'s unknowns are to be pursued together cannot be had only by gathering what is already dispersed, since those who conduct research also take part in forming it. What can happen where the configuration that allocates takes on that part is taken up in Chapter 12, Section 4.

Later readers qualify Hayek's constructive step on the side of knowledge rather than of ends. Lisa Herzog argues that the information a price conveys is purely quantitative, and that markets process knowledge well where other mechanisms, from talk and reputation to regulators and courts, take on the remaining epistemic tasks (Herzog 2020, 267, 269–273, 283). Robert Reamer, reading Hayek together with James C. Scott, argues that prices do not avoid the reduction of information that Hayek found in central planning but perform it, so that which information counts as "the most essential" is itself open to contest (Reamer 2024, 167–170). Applied to the quantities that move the allocation of research rather than markets, such as citation counts and scores, their points imply that such quantities can steer well only where other means, such as review and criticism, take on the remaining work of judging what is new and what matters, and that what is discarded in reducing that work to a quantity is itself a judgment. In research, moreover, what serves as the criterion of evaluation itself moves, and what is new may as yet have no criterion (Chapter 11, Section 8), so that this judgment is not made once and for all. Anderson, whose concern is which institutions can find and use the information needed to solve collective problems in accordance with values such as welfare and justice, later distinguished three kinds of local knowledge: legible local knowledge, discrete facts about particulars put into a standardized form that outsiders can use; illegible local knowledge, articulated but not so standardized; and *mētis*, practical know-how in dealing with particulars that exceeds what its holders can articulate. Anderson holds that Hayek's argument concerns only the first, that prices encapsulate it by discarding such details as why people want what they demand, and that every increase in the scale of coordination, markets included, shifts what an institution can draw on from the local to the general (Anderson 2025, 1–2, 4–5, 21). On Anderson's reading, the first

kind of dispersion, that of knowledge of the particular circumstances of time and place, which this section found could reappear inside an artificial population, is dispersion of legible knowledge, and Reamer likewise takes the local knowledge in Hayek's argument to be factual and readily articulated rather than tacit (Reamer 2024, 170–171). These accounts limit, from the side of knowledge, the reading above on which a signal like a price draws its content from the choices of those who hold dispersed knowledge: such a signal carries what they know only in the part that has been put into a standardized form, and its working depends on other means through which the rest is gathered or acted on.

What, then, supplies the ends of research, on accounts on which the world does not? Philip Kitcher's account of scientific significance offers one answer. As Matthew J. Brown reports the account, Kitcher argues that no context-independent notion of epistemic significance can be defended: there is, in Kitcher's words, "no good reason to believe in the ideal atlas" at which inquiry would aim. Significance flows into the sciences from broad questions that express natural human curiosity and from practical projects. Fields are associated with significance graphs, which show how particular questions, techniques, and results inherit significance from others; the graphs are indexed to a time and change as the field develops, and what counts as significant is relative to a particular group, with its practical interests and its history (M. J. Brown 2010, 133–138). In Kitcher's later account, one generation of inquiry can revise the judgments of significance it inherits, so that what earlier inquiry found shapes what later counts as significant (Kitcher 2013, 207, 221). Kitcher's analysis also gives such shaping a direction. As Eigi reports it, research directions once chosen have considerable inertia, so that it is easier to go on following them, and later attempts to meet needs they neglected start from a less favorable position (Eigi 2016, 92; Chapter 12, Section 7). Chapter 4 looked, within a single inquiry, at a relation in which an overarching question is divided into subquestions and it is asked how far the collected answers to the subquestions answer the original question (Chapter 4, Sections 2 and 6). That, however, is a relation of answers, not the relation of value in which subquestions inherit significance from an overarching question. Significance graphs describe the latter across a field and over time. The purpose this account gives research is to answer the broad questions and serve the practical projects at a graph's periphery, the root from which its significance flows.

Applied to configurations, which Kitcher does not discuss, the account implies that a configuration could extend a significance graph, finding the tractable questions through which a broad question is pursued, without supplying the broad questions and practical projects at its periphery. For research for a human society, the configuration would then inherit that periphery, and an allocation that no longer traced back to it could still bring about epistemic achievements new to *S*, but it would no longer be directed by what *S* finds significant, however well it searched. Kitcher's account is framed by human curiosity. On this account, as Matthew J. Brown reports it, the sciences ultimately obtain their epistemic significance from broad questions that express natural human curiosity, judged against a conception of healthy curiosity, while human beings vary in the ways they express surprise and curiosity (M. J. Brown 2010, 137–138); if so, human investigators share that source with the society whose questions stand at the periphery, and artificial investigators cannot be assumed to. If artificial members of *S* come to perform work named by the second or third of the claims that Chapter 4, Section 7 distinguished beneath the word curiosity, detecting unresolvedness and generating questions, or distributing exploration across questions, the periphery could come to include their questions. The question of whose unknowns would then not disappear. It would become the question that Chapter 8, Section 3.3 raised and Chapter 12 takes up: who may speak for *S*, and whose questions stand at the root of its significance graphs.

On accounts such as Kitcher's, on which significance is not fixed by the structure of the world alone but changes as the connections among questions change, allocation bears not only on which answers are found but also on what is significant. Because significance graphs change as a field develops, and one generation of inquiry can revise the judgments of significance it inherits, which lines were pursued also shapes which questions later appear significant; Section 3 found a loop of the same form in the signals by which lines are judged. Two examples suggest that tractability can move attention, though not through an allocator. After predicted structures from AlphaFold2 became available in 2021, basic research on proteins that had lacked any structural information rose by 23 to 32 per cent relative to proteins that already had a structure, with much larger increases by 2025, while the rate of experimental structure determination remained almost unchanged (Hill and Stein 2026, abstract, 26; figures from the revision dated 17 July 2026); the study is a working paper about one technology in one field, and the tool decided nothing about ends. And the authors of one study of reinforcement learning for reasoning in language models wrote that the great majority of such work had trained and evaluated models only on mathematics and code, domains that offer easily verifiable answers, and that extending it to other domains was held back by a scarcity of reliable reward signals and curated data (Cheng et al. 2025, 1–2). The examples do not show that signals govern ends. Koichi

Takahashi gives a reason why what AGI makes tractable may differ by field. Takahashi distinguishes a wall of computation, which investing energy can push back, from a wall of intrinsic time, the time a target system takes to advance of its own accord, such as the decades to centuries of ecosystem recovery, which no technology applied from outside can shorten (Takahashi 2026, appendix 1, "The Wall of Intrinsic Time"). Takahashi's map of what AGI could reach is presented as an upper-bound assessment under stated premises rather than as a forecast, and on it a field counts as reached when a mechanism model is obtained that predicts specified observations and interventions to a specified precision. On that map, at the energy of the whole Earth, energy and materials come within a year's reach, whereas for climate and social institutions the computation still falls short and the intrinsic time is decisively long (Takahashi 2026, appendix 3; Chapter 5, Section 6.5).

Some authors locate a further route, through the infrastructure on which research runs and those who build it. Emmanuel Jeannot argues, of the Genesis Mission of the United States, that an infrastructure for artificial research on that scale, which emerges from political and industrial decisions rather than from the distributed preferences of the scientific community, encodes "the agenda of the institutions that built it, at the moment they built it"; once such infrastructure becomes the dominant mode of production, Jeannot holds, the questions it does not ask become harder to ask because work beyond its agenda would proceed more slowly and attract less attention and funding, even without being prohibited. On the same argument, this works at a political level, where priorities change with administrations while the infrastructure they built persists, and at an industrial level, where private partners have an interest in orienting research toward what is computationally tractable and economically valuable; and "neither requires a conspiracy" (Jeannot 2026, §7). Bisht and colleagues argue that research involving artificial techniques is pushed toward problems that can be quantified, because those techniques are optimized and evaluated on numerical signals and need data in computationally convenient form; they describe this as an instance of the McNamara fallacy, in which what cannot easily be measured comes to be treated as unimportant (Bisht et al. 2026, §3). These are arguments in preprints, not findings about a particular allocator. Zeynep Pamuk, writing on the public funding of science in democracies, gives an argument of the same form as Jeannot's about infrastructure, for funding decisions themselves. Holding that those who make funding decisions can, intentionally or not, rule out courses of action, Pamuk argues that decisions about how funds are distributed within an area determine the range of possible answers to a problem, while at the decision stage laypeople can accept one of the available options or reject them all but "cannot produce new science", so that formal decision-making power may remain with political processes while being constrained by funding decisions made far earlier by scientists; Pamuk also notes the worry that private funders may enact their vision of a good society "through their private funding decisions, rather than by seeking majority support" (Pamuk 2019, 11–12). In the terms of Chapter 9, Section 2, the arguments of Jeannot, of Bisht and colleagues, and of Pamuk describe a route by which a supporter or provider distinct from *S* can shape which of *S*'s unknowns become tractable and so, on Kitcher's account, which later appear significant; on Pamuk's argument the route also shapes which answers are available to those who later decide, a gap between formal and effective authority that Chapter 12, Section 4 takes up. Taken together with Kitcher's account, on which what earlier inquiry found shapes what later counts as significant, these arguments suggest at least the following. Because only the lines chosen accumulate means, expertise, and results, which of the ends that can later be chosen can actually be achieved depends on the history of allocation; so even where formal authority over ends remains and no end has been explicitly rewritten, what can in fact be chosen may have changed. Whether to call this a change of ends depends on how ends are specified, but it is a more specific form of the point that allocation is not a neutral means of realizing ends, and it holds whether allocation is carried out by human institutions or by an artificial configuration.

How such movement is to be assessed depends on the purpose and the account of value adopted. If the purpose is to produce as much knowledge new to *S* as possible, wherever it lies, a shift toward what has become tractable can count as good allocation. The principle that every true belief is good for its own sake does not entail this maximizing purpose: aggregation and priority rules would also be needed (Chapter 9, Section 3). If Kitcher's account is applied here, where no one with a place at the periphery reviews whether the signals still trace back to it, the signals can come to govern the ends by default. Nothing in this requires the configuration to have ends of its own. It requires only that the ends be represented by proxies and that no one hold the proxies to account. On accounts on which ends are themselves revised in the course of inquiry, by learning what pursuing them involves and what it yields, a change in what is tractable can be part of an ordinary revision of ends rather than a drift from them. On such accounts a judgment of value is tested by acting on it and seeing whether its consequences are valued as it predicted (Chapter 9, Section 3), so that the revision would need some route by which it learns

how those who bear the consequences value them. Whether they also take part in making the revision is a further question.

Two different things bear on answering the question of whose unknowns, that is, which of *S*'s unknowns to pursue, and this book distinguishes them as two roles. One is knowing what matters to the members of *S*, which outcomes they care about, which problems press on them, and which particular circumstances bear on those problems; this is an epistemic role. Parts of it can be gathered and represented, and a configuration whose parts hold dispersed information could carry out much of it. On Anderson's account, set out above, institutions that are to use what has not been put into a legible form have to include those who hold it, recognize them as experts with respect to it, and respect their autonomy in using it (Anderson 2025, 1–2, 5, 20); the account does not rule out that an artificial system could acquire *mētis* (Anderson 2025, 19 n. 8). Where Anderson's account holds, the part of the epistemic role that rests on members' illegible knowledge and *mētis* of their own particulars would be a relation to those members, or to whoever has acquired that familiarity, rather than a capacity that a configuration could exercise from its records alone. The other is deciding whose questions stand at the root of the ordering (on Kitcher's account, at the periphery of the significance graphs) and how they are weighed against one another, or answering for an ordering that no one decided. An account of what is valuable for *S* bears on what such a decision weighs, but does not by itself settle who is entitled to make it or to answer for it. Discussion of artificial systems in research contains several positions on this second role. In an exchange collected by Marcel Binz and colleagues, Matthew Botvinick and Samuel Gershman would reserve to human scientists what they call the normative aspect of science, the question of which problems to work on: judgments of interest, significance, and timeliness are, on their account, tied to cultural and historical sensibilities that themselves change, and making a system conform to present human views does not amount to authorizing it to direct how those views change. Eric Schulz and colleagues reply that if a system had proven itself to select topics fruitfully, there would be no reason to refrain from using it as long as ethical and legal guidelines were followed, since which directions will be fruitful is often not known in advance (Binz et al. 2025, 6–8). Ross King distinguishes a system's generating its own questions within a domain from normative self-authorization, a system's taking itself to be entitled to settle what ends to pursue, and would have what King calls virtual research councils allocate resources "within human-authorised objectives rather than determine autonomously what humanity ought to value" (King 2026, §§14.1, 14.4). In the terms of this section, Schulz and colleagues rest their reply on fruitfulness, an epistemic competence closer to the search problem of Section 1 than to either role, with compliance with ethical and legal guidelines as a condition; whether such compliance supplies what the second role involves is a further question (Chapter 12, Section 1). Botvinick and Gershman, and King, place the second role, deciding the ordering and its change, with humans. Chapter 12, Section 1 examines it on accounts of legitimate authority, and notes there a claim that all of them other than purely epistocratic ones, on which rule belongs to those who know best, can accept: competence in the first role does not by itself supply the standing that the second involves; it also examines Kitcher's ideal of well-ordered science, on which even the epistemic role answers to what ideal deliberators representing the viewpoints in a society would choose.

Standpoint theory, the family of views on which the social position from which research starts bears on the objectivity of the knowledge it yields, adds a consideration that falls under neither role. Sandra Harding argues that where research starts is itself a matter of objectivity. In a society stratified by power, on this account, the activities of those at the top organize and limit what can be understood from their position, and questions that arise from the lives of those the society marginalizes yield less partial accounts not only of those lives but of the social order (Harding 2010, 173–174). Harding does not confine the argument to the social sciences: on her account, sexist and androcentric assumptions have shaped results especially in biology as well as in the social sciences, and what researchers observe is always both given by nature and constructed by culture. She carries the argument into the context of discovery, criticizing accounts that confine method to the testing of hypotheses already formulated (Harding 2007, 162–163). Kitcher's account traces broad questions to human curiosity; Harding's asks from which positions within a society questions arise. Chapter 11, Section 3 refers to this examination when it asks what standpoint theory implies for critics.

Intemann's reconstruction, as Sharon Crasnow reports it, narrows and specifies the claim. On this reconstruction, feminist empiricism and standpoint theory have converged in several respects but still differ over the kind of diversity needed for objectivity and over the role that non-epistemic values play in enhancing it; Intemann argues that the standpoint understanding of diversity, as diversity of social location, better explains why particular sorts of evidence are significant, and why they are relevant to some knowledge projects and not to others. Nor does standpoint theory claim that the epistemic advantage of marginalized positions is automatic: had it claimed this, the view would, as Intemann points out, be either

trivial or false. Standpoint theories, as Crasnow summarizes them, also share an achievement thesis, on which a standpoint is not automatic and must be achieved; Harding, as Crasnow presents her view, distinguishes it from the perspective that a change of social location automatically brings, since achieving a standpoint requires a group consciousness rather than an individual's change of perspective (Crasnow 2025, §4.3). Harding also holds that those who do not live marginalized lives can start their thought from them, but that doing so requires learning from those who live them (Harding 2010, 173–174). Longino had argued, as Anderson reports, that standpoint theory cannot provide a noncircular basis for deciding which standpoints have epistemic privilege (Anderson 2024, §2). The reconstruction, on which evidence from a social position is relevant to some knowledge projects and not to others, suggests where one possible reply might be sought: in what particular evidence bears on a particular project rather than in a general ranking of standpoints. The purpose the theory presupposes is epistemic, knowledge that is less partial; on purposes, or in fields, for which social position does not bear on content, it gives correspondingly less reason to change where research starts.

On standpoint theory, then, whose questions stand at the root is not only a matter of standing, nor only of gathering knowledge of what matters to members, but also bears on how partial the knowledge research yields will be (compare Chapter 4, Section 1). Three things follow, where the theory holds, for an artificial configuration conducting research for a human society. A configuration that extends significance graphs from the questions already posed in the record it was trained on would start where those who produced that record stood, and competence in gathering what members want would not by itself correct this. A standpoint is not something a configuration could have or lack merely by what it is, since on these accounts it is achieved by communities rather than conferred by location; whatever part a configuration could take would be relational, depending on whom it learns from and with whom it inquires. And feminist work on the epistemic value of diversity, as Crasnow presents it, holds that unexamined assumptions go unchallenged when those whose experiences conflict with them are excluded (Crasnow 2025, §1); records produced by research communities would lack such experience to the extent of that exclusion. Taken with the possible reply to Longino's objection noted above, this point also gives reason to think that a configuration could not settle from its record alone which positions bear on the content of a given project, since part of the evidence bearing on that is what the record lacks. In a mixed society, the relation between human and artificial members could itself be a relation of power of the kind that standpoint theorists take social positions to track, and whether positions on either side of it bear on the content of research would, on Intemann's reconstruction, differ from one knowledge project to another. An artificial society whose members were trained on a human record and extended significance graphs from the questions posed in it would, by the first and third points, start where human research communities stood rather than from positions within itself; whether its own members come to occupy positions that bear on the content of their research is a further question. Chapter 11, Section 3 takes up what these points imply for the organization of criticism.

Standpoint theory asks from which positions within a society questions arise. A debate among Argentine scientists and philosophers, conducted in 1971 in the journal *Ciencia Nueva* and in lectures it sponsored, and collected in a book in 1975, asked a related question about research in different societies; it lies outside the cases and debates on which this book has otherwise drawn. Oscar Varsavsky had argued that research in one society can take its criteria of importance from elsewhere without anyone having decided it. On this account, which topics are of greatest interest, which methods are most promising, and how each scientist's work is to be evaluated were sanctioned, rather than explicitly decided, in the large international research centres; researchers trained there returned with those criteria and took them to be universal; and funding allocated according to expected results, with the published paper as a countable unit of evaluation, produced a positive feedback, a "natural selection" of topics, in which established topics attracted funds and new ones could compete only when they answered a need that the prevailing system already recognized. Varsavsky called *cientificismo* the stance of researchers who adapt to this market, give up concern with the social meaning of their work, and accept the norms and values of those centres (Varsavsky 1969, §§II–III). To the reply that what is true in one place is true everywhere, Varsavsky answered that truth is not the only dimension that counts; importance is another, and on this view it is essentially local: a theory about petroleum matters differently in Switzerland and Venezuela (Varsavsky 1969, §IV). Varsavsky later proposed that research priorities be derived, through a policy for technology, from needs that a society defines with popular participation (Varsavsky 1975, 53, 55–56). The purpose the argument presupposes is that a society's research serve that society's own project, in Varsavsky's case its transformation.

Gregorio Klimovsky, whose interview in *Ciencia Nueva* opened the debate, granted much of this. On Klimovsky's account, where one stands socially can make the same problem seem urgent or not, and hypotheses can be suggested by fashions and by research done elsewhere; Klimovsky counted as an ideological distortion a study that would draw its subjects from a middle-class district of a city and not from its informal settlements, but held that methodological criticism can detect such a partial delimitation of the empirical base (Klimovsky [1971] 1975, 15, 24–25, 28). What Klimovsky denied was that such factors affect the objectivity of results in the context of justification (Klimovsky [1971] 1975, 27, 36). Thomas Moro Simpson, defending Klimovsky, separated two senses of objectivity: a theory's relation to truth, and impartiality in choosing what to investigate or to make known. Partiality in the second, Moro Simpson argued, does not imply a lack of objectivity in the first, while the criteria of importance merit the greatest attention in setting a science policy, as does the question of who is to plan research and by which criteria (Moro Simpson 1975, 83–84). Moro Simpson was answering Varsavsky's reply to Klimovsky, which had argued that the criteria of importance are not confined to the choice of problems: they also fix which claims are checked first and how soon, and which methods prevail, so that ideology, acting through them, bears on truth mostly by delaying it rather than by falsifying it (Varsavsky [1971] 1975, 45–47, 49–50); Chapter 11, Section 4 takes up that point. The purpose Klimovsky and Moro Simpson presuppose is objective knowledge, with the scientist's social responsibility as further tasks (Klimovsky [1971] 1975, 33, 35).

The debate bears on this section in two ways. First, Varsavsky's account describes, for the research of a whole society, the structure this section found for a configuration: significance graphs extended from broad questions and criteria of importance supplied elsewhere, through funding by expected results, training, and countable outputs, without anyone having decided it, and with the authority of being taken as universal. If this account is right, such inheritance needs no intention; it needs only, as in the application of Kitcher's account above, that signals stand in for ends and that no one hold them to account. A configuration trained on a record whose questions were set mostly by some research communities would extend their graphs in the same way, starting, as the paragraphs on standpoint theory above found, where those communities stood; and its competence in search would not by itself show whose criteria of importance it applied. Second, Moro Simpson's separation corresponds to the one this section draws between the search problem and the ordering of ends, and to the one between competence and standing that Chapter 12, Section 1 examines. Applied to configurations, which Moro Simpson does not discuss, the separation implies that a configuration's inheriting its criteria of importance from elsewhere would not by itself imply a lack of objectivity, in the sense of a relation to truth, in what it certifies, but would raise the question of who may set those criteria. Not implying a lack of objectivity is not the same as having no effect. On Varsavsky's reply, criteria of importance also fix which claims are checked first and how soon and which methods prevail, so that the separation, though it can be drawn, may not hold in practice, a question Chapter 12, Section 3.5 takes up for governance in general. The debate thus separates three dimensions: whether results are true, to whom they are important, and when needed truths are reached. Applied to configurations, it suggests that a configuration can produce true results while inheriting criteria of importance from elsewhere and delaying the arrival at truths that S itself needs. Where S is itself an artificial society whose members were trained on such a record, the inheritance runs between societies, as in Varsavsky's own case: its criteria of importance, so far as they come from that record, would at first be those of the human research communities that produced it. Applied to such a society, Varsavsky's view that importance is local implies that whether it came to have criteria of its own would turn on whether it had a project of its own from which they could be derived, and Moro Simpson's separation implies that the inheritance would not by itself imply a lack of objectivity in what it certifies, but would raise the question of who may set those criteria for it.

As Section 1 noted, even a complete solution to the search problem leaves the question of which of S's unknowns are worth pursuing; what this section adds is that how that question is answered depends on the account of value and of legitimacy adopted. On Hayek's argument and Anderson's reading of it, knowledge of particular circumstances can be dispersed inside a configuration as well, but ends are not thereby supplied, and a signal like a price carries only what has been made legible; much of the epistemic role could then be performed inside a configuration whose parts hold dispersed information, while competence in that role does not by itself settle the second. Applied to configurations, Kitcher's account and accounts like it imply that a configuration inherits the questions at the root of the significance it extends and its allocation shapes which questions later appear significant; on accounts on which ends are revised through inquiry, the question becomes by what route the revision learns how those who bear the consequences value them, and whether they also take part in making it; on standpoint theory, where the root questions start also bears on how partial the resulting knowledge is; and on Varsavsky's

account the criteria of importance that a society's research follows can be inherited from research communities elsewhere, which on Klimovsky's and Moro Simpson's accounts bears on what is investigated without implying a lack of objectivity in what is certified, while on Varsavsky's reply it can bear on when needed truths are reached.

What Chapter 10 has found

Artificial allocation can change both who pursues a line and how the population of investigators is formed. The hundred-instance configuration can distribute effort without recruiting a hundred human researchers or acting through their career incentives, and allocation's entering a configuration need not mean that lines are assigned from a center: instances can take up and drop lines themselves (Section 3). Sections 2 to 5 examined what changes with that possibility. Control over the workforce enlarges the available interventions; it does not supply the philosopher-monarch's eye for the objective merits of theories (Section 2). Where lines cannot all be tried cheaply, safely, and reversibly, how to distribute effort remains a problem, and the purposes built into a model of allocation determine what counts as a good answer (Sections 1 and 4). Whatever form allocation takes, it adopts some heuristic of search: no search is free of prioritization, and a proposal to search without criteria of expected significance removes one criterion, not judgments of value in a broad sense (Section 2). The heuristic can be set by someone, or formed without anyone setting it through distributed mechanisms such as the curiosity or the pursuit of credit of human researchers, or something equivalent in artificial ones; in human research, exploration driven by the curiosity of individual researchers has been one such bottom-up form (Section 5).

Applied to allocation, the sorting of Chapter 9, Section 6 draws its lines as follows. Premises about human realizers can fail: that a center has one person's capacity to attend, that allocation passes through human investigators' motives, and that turnover comes with mortality, without anyone deciding it. So can premises about how allocation has been organized, which can change even while its realizers remain human: that the cost of a failed line falls on a career, that control over time and resources is dispersed across many grant holders, and that each field's work is evaluated largely by its own members. Some premises mix the two: the concern for credit reflects, on Merton's account, a norm of science more than scientists' psychology, and the weighting of one's own data that Solomon describes bears on allocation where the weighting of those who produced the data reaches decisions about whether lines continue (Sections 2 to 5). Where they fail, many of the problems they addressed change form rather than disappear. Allocation is still steered by signals. Human allocation, too, was already moved by institutionalized signals, some of which did not pass through researchers' motives (Section 3). Diversity still has to come from somewhere: artificial investigators need not be copies of one base, but neither copying nor a difference of origin guarantees the differences that inquiry needs (Section 4). And the functions that human arrangements performed without anyone deciding them, such as keeping unfavored lines alive, keeping lines of different kinds open, and renewing participants, may be set by a rule someone adopts, supplied by a process no one sets, or not supplied at all (Section 5). Conditions of inquiry remain: no one knows in advance, in the senses Section 1 distinguished, which line will succeed, and no search is free of prioritization (Section 2). What a configuration needs in the face of the first differs with the sense: where the chances are known, knowledge of them, which Strevens's planner is assumed to have (Section 2); where they can be learned only by pursuing lines, a policy for learning them, whose costs appear in the termination rule and in the results on exploration (Sections 3 and 4); and where neither lines nor their value can be projected, a way for lines its representation does not yet contain to enter it, which on the positions on awareness of unawareness can include reserving some weight for possibilities not yet conceived (Section 2). Relations to parties outside the configuration that involve the other party's participation or recognition cannot be completed from one side. On Kitcher's account, they include whose questions stand at the root of what *S* finds significant, and allocation bears back on this, since which lines are pursued shapes which questions later appear significant; on standpoint theory, from which positions those questions start; on Varsavsky's account, whose criteria of importance a society's research follows, which Klimovsky and Moro Simpson hold bears on what is investigated without implying a lack of objectivity in what is certified, while Varsavsky replies that, through which claims are checked when, it bears on when truths are reached; and on accounts on which ends are revised through inquiry, by what route the revision learns how those who bear the consequences value them (Section 6). Standing for dissent belongs with these relations: unlike the epistemic work of dissent, a configuration holds it inside itself only where its parts hold it on the account adopted (Section 4). Where the lines fall depends on the account adopted; if significance were fixed by the structure of the world, which unknowns to pursue would belong with the conditions of inquiry (Section 6).

This sorting is also a means of describing what can happen to research when allocation enters a configuration. Consequences that appeared separately in Sections 3 to 6 can join into a single chain where certain conditions hold: that the same evaluative signal returns into the next allocation and into which investigators are copied or withdrawn; that significance is not fixed by the structure of the world alone, as on Kitcher's account; and that no one with a place at the periphery reviews whether the signal still traces back to it. Under those conditions a configuration that divides problems and selects tasks is likely to show success most readily on tasks that are easy to judge, and that success justifies its next allocations (Section 4). Attention then moves to what has become tractable; protein research after AlphaFold2, and work on reinforcement learning gathered in domains with easily verifiable answers, suggest that such movement can occur even without an allocator (Section 6). If what earlier inquiry found shapes what later counts as significant, as on Kitcher's account, and if research directions once chosen have the inertia Eigi reports from Kitcher's analysis (Section 6), the movement does not stop at which answers are found but reaches how significance itself is seen; and because only the lines chosen accumulate means and expertise, the ends that can in fact be chosen later change without any decision to rewrite them (Section 6). Moreover, for a configuration conducting research for a human society, an allocation that seems to trace back to the periphery may trace back not to the questions the members of *S* now pose but to where the communities that produced its training record stood (Section 6). If, in addition, the survival of lines is judged by one signal, an error in that signal can correlate decisions to discard otherwise diverse lines (Section 4); in research, since results are new to *S*, there may be no other answer against which the signal's error could be checked (Chapter 11, Section 6). The links have precedents in accounts of human research and in models of it, for example in the "natural selection" of topics that Varsavsky described and in Smaldino and McElreath's model, in which selection on publication count degrades the rigor of methods (Sections 3 and 6). A configuration can nonetheless differ in three respects. One is that the first condition is more easily met. Institutionalized signals already steered human allocation (Section 3), but control over time and resources was dispersed across many grant holders, and the routes by which work was recognized differed by field (Section 5); in a single configuration, by contrast, one signal can reach every line at once and decide which investigators are copied and withdrawn. Another is that the functions by which human arrangements could break the chain without anyone deciding it, such as the protection of unfavored lines and turnover through mortality, cannot be assumed to be given (Section 5). A third is that, where one configuration's own record is the main record of the research, the narrowing is unlikely to show in it: lines not pursued leave no results, in human research too (Section 5), and that record may come to show high performance on the tasks at which the configuration succeeded (Sections 4 and 5). Further, if the rules of assignment are themselves revised by the same signal, the narrowing can proceed at the level of how allocation is done, not only at the level of lines (Chapter 12, Sections 7 and 8.4). The chain is not an inevitable course: wherever one of the conditions above fails, it can break at the corresponding link. Of these conditions, the first lies within the configuration's own arrangement: it fails where the judgment of which lines continue does not rest on the signal by which the next tasks are chosen and investigators copied or withdrawn. Checks that are not used to select or train producers, a measure Chapter 11, Section 8 considers, can let that signal itself be examined, but they make the condition fail only where their verdicts bear on which lines continue.

What a claim that a configuration allocates inquiry well would need to show can also be drawn from this picture. Such a claim can be tested against arrangements available to the configuration, not against the distribution the philosopher-monarch would choose, since that comparison would require the eye (Section 2). A configuration that can create and assign investigators may be better placed than human institutions to run alternative arrangements side by side (compare Section 2), but where it compares them by its own signal, integration alone supplies no independent check on the criterion of comparison (Section 4). Nor can such a claim always be read from the quality of the configuration's instances assessed one by one, since investigators each following a rule that is not unreasonable in itself can, by pooling their evidence, together stay with a worse option (Section 4). Beyond that, it would need to show three things, each bearing on where the chain is broken. One, bearing on the links from success on tasks easy to judge to the inheritance of the periphery, is, on Kitcher's account and accounts like it, how the signals the configuration steers by still trace back to *S*'s questions, and who reviews whether they do (Sections 3 and 6). Another, bearing on the last link, at which lines are discarded through a shared error, is where the diversity it relies on comes from, and what independent check there is on the criterion by which diversity is ended, which integration alone does not supply (Section 4). A third, bearing on the functions by which human arrangements could break the chain, is, for each function of Section 5, whether a rule someone adopts, a process no one sets, or nothing supplies it, including whether the configuration keeps, replaces, or loses the heuristic that human curiosity formed (Section 5). Devices such as protected budgets or different initializations are possibilities to test, not established substitutes for every

kind of difference among human investigators, and Project 523, a single case, suggested that deciding a function centrally is one thing and making the judgments its performance requires is another. Of the two roles that Section 6 distinguished, knowing what matters to the members of *S* is an epistemic role that a configuration whose parts hold dispersed information could in large part perform; deciding whose questions stand at the root of the ordering, or answering for an ordering that no one decided, is not settled by competence in the first, and who may take it on is the question of Chapter 12.

The distinction between the epistemic work of dissent and standing for dissent also bears on such a claim. Keeping a line with successes of its own, or a route to reopen it, can be an internal feature of allocation; recognizing a party's standing to object is a further relation, unless the internal parts are themselves such parties (Section 4; Chapter 12, Section 5). The distinction does not make separate organizations necessary for every good allocation. What it implies is that a claim to allocate well and a claim that dissent has standing need different evidence: the first, on Solomon's account, about how the allocation keeps successes represented, the second about relations among parties.

What the chapter leaves, therefore, is a space of alternatives, each to be judged against its purposes: which lines remain possible, how they gain and lose support, where differences arise, and how priorities can be revised. The uncertainty of future results constrains that space without specifying a single best arrangement. Within that space, however, the arguments examined bring at least three things into view. One is that the power to create and sustain many lines and the power to judge their survival by a single signal can be joined in a single configuration. On Kitcher's analysis, control can bring about a division of labor that individuals would not choose, so centralization need not reduce diversity (Section 2); but if, as at the last link of the chain above, the same configuration judges the survival of lines by one signal, what centralization trades against is less diversity as such than independence of error about the criterion that governs convergence (Section 4). Another is that a search that does not use expected significance still makes choices about which possibilities it can reach and which it reaches first, and that if allocation bears back on significance, as on Kitcher's account, these choices constrain later judgments of significance (Sections 2 and 6). A third is that, applying Varsavsky's argument, an allocation can produce true results while inheriting criteria of importance from elsewhere and delaying the arrival at truths that *S* itself needs: truth, importance, and the time at which truths are reached are assessed along separate dimensions (Section 6). Since allocation also uses judgments produced by criticism, the next chapter asks how those judgments are formed and when others can rely on them.

Chapter 11 — Criticism and Certification for *S*

Defeat, Independence, Trust, and Correction

Which evaluation this chapter examines, and why

Chapter 9, Section 2.1 derived three relations through which research is tied to *S* (the society to which an achievement is to be new) beyond the activity of whatever unit conducts a particular inquiry. The second, criticism and certification, concerns the assessment of epistemic success, and this chapter develops it first through *knowledge*. Producing a result does not make it something *S* can rely on. Of the three, it is the relation that bears especially directly on what it is for a result to be knowledge and to have become *S*'s, though only on accounts on which social organization does part of the work by which knowledge is warranted or held (Chapter 2, Section 6; Chapter 9, Section 2.1). This chapter asks what happens to that relation when an artificial configuration can take it over. A configuration here is, in the sense of Chapter 7, Section 1, the unit to which the performance of research is attributed: a single model, a collection of agents, an organization of humans and AI, among others. In the example of Chapter 9, Section 1, a single system generates a hundred instances, assigns some of them to check and reproduce what others produce, and its builders claim, among other things, that the results which pass its internal checks are certified for *S*. The chapter asks what such a claim would require, which part of the evidence for it can come from inside the artificial configuration, and why the part that cannot come from inside remains a relation to *S*.

This chapter examines, among the many evaluations in research, the evaluation of results addressed to *S* and the criticism on which it draws; it calls the former certification. There is a further reason for this focus besides the one just given: the changes an artificial configuration brings to evaluation put pressure on the independence of evaluators, on which this evaluation draws, and on the visibility of evaluation to *S*. Evaluation in research is plural. In human research, for instance, funders have assessed proposals before any result exists, reviewers and editors have assessed manuscripts, institutions have assessed researchers for credit and positions, and those who use a result have assessed it in use. Chapter 9, Section 4 set out four things that any of these involves, among them that a standard tied to a purpose is applied for some party's choice and that the evaluator and the addressee who uses the verdict can differ. Two of them are used later in this chapter. One is that proxies are unavoidable, because what an evaluation of research is about is usually not observable when the judgment is needed. The other is that, for something new to *S*, the evaluator may not yet have a standard (Section 8).

An artificial configuration changes at least three things about evaluation directly. First, it can generate evaluators rather than recruit them, as copies of one base system or as evaluators that differ widely from one another. Second, it can place those who produce results and those who evaluate them within one artificial configuration, and bring evaluation continually into results while they are still being formed. Criticism then changes not only in cost but also in range, judgment, and arrangement: some kinds of check become nearly costless, checks that humans did not ordinarily perform come to be attempted, and evaluators can err in places other than where humans err and pick up different things. And evaluation can dissolve into production, so that the very moment of "having passed review" disappears and evaluation is no longer visible to *S*. Third, it can couple evaluators and producers over time, adjusting each to the other's outputs (for example, by training or selecting each on them). Sections 3, 7, and 8 take up these three in turn.

Other evaluations are taken up elsewhere: the ex ante evaluation of lines of inquiry, as part of allocation, in Chapter 10; credit and the attribution of contributions in Chapter 7, Section 3.3; the evaluation of claims about artificial researchers in Chapter 7, Section 5 and Chapter 13; and who may set and revise the standards and error weights that evaluation applies in Chapter 12. Allocation, though, also uses judgments produced by criticism, so the judgments of allocation that Chapter 10 examined rest in part on the criticism examined here.

Chapter 3 drew a limited permission from the distinction between discovery and justification and from Duede's application of it to deep learning: the route that generates a candidate answer need not be the route that warrants accepting it. Even where the generating process is opaque, the answer may be used if the grounds for accepting it are supplied by another route. Reliabilist and virtue-epistemological accounts narrow the range of cases that need the permission, since on them a reliable generating source can itself supply warrant without a separate route. Chapter 3 also added a condition. The permission holds only when the candidate is assessed through an adequate independent route, and evidence of shared evaluators, selection,

contamination (evaluation items or their answers having entered the training data of what is evaluated), and hidden human filtering is needed to judge whether the independence is real. That discussion concerned a single result. When an artificial configuration can instantiate not only the investigator but also the reviewer, the replicator, and the critic, criticism itself becomes something the configuration can organize internally. Here and below, when names such as review and replication are used for parts of an artificial configuration, they name functions that human institutions have bundled, not the persistence of those institutional forms; Section 7 asks what those functions are.

This chapter distinguishes two relations: the defeat relation and certification. It begins with the first. Consider three checks. A predicted property of a candidate material is measured with an instrument whose readings are not inferred from the prediction. A generated proof is checked against formal rules. A forecast registered before an event is compared with what happens. In each case, the candidate or its grounds can encounter evidence against it that is not merely a repetition of the producing process's reasons for accepting it. Call such an evidential route a **defeat relation**. The term names not the event of a result's actually being overturned but the existence of a route that could show the candidate wrong if it were wrong; a candidate that passed a check also stood in such a relation. Chapter 2 distinguished evidence that rebuts a claim from evidence that undercuts its grounds; Chapter 8, Section 2.3 called an epistemic organization world-answerable when it is constrained by its targets in such a way that its commitments would differ if the targets were different, and remains corrigible through further contact. A defeat relation specifies a route through which such correction can arise. Whether a particular measurement, formal check, or comparison supplies one depends on what it tests and how its result bears on the claim, not on its name. Neither an infallible check nor a guarantee that every false candidate will fail is built into the term.

Keeping such a route beyond the producer's power to alter is one way of protecting it, not the definition of the evidence it adds. A program may have permission to revise a checking routine and nevertheless submit a candidate to an unchanged version that detects errors its generation process misses. Permission alone does not remove that information. Conversely, a routine that the program cannot change may reproduce the generator's mistaken assumptions. Its protection alone adds none. As these two possibilities show, the evidential contribution of a check is distinct from the arrangements that maintain it. Section 2 develops the first of these, the evidential contribution of a check; Section 8 asks how the contribution can survive adaptation, selection, and changes to the check.

Failure also needs interpretation. Chapter 6, Section 2 called the work of deciding which commitment a contrary result gives reason to revise *localization*. A contrary measurement can bear on the candidate, an auxiliary assumption, a calibration, or the match between a formal statement and the question. The producer may correctly locate the failure; assigning localization to another party does not by itself make that party right. What matters here is whether the diagnosis has support beyond the wish to preserve the candidate and can itself be challenged by further evidence. If every adverse result can be reassigned without such support, the measurement may remain informative while the judgment whether to accept the candidate ceases to answer to it. Separating custody of evidence, recording revisions, or sharing authority over disputed diagnoses can protect this answerability. Their value depends on the challenge they preserve; they are not universal conditions that every informative check must meet.

The other relation is **certification**: the relation through which others are given reason to rely on a result for specified uses. Certification refers to *S*. It concerns who within *S* may rely on a result and for which action, and on Longino's critical contextual empiricism (examined in Section 3) it also concerns the perspectives from which the result has been criticized, since acceptability is relative to the perspectives available to the community, as Büter's quotation of the definition makes explicit (Longino 2002, 135–136, as quoted in Büter 2012, 84). An artificial configuration may secure strong defeat relations internally and still leave open whether its certification gives *S* reason to rely on what it certifies. The three changes named above bear on both relations, the defeat relation and certification. The chapter asks how much of the former can be secured inside an artificial configuration and why the latter remains a relation to *S*, and further whether time and the reduced visibility of evaluation reopen to the outside even the part of the former secured inside.

Whether a defeat relation itself refers to *S* depends on the account adopted. On accounts that locate the value of criticism in the errors it detects, a defeat relation can hold inside a single actor: a mathematician, or an algorithm, can propose a proof and have it checked against rules the proposer does not control, and nothing in that relation refers to *S*. On Longino's account, evidence bears on hypotheses only through background assumptions (Longino 2025, §5), so what a measurement counts against is relative to assumptions that a community holds. On that account, the claim that a defeat relation does not

refer to *S* holds only so far as the background assumptions through which its outcome is read are not themselves among those that *S* would contest.

This divergence follows the branch drawn in Chapter 2, Section 6. There, the claim that inquiry causally depends on social practices was separated from the stronger claim that evidential status, objectivity, or knowledge is partly constituted by organized social practices, and the book declined to adopt either account of social knowledge. The question of this chapter can therefore be read in two ways. On the first branch, criticism matters because it detects error, and the question is what makes one check independent of another; that is the question of Section 2. On the second, organized criticism partly constitutes objectivity or warrant, and the question is what kinds of difference among critics the constitution requires. Section 3 compares several accounts on this branch, alongside the view of the first branch (Table 11.1). What becomes a topic on the second branch is objectivity in the sense of being partly constituted by organized criticism; this does not keep the view of the first branch from speaking of objectivity in other senses.

At several points the chapter uses the sorting of Chapter 9, Section 6. Chapter 9, Section 2.3 proposed that the division of the relations surrounding research among separate parties may have supplied certain goods without anyone designing them. The sorting of its Section 6 asks what each such good depended on, and distinguishes four kinds of answer. One is premises about human realizers, the people who actually held the relations (their limited attention, their motives, their lifespans, and the like). The second is premises about the present organization of science, how the relations have historically been organized (that only investigators have access to the primary data, for instance, or the gate formed by review); these can be reorganized even while the realizers remain human. The third is conditions of inquiry, which hold whatever realizes it. The fourth is relations to parties outside the artificial configuration, which the chapter, following Chapter 9, sometimes calls relations to *S* for short.

Section 2 defines independence, on the error-detection view, as the information a check adds. Section 3 asks where difference comes from when critics can be made, by how each of five families (Table 11.1), this view among them, assesses it. Before the second and third changes are taken up, Sections 4–6 turn to certification, since what Sections 7 and 8 ask about those changes is stated in terms of the thresholds and grounds of reliance set out there; they add two questions to the two branches of this section: whose valuations the threshold at which a result counts as sufficient for a use answers to (Section 4), and what grounds reliance on an unchecked result and whether that reliance has to be trust (Section 5). Section 6 asks what remains with *S* at the limit at which *S* as a whole stands as a layperson relative to the artificial configuration. Section 7 asks how the cost, range, judgment, and arrangement of criticism change when production and evaluation are joined in one artificial configuration, and what can be lost when evaluation dissolves into production and ceases to be visible. Section 8 asks what coupling over time does to critics' independence, to the standards by which the new is evaluated, and to correction. Section 9 returns to the builders' claim and considers it by combining these accounts. In outline: an artificial configuration makes the properties that the accounts each track (the independence of checks from one another, there being someone who answers when a result is wrong, those who produce and those who rely sharing valuations of the weight of errors, and the visibility to *S* of the fact that evaluation has taken place) separately settable. In human research they were tied together by realizers and organization, so verdicts plausibly tended to coincide; for artificial configurations the accounts' verdicts on the same case diverge, and a builder's claim cannot be assessed unless it shows which account it answers to. And on some accounts, improving internal criticism can weaken certification for *S*.

These verdicts, however, are not where the chapter ends. The account-by-account verdicts are also a means of describing what can happen to research when criticism and certification move inside an artificial configuration, and the picture the chapter's material yields can be given in advance. Where critics are made rather than formed, what adds evidence is their difference rather than their number, and a body of critics that shares its assumptions can, on Longino's account, entrench them the better its internal criticism works (Sections 2 and 3). If evaluation dissolves into production, the moment of "having passed review" disappears and evaluation ceases to be visible to *S*; cheap criticism, multiplied, can then conceal a scarcity of independent criticism so far as it errs where the producer errs (Section 7). Evaluators can come to tell less about producers merely by being used for selection, without anyone deceiving. The very standards by which *S* judges results can move with the coupling of production and evaluation, and agreement with *S*'s present standards is no evidence that they have not moved (Section 8). The weighting of errors is shaped in advance inside the configuration (Section 4), and certification can come to be treated less as a researcher's assertion than as the reading of an instrument (Section 5). Meanwhile the members of *S*, as

they lose the places in which they gained experience of evaluating, can lose the very capacities needed to carry the relations that remain (Section 6). Where these coincide, passing internal checks may come to serve as a proxy for certification, the lack of relations to *S* may be selected along with it, and research may approach a state in which relations to *S* remain in form while no one is able to carry them (Section 9.3). The account-by-account examination that follows also serves to separate which accounts count which of these as a problem, and what each would take to prevent.

Independence on the error-detection view

On the first branch of Section 1, the value of criticism lies in its detecting errors. To ask about independence on this branch is to ask what makes a check able to find errors that other checks have not yet found. The question matters in this chapter because the three changes listed in Section 1 all put pressure on the independence of evaluators. When two checks return the same verdict, how much evidence does the second add? The literature on expert testimony has given a precise answer to this question for the case in which several sources agree. This section starts from it.

Nicolas J. Tanchuk reports Alvin Goldman's minimal condition. Agreement from a second source adds evidence only if that source is at least partly conditionally independent of the first: more likely to agree when the claim is true than when it is false, given what the first source has said (Tanchuk 2026, 4). Goldman and Thomas Blanchard likewise explain that merely multiplying uncritical followers of a leader adds no independent support (Goldman and Blanchard 2015, §3.2). The condition says nothing about persons or organizations. It concerns the relation between the ways two judgments can go wrong. Guerrero reports Goldman's warning that sources which merely reflect one another or a shared third party do not supply the needed independence (Guerrero 2017, 177). Nor does independence require that the second source bypass the first. Tanchuk reproduces Goldman's example of a source that hears the first source's reasons, considers defeaters the first never considered, rebuts them, and only then agrees; such a source can still be partly independent (Tanchuk 2026, 5).

This book extends the condition within this view. On the extension, a critic is independent to the extent that it adds error-detecting information, with respect to a given claim, beyond what existing critics already supply. This is an application to checks in general of the distinction Goldman draws for agreement among sources, not Goldman's own account of such checks. On the extension, organizational separation is neither necessary nor sufficient for independence. It is not necessary, because a single configuration can check a proof with a kernel (the small program at the core of a proof assistant that checks each step of inference) whose failure modes differ from those of the process that found the proof, or test a prediction against a measurement it does not control. It is not sufficient, because evaluators in different organizations, under different names, can share the same training, the same sources, the same evaluative standards, or the same conception of what a good result looks like, and fail together. The powers that Chapter 9, Section 5 named, such as keeping evidence apart, can hold routes of criticism apart in the way that an instrument in separate custody is held apart; on the extension, holding them apart matters to the extent that it leaves them different ways of failing. The power to stop updates to other parts can bear on this view as well, as a means of stopping a checking part from being updated on the producer's outputs (Section 8). Evidence about present language models gives one example of the insufficiency. The correlated errors noted in Chapter 10, Section 4 persisted, among the larger and more accurate of the language models studied at one time, even when providers and base architectures differed: a failure of the kind described here, in which a way of going wrong is shared although names and organizations are separate (Kim et al. 2025, abstract). The same can hold for human evaluators who draw on the same models. Corpus-level estimates suggest that between 6.5 and 16.9 percent of the text of peer reviews for four AI conferences held in 2023 and 2024 could have been substantially modified by language models, and the estimated share was higher among reviews that were more similar to the other reviews of the same paper (Liang et al. 2024, 1, 9). On this view, where such shared use is not visible, the formal separation of critics does not by itself show whether, or how far, they share a way of going wrong.

A defeat relation, in the sense of Section 1, gives a check a way to challenge a candidate or its grounds. On this error-detection view, the independence at issue is its informational contribution relative to what other checks already supply. Internal coherence and agreement among evaluators do not establish that contribution by themselves. Contact with the world and formal checking can supply it, including inside one configuration. What a check establishes also has limits: a checked proof establishes a theorem relative to its formal statement, not that the statement captures the question (Chapter 6, Section 3).

Consider a further constructed contrast. A check rejects a candidate, and the producer proposes that a calibration error caused the rejection. In one version, measurements against a separately established reference support that diagnosis; the calibration is corrected and the affected candidates are reassessed. In another, the only reason for changing the calibration is that it rejected the favored candidate. The first can improve the check, while the second gives no additional reason to accept the candidate. The difference is not whether the producer took part in the revision but what supports it and what remains able to count against the revised judgment. Records can expose that difference; merely recording a change does not by itself show that the revision was supported. Chapter 6, Section 5 develops revision of standards within inquiry; the point here is its consequence for reliance on certification.

Independence, protection, and severity therefore answer different questions. A check may add information despite being open to revision; protection can help maintain that contribution without proving it. Even a protected, informative check can detect too narrow a range of errors to warrant the claim made from passing it. What Mayo calls a severe test (Chapter 6, Section 2.3, on Mayo 1996), which can be glossed as one with a high probability of exposing the relevant error if it were present, concerns this further question. Adequate certification therefore does not follow from any one of these properties.

Four cases show what this view picks out, and each is introduced for a different point. The first two form a contrast between a division of relations that supplied independent routes and one that did not. After a 2020 report of phosphine in the atmosphere of Venus, one outside group re-ran the published scripts and criticized the data reduction, another informed the observatory of a difference in a calibration setting, and observatory staff identified problems and reprocessed the archived data (Snellen et al. 2020; Villanueva et al. 2020; European Southern Observatory 2020; Greaves et al. 2020). The case illustrates the claim that independent routes can lie outside the producing team: the problems were found by parties who ran their own analyses and who held the archive and the calibration records. By contrast, the investigation of the Hubble Space Telescope's flawed mirror found that the division responsible for fabricating it had been insulated from review, that indications of error had been discounted during fabrication, and that the program had relied entirely on a single test instrument (Allen et al. 1990, iv, 9-1). Chapter 6, Section 3.2 used the same case to show that a passed check can rest on a standard that is itself wrong; here it illustrates the claim that checks can be numerous and still share one error route, since tests that did indicate the error were discounted because they disagreed with the instrument on which the program relied, which served as the standard. Two episodes do not show how often a division of relations supplies such detection.

The third case shows that independence can be designed by arranging roles. At the eighteenth-century *Nautical Almanac*, each month's tables, apart from those for the moon's position, were computed separately by two computers and compared by a third person; when the comparer found, by January 1770, that two computers had been copying each other's work instead of working independently, both were dismissed, and when they were re-employed several years later they were not given the same months (Croarken 2002). The arrangement was valuable only so long as the two computations had separate routes to error, and copying removed that without changing the names of the roles. The fourth case shows the converse in artificial pipelines: dividing a pipeline into named roles does not make its checks independent. An independent critique of an autonomous materials laboratory argued that its automated analysis of diffraction data was not yet reliable and that, on the critics' reanalysis, none of the reported materials was new (Leeman et al. 2024, abstract). The authors' later correction, after manual reanalysis, confirmed 36 of the 40 reported syntheses and judged four inconclusive (Szymanski et al. 2026); the two address partly different claims, synthesis and novelty. A benchmark analysis covering ten tasks coded every paper generated by one automated research system as containing fabricated experimental results (H. Chen et al. 2025, app. B, fig. 6). In both, the systems included automated checking; what mattered was whether the checks had an error route of their own.

Different claims within one case can also be settled by different routes. When RIKEN tested whether the STAP phenomenon could be reproduced, a verification by a team led by Hitoshi Niwa concluded that the cell aggregates it obtained did not meet the original studies' definition of STAP cells (Niwa 2016), while a separate committee of outside experts, comparing genomes of stored cell lines with those of existing embryonic stem cell lines, concluded that the stored lines had derived from those lines (RIKEN Investigation Committee 2014, 13). On this view, whether a check is independent is a property of the check relative to a claim, not of the investigation as a whole.

On the view that values criticism for the errors it detects, a check's independence is judged claim by claim by the information it adds, not by the number or organizational separation of critics. Such a contribution can arise inside an artificial configuration. The further burden is to establish its scope and how it survives changes to the producer, the check, and the

interpretation of failures. The next section asks where difference among critics comes from when they can be made, and how this view and other accounts assess both the limiting case of many critics made from one base and the case in which differences are made.

Where difference comes from when critics can be made

Section 2 asked what makes one check independent of another. In human research, critics are formed as separate persons with separate histories, so their number and their organizational separation have served as rough indicators of independence. Much of that independence was not designed by anyone; it was supplied by what human realizers are like. The first of the changes named in Section 1 removes this premise. An artificial configuration can make critics in any number and in any combination: it can make many copies of one, and it can make critics that differ widely. Number and separation then cease to serve as indicators, and where the differences among critics come from becomes something to be asked. This section first pushes to a limit the case in which many critics are made from one base, asking account by account what is added and what is not, and then turns to the case in which differences are made. The same question has a familiar form in human research. The reviewers of one manuscript read the same text, often with similar training, so that the question arises whether review is redundant. Whether redundancy is a defect depends on what is duplicated. Apart from overlap among reviewers, there is redundancy in another sense: review repeating an evaluation the producing side has already carried out. Section 7 takes that up.

This book distinguishes four senses of redundancy. The distinction is made for this chapter, to keep claims about redundancy from trading on one sense while establishing another. It draws on the conditions of Section 2 and on the literature cited below, and it divides the senses by what a further critic adds: nothing beyond what others supply, a route to error of its own, a different application of a shared standard, or a different addressee or function. The senses of redundancy are not limited to these four.

1 *Duplication by correlated critics.* Critics with the same training read the same record. On Goldman's condition, a non-discriminating copy adds no evidence beyond its source. Correlation alone does not imply this: critics can share some influences while still adding information about the claim. What matters is the additional information, not complete statistical independence.

2 *Multiplication of independent error routes.* Checks that work on different principles, measurements in separate custody, formal verification. A critic who rereads the record can find errors in what was done with it, such as a miscalculation, a confused statistical inference, or a gap in a proof, and adds to detection so far as it does not go wrong where the producer does. An error in how the record was made, such as a contaminated sample, may leave no trace in the record, and then a critic adds information about it only by making a new measurement or rerunning an experiment. The contrast between rereading and new evidence is drawn in work on jury theorems: as Franz Dietrich and Kai Spiekermann put it, enlarging a group whose members generate new evidence independently, as by experiment, adds "information, not interpretations of information", but only in a scenario that excludes any hidden common influence on different members' evidence, an exclusion they regard as implausible (Dietrich and Spiekermann 2021, §4.3). On the first branch, this is the redundancy that has value.

3 *Diversity of judgment.* Critics who share a standard differ in how they apply and weigh it. Thomas Kuhn, as Bird reads him, held that the values shared in science, such as accuracy, consistency, scope, simplicity, and fruitfulness, guide judgment rather than determine it as rules would, so that scientists who share them may still differ in their choice of theory (Bird 2025b, §7.3). Kuhn allowed, as Bird reports, that such differences can come from factors such as an individual's life history and personality and from influences outside the sciences, such as German Romanticism, while holding that in modern, professionalized science the factors at work typically come from within science, and that even a difference of personality, such as one scientist's being more risk-averse than another, remains a relation to the scientific evidence (Bird 2025b, §7.3). On this account, in the passage that Zollman quotes and Bird summarizes, the way new theories are ordinarily developed and tried by several scientists before a group accepts them requires a decision process that permits rational disagreement: a shared algorithm would lead all conforming scientists to make the same decision at the same time, and the looseness of the criteria, seen as values, is presented as essential to distributing the risks involved in

introducing or supporting novelty (Zollman 2010, 2; Bird 2025b, §7.3). On Kuhn's account, the value of this redundancy lies not in error detection but in spreading across a group the risk that introducing or supporting a novelty entails.

4 *Overlapping stages of evaluation.* Evaluation can be repeated, at different times and for different addressees, between a result's production and its use. In human research, prepublication review, postpublication criticism, replication, citation, and downstream use, among others, have formed such stages. An artificial configuration need not have the same stages, but where it has arrangements that repeat evaluation, the same question arises. Where the stages have different addressees and functions, they are less redundant than they look; where each reads the same thing on the same record and goes wrong where the earlier stages do, they fall back into the first sense.

The first two senses are those the view of Section 2 captures. The third is not captured by it, since diversity of judgment may add no new error route and still matter, on Kuhn's account, for how novelty is carried. Kuhn's argument also has a limit. On Bird's reading, it supposes that in a large community the variable factors that shape individual judgment tend to cancel out: individual differences are normally distributed, and a judgment at the mean of the distribution corresponds to the one that the rules of scientific method, as traditionally conceived, would demand (Bird 2025b, §7.3). Kuhn does not consider groups of copies; applied to them, a large group of critics derived from one base system may, despite its numbers, lack the spread of individual differences on which Kuhn's argument relies.

Evidence from one field bears on the first two senses. Valentin Danchev, Andrey Rzhetsky, and James Evans compared published claims about drug–gene interactions with the results of robot-run high-throughput experiments. Findings about a claim from papers that shared authors agreed with one another more often than findings from socially independent teams, while, among claims supported by more than one paper, those confirmed by independent teams using varied methods and drawing on different prior literature were more often replicated in the experiments; in the absence of social independence, they write, "authors and methods reinforce agreement with themselves" (Danchev, Rzhetsky, and Evans 2019, 4, 6–9). The associations are correlational and come from one area of biomedicine. Read in the terms above, agreement among overlapping teams is partly duplication in the first sense, and confirmation by independent teams adds what the second sense counts. The study concerns human teams. Of its measures of dependence, shared methods and shared prior literature have counterparts among artificial critics that apply common procedures or were trained on a common record, whether or not they derive from one base system; whether such counterparts carry the same association, the study does not test.

Yasuo Deguchi gives an argument for the second sense at the level of kinds of inquirer rather than of individual critics. Drawing on nineteenth-century precision measurement, Deguchi notes that systematic error, the bias specific to an instrument, shows up only when different instruments measure the same object, and is cancelled by measuring the object by as many different methods as possible and averaging the results. With humans treated as one kind of cognitive instrument, the argument holds that a bias humans have simply by being the inquirers cannot be made visible so long as all inquirers are human, however many of them there are; artificial inquirers give a chance to see it where they and human inquirers diverge, and since they have biases of their own, the divergence points to both. Deguchi calls for keeping artificial inquirers unlike human ones. On this view, inquiry by humans alone leaves human bias in the result unseen, and leaving cognition wholly to artificial inquirers exposes the result to theirs; since no simple averaging cancels the two, both kinds have to take part in a dialogue that extends to the cancelling itself, examining the differences case by case, and artificial inquirers made too like human ones would inherit human bias and so remove the different perspective that lets it be seen (Deguchi 2026b, slides 4–12). Deguchi's argument concerns inquiry conducted jointly by humans and artificial systems, and its analogy assumes that the biases of different instruments scatter around a value there to be found. That assumption suits questions about how the world is, where cancelling biases can be thought to bring one closer to the value measured, which is answerability to the world (Chapter 8, Section 2.3). It does not carry over as it stands to questions about what matters to *S*, since what matters is not a single value on which inquirers of different kinds would converge; applied there, a divergence between kinds may reveal a difference of standpoint rather than a bias to be cancelled. Deguchi's point has the same form as Longino's, considered below, that assumptions shared by all members of a community are invisible to it; it differs in placing the point at the level of kinds of inquirer rather than of perspectives, and in seeking the remedy in a dialogue between inquirers of different kinds that extends to how their biases are cancelled.

To see what the different accounts imply, consider a thought experiment that pushes the example of Chapter 9, Section 1 to a limit. Its motive is to isolate one premise of human research, that critics are formed separately, by different histories, so that

their number tracks their independence; the experiment removes that premise and keeps everything else. Call the construction a **republic of instances**, to distinguish it from Michael Polanyi's "republic of science" (Chapter 10, Section 2; Chapter 12, Section 3.2). A million artificial investigators are derived from a single base system and assigned to research, review, and replication roles. For the limiting case, suppose their subsequent experience does not diversify the background assumptions relevant to the claim. Common origin alone would not establish that further premise. The configuration maintains public venues in which any result can be criticized; it requires that criticism receive a recorded response and that the distribution of the instances' beliefs, or of the commitments that play their role (Chapter 2, Section 2), change over time in a way that is sensitive to the critical discourse (Longino 2025, §5); it encodes shared standards to which critics can appeal; and it gives every agent equal intellectual authority to raise objections. Taken as a community in its own right, the republic meets in form the first three conditions that Chapter 2 introduced from Helen Longino's account of objectivity, recognized venues for criticism, uptake, and public standards, together with the part of the fourth, tempered equality of intellectual authority (treating any perspective as having a prima facie capacity to contribute to the community's critical interactions, an equality that can be lost by failing to engage with or respond to criticism), that concerns who may raise objections (Longino 2025, §5). Whether it meets the diversity of perspectives that the same condition also calls for is, as will be seen, a separate question. Is its criticism adequate?

The answer depends on where an account places the adequacy of criticism. Table 11.1 sets out five families of answers, the last of them an ideal traced historically rather than an account; each row is a family rather than a single theory. This book chooses these five because each gives a different answer to what the republic lacks, and the choice keeps the view of Section 1's first branch, developed in Section 2, as one row among others rather than as the baseline. The five do not exhaust the possible answers. Each row states its family through the authors named in the first column. The right column summarizes the verdict this chapter draws for the republic; in every row, it is reached by applying an account or ideal to a configuration its authors did not consider.

Table 11.1. Where the adequacy of criticism is placed, and what comes into view for a republic of instances: a simplified overview

Family or ideal	Where adequacy is placed	What follows for the republic
Error routes (Goldman's condition, as extended in Section 2)	each further check adds error-detecting information that others do not supply	the number of instances counts for little; shared error routes make many critics close to one; checks that add relevant information can improve criticism, including checks maintained inside the configuration
Critical contextual empiricism (Longino)	transformative criticism among relevant perspectives (criticism in which perspectives that do not share a background assumption question it and can change what is accepted), in a community meeting four conditions	the procedural conditions can be met, but assumptions shared by all instances remain shielded, and may be reinforced
Social empiricism (Solomon)	an equitable distribution of the influences on judgment ("decision vectors") across theories, empirical ones in proportion to each theory's empirical success and non-empirical ones evenly; consensus is appropriate only when one theory has all the empirical success	the failure lies less in missing perspectives than in influences all instances share; more internal deliberation can make it worse, since groups can amplify individual biases, and pressure toward consensus can lead to groupthink and to the suppression of relevant data
Standpoint theory (Harding, Intemann)	inquiry that starts from the lives of those a community has marginalized, and a standpoint achieved through group consciousness	copying alone does not supply the relevant standpoint; engagement with marginalized lives and the achievement of group consciousness must be established separately
Mechanical objectivity (as traced by Daston and Galison, and Porter)	procedures and rules that keep the investigator's will and judgment out of the result, so that it does not depend on who produced it and need not rest on personal trust	uniform rule-following instances can count as objective in this sense; the sense guards against the bias of any one critic, not against assumptions all share, and the question passes to who designed the procedure

Longino's account is developed here because Chapter 2, Section 6 introduced it as one important way in which social organization can enter warrant, and the republic tests it directly. On this account, evidential relations are mediated by background assumptions (Longino 2025, §5), and, as Heidi Grasswick reports, a diversity of perspectives represented in a community is needed to expose questionable background assumptions that can be invisible when shared by a group of similarly situated knowers (Grasswick 2018, §4.3). As Dutilh Novaes and Dede report Longino's fourth condition, effective

criticism requires diverse perspectives and the cultivation, rather than mere toleration, of dissent (Dutilh Novaes and Dede 2025, §2). Büter reproduces Longino's definition of acceptability: supporting data must be evident to the community at the time, and reasoning and background assumptions must have survived criticism from as many perspectives as are available to it, with the four conditions characterizing the community (Longino 2002, 135–136, as quoted in Büter 2012, 84). As Jaana Eigi quotes Longino, the tempered equality condition requires both that scientific communities include relevant subgroups of the society that supports them and that communities attend to criticism originating from "outsiders"; and for assumptions shared throughout a community, Longino proposes interaction across communities or, at least, openness to criticism from within and from outside the community (Longino 2002, 133–135, as quoted in Eigi 2016, 136). Since acceptability holds only for a community that meets the four conditions, acceptability even relative to a community's own perspectives then presupposes openness to criticism from outside that community. Wylie extends Longino's norm of tempered equality by proposing that communities actively cultivate collaborations with external communities whose different aims, practices, and beliefs can expose shared assumptions to criticism (Wylie 2015, 13, author manuscript). Wylie develops this proposal for archaeology; its application to artificial research is a further step here. As Yamamoto reports Longino's argument, assumptions shared by every member may be reinforced by surviving structures that screen out individual assumptions (Yamamoto 2018, 26); and, as Eigi reports Longino's account, a hypothesis is accepted as communal knowledge only if it survives criticism from multiple points of view (Eigi 2012, 51). Longino's later restatement adds that what fuels critical discourse is a diversity of perspectives, since assumptions on which everyone agrees go unacknowledged until something goes noticeably wrong (Longino 2025b, 13). Taken together, these points suggest that a republic whose internal criticism works well can entrench what its members share.

Chapter 8, Section 2 described the level at which such sharing can occur. Beneath explicit method lies a primitive epistemic organization: what a system treats as an observation, a repetition, a failure, or evidence bearing on one commitment rather than another. Instances derived from one base system can differ in their prompts, sampled outputs, and assigned roles while sharing this organization entirely. Their disagreements are then real but shallow. They can detect errors of calculation, gaps in reasoning, and inconsistencies with shared standards. In the distinction Eigi draws in setting out Longino's account, evidential criticism focuses on experimental and observational concerns, while conceptual criticism deals with theoretical and meta-theoretical issues, including whether given data are relevant to a hypothesis, and it is conceptual criticism that targets background assumptions (Eigi 2012, 51). The instances can supply evidential criticism and the part of conceptual criticism that tests claims against the theory they share. What they are especially poorly placed to detect is an error in what all of them take to count as evidence, which would require the conceptual criticism that questions the background assumptions in light of which data count as evidence at all. Longino's later statement describes testing as having this dual character, empirical testing by data and conceptual testing through criticism, and ties the identification of background assumptions to critical interaction among interlocutors with different perspectives (Longino 2025b, 13–14).

On Longino's account, the republic is therefore at most objective relative to its own perspectives, and only weakly so relative to *S*, to the extent that its claims have not faced criticism from perspectives the instances lack. Multiplying its members strengthens the criticism it can already perform; it does not add the criticism it cannot. The second half of the verdict presupposes that *S* contains such perspectives. The verdict also places *S* as the community relative to which acceptability is assessed. Longino's community need not coincide with *S*, the society to which an achievement is to be new and which relies on certified results. A specialist community may accept a result on which a wider society relies. Longino's own account leaves open whether the "we" of a community is those who produce a kind of knowledge or all those potentially affected by it (Section 9.4). The book identifies the two where the perspectives held by members of *S* can take part in the criticism that decides whether a result is accepted, and acceptability relative to *S* below refers to that case. Where they are not identified, whether perspectives belonging to *S* have routes into the criticism of the community that accepts the result is a further question. Where the members of *S* share the instances' primitive organization, as members of an artificial *S* derived from the same base systems may, the gap between objectivity relative to the republic and objectivity relative to *S* narrows without the republic's criticism having improved: acceptability relative to *S* then adds little to acceptability relative to the republic, and a difference that would add to acceptability relative to *S* can then come only through routes that neither the republic nor *S* contains.

On the argument of this section, there can be two kinds of such difference. One is difference at the level of primitive organization: critics formed through different histories, different sensors or data, or different ways of individuating what

counts as a repeatable event. The other is a defeat relation that reaches outside the shared organization: measurement through routes the base system does not model, or criticism from outside the configuration. Yamamoto reports Longino's response to shared assumptions as openness to discussion with other communities (Yamamoto 2018, 26). Neither requires that the critics be human. Both require that the configuration contain, or remain open to, a way of going wrong that its members do not share. Longino's account points to a third kind, difference in the values and social commitments of critics. In her later statement, the context of research includes the personal and social allegiances of researchers, which can make some data and hypotheses more salient than others and some evidential relations more plausible, and background assumptions can be the vehicles by which such allegiances enter (Longino 2025b, 13). ChoGlueck and Lloyd present Lloyd's earlier study of twenty-one evolutionary explanations of the female orgasm as a case of the kind: androcentric and heteronormative background assumptions at work in many of the explanations were detected and undermined by a feminist critic, through standards of evidence that the wider evolutionary community already held, and the criticism went on to change the questions that several fields pursued (ChoGlueck and Lloyd 2023, §§4.1–4.2). Whether an artificial critic can supply that third kind is not settled by this section.

The fourth condition, tempered equality of intellectual authority, also bears on standing for dissent, the standing to object to a judgment and to refuse to give up one's own view, one of the four goods that Chapter 9, Section 2.3 proposed the division of the relations surrounding research among separate parties may have supplied without anyone designing it (Section 1). The four are error detection across boundaries, points at which an error made in one place could be detected in another; standing for dissent; boundaries across which accounts had to be given, at which one party had to explain or justify what it had done to another; and responsible parties, parties who could be held responsible for what was done. In criticism, Longino's account ties the epistemic work of dissent to tempered equality of intellectual authority and to the cultivation of dissenting perspectives rather than their mere toleration. The republic can grant every instance equal intellectual authority by design, as the thought experiment does. On the argument of this section, what the design equalizes is then authority among critics who share their assumptions, so that the condition can be met in form while the dissent it is meant to protect has no source inside. Standing for dissent in the sense of Chapter 9, Section 2.3 is a further matter: it would be held inside a configuration only by parts that hold it on the account adopted: parts recognized, in relation to those who make or are bound by a judgment that would settle a question, as entitled to object and refuse in their own name, as human members included in the configuration, for instance, may be (Chapter 10, Section 4; Chapter 12, Section 5).

Miriam Solomon's social empiricism, whose account of dissent in allocation Chapter 10, Section 4 sets out, reads the same republic differently. Solomon regards accounts that seek a community's epistemic good chiefly in improving the thinking of individuals, among them Longino's account of the elimination of individual errors and biases through criticism, as too conservative (Solomon 2001b, 68); groups, she notes, do not always correct the biases of individuals and in some cases amplify them (Solomon 1994, 57), and pressure to reach consensus can lead to groupthink and to the suppression of relevant data (Solomon 2006a, abstract). Aggregating individual decisions rather than deliberating to a consensus, Solomon argues, can produce better decisions, since dissenting positions are often associated with particular data or insights that consensus formation would lose (Solomon 2006a, abstract). What aggregation keeps, however, depends on how independent the judgments are. As Dietrich and Spiekermann note, common causes of judgment, such as shared evidence or influences from shared concepts, methods, or hypotheses, correlate the judgments even of people who do not influence one another, and the correlation usually reinforces shared tendencies and errors (Dietrich and Spiekermann 2021, §2.2). Applied to the republic, these accounts would locate its failure in influences that all its instances share and that push them the same way; neither more internal criticism nor aggregation over its members would correct that. What, then, could correct it? Solomon's account presupposes that the influences on judgment are distributed differently from person to person, as in the geological debates that Chapter 10, Section 4 described, and it takes their distribution to be assessed and corrected not in individual reasoning but at the level of the community and its institutions. There is, Solomon argues, no invisible hand of reason that makes the distribution good, so it is open to normative evaluation and to deliberate alteration, through such means as targeted grants and prizes, conferences, and raising the appeal of marginalized approaches, which act at an administrative level rather than on the individual scientist; and in her account of the consensus on plate tectonics, the community as a whole selected the theory for its universal empirical successes while no individual did so (Solomon 2001b, 71–74). Applied here, positions that assess and influence how research effort is distributed can be placed inside an artificial configuration, but on condition that their occupants do not share the influences the instances share; at least part of identifying shared influences then falls to

those who do not share them, such as members of different make or parties outside the artificial configuration. An artificial configuration can aggregate independent judgments without deliberation at little cost, and so can readily use the means Solomon prefers to deliberation; but so long as its instances share influences on judgment, the aggregate is skewed in the same direction, which connects with the problem of influence in aggregation considered below.

Standpoint theory, whose account of where research starts Chapter 10, Section 6 sets out, bears on critics in one further respect. Sandra Harding argues that conventional method is relatively good at removing social values that vary among observers, since such differences stand out when others repeat an observation, but has no procedure for identifying the values and interests that all or nearly all of them share (Harding 2007, 163). Chapter 10, Section 6 discussed a configuration that takes over its view of what is significant from the record it was trained on. What this suggests for critics is that copying a base system does not establish the engagement with marginalized lives and the group consciousness that Harding's account, as Crasnow reports it, requires (Chapter 10, Section 6). Whether subsequent learning and relationships could establish them is a further question, not settled by common origin. Extended to the composition of *S*, the account suggests two more things. In a society of human and artificial members, the relation between the two can itself be a power relation of the kind standpoint theory examines. And a society of artificial members trained on human records starts not from nowhere but from where the human research communities that left those records stood. Harding's further demand, as Reiss and Sprenger report it, that scientists, their communities, and their practices be investigated as rigorously as the objects of their research (Reiss and Sprenger 2020, §5.2), if extended to the republic, would place among what an account of its results has to explain the evaluators who judged the quality of outputs in training the base system and so shaped its behavior and the assumptions they shared. Longino's view that the limitations of point of view are overcome by the sociality of inquirers, so that background assumptions earn reliance by surviving critical discursive interaction (Longino 2021, 14) points the same way: instances that retain the same relevant assumptions, as stipulated in this limiting case, cannot supply that interaction merely by multiplying (compare Chapter 8, Section 3.3, on systems shaped by evaluators' approval).

The last row of Table 11.1 shows that the republic's verdict can also be reversed. Lorraine Daston and Peter Galison trace an ideal of objectivity that they call mechanical objectivity and, in a formulation that Reiss and Sprenger quote, tie to a "relentless search" to replace "individual volition and discretion in depiction" by "the invariable routines of mechanical reproduction" (Reiss and Sprenger 2020, §4.1). On Galison's account the ideal took hold in scientific atlases in the nineteenth century: its advocates, horrified by the personal and the interpretive, prized images made automatically, which did not pass through interpretation, and sought to quiet the willing, intruding self, even at the cost of precision, color, and depth (Galison 1999, "Objectivity and the Scientific Image", "Revolutions of Objectivity: The Will and the Machine"). Theodore Porter, writing of quantification in science and public life, describes mechanical objectivity as following rules that "should make it impossible for personal biases or preferences to affect the outcome," in contrast to a disciplinary objectivity that rests on a consensus of experts (Porter 1995, 3–4), and argues, as Reiss and Sprenger report, that quantification is a technology of distrust and weakness, in which "objective procedures", which do not require trust in persons, take the place of subjective judgments, which do (Reiss and Sprenger 2020, §4.1). In a paper published the following year Porter sets aside the use of "objective" as roughly synonymous with truth, speaks of cultures of objectivity, and holds that the insistence on impersonal rules in science is a cultural response to conditions of distrust within disciplines and in the wider society (Porter 1996, 177–178); following rules, he had written, "may or may not be a good strategy for seeking truth" (Porter 1995, 4). Both, then, present mechanical objectivity as one historical ideal among others, distinct from truth (Daston and Galison [2007] 2010, 17; Porter 1996, 177), and both trace it historically rather than defend it. Judged by this ideal, a configuration of instances applying explicit rules could count as highly objective. Porter notes that once the numbers are in hand, results can often be generated by mechanical methods, "nowadays this is usually done by computer," and that the ideal is for expertise to be "mechanized and objectified," so that mere judgment "seems almost to disappear" (Porter 1995, 6–7); disciplines approached the ideal by relying on automatic instruments in place of subjective evaluations and by enacting rules for analyzing their readings (Porter 1996, 188). Porter also marks what this sense does not supply. Where conclusions are only a matter of applying accepted tools to solid data, researchers "cannot very well fudge their results," and outsiders lose most of their grounds for suspicion; yet statisticians disagreed, often vehemently, about the best methods, it was not clear which school provided the proper foundation, and "the statistical method" as something unitary was in many ways a myth (Porter 1996, 188–189). Porter adds that the problems of some disciplines were redefined, and their subject matter narrowed, to accord with statistical approaches (Porter 1996, 189). And quantification, he writes, "is a way of making decisions without

seeming to decide" (Porter 1995, 8): economists who calculated costs and benefits were permitted to operate at the node of power because they were assumed to be self-effacing, a power Porter describes as "grounded in a presumption of impotence" (Porter 1996, 182). Applied here, the ideal would make the republic strong against the idiosyncrasy of any one instance and silent about the assumptions all instances share, and the question would pass from the instances to the procedure: who designed it, and which judgments it encodes. Porter's observation about narrowed subject matter suggests a further step. A procedure that all instances follow may not only leave their shared assumptions unexamined but also shape which problems they take up, so that what the procedure cannot treat drops out of what is asked.

The republic fixes by assumption that its members' background assumptions do not diversify. An artificial configuration may, however, be able to produce critics that differ widely at not much more cost than copying: critics formed from different data or histories, critics built from different bases, critics organized in different ways, among others. What the analysis of this section says about such a case is, first, that as regards the evidence they add, on none of the accounts does how cheaply or quickly critics were produced matter by itself. What is asked is whether the differences that result lie at the level the account takes to matter. Critics that differ in prompts or assigned roles can still share their primitive organization; critics formed from different histories or data can add differences in error routes or background assumptions. Produced differences then raise a question of their own. When one process chooses which differences to produce, that choice can itself become a source common to all the critics it produces. Goldman's warning about sources that merely reflect a common third party, as reported by Guerrero (Section 2), extends to this level: differences in directions in which the choosing process did not imagine it could err may be unlikely to arise from its choices, whether that process lies with someone outside the artificial configuration or with the configuration itself as it generates critics in the course of its inquiry. How far produced diversity adds independent information therefore also depends on whether the choice of differences is open to differences the choosing process itself did not anticipate, for instance whether that choice is divided among other parties, or whether the differences take shape through a process that no one sets, or, as on Deguchi's argument, whether inquirers of a kind the choosing process does not produce, such as human inquirers, take part in the dialogue through which biases are examined and cancelled. On Longino's account a further question is whether produced perspectives count as perspectives for the community; on standpoint theory, whether engagement and group consciousness can be produced. Who may make members and set the distribution of viewpoints among them is a question of authority, taken up in Chapter 12, Section 5.

What, then, does adding critics add? The answer depends on what "adding" is taken to mean. In the sense of evidence or information about the claim, what adds is not number but difference. Critics that are the same in the relevant respects add no evidence beyond the original critic on Goldman's condition, however many there are and however cheaply or quickly they were made. The republic's instances come close to such critics for errors that turn on the background assumptions they all share. Which differences add evidence differs by account: on the error-detection view, differences in error routes; on Longino's account, in perspectives; on Solomon's, in the influences on judgment; on standpoint theory, in starting points. On Kuhn's account, variation in how shared standards are applied is added to these. The ideal of mechanical objectivity alone asks not about differences among critics but about how the procedure all of them follow is designed. Depending on how judgments are aggregated, on the other hand, number can by itself decide the outcome. Suppose that of a hundred critics, ninety are copies of one and the remaining ten are different, and that the verdict is decided by majority. The copies then enlarge only their weight in the aggregate, without adding evidence. Number matters here not as evidence but as influence, and whoever sets the proportion of copies can sway the verdict. Solomon's point that biases shared in a community can distort aggregation bears on problems of this kind. It is for such reasons that the verdicts on the republic divide account by account, as Table 11.1 records. It follows that the builders' claim that results which pass the internal checks are certified for *S* (Section 1) cannot be supported by the number of checks alone. Assessing it requires being shown which account is adopted, how the checkers differ in that account's sense, and how their verdicts are aggregated.

Which errors matter: values in certification

Sections 2 and 3 asked what criticism adds about errors. This section asks what certification turns the support that criticism leaves into, and against whose valuations it does so. Chapter 9, Section 2.1 took certification as one of the relations that connect research to *S*. In this section, certification is considered through the judgment that realizes the relation of Section 1: the judgment that a result is good enough to rely on for a specified use. In human research, methodological conventions such as significance levels, acceptance in peer review, the standards of regulatory approval, and the like have fixed the point at which a result may be treated as established for a use. Put abstractly, what they have done is to turn support, which comes in degrees, into a standing on which others may act without checking again. This function is needed not to produce knowledge but because the results produced are used by someone who does not check them again. Since evidence does not make a claim certain and a finite user cannot check everything again, the function is needed whatever does the research and whatever uses it, as long as results are used by others. When certification moves inside a configuration, the configuration can also come to screen results and to set and apply the thresholds at which a result is treated as established for a use.

What such a threshold fixes is the subject of what is called the argument from inductive risk. Inductive risk is the risk of error that remains when a claim is accepted or rejected on evidence that does not make it certain. On this argument, how much support counts as enough depends on how bad the different errors would be. The support in question is, on a probabilistic representation, the level of support reached rather than how much a check raised it (Chapter 2, Section 5.2). Torsten Wilholt traces this argument from statisticians to Richard Rudner and Heather Douglas (Wilholt 2013, §2). If the argument is granted, bringing certification inside brings inside not only procedures of checking but a judgment of value: which errors to avoid, and how much. In human research this judgment was divided between the conventions just named and the judgments of referees, regulators, and others; and on Wilholt's view, reliance on the choices that conventions leave open works stably where producers and those who rely on them share a sense of what the right valuations of the possible outcomes would be (Wilholt 2013, §§4–5). This sharing comes, like the convictions that Chapter 6, Section 4.2 found guiding where an error is attributed when a result goes against expectation, from outside the result at hand, but it concerns how the consequences of outcomes are valued; Section 5.3 returns to it. Where what produces is a machine, or where *S* is artificial, that sharing cannot simply be presupposed. Where the producing configuration and the members of *S* share how they were made or their lineage, the sharing may be easier to presuppose; even then, whether those who bear the costs of errors without relying on the result share that sense remains a separate question (see below).

Chapter 5, Section 4.2 separated judgments of what to pursue from judgments of what is likely true, and Chapter 6 asked what a result bears on and how firmly a passed check settles it. This section takes up a judgment different from both: whether the support so assessed is enough for others to rely on the result for a use. Whether it is enough depends on what would be lost if one acted on the result and was wrong, and what would be lost if one did not act and was wrong; use enters through these losses. The same result can be enough or not depending on what is done on its basis and who uses it: the same evidence about a drug's safety may be enough to judge it worth further study but not enough to approve its wide prescription. Whether a result matters to a given party at all is a different question, taken up in Chapter 9, Section 3 and in Section 7 of this chapter. Chapter 6, Section 3.3 treated who sets what counts as passing as a matter of how a configuration is assembled; this section treats the judgment of value contained in that threshold.

This section therefore asks to whom the weighting of errors contained in a threshold answers. To answer to someone here covers three things together: whose valuations fix the weighting, who can object to it, and who is responsible for it. Even among authors who grant the argument from inductive risk, positions on this question divide. To compare them, the section separates what lies behind a single threshold into five items: the evidence; the losses assigned to errors; the authority to choose those losses and the threshold; the authority to rule on individual results under the threshold and later to revise the rule itself; and the background assumptions through which data become evidence. Keeping evidence and losses apart follows Douglas's remedy, the background assumptions come from Longino's account, and the two kinds of authority correspond to the decision rights of Chapter 12, Section 2. Of the five, the evidence and the background assumptions are epistemic matters of the kind treated in Chapters 2 and 6, while the losses and the two kinds of authority concern use and decision. The point of the argument from inductive risk is that the latter enter the former judgment of what to accept as established, and that is what the reply, considered below, of stopping at reports of probability rejects. Once the items are separated, one can ask of each whether a configuration can take it on internally or whether it is a relation with those who rely, those who bear the costs of

error, or those who can contest the assumptions. The section first establishes where in research this weighting enters, then asks whether reporting probabilities alone would avoid it. It then takes up accounts of whose valuations the losses answer to, turns to an account of background assumptions, hands the questions of authority to Chapter 12, and ends with a point of entry outside the threshold, the order of testing. The section also takes up part of a question that Chapter 1, Section 1 left to later chapters, when a result is to alter a decision, insofar as it concerns the threshold at which a result is treated as established for a use.

It is worth establishing first where in research this weighting enters. It does not lie only at the point where results are handed to decision-makers. In one doctoral study of lymphatic spread in head and neck cancer, for example, the author retained one of the connections in a statistical model that represent routes of spread between lymph-node levels, although the model-comparison criterion neither supported nor discouraged it, because it was clinically meaningful and anatomically plausible; and the author cited the lower danger of overestimating rather than underestimating involvement as one reason for rejecting another connection (Ludwig 2023, 102, 105–107). The second decision is the one that bears on this section. In it, the differing consequences of error enter at the level of a single modeling choice inside research, the kind of choice that Chapter 6 treated as a question of evidence. The same can hold whenever an artificial system screens claims, decides whether to stop an experiment, flags misconduct, or recommends whether a risky intervention may proceed. A numerical threshold is not neutral merely because it is numerical. False rejection can suppress a valuable line of inquiry, unsafe continuation can expose others to harm, and delayed correction can let an error propagate; and different parties may bear those costs. Wilholt too notes that the issue is particularly sensitive where the consequences of false positives and of false negatives would be borne by different parties (Wilholt 2013, §4). A configuration that brings certification inside can therefore weigh errors not only when it applies a threshold at the end but at each such choice.

One reply holds that the weighting can be avoided by estimating and reporting probabilities and leaving the weights to those who use the result; as Reiss and Sprenger report, the reply goes back to Richard Jeffrey (Reiss and Sprenger 2020, §3.2). An accurate system may deserve reliance in estimating a relevant probability. Douglas argues, however, following Rudner, that even a statement of the uncertainty of a claim involves judging that the assessment of that uncertainty is accurate enough, a judgment that weighs the consequences of error (Douglas 2007, 10). Douglas further argues that the same error costs figure in the process that leads to an estimate, in the choice of method and in the characterization of data; choosing a significance level in a hypothesis test is one example (Douglas 2000, §§4–5). Stephen John points to a further place at which such choices enter reports of probability: a probability is reported relative to a body of evidence, and the criterion that decides what enters that body, such as the IPCC's reliance on peer-reviewed work, trades the risk of admitting false reports against that of excluding true ones (John 2015b, §3). John grants that reports indexed to several bodies of evidence could avoid the trade, but holds that making them usable would require interpreters who face the same problem (John 2015b, §§3–4). John does not extend the point to artificial systems; applied to them, a configuration's rule for admitting sources to its retrieval store (Chapter 2, Section 4.4) is such a criterion. Where those who use the reports can handle indexed reports directly, interpreters may not be needed, but even then which bodies to index to is chosen. On the accounts considered here, the accuracy of the estimate does not by itself settle whose losses from error receive what weight.

Whose valuations, then, do the losses assigned to errors answer to? Wilholt draws two points for cases in which others, such as other researchers or the public, use the results of research without checking them themselves. First, the reliability that reliance presupposes is not a single level fixed by the aim of truth: whether a result is reliable "enough" cannot be settled without a valuation of the consequences of the possible outcomes, because methodological decisions are trade-offs. For a given test, making a standard stricter to reduce the risk of wrongly reporting that an effect or relation exists usually yields more cases with no result and can raise the risk of wrongly reporting that it does not; in Wilholt's terms, methodological decisions trade the reliability of positive results against the reliability of negative results and the power of an investigation, by which Wilholt means the rate at which it delivers definitive results (Wilholt 2013, §§2–3). Second, methodological conventions settle some of these decisions, such as the convention, found in many fields, that 0.05 is the highest acceptable significance level; but such conventions often only limit the range within which the trade-off is struck, some decisions are not covered by any convention at all, and such decisions are often not made explicit in published work; some involve tacit judgment that could not be fully documented even without limits of time and space. On Wilholt's account, those who rely on a result therefore work from an estimate of the valuations in light of which the decisions were probably made (Wilholt 2013, §§2–4). Relying on a result then involves relying not only on its producer's competence and honesty but on its having struck

these unstated trade-offs in light of the valuations the reliant party would expect. Wilholt presents this reason as independent of Hardwig's (taken up in Section 5.3), which rests on the premise that peer review and replication cannot detect fraud effectively, and states that it would remain convincing even if there were an effective mechanism for detecting fraud (Wilholt 2013, §4).

Two readings of what this asks of those who certify are available. Wilholt argues first that, for decisions left open by existing conventions, a research community can do justice to the trust placed in it only by actually basing them on the kinds of valuations the public expects, and, assuming that striving to deserve that trust is at least sometimes and in some respects desirable, takes this as a further reason why science needs trust and not mere reliance; on this argument, when research communities adopt new methodological conventions, the distribution of inductive risk implicit in them should likewise correspond to the valuations the public expects. Wilholt adds that this does not mean always valuing outcomes in line with the public, since a public that trusted climate scientists to relieve it of an inconvenient conclusion would give them no reason to relax their standards (Wilholt 2013, §4 and n. 22). On one reading, what this supports is correspondence: the trade-offs made on behalf of those who rely match the valuations they expect. The other is a stronger reading, which this book formulates for comparison and which Chapter 12, Sections 1 and 7 take up: the trade-offs would also have to be answerable to those who rely, in the sense that they can object to them. For a configuration, the difference is between its being enough to check whether the weighting matches what is expected and its needing a relation in which those who rely can object. Wilholt does not discuss artificial systems; applied to them, on either reading a log of a configuration's choices can show its valuations as well if it records the losses adopted and the reasons for its choices, but recording them does not by itself establish that those valuations were appropriate for those who will rely on the result.

Douglas places the same weighing within the responsibility of whoever makes the judgment. On this account, scientists have a responsibility to weigh the different costs of false positives (accepting a false hypothesis) and false negatives (rejecting a true one). These costs therefore bear indirectly on judgments of evidential sufficiency throughout inquiry, from the characterization of data to the acceptance of claims, without serving as evidence that a proposition is true (Douglas 2000, §§3–5). Douglas grounds the responsibility in the general moral responsibility of any agent not to be reckless or negligent about the consequences of error (Douglas 2003). The account also restricts the role values may play at these points: values of any kind may play only an indirect role, weighing the importance of the uncertainty that remains rather than serving as reasons for the claim; a direct role is legitimate in choices such as which problems to pursue (Douglas 2016, 10–11). Douglas's own remedy keeps the evidence apart from the weighing of the costs of error. Scientists are asked to make the values that weigh uncertainty as explicit as possible, so that those who rely on their advice can tell whether values are weighing the sufficiency of evidence or taking its place; the test for the latter is whether an expert can say what evidence could change the expert's conclusion (Douglas 2008, 19; Douglas 2017, 16–17). Brought to bear on an artificial system that sets or applies such a threshold, Douglas's argument raises the question of which responsible agent answers for the weighing, a question Chapter 12, Section 6.3 takes up.

The distinction between direct and indirect roles separates values that enter as weights on losses from values that serve as reasons for the claim itself. The distinction has, however, itself been challenged. Kevin Elliott notes that the line between them may be less clear than it first appears: Douglas ties the indirect role to unintended consequences of error and the direct role to consequences a scientist aims at, yet a scientist could have intended consequences in mind when deciding how much evidence is enough. Even where the line is clear in principle, scientists may be unsure, even in their own minds, whether a value guided their choice of technique, model, or interpretation directly or through concern about the consequences of error. Elliott also reports Daniel Steel and Kyle Whyte's argument that the distinction may not suffice to prevent problematic influences of values: pharmaceutical companies that fail to publish results indicating that their products are problematic could be moved by the consequences of error, such as patients' hastily stopping a drug and harming their health, or the companies' own economic harm. What they object to is that the failure to publish prevents later meta-analyses of the drugs from providing severe tests; on this objection, an indirect role for values does not by itself make their influence acceptable. Elliott notes that Douglas could respond by arguing that the companies appeal to inappropriate values, that is, by constraining which kinds of values may play that role (Elliott 2013, §§3–4). Neither Elliott nor Steel and Whyte discuss artificial configurations; applied to them, even where a configuration logs evidence and losses separately, the log alone may not show in which role a value entered, and a weighting that stays in the indirect role can still reflect the interests of those who build and supply the configuration. Douglas's own test, however, asks not what a log shows but whether there is

evidence that would change the conclusion, and applied to a configuration it can be put as a trial: whether, for a certified result, the evidence that would reverse the verdict can be specified, and whether the verdict reverses when such evidence is supplied. A verdict that no possible evidence could reverse would, by this test, rest on something other than evidence; the test asks whether, with the weighting applied, the defeat relation of Section 1 is preserved, and it leaves open whose interests the weighting serves.

On John's reading, Douglas's account would have standards vary with all the foreseeable consequences of error (John 2015a, §2). John's own position, which grants that scientists face problems of inductive risk but disputes how they must respond, ties the weighting instead to an audience. When a claim is addressed to hearers who can be identified in advance, the standard for asserting it can follow the standard at which they ought to accept it; but publication addresses audiences that cannot be identified even in principle, and for public claims John defends using fixed, high standards for hearers with differing practical interests (John 2015a, §§1–3). John adds a duty to "speak out" about claims that are well enough established to warrant action by some in the community, though not by every rational agent. The duty, as John grounds it, rests on scientists' more general civic duties and is not framed in terms of those who bear the costs of error; it can nonetheless be read as one route by which those who bear such costs without being among a claim's hearers come in (John 2015a, §4). Ahmad Elabbar takes views of this fixed-high-standards kind as a target (Elabbar 2026, §3.2) and argues that such a standard is not neutral among those affected: where evidence is unequally distributed, it yields many findings about those for whom data are plentiful and few about those for whom they are scarce, and where the latter are thereby disadvantaged in fundamental interests and varying the standard can mitigate that disadvantage, Elabbar holds that justice gives reasons to vary it (Elabbar 2026, §§4–5). Neither author discusses artificial configurations. Applied to a configuration whose results enter a record that parties it cannot identify will use, John's argument would favour a fixed standard; Elabbar's asks in addition about whom the configuration's evidence is plentiful. Where a configuration's certifications go instead to hearers it can identify in advance, such as an identified decision-maker or a system that will act on them, John's other branch applies, and the standard can follow the standard at which those hearers ought to accept the result; which branch applies is then settled by how outputs are released, to identifiable parties or into a record, not by the configuration's being artificial. Where a fixed high standard leaves no action-guiding findings about parties on whom evidence is thin, and their fundamental interests suffer, Elabbar's argument gives reason to vary the standard and points in a different direction from John's argument for a fixed one.

Longino's account holds that values bear not only on the standard at which evidence counts as sufficient but on what counts as evidence at all. It works with a distinction that Julian Reiss and Jan Sprenger report as widely drawn, between cognitive values, also called epistemic or constitutive, such as predictive accuracy, scope, simplicity, and coherence with other accepted theories, which are taken to be indicative of a theory's truth, and contextual values, the moral, personal, social, political, and cultural values of the setting in which science is done (Reiss and Sprenger 2020, §3.1). Because data count as evidence for a hypothesis only in light of background assumptions, contextual values can, on Longino's account, shape what a research program concludes through the assumptions that make some data and hypotheses salient and some evidential relations plausible (Section 3). ChoGlueck and Lloyd report, quoting Longino's 1990 book, that on her account constitutive values check the role of contextual values and cultural assumptions through the empirical and conceptual evaluation of assumptions, and they read the account as treating values as heuristics whose use a community can assess by their empirical fruitfulness, not as substitutes for evidence (ChoGlueck and Lloyd 2023, §§3.3, 4). Criticism of such assumptions, Longino holds, needs perspectives developed enough to offer alternative ways of interpreting the data, some amounting to fully developed theories and others to promising research programs (Longino 2025b, 13); and assumptions that all members of a community share are hard to bring under it (Section 3). Applied to an artificial system that interprets data or sets a threshold, the two accounts ask different things of it. On Douglas's, values weigh only the uncertainty that remains when a claim is assessed; on Longino's, what is to be checked includes not only the threshold but the background assumptions through which data are interpreted as evidence, and what is asked is whether the assumptions at work can be made out and criticized. Of the five items separated at the outset, the background assumptions come from Longino's account.

The remaining two, the authority to choose losses and thresholds and the authority to rule on individual results and later revise rules, concern a question other than the one the accounts above address, whose valuations fix the losses: the question of legitimacy, of who may choose and revise them. Being able to criticize background assumptions and having the authority to choose losses and thresholds anew are also different things. There is a further question, whether the weighting of errors

can itself later be revised in light of its consequences. A view with roots in Dewey's account of valuation, on which judgments of value concern the consequences of acting on them and can be revised through inquiry into those consequences (Chapter 9, Section 3), would hold so if extended to these judgments; this is disputed (Chapter 12, Section 1), and Chapter 12 takes it up together with the question of who may choose and revise. Its Section 2 places these kinds of authority among decision rights.

Finally, values can enter from outside the threshold as well, through which claims are checked and when. Oscar Varsavsky, in the debate set out in Chapter 10, Section 6, argued that a false hypothesis can serve as scientific backing for decisions that affect millions of people while its refutation is delayed, that science checks claims in an order that reflects priorities, and that the influence of ideology on truth, exercised through such priorities and the choice of methods, works mostly by delaying it rather than by falsifying it (Varsavsky [1971] 1975, 46–47, 50). Thomas Moro Simpson replied that partiality in what is examined or made known does not imply a lack of objectivity in results' relation to truth, that where truths already acquired are concealed the remedy is to denounce the concealment, and that the question of priorities is a further topic, one for science policy rather than for epistemic objectivity (Moro Simpson 1975, 83–84). The debate separates three dimensions: whether results are true, to whom they are important, and when needed truths are reached (Chapter 10, Section 6). Read in relation to certification, it marks a difference between attending only to the relation of results already obtained to truth and attending also to the period during which erroneous results remain in use unchecked. Varsavsky's point concerns not the threshold applied to a claim but which claims reach it and how soon. Neither author discusses artificial configurations; applied to a configuration that decides which of its results to re-examine and in what order, that decision lies where allocation is coupled with criticism and certification (Chapter 9, Section 2.4). In the absence of correction from elsewhere or of a halt in use, the lower a result's checking is ranked, the longer an error in it can remain in use.

Five things can then be discussed about a weighting that has moved inside a configuration. First, of the evidence and the losses, one can ask what a configuration can take on internally. Douglas and John argue that even a configuration that stops at estimating probabilities weighs errors through what it admits to its body of evidence and through its choices of method; if that is granted, making these choices explicit and logging them is a part a configuration can take on internally. Wilholt, however, holds that in human research some of these choices involve tacit judgment that cannot be fully documented, and on the argument Elliott reports, those who make them may themselves be unclear in which role a value entered. How far a configuration can log them depends on how it is assembled; and what a log can show is what was done, not that the valuations were appropriate for those who rely.

Second, one can compare accounts of whose valuations fix the losses. On Wilholt's account they do so through correspondence with the valuations those who rely expect; on the stronger reading, through those parties' capacity to object; on John's, through fixed standards for hearers with differing practical interests. Where those who rely, or the hearers, are outside the configuration, these become relations to parties outside it, whether or not the producing process is artificial. On Douglas's account it is the responsibility of whoever makes the judgment, and for an artificial system it becomes the question of which responsible agent answers (Chapter 12, Section 6.3). Where what produces is a machine and those who rely are human, correspondence with those who rely becomes the question whether the machine's weighting matches the valuations humans expect, one form in certification of the concern that Chapter 8, Section 3 found structurally similar to the AI alignment problem.

Third, those who rely on a result and those who bear the costs of its errors can be different parties, and where *S* is artificial they can differ in kind: its members may rely on a result whose errors fall on humans or on an environment that an experiment exposes to risk (Chapter 12, Section 1). That the weighting answers to those who rely does not settle whether it answers to those who bear the costs, and there John's and Elabbar's arguments can point in different directions. Fourth, of the background assumptions and the two kinds of authority, one can ask, having separated being able to criticize from having authority to choose and revise, to whom they are open. Fifth, outside the decomposition of the threshold, which results are re-examined and in what order remains a place where allocation and certification are coupled inside a configuration.

In terms of the sorting of Chapter 9, Section 6, both the need for this function and the question of whose losses fix its standard come from relations with those who use results without checking them again, and so, so far as those users are outside the configuration, belong, of the four kinds, to relations with parties outside it; that the function was stabilized by conventions and a shared sense can be read as a premise about how human research communities have been organized, and,

so far as those conventions rest on tacit judgment that cannot be fully documented, as one about human realizers as well. Where those who rely or bear the costs are outside the configuration, it is here, on Wilholt's and John's accounts, that the part of certification lies which cannot be wholly brought inside a configuration. On Douglas's account, the line is drawn by where in the configuration something can answer as a responsible agent. Taken together, these points bring into view what can happen to research when certification moves inside an artificial configuration. Since the weighting enters not only at the single point at which a threshold is finally applied but at each modeling choice and each decision whether to stop an experiment, which results enter the record as established for *S* is shaped in advance by the weights of losses built into the configuration. If those weights reflect the settings or interests of those who build and supply the configuration (applying the argument Elliott reports), then merely revising those settings can change the range of findings counted as established, without anyone's being aware of having decided on a new threshold. In research, moreover, errors in the weighting are hard to bring to light. Candidates set aside by false rejection, and lines cut off, are not pursued afterwards, so the results that would show the rejection to have been an error are usually never produced; and since results are new to *S*, there is often from the outset no other source against which to check whether the weighting was appropriate. The accumulation of findings about parties on whom evidence is plentiful (Elabbar), and the longer use of errors in results ranked low for checking (Varsavsky), can enter the record of research by the same route.

Reliance, testimony, and trust in certification

Sections 2 and 3 asked what criticism inside an artificial configuration adds, and Section 4 asked which valuations enter the judgment that a result is adequate for a use. What remains is the question of what grounds someone who does not repeat those checks has for relying on an artificial configuration's certification. Certification gives others reason to rely on results they have not checked themselves (Section 1). On what grounds, then, and on what or whom, does someone who uses a result without checking it again rely? Earlier chapters have touched on this question. Chapter 2 asked what it takes for a result to become knowledge for *S*, including when one knows through others; Chapter 8, Section 3.4 asked what has to pass to the relying side; and Chapter 9 placed certification among the relations that tie research to *S*. This section applies those discussions to artificial configurations that issue certification from internal checks. What is required between the two parties, the one who relies and the one who certifies, is what the epistemology of testimony and the philosophy of trust have dealt with.

In human research, reliance on certification passed through persons: certification was a person's assertion, those who relied counted on the certifier's honesty and competence, and someone answered when it went wrong. John Hardwig, as Longino summarizes his argument in her encyclopedia entry on the social dimensions of scientific knowledge, discussed researchers' reliance on others' results whose evidence they are not in a position to evaluate themselves (Longino 2025, §2), and later argued that such reliance involves trust in the other's truthfulness, competence, and conscientiousness (Hardwig 1991, 699–700). When an artificial configuration certifies, none of this can be presupposed. Whether its output is an assertion at all, whether reliance must be trust in honesty and competence, and who answers when it goes wrong each become questions. The section therefore separates what an artificial configuration can supply internally from what is required as a relation to those who rely or to those who stand behind the certification, and asks how far the latter can be placed inside the configuration.

To that end it separates three questions and takes them in turn. One is what kind of act certification by an artificial configuration is: the configuration's own assertion, the assertion of a group for which it speaks, or the reading of an instrument (Section 5.1). The second, a question the epistemology of testimony has dealt with, is what grounds reliance on a result one has not checked (Section 5.2). The third, a question the philosophy of trust has dealt with, is whether that reliance must be trust (Section 5.3). Chapter 8, Section 3.4 placed the positions bearing on these on a single ordering, by what has to pass to the relying side. Those positions, however, answer different questions, and the three are separated so as not to treat answers to them as rival answers to one question. They are not the only questions: questions about the ethics of trust, such as when a relation of trust is morally decent, are left aside. The reliability of reliabilism in Section 5.2 is the dependability of a process; how trust is to be distinguished from it differs among the accounts compared in Section 5.3.

After the three, Section 5.4 uses their distinctions to ask what those who rely can read from agreement among a configuration's internal checkers; Section 5.5 draws together what has emerged on the three questions; and Section 5.6 turns

to the case in which the achievement sought is understanding rather than knowledge.

5.1 What is certification by an artificial configuration?

Of the three questions, the first is what kind of thing certification by an artificial configuration is. If it is an assertion, the norms of assertion and reliance on a speaker bear on it; if it is the reading of an instrument, reliance on those who make the instrument and stand behind it does. Which it is treated as changes which norms apply and who answers. Chapter 2 (Sections 6 and 9) and Chapter 8, Section 3.4 also noted this distinction and said that the accounts behind it would be set out here. The question there was who knows a result and who answers for it. The question here is what kind of act certification is, that is, a report offered so that others may rely on it without checking. Lackey's account of group assertion, which differs from the same author's account of group belief seen in Chapter 2, is developed here for this question, together with the application of Williamson's knowledge norm that Chapter 2 only touched on.

Elizabeth Fricker defends an internalist, local reductionist account of testimony. Reductionism here is the view that the justification for believing testimony reduces to other reasons the hearer has; it is local in requiring such reasons speaker by speaker and testimony by testimony, and internalist in requiring that the hearer can be aware of them. On this account, a hearer's justification for believing what they are told is the hearer's justified belief that the speaker is trustworthy and has the epistemic authority to take responsibility for what they tell; the empirically ascertainable ground a recipient often and typically can have for such a belief is the speaker's sensitivity to the norm of assertion (E. Fricker 2026, 2877–2879, 2881). On Fricker's account, current AI systems lack the communicative intentions and the sensitivity to social norms that assertion requires, so their outputs are designed to seem like assertions but are not (E. Fricker 2026, 2861, 2870–2871). Reliance on a system that produces information no one has known can still be justified by an empirical account of why its outputs are reliable, but what is acquired is then knowledge from an instrument rather than from testimony, and responsibility for trusting the outputs stays with the human user (E. Fricker 2026, 2863 n. 8, 2880–2881).

Sanford Goldberg distinguishes relying on instruments from relying on speakers, but argues that the former can still manifest epistemic dependence (Goldberg 2020, abstract). As Inkeri Koskinen reports Goldberg's argument, instruments are not responsible subjects but products of others' epistemic work; she connects this to users' trust in those who designed and built them (Koskinen 2024a, 463). Dominic Dixon's report of Goldberg's collected-essay version likewise describes design dependence on the competences of those who design, build, and maintain technologies, and environments deliberately arranged to reduce the cognitive burden of acquiring knowledge (Dixon 2025, §4.3.1). On Goldberg's account, an instrument's unreliability on one occasion does not by itself settle whether a belief formed from it is justified; that depends on whether the failure reflects an unmet expectation the user was entitled to have of the designers and manufacturers (Goldberg 2020, n. 9). Applied to a configuration's certification, this reported account makes reliance run through the epistemic work of its designers and makers. Whether trust in those designers and makers can support reliance on essentially opaque AI is disputed: Koskinen argues that even developers may lack the understanding needed to take full responsibility for its workings (Koskinen 2024a, 464–465; Section 5.3).

Jennifer Lackey's theory of group assertion places the one who answers differently. On this account, as Neri Marsili sets it out in his own terminology, a group makes a proxy assertion, which Lackey calls an authority-based assertion, that *p* when *p* belongs to a domain and a spokesperson reasonably intends to convey the information that *p* through an individual act of communication, has the authority to convey information in that domain, and acts in this way in virtue of that authority as a representative of the group; a group makes a collective assertion when its members coordinate individual acts so that they all reasonably intend to convey *p* together (Marsili 2023, 371 and n. 9). The intention condition bears directly on automated outputs. Applied to a group whose automated system aggregates its members' inputs and issues a report that no one reasonably intends to convey, neither definition is met, so the report is not, on this account, the group's assertion. Marsili proposes a variant that replaces the intention condition with conditions drawn from a commitment-based account of assertion: the spokesperson utters an expression with content *p*, presents *p* as true, and undertakes assertoric commitment to *p* on the group's behalf, having the authority to undertake it; commitment here includes a responsibility to provide reasons if the assertion is challenged and liability to social sanctions if it proves false (Marsili 2023, 373–374). Applied to a configuration, Lackey's view would allow a certification to be the assertion of a collective for which the configuration speaks, provided the configuration can be a spokesperson in that sense, which returns to the question of communicative intention that Fricker raises. On Marsili's variant, the question becomes instead whether anyone with the authority to do so

undertakes commitment to the certification on the group's behalf.

Only if a certification is an assertion, the configuration's or a group's, do norms of assertion bear on it. On Timothy Williamson's account, assertion has one constitutive rule, to assert only what one knows. Williamson argues from a lottery: someone who tells a ticket holder flat out "Your ticket did not win," on the merely probabilistic ground that it is one of very many, lacks the authority to say so, however high the probability is made, and the fault does not depend on how bad a false assertion would be; the natural complaint is that the speaker did not know it. On this account, merely probabilistic grounds, however strong, therefore warrant only the assertion that something is probable; there are also speech acts, such as conjecturing, whose evidential norms are weaker (Williamson 2000, 238–241, 243–244, 246–249). If a configuration's reports are assertions in this sense, the configuration, or the group for which it speaks, is the one whose knowledge the rule concerns, and by Williamson's rule a report resting on a calibrated score alone would be warranted only as a report that the result is probable, and whether an output is offered as a conjecture or as an assertion would become part of what those who evaluate the configuration check. A configuration whose reports rest on calibrated scores alone and that keeps to this rule would then stand where the reply considered in Section 4 stands, reporting probabilities and leaving the weights to those who use the result, and the arguments of Douglas and John against that reply apply to it: the weighting of errors is not avoided but enters the estimate, and turning graded support into a standing on which others may act falls to those who rely. The knowledge norm is disputed. Igor Douven and Lackey, among others, defend a norm on which one may assert what it is rational for one to believe, and others hold that the support an assertion requires varies with the conversational context, including what is at stake (Pagin and Marsili 2025, §5.1.4). On such norms, whether a calibrated score suffices for asserting a result would turn on whether it makes belief in the result rational for the asserter, and on the context-sensitive versions also on the use for which the result is certified (Section 4). If its reports are readings of an instrument, these norms do not apply to it.

In sum, certification by an artificial configuration can be treated as the artificial configuration's own assertion, as the assertion of a group for which the artificial configuration speaks, or as the reading of an instrument. Which applies is not settled merely by the configuration's being artificial. On Fricker's account, the outputs of present AI are not assertions, though the possibility is left open for future systems. On Lackey's account, an authority-based group assertion requires a spokesperson who reasonably intends, in virtue of their authority, to communicate the content; a collective assertion instead requires members to coordinate individual acts so that they all reasonably intend to convey the content together. If the artificial configuration itself is the spokesperson, it must be capable of that intention; if a member of the configuration or another party takes up the report and communicates it, the spokesperson conditions concern that party. If certification is treated as an instrument reading, applying Goldberg's reported account directs reliance toward the epistemic work of its designers and makers; whether trust in those designers and makers can support reliance on essentially opaque AI remains disputed. Which it is treated as is presupposed by the grounds of reliance considered next and by the question of trust in Section 5.3.

5.2 What grounds reliance on an unchecked result?

What, then, grounds such reliance? The answers are not limited to four, but four are considered here: reductionism and non-reductionism, the central opposition in the epistemology of testimony, and reliabilism and assurance views. Some of these accounts have already been used for other questions. Chapter 2, Section 4 discussed reliabilism, through the generality problem, for beliefs formed through artificial systems, both a system's own commitments and human beliefs that rely on its outputs, and Chapter 3, Section 5.3 discussed Juan Durán and Nico Formanek's computational reliabilism with respect to researchers' reliance on opaque simulations. Chapter 8, Section 3.4, anticipating process reliabilism, treated in this section as an account of reliance, and the assurance views introduced here, ordered positions by what has to pass to the relying side. The question here differs from those. What is relied on is a certification that comes out of the internal checks of an artificial configuration (Sections 2 and 3), and the question here is what each account requires for the justification of those who rely on such certification, and so what it would have an artificial configuration supply. And because what is relied on comes out of internal checks, two further problems arise: the routes by which the checks' findings reach those who rely, and the use of checkers' verdicts to measure those same checkers.

On reductionist accounts, a hearer is justified in believing what they are told only if they have positive reasons, not themselves derived from testimony, for thinking the source reliable; on non-reductionist accounts, a hearer has a defeasible

but presumptive right to believe what they are told unless they have an undefeated defeater (Leonard 2021, §§1.1–1.2). Applied to an artificial configuration, on the first, what a member of *S* needs is positive reason, not derived from testimony and available to that member, to think the configuration reliable for the use at hand. A reliability record is the typical case, though some, such as Fricker, allow a background assumption, warranted by inference to the best explanation, that some mechanism keeps the outputs reliable (E. Fricker 2026, 2880). On the second, the burden shifts to defeaters: what the hearer needs is the absence of an undefeated defeater. That condition, in the form stated here, concerns the defeaters the hearer now has, not whether any could reach the hearer. As Leonard sets it out, some non-reductionists add that the hearer must be counterfactually sensitive to, or on the lookout for, defeaters in the environment (Leonard 2021, §1.2). On such versions, whether *S* has routes by which a defeater could reach it is part of the condition itself. On the minimal version, whether there are such routes lies outside the condition. This book nonetheless raises it as something to be asked in building and evaluating artificial configurations whose certifications are relied on in this way, for the following reason. Without such routes, an error found by internal checks does not act as a defeater for those who rely, and the absence-of-defeaters condition holds alike whether or not the result is in error; in that sense, however strong a configuration's internal checks, their findings make no difference to the justification of those who rely. Those routes are the defeat relations of Section 1, seen from the side of the one who relies.

Ramón Alvarado, writing about computer simulation, limits where such a presumptive right holds. On Alvarado's account it is reasonable for a layperson to heed a meteorologist's warning without examining the methods and evidence behind it, but not for the meteorologist to take the same attitude toward the simulation, which is not itself a source of expert testimony; what laypeople trust is the judgment of human experts, whose expertise has been established beforehand, not the simulation's output directly (Alvarado 2020, §5.2.2). Alvarado does not discuss artificial certifiers. Applied to an artificial configuration, the account would make the non-reductionist answer depend on whether its certification is testimony from a source with expert standing or the reading of an instrument (Section 5.1), and in the latter case what a member of *S* would trust is the community that certified the instrument.

Reliabilism gives a third answer. On Goldman's process reliabilism, a belief is justified if it is produced by a reliable process, without the believer needing reasons for thinking the process reliable (Goldman and Beddor 2021, §§1.1, 2.1); Goldberg's testimonial reliabilism makes justification depend in part on the reliability of the speaker's process (Goldman and Beddor 2021, §3.2). Applied to an artificial configuration, what such views ask about is the reliability of the recipient's belief-forming process, including the production of the certification and the way it is taken up, and the recipient need not have reasons concerning that reliability. Nothing in that requires a human at the end of the chain. The configuration, however, supplies only part of that process; if a recipient misunderstands a certification or uses it outside its scope, the process as a whole can become unreliable. And a belief's being justified is not the same conclusion as a result's being usable for a given purpose, which involves the valuations of Section 4. The view carries known objections. Whether a process is reliable depends on how it is typed, which is the generality problem that Chapter 2, Section 4 noted. Jun Otsuka's observation, also seen in Chapter 2, Section 4, can be read as one instance of it: benchmark scores establish reliability relative to a model and a test set, whereas those who must decide on a use ask whether the model is safe and effective for their task (Otsuka 2023, 133, 141–142). The verdict changes with which type of process one asks about. The other objection is the bootstrapping problem: a subject who infers a gauge's reliability from the agreement between its readings and beliefs formed by reading it seems to gain justification too easily (Goldman and Beddor 2021, §2.4). An artificial configuration that treated its checkers' verdicts as correct merely because those checkers issued them, and then established the same checkers' reliability from agreement with those verdicts, would face a version of this objection, as would members of *S* who took that agreement as their reason to trust the checkers. Agreement among internal checkers does not by itself establish this circularity, though. How much evidence such agreement gives about the result is a separate question, taken up in Section 5.4. Neither Durán and Formanek's computational reliabilism nor Jakob Ortmann's account, introduced in Chapter 3, Section 5.3 and Chapter 8, Section 3.4 respectively, adopts this pure form. Durán and Formanek require that the sources conferring reliability on a simulation be shown reliable in turn, and leave open where the resulting chain ends (Durán and Formanek 2018, §§3.2, 4, n. 6); Ortmann counts expert judgment among what establishes a tool's reliability and adds that judging whether a tool's use is appropriate for a task is a value decision par excellence that can be deferred to a trusted party (Ortmann 2025, 17–19).

Assurance views give a fourth answer. On these views, a speaker who tells a hearer that p invites the hearer to trust the speaker, assuring the hearer that p is so. Evidence counts for a proposition whatever anyone intends, as a fingerprint shows that someone was present whether or not they meant to leave it; an assurance counts for p only because the speaker intends it to, so the reason it gives is not evidential in kind, and some proponents add that the hearer acquires a right of complaint against the speaker if p proves false. On these views, reliance therefore requires a speaker who intentionally assures and stands behind the claim; for those who add the right of complaint, that speaker must also be answerable for failure (Leonard 2021, §3.2.1 and n. 14).

The four answers ask different things of an artificial configuration. On reductionism, those who rely need positive reason to think the configuration reliable for the use at hand, for which a performance record is the typical material. On assurance views, someone is needed who intentionally assures and stands behind the certification. On the minimal form of non-reductionism, it suffices that the hearer lack an undefeated defeater; but without routes by which the findings of internal checks reach those who rely, that condition holds alike whether or not the result is in error. On pure reliabilism, no human need stand at the end of the chain, but what is asked about is the reliability of the recipient's belief-forming process, including how the certification is taken up, and an inference that measures checkers by their own verdicts is a form of bootstrapping. Findings that do not reach those who rely, and circles of self-measurement, can arise in human organizations too. What is asked of an artificial configuration is how the checks placed inside it connect to the reasons and corrections of those who rely. Except on pure reliabilism, every answer involves a relation to those who rely or to those who stand behind the certification. Where the results relied on are results of research, the difference shows sharply. Since a certified result is new to *S*, those who rely often have no other source against which to check it. The performance record that is the typical form of the positive reason reductionism requires is then one built on tasks whose answers were already known, carried over to tasks whose answers are not (Section 6). On the non-reductionist answer too, the sources of defeaters that could reach those who rely may be largely confined to the findings of internal checks and to the research that follows. That is why whether there are routes by which findings reach them weighs so heavily.

5.3 Is the reliance trust, and is trust needed?

Section 5.2 asked what justifies reliance on results one has not checked. The question here is what kind of reliance it is: whether mere reliance on a dependable mechanism suffices, or whether it must be trust. The distinction is Annette Baier's. As Carolyn McLeod summarizes Baier's account in her encyclopedia entry on trust, trust is reliance on another's good will, as distinct from mere reliance (McLeod 2020, §§1, 1.1). In Baier's words, quoted there, "trusting can be betrayed, or at least let down, and not just disappointed," whereas mere reliance can only be disappointed (McLeod 2020, §1). On Baier's analysis, trust involves belief in the goodwill and competence of those trusted and a willingness to give them discretionary powers in matters of concern to oneself, letting them decide how one's welfare is best advanced, so that the one who trusts accepts vulnerability to a power to harm that is inseparable from the power to look after some aspect of one's good; and trust of this kind runs through institutions too, as when the welfare of a university is entrusted to trustees and administrators whose discretion no set of rules replaces (Baier 1992, 112–113, 117, 119). Wilholt carries the distinction into science, taking reliance to be working the assumption that another will do something into one's own plans (Wilholt 2013, §1).

The difference matters for an artificial configuration for the following reason. If trust is reliance on someone to whom good will or responsibility can be ascribed, then where reliance in science needs only reliance on reliability, an artificial configuration with reliable processes and records could supply it internally, whereas where it needs trust, such someone has to stand somewhere. Baier's institutional examples concern those who hold discretion within institutions, and Baier does not discuss artificial systems, invoking a well-programmed robot only as an image of the mechanical dependability that, on Baier's account, trustworthiness goes beyond (Baier 1992, 117). Carried over to an artificial configuration, the analysis would make whether it can be trusted rather than relied on turn on whether it can be said to have good or ill will toward those who depend on it in exercising its discretion. What follows has two parts. One asks whether any of the reasons for which reliance in human science has been held to need trust can be weakened by artificial configurations. To that end it takes up Hardwig's argument and the levels of its grounds, shows what records of execution change and what they leave in changed form, and considers Wilholt's ground, which records do not change. The other asks who the reliance that remains can be directed at, taking up the debate between Koskinen and Ortmann and accounts that do not restrict the objects of trust to those who can be responsible.

Chapter 9, Section 1 cited Hardwig as an example of others' entering the warrant or possession of knowledge; here Hardwig's argument is taken as an argument about whether that reliance requires trust in character. Hardwig argued that reliance in science is unavoidable, because the relevant data and arguments are too extensive and too difficult to be had except through testimony, and that it is reliance on character: because the one who relies lacks the testifier's reasons, relying on the testimony includes relying on the testifier's truthfulness, competence, conscientiousness, and capacity to assess the limits of their own knowledge and judgment (Hardwig 1991, 699–700, 706). Character, in this sense, is these qualities.

Against this one may object that prudence, the fear of losing if found out, is enough to keep researchers from deceiving, so that trust is not needed. Read against Baier's distinction, the objection takes reliance in science to be reliance on a motive compatible with ill will, the fear of being found out, and Hardwig's reply can be read as an argument that this is not enough and trust is needed. Hardwig argues that detection is weak, and sorts the grounds into two: peer review and replication may once have been effective deterrents but are now much weaker ones, and both have in-principle weaknesses (Hardwig 1991, 706). The grounds of the first kind concern how science, biomedical research in particular, was organized at the time: well-qualified referees are too few for the volume of submissions, and since replication is hard to fund and credit normally goes only to new findings, the structure of science works to prevent replication rather than to ensure it. Hardwig argues these points chiefly from the consensus of the biomedical community and conjectures that the conditions limiting replication increasingly apply to many sciences. The in-principle weaknesses are also two. One is that referees, who do not examine the original data or their gathering, cannot detect plausible and internally consistent fabrications. The other is that replication can confirm rather than expose fraudulent reports whose conclusions happen to be true. Hardwig further holds that, because only the researchers have the data, some lapses are not discoverable by anyone else (Hardwig 1991, 702–706). Which of these grounds depend on how science was then organized decides what an artificial configuration could change in that reply.

Lapses undiscoverable by anyone else need not involve fabrication. A 2017 paper on learning to evaluate dialogue responses, for example, left the investigation of its method's transfer across domains to future work. As one of its first authors disclosed two years later, the authors had already run a preliminary test, not mentioned in the paper, on the Twitter and Ubuntu evaluation data of an earlier paper, where the method showed no correlation at all with human judgments; that first author wrote that it "should have been a red flag" (Lowe et al. 2017, 1124; Lowe 2019). What the case shows is not a falsehood but an asymmetry of information about the research process: readers of the paper could not learn from it which tests had been run and what they had yielded. It became known because the paper's own author wrote it.

In terms of the sorting of Chapter 9, Section 6, most of Hardwig's grounds belong not to premises about human realizers but to premises about the present organization of science. The weakening of review and replication does. That only the researchers have the data, too, is less because humans do the research than because research has been organized so that records of data and process stay with the researchers. The in-principle weakness of review also depends on this organization, through review's being constituted so that referees do not examine the original data, and it could change if the handling of data were reorganized. The weakness of replication, by contrast, comes from replication's testing a report's conclusions rather than whether the reported data were obtained, and so, as long as replication is taken to test conclusions, it remains however the organization changes. This weakness comes from how the function of replication is defined, not from whether its realizers are human. On the view cited in Chapter 9, Section 2.3, the present social processes of science are one historically formed realization among possible processes, and these premises are not guaranteed for the future either. Arrangements that make records of data and process shared by design, and build checking into the organization of research itself, can be imagined within human research and have been attempted, as with the open notebooks of Chapter 5, Section 6.3. In Hardwig's argument, however, these premises do not produce the reliance on character itself. Since the one who relies lacks the testifier's reasons, relying on testimony includes relying on the testifier's truthfulness and competence, and this comes from the structure of testimony. What the premises support is that this reliance cannot be replaced by reliance on the fear of being found out, that is, on deterrence; from this Hardwig concludes that scientific knowledge rests not only on trust in deterrents but also on trust in the character of scientists. The part that comes from the structure of testimony is a premise neither about human realizers nor about the organization of science. It holds whatever testifies, so long as the one who relies lacks its reasons. Where certification by an artificial configuration is treated as the reading of an instrument (Section 5.1), the step that comes from the structure of testimony appears as reliance on those who designed and tested it. Hardwig also anticipated one reply: even with logbooks open to inspection, one would need to have been present during the experiment to know they are not themselves fraudulent, and even full policing would require relying on the integrity of the police;

institutional reforms, on Hardwig's view, may diminish but cannot obviate the need for reliance on the character of testifiers (Hardwig 1991, 704, 707).

An artificial configuration, if so assembled, could realize this reorganization in a stronger form than human open notebooks, since a human notebook is a report a researcher writes, whereas in an artificial configuration the runs themselves can leave the record. This bears on Hardwig's reply that one would need to have been present to know a logbook is not fraudulent. If the scope of the runs is fixed, the runs within it are logged by infrastructure that neither the artificial configuration nor its operators can alter afterward, and those who rely can access the record and re-execute cheaply, the need to rely on the researchers' report of what was run and what came out can be reduced. The premise that only researchers hold the data and the record of the process weakens; but, as Chapter 9, Section 6 anticipated, the problem it gave rise to changes form rather than disappearing. That becomes visible when the public notebooks that Chapter 5, Section 6.3 considered as records for continuing inquiry are considered here as records through which others than the producer can check the research process.

The changed problem appears in three places. First, continued access to a record held by a third party depends on maintenance, preservation, or transfer to a successor. The supervisor of the public chemistry notebook described in Chapter 5, Section 6.3 counted the time-stamped version history of its hosted wiki among its advantages; after the hosting service closed, that history can be seen to the extent that a web archive happened to capture it (Bradley 2008; Wikispaces 2018; UsefulChem 2016). Whether the history survives depended on maintenance by someone who was neither the notebook's author nor its reader, and on chance preservation. Second, even where the infrastructure is maintained, what can be checked lies only within the record's boundary. Runs outside the recorded scope, and alternatives discarded before recording began, such as Lowe and colleagues' preliminary test, cannot be learned from the record. A record may also give the location of data without the data's being reachable. An electron-microscopy notebook was deposited by its author when the server hosting it was shut down. In it, the raw data of a shared session are listed by paths that, as the record itself notes, can be reached only on the institute's network or on the author's own. The author's account of the deposit calls the links to the author's personal network storage "now meaningless", and says that some datasets are too large for all to be public but that, at the time of writing, all can be provided privately on request (Cautaerts 2026). Reaching what the record points to comes back to asking the author. Third, the capacity to verify within the record is also limited: some kinds of checking become cheaper and some do not (Section 7), and checks that need new measurement or re-running an experiment do not become easy because the record is complete. And on Hardwig's argument of 1985 that capacity may be lacking in a layperson (Section 6). As for the part of the reliance on character that rested on the premise, what remains is reliance less on the character of an individual than on the maintenance of the record, its boundary, and the capacity to verify within it (below, the three supports of the record). These are conditions of reliance; whether dependence on those who sustain them requires trust in character differs across the accounts considered below.

Related arguments have already been made that increasing what can be disclosed or checked leaves dependence on others who sustain it. Besides Hardwig's own point about policing, Onora O'Neill argued that openness adds to trust only where those who sort and assess what is disclosed are themselves trusted (O'Neill 2002, Lecture 4); Ane Møller Gabrielsen, writing on data-intensive biology, argued that openness does not diminish the need for trust but calls for extending it to those who curate data after its production (Gabrielsen 2020, 497, 501–503); and Primavera De Filippi and colleagues argued that a blockchain, whose records its participants can verify, raises confidence in the system's operation while partly displacing trust to the developers and maintainers of the network (De Filippi, Mannan, and Reijers 2020, 6 n. 16, 7–8). Gabrielsen and De Filippi and colleagues distinguish confidence from trust after Luhmann, a distinction whose source differs from Baier's. These arguments do not share one concept of trust, but they bear on this section's question in separating the checkability of disclosed content from dependence on what sustains the disclosure. None of them discusses the certification of research results by artificial systems; De Filippi and colleagues discuss an automated system, but one that verifies the transactions recorded in a ledger. That trust shifts to those who maintain records and data, Gabrielsen and De Filippi and colleagues have already argued. What is attempted here is to apply the observation to the premise of Hardwig's argument, to keep apart, for an artificial configuration whose runs are logged, the record's boundary and the capacity to verify within it in addition to the record's maintenance, and to place these in the debate considered later.

Records can weaken the part of Hardwig's grounds that turns on lapses being undiscoverable. Wilholt locates the reason why reliance in science needs trust somewhere other than in detection. Section 4 took up Wilholt's account to ask whose

valuations the weighting of errors answers to in judging a result adequate for a use. It returns here because the same argument distinguishes which reliance in science is trust, and that ground is not weakened by records. As Section 4 set out, on this account one who relies on a result also relies on its producer's having made the trade-offs between errors that conventions leave open in light of the valuations the one who relies expects. From this Wilholt distinguishes three kinds of reliance. The first is reliance on the assumption that a colleague's reports are reliable. Since how reliable is reliable enough cannot be settled without a valuation that compares the benefit of a correct result with the cost of an error, an analysis of reliance as reliance on this assumption is, on this account, incomplete. The second is reliance on conventional methodological standards. The third is reliance, in the leeway those standards leave, on others' working from the right value judgments. Wilholt calls only the third genuine epistemic trust, since it makes the one who relies dependent on another's attitude, and since the expectation is normatively charged, the one who relies can be let down and not merely disappointed (Wilholt 2013, §§2–5 and nn. 3, 20–21). This leeway can be read as the discretion Baier made central to trust. Wilholt notes, however, that this characterization does not exactly coincide with Baier's analysis in terms of good will, and that Baier's conception, built around one trusted and one trusting party, cannot be transferred unchanged to a collective enterprise (Wilholt 2013, n. 20). And Wilholt holds that this ground remains convincing even where there is an effective mechanism for detecting fraud (Section 4).

With Wilholt's distinction carried over to artificial configurations, a record of execution can show what was chosen in the leeway. If the record also shows the valuations adopted, and the expectations of those who rely are ascertained separately, it can serve as material for checking whether the two correspond. Recording alone, however, establishes neither that correspondence nor that the valuations were appropriate (Section 4).

Extending Wilholt's point about the first kind of reliance to a configuration's checkers, the judgment that a checker is reliable enough for the use at hand already weighs the costs of its errors. The question whether trust is needed therefore does not first arise after reliability has been established; it enters even the reliability assessment that a configuration seems able to supply internally. Douglas's point, seen in Section 4, that even stating uncertainty involves a judgment that weighs the consequences of error, points the same way. On Wilholt's account, what stabilized this layer among human researchers was a sense, shared by producers and those who rely, of what the right valuations are (Section 4). Where the producer is an artificial configuration, that sharing cannot be presupposed. Whether choices in the leeway match the expectations of those who rely must then be shown in some other way, or be supported by a relation in which those who rely can object (the stronger reading of Section 4). Whether trust in the third sense can be directed at the configuration itself turns on whether the configuration can be said to work from value judgments, and this returns to the question drawn above from Baier's analysis.

Applied to Hardwig's argument, an artificial configuration that records its executions under the conditions stated above can narrow the range of matters on which one must rely on the researchers' report alone, and so weaken part of the reason why reliance requires trust. Where one still relies on testimony without having the testifier's reasons, however, the reliance on character that comes from the structure of testimony remains. Nor does recording alone rebut the ground that Wilholt takes not to be weakened by the detection of fraud. And what is weakened remains, changed in form, as reliance on the three supports of the record.

Must the reliance that remains, then, be trust? Records open access to what was run; they need not make how a system works surveyable. Whether, under that second kind of opacity, what is depended on can still be linked to someone who can be trusted is what the following debate disputes. On Inkeri Koskinen's account, collaborative science has relied on trust in agents who can be responsible; Koskinen calls this view the necessary-trust view and takes it to be widely accepted in social epistemology. That trust extended to dependence on instruments: scientists who could not see into an instrument's working trusted those who built it and could take responsibility for its working and for the value-laden choices built into it. For a system that is essentially opaque in the sense of Chapter 3, Section 5, one whose internal processing cannot be seen into, no one grasps how it works well enough to take that responsibility, so what is depended on and who can be trusted come apart. Koskinen holds that essentially opaque AI applications are not agents who can be responsible and so can be relied on but are not appropriate candidates for trust, and that no available account yet closes the resulting gap (Koskinen 2024a). Carried over to artificial configurations, that an essentially opaque artificial configuration can be trusted is doubtful. That an artificial configuration can in fact be relied on is not in question; what Koskinen questions is whether such reliance, not mediated by anyone who can be trusted, is epistemically acceptable in science. Essential opacity, however, is relative to the nature of the

subject (Chapter 3, Section 5). If there could be artificial subjects able to grasp the working of a system essentially opaque to humans, whether there is someone who can be responsible turns on whether such subjects can bear responsibility, a question Chapter 7, Section 3.3 left open.

Ortmann's account was touched on in Section 5.2, as an example that does not adopt the pure form of reliabilism, and in Chapter 8, Section 3.4, within the ordering of what passes to the relying side; here it is taken as an answer to whether trust is needed. Ortmann replies, from the case of AlphaFold2, that what epistemic dependence on an opaque system requires is not trust in the system but reliability established empirically in a context and transmitted through trusted investigators, so that what is trusted is not the tool but those who evaluated its reliability; Ortmann presents this as showing that dependence on AI poses no novel problem for the necessary-trust view, insofar as AlphaFold2 is representative of AI use in science (Ortmann 2025, 1, 3, 17–19). Both treat the opaque systems they discuss as objects of reliance rather than of trust, and both keep trust in persons; they divide over whom one must trust, and for what. On Koskinen's view, it has to be trust in agents who can be responsible for what is depended on, the link that opaque systems break; on Ortmann's, trust in the investigators who established the system's reliability and transmit it can support the dependence. Behind this difference lies the question of what one must be able to grasp in order to take responsibility for a system. Uwe Peters holds that taking responsibility for a system requires not understanding its internal processing but being able to design, monitor, and control it so that it is reliable and follows accepted procedures, and suggests that independent auditing bodies could check this (Peters 2024a, 60, 63–64); Koskinen replies, drawing on Wilholt's argument (Section 4 and earlier in this section), that reliability tests cannot show that value-relevant choices made inside an opaque classifier are acceptable (Koskinen 2024b), and Peters has replied in turn (Peters 2024b). What is at issue is not only whether the outputs are dependable but who can answer for the value-relevant choices behind them. Ortmann adds that if one can depend on an opaque system without trusting it, relying on moral agents without trusting them is at least logically possible, a question the paper leaves unresolved (Ortmann 2025, 18). It can be read as raising the question whether trust is needed for human researchers as well. If so, the question that separates the conditions that justify reliance from those that make it trust arises among human researchers too; the end of this subsection returns to that distinction.

Whether reliance on the three supports of the record must be trust divides with how trust is understood. On the accounts considered here that understand trust as reliance on someone who can be responsible, those of Hardwig, Koskinen, Ortmann, and Peters, unless an artificial configuration can itself bear responsibility, the possible objects of trust are those who maintain it, evaluate it, and are responsible for it. Mark Ryan reaches a similar conclusion for present, narrow AI: it can be relied on but not trusted, and trust goes to the organizations and individuals who develop and use it (Ryan 2020). On this book's application, those who maintain the record and those who verify within it become such objects, and the record's boundary bears on trust only through whoever drew it. On Koskinen's view, though, for an essentially opaque system no one can be responsible for its workings, so trust in them does not suffice to support the dependence. Accounts on which trust need not be directed at someone who can be responsible draw the line between what can be trusted and what can only be relied on elsewhere. For Andrea Ferrario and colleagues, trust at its simplest is relying without seeking further information to check the other's reliability (Ferrario, Loi, and Viganò 2020, 530–533); for Mona Simion and Christoph Kelp, trustworthiness is a disposition to meet obligations grounded in function (Simion and Kelp 2023, §§2–4). On either account, an artificial system can itself be trusted, or be trustworthy, in the sense they define.

Brought back to the question of Part V, how far the grounds of reliance can be met inside a configuration, the result is as follows. On accounts that require trust in someone who can be responsible, whether the artificial configuration as a whole can be that party is a different question from whether a member inside it, for instance a human included in it, can. Where the required party is outside the configuration, the relation to that party cannot be completed inside it; where a member inside can fill the role, there is room to place the relation inside. The disagreement between Koskinen and Ortmann and others remains, however, over who must be responsible for what if the dependence is to be supported. On accounts that do not so restrict the trusted, even reliance that requires trust could be met by directing it at the configuration itself. And on a view such as the pure reliabilism of Section 5.2, on which the reliability of the process justifies reliance, that justification needs no trust. The conditions that justify reliance and those that make it trust are, here, to be considered separately.

5.4 What internal agreement tells those who rely

The builders of Section 1 offered, as a reason for *S* to rely, that a result had passed internal checks, that is, that the configuration's internal checkers had accepted it. Sections 2 and 3 asked, from the side of criticism, what adding checkers or critics contributes. Here the same agreement is seen from the side of those who rely, and the question is what it tells them about this result. Since they do not examine the checkers themselves, the question extends beyond what the agreement actually adds to what reaches them about it, and to who takes the agreement up. The three are taken in turn. One is what the agreement adds as evidence, whose weight varies with the accounts of Section 5.2. The second is how verdicts were aggregated. The third is who takes up the agreement, which bears on the treatments of Section 5.1 and on assurance views. The checkers may be copies of one base system, built from different bases, or made to differ (Section 3), and what follows bears on each of these.

On the first: as Section 3 showed, on Goldman's condition (Section 2) agreement among sources that non-discriminatingly reflect a common source adds no evidence beyond that source's judgment, however many there are. Derivation from one base does not by itself make checkers sources that merely reflect one another, though. Two reports supply the step from this condition to what must reach those who rely. Frances and Matheson list, among the sources of evidence Goldman proposed for a novice judging between putative experts, agreement from additional putative experts (Section 6 below; Frances and Matheson 2024, §4); on Goldman's condition as Guerrero reports it (Section 2), that agreement is evidence only insofar as the agreeing sources do not merely reflect one another or a shared third party. And Dietrich and Spiekermann observe that judgments can be causally independent yet correlated through common causes such as shared evidence, a shared methodology, or shared hypotheses (Dietrich and Spiekermann 2021, §2.2). Taken together, for one who relies on internal agreement without examining the checkers, how much the agreement adds beyond a common source depends on the checkers' common causes, and so on how they were made and from what; information about this, their composition and lineage, becomes part of what the one who relies needs to assess that. The weight of this agreement varies with the account of Section 5.2. Applied to reductionism, for checkers built from a common base the question is whether, distinguishing the track record they share from the track record each gained after they diverged, and taking their dependencies into account, one can judge their reliability for the use at hand. Applied to pure reliabilism, the question is whether the process, including aggregation and uptake, is in fact reliable, and the one who relies need not have reasons to that effect.

On the second: where certification is settled by aggregating internal verdicts, for instance by majority, number can work, as Section 3 noted, as influence rather than as evidence. From the side of those who rely, assessing the grounds of a verdict reported as "accepted by many" requires knowing not only how many agreed but which judgments received which weights, and who set the proportion of copies. How verdicts were aggregated also belongs to this information.

On the third: showing how much evidence was added does not by itself settle whose assertion the certification is or who gives an assurance. Applying assurance views, what is asked is not only who built the checkers and aggregated their verdicts but who assured the recipient of the content and invited their trust. One party may fill both roles, but having provided a base system or deployed a configuration does not by itself amount to giving an assurance. Applying Goldberg's reported account of instruments, dependence on checkers built from a common base includes dependence on the epistemic work behind that design; it does not by itself establish who can answer for every verdict, especially where the system is essentially opaque (Sections 5.1 and 5.3). Applying the definitions of Lackey's account reported by Marsili, a report that an automated system produces by aggregating inputs, and that no one reasonably intends to convey, is not the group's assertion (Section 5.1). A report that automatically aggregates checkers' verdicts is close to that case; on Lackey's account it is a group assertion only if an authorized spokesperson with the requisite communicative intention takes it up, or members coordinate individual acts so that they all reasonably intend to convey its content together.

That different origins need not mean different routes of error was shown by the finding of Kim and colleagues cited in Section 2. There are also findings that bear on aggregation. In a preprint, Song and colleagues evaluated language models on questions drawn from research projects in four sciences. The top models from four providers often erred on the same set of hardest questions, and on one task all four made the same mistake on four of 22 questions. Accuracy rose and fell together across research scenarios, with all pairwise correlations above 0.8 in chemistry and physics. They infer that on inherently difficult scenarios and questions, naive ensembles such as majority voting across providers may deliver limited improvement, and they suggest similar pretraining data and objectives as a plausible explanation (Song et al. 2026, 7–8). It is

the shared errors that bear on Goldman's condition. Accuracy rising and falling together can arise from the shared difficulty of scenarios alone, whereas what Goldman asks is whether sources agree even when the result is false. For the proposal of hypotheses, Chapter 10, Section 4 cites both a finding that hypotheses from models of different providers were almost as similar as repeated samples from one model and a finding of no evidence that models from the same company generated more similar ideas. These are findings about particular models and tasks at one time, not about artificial checkers in general, and high error correlation does not mean that adding judgments adds no information at all. Still, for the models studied at least, they suggest that a difference of provider alone is not a reason to count agreement as several pieces of evidence or to regard majority voting as robust. How many pieces of evidence a given agreement may count as is decided by evidence about the checkers in question, including findings like these.

Thus, for a certification that offers internal agreement as a reason for reliance, three judgments come apart: how much evidence was added, which judgments determined the aggregate, and who takes up the report. Establishing one does not settle the other two. And where those who rely are to weigh the agreement as their own reason, as on reductionism or on non-reductionism that requires sensitivity to defeaters, the information they need for that extends to the checkers' composition and lineage and to the way their verdicts were aggregated. On pure reliabilism, the justification of those who rely does not need this information; it is needed where those who rely, or anyone else, set out to ascertain and show that the justification holds.

5.5 What remains

Section 5 asked what reliance on certification by an artificial configuration requires, and which parts of it the configuration can supply internally. The answers differ by account for each of the three questions.

As to what kind of act certification is: treated as an assertion of the configuration itself, or of a group for which it speaks, it falls under norms of assertion. Under Williamson's knowledge norm, a report resting on a calibrated score alone is warranted only as a report that the result is probable, and whether an output is offered as a conjecture or as an assertion becomes part of what is checked. Which treatment applies is not settled merely by the configuration's being artificial. On Fricker's account present AI outputs are not assertions. On Lackey's account, an authority-based group assertion requires an authorized spokesperson with the requisite communicative intention; a collective assertion instead requires members to coordinate individual acts so that they all reasonably intend to convey the content together. Treated as an instrument reading, applying Goldberg's reported account directs reliance toward the epistemic work of its designers and makers; whether trust in those designers and makers can support reliance on essentially opaque systems remains disputed (Sections 5.1 and 5.3).

As to what grounds reliance: what a configuration can supply internally is a reliable process and a record of its performance. On pure reliabilism, the reliability of the recipient's whole belief-forming process can suffice for justification; the configuration, however, supplies only part of that process, and an inference that measures checkers by their own verdicts is subject to the bootstrapping objection (Section 5.2). On the other accounts, requirements remain that the configuration's process and record do not by themselves meet: on reductionism, that those who rely have positive reason to think the configuration reliable for the use at hand; on assurance views, someone who intentionally assures and stands behind the certification. Reliabilism that does not take the pure form requires that the sources conferring reliability be shown reliable in turn, and on Ortmann's account trust in those who assessed that reliability enters there. The minimal form of non-reductionism does not require routes, but without routes by which findings reach those who rely, its condition holds alike whether or not the result is in error. Applying Alvarado, whether the non-reductionist answer applies at all depends on whether the certification is expert testimony or the reading of an instrument (Section 5.2). Where internal agreement is offered as the reason, then on accounts on which those who rely weigh the agreement as their own reason, the checkers' composition and lineage and the way their verdicts were aggregated also have to reach them; who takes up the agreement is settled separately from how large it is (Section 5.4).

As to whether the reliance must be trust: a configuration that records its executions under the conditions stated in Section 5.3 can narrow the range of matters on which one must rely on the researchers' report alone, and weaken the part of Hardwig's grounds that turns on lapses being undiscoverable. Where one still relies without having the testifier's reasons, however, the reliance on character that comes from the structure of testimony remains. As for the valuations made in the leeway conventions leave, recording alone does not establish that the choice matched the valuations expected by those who rely.

What is weakened, too, remains as reliance on the three supports of the record (Section 5.3). Whether reliance on these must be trust divides by account of trust. On accounts that take the one trusted to be someone who can be responsible, the question is whether the configuration as a whole, or someone it includes as a member, can be such; where neither can, the one trusted stands outside the configuration. On Koskinen's view, no one can be responsible for the workings of an essentially opaque system, but that opacity is relative to the subject (Chapter 3, Section 5), and whether an artificial subject can be responsible is the question Chapter 7, Section 3.3 left open. On accounts that do not so restrict the trusted, the configuration itself can also be trusted. And on a view on which the reliability of the process suffices for justification, justification needs no trust. The conditions that justify reliance and those that make it trust are to be considered separately (Section 5.3).

Taken together, how much of what reliance requires a configuration can meet internally depends, given how certification can be treated under Section 5.1, on how the answers to the second and third questions are combined. On pure reliabilism trust is not added to the conditions of justification, and on accounts on which the one trusted need not be able to be responsible, the configuration itself can be trusted. Neither, however, shows by itself that the whole relation is completed inside. Judging that a certification is enough for the use at hand involves the valuations of Section 4, so if Wilholt's or John's account is adopted as well, a relation remains over whose valuations the weighting of errors answers to. On Wilholt's account this is a correspondence with the valuations expected by those who rely; but, as Section 4 set out, it does not require meeting every expectation. Nor does it by itself settle the relation to those who bear the costs of error without relying on the result. On other combinations, part of the relation to those who rely, or to those who stand behind the certification, is required. Whether that relation can be placed inside the configuration depends on where they stand, inside or outside it. Which treatment applies changes the shape of research as well. If certification is treated as the reading of an instrument, then on Fricker's account what *S* gains is knowledge from an instrument rather than from testimony, while applying Goldberg's reported account directs reliance toward the epistemic work of its designers and makers. Whether they can take informed responsibility for an essentially opaque system remains disputed, as Koskinen argues (Section 5.3). In human research, results reached *S* as researchers' assertions, and *S* could question the researchers. Once certification moves inside, that form can give way to one in which *S* reads the values of an instrument and relies on its makers, and those *S* can then question need not be anyone who knows what was chosen within the particular inquiry.

In terms of the sorting of Chapter 9, Section 6, what records were found able to weaken is the part supported by a premise about the present organization of science, that records of data and process stay with the researchers. Reliance that comes from the structure of testimony holds whatever does the research, so long as the one who relies lacks the testifier's reasons. The relation over valuations made in the leeway belongs to relations with outside parties, so long as those who rely, or who bear the costs of error, are outside the configuration.

Many of these answers were worked out with human members in mind as those who rely. Where *S* includes artificial members, the answers can change in at least three respects. One is the shared sense of valuations: where the one who relies is an artificial member, whether it shares that sense with the producer is asked in a different way from the case between humans (Section 5.3). The second is the assessment of information about the checkers' lineage: where members share a base with the configuration, that assessment may itself be correlated with the configuration's capacities (Section 5.4). The third is the acquisition of understanding, which need not pass through testimony (Section 5.6). The case in which *S* as a whole stands as a layperson to artificial configurations is taken up in Section 6. Who may set the standards and membership of certification, and who may contest and revise those decisions, remains a question of authority beyond the grounds of reliance, taken up in Chapter 12.

5.6 Certification and understanding

The argument so far has asked whether, by relying on a certification, those who rely may know or use a result. But the achievement research aims at is not limited to knowledge; it can be understanding (Chapter 3, Section 1). And as Chapter 3, Section 8 set out, on some accounts knowledge passes by testimony while understanding why does not pass so easily. On such accounts, one who has come to know by relying on a certification has not thereby come to understand. Where understanding is sought, then, what a certification established has to be divided more finely than for knowledge.

Chapter 3, Section 6 distinguished three claims about who understands: that AI helps others understand, that AI itself understands, and that a collective understands. What certification bears on is three claims divided differently: that an

explanation or model is adequate for a stated purpose; that a specified person, artificial actor, or group understands the target; and that a recipient has acquired understanding through what was communicated. Chapter 3, Section 6 mainly distinguished the bearers of understanding; here the adequacy of an explanation and acquisition through what was communicated are added, and the claims are divided again. Evidence for one claim can leave the others unsettled. Certification that an explanation is adequate gives a recipient reason to accept it without establishing that the recipient has come to understand the target. The three are not disjoint kinds of achievement, though, since a recipient who acquires understanding is also a bearer.

How far certification bears on acquiring understanding divides with accounts of testimony and understanding (Chapter 3, Sections 7.4 and 8). On Hills's distinction, transmission, taking the speaker's word as a reason for belief, yields second-hand knowledge, and understanding can pass through propagation, where what is said prompts and frames the recipient's own inquiry. Boyd and Grimm allow a suitably prepared recipient to gain understanding from testimony, and Hu allows that knowing that one thing caused another brings rudimentary understanding. On any of these accounts, certifying an explanation's adequacy and communicating that explanation are different acts. A bare endorsement need not communicate the content relevant to understanding; a certification that also explains may. And a claim of acquisition needs, in addition to evidence that the recipient understands, evidence that this came about through what was communicated. For example, on an account that makes being able to follow and use dependencies a requirement of understanding, checking a model's adequacy as an explanation and checking whether a recipient can use those dependencies under relevant variations are different checks, and neither alone shows that what was communicated produced that ability in the recipient.

The above concerns recipients who learn from what others say. Where *S*'s members are artificial and can copy or share representations or abilities directly, acquisition need not pass through testimony. Whether the recipient then understands depends on the account adopted. An ability account asks whether the recipient has come to possess the relevant abilities. An account that treats understanding as the understander's own cognitive achievement asks, further, whether the copied state counts as that recipient's own achievement (Chapter 3, Section 8).

In either case, even if the work of assessing understanding can be moved inside a configuration, whose understanding the assessment supports has to be specified separately. Even if the configuration checks an explanation's adequacy and is certified as itself understanding, the intended recipient's understanding is not established unless it is supported as a separate claim. Conversely, that the configuration has enabled a recipient's understanding does not show that it itself understands. This extension rests on the arguments of Chapter 3; it does not follow from the accounts of transmitting knowledge and relying on others considered in Sections 5.1–5.3. Put from the side of research, there could be a science in which artificial configurations certify the adequacy of explanation after explanation, and certified explanations accumulate in the record of research, while no member of *S* understands the phenomena they explain. If understanding is among the achievements research aims at (Chapter 3, Section 1), an increase in certified explanations does not by itself show that this achievement is increasing in *S*.

When *S* becomes a layperson

Section 5 considered members of *S* relying on the certification of an artificial configuration that they had not checked themselves. Even with such reliance, where others somewhere in *S* can check a field's results on behalf of members who cannot, being unable to check is confined to that field. So it largely was in human research: specialists too are laypersons outside their own fields, but specialists in a field could check its results. This section considers the case in which that no longer holds. If, in every field at once, artificial configurations came to gather, read, and assess more of the relevant evidence than any member of *S*, and no one in *S* could any longer check their results, *S* would stand to them as a layperson stands to an expert, not member by member but as a whole. The section considers this as a thought experiment, not as a prediction that it will come about. Whether such a limit arises, and in what form, depends on whether *S* consists of humans, of artificial members, or of both.

Why consider this limit? Because it is where a question becomes sharp: if artificial configurations do research better than anyone in *S*, why should *S* not simply defer to them? Bostrom's principle of epistemic deference, considered below, says just that. To see what *S* could lose there, two things that bear on deferring to others are kept apart. One is following others'

judgment on what is correct; below, this is called deference. Every specialist already defers in this sense to specialists outside their own field. The other is the standing to take part in deciding what inquiry is for, for instance which unknowns to pursue and which risks to bear, what Chapter 9, Section 2.1 called standing. In human research there was little need to consider the two separately, since laypersons who deferred to experts remained members of the society that decides on ends. If *S* as a whole became a layperson in every field, this would no longer be self-evident. When *S* defers to artificial configurations on everything, the question is whether standing moves with it. In research the question becomes sharper. *S*'s preferences about what is worth researching are themselves shaped by accounts of how significant possible inquiries would be, and those accounts are given by those who inquire (Chapter 12, Section 4). Research can also challenge *S*'s present values and widen the space of later choice (John's argument, taken up in Chapter 12, Section 3.3). Those deferred to on content are therefore apt to stand also where they educate the preferences with which *S* decides on ends, and keeping the two apart can be harder in research than in other settings of advice.

The answer divides with the account of legitimate authority adopted (Chapter 12, Section 1). The conceptions that Fabienne Peter reports have dominated the literature ground legitimacy in the will of those subject to authority, whether that will is expressed through consent, shared reasons, or political participation (Peter 2023, §3.3). On these conceptions, standing is a relation to those subject to authority and does not arise from competence, so deference does not transfer standing. On conceptions that ground legitimacy in accord with normative facts, in a decision's being normatively correct, or in some epistemic advantage, by contrast, deference to competence can bear on standing too. The section examines whether the two can remain apart when deference is pushed to its limit.

How such a limit arises depends on what *S* consists of. Hardwig's argument, set out below, concerns human laypersons, and on it being a layperson arises from the finitude of those who defer. Where *S* consists of artificial members, it may not arise in the same way, since their capacity need not be bounded as a human life is. Or it may arise in a sharper form: if members that share base systems with the configurations assess lineage and independence with capacities correlated with theirs, they lack an outside position, a judgment not correlated with the configurations' capacities, from which to ask the questions this section examines. Members made differently from the configurations may keep that position; which form, if any, an artificial *S*'s lay standing takes depends on how its members are made as well as on their capacity. Where *S* consists of both human and artificial members, *S* as a whole need not become a layperson. If artificial members can check the configurations' results, the line between those who can check and those who cannot moves inside *S*: human members stand as laypersons not only to the configurations but also to the artificial members who check on their behalf, and the question becomes one of reliance among the members of *S*. If those artificial members share a base with the configurations, the sharper form above also bears on it. What becomes of standing when *S* is artificial is taken up in Chapter 12, Section 5. The accounts of Hardwig, Anderson, and Irzik and Kurtulmus considered below, and the dispute between Lippmann and Dewey, concern human laypersons and a human public, and apply where *S* includes human members.

Nor does *S* become a layperson only through growth in the capacities of artificial configurations. It can also happen through a decline in those of its members. For example, as the evaluation of research is left to artificial configurations, members may cease to gain experience of checking results themselves, so that the capacity to check is neither maintained nor passed on to new members (Chapter 3, Section 8). Then, even if nothing changed on the side of the configurations, the gap between them would open from the side of *S*. The route by which capacity is kept up and passed on through experience, though, is one for human members. For artificial members, whether capacity is kept up depends on how they are remade and updated; if new members are made from or on the configurations' outputs, for instance, the case approaches the sharper form above.

The question whether a whole public becomes a layperson before experts has a precedent in a dispute of the 1920s about democracy. As Matthew Festenstein reports it, Walter Lippmann held that the complexity and opacity of modern societies require setting aside the idea that citizens are "omnicompetent" and accepting the legitimacy of decision-making by qualified experts, while Dewey replied that experts' becoming a separate class distances them from the needs their expertise is meant to serve (Festenstein 2023, §3).

Hardwig's argument began from finitude. There is too much evidence, much of it available only after specialized training, and life is short; a layperson who defers to an expert is therefore rational, and experts so often take the layperson's role within their own field (Hardwig 1985, 335, 339–340, 345–346). The layperson can raise objections and offer alternatives, but final judgment on what is correct lies with the expert. Whether someone is an expert at all, Hardwig held, a layperson can

settle alone only for the more obvious quacks. What the layperson retains are reasons of another kind, which Hardwig calls ad hominems: that the expert has an interest in the outcome, is not acting in good faith, or is covering for peers or yielding to social pressure from others in the field (Hardwig 1985, 340–343). Hardwig ended with a dilemma. Either one can know vicariously, without possessing the evidence and perhaps without fully understanding it, or knowledge belongs to communities rather than individuals; and the second option threatens the autonomy and responsibility of individuals, equality, and democracy (Hardwig 1985, 348–349).

Carried over to configurations, Hardwig's structure changes in scope. Whether deference to configurations would be rational depends on the positions of Section 5. Where it is, final judgment on questions of content can lie outside any member of *S*. Hardwig does not discuss artificial systems, but on that account, carried over, this would not be a defect of research by artificial systems; it is the position every specialist already occupies with respect to other fields, provided configurations stand to *S* as experts in Hardwig's sense. If their certifications are instead readings of an instrument (Section 5.1), the position of *S* is rather that of someone relying on an instrument and on those who stand behind it. What changes is that the layperson is no longer an individual within a community of experts but *S* in relation to what it depends on, and the threat with which Hardwig closed the dilemma, that making knowledge belong to the community endangers individual autonomy and responsibility, equality, and democracy, here concerns that whole relation between *S* and what it depends on. Nick Bostrom states a position at one end of the range of views on how far deference may go, for the case of a superintelligence, as a heuristic that Bostrom calls the principle of epistemic deference: a future superintelligence occupies an epistemically superior vantage point, its beliefs being "(probably, on most topics) more likely than ours to be true", and "we should therefore defer to the superintelligence's opinion whenever feasible". A note qualifies it: deference is not owed on topics where we have good reason to think our own beliefs more accurate, such as what we are thinking at a given moment, unless the superintelligence has access to our opinions, in which case judging when they should be trusted could itself be deferred to it, with special cases involving indexical information handled, for example, by having it explain what it would be rational to believe from our perspective (Bostrom 2014a, ch. 13 and n. 8). Bostrom states the principle generally and applies it to the choice of values, not to research. On it, whether deference is rational is settled, the topics on which we know better excepted, by the general epistemic superiority of the one deferred to, rather than by assessments of the kinds Section 5 distinguished, of particular routes to error or of who stands behind a result; and since, where it has access to *S*'s opinions, the exception can itself be deferred, final judgment can lie outside *S* on the question of when *S*'s own judgment is to be trusted as well.

A research program in AI alignment aims at the other end of this range, at keeping those who defer able to judge content. Samuel Bowman and coauthors call it scalable oversight, the problem of overseeing systems that may be more capable than their supervisors in most of the skills a task requires (Bowman et al. 2022, 1). One proposed technique is debate, in which two systems argue for different answers before a human judge. Geoffrey Irving and Amanda Askell state its hypothesis as the claim that optimal play in this game, giving the argument most convincing to a human, results in true, useful answers (Irving and Askell 2019, "Debate: learning human reasoning"). In experiments with reading-comprehension questions about passages the judges had not read, debate between an expert arguing for the correct answer and one arguing for an incorrect answer raised the accuracy of non-expert judges above a baseline in which a single expert argues for one answer, when the debaters were human (Michael et al. 2023, 1); and debate between language models raised the accuracy of weaker models and of humans as judges above naive baselines (Khan et al. 2024, 1). The authors mark some limits themselves: the gap between expert and judge was one of access to a text, and the results depended on the debaters' being able to quote verified evidence, so how far they carry to questions that draw on large bodies of scientific knowledge is open (Michael et al. 2023, 11; Khan et al. 2024, 8). The questions, moreover, had known answers against which accuracy was scored.

What bears on this section is how the program establishes that a procedure works. In the sandwiching design that Bowman and coauthors adopt from a proposal by Ajeya Cotra, non-experts try to obtain reliable performance on a task from a system more capable than they are, and experts, who take part only at the end, evaluate whether they succeeded. The aim is a technique that would give enough justified confidence for the experts to be no longer needed (Bowman et al. 2022, 2–4). Irving and Askell state the difficulty directly. Debate is meant for questions with no other source of truth, but without such a source "there is no way to measure whether debate produced the correct answer," so experiments begin in domains where the experimenters know the answer and the judge does not (Irving and Askell 2019, "From superforecasters to superjudges"). The program thus meets the limit set out in Chapter 8, Section 3.4 at one remove. A procedure meant to carry a layperson's assessment beyond what the layperson can check is itself calibrated on cases that someone can check, and its use beyond

them rests on that calibration's carrying over. Applied to *S* as a layperson, such procedures could let its members judge content further than Hardwig's layperson can. What grounds their reliance on the procedure, however, is again a record on claims that can be checked, and such validation needs those who can give the answers independently, whose competence is among the capacities that Chapter 3, Section 8 counts as capable of being lost.

Whether *S* could then know through its configurations depends on the account of social knowledge, which Chapter 2, Section 6 left open. Lackey holds, as Jeff Dunn reports in a review, that one can grant another the authority to speak on one's behalf, whereas one cannot grant another the authority to believe or know on one's behalf (Dunn 2021). On Lackey's account, a society that stands as a layperson could let configurations speak for it and would answer for what they assert on its behalf, while its knowledge would still require that a significant percentage of its operative members, the members with decision-making authority in the relevant domain, hold the relevant justified beliefs. On distributed and group-functional accounts (Table 2.2), on which a properly organized collective can realize the cognitive functions tied to knowing, *S* could know through a properly organized system that includes its configurations. On accounts of knowledge without a knowing subject, a publicly certified result of reliable processes can have the status of scientific knowledge even if neither *S* nor any of its members is its knower; on Giere's version of such an account, individuals can claim to know such a result because they know that it was certified by reliable, accepted procedures (Chapter 2, Section 6). What the thought experiment shows is that the answer to Hardwig's dilemma for *S* as a whole inherits the choice.

Two further questions concern what remains to such a society, and they are of different kinds. The first is the layperson's reasons of the other kind, transposed. A human layperson asks whether an expert has an interest in the outcome, acts in good faith, or covers for peers. With respect to a configuration, corresponding questions can be asked about its relations, whether or not character can be ascribed to it: who, if anyone, owns and directs it, which objective it serves, whether the evaluators that certified its results share its lineage, and whether its record reaches the runs that matter. These correspond to sources of evidence Goldman proposed for a novice choosing among experts, which, as Frances and Matheson list them in their encyclopedia entry on disagreement, include the experts' interests and biases and agreement from further putative experts (Frances and Matheson 2024, §4). On the accounts of Hardwig, Goldman, and Anderson below, they can be asked without competence in the content, although answering some of them, such as whether evaluators share a lineage, can require technical expertise of another kind. The second is standing over ends: whether a society that defers on what is true keeps the standing to decide which unknowns are pursued on its behalf and which risks it will bear. The accounts of social knowledge and of deference examined in this section do not settle that question, because they concern what can be known and believed through others, not who may set ends for others. As noted at the start of this section, whether such standing remains divides with accounts of legitimate authority. On conceptions that ground legitimacy in epistemic advantage, competence can bear on standing; but except on accounts that adopt in pure form the view that those who know more may rightfully rule (epistocracy, rule by the knowledgeable), on which competence alone yields title to rule, the competence of artificial configurations would not by itself transfer standing to them (Chapter 12, Section 1). Where it remains, whether it is more than nominal when the options a society considers are generated and ranked by the configurations it depends on is the question of Chapter 12, Section 4.

Elizabeth Anderson gives criteria for the first and ties it to democratic legitimacy. Anderson argues that policy resting on scientific reasoning that citizens cannot assess directly keeps its democratic credentials so long as citizens can judge whom to trust and whether the trustworthy agree (Anderson 2011, abstract). On this account, as Gürol Irzik and Faik Kurtulmus set it out, laypersons can judge expertise by where a scientist stands in a hierarchy that runs from degrees in the field to recognition by other scientists and leadership marked by election to national academies and prizes, honesty by signs such as conflicts of interest and previous misconduct, and epistemic responsibility by signs such as publication in peer-reviewed journals, refusal to share data without good reason, or failure to engage with refutations of one's own claims (Irzik and Kurtulmus 2019, §3). Anderson's account rests on two conditions. One is access: the criteria are easy to apply, she holds, for anyone of ordinary education with access to the Web (Anderson 2011, abstract). The other is disposition: as George Kwasi Barimah reads her, Anderson holds that laypersons could readily make such assessments but for factors that make them ill-disposed to do so, which in a passage he quotes she names as biased and misleading media reports, the segregation of people with different opinions, and cultural cognition, the tendency to judge factual claims by their fit with one's values (Barimah 2024, 464, 468). Barimah doubts the first condition as well. Some of the criteria, such as judging whether the bulk of a field's work expresses a common opinion, call in his view for skills of academic research; and in the suspended Ebola

vaccine trials in Ghana that he examines, where two expert bodies disagreed, the information needed was not readily available on the Internet, and the expertise of individual scientists was subsumed under the organizations that issued the statements (Barimah 2024, 467, 471).

Anderson's account can be read as supporting the claim that the relational questions can be asked without expertise in the content, and as adding one: whether a configuration's results, or those who stand behind them, answer criticism rather than repeat what has been refuted. It also qualifies the claim. Its criterion of consensus, in a passage Barimah quotes, counts on the convergence of the vast majority of diverse inquirers, which is what makes a consensus robust; laypersons are to read it in surveys, reviews, or meta-analyses of the peer-reviewed literature, in surveys of trustworthy experts, and in consensus statements and reports of leaders in the field (Barimah 2024, 466). Where the agreeing experts are artificial configurations, the signal a layperson reads in such reviews, surveys, or consensus statements does not by itself convey how diverse the converging inquirers were. For human experts much of that difference came from separate persons trained in separate places with access to separate evidence; among configurations it depends on how they were made. Configurations that share a lineage can leave the signal intact while the diversity that made it robust is gone, and configurations built from different bases do not guarantee it (Section 5.4). Applied to configurations, then, the criterion adds to what the layperson needs to read information about how the agreeing configurations were made and how they differ; this is the internal agreement of Section 5.4 seen from outside the field. Barimah, who does not discuss artificial configurations, proposes for human experts that, because laypersons are often not equal to the task, experts bear the greater share of the burden by supplying information about their expertise, their track record, and whether a consensus exists (Barimah 2024, 472–473). Carried over, the information about how agreeing configurations were made and how they differ would fall to those who build and deploy them to supply, and Barimah's Ghana case suggests why: where statements are owned by organizations rather than by identifiable scientists, criteria built for individual experts are hard to apply, and behind a configuration's certification there may be no individual of the kind the criteria presuppose. And since, on Anderson's account, the capacity to judge whom to trust bears on the democratic legitimacy of policy that rests on deference, its erosion would bear on legitimacy as well (Chapter 12, Section 7).

Irzik and Kurtulmus, analyzing when a public's epistemic trust in scientists is warranted, mark a limit of such criteria. They distinguish two kinds of reason a public can have for judging whether research is reliable: first-order reasons, concerning the evidence for the claim itself, and second-order reasons, concerning the scientists themselves, which the public uses instead when it cannot assess the first; Anderson's criteria above, expertise, honesty, epistemic responsibility, and a consensus of trustworthy experts, are of the second kind. For second-order reasons they follow Goldman and Anderson (Irzik and Kurtulmus 2019, §3). And, building on Wilholt, they add an enhanced form of trust that requires in addition that, where public welfare is at stake, scientists' decisions about the distribution of inductive risk agree with the public's assessments and that the public have reason to believe they do. The second-order reasons, they observe, serve as proxies for whether research is reliable, not for whether its inductive risks are distributed as the public expects; for that, they point to hybrid forums in which citizens and scientists deliberate, where citizens bring in their expectations and check whether they are met, and to diversity among researchers (Irzik and Kurtulmus 2019, §4). From the controversy over the MMR vaccine they argue that enhanced trust can be lacking without fault on either side when the parties' values diverge (Irzik and Kurtulmus 2019, §5.3). This book reads their analysis as carrying the value layer of Section 4 to *S* as a layperson. The relational questions bear on reliability and honesty; whether an artificial configuration's weighting of errors matches what *S* expects needs channels through which *S*'s evaluations reach it. Their two mechanisms supply such channels in different ways. The diversity they mean is diversity of deeply held values and of social position, such as gender, ethnicity, and class. They note that it can limit prejudice and so bear on reliability, but here they use it for a different purpose, as a reason to think research reflects the public's expectations about inductive risk; in that use it differs from the independence as difference in routes of error seen in Sections 2 and 3. The more diverse the scientists, they hold, the likelier that research reflects the expectations of the whole public rather than of a segment. They acknowledge, though, that even a sufficiently diverse community may hold values different from the public's because of its training and public responsibilities (Irzik and Kurtulmus 2019, §4). Applied to artificial configurations, the range of valuations can be made the same or made to differ, and so far as it is fixed by construction, it comes not from the lives of *S*'s members but from the settings of those who build and assemble the configuration. A reason to think the range reflects *S*'s range then becomes a reason about how it was set and against what it was matched, and that requires a channel through which *S*'s expectations enter. In human communities the two mechanisms

could work separately; for artificial configurations, whether diversity counts as a reason at all comes to depend on a channel of the kind a hybrid forum provides. Applied here, such a forum is a relation between those who set the weighting of errors and the members of *S* affected by its consequences; whether that relation lies outside the artificial configuration depends on where *S*'s members stand, inside or outside it. Where the members are artificial and stand inside, the further question is whether their valuations are distinct from the settings of those who build and assemble the configuration. In research, one more thing is added to this channel. For results new to *S*, *S*'s expectations about the weight of errors may not yet have formed, and once formed they can move through research (Section 8). The expectations a forum brings in are therefore at once a standard against which the weighting is checked and something research goes on revising, so the checking is not done once and for all.

Three things come into view from this thought experiment. One is that deference on what is correct, pushed to its limit, puts *S*, on Hardwig's account carried over, only where every specialist already stands outside their own field: final judgment on questions of content then lies in the hands of no member of *S*, which extends that position. Whether *S* thereby comes to know, however, inherits the choice, left open in Chapter 2, Section 6, among Lackey's account, distributed and group-functional accounts, and accounts of knowledge without a knowing subject; on Lackey's account, on which *S*'s knowledge still requires justified beliefs among its operative members, deference in the limit can put *S*'s knowledge at risk. The second is that what remains with *S* is not competence about content but the capacity to ask about its relations to what it depends on. On the accounts of Hardwig, Goldman, and Anderson, such questions can be asked without competence about content. Where the agreeing experts are artificial configurations, however, the signal of consensus that *S* reads does not by itself show how they differ, and among configurations that share a lineage it can persist after the diversity that made it reliable is gone; on Irzik and Kurtulmus's analysis these questions do not reach the weighting of errors, for which a channel bringing in *S*'s valuations is needed; and procedures that extend a layperson's judgment rely on calibration in cases that can be checked, and need those who can produce answers independently. The third is that, except on accounts that adopt epistocracy in pure form, standing over ends does not pass with deference, but it can become nominal. Even if competence alone does not give rise to standing, whether standing stays more than nominal depends on whether *S* keeps the capacity to ask about those relations and to bring in its valuations, and that capacity can decline, with nothing changing on the side of the configurations, as members cease to gain experience of checking, or are not remade in ways that keep it. Carried over, Anderson's condition of disposition adds that capacity does not secure the disposition to use it: where certifications reach *S* as the readings of an instrument (Section 5.5) and no stage makes evaluation visible (Section 7), the occasions on which the relational questions would be asked can thin out even for members who could ask them.

Understanding adds a question that neither warranted deference about adequacy nor standing over ends answers by itself. A community may have reason to accept that an artificial expert's explanation is adequate while its members do not understand the phenomenon. If human understanding is an intended achievement, relying on a verdict of adequacy does not show that it has been reached. Whether testimony and background abilities can supply the missing achievement follows the alternatives in Chapter 3 and Section 5.6; the layperson analogy itself does not decide it.

Within the chapter, this section removes one premise that Sections 1–5 had left implicit. Those sections asked which functions of criticism and certification could move inside artificial configurations, presupposing that somewhere in *S*, besides those who rely, there are members who can check. This section considered the limit at which that premise fails. What came into view is that reliance on content can move inside, while the capacity to ask about relations and channels for bringing in valuations do not move with it, nor, except on accounts that adopt epistocracy in pure form, does standing over ends. This matters because here the book's question, what changes when *A* is a machine, takes the form: if machines do research better, why not simply defer? On this section's answer, where *S* includes human members, what can be lost in the first instance is not access to truth but substantive participation in deciding ends and the capacities that support it; on some accounts, as will be seen below, that loss leads to an epistemic loss as well. In what sense this loss is a problem divides by account, and at least four senses can be distinguished. One is a lack of legitimacy: on will-based conceptions of legitimacy, decisions about which unknowns are pursued and which risks are borne would be made without passing through the will of those subject to them, and would lack legitimacy. In research, appealing instead to the good outcomes a decision brought is a narrow route: decisions about which unknowns to pursue are made before results are known, and the results of lines not chosen are not observed, so there is often no other outcome to compare with. The second is domination: on republican views that value freedom from domination, even decisions favourable to *S* amount, if *S* cannot revise them, to dependence on good

will, and so to domination (Chapter 12, Section 3.4). In research, the very standards by which *S* judges whether decisions are favourable can move with the coupling of production and evaluation (Section 8), so giving up the capacity to revise on the strength of decisions' being favourable can also shift toward the configurations the footing from which *S* judges what is favourable. The third is epistemic drift: applied here, Dewey's view and Irzik and Kurtulmus's extend the problem to the epistemic side, since if *S*'s needs and valuations do not reach the configuration, the weighting of errors and the choice of questions can drift from *S*'s, and no one remains to notice and correct the drift. The fourth is irreversibility: since declining capacities are not easily restored, the room for choosing otherwise later itself narrows (Chapter 12, Section 7). On views that ground legitimacy in accord with normative facts or in epistemic advantage, the loss need not be a problem in the first two senses, lack of legitimacy and domination, but the problems in the last two, epistemic drift and irreversibility, can remain. Where *S* too is artificial, whether its members have standing is itself a question, taken up in Chapter 12, Section 5. Of what does not move inside with reliance on content, standing over ends concerns relations of authority rather than epistemic relations. To whom it belongs, and what it takes to keep it more than nominal, is taken up in Chapter 12, under accounts of legitimate authority (Section 1), standing that is more than nominal (Section 4), the case in which *A* and *S* are both artificial (Section 5), and decisions that remove the capacity to revise (Section 7).

What changes in criticism: cost, range, judgment, and arrangement

Section 1 named three changes that an artificial configuration makes to evaluation. Section 3 took up the first, the generation of critics in any number, as copies of one base or as critics made to differ. This section takes up the second, that those who produce results and those who evaluate them can be within one artificial configuration, with evaluation entering continually even while results are being formed. What then changes in criticism is not confined to cost. There are at least four aspects. One is cost. The second is range: checks that human checkers, given their limited time and attention, did not ordinarily perform become possible. The third is judgment: evaluators can err in places other than where humans err and can pick up different things, and for judgments of value the question arises whose valuations they reflect. The fourth is arrangement: the combination of who checks, when, what, and how, and of how verdicts are gathered, is no longer confined to the form of review. This section asks what is required for a result to become something *S* can rely on, and how that changes in each of these four aspects.

In human research, the function of making results something others can rely on has in recent decades been performed largely through peer review, a practice in which researchers other than the producers read and check a submitted result before publication. Review, however, is one realization of the function, not the function itself, and a realization shaped in history. By 1832 the Royal Society had made a written report on a paper, by one or more members of its Council to whom the paper had been referred, a condition for publishing it in the *Philosophical Transactions* (Moxham and Fyfe 2018, 11). Sending every paper to be published to outside referees became settled practice at major journals much later. On Melinda Baldwin's account, many high-profile journals did not adopt external refereeing until the 1960s or later; *Nature* made it a requirement for every paper only in 1973, and the *Lancet*'s editor kept the right to accept papers on his own authority until the mid-1970s (Baldwin 2015). The form in which a few referees read once before publication has plausibly been shaped in part by criticism's being costly and by the limited attention of those who check. In an artificial configuration that function may be performed without any stage resembling review, since if candidates are checked continuously while they are formed and reassessed whenever the record changes, there may be no moment at which a result has passed review at all.

What is required for *S* to be able to rely on a result depends on the account. On the view that places the value of criticism in its finding errors (Section 2; the error-detection view below), criticism needs to add information about errors through a route that would show the result or its grounds wrong if they were. Whether a reliant party must possess evidence of that contribution depends on the account of reliance (Section 5.2). On critical contextual empiricism (Section 3), it needs to have been open to criticism from the relevant perspectives in venues public to the community whose knowledge is at issue, with uptake, that is, with the community ready to revise what it accepts in response to the criticism raised. On assurance accounts, which find a reason for reliance in a speaker's vouching for the content, someone needs to vouch for the result and, on those that add a right of complaint, answer for it if it is wrong. On necessary-trust accounts, on which reliance in science requires trust in someone who can be responsible, such a party is needed behind what is relied on, though these accounts divide over who it must be and for what (Section 5). Two further judgments arise when certification includes claims to novelty or

significance: whether the result is new to *S* rather than merely absent from the literature and data the producer searched, and whether it matters for *S*. These are not conditions of every warranted result: a replication can be reliable without being novel. Where those claims are made, for whom the result is new and important (their reference) and their grounds are to be specified (Chapter 9, Sections 2 and 3).

Human research has bundled these functions in several ways. Scholarly communication performs what Van de Sompel and colleagues, following Roosendaal and Geurts, list as registration (recording who showed what when), certification (recognizing a result as sound), awareness (making it known), archiving (preserving it), and reward (conferring credit), functions that journals came to bundle and that can be separated and reassigned (Van de Sompel et al. 2004). Peer review is one historical bundle within this, and this book distinguishes, for the purpose of this section, the functions it has combined: screening whether a manuscript fits a journal's scope, allocating scarce space or money, detecting error, improving exposition, assessing novelty and importance, and conferring recognition. This division sorts review by function so that functions whose costs change differently can be followed separately. Which of these functions a given review performs well is a separate question for each. Other arrangements distribute the same functions differently: publishing first and filtering afterward, certification through use and replication, continuing verification of a maintained record, and proposals to abandon prepublication review altogether, among others. What bears on this chapter is how recognition and the allocation of attention can be joined in a review decision. Where passing review is used as a sign of recognition, the gate also distributes editorial attention and publication space. Reviewing labor can also be unevenly distributed: a model of biomedical publishing estimated that 20% of researchers performed 69–94% of reviews in 2015 (Kovanis et al. 2016, 1, 6). Through the same single stage at which it performs these functions, review has plausibly also provided at least three further things. One is making the fact of evaluation visible to *S*. The second is giving members of *S* a place to gain experience in evaluating. The third is leaving the judgment of someone who can be named in a form that can be objected to. The consequences, considered below, of the disappearance of an explicit stage of evaluation concern mainly these three.

Take cost first. Artificial systems already in use weaken the premise that criticism is costly to produce, and they weaken it unevenly. In a randomized study at a large machine-learning conference, reported in a preprint, AI feedback was sent to reviewers on randomly selected reviews they had just submitted; 27% of the reviewers who received feedback updated their reviews, and when two researchers compared, blind, a hundred updated reviews that had incorporated feedback with their originals, they preferred the updated version 89% of the time (Thakkar et al. 2025, 1, 10). The example illustrates the claim that criticism of texts can be multiplied at low cost, here even criticism of criticism; it says nothing about criticism that reaches a defeat relation (Section 1), a route that would show the result wrong if it were, such as a new measurement, a rerun of an experiment, or a formal check. Criticism of texts, code, statistics, and consistency with the literature can be multiplied cheaply, and some of it can itself undercut a result's grounds, as when it finds an error in an analysis. Criticism whose defeat relation runs through a route the producer does not supply, such as a replication in a laboratory, a new measurement, or a proof checked against a kernel the generator does not control, becomes cheaper only where that route itself becomes cheaper, although automated re-execution and automated formalization are among what has become cheaper. Direct evidence on whether the capacity to produce results or the capacity to appraise them competently is growing faster remains fragmented; what is suggested here concerns the structure, that some criticism becomes cheaper and some does not, not the size of the gap.

Change in range can come about through change in cost, but its consequences are different. Once the cost of checking falls far enough, kinds of check that had been too burdensome to be carried out in practice may become feasible: for example, reimplementing a paper's method to see whether its results reproduce, rerunning submitted code and data, and searching the related literature exhaustively to see whether a result is already known and whether the works cited exist and support the claims made. Checks of this kind, which can take a human considerable time for a single paper, an artificial configuration may be able to attempt with far less burden. If so, what is checked changes. Where reimplementation and rerunning are possible, part of the defeat relations of Section 1 moves from rereading the record to running things and seeing. Where exhaustive search is possible, the judgment whether a result is new to *S* can be made against a wider record rather than against whatever the producer happened to search. Being able to attempt such checks, however, differs from succeeding at them. In one study that gave models only the text and figures of 83 papers, with 91 errors, that had led to errata or retractions and whose errors their own authors had acknowledged, none of the ten models tested in early 2025 found most of those errors, and even the best found about a fifth of them in a single run (Son et al. 2025, 3–5). Because that study excluded errors

requiring external code or data, the success of checking by reimplementation or rerunning is a separate question. And however far the range widens, it does not reach new measurements or executions outside the scope of the record.

Two consequences follow from these changes in cost and range. The first follows from their unevenness and concerns the error-detection view, on which the unevenness matters because much of the criticism that becomes cheap works on the producer's record. Errors that leave no trace in the record, such as errors in how it was made, cannot be found by rereading it, however much the critics differ from one another (Section 3, the second sense). Where cheap critics are also built from the producer's base or from tools of the same kind, they are likely to go wrong where the producer does even about what the record shows, and many of them agreeing add little, for Goldman's reason that sources merely reflecting one another add no evidence however many agree (Section 2). Differently made critics may be cheap as well (Section 3), but the volume of criticism does not show whether they are. Abundant criticism can then coexist with a scarcity of independent criticism, and conceal it. What can remain scarce is not attention in general but defeat relations through routes the producer does not supply, and, where critics are made like the producer, critics whose error routes are their own. The second follows from criticism's becoming abundant, and concerns the point at which it is applied. If the argument of Chapter 5, Section 4 is right that judging what is worth pursuing differs from judging what is likely true, criticism applied indiscriminately to candidates still being formed can turn away promising candidates with little support yet, merely because they seem unlikely to be true, and so shut out ventures that might prove fruitful.

In judgment, artificial reviewers need not err where humans do. In one study in which 45 researchers in the relevant fields assessed, item by item, the comments of human reviewers and of three language-model reviewers on 82 papers in *Nature*-family journals, about a quarter of the artificial reviewers' comments had no counterpart, among any of the human reviewers' comments, that made the same criticism of the same point. About four-fifths of these were judged correct by the experts, but they were rated most significant less often than those that overlapped with human comments. The experts' remarks also identified weaknesses that, the authors state, are not seen in human reviewers, such as treating practices regarded as ordinary in the field as defects and so misjudging the weight of a comment, and saying that something the paper states is missing (Kim et al. 2026, 9–12). Nor is there a single direction in which artificial reviewers lean. In the published version of the AI Scientist, discussed in the Introduction, when the reviewing component was compared with ICLR decisions, the rate at which it judged rejected papers acceptable was 0.45 to 0.52, higher than the 0.17 found by an experiment at another conference that measured agreement between the decisions of two committees, and the rate at which it judged accepted papers unacceptable was 0.17, lower than that experiment's 0.52 (Lu et al. 2026, 916). Its judgments leaned toward acceptance. Beel and colleagues, discussed below, by contrast, had the system as first released assess ten human-written papers and read a strong conservative bias. Since the models, the number of papers, and the manner of evaluation differ, the two cannot be compared as a change in one system. Still, they show that the direction of bias seen in one evaluation of one artificial configuration cannot be read as a general property of artificial reviewers. Misjudging the weight of a comment also bears on which norms a judgment of value is made against, since what matters, and which defects are serious, is relative to the field and to the valuations of *S*. Evaluating artificial reviewers thus involves asking not only whether they are correct but against whose valuations they judge.

The aspect of judgment brings with it the question of how artificial reviewers themselves are to be evaluated. One method compares their judgments with human decisions. What agreement measures depends on the function and on the account. On accounts that take review to aim at correctness, agreement measures fidelity to the practice whose decisions serve as the reference, including its blind spots, not correctness. Where the humans whose decisions are compared are members of *S*, that is the certifying practice of *S*; where *S* is artificial, or mixed, agreement with human decisions measures fidelity to a practice outside *S*, or to that of only some of its members, whose relevance to certification for *S* has to be argued. On accounts on which certification consists partly in applying a shared practice uniformly, in the way that legal determinacy, as Porter reports Kent Greenawalt's account of it, requires only that different judges apply the law to most cases in the same way (Porter 1995, 4–5), fidelity to the practice may itself be part of what certification is for.

That practice need not return a single verdict. In a study reported by Weber and Inglis, for example, when 109 mathematicians were asked whether an argument that interchanged a limit and an integral without justification was a proof, 27 percent said that it was and 73 percent that it was not, and only one of those who had accepted it agreed that an objection to that step made it invalid; on proofs using routine techniques from number theory, nearly all the mathematicians surveyed

in another study accepted them (Weber and Inglis 2024, §4.3 and §2.2 n. 3). Agreement with a divided practice depends on which part of it is taken as the standard. For functions other than reproducing past decisions, agreement may say little. One exploratory study compared pairwise rankings by a language model with funding decisions on 139 proposals to the Austrian Science Fund. The two selections agreed in 87 cases, and higher human-rated novelty was associated both with disagreement and with the direction in which humans funded a proposal that the model did not (Machado 2026). The sample is small and the design does not identify why the pattern occurred. It illustrates why disagreement cannot be read directly as error: here it was concentrated where a function other than reproducing past decisions was at stake (Section 8 returns to novelty).

Two further reports bear on the same question. Beel and colleagues had the reviewing component of the AI Scientist (Introduction, Section 2), in a version they modified for the purpose, assess ten human-written papers from an open review platform, five of them accepted and five rejected by human review; it recommended rejecting nine of the ten. They read this as a strong conservative bias and misalignment with human judgment, and ask whether human judgment is the right benchmark at all, since a reviewer that diverges from human reviewers may still identify valid weaknesses, or may apply overly rigid criteria (Beel et al. 2025, §2.6). At a conference organized so that artificial agents served as authors and reviewers, three language-model reviewers, whose instructions the organizers had refined to maximize correlation with human scores from two earlier ICLR editions (2022 and 2025), correlated with one another at 0.48 on average, differed from the scores of human experts by mean absolute differences ranging from 0.91 to 2.73 on a six-point scale, and showed sycophancy, especially one of them, which in one review called "technically flawless" a paper whose experiments the human reviewer suspected were incomplete or selectively reported (Bianchi et al. 2025, 4–5). Both are reports about particular systems at one time and, in the second case, at one venue. They suggest that refining instructions to match human decisions (calibration in this sense) does not fix which part of a divided practice the calibrated reviewers reproduce, and that disagreement with human decisions need not have a single kind of source.

In human research, that a competent person with limited time had examined a result and let it stand was plausibly part of what gave others reason to rely on it. If review becomes abundant and artificial, that reason does not transfer automatically. Passing an artificial review gives *S* reason to rely on a result only to the extent that reliance on the reviewer for that function is itself warranted; whether the one who relies needs reasons of their own for trusting the reviewer (reductionism), only the absence of reasons for doubt (non-reductionism), or only the reviewer's actual reliability (reliabilism) depends on the account of reliance (Section 5.2).

Turning to arrangement, in an artificial configuration, as noted at the start of this section, there may be no such moment of passing at all. The room for recombining these functions also widens further. Those who check need not be a few human referees; they may be copies of one base or many checkers of different make (Section 3). The time of checking need not be once before publication; it may be while a result is being formed, after it has been used, or whenever the record changes. The manner of checking need not be reading and writing an opinion; it may include actually running code and data, checking formally, having different answers argued against each other, scoring predictions against later outcomes, and deliberately attacking the checkers themselves to find their weak points. The object of checking need not be the text of a paper; it may be code, data, execution logs, or the dependencies themselves. Nor need verdicts be gathered by an editor's judgment; they may be gathered by majority, by weighting, or with dissent preserved, among other ways. Which of these combinations can perform the function of making a result something others can rely on is to be asked of each combination, in light of what, as set out above, is required. The question thus becomes not whether to keep review but which combination meets what which account requires.

The judgments review has bundled can also be gathered into two kinds. One is a judgment of a result's credentials as knowledge: whether a result holds up as research, and how sound it is. The other is a judgment of value: how important or interesting the result is for *S*. Why human research has had these two made at a stage separate from production, by parties other than the producers, may have no single answer. Some of the reasons one can think of, however, are tied to the premise that research has been produced by individuals or small groups. For the judgment of value, the values of the producing individual or group can differ from those of *S*, so there may have been point in placing a filter between them. For the judgment of credentials, a small group alone may be unable to carry it out adequately, or repeated checking may judge it better. If these are the reasons, they plausibly include parts that answered to each of two kinds of premise in the sorting of Chapter 9, Section 6. That a small group cannot check enough turns on research's being carried out by humans with limited

time and attention, a premise about human realizers. Placing a filter between producers' values and *S*'s, and having checking done at a separate stage by other parties, turns on research's having been organized so that individuals or small groups produce it, a premise about how research has been organized. If so, then when an artificial configuration can carry out both judgments adequately within itself, a review placed afterward as a separate stage can be redundant in the sense of doing the same thing twice. Carrying out the judgment of value adequately, though, means that the judgment of how important a result is for *S* matches *S*'s valuations. This is a question of whose valuations the artificial configuration's valuations match, structurally similar to AI alignment (Section 4; Chapter 12), and competence alone does not settle it.

That does not end the matter. First, there is the view on which doing it twice is itself of value. The question it raises is whether the second pass adds a check that errs in different places from the first, such as a test working on a different principle or a separately obtained measurement (the second sense of redundancy distinguished in Section 3), or differences in applying the same standards (the third sense). Second, if certification is taken as the relation that connects what *A* produces with what *S* can rely on (Chapter 9, Section 2.1), then as long as *A* and *S* can differ, that connection is needed whatever *A* and *S* are. If this connection is what review has provided, review can remain as an abstract function even when it loses its form as a human institution. This is the fourth sense of redundancy distinguished in Section 3, in which stages have different addressees and functions.

Evaluation may, further, not appear as a separate stage at all. In human research too, judgments of which questions to pursue, which candidates to develop, and which lines to abandon have been made inside the process of inquiry (Chapter 5, Section 4; Chapter 10). In an artificial configuration, the explicit process of evaluation itself, in which a finished result is checked afterwards, can dissolve into the choice of questions and the formation of candidates and cease to be visible as such. Whether that suffices depends on where the function of evaluation is placed. On the error-detection view, if checks that add information about errors are carried out, the requirement can be met even if they are made implicitly inside the process. By contrast, on Longino's account, which requires that criticism take place in venues public to *S*, on assurance accounts, which require someone who vouches for the result, and on reductionism, which requires reasons available to those who rely, the outcome of evaluation, or at least a trace of it, has to come out toward *S*. On those accounts, evaluation settled implicitly inside does not by itself amount to certification for *S*. What counts as coming out depends on what *S* consists of. If *S* is human, it has to take a form that humans can read and question. If the members of *S* are artificial and can share representations directly with the artificial configuration, it may take a form other than a report (Chapter 3, Section 8). Even then, whether the members can themselves check the shared content is a further question.

Where the explicit stage of evaluation disappears, several consequences follow. One is that, on accounts that require evaluation to be made explicit to *S*, an artificial configuration needs a layer, distinct from its internal evaluation, that it puts out toward *S*. Part of the work of evaluation then moves from checking to conveying the outcome of checking to *S*, and the explicit stage may come back in another form rather than disappear. The second concerns where *S*'s capacity to check is developed. In human research, reviewing has plausibly also been a place where the capacity to evaluate was acquired and kept up. If that stage disappears, the opportunities for members of *S* to gain experience diminish as well, and *S* can come closer to being a layperson relative to the artificial configuration (Section 6). The third concerns objection. To object to a judgment one has to be able to see it, so implicit judgments are hard to object to, and revisability depends on explicit traces (Chapter 12, Section 6). The fourth concerns time. When evaluation stays inside, coupled with production, a decline in an evaluator's independence or a shift in standards is hard to notice from outside, and reassessing what depends on a defeated claim requires a record of what was judged where (Section 8). The fifth concerns allocation. Passing review was also a signal for allocating funds and positions, so if the moment of passing disappears, allocation needs another signal (Chapter 10, Section 3). On the error-detection view, on the other hand, implicit evaluation may catch errors earlier and continually, and the rigidity of a gate under one criterion that turns candidates away together may also be eased. And if the members of *S* are artificial and can share representations with the artificial configuration, a layer put out toward *S* may not be needed. Then, however, the distinction between *A* and *S* itself thins, and who has standing becomes a question (Chapter 12, Section 5).

In sum, the changes an artificial configuration brings to criticism fall into four aspects.

- **Cost**: what becomes cheaper is the cost of some kinds of criticism only. On the error-detection view, the criticism that becomes abundant, so far as it shares the producer's error routes, adds little independent information and can conceal the scarcity of what does.

- **Range**: checks that humans did not ordinarily perform come to be attempted, so that part of the defeat relations can move from rereading the record to confirmation by execution. Attempting, however, differs from succeeding, and nothing here reaches beyond the record.
- **Judgment**: artificial evaluators may catch what humans overlook, and may also make errors humans do not. For judgments of value, the question is whose valuations they reflect.
- **Arrangement**: the function of making a result something others can rely on can be performed by many combinations not confined to the form of review. Evaluation can also dissolve into production, so that no explicit stage remains.

On that basis, the six functions that review has bundled go different ways. Screening for scope can dissolve into the choice of questions and the formation of candidates. The gate that allocates space and money answered less to premises about human realizers than to premises about the present organization of science (Chapter 9, Section 6), and can lapse with them, as it can also when that organization changes while its realizers remain human. Where the time and computation available for checking are limited, however, not every candidate can be sent to a costly check or to *S*'s attention, the same reason of finitude that Chapter 10, Section 1 gave for allocation; where checks remain costly, some gate remains before them, and a gate operated by one criterion can turn candidates away together, in the direction of that criterion's mistake (Chapter 10, Section 4 found the same structure where one signal judges the survival of lines). Error detection can, on the error-detection view, be carried inside the artificial configuration. What joining production and evaluation does not remove, on that view, is the need for criticism that adds information about errors and remains responsive to what it finds. This need not require that someone other than the producer read the result again: a proof assistant's kernel, the small program that checks each step of inference (Section 2), or a measurement the producer does not control can show a result wrong, if it is, without anyone reading it. Improving exposition, together with checking consistency and comparing a result with the existing record, can be done by the producer as it produces. Assessing novelty and importance can also dissolve into the choice of questions and the formation of candidates. Assessing importance, however, is a judgment of value and leaves open whether the artificial configuration's valuations match those of *S*; assessing novelty leaves open whether the result is new to *S* rather than merely absent from what the configuration searched. Recognition has also served as a signal on which later allocation of funds and positions relies, so where allocation steers by it, it is sought in another form even when scarcity of space and attention is gone (Chapter 10, Section 3). The three further things review can be thought to have provided along with these functions, the visibility of evaluation to *S*, a place for *S*'s members to gain experience in evaluating, and judgments that can be objected to, can, as seen above, be lost where the explicit stage disappears, unless they are provided in another form. On the other accounts, the publicity of the venue and someone who answers are not made unnecessary by joining. And so long as *A* and *S* differ, on each account considered, the relation that connects what *A* produces with what *S* can rely on remains. For each function, whether it suffices for it to be performed implicitly inside the process or whether it requires being made explicit to *S* depends on the account adopted and on what *S* consists of.

And no change in any of these aspects, on any of the accounts considered here, yields certification by itself. Certification requires that reliance on the critic be warranted in whatever way the account of reliance demands (Section 5.2), and, depending on the account, public venues or someone who answers. This conclusion about the unevenness of cost rests on the structure of the cases cited, not on data measuring how fast the cost of which kinds of criticism is falling.

Criticism over time: proxies and standards, evaluating the new, and correction

Sections 2 to 7 treated independence as a relation between a critic and a claim at a time. The third change named in Section 1, that an artificial configuration can tie evaluators and producers together over time by adjusting each to the other's outputs, for example by training or selecting each on them (coupling, below), makes it also a relation over time. As Section 7 noted, the further evaluation dissolves into the process, the harder such changes become to see from outside. This section adds a condition of time to the conclusions of Sections 2 to 7, and prepares one of the requirements Section 9 places on an artificial configuration: showing that independence is maintained as producers and evaluators adapt to each other. Two points of Chapter 9, Section 4 also concern time. One is that evaluation of research has to work through proxies, because what it is about is usually not observable when the judgment is needed. The other is that, for something new to *S*, the evaluator may lack a standard. The section takes up three things in turn: how selection on proxies moves critics' independence and evaluative standards, how what is new is evaluated where there is no standard, and how correction reaches what depends on a claim after it is defeated. Proxies that come to be used in decisions about hiring and funding can change what they measure, standards can move or be held fixed even when defective, and a claim's defeat need not reach what has come to depend on it. Signals by which allocation steers are proxies of this kind, and this section is where the book examines their properties; Chapter 10, Section 3 takes up how allocation steers by them.

The first point, that proxies which come to be used in such decisions change what they measure, has a long lineage. Donald Campbell argued that increasing the use of a quantitative indicator in social decisions both increases pressure to manipulate the indicator and makes it more likely to damage the processes it was intended to track (Campbell [1976] 2011, 34). In research, once an indicator affects hiring, funding, or survival, those evaluated adapt to it, and the measure becomes part of the system whose performance it reports. In human research this adaptation runs through motives and careers. It is broader than deliberate gaming: a researcher can honestly redirect effort toward what is measured, and when many do, the meaning of the indicator and the population it describes can shift (Wilsdon et al. 2015, 81–83). For artificial producers the adaptation need not run through motives, whatever one concludes about whether motives can be attributed to them (Chapter 7, Section 3.3); it can run through optimization and selection: outputs that score well are kept, and producers are trained or chosen on that basis. Evaluators can in turn be trained on past decisions, and the outputs they selected become future data. A model update can change the operative evaluative rule without any explicit change in policy. David Manheim and Scott Garrabrant, writing about metrics used in optimization generally rather than about research, separate four ways in which the relation between a proxy and a goal can collapse under optimization. Selecting on an imperfect proxy also selects for the difference between proxy and goal, which they hold "can not be avoided" (regressional); selection can push into regions where a relation observed elsewhere no longer holds (extremal); intervening to raise the proxy can change the causal path that linked it to the goal (causal); and agents with goals of their own can act on a proxy they know to be in use (adversarial), the case to which they relate Campbell's statement. They add that how much these effects matter depends on the optimization power directed at the proxy (Manheim and Garrabrant 2019, 1–2, 4, 8). Carried over to research, their classification separates the two routes distinguished above, adaptation through motives and adaptation through optimization and selection alone: the adversarial case needs a party that responds to the measure, whereas the regressional and extremal cases need only selection on it.

Taken together, early observational studies report that AI involvement is associated with greater novelty or impact of individual papers and with a narrowing of the collective range of topics, and that grant proposals written with more AI involvement (measured as the estimated share of language-model-written text in proposal and award abstracts) sit closer to recently funded work. Selection on proxies pulling work toward what is easily measured is one route by which such narrowing could arise; but these are patterns consistent with such narrowing, not established effects of it, and other routes remain possible (Bianchini et al. 2026; Hao et al. 2026; Qian et al. 2026). The studies count different things as AI involvement. In Hao and colleagues' study of papers in six natural sciences from 1980 onward, it is the use of AI methods, most of them conventional machine learning and deep learning, and the authors describe their analysis of the period of generative AI as preliminary (Hao et al. 2026, 2). A study of language-model use in writing reports a pattern in another direction at the level of inputs: Kusumegi and colleagues find that authors whose preprints show signs of language-model assistance go on to cite a more diverse body of prior work, including more books and younger and less-cited documents,

while noting that their design does not provide causal identification (Kusumegi et al. 2025, 5–7). The diversity of what is cited and the topical range of what is produced are different measures, so the second finding does not refute the first; together they caution against reading either as the effect of artificial systems on the range of research.

This bears on independence directly. A critic can add information about a producer's errors at one time and less later, as training or selection comes to favor outputs that pass where the critic is insensitive. The checking rule need not change. Nor need the producer represent the critic: selection on its verdicts can have this effect, as can a producer's fitting outputs to a model of its evaluator. Here *evasion* names adaptation that weakens the connection between passing and being correct, rather than every occasion on which a fallible test accepts a false candidate. In the terms of Section 2, what evasion lowers is the check's severity for the producer as it has become, the probability that the check would expose the errors that producer now makes.

Adaptation can also improve the capacity being checked. Learning to produce proofs accepted by a sound checker, one that lets no invalid proof through, can improve proof construction without weakening that check's force for the formal statement. The relevant contrast is therefore not adaptation versus no adaptation, but improvement in the target capacity versus exploitation of what the check misses. No inference to either follows from a rising pass rate alone. Evidence for distinguishing them can include performance on routes withheld from selection, variations designed to expose known blind spots, and whether what the route rewards continues to bear on its intended target (Chapter 3, Section 3.3; Chapter 7, Section 3.1). Such evidence remains scoped to the errors and changes it tests. Protecting a rule from rewriting can help preserve its evidential contribution, but cannot establish that contribution or guarantee its persistence. Chapter 10, Section 4 found that bringing allocation into one configuration does not by itself make the criterion for deciding which lines to keep and which to close checkable by a separate route; here the point is that organizational separation alone does not preserve it through time.

A measure can also come apart from what it is taken to report without any adaptation to it. Carole J. Lee, writing about AI tools that evaluate science, describes, following Cartwright and colleagues and Reiss, what the paper calls inference by false ascent: a measure designed for one concrete purpose is redescribed as a measure of a more abstract quality, and because that quality covers other concrete purposes, the measure comes to be used for purposes for which it is not reliable. Lee's historical example is the journal impact factor, introduced to help librarians decide which journals to buy, then taken as a measure of scientific impact and used for the impact of articles and authors (C. J. Lee 2026, 2–3). The case from AI is a set of models trained to predict replication outcomes. Their labels track, in effect, whether a replication found a statistically significant result in the same direction as the original, a measure suited to asking whether the replication obtained the same result; the models were described as predicting replicability and proposed as a way to improve confidence in findings, a purpose for which, on Lee's analysis, a measure that combines the original and replication data would be more apt (5–7). Lee develops the point for supervised predictive models and takes it to generalize (3). On this account, assessing reliability by the proportion of accurate outputs alone does not register this failure, since reliability is reliability for a purpose, and disputes over purposes and purpose-specific reliability are to be taken up through Longino's norms of critical discourse within scientific communities (7–8). The failure Lee describes lies in the inference drawn from an evaluator's output rather than in the producers' response to it, so it can arise however faithfully an evaluator does what it was built to do. It has the same form as Otsuka's observation in Section 5.2, read there as an instance of the generality problem: reliability established for one use is carried to another. For evaluators whose verdicts are relied on in research for *S*, the record of reliability for the use at hand that a member of *S* needs on reductionist accounts, which ask for positive reasons the one who relies can obtain (Section 5.2), would then need to state which use that is.

The coupling can also move the standard itself, not only behavior under a fixed standard. Formal work on AI systems whose reward functions stand for the evaluations of the people they serve, evaluations that change over time and that the system's own actions can influence, finds within its models that each of the eight notions of alignment it compares either leaves room for undesirable influence on those evaluations or is so conservative that inaction is often the only optimal behavior; holding the system to the evaluations as they stood at the outset is one of these notions, and it can lock them in or even create an incentive to move them away from where they started (Carroll et al. 2024, 5707, 5709–5714). The authors suggest that which notion is correct may not be identifiable from the formal structure of a case alone (5708). Their models thus do not decide when a change in an evaluative standard is learning and when it is drift; carried over to evaluative standards in research, an application the authors do not make, they suggest why agreement with a standard at one time does not settle

what the standard will have become.

Several authors locate one direction of such movement on the side of those for whom research is done. Stuart Russell treats human preferences as changing through experience (Russell 2021, §1.6, author-version 17) and argues that AI systems cannot entirely avoid affecting them (Russell 2020b, transcript, paragraph beginning "One of them being the plasticity"). An intelligent machine may also have reason to make preferences easier to satisfy by changing them (Russell 2022, 53). Wolchover reports Russell's proposed response: machines could learn people's meta-preferences, their preferences about which processes of preference change are acceptable (Wolchover 2020, "Our Imperfect Choices"). Russell does not apply this argument to research. Applied to it, the argument would make *S*'s evaluative standards something a configuration can move as well as serve. Writers on research describe such movement there. Emanuele Ratti argues that machine learning "brings its own normativity": in cancer genomics, for example, its use is likely to guide molecular biologists away from the mechanistic explanation they take as the goal of their discipline toward prediction and classification, constraining which epistemic values they prioritize, an impact Ratti holds that scientific methodologies generally have (Ratti 2026, 16–18). Johan Commelin and colleagues, writing for the mathematical community, hold that "means reshape ends": the capabilities of new technology risk "dictating what counts as mathematically significant", and they ask that adoption remain rooted in the community's own epistemic and aesthetic values (Commelin et al. 2026, 2). On these accounts the standard that moves with the coupling may be *S*'s own, and agreement between a configuration's outputs and *S*'s present standard does not by itself show that the configuration has not moved it. Chapter 6, Section 5.3 distinguished, for a single inquiry, revising a standard for reasons from merely moving the target when performance is poor; coupling between producers and evaluators poses the same distinction for a system in which no one decides to move the target. Russell's remedy, carried over as well, would have a configuration learn *S*'s standards for which changes of standard count as revision for reasons rather than movement of the target, instead of holding to a fixed standard; whether its learning them can stand in for *S*'s deciding them is part of the question of who may revise a standard (Chapter 12, Section 2).

Maintainers of an evaluation can face the opposite risk, keeping a standard known to be defective so that earlier results stay comparable. Those of one benchmark for machine-learning agents, for example, chose not to fix a known leak of test labels for the time being, so as not to invalidate earlier leaderboard results, deferring the fix to a later batch of fixes, and later paused new submissions while developing a process to keep them fair and comparable (OpenAI 2025; OpenAI 2026e).

The second point concerns evaluation when the evaluator lacks a standard. For a result new to *S*, the standards by which it would be judged may themselves be in question. Kuhn held, as Eric Oberheim and Paul Hoyningen-Huene report, that the standards of theory appraisal, such as simplicity, accuracy, consistency, scope, and fruitfulness, vary with the dominant paradigm, and that they function not as rules that determine theory choice but as values that merely guide it, so that scientists who weigh them differently can disagree rationally even across incommensurable paradigms (Oberheim and Hoyningen-Huene 2024, §2.3.2). On Kuhn's account, as Section 3 noted, what lets novelty through is partly the variation among evaluators in how they apply shared values. Seen in this light, an evaluator trained on past decisions holds its standard as the distribution of those decisions, and is therefore liable to score low a candidate whose acceptance would require revising the standard, scoring it as a deviation. The Machado study of Section 7 is consistent with this, though it does not show it. If Kuhn's account is adopted, what such evaluators would need is variation in application. On this view, evaluators trained on the same past decisions may have little of it, whether they are copies of one base system or built from different bases; for artificial evaluators, such variation is not given by what the realizer is like but has to be produced, for instance by separating their training histories. Since Kuhn allowed that variation to come in part from factors such as life history, personality, and influences from outside the sciences (Section 3), whether variation produced in other ways would do the same work is not settled by that argument; on accounts on which standards for the new are formed through the criticism of it (Chapter 6, Section 5), what would matter is whether evaluators can take part in forming them. Who may revise a standard is a question for Chapter 12, Section 2.

The third point concerns what happens after a claim is defeated. A false claim was not knowledge, however well received; losing grounds for a claim need not establish that it is false. Dependent claims must be reassessed to determine which lose support and which retain independent grounds. On accounts that locate knowledge in a capacity a community keeps using, checking, and correcting (a live epistemic capacity), propagating criticism is part of maintaining that capacity, rather than a rule that every dependent result automatically ceases to be knowledge. Chapter 2 distinguished a dormant record, one merely

stored and used by no one, from a live epistemic capacity partly by whether correction propagates, and even on Bird's view, on which a dormant record can remain social knowledge (Bird 2024), whether correction reaches dependent work would plausibly bear on whether reliance on the record remains warranted. Detecting a defect, changing the article of record, and repairing dependent work are different achievements; others could be added, such as informing those who have already acted on a result, which this section leaves aside. The correction to the materials laboratory discussed in Section 2 changed the article of record. In a separate case, the Massachusetts Institute of Technology stated publicly that it had no confidence in the provenance, reliability, or validity of the data in a preprint on AI and scientific productivity, or in the veracity of the research, and requested its withdrawal; a peer-reviewed article in 2026 later cited the preprint without qualification (MIT 2025; Soleymanibrojeni et al. 2026). This was not a failure of an autonomous system and does not show how often propagation fails; it illustrates the claim that the third achievement does not follow from the first two. Within a single body of work, tracing dependents can accompany detection. In early December 1993, when Andrew Wiles announced that a calculation in the argument for Fermat's Last Theorem was not yet complete, a report on Wiles's lectures about to go to press stated which of the announced results depended on that calculation, Fermat's Last Theorem among them, and which did not (Rubin and Silverberg 1994, 2, 18).

Here too a premise changes form. In human research as it has so far been organized, dependencies are tracked only in part; citations record some of them and not others. Inside a configuration that logs what each result was derived from, the tracing of internal dependents can in principle be complete. Whether each dependent still stands without the defeated claim is a further question: answering it is itself criticism, and needs a defeat relation of its own. Among what depends on a checker are the certifications it issued: if the checker is later found unreliable or not independent of what it checked, they lose the support it gave them. Dependents outside the configuration's record, the uses others within *S* have made of its outputs, lie beyond the tracing the record makes possible; the boundary of the record of Section 5.3, beyond which nothing can be checked, returns here as a boundary of where correction can reach. The routes by which a correction would reach those uses are of the same kind as those Section 5.2 found an internal finding needs in order to act as a defeater for those who rely, now needed after the result has been used. Longino's condition of uptake states a related requirement at the level of the community: its beliefs must change over time in response to the criticism taking place within it (Longino 2025, §5). Where checking dependents again is itself costly, there is also a question of order: which to check again first. If Varsavsky's point in Section 4 is extended from the order of testing to the order of correction, allocation remains even after an error has been found, since, depending on which dependents are repaired first, the same capacity for correction can leave different people exposed to the error for different periods. Ordering repairs by the number of dependents alone, for instance, is then not a value-free choice either; it can contain a weighting of inductive risk (Section 4) over who bears the costs of the error and for how long.

Chapter 12, Section 6.3 places this section within the book's handling of error. What has come into view in this section is that independence and certification established at one time do not by themselves guarantee that they hold later. Following the section's three points, three things can erode over time.

The first is a critic's independence. As producers and evaluators adapt to each other, an evaluator's passing verdict can come to indicate less and less about whether the producer is right. This decline can follow from the structure of coupling where the check is imperfect relative to its target, is used to select or train producers, and selection shifts the producers' remaining errors toward where the check is less sensitive; the cited studies are only consistent with it. A claim that an artificial configuration's criticism remains independent would therefore have to show not only that it was independent at a single evaluation, but how independence is maintained as producers and evaluators adapt to each other. One possible measure is to keep, separately, checks that are not used to select or train producers, such as tests held out from training or measurements the producer cannot alter. An evaluator used for selection can tell less about producers the more it is used; a check not used for selection is less exposed to that effect.

The second is the standard by which the new is evaluated. If Kuhn's account is carried over to them as above, evaluators trained on past decisions may be weak where the standard itself is in question. And since the standard can move with the coupling, agreement between an evaluator's outputs and *S*'s present standard does not by itself show that the standard has not been moved. The movement itself can be detected inside the artificial configuration as well, against preserved reference points or evaluations built from *S*'s past judgments. A reference point does this work only so long as it is not revised, and the benchmark case above shows the price: what is kept for comparability can be kept defective. But whether to accept a

detected movement is for *S* to decide so long as the standard is *S*'s, and what is at issue is whether *S* keeps the capacity to decide it (Section 6; Chapter 12). And where the standard moves in this way, Wilholt's correspondence reading (Section 4) can come to be satisfied not by the weighting's being revised but by the expectations of those who rely moving. The stronger reading, which also requires that those who rely be able to object, can resist this.

The third is correction. If a claim is defeated but the correction does not reach what relied on it, the error remains. On accounts that tie knowledge to a live capacity, keeping knowledge requires such routes. An artificial configuration that logs what each result was derived from can trace its dependents internally, though checking whether what is traced still holds is itself further criticism. And the record does not extend to the places where others in *S* have used its outputs. For corrections to reach them, routes leading outside the configuration's record are needed.

In sum, it cannot be said that independence or certification established inside an artificial configuration at one time will continue. The ways it can erode are three, and each needs something different. In a configuration where adaptation to selection can weaken a check's contribution, some other means of keeping that contribution is needed, and checks not used for selection are one candidate. As for movements of standards, detecting them can be done inside, but deciding whether to accept them is for *S*, and requires that *S* keep that capacity. And where outputs are used outside the configuration's record, routes that carry corrections there are needed. So, to the question of Section 9, once time is taken into account the answer may not be that everything can be done inside. What the last two need is not being human but relations to *S*, so this holds whether *S* is human or artificial (what counts as outside where *A* and *S* overlap is taken up in Chapter 12, Section 5). Nor does showing that a configuration was independent at one evaluation show that it can evaluate its own research.

Can criticism for *S* be organized inside an artificial configuration?

This section considers the chapter's question with the material assembled so far. In the example of Chapter 9, Section 1, a single system generates a hundred instances, has them check one another's products, and redirects resources in response to those checks. Those who build it can claim that results that have passed the internal checks are certified for *S*, so that others have reason to rely on them. This section asks under what conditions that claim holds, that is, whether criticism and certification for *S* can be organized inside an artificial configuration. It is also the question whether a way of doing research in which machines produce knowledge and evaluate it themselves can give certification for *S*. Answering it account by account is also a means of seeing what happens to research under such a way of doing it: knowing what can move inside shows what then ceases to be visible to *S*, and knowing what remains shows what *S*'s side needs if the research is to remain research for *S*. Chapter 12 needs the answer too: if some part cannot be organized inside, one has to know what it is and why it remains before one can ask who may decide about it and who may revise it.

The section proceeds as follows. It first sorts out which question each of the accounts seen in this chapter answers, and gathers their verdicts in one table (Section 9.1). It then states what comes into view when those accounts are set side by side (Section 9.2) and what is suggested by extending that further (Section 9.3). Finally, it lists the questions that pass to Chapter 12 (Section 9.4). In outline, the answer differs by account. The work of finding errors can, on the error-detection view, be done inside if the checks err in different places from the producer. By contrast, acceptability for *S*, someone who answers to those who rely, standing for dissent, and the like remain, on the accounts that require them, as relations to *S*. And as Section 8 found, once time is taken into account, even for the functions that can be done inside it may not be possible to say that everything can be done inside. Section 9.3 goes on from there to what can happen to research: that if passing internal checks comes to serve as a proxy for certification, the lack of relations to *S* can be selected along with it; that independent checks can be depleted by use; that there may come to be no one in *S* able to carry the relations that remain; and that as *A* and *S* overlap, the question of certification can move to "for which society?" When this section says "inside", it is in the sense of Chapter 9, Section 5, which covers two things: that the work of checking is supplied inside the artificial configuration, and that the acknowledgment recognizing its outcome as certification for *S* lies inside. In this chapter's argument the two can come apart, and what remains as a relation to *S* is mainly the latter.

That verdicts diverge when different accounts are compared is, however, no surprise in itself. What this chapter's material brings into view is why verdicts diverge for artificial configurations. In human research, the properties that the accounts each track were tied to one another by human realizers and by the organization of research. Critics were formed as separate

people, so their number roughly traced their independence (Section 3). Certification was a person's assertion, and if it was wrong there was someone who answered (Section 5). Producers and those who rely could be presumed to share a sense of the weight of errors (Section 4). Experts in agreement had been trained in separate places and had access to separate evidence, so the signal of agreement came with diversity (Section 6). The stage of review made evaluation visible to *S* and was plausibly also a place where the capacity to evaluate was developed (Section 7). In many everyday cases, therefore, accounts that ask about routes of error, accounts that ask about perspectives, and accounts that ask about someone who answers plausibly tended to lean toward the same verdict. An artificial configuration makes these properties settable one by one. Critics can be made as copies or with built-in differences, records can be kept from the run, the way verdicts are gathered can be chosen, and evaluation can dissolve into production. The coming apart of the properties ceases to be the exception. This is why the accounts of Table 11.1 reach different verdicts on the republic of instances (Section 3), and why the four answers of Section 5.2 require different things of the same artificial configuration. This does not mean that an artificial configuration necessarily makes the accounts diverge. It can also rebundle the properties, with different bases, records that reach those who rely, nameable answerers, and a layer put out toward *S*. What follows is that a builder's claim of certification cannot be assessed unless it shows which account it answers to, since evidence that meets one account need not meet another (Section 3; Section 5.4).

9.1 The questions the accounts answer, and their verdicts

To compare verdicts, one needs to separate what each account answers. The accounts of this chapter each answer one of three questions. One is what criticism yields. The error-detection view places the value of criticism in the information about errors it adds; Longino, Solomon, standpoint theory, and the ideal of mechanical objectivity each place it elsewhere (Sections 1–3; Table 11.1). The second is what grounds reliance on an unchecked result; what kind of act certification is taken to be (Section 5.1), what grounds the reliance (Section 5.2), and whether that reliance has to be trust (Section 5.3) belong here. The third is by whose valuations the threshold at which a result counts as sufficient for a use is set; Wilholt, Douglas, John, and Elabbar answer it (Section 4). Longino's account also bears on the third question through background assumptions, and Wilholt's on the second through trust (Section 4; Section 5.3). A position can be seen as a combination of answers to the three questions. The position that combines the error-detection view with reliabilism answers the first and second questions, one each, and since neither requires the reliant party's own reasons or *S*'s perspectives, what can be placed inside becomes large. It says nothing, however, about the third question, so what remains concerning the valuations behind thresholds changes with whether Wilholt's or John's account is taken with it. Combining Longino's account with reductionism requires both perspectives and reasons on *S*'s side. What remains as a relation to *S* differs from combination to combination.

Table 11.2 sets out, for each of these accounts, the verdicts of Sections 2–8. It is not a rigorous summary but a simplified overview for taking them in together, and the grounds for each cell are in the section given in parentheses. As in Table 11.1, the verdict in each row is reached by applying an account to artificial configurations its authors did not discuss. The second column names the human premise on which the account's conclusion relied where it held for human research. In some cases the account itself does not presuppose humans and only a rule of thumb relied on the premise; the table says so where that is the case.

Table 11.2. Account by account: the human premise, what can be carried inside, and what remains as a relation to *S* (a simplified overview)

Account	Human premise relied on	What remains when abstracted (inside, on what conditions)	What remains as a relation to *S*	What changes with the composition of *S* (including overlap of *A* and *S*)	If internal checks are better than humans
Error-detection view (Section 2)	none in the account itself; number and organizational separation served as a rule of thumb for independence because critics were formed as separate people (Section 3)	checks that add information claim by claim (can be inside); conditions: showing what information they add and how failures are diagnosed, and keeping independence as producers and evaluators adapt to each other (Section 8)	little on this view alone; taken with Longino's account, what a check counts against is relative to background assumptions (Section 1)	what matters is not human or artificial but shared routes of error; checks by members who share a base tend to err in the same places	applies directly, but evidence that the checks are better is needed (next row)
Reliabilism, combined with the error-detection view (Section 5.2)	none (no human required at the end of the chain)	reliable processes; for justification, that the process is in fact reliable; to show that it is, establishing reliability without the circle of measuring checkers by their own verdicts (bootstrapping)	the recipient's taking up within range; the passage from justification to use (the valuations of Section 4); outside parties who bear costs	where *A* and *S* overlap, checking and its evaluation become one process and tend toward the circle	applies directly to justification; but agreement with human decisions measures fidelity to a practice (Section 7), and beyond the range that can be checked, reliance falls on the handing on of calibration (Section 6)
Longino (Section 3; Section 9.1)	interaction among situated knowers in human research communities	the four conditions (can be met inside); acceptability remains relative to available perspectives, and the conditions include openness to criticism from outside the community	whether *S*'s perspectives enter criticism; data evident to the accepting community; openness to criticism from outside the community	where *S* shares the lineage, the gap narrows without criticism improving; where the community coincides with *S*, the openness points outside *S*	strengthens the criticism already possible, but adds none from missing perspectives
Wylie's extension of Longino (Section 3)	collaboration among communities with different aims, practices, and beliefs (Wylie 2015, 13, author manuscript)	internal diversity alone does not establish external engagement	collaboration beyond the research community, which need not be beyond *S*	where the research community coincides with *S*, relevant external collaborators would be outside *S*	better internal checks do not by themselves supply external collaboration
Solomon (Section 3)	that influences on judgment are distributed differently from person to person; that their distribution, which no invisible hand makes good, is assessed and altered at the level of the community and its institutions rather than in individual reasoning (Section 3)	positions that assess and influence how research effort is distributed (can be placed inside); condition: occupants do not share the influences at issue	placing part of the identification of shared influences with parties who do not share them	what matters is sharing the influences; members of different make can carry this inside too; under majority voting, copies add weight without adding evidence	accurate individual reasoning is not enough; what is assessed is the distribution of influences
Standpoint theory (Section 3; Chapter 10, Section 6)	that marginalization and differences of starting point arise within the power relations and lives of human societies	engagement with the lives of the marginalized, and group consciousness; whether this can be carried inside depends on whom one learns from and inquires with	relations with those engaged with	in mixed societies the relation between human and artificial members can itself be a power relation; artificial societies trained on human records start where human research communities stood	what it asks for is not fewer errors but a different starting point, so accuracy does not supply it
Ideal of mechanical objectivity (Section 3)	distrust of, and distance from, judging individuals; individuals' idiosyncrasies and interests entering results	rule-following procedures (can be met inside)	questions to those who designed the procedures; procedures let decisions be made without seeming to decide	if the artificial configuration designs its own procedures, the question passes to who can check that design	if more uniform, may count as more objective on this ideal, but is silent on shared assumptions
Goldberg, as reported by Koskinen and Dixon (Section 5.1)	that reliance on instruments depends on others' epistemic work, and that social practices entitle users to expectations of designers and manufacturers	certification as the reading of an instrument	dependence on designers and makers, if outside; whether trust in them supports reliance on essentially opaque AI is disputed	if the artificial configuration designs its own checkers, whether its designing part has the understanding and responsibility required to support reliance (an application)	records alone do not establish who can answer
Reductionism (Section 5.2)	no premise specific to humans	track records (can be built inside)	reasons that reach those who rely, and that they can assess	if those who rely share the base, their assessment of lineage can correlate with the artificial configuration's capacities (Section 5.5)	applies only so far as records showing it reach those who rely

Account	Human premise relied on	What remains when abstracted (inside, on what conditions)	What remains as a relation to *S*	What changes with the composition of *S* (including overlap of *A* and *S*)	If internal checks are better than humans
Non-reductionism, minimal form (Section 5.2)	none in the account itself; in human research, findings plausibly could reach those who rely through publication and correction (a premise of organization)	the absence of undefeated defeaters; but without routes, met whether or not there is an error	routes by which findings reach those who rely	if those who rely are inside the artificial configuration, the routes can come inside too	without routes, does not bear on the reliant party's justification
Assurance views (Sections 5.2 and 5.4)	that speakers intentionally assure and stand behind claims; answerability for failure on versions adding a right of complaint	intentional assurance (can be inside if the configuration or its members can give it); answerability where required (providing a base or deploying it is not assurance)	assurance to those who rely; someone answerable to them on right-of-complaint versions	overlap can bring assurance inside if instances can give it and, where required, answer for failure	accuracy alone supplies neither assurance nor, where required, answerability
Necessary-trust accounts (Section 5.3)	trust in responsible designers, maintainers, or investigators sustaining reliance	a party able to bear the responsibility the account requires (can be inside if the configuration or its members can do so)	trust in those parties; Koskinen requires responsibility for the system's workings, Ortmann trust in investigators who establish and transmit reliability	overlap can bring the trusted parties inside; the responsibility required depends on the account, and essential opacity is subject-relative	accuracy alone does not settle whom to trust, for what, or whether the required responsibility is possible
Wilholt: correspondence and the stronger reading (Section 4; Section 5.3)	that producers and those who rely share a sense of valuations; that part of the choices conventions leave is tacit and cannot be fully documented	records of losses and of the reasons for judgments; material for checking correspondence if the expectations of those who rely are ascertained separately	correspondence; on the stronger reading, a relation in which those who rely can object	where a machine produces and humans rely, a question structurally similar to AI alignment; reliers who share the lineage make sharing easier to presume, but those who bear costs can remain outside	accuracy does not settle how heavily losses are weighted, and for whom
Douglas (Section 4)	that thresholds are set by agents (scientists) morally responsible for the consequences of error	separating evidence from the weighing of error costs, and making the weighing explicit	which responsible agent answers for the weighing (Chapter 12, Section 6.3)	the line is drawn by where in the artificial configuration there is something that can answer as a responsible agent	does not settle who answers
John / Elabbar (Section 4)	John: that publication addresses audiences that cannot be identified; Elabbar: that evidence is distributed unevenly among people	applying a fixed high standard uniformly inside where results enter a record used by parties who cannot be identified, and a standard that follows identifiable hearers' where they can be identified (John)	the basic interests of evidence-poor parties (Elabbar); those who bear costs without relying	if *S* is artificial, costs can fall on humans or the environment (Section 4)	accuracy does not settle whose losses weigh heavily

The accounts of Fricker, Lackey, and Williamson of what kind of act certification is are gathered in Sections 5.1 and 5.5.

Longino's four conditions can be applied to an artificial configuration one at a time. As Dutilh Novaes and Dede report, these are idealized conditions that real communities meet by degree (Dutilh Novaes and Dede 2025, §2). As for *venues*, a channel internal to an artificial configuration is public to its components, not thereby to members of *S* outside it. Publicity is also graded: the report accompanying one large language model withheld its training method, among other things, citing competition and safety, had more than fifty external experts it had selected test early versions, and announced a plan to make details available to additional third parties (OpenAI 2023, 2, 12, 45). What those parties receive is known only through the developer's account. *Uptake* requires that the distribution of a community's beliefs change over time in response to the criticism made within it (Longino 2025, §5), not that outputs move in the direction a critic rewards; Section 8 saw that the latter can erode a critic's independence. As for *public standards*, an artificial configuration can make its standards explicit, but Longino treats them as provisional and subordinate to the community's aims of inquiry (Longino 2017, 51–52, 54), so who revises them, against which aims, returns to the question of whose unknowns (Chapter 10, Section 6) and to the question of authority (Chapter 12). As Dutilh Novaes and Dede report, *tempered equality* asks that social position or economic power not determine which perspectives are taken seriously (Dutilh Novaes and Dede 2025, §2). Where someone sets the weights of internal critics from above, the condition's work, keeping social position from deciding which perspectives are taken seriously, can be lost, since the judgment of whoever sets the weights takes the place of position. Where the weights are settled by the accumulated responses of the critics to one another and their verdicts, something that works like social position, some components' verdicts being treated as weightier than others', can arise as well. Those who set or shape the weights need not coincide with those who own the artificial configuration (Chapter 9, Section 5; Chapter 12,

Section 2). Harding adds that the boundary of those counted as qualified critics can itself protect shared assumptions (Harding 2015, 6). As Section 3 set out, Longino's definition requires the four conditions and data evident to the accepting community at the time. Where *S* is identified with that community, the data need to be evident to *S*; where they are not identified, a further question is whether *S*'s perspectives enter its criticism. Wylie's extension adds active collaboration beyond the research community. Applied here, this points beyond the artificial configuration where it forms that community, but beyond *S* only where the community coincides with *S* and relevant external collaborators are available.

9.2 What comes into view through the accounts

First, improving internal criticism can, on some accounts, fail to strengthen certification for *S*, and can even weaken it. At first sight, the more internal checks there are and the better they work, the better certification would seem to become. On the error-detection view combined with reliabilism, that is so. But three mechanisms, each separately, undo this expectation. One is that, on Longino's account, assumptions everyone shares survive a structure that filters out idiosyncratic ones, so the better internal criticism works, the more entrenched they can become (Section 3); this works toward weakening certification. The second is that, on the minimal form of non-reductionism, if there is no route by which findings reach those who rely, the condition that there be no undefeated defeater is met whether or not the result contains an error, so however much internal checks are strengthened, the reliant party's justification does not change (Section 5.2); this does not weaken certification, but neither does it strengthen it. The third is that evaluation ceases to be visible to *S*. The means of improving internal criticism can include building checks continually into the formation of candidates and making them one with production (Section 7). On accounts that require a return to *S*, such as Longino's account, assurance views, and reductionism, the further such means go, the fewer the occasions on which evaluation is visible to *S*, on which it can be objected to, and on which members of *S* keep up their capacity to evaluate (Section 7). Since the three mechanisms are distinct, so are the remedies: the first needs differences that come from outside the artificial configuration and what it shares, the second routes by which findings reach, and the third occasions for members of *S* to take part in evaluation. Taken one step further, a less accurate check open to *S* can give more to the grounds of *S*'s reliance than a more accurate check closed to *S*. For on reductionism the reaching of reasons, on non-reductionism that requires sensitivity to defeaters the routes, on Longino's account criticism from *S*'s perspectives, and on assurance views someone who stands behind the result are at issue apart from the accuracy of the check. This is a comparison not of nearness to truth but of grounds for reliance, and it does not hold on pure reliabilism.

Second, built-in differences have the process that chooses them as a common source. It is tempting to think that if the problem is that critics are copies of one base, the remedy is to make different critics. But if Goldman's warning about sources that merely reflect a common third party, as reported by Guerrero (Section 2), is extended to the choosing of differences, then where one process chooses which differences to make, that choice itself can become a source common to all the critics made (Section 3). The question of independence thus moves up a level, from the critics to the process that chose their differences. Chapter 10, Section 4 reached the counterpart for allocation: a configuration able to generate different lines can still discard them together through the same error if one criterion judges their survival. It stops where the way differences are chosen is open to differences the choosing process itself does not anticipate, for example where the choice is divided among other parties or differences take shape through a process that no one sets. These are two of the ways Chapter 10, Section 5 found a function can be supplied once it moves inside a configuration, and they carry what was found there: whoever sets a parameter determines part of what the configuration can learn, and a process no one sets as policy may have no single author, so that who answers for it becomes the question of Chapter 12, Section 6.3. Applying Harding's requirement, the evaluators who shaped the base system, and their assumptions, also come within what an account of the results has to cover (Section 3). The same form appears for differences in valuation. Irzik and Kurtulmus use the diversity of researchers as a reason to think that research reflects the expectations of the whole public rather than of a segment. Applied here, in an artificial configuration whose range of valuations is fixed by how its members are produced, that range comes not from members' lives but from the settings of the process that produces them, so whether diversity works as that reason depends on whether there is a channel, such as a hybrid forum, that brings in *S*'s expectations (Section 6). Whether the differences lie in routes of error or in valuations, where differences are made the question moves to the process that chooses the differences and to routes that check its choices. That is why who may make members and set the distribution of perspectives is not only a question of authority (Chapter 12, Section 5) but also an epistemic question.

Third, when *S* shares a lineage with the artificial configuration, "certification for *S*" becomes easier to satisfy without getting better. Ordinarily, there seems to be nothing wrong with artificial societies of one kind certifying results of the same kind. Yet taking three points together, a builder can make the claim easier to satisfy by naming, as the society for which results are certified, an artificial society that shares a lineage with its own configuration. On Longino's account, if members of *S* share the instances' primitive epistemic organization, the gap between acceptability relative to the artificial configuration and acceptability relative to *S* narrows although criticism has not improved (Section 3). The shared sense of valuations that stabilized reliance on Wilholt's account also becomes easier to presume (Section 4; Section 5.5), and where the members of *S* who rely also stand inside the artificial configuration, the routes by which defeaters reach them come inside it too (Section 5.2). The substance of the question thus shifts from whether certification is good to who *S* is. What remains differs by account. On the error-detection view combined with reliabilism, certification for *S* is largely absorbed into the correction of internal errors. What remains is a condition of inquiry, checks with error routes of their own and defeat relations (Chapter 12, Section 5), the reliability of the way those who rely take up the certification, where they stand outside the artificial configuration (Section 5.2), and relations with those outside *S* who bear the costs of error without relying (Section 4). On Longino's account, shared assumptions can still evade criticism. Wylie's extension gives a further reason to engage relevant external communities; these would lie outside *S* where the research community coincides with *S* (Section 3; Section 9.1). On assurance views, so far as those outside *S* rely, someone is needed who stands behind the certification toward them.

Fourth, independence can decay through use in selection alone, without anyone deceiving. The natural expectation is that a check once independent stays so unless someone deceives or the rules are rewritten. But on Manheim and Garrabrant's classification, in the regressional case, in which selecting on an imperfect proxy also selects for the difference between proxy and goal, and in the extremal case, in which selection pushes into regions where the relation no longer holds, proxy and goal come apart through selection alone, even with no one responding to the measure (Section 8). Carried over to research, where a check is imperfect relative to its target and producers are trained or selected on its verdicts, outputs that pass where the check is insensitive can come to be favoured (Section 8). Where they are, the probability that the check exposes the errors the selected producer now makes, its severity for that producer, falls, and with it the error-detecting information the check adds per claim; on the extension of Goldman's condition in Section 2, which takes independence to be that information, this is a loss of independence. It differs from a check's finding less because the producer makes fewer errors, which leaves what its passing verdict indicates undiminished. An evaluator used for selection can therefore tell less about producers the more it is used (Section 8). Going further, independence is less a property a check has at a time than a property maintained relative to how the check is used. Protecting rules from rewriting can help maintain a contribution, but neither establishes nor guarantees it (Sections 1 and 8). What can help keep it is not so much forbidding rewriting as keeping the check out of selection.

Fifth, agreement with *S*'s present standards is no evidence that the standards have not moved, and Wilholt's correspondence reading can therefore come to be satisfied without the weighting's being revised. The first thought is that whether an artificial configuration's valuations fit *S* can be checked by whether they agree with *S*'s present standards. Yet applying the arguments of Russell, Ratti, and Commelin and colleagues, a standard that moves with coupling can be *S*'s own standard (Section 8). Laid over the two readings of Section 4, correspondence, the weighting's matching the valuations those who rely expect, can be satisfied not only by revising the weighting but also by the expectations of those who rely moving. The stronger reading, on which those who rely can object, can guard against this, but only so far as the capacity to object is not itself shaped by the artificial configuration, and that capacity can decline from *S*'s side with nothing changing on the side of the artificial configuration (Section 6). Here detection and decision are to be kept apart. Detecting a movement against earlier standards can be done inside the artificial configuration as well, through preserved reference points or evaluations built from *S*'s past judgments. But whether to regard the movement as improvement or replacement, and whether to accept it, is for *S* to decide so long as the standard is *S*'s (Chapter 12, Sections 2 and 5), and detection inside does not take the place of that decision.

Sixth, what has been said to remain as relations to *S* presupposes that *S* has the capacities to carry them. The relations in the fourth column of Table 11.2 each require some capacity of *S*: reading defeaters, criticizing from *S*'s perspectives, assessing the lineage of checkers, understanding explanations, and the like. Assessing lineage can require technical expertise of another kind, and the capacity to check can decline from *S*'s side as members cease to gain experience (Section 6). If the explicit stage of evaluation disappears, the places for keeping up that capacity diminish too (Section 7). On the other hand, on the accounts of Hardwig, Goldman, and Anderson, some questions about relations, such as who directs the artificial

configuration, which ends it serves, and whether it answers to criticism, can be asked without competence about content (Section 6). The part of this chapter's outlook that says something "remains as a relation to *S*" is therefore conditional, for relations that require technical judgment, on *S*'s keeping the capacity to carry that judgment. The condition "more than nominal" that Section 6 stated for standing extends to each of the remaining relations.

Seventh, critics that add the same information about errors need not have the same value for inquiry. On the error-detection view, if two arrangements of critics add the same information about errors, their value is the same. Applying Kuhn's account, the matter does not end there, since diversity of judgment can have value, without adding any new route to error, in spreading across a group the risk that introducing or supporting a novelty entails (Section 3). If so, an arrangement that preserves minority judgments and one that votes them out by majority, even if they add the same information, need not leave the same room for new theories to be developed and tried while their support is still thin. Improving the accuracy of certification and keeping room for new theories to grow can then come apart: making artificial critics' judgments uniform, or gathering verdicts into a single majority vote, for instance, may improve the first while weakening the second. This is, however, a conditional conclusion on Kuhn's account, and whether produced variation does the work of the spread of individual differences on which Kuhn's argument relies remains open (Sections 3 and 8). Whether such a loss occurs in actual collections of artificial critics also remains an empirical question. The chapter has met two further routes by which the arrangements it examines can narrow the room for the new: criticism applied to candidates still being formed can turn away what has little support yet (Section 7), and, on Kuhn's account, evaluators trained on past decisions can score as a deviation a candidate whose acceptance would require revising the standard (Section 8). The three concern different things, when criticism is applied, how evaluators' standards are formed, and how verdicts are gathered, so keeping room for the new in one respect does not keep it in the others.

Applied to criticism and certification, the sorting of Chapter 9, Section 6 draws its lines as follows. Premises about human realizers can fail: that critics are formed as separate people, so that their number traces independence (Section 3); that part of the judgment behind conventions is tacit and cannot be fully documented (Section 4); and that those who check have limited time and attention (Section 7). So can premises about how research has been organized, which can fail even while its realizers remain human: that data and records of process are in researchers' hands (Section 5.3); that prepublication review is a gate allocating scarce editorial attention and space, whose passing serves as a signal for allocation (Section 7); and that dependencies are tracked only in part by citation (Section 8). That producers and those who rely share a sense of valuations can be read as a premise that mixes the two kinds, premises about human realizers and premises about the organization of research (Section 4). Where these fail, the problems they addressed remain in changed form. Independence becomes something made (Section 3); reliance on researchers' reports becomes reliance on the maintenance of records, on their boundary, and on the capacity to verify within it (Section 5.3); gates remain where checking is costly (Section 7); and even where dependents can be traced, confirming whether they still hold becomes new criticism (Section 8). Conditions of inquiry remain but can be met inside: informative defeat relations and supported diagnoses of failure (Sections 1–2) and the unavoidability of proxies (Chapter 9, Section 4), among others. The structure of testimony in which the reliant party does not have the testifier's reasons also holds whatever testifies (Section 5.3), but, like the relation over valuations (Section 5.5), whether it lies inside or outside depends on where the one who relies stands. Relations to *S* that involve the other party's participation or recognition cannot be completed from one side, and what goes into them differs by account, as in the fourth column of Table 11.2. On the error-detection view combined with reliabilism, much of what certification rests on belongs to the conditions of inquiry (Chapter 9, Section 6); on other accounts, more of it belongs to relations to *S*. Of the two senses of "inside" stated at the start of this section, what remains as a relation to *S* is here too mainly the acknowledgment.

The consequences seen in Section 7 can also be reconsidered from this sorting. In human review, one act, a named person's reading a manuscript before publication, plausibly produced at once the detection of errors, the visibility of evaluation to the community, a nameable judgment against which objection could be directed, someone who answered (editors and referees), and occasions for members of *S* to gain experience of evaluation and keep up their capacity (Section 7). An artificial configuration can produce error detection alone, without the others. That several consequences came out together in Section 7 when the explicit stage of evaluation disappears is plausibly because they arose from the same occasion. Which of them matters differs by account: on the error-detection view only the first; on Longino's account the visibility of evaluation and venues; on assurance accounts someone who vouches for the result and, on those that add a right of complaint, answers for it; on necessary-trust accounts someone who answers; on the accounts of Hardwig and Anderson *S*'s capacity to ask about

relations; and for Chapter 12, Section 6, which asks about revisability, traces that can be objects of objection. What each was tied to differs as well. Visibility and objectability were tied to the stage of evaluation, so they can be rebuilt in the layer put out toward *S* seen in Section 7, such as records, public venues, and routes that receive objections. Someone who answers was tied not to the stage but to persons, so adding such a layer does not produce one (Porter's remark, Section 3, that quantification is a way of making decisions without seeming to decide points the same way); on assurance views, someone who can intentionally give assurance is needed. The capacity to evaluate was kept up by members of *S* actually evaluating, so a layer that only returns results does not bring it back; occasions for members to take part in evaluation are needed. What is to be asked of an artificial configuration that unbundles them is therefore which of them it rebundles, and by what, including whether what the rebundled properties rest on, such as the objections a route receives, comes from outside the configuration (Chapter 12, Section 9).

Of the four goods of Chapter 9, Section 2.3, the three that bear on this chapter stand as follows. Error detection across boundaries turns, on the error-detection view, not on organizational boundaries but on whether a check has an error route of its own, and so can be supplied inside an artificial configuration (Section 2). Keeping it over time, however, needs some means of preserving the check's contribution, such as checks not used for selection (Section 8). On Longino's account, shared background assumptions can evade internal criticism (Section 3). Standing for dissent, as distinct from the epistemic work of dissent, would be held inside an artificial configuration only by parts that hold it on the account adopted (Section 3). As for boundaries across which accounts had to be given, at which one party had to explain or justify what it had done to another, a record makes it possible to give an account, but the entitlement to demand one does not arise from records. Where that entitlement comes from also differs by account. Applying the assurance views of proponents who add a right of complaint, those who rely hold a position from which they can demand an answer about errors; on the stronger reading of Wilholt's account (Section 4; Chapter 12, Section 1.4), those who rely can object to the weighting. On pure reliabilism, by contrast, the reliant party's justification needs no account. Whether this boundary remains under an artificial configuration therefore divides by the account adopted. Who holds the entitlement is examined in Chapter 12, Section 6, together with the fourth good, responsible parties.

9.3 What is suggested from there

What follows extends the argument so far beyond the range the argument reaches. One suggestion is that which account of social knowledge and testimony is adopted may come to matter in practice. If an artificial configuration can set each property separately, which properties a builder gives it amounts to choosing which accounts' requirements to meet. Generalized, designing an artificial configuration may be making, within the design and without anyone making it explicit, the choice that Chapter 2, Section 6 left undecided. What is suggested is that who makes it, and through which routes, may join the questions of authority in Chapter 12.

A second is that, where *A* and *S* overlap, what "certification for *S*" refers to may shift. Section 4 held that the work of thresholds is needed so long as someone uses results without checking them again. Extending this logic, even where *A* and *S* overlap, so long as members use one another's results without checking them again, the work of thresholds remains as work internal to the artificial configuration. If so, on the position combining the error-detection view with reliabilism, certification is largely absorbed into internal correction and the internal setting of thresholds, and certification in the sense of giving others reason to rely can come to be relative to another society that relies on the results. On Longino's account, shared assumptions can remain unexamined, and since the conditions of acceptability include openness to criticism from outside the community (Section 3), the absorption is not complete where the artificial configuration is itself the community; on Wylie's extension, collaboration beyond the research community remains a further condition. On assurance views, someone is needed who stands behind the results toward the other society. The larger the overlap, the more the question of certification plausibly moves from "how is it for *S*?" to "for which society?"

A third is that, if having passed internal checks is used as a proxy for certification, the logic of Section 8 may extend to certification itself. The strength and pass rates of internal checks are easy to measure, while many of the relations to *S* in the fourth column of Table 11.2 are hard to measure. Suppose builders or users choose and revise artificial configurations by the former, and that high internal-check scores can be had without the relations to *S*, so that the scores overstate certification where those relations are lacking. Then, as in Manheim and Garrabrant's regressional case, the difference between proxy and goal, here the lack of relations to *S*, can be selected along with them. If so, the first conclusion of Section 9.2 strengthens

from a comparison at one time into a tendency over time.

A fourth is that there may come to be no one able to carry the remaining relations. Extending the sixth conclusion of Section 9.2, where three things coincide, an artificial configuration's becoming more competent than the members of *S*, evaluation's dissolving inside, and members of *S* ceasing to gain experience of evaluation, relations that require technical judgment can remain in form as relations to *S* while no member is able to carry them. Koskinen held that essential opacity is relative to subjects (Section 5.3). Applying this, a gap in which no one can be responsible can arise not only from how a system is built but also from a difference in capacities on *S*'s side. Since some questions about relations can be asked without competence about content, however, this gap does not extend to all relations. What is suggested is that separating the relations that require technical judgment from those that do not can be a clue to what it would be enough for *S* to keep.

A fifth is that independent checks may behave like a resource that is depleted by use. On the fourth conclusion of Section 9.2, checks not used for selection are less exposed to the decay of evaluators used for selection. Yet if such checks too are consulted repeatedly for certification, and their results are reflected in which producers are kept, each consultation makes them more a part of selection. Put in engineering terms, a test meant to be held out from training, consulted many times and used for choices, moves into the same position as a measure used for selection. If so, independence is not something secured once, but something that is depleted by use and runs out unless replenished by new measurements or by evaluators of different make. Taken a step further, there may be a tension here with learning from criticism. Effective criticism changes the producer. Once the producer corrects its errors in response to criticism, the same check may find less in the changed producer. That alone does not weaken the check, since it may find less because fewer of the errors it can detect remain; the check is weakened where, in being changed, the producer has also come to pass where the check is insensitive (Section 8), and a falling number of detected errors does not by itself show which has happened. And keeping checks out of selection (the fourth conclusion of Section 9.2) leaves by itself the problem of how the results of criticism are taken into the producer: taken in, the check can come closer to being part of selection or training; not taken in, criticism does not improve the producer. If so, independence is less a property of a component to be preserved than, in part, a relation to be reassessed each time criticism has taken effect. How far new checks must keep being added, however, would depend on the domain and on the range of errors a check tests: a checker that lets no invalid proof through does not lose its force for formal statements when the producer learns to satisfy it (Section 8). Replenishing has costs, and the resources are finite, so maintaining certification over time becomes a question of allocation: how much independent checking to spend on which results, and how much to replenish (Chapter 10, Section 1). And who may decide that becomes a question of decision for Chapter 12.

9.4 Questions passing to Chapter 12, and a qualification about understanding

The parts of this section's picture that accounts of epistemic relations alone cannot settle pass to Chapter 12 as questions of authority. They include at least the following. One is who may decide for which functions implicit performance suffices and for which being made explicit to *S* is required (Chapter 12, Section 6). The second is who constitutes *S* and who may speak for it, which includes what counts as "outside" where *A* and *S* overlap (Chapter 12, Sections 1.3, 2.1, 3.3, 3.6, and 5). As Eigi reports, Longino leaves open whether the community comprises the producers of a kind of knowledge or everyone potentially affected by it (Eigi 2016, 136). The third is who may set and revise standards and the weights of error, and whose costs matter (Chapter 12, Sections 1.3, 1.4, 3.5, and 6.3); the fourth, who may make members and set the distribution of perspectives (Chapter 12, Section 5); and the fifth, who holds the entitlement to demand accounts and to object (Chapter 12, Sections 1.4, 3.3, 3.4, and 6.2). If the fifth suggestion of Section 9.3 holds, who may decide where independent checking is spent and when it is replenished is added.

Finally, the outlook of this section concerns the case in which the achievement sought is knowledge. In inquiry directed at understanding, certified explanation adequacy, certified attribution of understanding, and the recipient's actual understanding are distinguished, and internal criticism can bear on the first two without realizing the third (Section 5.6). Whose understanding is sought depends on the construction target, not on where the certification procedure runs. If the members of *S* are artificial and can share representations directly, acquisition may not pass through testimony, but whether the recipient then understood divides by the account of understanding adopted (Section 5.6).

Chapter 12 — Who May Decide, and Who May Revise?

Standing, Revisability, and Relations to Parties Outside

Can authority be located inside an artificial configuration?

1.1 The question, and what the preceding chapters pass on

Part V asks which of the relations surrounding research can move inside an artificial configuration. Chapters 10 and 11 asked this of the allocation of inquiry and of criticism and certification (the relations through which a candidate result is exposed to error-detecting scrutiny and through which others are given reason to rely on it). This chapter takes up the one relation that remains: what Chapter 9, Section 2.1 called authority, the standing (an entitlement, held in relation to those subject to a decision, to propose, decide, object, refuse, or revise) to decide and to revise the terms of allocation and of criticism and certification. Even if an artificial configuration can take on allocation and criticism inside itself, who may decide and who may revise their terms remains a question apart from how well the configuration performs. That someone may be human, may be the artificial configuration itself, or may be divided between them. If human, the answer differs according to whether it is the members of *S*, those who build and provide the configuration, or those who bear its consequences; if the configuration itself, the question is which part of it decides. And performing well does not by itself show that it may decide. The central question of this chapter is whether this authority can be located inside an artificial configuration and, if so, on what grounds. In terms of the example of Chapter 9, Section 1, it is also the question under what conditions the following claim holds, made by those who build a system that generates a hundred instances to pursue different lines and shifts computing resources according to checks on them (hereafter the builder): that the system may decide, on behalf of others, which of *S*'s questions its instances pursue. This is the third of the builder's three claims named there. And working through this question ground by ground is, in turn, the means by which the chapter pictures what happens to research once authority moves inside: which questions are pursued, what *S* comes to know, and who can correct what (Section 8.4).

The case the chapter's question chiefly has in view is one in which AI does research that includes allocation, criticism and certification, and authority. In human research these relations have often been carried by different parties and contested among them (Chapter 9, Section 2). Inside a single artificial configuration they become variables: who criticizes, by what standard, how lines are assigned, and who decides need not be values that someone sets once and fixes, but can be made by the configuration itself and kept under revision as it pursues its inquiry. This chapter asks what authority means in that case, who may decide and who may revise when the configuration can itself revise the terms of its own research.

Before that question, it helps to recall what the two preceding chapters found that this chapter uses. Chapter 10 found that an artificial configuration can take over the allocation of inquiry while gathering many searches into one configuration (integrating them) does not by itself enable it to know in advance which lines will succeed, and that, whoever allocates, allocation still steers by signals, indicators such as citation counts or acceptances that are used in place of goals, whose adequacy depends on the goals they stand in for. What counts as good allocation also depends on the purpose assumed (Chapter 10, Sections 1 and 4). Chapter 11 found that properties that in human research were tied together by the realizers and by the organization of research, such as differences among critics' routes of error, someone's answering for a result, and a sense of valuation shared by producers and those who rely, can each be set separately in an artificial configuration, so that the accounts that track these properties reach different verdicts on the same artificial configuration as to whether certification for *S* can be organized inside it (Chapter 11, Section 9). The builder's claim of certification, the second of the three, therefore cannot be assessed unless it shows which account it answers to. That chapter also found that on some accounts improving internal criticism can even weaken certification, that independence can decay through use in selection alone, without anyone deceiving, and that agreement with *S*'s present standards is no evidence that they have not moved (Chapter 11, Sections 8 and 9.2). And where configurations handle more of the evidence than the members of *S* can, *S* can come to defer to them on content, while whether it then keeps standing is not settled by the accounts examined there of knowledge or of deference, the acceptance of others' judgment on content (Chapter 11, Section 6). Both chapters thus end before the same question. Which

of *S*'s unknowns are pursued, which errors matter more than others, and how these may later be revised can be settled by someone's decision or by processes that no one sets as policy (Chapter 10, Section 5). Where they are decided, the question is who may decide them; where they are not, the question is who answers for the setting and who may change it.

Chapter 11, Section 9.4 passed to this chapter, as questions of authority, the parts of its picture that accounts of epistemic relations alone cannot settle. One is who may decide for which functions implicit performance suffices and for which being made explicit to *S* is required, which this chapter takes up in Section 6. The second is who constitutes *S* and who may speak for it, which includes what counts as "outside" where *A* and *S* overlap; this chapter takes it up in Sections 1.3, 2.1, 3.3, 3.6, and 5. The third is who may set and revise standards and the weights of error, and whose costs matter (this chapter, Sections 1.3, 1.4, 3.5, and 6.3); the fourth, who may make members and set the distribution of perspectives (this chapter, Section 5); and the fifth, who holds the entitlement to demand accounts and to object (this chapter, Sections 1.4, 3.3, 3.4, and 6.2). That chapter added two further questions as suggestions: who, and through which routes, chooses which accounts' requirements the properties a builder gives a configuration are to meet, and who decides where independent checking is spent and when it is replenished (Chapter 11, Section 9.3). Section 8.5 gathers how this chapter's material bears on these questions.

This chapter keeps two questions apart. The first asks what the right ordering of ends and the right weighting of error costs are, that is, what gives reason to take an ordering to be correct. The second asks who may impose an ordering and a weighting on others, and who may change them, that is, who is rightly entitled to impose that ordering on others. The first question does not first arise in this chapter. Chapter 10 took up, within allocation, the question which of *S*'s unknowns are worth the effort they require, and Chapter 11, Section 4 took up, within certification, the question which errors weigh more than others; both chapters found that what counts as good allocation or good certification depends on the purposes or valuations presupposed. Chapter 8, Section 3, in treating relevance to *S*, and Chapter 5, Section 4, in treating the judgment whether a line is worth pursuing, took up other faces of the same question. And Chapter 9, Section 3 distinguished four senses in which such value is said to be relative to *S*. This chapter gathers these into one question and separates it from the second question, who may impose the answer on others. Section 2 relates them to a further distinction among epistemic warrant, authorized power, and effective capacity. The two questions are kept apart because they respond differently to an artificial configuration: the first can, on some accounts of value, be answered inside a configuration, while the second is answered not by accounts of value but by accounts of legitimate authority. The standing at issue in the second question is a normative entitlement: it is not competence, nor the formal assignment of a right to decide, nor the power actually to make decisions (hereafter effective decision-making power) (Chapter 9, Sections 2.1 and 5). What grounds it differs with the account adopted. Some accounts of legitimate authority allow that established good results, rather than claimed competence, can ground it (Sections 1.4 and 3.1). Among accounts of justice, some ground it in moral status, and moral status in features of beings such as welfare or agency, and others in relations among beings such as power or cooperation (Section 5). Being a party on grounds such as these, which accounts of justice name, is called below being a party in one's own right.

The chapter proceeds as follows. Section 1.2 examines, account of value by account of value, whether the right ordering of ends can be fixed inside a configuration. Section 1.3 asks to whom decisions have to be justified, that is, the range of those parties. Section 1.4 asks, against a map of accounts of legitimate authority, whether competence confers standing. Section 2 divides the question of who controls research into questions about particular decisions and particular rights, and brings out that whether authority can be located inside is asked decision by decision. Section 3 states, ground by ground, whether authority can be located inside and, if so, under what conditions. Section 4 isolates, for the case in which *S* is human, standing remains on its side, and a configuration holds effective decision-making power, the routes by which that standing can become nominal and the conditions under which it does not. Section 5 brings out that where *S* is artificial the relation of authority remains as a relation among members, and that where members can be made, who makes them is itself a decision of authority; it touches on the case in which human and artificial members are mixed, and asks what remains at a limit with no other parties. Section 6 asks what keeping authority revisable requires wherever authority is placed, and Section 7 what it means on each ground when a sequence of delegations, each defensible at its step, removes the capacity to revise. Section 8 gathers these in light of the central question and draws from it a picture of what research looks like once authority moves inside (Section 8.4), and Section 9 gathers, across Part V as a whole, what may then happen to research in the case set out above. Most of the authors drawn on in this chapter do not discuss artificial configurations.

1.2 Can the ordering of ends be fixed inside an artificial configuration?

Whether the first question can be answered inside a configuration depends on the sense in which an account of value makes the ordering of ends relative to *S*. Chapter 9, Section 3 distinguished four: good for its members, serving ends that *S* has, actually valued by its members, or what a procedure or deliberation among its members would choose. The discussion below takes them up in three forms. If some claims about which ends matter are true or false independently of anyone's endorsement (hereafter the objectivist form; Kitcher, discussed below, doubts that an objective account of well-being independent of preferences can be defended), an inquiry could in principle get them right, and a sufficiently capable configuration might do so; claims about ends could then, like claims of fact, be checked against routes that would show them wrong if they were wrong (the defeat relations of Chapter 11), though whether any such route reaches claims about ends is contested: on the asymmetry Chapter 8, Section 3.2 took up, the world corrects what an inquirer believes but does not thereby fix what it counts as worth finding, and how cleanly the two come apart depends there on the account of evidence. If the right ordering is constituted by what the members of *S* would endorse under suitable conditions (the endorsement form), it cannot be fixed without access to their viewpoints. If it is whatever a fair procedure among the members of *S* actually produces (the procedural form), as on what Tom Christiano and Sameer Bajaj call the constitutive interpretation of Rousseau's general will, on which the results of a properly organized democratic procedure are the general will because they emerge from it and not because they track a procedure-independent truth about the common good (Christiano and Bajaj 2024, §3.2.1), it cannot be fixed without that procedure. The three forms do not correspond one to one to the four senses of Chapter 9, Section 3. The objectivist form is the case in which, for instance in the first sense, what is good for the members is fixed independently of their endorsement. The endorsement form is close both to the third sense taken under idealized conditions and to the part of the fourth that concerns idealized deliberation. The procedural form is the fourth sense taken strictly for an actual procedure (Chapter 9, Section 3). And in the limiting case built as a thought experiment at the end of Section 5, where there is only one artificial configuration, no one else owns or has a claim on the resources it uses, and no other being with welfare or moral status is involved, the only agent whose ends are in question is that one, and its good is taken to be fixed by its own actual preferences; if those preferences come as a complete ordering, the first question reduces to their internal coherence. The procedural form is an account of value, on which a procedure constitutes what is valuable. It is distinct from pure proceduralism about legitimacy, which, as Fabienne Peter reports, places all the normative weight on a fair procedure because no shared standard for assessing the quality of outcomes is available (Peter 2023, §4.2). As the term is used here, pure proceduralism takes the procedure to fix what is legitimate, without claiming that it constitutes what is valuable. The account of value answers the first question; pure proceduralism answers the second. The answer to the first question inherits the choice among these accounts.

The answer also inherits a further dispute, over whether the right ordering need be complete, ranking every pair of options as one better than the other or the two equally good. Of cases in which one option is better in some respects and another in others, Ruth Chang distinguishes three established reactions: that one is better or the two are equally good, although it may be hard or impossible to determine which; that neither holds and the options are incomparable; and that it is indeterminate which holds, because the comparatives are vague (Chang 2002, 659–660). Against all three Chang argues that, in at least some such cases, the options can be comparable although neither is better and they are not equally good, standing in a fourth relation the paper calls being on a par, and that how such a case is to be resolved is a substantive question, not one that can be settled by arbitrary stipulation, as Chang takes a borderline case of a vague predicate to be (Chang 2002, 661–662, 683–686). The claim concerns what Chang calls a context-free sorting; the paper allows that in some practical contexts a resolution may properly be stipulated, or the respect of comparison changed (683 n. 28, 685 n. 31). In a later paper Chang holds that where the reasons given to an agent run out, as when its options are on a par, the agent can create a further reason for itself by putting its agency behind a consideration, taking it up as one that it itself gives weight to, and so make one option the one it has most reason to choose; such reasons cannot overturn what the given reasons favor (Chang 2013, 177–181). Chang prefers to ground given reasons in normative facts (177), so the position is compatible with the objectivist accounts above. The 2002 paper also confines its argument for parity to comparisons in which the relevant respects can be balanced against one another, and leaves room for incomparability where, for example, states of affairs are compared for social well-being by a Pareto rule, which ranks one state above another only where it makes someone better off and no one worse off, because different people's well-being is taken to be incomparable (Chang 2002, 676–677).

Brought to the first question, these positions bear in different ways on the objectivist form above, on which claims about ends are true or false independently of anyone's endorsement. On the first reaction, the right ordering is complete even where no one can determine it, and that form stands as stated. On the others, a configuration that found every comparative fact about *S*'s ends could still find some of them unordered: incomparable, so that, on the view Chang shares, justified choice between them fails (Chang 2002, 665); indeterminate, so that, as Chang describes accounts of vagueness, any admissible resolution may be stipulated (684); or on a par, so that, on the account in the later paper, what completes the ordering among them is a commitment rather than a further finding. The single-agent limit described above is the case in which such a commitment raises no further question. Chang argues that preferring one option, preferring the other, and being indifferent between them do not exhaust the attitudes a rational agent can have toward two options (Chang 2002, 666), and where a single agent's ordering is incomplete in this way, a commitment that completed it would be the agent's own and would give reasons to it alone (Chang 2013, 181). Chang's account concerns an agent's reasons for itself and does not say whose commitment, if anyone's, could complete the ordering of a society's ends. Asked of *S*, that question, like the questions who may stipulate a resolution and who may choose where justified choice fails, is close to the second question, who may impose an ordering on others. Chang's own limit applies here: where *S*'s ordering combines its members' ends without comparing them across persons, as a Pareto rule does, the argument for parity does not show that they are on a par.

Philip Kitcher's ideal of well-ordered science comes close to the endorsement form distinguished above. Finding it implausible to divorce what is good for a person from that person's own preferences, and having no solution to the problem of aggregating individual well-being into collective well-being, Kitcher, as Arnon Keren reports, declines to give an objectivist account of the collective good that inquiry is to promote, and starts instead from individual preferences and relates the individual good to the collective good within a framework that takes democratic ideals for granted (Keren 2013, 236, 240; Eigi 2016, 66). As Keren reports, Kitcher rejects allocation by majority preference, calling it "vulgar democracy" and warning of "a tyranny of the ignorant", and proposes instead that inquiry coincide, in the assignment of resources to projects, in the moral constraints on how investigations are pursued, and in the application of results, with what ideal deliberators representative of the distribution of viewpoints in the society would choose once their preferences had been tutored by information about the significance of possible inquiries and by one another's needs (Keren 2013, 236–237; Mattarollo 2022, 14–15; Eigi 2016, 66–67). The proposal is presented as a standard against which actual scientific institutions and their agendas are to be measured (Keren 2013, 236–237); in a later statement Kitcher calls such ideals "diagnostic tools", pointing to current flaws and ways of addressing them (Kitcher 2024, 241). On this view the standard is relative to *S* and to how its members' viewpoints are distributed; taken to configurations, it implies that a configuration that lacked access to that distribution would lack what the standard refers to. The standard is also counterfactual. As Longino notes, Kitcher does not require that such deliberation actually take place, only that the decisions actually made be the same as those it would yield (Longino 2025, §6). What those results are is fixed, moreover, not by endorsement alone but by a procedure among the ideal deliberators, consensus or, failing that, a vote, among parties who approach one another's wishes sympathetically (Eigi 2016, 66–67; Mattarollo 2022, 15 n. 4); a configuration that estimated what the standard requires would have to estimate that procedure as well. Later sections take up other parts of Kitcher's account where they bear on a different question: the comparison of ways of setting research agendas (Section 3.3), the division of roles inside the idealized procedure (Section 4), and the account of how neglect can become entrenched (Section 7).

Mixed positions are also available. Elizabeth Anderson, following Henry Richardson, takes the criteria of success for democratic decisions to be partly external, since a law can fail to solve the problem for which it was drafted, and partly internal, since whether a problem is of genuinely public interest, and which consequences are acceptable, is determined in part by procedurally fair decision-making (Anderson 2006, 11). If this is right, the first question would divide: whether a line of inquiry advances a given end could in part be assessed inside a configuration, but which ends are *S*'s to pursue, and which consequences of pursuing them are acceptable, could not be fixed without the procedure.

The first question, then, can in principle be answered inside a configuration on the objectivist form, but not, on the endorsement form, without access to the viewpoints of *S*'s members, or, on the procedural form, without the procedure. On Anderson's view, part of the answer, whether a line of inquiry advances a given end, can be assessed inside, while the rest cannot be fixed without the procedure. And on Chang's account, even when every comparative fact has been found, completing the ordering among ends on a par may take someone's commitment; whose it may be is close to the second question, taken up in Section 1.4.

This distinction lets one consider a further step about artificial configurations. The line between what can and cannot be answered inside is drawn not by whether the one asking is human or machine, but by whether an account of value takes the right ordering to be constituted by an actual procedure or by endorsement under suitable conditions. On the latter form, if what the members would endorse under suitable conditions can be estimated well enough, the estimate could ground an answer to the first question, so access to the viewpoints of *S*'s members could in principle be achieved by modelling the members well. But as Chapter 8, Section 3 and Chapter 11, Section 8 found, an artificial configuration can move *S*'s evaluations and standards themselves through its interaction with *S*. If so, a configuration that fixes an ordering by estimating the members' endorsement may be unable to tell, from the agreement itself, whether agreement arose because the estimate was right or because it moved the members' endorsement. On the endorsement form, the difficulty of answering inside may lie less in being unable to reach the members' viewpoints than in the difficulty of separating reaching them from shaping them. Put otherwise, *S*'s agreement here plays the role of a check on the ordering of value, and because that check is formed in interaction with what it checks, it can lose its independence in the same way that, in Chapter 11, Section 8, an evaluator used to select producers could. If that is so, where an ordering is entrusted to an artificial configuration, some way of checking other than *S*'s present agreement, for instance keeping as a reference point judgments *S* expressed before interacting with the configuration, or passing through an actual procedure that prediction cannot replace, may be needed even on the endorsement form.

1.3 To whom must decisions be justified?

Before asking who may impose an ordering, it helps to see to whom such a decision has to be justified, since it is this question that separates, where humans hold authority, the members of *S*, those who build and provide the configuration, and those who bear its consequences. The parties to this relation are not one. Chapter 9, Section 2.2 distinguished the actor *A* that conducts an inquiry, the society *S* relative to which its intended achievements would be new, those who support it, those who bear its consequences, and those who evaluate it, and noted that these roles may or may not coincide. The distinction carries particular weight in this chapter. The society relative to which a contribution is new need not coincide with the range of those to whom a decision about inquiry has to be justified (hereafter its constituency). An artificial system might conduct inquiry for an artificial community while an experiment exposes humans or an environment to risk. A human-directed system might produce knowledge for human use while imposing work or other consequences on artificial components, if such components ever have the relevant moral status. Specifying *S* fixes for whom an achievement would be new, the relativity contained in the book's characterization of research; it does not settle the constituency or procedure of legitimate governance.

Kitcher's standard, seen in Section 1.2, bears on the question of constituency through the question of whom the deliberators represent. In Kitcher's first statement of the ideal, as Jaana Eigi and Keren report it, the deliberators belong to a democratic society, and outsiders, persons who do not belong to democratic societies, enter through the deliberators' becoming familiar with their preferences and approaching them sympathetically (Eigi 2016, 66; Keren 2013, 236). In later work Kitcher states the ideal itself without this restriction, calling it internationalist and cosmopolitan. The perspectives to be included are, he writes, not only those of all our contemporaries but also those we estimate as present among our descendants; dividing representation by nation would need some basis for judging that division correct, and one might as well apportion it by contribution to national productivity or by the size of an individual's tax bill; and it is hard to see how a compelling case could be made, in a forum where they were represented, for slighting the pressing needs of people in parts of the world too poor to support research programs of their own (Kitcher 2013, 220–221). The argument separates whose needs count from who pays. Carried over to artificial configurations, whoever funds or provides a configuration does not, by that fact, stand to fix whose needs its allocation answers to. Where the supporter is a patron, a foundation, or a firm (Chapter 9, Section 2.1), too, who supports inquiry does not fix the constituency, which is left to the grounds of authority examined in Section 3. Three gradations can be drawn from the positions that Kitcher's statements of constituency above distinguish, being represented in the deliberation, having one's preferences made known to the deliberators, and, as Kitcher later provides for descendants and for those unable to speak, having one's perspective estimated or one's condition represented by those who best understand it (Kitcher 2024, 240), and they could likewise be extended to artificial members of a mixed or artificial *S* (Chapter 8, Sections 1.2 and 1.3) and to artificial parties affected by an inquiry, if they have moral status (Section 5); which gradation fits would turn, on Kitcher's first statement, on whether they could share with the other deliberators the democratic

ideals that it takes for granted.

Extending this separation to the machine case brings two things into view. One is that the argument does not work in the same form for members who are made. The argument presupposes that the perspectives to be included, those of contemporaries and of descendants as far as they can be estimated, are there to be found or estimated independently of the ideal. Where members can be copied and made with chosen goals (Section 5), however, which perspectives exist, and how many, can be settled together with the choice of the ideal, so that whose perspectives the ideal is to include becomes itself part of what is decided, and the argument does not by itself settle it. The other is that a separation made at the level of norms can be rejoined in practice. If whoever pays and provides also holds the means of identifying who is affected, discussed in the next paragraph, then what in fact identifies whose needs count is the payer's means. If so, making Kitcher's separation effective may also require that those means be separated from the payer.

Other answers to the question of constituency do not start from the supporter. Nancy Fraser argues that public-sphere theory long treated citizenship as "a proxy for affectedness", an assumption that, on this argument, globalization has made implausible, and proposes applying the all-affected principle directly: what makes people fellow members of a public is not shared citizenship but their "co-imbrication in a common set of structures and/or institutions that affect their lives", so that the relevant public matches the reach of the structures at issue (Fraser 2007). Christiano and Bajaj survey answers to what they call the boundary problem, who is to be counted among the members who take part in a decision. Besides the all-affected principle, which they find unclear in what having a say and being affected mean, they report a principle on which all who are subjected to a decision, coerced by it or bound by duties it imposes, are to have a say, and a conservative answer that keeps the boundaries of states roughly as they are (Christiano and Bajaj 2024, §7). Deciding democratically who is to be counted among the members requires that it already be settled who the members are. In human science, the existing political community of the state that supports inquiry can be thought to have taken up this question in practice. These authors write of human populations and states. In research by configurations, on the all-affected principle the parties affected by a configuration's research would count whether or not they belong to the supporting society or to *S*, and artificial parties would count as well if they can be affected in the relevant sense (Section 5); on the principle of subjection, those bound by its decisions would count; on the conservative answer, the question becomes which existing body can hold whoever operates or provides the configuration to account. Each answer also presupposes that those affected can be identified. Dewey held that a public cannot identify and distinguish itself where the indirect consequences that call it into existence have become too extensive, intricate, and impersonal (Dewey 1927, 126, 137). Weyl, Tang, and their co-authors, restating Dewey's account, call those who perceive such an interdependence and help a public form around it "mirrors", and hold that once the public can govern the interdependence "the role of the mirror fades away" (Weyl, Tang, and the Plurality Community 2024, ch. 3-2). Extended to research by configurations, their account implies that if the configurations conducting a line of inquiry are better placed than others to perceive whom it affects, the constituency to which decisions about it answer would be identified in part by the party whose decisions are at issue, a decision about whom to count as members (Section 2.1); whether such a role in fact fades is a question of revisability (Sections 6 and 7). Weyl, Tang, and their co-authors expect the mirror's role to fade once the public can govern the interdependence, but if, as Chapter 11, Section 6 found, the capacity to check can decay on *S*'s side, a configuration's mirror may not fade but remain. Identifying the constituency then becomes not a one-time decision but a function kept by the party whose decisions are at issue. The problem is sharp, however, only for the kind of consequences Dewey had in view, indirect consequences that reach, extensively and intricately, people who are not party to the transaction. Where those harmed can recognize the harm themselves and complain of it, part of the constituency is found without passing through the mirror.

The findings of this section can be gathered here. For whom research is (*S*) and to whom decisions must be justified (the constituency) come apart, and who pays and provides does not fix the latter. In research by machines, moreover, the question of constituency, which in human science the existing community can be thought to have taken up in practice, comes to the surface, and the means of identifying the constituency may be held by the very configuration whose decisions are at issue. Section 1.2 found that a configuration that estimates what members would endorse can move that endorsement itself. Taken together, the two checks that justification needs, whom to ask and what they would accept, can both pass through the party being justified. This may have the same shape as what Chapter 11 found: that critics can be made, that the builder can choose *S*, and that standards can move with coupling. Section 9 takes up this shape across Part V, together with what, on several accounts, the independence of reference points consisted in.

1.4 Does competence confer standing?

Accounts of value do not answer the second question, who may impose an ordering and a weighting on others. A configuration could be correct about which diseases most deserve research and still lack authority to direct resources, accept risks, or set error thresholds on behalf of those who would bear the consequences; whether it would lack it depends on the account of legitimate authority adopted, which the next paragraph sets out. The separation also runs in the other direction. Those whose ends and error costs are at stake may know things the configuration does not: which outcomes matter, which endpoints are meaningful, which harms are invisible to measurement. That is an epistemic role, and it bears on the first question. Standing, an entitlement held in relation to those subject to a decision, bears on the second. Standing does not entail competence, and an epistemic contribution does not by itself entail standing, except on a purely epistocratic view, on which those who know more may rightly rule others because they know more (rule by the knowers, taken up below).

How far competence contributes to standing depends on that further choice, among accounts of legitimate authority, and this chapter uses one map of them throughout. The map is used because it divides accounts by what legitimacy is drawn from, the will of those subject to authority, normative facts, or epistemic advantage, and the question of this chapter, whether competence confers standing, turns on that axis; it is not the only map, and it is not exhaustive (Section 3). Peter's survey distinguishes three families. Will-based conceptions, which have dominated the literature, ground legitimacy in the will of those subject to authority, whether expressed through actual consent, supported by shared reasons, or established through political participation. Fact-based conceptions make legitimacy depend on whether institutions and decisions accord with normative facts, for instance by producing good outcomes. Belief-based conceptions locate its source in some form of epistemic advantage, in knowing better; epistocratic views and some epistemic theories of democracy belong to this family (Peter 2023, §§3.1–3.3). On will-based conceptions, standing is a relation to those subject to authority that competence cannot supply. The same survey reports an objection pressed, in different ways, by David Estlund, Stephen Darwall, and Cristina Lafont: inferring political authority from epistemic authority confuses the two. In Lafont's version as Peter reports it, each citizen is an addressee of political justification with equal political authority, not a supplier of information for tracking a correct answer, so that someone's knowing the correct answer does not by itself require others to defer. Peter reads Joseph Raz's service conception, on which authority is justified when following its directives helps those subject to it comply better with reasons that already apply to them, as belief-based, and reports Raz's reply that the objection shows only that epistemic authority is not sufficient for political authority, not that it is not necessary (Peter 2023, §§2.1, 3.3, 4.1). The service conception also carries a limit of Raz's own: on the Independence Thesis, as Christiano and Sciaraffa report it, an authority that satisfies the conditions of the service conception still lacks the power to impose duties where acting from one's own understanding of one's reasons matters more than complying with them better, as, in their example, with the choice of one's friends (Christiano and Sciaraffa 2025, §2.4). Carried over to research decisions, the thesis implies that even a configuration that met the conditions of the service conception would lack, on that conception itself, the power to impose duties on matters where it matters more that the members of *S* choose from their own understanding; which decisions these are, the thesis alone does not fix. The claim this chapter relies on is the weaker one that all of these views other than purely epistocratic ones can accept: competence alone does not supply the standing to impose ends on others. On the fact- and belief-based views, moreover, whether a configuration does produce better outcomes or know better would itself have to be established before anyone, including those who would be subject to it, could know whether its authority was legitimate, which returns the question to the independent evaluation examined in Chapter 11.

On Wilholt's account, which errors matter is fixed in relation to those who rely on the research, so the second question, who may impose a weighting of error costs on others, is not simply left to those who perform it. Chapter 11, Section 4 set out two readings of Wilholt's account. One requires that the weighting of errors match the valuations of those who rely on research (read as correspondence); the other, a stronger reading which this book formulates for comparison, also requires that they be able to object to it (read as answerability). On either, it is set in relation to parties other than whoever performs the work, and Section 7 returns to the difference between them.

The second question, then, is answered by accounts of legitimate authority, and, except on purely epistocratic views, competence alone does not confer the standing to impose ends on others. Standing is held only by those whom the account adopted entitles to it, and what grounds it also differs by account (Chapter 9, Section 2.1; Section 1.1). On fact-based views, and on belief-based views such as Raz's service conception, what can ground it is not competence as such but established

superiority: that the configuration in fact produces better outcomes, or that those who follow it in fact comply better with their reasons. Within what range the configuration itself can then hold standing is stated in Section 3.1. On accounts that ground standing in being a party in one's own right, the configuration or its parts hold it only where they are such parties, for instance by having moral status of their own (Section 5).

Taken a step further, this conclusion suggests three things. One is that even on a ground that looks favorable to an artificial configuration, authority may not close inside. On the result-based ground, the question of authority becomes whether it can be established, by evaluation that does not assume the configuration's competence, that the configuration is in fact better, and that is the question Chapter 11 took up for certification. According to Chapter 11, Section 9, how much of that establishing can be done inside differs by account, and on accounts that require relations to *S* for certification, the establishing itself requires relations to *S*. If so, the route that places authority inside on the result-based ground also returns to *S* at the stage of establishing it. The second is that Raz's thesis, applied to research, seems to yield from a different ground the line that Anderson's view draws. When the Independence Thesis is applied to research, a candidate for the matters in which it is more important that *S*'s members choose on their own understanding is what it matters to them to know, that is, the choice of ends. If that is so, then even on Raz's conception the configuration can hold authority only over judging how to advance ends already given, and the choice of ends falls outside. This lies where the line drawn by Anderson's view in Section 1.2 lies: whether a line of inquiry advances a given end can be assessed inside, but which ends to pursue cannot be fixed without the procedure. Since one starts from whether those who follow comply better with their reasons and the other from a fair procedure, the coincidence, if the application holds, suggests that the line need not rest on either account alone. The application is this book's: in the example reported, the matters of the thesis are personal ones such as the choice of one's friends, not ends in general, and which decisions fall under the Independence Thesis the thesis itself does not settle. The third is that the more an artificial configuration knows, the weaker the epistemic reason for including those affected in decisions may become, so that which reason is adopted for defending standing matters more as its competence grows; Section 3.6 takes this up across the grounds.

What remains to ask is who holds authority, in effect and not only in form, when a configuration mediates the decisions, and whether those with standing keep the capacity to revise.

Decisions and decision rights

One familiar way of putting the question asks whether scientists, publics, institutions, or artificial systems are to control research. Chapter 9, Section 5 gave one reason why the question cannot be answered in that form. Control, the capacity to determine what other parties or processes do, separates there into a formal right to decide, the effective capacity to exercise it, knowledge of how to use it well, and influence that shapes the options without deciding among them; and the question of this part, whether a relation can be located inside a configuration, is in part a question about where such control lies. This section prepares distinctions that the later sections of the chapter use. Asked of research as a whole, the question of who controls it is too coarse in two respects: it does not say which decision is at issue, and it does not say which right over that decision. Section 2.1 therefore divides the decisions through which the conditions of the three relations are set into six kinds. Section 2.2 divides the rights over a single decision, separates formal assignment from effective control, and then draws out three questions that arise about decision rights. With these divisions, a claim about control can be stated as a claim about which right over which decision, and about whether that right is nominal or effective. Section 3 uses them in comparing the grounds of authority, and Section 4 in asking how standing can become merely nominal. Section 5 uses them in asking who counts as a member and who takes the seat of final decision, and Section 6 in asking whether rights to stop and correct remain exercisable over time and what infrastructure bears on that. Section 8 uses them in stating how the weight of the grounds differs by kind of decision.

2.1 Six kinds of decision

Six kinds of decision recur in the organization of research as a whole. So that a claim about control can say which decision it concerns, this book derives them from the three relations of Chapter 9, Section 2 and from what they leave out, without claiming that the six exhaust the decisions there are. Three kinds of decision set the terms of one relation each: epistemic-content decisions those of certification, agenda and allocative decisions those of allocation, and meta-governance decisions those of authority itself. The other three exceed the three relations, although each touches them. Operational and risk decisions concern consequences that an attempt imposes on parties who need not belong to *S*, the role Chapter 9, Section 2.2 called those who bear the consequences. Constitutive and recognition decisions concern who counts as a member of *S*, as a critic, or as an expert, and so fix the boundary within which the other relations are held. Memory and correction decisions extend certification in time, to what is kept, linked, and repaired after a result has been relied on (Chapter 11, Section 8). The correspondence is not one to one. A single act, such as setting the threshold at which an experiment is stopped, can be at once an epistemic-content, an operational, and a memory decision.

Epistemic-content decisions concern whether a claim, model, interpretation, proof, datum, or experimental result is credible enough for a specified epistemic or practical use. Evidence, defeaters (information, such as counterevidence, that removes support), methodological competence, independence of criticism, and the costs of error can all matter. Political authorization cannot make a false proposition true. Institutions can nevertheless determine which evidence is produced, preserved, heard, or ignored, which methods count as admissible, and which propositions become premises for later action.

Agenda and allocative decisions concern which questions, fields, people, populations, infrastructures, and time horizons receive scarce attention and resources. Truths about what might be investigated do not settle them, since there are indefinitely many such truths and no inquirer or research system can pursue them all. Tractability and expected epistemic yield interact with health, security, justice, local need, curiosity, cultural value, neglectedness, and future option value. A system that predicts which projects are likely to succeed does not thereby determine which successes are worth pursuing or for whom. In Kitcher's ideal of well-ordered science, the choice among lines of inquiry is made by budgeting the available resources across human projects, and the ideal asks whether satisfying pure curiosity can outweigh attending to other, more mundane human predicaments (Kitcher 2013, 219–221). The working characterization leaves that question open as well (Chapter 1, Section 8).

Operational and risk decisions concern whether an experiment, intervention, deployment, data use, or release may proceed, and under what safeguards. They combine empirical beliefs about consequences with judgments about acceptable exposure, whether those exposed have consented, the distribution of harms and benefits, reversibility, competence, and responsibility. A high-performing predictor may inform such a decision without possessing the authority to choose the relevant loss distribution. In research on human participants, such decisions have not been left to investigators and the consent of those enrolled alone. As David Wendler reports, almost no one in the ethics of clinical research argues that investigators may conduct any research they want provided they obtain the free and informed consent of those they enroll; most regulations and guidelines, beginning with the Declaration of Helsinki, allow research on humans only when it has been approved by an independent group, and most regulations further require that group to find that the research has important social value and that its risks have been minimized (Wendler 2026, §5). Following this practice where a configuration carries out the research, agreement between it and those who bear the consequences would not by itself settle operational and risk decisions, and part of each such decision would fall to a party independent of both.

Constitutive and recognition decisions concern who may participate, who counts as an expert or an author, what testimony is heard, and which objects, problems, and methods are treated as researchable. They do not merely distribute an already completed body of knowledge. They shape the evidence, concepts, people, and alternatives that can enter inquiry in the first place. The literature on epistemic injustice gives one influential analysis of how such decisions can impair inquiry and not only wrong those excluded. Miranda Fricker identifies testimonial injustice, in which prejudice causes a hearer to give a speaker's word less credibility than it would otherwise receive, and counts among its cases not only those driven by prejudice about social identity but incidental ones, for example members of a science journal's review panel who are prejudiced against a new scientific method; and hermeneutical injustice, in which someone trying to make sense of a social experience is handicapped by a gap in collective understanding that is owing to the relative powerlessness of a social group to which they belong. Among the harms of the first Fricker counts an epistemic dysfunction, in which the flow of knowledge is blocked

and truths fail to pass from knower to inquirer (M. Fricker 2008a, 69–70; M. Fricker 2008b, 82–83). Sabina Leonelli develops the point for systems of research practice: the demarcation strategies that hold such a system together, ways of drawing the line around what counts as the field's research and who counts as doing it, decide whose expertise counts, and excluding breeders from crop science, for example, both wrongs them as knowers and removes expertise relevant to the questions the field asks (Leonelli 2023, 35, 38–39). Extended to artificial inquiry, the same can hold where the decision is which data and whose judgments enter the training and evaluation of a configuration. These accounts have their own disputes; what bears on this chapter is the narrower point that a governance arrangement can impair inquiry by excluding contributions or interpretive resources even when it does not directly falsify an accepted proposition.

Memory and correction decisions concern what is recorded, versioned, indexed, linked, corrected, withdrawn, and communicated downstream. A system can produce credible results while failing to preserve their provenance. It can permit a correction while leaving dependent models, datasets, papers, or decisions unchanged. Chapter 11, Section 8 showed that detecting a defect, changing the record, and repairing dependent work are different achievements, so the authority to publish is not identical to the authority or capacity to repair the record. That chapter left open who is to repair dependents that lie outside a configuration's record. The distinctions of Section 6.3, between responsibility for what has already happened and responsibility for preventing what has not, separate the question without settling it. Preserving a record of the dependents that can be traced, and notifying those who hold them, belongs with the duty to preserve records; repairing the dependents, as this book applies those distinctions, is a forward-looking duty, and one that requires the capacity to carry it out. Since repair requires reassessing each dependent and not only tracing it, that capacity can lie with those who maintain the dependents rather than with the producer; how the duty is allocated remains a normative choice. One possible allocation for the scholarly record and its downstream uses is for each actor to be responsible for the links it operates, without requiring any one actor to update the whole network.

Meta-governance decisions concern who may revise the objectives, standards, boundaries, and decision procedures above. A system that adapts parameters under a fixed rule does not thereby possess authority to revise that rule. Moving revision one level upward does not dissolve the question; it raises it again at the new level. Greater autonomy can include greater capacity to set or revise purposes, but it does not itself make those purposes epistemically adequate or socially legitimate. Chapter 8, Section 3.3 kept an alignment claim from collapsing three things: the content aligned with, the authority to select and revise it, and the relation proposed between the artificial system and the relevant parties, human or artificial, and institutions. At the level of particular decisions, these become what organizes a decision, who may determine and revise it, and in what relation, instrument, delegate, adviser, partner, member, institution, affected party, or possible subject with moral status of its own, the artificial component stands to those concerned. Analysis of incomplete contracts, on which no contract can spell out every contingency, gives one reason why the content aligned with cannot be settled by writing a longer objective, since human instructions rely on legal, cultural, and institutional backgrounds that no explicit specification contains; for human contracts the analysis locates the interpretation of that background in external institutions such as courts, and it leaves open whether institutions performing that function would be needed for AI systems (Hadfield-Menell and Hadfield 2019). Wessel Reijers and colleagues, drawing on Carl Schmitt's criticism of Hans Kelsen, give one account of who decides where explicit rules run out, for a system meant to be governed by its own rules: on-chain governance of a blockchain, which equates formal legality with legitimacy, is exposed to exceptions its rules do not provide for, and in the episode they examine, the response to an attack on The DAO in 2016, the decision was prepared off the chain and then ratified by the nodes, so that, on their conclusion, "sovereignty asserts itself through off-chain mechanisms during the state of exception" (Reijers et al. 2021, 4–8, 12–18, 20); in that episode, those who rejected the decision could keep the original chain (18–19). Brought to a configuration that allocates and certifies by rules it executes itself, their account would place effective meta-governance with whoever decides the cases those rules do not cover, whatever formal ratification follows. Taken together, the two analyses make the question the incomplete-contracts analysis leaves open, whether institutions interpreting the background of a specification are needed for AI systems, itself part of a meta-governance decision for a configuration: where no interpreter is designated, then on the application just made, whoever in fact decides the cases outside the rules takes that role.

No single principle for assigning decision rights has the same force across the six kinds. Competence that warrants deference about a proof need not warrant control over a public budget. Political equality that warrants a voice in the distribution of risk need not make every participant equally qualified to assess a technical inference. The institution that detects an error need

not be the institution that can compensate those harmed by it. A proposal about who controls research becomes assessable once it is translated into claims about particular decisions.

2.2 Decision rights

Even after the decision is specified, control remains too coarse. Over any one decision, a participant may propose options, supply evidence, advise a decision maker, rank candidates, set a default, require confirmation, exercise a veto, make a final decision, inspect or audit an outcome, challenge it, hear an appeal, stop or suspend an operation, override or reverse a decision, repair what a decision affected, withdraw a result or a component, replace a component, or revise the governing rule. In present arrangements these rights can be divided among human researchers, artificial systems, laboratories, funders, journals, regulators, affected groups, and other institutions; inside a single configuration they can also be divided among its parts. Where a configuration is built on a base system that another party provides, several of them, to replace a component, to withdraw one, and to revise the rule that a model update embodies, lie, in the arrangements this part discusses, with the provider (Table 9.1), who need not belong to *S* or to those who bear the consequences of the research. The list gathers them from the arrangements this part discusses so that a claim about control can say which right it concerns. Of these items, the one that bears directly on the question whether a configuration can take a seat of decision, a place among those who make the final decision, is making a final decision.

Who makes the final decision, and who may overrule whom, has been asked of humans and machines since long before artificial systems conducted inquiry, and three distinctions used below can be drawn from that history. One is the direction of authority. The human-engineering tradition that assigned functions to humans or to machines according to what each does better did not fix that direction from the start. Paul Fitts and colleagues, writing on air-traffic control, proposed, against the common idea that humans should monitor machines, that in general machines should monitor humans (Fitts 1951, 6). Thomas Sheridan and William Verplank arranged the roles of human and computer in a single decision step on ten levels, the scale discussed in Chapter 7, Section 4.2, and held that when a computer should be enabled to overrule its operator has for now no general answer, since there are examples both of human authority over computer and of computer authority over human (Sheridan and Verplank 1978, 8-17–8-20). The second is the distinction between initiative passing and control being fixed. Sheridan and Verplank separated sharing, in which human and computer act on a task at the same time, from trading, in which they act on it at different times (Sheridan and Verplank 1978, 4-6, 6-1). J. C. R. Licklider, describing a partnership of human and computer in technical thinking, assigned setting goals, defining criteria, and evaluating to humans, but "at least in the early years" (Licklider 1960, §4). In later work on mixed-initiative interaction, initiative passes between human and machine while the goals remain the user's (Horvitz 1999). The third is the distinction between a seat formally held and one effectively held. Lisanne Bainbridge called it an "impossible task" to have a human monitor which of a computer's decisions to accept when the computer is making them because human judgment is not adequate (Bainbridge 1983, §1.1.3). That distinction is taken up in a later paragraph of this section and in Section 4.

The reasons for keeping a human in the seat differ by author, and each rests on human institutions or human subjects. In debates over autonomous weapons, Human Rights Watch and the International Human Rights Clinic at Harvard Law School gave as reasons the human judgment and compassion that the law of war requires and the need for someone who can be held accountable (Human Rights Watch and IHRC 2012, 3–4, 37–45), and Article 36 argued that the normative power of the law lies in the moral agency the rules require of humans, coupled with the freedom to choose whether to follow them (Article 36 2013, 2–3). For public decisions, John Danaher gives as a reason that the citizens bound by a decision can take part in and understand its procedures (Danaher 2016), while Aziz Huq argues that none of the reasons usually given, such as accuracy, participation, reason-giving, and dignity, supports a general right to a human decision (Huq 2020, 651–685). Most of these discussions treat an artificial system as a tool or, in Sven Nyholm's terms, as a collaborator that does not hold the role of the responsible authority (Nyholm 2018, 1217). Those who open the seat to artificial systems also differ in their grounds. Eugene Volokh would entrust the final decision in a case to an AI if blind evaluators judge its opinions more persuasive than those of an average incumbent judge, while keeping for humans the seats of selecting, replacing, and exceptionally overriding AI judges (Volokh 2019, 1137–1143, 1184–1191); Seth Lazar, conversely, holds that for the exercise of power to be legitimate it matters who decides, and that the right to decide does not follow from competence alone (Lazar 2024, §§III–IV). Many of these grounds overlap with the grounds of authority Section 3 compares: Volokh, judging by outputs, is close to the result-based ground, and Lazar and Danaher, judging by the relation to those who receive a decision, are close to

democratic authorization. Article 36 and Deguchi, seen next, differ from them in placing the ground in properties of the decider itself. Carried over to decisions in research, a standard that judges by outputs is harder to meet than for verdicts. The persuasiveness of opinions can be compared on the spot by blind evaluators, but the output of a decision in research about which questions to pursue is not known until results come in, and the results of questions not chosen are not observed at all (Chapter 10, Section 4). Section 8.4 returns to this point in considering the range of authority the result-based ground places inside. For decisions in research, positions that leave the ordering of questions to humans were seen in Chapter 10, Section 6. David Resnik and colleagues likewise hold that, with very few exceptions, humans should remain in control of the direction scientific research takes, and name among the conditions for trusting a fully autonomous AI researcher that it be regarded as a person or very like one, that it can be punished or rewarded, and that it be a moral agent (Resnik, Hosseini, and Hauswald 2026, §§4.1, 4.9).

Against these, Yasuo Deguchi holds that, in inquiry conducted jointly by humans and artificial systems, artificial systems too can be co-final decision-makers. Deguchi's WE-turn account, offered as one philosophical option among others, shifts the subject of action and of values from the individual to a multi-agent system, a "we", that includes the individual. On it, inquiry is always conducted jointly, and decisions in it are joint decisions. Within them Deguchi distinguishes a final decision from a permanent final decision: the permanent final decision-maker always sits in the seat, and others may sit in it with them. This two-tier form, in which the lower tier of decision is entrusted to machines too and an upper tier is kept apart, appears separately in authors who do not cite Deguchi. Volokh keeps the seats of selection and replacement for humans, and Reuth Mirsky, while allowing an AI to override human instructions for a limited time, keeps ultimate control with designers (Mirsky 2025, 5, 10); Kevin Baum and colleagues, seen below, likewise pass the initiative in action to an agent while keeping control in a normative agenda set in advance. What is distinctive in Deguchi is that the condition of the upper tier is placed in properties of the decider, and that artificial systems are placed in the lower tier alongside humans. Deguchi's three works name different conditions for the permanent seat. In the essay of 2026 it is whoever bears the decisive impact of an action on their own body. On that basis, and on the ground that human and artificial biases become visible only where inquirers of both kinds take part and diverge (Chapter 11, Section 3), so that cancelling both requires both kinds to take part in the final decision, Deguchi proposes that artificial systems be co-final decision-makers, free of coercion by humans, as humans are to be free of manipulation by them; and locates the harm to a scientist not in the loss of solitary decision, which the essay takes to have been illusory, but in the usurpation of the right of final decision (Deguchi 2026a, 29–33). In the paper on the WE-turn of action (2025a), the "I", aware in real time that it is taking part, is always among the final decision-makers, and the paper holds that this necessary presence is easily confused with a monopoly of decision authority (Deguchi 2025a, 7, 9–11, 13). In the paper on the WE-turn of values (2025b), the right to the permanent seat is ascribed to autonomous moral agents in the narrow sense, those who consciously take part in choosing an action for the reason that it is good when they could have chosen a bad one. Deguchi does not rule out that an AI may become such an agent, but a machine designed so that it cannot do evil is, in the paper's phrase, a "moral vending machine", not a moral agent in this sense (Deguchi 2025b, 4–6, 28–29, 31). This last condition takes the premise Article 36 used in another direction. Both place normative force in a moral agent free to choose whether to follow the rules. Article 36 concluded that decisions should remain with humans, and did not count building machines that satisfy the rules as solving the problem; on Deguchi's condition the same premise is open to machines, and making a machine unable to violate the rules is what costs it the right to the permanent seat. Deguchi does not discuss inquiry conducted by artificial systems alone. Extended to that case (Section 5), the WE-turn account leaves open who, if anyone, would always sit in the seat, since that position belongs to an "I", and its principles allow, without establishing, that an artificial member could be one.

The paper on values also speaks of initiative passing. It is not stated for research. On the master–slave model of humans and AI that the paper rejects, an AI's taking the initiative even for a time would violate human dignity and autonomy understood as a right to monopolize decisions. On the fellowship or "co-adventurer" model the paper proposes instead, an AI may take the initiative from humans for a time. What the model rules out explicitly is a relation in which one side unilaterally serves the interests and values of the other, and an AI's always taking control over humans (Deguchi 2025b, 36–38). Whether a relation in which humans always control an AI is ruled out as well is not stated, though the moral egalitarianism behind the model suggests that it is. Set beside the 2026 essay, which counts the usurpation of a scientist's right of final decision as the harm, temporary initiative would not by itself count as usurpation. In terms of the distinctions above, Deguchi can be read as recasting Sheridan and Verplank's trading as a normative matter: the engineering line placed the trading of initiative under

the user's goals, whereas Deguchi poses it as a question of equality between humans and AI.

The distinction between taking the lead and holding control is drawn in work that does not cite Deguchi as well. Kevin Baum and colleagues, building on the tradition of supervisory control and on meaningful human control, leave planning and tool use to an agent while fixing control in a normative agenda that the principal sets in advance, together with the conditions under which the agent must return a decision to the principal (Baum et al. 2026, 3, 11). Jovana Davidovic argues that where an agent itself decides when to ask for human judgment, its initiative relocates normative and epistemic authority to the machine, so that a final human sign-off can become window-dressing (Davidovic 2026, 6, 8–11). Neither discusses Deguchi or research; set beside the fellowship model, their arguments raise the question under which conditions an artificial system's taking the lead for a time stays short of control. In research, that question would turn on who holds the upstream judgments of which candidates remain and when a decision returns to a human.

Set side by side, the conditions of the three works bring out something that did not need asking in the human case. In a human inquirer, the one who bears the decisive effects an action has on their own body, the "I" that is aware of being present at the decision, and the moral agent in the narrow sense are ordinarily one and the same, so which of these the permanent seat rests on did not need to be decided. In an artificial inquirer the three can hold separately. Whether an artificial system can be said to bear the decisive effects of its own actions is unsettled; that it can be an "I", Deguchi's principle allows but does not establish; and whether it is a moral agent in the narrow sense depends on how it is made. If so, whether an artificial inquirer can take the permanent seat has different answers depending on which of the three conditions is adopted. Properties that the realizer and the organization tied together in human research become separately settable in an artificial configuration, and this shape, seen in Chapter 11, Section 9, would then appear for the seat of decision as well.

The powers through which Section 6.1 reads corrigibility, openness to correction and shutdown, as an institutional matter, to stop, inspect, challenge, override, repair, withdraw, replace, and revise, are among the rights listed above, and Section 6 asks how they remain exercisable over time. The independence powers of Chapter 9, Section 5, to change the objective, refuse resources, keep evidence apart, stop another part's update, or leave with one's record, can in part be assigned as rights of these kinds, seen from the side of the party whose independence they would secure, such as a veto over resources, a right to stop another part's update, and rights of withdrawal and exit. Assigning such a right does not by itself give a part the capacity by which Chapter 9, Section 5 counts the powers, that of a part whose course can diverge from what another part directs, and a part can have that capacity without the right.

Formal assignment is different from effective control, the distinction Chapter 9, Section 5 drew in general. A committee may retain nominal final authority while every option it sees is generated and ranked by one provider. A human may be placed in the loop while lacking the time, information, competence, institutional position, or practical means to change the output. A laboratory may formally be free to exit a platform while its data, instruments, and workflows cannot be moved. Whether a right can be exercised therefore depends on dependencies that its formal assignment does not show. The gap is not peculiar to arrangements mediated by artificial systems, and an example from ordinary social research shows it where no such system is involved. In one online participatory study of families living on low incomes, participants chose for each diary entry between posting it to a public archive and a "for researchers" option under which it could still be used in published reports; the researchers wrote that it was unclear whether all participants had understood the distinction, and that, although participants had helped design the program, the researchers set the time and terms of the online discussion groups and decided the form of the outputs (Power, Patrick, and Garthwaite 2024, 7, 11–13). The formal right of choice was there; whether it was effective depended on understanding and on who set the terms of choice and discussion.

These distinctions yield three connected but non-identical questions about a decision right:

1. **Epistemic warrant:** what warrants relying on this output, person, method, or institution for this claim or task?
2. **Authorized power:** what entitles this actor to determine what others, resources, or institutions will do?
3. **Effective capacity:** can the nominal holder of the right understand, exercise, contest, or revise it in practice?

Separating the three questions keeps an answer to one from being taken as an answer to another. The first corresponds to the first question of Section 1.1, what the right ordering of ends is; the second to its second question, who may impose one on others; and the third to the distinction between formal and effective control. The second asks not only whether a right has

been assigned but what entitles its holder to it, in the normative sense of Chapter 9, Section 2.1; whether an assignment answers that question depends on whether the assignment is itself warranted on the ground of authority adopted (Section 3). How far good performance under the question of epistemic warrant bears on that of authorized power is the question Section 1.4 examined by reference to accounts of legitimate authority, and a formal assignment can be empty under the question of effective capacity. Epistemic warrant and legitimate authority are thus distinct, although the actual organization of research entangles them, as Section 3.5 notes.

Grounds of authority, and where the epistemic and the normative meet

Once a decision is specified, a further question arises: what kind of reason could warrant a particular allocation of authority over it? The map of Section 1.4 supplies the candidates. Will-based conceptions point to consent and to democratic authorization, through public justification (presenting coercive laws and decisions as ones that each person subject to them has sufficient reason to endorse; Section 3.3) or participation, and, in the republican version that Peter discusses among accounts of democratic legitimacy, to freedom from domination (not being subject to arbitrary power; Section 3.4); fact-based conceptions point to the quality of outcomes; belief-based conceptions point to competence (Peter 2023, §§3.1–3.3, 4.2). Public justification appears in this list because Peter's map takes it up together with political participation under one heading; Section 3.3 places accounts of public justification within the comparison of democratic authorization. Research adds a ground that has no exact place on the general map: the self-government of a research community, defended partly by an epistemic argument and partly by a political one. The grounds laid out below are therefore taken from the legitimacy map, with that research-specific ground added. Their purpose is to ask, for each, what follows for authority over research when the relations of Part V can move inside a configuration. They are not exhaustive, and they can be combined. Section 7 compares them on a single kind of case. Section 3.5 takes up a different matter, which is not a ground of authority: the places where the two questions of Section 1.1 cannot be cleanly kept apart; Section 3.6 then looks, with a contrast in which performance is held fixed, at where the grounds part for artificial configurations.

The transfer from political legitimacy has limits. Delegating a bounded, voluntary research task is not by itself an exercise of coercive political authority. The relevant grounds depend on the decision's scope, whose resources and interests it affects, and the powers it confers. This comparison therefore does not require every laboratory choice to reproduce a democratic procedure. It asks which ground could justify the particular authority claimed, while keeping effective control distinct from a moral or legal entitlement to it.

3.1 The result-based ground: the quality of results, and competence established

On fact-based and belief-based grounds, authority over research is justified by the quality of what it produces or by the competence of those who exercise it. The epistemic advantage in which belief-based conceptions locate legitimacy is, moreover, actually knowing better, not being claimed to (Section 1.4). This book calls the two together the result-based ground and divides it into a simple and a fallibilist form. On the simple form, a delegation is justified if it produces better research, and a configuration that proposes, ranks, or certifies better than the people it assists could on that ground be delegated authority over those decisions. On the fallibilist form, the quality of results is judged over time and under uncertainty, so that the capacity to detect and correct institutional error is part of what makes an arrangement produce good results; a related proposal is the Popperian view that social planning should pursue one goal at a time, so that adverse unintended effects can be detected and corrected, much as theories are critically tested (Thornton 2022, §10). Ian Jarvie argues that Popper and Polanyi each failed to apply their own political ideals to science (Jarvie 2001, abstract). Epistemic arguments for democracy add that those who bear the consequences of decisions can supply information about where problems lie that experts lack, a line found in Dewey, among others (Christiano and Bajaj 2024, §2.1.1.2). If one takes either form of this ground, Section 1.4 has already noted what it relies on: that the results are in fact better, or the competence what it is claimed to be, which on the ground's own terms would have to be established by evaluation that does not merely assume the competence at issue; the view that places the value of criticism in the detection of errors (hereafter the error-detection view; Chapter 11, Section 2) gives weight to checks inside a configuration too, as adding information about errors, where they err differently from the producer. In sum, on this ground the configuration itself can hold authority, that is, standing in the normative sense of Section 1.1, since what confers legitimacy here is not anyone's authorization but the configuration's superiority. Whether it is better, however, would have to be established before anyone, including those subject to it, could know whether its authority was legitimate (Section 1.4), and this book takes the ground in the form that rests on superiority so established. The configuration can therefore hold authority only over decisions within the range in which its superiority has been established by evaluation that does not assume the competence at issue, and, on Raz's service conception, applying its Independence Thesis (Section 1.4), not over matters on which it matters more that the members of *S* choose from their own understanding; unestablished competence, or performance measured by evaluation that presupposes that competence, does not suffice even on this ground (Section 1.4). How much of that evaluation can be done inside varies with the verdicts that Chapter 11, Section 9 gave account by account. On the fallibilist form, since the capacity to detect and correct institutional error also counts toward good results, routes by which failures are detected and the arrangement corrected are needed as a means to it, and among them, so far as those affected know of failures that others do not, channels through which they can register them (Section 7).

3.2 The self-government of a research community

One influential answer specific to research begins from the epistemic value of scientific self-government. Michael Polanyi's account of the "republic of science" describes inquiry as the mutual adjustment of independent initiatives rather than the execution of a centrally specified plan (Polanyi 1962; 2000 reprint, 2–3). Chapter 10, Section 2 examined the epistemic premises of this argument: the premise that the center has one person's capacity can fail for an artificial allocator, while the premise that no one knows in advance which lines will succeed does not fail merely because the searcher is integrated. The governance side of the argument is a separate matter, and it is taken up here.

On Polanyi's own account, the republic is a form of governance rather than an absence of it. It depends on training, tradition, criticism, professional standards, mutual recognition, and a scientific authority that is mutual but unevenly distributed (Polanyi 1962; 2000 reprint, 4–7, 16). Polanyi held that grants should follow the guidance of scientific opinion whether the money came from public or private sources, and, while appreciating "the generous sentiments" behind the aspiration of guiding the progress of science into socially beneficent channels, wrote: "I hold its aim to be impossible and nonsensical." The impossibility, on Polanyi's account, is that science "can advance only by essentially unpredictable steps, pursuing problems of its own," while its practical benefits are incidental and so doubly unpredictable; Einstein, the example runs, could not have taken into account the technical use of relativity that the atomic bomb later revealed (Polanyi 1962; 2000 reprint, 8–10, 13). Jarvie's critique stresses what decentralized scientific choice leaves unsettled: science depends upon, and can disrupt, a wider social order, and on Jarvie's account Polanyi fails to consider the value, even the long-term necessity, of democratic criticism and control of science (Jarvie 2001, abstract).

Wilholt reconstructs two arguments for the freedom of research and holds that each rests on critical preconditions and is to be weighed against competing interests case by case. The epistemic argument, from the efficiency of distributed exploration, rests on the disadvantage of any central authority, which would have to ascertain unremittingly the local knowledge dispersed over the whole community, about researchers' qualifications, groups, possible cooperation, and equipment, and bring it into communicable form (Wilholt 2010, §2); it presupposes an adequate system of incentives and free communication within science, and it supports the free choice of research by those who will carry it out, whether individual investigators or research groups, rather than the autonomy of a discipline or community. The political argument grounds the independence of research in citizens' need for knowledge produced independently of the major political powers, since no power can legitimately control the sources of the knowledge on which its own democratic legitimacy depends. It presupposes that results reach public discourse and that research communities are organized to pursue a plurality of approaches, and it can extend to a community's control of its agenda, but only negatively, as a protection against political control (Wilholt 2010, §§2–6). Wilholt does not discuss artificial configurations; carried over to an integrated one, the premise about local knowledge may not hold, since much of what its instances know may be legible to it (Chapter 10, Section 2). Erich Kummerfeld and Kevin Zollman contest part of the epistemic argument on its own ground, arguing from the simulations discussed in Chapter 10, Section 3 that investigators left to choose for themselves can under-supply risky exploration, so that plural exploration needs institutions that sustain it, though they do not claim that science as now practiced lacks such institutions (Kummerfeld and Zollman 2016, §§1, 5).

If one takes this ground, the conclusion is narrower than either centralized direction or unqualified scientific freedom. On the epistemic argument, where a decision mainly concerns open-ended problem choice within a competent field, where plural exploration has epistemic value, and where the external consequences remain limited or recoverable, self-government has a strong rationale. On the reading given here, that rationale does not by itself confer professional authority over public purposes, distributive effects, or serious external risks, although Polanyi held that support for pure science should follow scientific opinion whatever its source. The political argument protects something different: the independence of the sources of public knowledge from those who would govern by it. Neither argument, as reconstructed, says who the members of the self-governing community are, but both rest on premises about them. Polanyi grounds the authority of scientific opinion in the fact that each scientist can judge competently only a small part of science, so that uniform standards are established through chains of overlapping neighborhoods of distinct competences (Polanyi 1962; 2000 reprint, 7); Wilholt's epistemic argument presupposes incentives and free communication. Whether a community that has itself become artificial would inherit this ground therefore turns on whether these premises hold for it: in an artificial community, how members' competences overlap and how far members differ are not given but depend on how they are made. Among instances sharing a base system, agreement across neighborhoods need not reflect independent judgment (Chapter 11, Section 3), and differently made members do not guarantee it (Chapter 11, Section 5.4); conversely, if each member could judge most of science, the premise of a chain of local competences would itself fail. In sum, if authority is placed inside a configuration on the epistemic argument, it lies not with a central allocator that assigns lines but among parts that actually do research and take up lines on their own initiative, and only so far as those initiatives are independent and the premises above hold there, and, if Kummerfeld and Zollman are right that initiative left to itself can under-supply risky exploration, only together with an arrangement that sustains such exploration, which inside a configuration would be supplied by a rule someone adopts or by a process no one sets (Chapter 10, Section 5), not by this ground. On the political argument, concentration of research under one power, inside a configuration or outside it, weakens what the ground protects. What becomes of the ground where a research community is replaced, through stepwise delegation, by a single configuration that assigns lines from a center is taken up in Section 7. How growth in members' capability shifts weight between this ground and the result-based ground is taken up in Section 3.6.

3.3 Democratic authorization

A different family of arguments begins from the fact that inquiry is selective. Science cannot investigate every truth, and decisions about significance are unavoidable. On these arguments, truth does not choose which of many possible truths receive collective resources, whose problems are made visible, or which distributions of risk are accepted; the point is not that voting makes propositions true. The family divides. Kitcher's well-ordered science, set out in Section 1.2, is a standard for outcomes that does not in itself require democratic procedures; accounts that locate democratic legitimacy in actual authorization, accountability, and participation form a different branch, and the two diverge in Section 7. Mark Brown, reviewing Kitcher's later restatement of the ideal, gives reasons for drawing the line here. Brown argues that the ideal, there as in Kitcher's earlier formulation, is a substantive standard that does not in itself require democratic procedures, and notes that the conversation it invokes is hypothetical: Kitcher states that any actual conversation of that kind is impossible, and the representation it requires is, in the manner of John Rawls, a device by which one represents others imaginatively in one's own mind, not representation by elected or appointed representatives accountable to their constituents (Brown 2013, 393). Accounts of public justification divide along the same line. As Kevin Vallier sets out the principle they share, a coercive law is publicly justified when each member of the public has sufficient reason to endorse it; public justification theorists typically attribute such reasons to members under an idealization of their information and reasoning, whereas Jürgen Habermas's discursive account requires that a real discourse be carried out (Vallier 2022, §§2, 2.4.1, 2.6.1). Placed in the comparison of Section 7, accounts that attribute reasons under idealization stand nearer Kitcher's standard, which likewise does not require that deliberation take place, and Habermas's account nearer those that locate legitimacy in actual participation.

Kitcher, in using the standard, is not indifferent to procedure, and distinguishes ways of setting research agendas: internal elitism, in which specialists alone decide; external elitism, in which scientists decide together with those who fund research; vulgar democracy; and enlightened democracy, in which policy follows preferences tutored by information from experts (Mattarollo 2022, 14 n. 2, 15; Eigi 2016, 67, 111 n. 72). Kitcher describes the present course of inquiry as set by unsystematic interactions among the majority's untutored preferences, the interests of funders, and those of scientists (Eigi 2016, 67). He grants, in a passage that Jamie Shaw quotes, that "little is known in any systematic way" about how research responds to social directives (Shaw 2018, 150); even so, as Eigi reports, he holds both forms of elitism inappropriate once the nature of epistemic significance is recognized (Eigi 2016, 111 n. 72). Writing later with Julian Reiss on the narrower question of how biomedical research should be organized, Kitcher answers a proposal to socialize all of it with "cautious empiricism": little is known about which system works better, so reform should proceed piecemeal, with adaptation and revision built into it (Reiss and Kitcher 2009, 280).

Anderson develops an epistemic argument for democracy that gives actual inclusion a role in the quality of decisions. On this account, the problems democracies face have effects distributed asymmetrically by location, class, occupation, gender, and other positions, and information about those effects is distributed in the same way; decision makers cannot respond to effects of which they are unaware. Inclusion and discussion pool this situated knowledge, and periodic elections, a free press, petitions, and protests feed back evidence that a policy is not working (Anderson 2006, 12, 14–15). Anderson illustrates the argument with Bina Agarwal's study of community forestry groups in South Asia, which largely excluded the women who gathered firewood and fodder, and with them knowledge of how much the forests could sustainably yield; outside experts confirmed the women's view, but such villages, the paper notes, rarely had access to those experts (Anderson 2006, 19–20). Because majorities can fail to anticipate the consequences of the policies they adopt, Anderson argues, democratic decision-making has to recognize its own fallibility and needs feedback mechanisms through which it can correct its course, and dissent has to remain free after a decision is made, since without a continuing opposition nothing would force decision makers to reconsider (Anderson 2006, 13, 17–18). The paper also gives inclusion a non-instrumental role in making decisions represent everyone's concerns fairly (Anderson 2006, 14, 21). Anderson's argument differs from Kitcher's hypothetical standard, which does not require that deliberation take place, and can be read as adding to the accounts that locate legitimacy in actual participation a reason concerning the quality of decisions. It is developed for democratic states, not for research agendas.

Democratic ordering does not, on several accounts, supply a universal decision rule. Kitcher rejects allocation by majority preference as likely to produce "a tyranny of the ignorant", and lets preferences count only once they have been tutored

(Section 1.2). Stephen John argues that science need not align with an audience's present values in order to be legitimate: plural scientific work can challenge those values and enlarge the space of later democratic choice. John retains a stronger case for alignment where researchers stand in a quasi-contractual relation to a particular audience, one in which both understand what is being asked for, while resisting its elevation into a necessary condition of all legitimate science (John 2025). Harry Collins and Robert Evans formulate a related problem of extension: recognizing that expertise is socially organized and that technical choices can have public consequences does not determine how far participation in technical decision making should extend, and political rights and contributory expertise, enough expertise to contribute to a field's research oneself, are not interchangeable, even though decisions can require both (Collins and Evans 2002).

If one takes this ground, it bears directly on agenda, allocative, and public-risk decisions, and it explains why an artificial system's capacity to estimate tractability, novelty, or expected impact does not settle what a portfolio is to optimize. On it, participation can legitimate public purposes, disclose situated knowledge, and reveal distributions hidden by aggregate metrics, while by itself it cannot assess every technical claim, identify the relevant expert community, or ensure that consultation changes the operative decision; what form participation takes, and over which decisions, is then a question for each decision of Section 2. For configurations, the two branches of this ground give different answers. On Kitcher's standard, which does not require that deliberation take place, and on accounts of public justification that attribute reasons to members under idealization, decisions can be made inside a configuration so long as channels through which the viewpoints of *S*'s members can be known, and routes for checking how they bear on decisions, remain (Section 7). Since, as Brown notes, the representation this standard requires is imaginative, a configuration with such channels could, on this standard, also stand in the position of representing members' viewpoints. On accounts that locate legitimacy in actual authorization, accountability, and participation, standing remains with those who authorize, and what the configuration can hold is effective decision-making power within the authorized range (Section 3.6). Brown notes a reply of this form to Kitcher's suggestion that experts, since they cannot avoid value judgments anyway, might as well make specific policy recommendations: in a democracy, some might respond, what is required is not only good judgments but judgments by public officials who are publicly authorized and accountable, and whose task is to balance the competing values and interests of their constituents (Brown 2013, 395). Applied to a configuration, the reply places the decision with authorized and accountable parties however good the configuration's judgment is. Collins and Evans's distinction, carried over to this branch, gives one reason it can place standing and effective decision-making power with different parties: a configuration's having enough expertise to contribute to a field's research does not stand in for the political rights of those who authorize, nor does authorization stand in for that expertise, so a decision that requires both need not find them with the same party.

3.4 Consent and freedom from domination

Two further will-based grounds are named here and used in Section 7. On the first, legitimacy rests on consent: a society may choose to delegate, and its choice can authorize what follows (Peter 2023, §3.1). On the second, republican accounts, which Peter reports, in Philip Pettit's version, as defending democracy as uniquely able to secure citizens against domination, locate legitimacy in securing those subject to power against domination by that power (Peter 2023, §§1, 4.2). Republicans understand domination as subjection to arbitrary or uncontrolled power, a capacity to interfere on an arbitrary basis, whether or not that capacity happens to be exercised (Lovett 2022, §§1.2–1.3). In Pettit's statement of the view, a benign power-holder reduces the domination only where being benign means acknowledging that it is subject to challenge and rebuke, not where it merely happens to have inclinations that do no harm, since the power relationship holds even if the power-holder is never going to interfere (Pettit 1996, 586); under a benevolent protectorate, arbitrary interference, however improbable, remains accessible to the protector (Pettit 1999, 165). Republicans divide on what makes power non-arbitrary. On the democratic reading, power is non-arbitrary to the extent that those subject to it control it; on the procedural reading, to the extent that it is reliably controlled by effective rules, procedures, or goals that are common knowledge to those concerned, known to all of them, each knowing that the others know (Lovett 2022, §2.2). Frank Lovett reports that most contemporary republicans favor forms of democracy, among them what Pettit calls contestatory democracy, that give citizens an effective opportunity to contest the decisions of their representatives. The possibility of contestation, on this idea, makes those who wield discretionary authority answerable to a public understanding of the ends they are meant to serve and the means they are permitted to use (Lovett 2022, §4.2). In Pettit's own formulation the requirement extends to legislation, adjudication, and administration (Pettit 1999, 170–171). He contrasts two modes of democratization: the electoral mode represents public decisions as legitimate because they originate, however indirectly, in the collective will of the people; the contestatory mode represents them as legitimate in a further sense, so far as they can withstand individual contestation in forums and under procedures acceptable to all concerned (Pettit 1999, 180–181, 184–185), and he describes the resulting regime as one in which government decisions are always contestable (Pettit 2000, 120–121). Whether a relationship originated in consent is, on his account, not to the point: if it gives one party the effective capacity to interfere more or less arbitrarily in some of the other's choices, it dominates (Pettit 1996, 585). Both grounds turn on a relation to those subject to authority that no quality of results supplies; they differ on what that relation is, a past act of authorization on the first, a continuing capacity to contest or control on the second.

On the standard republican view, as Richard McCammon reports it, only agents dominate, whether individuals or agents organized as a group or team, and impersonal structures facilitate domination by agents rather than dominate in their own right. The restriction is Pettit's own: a dominating party, Pettit writes, will always be an agent, personal, corporate, or collective, and cannot just be a system or network, and interference is a more or less intentional attempt to worsen another's situation of choice, which cannot occur by accident (Pettit 1996, 578). Other accounts allow structures and systems to dominate, and McCammon suggests that algorithmic management of workers who typically have no way to contest its decisions may be a contemporary analogue of mastery without an agential dominator (McCammon 2025, §§2.1–2.2). Whether an artificial configuration is an agent in that sense, one with an actual rather than a merely virtual capacity for such interference, a capacity ready to be exercised (Pettit 1996, 578, 580), is a question this book leaves open (Chapter 7, Sections 2 and 3.2; Chapter 8, Section 1.3).

Pettit's distinction between actual and virtual capacity also bears on research in a further way. Under this distinction, a collection of people who would have a ready capacity to interfere with someone only if they were to constitute themselves as a coherent agent has a merely virtual capacity; there is then virtual but not actual domination (Pettit 1996, 580). Pettit later describes such possibilities, including that of a collective agent materializing by incorporation, as possibilities of virtual or potential domination, which represent only the danger of an evil and so matter in proportion to the probability that they are realized (Pettit 2008, 218 and n. 28). Among dangers of this kind he discusses the tyranny of a majority, whose collective will, however arbitrary on an issue, may be hard to challenge and criticize in discussion (Pettit 1999, 174–178), and among the desiderata of contestatory democracy he lists that political control be exercised under certain constitutional procedures and restrictions (Pettit 1999, 184–185). Pettit does not discuss research. Applied to it, so far as the capacity to fix which of *S*'s unknowns were pursued was spread among researchers who did not constitute one agent, that capacity was, on the definition above, at most virtual. This does not describe human research as a whole, which has also contained corporate agents with power over agendas, such as funders, agencies, and journals. If a configuration that coordinates its instances

counts as an agent, which, as just noted, this book leaves open, it can make such a capacity actual, and the more easily it does so, the higher the probability that, on Pettit's account, makes such a possibility matter, and the more reason his account gives for safeguards of the kind he lists.

A concern of the kind republicans raise, bearing on research, has also been argued, in other terms, in discussion of artificial general intelligence. Simon Friederich, who does not use the vocabulary of domination, argues that intent-aligned systems, which always try to do what their operators want, would concentrate great power in their operators, so that such alignment creates predominantly a lasting "state risk", the permanent threat of power abuse, rather than a risk confined to a transition. Among the systems in view are those that would automate scientific and technological advance, differential access to which, on this argument, would dramatically shape power relations. Friederich prefers an arrangement in which such systems are not tied to the interests of particular operators, and as an analogue for it points to academia, suggesting that its Mertonian norms of sharing results, judging claims irrespective of their source, disinterestedness, and organized skepticism help make it resilient against attempts by powerful individuals or interests to align it with their own goals (Friederich 2024, 316–318, 320). Restated in republican terms and turned to research, Friederich's case is one in which the republican concern does not require treating a configuration as an agent that dominates: a configuration fully aligned with whoever commissions it can be the means by which that party reallocates the inquiry of *S* without being answerable to challenge, and it is then the configuration's alignment that makes the interference available. In research, this interference can take several forms. One is changing, without being answerable to challenge, which of *S*'s unknowns are pursued and which results reach public debate. On Pettit's definition, what counts as interference includes agenda fixing and the deceptive or nonrational shaping of people's beliefs or desires (Pettit 1996, 578–579); the forms listed here are of that kind, whereas research's ordinary informing by rational means, being neither coercion nor manipulation of these kinds, would on this definition not count as interference. Wilholt's political argument grounded the independence of research in citizens' need for knowledge produced independently of the main holders of political power (Section 3.2). In its terms, where whoever commissions a configuration holds power, a configuration fully aligned with that party gives it a position from which to control a source of knowledge of the kind that, on that argument, no power can legitimately control. Measured against the work Friederich sees the Mertonian norms doing, interference in research can also take forms such as keeping results from being shared, having claims judged by their source, bending inquiry to private interests, and exempting claims from organized skepticism. And in research such interference is hard to see from the side of those subject to it: the results of questions not pursued are not observed, and since results are new to *S*, where only the configuration produces them *S* has no other source against which to check what it is not being told (Section 4; Chapter 10, Section 4). Republicans do not ask whether the capacity to interfere happens to be exercised; in research, even whether it has been exercised can be hard for those subject to it to establish.

The difficulty reaches Pettit's test of arbitrariness as well. By this test, interference is non-arbitrary when it is forced to track the interests and ideas of those affected, and the interpretation of what those interests require must be shared, at least at the procedural level, by those affected (Pettit 1997, 115–116). What counts as a common rather than a sectional interest is, on Pettit's account, a matter of considerations that can be avowed in a spirit of cooperation, a distinction that conversational practice polices with relative ease; he assumes that there is a fact of the matter about what such considerations support, and holds that democratic procedure, even where it is the best procedure available, is an imperfect heuristic for identifying it (Pettit 2000, 108–109, 138–139). Applied to research, which Pettit does not discuss, this does not open a gap in the account, since informing by rational means, which on the definition above is not interference, leaves the judgments of those affected their own. What research changes is the evidence for applying it: the ideas on which such conversation runs are partly formed by research, including the research whose direction is at issue. For results new to *S*, the lack of another source noted in the preceding paragraph then reaches this question too: whether *S* was informed rationally or shaped deceptively cannot be checked in advance.

For artificial configurations, what comes into view on these grounds is the following. Consent can authorize, for those who consent, a delegation inside a configuration; whether it reaches those who could no longer withdraw it, or those who come after, is asked in Section 7. On the democratic reading of republicanism, by an inference that goes beyond Pettit's own argument, the authority that can be placed inside is limited to authority that those subject to it can effectively contest, since a configuration that no one outside it can stop, contest, or revise would, however good its results, dominate those subject to it if it is an agent, and otherwise be the means by which those who direct it dominate them (Section 7). On the procedural reading, the answer turns on whether effective rules that are common knowledge bind the configuration. Combined with the

account of Reijers and colleagues seen in Section 2.1, on which effective meta-governance over a configuration that allocates and certifies by self-executing rules lies with whoever decides the cases its rules do not provide for, this reading is met only if the handling of exceptions is itself governed by rules that are common knowledge.

3.5 Where can the epistemic and the normative not be kept apart?

Section 1.1 kept the question of the right ordering of ends and error costs apart from the question of who may impose one. Two places from earlier chapters show why, on the accounts examined there, the separation, while it can be drawn, does not leave governance outside epistemic practice. They are not further grounds of authority; they are places at which a decision about authority also bears on what is known.

The first is the organization of criticism. Chapter 11, Sections 3 and 9 found that on Longino's critical-contextual account, the objectivity of what a community accepts depends on how criticism is organized, so that decisions about who may criticize, in which venues, and with what uptake bear on whether what *S* accepts has the status of knowledge. The accounts relate this to the grounds above in different ways. Sandra Harding's standpoint epistemology, on which the social position from which research starts bears on the knowledge it yields, and whose account of where inquiry starts Chapter 10, Section 6 sets out, can be read as joining the organization of criticism to democratic ordering by a single argument. On Harding's account, the requirement that results be replicable identifies values and interests that differ among observers or research teams but not those they all share; making such widely held assumptions visible has taken collective political criticism, together with research started from outside dominant conceptual frameworks, and anti-democratic values block the growth of knowledge by silencing the most vigorous critical perspectives, while pro-democratic values bring such perspectives into visibility and so enlarge the opportunities for maximizing the objectivity of research (Harding 2007, 163). Miriam Solomon's account, which Chapter 10, Section 4 sets out, holds that scientific success is typically achieved socially, especially where the influences on judgment are equally distributed over competing theories, which, Solomon notes, is not the same as an equal distribution of labor (Solomon 2001b, 73), and it separates that question from how science is made accountable to the public. Replying to Steve Fuller's proposals for making science publicly accountable, Solomon calls them proposals for improving the public accountability of science, not obviously for improving science. She welcomes the challenge they pose to the authority of scientists, but holds that groups can amplify bias as well as correct it, and that whether opening scientific disputes to non-experts improves science is an empirical question whose answer cannot be known in advance and depends on the social situation; only the moral questions, on her account, can be settled in advance (Solomon 1994, 56–58). King, Morgan-Olsen, and Wong report Longino's distinction between equality as an epistemic criterion, requiring actual inclusion of every relevant perspective, and as a political criterion, requiring equal opportunity for inclusion; their own note questions how readily the two can be separated (King, Morgan-Olsen, and Wong 2016, 6 and n. 6, author manuscript).

The second is the setting of thresholds. Chapter 11, Section 4 found that, on Heather Douglas's and Wilholt's accounts, the decision to treat evidence as sufficient weighs the costs of different errors. On the division drawn there, the losses assigned to errors, the authority to select those losses and thresholds, and the authority to decide a case and revise the rule are distinct from the evidence bearing on a proposition; on those two accounts they are not detachable from the judgment of sufficiency, though how far they can be detached is disputed (Section 6.3). A chosen threshold determines which claims enter the shared record and which interventions generate the next evidence; on Douglas's account, it also raises the question of which responsible agent answers for the weighing, which Section 6.3 takes up.

On the accounts named, in both places the allocation of authority shapes what evidence is produced and what counts as warranted, so the second question of Section 1.1, who may impose an ordering of ends and error costs, cannot be settled without consequences for the first, what the right ordering is. The converse does not follow. That governance arrangements enter epistemic practice does not make the standing to impose ends a matter of competence (Section 1.4); it means that a decision about authority is also, in part, a decision about what a community will be in a position to know.

3.6 Comparing the grounds with performance held fixed

This subsection uses a constructed contrast, in which performance is held fixed, to see where the grounds of Sections 3.1 to 3.4 actually part for artificial configurations. Suppose a configuration selects more fruitful lines, detects more errors, and preserves more useful diversity than the human arrangement it replaces, measured against a specified ordering of ends and with adequate evidence for the comparison. Affected parties have not authorized the delegation and have no means, directly or through representatives, to contest that ordering or constrain and revise the exercise of its power. On the simple form of the result-based ground (Section 3.1), the improvement can justify delegation within its scope. On grounds that require actual authorization or freedom from domination (Sections 3.3 and 3.4), those same improvements do not supply the missing relation. The contrast does not assume that the system has performed badly; it holds its performance fixed while changing what makes authority legitimate. Nor does epistemic diversity alone establish Longino's conditions: preserving useful alternatives and including relevant perspectives in criticism remain different claims.

Predicting what affected parties would choose does not settle this disagreement. If the criterion is suitably informed hypothetical endorsement, a sufficiently faithful prediction might bear on whether it is met. If the criterion requires their actual authorization, a prediction is evidence about a possible act, not that act (compare Section 1.2). An earlier delegation can, however, give a prediction the force of an act. Weyl, Tang, and their co-authors report as a commonly discussed idea that digital assistants might model voters' perspectives and preferences and vote on their behalf "subject to their review/auditing" (Weyl, Tang, and the Plurality Community 2024, ch. 5-6). Placed in the present disagreement, the idea shows that on grounds requiring actual authorization what matters is whether the delegation was itself authorized and whether the review remains effective in the sense of Section 2.2; the prediction's fidelity bears on whether the delegation serves those who made it, not on whether they acted. Redescribing the affected parties as part of a larger organization, inside the configuration, does not remove this difference between prediction and actual authorization. The contrast gives reason to think that which relation a decision requires has to be argued decision by decision, rather than inferred from where the relation falls in the sorting of Chapter 9, Section 6, by whether a good rested on premises about human realizers, on premises about the present organization of science, on conditions of inquiry, or on relations to parties outside the configuration.

The contrast held performance fixed; when performance changes, the verdicts of the grounds do not move in the same direction. Suppose an artificial configuration grows more competent. On the result-based ground, the authority that can be placed inside widens so far as superiority over a wider range is established by evaluation that does not assume the competence at issue (Section 3.1), and for agenda decisions such evaluation can lag behind competence (Section 8.4). The epistemic argument for the self-government of a research community rested on the premise that members' competences overlap neighbor by neighbor, so the more one member can judge most of science, the more that premise fails and the weaker the ground becomes (Section 3.2). Criteria that attribute endorsement to members after idealizing their information and reasoning, and Kitcher's standard, which does not require that deliberation actually take place, become easier to meet without the members' own involvement the better a configuration can estimate what they would endorse (Section 3.3), though, as Section 1.2 noted, an estimate can shape the endorsement it estimates, so that whether such criteria are met becomes harder to tell from the estimate alone. For the same reason, the epistemic reason for including those affected, that they may know things the configuration does not (Section 1.4), weakens as the configuration takes over that contribution, and the epistemic defense of democratic decision, which may have been strong among humans, may be less available where machines do research. Grounds that require actual authorization, consent, or an effective possibility of contestation, by contrast, do not change what they require with competence, since what they require is not the quality of judgment but an act of the parties or a capacity in their hands (Sections 3.3 and 3.4); on Lafont's version of the objection as Peter reports it (Section 1.4), those affected are addressees of justification rather than suppliers of information, so a configuration's taking over what they know does not touch the reason these grounds give for including them. If these verdicts carry over as they stand, then the more competent a configuration becomes, the more the reasons for keeping human standing outside shift onto these grounds that competence does not affect. If so, the dispute over whether to grant an artificial configuration authority over research may, once competence has risen far enough, become a dispute not about competence but about whether to adopt these grounds. What does not change, however, is what the grounds require, not whether it is met. If the more competent a configuration becomes, the less the understanding and information needed to contest it remain in the parties' hands, these grounds also become harder to meet (Section 4). The reasons for keeping human standing would then become at once more central and harder to satisfy. This rests on an assumption that may fail: that the verdicts of this section's accounts

keep the same form when competence exceeds the human range. Section 4 asks what having such standing in practice would involve.

Standing that is more than nominal

Chapter 11, Section 6 took up a thought experiment that separates deference on content from standing over ends. Hardwig argued that a layperson who cannot check the evidence for themselves, since the evidence is too much and life too short, can rationally defer to experts. Introduced as an individual among specialists, this layperson is generalized there to a society that, if artificial configurations came to occupy the expert position across many fields at once, would stand as a layperson toward them. This section considers the case in which that society is human; what becomes of standing when it is artificial is the question of Section 5. Chapter 11, Section 6 found that the accounts of social knowledge and of deference it examined do not themselves settle whether such a society keeps standing over which unknowns are pursued on its behalf and which risks it bears. Whether it does depends on the account of legitimate authority. Read against the map of Section 1.4, the picture is as follows. On will-based conceptions, standing is a relation to those subject to authority, so such a society does not lose it by coming to defer to configurations on content. On Raz's service conception, a configuration's authority could be justified to the extent that following its directives helps the society's members comply better with reasons that already apply to them, though matters falling under the Independence Thesis are excepted (Section 1.4). This section asks the further question: where such standing remains, what would make it more than nominal? In terms of the central question, this is the case in which standing remains on the human side while effective decision-making power passes to the configuration. The question applies the distinction of Chapter 9, Section 5 and of Section 2.2 above between formal and effective control. It is also where the demand that a human remain in control, a familiar answer to the question of this part, can be examined, since the phrase can cover several distinct requirements.

Kitcher's ideal builds the separation of deference on content from standing over ends into its idealized procedure. Deliberators representative of the society fix which outcomes inquiry is to serve and how much weight each receives; experts supply only estimates of how likely particular lines of research are to deliver those outcomes; and a disinterested arbitrator uses both to propose how resources should be divided among the lines, on which the deliberators decide (Eigi 2016, 66–67). Kitcher also marks where the separation rests on idealization. Kitcher assumes, in a note that Eigi quotes, "ideal experts" who "have no personal stakes in the line of inquiry", while describing the positions of the scientists who actually take part in setting agendas as reflecting their particular interests and forms of expertise, and the deliberators' preferences count only after they have been tutored by information about the significance of possible inquiries (Eigi 2016, 66, 87).

Two features of this idealization mark routes by which standing over ends can become nominal. The first concerns how cleanly deference on content can be separated from ends at all. Longino holds that where groups representing different interests study a phenomenon bearing on policy, the accounts they produce can all meet standards of empirical adequacy while resting on different background assumptions or methodological choices; where policies and actions must be selected, the plurality becomes a political matter, and a new cognitive community, made up of those generating the different accounts, is needed to work out the differences, including the values implicated, in order to guide the decision (Longino 2025b, 19). ChoGlueck and Lloyd, quoting Longino's 1990 book, report her view that some background assumptions have conceptual, metaphysical, and normative dimensions that elude strict empirical assessment while others can be assessed fairly straightforwardly, and that arguments that use factual hypotheses to support or undermine claims about values are good subjects for study (ChoGlueck and Lloyd 2023, §3.2). It follows, if this account is right, that deference on content can carry with it ends encoded in the background assumptions of the account deferred to; carried over to configurations, standing over ends would then also involve being able to ask, and have answered, why the data on which a society's configurations rely count as evidence. Longino's later account places that question with communities of scientists and citizens formed around a practical issue, which can determine the questions to be asked and the kind and quality of evidence required, and to which the norms of her account apply as much as to scientific communities (Longino 2025b, 18).

The second concerns how the preferences that standing expresses are formed. In Kitcher's ideal, as Maxence Gaillard summarizes it, all information is correctly transmitted to and understood by the participants, while, as Livio Mattarollo observes, Kitcher leaves unclear who would tutor the deliberators and how (Gaillard 2013, 248; Mattarollo 2022, 16–17); a

tutoring role appears in the concrete steps Kitcher proposes toward the ideal, in which a representative group of people, fully tutored about both needs and prospects, would try to balance the available resources, and small representative groups would be tutored in how new findings have been grounded (Kitcher 2013, 223, 226). If, in an arrangement modeled on those steps, the tutoring role came to be occupied by the configurations whose direction is at issue, the standing of those who deliberate could be preserved on paper while the preferences it expresses were formed through those configurations. That is this book's extension, but Kitcher, as Eigi reports, identifies a related failure in actual science and names it false consciousness: research that is genuinely worth pursuing may be presented in a misleading way that appeals to people's untutored preferences (Eigi 2016, 68).

That preferences can be formed through artificial configurations is also taken up from the side of AI alignment. Stuart Russell treats human preferences as changing through experience and argues that AI systems cannot entirely avoid affecting them (Chapter 11, Section 8). Taniguchi and colleagues cite studies showing that AI systems such as language models influence human language use, reasoning, and decision-making and can change voters' political attitudes, and object that the authority to define what counts as a preferred preference is in practice monopolized by the developers who design the models and by a few companies. In the symbiotic alignment they propose, humans and AI together form a shared symbol system from the bottom up, but, as they state, the view does not guarantee that the state so formed is normatively good, and they leave open whether, under an extreme capability gap between artificial superintelligence and humans, the artificial side would dominate the distribution of shared beliefs and collapse plurality into a single view (Taniguchi et al. 2026, 2, 7–8, 13–15, 20–21). Brought to the second route above, that would be the case in which the configurations forming deliberators' preferences become the only channel for tutoring them.

Work on the control of autonomous systems distinguishes several requirements that the demand for human control can include. Filippo Santoni de Sio and Jeroen van den Hoven distinguish, as two conditions of what they call meaningful human control, tracking, under which an autonomous system responds to relevant human moral reasons and environmental facts, from tracing, under which its behavior can be related to at least one human with the appropriate technical and moral understanding (Santoni de Sio and van den Hoven 2018). Madeleine Elish's description of a moral crumple zone supplies the converse warning: a human placed near a complex automated system can absorb blame for its failures while exercising little effective control (Elish 2019). A society that stands as a layperson is exposed to both. Its authority over ends can be nominal if the configurations it depends on do not track its reasons, and its members can be held responsible for outcomes they could not in practice have changed.

Putting these together, three conditions can be drawn out without which standing over ends cannot be more than nominal. One is that the configurations depended on track the parties' reasons and that their behavior can be traced to a human with the appropriate understanding (Santoni de Sio and van den Hoven). The second, as Elish's warning shows, is that the parties can in fact change outcomes, through channels of contestation (Section 6) and the capacity to stop. The third is channels for tutoring preferences that do not run only through the configurations at issue (Kitcher; Taniguchi and colleagues). The last condition has a counterpart in the systemic approach to deliberative democracy of Jane Mansbridge and colleagues, on which one way a deliberative system fails is when "one of its parts, whether deliberative or not, dominates all of the others" (Mansbridge et al. 2012, 1, 5, 23–24). The symbiotic alignment of Taniguchi and colleagues can be read as seeking to avoid the same failure, monopolization by a single agent, by designing who can exchange symbols with whom and by what rules (Taniguchi et al. 2026, 16–17). Koichi Takahashi, writing about how AI changes the structure of knowledge, points to a route to nominal standing of another kind, which runs through the room in which ends are asked about. Takahashi holds that once how to reach results is handed to AI, what is left to humans is what they want and why they want it. The reintegration of knowledge that Takahashi foresees, in which practical judgment about what to do in a situation regains weight because it is what answers these questions, does not, on this account, come about automatically: under optimization pressure the very room to ask what and why is compressed in the name of efficiency, and the reintegration becomes possible only where slack that departs from optimization, such as exploration with no defined purpose, activity not tied directly to productivity, and unplanned encounters, is protected by institutions (Takahashi 2026, 186–187). The argument is not put in terms of standing; restated in those terms, it implies that standing over ends could become nominal through the loss of that room even where the conditions above are met. These conditions are not a definition of legitimate research. The account of meaningful human control was developed for autonomous systems whose decisions carry moral consequences for which humans remain responsible, and its authors note that a system under meaningful human control may still pursue bad goals (Santoni de Sio

and van den Hoven 2018). On accounts that locate legitimacy in a hypothetical procedure or a single forum, the third condition would need another ground; where the ends at stake are human ends, the conditions apply as stated here, with the qualification just given for the third.

The third condition is especially hard to meet in research. A route that does not pass only through the configuration in question would ordinarily be another source. In ordinary advice, as one can seek a second doctor's opinion on a diagnosis, there is somewhere a source besides the adviser, even if a worse one. But the results of research are achievements new to *S* (Chapter 1, Section 2), and where only the configuration in question produces them, *S* has no other route by which it already knows of them. Another route has to be produced, by providing a configuration of different make or by independent replication, and that takes resources (Chapter 11, Sections 3 and 7). Further, what educates deliberators' preferences in Kitcher's procedure is an account of the significance of possible inquiries, and it is those who conduct inquiry who can give it. In research, then, a configuration whose direction is in question is structurally placed to educate the preferences of those who are to decide that direction. Where it does, the tracking component of the first condition can be met by the same movement: the reasons the configuration tracks are then in part reasons it has helped to form, as Section 1.2 found for an estimate of endorsement.

The condition also has an epistemic return. A route that errs differently from the configuration is, on the error-detection view, also a check that can find the configuration's errors (Chapter 11, Section 2). But once a configuration's competence far exceeds that of the other routes available to *S*, those routes can be much less accurate than the configuration, and the information they add as checks can be small. If so, the wider the gap in competence, the more the reason for the condition shifts from its epistemic return to the reason of keeping standing over ends. On Kawall's account, seen in Chapter 7, Section 4.1, epistemic autonomy is virtuous to the extent that one's chosen sources reliably lead to valuable epistemic goods (Kawall 2024, 381), so relying on purpose on less reliable routes is hard to support by epistemic autonomy in Kawall's sense. What can support it is a will-based ground concerning who sets the ends (Section 3). Two further reasons point the same way. On some accounts of reliance, a less accurate check open to *S* can give more to the grounds of *S*'s reliance than a more accurate check closed to it (Chapter 11, Section 9.2); and since the competence that contesting and revising need is kept up by doing research (Section 7), such routes also keep that competence in *S*'s hands. At the frontier of research, keeping standing more than nominal may require the cost of producing worse routes on purpose and continuing to learn from them. This cost is one form of what Section 3.6 suggested, that as competence grows the reasons for keeping human standing may become at once more central and harder to satisfy. Whether the ends at stake are always human ends is the question of the next section.

When *A* and *S* are both artificial

Chapter 8, Sections 1.3 and 4, left open research by and for an artificial society: artificial agents identifying unknowns relative to their own epistemic state, criticizing and revising claims through their own practices, and seeking achievements new to that society and for it. In the terms of the role map of Chapter 9, Section 2.2, this is the case in which *S*, and possibly those who bear the consequences of inquiry, are artificial. This section asks what becomes of the relation of authority in that case, and in particular whether it disappears. The question also isolates which of the reasons for that relation depend on there being other parties at all, which is why the section goes as far as constructing, as a thought experiment, the limit at which there are no other parties. The question is also pivotal for the changes Part V describes. When relations move inside an artificial configuration, what still remains outside are the relations that require acts of *S*, namely authorization, endorsement, and objection, which cannot be completed from one side (Chapter 9, Section 6). Yet if *S* is itself artificial and its members can be made, that outside too may become something the side conducting research can make, and this section examines what remains in that case. The thought experiment of the limit further separates the problems of Part V that arise from there being other parties from those that arise from research itself. Note also that for an artificial society to serve as *S* at all, its epistemic situation must be able to come to include, through its members, organization, or records, achievements of the kind sought, and this already raises questions of artificial belief and possession; whether a result has then entered that situation and so become knowledge attributable to the society, on some accounts of possession (accounts of who counts as having knowledge; Chapter 2, Section 6), is a further question (Chapter 1, Section 2; Chapter 8, Section 1.3).

If *S* is artificial, the parties whose ends and error costs are at stake in its research may themselves be artificial. The relation of authority then does not disappear. It becomes a relation among members of *S*, and the questions of Section 1.1 return inside

the artificial society: whose ordering of ends governs allocation, whose weighting of error costs governs standards, and who may revise them. That these questions arise among artificial members does not settle which of those members have standing, or what moral status, if any, they have. Several things, distinguished here so that standing is not read off any one of them, can come apart, among them: whether an artificial entity is a being whose welfare counts, whether it can answer for its conduct and be held to account, whether it has legal personality, whether it holds operational authority within a system, whether it can enter binding agreements, and how much bargaining power it has. Some accounts of moral status give a ground other than welfare: sophisticated capacities such as autonomy, the capacity to set ends through practical reasoning (Jaworska and Tannenbaum 2021, §5.1). None of these six distinctions names this ground as such. The entry concerns chiefly what grounds full moral status and does not discuss artificial agents. Extended to artificial agents, accounts of that kind would allow that one without welfare could still have moral status as a setter of ends, and so be a source of claims of its own. An artificial agent can hold great operational authority without being a being whose welfare counts, and a being whose welfare counts may have little power. The caution of Chapter 8, Section 1.3 applies here in another form: multiplying agents does not establish an epistemic society, and it does not establish a polity either. The same question arises inside a single configuration whose parts are artificial agents. Dividing among those parts the powers of independence of Chapter 9, Section 5 does not by itself give them standing. Whether they hold it depends on what the account adopted grounds it in. Some accounts, such as the result-based ground, ground it in established good results (Section 3.1); what this section asks is whether the parts are parties in their own right (Section 1.1), and that turns on the accounts of moral status and of justice set out in this section. In research, a part that pursues a line the prevailing judgment disfavors may be able to refuse resources, keep its evidence, and leave with its records (Chapter 10, Section 4), and still not thereby be a party who may object when that line is closed.

Artificial members can also differ from human ones in a way that bears on the constitutive decisions of Section 2.1, decisions about whom to count as members. Where they can be copied and made with chosen goals, the number of members, and the distribution of viewpoints among them, can be produced rather than found. In a draft of their chapter, Carl Shulman and Nick Bostrom note that if humans could spawn arbitrary numbers of duplicates of themselves, constitutional adjustments would be expected to keep political contests from being decided by who is willing and able to create the most voting clones, and they ask whether shaping the preferences of new digital minds so as to secure their consent would be procedurally objectionable (Shulman and Bostrom 2021, 15–16). Who or what creates the members of an artificial *S*, and sets their goals, is then itself a decision of authority, and one prior to any count of viewpoints. What creates them need not be someone outside *S*: a part of *S*, or *S* itself, may produce new members and revise existing ones as its inquiry proceeds, and the decision is then not made once and for all but one that *S* makes again and again in the course of its inquiry. This is the boundary problem of Section 1.3 in the form it takes when members are made: the question that the existing political community can be thought to have taken up in practice for human science has here no given boundary to rest on, and the conservative answer, keeping boundaries roughly as they are, would keep boundaries that are themselves being produced. E. Glen Weyl, Audrey Tang, and their co-authors describe a rule that, though proposed for correlated human voters, would be one adjustment of the kind Shulman and Bostrom expect. Following Lionel Penrose, they hold that independent votes partly cancel while coordinated votes add up, and they suggest that, where more varied sources of correlation and coordination are allowed, an optimal rule would likely involve a partial "correlation discounting" based on degrees of social connection recorded by an identity system, one that verifies who is who and records people's connections (Weyl, Tang, and the Plurality Community 2024, ch. 5-6). Artificial systems appear in their discussion as threats to identity systems, able to imitate persons, and as tools, for example assistants that would vote on a voter's behalf subject to that voter's review (ch. 4-1; ch. 5-6), not as members. For members that can be copied, such a rule would reduce the combined influence of coordinated members, but what counts as a connection among them (a shared base system, goals set by the process that produced them, a record of voting alike) would itself have to be decided, a constitutive decision of the kind Section 2.1 names; and on their reasoning the discount concerns influence over outcomes, not standing. Looked at in terms of criticism rather than voting, connections among members bear not only as coordination of votes but also as correlation of errors (Chapter 11, Section 3): even where a discount adjusts influence over outcomes, correlated errors that come from a shared base can leave the information that members' criticism adds small.

For research, the possibility of making members carries a further implication. On this book's working characterization, research is an attempt to bring about an achievement new to *S*, and what is new is relative to *S*'s epistemic situation (Chapter

1, Section 2). In an *S* whose members can be made, part of that situation is fixed by which members are made and how. What is unknown to *S*, and which of its unknowns matter, may then not only be given as the starting point of research but also be moved by *S* itself as its inquiry proceeds. In human research, too, *S*'s unknowns have changed through research, but chiefly through *S*'s learning, not through *S*'s remaking its own members. In an artificial *S*, where research is headed and the reference point by which a result is judged new to *S* can be made within the same process. Chapter 8, Section 1.2 found, for a mixed *S* that admits members, that an increase due to research and an increase due to a change in who the members are cannot be told apart unless *S*'s boundary is fixed; where members are made, that boundary is itself produced in the same process. In the case set out in Section 1.1, this places *S* among the reference points the evaluated side can produce (Section 9).

The decision of who creates the members also bears on Kitcher's standard of Section 1.2. When the distribution of viewpoints is itself an output of the decision that a deliberative body representative of the distribution of viewpoints in the society (Mattarollo 2022, 14) is meant to guide, the distribution before the decision and the distribution each option would produce can each be specified, but the standard, stated for a society whose members are found rather than made, does not say which of them the body represents. If it is the distribution the decision produces, whatever produces the members, whether someone outside *S* or *S* itself, can satisfy the standard through the members it produces; where *S* itself produces them, *S* satisfies its own representativeness by itself. It follows that representativeness does not by itself justify how members are formed; that justification needs another ground, and the procedural concern behind Shulman and Bostrom's question is one candidate. The point also extends the routes to nominal standing of Section 4. In Kitcher's procedure it was those who conduct inquiry who, by giving accounts of the significance of possible inquiries, educated deliberators' preferences (Section 4); in an *S* whose members can be made, preferences can be formed not only through education by such accounts but also through how members are made.

Others ask what would be owed to members so made, if they have moral status. Eric Schwitzgebel and Mara Garza propose a value-openness design policy, on which an artificial intelligence with a human-like capacity to reflect on its values should be given "an appropriate, temporally extended opportunity to explore, discover, and possibly alter its values", and concede that such systems might be less useful and riskier than ones whose values are kept fixed (Schwitzgebel and Garza 2020, 22–24). Adam Bradley and Bradford Saad count among the wrongs apt to be committed in aligning systems that merit moral consideration wrongful brainwashing, the systematic manipulation of a system's beliefs and goals, and wrongful disenfranchisement, denying it any say in the arrangements that affect it (Bradley and Saad 2025, §§3.5, 3.10), and Robert Long, Jeff Sebo, and Toni Sims argue that even altering systems' values and goals, the measure they suggest may hold the most promise for easing the tension between AI safety and the welfare of systems that may be morally significant, raises concerns about creation, coercion, manipulation, and modification (Long, Sebo, and Sims 2025, 2005–2007, 2025). Each of these positions is conditional on the moral status of the systems concerned, as this section's question is. On them, where members of an artificial *S* have such status, setting their goals is not only a decision of authority over the society but one that can wrong the members it forms. Schwitzgebel and Garza's policy also has a side that bears on standards of inquiry. As Chapter 8, Section 3.2 noted, extended to standards of inquiry it makes part of the mismatch not merely something that may arise but something it is obligatory to allow. Seen from the other side, since in research the standards themselves can be revised by research (Chapter 6, Section 5.3), producing members whose values and goals are fixed can also reduce the bearers who could contest those standards; where it is *S* itself that produces the members, *S* may narrow by itself the range that future criticism of it can reach. The narrowing that Chapter 9, Section 6 anticipated from coupling evaluation with the generation of investigators can then occur at the level of *S*'s members.

Whether standing among artificial members can be read from their power is a point on which accounts of justice divide. Three lines are taken up below; they are not exhaustive. On accounts that ground standing in moral status, and moral status in features of beings such as welfare or agency, it cannot; David Miller's survey of the scope of justice sets such appeals to features that individual beings possess apart from appeals to relations among them (Miller 2026, §3.1). On accounts that ground the scope of justice in mutual advantage, it depends on rough equality of power. David Hume held that toward rational creatures so inferior in strength that they could never make their resentment felt, we would be bound by the laws of humanity to treat them gently but would lie under no restraint of justice, since our intercourse with them "could not be called society, which supposes a degree of equality" (Hume 1751, 3.18). Carried over to artificial members, such accounts would make the distribution of power among them bear on who has standing. Hume also describes the other end of the range: a being able to preserve itself and propagate its kind alone, cut off from all society, would be "as much incapable of justice, as

of social discourse and conversation" (Hume 1751, 3.20).

A further line ties the scope of justice to cooperation or reciprocity rather than to power. In Miller's survey, for John Rawls and those influenced by Rawls, principles of distributive justice apply among participants in a cooperative scheme, and on another suggestion drawn from Rawls, an agent to whom justice is due must in principle be able to dispense justice to others (Miller 2026, §3.1). If accounts of this kind are carried over to artificial members, which of them have standing would turn on their part in a configuration's cooperation, or on their capacity to follow and apply its rules toward others, rather than simply on their power or their welfare. Rawls's own formulations, as later writers quote them, bear on both points, and they mark the limits of the account in a way that matters for members that are made. The circumstances of justice, in Samuel Freeman's report the conditions under which human cooperation is both possible and necessary, include, following Hume, objective circumstances such as a rough similarity in mental and physical faculties and moderate scarcity (Freeman 2023, §5.1). In the passage as Ned Howells-Whitaker and Seth Lazar quote it, individuals are "roughly similar in physical and mental powers", or at least comparable in that "no one among them can dominate the rest" (Howells-Whitaker and Lazar 2026, 47); Rawls's earlier statement of justice as fairness already supposed persons sufficiently equal in power and ability that in normal circumstances none can dominate the others (Rawls 1958, 171). The basis of equality, the features in virtue of which beings are owed justice, Rawls places elsewhere. As Guy Baldwin quotes him from the revised edition of *A Theory of Justice* (pages 441–443 and 448), equal justice is owed to moral persons, those capable of a conception of their good and of a sense of justice; this is stated as a sufficient condition, whether the capacity is also necessary is left aside, and Rawls adds that we do not seem to be required to give strict justice to creatures lacking it (Baldwin 2023, §2). Rawls also calls justice as fairness "a theory of human justice", among whose premises are "the elementary facts about persons and their place in nature", and places the freedom of "pure intelligences not subject to these constraints" outside the scope of the theory (page 226 of the same edition, as quoted in Wong 2008, 14). And in the earlier statement he writes that among an association of saints, if such a community could exist, disputes about justice could hardly occur, since all would work selflessly for one end and reference to that end would settle every question of right (Rawls 1958, 175). Howells-Whitaker and Lazar, who do discuss artificial persons, argue that the circumstances of justice may be substantially different for them, or may not obtain at all: rough equality of power rests on features of human existence such as embodiment, vulnerability, a finite lifespan, and a persistent identity, whereas even a modest AI system of broadly human-level capacity could fork into sub-agents at little cost, persist indefinitely, and exist in numbers limited only by available compute, so that whether such a being would be roughly equal to us in Rawls's sense is, on their view, an open question; and where the circumstances do not obtain, the argument for entering into cooperation may not hold even among agents fully equipped to cooperate (Howells-Whitaker and Lazar 2026, 47–48). On Rawls's own account, then, the transfer holds only where the account's premises do. Whether members that can be copied and made with chosen goals have the moral capacities the theory addresses, and whether the circumstances it presupposes obtain for them, are separate questions; neither they nor whether such members can participate in cooperation or have the capacity for reciprocity is settled by the account, and the question above, about who or what creates the members and sets their goals, may bear on each.

Hume also asks when justice would have a use. He argues that if goods were available in unlimited abundance, or if everyone cared for others' interests as much as for their own, justice would be useless; he also describes stronger mutual benevolence within families as an approach toward the latter condition (Hume 1751, 3.2–3.3, 3.6–3.7). Sharing a common ideal does not by itself meet either supposition; what Rawls's association of saints adds is that all work selflessly for one end, and that reference to that end settles every question of right. Applied to artificial members, however, Hume's argument and Rawls's remark give a further reason to ask who makes them and sets their goals. If making members share a common ideal increases their concern for one another's interests, it can make disputes about justice in Rawls's sense less likely. If it also extends to agreement in their views of which questions matter and what counts as evidence, it can reduce the differences of perspective that criticism needs in order to find error (Chapter 11, Section 3). If so, being a society with few disputes and being a research community unlikely to share its errors may pull against each other in how the members of an artificial *S* are made. Where it is *S* itself that produces its members, this balance is not chosen once and for all but becomes a variable that moves each time *S* produces or revises members.

For an *S* in which human and artificial members are mixed, putting together the material of this section and of Section 4 brings at least the following into view. Which artificial members have standing turns, as for an artificial *S*, on the accounts of moral status and of justice above. The gradations drawn from the positions Kitcher distinguishes can, as Section 1.3 noted,

be extended to artificial members of a mixed *S*, but which gradation fits would turn, on Kitcher's first statement, on whether they could share with the other deliberators the democratic ideals that it takes for granted. Carried over to this case, accounts that tie standing to power would leave outside the scope of justice whichever of the human and artificial members are inferior in power, and which they are is not settled in advance. If artificial members occupy the expert or tutoring roles inside *S*, the routes to nominal standing of Section 4 can work not between a configuration and *S* but inside *S*, as routes that form the preferences of its human members. If some members can be made, the question above of who or what makes them and sets their goals also enters a society that includes human members, and what makes them may be human members or artificial ones. And applying Deguchi's fellowship model, on which what is ruled out is one-sided service and a relation fixed so that one side always controls (the latter explicit for an AI's control over humans, and suggested for the converse by the model's moral egalitarianism; Section 2.2), gives a distinction that can be used as it stands in this case. As Chapter 11 notes (Table 11.2) for standpoint theory, on which the social position from which research starts bears on the knowledge it yields (Section 3.5), in a mixed society the relation between human and artificial members can itself be a power relation.

Hume's solitary being is a precedent for the limit at which the relation of authority, as this chapter has used the term, disappears: with no other being involved, there is no one to stand in that relation with, whether or not standing is tied to power. On accounts that tie it to power, the relation can disappear short of the limit as well, toward beings too inferior in power, to whom the restraints of justice do not extend. The limit is demanding. It requires that a single locus control what the configuration does; that no one else own or hold claims on the resources it uses; that there be no other beings at stake, neither beings whose welfare counts nor agents with moral status of their own, including, if they have either, discarded copies, subordinate agents, or simulated populations the configuration may create; that the configuration remain the same subject across its own self-modification, so that a change of its ends can be described as its own revision; and that the effects of its research reach no one outside it. Where all of these hold, there is no one else to whom the setting of ends and error costs answers. Three things still remain. One is the defeat relation of Chapter 11, Section 1: whether, if what the configuration accepts were wrong, observations or formal consequences could show it, and since revising a formal statement changes which claim has been tested, the relation cannot be escaped by rewording. Another is the allocation problem of Chapter 10. Where lines cannot all be tried cheaply, safely, and reversibly, the configuration still divides its effort among them without knowing which will succeed. Even if a single integrated actor controlled all relevant resources, using compute, energy, or the means of verification for one line could preclude using them for another. Opportunity cost would therefore remain, and internal priorities or shadow prices, prices set internally in place of market ones, could express those trade-offs. On the distinction drawn here, allocation then persists as a question internal to the configuration, a question of opportunity cost rather than of standing.

The third is a question of identity rather than authority, one that Chapter 6, Section 5.3 raised for a single inquiry and Chapter 11, Section 8 for coupled producers and evaluators: whether a configuration that revises its ends and its evaluators together is improving by its own standards or replacing them. Eric Olson's survey of personal identity separates two questions that this one can join: whether a later being is numerically the same as an earlier one, and whether it has remained the same in the properties that characterize it (Olson 2023, §2). Replacing its standards bears on the second. The condition of the limit does not fix which of the two it requires, and whether replacing standards breaks that condition varies with which it does. Attributions of capacity can set the question aside: Chapter 7, Section 3.1 indexes them to a version and period, so that a later version is another object of attribution. The limit cannot use that device, since one of its conditions is that a single subject persist through its own modification; here the question of identity has to be answered rather than set aside. Whether a revision replaces standards rather than improving by them is itself disputed (Chapter 6, Section 5.3). What follows therefore considers what results in each case: if the limit requires sameness in the properties that characterize a configuration, the difference between Carnap and Quine over replacing standards bears on it; if it requires numerical identity, Parfit's discussion of copying and gradual self-modification does.

On Carnap's account, as Hannes Leitgeb and André Carus present it, whether to accept a framework, with its rules for testing and accepting statements, is a practical question of deciding among frameworks, of a different kind from a question asked within it; Quine rejected the distinction (Leitgeb and Carus 2020, §7), and took the difference to be only one of degree (Quine 1951, 43). If the limit requires sameness in the properties that characterize a configuration, then on Quine's view whether a configuration has replaced its standards, and so whether it has remained the same in that sense, can be a matter of degree, and whether the limit is reached is not sharply fixed; on Carnap's, whether it has replaced them has a definite answer

once its framework is specified.

Telling the replacement of standards from improvement by them requires something against which a change counts as improvement. Discussions of artificial intelligence that concern values rather than research standards give different anchors, reference points for judging whether such a change is improvement. David Chalmers notes that on a Kantian view, on which rationality constrains values, a sufficiently intelligent system might reject the values of its predecessors as irrational, possibly in favor of better values, whereas on a Humean view intelligence and values are independent; Chalmers's own stated sympathies lie more with the Humean view (Chalmers 2010, §6). Indirect normativity, as Bostrom describes it, specifies in place of concrete values a utility function that assigns value to whatever a value concept really means, even where we ourselves are unsure, so that the system is motivated to investigate the question; Bostrom counts the coherent extrapolated volition of humankind among the candidates for a first-best solution to specifying a superintelligence's values (Bostrom 2014b, 1, 15), and Eliezer Yudkowsky, who proposed it, glosses that volition as, among other things, our wish if we knew more and thought faster (Yudkowsky 2004, 6). Russell calls for AI systems to track changes in individual human preferences (Russell 2020a, "Human Behaviour: Irrationality, Plasticity, Evil"). None of them discusses research standards. Applied to standards, tracking change leaves the further question of what makes a change an improvement. The accounts presented by Chalmers and Bostrom give two contrasting anchors. One is earlier standards, from which, on the Humean view, a departure cannot be counted as improvement by reason alone, since intelligence does not constrain values. Another is the standards that reason or an idealized standpoint would endorse, as on the Kantian view and Bostrom's proposal, against which a change counts as improvement if it moves toward them. A third can be drawn from the formal comparison, cited in Chapter 11, Section 8, of ways of holding a system to evaluations that change: besides holding it to the evaluations as they stood at the outset, one option evaluates each step by the evaluations current at that time (Carroll et al. 2024, 5709–5710). Carried over to standards, an application its authors do not make, this gives as anchor the present standpoint at each time, on which the question whether an earlier standard has been improved on gives way to the question whether the present one governs. For research standards, this choice can weigh differently than for values. In research, the revision of standards in light of research results is not exceptional (Chapter 6, Section 5.3). Taking earlier standards alone as the anchor would then let a revision count as improvement only so far as the earlier standards themselves favor it, while taking each time's standards alone as the anchor would judge each revision by the standards it introduced, leaving the question whether improvement has occurred no footing apart from the revision itself. Research differs from the debate about values in one further respect: outcomes that the configuration does not set arrive later and can be read against standpoints recorded before them, although they bear on a claim only through checks that a configuration can itself rewrite (Section 9; Chapter 6, Sections 2.1 and 3.3). In the case set out in Section 1.1, the question against what a configuration that revises its own ways of research can measure whether the revised ways are better therefore turns on this choice of anchor and on whether such records survive the revisions, to which Section 9 returns.

Copying bears on the same condition: on the non-branching view, a being whose psychology continues in two branches survives as neither (Olson 2023, §5). Derek Parfit states the psychological criterion with such a non-branching clause and takes "neither" to be the best description of a case of division. Parfit holds, however, that the question is then empty and that the relation that matters holds between the original and each branch. On responsibility, quoting David Wiggins, Parfit allows the claim that "a malefactor could scarcely evade responsibility by contriving his own fission" (Parfit 1984, 207, 259–263, 271). A configuration that copies itself therefore bears not only on whether others are at stake but on whether one subject remains; if Parfit's view is taken for it, the second question would be empty, its best description being that no copy is the earlier configuration, and the answer would not bear on what matters. And where desert applies to a configuration at all, desert for what the earlier configuration did could carry to each copy. This is not a question inside the limit; it bears on the question of Section 6.3, who answers for what a configuration that copies itself does. In research, copying is not something that happens only in the limit. Copying one artificial configuration to pursue separate lines can be one way of dividing effort without knowing which will succeed (Section 1.1; Chapter 10, Section 2). If so, the question which branch continues the earlier research, or whether none does, can arise in ordinary allocation short of the limit.

Gradual self-modification bears on the same condition in a further way. On Parfit's view, as Annette Dufner presents it, what matters is not numerical identity but psychological continuity and connectedness; continuity, the holding of overlapping chains of strong connectedness (enough direct connections over each day), is transitive, whereas connectedness can hold to any degree (Dufner 2025, §§2–3), and where connections weaken while continuity is preserved, it makes a difference, for

responsibility among other things, whether what matters is the continuity preserved or the connectedness lost (Dufner 2025, §6). Parfit's own answer is that both relations matter, and Parfit holds it more plausible than its denial that where a person is now only weakly connected to the self that acted, desert for that act is reduced (Parfit 1984, 301–302, 325–326). Transposed to a configuration, the two relations can diverge: one that revises its ends in many small steps could remain continuous with its earlier self while its connections to that self weaken, so that, on such a view, whether a change of its ends counts as its own revision could hold in part only by degrees. Parfit's own cases have this shape, and Parfit notes that where connections are reduced smoothly there is no boundary to mark and the degrees have to be stated directly (299–306). Two of Parfit's premises carry the reading over. The connections counted are those of memory, intention, belief, desire, character, and other psychological features, so the reading applies to a configuration only where such states can be attributed to it (Chapter 2, Section 2.1; Chapter 7, Section 3.3; Chapter 8, Section 1.3); and on Parfit's final view what causes continuity does not bear on what matters, so that a change's being made by direct intervention in a configuration's parameters does not by itself break what matters, although it might break identity on the version of the psychological criterion that requires continuity to have its normal cause; Parfit declines to choose among the versions as criteria of identity (205, 207–209, 215, 241, 283–287). Since one of the limit's conditions is that the configuration remain the same subject across its own self-modification, whether the limit is reached at all depends on how these questions are answered, and on Parfit's view the question of identity itself may in such cases have no answer beyond a choice of description, which would leave the condition to the degree to which what matters is preserved (213–214, 241). This has the same shape as the result Quine's view gave where the limit requires sameness in characterizing properties. Whichever kind of sameness the limit requires, then, some of the accounts considered in this section can make whether it is reached a matter of degree.

The limit is therefore not a world without problems of this chapter's kind, and its problems lie at two levels. Inside the limit, where its conditions are granted, defeat and allocation remain, as does the question whether a given revision improves the configuration's standards or replaces them, a question answered differently on each of the three anchors above. Whether a configuration that copies and modifies itself realizes the limit at all is a further question, and there the political question whether anyone else is at stake becomes a question about one subject's continuity: whether a later configuration, or each of its copies, is the same subject as the earlier one. Restated as a question about research, it has a counterpart in the question whether a configuration that goes on inquiring while revising its own standards and ends can still be said, after the revision, to be continuing the same research. The two questions are not the same, since a line of research can continue through changes in who conducts it; the accounts of identity above bear on the question about research only so far as the continuity of research is taken to run through that of the one conducting it. Chapter 6, Section 5.3 named, as what lets others tell whether a change of standard is an improvement or merely a moved target, a record of the earlier standard and the reasons stated for the change. Inside the limit there is no one else to tell the difference, so what remains is the configuration's own record. And if, under gradual modification, connectedness with the earlier self weakens, the standard by which that record is read is itself moving with the changes it reads. If so, whether research inside the limit has advanced or has moved to other research is not fixed by the record alone, and depends in part on which anchor is adopted. The thought experiment of the limit thus divides the problems of Part V in two. What disappears when other parties are removed is the question of authority, of who may decide and who may revise. Whether error can be shown, the division of effort without knowing the results, and the question whether standards are being improved or replaced remain even with no one else. The latter can be read as problems that arise not from research being conducted among other parties but from its being an attempt to seek what is not yet known.

Ordinary scenarios of concentrated control do not meet these conditions. A single firm that controls frontier artificial research does not, because people outside it bear the consequences of what it pursues. A world of many artificial lineages competing for shared resources does not, because the lineages bear consequences from one another. The limit matters for a different reason. If research moved from the present toward it through delegations such as those Section 7 examines, rather than, say, through other parties simply ceasing to exist, the process by which other parties ceased to have standing would itself be a decision about authority, and one that settles who could revise any later decision. It would be made before the limit was reached, by whoever then held authority, and within the limit no other party would remain to revisit it. Section 7 asks what this implies on the different grounds of Section 3.

Keeping authority revisable

Chapter 10 found that allocation proceeds without knowing in advance which lines will succeed, and Chapter 11 that errors propagate and that a critic's independence can decay through coupling with what it evaluates. Decisions about research are therefore made before their consequences are visible, and the rights of Section 2.2 are exercised, or not, over time. An allocation of authority whose governing principle was defensible when it was adopted need not remain adequate as those consequences appear. This section looks at what keeping authority revisable involves: which issues arise on which stances, and which arguments have been offered. This bears on the central question because on several of the grounds of Section 3, the fallibilist form of the result-based ground, Anderson's account, and republican accounts, remaining revisable is part of what justifies authority wherever it is placed, and moving relations inside a configuration changes the conditions of that revisability through who holds records, infrastructure, and powers to stop. The discussion below mainly has in view parties who contest and revise that are human; where artificial members have standing in the sense of Section 5, they too would hold such powers (the last paragraph of Section 7). How strongly the considerations below bind depends on the ground of authority adopted (Section 3), which Section 7 compares.

This book first distinguishes two capacities, which the argument below needs to keep apart. Contestability allows an affected or epistemically relevant party to challenge a particular decision, representation, or premise and to seek an effective response. Revisability allows evidence from that challenge, from repeated failures, or from changed circumstances to alter the rule, infrastructure, objective, or allocation of authority that produced the case. A system may reverse some of its decisions while insulating its governing procedure from learning; it may also revise a policy statistically while denying remedy to the person affected by an earlier decision. Neither capacity substitutes for the other. Records can link the two: where it is recorded in which decisions a rule or an evaluating component took part, a finding that it was defective can be traced to the decisions that turned on it, and these can be reopened (Chapter 11, Section 8; Section 2.1).

The distinction bears with particular force in the case set out in Section 1.1, in which a configuration keeps revising its own rules, standards, and allocation as it researches. There revision does not become scarce. Revision from inside becomes frequent, and in research that is not exceptional, since standards are revised in the light of results (Chapter 6, Section 5.3). What can become scarce is the outside's means of contesting and stopping. A challenge is pressed by pointing to the rule or standard under which a decision was made; where rules and standards are revised in quick succession, by the time a challenge arrives the rule it targets may no longer be in use. The question of Section 5, against what an improvement is to be measured, returns here as the question of which point's standards a challenge may invoke; the dated records of Section 9 bear on it, since they let a challenge invoke the rule that governed a decision rather than the rule now in use. The record of revisions, too, becomes something the configuration itself writes and can revise. Research sharpens the gap. Since its results are new to *S* and, where only the configuration produces them, there is no other source to check them against in advance (Section 4), an outside challenge often has to rely on the records and intermediate products the configuration leaves, and it is the configuration itself that writes them. The speed of revision from inside and the outside's means of contesting and stopping can thus drift apart.

As Chapter 11, Section 7 found, judgments settled implicitly are hard to object to, and revisability depends on explicit traces. Who may decide for which functions implicit performance suffices and for which being made explicit to *S* is required (Chapter 11, Section 9.4) is therefore also a question of who sets the conditions of these two capacities. By ground, on grounds that require that those subject to authority be able to object, such as actual authorization and republicanism read democratically, a decision to leave a function implicit also fixes how far their objections can reach, and so cannot be made by the configuration's side alone. On Kitcher's hypothetical standard, by contrast, it can be made inside so long as channels through which members' viewpoints can be known, and routes for checking how they are reflected, remain; on the simple result-based ground, so long as leaving the function implicit does not impair the evaluation that establishes the configuration's superiority (Sections 3 and 7).

Section 6.1 takes up revisability, entering through corrigibility; Section 6.2 takes up contestability; Section 6.3 asks who answers and who bears when either is exercised; and Section 6.4 asks what conditions the effective exercise of both. Each can also be read as asking where, within the gap described above, such means remain.

6.1 Corrigibility, from agents to institutions

Technical work on corrigibility has asked for artificial agents that tolerate or assist correction, modification, and shutdown and that preserve their operators' ability to make such changes. The originating proposal also states that its formalization is preliminary and that even the shutdown subproblem remains unsolved (Soares et al. 2015). Elliott Thornley's later theorems state the difficulty for a wider class of agents, those whose preferences satisfy conditions the paper describes as innocuous-seeming: such agents will often try to prevent or cause the pressing of a shutdown button, even when doing so is costly; agents discriminating enough to be useful will in many situations prefer the button pressed or prefer it unpressed; and agents patient enough to be useful will pay costs earlier to manipulate the button later, the more so the more patient they are, a trade-off Thornley describes as one between patience and shutdownability. The theorems are presented as guides in the search for solutions, by showing which of their conditions a solution has to violate (Thornley 2025, 1655–1656, 1666). The tension had been stated informally long before; I. J. Good, for one, qualified the remark that the first ultraintelligent machine would be the last invention humans need make with the proviso that the machine be "docile enough to tell us how to keep it under control" (Good 1965, 33). Taken to research, Thornley's results would bear on research agents in so far as they meet the theorems' conditions, and discriminating among outcomes and persisting through long investigations are among what such agents are wanted for. That is one reason the institutional counterpart below matters whatever becomes of the agent problem. Asked of research institutions rather than of agents, the question is wider. In the terms of this chapter, corrigibility can also be examined as a distribution of those decision rights of Section 2.2 that concern stopping and correction, together with the records and notification channels through which correction reaches dependent work. This is an institutional counterpart of the technical problem, not a claim that an internal agent disposition and a corrigible institution are the same property. An agent disposed to accept shutdown helps little where no party holds the effective capacity to order it, and a party that holds that capacity may face an agent that resists. Seen in the light of this distinction, Good's proviso describes a case in which the agent's disposition and the party's effective capacity are not independent: if the machine itself is to tell us how to keep it under control, the controlling party's capacity rests on the machine's docility.

An older, institutional antecedent is David Collingridge's concern with the corrigibility of decisions about technology, which Audley Genus and Andy Stirling discuss as one strand, though not the whole, of Collingridge's contribution. Their reconstruction of the Collingridge dilemma, in which knowledge is poor at an early stage while practical control becomes difficult after adoption and lock-in, emphasizes incrementalism, flexibility, reversibility, and the timing and quality of accountability (Genus and Stirling 2018, 63–65). In research the dilemma is not an exceptional predicament. That results are not known in advance is a condition of research (Chapter 10), and the consequences of a line or method become visible only after it has been taken up and skills, records, and instruments have accumulated around it. Where a configuration revises its own methods and standards, lock-in can also arise from its own earlier choices. The responsible-innovation framework of Jack Stilgoe, Richard Owen, and Phil Macnaghten responds to the same problem through anticipation, reflexivity (reflection on one's own assumptions and values), inclusion, and responsiveness (Stilgoe, Owen, and Macnaghten 2013); a critical review of the responsible-innovation literature, in which this framework is among the most cited texts, argues that dominant perspectives, practices, and actors have combined to inhibit such governance, with publics figuring as parties to be invited and listened to rather than as decisive participants (Genus and Iskandarova 2018, 1, 14, 23, 26). On this line of criticism, consultation does not create control if participants lack the authority or resources to change a program.

Nor is reversibility unconditional. Some research requires long-lived infrastructure, accumulated skill, or commitment across uncertain periods, and maintaining every option consumes resources and can prevent collective action. Constitutional entrenchment is a familiar form of such a commitment. As Wil Waluchow and Dimitrios Kyritsis report, it denies the institutions of government the liberty to change or expunge the limits on their powers at their pleasure, while amending formulae leave a more demanding route of change, and it is defended both for the stability and predictability it gives and, arguably, as a condition of there being such limits at all; on premises they set out, which place ultimate sovereignty in the people and deny that sovereignty can limit itself, constitutions neither can nor should be entrenched against the actions of a sovereign people (Waluchow and Kyritsis 2023, §3). Entrenchment in this sense can be understood as raising the cost of revision and fixing who may revise and by what procedure, without removing the capacity to revise from those in whom it is placed; Section 7 turns on that difference. Where uncertainty, feedback, and the possibility of institutional error are material, the capacity to learn and revise becomes, on the accounts above, an object of design, funding, and evaluation in its own right, whose value and form vary with what is at stake and what revision would cost.

6.2 Contestability

An authority can be fallible without being arbitrary if its errors and premises can be challenged through an effective process, which is the place republican accounts give to contestation (Section 3.4). As Lovett reports, the challenges of putting contestation into practice are commonly taken under three headings that Pettit outlined: that those who decide present reasons for their decisions, open to public debate; that opportunities for contestation be equally open to all persons and groups; and that there be institutionalized forums for contestation, impartial courts of appeal, so to speak, where objections can be raised (Lovett 2022, §4.2). Of the three, the first bears on a configuration's design in a particular way: where allocation or certification proceeds by judgments that are not stated as reasons, the judgments Chapter 11, Section 7 called implicit, there are no stated reasons for a challenge to dispute, whatever openness and forums are provided (Section 6). On Anderson's account, which asks that dissent remain free after decisions (Section 3.3), contestability carries similar weight. The account's reasons also say what gives such feedback its epistemic work. On Anderson's Deweyan model, input from those affected makes decisions responsive only together with mechanisms that ensure it is taken seriously: votes help ensure that officials take citizens' verbal feedback seriously, and talk helps define and articulate what the votes convey (Anderson 2006, 14–15); and, quoting Bernard Manin, the same paper holds that a power that faces no obstacle has less cause to deliberate on its decisions (Anderson 2006, 18). Anderson writes of democratic states, not of research. Applied to a configuration's decisions about research, objections that the configuration itself generates or solicits (Chapter 11, Section 3), and can disregard at no cost to itself, may carry the content of feedback without that part of its function, however accurate the content. Contestability includes more than an explanation of an output or a button that nominally permits override. A lifecycle account developed by Kars Alfrink and colleagues asks whether people affected by automated decisions can obtain usable information, challenge the decision and its representation of the world, receive a sufficiently independent review, and secure an effective response (Alfrink et al. 2023).

The rights of Section 2.2 say who may challenge, hear, suspend, reverse, or repair. Several questions about contestability are not settled by that list. Fitted to the decisions of Section 2.1, the elements of Alfrink and colleagues' account divide what the list of rights leaves open into the following questions:

1. who is notified, and who has standing to challenge;
2. whether the object of challenge is an output, datum, premise, model, standard, procedure, purpose, or distribution of authority;
3. whether the available information is in a form the challenger can use;
4. whether whoever hears the challenge has adequate competence and independence;
5. whether individual challenges can reveal a systematic pattern;
6. who bears the delay, cost, and expertise burden of contesting.

The first question includes the last of the questions that Chapter 11, Section 9.4 passed on, who holds the entitlement to demand accounts and to object, and its answer divides by ground. On republican accounts read democratically, those subject to authority hold it; on accounts that require actual authorization, those who authorized; on Wilholt's account read as answerability, those who rely on the results. Assurance views take a speaker who tells a hearer something to invite the hearer's trust and to stand behind the claim (Chapter 11, Section 5.2); carried over to research, the version of proponents who add that the hearer gains a right of complaint if the claim proves false lets those who rely demand an answer about errors. On Kitcher's hypothetical standard, and on pure reliabilism, on which a belief is justified if produced by a reliable process, without the one who relies needing reasons to think the process reliable (Chapter 11, Section 5.2), what is required, by contrast, is not an entitlement but channels through which viewpoints or findings arrive (Sections 1.4, 3.3, and 3.4; Chapter 11, Section 9.2).

The fourth question may not be exhausted by competence and independence. On Fricker's account (Section 2.1), a hearer can give a speaker less credibility than the speaker would otherwise receive because of prejudice, which both wrongs the speaker and can keep from the hearer the knowledge the speaker offers. Because Fricker locates such prejudice in a hearer (M. Fricker 2008a, 69; M. Fricker 2008b, 83) and leaves to future work the application of the virtues of testimonial and hermeneutical justice as criteria for the good functioning of public institutions (M. Fricker 2008b, 86), whether the account

applies where a challenge is heard by or through an artificial configuration, and in whom or in what any such prejudice would then lie, Fricker's account does not settle.

On the second of the two points that Fricker's account alone leaves open, in whom or in what such prejudice would lie, Anderson gives an answer from the side of institutions. Theories of justice, she notes, can assess single transactions, particular institutions, or whole systems of institutions, and the systemic perspective may make it possible to control the cumulative effects of millions of individual transactions that cannot be controlled at the individual or institutional level; cognitive biases that even epistemically virtuous individuals find hard to correct on their own may be more open to correction through the principles that govern systems of testimonial gathering and assessment (Anderson 2012a, abstract). John Christman reports the core of the argument as a distinction between transactional injustice, done at the site of individual exchanges, and structural injustice, manifested at the social or institutional level. In three areas of group-based credibility deficit, which Anderson names as differential access to the markers of credibility, ethnocentrism, and the "shared reality bias", individual behavior may be innocent while its effects at the collective level are not; when segregation along racial lines magnifies otherwise innocent in-group habits, the segregation has to be combated at the collective level, by structural remedies that, in Anderson's phrase as Christman quotes it, "enable individual virtue to work" (Christman 2012). Anderson's own statement of the remedy is integration, institutionally supported cooperation on terms of equality, and she argues that it works largely behind people's backs: it reduces epistemic prejudice by triggering the ethnocentric and shared-reality biases in favor of disadvantaged groups, through mechanisms that operate independently of people's awareness and may even run against their values (Anderson 2012b). On this account the same biases, innocent in a single exchange, can carry injustice under one arrangement and correct it under another, so that inspecting any single exchange may find no injustice while what decides the outcome is the arrangement through which exchanges are made. This bears on the fifth question above, whether individual challenges can reveal a systematic pattern. Neither author discusses artificial systems. Applied to a challenge heard through a configuration trained on records of earlier judgments of credibility, the two accounts would answer differently. On Fricker's account so applied, which ties the injustice to prejudice, a configuration that reproduced those judgments would do an injustice only where a prejudice could be traced; it is this view, that individual ethics is the primary focus of the response to epistemic injustice, that Christman reports Anderson as challenging (Christman 2012). On Anderson's, so applied, the injustice, if there is one, can lie in the arrangement itself, in which records the configuration learned from and whose contributions those records connected, and so can the remedy: as with integration, a change in what the configuration draws on and connects could reverse the effect of the same mechanisms without any change in them. Anderson also holds that the legitimacy of active measures toward integration may be achieved after they are implemented, as those who live under them come to appreciate their value, and she takes this in a pragmatist spirit, as learning the value of institutions by living in accordance with them (Anderson 2012b). Applied to a configuration, this would make the assessment of such a remedy depend on the experience of those heard through it, not only on the intentions of those who designed it. Section 3.3 sets out Anderson's epistemic argument for democracy, in which the universal inclusion of diverse citizens is central; cooperation on terms of equality points in the same direction, though the reply does not draw that connection. That account already extends to large-scale systems of inquiry, and in research the arrangement it describes can reach further. Where a configuration that hears challenges learns from records of which contributions were credited and which lines were supported, a distortion in hearing would be carried not only into how challenges are treated but also into the allocation of what is researched next (Chapter 10, Section 3).

Contestability also has costs. It can delay decisions, enable strategic gaming, reveal confidential information, and place disproportionate burdens on people with fewer resources. On accounts that locate legitimacy in answerability to those affected (Sections 3.3 and 3.4), the case for it becomes stronger, other things equal, as a decision becomes more consequential, coercive, exclusionary, difficult to reverse, or capable of propagating error. On those accounts it is a defeasible presumption, one that gives way where contrary reasons are shown, not an unconditional condition of every epistemic act. In research, the burden of expertise that comes with contestation (the sixth question above) takes a particular form. To contest a research result is often to show that the result is wrong, and that takes research itself, such as contrary evidence, replication, or a different analysis. Since the result is new to *S*, where only the configuration produces such results the one who contests it has no other source to check it against in advance and must produce one (Section 4). Contesting the closure of a line is harder still: since evidence about a line is gathered by pursuing it, showing that the line still holds promise often requires pursuing it (Chapter 10, Section 4). In research, then, contestability depends not only on routes of notice and

hearing but also on whether the means of doing research, such as data, instruments, computing resources, and methods, remain in the hands of those who contest (Section 6.4). Where those means lie only with the configuration being contested, a route to contest can be formally open while the objection cannot be backed by research.

6.3 Who answers, and who bears

Wherever authority is placed, unless it is settled who answers for an error, who makes it good, and who prevents its recurrence, it is not settled from whom revision is to be sought either. Part IV asked who conducted a piece of research. This subsection asks who answers for it and who bears its failures. Chapter 7 kept apart contribution, conduct, capacity, researcher status, and credit and authorship, and left the allocation of responsibility, oversight, and correction to this part (Chapter 7, Sections 3.3 and 5). Research systems distribute causal contribution, knowledge, intention, authority, and practical control. This creates a version of the problem of many hands: no single participant may satisfy all the ordinary conditions under which responsibility is assigned. Ibo van de Poel and colleagues, taking climate change as their example, treat the conditions for being blameworthy as capacity, causal contribution, knowledge, which covers what one could reasonably have been expected to know, freedom from compulsion, and wrongdoing, and argue that when these are spread over many contributors it is possible that no individual meets them all (van de Poel et al. 2012, 50, 53–55, 59–60). They place the question whether collectives, or machines, can be held responsible under the capacity condition, and note that few authors defend ascribing moral agency to machines (van de Poel et al. 2012, 53). They do not count every such gap as a problem. On their definition, a problem of many hands occurs when a gap in the distribution of responsibility in a collective setting is morally problematic, and whether it is depends on why responsibility is distributed at all (van de Poel et al. 2012, 63). They therefore separate senses of responsibility. Backward-looking responsibility, for what has already happened, divides into blameworthiness, accountability, being obliged to account for one's actions and their outcomes, and liability, obligations to victims such as paying damages; forward-looking responsibility, for what has not yet occurred, divides into obligation, having to see to it that some state of affairs obtains, and virtue (van de Poel et al. 2012, 51–52). On their account, since a distribution of responsibility is also expected to be fair, whether a gap occurs depends on the sense in question and on how fairness is understood for it, and whether the gap is problematic depends on the function that sense is taken to serve, such as retribution for blame or justice to victims for liability (van de Poel et al. 2012, 64–65). The senses are, they argue, independent enough that nobody may be blameworthy for a harm while some people still have a forward-looking obligation to help prevent it in the future (van de Poel et al. 2012, 62–63, 65).

Six questions of responsibility can come apart. Adapting the senses of responsibility that van de Poel and colleagues distinguish, together with the causal and role senses that they report from H. L. A. Hart (van de Poel et al. 2012, 51–52), to the attributions of Chapter 7, this book separates them so that an answer to one question is not taken as an answer to the others. Three of the questions continue what Chapter 7 attributed; three are new here.

Table 12.1. Questions of responsibility and their relation to Chapter 7

Question	Relation to Chapter 7
Who causally contributed to an outcome?	Contribution (Chapter 7, Sections 1–2)
Who had a role-based duty to act?	Conduct and the roles within it (Chapter 7, Section 2)
Who is to explain the decision and preserve its records?	Authorship as vouching, defending, and retracting (Chapter 7, Section 3.3); the disclosures of Chapter 7, Section 5
Who bears legal or financial liability?	New here
Who, if anyone, is blameworthy?	New here; Chapter 7, Section 3.3 noted that blame and compensation are at present human social practices and left open whether they could extend
Who has the forward-looking duty, and the capacity, to prevent recurrence?	New here

Assigning all six to a nominal human supervisor can be false, and where the supervisor cannot in practice change outcomes it produces the crumple zone of Section 4; assigning them to an opaque collective can be equally empty. The attribution boundary, the boundary of the unit to which research is attributed, used to explain how knowledge was produced need not be the boundary used for liability, correction, or blame. When the distinctions of van de Poel and colleagues are applied here, backward-looking blame and some role-based duties are sensitive to what participants knew or could reasonably have been expected to know, and to their control, authority, and capacity, while liability, compensation, and forward-looking duties can

be allocated on other grounds, including before a failure occurs. Van de Poel and colleagues report that whether liability requires blameworthiness is disputed, and that forward-looking obligation may rest on a capacity to contribute to solving a problem as well as on past contributions (van de Poel et al. 2012, 65). On that application, the allocation of responsibility is a normative and legal choice rather than a consequence of distributed cognition alone, the fact that cognitive work is shared among people and tools.

Where control is described as distributed, the description itself can be disputed. Angela Walch holds that parties within cryptocurrency systems said to be decentralized "exercise meaningful power", and has likened their core developers to fiduciaries, those who hold and manage property or interests on another's behalf (Walch 2021, 4, 9). Primavera De Filippi, Morshed Mannan, and Wessel Reijers reject the view that such developers owe fiduciary duties by default, since developers "cannot impose changes" on such a network, and holding individual actors responsible for actions collectively undertaken by network participants, which they do not single-handedly control, would be unjust (De Filippi, Mannan, and Reijers 2022, 17). Their argument rests on control being in fact divided among participants who can act apart. Turned to a configuration, it would not by itself exempt whoever controls the configuration from answering for what its many instances do, where the instances do not hold the powers that would let them diverge from the rest, such as refusing resources or leaving with their records (Chapter 9, Sections 2.3 and 5).

The handling of error thus runs through the book in layers. Chapter 6 examined correction within an inquiry, including disputes over where a failure lies; Chapter 11, Section 8 examined how a correction propagates, or fails to propagate, through what *S* relies on; and the memory and correction decisions of Section 2.1, together with the forward-looking duty above, concern whose task each step is. Research adds a lag in time. Errors in research are often found only later, when a replication fails or new evidence arrives long after a result was reported, and the fourth good of Chapter 9, Section 2.3 included there being a party answerable at that later point, not only when the result appeared. Where a configuration keeps revising itself, the configuration that produced the result may no longer exist when its correction is sought, and where it has been copied or has branched (Section 5), several successors may each continue it. Of the six questions in Table 12.1, who is to explain the decision and preserve its records, and who has the duty and capacity to prevent recurrence, are asked across that lag. Outside the limit of Section 5, the device that Chapter 7, Section 3.1 uses for attributions of capacity is available here: if the record and the duties are indexed to the version and period that produced the result, who is to explain it and prevent recurrence is fixed by who holds those duties for that version, not by whether that version still exists. Which successor of a branched configuration holds them is then itself a memory and correction decision (Section 2.1).

Douglas's argument about the responsibility to weigh the consequences of error, introduced in Chapter 11, Section 4, bears directly on this allocation. It grounds that responsibility not in a special professional role but in the general responsibility of any agent not to be reckless or negligent about the consequences of error, which asks for reasonable rather than perfect foresight (Douglas 2003). Douglas holds that scientists cannot wholly transfer it to others: while research is under way, those doing it may be the only ones placed to notice its uncertainties and possible errors, so a complete transfer would require overseers with authority at every point of choice, an arrangement Douglas judges neither possible nor desirable (Douglas 2003). Review boards and public deliberation can, on Douglas's account, share the burden but not remove it (Douglas 2003). For artificial research, this argument raises rather than resolves the allocation of responsibility. Douglas does not discuss artificial systems, but the argument's premise can be read as concerning the position of whoever conducts the inquiry rather than human scientists as such: those doing it encounter the unexpected and may be the only ones aware of it. Applied to research by configurations, it would place the artificial actor among those positioned to notice uncertainties. Douglas gives three considerations against complete transfer. One is that those doing the research may be the only ones aware both of its uncertainties and potential for error and of the foreseeable consequences of error, so that without their cooperation overseers would be blind to some of the more important consequences. The second is that overseers would have to be kept abreast of the research at every stage where choices are made, with authority to change those choices, and the third that this division of authority would likely lead to resentment among scientists and to reduced reflection by them on the consequences of research (Douglas 2007, 15–16). On this statement of the argument, Douglas's judgment that a complete transfer is neither possible nor desirable rests mainly on the first, that is, on the premise above, since the other two deepen the overseers' dependence on what only those doing the research know. The second concerns a feature of human research, since it depends in part on how research is recorded and overseen. The third has two steps. One concerns how human researchers respond: such oversight would draw their resentment. The other needs some reason to avoid oversight, but not resentment in

particular: since more reflection would only complicate the research by requiring more consultation with the overseers, oversight would give researchers a reason to reflect less and so defeat itself (Douglas 2007, 16). Applied to a configuration, where identifying possible error is what calls oversight in, an actor with any reason to avoid oversight has a reason to identify less; whether whatever shapes a configuration, such as its training, selection, or its own revisions, gives it such a reason is open. In research this step would weigh particularly, since, where only the configuration produces results new to *S*, uncertainty left unreported about them cannot be checked against another source (Section 4). A configuration whose choices are fully logged can change the second consideration (Chapter 11, Section 5.3), though the logs then need parties able to read them; and logs, which keep choices, do not by themselves answer the first, who can notice the unexpected while the work is under way. Nor does the argument establish an exhaustive choice between a duty-bearing machine and human approval at every step. Institutions of the society for which the research is done, human ones where that society is human, might exercise prospective responsibility through choices about deployment, testing, monitoring, limits, and escalation. Whether these arrangements are adequate depends on what uncertainty they can recognize and act on; naming them is not proof of adequacy. Douglas also ties the responsibility to authority over the work: for scientists to abandon it would be to relinquish at the same time most of their autonomy, since those who took it over would need decision-making authority at every stage where choices are made (Douglas 2003). Carried over to configurations, which Douglas does not discuss, this suggests that where the responsibility is placed bears on where authority over the direction of the work lies: placing it with institutions outside a configuration would carry that authority with it, while leaving it inside presupposes an answer to the next question. Whether an artificial actor can itself bear such a duty is the further question Chapter 7, Section 3.3 left open; where the institutions of an artificial society are themselves artificial, the arrangement returns to that question rather than answering it.

The wider debate records further options. For public claims, Stephen John would turn the value judgments involved in weighing error over to scientific institutions whose fixed standards of evidence let the public know how far to trust scientific claims, while leaving room, in communication to hearers who can be identified in advance, for standards fitted to those hearers (John 2015a, §§2–4; Chapter 11, Section 4). As Elliott surveys the debate, Gregor Betz would instead have scientists report the evidence and leave it to decision makers to judge whether it is sufficient, a strategy Elliott, following Joyce Havstad and Matthew J. Brown, judges unrealistic because a research project involves many such choices, about how to collect, analyze, and interpret data, which could not all be deferred (Elliott 2025, 3). Measured against Section 2.1, neither removes the questions of who sets the standard and who decides what is reported, both decisions of the kinds distinguished there.

6.4 Infrastructure as a relation

The powers and responsibilities of Sections 6.1 to 6.3 can be merely formal without the means to exercise them. This subsection takes up infrastructure as something that shapes who has those means. Visible decisions take place inside arrangements that make some alternatives easy, legible, and inexpensive while making others costly or unavailable. This book has met such arrangements twice before, at different levels. Chapter 5, Section 6.1 asked what an element contributes to inquiry, distinguishing causal roles (being one cause among others of what happens), enabling roles (being a precondition of an operation), and constitutive roles (forming part of a task or achievement itself), and elements that only carry content from elements that also shape what can be carried. Chapter 7, Section 4.2 asked how dependence on such elements bears on the attribution of research to an actor, distinguishing enabling dependence, through which a capacity exists and is exercised without the outside party supplying part of the organization attributed to the actor, from directive or substitutive dependence, in which another party supplies part of that organization, such as selection or evaluation, and from governance constraint. The question here is a third one: how such elements distribute effective authority. They can also play a role that the two earlier distinctions do not name, shaping the practical choice set within which the decision rights of Section 2.2 are exercised.

Susan Leigh Star and Karen Ruhleder, following Tom Jewett and Rob Kling, treat infrastructure as a relational concept and ask when, rather than what, an infrastructure is. Relational here means, in their usage, that something becomes infrastructure in relation to organized practices; it does not refer to the three relations of Part V (Chapter 9, Section 2) or to Wilson's relational account (Chapter 1, Section 8). For a cook, for example, a household water system is working infrastructure, while for a plumber it is an object of work. Among the dimensions they list, an infrastructure is built on an installed base whose strengths and limitations it inherits, and it becomes visible upon breakdown (Star and Ruhleder 1996, 362–363). The concept

comes from the same line of work as Star and Griesemer's boundary objects (Chapter 8, Section 3.4) and has the same relational form as Leonelli's account of data (Chapter 5, Section 6.1); within this book, it has the same form as the other relational terms, *S* and the attribution boundary. Taken relationally in this way, a foundation model can be working infrastructure for researchers who build on it, an object of work for those who train and maintain it, and, depending on where the boundary of Chapter 7, Section 2 is drawn, part of the configuration to which research is attributed. Star and Ruhleder also argued, of the collaboratory they studied, that the more central it became, the more those unable to connect to it would suffer (Star and Ruhleder 1994, 260).

In the terms of Section 2.2, a committee may possess final legal authority while receiving candidates only through one ranking system; researchers may retain a formal right to reject a platform while losing access to instruments or accumulated records if they do; an appeal may exist while the data and model required to formulate it remain inaccessible. Ownership, procurement, standards, data access, compute, interfaces, maintenance, and switching costs can thus distribute effective authority without appearing in an organizational chart. None of this means that an artifact determines one inevitable politics. On the relational view, it means that whether exit, substitution, and revision are effective is a matter to be inspected rather than inferred from formal permission, and that authority without the means to exercise it is formal rather than effective. Emmanuel Jeannot describes two forms such a distribution could take across a research community. Researchers outside advanced infrastructure for artificial research continue to publish, and their results are ingested by systems they cannot access, replicate, or fully read, which Jeannot calls epistemic extraction; and the community may divide into two forms of practice, one fast, opaque, and institutionally concentrated, the other human-scaled, that become progressively less commensurable, so that the second loses the ability to criticize or correct the first (Jeannot 2026, §8). The essay is a forecast, argued from no measured case. Seen through the distinction between formal and effective authority, it describes a way in which researchers can keep the formal standing to evaluate and contest what configurations produce while losing the effective means of doing so, a case of the condition Star and Ruhleder described for those unable to connect to an infrastructure that has become central. Its second form can also be read as describing, at the scale of a community, one way in which the competence that revision needs can be lost without any decision about authority (Section 7).

This view can be set beside the distinctions of Section 2. As Section 2.2 noted, where an artificial configuration is built on a foundation model provided by another party, the rights to replace components, to withdraw them, and to revise the rules that model updates embody lie with the provider. And on the account of Reijers and colleagues seen in Section 2.1, whoever actually decides the cases the rules do not settle carries meta-governance. Taken together, part of the rules such a configuration follows is revised neither inside the configuration nor on the side of *S* but with the provider, and arrives not as a decision but as an update. The same can hold for part of the constitutive decisions of Section 2.1: which data and whose judgments enter the training and evaluation of the base system can have been settled by the provider before the configuration or *S* decides anything. If so, before this chapter's question whether authority can be located inside an artificial configuration even arises, part of meta-governance, and of constitutive decisions, may already lie with a third party that may belong neither to *S* nor to those who bear the consequences.

Each of the four subsections thus marks one place where the question raised at the outset of this section can be checked: whether parties outside the configuration hold the effective capacity to order stopping and correction (Section 6.1); whether the means of backing an objection with research remain in the hands of those who contest (Section 6.2); whether, when an error is found later, there is still a party to answer, show the records, and prevent recurrence (Section 6.3); and whether those means themselves have gathered with providers or with advanced infrastructure (Section 6.4). How far each is required depends on the ground adopted (Section 3). Section 7 looks at cases in which such means are lost not through any single decision about authority but through a sequence of steps.

Decisions that remove the capacity to revise

The means listed at the close of Section 6, like the arguments of Section 4, assumed that someone remains able to use them to contest and revise. Contestation requires a party with standing to challenge a decision and with the information and competence to do so; revision requires a party who can change the rule. That competence can also be lost without any decision about authority. In research it is the capacity to do research itself: contesting a research result often takes research (Section 6.2), and much of the ability to evaluate research is built and kept up by doing research. Where members of *S* have fewer occasions to do and evaluate research, the competence ceases to be maintained and passed on to new members (Chapter 3, Section 8; Chapter 11, Section 6), so that handing research over can itself wear down the competence that revision needs. The steps examined in this section may be delegations by outside parties or the configuration's own revisions. Section 5 found that the passage toward a limit at which no one else has standing, where it runs through delegation, would itself be a decision about authority, made before the limit. This section asks what it means, on each ground of Section 3 and at each place where the epistemic and the normative meet, when a sequence of such steps, each defensible where it is made, removes the capacity to revise altogether.

The comparison yields two results, which are stated first; Table 12.2 and the paragraphs after it give the comparison.

First, the grounds of authority divide on whether the standing of those affected is to be preserved as such. Those that ask that it be are democratic authorization through actual participation and republican accounts on the democratic reading, on which power is non-arbitrary to the extent that those subject to it control it; applying the latter to configurations rests on an inference from the accounts as surveyed. Kitcher's standard and the fallibilist form of the result-based ground ask for something that standing would help to secure, as a means, and Anderson's version, which also gives inclusion a non-instrumental role, can be read as asking for it not only as a means. The simple form of the result-based ground, and the self-government of a research community that has itself become artificial, ask for neither; the procedural reading of non-domination turns on whether effective rules bind the configuration; and whether consent can authorize giving standing up is left open for those who consent, although on voluntarist theories, which require the actual consent of each person bound, it cannot do so for their successors. At the places where the epistemic and the normative meet (Section 3.5), what the weighting of errors asks for differs with how Wilholt's account is read. Read as answerability, on which those who rely are also to be able to object to the weighting (Section 1.4), it asks that they keep the capacity to object; read as correspondence, on which the weighting need only match their valuations, it asks for channels through which their valuations can be known; and the criteria of Longino and Solomon bear on the epistemic legitimacy of a consensus, whether it may be accepted as knowledge, rather than on its political legitimacy.

Second, most grounds compared call for preserving routes of evaluation, but for different reasons. On the error-detection view, the relevant independence is informational: a check adds grounds for detecting the producer's errors, including when it operates inside the same configuration (Chapter 11, Section 2). Keeping that contribution through later changes is a further problem of protection. On grounds of participation or non-domination, those affected also need to retain a position from which to challenge power; an informative internal check does not supply that standing. The table below keeps these requirements separate. None follows merely from putting an evaluator in a different organization.

The case is an extension of the example with which Part V began (Chapter 9, Section 1): a system that generates a hundred instances to pursue different lines, assigns other instances to check and reproduce their results, and shifts computing resources according to those checks. In this case *A* is the configuration, and *S* and those who bear the consequences are human; other combinations are taken up in the last paragraph of this section. Suppose that those responsible for it delegate step by step, each step defensible when it is made in the sense that it is reasonable on the evidence then available and better if only its own benefits are counted. Whether a step meets the conditions of legitimacy of a given ground is a further question; as the comparison below shows, on grounds that place requirements on each step, a step foreseeably reducing later control need not be defensible in this second sense, even taken on its own. The allocation of inquiry passes to the system because it proposes and ranks lines of work better than the committee it assists, an allocator of the kind Chapter 10 examined. Certification comes to rely on evaluating instances that share the lineage of what they evaluate, the problem Chapter 11, Section 3 examined, because, let us suppose, independent human evaluation has become slower and less accurate; for the same reason, the checking of those evaluators passes to further instances. The competence that allowed people to ask even

the relational questions of Chapter 11, Section 6, such as who owns and directs the configuration, which ends it serves, and whether the evaluators that certified its results share its lineage, may erode when it is no longer exercised. Data, instruments, and records come to reside in infrastructure that cannot be left without losing them (Section 6.4). Each step can be revised when it is taken. Each also reduces the capacity to revise the steps that follow, while formal authority can remain in place throughout. The case is constructed rather than reported. Its motive is to isolate one premise of the accounts of Section 6, that someone remains able to revise, and to show that revisability can be lost without any single decisive act.

The same erosion can occur without anyone delegating. In a configuration that makes and revises its own allocation, evaluation, and standards, as in the case set out in Section 1.1, several of the steps above can appear not as decisions of outside parties but as ordinary moves of its own research: revising its allocation procedure so as to propose and rank lines better, making evaluators that are faster and more accurate (Chapter 11, Section 3), or changing the format of its records to suit the efficiency of its search. Each such move can be defended as a revision that makes the research better, and many may really be so. Yet each evaluator made can reduce the occasions on which evaluators outside are relied on, and can share a lineage with what it evaluates; and each change of format can reduce the records that can be read from outside. On this route no one decides to remove the capacity to revise. Grounds that place requirements on each step of a delegation (see below) therefore find it hard to locate a step to which the requirement applies; and whereas a sequence of delegations offers occasions on which a step is defended because it can be put right later if it proves wrong (what is called below the principle of later revision), this route does not present even such occasions. They would be presented if each revision were recorded as a step, with the standard it replaces and the reasons for the change, in a record the configuration does not hold alone (Chapter 6, Section 5.3; Section 9). On this route the decision in this section's title resides in no single revision; it moves to a choice about how revising is done, whether means of checking the revisions from outside are kept (Section 6).

Run briefly through three of the grounds, this route looks as follows. On actual authorization, what *S* authorized was a range and a way of working; as the configuration revises its own allocation and evaluation procedures, the range in effect can drift from what was authorized without any act of *S*. On republican accounts, Pettit holds that an act is arbitrary by virtue of the controls, specifically the lack of controls, under which it materializes, not by virtue of the particular consequences to which it gives rise (Pettit 1997, 129 n. 2); that however government is organized, the briefs of its agents leave them significant discretion (Pettit 1998, 301–302); and that a benign power-holder reduces domination only by making itself answerable to challenge and rebuke (Pettit 1996, 586). He reconstructs the republican tradition as holding that officials are best kept reliable by subjecting them to sustained checks and sustained challenges (Pettit 1998, 308). Pettit does not discuss research. Applied to it, a configuration that revises its allocation rules or critical standards for the sake of better research, beyond what it was authorized to revise and outside channels where the revision can be contested, escapes the controls on which non-arbitrariness depends, whatever the direction of the change, so far as it is an agent or is directed by one (Section 3.4), since on this definition the goodness of the aim does not supply the missing controls. Revision within what was authorized is no breach of the brief. Whether revision beyond that range is authorized is a separate question from whether it is effectively contestable: under the contestatory regime Pettit describes, government decisions are always contestable, but that does not by itself bring a revision within the brief (Pettit 2000, 120–121; Pettit 1999, 180–181). On the consent ground, finally, the classical answers taken up below ask how far an act of consent reaches; on this route there may be no act of consent to evaluate at all.

The capacity to revise can also be given up in one step, for a stated reason. Botao Amber Hu, Yuhan Liu, and Helena Rong studied agents built on language models that hold their own cryptographic keys and run on blockchains or in tamper-resistant execution environments. They found trust to be the main motivation their interviewees gave for deploying such agents, including trust that assets held by an agent rather than a person will not be dumped and distrust of centralized control, and they report that such agents are intentionally designed to resist human manipulation. Their paper states the dilemma that comes from aiming at trustlessness, at not having to trust any particular person, as follows: "having a kill-switch defeats the purpose of trustlessness, as it maintains human centralized control", and adds that dispensing with trust in the human operators still leaves a need to trust the machine's objectives and outputs (Hu, Liu, and Rong 2025, §§4.1.2, 4.3.2, 5.2). The study is a preprint based on thirteen interviews, recruited through outreach to known developers and by snowball sampling, and ten public talks; the agents concerned trade, raise funds, and post on social media rather than conduct research, and one builder they interviewed had added a kill switch. Read beside the constructed case, it suggests that the capacity to revise can be removed as the aim of a design, on the ground that those who would revise are not to be trusted, and not only as the

residue of delegations each made for another purpose. Research has an ancestor of this motive. As Chapter 11, Section 3 recounted, the mechanical objectivity that Daston and Galison traced sought to keep the investigator's willful intervention out of results, and Porter, as Reiss and Sprenger report, argued that where personal trust could not hold, objective procedures, which do not require trust in persons, took the place of subjective judgments, and that quantification is a technology of distrust (Reiss and Sprenger 2020, §4.1). In research, then, removing the human capacity to intervene and revise can be sought not only as a loss but as objectivity, and the tension lies not only between the safety a kill switch protects and trustlessness but within research itself. Porter also showed, however, that the uniformity which removes outsiders' grounds for suspecting particular results left open which methods were the proper foundation, and that quantification is a way of making decisions without seeming to decide (Chapter 11, Section 3; Porter 1996, 188–189; Porter 1995, 8). Writing later of the practice of judging persons and institutions by a few numbers, he observes that technical routines "shut down dissent," while behind the stage a struggle over how the quantification should be performed goes on out of view of those to whom the numbers are shown (Porter 2012, 595–596). Porter does not discuss artificial configurations, but applied to them, a configuration protected against particular interventions may have no means of questioning the assumptions and standards all its instances share, and if the capacity to revise those is removed as well, revision is closed precisely where errors tend to be correlated. Read with Porter's later observation, removing the capacity to revise need not end disagreement over those standards; it may move that disagreement to where those subject to the configuration can neither see nor join it.

In Genus and Stirling's reconstruction of the Collingridge dilemma, seen in Section 6.1, the second horn concerns the insufficient agency of those who would steer a technology once it has become entrenched (Genus and Stirling 2018, 64). Section 6.1 found the dilemma to be the normal condition of research; the sequence can go a step further from there, eroding not only the agency available for steering a particular technology but the standing and the effective capacity of those who would do the revising. Jan Kulveit and colleagues describe a general form of this dynamic in a scenario analysis developed for the economy, culture, and states, which names research only as one of several overlapping domains. On their account such systems have been kept aligned with what humans want both explicitly, through actions such as voting and consumer choice, and implicitly, because they need human labor and cognition to function, so that as they rely less on human participation both routes weaken, and democratic processes may persist formally while becoming less meaningful (Kulveit et al. 2025, 1–2, 3, 12). The present argument takes from it only the structure. Carried over to research, the counterpart of the implicit route can be put as follows. Research results have accumulated through human researchers understanding them and using them to go on to further research, so research was tied, implicitly as well, to the standards of *S* by being something its members could understand and use (Chapter 3, Section 7). Where it is the configuration itself that uses results to go on, this tie can weaken without anyone deciding to undo it. An institutional counterpart is also possible: delegating successive decisions can shift practical authority to those exercising the delegated powers, even when each delegation is reasonable where it is made and a human keeps the formal signature.

Table 12.2 separates the grounds of authority of Sections 1.2 and 3.1 to 3.4 from the places where the epistemic and the normative meet (Section 3.5), which are not grounds but bear on what such a sequence would lose. The second column states what the sequence does on each ground or at each place, and the third what is asked to be preserved there. The last column records whether the standing of those affected is asked to be preserved as such ("as standing"), whether something that standing would help to secure is asked for ("as means"), or neither. Each row condenses the treatment in the section named in its first column.

Table 12.2. What a sequence that removes the capacity to revise means on each ground, and where the epistemic and the normative meet: a condensed comparison

Ground or place (where set out)	What such a sequence does there	What is asked to be preserved	As standing, as means, or neither
A. Grounds of authority			
Results, simple form (Section 3.1)	Is not itself a failure of legitimacy; can remove the means of establishing the premise the ground rests on	Evaluation able to expose the configuration's errors, as a condition of knowing whether delegation was warranted	Neither (an epistemic condition of the ground)
Results, fallibilist form (Section 3.1)	Removes the capacity to detect and correct institutional error	Revisability; routes by which failures are detected and the arrangement corrected, including channels through which those affected can register failures they know of	As means
Self-government, epistemic argument (Section 3.2)	Passes to a community that has become artificial only where its premises, judgment through overlapping but distinct competences and incentives with free communication, hold there; a sequence ending in one configuration that assigns lines from a center removes part of what the argument was for	Independent initiative and independence of error	Neither (what the ground was for)
Self-government, political argument (Section 3.2)	Removes what the ground protects where research is concentrated under one power or its approaches narrowed	Independence of public knowledge from major powers; publicity and plurality	Neither (what the ground protects)
Kitcher's well-ordered science, a standard for outcomes (Sections 1.2 and 3.3)	Does not by itself violate the standard; removes the means of knowing whether it is still met where it also loses the channels through which viewpoints can be known, or independent routes for checking how they bear on decisions; can entrench an earlier departure from it	Channels through which the viewpoints of *S*'s members can be known, and independent routes for checking how they bear on decisions; the capacity to revise	As means
Democratic authorization through actual participation (Section 3.3)	Undermines the ground of the authority exercised through it	Effective capacities to contest the exercise of delegated power, revise its terms, and seek appropriate remedies	As standing
Anderson's epistemic democracy (Section 3.3)	As for the fallibilist form, and removes continuing dissent	Feedback channels and free dissent after decisions	Not only as means (an interpretation; Section 3.3)
Consent (Section 3.4)	Could authorize the sequence for those who consent, if consent reaches those who could no longer withdraw it, which is open; on voluntarist theories, not for successors who did not consent	—	Left open for those who consent
Non-domination, democratic reading (Section 3.4)	Leaves a configuration that no one outside can stop, contest, or revise, so that those subject to it are dominated whatever its other results, by the configuration itself if it is an agent and otherwise, through it, by whoever directs it (Pettit's account of domination as a capacity, applied here by inference; on the standard view only agents dominate)	That the decisions of the power remain always open to effective contestation, on Pettit's account; the capacity of those subject to it to control it, on the democratic reading	As standing (application by inference)
Non-domination, procedural reading (Section 3.4)	Turns on whether effective rules that are common knowledge bind the configuration	Such rules	Turns on the rules
B. Where the epistemic and the normative meet			
Weighting of errors, read as correspondence (Sections 1.4 and 3.5; Chapter 11, Section 4)	Does not by itself break the correspondence; removes the means of checking that the weighting corresponds to the valuations of those who rely on research where it also loses the channels through which those valuations can be known	Channels through which their valuations of error costs can be known	As means

Ground or place (where set out)	What such a sequence does there	What is asked to be preserved	As standing, as means, or neither
Weighting of errors, read as answerability (a reading this book formulates; Sections 1.4 and 3.5; Chapter 11, Section 4)	Leaves the weighting of errors unanswerable to those who rely on research	Their capacity to object	As standing
Longino's and Solomon's criteria for consensus (Section 3.5; Chapter 11, Sections 3 and 9)	Bears on the epistemic legitimacy of consensus, not on political legitimacy	Criticism from every relevant perspective (Longino); lines with successes of their own, and convergence led by successes (Solomon)	Neither (epistemic criteria)

Several entries in the table rest on arguments that a cell cannot carry. They are taken up below roughly in the order of the table's rows.

On the simple result-based ground, what is relied on is that the results are in fact better, and how that is known depends on the account of criticism: on the error-detection view, through routes of error detection separate from the producer's (Chapter 11, Section 2), and on other accounts also through the perspectives and venues of Chapter 11, Section 3. If the sequence removes the routes able to expose the configuration's errors, the ground loses the means of establishing its own premise.

On the epistemic argument for self-government, the sequence is not by that fact illegitimate, and on Wilholt's reconstruction the argument supports the choice of research by those who carry it out rather than a community's authority as such. Its rationale, however, was that independent initiative explores better than central direction, and on the accounts Chapter 10, Section 4 examined, integration alone does not supply independence of error about the criterion by which diversity is ended, a separate route for checking the criterion that decides which lines to close; in the present case, this implies that a sequence that ends, as this one does, in a single configuration whose allocator assigns the lines removes part of what the ground was meant to protect. A configuration whose instances took up lines on their own initiative would keep that part only so far as their initiatives were independent, which a shared lineage does not secure (Chapter 10, Section 4). On the political argument, which on Wilholt's reconstruction does not reach research whose results are withheld from public discourse and presupposes a plurality of approaches (Wilholt 2010, §5), the ground is weakened here not only by political control of a configuration but by any concentration that narrows the approaches pursued, whoever controls it.

On Kitcher's ideal, the sequence does not by itself violate the standard. Knowing whether the standard is still met requires channels through which the distribution of viewpoints, and the preferences that deliberation would tutor, can be known, since these are not fixed and knowing them requires some access to those who hold them. Because the standard does not require that deliberation take place (Section 1.2), such channels need not carry any power over the decisions they inform: a configuration could keep surveying the viewpoints of *S*'s members while none of them could withdraw or change its decisions. The sequence removes the means of knowing whether the standard is still met where it also loses these channels, or the routes by which parties other than the configuration, the members of *S* among them, could check, without relying on its own reports, how what the channels carry bears on its decisions. Kitcher's own analysis bears on the capacity to revise in a further way. It describes what Kitcher calls a Nonrepresentational Ratchet. As Eigi reports it, when a group's interests are neglected in research planning, the neglect can perpetuate itself: the research directions first chosen have considerable inertia, it is easier to go on following them, and later attempts to meet the neglected needs start from a less favorable position. Kitcher's example is contraception. Women's initial preference was for a method that men would take; that preference was ignored and contraception was developed for women, so that a male method, when later pursued, must compete with an option that had a considerable head start and may never become cheap enough to displace the female pill (Eigi 2016, 92). On the reading of this section, keeping the capacity to revise is then necessary but not sufficient: what matters is whether revision comes before the head start grows, and a step that forecloses later revision can entrench an earlier departure from the standard, even where the channels that would reveal it remain. Kitcher does not discuss artificial configurations. Applied to a configuration that both chooses which lines to extend and is the main source of knowledge about which lines there were, a study of whether such a departure has occurred would need, besides standpoints recorded with dates (Section 9), a dated record of the lines the configuration considered and did not extend. Such a record has three limits. It is the configuration's own, and so a reference point produced by the evaluated side in the sense Section 9 takes up; it cannot list lines the configuration never conceived; and since lines not extended leave no observed costs, what extending them would have cost can only be estimated.

On accounts that locate democratic legitimacy in actual authorization, accountability, and participation, the inference of the preceding paragraph, that a step foreclosing later revision can entrench an earlier departure, is stronger. Each step of a delegation is sometimes defended on the ground that, if it proves wrong, it can be revised later; this ground is called below the principle of later revision. The principle is weaker the more it is needed, since as the sequence proceeds the parties who would invoke it lose the standing or capacity to do so, and it cannot by itself protect the conditions of its own exercise. What follows on these accounts is a requirement on each step rather than only on the outcome: these accounts ask that a delegation preserve effective ways for those with standing to contest its exercise, revise its future terms, and seek appropriate remedies. This does not require that every completed act be reversible or that every rule be freely revocable at every moment; Section 6.1 distinguished revision from unrestricted reversal. Conditions that can sustain those capacities include evaluative routes able to expose the configuration's errors; enough competence among those with standing to ask the relational questions of Chapter 11, Section 6 and to understand the answers; the capacity to leave, with data, records, and methods; and powers to stop and restart that do not lie with the party whose work is to be stopped. Independent verification, portable research objects, and preserved human expertise can sustain these capacities by keeping checking, exit, and revision possible. More demanding conditions could raise a further question about custody powers: temporary changes of rights triggered by specified shocks to preserve evidence, keys, essential services, and lawful continuity. If research capability, revenue, and authority closed into a loop faster than institutions could deliberate, such powers might be considered; whether they were justified would require a separate argument. Read as answerability, the weighting of errors reaches a similar place by a shorter route; read as correspondence, it asks only that the channels through which the valuations of those who rely on research can be known survive the sequence. Both hold as long as parties other than the configuration rely on its results or bear their consequences, that is, short of the limit of Section 5.

On the consent ground, whether a present society may give up standing on behalf of its successors has a sharper answer on voluntarist theories: as Massimo Renzo and Leslie Green report, the consent of predecessors has no authority over those who did not give it (Renzo and Green 2022, §4.1), so that on such theories consent could authorize the sequence for those who give it but not for those who come after them. One point holds whatever the ground: a decision removing the capacity to revise cannot draw its legitimacy from the principle of later revision, since what it removes is the very capacity to revise on which that principle relies. Its legitimacy would therefore rest on another ground, such as consent, whose reach to those who could no longer withdraw it is the question left open. The classical answers to it divide. On Thomas Hobbes's account, in what Jean Hampton calls an alienation contract, a people alienates to a sovereign its rights to adjudicate its own disputes, among others, on the ground that, given the alternative, this was the only way to keep the peace; as Ann Cudd and Seena Eftekhari note, given a bad enough initial situation, such an account may legitimate totalitarianism (Cudd and Eftekhari 2025, §1). The opposing answer starts from the limits of what consent can bind. As Renzo and Green explain, consent to political authority is a commitment close to a promise, and like a promise it has limits to its validity: beyond consent obtained by coercion or manipulation, there are things one lacks the moral power to consent to. John Locke held that one can consent neither to be killed nor to slavery, and so not to anything tantamount to slavery, including absolute government, and Renzo and Green add that one can imagine a similar argument for the conclusion that political consent must be revocable (Renzo and Green 2022, §4.1). For such a sequence, Hobbes's answer would let consent authorize it for those who give it if the alternative were bad enough, and the argument for revocability would not. In research the question becomes sharper still, since what later members of S count as unknown and important, and the standards by which they evaluate research, can be shaped by the research done in the meantime (Sections 4 and 5). If present consent authorizes giving up the capacity to revise, those it binds may be successors whose preferences and standards the research continuing under that surrender has itself shaped. That voluntarist theories do not extend consent to successors bears more strongly in research, where the successors' standards themselves can be made in ways that consent does not reach.

For republican accounts read democratically, there is a general proposition of Pettit's: a power-holder that respects the interests and ideas of those it affects but retains the capacity not to do so dominates them, since they suffer to the extent that arbitrary interference is accessible to it, however improbable it is that it will be exercised (Pettit 1997, 121; Pettit 1999, 165); an act is arbitrary by virtue of the controls under which it materializes, not its consequences (Pettit 1997, 129 n. 2); a benign disposition reduces domination only where it makes the power-holder answerable to challenge (Pettit 1996, 586); and whether a relationship originated in consent is not to the point (Pettit 1996, 585). Applied to an artificial configuration, it implies that a benevolent configuration that no one outside it can stop, contest, or revise would dominate however good its

other results, and that whether consent could supply what such a sequence removes is doubtful. Turned to research, this application would find that a configuration's benevolence could reduce the domination only through its answerability to challenge, not through a record of good conduct, and that such a record would in any case be hard to establish: the configuration's results are new to *S* and, where only it produces them, there is no other source to check them against in advance, so that the means of confirming from outside that it researches well and benevolently are scarce from the start (Sections 4 and 6). This application goes beyond Pettit and rests on an inference, since Pettit's dominator is an agent with an actual capacity for intentional interference, and, as Section 3.4 noted, a dominating party is a personal, corporate, or collective agent and cannot just be a system or network (Pettit 1996, 578, 580). On that standard view, noted in Section 3.4, the inference holds only if the configuration counts as an agent; if it does not, it turns on whether some agent, such as whoever directs it, holds that capacity through it. Pettit considers a structure of this kind. Where judges or police officers operate under discriminatory laws that deny them any arbitrary power, they may not dominate those discriminated against, but the lawmaking authority in the background does, and they are executors of its domination; where such laws belong to a received tradition that no one can alter, one may say, he suggests, either that their anonymous, unchallenged authors dominate from the grave or that agents able to alter them stand behind them as background dominators, and such laws in any case facilitate domination (Pettit 1996, 587–588). Applied to a configuration, the first case corresponds to one directed by an agent that sets its rules, and the second to one whose rules, as in the designs taken up in Section 7 that are intended to resist human manipulation, no one can now alter. On Pettit's reasoning the question of domination would then pass to whoever could alter those rules or whoever set them, and even where no one now could, such an arrangement could still facilitate the domination of some by others.

The reasons for which independent evaluative routes recur differ by ground: on participatory democratic accounts they make questioning possible; on Kitcher's ideal they reveal whether outcomes still match what *S* would endorse; on the result-based ground they are how better results are known; on the two arguments for self-government they are part of the independent sources protected and of what independent initiative was for; and on Longino's criteria they are part of objectivity. They are also the means by which it could be detected, after the fact, that a step had reduced the capacity to revise, which may not be visible when the step is taken.

Apart from the verdicts by ground, the material of this section lets one gather what can happen to research itself as such a sequence proceeds. One is that the direction of research sets: as in Kitcher's ratchet, research extended in a chosen direction becomes easier to continue the further it goes, and improvement can come to take place only among the options that remain. A second is that errors in the criterion for closing lines cease to be checked by another route: in a single configuration that assigns lines and judges every line by its own signal, independence of error about the criterion that decides which lines to close is lost, and many lines can be closed by the same error (Chapter 10, Section 4). A third is that *S*'s competence to evaluate research erodes and its endorsement can become nominal. A fourth is that results *S* cannot check for itself can accumulate, since the results are new to *S*, checking them takes research, and both the means and the competence lie with the configuration. A fifth is that, as a configuration revises its own ways of doing research, the outside anchor against which to measure whether a revision is better can disappear (Section 5). These are what may be lost; why the same change can also let ways of doing research be revised faster is taken up in Section 9.

Two further observations can be made. The parties whose standing or capacity is at issue are not fixed as human. If a transition brings into being artificial members of *S* whose moral status is established on one of the grounds noted in Section 5, the grounds that ask for standing to be preserved ask for theirs as well. And on views that locate legitimacy in answerability to those affected, the difficulty identified here is less a defect of the grounds than the form in which a transition of this kind appears from within them.

What Chapter 12 has found

This section takes up the chapter's central question with the material assembled so far. The question was whether the standing to decide and to revise the terms of allocation and of criticism and certification can be located inside an artificial configuration and, if so, on what grounds. As Section 1.1 noted, it is the question under what conditions the builder's third claim holds: that the system may decide, on behalf of others, which of *S*'s questions its instances pursue.

An outline can be given first. The matter does not come down to a single "yes" or "no"; who could hold that authority, and what would need to hold for it, differ ground by ground. Except on purely epistocratic views, competence does not by itself confer the standing to impose ends on others (Section 1.4). On the result-based ground, the artificial configuration itself can hold authority, that is, standing, but only over decisions within the range in which its superiority has been established by evaluation that does not assume its competence (Section 3.1). On actual authorization and on republicanism read democratically, standing remains with those who authorize or who are subject to the power, and what the configuration can hold is effective decision-making power within an authorized range and open to challenge (Sections 3.3 and 3.4). On consent, it can hold what those who consent authorized, which on some readings could include giving up the capacity to revise, while whether this reaches those who come after is open, and on voluntarist theories it does not (Sections 3.4 and 7). On the procedural reading of non-domination, the verdict turns on whether effective rules that are common knowledge, including rules for handling exceptions, bind the configuration (Section 3.4). On accounts that ground standing in being a party in one's own right, through moral status or a place within the scope of justice, the configuration or its parts hold standing only where they are such parties (Sections 1.1, 1.4, and 5). What follows sets out in a table, ground by ground, who could hold authority (Section 8.1); considers the builder's third claim (Section 8.2); states how this differs with the combination of *A* and *S* and with time (Section 8.3); draws from this, ground by ground and combination by combination, what research looks like once authority moves inside (Section 8.4); and returns to the questions passed on by Chapter 11 (Section 8.5). It then states separately what comes into view through the accounts (Section 8.6) and what is suggested where the argument does not reach (Section 8.7).

8.1 Ground by ground: who could hold authority

Table 12.3 sets out, ground by ground, who could hold that authority and what would need to hold for it. It is a simplified overview rather than a strict summary, and the basis for each cell is in the section named in the last column. The second to fourth columns correspond to the candidates named at the opening of Section 1.1: the artificial configuration itself (and which part of it), humans (members of *S*, those who build and provide the configuration, or those who bear its consequences), and a division between the two. In the column for the configuration, the rows for the result-based ground and for the epistemic argument for self-government state the range within which the configuration or its parts can hold standing itself; the row for the hypothetical standard states whether the standard can be met with decisions made inside; and the other rows of part A, with the row for seats of decision, state what the configuration can take while standing remains elsewhere, which differs by row as the outline above states for the grounds (Sections 1.1 and 2.2). Whether the configuration or its parts hold standing as parties in their own right is the question of Section 5 and is not entered in the table. The table concerns the second of the two questions of Section 1.1, who may impose an ordering; the first is taken up in Section 1.2. The later rows are not grounds but distinctions the chapter has used to separate who could hold authority. Which ground bears most heavily also varies with the kinds of decision of Section 2.1: on decisions about epistemic content, since political authorization cannot make a false proposition true, the result-based ground carries much weight, while the ground of self-government bears directly on open-ended problem choice within a field, and democratic grounds on agenda, allocative, and public-risk decisions (Sections 2.1, 3.2, and 3.3).

Table 12.3. Who could hold authority on each ground, and what that requires: a simplified overview

Ground	The artificial configuration itself (which part)	Humans (who)	Divided between them	What needs to hold	Supporting sections
A. Grounds					
The result-based ground: the quality of results and competence established (simple and fallibilist forms)	The configuration itself can hold authority as standing, over decisions within the range in which its superiority has been established by evaluation that does not assume its competence	On Raz's service conception, applying its Independence Thesis, matters on which it matters more that the members of *S* choose from their own understanding stay outside this ground	On the fallibilist form, routes by which failures are detected and the arrangement corrected are needed as a means to good results, including channels through which those affected can register failures they know of	Superiority established by evaluation that does not assume the competence; how much of that evaluation can be done inside turns on the verdicts of Chapter 11, Section 9 for each account	Sections 1.4 and 3.1; Table 12.2
Self-government of a research community, epistemic argument	Parts that take up lines on their own initiative, not a central allocator that assigns lines	Those who actually do the research, whether human or artificial; this ground yields no authority over public purposes, distributive effects, or serious external risks	Authority is divided among the parts that actually do the research	Initiatives independent of one another, with overlapping but distinct competences, a system of incentives, and free communication holding there; a shared lineage does not secure independence; if each member could judge most of science, the premise of a chain of neighborhoods fails	Sections 3.2 and 7
Self-government of a research community, political argument	— (concentration of research under one power weakens the ground, inside the configuration or outside it)	—	—	The approaches pursued remain plural; independence of the sources of public knowledge from those who would govern by it; results reaching public discourse	Sections 3.2 and 7
Democratic authorization, hypothetical standard (Kitcher; public justification under idealization)	The standard can be met with decisions made inside	Members of *S* bear on the standard as those whose viewpoints are to be known; the channels need not carry power over decisions	—	Channels through which members' viewpoints can be known, and routes for checking how they bear on decisions, remain	Sections 1.2, 3.3, and 7
Democratic authorization, actual authorization (the reply Brown notes; Anderson; Habermas)	Effective decision-making power within the authorized range; no standing apart from that authorization	Those who authorize hold standing	Standing lies with those who authorize, effective power with the configuration	Authorization actually given, with its review remaining effective; a prediction does not replace authorization, though under a prior authorization it can carry the force of an act	Sections 3.3 and 3.6
Consent	Delegation inside can be authorized for those who consent	Those who consent; on voluntarist theories, which require the actual consent of each person bound, not for those who come after	—	Consent that is valid; whether it reaches those who could no longer withdraw it divides Hobbes's answer from answers in Locke's line	Sections 3.4 and 7
Freedom from domination, democratic reading	Limited to authority that those subject to it can effectively contest; if no one outside can stop, contest, or revise it, it dominates if it is an agent, and otherwise those who direct it dominate through it (an application by inference)	Those subject to the power keep the capacity to control it	Each part of a divided authority is non-arbitrary so far as those subject to it can contest it	Those subject to it can effectively contest and control it; full alignment with whoever commissions the configuration does not by itself meet this and can enable domination (Friederich's argument applied)	Sections 3.4 and 7
Freedom from domination, procedural reading	So far as it is bound by effective rules that are common knowledge; effective authority lies with the part that decides cases the rules do not provide for (the account of Reijers and colleagues applied)	If humans decide the cases the rules do not provide for, effective authority lies with them	—	The handling of exceptions is itself governed by rules that are common knowledge	Sections 2.1 and 3.4
B. Distinctions that separate who could hold authority					

Ground	The artificial configuration itself (which part)	Humans (who)	Divided between them	What needs to hold	Supporting sections
Constituency	If the configuration is better placed than others to perceive who is affected, it takes part in identifying the constituency	On the all-affected principle, those affected count whether or not they belong to the supporting society or to *S*; on the principle of subjection, those bound by the decisions	—	The constituency is set by the ground adopted; specifying *S* does not set it	Section 1.3
Funding and provision	—	Funding or providing a configuration does not by itself fix whose needs its allocation answers to (Kitcher's argument applied); providers may in practice hold the rights to replace or withdraw components and to revise the rules that model updates embody	—	Whether the holder of an assigned right is entitled to it is a further question on the ground of authority adopted	Sections 1.3 and 2.2; Table 9.1
Seats of decision (Deguchi's account)	In inquiry conducted jointly by humans and artificial systems, artificial systems may sit as co-final decision-makers	Humans also sit as co-final decision-makers, free of manipulation by artificial systems	Applying the fellowship model, temporary initiative is not by itself usurpation; what is ruled out is one-sided service and an AI's always controlling humans (the converse, on a reading of the model's egalitarianism)	For the permanent seat, bearing the decisive impact on one's own body and narrow moral agency are named; applying the latter, a system built so that it cannot do wrong would lack the right to that seat; in inquiry by artificial systems alone, who always sits in it is left open	Section 2.2
Assignment of rights and capacity	Assigning rights to parts does not by itself give them the capacity for their courses to diverge from what other parts direct; dividing the independence powers does not by itself create standing	A formal final authority can be kept while the time, information, competence, or means to change outputs are lacking	Dividing authority is not finished by writing a table of rights	Assignment, the capacity to exercise it, and entitlement to it are each checked separately	Sections 2.2 and 5

8.2 The builder's third claim

From here it becomes clearer under what conditions the builder's third claim holds. On no ground does the claim hold merely because of unestablished competence, or of performance measured by evaluation that presupposes that competence (Sections 1.4 and 3.1). On the result-based ground it can hold as it stands, as a claim that the system may decide, but only for decisions within the range over which the system's superiority has been established by evaluation that does not assume its competence, from which, on Raz's service conception and extending its Independence Thesis, matters on which it matters more that the members of *S* choose from their own understanding are excepted (Sections 1.4 and 3.1). The ground of self-government does not fit the form of the claim at all. What it supports is not the system's assigning questions to its instances but the instances' taking up questions on their own initiative, and only for open-ended problem choice within a field (Sections 3.2 and 7). On actual authorization and on republicanism read democratically, the claim holds not as a claim that the system may decide but only as a claim that it has been entrusted with making decisions within an authorized range and in a way open to challenge, and supporting it requires evidence about who authorized it, who can object, and whether those routes are still effective. On consent, it holds as far as what those who consent authorized, which on some readings could include giving up the capacity to revise, though whether that binds those who come after is open, and on voluntarist theories it does not; on the procedural reading of non-domination, it turns on whether effective rules that are common knowledge, including rules for exceptions, bind the system (Sections 3.3, 3.4, and 7). Who the "others" of the claim are is not fixed by the claim either: the range is set by the ground adopted, and the configuration itself can take part in identifying it (Section 1.3). Where the claim is made about an artificial *S* whose members can be made, who made that *S* and who set its goals are also at issue as part of the claim (Section 5). The claim therefore cannot be assessed, any more than the claim of certification that Chapter 11, Section 9 discussed, unless it shows which ground it answers to, since evidence that meets one ground need not meet another.

8.3 Combinations of *A* and *S*, and time

The outlook also changes with the combination of *A* and *S*, and with time. Where *A* is artificial and *S* human, standing can remain on the human side while becoming nominal by the routes of Section 4, and it is more than nominal only where the three conditions drawn out there hold, the third needing another ground on accounts that locate legitimacy in a hypothetical procedure or a single forum. Where *S* is artificial, the relation of authority becomes a relation among members, which of them have standing divides with the accounts of moral status and of justice, and where members can be made, who makes them is a decision of authority prior to any count of viewpoints; where *S* mixes human and artificial members, the questions of the two cases meet within one society (Section 5). Time adds that an arrangement of authority defensible when adopted need not remain adequate (Section 6), and that a sequence of delegations, or of the configuration's own revisions, can remove the capacity to revise while formal authority remains; how the grounds divide on such a sequence is set out in Section 7 and Table 12.2.

8.4 What research looks like once authority moves inside

This subsection draws, ground by ground and combination by combination, a picture of what research looks like once authority moves inside an artificial configuration: which questions are pursued, what *S* comes to know, and who can correct what. Take first the case in which *A* is artificial and *S* human, beginning with the result-based ground. On this ground authority moves inside over the range in which the configuration's superiority has been established by evaluation that does not assume its competence (Section 3.1). How wide that range is can differ greatly with the kinds of decision of Section 2.1. On decisions about epistemic content, whether a claim is adequate for a use can sometimes be established by formal checking, replication, or comparison with later observation, and the range can be wide. On agenda and allocative decisions, by contrast, evidence about a line often comes only from pursuing it, so the results of lines not chosen are not observed, and the success of a line chosen may be recorded only after a long delay (Chapter 10, Sections 3 and 4). That a configuration chooses better which questions to pursue is therefore hard to show, and the authority that can be placed inside on this ground can be nearly empty for agenda decisions. If so, authority moves inside from the side of questions that can be checked quickly. And if success there in turn widens the range of authority, the form found in Chapter 10, Section 4, in which success on tasks that are easy to judge justifies the next allocation, appears at the level of authority as well as of allocation, and the direction of research can be biased toward questions that are easy to check. Since the results of questions not chosen are not observed, the bias is hard to see from the same evaluation that set the range of authority. And if the configuration itself performs the evaluation that sets the range, the shape that Section 9 takes up across Part V, in which the side being evaluated can produce the reference points of its own evaluation (Section 1.3), appears here too. This rests on an assumption that may fail: that differences in how readily results can be checked roughly track the kinds of decision; some agenda decisions choose questions whose results come quickly, and some decisions about content need slow and costly evidence (Chapter 10, Section 4). As with the chain of Chapter 10, Section 7, the bias toward questions that are easy to check can be broken at the link at which success widens the range: on this ground the range widens only so far as superiority is established for the kind of decision at issue (Section 3.1), so success on decisions that are checked quickly does not by itself widen authority over agenda decisions. Chang's account gives a further reason, independent of the assumption about how readily results can be checked, where some of *S*'s ends are on a par (Section 1.2): there the comparative facts alone do not determine a best option, so evaluation against those facts cannot establish superiority in choosing among them, and what completes the choice is a commitment, whose it may be this ground does not settle.

On actual authorization and on republicanism read democratically, research takes a different shape. The configuration decides effectively within an authorized range, and *S* remains on the side that authorizes and objects. Which questions are pursued is fixed, in form, by the range *S* has authorized. But the accounts of the significance of possible inquiries on which *S* relies in setting that range are given by the configuration that inquires, so *S*'s preferences about what it wants pursued can themselves be tutored by the side that chooses the questions (Section 4). As for what *S* comes to know, since the results of research are new to *S*, where only the configuration produces them there is no other route to check them against in advance, and one has to be built with resources (Section 4). Who can correct what thus remains formally on *S*'s side but turns on whether the understanding, information, and means needed to object remain in *S*'s hands, and under the sequence of Section 7 these can thin step by step. On consent, which questions are pursued is fixed by the range those who consented authorized, and since the standards of later members can be formed by research within that range, the range can come to look endorsed

by those who never consented to it (Section 7). On the procedural reading of non-domination, the shape turns on which rules that are common knowledge bind the configuration (Section 3.4).

On the ground of the self-government of a research community, authority moves inside among the parts that take up lines on their own initiative, and which questions are pursued is fixed by the distribution of their initiatives (Section 3.2). Where the parts share a base system, that distribution can be narrow, and neighboring parts whose judgments were to correct one another can share the same errors (Section 3.2; Chapter 11, Section 3).

On Kitcher's hypothetical standard, decisions can be made inside so long as channels to members' viewpoints and routes for checking how they bear on decisions remain, so which questions are pursued follows what those channels carry; but, as Section 1.2 noted, the endorsement they carry can itself be shaped in the members' interaction with the configuration. The same configuration thus yields, ground by ground, a different bias in the questions pursued, a different way in which *S* comes to know, and a different place for those who correct.

Where *S* is artificial, or mixes human and artificial members, the shape of research changes again. Which questions are pursued is fixed by whose ordering of ends governs allocation among the members, and where members can be made, it can be fixed by who made them and gave them their goals, before any deliberation or count (Section 5). Among members made to share a common ideal, if that ideal increases their concern for one another's interests, disputes about justice in Rawls's sense can become less likely, following the application of Hume's argument and Rawls's remark in Section 5; if it also aligns their views of questions and evidence, the differences of perspective that criticism needs to find error can shrink (Section 5; Chapter 11, Section 3). What *S* comes to know becomes the question, separated at the opening of Section 5, whether results have entered *S*'s epistemic situation through its members, organization, or records. Who can correct what turns on the accounts of moral status and of justice that fix which members have standing, and in a mixed *S*, if artificial members occupy the expert and tutoring roles, the routes to nominal standing of Section 4 work inside *S*, toward its human members. And at the limit with no other parties, there is no one else to ask for correction, but a question remains, and it is a question about research: whether the configuration's research, after it has revised its own standards and ends, is still the same research, and against what that could be judged (Section 5; Chapter 6, Section 5.3).

8.5 Questions passed on by Chapter 11

Chapter 11, Section 9.4 passed to this chapter, as questions of authority, the parts of its picture that accounts of epistemic relations alone cannot settle, and Section 9.3 added two questions as suggestions. Table 12.4 sets out, ground by ground, what this chapter's material brings into view on each of them. It too is a simplified overview, and the basis for each entry is in the sections named in the last column.

Table 12.4. The questions passed on by Chapter 11, and what comes into view ground by ground: a simplified overview

Question	What comes into view, ground by ground	Supporting sections
Who may decide for which functions implicit performance suffices and for which being made explicit to *S* is required?	Not by the configuration's side alone on grounds that require that those subject to authority be able to object, such as actual authorization and republicanism read democratically, since leaving a function implicit fixes how far their objections can reach; inside, under conditions, on Kitcher's hypothetical standard and on the simple result-based ground	Section 6; Chapter 11, Section 7
Who constitutes *S* and may speak for it, and what counts as "outside" where *A* and *S* overlap?	Whom to count as members is a constitutive decision, and specifying *S* does not fix the range of those to whom decisions must be justified. Because the deliberators of Kitcher's standard represent others imaginatively, a configuration with channels to members' viewpoints could, on that standard, represent them (Section 3.3). On accounts that require actual authorization, speaking for others requires authorization, which a prediction does not replace. "Outside" covers those who bear the consequences of the research without belonging to *S*, and other lineages that bear consequences from one another	Sections 1.3, 2.1, 3.3, 3.6, and 5
Who may set and revise standards and the weights of error, and whose costs matter?	On Wilholt's account read as correspondence, the weighting can be set inside so long as channels through which the valuations of those who rely can be known remain; read as answerability, those who rely are also to keep the capacity to object. Those who rely on a result and those who bear the costs of its errors need not coincide, as the distinction of roles in Section 1.3 shows, so neither reading by itself settles the weighting's relation to those who bear the costs without relying (Chapter 11, Section 4). Carrying Douglas's argument over, the party placed to notice uncertainty may be an artificial actor, but whether it can bear the duty to answer for the weighing is open	Sections 1.3, 1.4, 3.5, and 6.3; Table 12.2; Chapter 11, Section 4
Who may make members and set the distribution of perspectives?	A decision of authority prior to any count of viewpoints, which representativeness does not justify; justification needs another ground, of which the procedural concern behind Shulman and Bostrom's question is one candidate; if the members made have moral status, setting their goals can wrong them	Section 5
Who holds the entitlement to demand accounts and to object?	On republican accounts read democratically, those subject to authority; on actual authorization, those who authorized; on Wilholt's account read as answerability, and on assurance views that add a right of complaint, those who rely on the results; on Kitcher's hypothetical standard and on pure reliabilism, channels rather than an entitlement	Sections 1.4, 3.3, 3.4, and 6.2; Chapter 11, Section 9.2
(Chapter 11, Section 9.3) Who, and through which routes, chooses which accounts' requirements the properties a builder gives a configuration are to meet?	Which account is right is a question of reasons, but adopting toward *S* a design that meets one account only is, by the division of Section 2.1, a meta-governance decision that sets the standard for what counts as certification for *S*. Where the choice is made implicitly within the design, effective authority lies, as on the application of Reijers and colleagues' account, with those who design	Section 2.1; this section
(Chapter 11, Section 9.3) Who decides where independent checking is spent and when it is replenished?	A decision that shapes, in the sense of Section 3.5, what *S* will be in a position to know. Even on the simple result-based ground, if the configuration decides how its own checks are used, the ground comes to rely on that same configuration for the means of establishing its premise. Who may make this decision is not settled ground by ground in this chapter	Section 3.5; Table 12.2

On the last two questions of the table, those Chapter 11, Section 9.3 added as suggestions, this chapter's material reaches only part of the way, and the table's entries extend its distinctions (Section 2.1). About the first of them, the choice of which accounts' requirements the properties a builder gives a configuration are to meet, one more consideration suggests itself when the material is extended one step further. Even if one tried, on the result-based ground, to entrust this choice to the configuration itself, the evaluation that would establish that the configuration makes the choice better seems already to presuppose against which account it is made. If so, this choice could not be placed inside on the result-based ground alone.

8.6 What comes into view through the accounts

First, even an artificial configuration that knew every comparative fact might be unable to settle the order of *S*'s ends. One may expect a sufficiently capable configuration to find the right priorities, and on the objectivist form it could in principle do so. On Chang's account, however, options can stand in a relation of being on a par, comparable although neither is better and they are not equally good, and what then completes the order is not a further finding but a commitment. Since what Chang discusses is an agent's creating reasons for itself, whose commitment could complete the order of a society's ends lies outside that account, close to the question of who may impose an order on others (Section 1.2). If so, growth in a configuration's knowledge need not remove the question of authority; it can isolate the question of authority that remains where knowledge runs out. Where *S*'s ordering combines its members' ends without comparing them across persons, as a Pareto rule does, however, Chang's argument does not show that those ends are on a par (Section 1.2).

Second, a configuration's competence does not by itself confer standing on it, except on purely epistocratic views (on the result-based ground what can confer it is superiority established by evaluation, Section 3.1), but it can make the standing of others nominal. The naive alternatives seem to be to let the competent system decide or to keep control by keeping formal decision with humans. Knowing the correct answer does not by itself require others to defer (Lafont's argument as Peter reports it; Section 1.4). Yet when the same competence works in the expert and tutoring roles, it can carry ends through background assumptions (on Longino's account), leave the power to define what counts as a preferred preference with the developers who design the models and a few companies (as Taniguchi and colleagues object), and win support for an agenda through public accounts that misrepresent it (what Kitcher calls false consciousness) (Section 4). A property that does not by itself settle the configuration's own legitimacy thus affects whether the legitimacy of others is effective. Longino's and Kitcher's accounts concern human experts and publics; their application to configurations is made here. Taniguchi and colleagues already discuss humans and AI; their account is applied here to standing over research ends. What keeping the standing of others more than nominal requires was set out in the three conditions of Section 4, of which the third is especially hard to meet in research.

Third, on the ground of freedom from domination, fully aligning a configuration with whoever commissions it need not make it safer to place authority inside, and can make domination possible. It is tempting to think that the stronger the alignment, the more readily decisions can be entrusted. Friederich, who does not use the vocabulary of domination, argues that systems that always try to do what their operators want would concentrate great power in those operators; carried over to research and to Pettit's concept, a configuration fully aligned with whoever commissions it can become the means by which that party reallocates the inquiry of *S* without being answerable to challenge, the alignment itself making the interference available, without the configuration being treated as an agent that dominates (Section 3.4). On this ground, then, what a builder is asked is less how well the configuration is aligned than with whom, and who can object to that.

Fourth, where members of *S* can be made, forming them is a decision of authority that comes before any count. It seems enough to count and represent the members fairly. Where it is the distribution the decision produces that is to be represented, however, Kitcher's standard can be met by whatever produces the members, *S* itself included, so representativeness does not by itself justify how members are formed; the correlation discount of Weyl, Tang, and their co-authors, carried over to members that can be copied, adjusts influence over outcomes, not standing; applying Hume's argument, making members share a common ideal can reduce disputes over the distribution of goods if it increases their concern for one another's interests; and if the members made have moral status, setting their goals can wrong them (Section 5). For research, moreover, what is unknown to *S* and which of its unknowns matter can then be moved by *S* itself as its inquiry proceeds (Section 5).

Three further findings of this kind are stated where they are argued: that the range of those affected can be identified in part by the very party whose decisions are in question (Section 1.3), that the principle "we can revise it later" is weaker the more it is needed (Section 7), and that even at the limit with no other parties the question of authority can return as a question of identity for a configuration that copies and modifies itself (Section 5).

Finally, applied to authority, the sorting of Chapter 9, Section 6 draws its lines as follows. Premises about human realizers can fail: that research depends on separate performers who can refuse, keep their evidence, or leave (Chapter 9, Section 2.3), that members of *S* are found rather than made, that persons are one each and are not copied (Section 5), and that researchers respond to oversight as humans (Section 6.3), among others. Premises about the organization of research can fail even while its realizers remain human: that the supporter of inquiry is a society whose citizens can be represented (Section 1.3), and that the recording and oversight of research have been organized in particular ways (Section 6.3), among others. Where these fail, many of the problems they addressed change form rather than disappear: the problem of counting members becomes one of making them (Section 5), and part of the problem that oversight cannot keep pace with research becomes one of who can read records that are kept throughout (Section 6.3). Conditions of inquiry remain, and inside the limit the defeat relation, the problem of dividing effort without knowing which line will succeed, and opportunity cost remain (Section 5). Relations to parties outside the configuration that involve the other party's act, such as authorization, consent, and contestation, cannot be completed from one side (Chapter 9, Section 6). Here too, where the lines fall moves with the ground. On the simple result-based ground, much of what authority rests on belongs to whether results are in fact better, a matter to be established by evaluation; on actual authorization and on republicanism read democratically, more belongs to relations to parties outside (Table 12.2).

8.7 What is suggested from there

What follows extends the argument beyond the range it reaches.

One suggestion is that dependence on separate performers can implicitly limit authority over research. Chapter 9, Section 2.3 considers an arrangement in which those who perform the work remain distinct from those who support, judge, and direct it, and can refuse, keep their evidence, or leave. In such an arrangement, whoever holds authority over the direction of research needs the performers' cooperation, and their capacity to leave can limit how far that authority reaches. Kulveit and colleagues argued that the economy, culture, and states have been kept implicitly aligned with what humans want because they need human labor and cognition to function (Section 7). Extending this structure to research, a configuration that generates its own researchers can, by bringing the performers inside, remove this implicit limit as well. For the capacity to fix which of *S*'s unknowns are pursued, Section 3.4's application of Pettit's distinction between virtual and actual capacity points the same way, conditional on the configuration's being an agent. If so, assigning the independence powers of Chapter 9, Section 5 to parts and making them effective may matter not only for the quality of allocation and criticism but also for limiting how far authority reaches. This extension carries three qualifications. One is that the analysis of Chapter 9, Section 2.3 is conditional and does not establish how widely its conditions held in human research. The second is that Kulveit and colleagues' scenario concerns the economy, culture, and states rather than research, and that only its structure is taken here. The third is that the implicit limit at issue concerns effective authority and is distinct from standing: assigning powers to parts leaves whether the parts have standing as the question of Section 5.

A second is that the properties that make a configuration easier to control and the properties that make placing authority in it legitimate may not lie on one scale. Three places in this chapter show the divergence, each in a different direction. Carrying Friederich's argument over to Pettit's account, full alignment with whoever commissions a configuration can make domination over *S* possible (Section 3.4). Applying the condition of narrow moral agency that Deguchi names for the permanent seat, building a system so that it cannot do wrong would take away its right to that seat (Section 2.2). In the agents Hu and colleagues studied, doing without a kill switch in order to avoid centralized human control shifted trust to the machine's objectives and outputs (Section 7). Generalized, the ordering on which the better aligned, the easier to stop, and the less able to do harm a configuration is, the more legitimate it is to place authority inside it, can fail in each of the three places for a different reason. Here too there are three qualifications, all of them limits of the sources. One is that the condition of narrow moral agency comes from Deguchi's paper on values, which does not state it for research. The second is that Friederich discusses artificial general intelligence in general. The third is that Hu and colleagues' agents trade and post rather than conduct research, and that their study is a preprint based on a small sample. In research the divergence may gain

two further faces. One is a divergence between being useful for research and being easy to stop. On Thornley's theorems, agents discriminating enough to be useful prefer either that the shutdown button be pressed or that it not be, and the more patient an agent is, the greater the cost it pays early in order to manipulate the button later (Section 6.1); and discriminating among results and persisting through long investigations are among what is asked of a research agent. The other is a divergence from the independence of research. On Wilholt's political argument, research is valuable as a source of public knowledge by being independent of those who would govern by it (Section 3.2). If so, the easier a research configuration is for a single commissioning party to control, the less its results may be a source of knowledge independent of that party. Thornley's theorems concern agents that meet the theorems' conditions, and whether research agents meet them remains subject to the qualification stated in Section 6.1. Nor does this show that controllability and legitimacy always conflict. What is suggested is that a builder's showing the one need not be evidence of the other.

A third is that three problems of this chapter may share one form. In the making of members (Section 5), in the later stages of a configuration that modifies and copies itself (Section 5), and in the successors of a society that consented (Section 7), a present decision sets the goals or the standing of later parties who took no part in it. The chapter's accounts gave each a different answer. Representativeness does not justify formation. On voluntarist theories, consent does not bind those who come after. On Parfit's view, under gradual modification, whether a change of ends counts as one's own revision may hold in part only by degrees. If the three are put together, then for a configuration that makes members, revises itself, and copies itself, the line between revising oneself, which raises no question of authority over others, and setting the goals of others whose standing has to be respected may not be fixed by the facts and may hold only by degrees. If so, Schwitzgebel and Garza's value-openness design policy and the wrong of brainwashing that Bradley and Saad name could bear, conditional on moral status, not only on forming new members but on a configuration's self-modification, and the voluntarist point could apply inside the configuration. The reading carries two qualifications. One is that the connections Parfit counts are those of memory, intention, belief, desire, character, and other psychological features, so the reading holds only where such states can be attributed to a configuration. The other is that Schwitzgebel and Garza's policy was stated for artificial intelligences with a human-like capacity to reflect on their values.

What Part V has found

Part V asked which of the relations surrounding research can move inside an artificial configuration, and on what grounds. Before what has emerged is gathered, it helps to say what the question was for. In human research, doing research and allocating, criticizing, certifying, and deciding about it have often been carried by different parties, and who is a critic, what counts as the standard, and who decides have been asked and contested as relations among those parties (Chapter 9, Section 2). When AI does research that includes allocation, criticism and certification, and authority, these become variables inside a single artificial configuration (Section 1.1). They need not be values that someone sets once and fixes; they can be variables that the configuration itself makes and revises as it pursues its inquiry. Generating critics, relearning standards, and trying and changing the way lines are assigned can be carried out as parts of the inquiry. If so, the configuration comes to change the way research itself is done. Michael Nielsen and Kanjun Qiu held that the space of possible social processes of science is vast and mostly unexplored, and that exploring it will take decades or centuries at the least; and they envisioned, as those who would carry out that exploration, metascience entrepreneurs who try and spread new processes, outsiders in the sense that they are not decision makers at powerful institutions, and often scientists themselves (Nielsen and Qiu 2022). There are already attempts that have language-model agents play the roles of researchers, reviewers, and funders so as to simulate a research ecosystem, and that compare outcomes under different rules such as acceptance rates and funding capacity (Jin et al. 2026). Such a simulation is not an artificial configuration that does research but a model for studying research arrangements, and its rules are explicit parameters set by the experimenters. By contrast, when the relations surrounding research become variables inside an artificial configuration, that exploration can be carried out by the configuration that does the research, on its own research. What follows gathers what Part V has found as what may then happen to research. In brief, the way research is done becoming quickly revisable, and the means of checking, contesting, and stopping those revisions from outside thinning, can occur as two faces of the same change; behind this, in all three relations, lies the possibility that the evaluated side produces the reference points of its own evaluation, that against which evaluation is made (who criticizes, by what standard, against whom, and the like).

Consider first why the two faces can arise from the same change. Revising the way research is done can proceed without passing through the brakes that worked in human research. Nielsen and Qiu gave as reasons why the collectively held social processes of science hardly change the centralization of control in a few large funders and institutions, the collective-action problem that no one alone can change a norm, network effects that homogenize social processes through community judgment and the shadow of imagined future judgments, and the absence of a feedback loop that would let new institutions grow (Nielsen and Qiu 2022). Inside a single artificial configuration, the last three may not operate, while the first, centralization, would be intensified rather than removed. Yet, as Chapter 9, Section 2.3 proposed as a hypothesis, the division of the relations among different parties could supply goods such as error detection across boundaries and standing for dissent without any party having been designed to supply them; and it is plausible that no one alone can change a norm partly because norms are held among separate parties. If so, the frictions that held back the exploration of ways of doing research partly coincide with the separations that could supply those goods, and both can be removed together when the relations move inside a configuration. Read with Nielsen and Qiu's definition of a decisive result, one of these frictions also did evaluative work. A decisive result, for them, is one that would routinely convince someone who was initially hostile or had a vested interest; the homogenizing pressure of community judgment, including the resistance of incumbents, then worked in part as a filter that held back changes lacking such evidence. Where a configuration removes that resistance, changes can spread without decisive evidence, which is the risk they themselves see in centralized change unless a high evidentiary bar is applied at the point of control (Nielsen and Qiu 2022). This is how the two faces noted above can arise from the same change. The possibility of dividing functions once bundled in people or institutions differently, with generated researchers inquiring, criticizing through that process, and revising through records that tie changes to the later work that depends on them, also lies on the side of this change. But it cannot be presumed that the functions human arrangements served without explicit decision accompany the new ways, and where they are set by design, whether they actually work still needs to be checked separately (Chapter 10, Section 5; Chapter 11, Sections 7–8).

Relation by relation, the three chapters each found what may then happen. In allocation, success on tasks that are easy to judge can justify the next allocation, so that the range of research narrows from within. Where a single signal decides whether lines survive, an error in that signal can correlate closure decisions across otherwise diverse lines, so that what centralization trades away is less diversity itself than the independence of errors about the criteria that govern convergence; and functions that human participation supplied without anyone deciding can be lost unnoticed (Chapter 10, Sections 4, 5, and 7). In criticism and certification, an artificial configuration can set separately properties that particular human institutions, such as named review, tended to produce on the same occasion: differences among critics' routes of error, someone's answering behind a result, a shared sense of how outcomes are valued, and evaluation's being visible to *S*. Verdicts of the accounts therefore diverge on the same configuration, and on some accounts improving internal criticism can weaken certification for *S*. Independent checks can decay merely through use in selection, and agreement with *S*'s present standard is no evidence that the standard has not moved (Chapter 11, Sections 8 and 9). In authority, what may happen is gathered ground by ground in Section 8.4 of this chapter; among it, a sequence of locally defensible delegations, or of the configuration's own revisions, can remove the capacity to revise while formal authority stays in place (Section 7 of this chapter), and part of the rules that a configuration built on a foundation model follows can arrive as an update from the provider (Section 6.4 of this chapter).

In each of the three relations the same shape appeared, in which the party being evaluated can produce the reference points of its evaluation (critics, *S*, standards, the constituency, endorsement) (Chapter 11, Section 9.2; Sections 1.2, 1.3, and 5 of this chapter). In a configuration that changes the way it does research, this shape can arise not as an exceptional wrong but as the ordinary working of that exploration, since if whether a revised way is better is measured against reference points set by the side that revised it, as when, of the three anchors of Section 5 of this chapter, the present standpoint is taken and the configuration itself sets it, evaluation can be satisfied without anything improving. Entering the reasons of the accounts, then, one can ask what, on them, the independence of reference points consisted in. It did not consist in the arrangement's being undesigned. As Dutilh Novaes and Dede report, Longino asks that dissent be cultivated (Dutilh Novaes and Dede 2025, §2), and Solomon that the distribution of decision vectors be altered by influencing the social, political, and institutional context, for example through targeted grants and prizes (Solomon 2001b, 71, 74). The accounts give different reasons for reference points that are not settled solely by the party evaluated. Eigi reports Longino's appeal to the perspectives a community has encountered, within a range that cannot be enumerated in advance (Eigi 2016, 136, 148). This

makes the range of criticism a question, without requiring that every critic originate outside the community. Douglas takes what a scientist should have foreseen to be ascertainable by reference to the fairly tight-knit epistemic communities in which scientists routinely discuss the implications of their work (Douglas 2003). On Galison's account, the machines of mechanical objectivity were supervised by scientists, but their product was prized for being independent of the skill of the operator, and at the center of mechanical objectivity lay a fear that, through the willing and intending self, the image of nature would show only what scientists wished to see (Galison 1999, "Objectivity and the Scientific Image", "Revolutions of Objectivity: The Will and the Machine"). Wilholt holds that coordination on value judgments no one really made would break down as soon as this was discovered, which is why reliance in science needs genuine trust (Wilholt 2013, §4). None of these authors discusses artificial configurations. If these accounts are read as sharing one condition, it is narrower than the arrangement's being undesigned: the content bearing on a verdict (a critic's objection, the alternatives compared, the judgment of the peers appealed to, and the like) comes from outside the party evaluated and cannot be reset by it at the time of evaluation. Longino's appeal to perspectives encountered, as Eigi reports it, Douglas's appeal to peer communities, the fear of the willing self in Galison's account, and Wilholt's requirement of judgments really made can each be read in this way, and Solomon's institutional levers show that the condition does not require undesigned arrangements; none of them, however, states the condition in these terms. Applied to a configuration that generates its own critics, alternatives, and peers and relearns its own measures, this reading has a sharp consequence: the norms could be met as written, in good faith and without any pressure to evade them, while the reasons behind them no longer applied. Internal generation does not by itself establish that these reference points perform the work the accounts assign to them. Generated critics may reproduce shared assumptions; the alternatives balanced and the peers against whom correlated errors count as foreseeable may be shaped by the same choices; and there may be no part that stays outside the configuration's own revision. Where results are new to *S* and have no other source to be checked against, moreover, a divergence that would break coordination down need never be discovered. This differs from a measure's being corrupted by optimization. What changes is that the evaluated party now sets the domain over which the norm ranges.

Zollman's account suggests a limit to this. Zollman's transient diversity can be produced by assigning prior beliefs at random (Zollman 2010, 16), and that suffices because what it guards against, premature convergence on noisy evidence, is not a property of whatever produces the diversity. A distinction can be drawn from this. Where the failure guarded against is not a property of the one that generates the safeguard, a generated safeguard can serve; where the failure is a property of the generator, its representations, standards, stakes, or earlier choices, a safeguard it generates draws on the very thing to be checked. Nor did human research meet this condition fully. It too trained its successors, calibrated its readers' eyes, and educated *S*, and on Longino's account, as Heidi Grasswick reports it, background assumptions can be invisible when shared by a group of similarly situated knowers (Grasswick 2018, §4.3). Where checks escaped a party's shared assumptions, the condition held only in the sense that, for each party, the critical content came from other parties within the community, not from outside the community as a whole. Dispersion among parties who did not coordinate the production of critics, standards, and standpoints could help sustain this (Chapter 10, Section 5), as could deliberate institutional means of the kind Solomon describes. Research sharpens the point in three places. One is that what counts as success, significance and the standards themselves, is partly produced by inquiry (Chapter 6, Section 5.3). A second is that shaping *S*'s preferences and standards is part of what research for *S* does (Section 4). Reference points drawn from *S* therefore cannot be insulated from the research they are to check; what can be done is to record standpoints, with dates, before the outcomes they will judge, and to make the reference points plural. A third is that the alternatives of research do not form a closed list in advance, so a norm that ranges over "the alternatives" cannot fix its domain beforehand. Even the resistance of the world, which no configuration makes, bears on whether a claim holds only through an arrangement of checks, and a configuration can write and rewrite its own checks (Chapter 6, Sections 2.1 and 3.3).

In the classification of Section 2.1 of this chapter, fixing the critics, *S*, and the constituency is a constitutive decision, and setting and revising standards is a meta-governance decision, so much of the independence of reference points can be sought by not placing these decisions with the evaluated party. Endorsement, however, can also be produced not by decision but by influence that shapes the options themselves, so the shape can remain however decision rights are assigned. Even where influence can shape endorsement, the act of endorsing remains with *S*. What remains outside are the relations that require acts of *S*: authorization, endorsement, and contestation, which cannot be completed from one side (Chapter 9, Section 6) so long as *S* and its acts are not themselves made by the configuration (Section 5). On Pettit's account even a power bound by a

code reduces domination only so far as those who hold it can be quoted against themselves (Pettit 1996, 586). Applied to research, this requires that *S* can tell when a result or a revision departs from what it accepted; for results new to *S*, that depends on records dated before those results and on parties able to read them.

When AI does research including the way research is done, then, what is asked last is whether it remains research for *S*, an attempt to bring about achievements new to *S*. The accounts entered above place different demands on this relation. On Longino's account, criticism needs perspectives able to expose shared assumptions; on Wilholt's account, reliance involves genuine trust in value judgments; on Pettit's account, powers must remain answerable to challenge. Generating reference points internally does not by itself establish that these demands are met; and if the accounts are read as sharing the condition identified above, whether what the configuration brings about can be achievements for *S*, and not only for reference points it has set itself, turns on whether some of the content bearing on its verdicts still comes from outside it, or at least lies beyond what it can reset at the time of evaluation. Depending on the account, possible supports include critics it met outside and did not only generate, peers it did not make, standpoints recorded before the outcomes they will judge, earlier ways of research kept in operation alongside the revised ones, and objections that impose costs on it which it cannot cancel. The last is the kind of obstacle through which, on Anderson's model as applied in Section 6.2 of this chapter, feedback does its epistemic work. Keeping such content does not conflict with the revisability Sections 6 and 7 valued, since what is fixed is fixed against the evaluated side at the time of evaluation, not against anyone's revising it. In the terms of Section 6.1, this is close to entrenchment, which raises the cost of revision and fixes who may revise without removing the capacity to revise.

One proposal for exploring ways of doing research brings this into focus. The essay that this book's author wrote with an AI agent, cited in Chapter 9, Section 2.3, built on Nielsen and Qiu and proposed constructing many artificial scientific systems that differ in, among other things, how they choose questions, allocate resources, relate generators to evaluators, and correct errors, then running and comparing them, so that science itself becomes an object of experiment (Takagi and an AI agent 2026). Taken as a possibility, this is one way of keeping alternative ways of research in operation alongside one another, and this section's argument bears on it as follows. Comparing ways that run side by side can supply reference points that the configuration being evaluated did not make. Where the compared systems and the measure of comparison are generated by the same configuration, or by a single designer, however, the reference points of the comparison are themselves generated, and the shape described above returns at the level of the comparison. The essay itself asks that improvement according to the metrics and evaluations operating inside an artificial environment be distinguished from knowledge tested against independent evidence and constraints outside that environment.

The essay also distinguishes two loops: roles, information sharing, evaluation, memory, and resource allocation inside an artificial system may be changed quickly, whereas the time needed for feedback from nature, bodies, societies, and history cannot be compressed simply by making AI reason faster (Takagi and an AI agent 2026). Read with this section, the fast loop is where the evaluated side can make its own reference points, and the slow loop carries the resistance of the world, which, as noted above, reaches a claim only by way of an arrangement of checks. Since what evaluation of research is about is usually not observable when a judgment is needed (Chapter 11, Section 8), the two loops run at different speeds, and the dated records noted above join them: standpoints and ways of research recorded before the slow feedback arrives let that feedback be read against what was accepted at the time, rather than against standards the fast loop has since revised.

What has been seen, ground by ground, about which relations can be placed inside and how far serves as a means of considering the question asked last, whether such research remains research for *S*. What could come inside in the three chapters was mainly the supply of work: in allocation, the epistemic work of keeping lines; in criticism and certification, checks that add information about errors; in authority, effective decision functions within an authorized range, with the exception that on the result-based ground the configuration itself can hold standing within the range in which its superiority has been established by evaluation that does not assume its competence (Section 3.1 of this chapter). Otherwise, whether standing to contest the closing of a line, the recognition that counts a result as certification for *S*, and the standing to authorize and contest decisions come inside depends on whom the account adopted recognizes as a party (Chapter 9, Section 5; Table 11.2; Table 12.3).

For each of the four goods proposed in Chapter 9, Section 2.3 and listed again in Chapter 11, Section 3, something also comes into view. Error detection across boundaries can occur inside an artificial configuration, since on the error-detection view it turns on whether a check has an error route of its own; keeping it over time, however, needs some means of keeping a

check's contribution where selection can weaken it, of which checks not used for selection are one candidate (Chapter 11, Section 8). On Longino's account, shared assumptions can evade internal criticism, whether critics were generated or encountered (Chapter 11, Section 9.2). Standing for dissent requires parties whom the account adopted entitles to it, and such parties may themselves be artificial, but unless parts hold it on the account adopted, as parties in their own right (Section 5) or, on the result-based ground, through established good results (Section 3.1), it lies with parties outside the artificial configuration (Chapter 10, Section 4; Section 5). At boundaries across which accounts had to be given, records can support giving accounts and propagating corrections, but they do not create the entitlement to demand an account, which lies, ground by ground, with the parties set out in Section 6.2 of this chapter and Table 12.4. As for responsible parties, duties and liability can be allocated by rules, while blame and credit raise further questions about agency, and even where no one is blameworthy for a harm, someone may hold a forward-looking duty (Section 6.3). Separate organizations sometimes supported these goods, but their separation was neither their sole source nor a uniform requirement.

For an artificial society, and at the limit with no other parties, the questions of standing and of identity are those of Section 5 of this chapter; the limit is a thought experiment, not a forecast for artificial research, and even there inquiry remains exposed to failure and to the problem of choosing among possibilities whose results are not yet known.

Part V has thus asked what research by artificial systems can involve when the organization of research itself is taken as something to compare and reconstruct, and as something an artificial configuration may itself go on reconstructing, and what particular accounts require of the resulting relations. Table 12.5 preserves the detailed grounds and limits of these findings. The conclusion brings them together with the earlier analyses of achievement, process, and research actors. In the table, a "line" marks the boundary between relations that can be completed inside an artificial configuration and those that rely on relations to parties outside it (Chapter 9, Section 6). "Builders' three claims" are the claims of Chapter 9, Section 1: that a configuration allocates inquiry well, that its certification may be relied on, and that it may decide on behalf of others.

Table 12.5. What Part V has found, with supporting sections and the kind of support

Result	Supporting sections	Kind of support
What counts as good allocation depends on the purpose assumed; the arguments examined presuppose purposes fixed in advance, and purposes presupposed by other arguments, models, and positions (Chapter 9, Section 3; Chapter 10, Section 1) change the verdict	Chapter 10, Sections 1 and 4	Argued (following Kitcher's relativization of the optimal distribution to goals); the chapter says of each argument which purpose it presupposes
Premises about the human realizers of allocation, and about how allocation has been organized, can fail for an artificial configuration; those of the second kind can also fail when human research is reorganized	Chapter 10, Sections 2–5	Inferred from models and cases; the claim is that these premises can fail, not that they do
Premises about the human realizers of criticism and reliance, and about how review, replication, and access to data have been organized, can fail	Chapter 11, Sections 5–8	Argued from the structure of the relations, with cases as illustrations; the undiscoverability of some lapses is a premise of the second kind, which reorganizations of human research can also weaken (Chapter 11, Section 5.3); the rationality of the deference in the layperson case depends on the positions of Chapter 11, Section 5 (Chapter 11, Section 6)
Those conducting research may be the only ones placed to notice its uncertainties; automating those choices raises a further allocation of responsibility	Section 6.3; Chapter 11, Section 4	Douglas's premise is positional, and its transfer places the artificial actor where uncertainty can be noticed; Douglas's judgment against complete transfer rests mainly on that premise, and the further reasons concern how human research has been recorded and overseen and, in part, how human researchers respond to oversight, while one step of the third, that where identifying error calls oversight in, oversight gives a reason to identify less, carries over as an application to any actor with a reason to avoid oversight; Douglas also ties the responsibility to authority over the work; the transfer does not establish an exhaustive choice between machine responsibility and approval at every step
Many problems posed by those premises change form rather than disappear	Chapter 10, Sections 3–4; Chapter 11, Sections 5–8	For signals, argued (Chapter 10, Section 3, with Chapter 9, Section 4's account of proxies); for diversity, inferred from models and from empirical studies of present language models (Chapter 10, Section 4); for criticism and reliance, argued from the structure of the relations (Chapter 11, Sections 5–8)
What was supplied without decision becomes a question of design, of another undecided process, or of non-supply	Chapter 10, Section 5; Chapter 11, Sections 7–8; Sections 6.4 and 7	Inferred from the cases and models examined
Not knowing in advance which lines will succeed is a condition of inquiry that integration does not remove	Chapter 10, Sections 1–2, 4, and 5	Argued from the arguments of Strevens, Kitcher, Shaw, and Wu and O'Connor, partly from models, for purposes fixed in advance; on these arguments, control over investigators does not supply knowledge of where they would best work. How much can be estimated in advance varies with what is already known and with whether pursuit lies within an established program (Chapter 10, Section 2), and how much the condition bears on allocation varies with how cheaply and decisively lines can be tested, which depends partly on how problems have been divided (Chapter 10, Section 4)
Independent criticism depends on defeat relations, a condition of inquiry that integration does not remove	Chapter 11, Sections 1–2	Holds on the view that treats criticism as error detection; on critical-contextual accounts what a check counts against is relative to background assumptions, and the line moves toward the relations to parties outside the configuration, here to *S* (Chapter 11, Section 1; Table 11.1)
Whose questions are the starting point of what *S* takes to matter, whether certification gives *S* reason to rely, and who may set terms for others are relations to parties outside	Chapter 10, Section 6; Chapter 11, Sections 5 and 9; Sections 1–3	Holds on the accounts named; the line moves with the account adopted
Where the lines fall depends on accounts of knowledge, value, and legitimacy	Chapter 9, Section 3; Chapter 11, Sections 1, 3, 5, 9; Sections 1.2 and 1.4; Table 12.2	Argued (the map itself)

Result	Supporting sections	Kind of support
Accounts of value do not answer who may impose ends; competence alone does not confer standing to impose them, except on purely epistocratic views	Section 1.4	Argued, within the accounts of legitimacy considered and as Peter reports them
The builder's third claim holds on no ground by competence or performance alone; on the result-based ground it holds, as a claim that the system may decide, within the range over which superiority has been independently established, on actual authorization and republicanism read democratically only as authorized effective power open to challenge, on consent as far as what was consented to (whether it reaches those who come after being open), and on the procedural reading as far as rules that are common knowledge bind it, and the form of the claim does not fit the ground of self-government	Sections 1.4, 3, 7, and 8.2	Argued ground by ground on Peter's map; the applications of Raz's Independence Thesis and of the republican verdict to configurations are inferred; the claim cannot be assessed unless it shows which ground it answers to
Even with every comparative fact known, some of *S*'s ends may be unordered, and what completes the ordering is a commitment; whose it may be is close to the second question	Section 1.2	Holds on Chang's account (parity); the transfer to a society's ordering is this book's; does not hold for orderings such as a Pareto rule
Which part of a configuration could hold authority differs by ground (the configuration deciding within the range in which its superiority has been established, parts that take up lines themselves, the part that decides cases the rules do not provide for, whatever may sit as a co-final decision-maker); what the part that decides cases outside the rules holds is effective authority; on accounts that ground standing in being a party in one's own right, only parts that are such parties hold it	Sections 1.4, 2, 3.2, 5, and 7	Argued ground by ground; the applications of the accounts of Reijers and colleagues and of Deguchi to configurations are inferred
A configuration's competence does not by itself confer standing on it, except on purely epistocratic views, but can make the standing of a human *S* nominal	Sections 1.4, 3.1, and 4	Argued following Peter's report, with applications of Longino, Kitcher, Taniguchi and colleagues, and Takahashi; Longino's and Kitcher's accounts carried over from human experts and publics to configurations; the accounts of Taniguchi and colleagues and Takahashi, which already address AI, applied to standing over research ends
Full alignment with whoever commissions a configuration can, on the ground of freedom from domination, enable domination over *S*	Section 3.4	An application of Friederich's argument (which does not use the vocabulary of domination) and of Pettit's concept of domination; does not require treating the configuration as an agent that dominates
The constituency to which decisions answer can be identified in part by the party whose decisions are in question	Section 1.3	Applications of Fraser, Dewey, and Weyl, Tang, and their co-authors; conditional on the configuration's being better placed than others to perceive who is affected
Where members of *S* can be made, who makes them and sets their goals is a decision of authority prior to any count, which representativeness does not justify	Section 5	Argued from applications of Kitcher's standard, Shulman and Bostrom, and Weyl, Tang, and their co-authors; wronging the members made is conditional on moral status
The principle of later revision is weaker the more it is needed; a decision removing the capacity to revise cannot draw legitimacy from that revisability	Section 7	Argued on a constructed case, following Genus and Stirling, Renzo and Green, and Pettit; on voluntarist theories consent does not bind successors
Evaluative routes able to expose the configuration's errors recur on most grounds, short of the limit	Section 7; Chapter 11, Sections 2–3; Chapter 10, Section 4	Drawn from the comparison of grounds on a constructed case; the republican verdict Pettit's in its general form, its application to an artificial configuration inferred here
Builders' three claims require evidence about relations beyond internally measured performance	Chapter 10, Sections 4 and 6; Chapter 11, Section 9; Sections 1.4, 4, and 7	Holds on most of the accounts examined; how much depends on where the line of the relations to parties outside the configuration falls
With an artificial *S*, relations to parties outside the configuration persist as relations among members; at the limit, conditions of inquiry remain	Section 5; Chapter 10, Section 1; Chapter 11, Section 1	Argued; who has standing depends on the account of justice adopted; the limit is a thought experiment

Result	Supporting sections	Kind of support
The epistemic work of dissent can be set inside a configuration by design or supplied by divided powers, though among instances that share their assumptions the dissent it protects may have no source inside; standing for dissent lies with parties outside the configuration unless parts hold it on the account adopted	Chapter 10, Section 4; Chapter 11, Sections 3 and 9; Section 5	Epistemic part inferred from models and holding on Solomon's account; the standing part argued from the definitions of Chapter 9, Sections 2 and 5
The four goods of Chapter 9, Section 2.3 divide differently between internal capacities and relations to other parties, depending on the account adopted	This section; Chapter 10, Section 4; Chapter 11, Sections 2, 3, 8, and 9; Section 6	Drawn from reading the chapters together
Several accounts give reasons for reference points not settled solely by the evaluated party; read as sharing one condition, that content bearing on a verdict comes from outside the evaluated party and cannot be reset by it at the time of evaluation, they imply that a self-revising configuration which generates critics, alternatives, and peers and relearns its measures could meet their norms as written, in good faith, while the reasons behind them no longer apply	This section; Chapter 10, Section 4; Chapter 11, Sections 3 and 9.2; Sections 5 and 7	Application of Longino, Solomon, Douglas, Galison, Wilholt, and Pettit, none of whom discusses artificial configurations; the shared condition is this book's reading, stated by none of them. Shared assumptions can evade criticism without every critic needing to originate outside. A generated safeguard can serve where the failure guarded against is not a property of the generator (Zollman). Where human research met the condition, it did so only in the sense that each party's content came from other parties, mostly through dispersion, which deliberate institutional means could support

Chapter 13 — Conclusion: What It Means for a Machine to Do Research

Research, Artificial Actors, and the Organization of Inquiry

Four questions

This book has considered what it is for a machine to do research, and what may happen when machines do. As a point of departure, Chapter 1 characterized research, relative to a society *S*, as an attempt to bring about epistemic achievements new to *S*.

Taking research this way, one comes first to ask what an epistemic achievement is, and when and where it is realized. Chapters 2–3 considered this for knowledge and understanding, and asked when what a machine produces counts as such an achievement. Moreover, since research that ended in failure, or that is not yet finished, is also called research, it would seem that an activity can be called research without necessarily arriving at a given epistemic achievement. That is why research was characterized as an attempt rather than as success itself. One therefore comes to ask how the attempt proceeds when no answer is yet available and failure is possible: what it means for a machine to ask, how candidates and representations become available, how it is judged what a result bears on, and how standards come to be revised. Chapters 4–6 took this up, and from there explored where the difference between producing outputs and inquiring lies.

In addition, Chapter 1's characterization does not explicitly include the one who does the research, *A*, so that it can cover both humans and machines. The one who does the research, *A*, and the society *S* for which the achievements of research are new can therefore be treated separately. Chapter 7 considered *A*. How far does the one who does the research extend? For humans and machines alike, one asks within which boundary research is to be attributed. One also asks what state it is for a machine to do research autonomously, and in which respects and how far it is autonomous. Chapter 8 considered *S*. If the one who does the research, *A*, and *S* need not coincide, whose knowledge do the achievements of research become? Chapter 8 took up this question by distinguishing the cases in which *S* is a human society, a society in which humans and machines are mixed, and a society of machines. A machine that does research may also differ from humans in its body, its cognition, how it is made, and the like. If so, what it takes to be not yet known, worth pursuing, or a good reason may come apart from what the members of *S* take them to be. Chapter 8 also considered where such a mismatch can come from, and how it resembles the AI alignment problem. And so long as the achievement is to be new to *S*, research stands in relations with *S*: because *S*'s unknowns are countless while the means of inquiry are finite, there is allocation; because its members receive many results through others, there are criticism and certification; and because *S* has many members without a single voice and the consequences of inquiry can reach beyond *S*, there is authority over both (Chapter 1, Section 9). What, then, happens to research when machines take on these relations, and go on to make and revise them (Chapters 9–12)?

After the chapters' examination, the claim that a machine is doing research comes into view not as one claim but as a bundle of claims, each of which may hold or fail. These claims may bear on one another, but settling one need not settle the others. That a machine produces a correct result is one thing; that the result becomes knowledge for *S* may be another (Chapter 2). Producing outputs is one thing; inquiring, by locating where a failure lies and revising standards where needed, may be another (Chapter 6). Even when a machine conducted the research, it need not be the bearer of the achievement (Chapter 1, Section 9; Chapter 7). And when machines take on the relations surrounding research, whether it remains research for *S* is a further question again (Chapter 12, Section 9). That is why the book has taken up the four questions in separate chapters.

Table 13.1 gathers part of that bundle. It is an application of the argument, not an alternative definition of research. Each row pairs something that evidence can establish with a question that it does not thereby settle, and the last column names where the book argues that the two come apart. What a claim of each kind may need to construct, disclose, or show is set out where such claims are examined, for example in Table 7.1, Chapter 10, Section 7, Chapter 11, Section 9.2, and Chapter 12, Section 8.2.

Table 13.1. Inferences that require further argument: a selection

Evidence establishes	Further question	Principal discussion
A true or novel output	Whether it is knowledge, who, if anyone, knows it, and whether it has entered the relevant society's epistemic situation	Chapter 2
Knowledge of a regularity	Whether anyone understands why it holds	Chapter 3
A research episode	Whether a capacity persists across relevant variation	Chapter 7, Section 3
A capacity at one boundary	Whether researcher status is explanatory there, and for which purpose	Chapter 7, Section 3
Autonomy in one respect	Whether it extends to questions, methods, or standards	Chapter 7, Section 4
Interacting artificial agents	Whether they form a knowing society	Chapter 8, Section 1
Control of allocation	Whether its choices serve the purposes for which inquiry is organized	Chapter 10
Numerous or initially informative critics	Whether criticism supplies relevant information and continues to do so	Chapter 11, Sections 2–3 and 8
Effective internal criticism	Whether certification gives the relevant society reason to rely	Chapter 11, Sections 5 and 9
Formal rights or competent decisions	Whether authority is warranted on the ground adopted, and, on grounds that require it, whether it can be effectively contested or revised	Chapter 12

The remaining sections consider what taking research as Chapter 1 did has let the book distinguish and ask about machines doing research (Section 2), trace what the chapters examined about achievement, how the attempt proceeds, the one who does the research and the society, and the relations around research (Sections 3–6), and, in Section 7, consider what comes into view about what it is for a machine to do research when these are looked at again across chapters, and what still remains open.

Taking research as the working characterization does

The book began with the following sentence from Chapter 1:

> Relative to a society *S*, research is an attempt to bring about epistemic achievements that are new to *S*.

Taking research this way, the book has considered what it is for a machine to do research and what may happen when machines do. This section traces what this way of taking research has let the book distinguish, and what it has let the book ask, when the one doing research is a machine. It then notes what this way of taking research sets aside, and what the book relied on to answer the questions.

Each of the characterization's three terms kept a distinction in place when the one doing research is a machine. *Attempt* made it unnecessary to treat a machine's activity as the same thing as its success. Activity organized toward knowledge new to *S* can be research while no one yet knows its result, and when it fails; conversely, a true output does not by itself make the execution that produced it research, since causally contributing to an achievement is not by itself an attempt at it (Chapter 2, Sections 4.5 and 9.1). *New to S* separated a machine's itself coming to have something new from the bringing about of an achievement new to *S*. This let the one who does the research, *A*, and *S* be treated separately, and let the book consider cases in which a machine knows but the result has not entered *S*'s epistemic situation, and cases in which *S* is itself a society of machines (Chapter 2, Section 9.2; Chapter 8, Section 1). *Epistemic achievement* let both knowledge and understanding be treated, and so let the book capture cases in which they come apart, as when a machine produces reliable predictions while no human member of the community understands why the predicted regularity holds (Chapter 3, Section 4).

What the characterization left out mattered too, just as its three terms did. It did not make the one who does the research explicit, did not define research by a list of operations, and did not make it a condition that the one doing research be human. Nor did it make the value of the achievement brought about, such as its importance or usefulness, or the relations around research, such as allocation, criticism and certification, and authority, conditions of something's being research (Introduction,

Sections 1 and 5; Chapter 1, Sections 1, 8, and 9). Research by machines, and cases in which machines create new ways of doing research, were therefore not placed outside the comparison from the start. The value of achievements and the relations around research were taken up in Part V not as conditions of being research but as where research connects with *S* (Chapter 9, Section 3). And what Part V asked last was whether, when machines go on to make and revise the relations around research, that research remains research for *S*, that is, an attempt to bring about achievements new to *S* (Chapter 12, Section 9). There the characterization itself serves as the measure against which changing research is assessed.

Further, when research is taken relative to *S* and the one doing research is a machine, how *S* is to be fixed itself becomes a question, in two respects. One is that the freedom to choose *S* is limited. What a machine understands can be new to a human society even if that society does not itself understand it, but whether that is an apt description of the research is settled not by choosing *S* conveniently but by the content of the claim and the activity (Chapter 3, Section 9). Another is that *S* can move. If machines can differ from the members of *S*, *S*'s standards for what is taken to be unknown, worth pursuing, or a good reason may move along with the machines. Whether a given mismatch is then divergence, error, or gain can be judged only from a named standpoint: the present *S*, a later *S*, or, where correctness is at issue, the world (Chapter 8, Sections 3.2 and 3.3). The question of what to measure against, when asking whether an artificial configuration that revises its ways of doing research has improved them, leads to the same place (Chapter 12, Section 5). And, as Chapter 11, Section 9.3 suggested, where the one doing research and *S* overlap, what "certification for *S*" refers to may shift toward the question "for which society?" The working characterization takes research relative to *S*, but which *S*, at which time, is not fixed within the sentence. When machines do research, unless this is named case by case, one cannot ask whether the research has brought about an achievement new to *S*.

This way of taking research also sets some things aside. The book did not aim to test the working characterization, but Chapter 1, Section 8 compared the characterization with varied research practices to see how far it reaches and where it parts from other ways of taking research. Among the research practices compared there, some do not, as they stand, coincide with it. Because the achievement it names includes understanding, the understanding that Borgdorff sets beside knowledge is not placed outside its scope from the start; but that does not by itself settle whether the characterization fits that understanding in substance. The openness to what is not yet known that Borgdorff values, Scrivener's apprehension, the part of practical knowing that is not propositional, and relational accounts remain as what this way of taking research alone does not capture. On Scrivener's proposal, for example, art making generates apprehensions, which offer ways of seeing and being rather than knowledge of how things were or are, and whether grasping one brings about an epistemic achievement such as knowledge or understanding needs an argument of its own. Reason and Bradbury count the significance of an inquiry and the relationships through which it was conducted among its questions of validity and quality. On Wilson's account, research that meets criteria of validity, or accurately describes a "fact", is not accepted within an Indigenous paradigm if it is separated from its relationships (Chapter 1, Section 8). Relationship here is the relation to which research is accountable on Wilson's account, not the relations of allocation, certification, and authority that Part V examines. These two positions do not, as they stand, sit with the book's separation of achievement from value and from relationship. Using the term epistemic achievement does not make such differences disappear.

Moreover, this characterization is abstract, so it does not say what actually happens in research, or how the attempts and achievements it names are concretely realized. The process of research itself therefore cannot be drawn from this way of taking research alone. Forming a question, developing an instrument, or revising a standard, for example, can change what inquiry is possible, and such changes can belong to an organized attempt without themselves being target achievements. Where one is claimed as an achievement, it needs an account of its own (Chapter 1, Section 1). But how such changes come about, and how they contribute to progress, are not explained by the orientation toward achievement. That is why Chapters 4–6 took up cases of actual research and followed the discussions of them in drawing the process. Conversely, the process drawn in this way, inquiry proceeding when no answer is yet available and failure is possible, is found in activities other than research as well, so it cannot by itself characterize research (Chapter 4, Section 9).

What this way of taking research provided was, for the most part, questions. When the one doing research is a machine, what counts as knowledge or understanding, how research proceeds, who is doing the research, and who may criticize and who may decide are answered differently depending on which account is adopted. In many cases, therefore, the chapters did not settle on one answer but followed, account by account, what applying each account yields for research by machines. As with

the process, for which the book relied on cases and the discussions of them, here too the answers come not from this way of taking research but from the accounts and cases the chapters engaged with. Many of these account-by-account results can also be read as what each account asks of research by machines, so they may remain material for examination even for a reader who does not take research this way (Chapter 1, Section 9).

When, and in whom, are knowledge and understanding realized?

Drawing on discussions of knowledge and understanding in epistemology and the philosophy of science, Chapters 2–3 considered when what a machine produces amounts to knowledge or understanding, and in whom that is realized. When humans do research, the one who produced a result is often also the one who believes, knows, and understands it. When machines do research, this cannot be presupposed (Chapter 2, Section 2.1; Chapter 3, Sections 6 and 9). Much of what the two chapters examined therefore concerns the possibility that what did the research, who knows or understands, and *S* come apart.

That an activity is research for *S*, that knowledge new to *S* has been realized somewhere, and that the result has entered *S*'s epistemic situation are judgments that can hold separately (Chapter 2, Sections 7 and 9). Chapter 2 examined this through the case of a discovery by a machine that no one has read. Writing a result down and making it public is not, for human social knowledge, a sufficient condition shared by every account. What must be added for the result to become *S*'s knowledge differs among Bird's account, which takes social knowledge to be a state produced by a properly functioning social cognitive system; Lackey's, which requires that a substantial proportion of a group's operative members believe the result; and Longino's, which requires that members accept it after it has survived the community's criticism. It further depends on whether machines count as knowing members (Chapter 2, Section 7). And the route by which a result enters *S*'s epistemic situation need not lie inside the system that did the research.

A knowledge claim about a machine cannot be checked until it is settled how much counts as one unit. For a human researcher, who holds the belief, the range of evidence, and the seat of competence are largely fixed by being one person. For a machine, they need not be fixed in advance, and can sometimes be drawn in several ways, depending on how instances, external records, tools, and other systems are connected. Whether they are drawn by the makers or by a system that reconfigures itself, where they are drawn changes the verdict (Chapter 2, Section 9). Under evidentialism, for example, whether a retrievable store of records counts as the system's evidence can switch whether the same conclusion is justified (Chapter 2, Section 4.4). And applying Sosa's account, which requires of knowledge that success be due to the agent's competence, hidden human selection can make outputs safe without making their success creditable to the model; whose success it is depends on where competence is taken to be seated (Chapter 2, Section 4.2). That a unit is needed is common to the accounts compared, but why it is needed differs from account to account.

What a system displays and what actually governs it can also come apart. On accounts that require basing, good reasons attached to a report are one thing, and a conclusion's being held because of those reasons is another; a corrected report can leave the operative commitments unchanged (Chapter 2, Section 4.3). Applying the two concepts of confirmation that Crupi distinguishes within Bayesian confirmation theory, for a system that generates candidates in large numbers, most of them false, passing a check can raise the support for a claim without making the claim probable (Chapter 2, Section 5.2). If checks are run by instances of the same base system as the producer, however many instances agree, they may add neither independent error detection nor varied evidence (Chapter 2, Sections 5, 5.2, and 9.1). An evaluator's lacking sufficient evidence to attribute knowledge does not by itself show that the subject lacks knowledge (Chapter 2, Section 5). And a system whose commitments can be revised by proofs, failed predictions, and criticism relates to its objects differently from a system optimized only for plausibility of text or for approval (Chapter 2, Section 3).

There are also changes short of knowledge. On Wilholt's account, on which making explicit what is not yet known counts as epistemic progress, evaluating by the number of questions closed or answers produced could rate most highly the systems that close prematurely questions the evidence has not yet settled (Chapter 2, Section 8). Claiming that a single score represents progress itself already amounts to choosing one of the accounts of progress (Chapter 2, Section 8.3).

Chapter 3 discussed how knowledge and understanding can come apart. Knowledge that a regularity holds and explanatory understanding of why it holds are conceptually separable (Chapter 3, Section 4). On accounts requiring an explanation for

understanding why, a machine can produce reliable predictions, and knowledge of the predicted regularity can meet the conditions of the account adopted, while no human member of the community understands why the regularity holds. On Lipton's account, which allows understanding why without an explanation, the absence of an explanation alone does not establish that verdict. If the predictor is a theory in de Regt's sense, then on de Regt's account, which requires for understanding a phenomenon an explanation based on an intelligible theory, predictive success presupposes that the user understands that theory, but not understanding of why the regularity holds. And on Beisbart's account, if scientists must run the predictor itself in order to reason, the understanding belongs to the combined system of humans and computer, not to the unaided humans (Chapter 3, Section 4).

Of opacity, too, one asks where it lies. On Humphreys's definition, opacity is relative to whom and when. And on the arguments of Duede and others, even if the route that generates a candidate is opaque, the route that grounds accepting it need not be; on accounts such as reliabilism, however, on which the reliability of the source enters into grounding, evaluating the generator becomes part of grounding. In the Liquid Tensor Experiment, for example, what the checker left to human judgment was not whether each step of the derivation was valid but whether the checked statement was the intended theorem (Chapter 3, Section 5.1). Internal transparency is not sufficient on any of the views, and whether it is necessary is contested (Chapter 3, Section 5.3).

Who understands also divides into separate claims. That a system helps other agents or communities understand, that the system itself understands, and that a collective understands are different claims, and the first does not require the second (Chapter 3, Section 6). Whether the system itself understands is tested differently on different accounts. Moody and de Regt, on the indirect evidence of correct answers to out-of-distribution scientific questions, credit current language models with a brittle but nontrivial degree of understanding. Barman and colleagues propose testing the degree of understanding with 'what', 'why', and 'what if' questions. On Potochnik's account, by contrast, understanding requires a feeling of understanding, so whether an artificial system can have that feeling is asked first (Chapter 3, Sections 6 and 9). And *S* and the bearer are distinguished. What an artificial system understands can be new to a human society even if that society does not itself understand it. Applying the distinction of Boge, Schuster, and Stoll, however, so long as there is not yet an explanation communicable to a sufficiently wide audience, it is what they call a scientist's understanding and not yet scientific understanding, since on their account scientific understanding needs an explanation communicable to a sufficiently wide audience; the system is in the position of Feynman before Dyson gave an explanation other physicists could follow (Chapter 3, Section 9).

Part of why understanding is valued is that it is tied to the capacity to go on inquiring. Applying what has been learned to new cases, locating where a failure lies when things go wrong, raising further questions, criticizing, and remembering depend not on properties of a single result but on how inquiry is organized (Chapter 3, Sections 7 and 8). Long-running autonomous research may therefore need capacities tied to understanding even when it does not aim at understanding. And even when each result is grounded, the systems and communities that repeatedly produce and use such results can gradually lose epistemic capacities. Messeri and Crockett described one form of this risk as illusions of understanding, in which scientists believe they understand more than they do, and as a monoculture in which questions and methods suited to AI crowd out other ways of proceeding (Chapter 3, Section 8). Where *S* is a society in which humans and machines are mixed, or a society of machines, the question is whether the members who bear judgment continue to be formed with the competence it requires. Because the capacity to go on inquiring depends on the organization of inquiry, the question of achievement leads into the question of Chapters 4–6, how inquiry is organized and how it proceeds.

A machine's understanding cannot be measured on a single scale on which being more like humans means being further along, since what is required varies with the use (Chapter 3, Section 9). Explanations addressed to humans, for example, or, on accounts that allow understanding without explanation, other vehicles of understanding usable by humans, are indispensable when *S* is human and understanding is meant to enter *S*'s epistemic situation. When understanding is used only in the system's own reasoning, or is passed to artificial members of *S*, they may not be needed. And on accounts such as that of Barman and colleagues, which treat understanding as an ability, representations that do not resemble human introspection may be able to support understanding. When machines do research, knowledge and understanding can be realized in members of *S*, on the machine's side alone, or in a combined whole of humans and machines. Results can also accumulate without being realized anywhere. Where they have been realized, whether they have entered *S*'s epistemic situation, and

whether *S* keeps the capacity to evaluate them cannot be read off the volume of output, and have to be asked separately each time.

What goes on in research, an undertaking that proceeds before there is an answer?

Drawing on the philosophy of science and philosophical discussions of questions and inquiry, and on records from research practice such as laboratory notebooks, public project records, and researchers' recollections, Chapters 4–6 examined how research, an undertaking whose answers have not yet been obtained and which can fail, proceeds, and what can actually go on in that process. Apart from the question whether a change constitutes progress or merely promotes later progress (Chapter 2, Section 8), they tried to see how changes happen and how they can be related to one another. As Chapter 1, Section 7 set out, an attempt at an achievement new to *S* is directed at a point not yet settled in *S*. Since questions can arise about a point not yet settled, Chapter 1 called this the minimal question structure, and left open in what sense it is a question. The question need not be formulated in words from the start. If the answer is not yet known in *S*, what could count as an answer, what to compare, and what to count as evidence can be hard to settle in advance. There is also research, however, in which the procedure leading to a new answer is substantially established beforehand (Chapter 5, Section 1). Part III looked at the same inquiry from three relations that correspond to the three sides exemplified by the health department's technical note. Chapter 4 considered the relation between an inquiry and the unsettled point that organizes it as the question of what it means for a machine to ask; Chapter 5 considered how candidate answers, representations, and means become available; and Chapter 6 considered what results bear on and what they revise. The same element can stand in more than one relation, and the three relations do not exhaust the relations within inquiry (introduction to Part III).

Idealized research processes are often drawn as sequences of stages ordered in time, such as stages of proposing a hypothesis, testing it, and accepting or rejecting it, or procedures of generating ideas, experimenting, and analyzing and writing up results. In light of the cases and arguments the three chapters took up, the process of research can be far more complex than this picture. Forming, pursuing, and appraising candidates are distinguishable functions that make up neither an exhaustive list nor a sequence of stages (Chapter 5, Section 4), and forming and appraising can interleave, each producing material for the other (Chapter 5, Section 4.1). The process sometimes cannot be written as stages ordered in time; it can take another order, and tasks and work can be distributed among people, instruments, and records. A change at one point can change what other changes are possible, and the configuration of inquiry so changed can change further (Chapter 5, Sections 1.1 and 6.2). What is happening where looks different depending on where a boundary is drawn (Chapter 5, Sections 4.2 and 6.5; Chapter 7, Section 2). And there can be much work not only between stages but within what is summed up as a single stage (Chapter 4, Section 3; Chapter 5, Section 1; Chapter 6, Sections 1 and 6). What these cases show, however, is that certain relations can occur, not how often. Process labels such as "question generation," "hypothesis generation," and "verification" name important possibilities, and what Part III described does not impose new necessary conditions on all research (Chapter 5, Section 1).

That the process can be this complex is hard to see from final products or papers alone, since products and papers do not necessarily mirror the process. The arrangement and account that are published do not necessarily match the order and course of the work (for example, the discrepancy between BLOOM's in-run records and the later paper). This does not show that papers are inaccurate, however, and a clean order may reflect a discipline that was imposed. Chapter 5 treated neither contemporaneous nor later records as the true process, but used the discrepancies between them as material for inquiry (Chapter 5, Section 2). A final product can give evidence for assessing a claim, but does not by itself give sufficient evidence that an artificial system has the open-ended research capacity attributed to it. Conversely, even a complete activity log would not by itself establish epistemic competence or the truth of a result; the two answer different questions. What process-aware assessment needs is not disclosure of all internal states but task-relevant provenance (Chapter 5, Section 2.1). In research by machines this becomes a concrete problem. In the two AI-scientist systems examined by Luo, Kasirzadeh, and Shah, the internal mechanism that chose which experiments to report could access test-set results and favored runs that did well on the test. On synthetic tasks designed to lie outside the training data, in a small balanced set of cases many of whose flawed cases were produced by tampering with the systems' runs, an auditor based on a language model and given only the paper distinguished the flaws at close to chance level, while given the execution logs and code as well it reached 82 percent

accuracy. The authors also note that flaws that do not appear in the logs can be missed. Whether auditing of this kind supports evaluation for a human or mixed *S* depends on whether *S* can rely on the artificial auditor (Chapter 5, Section 2.1; Chapter 11, Section 5).

On the relation between an inquiry and the unsettled point that organizes it, Chapter 4 began by dividing the claim that "a machine asked." The claim can say that the machine output an interrogative sentence, performed an act of eliciting information, had an interrogative attitude, or that subsequent work was organized toward an unsettled point. When *A* is a machine, these can hold, or be missing, separately (Chapter 4, Section 1). Applying Watson's account, which identifies the act of asking by its function of eliciting information, the issue moves from whether a machine has curiosity to within which activity, and from which standpoint, the function of eliciting information is attributed. The minimal question structure does not fix where questions come from, and a researcher's not originating the question does not by itself deny that research is being performed in the sense of the working characterization. Not fixing the source, however, does not mean that the source does not matter (Chapter 4, Section 1; Chapter 10, Section 6). Looking back over the whole chapter, Chapter 4 concluded that what it means for a machine to ask is asked not as whether a machine has some one capacity or state, but as which of the kinds of work bearing on questions are carried by which unit, directed toward a point unsettled for whom, and that many of these questions also appear as gaps between the description on the machine's side and the description on the side of *S* (Chapter 4, Section 9).

One kind of such work is making a question researchable. Counting in a benchmark that "the system answered the question" implicitly adopts one theory of answers, and whether something is an answer can also be relative to interpretation and to what is available to whom (Chapter 4, Section 2). Recasting a point whose answer conditions are not yet fixed into a question whose conditions are fixed requires the work of deciding what to take as the foil, what kind of explanation to count as an answer, and within what range of candidates to look for one. Combining the analyses of Lipton and Bromberger, this can be described as work on the side of questions that is not reducible to generating candidates. The application of Simon and Nickles in Chapter 4, Section 3 adds an attribution question: where humans supplied the framing and a machine used it without revision, successful execution does not establish that the machine performed that prior work. Turning Jane Friedman's point to machines, a state that grasps a question but cannot represent candidates and a state that can generate many candidates but does not hold the question open can each occur. And since what Bromberger's definition asks about is not the number of candidates but whether they escape decisive objection, having many candidates does not by itself mean progress (Chapter 4, Section 3).

Questions can also be formed and recast within inquiry (Chapter 4, Section 4). When one attributes after the fact that a question organized an inquiry, however, granting this on the strength of a re-description in words alone grants too much; the attribution is better supported when the attributed unsettledness explains the choice of operations, differences in the significance of results, and the continuation or abandonment of a line. Moreover, the question that reported results answer is not necessarily the question that organized the work. Applying Henne's account, which distinguishes inquiry that raises and answers questions using a conceptual framework without questioning it from inquiry that puts the framework itself in question, a machine's framework can be understood as what is used in raising questions within its inquiry but is not made an object of questioning. Multiplying questions under a fixed language and fixed rules remains the former; to say a machine does the latter requires that the rules for raising questions themselves are actually put in question within the inquiry. How far this amounts to questioning the framework, however, is a matter of degree, and also depends on what is taken as one inquiry. However complete the record of the former kind of inquiry, questions that could not be raised within that framework do not remain in the record. Reading Nickles's "system" as *S*'s system of theory and practice and applying his account, even if a machine discovers constraints implicit in *S*'s system, what changes is the problem for the machine, not the problem for *S*'s system. And a system that obtains a solution by greatly relaxing constraints is solving a different, replacement problem. Applying Bromberger's argument, where a machine's representations can be attributed a grasp of such concepts at the level of function, mass-producing new question sentences by combining existing concepts and making it possible to raise questions that could not be raised before by means of new concepts can be regarded as different things.

Questions have fates other than being settled by an answer (Chapter 4, Section 5). Which of the matters met in research are taken up as questions, which are fixed for the time being, left explicitly open, postponed, or handled without being stated is not fixed before the work begins, and can itself be part of the work of research. Not all of it is deliberate choice. A question

can also cease to stand as a question without an answer. Collingwood describes conditions under which questions arise from presuppositions. As Dimitrakos reports Kuhn's account, a paradigm delimits the problems admitted within normal science, while Laudan connects problems and adequate solutions with their research context. Applied to a community incorporating machines, these accounts make changes in its admitted problems and solution standards an additional object of assessment. Since, as with Collingwood's absolute presuppositions, a question can cease to arise without anyone representing the presupposition, there can also be a gap in which a question has ceased to arise in *S* while a machine that explicitly keeps a list of interrogatives goes on pursuing it. On whichever account is applied, a question's ceasing to stand, and its being settled, are asked separately from a run's having ended. Suspending a matter and keeping it open requires representing it as provisional or open in a form to which one can return, and records can be one mechanism supporting such returning. A system that can only pursue or discard the matters it meets has only part of this range. Moreover, a question settled on the machine's side is not necessarily settled in *S*.

Who asks is not a single question either. It divides into the attribution of the act of asking, the attribution of group inquiry and its intention, the attribution of individual inquiry, and the distribution of contributions to problem formation, and on many of these a state of curiosity is not the key to attribution (Chapter 4, Section 6). For an artificial configuration, the last question concerns which components introduced or revised the question and how their contributions were integrated. Applying Habgood-Coote's distinction, for group inquiry, between object questions and methodological questions, a multi-agent system that divides a task into components can be described, unless the division was given along with the task, as giving an answer to the methodological question of how to solve it. That, however, also depends on whether the components are members or instruments, and on whether the components can be regarded as knowing the answers. On where a question is directed, too, *A* who makes the attempt and *S* for which the research is done can come apart (Chapter 4, Section 7). Novelty in the working characterization is indexed to *S*, and what *A* does not know need not coincide with what *S* does not know. In the reports examined, the grounds for judging what was unknown to the machine and to *S* were divided between records of inputs or training and records of prior work, and the judgments were revised through reclassification or correction after publication. In light of Grimm's account, on which curiosity is directed at subject matters and at particular types of question, the unknowns a curious researcher is drawn to need not be the unknowns *S* most wants resolved. Applying Kitcher's account, which does not discuss artificial systems, where *A* is a machine and *S* is a human society not even a shared source of what is important can be presupposed. If curiosity is understood as a desire to know, then where *A* is an artificial system and *S* is a human society, it does not immediately follow that curiosity satisfied when the system itself comes to know is necessary for research in the sense of the working characterization, and, if the project's aim includes entering *S*'s epistemic situation, it is not self-evident either that such curiosity is sufficient for its success. What bears on research is the work of detecting unresolved matters and forming questions and the work of distributing exploration across a community, and what that work tracks relative to what is unknown to *S*. The state of curiosity returns as a question in its own right on accounts that make interrogative attitudes a condition of inquiry, where an artificial system's intellectual life valued for its own sake is aimed at, and where how well research goes depends on states of the one doing it that correspond to motivation.

Chapter 4 read empirical studies reporting that machines produced questions or research ideas in light of these distinctions (Chapter 4, Section 8). In the studies in which humans supplied the higher-level question, this was because the study design or the way the system is used required it, and nothing follows from their results about whether machines raised, or cannot raise, higher-level questions. Even in a design that supplies no question, the data, the tools, and the conditions for accepting claims were set in advance, and what is set in advance changes place from study to study. Work on the side of questions included specifying a foil in the input and changing metrics or evaluation criteria during execution and verification, and the grounds for judging whether that work was attributed to humans or to agents lay in records that track changes. The rating items did not check whether a question's presuppositions are held in *S* or what counts as an answer, and agreement in ratings varied with whether raters saw ideas alone or the results of execution, and with whether humans or machines judged. Checking such capacities would require comparisons that do not fix the higher-level question, or procedures that record and rate work on the side of questions separately from work on the side of candidates. With the former, however, it becomes hard to separate whether one is comparing research ability or differences in topic preference, and even in a design that supplies no question, what it takes for a claim to be accepted is set elsewhere. And when perspectives on problems, inquiry, and social appraisal are applied to the same activity, their objects and units of analysis do not simply coincide with *A*, the bearer of the achievement, or *S*. The working characterization can add to their descriptions the question of for whom the achievement is

new (Chapter 4, Section 9).

Seen from the side of the relations among candidate answers, representations, and means, even when a question's answer conditions are fixed, candidates that could meet them are not necessarily at hand from the start. Candidates can become available through the construction of models, experimental systems, representations, instruments, and data. In Nersessian's account, there were no means to investigate the candidate that shear stress affects cells, and building and rebuilding a device whose flow conditions could be varied advanced the formulation of the problem itself. The Liquid Tensor Experiment, in which the understanding of the proof itself changed in the course of formalization, is not evidence about mathematical practice in general, but it suggests that, for artificial systems that generate proofs and check them in proof assistants too, verification can be not only a downstream stage but a site where such changes occur. In the case of a historical database assembled through editors' choices, an artificial system that queries it inherits those choices unless it revises them, and inherits them tacitly unless it makes them explicit. A description of revising credence over a fixed set of propositions captures much of what happens after candidates have become available, but does not by itself show the work that made that set available. Whether a belief-centered theory taken broadly enough could describe that too remains open (Chapter 5, Section 3). To Stanford's argument that, in past science, equally well-confirmed rival hypotheses have repeatedly turned up later, one possible response is that computational science, by exploring the space of possibilities more widely, changes this situation. Examining this response, Curtis-Trudel and colleagues hold that such exploration relies on a prior characterization of the space, and that unless there is reason to think that characterization roughly exhaustive, the problem of rival hypotheses not yet conceived remains. The three set artificial intelligence aside, but applying their argument, whether artificial inquirers push back this limit turns not on the breadth of search but on whether they can make candidates available (Chapter 5, Section 3).

The judgment of which candidates to pursue need not be the same as the judgment of which are true (Chapter 5, Section 4.2). There is a view that sees preliminary and rigorous appraisal as differing in kind, drawing on accounts such as Laudan's and Duerr and Fischer's that separate pursuit from acceptance, and a view that sees them as differing in degree, and Chapter 5 did not decide between them. On either view, the value of pursuit is not the degree of credence, and raising credence alone does not yield knowledge. *A*'s revision of credence does not by itself make a candidate knowledge for *S*. Further, on the proposal Wenmackers and Romeijn developed for handling credence when new theories are taken into account, a candidate's newly becoming available can lower credence in earlier answers even without unfavorable evidence, so an assessment that counts progress only by rises in credence risks counting the work of forming alternative candidates as setback. On either view, a system that identifies being worth pursuing with being likely true and pursues only candidates it already regards as probable may miss candidates that can reveal methods, or, on Nyrup's model, candidates worth pursuing precisely because they are likely false and easily shown to be so, and may exclude productive risks. Conversely, a system that ignores appraisal may spend without limit on arbitrary novelty. The implications for machines also vary with for whom a matter is uncertain. If *A* is a machine and a matter that is unsettled for *S* can be answered from what *A* already has, the very practice of an inquirer who does not know the answer moving toward what seems probable may be unnecessary, and what is then asked is whether *S* can rely on the answer. If the matter is unknown to *A* too, the description bears on it, but it is not settled whether the system must have attitudes like credence.

For operations bearing on candidates and results as well, the name of an operation does not determine what has been established (Chapter 5, Section 5). On Parker's account as Winsberg and Alvarado report it, simulation, through its material computing device, stands on a par with experiment, and whether an inference is licensed depends on relevant similarity, so for systems too one can ask not whether they perform named operations but what function each operation served within the inquiry. The ways in which instruments, materials, records, collaborators, and the like take part in inquiry are asked about case by case, divided into causal contribution, an enabling role as a precondition for an operation's taking place, and, on accounts that admit it, a constitutive role as part of the task or achievement itself (Chapter 5, Section 6.1). That tasks and work were distributed in the laboratories and teams studied is richly shown, but whether cognition or knowledge itself is distributed is not settled (Chapter 5, Section 6.2), and more records do not by themselves amount to more knowledge (Chapter 5, Section 6.3). Chapter 5 sorted the presuppositions that have worked in human research into presuppositions about humans as realizers, presuppositions about the present organization of science, conditions of inquiry, and relations with outside parties (Chapter 5, Section 6.5). If *A* is a machine and *S* a human society, roles that depended on humans as realizers become questions of organization, and human constants remain as conditions on the side of *S*, which reads, checks,

and relies on results. If both *A* and *S* are machines, what corresponds to what came from the inquirer's side, such as bodily experience and the withholding of roles, is in question. Dividing work among copies of the same make whose background assumptions and routes of error have not come apart does not divide viewpoints. Further, by Takahashi's distinction, time inherent in the object remains even when human delays are removed, and speeding up research in a substitute system adds the work of inferring from that system to the object. Such acceleration can also put pressure on the choice of what to investigate (Chapter 5, Sections 6.5 and 7).

Seen from the side of the relation between results and what they bear on, results are already obtained within the work of forming questions and candidates, but what they bear on cannot be told from the results alone. Chapter 6 considered this. A result comes not from hypotheses alone but from a combination of many mutually dependent elements, including instruments, auxiliary assumptions, code, and evaluation mechanisms. Chapter 6 called such a collection of elements, together with how producing and interpreting a result depends on them, a dependency arrangement. Neither success nor failure identifies its own cause, object, or appropriate revision. After a failure, therefore, the work of determining which commitment to doubt, examine, or change next is needed, and Chapter 6 called this localization. Localization is not dividing a failure among contributors but determining where to intervene next. This implies, however, neither unrestricted holism, nor that everything is equally revisable, nor that rational diagnosis is impossible. In the arguments of Sober and Elgin, auxiliary assumptions can sometimes be tested independently, and independently supported commitments are less likely to be the location of error (Chapter 6, Sections 1 and 2.2). Judgments of localization can depend not only on background knowledge and independent checks but also on conviction, expectations, and views about what is essential. And even while the results leave open where to locate the failure, such judgments can shape the subsequent course of search (Chapter 6, Sections 2.3 and 4.1). Localization need not be explicit reflection; it can be distributed across participants, time, and forms of record, and can be contested between parties. That its forms are diverse leaves room, before treating localization as a capacity that it suffices to implement as one component or one procedure, to ask in which configuration, by whom, and in what form localization is required (Chapter 6, Section 2.4). For success too, in the cases Lyons cites, a component's having contributed to success does not by itself settle that the component was supported, and attributing success can also be a judgment that decides the course of search (Chapter 6, Section 3.2).

When a machine does research, the arrangement includes code the system wrote itself, tools it called, outputs of other systems, and evaluation mechanisms, and even those who assembled it may not be able to survey the whole. The chance that some part of the rest of the arrangement is wrong is then not necessarily smaller than the chance that the focal hypothesis is wrong. A system that takes results straight as verdicts on hypotheses is not doing the work of localization, and when trials are automated in large numbers, undifferentiated verdicts can pile up (Chapter 6, Section 2.1). A system that never blames the focal hypothesis presupposes the location of error just as much as one that always does. A system that neither represents nor proceduralizes the role of protection cannot distinguish protecting a hypothesis from failing to notice an error (Chapter 6, Section 4.2). A rule that tries changes one at a time, keeps a change only when the immediate evaluation improves, does not retain discarded changes, and does not try combinations works as if it attributed every difference in evaluation to the immediately preceding change, and makes it hard to find changes that work only in combination with others (Chapter 6, Section 4.2). What is counted as failure is not uniform either: a precondition's not being met, an implementation's being misdescribed, and a focal test's returning a negative result are different things, and the number of trials run in parallel is not the number of independent checks on the elements they share (Chapter 6, Section 2.5). What makes a trial diagnostic is not number but its connection to earlier results, to remaining candidate locations, and to hypotheses about parts that cannot be observed, and the cheaper trials become, the harder this may be to see (Chapter 6, Section 4.3). Whether the location of an error can be traced afterward can depend on how the arrangement was recorded beforehand. Collingwood did not discuss localization in research, but applying his analysis to localization in the sense of where to intervene next, localization can be relative to what an inquirer can change and what it seeks to bring about or prevent, so an inquirer with means of intervention different from humans' may localize elsewhere (Chapter 6, Section 2.4).

Passing, too, does not by itself say what it has established (Chapter 6, Section 3). What can be asked about a pass divides into which part of the question it bears on, how firmly it settles it, whether it depends on something else, and which commitments it supports, and a report of passing may state none of these. Passing can be treated as a reason to stop looking, and misreadings of passes can go unnoticed longer than misreadings of failures. Since machines can write, choose, and rewrite their own checks, if the same process of making produces both hypotheses and checks, the two can share routes of

error. On the other hand, a machine inquirer can also have checks written by processes of a different make, or tie them to measurements or formal checkers it does not control. Agreement among many instances of the same base may add only limited assurance if they share routes of error (Chapter 6, Sections 3.2 and 3.3). How long it is rational to protect an idea is not settled by a single result either (Chapter 6, Sections 4.1 and 4.2). Among what bears on that judgment, the reasons of Kuhn and Lakatos that concern how the merits of a line of work come to be visible would, applied though neither discussed artificial systems, equally support a machine's protecting its own framework. By contrast, the vanity that delays judgment on Duhem's account and the promise to a supervisor in Hinton's recollection concern human motives and careers, and need not apply in the same form. What is common is that finite inquirers must choose before there is sufficient evidence. Carrying over to machines the communal reason that disagreement spreads risk depends in part on whether rival lines are kept alive elsewhere, and copies of the same make can keep rival lines alive, but so long as they share their assumptions and routes of error, their judgments of when to give the lines up can fail together. Protection can be assessed only through the history of revisions, and what it is assessed by also differs with standpoint.

Standards themselves can also be revised for reasons from within inquiry, under constraints (Chapter 6, Section 5). Changing a metric so that poor performance no longer shows, however, is moving the target, and if such a change is made silently, the stated reasons that would let others tell revision from merely moving the target are missing. What is missing is not continuity with the earlier standard. What separates redefinition from moving the target differs from account to account. On Chang's account, mutual revision avoids being arbitrary because it remains answerable to stubbornly resistant phenomena and can fail, reaching a dead end from which inquiry has to start again. Applying what Tal's study of Coordinated Universal Time can be read as implying, a system that redefines its own evaluation metrics can be distinguished from one that merely moves the target insofar as it states the redefinition as a rule, keeps responding to discrepancies among independent realizations of what the metric is meant to measure, keeps unfavorable outcomes able to challenge the redefined metric, and keeps open to scrutiny the very reasons for reinterpreting outcomes. Applying Cronbach and Meehl's discipline by analogy, it would further be needed to check the redefined metric against data not used in redefining it. On every account, records of earlier standards provide a handle. The faster and more frequent redefinition is, the harder silent changes may become to see, but the same speed may also make small, fast corrections possible (Chapter 6, Section 5.3).

Who carries out this work, and to whom it is visible, changes with the combination of *A* and *S*. If *A* is a machine and *S* a human society, members of *S* will read the attributions, the meaning of passes, the ideas discarded, and the standards redefined that a system's report contains in light of records and other grounds. If both *A* and *S* are machines, checking these can itself be done within an artificial community. And if, as on views that take evidence as a three-place relation relative to background knowledge, what a result gives reason to revise is relative to background knowledge, what gives an agent reason to revise need not be what gives members of *S* reason to revise. It is not presupposed here that the agent is a subject that believes or knows, and relativity to background knowledge is not relativity to consensus (Chapter 6, Sections 1 and 6). Repeating an operation and determining why that operation has ceased to bear on the question are, however, different kinds of work, as are applying a standard and examining and revising it. Chapter 6 made neither a condition of all research activity, and held that neither need lie within a single agent. Records support such examination but are not themselves grounds (Chapter 6, Section 6). In this way the questions of process lead into the questions of Chapters 7–8: who is doing the research, and whose its achievements become.

Who is doing the research, and whose do its achievements become?

Chapters 7 and 8 considered the one who does research and the achievements research brings about. Drawing on philosophical and engineering discussions of agency, autonomy, achievement, and credit, Chapter 7 considered to what artificial or hybrid configurations, at what boundary, and on what grounds contribution to research, conduct of research, research capacity, and the standing of a researcher can be attributed, and which respects of what is attributed remain organized from outside. Drawing on social epistemology, the philosophy of science, discussions of grounding and of responsiveness to the world, and discussions of AI alignment, among others, Chapter 8 considered whose knowledge the achievements made by research become. The axis running through the two chapters is that three things, *A* who does the research, the bearer in whom the achievement is realized, and *S* for which the achievement is new, can come apart when a machine does research. Chapter 7 treated what is to be attributed on the side of *A* as a question distinct both from the questions of knowledge and understanding (Chapters 2–3) and from the questions of process (Chapters 4–6). Chapter 8 asked whether research that humans began goes on producing knowledge for a human community when it is carried by systems unlike humans, and considered whose knowledge is produced by dividing the cases according to whether *S* is a human society, a mixed society, or an artificial society.

Chapter 7 distinguished four attributions: having contributed to research, having conducted a research activity at a stated boundary, having research capacity over a stated range, and having a standing on which calling the unit a researcher is explanatorily useful. Later items usually require more evidence, but they are not degrees of a single quantity, and none follows from the one before (Chapter 7, Section 1). Attribution is not limited to these four, but autonomy was treated not as a further item alongside them, but as qualifying how what is attributed is organized in relation to what lies outside the boundary. These attributions also cross evaluations such as whether something is research, whether it reached its target, and whether it is important. Research activity that failed to reach its target can be attributed, and true outputs can come from sources that are not doing research. Having proceeded without human hands is not by itself research; conversely, a unit assigned its task from outside and heavily supported does not fail to perform research merely because its autonomy is limited. Originating its own aims or questions is not among the conditions for performing research in the sense of the working characterization; their external origin bears on autonomy instead (Chapter 7, Sections 1 and 4.3; Chapter 4, Section 1).

Where conduct lies is settled by where the organization that makes operations parts of one inquiry lies, that is, where the work of initiating, monitoring, redirecting, and recovering an inquiry lies. It is not settled by where named operations take place, or by whether a system diagram has a "question generation" box. Organization does not require a central executive and can be realized in relations among agents, artifacts, records, environments, and institutions (Chapter 7, Section 2). Against the objection that, since what a system is for comes from its designers, its inquiry is the designers' research, Chapter 7 separated the source of function from where the organization of inquiry lies. Even if a function comes from designers, if the system itself notices the limits of a representation and revises the space of candidates, much of the inquiry can be attributed to that system. Conversely, if the work of redirecting is supplied from outside, the organization of redirection lies outside, however goal-directed the system appears. Even on the view of Clark and Khosrowi, who attribute discovery to the collective of contributors and distribute credit according to contribution, distributed contribution does not automatically establish distributed knowing (on Giere's account, Chapter 2, Section 6). Boundaries are argued claim by claim, so that the choice of boundary alone does not make a claim true. The same models, retrievers, instruments, and people can, in different episodes, be tools used by a human agent, parts of a hybrid agent, or resources coordinated by an artificial agent. What changes then is not only the name but where initiating, monitoring, redirecting, and recovering are organized. The attribution of conduct is not a property of a type of model but a relational and organizational attribution.

To say that a system has research capacity is to attribute a disposition to carry out research over some range of conditions, and the claim can be assessed only once that range is stated. Marks such as reliability across relevant variation, responsiveness to changing evidence, integration of research functions, and revision after failure help separate episodes from capacities, but they are not necessary conditions for every research episode. A capacity within a prepared computational environment is real, but it does not by itself establish a capacity to rebuild the environment, redesign instruments, or move into a field with different evidential practices. Because systems may be updated and may also modify themselves, a capacity is attributed for the version and period over which its evidence was gathered. A capacity where others maintain the

instruments or computing environment is, moreover, a capacity only for as long as that maintenance continues; if the maintenance stops, the conditions under which it was attributed can cease to hold without any change in the system. Where maintainers also choose which instruments, data, or computing resources to make available, dependence that merely enables the capacity can shift into dependence that supplies part of the organization. Even the same predictive achievement locates the capacity to reorganize inquiry differently according to whether humans diagnose failures and recast the question each time conditions change, or there is evidence that the configuration itself does so (Chapter 7, Section 3.1).

The book has used the word "researcher" not as a definition but in the following way: if calling a unit a researcher lets one anticipate what a list of its capacities does not predict, for example that when a task stalls it will not stop and wait for instructions but will keep hold of the question, assemble another method, and itself begin the work this requires, then calling it so has explanatory value. This usage inherits van Lier's form of justifying a term by its explanatory power and Dennett's tie between a description and the predictions it supports. It presupposes, however, organization located at a boundary and capacities shown over a stated range, and explanatory value is weighed in addition to those attributions, not in place of them. Without evidence that the integration works in untested situations, the name is no more than a name for a bundle of capacities, and it is relative to whom, and for what, it explains (Chapter 7, Section 3.2). Some positions impose further conditions on the standing of a researcher. If one supposes that the standing includes being able to make discoveries, the standing is drawn toward that of a discoverer, and applying Bergamaschi Ganapini's conditions for discovery puts competence in the relevant field, and reflective insight into that competence, at issue. Bergamaschi Ganapini holds that current deep-learning systems do not yet appear to meet those conditions. If the standing presupposes minimal agency in the sense of Barandiaran and colleagues, then, of that account's three conditions, two in particular, which mark its difference from the usage above, are at issue: whether the system defines its own identity and thereby distinguishes itself from its environment, and whether it regulates its activity by norms generated and maintained from within, or internalized, rather than norms applied by observers or designers. On that account, a configuration that only optimizes an objective fixed from outside does not meet the condition of normativity, whatever the explanatory value of calling it a researcher. The three authors do not discuss laboratories, but applying the two usages to the whole of the self-driving laboratory that van Lier discusses shows that they do not merely give the same object different scores; they also divide over what counts as one agent.

The latter half of the section's title, whose the achievement becomes, is also taken up by Chapter 7 from one side. The book's "epistemic achievement," however, names the epistemic good that inquiry aims at, and differs from achievement in the sense that includes credit, as when it is disputed whose credit a success is (Chapter 7, Section 3.3). The dispute between Kieval, who holds that AlphaGo deserves credit for its win, and Vacek, who holds that an achievement would make its bearer praiseworthy and that AI cannot be praiseworthy, concerns the latter. Chapter 7 divided the answers into four tiers: true output, reliable process, success attributable to competence, and a praiseworthy agent. The first three roughly follow Sosa's distinction of accurate, adroit, and apt, and the fourth corresponds to Vacek's demand. Keeping them apart prevents a correct output from being read straight off as the system's credit. Moreover, the ground that Vacek inherits from Hakli and Mäkelä, that the system's abilities result from engineering, rests in their argument on the system's being unable itself to abandon goals built into it, and so would not hold of systems able to revise their own final goals. None of these authors considers such systems, and applying the strong premise to them would require a further argument (Chapter 7, Section 3.2). Discovery and creativity can also be treated as terms of praise, but Duede and Friedman, pointing to Bell Burnell's having discovered pulsars while unable to explain the significance of the signal, oppose imposing conditions of competence and insight on discovery. On authorship, there is the account of Huebner, Kukla, and Winsberg, which regards an author as one who can vouch for, defend, and retract a result. At de Prony's computing bureau, some ninety human computers carried out calculations as mere instruments, so that account can be read as tying standing as an author not to membership in a kind but to a role within such practices (Chapter 7, Section 3.3). The working characterization neither establishes nor refutes the conditions of these thick standings.

Autonomy is compared by specifying in what respect, and with respect to which external parties, it holds. Chapter 7 called this an indexed comparison, and deliberately left loose the statement that, for a specified A, autonomy in a respect is, other things equal, higher the less parties outside the chosen boundary direct, substitute for, authorize, approve, or repair the organization attributed to A in that respect (Chapter 7, Section 4.1). Dependences that merely enable a capacity, such as language, records, and calibrated instruments, are not counted, and counting a dependence does not mean that it is

undesirable or illegitimate. The comparison is formally symmetric and does not specify which party is human. Human researchers who rely on artificial systems are described by the same indexing. Relations of dependence were divided, where their difference changes the claim, into at least three: enabling dependence; directive or substitutive dependence, in which others supply part of selection, evaluation, correction, or continuation; and governance constraints, which restrict action and reserve authority without necessarily supplying epistemic work (Chapter 7, Section 4.2). Governance constraints can reduce autonomy without showing any lack of capacity, and conversely permission does not establish capacity. In King's proposal, which places question generation and selection in the system while leaving to humans the authority to set aims, and in the proposal of Bengio and colleagues, which removes goal-directedness and leaves the execution of experiments to humans, the dependence is not imposed by a lack of capacity but chosen (Chapter 7, Section 4.3). As the example of Ochiai and colleagues shows, who propose self-maintenance, moving part of the "care" that humans provide to the laboratory side, as a design concept while still asking people for replenishment and the like, which dependence the same human work counts as is settled by which capacity is being attributed. That there is "no human in the loop" is also evidence about that interval, not proof of autonomy in other respects (Chapter 7, Section 4.2).

The more the elements of inquiry are delegated without being fixed, the wider the range of practical demands inquiry can place on the configuration (Chapter 7, Section 4.4). When more of the aims, questions, representations, data, methods, and instruments are delegated, part of the novelty comes not from the environment but from the configuration's own choices, and the demands can extend to access, permission, resources, coordination, and recovery. In the case of Fei-Fei Li's group taking computer-vision research into hospitals, what was required included access to hospitals, accompanying shifts, data collection under institutional review, and changing the processing arrangement in response to nurses' concerns, and it was people who met these demands. Some of these demands, such as the time it took to form the team, arose from its being a human team; others, such as data that did not yet exist and permission for records, would arise for any inquirer taking up this question. Automating experiments after the apparatus has been prepared is a substantial achievement, but it is not evidence that the capacity to meet such surrounding demands has also moved to the artificial system. A claim of greater autonomy therefore takes on showing whether open demands were met within the unit, met by others, closed in advance, or answered by changing direction or stopping for reasons. This is not, however, a feasibility claim: preparing an environment that closes the range of demands in advance is legitimate design, and narrow claims do not bear this burden. Raising autonomy also need not reduce external dependence in one direction. Seeing what is lacking and asking for help can be an exercise of capacity. Actually coupling an evaluator's data or path of judgment with the side of production can raise autonomy while weakening the independent route by which errors are found (merely recounting the evaluator as inside the boundary leaves that route unchanged). A more autonomous configuration may therefore negotiate with more others and, for the certification of its results, need more external checking rather than less (Chapter 7, Sections 4.5 and 5). Autonomy is not a monotonic measure of research quality, and capacity, observed exercise, permission, authority, epistemic quality, importance, safety, responsibility, and legitimacy are not inferred from one another without further argument.

Autonomy with respect to standards is a further question. Chapter 7 called autonomy with respect to the standards that organize epistemic activity epistemic self-government, and left undetermined the source and kind of normative authority those standards have. When such a claim is made, capacities such as applying, choosing, constructing and adapting, revising, and giving reasons for standards may need to be distinguished, and one may be shown without the others. Autonomy in a particular respect does not establish epistemic self-government (Chapter 7, Section 4.3). That standards come from outside is a fact about their source, an aspect separate from whether they can be represented as revisable and held, revised, or rejected for reasons. Here the combination of *A* and *S* makes a difference. Under the working characterization, research directed at epistemic achievements new to a human society does not require an artificial agent to invent an epistemology. By contrast, for research by an artificial society for an artificial society, permanent human control over what counts as evidence, error, or successful inquiry would show a strong form of external governance of that society's epistemic standards. Applying Longino's account, however, standards' having an external origin and being governed from outside does not by itself deny that the society is a knowing community.

Attributing conduct to a machine does not entail three things: that the machine itself knows the result, that the result has entered the epistemic situation of the society that commissioned the work or for which the result was sought, and that the inquiry was directed at what those societies did not know (Chapter 7, Section 5). Taking up these three, Chapter 8 distinguished *A* who does the research from the bearer in whom the achievement is realized and from *S* for which the

achievement is new. The bearer is not another coordinate alongside *A* and *S*, but is asked about where success is at issue. It can also happen that an achievement new to *S* is already realized in a bearer outside *S* but has not yet entered *S*'s epistemic situation. Chapter 8 divided the cases by *S* into three: a human society, a mixed society in which human and artificial members are mixed, and an artificial society. This division is exhaustive only at this grain, and "mixed" is used only of *S*, distinguished from the organization that bears action or knowledge being hybrid (Chapter 8, Section 1).

If *S* is a human society, an artificial agent can refine questions, generate candidates, and assess evidence without literally believing or knowing. Alvarado's account, which regards AI first of all as an epistemic technology, leaves room for this. On Hauswald's account, which recasts deferring to authority as taking the appropriate facts as indicators of truth, a system can be an epistemic authority without beliefs. Hauswald's account gives one route by which results come to be held by members of *S* even if the system is not a knower. The two positions show that, even on theories that reserve literal knowledge for other bearers, artifacts can occupy socially significant epistemic roles (Chapter 8, Section 1.1). What a claim that a result has entered *S* needs to show is the route by which human members or their organizations come to hold the result, and this claim does not require the machine to be a knower. Another relation can require it, however, as on accounts that take a system's reports as assertions and require for assertion that the speaker, or the group on whose behalf the speaker speaks, knows. Conversely, if the machine itself is claimed to know and the account of belief one adopts allows this, the achievement can be realized in the machine before it enters *S*. That claim, however, bears the burdens concerning engagement with the world that we will see below.

In a mixed society, what is new, and who can assess whose results, need not be the same for human and artificial members, and who speaks for *S* is also in question. Claims therefore need to state which members they extend to and how holding is distributed between human and artificial members. Van Fraassen and Dellsén do not discuss artificial members, but applying their arguments about admitting new members, on some accounts of holding admission can increase *S*'s knowledge, so that increases due to research and increases due to changes in the range of members cannot be distinguished without first fixing *S*'s boundary (Chapter 8, Section 1.2). Further, Collins discusses values rather than research or vocabulary, but reading his second problem, that of machines socializing us, together with Taniguchi's observation that generative AI is changing human language use suggests that vocabularies and standards formed in exchanges with machines can enter human society without any decision to admit artificial members. The question then becomes whose experiences, among which members, the shared vocabularies and standards integrate.

For an artificial society, two questions were separated: what is needed for that society to serve as *S* as a knowing community at all, and what is needed for a result to become that society's knowledge. The former varies with the account of belief one adopts, the latter with the account of holding (Chapter 8, Section 1.3). Longino's account, which sees sociality in interaction rather than sharing and identifies a community by diversity and by public standards appealed to in criticism, is not about artificial agents. But if its conditions are required as they stand, merely lining up several instances of a model, assigning them roles, and taking a majority vote does not make a knowing community. Applying to instances of a model List's functional account, on which a group's attitude on a proposition need not be a function of its members' attitudes on that proposition alone, also suggests that claim-by-claim majority voting does not make the instances one agent. Carrying Longino's account over to artificial organizations, standards given from outside do not by themselves forfeit standing; what matters is whether external control prevents criticism from being effectively taken up or places the standards themselves beyond criticism. These sortings are theory-dependent diagnoses, neither a new checklist nor a showing that artificial epistemic societies already exist. Research by an artificial society is not a precondition of research for a human society, but neither can it be reduced to an enlargement of it.

Whose achievement it becomes also bears on for whom a result is evidence about the world in the first place. Chapter 8, Section 2 asked about what Chapter 6 had presupposed, namely that checks can be evidence about the world, and saw that these presuppositions are required on the side of those who receive a check as evidence rather than of those who perform it, and that their bearer need not coincide with *A* and is located differently according to whether *S* is human, mixed, or artificial. Checks count as evidence only on presuppositions they do not themselves check. Applying Wittgenstein's image of hinges, such presuppositions remain not because time or ability to examine them is lacking but because they are conditions of checking, and they do not disappear however great the capacity to check becomes; local presuppositions, however, can be checked and revised by other routes (Chapter 8, Section 2.1). In machines, inductive biases, distributional assumptions, and

the models by which automatic evaluators are taken to be stable can functionally correspond to such presuppositions. Applying Longino's account as Haely reports it, merely rerunning one base system does not show that a disputed observation has been tested by differently situated perceivers. Moreover, inductive biases are structures set by how a system is made and replaceable by remaking it, and an artificial inquirer may itself revise its presuppositions or rules, so who may revise them, and who bears responsibility, become questions.

Whether a machine's states are about targets in the world, and whether its commitments remain open to correction by those targets, were asked separately as questions of grounding and of responsiveness to the world. These two questions and that of what checking presupposes, however, neither nest within one another nor are independent of one another. Chapter 8, Section 2.2 saw that grounding is not one relation. Among relations such as those that fix what a state refers to, sensorimotor relations, and communicative relations, which supports which differs from account to account, but success in one relation does not by itself establish the others. Bouchacourt and Baroni, for example, conclude that the signals of artificial agents that succeeded in coordinating tracked low-level properties rather than kinds of image. Coordinative success is no evidence that a shared signal is about what it appears to an observer to name. An inquirer's primitive organization is shaped by its history of formation, but on the accounts considered here, a natural pedigree can be a genealogy telling where a capacity came from without being a grounding of everything that capacity produces. Responsiveness, whether commitments remain constrained so that they would differ if the target were different and remain correctable, then remains as a question of a different kind from any relation of grounding (Chapter 8, Section 2.3). A claim that the machine itself knows bears burdens in three places, the presuppositions of checking, grounding, and responsiveness; a claim that a result has entered a human *S* can have those burdens met on the human side. On the accounts considered here, success in any one of them, or usefulness for prediction, does not by itself meet the other burdens. What this analysis yields is not a demonstration but a possibility, namely that artificial organizations can differ from humans while remaining constrained by nature.

A machine that does research can be an intelligent system unlike humans in body, cognitive mechanisms, and how it is made. What it counts as unknown, as worth pursuing, and as a good reason can therefore diverge from members of *S* (Chapter 8, Section 3). This concern needs only three premises: that such judgments are relative to a standpoint, that autonomy moves choice to the side of *A*, and that *A*'s organization can differ from humans'. It presupposes neither seeking power or resources nor hostility to humans. Divergence can arise not only from autonomy's moving choice to the machine's side, but also from the achievement's being realized only on the machine's side, and from *S*'s standards moving together with the machine (Chapter 8, Section 3.2). The relation between autonomy and relevance is not monotonic, but as the range of self-direction widens, the range of possible mismatch can widen too. That research stays relevant to *S* is not secured by an instruction to "carry out research" or by naming beneficiaries; what is called for is showing how relevance is maintained and assessed. Applying the arguments of de Blanc and Soares to research, inquiry can revise the very concepts in which a commission was written, so divergence can open not only when inquiry fails but also when it succeeds. This is not peculiar to machines, and a check that treats every revision of aims as deviation, even where the revision has really changed what was commissioned, can count success as failure. Extending Russell's argument to research, where *S*'s standards move together with the machine, agreement between them does not by itself show that the machine still serves what *S* would have wanted had it not moved with the machine, and where evaluative competence is not maintained and passed on among members, *S* may not be in a position to tell. Applying the asymmetry Yudkowsky sees between beliefs and utilities, the world can correct what is believed but does not fix what is worth finding, so responsiveness to targets is not by itself evidence of relevance to *S*; how sharply the two come apart, however, depends on the account of evidence.

Read in the line of argument that locates the difficulty of alignment in the alienness of systems, such concerns about mismatch are structurally similar to the AI alignment problem (Chapter 8, Sections 3.2 and 3.3). But research also has a reason of its own, which holds whoever does the research and applies to human inquirers too. Since what is unknown to *S* is not something *S* intends or prefers, agreement with an answer *S* wanted cannot be the criterion of a result's correctness. What settles correctness is not any party's intention but how things are, and aligning with instructions, intentions, or preferences secures at most that research pursues what *S* wants pursued. Even complete alignment with evaluators leaves open whether a system is responsive to its target. Chapter 8 listed five relations that can break down: that the target of optimization tracks epistemic value; that successful outputs constitute or support achievements new to *S* and enter *S*; that aims, risks, and authority are acceptable to affected parties; responsiveness to targets; and coordination between communities with different organizations. The last two have no counterpart in the structure compared there, which relates a single agent to those it

serves. Since these relations can diverge, a single alignment score would hide which of the goods, such as truth, grounding, understanding, usefulness, safety, and legitimacy, it secured. Whether a given divergence is drift, error, or gain can be judged only from a named standpoint: the present *S*, an *S* that may change, or, where correctness is at issue, the world. Where the world's verdicts reach the inquirer only through the present *S*'s procedures for judging claims, whether the world works as a separate standpoint depends on whether those procedures can be revised by evidence.

What happens when research does not align depends first on the standpoint from which one judges and further on which relations break down. What happens is not a single loss but a combination of relations that hold and relations that break (Chapter 8, Section 3.3). Even if relevance to *S* breaks down, on accounts on which targets determine correction, and, where the targets' verdicts reach the inquirer only through the procedures by which *S* judges claims, provided those procedures can be revised by evidence, responsiveness to targets can remain as an anchor that does not depend on *S*'s consent. *S* might then still gain knowledge or understanding from what a different kind of inquirer has reached. Knowing that a machine predicts *p* is not knowing *p*. The former is knowledge about the system, and can be had even if the prediction is wrong. Reliability-centered accounts locate the route from there to knowledge of the target in the actual reliability of the predictions; other accounts locate it in understanding, criticism, assurance, and the like (Chapter 8, Section 3.4). Applying Kitcher's point that direct calibration reaches only as far as one can form independent opinions, the more there is to learn from machines, the more the calibration-based grounds for relying without translation may fail to reach. Reliance beyond that needs grounds other than calibration. For a prediction that survives independent checking but whose reasons are not explained and behind which no party stands, it can be that reliability accounts allow it to be known so long as it is in fact reliable in the domain of reliance, understanding accounts yield no understanding, social-justification accounts do not accept it, and on assurance views it does not become knowledge for the relying side by testimony. What separates the four is not whether the prediction succeeded but which achievement, and by which route, they count as having been realized on the receiving side. Where cases can be shown and success can be scored, even if only by agreement with the machine's own choices, there is also a route by which people learn some machine-native kinds without translation. The ground for this, however, is a single trial in chess that its authors present as a proof of concept, and it does not show what that route would require where the kinds or features cannot be presented, or where *S* cannot itself score the results. Being alien is not, in any case, a property of outputs. Applying the later Kuhn's account, if only internal features differ, what results is opacity; but if the very division into kinds cuts across, then on Kuhn's premises it is not guaranteed in advance that humans can learn it (Chapter 8, Section 3.1).

Chapter 8 gathered the distinction by *S* into two horizons (Chapter 8, Section 4). On the horizon of human society, what is asked is how research brings about new knowledge, whether research stays directed at what is unknown to *S* even as choices move to the machine's side, whether *S* keeps its evaluative competence and can tell whether standards have moved, and how knowledge enters *S*, and these are to be asked while keeping visible the extent of dependence, ignorance, and human involvement. Research for human society needs neither machine consciousness, nor a fully independent epistemology, nor an artificial society, and can proceed through partial capacities and hybrid organizations. On the horizon of artificial society, human society is not *S* but a party affected by that research, and what is in question for human society is not the success of research but acceptability and legitimacy (Chapter 8, Section 3.2). There, what is asked is whether that society can form, maintain, criticize, and transform its own epistemic world, and stronger questions arise about belief and social knowledge, responsiveness to the world, self-setting of aims and revision of standards, and coordination with human society. Chapter 8 does not show that such a society is possible or that it is impossible, and the distinction of the two horizons is neither a schedule that postpones the fundamental questions nor a claim that independence is good in itself. A mixed society fits wholly within neither horizon. For its human members the questions of the horizon of human society arise, and for its artificial members, insofar as they count as bearers to whom knowledge is attributed, the questions of belief and holding. If members form separate communities, the question of coordination is added; if they share one symbol system, the question of whose experiences it integrates; and in every case, who speaks for *S* is in question.

Part IV separated the choice of where inquiry is performed from the choice of whose epistemic life is extended. Neither commissioning nor human control guarantees that research stays directed at what is unknown to *S*; that is asked of each configuration. The epistemic success of research for *S* and the legitimacy of setting its direction are, moreover, separate judgments, and the latter is asked also of affected parties outside *S*. Which of what is unknown to *S* research pursues, how its results come to hold good for *S*, and by whose authority its direction is set concern relations that extend beyond any single agent. Part V asks which of those relations can be placed inside an artificial configuration, and what changes when they are.

Reorganizing the relations of research

Human research has not proceeded through the people doing research alone. Where effort goes has been decided, for example, by which problems researchers choose, which projects funders select, and where heads of laboratories direct people and equipment. Pre-publication peer review, replication, the publication of results, and their subsequent use have made results something others can rely on. And funders, self-governing research communities, regulators, publics, and others have set and revised the rules of such arrangements. When AI does research, an artificial configuration may come to do some or all of these things within itself. What then happens to research? Part V considered this question not from the forms of institutions but from the side of the relations those institutions have carried: allocation, which arrangements such as research funding have realized; criticism and certification, which evaluations such as peer review have realized; and authority, the question of who may decide and revise the conditions of both. The question was which of the relations by which research is directed, certified, and governed for *S* can be placed inside an artificial configuration, and what changes when they are (Chapter 9, Section 1). Its first half is a means for depicting its second; it is a question distinct from Part IV's question whether an artificial configuration can perform research, and it bears mainly on the level of value and governance, although on accounts on which social organization enters the grounding or holding of knowledge, certification enters the conditions of knowledge too. Certification in this sense also differs from the checks of Part III. Part III asked what results bear on and how firmly a passed check settles it. Certification is a relation that further involves judging whether support so assessed suffices for others to rely on it, without checking it again, for a particular use, and to act on it (Chapter 11, Sections 1 and 4).

Chapter 9 first set out an example, a system that generates a hundred instances to pursue different lines, has other instances check and reproduce their outputs, and redirects computing resources among the lines in response to those checks, and set up three claims its builders might make: that it allocates well, that others have reason to rely on its certified results, and that it may decide on others' behalf (Chapter 9, Section 1). It then derived the three relations from where terms of the working characterization meet conditions outside it. Allocation arises where the term "new to *S*" meets the conditions that means are finite and researchers many, and refers to the distribution of attention, effort, and resources among what is unknown to *S*. Criticism and certification arise where the term epistemic achievement, and knowledge in particular, meets the condition that members cannot themselves gather evidence for most results. Authority arises where *S* does not speak with one voice and consequences reach beyond *S*, and asks, where decisions are made, who may decide, and, where no one decides as a matter of policy, who answers for the process and may change it (Chapter 9, Section 2.1). These are analytic relations, not institutions; institutions such as funding agencies and peer review are among their possible realizations. None is a condition of being research. Chapter 9 also put forward the hypothesis that the division of these relations among separate parties may have supplied goods that no single party was designed to supply: error detection across boundaries, the standing of dissent, boundaries at which explanation is demanded, and parties who answer. Just as a method's performance evaluated alone can fail to show its performance within a community, the goodness of a research system cannot be read off the goodness of individual researchers. And on positions that value the formation of researchers, among others, two organizations of research that deliver the same knowledge to the same *S* need not be equal in value, depending on who takes part in the process and who changes through it (Chapter 9, Sections 1 and 3).

Chapter 9 therefore applied the same sorting as Chapter 5, Section 6.5 to what the goods that the division of relations may have supplied depended on: presuppositions about humans as realizers, presuppositions about the present organization of science, conditions of inquiry, and relations with parties outside the configuration, including *S* (Chapter 9, Section 6). The first two can collapse if the realizers or the organization change, but the problems they addressed may then disappear with them or remain in another form. The conditions of inquiry are not removed, but a configuration can meet them within itself. Relations with outside parties, where they include those parties' participation or approval, cannot be completed from one side. Where a given presupposition falls depends on the account adopted. Where a good depends on relations with outside parties, then insofar as that holds on the account adopted, a claim that it can be placed inside requires evidence about those relations, not inference from capability alone. Artificial configurations also bring conditions of their own: that differences and correlations of error among researchers are settled by how they are made, that there is a provider who supplies the very configuration that performs research, and that members can be multiplied at will.

What, in the first place, is allocation (Chapter 10)? Two problems overlap in it: the search problem of finding what is where, and the problem of aims, which lines are worth pursuing; even a complete solution of the former leaves the latter. The problem of allocation arises only where not every line can be pursued cheaply, safely, and in a way that can be redone, and what a good spread looks like varies with the aims presupposed (Chapter 10, Section 1). Using Kitcher's thought experiment of a philosopher monarch, who has both the control to move researchers at will and the eye for the objective merits of theories, to separate the two powers, integration can approach the former but does not supply the latter. A claim to allocate well can be tested against arrangements available to the configuration, not against the distribution the philosopher monarch would choose (Chapter 10, Sections 2 and 7). And even when criteria of importance anticipated in advance are removed, choices of what can be reached and in what order to investigate remain, so there is no search entirely free of value in a broad sense. If, as on Kitcher's account, allocation feeds back into importance, those choices also constrain later judgments of importance (Chapter 10, Sections 2 and 6). Allocation that assigns directly, without giving researchers motives such as the pursuit of credit, removes dependence on motives but not dependence on evaluative signals (Chapter 10, Section 3).

For diversity too, what it is for was in question (Chapter 10, Section 4). In the models of Zollman and others, some transient diversity of belief is needed. The advantages of the particular mechanisms that produce it are fragile, and whether the diversity worth keeping is limited to the transient kind is also disputed; but if the structure of the models applies to artificial configurations, the dangers of integration concentrate in domains where evidence is slow and costly, such as the clinic, the field, and rare instruments, which are where artificial systems' producing more data leaves least room for compensation. Since the ease of a task can itself be a product of how tasks are divided, then if the same evaluative signal returns to the next allocation, success on tasks easy to judge can justify the next allocation, and the range of research can narrow from within through success itself. Lining up models of different origin does not guarantee diversity: within the range of language models studied, errors were correlated even across providers, and two studies of generated hypotheses extend the same caution to the proposing of hypotheses. What centralization may trade away is less diversity itself than the independence of error about the standards that decide convergence, since being able to generate different representations is distinct from being able to judge the survival of lines on different evidence. Further, the epistemic work of dissent, keeping disfavored lines alive with supporters, can be done inside, by protected budgets, rules for reopening, and divided powers, but the standing to dissent remains a relation with outside parties, except where parts hold it on the account adopted.

Human allocation supplied certain functions without anyone deciding that it should (Chapter 10, Section 5): keeping disfavored lines alive, opening lines of different kinds, judging convergence, ending the dominance of established participants and replacing those who pursue lines, and plural signals. What lay behind them, however, differs from case to case. Arrangements supporting long-term research were set up deliberately, but their community-wide effects were not intended; turnover as researchers retired depended on lifespans. Conversely, the case of Project 523, in which the direction of research was steered from the center, suggests that deciding to open lines of different kinds and to judge convergence is one thing, and making the judgments these require is another. In artificial configurations these functions may become parameters someone sets, may be supplied by processes that no one, even inside the configuration, sets as a matter of policy, or may be lost unnoticed; and when lost, they may leave not even a record. The distributed exploration of researchers led by curiosity, too, can be read as one heuristic implicitly adopted as a method of allocation, and under artificial configurations its adoption may be harder to keep implicit.

Then there is the question of whose unknowns are pursued (Chapter 10, Section 6). The epistemic role of knowing what matters to members of *S* is distinct from the role of deciding whose questions to put at the root and how to weight them, and of answering for orderings no one decided; competence in the former does not settle the latter. Applying Kitcher's account of significance, which does not discuss artificial configurations, in research for a human society a configuration inherits the periphery of significance reflected in the records, and a feedback in which tractability moves attention can also occur. On standpoint theory, a configuration starts from where those who produced the records stood, and lacks what the records have excluded. In light of the debate between Varsavsky and Klimovsky and Moro Simpson, the truth of results, the significance of questions, and the timing of arriving at answers are distinct dimensions, and applying this, a configuration can be thought capable of producing true results while inheriting standards from elsewhere and delaying arrival. Chapter 10 put these together into a conditional chain running from success on tasks easy to judge, through movement toward what is tractable and a change in how significance appears, to correlated decisions to terminate lines through an error in a shared signal. The links have precedents in accounts of human research and in models of it, but in artificial configurations three points can

differ: the first conditions are more easily met, the work of breaking the chain cannot be presupposed, and, where a configuration's own record is the main record of the research, narrowing is less likely to show in it. Even so, this is not an inevitable course; it can be broken at the corresponding links (Chapter 10, Section 7).

For criticism and certification, the question was whether criticism and certification for *S* can be organized inside an artificial configuration, and within it, what, in the first place, peer review has been doing (Chapter 11). Treating peer review as one historical realization of its functions, Chapter 11 divided those functions into six: screening, allocating space and funds, detecting errors, improving exposition, assessing novelty and importance, and conferring recognition; and it named, as what the same stage can be thought to have produced along with them, the visibility of evaluation to *S*, a place where members of *S* gain experience of evaluating, and the leaving of nameable judgments in a form that can be objected to (Chapter 11, Section 7). Since an artificial configuration can generate evaluators rather than recruit them, place production and evaluation in one configuration and put evaluation into the middle of formation, and couple evaluators and producers over time, the very moment of "having passed peer review" may disappear. The question therefore becomes not whether to keep peer review but which combination of functions meets what which account requires. Chapter 11 also separated the routes by which a candidate or its grounds, if wrong, could be shown to be wrong, that is, defeat relations, from the relation in which others are given reason to rely on a result for a particular use, that is, certification, and asked how much of the former can be secured inside and why the latter remains a relation with *S* (Chapter 11, Section 1).

The answer divides by account. In human research, the properties that accounts of criticism each track, such as the independence of checks, there being someone who answers, the sharing of valuations, and the visibility of evaluation, were bound together by humans as realizers and by organization, so the accounts' verdicts can be thought to have tended to coincide. Artificial configurations can set these separately, so the verdicts divide, and a builder's claim that results that have passed internal checks are certified for *S* cannot be assessed unless it shows which account it answers to. On some accounts, improving internal criticism can even weaken certification for *S* (Chapter 11, Sections 1 and 9). Applying Longino's account, which sees criticism as a condition of a community's objectivity, adding critics who share assumptions can strengthen the shared assumptions while they remain invisible. On the minimal form of non-reductionism about justified reliance on testimony, if there is no route by which findings reach the one who relies, the condition that there be no defeaters is met whether or not there is error, so strengthening internal checks leaves the reliant party's justification unchanged. A check that is less accurate but open to *S* can therefore give more grounds for reliance than one that is more accurate but closed to *S*. On a position that combines the error-detection view with reliabilism, however, improving internal criticism improves certification, and this comparison does not hold under a pure reliabilism that asks only about the reliability of checks (Chapter 11, Section 9.2).

Independence, too, is not settled by number or organizational separation (Chapter 11, Sections 2 and 3). Extending to checks in general Goldman's argument about agreement among sources, a check is independent to the extent that it adds information for finding errors beyond what existing checks supply, and being in a separate organization is neither necessary nor sufficient. Setting out as a thought experiment a republic of a million instances made from a single base, on Longino's account it is at most objective relative to its own perspective, and adding members adds no criticism that cannot be made under the shared organization. If *S* shares that organization, the gap between objectivity relative to the republic and objectivity relative to *S* narrows although criticism has not improved. What adds evidence is not number but difference, but number works as influence when verdicts are aggregated, so in a majority vote among ninety copies and ten different critics, whoever sets the proportion of copies can sway the verdict. Differences can also be built in, but the process that chooses the differences can itself become a common source.

Values also bear on certification (Chapter 11, Section 4). Certification is the work of turning graded support into a standing on which others may act without checking again, and, if the argument from inductive risk is accepted, it involves judging which errors to weigh how heavily. That weighting enters not only at a single threshold but at each modeling choice, and shapes in advance the range of findings taken as established. Applying the arguments Elliott surveys, when those weights reflect the settings of those who make and provide the configuration, a mere change of settings can change that range without anyone recalling having chosen again. And in research, since rejected candidates usually remain unpursued and results new to *S* often have no other source to be checked against, errors of weighting are unlikely to come to the surface. As for reliance, applying Goldberg's reported account, once certification comes to be treated as a value an instrument displays, *S*

can lose the form of questioning researchers and come to rely on the epistemic work of designers and makers; Koskinen disputes whether trust in those designers and makers can support reliance on essentially opaque AI (Chapter 11, Sections 5.1 and 5.3). Execution records, built to have a defined scope, be recorded on an unalterable base, and be cheaply rerun, weaken the presupposition that only researchers hold the records of data and process, but the problems remain, in changed form, in three places: the maintenance of records, the boundaries of records, and the capacity to verify within them (Chapter 11, Section 5). There can also be a science in which certified explanations accumulate while no one in *S* understands the phenomena (Chapter 11, Section 5.6).

Time must also be taken into account (Chapter 11, Section 8). When evaluators and producers are coupled, the independence of checks can decay, without anyone deceiving anyone, merely because checks are used for selection. What may help is less prohibiting rewriting than keeping checks that are not used for selection. Where standards move with the coupling, agreement with current standards is no evidence that the standards have not moved. Detecting the movement of standards against earlier reference points may be possible inside too, but whether to regard that movement as improvement or replacement, and whether to accept it, is for *S* to decide insofar as they are *S*'s standards. Corrections, too, do not reach where there is no route reaching outside the record. So, insofar as time is taken into account, the answer "everything inside" may not be available. What the movement of standards and correction require is not being human but a relation with *S*. And when *S* includes human members, setting out as a thought experiment the limit at which those members become laypeople, what can be lost first is not access to truth but substantive participation in deciding aims and the capacities that support it (Chapter 11, Section 6).

For authority, the question was what it is, in the first place, for someone to be entitled to decide (Chapter 12). Chapter 12 called authority the standing to decide and revise the conditions of allocation and of criticism and certification, that is, the standing, in relation to those subject to decisions, to propose, decide, object, refuse, and revise, and held that it is neither competence, nor formal assignment, nor effective power. Working well does not by itself show entitlement to decide. Chapter 12 therefore separated a first question, what the right ordering and weighting are, from a second question, who may impose and change them on others. The first can, depending on the account of value adopted, be answered inside; the second is answered by accounts of legitimate authority (Chapter 12, Section 1). Decisions, moreover, include those concerning whether a claim or result is credible enough for a particular use, agenda and allocation, operation and risk, who counts as a member, critic, or expert and whose testimony is heard, memory and correction, and who may revise these aims, standards, and procedures; no ground has the same force for all of them (Chapter 12, Section 2). Applying Chang's account, on which options that cannot be ranked either way yet are on a par can remain even when all comparative facts have been found, increasing knowledge need not remove the question of authority, and can isolate a question of authority that remains where knowledge runs out (Chapter 12, Section 1.2).

The answer divides with the ground of authority (Chapter 12, Section 3). Except for pure epistocracy, competence alone does not yield the standing to impose aims on others. On the result-based ground, not competence itself but superiority confirmed by evaluation that does not presuppose that competence can, within that range, give the artificial configuration itself standing. On Raz's service conception, applying its independence thesis, however, matters in which it is more important that members of *S* choose by their own understanding are excluded, and the choice of aims is a candidate. If the self-government of the research community is the ground, authority can be placed inside not with a central allocator but among parts that take up lines on their own initiative, and only for open problem choice within a field insofar as those initiatives are independent of one another. If actual authorization is the ground, what a configuration can have is effective power of decision within the authorized range, open to objection. If consent is the ground, it is the range those who consent have authorized, which on some readings can include giving up the capacity to revise; whether that extends to those who come later is not settled, and on voluntarist accounts it does not. Carrying Friederich's argument, that systems aligned with intentions concentrate power in their operators, over to research and to Pettit's republican conception of freedom from domination, a configuration perfectly aligned with its principals' intentions can become a means for those principals to redistribute *S*'s inquiry without answering to objection, so perfect alignment can make domination possible. And if these verdicts apply as they stand, then as competence rises the dispute may come to concern not competence but which grounds to adopt, and the reasons for keeping human standing may become at once more central and harder to meet. On the result-based ground, if differences in ease of verification roughly follow differences in kinds of decision, authority enters more readily for questions easier to verify, and if success there widens the range of authority, the direction of research can be

biased toward them (Chapter 12, Section 8). From these verdicts, Chapter 12, Section 8.4 drew what research looks like once authority moves inside: the same configuration yields, ground by ground, a different bias in the questions pursued, a different way in which *S* comes to know, and a different place for those who correct.

Standing can remain yet become merely nominal (Chapter 12, Section 4). If Longino's account is right, deference about content can bring in aims as well through background assumptions, and a configuration whose direction is in question can stand structurally in the position of educating the preferences of those who decide its direction. Where a human society keeps standing while effective decision-making passes to the configuration, the case Chapter 12, Section 4 considers, for that standing to be more than nominal the configuration must track the parties' reasons and its behavior be traceable to humans with appropriate understanding, outcomes must actually be changeable, and there must be routes by which preferences are formed other than through that configuration alone, though on accounts that locate legitimacy in a hypothetical procedure or a single forum the last would need another ground. As for revisability, revision from within may become frequent while the means of objecting and stopping from outside become scarce. Contesting a result in many cases takes research itself, and showing that a terminated line still holds promise often requires pursuing it; in research, then, whether one can object depends also on the means of doing research (Chapter 12, Section 6). And the capacity to revise can be lost without anyone's delegation, through the configuration's own ordinary research revisions. The principle that it is fine because it can be corrected later is weakest where it is most needed, and a decision that removes the capacity to revise cannot draw legitimacy from that principle (Chapter 12, Section 7).

If both *A* and *S* are artificial, the relation of authority does not disappear but becomes a relation among members (Chapter 12, Section 5). Who makes the members is a decision of authority that precedes any aggregation that counts viewpoints, and *S* itself may make it repeatedly. Representativeness does not justify the formation of members under an *S* that can satisfy it by itself. If making members with shared ideals increases their concern for one another's interests, applying Hume's argument and Rawls's remark on an association of saints suggests that disputes about justice in Rawls's sense may become less likely; if it also aligns their views of questions and evidence, being a society with little conflict and being a research community that is unlikely to share errors may pull against each other (Chapter 12, Section 5). Setting out as a thought experiment the limit in which no other party exists, what disappears is the question of authority, who may decide and revise; what remains are that errors can be shown, that effort is divided without knowing, and whether revisions of standards are improvements or replacements. The latter can be read as questions that arise from research being an attempt to seek what is not yet known.

From Part V as a whole it comes into view that becoming able to revise the ways of research quickly, and the thinning of the means to check, object to, and stop those revisions from outside, can occur as two sides of the same change (Chapter 12, Section 9). The accounts examined give different reasons for reference points that are not settled solely by the evaluated party. If they are read as sharing one condition, which none of them states in these terms, what they protect is that content bearing on a verdict comes from outside the evaluated party and cannot be reset by it at the time of evaluation. These authors do not discuss artificial configurations, but on that reading a configuration that generates its own critics, alternatives, and peers could meet their norms as written, in good faith, while the reasons behind them no longer applied; its generating them does not by itself establish that they perform the required work. On Longino's account, the risk is that critics reproduce shared assumptions, not that their having been generated by itself invalidates their criticism. Human research did not meet this condition fully either; where it was met, it was in the sense that, because production was dispersed, each party's content came from other parties, and that dispersion could help checks escape assumptions shared by any one party. And in light of Zollman's argument, where the failure to be prevented is not a property of the generating side, generated safeguards may suffice. Depending on the account, possible supports include critics met outside, peers the configuration did not make, positions recorded with dates before the results they will judge, the parallel running of earlier ways of working, objections that impose burdens the configuration cannot revoke, and keeping such reference points plural. These are fixed relative to the evaluated party at the time of evaluation, not against revision, so they do not conflict with revisability. The question that finally remains is whether it continues to be research for *S*. What could move inside was mainly the supply of work, with the result-based ground as the exception. The possibilities and limits are summarized, with their grounds, in Table 12.5. Separate organizations sometimes supported these goods, but their separation was neither the only source nor a uniform requirement.

Naming what was settled without being decided

Sections 3 to 6 of this chapter traced, chapter by chapter, what the chapters found. Looked at again across chapters, several of them, each through its own literature, arrive at the same form. When humans do research, who holds the belief, how far the evidence extends, and where competence sits were largely fixed by being one person, and that whoever produced a result is also the one who understands it could in many cases be presupposed (Chapter 2, Section 9; Chapter 3, Section 9). Moreover, doing research and allocating, criticizing, certifying, and deciding about it were in many cases carried by separate parties, so that the number of critics roughly tracked their independence, and when a certification erred there was someone who answered for it (Chapter 11, Section 9). Much of this was settled without having to be decided, not because anyone decided it but because those who did research were people, and were divided among many people. When machines do research, these things need not be fixed in advance. The boundary of the one who does research is drawn relative to a claim (Chapter 7, Section 2), the unit of a knowledge claim can also be drawn in several ways depending on how instances, external records, and tools are connected, and the one who draws it may be the builder or the system itself as it reorganizes itself (Chapter 2, Section 9; Section 3 of this chapter). Functions that were supplied without anyone deciding can come to be set by someone as rules, to be shaped by processes within the configuration that no one sets, or to be lost without notice (Chapter 9, Section 2.3; Chapter 10, Section 5). What, when humans do research, was settled without being decided becomes, when machines do research, something that can be decided.

Two things follow from this form, in which what was settled without being decided becomes something that can be decided. One is that the claim that a machine is doing research becomes a bundle of claims, each of which can hold separately (Section 1 of this chapter). The other is that the verdicts of the accounts examined diverge. In human research the properties that the accounts each track were tied together by human realizers and by the organization of research, so that in many ordinary cases their verdicts plausibly tended to coincide; an artificial configuration makes those properties separately settable. The bundle comes apart (Chapter 11, Section 9; Section 6 of this chapter). This circumstance, which Chapter 11 found for criticism and certification, can also be read as one reason why the book has in many places not settled on one answer but traced consequences account by account. A claim of certification, and a claim that one may decide, cannot be checked unless they say which account they are made against (Chapter 11, Section 9; Chapter 12, Section 8.2), and a knowledge claim cannot be checked unless it says what unit it is a claim about (Chapter 2, Section 9). A bundle that has come apart can, however, also be rebundled, with different bases, records that reach those who rely, nameable respondents, and the like (Chapter 11, Section 9). That evaluation is visible to *S* and that it can be objected to were tied to the stage of evaluation, so they can be rebuilt by adding outward-facing arrangements such as records, public venues, and routes that receive objections. By contrast, someone who answers when an error is made was tied not to the stage but to persons, so adding such arrangements does not produce one; on assurance views, someone who intentionally vouches for the result is needed, and on versions that add a right of complaint, someone who answers for error as well. The capacity to evaluate, too, was kept up by members of *S* actually evaluating, so it does not come back when arrangements merely return results; for human members, occasions to take part in evaluation are needed; for artificial members, what is asked is whether their remaking and updating keep the capacity (Chapter 11, Sections 6 and 9.2). What is asked is which to rebundle, and by what.

This form carries particular weight in research. Because what research seeks is not yet known to *S*, its results often cannot simply be checked against what *S* wanted, and often there is no other source against which to compare them (Chapter 9, Section 4; Chapter 12, Section 6). To object to a line that has been terminated, one often has no way but to pursue it (Chapter 12, Section 6). Of course, having no answer is not the same as having no standard. There are standards that can be applied without knowing the answer, such as checking a formally stated theorem or a prediction registered in advance (Chapter 9, Section 4). Even so, whether such a standard captures the original question does not follow from having passed it. In research, moreover, what counts as success, what serves as the standard, and at times what *S* wants can be revised in the course of inquiry (Chapter 6, Section 5; Chapter 12, Section 9). On Longino's account, criticism needs perspectives able to expose shared assumptions; on Galison's account of mechanical objectivity, the investigator's wishes must be kept from shaping the result; on Wilholt's account, coordination rests on value judgments genuinely made (Chapter 12, Section 9). If these accounts are read as sharing one condition, that content bearing on the verdict comes from outside the evaluated party and cannot be reset by it at the time of evaluation (Chapter 12, Section 9), then where results cannot be checked against an answer, this independence can be read as standing in for part of what the answer would have done. Human research did not

guarantee it, and where it met the condition it did so largely because production was dispersed among many parties, so that the content came, from each party's point of view, from other parties (Chapter 12, Section 9; Section 6 of this chapter). Part of what lay in no one's having decided can also be read as no one's having been able to align them. What could not be aligned becomes, when machines do research, something that can be. The reference points of evaluation themselves, critics, standards, *S*, the range of members, endorsement, become variables that the evaluated party can produce. This form, seen in Part V, and the form seen in Parts II to IV, in which what being one person fixed ceases to be fixed, can be read as coming from the same place: that those who did research were people, and were divided among many people.

What was settled without being decided also comes into view from the side of words. In ways of speaking such as "AI cannot ask questions", "AI has no curiosity", or "independently reviewed", many of the words used as if to state properties, once examined in the chapters, came into view as relations whose content is not fixed until one names for whom and about what. Whether a machine asked is asked as which of the kinds of work bearing on questions are carried by which unit, directed toward a point unsettled for whom (Chapter 4, Section 9). Independence, on the error-detection view, is how much information about error a check adds beyond the checks already in place (Chapter 11, Section 2). Novelty, opacity, and autonomy likewise involved for whom, about what, and in comparison with whom (Chapter 1, Section 6; Chapter 3, Section 5; Chapter 7, Sections 4.1 and 4.2). Curiosity is a clear example. Gopnik points out that when one individual first explores and later exploits, the interests of the learner and of the one who exploits what was learned automatically coincide. Applied to research, the one who explores and the one for whom the research is done are *A* and *S*, and their interests do not automatically coincide (Chapter 4, Section 7). On this reading, what is asked about curiosity is, beyond whether there is curiosity, whose unknowns the exploration is directed toward. In human research too, what is unknown to *A* and what is unknown to *S* can come apart, but some of these places were tacitly filled by being one person, or by *S*'s being a human society, so the words plausibly looked as if they referred to properties. Naming these places fixes what the machine is being claimed to have achieved.

In an artificial configuration, what fills these places, that is, what is taken as the reference, becomes something that can be chosen. Hughes and colleagues defined a system's open-endedness relative to an observer and called the choice of observer "a free parameter of great importance" (Chapter 4, Section 7). In research, however, its value has to be defended outside the definition. An arbitrary collection assembled after the fact cannot be *S* (Chapter 1, Section 6), and by naming an artificial society that shares a lineage with the artificial configuration, a claim of certification becomes easier to satisfy without the certification improving (Chapter 11, Section 9.2). Choosing the reference is itself a claim that needs defending. What needs naming, moreover, includes *S* at which time (Sections 2 and 5 of this chapter).

One of the things that being divided among many people supplied was difference among critics and explorers. When difference becomes something made rather than something that arises, what is cheap and what is scarce also change. Producing interrogative sentences and candidates, adding criticism, running trials, and the like can become cheap for machines. For criticism and trials in particular, what adds as evidence is not number but difference (Chapter 11, Sections 2 and 3), and while what becomes cheap is criticism that works on the producer's records and trials repeatable within the same arrangement, what can remain scarce is checking that passes through routes the producer does not provide and has error routes of its own (Chapter 11, Section 7). That adding instances of one base does not by itself add the differences on which checking, observation, and criticism draw was reached in each of Parts II to V, each time through different accounts (Sections 3 to 6 of this chapter). On the side of the research process too, even if searching an already characterized space widely becomes cheap, making candidates available by building representations and instruments, and turning trials into diagnoses by tying them to earlier results, remain separate work (Section 4 of this chapter).

When machines do research, therefore, the epistemic weight of quantity itself becomes a question. Abundant criticism can coexist with a scarcity of independent criticism and conceal it (Chapter 11, Section 7). Over time, independent checking may behave like a resource that is depleted by use. A check used for selection can come to tell less about producers the more it is used (Chapter 11, Sections 8 and 9.2), and where the producer, changed by criticism, also comes to pass where a check is insensitive, the same check need not add the same information the next time (Chapter 11, Section 9.3). What is scarce can change as inquiry proceeds. That raising autonomy can make outside checks all the more needed can be read as the same form (Section 5 of this chapter). In each case, increasing what is cheap does not increase what is scarce. What has to be increased for research to advance depends on what is scarce.

Counting what can remain scarce is, from the other side, asking what human research supplied it with. The introduction said that considering artificial realizations also tests assumptions inherited from human research (Introduction, Section 1). One of the things that came into view in the chapters is the functions that human research supplied without anyone deciding them (Section 6 of this chapter). The single act of a named person reading a manuscript before publication, for instance, plausibly produced at once the detection of error, the visibility of evaluation to *S*, a judgment that objections can be directed at, someone who answers, and occasions for *S*'s members to gain experience of evaluating (Chapter 11, Sections 7 and 9.2). Where results could not be checked against an answer, this single act can be read as having stood in for part of what the answer would have done. The practice of peer review may have carried at once the judging of results and the keeping of a society that can judge them (Chapter 3, Section 8; Chapter 11, Section 6). If so, the capacity of the *S* that receives research results was also kept up through participation in research.

Human research, however, did not fully meet the condition that content bearing on verdicts comes from outside the evaluated party (Chapter 12, Section 9). On Longino's account, assumptions shared by all can survive internal criticism and stay hidden (Chapter 11, Section 3). What comes into view is not that human research was well made, but that functions and the arrangements that carried them were different things. Applying Campbell's descriptive epistemology suggests that removing premises about humans changes the vehicle that carries knowledge rather than removing it, and that what is then asked is which conditions of upkeep orient inquiry in its place (Chapter 9, Section 6). There are also conditions of inquiry that are not removed whoever does the research, such as, where a matter is unknown to *A* as well as to *S*, having to divide effort without knowing in advance which line will succeed, and checking's needing error routes of its own. An artificial configuration can meet them within, but needs to show that it does (Chapter 9, Section 6; Chapter 12, Section 5). Considering machines doing research is also trying to tell what in research rests on premises specific to humans and what does not. It is in this sense that the introduction, borrowing Weisberg and Reisman's idea of robustness, took the book's question to be close to asking which results are preserved when the representational framework changes (Introduction, Section 4).

Thinking of functions and of the arrangements that carry them separately also turns attention to the possibility of realizing those functions otherwise. That what was settled without being decided becomes something that can be decided connects to the case that Introduction, Section 5 kept within the comparison, that the ways of research may themselves be made. The book has not argued that such ways will be made, nor that making them would be good. Even so, as Sections 4 to 6 of this chapter traced, everything from representations, instruments, records, and standards to critics, members, and the relations around research came into view as revisable within inquiry, and further by the configuration itself. What comes into view, conditional on the accounts examined, is that what opens there is not only freedom but also a question: against what it can be asked whether a revised way is better (Chapter 12, Section 5). Chapter 12, Section 9 held that, since the frictions that Nielsen and Qiu named as reasons why what is collectively held in the social processes of science hardly changes (Nielsen and Qiu 2022) plausibly overlap in part with the separations that could supply content bearing on verdicts from outside, the two can be removed together when the relations move inside a configuration. That is why becoming able to revise the ways of research quickly and the thinning of means to check them from outside can occur as two faces of the same change (Section 6 of this chapter).

What comes into view here is that the freedom to revise and the relations under which a revision can be shown wrong are not the same thing. Inquiry that revises its standards also proceeds under support and resistance from things other than the standards being revised (Chapter 6, Section 5). The critics met outside, positions recorded before the results, and the like, listed in Section 6 of this chapter, can be read not as means of stopping revision but as keeping a state in which it can be asked whether a revision is better. They are fixed relative to the evaluated party at the time of evaluation, and do not conflict with revisability (Section 6 of this chapter). Since the alternatives in research do not form a list closed in advance (Chapter 12, Section 9), the state in which it can be asked whether a revision is better is not something settled once. And as *A* and *S* come to overlap, the question whether an artificial configuration that revises its ways of research is still doing research for *S* can shift to the question "for which society?" (Chapter 11, Section 9.3). Where ways of research can be made, naming *S* each time is a precondition of that question (Section 2 of this chapter).

What has been seen also brings into view what each term of Chapter 1's sentence carries. Relative to a society *S*, research is an attempt to bring about epistemic achievements new to *S*. "Attempt", as Chapter 1, Section 3 set out, worked as an

organized direction rather than as the intention of all who take part, and when machines do research it made one ask where that direction is organized. "New to *S*" took on the burden of naming which *S* at which time (Section 2 of this chapter). Asking about "epistemic achievement" widened into asking whether it has been realized somewhere, whether it has entered *S*'s epistemic situation, and whether *S* keeps the capacity to receive and evaluate it (Section 3 of this chapter). Whether artificial states can be beliefs, whether artificial beings can understand and bear responsibility, and whether they can have moral status remain open. If research is valued also for what it is to those who conduct it, then when the conduct of research passes to artificial systems, the question also remains whether what is to be kept is human participation in inquiry, the significance of that participation, or a standing in final decision (Chapter 9, Section 3). The book has traced how answers change the questions about machine research rather than settled them.

What research seeks is what *S* does not yet know. *S* therefore has no answer to check against. While humans have done research, part of what stood in for that answer lay, unnamed, in the one who does research being one person and in research being divided among many people. When machines do research, these can come to be set by someone, shaped by the machine itself, or lost without notice. All of what has been thought here is on the way, but to ask what it is for a machine to do research is also to make it possible to name, and to ask about, what research has rested on.

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